

ADVANCED JEE PHYSICS

for both Main & Advanced levels of JEE

OPTICS & MODERN PHYSICS



Rahul Sardana

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M.Sc (Hons). Physics



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ADVANCED JEE PHYSICS, Optics & Modern Physics
Rahul Sardana

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Preface

text overview, goals & focus

In the past few years, the IIT-JEE has evolved itself as an examination designed to check out true scientific skills. The examination pattern wants us to see those little details which, others fail to see. Those details which tell us how much in depth we should know to describe as much as possible. Keeping the present day scenario in mind, this book is written for students, to allow them not only to learn the tools that "Optics & Modern Physics" provides but also to see why they work so nicely in explaining the beauty of ideas behind the subject. The central goal of this text is to help the students develop a thorough understanding of the principles of "Optics & Modern Physics". This book stresses on building a rock solid technical knowledge based on firm foundation of the fundamental principles than on a large collection of formulae. The primary philosophy of this book is to act as a guide who creates a careful, detailed groundwork for strong conceptual understanding and development of problem solving skills like mature and experienced physicists.

features of book

theory with illustrations

"Optics & Modern Physics" are important topics, and in this book I have tried to make these topics lively, clear and precise to the greatest levels. I have generally seen students not stressing on the theoretical details. They always feel that doing more numerical problems will solve their purpose. But let me tell you here, that numerical problems are just the special cases of the theoretical concepts. The entire Physics is based on a simple program "IF $\rightarrow$ THEN $\rightarrow$ ELSE". Try to follow this and see how you get to your ultimate goal i.e., IIT-JEE. So, keeping this in mind, the entire theory part of all the chapters has been kept elaborative, simple to understand with supportive Illustrations at all the places. DO NOT TRY TO ATTEMPT ILLUSTRATIONS WITHOUT GOING THROUGH THE THEORY.

conceptual notes, remarks, words of advice, misconception removal

Throughout the text, the Conceptual Notes and Remarks are highlighted which focus on the principal ideas and concepts that a student must take care of. Places where students commonly develop a misconception have been supported by Misconception Removal, highlighted in grey and supported by Words of Advice. Throughout my teaching career of 18 years I have always found my students getting benefitted from these Conceptual Notes, Remarks, Words of Advice, Misconception Removals. All these are actually used to provide warnings to the students about common errors and ways to avoid them.

problem solving techniques

These techniques, highlighted in grey, always ensure that the students become capable of solving a variety of problems in an easy way. Wherever necessary, the text is supplemented with them for having a thorough understanding to the application processes.

in chapter exercises (ice) : topic wise

After you study the theory and apply it to the Illustrations, its time you practice something on your own and that too topic wise. For this purpose I have created In Chapter Exercises (ICE) (except for Mathematical Physics). Each ICE has the name of the topic(s) clearly mentioned on it. Please note that ICE are based on simple, single concept classification technique. They are fully solved, so that if you come across only problem, then you just refer to the solutions.

solved problems

After you have gone through the entire Theory (with Illustrations) and all the supplements (ICE, Conceptual Notes, Remarks, Words of Advice, Misconception Removal, Problem Solving Techniques), its high time to do problems that are a true mix of concepts studied. This section has problems that involve **multiple concept usage** so that your brain is exposed to the ultimate throttle required to extract the best from you at IIT-JEE.

practice exercise sets (fully solved)

Now comes the time when you are very much ready to do the practice as per the IIT-JEE pattern. This section contains all the variety of questions that have been asked in the IIT-JEE. In this section you will come across the following variety of questions.

single correct choice type (SCCT)

Each question, in this section, has four choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

multiple correct choice type (MCCT)

Each question, in this section, has four choices (A), (B), (C) and (D), out of which **ONE OR MORE** is/are correct.

reasoning based questions / assertion-reason type (ART)

This section contains Reasoning type questions, also called Assertion-Reason type question, each having four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. Each question contains **STATEMENT 1** (Assertion) and **STATEMENT 2** (Reason). You have to mark your answer as

Bubble (A) If both statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1.

Bubble (B) If both statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1.

Bubble (C) If STATEMENT 1 is TRUE and STATEMENT 2 is FALSE.

Bubble (D) If STATEMENT 1 is FALSE but STATEMENT 2 is TRUE.

linked comprehension type (LCT) / paragraph type

This section contains Linked Comprehension Type Questions or Paragraph based Questions. Each set consists of a Paragraph followed by questions. Each question has four choices (A), (B), (C) and (D), out of which only one is correct. (For the sake of competitiveness there may be a few questions that may have more than one correct options).

matrix match type (MMT) / column matching

Each question in this section contains statements given in two columns, which have to be matched. The statements in **COLUMN-I** are labelled A, B, C and D, while the statements in **COLUMN-II** are labelled p, q, r, s (and t). Any given statement in **COLUMN-I** can have correct matching with **ONE OR MORE** statement(s) in **COLUMN-II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following examples:

If the correct matches are $A \rightarrow p, s$ and t ; $B \rightarrow q$ and r ; $C \rightarrow p$ and q ; and $D \rightarrow s$ and t ; then the correct darkening of bubbles will look like the following:

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

integer answer type questions (IATQ) / numerical type questions

In this section the answer to each of the question is a four digit integer, ranging from 0 to 9999. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answer to question number X (say) is 6092, then the correct darkening of bubbles will look like the following :

X.

0	●	0	0
1	1	1	1
2	2	2	●
3	3	3	3
4	4	4	4
5	5	5	5
●	6	6	6
7	7	7	7
8	8	8	8
9	9	●	9

answers & solutions

Each chapter contains answers followed by solutions to the problems. The solutions are exhaustive with complete methods and reasons which will help you a lot to understand a particular concept. Short cuts are also included (wherever necessary) for enhancing your problem solving skills.

This book, I hope, will nourish you with the concepts involved such that you get a great rank at IIT-JEE. To conclude, I apologise in advance for the errors (if any) that may have inadvertently crept in the text. I would be grateful to the readers who bring errors of any kind to my attention. I truly welcome all comments, critiques and suggestions at sardanaeducation@gmail.com. PRAYING TO GOD FOR YOUR SUCCESS AT IIT-JEE, GOD BLESS YOU!

The Author

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CHAPTER 1**Ray Optics**

(a) Reflection at Plane & Curved Surfaces	1.1
(b) Refraction at Plane Surfaces	1.27
(c) Refraction at Curved Surfaces	1.55
Solved Practice Problems	1.96

Practice Exercise Sets

➤ Single Correct Choice Type Questions	1.108
➤ Multiple Correct Choice Type Questions	1.135
➤ Reasoning Based Questions (Assertion Reason Type)	1.141
➤ Linked Comprehension Type Questions (Paragraph Type)	1.144
➤ Matrix Match Type Questions (Column Matching Type)	1.154
➤ Integer Answer Type Questions	1.160

Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	1.163
--	-------

Solutions to In Chapter Exercises (ICE)	1.169
---	-------

Solutions to Practice Exercise Sets	1.193
---	-------

CHAPTER 2**Wave Optics**

Wave Optics	2.1
Solved Practice Problems	2.41

Practice Exercise Sets

➤ Single Correct Choice Type Questions	2.52
➤ Multiple Correct Choice Type Questions	2.66
➤ Reasoning Based Questions (Assertion Reason Type)	2.69
➤ Linked Comprehension Type Questions (Paragraph Type)	2.71
➤ Matrix Match Type Questions (Column Matching Type)	2.77
➤ Integer Answer Type Questions	2.80

Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	2.83
--	------

Solutions to In Chapter Exercises (ICE)	2.86
---	------

Solutions to Practice Exercise Sets	2.92
---	------

CHAPTER 3**Dual Nature of Radiation & Matter**

Dual Nature of Radiation & Matter	3.1
Solved Practice Problems	3.15

Practice Exercise Sets

➤ Single Correct Choice Type Questions	3.21
➤ Multiple Correct Choice Type Questions	3.33
➤ Reasoning Based Questions (Assertion Reason Type)	3.36
➤ Linked Comprehension Type Questions (Paragraph Type)	3.38

➤ Matrix Match Type Questions (Column Matching Type)	3.40
--	------

➤ Integer Answer Type Questions	3.43
---------------------------------------	------

Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	3.45
--	------

Solutions to In Chapter Exercises (ICE)	3.48
---	------

Solutions to Practice Exercise Sets	3.52
---	------

CHAPTER 4**Atomic & Nuclear Physics**

Atomic Physics	4.1
----------------------	-----

Nuclear Physics	4.21
-----------------------	------

Solved Practice Problems	4.48
--------------------------------	------

Practice Exercise Sets

➤ Single Correct Choice Type Questions	4.58
--	------

➤ Multiple Correct Choice Type Questions	4.73
--	------

➤ Reasoning Based Questions (Assertion Reason Type)	4.78
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➤ Linked Comprehension Type Questions (Paragraph Type)	4.81
--	------

➤ Matrix Match Type Questions (Column Matching Type)	4.89
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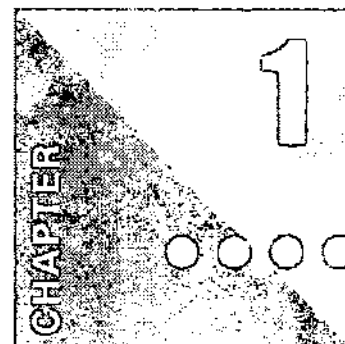
➤ Integer Answer Type Questions	4.92
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Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	4.94
--	------

Solutions to In Chapter Exercises (ICE)	4.99
---	------

Solutions to Practice Exercise Sets	4.106
---	-------

Ray Optics



Contents

(a) REFLECTION AT PLANE & CURVED SURFACES	1.1
(b) REFRACTION AT PLANE SURFACES	1.27
(c) REFRACTION AT CURVED SURFACES	1.55
Solved Practice Problems	1.96
Practice Exercise Sets	
➤ Single Correct Choice Type Questions	1.108
➤ Multiple Correct Choice Type Questions	1.135
➤ Reasoning Based Questions (Assertion Reason Type)	1.141
➤ Linked Comprehension Type Questions (Paragraph Type)	1.144
➤ Matrix Match Type Questions (Column Matching Type)	1.154
➤ Integer Answer Type Questions	1.160
Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	1.163
Solutions to In Chapter Exercises (ICE)	1.169
Solutions to Practice Exercise Sets	1.193

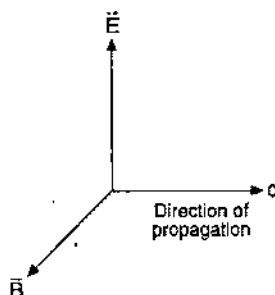
Ray Optics

(a) Reflection at Plane & Curved Surfaces

NATURE OF LIGHT : An Introduction

Light is a form of energy that makes object visible to our eyes or light is the form of energy that produces in us the sensation of sight. In Seventeenth century Newton and Descartes believed that light consisted of a stream of particles, called corpuscles. Huygens proposed wave theory of light and proposed that light is a disturbance in a medium called Ether. This theory could explain the phenomena of interference, diffraction, etc. Thomas Young, through his double slit experiment, measured the wavelength of light.

Maxwell suggested the electromagnetic theory of light. According to this theory, light consists of electric and magnetic fields, in mutually perpendicular directions, and both are perpendicular to the direction of propagation. Heinrich Hertz produced in the laboratory the electromagnetic waves of short wavelengths. He showed that these electromagnetic waves possessed all the properties of light waves.



Light travels in vacuum with a velocity given by

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ ms}^{-1}$$

where μ_0 and ϵ_0 are the permeability and permittivity of free space (vacuum).

The magnitudes of electric and magnetic fields are related to the velocity of light by the relation

$$\frac{E}{B} = c$$

In 1905, Albert Einstein revived the old corpuscular theory using Plank's Quantum Hypothesis and through his photoelectric effect experiment showed that light consists of discrete energy packets, called photons. The energy of each photon is

$$E = hf = \frac{hc}{\lambda}$$

So, in view of these developments, light must be regarded to have a dual nature i.e., it exhibits the characteristics of a particle in some situations and that of a wave in other situations. So the question "Is light a particle or a wave?" is purely inappropriate to be asked. At present, it is believed that light has dual nature, i.e., it has both the characters, wave-like and particle-like.

OPTICS : An Introduction

Optics is the study of the properties of light, its propagation through different media and its effects. In most of the situations, the light encounters objects of size much larger than its wavelength. We can assume that light travels in straight lines called rays, disregarding its wave nature. This allows us to formulate the rules of optics in the language of geometry, as rays of light do not disturb each other on intersection. Such study is called geometrical (or ray) optics. It includes the working of mirrors, lenses, prisms, etc.

When light passes through very narrow slits, or when it passes around very small objects, we have to consider the wave nature of light. This study is called wave (or physical) optics.

DOMAINS OF OPTICS

The study of light can be categorized into three broad domains.

- Geometrical Optics (Ray Optics)
- Physical Optics (Wave Optics)

c) Quantum Optics

Please note that these domains are not strictly disjoint as the transitions between them are continuous and not sharp. However for convenience we consider them as distinct. These domains are distinguished as follows.

A. GEOMETRICAL OPTICS (RAY OPTICS)

This branch involves the study of propagation of light based on the assumption that light travels in fixed straight line as it passes through a uniform medium and its direction is changed when met by a surface of a different medium or if the optical properties of the medium are non uniform either in time or in space. The ray approximation is valid for the wavelength λ very small compared to the size of the obstacle (d) or the size of the opening through which the ray passes.

This approximation $\lambda \ll d$ proves to be very good for the study of mirrors, lenses, prisms and associated optical instruments such as microscope, telescope, cameras etc.

B. PHYSICAL OPTICS (WAVE OPTICS)

This branch involves the study of propagation of light in the form of a wave and it deals with the phenomenon of interference, diffraction, polarization etc. This nature of light has to be taken when the light passes through very narrow slits or when it goes past very small objects. So this branch works effectively when $\lambda \gg d$.

C. QUANTUM OPTICS

This branch involves the study of propagation of light as a stream of particles called as **Photons**. This concept of light behaving as particles called photons is of utmost importance while studying the origin of spectra, photoelectric effect, concept of radiation pressure, Compton effect etc.

FUNDAMENTAL LAWS OF GEOMETRICAL OPTICS

To a first approximation, we can consider the propagation of light disregarding its wave nature and assuming that light propagates in straight lines called **rays**. This allows us to formulate the laws of optics in the language of **geometry**. Thus, the branch of optics where the wave nature of light is neglected is called **geometrical (or ray) optics**.

Geometrical optics is based on five fundamental laws.

1. **Law of Rectilinear Propagation of Light.** It states that light propagates in straight lines in homogenous media.
2. **Law of Independence of Light Rays.** It states that rays do not disturb each other upon intersection.
3. **The Law of Reversibility of Light.** According to this law, if a ray of light, after suffering a number of reflections and refractions, has its path reversed at any instant, then the ray retraces its path back to the source.
4. **The Laws of Reflection.** The Laws of Reflection govern the bouncing back of the incident ray after striking a surface to the medium from which it was coming.

5. **The Laws of Refraction (discussed later).** The Laws of Refraction govern the bending of light when the light goes from one medium to the other (rarer to denser or denser to rarer) medium.

BASIC TERMS & DEFINITIONS**SOURCE**

A body which emits light is called **source**. The source can be a point one or an extended one. A source is of two types.

- a) **Self luminous :** The source which possess light of its own.

EXAMPLE : sun, electric arc, candle etc.

- b) **Non-luminous :** It is a source of light which does not possess light of its own but acts as source of light by reflecting the light received by it.

EXAMPLE : moon, objects around us, book etc.

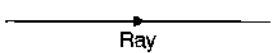
REMARK(S)

Sources are also classified as isotropic and non-isotropic. Isotropic sources give out light uniformly in all directions whereas non-isotropic sources do not give out light uniformly in all directions.

RAY

The straight line path along which the light travels between two points in a homogeneous medium or in a pair of media is called a **Ray**. It is represented by an arrow head on a straight line, the arrow head represents the direction of propagation of light. A ray of light will always follow a path along which the time taken is the minimum.

Ray


REMARK(S)

A single ray cannot be isolated from a source of light.

MEDIUM

Substance through which light propagates or tends to propagate is called a **medium**. It is of following three types.

- a) **Transparent :** It is a medium through which light can be propagated easily.

EXAMPLE : glass, water etc.

- b) **Translucent :** It is a medium through which light is propagated partially.

EXAMPLE : oil paper, ground glass etc.

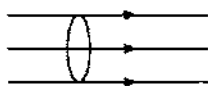
- c) **Opaque :** It is a medium through which light cannot be propagated.

EXAMPLE : wood, iron etc.

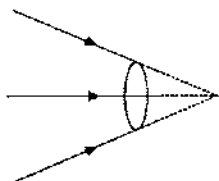
BEAM

A bundle or bunch of rays is called a **beam**. It is of following three types.

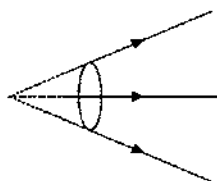
- a) **Parallel beam** : It is a beam in which all the rays constituting the beam move parallel to each other and diameter of beam remains same. A very narrow beam is called a Pencil of Light.



- b) **Convergent beam** : In this case diameter of beam decreases in the direction of ray.



- c) **Divergent beam** : It is a beam in which all the rays meet at a point when produced backward and the diameter of beam goes on increasing as the rays proceed forward.

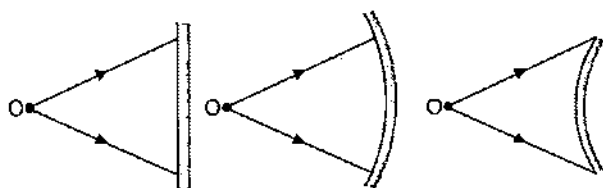


OBJECTS

The object for a mirror can be real or virtual.

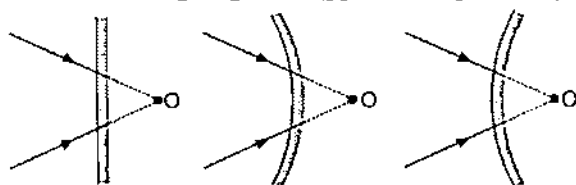
A. REAL OBJECTS

If the rays from a point on an object actually diverge from it and fall on the mirror, the object is said to be real. In simple language the incident rays are diverging and the point of divergence is the position of the real object. The following diagrams support the arguments given.



B. VIRTUAL OBJECTS

If the rays incident on the mirror appear to converge to a point, then this point is said to be virtual point object for the mirror. In simple language the incident rays are converging and the point of convergence is the position of the virtual object. The following diagrams support the arguments given:



CONCEPTUAL NOTE(S)

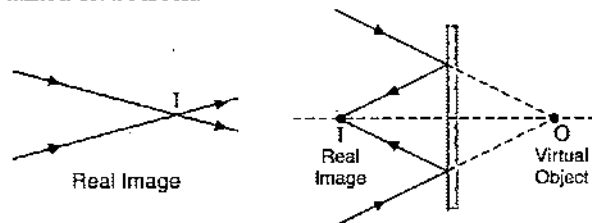
Virtual object cannot be seen by human eye, because for an object or an image to be seen by the eyes, the rays received by the eyes must be diverging.

IMAGES

An **optical image** is a point where reflected or refracted rays of light either intersect or appear to intersect. Thus, the image of an infinite object is actually an assembly of the image points corresponding to various parts or the points of the object. The images formed can again be real or virtual.

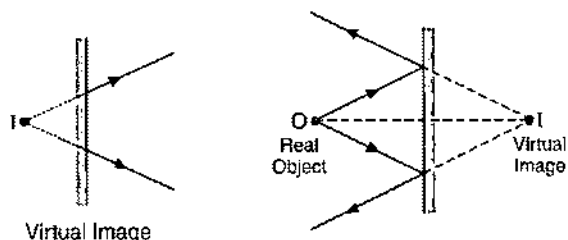
A. REAL IMAGES

If the rays after reflection or refraction actually converge (or meet) at a point then the image is said to be **real** and it can be obtained on a screen.



B. VIRTUAL IMAGES

However, if the rays do not actually converge but appear to diverge from a point (or appear to meet at a point), then the image so formed is said to be **virtual image**. A virtual image cannot be obtained on a screen.



CONCEPTUAL NOTE(S)

- The real images can be obtained on a suitably placed screen, but virtual images cannot be obtained on a screen.
- Human eye cannot distinguish between the real image and the virtual image because in both the cases the rays are diverging.

REFLECTION OF LIGHT

When light strikes the surface on an object, some part of the light or the complete light is sent back into the same medium. This phenomenon is called as reflection. The surface, which reflects light, is called mirror. A mirror could be plane or curved.

CONCEPTUAL NOTE(S)

In reflection, the frequency, speed and wavelength remain unchanged, but a phase change may occur depending on the nature of reflecting surface.

The reflection from a denser medium causes an addition phase change of π or a path change of $\frac{\lambda}{2}$ (by Stoke's Law) while reflection from rarer medium does not cause any phase change.

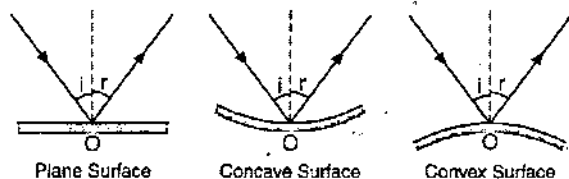
Diffused (irregular) reflection takes place from a rough surface where as Specular (regular) reflection takes place from an extraordinarily smooth surface. However, the Laws of Reflection are applicable for both kinds of surfaces.

LAWS OF REFLECTION

- The incident-ray, the reflected-ray and the normal to the reflecting surface at the point of incidence, all lie in the same plane.
- The angle of reflection is equal to the angle of incidence ($i = r$).

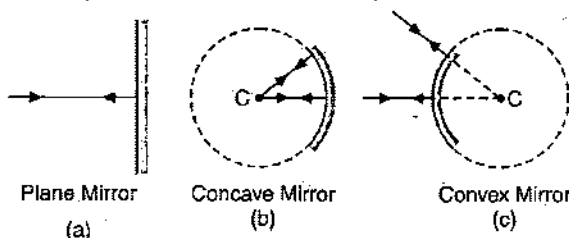
The angle of incidence i is the angle made by the incident ray with the normal.

The angle of reflection r is the angle made by the reflected ray with the normal.



SPECIAL CASES

- If $i = 0$, then $r = 0$. It means a ray incident normally on a boundary, after reflection it retraces its path.



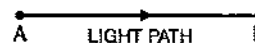
- The angle made by the incident ray with the plane reflecting surface is called glancing angle. Thus, the glancing angle $= 90^\circ - i$.
- For grazing incidence, the incident ray grazes the reflecting surface, so $i \rightarrow \frac{\pi}{2}$ and hence $r \rightarrow \frac{\pi}{2}$ as shown in the figure.



FERMAT'S PRINCIPLE OF LEAST TIME

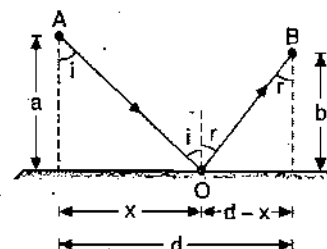
According to this theorem, light will always follow the path taken between any two points by a ray of light is the path along which the time taken is the minimum. This principle is sometimes taken as the definition of a ray of light.

To understand this theorem, let us consider two points A and B in the same medium. Since, we know that between these two points light travels in a straight line, so the time taken by the light to go from A to B must logically be the minimum.



LAWS OF REFLECTION USING FERMAT'S THEOREM

Consider a plane mirror on which light is incident as shown.



Let the incident light start from A, hit the mirror at O and get reflected to point B. Let the points A and B be at perpendicular distances a and b from the mirror and let A and B have a separation d between them as shown in figure. The time taken by the light to go from A to O to B is given by

$$t = t_{A \rightarrow O} + t_{O \rightarrow B}$$

$$\Rightarrow t = \frac{AO}{c} + \frac{OB}{c}$$

$$\Rightarrow t = \frac{1}{c} (\sqrt{a^2 + x^2} + \sqrt{b^2 + (d-x)^2})$$

Now, according to Fermat's Principle, t is MINIMUM, so

$$\frac{dt}{dx} = 0$$

$$\Rightarrow \frac{d}{dx} (\sqrt{a^2 + x^2}) + \frac{d}{dx} (\sqrt{b^2 + (d-x)^2}) = 0$$

$$\Rightarrow \frac{1}{2} \left(\frac{2x}{\sqrt{a^2 + x^2}} \right) + \frac{1}{2} \left(\frac{2(d-x)(-1)}{\sqrt{b^2 + (d-x)^2}} \right) = 0$$

$$\Rightarrow \frac{x}{\sqrt{a^2 + x^2}} = \frac{(d-x)}{\sqrt{b^2 + (d-x)^2}}$$

From the figure, we observe that

$$\frac{x}{\sqrt{a^2 + x^2}} = \sin i \text{ and } \frac{(d-x)}{\sqrt{b^2 + (d-x)^2}} = \sin r$$

$$\Rightarrow \sin i = \sin r$$

$$\Rightarrow i = r.$$

[The Law of Reflection]

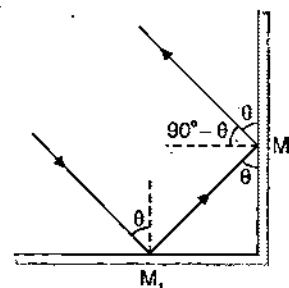
PROBLEM SOLVING TRICK(S)

- Basic Problems in Optics :** Most of the problems asked in optics expect us to find the position and nature of the final image formed by certain optical systems for a given object. The optical system may be just a mirror, or a lens or a

combination of several reflecting and refracting surfaces.

- b) **Basic Strategy for Solving the Problems :** To handle these kinds of problems, first of all, we identify the sequence in which the reflection and refraction are taking place. The several events of reflection or refraction can be named as Event 1, Event 2 and so on following the sequence in which they occur.

Now, the image of Event 1 would be object for Event 2, image of Event 2 will be object of Event 3 and so on. This way one can proceed to find the final image.

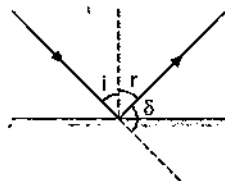


CONCEPTUAL NOTE(S)

The same is found to hold good for three plane mirrors arranged mutually perpendicular to each other thus forming the corner of a cube such that the light incident on this arrangement suffers one reflection from each of the mirrors so as to emerge out anti-parallel to the incident light. This arrangement of three mutually perpendicular plane mirrors forming the corner of a cube is called the **CORNER REFLECTOR**.

ANGLE OF DEVIATION (δ)

Deviation (δ) is defined as the angle between the initial direction of the incident ray and the final direction of the reflected ray or the emergent ray.

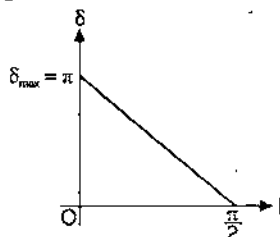


Deviation produced in Reflection is $\delta = 180^\circ - (i + r)$

Since $r = i$

$$\Rightarrow \delta = 180^\circ - 2i$$

The variation of deviation (δ) with the angle of incidence (i) is shown in figure.



PROBLEM SOLVING TRICK(S)

- The deviation is maximum for normal incidence i.e., when $i = 0$ then, $\delta = \delta_{\max} = 180^\circ$.
- The deviation is minimum for grazing incidence i.e., when $i \rightarrow \frac{\pi}{2}$, then $\delta = \delta_{\min} = 0^\circ$.
- While dealing with the case of multiple reflections suffered by a ray, the net deviation suffered by the incident ray is the algebraic sum of deviation due to each single reflection. So,

$$\delta_{\text{total}} = \sum \delta_{\text{individual reflection}}$$

DO NOT FORGET TO TAKE INTO ACCOUNT THE SENSE OF ROTATION WHILE SUMMING UP THE DEVIATIONS DUE TO SINGLE REFLECTION.

TWO IDENTICAL PERPENDICULAR PLANE MIRRORS

If two plane mirrors are inclined to each other at 90° , the emergent ray is always antiparallel to the incident ray if it suffers one reflection from each (as shown in figure) whatever be the angle of incidence.

Illustration 1

Two plane mirrors are inclined to each other at an angle θ . A ray of light is reflected first at one mirror and then at the other. Find the total deviation suffered by the ray.

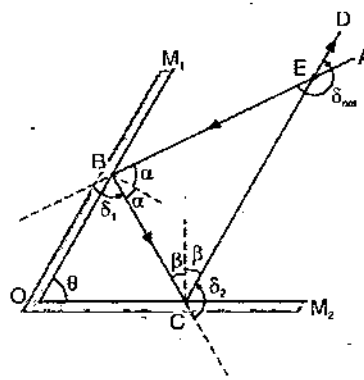
Solution

α be the angle of incidence for mirror M_1

β be the angle of incidence for mirror M_2

δ_1 be the deviation due to mirror M_1 and

δ_2 be the deviation due to mirror M_2



From figure, we observe

$$\delta_1 = \pi - 2\alpha, \quad \delta_2 = \pi - 2\beta$$

Also ray is rotated in same sense i.e., anticlockwise, so

$$\delta_{\text{net}} = \delta = \text{Total deviation} = \delta_1 + \delta_2$$

$$\Rightarrow \delta = 2\pi - 2(\alpha + \beta)$$

Now in $\triangle OBC$, $\angle OBC + \angle BCO + \angle COB = 180^\circ$

$$\Rightarrow (90^\circ - \alpha) + (90^\circ - \beta) + \theta = 180^\circ$$

$$\Rightarrow \alpha + \beta = \theta$$

$$\Rightarrow \delta = 2\pi - 2\theta = 360^\circ - 2\theta$$

Alternative Method :

$$\delta = \angle BEC + \angle CEA + \angle AED$$

Now, $\angle BEC = \angle AED$ (vertically opposite angle)

$$\Rightarrow \angle BEC = 180^\circ - 2(\alpha + \beta)$$

$$\Rightarrow \angle BEC = 180^\circ - 2\theta \quad \{\because \theta = \alpha + \beta\}$$

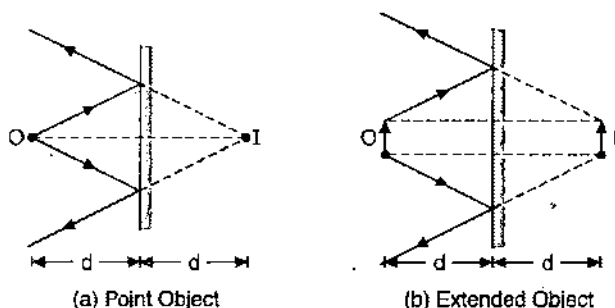
$$\text{Also, } \angle CEA = 2\alpha + 2\beta \Rightarrow \angle CEA = 2(\alpha + \beta) = 2\theta$$

$$\Rightarrow \delta = (180^\circ - 2\theta) + 2\theta + (180^\circ - 2\theta)$$

$$\Rightarrow \delta = 360^\circ - 2\theta$$

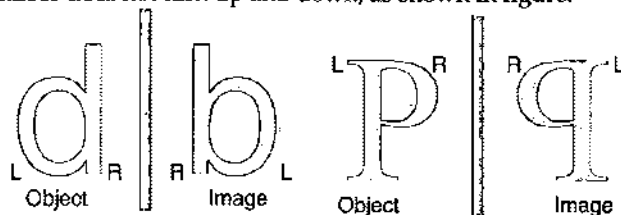
REFLECTION FROM A PLANE SURFACE OR PLANE MIRROR

When a real object is placed in front of a plane mirror, the image is always erect, virtual and of same size as the object. It is at same distance behind the mirror as the object is in front of it.



LATERAL INVERSION

The image formed by a plane mirror suffers **lateral-inversion**. That is, in the image the left is turned to the right and vice-versa with respect to object. However, the plane mirror does not turn up and down, as shown in figure.



Actually, the plane mirror reverses forward and back in three-dimensions (and not left into right). If we keep a right-handed coordinate system in front of a plane mirror, only the z-axis is reversed. So, a plane mirror changes right-handed co-ordinate system (or screw) to left-handed.

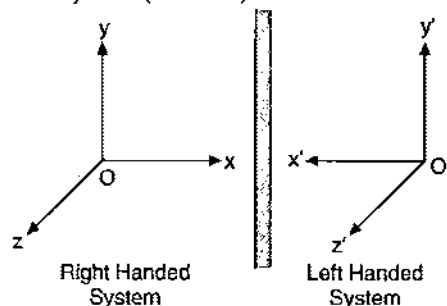
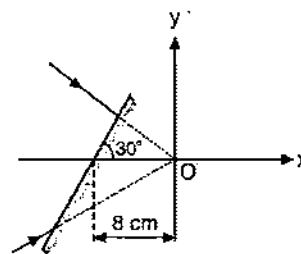


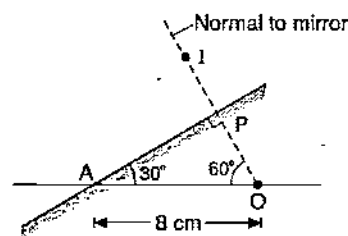
Illustration 2

Find the co-ordinates of the location of the image formed for an object kept at origin as shown in figure.



Solution

The first thing we observe is that the object is virtual, because the ray of light is converging on plane mirror. Also, the co-ordinates of object are $(0, 0, 0)$ and the image co-ordinates are the reflection of object coordinates in the mirror as shown in figure.



The image lies on normal of mirror at I. From $\triangle AOP$, we have

$$\sin(30^\circ) = \frac{PO}{8}$$

$$\Rightarrow PO = 4 \text{ cm}$$

$$\Rightarrow OI = 2(PO) = 8 \text{ cm}$$

So co-ordinates of I are

$$x = -8 \cos(60^\circ) = -4 \text{ cm},$$

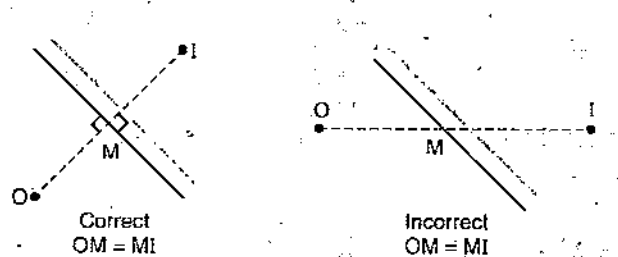
$$y = 8 \sin(60^\circ) = 4\sqrt{3} \text{ cm and}$$

$$z = 0$$

So, the co-ordinates of image are $(-4, 4\sqrt{3}, 0)$

PROBLEM SOLVING TRICK(S)

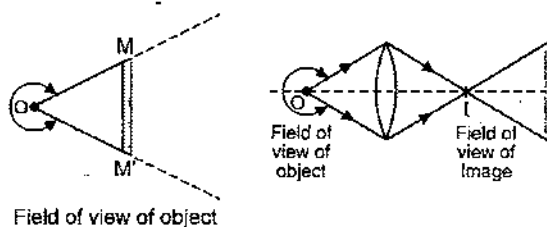
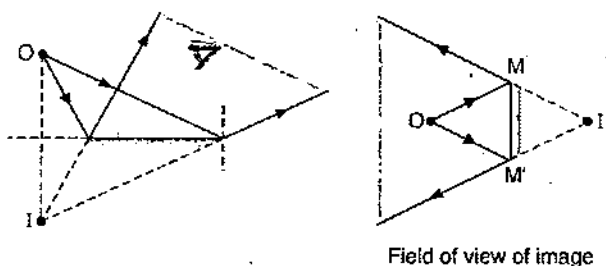
For finding the location of an image of a point object placed in front of a plane mirror, we must see the perpendicular distance of the object from the mirror.



FIELD OF VIEW OF AN OBJECT

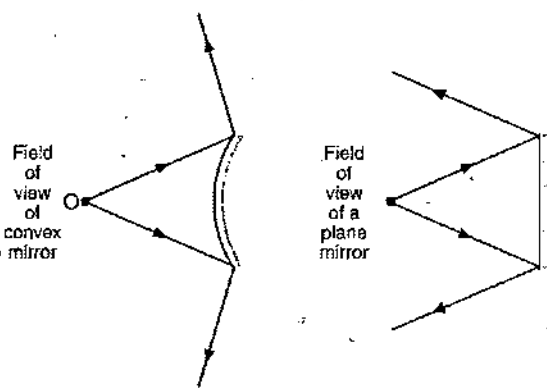
Suppose a point object O is placed in front of a mirror, then a question arises in mind whether this mirror will form the image of this object or not. The answer is yes, it will form. A mirror, irrespective of its size, forms the images of all objects lying in front of it. But every object has its own field of view for a given mirror.

Field of view is the region where diverging rays from object or image are present. If our eyes are present in field of view then only we can see the object or an image as the case may be. Field of view of image is decided by rays which get reflected or refracted from the extremities or the extreme ends of the mirror or a lens and depends on the location of the object in front of mirror or lens.



CONCEPTUAL NOTE(S)

It has been observed that a convex mirror gives a wider field of view than a plane mirror. Therefore, the convex mirrors are used as rear view mirrors in vehicles. Though they make the estimation of distances more difficult but still they are preferred because for a large movement of the object vehicle there is only a small movement of the image.

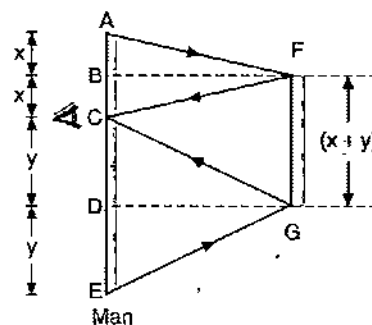


MINIMUM SIZE OF A PLANE MIRROR TO SEE A COMPLETE IMAGE

CASE-I:

To find the minimum size of mirror to see a full image we use the fact that light rays from extreme parts of object should reach eye after reflection from mirror. Let us consider following two situations

- The minimum size of mirror to see one's full height is $\frac{H}{2}$ where H is the height of man. To see full image mirror is positioned in such a way so that rays from head and foot reach eye after reflection from mirror, as shown in the figure.



- A ray starting from head (A) after reflecting from upper end of the mirror (F) reaches the eye at C. Similarly the ray starting from the foot (E) after reflecting from the lower end (G) also reaches the eye at C. In similar triangles ABF and BFC

$$AB = BC = x \text{ (say)}$$

Similarly in triangles CDG and DGE, we have

$$CD = DE = y \text{ (say)}$$

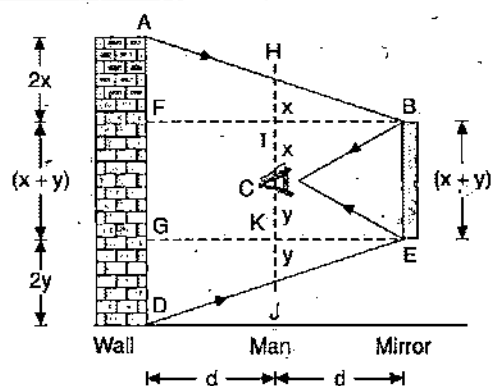
Now, we observe that height of the man is $2(x+y)$ and that the length of mirror is $(x+y)$, i.e., the length of the mirror is half the height of the man. Please note that the mirror can be placed anywhere between the centre line BF (of AC) and DG (of CE).

CONCEPTUAL NOTE(S)

- In order to see full image of the man, the mirror is positioned such that the lower edge of mirror is at height half the eye level from the ground.
- Minimum size is independent of the distance between man and mirror.

CASE-II:

The minimum length of the mirror required to see the full image of a wall behind the man who is standing at the middle of the mirror and the wall is $\frac{H}{3}$, where H is the height of wall. The ray diagram for this situation is shown in figure.



In triangles HBI and IBC let $HI = IC = x$. Now, in triangles HBI and ABF , we have

$$\frac{AF}{HI} = \frac{FB}{BI}$$

$$\Rightarrow \frac{AF}{x} = \frac{2d}{d}$$

$$\Rightarrow AF = 2x$$

Similarly if, $CK = KJ = y$, then $DG = 2y$. Now, we observe that height of the wall is $3(x+y)$ while that of the mirror is $(x+y)$.

NUMBER OF IMAGES IN INCLINED MIRRORS

Let θ be the angle between two plane mirrors and 'n' be the number of images formed:

$$\text{Then } n = \begin{cases} \frac{360}{\theta}, & \text{if } \frac{360}{\theta} \text{ is odd} \\ \left(\frac{360}{\theta} - 1\right), & \text{if } \frac{360}{\theta} \text{ is even} \end{cases}$$

Further when $\frac{360}{\theta}$ is odd, then

$$n = \begin{cases} \left(\frac{360}{\theta} - 1\right), & \text{if object lies symmetrically on the angle bisector of two mirrors} \\ \frac{360}{\theta}, & \text{if object lies unsymmetrically} \end{cases}$$

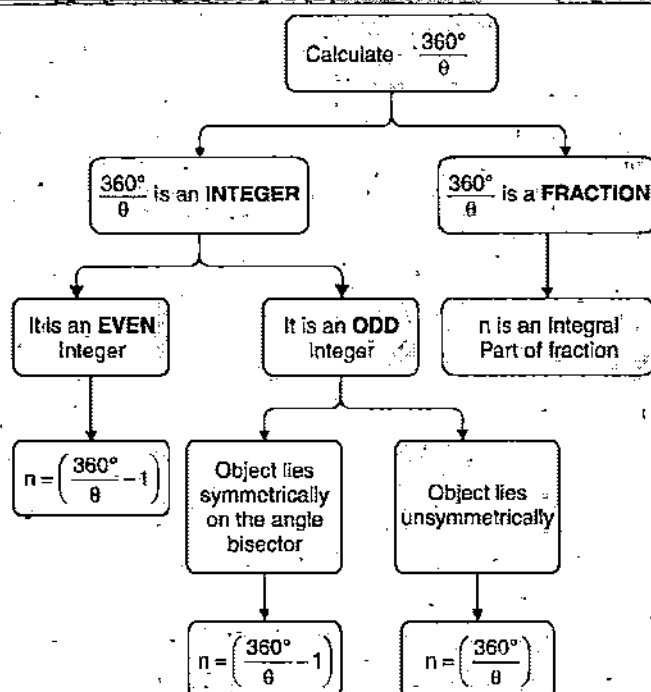
Further if $\frac{360}{\theta}$ is a fraction, then the number of images

formed will be integral part of the fraction e.g. if $\frac{360}{\theta}$ is 4.8, then $n=4$. Following diagram shows the process to calculate n.

Net deviation produced by two plane mirrors inclined at an angle θ is

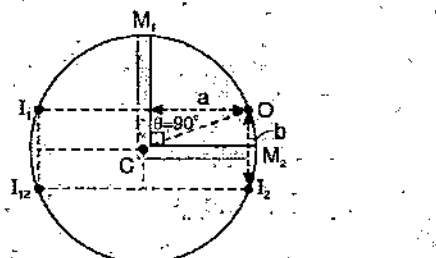
$$\delta = 360^\circ - 2\theta$$

Clearly δ is independent of the angle of incidence of the ray of light.



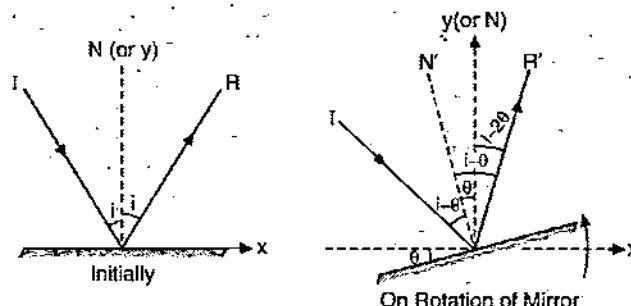
CONCEPTUAL NOTE(S)

- If an object is placed between two parallel mirrors ($\theta = 0^\circ$), the number of images formed will be infinite.
- All the images lie on a circle with radius equal to the distance between the object O and the point of intersection of the mirrors C.
- The number of images formed may be different from the number of images seen (which depends on the position of the observer).



ROTATION OF A PLANE MIRROR

When a mirror is rotated by an angle θ (say anticlockwise), keeping the incident ray fixed, then the reflected ray rotates by 2θ along the same sense, i.e., anticlockwise.



Let I be the incident ray, N the normal and R the reflected ray, then on rotation, I remains as it is, N and R shift to N' and R' .

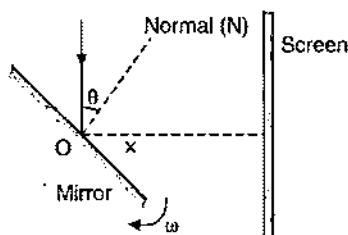
From the two figures we can observe that the reflected ray earlier made an angle i with y -axis while after rotating the mirror it makes the angle $(i - 2\theta)$. So, we conclude that the reflected ray has been rotated by an angle 2θ .

CONCEPTUAL NOTE(S)

If a plane mirror rotates with angular velocity ω , then the reflected ray rotates with angular velocity 2ω (excluding rotation of mirror with normal as the axis).

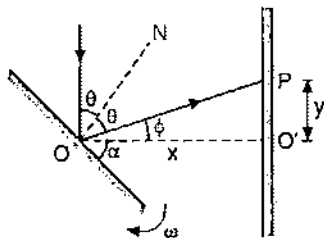
Illustration 3

A plane mirror hinged at O is free to rotate in a vertical plane. The point O is at a distance x from a long screen placed in front of the mirror as shown in figure. A laser beam of light incident vertically downward is reflected by the mirror at O so that a bright spot is formed at the screen. At the instant shown, the angle of incidence is θ and the mirror is rotating clockwise with constant angular velocity ω . Find the speed of the spot at this instant.



Solution

Let P be the bright spot, shown on the screen. Let the distance of point P from O' be y at this instant shown in figure. Then according to the problem we need to calculate $\frac{dy}{dt}$.



From the figure

$$\theta + \theta + \phi = 90^\circ \quad \dots(1)$$

$$\theta + \phi + \alpha = 90^\circ \quad \dots(2)$$

$$\Rightarrow \alpha = 0$$

$$\Rightarrow \phi + 2\theta = 90^\circ$$

$$\Rightarrow \phi + 2\alpha = 90^\circ$$

$$\Rightarrow \frac{d\phi}{dt} + 2\frac{d\alpha}{dt} = 0$$

$$\Rightarrow \frac{d\phi}{dt} = -2\frac{d\alpha}{dt}$$

So, the angular speed of the reflected ray is double the angular speed of the mirror.

Since, $y = x \tan \phi$

$$\Rightarrow \frac{dy}{dt} = x \sec^2 \phi \frac{d\phi}{dt}$$

$$\text{Since } \left| \frac{d\phi}{dt} \right| = 2\omega$$

$$\Rightarrow \left| \frac{dy}{dt} \right| = (x \sec^2 \phi)(2\omega)$$

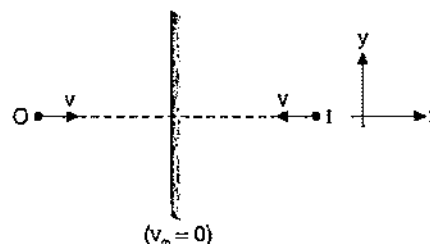
$$\text{So, the speed of the spot is } \left| \frac{dy}{dt} \right| = 2x\omega \sec^2 \phi$$

VELOCITY OF IMAGE IN A PLANE MIRROR

To understand and interpret the moving images of moving objects in front of plane mirror, we must understand the following cases.

CASE I :

Object moving along the normal to the plane mirror which is at rest. All velocities measured w.r.t. ground frame.



Velocity of object with respect to mirror is

$$\vec{v}_{Om} = v\hat{i}$$

Velocity of image with respect to mirror is

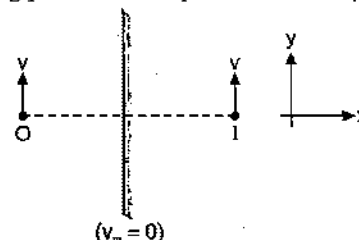
$$\vec{v}_{Im} = -v\hat{i}$$

Velocity of object with respect to image is

$$\vec{v}_{OI} = \vec{v}_O - \vec{v}_I = (2v)\hat{i}$$

CASE II :

Object moving parallel to the plane of mirror (at rest)



Velocity of object w.r.t. mirror is

$$\vec{v}_{Om} = v\hat{j}$$

Velocity of image w.r.t. mirror is

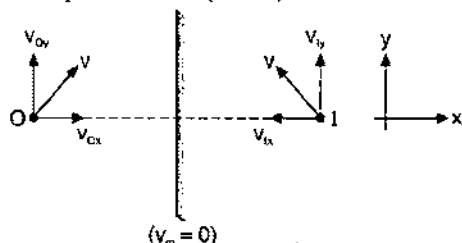
$$\vec{v}_{im} = \vec{v}\hat{j}$$

Velocity of object w.r.t. image is

$$\vec{v}_{oi} = \vec{0}$$

CASE III:

Object moving neither along the normal nor along the parallel to the plane mirror (at rest).



Clearly, we observe this case to be a combination of Case I and Case II. So, here

$$(v_{oi})_x = 2v_{ox} \text{ and } (v_{oi})_y = 0$$

PROBLEM SOLVING TRICK(S)

While solving problems that involve the calculation of image of an object w.r.t. any observer, then

Step 1 : Firstly, calculate the velocity of image w.r.t. mirror keeping in mind that

$$(\vec{v}_{im})_{\text{along mirror}} = (\vec{v}_{om})_{\text{along mirror}}$$

$$\Rightarrow (\vec{v}_{im})_{\parallel} = (\vec{v}_{om})_{\parallel}$$

Since, both the object and the image approach the mirror with equal and opposite speed, so we have

$$(\vec{v}_{im})_{\text{normal to mirror}} = -(\vec{v}_{om})_{\text{normal to mirror}}$$

$$\Rightarrow (\vec{v}_{im})_{\perp} = -(\vec{v}_{om})_{\perp}$$

$$\Rightarrow \vec{v}_i - \vec{v}_m = -(\vec{v}_o - \vec{v}_m)$$

$$\Rightarrow \vec{v}_i = 2\vec{v}_m - \vec{v}_o$$

Step 2 : Then the velocity of image w.r.t. mirror is

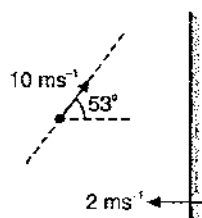
$$\vec{v}_{im} = (\vec{v}_{im})_{\parallel} + (\vec{v}_{im})_{\perp}$$

However, velocity of image w.r.t. any other observer, say A is then given by

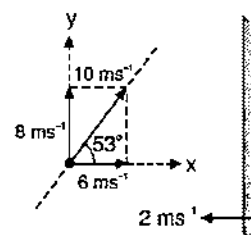
$$\vec{v}_{ia} = \vec{v}_i - \vec{v}_a$$

Illustration 4

Find the velocity of image of a moving particle shown in figure.



Solution



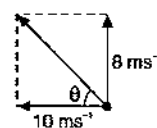
The component of velocity of image perpendicular to mirror is

$$\vec{V}_i = 2\vec{V}_m - \vec{V}_o$$

$$\Rightarrow (\vec{V}_i)_{\perp} = 2(-2) - (6) = -10 \text{ ms}^{-1}$$

For component of velocity of image parallel to the mirror

$$(\vec{V}_i)_{\parallel} = 8 \text{ ms}^{-1}$$

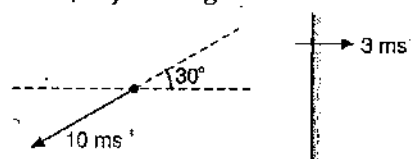


$$\text{Velocity of image } (V_i) = \sqrt{(V_i)_{\perp}^2 + (V_i)_{\parallel}^2}$$

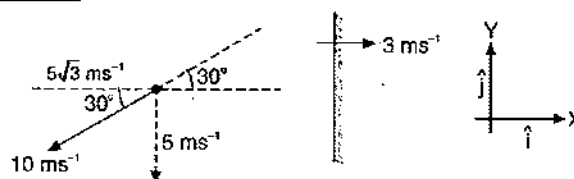
$$\Rightarrow V_i = \sqrt{100 + 64} = \sqrt{164} \text{ ms}^{-1} \text{ and } \theta = \tan^{-1}\left(\frac{4}{5}\right)$$

Illustration 5

A point object is moving with a speed of 10 ms^{-1} in front of a mirror moving with a speed of 3 ms^{-1} as shown in figure. Find the velocity of image of the object with respect to mirror, object and ground.



Solution



$$\text{Velocity of object, } \vec{v}_o = (-5\sqrt{3}\hat{i} - 5\hat{j}) \text{ ms}^{-1}$$

$$\text{Velocity of mirror, } \vec{v}_m = 3\hat{i} \text{ ms}^{-1}$$

For component of velocity perpendicular to mirror, we have

$$(\vec{v}_{im})_{\perp} = -(\vec{v}_{om})_{\perp} = -(\vec{v}_o - \vec{v}_m)_{\perp}$$

$$\Rightarrow (\vec{v}_{im})_{\perp} = -(-5\sqrt{3}\hat{i} - 3\hat{i}) = (5\sqrt{3} + 3)\hat{i} \text{ ms}^{-1}$$

For component of velocity parallel to mirror, we have

$$(\vec{v}_{im})_{\parallel} = (\vec{v}_{om})_{\parallel} = \vec{v}_o - \vec{v}_m = -5\hat{j} - 0 = -5\hat{j}$$

Since, $(\vec{v}_{IM}) = (\vec{v}_{IM})_{\perp} + (\vec{v}_{IM})_{\parallel}$

$$\Rightarrow (\vec{v}_{IM}) = (5\sqrt{3} + 3)\hat{i} - 5\hat{j}$$

Also, $\vec{v}_{IM} = \vec{v}_I - \vec{v}_M$

$$\Rightarrow (5\sqrt{3} + 3)\hat{i} - 5\hat{j} = \vec{v}_I - (3\hat{i})$$

$$\Rightarrow (5\sqrt{3} + 3)\hat{i} + 3\hat{i} - 5\hat{j} = \vec{v}_I$$

$$\Rightarrow \vec{v}_I = [(5\sqrt{3} + 6)\hat{i} - 5\hat{j}] \text{ ms}^{-1}$$

Further $\vec{v}_{IO} = \vec{v}_I - \vec{v}_O$

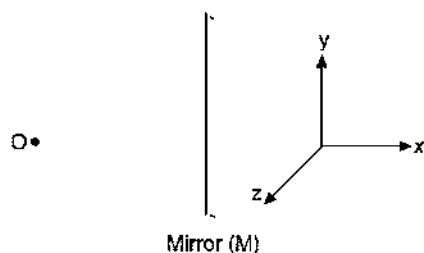
$$\Rightarrow \vec{v}_{IO} = (5\sqrt{3} + 6)\hat{i} - 5\hat{j} - (-5\sqrt{3}\hat{i} - 5\hat{j})$$

$$\Rightarrow \vec{v}_{IO} = (5\sqrt{3} + 6 + 5\sqrt{3})\hat{i} + (5 - 5)\hat{j}$$

$$\Rightarrow \vec{v}_{IO} = (10\sqrt{3} + 6)\hat{i} \text{ ms}^{-1}$$

Illustration 6

A plane mirror in y - z plane moves with a velocity $-3\hat{i}$ as shown in figure. An object O starts moving with a velocity $4\hat{i} + \hat{j} - 4\hat{k}$. Find the velocity of the image.



Solution

Since the mirror is placed in y - z plane, so the y and z components of the velocity of the image remain the same as that of the object. However, **perpendicular to the mirror**, the velocity of approach of object towards the mirror is always equal and opposite to the velocity of approach of the image towards the mirror, so, we have

$$(\vec{v}_{OM})_x = -(\vec{v}_{IM})_x$$

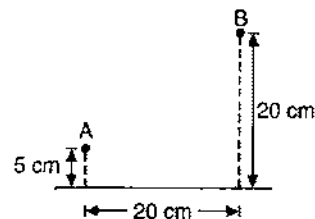
$$\Rightarrow (\vec{v}_O)_x - (\vec{v}_M)_x = -(\vec{v}_I)_x + (\vec{v}_M)_x$$

$$\Rightarrow (\vec{v}_I)_x = 2(\vec{v}_M)_x - (\vec{v}_O)_x$$

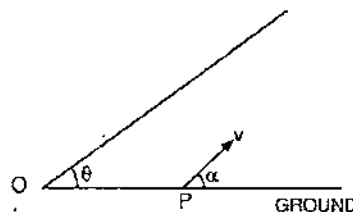
$$\Rightarrow (\vec{v}_I)_x = 2(-3\hat{i}) - 4\hat{i} = -10\hat{i}$$

$$\text{So, } \vec{v}_I = -10\hat{i} + \hat{j} - 4\hat{k}$$

2. A ray of light travels from point A to a point B after being reflected from a plane mirror as shown in figure. From where should it strike the mirror?

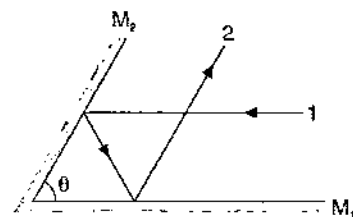


3. A plane mirror is inclined at an angle $\theta = 60^\circ$ with horizontal surface. A particle is projected from point P on the ground (see figure) at $t=0$ with a velocity v at an angle α with horizontal. The image of the particle is observed from the frame of the particle projected. Assuming the particle does not collide the mirror. Find the time when image will come momentarily at rest with respect to particle.

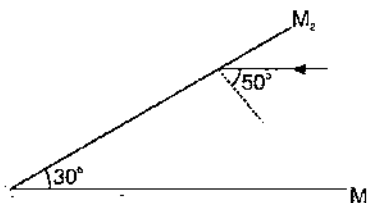


4. Two plane mirrors are inclined to each other such that a ray of light incident on the first mirror and parallel to the second is reflected from the second mirror parallel to the first mirror.
(a) Find the angle between the two mirrors.
(b) Also calculate the total deviation produced in the incident ray due to the two reflections.

5. Two plane mirrors M_1 and M_2 are inclined at angle θ as shown in figure. A ray of light 1, which is parallel to M_1 strikes M_2 and after two reflections, the ray 2 becomes parallel to M_2 . Find the angle θ .



6. Calculate the deviation suffered by an incident ray in the situation shown in figure after it suffers three successive reflections.



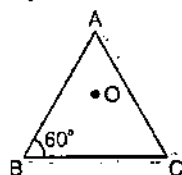
7. Two plane mirrors are placed parallel to each other and 40 cm apart. An object is placed 10 cm from one mirror.

ICE I BASED ON REFLECTION AT PLANE SURFACES (Solutions on page 1.169)

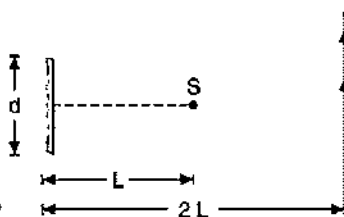
1. A ray of light travelling in the direction $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ is incident on a plane mirror. After reflection, it travels along the direction $\frac{1}{2}(\hat{i} - \sqrt{3}\hat{j})$. Find the angle of incidence.

Find the distance from the object to the respective image for each of the five images that are closest to the object.

8. Find the number of images formed of an object O enclosed by three mirrors AB , BC , AC having equal lengths in situation shown in figure.

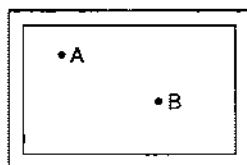


9. A point source of light S , placed at a distance L in front of the centre of a mirror of width d , hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it as shown. Find the greatest distance over which he can see the image of the light source in the mirror.

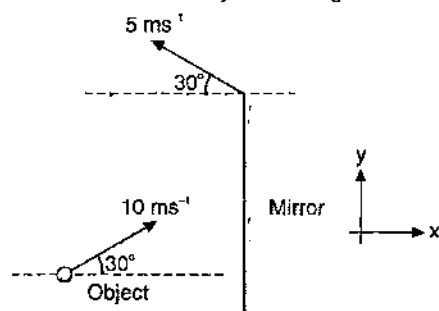


10. Find the smallest size of a looking glass which a man with a face $24 \text{ cm} \times 16 \text{ cm}$ should purchase that will enable him to see his whole face completely, if the
(a) man is one eyed.
(b) man is two eyed.
Given that the separation between his eyes is 8 cm .

11. In what direction should a beam of light is to be sent from point A (shown in figure) contained in a mirror box for it to fall onto point B after being reflected once from each of the four walls. If the points A and B are in one plane perpendicular to the walls of the box (i.e., in the plane of the drawing) then in what direction should the beam be sent from B to A ?



12. The object and the mirror move with velocity shown in figure. Calculate the velocity of the image.



13. A ray of light is incident on an arrangement of two plane mirrors inclined at an angle θ with each other. It suffers two reflections one from each mirror and finally moves in a direction making angle α with the incident ray (α is acute). Find the angle α and show that it is independent of angle of incidence.

14. A ray of light is incident at an angle of 30° with the horizontal. At what angle with horizontal must a plane mirror be placed in its path so that it becomes vertically upwards after reflection?

15. Two plane mirrors are inclined to each other at an angle of 70° . A ray is incident on one mirror at an angle θ . The ray reflected from this mirror falls on the second mirror from where it is reflected parallel to the first mirror. Find the value θ .

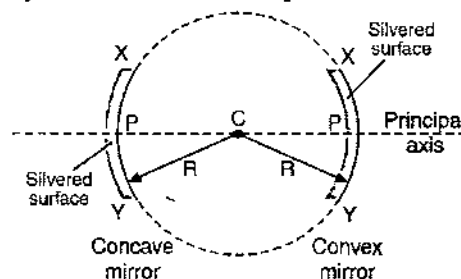
16. A ray of light is incident on a plane mirror along a vector $\hat{i} + \hat{j} - \hat{k}$. The normal at the point of incidence is along $\hat{i} + \hat{j}$. Find a unit vector along the reflected ray.

REFLECTION FROM CURVED SURFACES

A small curved reflecting surface can be considered to be a part of a sphere. Hence, such surfaces are called spherical mirrors. Depending upon the surface silvered, these are of two types—concave and convex, as shown in figure. Some important terms are described below.

- Pole or Vertex** : Centre P of the surface of the mirror.
- Centre of Curvature** : Centre C of the sphere.
- Radius of Curvature** : Radius R of the sphere.
- Principal Axis** : Line PC , joining the pole and the centre.
- Linear Aperture** : Distance XY between the extremities of the mirror surface.

Note that since lenses are also made of spherical surfaces, the above terms also apply to lenses, except that the pole is replaced by a new term called as Optical Centre.



Important Terms and Definitions

- Centre of curvature** : It is the centre of the sphere of which the mirror/lens is a part.
- Radius of curvature** : It is the radius of the sphere of which the mirror/lens is a part.
- Pole** : It is the geometrical centre of the spherical reflecting surface of which the mirror/lens is a part.

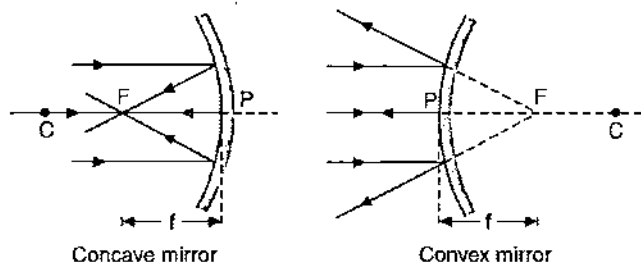
- d) **Principal axis (for a spherical mirror) :** It is the straight line joining the centre of curvature to the pole.
- e) **Focus :** When a narrow beam of rays of light, parallel to the principal axis and close to it, is incident on the surface of a mirror (lens), the reflected (refracted) beam either converges to a point or appears to diverge from a point on the principal axis. This point is called the focus (F).
- f) **Focal length (for a mirror) :** It is the distance between pole and the principal focus (F).
- g) **Real image :** If reflected (or refracted) rays converge to a point (i.e. intersect there), then the point is a real image.
- h) **Virtual image :** If reflected (or refracted) rays appear to diverge from a point, then the point is a virtual image.
- i) **Real object :** If the incident rays diverge from a point, then the point is a real object.
- j) **Virtual object :** If incident rays converge and appear to intersect at a point behind the mirror (or lens), then the point is a virtual object.

PARAXIAL RAYS

Paraxial rays are the rays which are either parallel to the principal axis or make small angles with it i.e., these rays are nearly parallel to the principal axis. Our treatment for the spherical mirrors has been restricted to these rays and due to this we shall be considering the **curved mirrors** that have **smaller aperture**. However, for the sake of convenience, comfort and clarity, we shall be drawing the diagrams of larger size.

FOCUS AND FOCAL LENGTH

When a narrow beam of light, parallel to the principal axis and close to it, is incident on the surface of a mirror (lens), the reflected (refracted) beam is found to converge to or appears to diverge from a point on the principal axis. This point is the focus also called Principal focus in case of mirror(s). The plane passing through the focus and perpendicular to the principal axis is called **focal plane**.



Focal length (f) is the distance of focus (F) from the pole (P) of the mirror or the optical centre for a lens.

SIGN CONVENTIONS FOR MIRRORS

While solving problems, we must follow a set of sign conventions given for convenience. According to this sign convention

- a) Origin is placed at the pole (P).
- b) All distances are to be measured from the pole (P).
- c) Distances measured in the direction of incident rays are taken as positive.
- d) Distances measured in a direction opposite to that of the incident rays are taken as negative.
- e) Distances above the principal axis are taken as positive.
- f) Distances below the principal axis are taken as negative.
- g) This sign convention is used to find the position and nature (virtual or real, erect or inverted) of the image formed by the mirror (or lens).
- h) Object distance is denoted by u , image distance by v , focal length by f and radius of curvature by R .
- i) Note that generally we keep the object to the left of the mirror (or lens), so that the ray of light starting from object must go from left to the right i.e., towards positive direction of x-axis. Now since the distances have to be measured from the pole consequently,

u must always be negative,

v is positive (for a virtual image) and negative (for a real image).

f is positive (for a convex mirror) and negative (for a concave mirror).

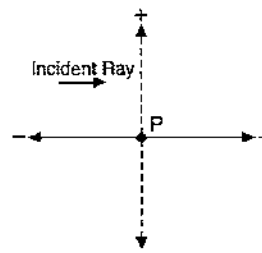
For both the mirrors and lenses.

Magnification for a real image is negative i.e., $m_{\text{real}} = \ominus$

Magnification for a virtual image is positive i.e., $m_{\text{virtual}} = \oplus$

CONCEPTUAL NOTE(S)

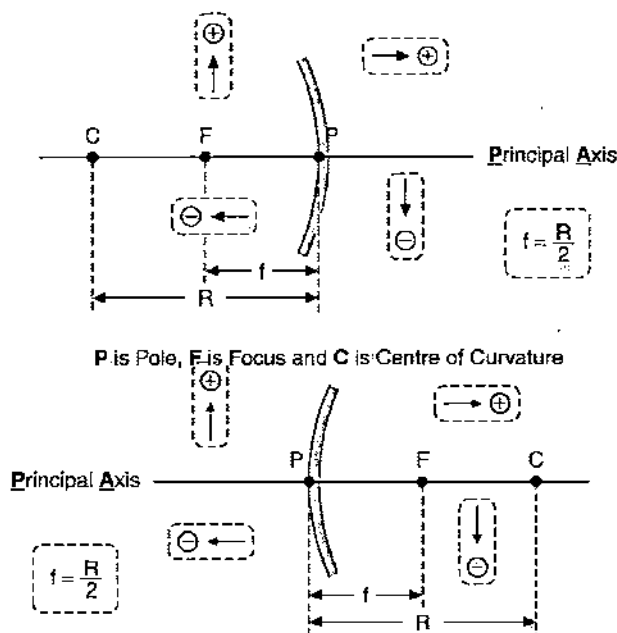
The convention that all distances measured along the ray of light are positive and all distances measured opposite to the ray of light are negative matches exactly with the **Cartesian coordinate system**, where we can simply place the origin at the pole P and say that all distances to the left of the pole are negative, all distances to the right of the pole are positive, all distances above the pole are positive and all distances below the pole are negative.



- j) For solving problems in which any of u , v , f (or R) is to be found, we must make sure that no convention should be applied on the quantity to be found. The

unknown quantity will automatically take up its sign from which we shall make obvious conclusion.

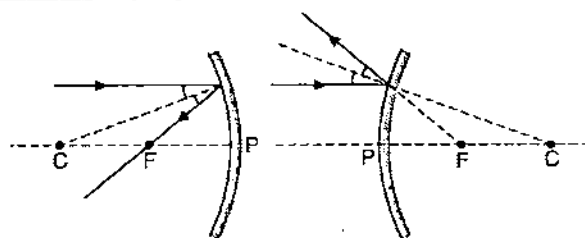
- k) The diagrams show the application of sign convention to curved mirrors.



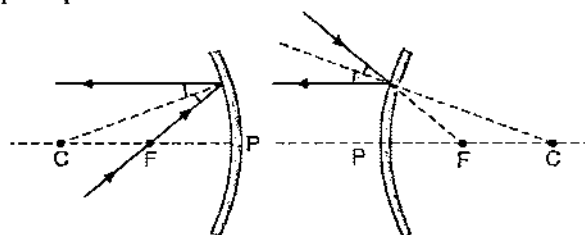
RULES FOR OBTAINING IMAGE

These rules are based on the laws of reflection, i.e. the angle of incidence equals the angle of reflection, $i = r$ and are used to find the location, nature (real or virtual, inverted or erect) and size of the image formed by a spherical mirror. Take any two rays coming from any given point on the object. Find out at which point these rays actually meet (or appear to meet) after reflection from the mirror. This point is the real (or virtual) image. In this way, taking one point after another on the object, the entire image can be constructed.

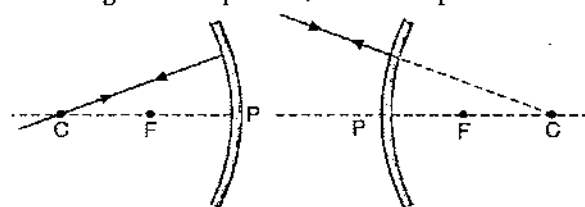
- a) A ray of light coming parallel to principal axis, after reflection passes through the focus (in case of concave mirror) or appears to come from the focus (in case of convex mirror).



- b) A ray of light passing through the focus (in case of concave mirror) or appearing to pass through the focus (in case convex mirror) is reflected parallel to the principal axis.



- c) A ray of light passing through the centre of curvature falls normally on the mirror and is therefore reflected back along the same path i.e., retraces its path.



- d) Incident and reflected rays at the pole of a mirror are symmetrical about the principal axis. (Because for the pole principle axis acts as normal and by Laws of Reflection $i = r$). So by observing the size of erect image in a mirror we can decide the nature of the mirror i.e., whether it is convex, concave or a plane mirror.

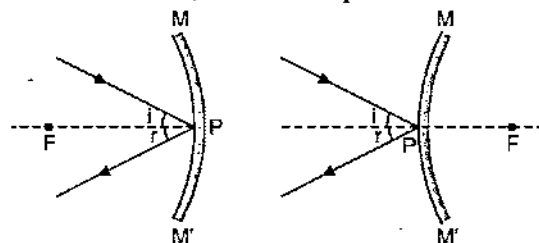


IMAGE FORMATION BY CONCAVE MIRROR

OBJECT POSITION	DIAGRAM	POSITION OF IMAGE	NATURE OF IMAGE
At infinity		At the principal focus (F) or in the focal plane	Real, inverted and extremely diminished

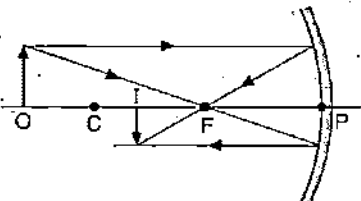
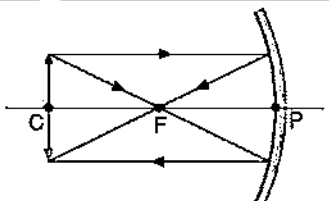
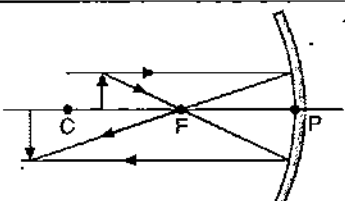
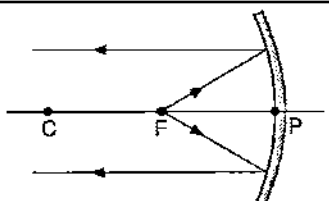
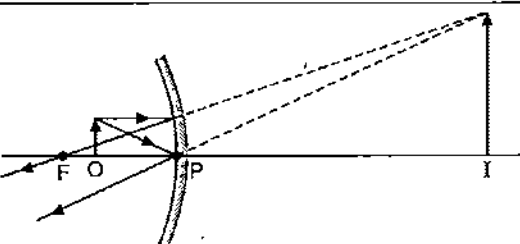
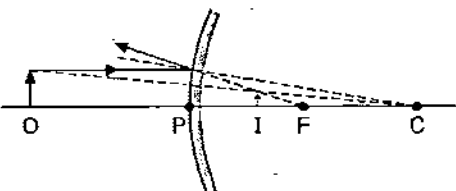
Beyond C		Between F and C	Real, inverted and diminished
At C		At C	Real, inverted and of same size as the object
Between F and C		Beyond C	Real, inverted and magnified
At F or in the focal plane		At infinity	Real, inverted and highly magnified
Between F and P		Behind the mirror	Virtual, erect and magnified

IMAGE FORMATION BY CONVEX MIRROR

OBJECT POSITION	DIAGRAM	POSITION OF IMAGE	NATURE AND SIZE OF IMAGE
For all positions of object		Images formed between the Pole and the focus (F).	Always forms a Virtual, Erect and Diminished Image

RELATION BETWEEN FOCAL LENGTH (f) AND RADIUS OF CURVATURE (R)

A ray parallel to the principal axis passes through the focus (as in concave mirror) or appears to pass through the focus (as in convex mirror). The normal to the mirror(s) at the point of reflection i.e., A must pass through the centre of curvature. In triangle CAN, we have

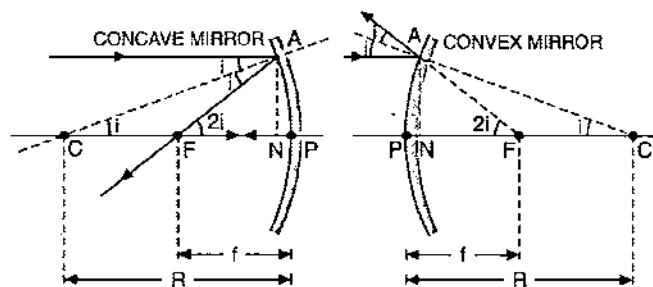
$$\tan i = \frac{AN}{NC}$$

For paraxial rays and mirrors of small aperture, we have

$$\tan i \cong i = \frac{AN}{NC} \quad \dots(1)$$

In triangle FAN, we have

$$\tan(2i) = \frac{AN}{NF}$$



Again for paraxial rays and mirror of small aperture, we have

$$\tan(2i) \cong 2i = \frac{AN}{NF} \quad \dots(2)$$

From (1) and (2), we get

$$2\left(\frac{AN}{NC}\right) = \frac{AN}{NF} \quad \dots(3)$$

Since, aperture is small, so N coincides with P, so we have

$$NC \cong PC \text{ and } NF \cong PF$$

For convex mirror, we have

$$PC = +R \text{ and } PF = +f$$

$$\Rightarrow f = \frac{R}{2}$$

For concave mirror, we have

$$PC = -R \text{ and } PF = -f$$

$$f = \frac{R}{2}$$

So, for a curved mirror of small aperture, focal length is half the radius of curvature.

MIRROR FORMULA

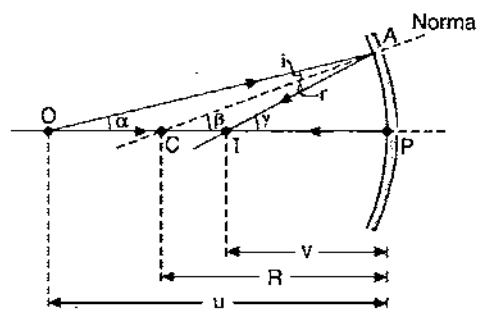
For Concave Mirror

Consider a point object O placed on the principal axis of a concave mirror. A ray of light, incident on the point A at an angle of incidence i on the mirror makes an angle r with the normal as shown in the figure. From the Laws of Reflection we know that $i = r$. Further to find the location of the image let us take another ray along the principal axis so that it hits the mirror normally at the point P to reverse its path and meet the other ray at I. This point of intersection of the two rays happens to be the place where the image is formed.

Since from geometry we know that in a triangle, external angle equals sum of internal opposite angles, so for triangle CAO and triangle CAI, we have

$$\beta = \alpha + i$$

$$\text{and } \gamma = r + \beta$$



Since by Laws of Reflection, we have

$$i = r$$

$$\Rightarrow \alpha + \gamma = 2\beta$$

Applying paraxial ray approximation, we get

$$\tan \alpha \approx \alpha = \frac{AP}{PO}, \tan \beta \approx \beta = \frac{AP}{PC} \text{ and } \tan \gamma \approx \gamma = \frac{AP}{PI}$$

$$\Rightarrow \tan \alpha + \tan \gamma = 2 \tan \beta$$

$$\Rightarrow \frac{AP}{PO} + \frac{AP}{PI} = 2 \left(\frac{AP}{PC} \right)$$

Using sign conventions, we have

$$PO = -u, PI = -v \text{ and } PC = -R$$

$$\Rightarrow \frac{1}{(-u)} + \frac{1}{(-v)} = \frac{2}{(-R)}$$

Since we know that $f = \frac{R}{2}$, so we get

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} = \frac{1}{f} \quad \text{[Mirror Formula]}$$

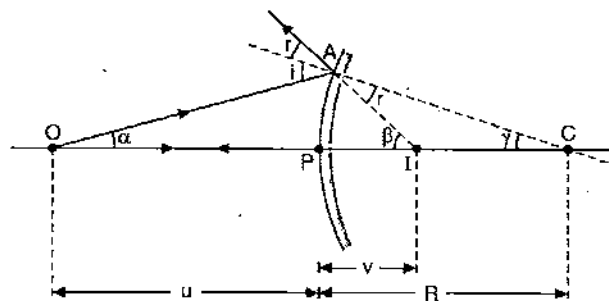
For Convex Mirror

Similarly we can drive a formula for a convex mirror.

Since from geometry we know that in a triangle, external angle equals sum of internal opposite angles, so for triangle CAO and triangle CAI, we have

$$i = \alpha + \gamma$$

$$\text{and } \beta = r + \gamma$$



Since $i = r$, so we get

$$-\alpha + \beta = 2\gamma$$

Applying paraxial ray approximation, we get

$$-\tan \alpha + \tan \beta = 2 \tan \gamma$$

$$\Rightarrow -\frac{AP}{PO} + \frac{AP}{PI} = 2\left(\frac{AP}{PC}\right)$$

Using sign conventions, we have

$$PO = -u, PI = +v \text{ and } PC = +R$$

$$\Rightarrow -\frac{1}{(-u)} + \frac{1}{v} = \frac{2}{R}$$

$$\Rightarrow \frac{1}{u} + \frac{1}{v} = \frac{2}{R} = \frac{1}{f} \quad \text{[Mirror Formula]}$$

Interestingly, the mirror formula is the same irrespective of the mirror used.

NEWTON'S FORMULA

If instead of measuring the object distance and the image distance from the pole, the distances are measured from the focus, then we get a modified mirror formula. This modified mirror formula is called the **Newton's Formula**. Let

x_1 be the distance of object from focus and

x_2 be the distance of image from focus, then

$$u = f + x_1$$

$$\text{and } v = f + x_2$$

According to the mirror formula, we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{(f+x_2)} + \frac{1}{(f+x_1)} = \frac{1}{f}$$

$$\Rightarrow (2f+x_1+x_2)f = (f+x_1)(f+x_2)$$

$$\Rightarrow 2f^2 + (x_1+x_2)f = f^2 + (x_1+x_2)f + x_1x_2$$

$$\Rightarrow x_1x_2 = f^2$$

This is known as Newton's formula.

LINEAR MAGNIFICATION OR LATERAL MAGNIFICATION OR TRANSVERSE MAGNIFICATION

To have an idea of the relative size of the image and the object, we define linear magnification also called as lateral magnification as

$$m = \frac{\text{size of the image}}{\text{size of the object}} = \frac{h_2}{h_1}$$

For both concave and convex mirrors, it can be shown that

$$m = -\frac{v}{u}$$

Since we know that $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ so we get

$$m = -\frac{v}{u} = -\frac{f}{f-u} = \frac{f-v}{f}$$

For spherical mirrors positive value of m means v and u are having opposite signs i.e. when u is negative v is positive and vice versa.

So for a real object if the image formed is virtual, erect and three times the size of the real object then, we have $m = +3$.

Similarly for a real object if the image formed is real, inverted and one third the size of the real object then

$$m = -\frac{1}{3}$$

Illustration 7

An object is placed at a distance of 15 cm from a concave mirror of focal length 10 cm. Describe the size, nature and position of the image formed.

Solution

The rough figure indicating the pole of the mirror, focus, and the given distances is shown. The sign convention is also given.

Since, $u = -15$ cm (negative since it lies to the left of O)

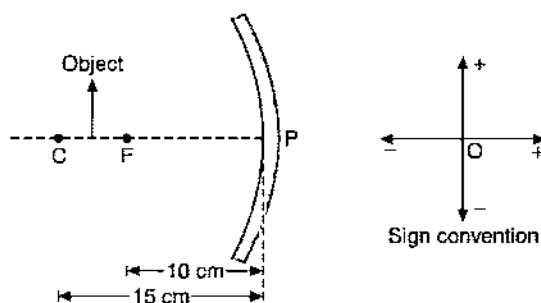
$f = -10$ cm (negative since it lies to the left of O)

Since we have, from mirror formula that

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{(-10)} - \frac{1}{(-15)} = -\frac{1}{10} + \frac{1}{15} = \frac{-15+10}{150} = -\frac{5}{150}$$

$$\Rightarrow v = -\frac{150}{5} = -30 \text{ cm}$$



The negative sign for v shows that the image lies to the left of O .

Now, the magnification is given by

$$m = -\frac{v}{u} = -\frac{-30}{-15} = -2$$

The negative sign for m indicates that the image is inverted, and hence real and is double the size of the object.

Thus, we find that the image is real, inverted, twice the size of the object, and is formed 30 cm in front of the mirror. The ray diagram is shown in figure.

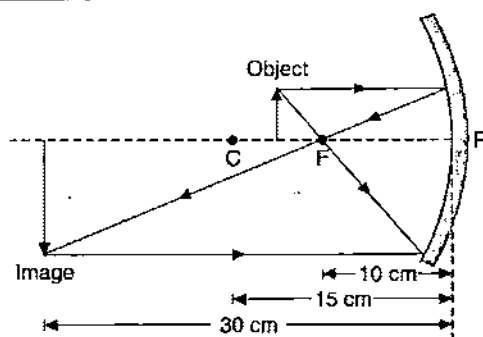


Illustration 8

A beam of light converges towards a point O , behind a convex mirror of focal length 20 cm . Find the nature and position of image if the point O is

- 10 cm behind the mirror
- 30 cm behind the mirror

Solution

- Here, in this case the object is virtual. So, for this we have

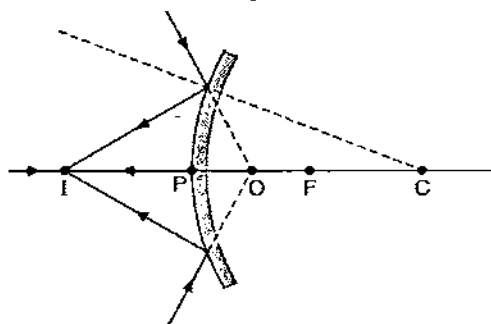
$$u = +10\text{ cm}, f = +20\text{ cm}$$

Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{10} = \frac{1-2}{20} = -\frac{1}{20}$$

$$\Rightarrow v = -20\text{ cm}$$

$$\text{Magnification, } m = -\frac{v}{u} = -\frac{(-20)}{10} = 2$$



Hence, the image formed will be real, erect and enlarged, and at a distance of 20 cm in front of the mirror.

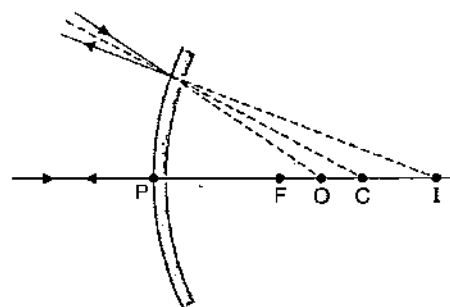
- Again, in this case too the object is virtual. So, we have

$$u = +30\text{ cm}, f = +20\text{ cm}$$

Using mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$$

$$\Rightarrow v = 60\text{ cm}$$



$$\text{Magnification, } m = -\frac{v}{u} = -\frac{60}{30} = -2$$

Hence, the image formed will be virtual, inverted and enlarged, and at a distance of 60 cm behind the mirror.

CONCEPTUAL NOTE(S)

Note that for the real objects, a convex mirror always gives virtual and diminished image, but for virtual objects it gives real image if $u < f$ and virtual image if $u > f$.

LONGITUDINAL MAGNIFICATION OR AXIAL MAGNIFICATION

When an object is placed with its length along the principal axis, then instead of defining the linear magnification we define the axial magnification. Mathematically we define axial magnification, for small objects as

$$m_{\text{axial}} = \frac{\text{Size of image along principal axis}}{\text{Size of object along principal axis}}$$

$$m_{\text{axial}} = -\left(\frac{v_2 - v_1}{u_2 - u_1}\right) = -\frac{dv}{du}$$

Further since we know that $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ i.e., $v^{-1} + u^{-1} = f^{-1}$.

Taking the derivative of this equation with respect to u , we get

$$\frac{d}{du}(v^{-1}) + \frac{d}{du}(u^{-1}) = \frac{d}{du}(f^{-1})$$

$$\Rightarrow -v^{-2} \frac{dv}{du} - u^{-2} = 0$$

$$\Rightarrow m_{\text{axial}} = \frac{dv}{du} = -\frac{v^2}{u^2} = -\left(\frac{f}{f-u}\right)^2 = -\left(\frac{f-v}{f}\right)^2$$

AREAL MAGNIFICATION

Areal magnification is simply defined as the ratio of area of image to the area of object, so

$$m_{\text{areal}} = \frac{\text{Area of image}}{\text{Area of object}} = \frac{A_i}{A_o} = \frac{v^2}{u^2} = \left(\frac{f}{f-u}\right)^2 = \left(\frac{f-v}{f}\right)^2$$

RELATION BETWEEN OBJECT AND IMAGE VELOCITY FOR CURVED MIRRORS

According to the mirror formula, we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow v^{-1} + u^{-1} = f^{-1} = \text{constant}$$

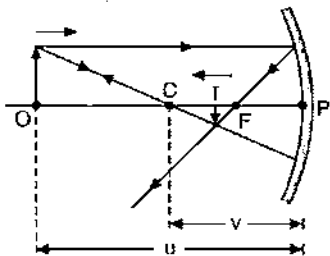
Differentiating with respect to time, we get

$$-v^{-2} \frac{dv}{dt} - u^{-2} \frac{du}{dt} = 0$$

$$\Rightarrow \frac{dv}{dt} = -\left(\frac{v^2}{u^2}\right) \frac{du}{dt} \quad \dots(1)$$

Here $\frac{du}{dt}$ is the rate at which the object distance u is changing i.e., it is the object speed if the mirror is stationary.

Similarly, $\frac{dv}{dt}$ is the rate at which v (distance between image and mirror) is changing i.e., it is image speed if the mirror is stationary. So if at a known values of v and u , the object speed is given, we can find the image speed from the above formula.



Let us take the example for a concave mirror.

Suppose the object is moved from infinity towards focus, then since u is decreasing therefore,

$$-\left(\frac{du}{dt}\right) = \text{rate of decrease of } u \quad \{\text{object speed}\}$$

$$\Rightarrow \left(\frac{dv}{dt}\right) = \text{rate of increase of } v \quad \{\text{image speed}\}$$

Further, when the object lies between ∞ and C , then $v < u$,

$$\Rightarrow \left(\frac{dv}{dt}\right) < \left(\frac{du}{dt}\right) \quad \{\text{from equation (1)}\}$$

Hence, when the object is moved towards the mirror, its image (which is real) will recede from the mirror with speed less than the speed of object.

When the object is at C , image is also at C

$$\Rightarrow v = u$$

$$\Rightarrow \left(\frac{dv}{dt}\right) = \left(\frac{du}{dt}\right)$$

Hence, when the object is at C speed of image is equal to the speed of object.

When the object lies between C and F then $v > u$

so, the image speed is more than the object speed.

When the object lies between F and P , then the image becomes virtual i.e., u and f are negative while v is positive. So from mirror formula we get,

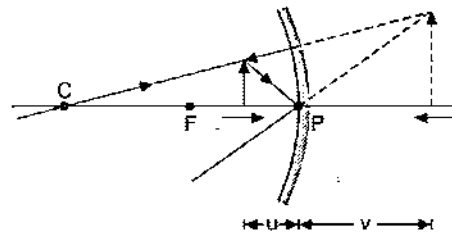
$$\frac{1}{v} + \frac{1}{(-u)} = \frac{1}{(-f)} \Rightarrow \frac{1}{u} - \frac{1}{v} = \frac{1}{f}$$

$$\Rightarrow -u^{-2} \left(\frac{du}{dt}\right) - v^{-2} \left(\frac{dv}{dt}\right) = 0$$

$$\left(-\frac{dv}{dt}\right) = \left(\frac{v^2}{u^2}\right) \left(-\frac{du}{dt}\right)$$

Now, when u is further decreased, v also decreases to keep $\frac{1}{f}$ constant. So, $-\frac{du}{dt}$ is the rate at which object is

approaching towards mirror and $\left(-\frac{dv}{dt}\right)$ is rate at which the image is approaching towards the mirror.



Further in this case we observe that the image is always enlarged i.e., $v > u$. Therefore, image speed is more than the object speed. Thus, the above entire discussion can simply be concluded as follows.

CONCLUSION

When an object is moved from $-\infty$ to F , the image (real) moves from F to $-\infty$ and then when the object is further moved from F to P image (now virtual) moves from $+\infty$ to P .

CONCEPTUAL NOTE(S)

When the object is either at centre of curvature C or at pole P , the two speeds are equal. When the object is at pole, then due to the small aperture of the mirror, it can be assumed, as if the image is being formed by a plane mirror.

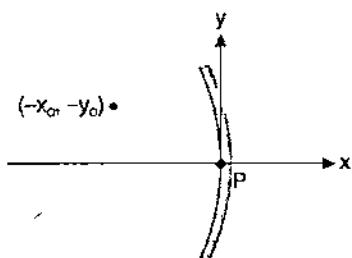
FINDING COORDINATES OF IMAGE OF A POINT

If the coordinates of a point object $(-x_o, -y_o)$ with respect to the coordinate axes shown in figure are known to us and the coordinates of image be (x_i, y_i) then for finding the x -coordinate, we use the mirror formula, according to which

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{x_i} + \frac{1}{x_o} = \frac{1}{f} \quad \left\{ \because u = -(-x_o) = x_o \right\}$$

$$\Rightarrow x_i = \frac{fx_o}{x_o - f}$$



For finding the y -coordinate, we apply the concept of magnification (m), according to which, we have

$$m = \frac{y_i}{y_o} = -\frac{v}{u} = -\frac{x_i}{x_o}$$

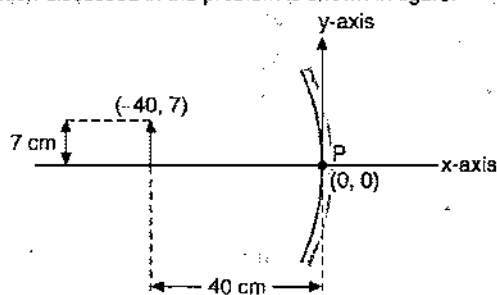
$$\Rightarrow y_i = \frac{fy_o}{f - x_o}$$

EXAMPLE

A point object is placed at $(-40, 7)$ cm in front of a concave mirror of focal length 5 cm having its pole at the origin $(0, 0)$. Assuming the principal axis to be along x -axis, find the position of the image formed.

SOLUTION

The situation discussed in the problem is shown in figure.



$$\text{Since, } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{(-40)} = \frac{1}{(-5)}$$

$$\Rightarrow v = -\frac{40}{7} \text{ cm}$$

$$\text{Since, } m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$\Rightarrow \frac{h_i}{h_o} = -\frac{(-40/7)}{-40}$$

$$\Rightarrow \frac{h_i}{h_o} = -\frac{1}{7}$$

$$\text{But } h_o = 7 \text{ cm}$$

$$\Rightarrow h_i = -1 \text{ cm}$$

$$\text{So, image coordinates are } \left(-\frac{40}{7}, -1\right) \text{ cm}$$

PROBLEM SOLVING TRICK(S)

- Place the object to the left of the mirror (or lens), so that sign convention matches with the familiar sign convention in the coordinate geometry.
- Both for concave as well as convex mirrors, use the same mirror formula i.e.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{and} \quad m = -\frac{v}{u}$$

- Substitute the numerical values of the given quantities with proper sign (+ve or -ve) as per sign convention.
- Though the SI unit of distance is metre, it may be more convenient in some problems to take the given distances in cm rather than in m. But then your answer too will be in cm.
- Do not give any sign to the quantity to be determined. In your answer, the unknown quantity will be obtained with its proper sign.

In addition to the above hints, if you remember the following facts, it will help you.

- Since the object is always placed to the left of the mirror so, u is always negative.
- For a concave mirror, f is negative.
- For a convex mirror, f is positive.
- A real image is formed in front of the mirror, so for a real image v is negative.
- A virtual image is formed behind the mirror, so for a virtual image v is positive.
- A real image is always inverted, so for a real image h is negative.
- A virtual image is always erect, so for a virtual image h is positive.
- For the real image of a real object and the virtual image of a virtual object, m is negative.
- For the virtual image of a real object and the real image of a virtual object, m is positive.

GRAPH BETWEEN $\frac{1}{v}$ VERSUS $\frac{1}{u}$

Let us first take the case of a concave mirror. Here, two cases are possible.

Case 1 : When the Image formed is Real.

When the image is real, i.e., object lies between F and infinity. In such a situation u , v and f are negative.

Hence, the mirror formula i.e., $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ becomes

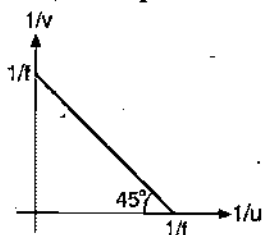
$$-\frac{1}{v} - \frac{1}{u} = -\frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = -\frac{1}{u} + \frac{1}{f}$$

Comparing with $y = mx + c$, the desired graph will be a straight line with slope -1 and intercept on y -axis is equal to $\frac{1}{f}$.

Do not confuse here, the slope m with magnification.

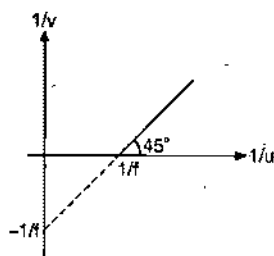


Case 2 : When the Image formed is Virtual.

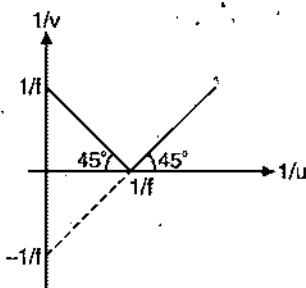
When the image is virtual, i.e., object lies between F and P . Under such situation u and f are negative while v is positive. The mirror formula thus becomes

$$\frac{1}{v} = \frac{1}{u} - \frac{1}{f}$$

Comparing it with $y = mx + c$ the desired graph is a straight line with slope $m = 1$ and intercept on y -axis is equal to $-\frac{1}{f}$.



The graph is thus shown in figure. The two graphs can be drawn in one single graph as in figure.



CONCEPTUAL NOTE(S)

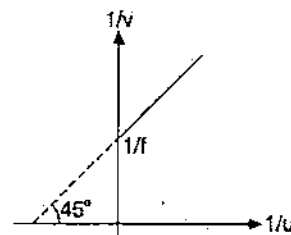
Please note that $\frac{1}{u}$ and $\frac{1}{v}$ are actually the magnitudes of $\frac{1}{u}$ and $\frac{1}{v}$ (i.e., without sign)

For a **convex mirror**, the image formed is always virtual, i.e., u is always negative while v and f are always positive. Hence, the mirror formula becomes,

$$\frac{1}{v} + \frac{1}{(-u)} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

Comparing with $y = mx + c$, the desired graph is a straight line of slope $m = 1$ and intercept on y -axis equal to $\frac{1}{f}$. The graph is thus shown in figure.



PROBLEM SOLVING TRICK(S)

a) As focal-length of a spherical mirror $f = \frac{R}{2}$ depends only on the radius of mirror and is independent of wavelength of light and refractive index of medium so the focal length of a spherical mirror in air or water and for red or blue light is same. This is also why the image formed by mirrors do not show chromatic aberration.

b) In case of spherical mirror if $R \rightarrow \infty$ (i.e., it becomes plane), $f = \frac{R}{2} = \infty$, the mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \text{ reduces to } \frac{1}{v} + \frac{1}{u} = 0 \text{ i.e., } v = -u$$

i.e., image is at same distance behind the mirror as the object is in front of it. This in turn verifies the correctness of mirror formula.

c) Every part of a mirror forms complete image. If some portion of a mirror is obstructed (say covered with black paper), then complete image will be formed but intensity will be reduced.

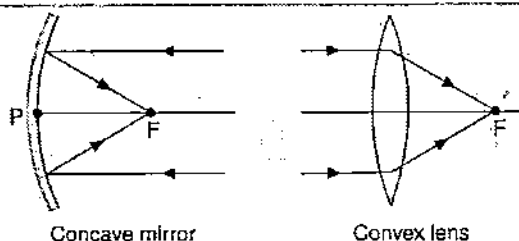
d) In case of spherical mirrors if object distance (x_1) and image distance (x_2) are measured from focus instead of pole, $u = (f + x_1)$ and $v = (f + x_2)$ so the mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \text{ reduces to } \frac{1}{(f + x_2)} + \frac{1}{(f + x_1)} = \frac{1}{f} \text{ which on}$$

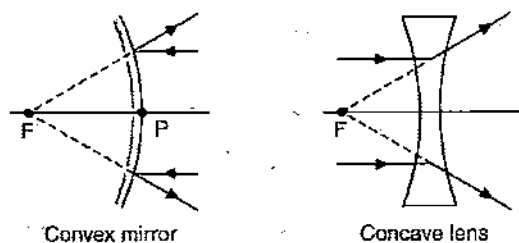
simplification gives $x_1 x_2 = f^2$. This result is called 'Newton's formula'.

e) If an object is moved at constant speed towards a concave mirror from infinity to focus, the image will move (slower in the beginning and faster later on) away from the mirror. This is because, during the time the object moves from ∞ to C the image will move from F to C and when the object moves from C to F the image will move from C to ∞ . At C the speed of object and image will be equal.

f) Concave mirror behaves as convex lens (both convergent) while convex mirror behaves as concave lens (both divergent). This is shown in figure.



(a) Convergent behaviour

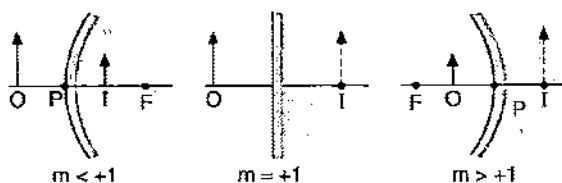


(b) Divergent behaviour

- g) As convex mirror gives erect, virtual and diminished image, field of view is increased. This is why it is used as rear-view mirror in vehicles. Concave mirrors give enlarged erect and virtual image (if object is between F and P) so are used by dentists for examining teeth. Further due to their converging property concave mirrors are also used as reflectors in automobiles head lights and search lights and by ENT surgeons in ophthalmoscope.

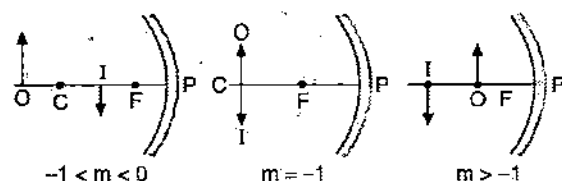
- h) For real extended objects, if the image formed by a single mirror is erect it is always virtual and in this situation if the size of the image is

- | | | |
|--|--|---|
| smaller than object
• the mirror is convex. | equal to object
• the mirror is plane | larger than object
• the mirror is concave |
|--|--|---|



- i) For real extended objects, if the image formed by a single mirror is inverted, it is always real (i.e., m is -ve) and the mirror is concave. In this situation if the size of image is

- | | | |
|--|--|---|
| smaller than object
• object is between ∞ and C
• and image between F and C | equal to object
• object is at C
• and image is at C | larger than object
• object is between C and F
• and image between C and ∞ |
|--|--|---|



- j) In case of spherical mirrors if we plot a graph between u and v the graph will be a hyperbola as for $u=f$, $v=\infty$ and for $u=\infty$, $v=f$. A line $u=v$ will cut this hyperbola at

$(2f, 2f)$. This all is shown in figure.

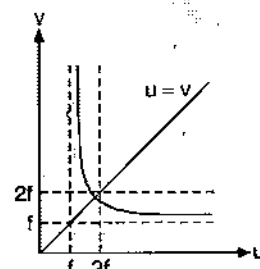
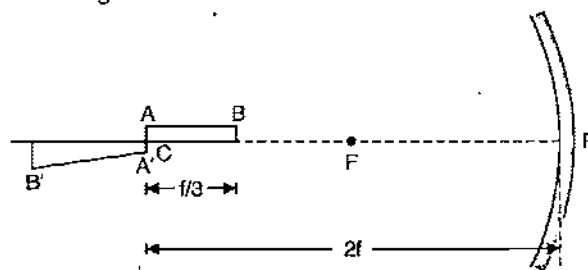


Illustration 9

A thin rod of length $\frac{f}{3}$ is placed along the principal axis of a concave mirror of focal length f such that its image, which is real and elongated, just touches the rod. What is the magnification?

Solution

According to the problem, the image is real and enlarged, the object must have been placed between C and F. Since one end of the image just touches one end of the object so, this end must lie on C. Let AB be the object and A'B' be its image, such that A and A' both lie at C, as shown in figure.



Now, as the length of the object AB is $\frac{f}{3}$, so the distance of end B of the object from the pole P is

$$u_B = -\left(PA - \frac{f}{3}\right) = -\left(2f - \frac{f}{3}\right) = -\left(\frac{5}{3}\right)f$$

The distance of the image of end B, v_B , is calculated by using the mirror formula,

$$\begin{aligned} \frac{1}{v} + \frac{1}{u} &= \frac{1}{f} \\ \Rightarrow \frac{1}{v_B} + \frac{1}{-\frac{5}{3}f} &= \frac{1}{f} \\ \Rightarrow v_B &= -\frac{5}{2}f \end{aligned}$$

Therefore, the size of the image A'B' is

$$A'B' = |v_B| - |v_A| = \frac{5}{2}f - 2f = \frac{1}{2}f$$

$$\text{Now, magnification, } m = \frac{A'B'}{AB} = -\frac{\left(\frac{f}{2}\right)}{\left(\frac{f}{3}\right)} = -\frac{3}{2}$$

Illustration 10

A gun of mass m_1 fires a bullet of mass m_2 with a horizontal speed v_0 . The gun is fitted with a concave mirror of focal length f facing towards the receding bullet. Find the speed of separations of the bullet and the image at the instant just after the bullet is fired from the gun.

Solution

Let v_1 be the speed of gun (or mirror) just after the firing of bullet. By Law of Conservation of Linear Momentum, we have

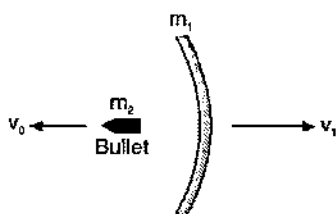
$$m_2 v_0 = m_1 v_1$$

$$\Rightarrow v_1 = \frac{m_2 v_0}{m_1} \quad \dots(1)$$

Now, $\frac{du}{dt}$ is the rate at which distance between mirror and bullet is increasing, so

$$\frac{du}{dt} = v_1 + v_0 \quad \dots(2)$$

$$\text{Since, we know that } \left| \frac{dv}{dt} \right| = \left(\frac{f}{f-u} \right)^2 \left| \frac{du}{dt} \right| \quad \dots(3)$$



Since, at the instant just after the bullet is fired from the gun, the bullet is actually very close to the pole of the mirror, so $u \rightarrow 0$ and hence we get at that instant

$$\frac{v^2}{u^2} = \left(\frac{f}{f-u} \right)^2 = m^2 = 1$$

So, from (2) and (3), we get

$$\frac{dv}{dt} = \frac{du}{dt} = v_1 + v_0 \quad \dots(4)$$

where $\frac{dv}{dt}$ is the rate at which distance between image (of bullet) and mirror is increasing.

Therefore, speed of separation of bullet and image will be,

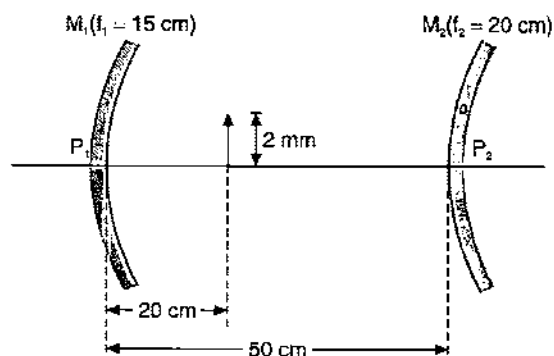
$$v_r = 2(v_1 + v_0)$$

Substituting value of v_1 from equation (1) we get

$$v_r = 2 \left(1 + \frac{m_2}{m_1} \right) v_0$$

Illustration 11

Find the location, size and the nature of the image of an object of height 2 mm kept between two mirrors (as shown in figure) after two successive reflections, considering the first reflection at the concave mirror and then at the convex mirror.



Solution

As asked in the problem, let us first consider the reflection at mirror M_1 . Before executing the mirror formula, we must keep two things in mind.

1. The incident light must go from the object to the mirror and we preferably take it parallel to the principal axis.
2. All distances have to be measured from the pole of the respective mirror for which reflection is being considered.
3. All distances measured along the incident ray are positive and all distances measured opposite to the incident ray are negative.

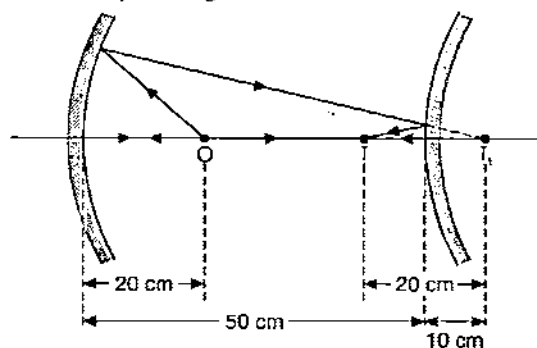


Figure is just representative and not to scale

Now, for reflection at concave mirror M_1 , the incident ray from the object goes to left of object and object distance is measured towards right of pole P_1 , so

$$u = -20 \text{ cm}$$

Similarly, $f_1 = -15 \text{ cm}$

Now, according to mirror formula, we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{(-20)} = \frac{1}{(-15)}$$

$$\Rightarrow v = -60 \text{ cm}$$

Negative sign with v means that it is formed to right of pole P_1 at a distance of 60 cm from P_1 (10 cm behind M_2).

$$\Rightarrow m_1 = -\frac{v}{u} = -\frac{(-60)}{-20} = -3$$

So, image (I_1) formed is real, inverted and three times size of object i.e., 6 mm.

This image (I_1) formed now acts as object for the convex mirror. Further, this image formed is 10 cm to the left of P_2 and the incident ray from the original object goes to the right for reflection at M_2 to take place, so

$$u = +10 \text{ cm}$$

Similarly, $f = +20 \text{ cm}$

Applying the mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} + \frac{1}{10} = \frac{1}{20}$$

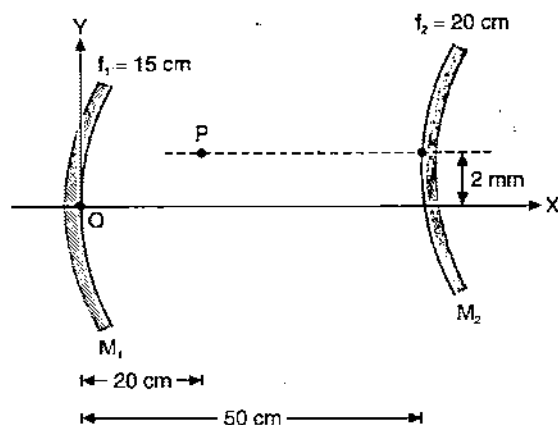
$$\Rightarrow v = -20 \text{ cm}$$

$$\Rightarrow m_2 = -\frac{v}{u} = -\frac{(-20)}{10} = 2$$

So, image (I) formed is virtual, erect and two times the size of object (here I_1). Hence the size of I is 12 mm. So, finally I is formed at 20 cm in front of convex mirror M_2 , with size 12 mm, virtual and erect.

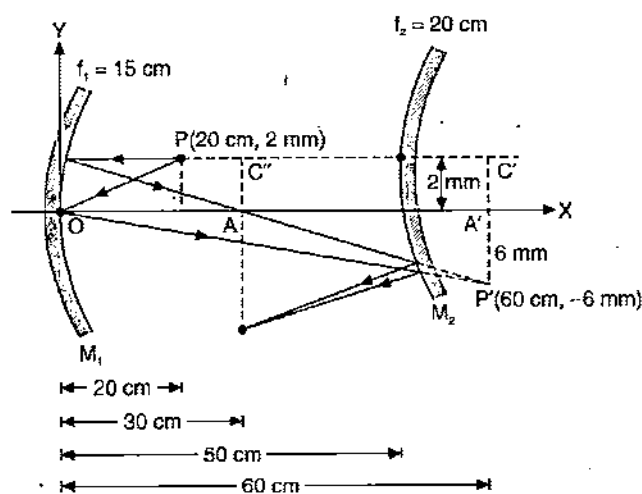
Illustration 12

Find the co-ordinates of image of point object P formed after two successive reflections in figure, considering the first reflection at concave mirror and then at convex mirror.



Solution

The ray diagram for the situation is drawn in figure (but not to scale).



For reflection at concave mirror M_1 , we have

$$u = -20 \text{ cm}$$

$$f_1 = -15 \text{ cm}$$

$$\text{Since, } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v_1} + \frac{1}{(-20)} = \frac{1}{(-15)}$$

$$\Rightarrow v_1 = -60 \text{ cm}$$

$$\text{So, magnification } m_1 = -\frac{v_1}{u} = -\frac{-60}{-20} = -3 \text{ (Inverted)}$$

$$\Rightarrow A'P' = m_1(AP) = 3 \times 2 = 6 \text{ mm}$$

For reflection at convex mirror M_2 , we have

$$u = +10 \text{ cm}$$

$$f_2 = +20 \text{ cm}$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v_2} + \frac{1}{10} = \frac{1}{20}$$

$$\Rightarrow v_2 = -20 \text{ cm}$$

$$\text{Again, magnification, } m_2 = -\frac{v_2}{u} = -\frac{(-20)}{10} = 2$$

$$\Rightarrow C'P' = m_2(CP) = 2 \times 8 = 16 \text{ mm}$$

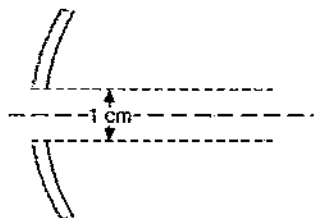
So, the co-ordinate of image of point object P as measured from the origin O is (30 cm, -14 mm)

ICE II

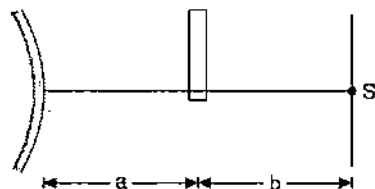
BASED ON REFLECTION AT CURVED SURFACES

(Solutions on page 1.173)

1. An object of height 2.5 cm. is placed at a $1.5f$ from a concave mirror, where f is the magnitude of the focal length of the mirror. The height of the object is perpendicular to the principal axis. Find the height of the image. Is the image erect or inverted?
2. A mirror (in a laughing gallery) forms an erect image four times enlarged, of a boy standing 2.5 m away. Is the mirror concave or convex? What is its radius of curvature?
3. A concave mirror forms the real image of a point source lying on the optical axis at a distance of 50 cm from the mirror. The focal length of the mirror is 25 cm. The mirror is cut in two halves and these halves are drawn apart at a distance of 1 cm in a direction perpendicular to the optical axis. How will the images formed by the halves of the mirror be arranged?



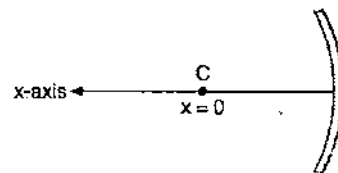
4. Find the distance of object from a concave mirror of focal length 10 cm so that image size is four times the size of the object.
5. A concave mirror has a radius of curvature of 24 cm. How far is an object from the mirror when the image formed is
 - (a) virtual and 3 times the size of the object.
 - (b) real and 3 times the size of the object and
 - (c) real and $\frac{1}{3}$ the size of the object?
6. A thin flat glass plate is placed in front of a convex mirror. At what distance b from the plate should a point source of light S be placed so that its image produced by the rays reflected from the front surface of the plate coincides with the image formed by the rays reflected from the mirror? The focal length of the mirror is $f = 20$ cm and the distance from the plate to the mirror $a = 5$ cm. How can the coincidence of the images be established by direct observation?



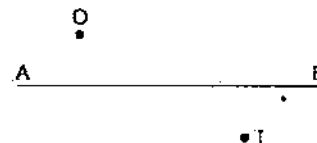
7. A ball swings back and forth in front of a concave mirror. The motion of the ball is described approximately by the equation $x = f \cos(\omega t)$, where f is the focal length of the mirror and x is measured along the axis of mirror. The origin is taken at the centre of curvature of the mirror.
 - (a) Derive an expression for the distance from the mirror

to the image of the swinging ball.

- (b) At what point does the ball appear to coincide with its image.
- (c) What will be the lateral magnification of the image of the ball at time $t = \frac{T}{2}$, where T is time period of oscillation?



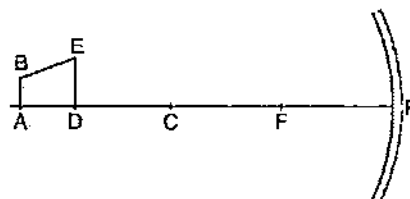
8. An image I is formed of a point object O by a mirror whose principal axis is AB as shown in figure.



- (a) State whether it is a convex mirror or a concave mirror.
- (b) Draw a ray diagram to locate the mirror and its focus. Write down the steps of construction of the ray diagram. Consider the possible two cases :
 - (i) When distance of I from AB is more than the distance of O from AB and
 - (ii) When distance of O from AB is more than the distance of I from AB

9. Convex and concave mirrors have the same radii of curvature R . The distance between the mirrors is $2R$. At what point on the common optical axis of the mirrors should a point source of light A be placed for the rays to converge at the point A after being reflected first on the convex and then on the concave mirror?

10. An object $ABED$ is placed in front of a concave mirror beyond centre of curvature C as shown in figure. State the shape of the image.



11. An object is 30 cm from a spherical mirror, along the central axis. The absolute value of lateral magnification of an inverted image is $\frac{1}{2}$. Find the focal length of the mirror?

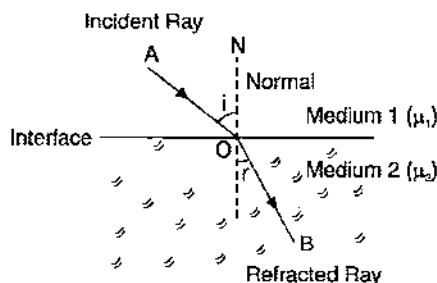
12. A thin rod of length $\frac{f}{3}$ is placed along the principal axis of a concave mirror of focal length f such that its image just touches the rod. Calculate magnification.

Ray Optics

(b) Refraction at Plane Surfaces

REFRACTION OF LIGHT AT PLANE SURFACES

The phenomenon of the bending of light rays as they travel from one medium to the other is called Refraction. The surface separating two media is called an Interface. In other words, the phenomenon of bending of light rays at the boundary between two media is called refraction.



LAWS OF REFRACTION

- a) The incident ray, the refracted ray and normal at the point of incidence to the surface separating the two media all lie in the same plane.

- b) **Snell's Law**

For two media, the ratio of sine of angle of incidence i to the sine of the angle of refraction r is constant (for a beam of particular wavelength). For a given set of media this constant is called the **refractive index** of the medium 2 with respect to medium 1 (represented as ${}^1\mu_2$) i.e.,

$$\frac{\sin i}{\sin r} = \text{constant} = \frac{\mu_2}{\mu_1} = {}^1\mu_2 \quad (\text{SNELL'S LAW})$$

$$\text{OR } \mu_1 \sin i = \mu_2 \sin r$$

where μ_1 and μ_2 are Absolute Refractive Indices of Medium 1 and 2 respectively and ${}^1\mu_2$ is the refractive index of medium 2 with respect to medium 1. If medium 1 happens to be the vacuum, then the constant

is simply called as the **Absolute Refractive Index** of medium 2, expressed as μ_2 or simply μ .

REFRACTIVE INDEX (RI)

The refractive index of a medium is not determined by its density. It is governed by the velocity of light in the medium. The lesser the value of the velocity of light, the more is the refractive index of the medium, and the denser is the medium. A medium having greater refractive index is called denser medium whereas the other medium is called rarer medium.

ABSOLUTE REFRACTIVE INDEX

The absolute refractive index of a medium is defined as the ratio of the speed of light in vacuum to the speed of light in the medium,

$$\mu = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} = \frac{c}{v} > 1$$

Absolute refractive index is more than one because the speed of light is maximum in vacuum/air.

RELATIVE REFRACTIVE INDEX

The relative refractive index of medium 2 with respect to medium 1 is denoted by ${}^1\mu_2$ and is given by

$${}^1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\left(\frac{c}{v_2}\right)}{\left(\frac{c}{v_1}\right)} = \frac{v_1}{v_2}$$

The relative refractive index of medium 1 with respect to medium 2 is denoted by ${}^2\mu_1$ and is given by

$${}^2\mu_1 = \frac{\mu_1}{\mu_2} = \frac{\left(\frac{c}{v_1}\right)}{\left(\frac{c}{v_2}\right)} = \frac{v_2}{v_1}$$

CONCEPTUAL NOTE(S)

- a) The velocity of light in air is not much different from that in vacuum. Hence, while defining the refractive index of a medium we often take velocity of light in air rather than that in vacuum.

Medium	Refractive Index (μ)
Water	$\frac{4}{3} = 1.33$
Glass	$\frac{3}{2} = 1.50$

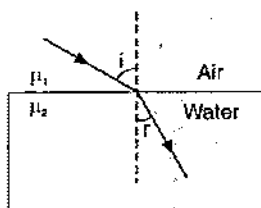
- b) Relative refractive index can be less than one. If we calculate the refractive index of water with respect to glass, then

$${}^a\mu_w = \frac{\mu_w}{\mu_g} = \frac{\left(\frac{4}{3}\right)}{\left(\frac{3}{2}\right)} = \frac{8}{9} < 1$$

- c) Refractive index is different for different wavelengths for a pair of media, because $\mu_1 \lambda_1 = \mu_2 \lambda_2$.

BENDING OF A LIGHT RAY

- a) When light passes from rarer to denser medium, it bends towards the normal i.e., a light ray passing from air to water bends towards the normal as shown in the figure.



According to Snell's Law

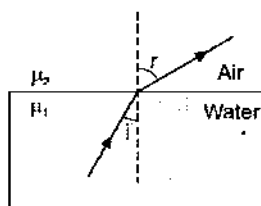
$$\mu_1 \sin i = \mu_2 \sin r$$

$$\Rightarrow \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} > 1$$

$$\Rightarrow \sin i > \sin r$$

$$\Rightarrow i > r$$

- b) When light passes from denser to rarer medium it bends away from the normal as i.e., a light ray passing from water to air bends away from the normal shown in the figure.



According to Snell's Law

$$\mu_1 \sin i = \mu_2 \sin r$$

$$\Rightarrow \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} < 1$$

$$\Rightarrow \sin i < \sin r$$

$$\Rightarrow i < r$$

REFRACTION : IMPORTANT POINTS

- a) Whenever light goes from one medium to another, the frequency of light (f) remains unchanged. Since

$$\mu = \frac{c}{v} = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

$$\Rightarrow \mu = \frac{f \lambda_{\text{air}}}{f \lambda_{\text{medium}}} = \frac{\lambda_{\text{air}}}{\lambda_{\text{medium}}}$$

where λ_{air} and λ_{medium} being wavelengths of light in air and medium respectively.

$$\Rightarrow \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \frac{v_2}{v_1} = \frac{v_2}{c} = \frac{\lambda_1}{\lambda_2}$$

(MODIFIED FORM OF SNELL'S LAW)

From above we conclude that

$$\mu_1 \lambda_1 = \mu_2 \lambda_2$$

$$\Rightarrow \mu \lambda = \text{constant}$$

Also, we conclude that

$$\mu_1 v_1 = \mu_2 v_2$$

$$\Rightarrow \mu v = \text{constant}$$

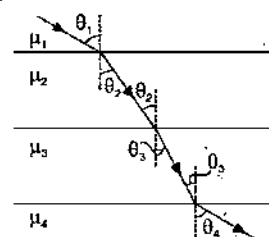
- b) ${}^2\mu_1 \times {}^1\mu_2 = 1$

$$\Rightarrow {}^2\mu_1 = \frac{1}{{}^1\mu_2}$$

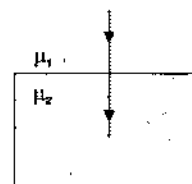
- c) When light propagates through a series of parallel layers of different medium as shown in the figure, then the Snell's Law may be written as

$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2 = \mu_3 \sin \theta_3 = \mu_4 \sin \theta_4 = \text{constant}$$

In general, $\mu \sin \theta = \text{constant}$



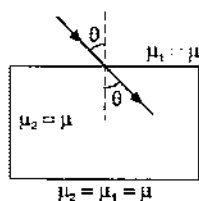
- d) If light is incident normal to a boundary (i.e. $i=0$), then, it passes undeviated from the boundary as shown in the figure.



$$\mu_2 \neq \mu_1$$

Condition for no refraction

- e) If the refractive indices of the two media are equal as shown in figure, then also the light ray is not refracted and the boundary between the two media is not visible. This is why a transparent solid is invisible in a liquid of same refractive index.



Condition for no refraction

- f) Note that for sound waves,
speed in air, $v_i = 330 \text{ ms}^{-1}$
speed in water, $v_w = 1500 \text{ ms}^{-1}$

Therefore, the refractive index of water with respect to air, for sound waves is

$$\mu_w = \frac{v_a}{v_w} = \frac{330}{1500} = 0.22$$

Thus, we find that for the refraction of sound waves, water is rarer than air.

Illustration 13

A ray of light falls on a glass plate of refractive index $n = \sqrt{3}$. What is the angle of incidence of the ray if the angle between the reflected and refracted rays is 90° ?

Solution

According to Snell's Law

$$n = \frac{\sin i}{\sin r}$$

Since $i + r = 90^\circ$

$$\Rightarrow r = 90^\circ - i$$

$$\Rightarrow \sqrt{3} = \frac{\sin i}{\sin(90^\circ - i)} = \tan i$$

$$\Rightarrow i = \tan^{-1}(\sqrt{3}) = 60^\circ$$

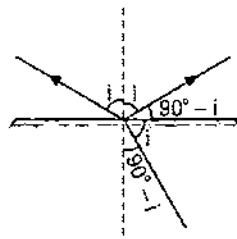


Illustration 14

A ray of light passes through a medium whose refractive index varies with distance as $n = n_0 \left(1 + \frac{x}{2a}\right)$. If ray enters the medium parallel to x-axis, what will be the time taken for ray to travel between $x = 0$ and $x = a$?

Solution

Since, we know that $\mu = \frac{c}{v}$

$$\Rightarrow v = \frac{c}{\mu}$$

So, if v be the speed at a distance x from y-axis, then

$$v = \frac{c}{n_0 \left(1 + \frac{x}{2a}\right)}$$

Since, $v = \frac{dx}{dt}$, so we have

$$\frac{dx}{dt} = \frac{c}{n_0 \left(1 + \frac{x}{2a}\right)}$$

$$\Rightarrow \int_0^a dt = \frac{n_0}{c} \int_0^a \left(1 + \frac{x}{2a}\right) dx$$

$$\Rightarrow t = \frac{5n_0 a}{4c}$$

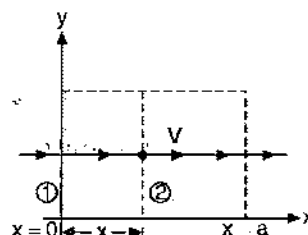
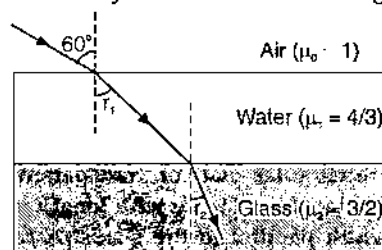


Illustration 15

For the arrangement shown in the figure, a light ray is incident at an angle of 60° on the layer of water. Find the angle between this ray and the normal to the glass.



Solution

According to Snell's Law, we have

$$\mu_0 \sin(60^\circ) = \mu_1 \sin r_1 = \mu_2 \sin r_2$$

$$\Rightarrow \sin r_2 = \frac{\mu_0 \sin(60^\circ)}{\mu_2} = \frac{1 \left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{3}{2}\right)} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow r_2 = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 35^\circ$$

Illustration 16

A ray of light goes from air to medium of refractive index μ . If i be the angle of incidence, r be the angle of refraction and δ be the angle of deviation, then prove that

$$\tan\left(\frac{\delta}{2}\right) = \left(\frac{\mu-1}{\mu+1}\right) \tan\left(\frac{i+r}{2}\right)$$

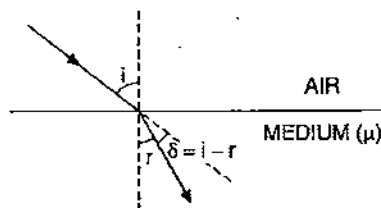
Solution

The angle of deviation δ is given by

$$\delta = i - r \quad \dots(1)$$

According to Snell's Law,

$$\mu = \frac{\sin i}{\sin r}$$

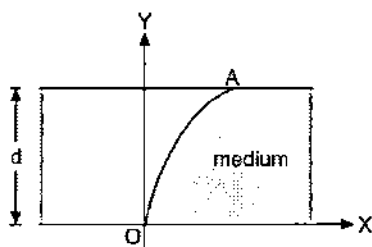


Applying componendo and dividendo, we get

$$\begin{aligned} \frac{\mu-1}{\mu+1} &= \frac{\sin i - \sin r}{\sin i + \sin r} \\ \Rightarrow \frac{\mu-1}{\mu+1} &= \frac{2 \cos\left(\frac{i+r}{2}\right) \sin\left(\frac{i-r}{2}\right)}{2 \sin\left(\frac{i+r}{2}\right) \cos\left(\frac{i-r}{2}\right)} \\ \Rightarrow \frac{\mu-1}{\mu+1} &= \frac{\tan\left(\frac{\delta}{2}\right)}{\tan\left(\frac{i+r}{2}\right)} \\ \Rightarrow \tan\left(\frac{\delta}{2}\right) &= \left(\frac{\mu-1}{\mu+1}\right) \tan\left(\frac{i+r}{2}\right) \end{aligned}$$

Illustration 17

A long rectangular slab of transparent medium of thickness d is placed on a table with length parallel to the x -axis and width parallel to the y -axis. A ray of light is travelling along y -axis at origin. The refractive index μ of the medium varies as $\mu = \frac{\mu_0}{1 - \left(\frac{x}{r}\right)}$, where μ_0 and $r(>1)$ are constants. The refractive index of air is 1.



- Determine the x -coordinate of the point A, where the ray intersects the upper surface of the slab-air boundary.
- Write down the refractive index of the medium at A.
- Indicate the subsequent path of the ray in air.

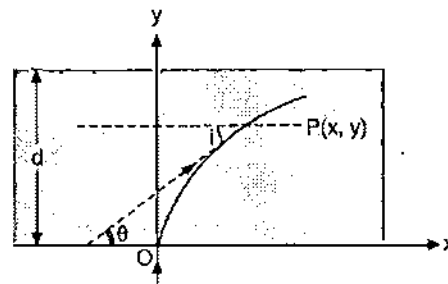
Solution

- Refractive index is a function of x , i.e., the plane separating the two media is parallel to y - z plane or normal to this plane at any point is parallel to x -axis. Further refractive index increases as x is increases. So, the ray of light will bend towards normal and the path is shown in figure. Let at the point $P(x, y)$ the angle of incidence be i . Then

$$\theta = i$$

$$\Rightarrow \tan \theta = \tan i$$

$$\Rightarrow \frac{dy}{dx} = \tan i \quad \dots(1)$$



Applying Snell's Law at O and P, we get

$$\mu_0 \sin i_0 = \mu_P \sin i_P$$

$$\Rightarrow (1)(\sin 90^\circ) = \left(\frac{\mu_0}{1 - \frac{x}{r}}\right) \sin i$$

$$\Rightarrow \sin i = \left(\frac{1 - \frac{x}{r}}{\mu_0}\right)$$

$$\Rightarrow \tan i = \frac{\left(1 - \frac{x}{r}\right)}{\sqrt{\mu_0^2 - \left(1 - \frac{x}{r}\right)^2}} \quad \dots(2)$$

From equations (1) and (2), we get

$$dy = \frac{\left(1 - \frac{x}{r}\right)}{\sqrt{\mu_0^2 - \left(1 - \frac{x}{r}\right)^2}} dx$$

Integrating, we get

$$\int_0^d dy = \int_0^x \frac{1 - \frac{x}{r}}{\sqrt{\mu_0^2 - \left(1 - \frac{x}{r}\right)^2}} dx$$

$$\Rightarrow x = r \left[1 - \sqrt{\mu_0^2 - \left(\frac{d}{r} + \sqrt{\mu_0^2 - 1}\right)^2} \right]$$

- At point A, $1 - \frac{x}{r} = \sqrt{\mu_0^2 - \left(\frac{d}{r} + \sqrt{\mu_0^2 - 1}\right)^2}$

$$\Rightarrow \mu = \frac{\mu_0}{\sqrt{\mu_0^2 - \left(\frac{d}{r} + \sqrt{\mu_0^2 - 1}\right)^2}} \quad \left\{ \because \mu = \frac{\mu_0}{1 - \frac{x}{r}} \right\}$$

- After A, medium is again air. Hence, from Snell's Law, angle of incidence will again become 90° or it will move parallel to y -axis as shown.

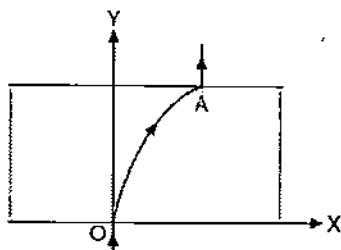
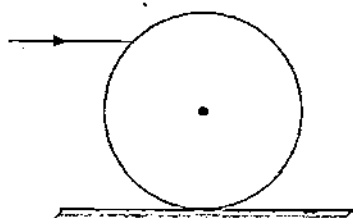


Illustration 18

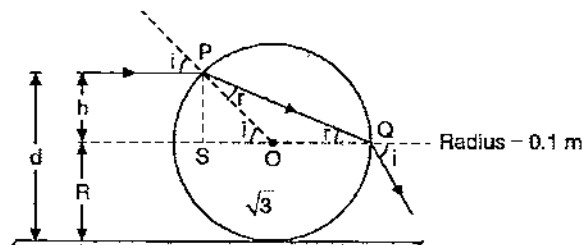
A cylindrical glass rod of radius 0.1 m and refractive index $\sqrt{3}$ lies on a horizontal plane mirror. A horizontal ray of light moving perpendicular to the axis of the rod is incident on it.

- At what height from the mirror should the ray be incident so that it leaves the rod at a height of 0.1 m above the plane mirror?
- At what distance a second similar rod, parallel to the first, be placed on the mirror, such that the emergent ray from the second rod is in line with the incident ray on the first rod?



Solution

Let us first draw the ray diagram for the situation.



- Since, $PO = OQ$
 $\Rightarrow \angle OPQ = \angle OQP = r$ (say)

Also, $i = r + r = 2r$

In $\triangle POS$, we have

$$h = OP \sin i = 0.1 \sin i$$

$$\Rightarrow h = 0.1 \sin 2r$$

$$\Rightarrow h = 0.2 \sin r \cos r$$

...(1)

Applying Snell's Law at P, we get

$$\sqrt{3} = \frac{\sin i}{\sin r} = \frac{2 \sin r \cos r}{\sin r} = 2 \cos r$$

$$\Rightarrow r = 30^\circ$$

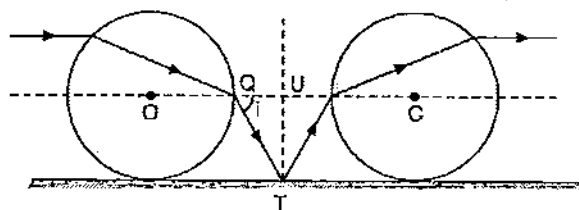
Substituting in equation (1), we get

$$h = (0.2) \left(\frac{1}{2} \right) \left(\frac{\sqrt{3}}{2} \right) = 0.086 \text{ m}$$

Hence, height from the mirror is

$$d = h + R = 0.1 + 0.086 = 0.186 \text{ m}$$

- Using the principle of reversibility of light, we get
 $i = 2r = 60^\circ$



$$\text{Now, } \frac{QU}{TU} = \cot i = \cot(60^\circ) = \frac{1}{\sqrt{3}}$$

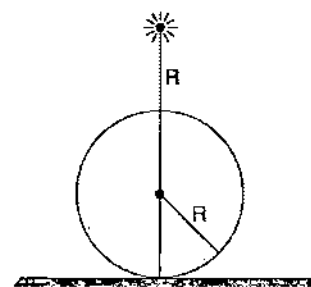
$$\Rightarrow QU = \frac{TU}{\sqrt{3}} = \frac{0.1}{\sqrt{3}}$$

So, the desired distance is

$$OC = 2(0.1) + 2 \left(\frac{0.1}{\sqrt{3}} \right) = 0.315 \text{ m}$$

Illustration 19

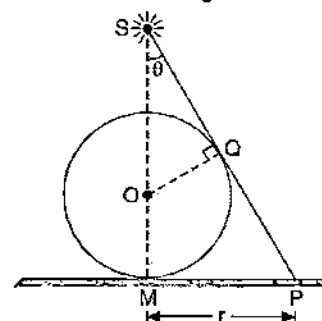
An opaque sphere of radius R lies on a horizontal plane. On the perpendicular through the point of contact there is a point source of light a distance R above the sphere.



- Find the area of the shadow on the plane.
- A transparent liquid of refractive index $\sqrt{3}$ is filled above the plane such that the sphere is just covered with the liquid. Show that new area of the shadow.

Solution

- The situation is shown in the figure



Since, we observe that

$$\sin \theta = \frac{OQ}{OS} = \frac{R}{2R} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Further, radius of shadow is given by

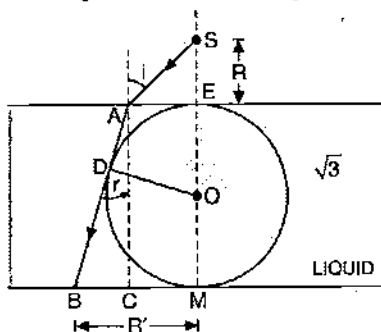
$$r = MP = MS \tan 30^\circ$$

$$\Rightarrow r = (3R) \left(\frac{1}{\sqrt{3}} \right) = \sqrt{3}R$$

So, area of the shadow is

$$A = \pi (\sqrt{3}R)^2 = 3\pi R^2$$

(b) The situation is again shown in the figure.



$$AB = AD + BD$$

$$AB = AE + BD$$

(by geometry)

Further $AB = R' + R \tan i$

$$\Rightarrow \cos r = \frac{AC}{AB} = \frac{2R}{R' + R \tan i}$$

$$\Rightarrow R' = 2R \sec r - R \tan i \quad \dots(1)$$

Also, we observe that

$$R' = BC + CM = 2R \tan r + R \tan i \quad \dots(2)$$

From equations (1) and (2), we get

$$2 \sec r - \tan i = 2 \tan r + \tan i$$

$$\Rightarrow \sec r - \tan r = \tan i$$

Using the concepts of trigonometry, we get

$$\tan i = \frac{1 - \sin r}{\cos r} = \frac{\left(\cos \frac{r}{2} - \sin \frac{r}{2} \right)^2}{\cos^2 \frac{r}{2} - \sin^2 \frac{r}{2}}$$

$$\Rightarrow \tan i = \frac{\cos \left(\frac{r}{2} \right) - \sin \left(\frac{r}{2} \right)}{\cos \left(\frac{r}{2} \right) + \sin \left(\frac{r}{2} \right)} = \frac{1 - \tan \left(\frac{r}{2} \right)}{1 + \tan \left(\frac{r}{2} \right)}$$

$$\Rightarrow \tan i = \tan \left(\frac{\pi}{4} - \frac{r}{2} \right)$$

$$\Rightarrow 2i = \frac{\pi}{2} - \frac{r}{2} \quad \dots(3)$$

Also, from Snell's Law, we get

$$\sin i = \sqrt{3} \sin r \quad \dots(4)$$

Solving the above equations, we get

$$R' = \sqrt{2}R$$

So the new area of shadow is

$$A' = \pi R'^2 = 2\pi R^2$$

CONCEPT OF OPTICAL PATH LENGTH (OPL) AND REDUCED THICKNESS

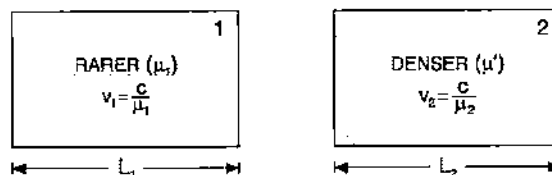
If a distance L separates two buildings, then the measured distance has nothing to do with the medium between the buildings. If this separation is filled with water, then too the distance between the buildings is L . However the time taken by the light to travel between the buildings is different for different media between the buildings. This time difference is due to the interaction of the light with the molecules of the medium which impede (slow down) the light's velocity and this cause the light to take more time to travel the same physical distance for different media.

Due to this a new concept of distance needs to be introduced that accounts for the delay in the travelling time of the light in water (or a denser medium) in comparison to air (or a rarer medium). This new distance is called the **Optical Path Length (OPL) or Optical Path** and takes into account the slower velocity of light within a denser medium and it is simply the product of the distance with the refractive index i.e.,

$$OPL = \mu L$$

Thus, light passing through a denser medium seems to travel a longer distance than the light propagating in free space/vacuum, during the same time intervals for both the media.

Let me illustrate this thing to you. For that let me take two media, one rarer of length L_1 , refractive index μ_1 and other denser of length L_2 and refractive index μ_2 , as shown.



Time taken by light to travel a distance L_1 in rarer medium

with speed $v_1 = \frac{c}{\mu_1}$ is

$$t_1 = \frac{L_1}{v_1} = \frac{L_1}{\left(\frac{c}{\mu_1} \right)} = \frac{\mu_1 L_1}{c} \quad \dots(1)$$

The time taken by light to travel a distance L_2 in denser

medium with a speed $v_2 = \frac{c}{\mu_2}$ is

$$t_2 = \frac{L_2}{v_2} = \frac{L_2}{\left(\frac{c}{\mu_2} \right)} = \frac{\mu_2 L_2}{c} \quad \dots(2)$$

Now if both times are equal, as said above, then

$$t_1 = t_2 \Rightarrow \mu_1 L_1 = \mu_2 L_2 \quad \dots(3)$$

Since, Optical Path Length (OPL) is the distance travelled by light in vacuum/air/rarer medium during the same time it travels a distance L_2 in medium. So, from (3), we get

$$\mu_1 L_1 = \mu_{\text{air}} L_{\text{air}} = \mu_{\text{medium}} L_{\text{medium}} = \mu_2 L_2$$

Since, $\mu_{\text{air}} = 1$, so, we get

$$OPL = L_{\text{air}} = \mu_{\text{medium}} L_{\text{medium}}$$

[for same time in air and medium]

Since light always travels slower in denser medium, so the OPL (the distance in air corresponding to same time in both) is always longer than the actual thickness L of the medium.

CONCEPTUAL NOTE(S)

Also, note that $\mu_1 L_1$ is OPL in air/rarer medium and $\mu_2 L_2$ is OPL in denser medium. However for standard purposes OPL is the distance travelled by light in vacuum/air to travel a distance L in a medium during the same time in either air or medium.

So, from equation (3), we conclude that for a pair of media,

$$\left(\begin{array}{c} \text{Optical Path Length} \\ \text{in Air/Rarer Medium} \end{array} \right) = \left(\begin{array}{c} \text{Optical Path Length} \\ \text{in Denser Medium} \end{array} \right)$$

$$\Rightarrow \mu_1 L_1 = \mu_2 L_2$$

$$\Rightarrow L_2 = \frac{\mu_1}{\mu_2} L_1$$

Since $\mu_1 < \mu_2$, so we get

$$L_2 < L_1$$

Due to this reason, L_2 is also called the **Reduced Thickness**.

So, in general, we get

$$\left(\begin{array}{c} \text{Reduced} \\ \text{Thickness} \end{array} \right) = \left(\frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} \right) L_{\text{rarer}} = \frac{\text{OPL in air}}{\mu_{\text{denser}}}$$

Illustration 20

A light beam of wavelength 600 nm in air passes firstly through film 1 of thickness $1 \mu\text{m}$ and refractive index $n_1 = 1.2$, then through an air film 2 of thickness $1.5 \mu\text{m}$ and finally through film 3 of thickness $1 \mu\text{m}$ and refractive index $n_3 = 1.8$.

- Which film does the light cross in the least time and what is that least time?
- Calculate the total number of wavelengths (at any instant) across all three films together.

Solution

$$(a) \text{ Since, } t_1 = \frac{d_1}{v_1} = \frac{10^{-6}}{\frac{(3 \times 10^8)}{1.2}} = 4 \times 10^{-15} \text{ s} \quad \left(v = \frac{c}{n} \right)$$

$$\text{Similarly, } t_2 = \frac{d_2}{c} = \frac{1.5 \times 10^{-6}}{3 \times 10^8} = 2 \times 10^{-15} \text{ s}$$

$$\text{and } t_3 = \frac{d_3}{v_3} = \frac{10^{-6}}{\frac{(3 \times 10^8)}{1.8}} = 6 \times 10^{-15} \text{ s}$$

$$\text{So, } t_{\text{min}} = 2 \times 10^{-15} \text{ s}$$

- The total number of wavelengths in a film of refractive index μ , thickness d is

$$n = \frac{\text{Optical Path Length}}{\text{Wavelength of Light}}$$

$$\Rightarrow n = \frac{\mu d}{\lambda}$$

So, total number of wavelengths, is

$$n = \frac{\mu_1 d_1}{\lambda} + \frac{\mu_2 d_2}{\lambda} + \frac{\mu_3 d_3}{\lambda}$$

$$\Rightarrow n = \frac{1}{\lambda} (\mu_1 d_1 + \mu_2 d_2 + \mu_3 d_3)$$

$$\Rightarrow n = \frac{10^{-6}}{600 \times 10^{-9}} ((1.2)(1) + (1)(1.5) + (1.8)(1))$$

$$\Rightarrow n = \frac{1000}{600} (4.5) = \frac{4500}{600} = 7.5$$

Illustration 21

A light ray enters the atmosphere of a planet and descends vertically 20 km to the surface. The index of refraction where the light enters the atmosphere is 1 and it increases linearly to the surface where it has a value 1.005. How long does it take the ray to traverse this path.

Solution

Since variation is linear, so we have

$$\frac{x-0}{\mu-1} = \frac{2 \times 10^4 - 0}{1.005 - 1}$$

$$\Rightarrow \mu = 1 + \frac{0.005x}{2 \times 10^4}$$

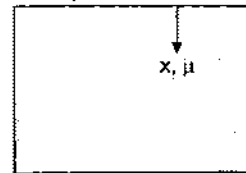
Now, by definition, we have

$$\mu(x) = \frac{c}{v(x)}$$

$$\Rightarrow v(x) = \frac{c}{\mu(x)}$$

$$\text{where } \mu(x) = \mu = 1 + \frac{0.005x}{2 \times 10^4} \text{ and } c = 3 \times 10^8 \text{ ms}^{-1}$$

$$x = 0, \mu = 1$$



$$x_0 = 2 \times 10^4 \text{ m, } \mu = 1.005$$

$$\Rightarrow v(x) = \frac{c}{\mu(x)} = \frac{3 \times 10^8}{1 + \frac{0.005}{2 \times 10^4} x} = \frac{3 \times 10^8}{1 + 2.5 \times 10^{-7} x}$$

$$\Rightarrow \frac{dx}{dt} = \frac{3 \times 10^8}{1 + 2.5 \times 10^{-7} x}$$

$$\Rightarrow dt = \frac{1}{3 \times 10^8} [(1 + 2.5 \times 10^{-7} x) dx]$$

$$\Rightarrow \int_0^t dt = \frac{1}{3 \times 10^8} \int_0^{2 \times 10^4} (1 + 2.5 \times 10^{-7} x) dx$$

$$\Rightarrow t = \frac{1}{3 \times 10^8} \left[2 \times 10^4 + \frac{(2 \times 10^4)^2 (2.5 \times 10^{-7})}{2} \right]$$

$$\Rightarrow t = 6.68 \times 10^{-5} \text{ s}$$

LAWS OF REFRACTION USING FERMAT'S PRINCIPLE

Consider a refracting surface / interface separating medium 1 from medium 2. Let the incident light start from A, in medium 1, hit the surface at O and get refracted to a point B, in medium 2. Let the points A and B be at perpendicular distances a and b from the interface. Further, let A and B be at a separation d as shown in figure. The time taken by the light to go from A to O to B is

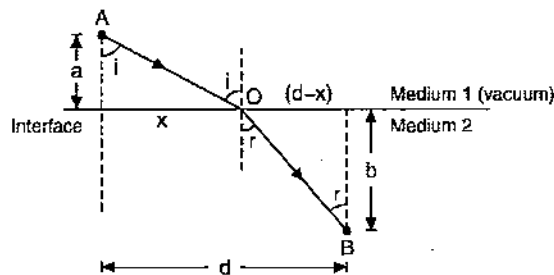
$$t = t_{A \rightarrow O} + t_{O \rightarrow B}$$

$$\Rightarrow t = \frac{AO}{c} + \frac{OB}{v}$$

$$\Rightarrow t = \frac{\sqrt{a^2 + x^2}}{c} + \frac{\sqrt{b^2 + (d-x)^2}}{v}$$

Now, according to Fermat's Theorem, t is MINIMUM, so

$$\frac{dt}{dx} = 0$$



$$\Rightarrow \frac{1}{c} \frac{d}{dx} (\sqrt{a^2 + x^2}) + \frac{1}{v} \frac{d}{dx} (\sqrt{b^2 + (d-x)^2}) = 0$$

$$\Rightarrow \frac{1}{2c} \frac{2x}{\sqrt{a^2 + x^2}} + \frac{1}{2v} \frac{2(d-x)(-1)}{\sqrt{b^2 + (d-x)^2}} = 0$$

$$\Rightarrow \frac{1}{c} \frac{x}{\sqrt{a^2 + x^2}} = \frac{1}{v} \frac{(d-x)}{\sqrt{b^2 + (d-x)^2}}$$

From the figure, we observe that

$$\frac{x}{\sqrt{a^2 + x^2}} = \sin i \text{ and } \frac{d-x}{\sqrt{b^2 + (d-x)^2}} = \sin r$$

$$\Rightarrow \frac{1}{c} \sin i = \frac{1}{v} \sin r$$

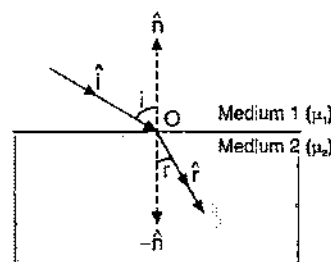
$$\Rightarrow \frac{\sin i}{\sin r} = \frac{c}{v} = \mu \quad \text{[The Law of Refraction]}$$

VECTOR FORM OF SNELL'S LAW

$$\mu_1 \sin i = \mu_2 \sin r$$

Using our knowledge of cross product of vectors, we have

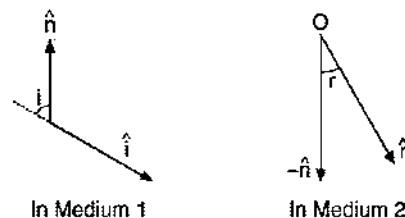
$$\hat{i} \times \hat{n} = (1)(1) \sin i, \odot \text{ outwards}$$



$$\text{and } -\hat{n} \times \hat{r} = \hat{r} \times \hat{n} = (1)(1) \sin r, \odot \text{ outwards}$$

So, from above we conclude that

$$\mu_1 (\hat{i} \times \hat{n}) = \mu_2 (\hat{r} \times \hat{n})$$



REFRACTION THROUGH A COMPOSITE SLAB

Consider the refraction of light ray through a series of media as shown in figure. The ray AB is incident on interface X_1Y_1 at an angle i . The ray is deviated in medium 2 along BC towards the normal. Then it falls on interface X_2Y_2 and is again deviated towards normal along CD. If the last medium is again Medium 1, the ray emerges parallel to the incident ray. Let r_1 and r_2 be angles of refraction in Medium 2 and Medium 3 respectively. Then from Snell's Law,

$$\frac{\sin i}{\sin r_1} = \frac{\mu_2}{\mu_1} = {}^1\mu_2 \quad \dots(1)$$

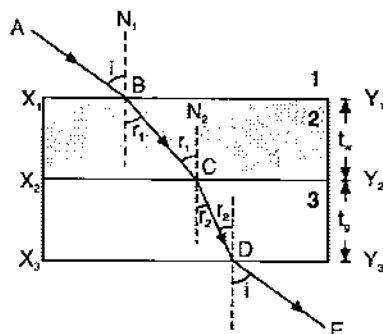
$$\frac{\sin r_1}{\sin r_2} = \frac{\mu_3}{\mu_2} = {}^2\mu_3 \quad \dots(2)$$

$$\frac{\sin r_2}{\sin i} = \frac{\mu_1}{\mu_3} = {}^3\mu_1 \quad \dots(3)$$

μ_1 = refractive index of medium 1

μ_2 = refractive index of medium 2

μ_3 = refractive index of medium 3



Multiplying (1), (2) and (3), we get

$${}^1\mu_2 \times {}^2\mu_3 \times {}^3\mu_1 = 1$$

$$\Rightarrow {}^1\mu_2 \times {}^2\mu_3 = {}^1\mu_3$$

In general if a ray passes through a number of composite parallel plate glass slabs, then

$${}^1\mu_2 \times {}^2\mu_3 \times {}^3\mu_4 \times {}^4\mu_5 = {}^1\mu_5$$

Illustration 22

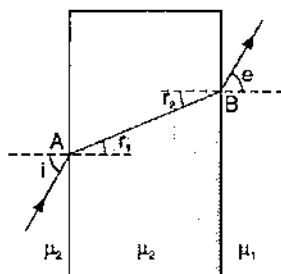
A light beam passes from a parallel plate glass slab of refractive index μ_2 placed in a medium of refractive index μ_1 . Show that the emerging beam is parallel to the incident beam.

Solution

Applying Snell's Law at A, we get

$$\mu_1 \sin i = \mu_2 \sin r_1$$

$$\Rightarrow \frac{\mu_1}{\mu_2} = \frac{\sin r_1}{\sin i} \quad \dots(1)$$



Applying Snell's Law at B, we get

$$\mu_2 \sin r_2 = \mu_1 \sin e$$

$$\Rightarrow \frac{\mu_1}{\mu_2} = \frac{\sin r_2}{\sin e} \quad \dots(2)$$

From equation (1) and (2), we get

$$i = e$$

i.e., the emergent ray is parallel to incident ray.

Illustration 23

Refractive index of glass with respect to water is $9/8$. Refractive index of glass with respect to air is $3/2$. Find the refractive index of water with respect to air.

Solution

$$\text{Given, } {}^w\mu_g = \frac{9}{8} \text{ and } {}^a\mu_g = \frac{3}{2}$$

$$\text{Since, } {}^a\mu_g \times {}^g\mu_w = {}^a\mu_w$$

$$\Rightarrow {}^a\mu_w = \frac{{}^a\mu_g}{{}^g\mu_w} \quad \left\{ \because {}^g\mu_w = \frac{1}{{}^w\mu_g} \right\}$$

$$\Rightarrow {}^a\mu_w = \frac{\left(\frac{3}{2}\right)}{\left(\frac{8}{9}\right)} = \frac{4}{3}$$

LATERAL SHIFT ON PASSING THROUGH A GLASS SLAB

Consider a ray AO incident on the slab at an angle of incidence i through the glass slab EFGH of thickness t . After refraction the ray emerges parallel to the incident ray.

Let PQ be perpendicular dropped from P on incident ray produced as OQ.

The lateral displacement caused by plate,

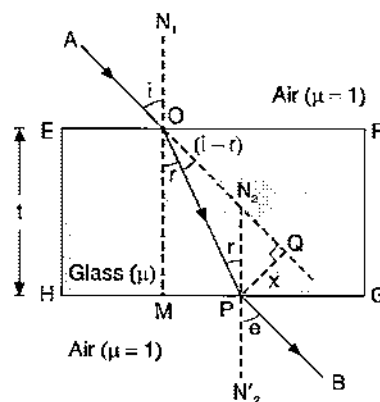
$$x = PQ = OP \sin(i - r)$$

$$x = \frac{OM}{\cos r} \sin(i - r) \quad \left\{ \because OP = \frac{OM}{\cos r} \right\}$$

$$\Rightarrow x = \frac{t}{\cos r} \sin(i - r)$$

$$\Rightarrow x = \frac{t}{\cos r} (\sin i \cos r - \cos i \sin r)$$

$$\Rightarrow x = t(\sin i - \cos i \tan r)$$



$$\text{Since } \mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sin r = \frac{\sin i}{\mu}$$

$$\Rightarrow \tan r = \frac{\sin i}{\sqrt{\mu^2 - \sin^2 i}}$$

$$\Rightarrow y = t \sin i \left(1 - \frac{\cos i}{\sqrt{\mu^2 - \sin^2 i}} \right)$$

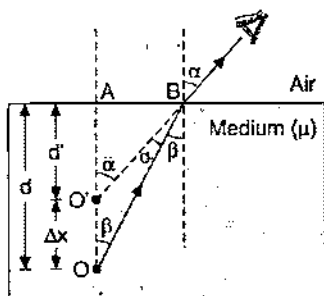
If i is very small then r is also very small, hence
 $\sin i \rightarrow i$, and $\cos i \rightarrow 1$

Then expression for lateral displacement takes the form

$$x = ti \left(1 - \frac{1}{\mu} \right)$$

APPARENT DEPTH

An object O placed in a medium of refractive index μ is observed from air at a small angle α to the normal to the interface (in figure, angle α is shown exaggerated for clarity) i.e., for near normal incidence.



If the object O is at a real depth d from the interface, its apparent depth d' can be calculated. From $\Delta s ABO$ and ABO' ,

$$\frac{\tan \alpha}{\tan \beta} = \frac{d}{d'}$$

Since angles α and β are small, so $\sin \alpha \approx \tan \alpha$ and $\sin \beta \approx \tan \beta$.

Therefore, from Snell's Law, we get,

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin \alpha}{\sin \beta} \approx \frac{\tan \alpha}{\tan \beta} = \frac{d}{d'}$$

$$\Rightarrow \text{Apparent depth, } d' = \frac{d}{\mu}$$

The **apparent shift** in normal direction (or the **normal shift**) in the position of the object is

$$\Delta x = d - d' = d \left(1 - \frac{1}{\mu} \right)$$

In case the object is seen through n number of slabs with different refractive indices, the total apparent shift is simply the sum of individual shifts, so

$$\Delta x = \Delta x_1 + \Delta x_2 + \Delta x_3 + \dots + \Delta x_n$$

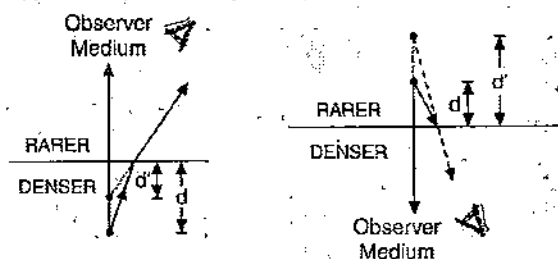
$$\Rightarrow \Delta x = d_1 \left(1 - \frac{1}{\mu_1} \right) + d_2 \left(1 - \frac{1}{\mu_2} \right) + d_3 \left(1 - \frac{1}{\mu_3} \right) + \dots + d_n \left(1 - \frac{1}{\mu_n} \right)$$

CONCEPTUAL NOTE(S)

- a) If the medium in which the object is placed is rarer ($\mu < 1$) and it is seen from the denser medium, the apparent shift calculated will be **negative**. It means that the object apparently shifts away from the observer.

If the shift comes out to be positive, the image of the object shifts towards the observer.

- b) At near normal incidence (small angle of incidence i) apparent depth (d') is given by



$$d' = \frac{d}{\mu_{\text{relative}}} \quad \text{and} \quad v' = \frac{v}{\mu_{\text{relative}}}$$

where $\mu_{\text{relative}} = \frac{\text{RI of medium of incidence object}}{\text{RI of medium of refraction observer}}$

$$= \frac{\mu_{\text{object}}}{\mu_{\text{observer}}}$$

d = distance of object from the interface = real depth.
 d' = distance of image from the interface = apparent depth.
 v = velocity of object perpendicular to interface, relative to surface.
 v' = velocity of image perpendicular to interface relative to surface.

Illustration 24

A fish rising vertically to the surface of water in a lake at a uniform speed of 3 ms^{-1} . It observes that a bird diving vertically towards the water at a uniform speed of 9 ms^{-1} . If the refractive index of water is $\frac{4}{3}$, find the actual speed of dive of the bird.

Solution

Let x be the depth of the fish F below the surface of water, and y be the height of the bird B above the surface at an instant.

To the fish, the bird will appear to be farther away, at an apparent height y' given by

$$y' = \frac{y}{\mu_{\text{relative}}} = \frac{y}{\left(\frac{\mu_{\text{object medium}}}{\mu_{\text{observer medium}}} \right)} = \frac{y}{\left(\frac{\mu_{\text{bird medium}}}{\mu_{\text{fish medium}}} \right)}$$

$$\Rightarrow y' = \frac{y}{\left(\frac{1}{\mu} \right)} = \mu y$$

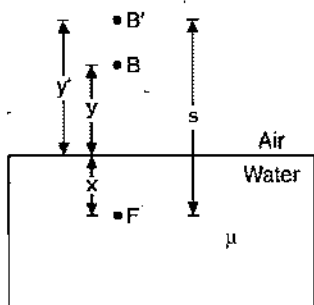
The total apparent distance of the bird from the fish is

$$s = x + y'$$

$$\Rightarrow s = x + \mu y$$

Differentiating w.r.t. time t , we get

$$\frac{ds}{dt} = \frac{dx}{dt} + \mu \frac{dy}{dt}$$



Substituting the values, we get

$$9 = 3 + \left(\frac{4}{3}\right) \frac{dy}{dt}$$

Therefore, the actual speed of dive of the bird is given by

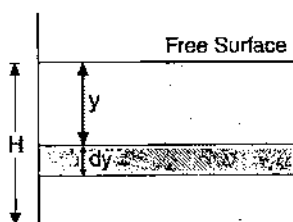
$$\frac{dy}{dt} = (9 - 3) \left(\frac{3}{4}\right) = 4.5 \text{ ms}^{-1}$$

Illustration 25

A vessel is filled with a non-homogeneous liquid whose refractive index varies with the depth y from the free surface of liquid as $\mu = \left(1 + \frac{y}{H}\right)$. Calculate the apparent depth as seen by an observer from above, if H is the height to which the liquid is filled in the vessel.

Solution

Let us consider a thin layer of liquid of thickness dy at a distance y below the free surface of liquid. The apparent depth of this layer having real depth dy is $dH' = \frac{dy}{\mu}$.



$$\Rightarrow dH' = \frac{dy}{\mu}$$

$$\Rightarrow dH' = \frac{dy}{\left(1 + \frac{y}{H}\right)} \quad \left\{ \because \mu = 1 + \frac{y}{H} \right\}$$

Total apparent depth is obtained by integrating this expression within appropriate limits. So,

$$H' = \int dH' = \int_0^H \frac{dy}{\left(1 + \frac{y}{H}\right)} = H \log_e \left(1 + \frac{y}{H}\right) \Big|_0^H$$

$$\Rightarrow H' = H \log_e 2$$

SHIFT OF POINT OF CONVERGENCE OR DIVERGENCE

If a glass slab of thickness t , refractive index μ is placed in the path of a convergent (or divergent) beam of light, the point of convergence (or divergence) gets shifted by

$$\Delta s = t \left(1 - \frac{1}{\mu}\right)$$

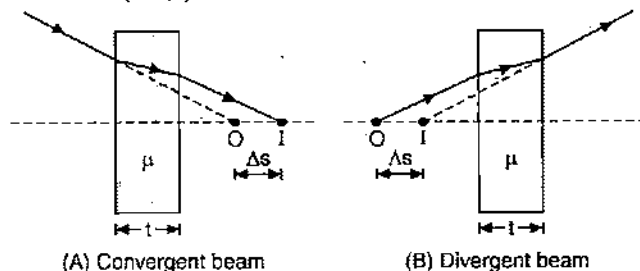
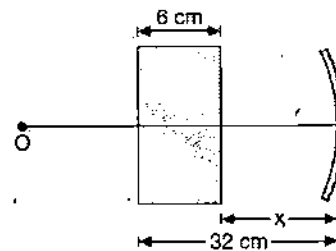


Illustration 26

A point object O is placed in front of a concave mirror of focal length 10 cm. A glass slab of refractive index $\mu = \frac{3}{2}$ and thickness 6 cm is inserted between object and mirror. Find the position of final image when the distance x shown in figure is

- (a) 5 cm
(b) 20 cm



Solution

The normal shift produced by a glass slab is given by

$$\Delta x = \left(1 - \frac{1}{\mu}\right)t = \left(1 - \frac{2}{3}\right)(6) = 2 \text{ cm}$$

i.e., for the mirror the object is placed at a distance $(32 - \Delta x) = 30 \text{ cm}$ from it. Applying mirror formula i.e.

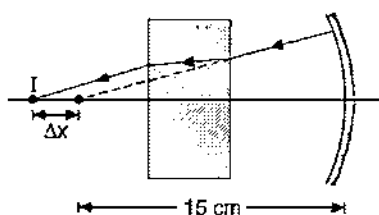
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}, \text{ we get}$$

$$\frac{1}{v} - \frac{1}{30} = -\frac{1}{10}$$

$$\Rightarrow v = -15 \text{ cm}$$

- (a) When $x = 5 \text{ cm}$

The light falls on the slab after being reflected from the mirror as shown. But the slab will again shift it by a distance $\Delta x = 2 \text{ cm}$. Hence, the final real image is formed at a distance $(15 + 2) = 17 \text{ cm}$ from the mirror.



(b) When $x = 20$ cm

This time too the final image is at a distance 17 cm from the mirror but it is virtual as shown.

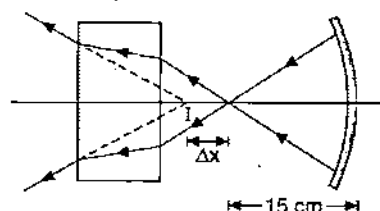
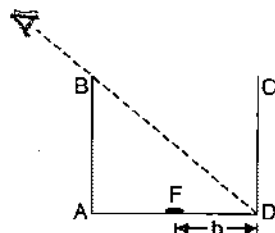


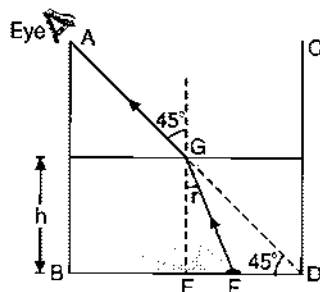
Illustration 27

A cubical vessel with non-transparent walls is so located that the eye of an observer does not see its bottom but sees all of the wall CD. To what height should water be poured into the vessel for the observer to see an object F arranged at a distance of $b = 10$ cm from corner D? The face of the vessel is $a = 40$ cm and refractive index of water is $\frac{4}{3}$.



Solution

Since, the vessel is cubical, $\angle GDE = 45^\circ$ and $GE = ED = h$ (say) then $EF = ED - FD$



$$\text{But } \tan(45^\circ) = 1 = \frac{GE}{ED}$$

$$\Rightarrow ED = GE = h$$

$$\Rightarrow EF = ED - FD = h - 10$$

$$\text{Further, } \frac{4}{3} = \frac{\sin(45^\circ)}{\sin r}$$

$$\Rightarrow r = 32^\circ$$

$$\text{Now } \frac{EF}{GE} = \tan r = \tan(32^\circ)$$

$$\Rightarrow \frac{h-10}{h} = 0.62$$

Solving this, we get

$$h = 26.65 \text{ cm}$$

MULTISLABS

If a number of slabs (or immiscible liquids) of depth $d_1, d_2, d_3, \dots$ and refractive index $\mu_1, \mu_2, \mu_3, \dots$ are placed one over the other, the real depth is

$$d = d_1 + d_2 + d_3 + \dots$$

The apparent depth is given as

$$d' = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} + \frac{d_3}{\mu_3} + \dots$$

Therefore, for the combination, the effective μ is

$$\mu = \frac{d}{d'} = \frac{d_1 + d_2 + d_3 + \dots}{\left(\frac{d_1}{\mu_1}\right) + \left(\frac{d_2}{\mu_2}\right) + \left(\frac{d_3}{\mu_3}\right) + \dots} = \frac{\sum d_i}{\sum \left(\frac{d_i}{\mu_i}\right)}$$

If there are only two slabs, of equal thickness, $d_1 = d_2 = d$,

$$\mu = \frac{d + d}{\left(\frac{d}{\mu_1}\right) + \left(\frac{d}{\mu_2}\right)} = \frac{2\mu_1\mu_2}{\mu_1 + \mu_2} = \text{harmonic mean of } \mu_1 \text{ and } \mu_2$$

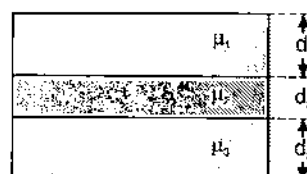


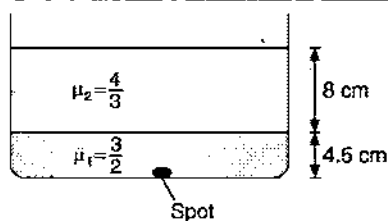
Illustration 28

The bottom of a tub has a black spot. A glass slab of thickness 4.5 cm is placed over it and then water is filled to the height of 8 cm above the glass slab. Looking from top, what shall be the apparent depth of the spot below the water surface? Also find the effective refractive index of the combination of glass slab and water layer. (Refractive index of glass is $\frac{3}{2}$ and of water is $\frac{4}{3}$).

Solution

The apparent depth is given as

$$d_s = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} = \frac{4.5}{\left(\frac{3}{2}\right)} + \frac{8}{\left(\frac{4}{3}\right)} = 3 + 6 = 9 \text{ cm}$$



The effective refractive index is given as

$$\mu_{\text{effective}} = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

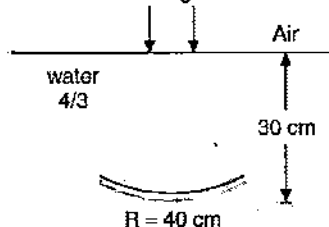
$$\Rightarrow \mu_{\text{effective}} = \frac{d_r}{d_a} = \frac{d_1 + d_2}{d_a} = \frac{4.5 + 8}{9} = \frac{12.5}{9} = 1.39$$

ICE III

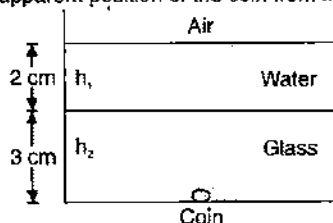
BASED ON GENERAL REFRACTION

(Solutions on page 1.176)

1. An object lies 100 cm inside water. It is viewed from air nearly normally. Find the apparent depth of the object.
2. A concave mirror is placed inside water with its shining surface upwards and principal axis vertical as shown. Rays are incident parallel to the principal axis of concave mirror. Find the position of final image.

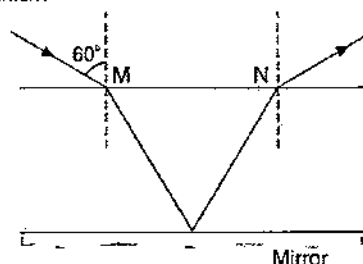


3. A small object is placed on the principal axis of a concave spherical mirror of radius 20 cm at a distance of 30 cm. By how much will the position and size of the image alter, when a parallel-sided slab of glass of thickness 6 cm and refractive index 1.5 is introduced between the centre of curvature and the object? The parallel sides are perpendicular to the principal axis.
4. The velocity of light in air is $3 \times 10^8 \text{ ms}^{-1}$. If yellow light of wavelength $6000 \text{ \AA}$ is passed from air to glass of refractive index 1.5, determine the velocity, the wavelength and the colour of light in glass.
5. A 2 cm thick layer of water covers a 3 cm thick glass slab. A coin is placed at the bottom of the slab and is being observed from the air side along the normal to the surface. Find the apparent position of the coin from the surface.



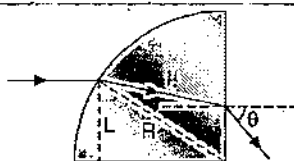
6. A point source of light is arranged at a height h above the surface of water. Where will the image of this source be if the depth of the vessel full of water is d ? Refractive index of water is $n = \frac{4}{3}$. Consider only two steps.

7. A plate with plane parallel faces having refractive index 1.8 rests on a plane mirror. A light ray is incident on the upper face of the plate at 60° . How far from the entry point will the ray emerge after reflection by the mirror of the plate is 6 cm thick?



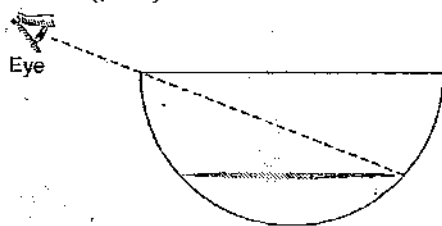
8. A pole 4 m high is driven into the bottom of a lake and happens to be 1 m above the water. Determine the length of the shadow of the pole at the bottom of the lake if the sunrays make an angle of 45° with the water surface. The refractive index of water is $\frac{4}{3}$.
9. A ray of light falls onto a plane-parallel glass plate 1 cm thick at an angle of 60° . The refractive index of the glass is $\sqrt{3}$. Some of the light is reflected and the rest, being refracted, passes into the glass is reflected from the bottom of the plate, refracted a second time and emerges back into the air parallel to the first reflected ray. Determine the distance ℓ between the rays.
10. A ray of light is refracted through a sphere whose material has a refractive index μ in such a way that it passes through the extremities of two radii which make an angle β with each other. prove that if α is the deviation of the ray caused by its passage through the sphere,

$$\cos\left(\frac{\beta - \alpha}{2}\right) = \mu \cos\left(\frac{\beta}{2}\right)$$
11. A vertical beam of light of cross-sectional radius r is incident symmetrically on the curved surface of a glass hemisphere ($\mu = \frac{3}{2}$) of radius $2r$ placed with its base on a horizontal table. Find the radius of the luminous spot formed on the table.
12. A material having an index of refraction μ is surrounded by vacuum and is in the shape of a quarter circle of radius R . A light ray parallel to the base of the material is incident from the left at a distance of L above the base and emerges out of the material at an angle θ . Determine an expression for θ .



13. A circular disc of diameter d lies horizontally inside a metallic hemispherical bowl of radius a . The disc is just visible to an eye looking over the edge. The bowl is now filled with a liquid of refractive index μ . Now, the whole of the disc is just visible to the eye in the same position. Show that

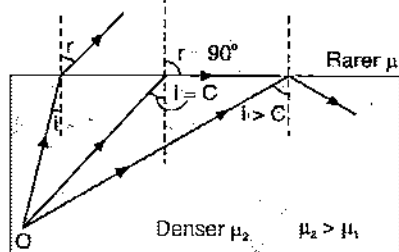
$$d = 2a \left(\frac{\mu^2 - 1}{\mu^2 + 1} \right)$$



TOTAL INTERNAL REFLECTION (TIR)

When a ray of light goes from a denser to a rarer medium, it bends away from the normal. If the angle of incidence in the denser medium is increased the angle of refraction in the rarer medium also increases. At a particular angle of incidence in the denser medium (called as the Critical angle C), the angle of refraction in the rarer medium is 90° (i.e., the refracted ray grazes the interface). This angle of refraction in the denser medium for which the refracted ray grazes the interface is called the critical angle for the pair of interface.

Please note that for small angles of incidence, both reflection and refraction occur, however we shall be neglecting the reflection at the interface as most of the light is refracted. However, when $i > C$, no part of light is refracted and the entire light is reflected back to the denser medium itself. This phenomenon is called total internal reflection (TIR) and was first noted by Kepler in 1604.



Images formed by TIR are much brighter than those formed by the mirrors (or lenses). Some loss of intensity always takes place, when light is reflected from a mirror (or refracted through a lens).

CRITICAL ANGLE

According to Snell's Law, we have

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \frac{\sin C}{\sin 90} = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}}$$

$$\Rightarrow \sin C = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} = \frac{1}{\mu_{\text{denser}}}$$

$$\text{Thus } C = \sin^{-1} \left(\frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} \right) = \sin^{-1} \left(\frac{1}{\mu_{\text{denser}}} \right)$$

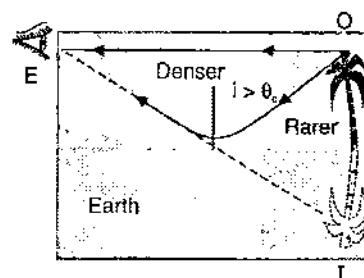
where μ_{denser} is the refractive index of the denser medium w.r.t. the rarer medium. The lesser the value of μ_{denser} , the greater is the critical angle C .

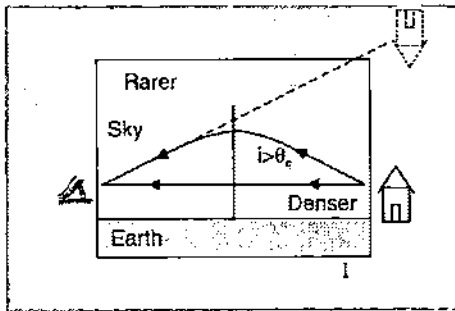
For a given pair of media, since μ depends on the wavelength of light the critical angle also depends on the wavelength. The greater the wavelength, the greater will be the critical angle.

Media Pair	μ_{denser}	Critical angle $C = \sin^{-1} \left(\frac{1}{\mu_{\text{denser}}} \right)$
Water-Air	$\mu_d = \frac{\mu_w}{\mu_a} = \frac{4/3}{1} = \frac{4}{3}$	49°
Glass-Air	$\mu_d = \frac{\mu_g}{\mu_a} = \frac{3/2}{1} = \frac{3}{2}$	42°
Glass-Water	$\mu_d = \frac{\mu_g}{\mu_w} = \frac{3/2}{4/3} = \frac{9}{8}$	63°

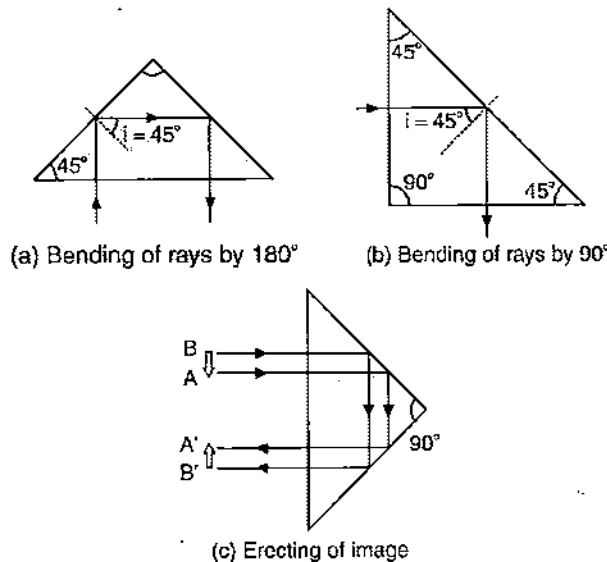
EXAMPLES OF TOTAL INTERNAL REFLECTION

- a) **Mirage** : Mirage is an optical illusion observed in deserts and roads on a hot day. When the air near the ground is hotter (and hence rarer) than the air above, there occurs a continuous decrease of refractive index of air towards the ground. A ray of light from a point O of a tree is, therefore, refracted more and more away from the normal. Ultimately it gets totally reflected to reach the eye E . To the observer it appears to come from I , which is the image of O . This image gives the impression of reflection from a pond of water.
- b) **Looming** : Similarly, in extremely cold regions (near polar regions), the refractive index decreases with height. Due to TIR (shown in figure), the image of a hut appears hanging in the air. This is called looming.





- c) The μ of diamond is 2.5, for which C is only 24° . Diamonds are cut such that $i > C$, so TIR takes place again and again inside it. The light coming out from few meticulously cut surfaces makes it sparkle.
- d) Air bubbles in water shine due to TIR.
- e) The working of an optical fibre is due to multiple TIR inside it.
- f) **Porro prisms** used in periscopes or binoculars bend the ray due to TIR. Some examples are shown in figure.



FIELD OF VISION OF A FISH

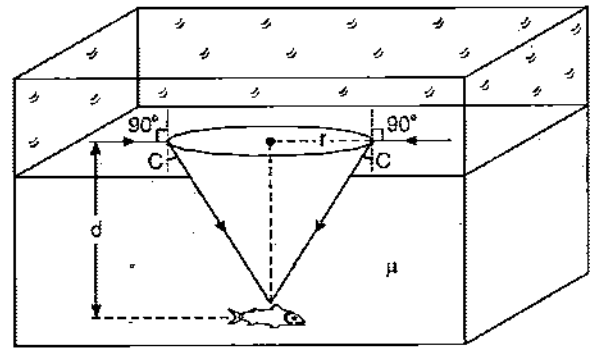
A fish inside a pond does not see the outside world through the entire surface of water. The light from outside can reach the fish only through a circular patch, which forms a cone of half angle equal to the critical angle.

If r is the radius of the circular patch, d is the depth of the fish and μ is the refractive index of water, then

$$r = d \tan C = d \frac{\sin C}{\cos C} = d \frac{\sin C}{\sqrt{1 - \sin^2 C}}$$

Since, $\sin C = \frac{1}{\mu}$

$$\Rightarrow r = \frac{d}{\sqrt{\mu^2 - 1}}$$

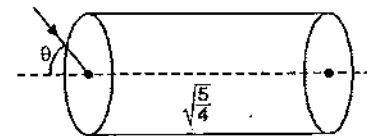


Fish in Glass Tank

Similarly, if a source of light is kept in a pond, its light will come out only through a circular region. For any incident angle i greater than C , the light will be totally reflected back into the water, making corresponding region on the surface of water appear dark.

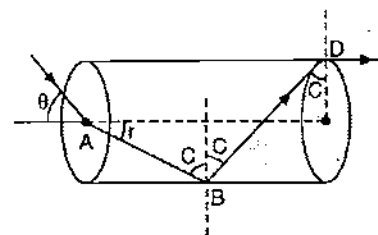
Illustration 29

Light is incident making an angle θ with the axis of a transparent cylindrical fiber of refractive index $n = \sqrt{\frac{5}{4}}$ as shown in figure. Determine the maximum value of θ so that the light entering the cylinder does not come out of the curved surface.



Solution

The ray of light is incident at A and it just gets reflected totally at B . Therefore incident angle at B is equal to the critical angle given as $C = \sin^{-1}\left(\frac{1}{n}\right)$



Snell's Law of refraction at A gives

$$\frac{\sin \theta}{\sin r} = n$$

$$\Rightarrow \sin r = \frac{\sin \theta}{n} \quad \dots(1)$$

Since $r + C = 90^\circ$

$$\Rightarrow \sin r = \sin(90^\circ - C) = \cos C$$

For a ray not to come through the curved surface,

$$r \leq 90^\circ - C$$

$$\Rightarrow \sin r \leq \sqrt{1 - \sin^2 C} \leq \sqrt{1 - \frac{1}{n^2}} \quad \dots(2)$$

Eliminating $\sin r$ from (1) and (2), we get

$$\frac{\sin \theta}{n} \leq \sqrt{1 - \frac{1}{n^2}}$$

$$\Rightarrow \sin \theta \leq \sqrt{n^2 - 1}$$

$$\Rightarrow \sin^2 \theta \leq 1.25 - 1$$

$$\Rightarrow \sin^2 \theta \leq 0.25$$

$$\Rightarrow \sin \theta \leq \frac{1}{2}$$

$$\Rightarrow \theta \leq 30^\circ$$

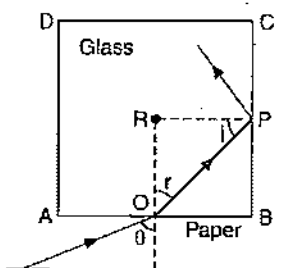
$$\Rightarrow \theta_{\max} = 30^\circ$$

Illustration 30

A rectangular block of glass is placed on a printed page lying on a horizontal surface. Find the minimum value of the refractive index of glass for which the letters on the page are not visible from any of the vertical faces of the block.

Solution

Light will not emerge out from the vertical face BC, when



$i > \text{Critical Angle } (C)$

$$\Rightarrow \sin i > \sin C$$

$$\Rightarrow \sin i > \frac{1}{\mu} \quad \left\{ \because \sin C = \frac{1}{\mu} \right\}$$

Applying Snell's Law at O, we get

$$1 \sin \theta = \mu \sin r$$

$$\Rightarrow \sin \theta = \mu \sin(90^\circ - i) = \mu \cos i$$

$$\Rightarrow \cos i = \frac{\sin \theta}{\mu}$$

$$\Rightarrow \sin i = \sqrt{1 - \cos^2 i} = \sqrt{1 - \frac{\sin^2 \theta}{\mu^2}}$$

Therefore, the condition for no light to emerge from vertical face BC becomes,

$$\sqrt{\frac{\mu^2 - \sin^2 \theta}{\mu^2}} > \frac{1}{\mu}$$

$$\Rightarrow \mu^2 > 1 + \sin^2 \theta$$

$$\Rightarrow \mu^2 > 1 + 1 \quad [\because \text{maximum value of } \theta \text{ can be } 90^\circ]$$

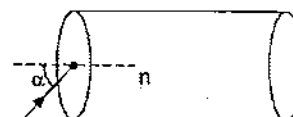
$$\Rightarrow \mu > \sqrt{2}$$

So, the minimum value of refractive index is

$$\mu_{\min} = \sqrt{2}$$

Illustration 31

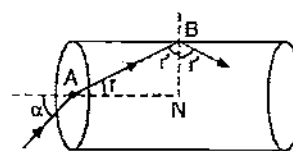
Light is incident at an angle α on one planar end of a transparent cylindrical rod of refractive index n . Determine the least value of n so that the light entering the rod does not emerge from the curved surface of the rod irrespective of the value of α .



Solution

$$\text{Since, } \sin C = \frac{1}{n}$$

$$\text{In triangle } ABN, r' + r + 90^\circ = 180^\circ$$



$$\Rightarrow r' = 90^\circ - r$$

$$\Rightarrow (r')_{\min} = 90^\circ - (r)_{\max}$$

$$\text{and } n = \frac{\sin(i)_{\max}}{\sin(r)_{\max}} = \frac{\sin 90^\circ}{\sin(r)_{\max}} \quad (i_{\max} = 90^\circ)$$

$$\text{Then, } \sin(r)_{\max} = \frac{1}{n} = \sin C$$

$$(r)_{\max} = C$$

$$\Rightarrow (r')_{\min} = (90^\circ - C)$$

Now, if minimum value of r' i.e., $90^\circ - \theta_c$ is greater than θ_c , then obviously all values of r' will be greater than θ_c i.e., total internal reflection will take place at face AB in all conditions. Therefore, the necessary condition is

$$(r')_{\min} \geq C$$

$$\Rightarrow (90^\circ - C) \geq C$$

$$\Rightarrow \sin(90^\circ - C) \geq \sin C$$

$$\Rightarrow \cot C \geq \sin C$$

$$\Rightarrow \cos C \geq 1$$

$$\Rightarrow \sqrt{n^2 - 1} \geq 1$$

$$\Rightarrow n^2 \geq 2$$

$$\Rightarrow n \geq \sqrt{2}$$

Therefore, minimum value of n is $\sqrt{2}$

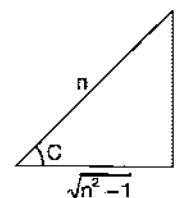


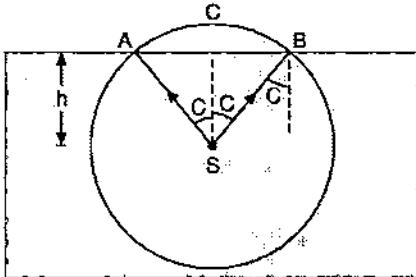
Illustration 32

A point source of light is placed at a distance h below the surface of a large and deep lake. Show that the fraction f of light that escapes directly from water surface is

independent of h and is given by, $f = \frac{1 - \sqrt{1 - \frac{1}{\mu^2}}}{2}$.

Solution

Due to TIR, light will be reflected back into the water for $i > C$. So only that portion of incident light will escape which passes through the cone of angle $\theta = 2C$.



So, the fraction of light escaping is given by

$$f = \frac{\text{Area of Surface ACB}}{\text{Total Area of Sphere}}$$

$$\Rightarrow f = \frac{2\pi R^2(1 - \cos C)}{4\pi R^2} = \frac{1 - \cos C}{2}$$

Now, as f depends on C , which depends only on μ , hence f is independent of h .

Since, we know that

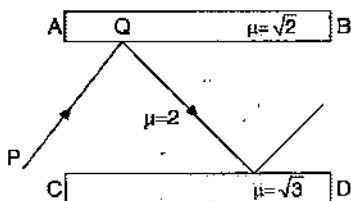
$$\sin C = \frac{1}{\mu}$$

$$\Rightarrow \cos C = \frac{\sqrt{\mu^2 - 1}}{\mu} = \sqrt{1 - \frac{1}{\mu^2}}$$

$$\Rightarrow f = \frac{1 - \sqrt{1 - \frac{1}{\mu^2}}}{2}$$

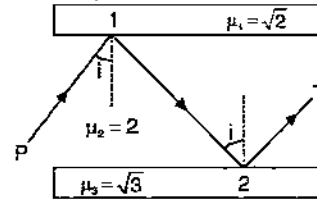
Illustration 33

AB and CD are two slabs. The medium between the slabs has refractive index 2. Find the minimum angle of incidence of Q , so that the ray is totally reflected by both the slabs.



Solution

Let the critical angles at 1 and 2 be C_1 and C_2 respectively. Then



$$C_1 = \sin^{-1}\left(\frac{\mu_1}{\mu_2}\right) = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

$$\text{and } C_2 = \sin^{-1}\left(\frac{\mu_3}{\mu_2}\right) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$$

For TIR, $i > C_2$

Therefore, minimum angle of incidence, for total internal reflection to take place on both slabs must be 60° .

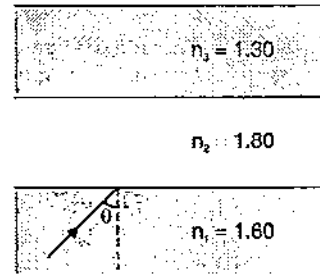
$$i_{\min} = 60^\circ$$

ICE-IV

BASED ON TOTAL INTERNAL REFLECTION (TIR)

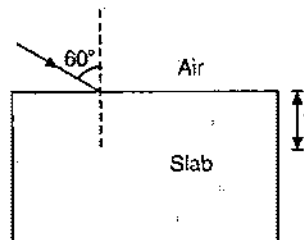
(Solutions on page 1.179)

- Light refracts from medium 1 into a thin layer of medium 2, crosses that layer and then is incident at the critical angle on the interface between media 2 and 3 as shown in figure.



- Find the angle θ .
- If θ is decreased, will the light be refracted to medium 3?

- A ray of light enters into a glass slab from air as shown in figure. If refractive of glass slab varies with t , the thickness of the slab measured from the top as $\mu = A - Bt$ where A and B are constants. Find the maximum depth travelled by ray in the slab. Assume thickness of slab to be sufficiently large.



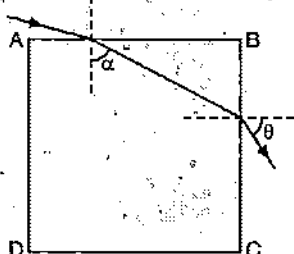
3. A container contains water upto a height of 20 cm and there is a point source of light at the centre of the bottom of the container. A rubber ring of radius a floats centrally on the water. The ceiling of the room is 2 m above the water surface.

- (a) Find the radius of the shadow of the ring formed on the ceiling if $a = 15$ cm.
(b) Find the maximum value of a for which the shadow of the ring is formed on the ceiling.

Refractive index of water = $\frac{4}{3}$.

4. ABCD is the plane of glass cube of refractive index μ . A horizontal beam of light enters the face AB at the grazing incidence.

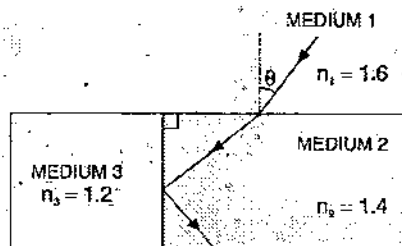
- (a) Show that the angle θ which any ray emerging from BC would make with normal to BC is given by $\sin \theta = \cot \alpha$, where α is the critical angle.



- (b) What is the greatest value that the refraction index of glass may have if any of the light is to emerge from BC?

5. Plot the deviation (δ) versus the angle of incidence (i) graph for a ray travelling from denser to rarer medium.

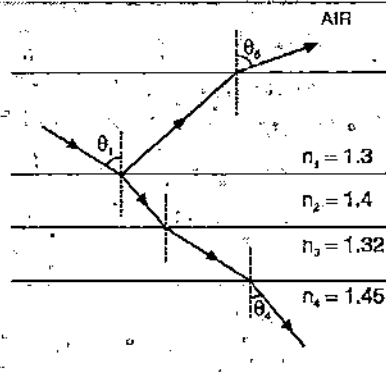
6. In figure, light refracts into material 2, crosses that material and is then incident at the critical angle on the interface between materials 2 and 3.



- (a) What is angle θ ?
(b) If θ is increased, is there refraction of light into material 3?

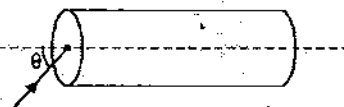
7. An isotropic point source is placed at a depth h below the water surface. An opaque disc capable of floating on water surface is placed on the surface of water so that the bulb is not visible from the surface. Find the minimum radius of the disc for the bulb not to be visible. Take refractive index of water = μ .

8. In figure, light begins from medium of refractive index $n_1 = 1.3$, undergoes three refractions as it heads downward and a reflection and then a refraction to reach the air. The initial angle $\theta_1 = 30^\circ$. Find the values of the angles



- (a) θ_2 and
(b) θ_4 .

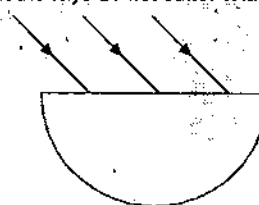
9. Determine the maximum angle θ for which the light ray incident on the end of pipe shown in figure are subject to TIR along the walls of the pipe. Assume that the pipe has an index of refraction of 1.36 and the outside medium is air.



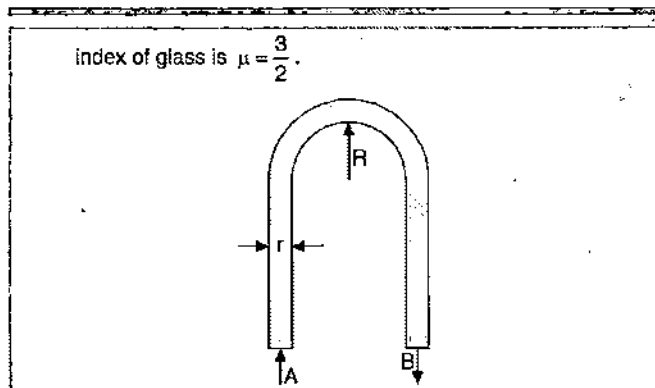
10. A point source of light S is placed at the bottom of a vessel containing a liquid of refractive index $\frac{5}{3}$. A person is viewing the source from above the surface. There is an opaque disc of radius 1 cm floating on the surface. The center of the disc lies vertically above the source S . The liquid from the vessel is gradually drained out through a tap. What is the maximum height of the liquid for which the source cannot at all seen from above.

11. A rectangular glass block is placed on top of a sheet of paper on which there is a small cross. When the paper is soaked in alcohol and a sodium lamp is placed opposite to one vertical of the block the cross can be seen through the opposite vertical face up to a point where the angle of emergence of the light is 30° . If the refractive index of the glass is 1.5, find the refractive index of alcohol. Why can't the black cross be seen through the face when the paper is dry.

12. Rays of light fall on the plane surface of a half cylinder at an angle 45° in the plane perpendicular to the axis (see figure). Refractive index of glass is $\sqrt{2}$. Discuss the condition that the rays do not suffer total internal reflection.



13. Find the maximum value of $\frac{r}{R}$, so that the beam of light incident normally at the face A of a U shaped glass tube emerges through B as shown in the figure. The refractive

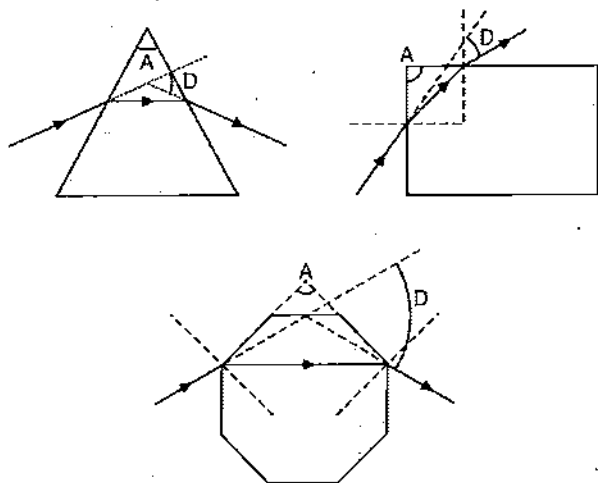


PRISM

Prism is a transparent medium bounded by any number of surfaces in such a way that the surface on which light is incident and the surface from which light emerges are plane and non-parallel.

Refracting angle of prism, or simply the angle of prism is the angle between the faces on which light is incident and from which light emerges. In all the prisms shown in figure above, angle A is the angle of prism.

Angle of deviation (D) is the angle between the incident ray and the emergent ray. Sometimes the angle of deviation is also denoted by δ .

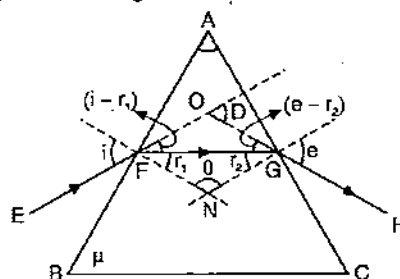


Please note that, for a glass-slab, the angle of prism is zero, and the incident ray emerges parallel to itself, i.e., there is no deviation. If μ of the prism material is same as that of its surroundings, no refraction takes place and light passes through undeviated.

REFRACTION THROUGH A PRISM

Consider a monochromatic ray EF to be incident on the face AB of prism ABC of refracting angle A at angle of incidence i . The ray is refracted along FG , r_1 being angle of refraction. The ray FG is incident on the face AC at angle of incidence r_2 and is refracted in air along GH . Thus GH is the emergent ray and e is the angle of emergence. The angle

between incident ray EF and emergent ray GH (produced backwards) is called angle of deviation D .



In triangle OFG ,

$$D = (i - r_1) + (e - r_2)$$

$$\Rightarrow D = (i + e) - (r_1 + r_2) \quad \dots(1)$$

Also in quadrilateral $AFNG$,

$$A + 90^\circ + \theta + 90^\circ = 360^\circ$$

$$\Rightarrow A + \theta = 180^\circ \quad \dots(2)$$

And in triangle FGN ,

$$r_1 + r_2 + \theta = 180^\circ \quad \dots(3)$$

Comparing equations (2) and (3), we get

$$A = r_1 + r_2 \quad \dots(4)$$

From (1), we get

$$D = i + e - A$$

$$\Rightarrow i + e = A + D \quad \dots(5)$$

If μ is the refractive index of material of prism, then from Snell's Law

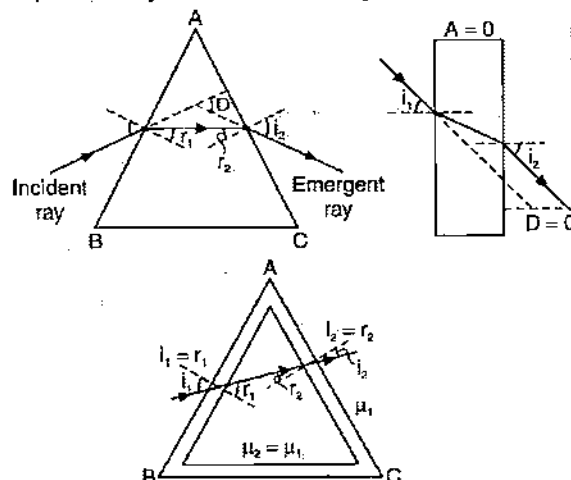
$$\mu = \frac{\sin i}{\sin r_1} = \frac{\sin e}{\sin r_2} \quad \dots(6)$$

For a prism with small refracting angle, we have

$$D = (\mu - 1)A$$

CONCEPTUAL NOTE(S)

- a) Angle of deviation (D) means the angle between emergent and incident rays i.e., the angle through which incident ray turns in passing through a prism. It is represented by D and is shown in figure.



- b) If the faces of a prism on which light is incident and from which it emerges becomes parallel (as in figure), angle of prism will be zero and as incident ray will emerge parallel to itself, deviation will also be zero i.e., the prism will act as a slab.
- c) If μ of the material of the prism becomes equal to that of surroundings, no refraction at its faces will take place and light will pass through it undeviated. So, deviation is zero. i.e., $D = 0$

CONDITION OF NO EMERGENCE

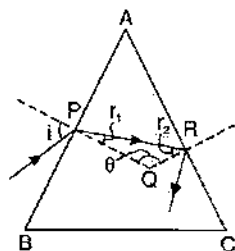
The light entering the prism at surface AB , will not be able to come out from the surface AC , if TIR takes place at this surface. For any angle of incidence, this condition will be satisfied, provided we have at surface AC ,

$$(r_2)_{\min} > C$$

$$\text{Since, } r_1 + r_2 = A$$

$$\Rightarrow r_2 = A - r_1$$

$$\Rightarrow (r_2)_{\min} = A - (r_1)_{\max} \quad \dots(1)$$



But $(r_1)_{\max}$ is possible when $i = i_{\max} = 90^\circ$ i.e., incident ray grazes the interface AB .

Now, applying Snell's Law at AB ,

$$1 \times \sin i = \mu \sin r_1$$

$$\Rightarrow \sin(90^\circ) = \mu \sin r_1$$

$$\Rightarrow r_1 = \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$\Rightarrow r_1 = C \quad \dots(2)$$

From equations (1) and (2), we get

$$r_1 + r_2 > 2C \quad \dots(3)$$

$$\text{Since, } r_1 + r_2 = A$$

Therefore, the condition becomes

$$A > 2C$$

$$\Rightarrow \sin\left(\frac{A}{2}\right) > \sin C$$

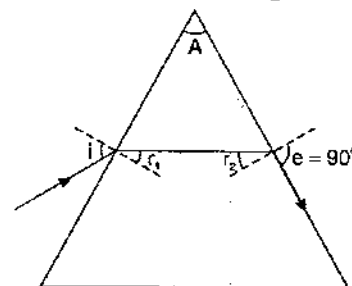
$$\Rightarrow \sin\left(\frac{A}{2}\right) > \frac{1}{\mu}$$

$$\Rightarrow \mu > \operatorname{cosec}\left(\frac{A}{2}\right)$$

Thus, a ray of light will not emerge out of a prism (whatever be the angle of incidence) if $A > 2C$, that is, if $\mu > \operatorname{cosec}\left(\frac{A}{2}\right)$.

CONDITION FOR GRAZING EMERGENCE

A ray can enter a prism in such a way that the angle of emergence, $e = 90^\circ$, as shown in the figure.



We can determine the angle of incidence i for such grazing emergence. We should have

$$r_2 = C$$

$$\text{Since, for a prism, } r_1 + r_2 = A$$

$$\Rightarrow r_1 = A - r_2 = A - C$$

Using Snell's Law,

$$1 \sin i = \mu \sin r_1 = \mu \sin(A - C)$$

$$\Rightarrow \sin i = \mu (\sin A \cos C - \cos A \sin C)$$

$$\Rightarrow \sin i = \mu \left[(\sin A) \sqrt{1 - \sin^2 C} - (\cos A) (\sin C) \right]$$

$$\Rightarrow \sin i = \mu \left[\sin A \sqrt{1 - \frac{1}{\mu^2}} - \cos A \left(\frac{1}{\mu} \right) \right]$$

$$\Rightarrow \sin i = \sin A \sqrt{\mu^2 - 1} - \cos A$$

$$\Rightarrow i = \sin^{-1}(\sin A \sqrt{\mu^2 - 1} - \cos A)$$

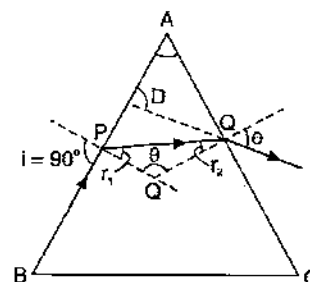
The light will emerge out of the prism only if the angle of incidence i is greater than the above value.

MAXIMUM DEVIATION

The angle of deviation D is maximum when the angle i is maximum, i.e., $i = 90^\circ$.

$$D_{\max} = (i + e) - A = (90^\circ + e) - A$$

Under such conditions of grazing incidence, $r_1 = C$



And at the second surface,

$$\mu \sin r_2 = 1 \sin e$$

$$\Rightarrow \sin e = \mu \sin r_2$$

$$\text{Since } r_1 + r_2 = A$$

$$\Rightarrow \sin e = \mu \sin(A - r_1) = \mu \sin(A - C)$$

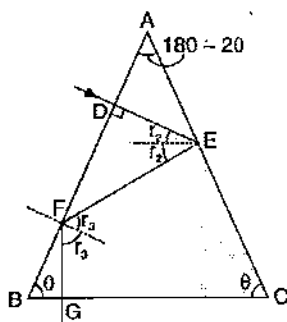
$$\Rightarrow e = \sin^{-1}(\mu \sin(A - C))$$

Illustration 34

An isosceles glass prism has one of its faces coated with silver. A ray of light is incident normally on the other face (which is equal to the silvered face). The ray of light is reflected twice on the same sized faces and then emerges through the base of the prism perpendicularly. Find angles of prism.

Solution

As the ray is incident normally at the face AB , so $r_1 = 0$



Since, we know that $r_1 + r_2 = A$, so we get

$$r_2 = A = 180^\circ - 2\theta \quad \dots(1)$$

$$\text{Now, } \angle DFE = 180^\circ - 90^\circ - 2r_2$$

$$\Rightarrow \angle DFE = 180^\circ - 90^\circ - 360^\circ + 4\theta \quad \{\because r_2 = 180 - 2\theta\}$$

$$\Rightarrow \angle DFE = 4\theta - 270^\circ \quad \dots(2)$$

$$\text{Since, } r_3 = 90^\circ - \angle DFE \quad \dots(3)$$

$$\Rightarrow r_3 = 360^\circ - 4\theta$$

$$\text{Again } \angle BFG = 90^\circ - \theta = 90^\circ - r_3$$

$$\Rightarrow r_3 = \theta \quad \dots(4)$$

From equations (3) and (4), we get

$$5\theta = 360^\circ$$

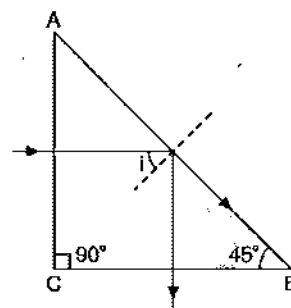
$$\Rightarrow \theta = 72^\circ \text{ and } 180^\circ - 2\theta = 36^\circ$$

So, the angles of prism are 72° , 72° and 36° .

Illustration 35

A ray of light incident normally on one of the faces of a right angle isosceles prism is found to be totally reflected as shown. What is the minimum value of the refractive index of the material of the prism? When prism is immersed in water ($\mu = 1.33$) trace the path of the

emergent ray for the same incident ray, indicating the values of all the angles.



Solution

For total internal reflection to take place at surface AB , we have

$$i > C$$

$$\Rightarrow \sin i > \sin C$$

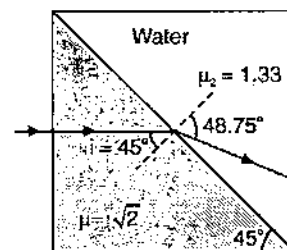
$$\text{Since, } \sin C = \frac{1}{\mu}$$

$$\Rightarrow \sin 45^\circ > \left(\frac{1}{\mu}\right)$$

$$\Rightarrow \mu > \sqrt{2}$$

$$\Rightarrow \mu_{\min} = \sqrt{2}$$

When the prism is immersed in water, the boundary AB now separates glass from water.



$$\Rightarrow C = \sin^{-1}\left(\frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}}\right) = \sin^{-1}\left(\frac{\sqrt{2}}{1.33}\right)$$

$$\Rightarrow C = 70.12^\circ$$

Since $i = 45^\circ$ and also, we observe that $i < C$

Hence, TIR will not take place.

From Snell's Law, we get

$$\mu_1 \sin i = \mu_2 \sin r$$

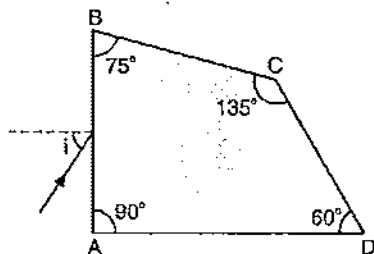
$$\Rightarrow \sqrt{2} \sin(45^\circ) = 1.33 \sin r$$

$$\Rightarrow \sin r = \frac{\sqrt{2} \left(\frac{1}{\sqrt{2}}\right)}{1.33} = 0.752$$

$$\Rightarrow r = \sin^{-1}(0.752) = 48.75^\circ$$

Illustration 36

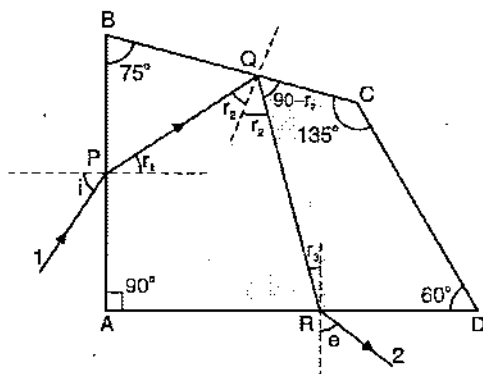
A ray of light is falling on face AB of a tetrahedral of refractive index μ at angle of incidence i . The ray after getting internally reflected on face BC emerges from AD perpendicularly to the incident beam. Find the range of μ and i .



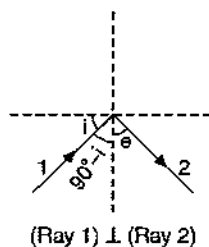
Solution

Since, $r_1 + r_2 = \angle B = 75^\circ$

...(1)



From figure, we observe that $e = i$, because 1 and 2 are perpendicular



In quadrilateral $QCDR$, we have

$$(90^\circ - r_2) + (90^\circ + r_3) + 60^\circ + 135^\circ = 360^\circ$$

$$r_3 = 360^\circ - 60^\circ - 135^\circ - (90^\circ - r_2) - 90^\circ \quad \dots(2)$$

$$r_3 = r_2 - 15^\circ$$

$$\text{Further, } \mu = \frac{\sin i}{\sin r_1} = \frac{\sin e}{\sin r_3}$$

$$\Rightarrow r_3 = r_1 \quad \text{(because } i = e \text{)} \quad \dots(3)$$

Solving equations (1), (2) and (3), we get

$$r_2 = 45^\circ \text{ and } r_1 = 30^\circ$$

Now, for TIR (total internal reflection) to take place at the face BC ; we have

$$r_2 > C$$

$$\Rightarrow \sin r_2 > \sin C$$

$$\Rightarrow \sin(45^\circ) > \frac{1}{\mu}$$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{\mu}$$

$$\Rightarrow \mu > \sqrt{2}$$

$$\left\{ \because \sin C = \frac{1}{\mu} \right\}$$

$$\text{Further, we have } \mu = \frac{\sin i}{\sin r_1} = \frac{\sin i}{\sin(30^\circ)} = 2 \sin i$$

$$\text{Since, } \mu > \sqrt{2}$$

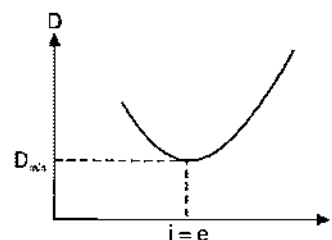
$$\Rightarrow 2 \sin i > \sqrt{2}$$

$$\Rightarrow \sin i > \frac{1}{\sqrt{2}}$$

$$\Rightarrow i > 45^\circ$$

MINIMUM DEVIATION

The angle of deviation D varies with the angle of incidence i , as shown in the figure.



It is found that D is minimum when $i = e$. Thus,

$$D_{\min} = (i + e) - A = 2i - A$$

Using Snell's Law,

$$1 \sin i = \mu \sin r_1$$

$$\text{and } \mu \sin r_2 = 1 \sin e = \sin i$$

$$\Rightarrow \mu \sin r_1 = \mu \sin r_2$$

$$\Rightarrow r_1 = r_2 = r \text{ (say)}$$

$$\text{Since, } r_1 + r_2 = A$$

$$\Rightarrow r = \frac{A}{2}$$

$$\Rightarrow \mu = \frac{\sin i}{\sin r_1} = \frac{\sin \left(\frac{A + D_{\min}}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

Note that if the prism is equilateral or isosceles, then the ray inside the prism is parallel to its base.

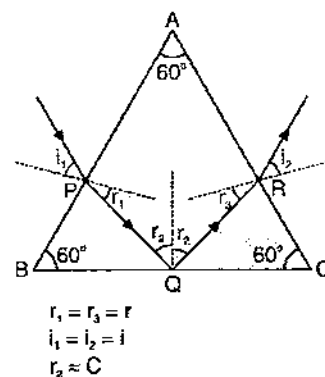
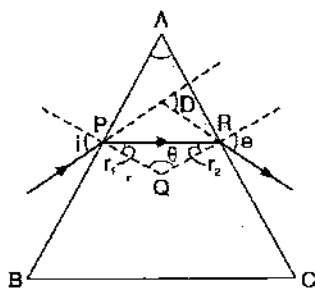


Illustration 37

The angle of minimum deviation for a glass prism with refractive index $\sqrt{3}$ equals the refracting angle of the prism. What is the angle of the prism?

Solution

Since we know that

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Since $\delta_{\text{minimum}} = \delta_{\text{min}} = A$, so we get

$$\sqrt{3} = \frac{\sin A}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \sqrt{3} = 2 \cos\left(\frac{A}{2}\right)$$

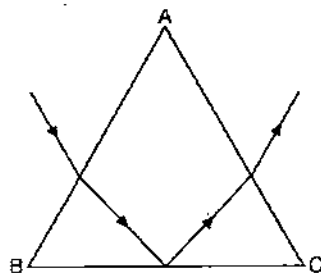
$$\Rightarrow \cos\left(\frac{A}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{A}{2} = 30^\circ$$

$$\Rightarrow A = 60^\circ$$

Illustration 38

The path of a ray of light passing through an equilateral glass prism ABC is shown in the figure.



The ray of light is incident on face BC at an angle just greater than the critical angle for total internal reflection to take place. The total angle of deviation after the refraction at face AC is 120° . Calculate the refractive index of the glass.

Solution

The ray diagram is drawn for the sake of convenience.

Since, $r_1 + r_2 = r_2 + r_3 = 60^\circ$

$$\Rightarrow r_1 = r_3 = r \text{ (say)}$$

Similarly by symmetry, we have $i_1 = i_2 = i$ (say)

Also, $r_2 \approx C$

$$\Rightarrow r = 60^\circ - C$$

Given, $\delta_{\text{Total}} = 120^\circ$

$$\Rightarrow \delta_p + \delta_Q + \delta_R = 120^\circ$$

$$\Rightarrow (i - r) + (180 - 2C) + (i - r) = 120^\circ$$

$$\Rightarrow 2i - 2(60^\circ - C) + 180^\circ - 2C = 120^\circ$$

$$\Rightarrow 2i = 60^\circ$$

$$\Rightarrow i = 30^\circ$$

Now, according to Snell's Law, we have

$$\mu = \frac{\sin i_1}{\sin r_1} = \frac{\sin i}{\sin r}$$

But $r_1 + r_2 = 60^\circ$

$$\Rightarrow r + C = 60^\circ$$

$$\Rightarrow r = 60^\circ - C$$

$$\Rightarrow \mu = \frac{\sin(30^\circ)}{\sin(60^\circ - C)}$$

$$\Rightarrow \mu = \frac{0.5}{\frac{\sqrt{3}}{2} \cos C - \frac{1}{2} \sin C}$$

$$\text{But } \sin C = \frac{1}{\mu}$$

$$\Rightarrow \mu = \frac{0.5}{0.5 \left(\sqrt{3} \sqrt{1 - \frac{1}{\mu^2}} - \frac{1}{\mu} \right)}$$

$$\Rightarrow \mu = \frac{\mu}{\sqrt{3} \sqrt{\mu^2 - 1} - 1}$$

$$\Rightarrow \sqrt{3} \sqrt{\mu^2 - 1} = 2$$

$$\Rightarrow 3(\mu^2 - 1) = 4$$

$$\Rightarrow 3\mu^2 = 7$$

$$\Rightarrow \mu = \sqrt{\frac{7}{3}}$$

$$\Rightarrow \mu = 1.52$$

WHITE LIGHT

White light consists of infinite number of continuous wavelengths (colours) ranging from 4000 Å to 7800 Å. For convenience it is divided into seven colours.

Violet, Indigo, Blue, Green, Yellow, Orange, Red called as 'VIBGYOR' pattern.

The Violet having least wavelength (maximum frequency) and Red having maximum wavelength (minimum frequency).

VARIATION OF REFRACTIVE INDEX WITH COLOUR (CAUCHY'S FORMULA)

The refractive index (μ) of a medium varies with wavelength (λ) according to Cauchy's formula

$$\mu = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + \dots$$

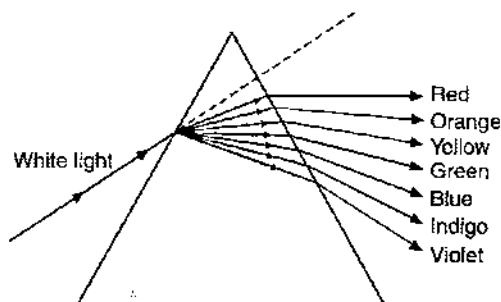
where A , B and C are constants.

From above we observe that refractive index decreases with increase of wavelength. It is maximum for violet and minimum for red colour and due to this variation of the refractive index with the wavelength or the colour, a composite beam of light entering a prism splits into constituent colours.

DISPERSION

It has been observed that when a beam of composite light (consisting of several wavelengths) passes through a prism, it splits into its constituent colours. This phenomenon is called **dispersion**. The band of colours thus obtained on a screen is called the **spectrum**.

If white light is used, seven colours are obtained as shown in the figure. The sequence of colours is VIBGYOR, from bottom to top.



The dispersion of light takes place because the refractive index μ of the medium depends on the wavelength of light as given by **Cauchy's formula**, according to which

$$\mu = A + \frac{B}{\lambda^2}$$

where A and B are constants. The smaller the value of λ , the larger is the value of μ . Thus, μ is maximum for violet colour and minimum for red. The deviation of a ray depends on μ it is larger for higher μ . Hence, violet suffers the maximum deviation and red the minimum.

If light from sodium lamp falls on a prism then it disperses (breaks) into two lines called D_1 (5890 Å) and D_2 (5896 Å) lines. Thus we observe that a prism causes deviation as well as dispersion.

If D_V , D_R and D_Y are the deviations caused by prism for violet, red and mean yellow rays, then for prism with small refracting angle (A), we have

$$\text{Angular Dispersion } D = D_V - D_R = (\mu_V - \mu_R)A$$

DISPERSIVE POWER OF A PRISM

The ratio of angular dispersion to the mean deviation is called dispersive power, so Dispersive Power is

$$\omega = \frac{\text{Angular Dispersion}}{\text{Mean Deviation}} = \frac{D}{D_Y} = \frac{D_V - D_R}{D_Y}$$

where D_Y is the deviation of mean light i.e., yellow light, whose wavelength is considered as mean of all the wavelengths present. Further for a prism of small refracting angle A , we have

$$D = (\mu - 1)A$$

So, we have

$$D_V = (\mu_V - 1)A, D_R = (\mu_R - 1)A \text{ and } D_Y = (\mu_Y - 1)A$$

So the dispersive power ω becomes

$$\omega = \frac{(\mu_V - \mu_R)A}{(\mu_Y - 1)A} = \frac{(\mu_V - \mu_R)}{\mu_Y - 1} = \frac{d\mu}{\mu - 1}$$

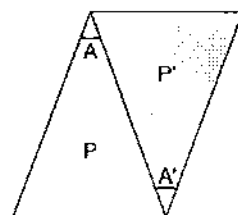
where $d\mu = \mu_V - \mu_R$ and $\mu = \mu_Y$

The dispersive power ω has no units and no dimensions. Its value depends on the material of the prism.

COMBINATION OF TWO PRISMS

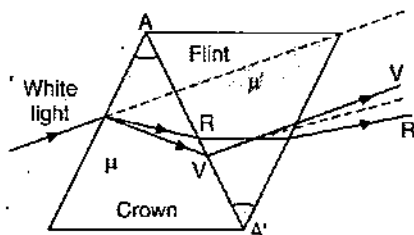
From a single prism, it is not possible to get deviation without dispersion, or to get dispersion without deviation. However, two small angled prisms may be combined to produce Dispersion without Deviation or Deviation without Dispersion. The prism

placement for both is shown here. The placement remains the same. It is just that we are to decide the relation between their refractive indices such that required condition may be achieved.



A. DISPERSION WITHOUT DEVIATION

Two prisms can be combined in such a way that the deviation of the mean ray produced by one is equal and opposite to that produced by the other. Such a combination is called a **direct vision prism**.



So, in this arrangement of prisms, the mean deviation (D) caused by one prism is cancelled by the mean deviation (D') caused by the other prism i.e.

$$D - D' = 0$$

$$\Rightarrow (\mu - 1)A - (\mu' - 1)A' = 0$$

$$\text{or } A' = \left(\frac{\mu - 1}{\mu' - 1} \right) A$$

The net dispersion produced is $= (D_V - D_R) - (D'_V - D'_R)$

$$\text{Net Dispersion} = (\mu_V - \mu_R)A - (\mu'_V - \mu'_R)A'$$

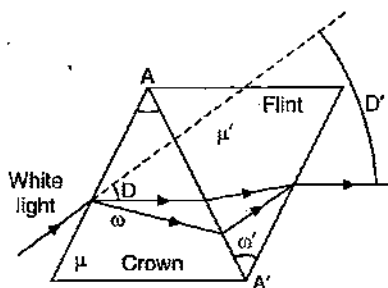
$$\left(\begin{array}{c} \text{Net} \\ \text{Dispersion} \end{array} \right) = \left(\frac{\mu_V - \mu_R}{\mu - 1} \right) (\mu - 1)A - \left(\frac{\mu'_V - \mu'_R}{\mu' - 1} \right) (\mu' - 1)A'$$

$$\Rightarrow \text{Net Dispersion} = \omega D - \omega' D'$$

where ω and ω' are dispersive powers of prisms P and P' .

B. DEVIATION WITHOUT DISPERSION (ACHROMATIC PRISM)

It is possible to combine two prisms of different materials in such a way that each cancels the dispersion due to the other. Thus, the net dispersion is zero but a deviation is produced. So, in this arrangement of prisms, the dispersion ($D_V - D_R$) caused by one prism is cancelled by dispersion ($D'_V - D'_R$) produced by the other prism.



$$\text{i.e., } (D_V - D_R) - (D'_V - D'_R) = 0$$

$$\text{or } (\mu_V - \mu_R)A - (\mu'_V - \mu'_R)A' = 0 \quad \dots(1)$$

This gives

$$A' = \left(\frac{\mu_V - \mu_R}{\mu'_V - \mu'_R} \right) A$$

Also from (1) we get

$$(\mu_V - \mu_R)A = (\mu'_V - \mu'_R)A'$$

$$\Rightarrow \left(\frac{\mu_V - \mu_R}{\mu - 1} \right) (\mu - 1)A = \left(\frac{\mu'_V - \mu'_R}{\mu' - 1} \right) (\mu' - 1)A'$$

$$\Rightarrow \omega D = \omega' D'$$

is the condition for Deviation without Dispersion.

The net mean deviation is

$$D - D' = (\mu - 1)A - (\mu' - 1)A'$$

Illustration 39

The refractive indices of the crown glass for blue and red light are 1.51 and 1.49 respectively and those of the flint glass are 1.77 and 1.73 respectively. An isosceles prism of angle 6° is made of crown glass. A beam of white light is incident at a small angle on this prism. The other flint glass isosceles prism is combined with the crown glass prism such that there is no deviation of the incident prism.

- Determine the angle of the flint glass prism.
- Calculate the net dispersion of the combined system.

Solution

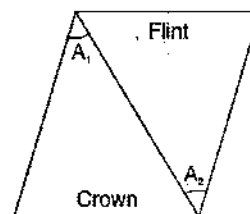
- When angle of prism is small and angle of incidence is also small, the deviation is given by $\delta = (\mu - 1)A$

Net deviation by the two prisms is zero, when deviation due to one cancels the deviation due to the other. So,

$$\delta_1 - \delta_2 = 0$$

$$\Rightarrow (\mu_1 - 1)A_1 - (\mu_2 - 1)A_2 = 0 \quad \dots(1)$$

Here, μ_1 and μ_2 are the refractive indices for crown and flint glasses respectively, where



$$\mu_1 = \frac{1.51 + 1.49}{2} = 1.5 \text{ and } \mu_2 = \frac{1.77 + 1.73}{2} = 1.75$$

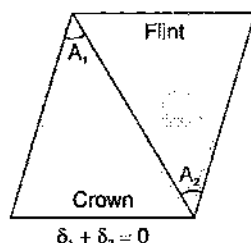
Angle of prism for crown glass is $A_1 = 6^\circ$

Substituting this values in equation (1), we get

$$(1.5 - 1)(6^\circ) - (1.75 - 1)A_2 = 0$$

This gives $A_2 = 4^\circ$

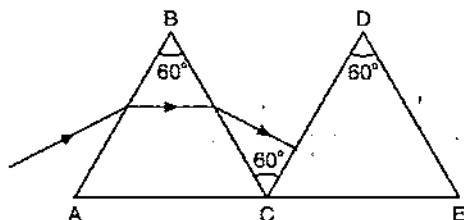
Hence, angle of flint glass prism is 4°



- (b) Net dispersion due to the two prisms is given by
 Net Dispersion = $(\mu_{s_1} - \mu_{e_1})A_1 - (\mu_{s_2} - \mu_{e_2})A_2$
 $\Rightarrow$ Net Dispersion
 $= (1.51 - 1.49)(6^\circ) - (1.77 - 1.73)(4^\circ) = -0.04^\circ$
 $\Rightarrow$ Net dispersion = -0.04°

Illustration 40

A ray of light is incident on a prism ABC of refractive index $\sqrt{3}$ as shown in figure.



- (a) Find the angle of incidence for which the deviation of light ray by the prism ABC is minimum.
 (b) By what angle the second prism must be rotated, so that the final ray suffer net minimum deviation.

Solution

- (a) At minimum deviation, we have $r_1 = r_2 = 30^\circ$

According to Snell's Law, we have

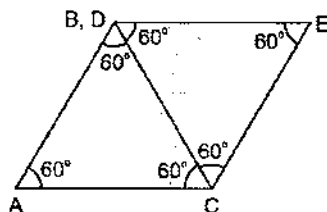
$$\mu = \frac{\sin i_1}{\sin r_1}$$

$$\Rightarrow \sqrt{3} = \frac{\sin i_1}{\sin(30^\circ)}$$

$$\Rightarrow \sin i_1 = \frac{\sqrt{3}}{2}$$

$$\Rightarrow i_1 = 60^\circ$$

- (b) In the position shown, net deviation suffered by the ray of light should be minimum. Therefore, the second prism should be rotated by 60° (anticlockwise).



COLOURS OF OBJECTS AND COLOUR TRIANGLE

The colours of objects are due to a number of phenomena.

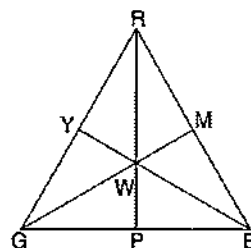
The colours of opaque bodies are due to **Selective Reflection**. For example grass appears green because when white light is incident on grass, it absorbs all colours except green which is reflected. Black appears black because it absorbs all colours falling on it and reflects nothing. Similarly white appears white because it reflects all colours falling on it and absorbs nothing.

The colours of transparent bodies are due to **Selective Transmission**. For example a glass appears blue, because it absorbs all colours except blue, which it transmits.

The colours of sky, rising and setting sun are due to **scattering** while the colours of soap bubble and kerosene oil film are due to **interference**.

COLOUR TRIANGLE

If, Red (R), Green (G) and Blue (B) are primary colours. If P denotes Peacock Blue also called Cyan, M denotes Magenta, Y denotes Yellow and W denotes White, then from colour triangle we observe that



$$R + G + B = W$$

$$R + G = Y$$

$$G + B = P$$

$$R + B = M$$

$$B + Y = W$$

$$R + P = W$$

$$G + M = W$$

RAYLEIGH LAW

According to Lord Rayleigh, intensity (I) of scattered light is inversely proportional to the fourth power of the wavelength λ . So,

$$I \propto \frac{1}{\lambda^4}$$

It can also be concluded that the amplitude (a) of the scattered light is inversely proportional to the square of the wavelength.

$$\text{So, } a \propto \frac{1}{\lambda^2} \quad \left\{ \because I \propto a^2 \right\}$$

COLOUR OF THE SKY

When light from the sun travels through earth's atmosphere, it gets scattered by the large number of molecules of various gases. It is found that the amount of scattering by molecules, called Rayleigh scattering, is inversely proportional to the fourth power of the wavelength. Thus light of shorter wavelength is scattered much more than the light of longer wavelength. Since blue colour has relatively shorter wavelength, it predominates the sky and hence sky appears bluish.

COLOUR OF CLOUDS

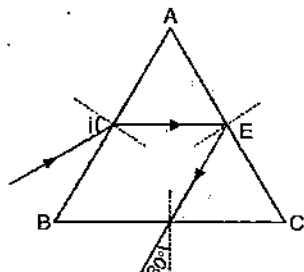
Large particles like water droplets and dust do not have this selective scattering power. They scatter all wavelengths almost equally. Hence clouds appear to the white.

ICEV

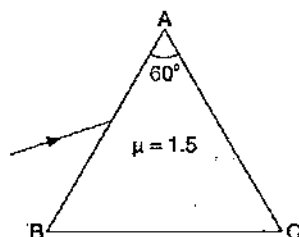
BASED ON PRISM

(Solutions on page 1.183)

- The path of a ray undergoing refraction in an equilateral prism is shown in figure. The ray suffers refraction at the face AB and the refracted ray is incident on the face AC at an angle slightly greater than the critical angle and hence, totally reflected. After refraction at the face BC the emergent ray makes an angle of 30° with normal at BC at the point of emergence. Find the



- corresponding angle of incidence i .
 - refractive index of the prism.
- A beam of light enters a glass prism at an angle α and emerges into the air at an angle β . Having passed through the prism, the beam is deflected from the original direction by an angle δ . Find the refracting angle of the prism and the refractive index of the material of the prism.
 - In a prism of refractive index $\mu = 1.5$ and refracting angle 60° , the condition for minimum deviation is fulfilled. If face AC is polished



- Find the net deviation.

- If the system is placed in water what will be the net deviation? Refractive index of water $= \frac{4}{3}$

- A ray of light incident on the face of a prism is refracted and escapes through an adjacent face. What is the maximum permissible angle of refraction of the prism, if it is made of glass with a refractive index of $\mu = 1.5$.
- In an isosceles prism of angle 45° , it is found that when the angle of incidence is same as the prism angle and the emergent ray grazes the emergent surface.
 - Find the refractive index of the material of the prism.
 - For what angle of incidence the angle of deviation will be minimum?
- A prism of flint glass ($\mu_g = \frac{3}{2}$) with an angle of refraction 30° is placed inside water ($\mu_w = \frac{4}{3}$).
 - At what angle should a ray of light fall on the face of the prism so that inside the prism the ray is perpendicular to the bisector of the angle of the prism.
 - Through what angle will the ray turn after passing through both faces of the prism?
- Light rays from a source are incident on a glass prism of index of refraction μ and angle of prism α . At near normal incidence, calculate the angle of deviation of the emerging rays.
- One face of a prism with a refracting angle of 30° is coated with silver. A ray incident on another face at an angle of 45° is refracted and reflected from the silver coated face and retraces its path. What is the refractive index of the prism?
- A ray of light is incident at an angle of 60° at one face of a prism having refracting angle 30° . The ray emerging out of the prism makes an angle of 30° with the incident ray. Find the angle of emergence and calculate the refractive index of the material of the prism.
- A ray of light is incident upon one face of a prism (angle of prism $< \frac{\pi}{2}$) in a direction perpendicular to the other face. Prove that the ray will fail to emerge from the other face if $\cot A < \cot C - 1$, where C is critical angle for the material of prism.
- The index of refraction for violet light in silica flint glass is 1.66 and that for red light is 1.62. Find the angular dispersion of visible light passing through a prism of apex angle 60° , if the angle of incidence is 50° .
- A light ray is passing through a prism with refracting angle $A = 90^\circ$ and refractive index $\mu = 1.3$. Find the minimum and maximum angle of deviation.
- A ray of light is incident at an angle of 60° on the face of a prism having refracting angle 30° . The ray emerging out of the prism makes an angle 30° with the incident ray. Find the angle of emergence of the ray.

14. The refracting angle of a glass prism is 30° . A ray is incident onto one of the faces perpendicular to it. Find the angle δ between the incident ray and the ray that leaves the prism. The refractive index of glass is $n = 1.5$.
15. The refractive index of the material of a prism is 1.6 for a certain monochromatic ray. What should be the maximum angle of incidence of this ray on the prism so that no total internal reflection occurs when the ray leaves the prism? The refracting angle of the prism is 45° .
16. A ray of white light falls onto the side surface of an isosceles prism at such an angle that the red ray leaves the prism normally to the second face. Find the deflection of the red and violet rays from the initial direction if the refraction angle of the prism is 45° . The refractive indices of the prism material for red and violet rays are 1.37 and 1.42, respectively.
17. A parallel beam of light falls normally on the first face of a prism of small refracting angle. At the second face it is partly transmitted and partly reflected, the reflected beam

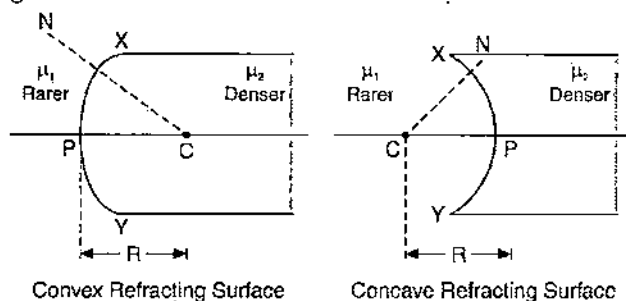
striking at the first face again and emerging from it in a direction making an angle of 4° with the reversed direction of the incident beam. The refracted beam is found to have undergone a deviation of 1° from the original direction. Calculate the refractive index of the glass and the angle of the prism.

Ray Optics

(c) Refraction at Curved Surfaces

SINGLE REFRACTING SURFACE

A spherical surface which separates two media of different refractive index is called a single refracting surface. The convexity or concavity of the surface is decided by looking at it from rarer medium as shown in figure.



SOME TERMS CONNECTED WITH SINGLE REFRACTING SURFACE

- Pole (P)** : It is a point which bulges out most (in case of convex surface) or is depressed most (in case of concave surface) as seen from the rarer medium.
- Centre of Curvature (C)** : It is the centre of the sphere of which the surface forms a part.
- Radius of Curvature (R)** : It is the radius of the sphere of which the surface forms a part.
- Aperture (XY)** : The diameter of the refracting surface is called the aperture of the surface.
- Principal axis** : The line joining the pole and centre of curvature and extended on either side of the surface is called the principal axis.

SIGN CONVENTIONS

Following sign conventions must be used while dealing with ray diagrams.

- All the distances will be measured from the pole of the surface.

- The distances measured against the incident ray are taken as negative.
- The distances measured along the incident ray are taken as positive.
- All transverse measurements done above the principal axis are taken as positive while the ones done below the principal axis are taken as negative.

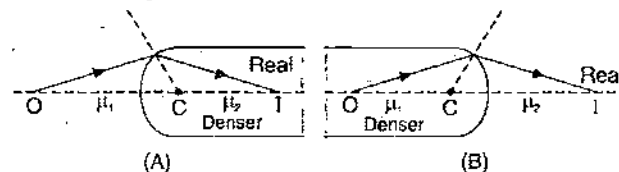
ASSUMPTIONS

While obtaining some relations, in ray optics, we make some assumptions given below. All those formulae will hold good only if these conditions are satisfied.

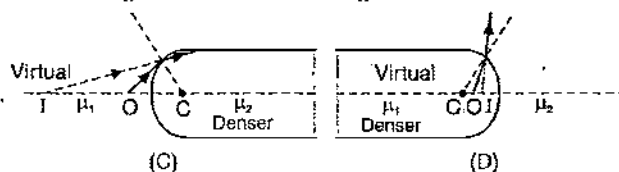
- The object/source is considered to be point object/source placed on principal axis.
- The aperture of the surface/lens is small.
- Rays of light make smaller angles with the principal axis i.e., are paraxial in nature.

REFRACTION OF LIGHT AT CURVED SURFACES

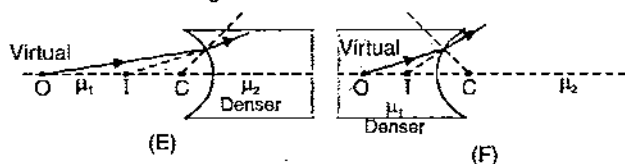
For the curved surfaces the same law of refraction are applicable. When a light-ray enters a denser medium, it bends towards the normal. The figures show six situations. The shaded region is denser.



In Figs. (A) and (B), the object O is kept relatively far from the refracting surface, and the image formed is real.



In Figs. (C) and (D), the object is nearer the refracting surface, and the image is virtual.



In Figs. (E) and (F), the refraction always directs the ray away from the central axis, and hence virtual images are formed.

Note the major difference from the images formed due to reflection from a spherical mirror. Here, real images are formed on the other side of the refracting surface, and virtual images are formed on the same side as the object.

REFRACTION AT CONVEX SURFACE

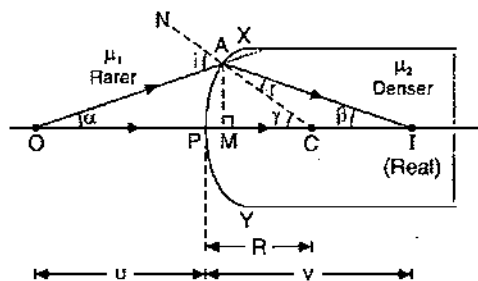
CASE 1 : When the object lies in the rarer medium and the image formed is real.

Consider a spherical surface of radius R separating the two media 1 and 2 ($\mu_2 > \mu_1$). A point object O is placed on the principal axis to the left of the pole P at a considerable distance from it. The incident ray from O falls on point A and is refracted according to

$$\mu_1 \sin i = \mu_2 \sin r \quad \dots(1)$$

Since the rays are paraxial, so the angle α is small and hence the angles i and r will also be small. Thus, applying such paraxial approximation, then $\sin i \approx i$ and $\sin r \approx r$, so we get from (1)

$$\mu_1 i = \mu_2 r \quad \dots(2)$$



Using the geometrical property that an exterior angle of a triangle is equal to the sum of the two internal opposite angles, we get from triangles AOC and AIC ,

$$i = \alpha + \gamma \quad \dots(3)$$

$$\text{and } r = \beta + \gamma \quad \dots(4)$$

Substituting the value of i and r from Equations (3) and (4) in Equation (2), we get

$$\mu_1 (\alpha + \gamma) = \mu_2 (\beta + \gamma) \quad \dots(5)$$

$\Rightarrow \mu_1 \alpha + \mu_2 \beta = (\mu_2 - \mu_1) \gamma$

Now, since the aperture of the refracting surface is small, so M and P are very close to each other and hence, we have

$$\alpha \approx \tan \alpha = \frac{AM}{MO} \approx \frac{AM}{PO}$$

$$\beta \approx \tan \beta = \frac{AM}{MI} \approx \frac{AM}{PI} \quad \text{and}$$

$$\gamma \approx \tan \gamma = \frac{AM}{MC} \approx \frac{AM}{PC}$$

Therefore (5) becomes

$$\Rightarrow \mu_1 \left(\frac{AM}{PO} \right) + \mu_2 \left(\frac{AM}{PI} \right) = (\mu_2 - \mu_1) \left(\frac{AM}{PC} \right)$$

$$\Rightarrow \frac{\mu_1}{PO} + \frac{\mu_2}{PI} = \frac{\mu_2 - \mu_1}{PC}$$

Since $PO = -u$, $PI = +v$, $PC = +R$ so we get

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

If the object is in air, then $\mu_1 = 1$ and $\mu_2 = \mu$, so we get

$$\frac{1}{-u} + \frac{\mu}{v} = \frac{\mu - 1}{R}$$

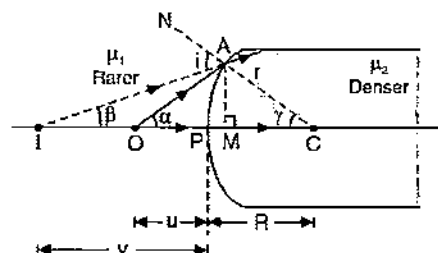
CASE 2 : When the object lies in the rarer medium and the image formed is virtual.

Consider a spherical surface of radius R separating the two media 1 and 2 ($\mu_2 > \mu_1$). A point object O is placed on the principal axis to the left of the pole P . The incident ray from O falls on point A and is refracted according to

$$\mu_1 \sin i = \mu_2 \sin r \quad \dots(1)$$

Since the rays are paraxial, so the angle α is small and hence the angles i and r will also be small. Thus, applying such paraxial approximation, then $\sin i \approx i$ and $\sin r \approx r$, so from (1), we have

$$\mu_1 i = \mu_2 r \quad \dots(2)$$



Using the geometrical property that an exterior angle of a triangle is equal to the sum of the two internal opposite angles, we get from triangles AOC and AIC ,

$$i = \alpha + \gamma \quad \dots(3)$$

$$\text{and } r = \beta + \gamma \quad \dots(4)$$

Substituting the value of i and r from Equations (3) and (4) in Equation (2), we get

$$\mu_1 (\alpha + \gamma) = \mu_2 (\beta + \gamma) \quad \dots(5)$$

$\Rightarrow \mu_1 \alpha - \mu_2 \beta = (\mu_2 - \mu_1) \gamma$

Now, since the aperture of the refracting surface is small, so M and P are very close to each other and hence we have

$$\alpha \cong \tan \alpha = \frac{AM}{MO} \cong \frac{AM}{PO},$$

$$\beta \cong \tan \beta = \frac{AM}{MI} \cong \frac{AM}{PI} \text{ and}$$

$$\gamma \cong \tan \gamma = \frac{AM}{MC} \cong \frac{AM}{PC}$$

$$\Rightarrow \mu_1 \left(\frac{AM}{PO} \right) - \mu_2 \left(\frac{AM}{PI} \right) = (\mu_2 - \mu_1) \left(\frac{AM}{PC} \right)$$

$$\Rightarrow \frac{\mu_1}{PO} - \frac{\mu_2}{PI} = \frac{\mu_2 - \mu_1}{PC}$$

Since $PO = -u$, $PI = -v$, $PC = +R$ so we get

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

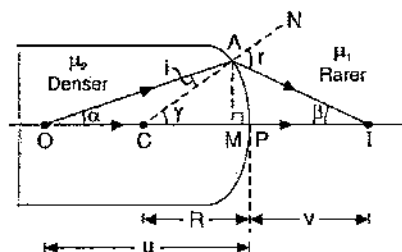
CASE 3 : When the object lies in the denser medium and the image formed is real.

Consider a spherical surface of radius R separating the two media 1 and 2 ($\mu_2 > \mu_1$). A point object O is placed on the principal axis to the left of the pole P . The incident ray from O falls on point A and is refracted according to

$$\mu_2 \sin i = \mu_1 \sin r \quad \dots(1)$$

Since the rays are paraxial, so the angle α is small and hence the angles i and r will also be small. Thus, applying such paraxial approximation, then $\sin i \cong i$ and $\sin r \cong r$, so from (1), we have

$$\mu_2 i = \mu_1 r \quad \dots(2)$$



Using the geometrical property that an exterior angle of a triangle is equal to the sum of the two internal opposite angles, we get from triangles AOC and AIC

$$\gamma = \alpha + i \quad \dots(3)$$

$$\text{and } r = \beta + \gamma \quad \dots(4)$$

Substituting the value of i and r from Equations (3) and (4) in Equation (2), we get

$$\begin{aligned} \mu_2(\gamma - \alpha) &= \mu_1(\beta + \gamma) \\ \Rightarrow \mu_2\gamma - \mu_2\alpha &= \mu_1\beta + \mu_1\gamma \end{aligned} \quad \dots(5)$$

Now, since the aperture of the refracting surface is small, so M and P are very close to each other and hence we have

$$\alpha \cong \tan \alpha = \frac{AM}{MO} \cong \frac{AM}{PO},$$

$$\beta \cong \tan \beta = \frac{AM}{MI} \cong \frac{AM}{PI} \text{ and}$$

$$\gamma \cong \tan \gamma = \frac{AM}{MC} \cong \frac{AM}{PC}$$

$$\Rightarrow \mu_2 \left(\frac{AM}{PO} \right) + \mu_1 \left(\frac{AM}{PI} \right) = (\mu_2 - \mu_1) \left(\frac{AM}{PC} \right)$$

$$\Rightarrow \frac{\mu_2}{PO} + \frac{\mu_1}{PI} = \frac{\mu_2 - \mu_1}{PC}$$

Since $PO = -u$, $PI = +v$, $PC = -R$ so we get

$$\frac{\mu_2}{-u} + \frac{\mu_1}{v} = \frac{\mu_1 - \mu_2}{R}$$

Simply replace subscript 2 by 1 and 1 by 2 in the formula derived in Case 1 or Case 2.

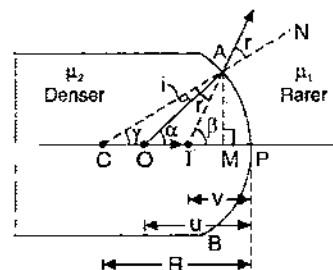
CASE 4 : When the object lies in the denser medium and the image formed is virtual.

Consider a spherical surface of radius R separating the two media 1 and 2 ($\mu_2 > \mu_1$). A point object O is placed on the principal axis to the left of the pole P . The incident ray from O falls on point A and is refracted according to

$$\mu_2 \sin i = \mu_1 \sin r \quad \dots(1)$$

Since the rays are paraxial, so the angle α is small and hence the angles i and r will also be small. Thus, applying such paraxial approximation, then $\sin i \cong i$ and $\sin r \cong r$, so from (1), we have

$$\mu_2 i = \mu_1 r \quad \dots(2)$$



Using the geometrical property that an exterior angle of a triangle is equal to the sum of the two internal opposite angles, we get from triangles AOC and AIC

$$\alpha = i + \gamma \quad \dots(3)$$

$$\text{and } \beta = r + \gamma \quad \dots(4)$$

Substituting the value of i and r from Equations (3) and (4) in Equation (2), we get

$$\begin{aligned} \mu_2(\alpha - \gamma) &= \mu_1(\beta - \gamma) \\ \Rightarrow \mu_2\alpha - \mu_2\gamma &= \mu_1\beta - \mu_1\gamma \end{aligned} \quad \dots(5)$$

Now, since the aperture of the refracting surface is small, so M and P are very close to each other and hence we have

$$\alpha \cong \tan \alpha = \frac{AM}{MO} \cong \frac{AM}{PO},$$

$$\beta \cong \tan \beta = \frac{AM}{MI} \cong \frac{AM}{PI} \text{ and}$$

$$\gamma \cong \tan \gamma = \frac{AM}{MC} \cong \frac{AM}{PC}$$

$$\Rightarrow \mu_2 \left(\frac{AM}{PO} \right) - \mu_1 \left(\frac{AM}{PI} \right) = (\mu_2 - \mu_1) \left(\frac{AM}{PC} \right)$$

$$\Rightarrow \frac{\mu_2}{PO} - \frac{\mu_1}{PI} = \frac{\mu_2 - \mu_1}{PC}$$

Since $PO = -u$, $PI = -v$, $PC = -R$ so we get

$$\frac{\mu_2}{-u} + \frac{\mu_1}{v} = \frac{\mu_2 - \mu_1}{R}$$

REFRACTION AT CONCAVE SURFACE

For a concave refracting surface the image formed is always virtual irrespective of the placement of the object.

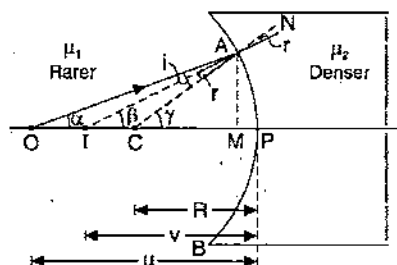
CASE 1 : When the object lies in the rarer medium.

Consider a spherical surface of radius R separating the two media 1 and 2 ($\mu_2 > \mu_1$). A point object O is placed on the principal axis to the left of the pole P . The incident ray from O falls on point A and is refracted according to

$$\mu_1 \sin i = \mu_2 \sin r \quad \dots(1)$$

Since the rays are paraxial, so the angle α is small and hence the angles i and r will also be small. Thus, applying such paraxial approximation, then $\sin i \cong i$ and $\sin r \cong r$, so from (1), we have

$$\mu_1 i = \mu_2 r \quad \dots(2)$$



Using the geometrical property that an exterior angle of a triangle is equal to the sum of the two internal opposite angles, we get from triangles AOC and AIC

$$\gamma = \alpha + i \quad \dots(3)$$

$$\text{and } \gamma = \beta + r \quad \dots(4)$$

Substituting the value of i and r from Equations (3) and (4) in Equation (2), we get

$$\begin{aligned} \mu_1 (\gamma - \alpha) &= \mu_2 (\gamma - \beta) \\ \Rightarrow \mu_1 \alpha - \mu_2 \beta &= (\mu_1 - \mu_2) \gamma \quad \dots(5) \end{aligned}$$

Now, since the aperture of the refracting surface is small, so M and P are very close to each other and hence we have

$$\alpha \cong \tan \alpha = \frac{AM}{MO} \cong \frac{AM}{PO},$$

$$\beta \cong \tan \beta = \frac{AM}{MI} \cong \frac{AM}{PI} \text{ and}$$

$$\gamma \cong \tan \gamma = \frac{AM}{MC} \cong \frac{AM}{PC}$$

$$\Rightarrow \mu_1 \left(\frac{AM}{PO} \right) - \mu_2 \left(\frac{AM}{PI} \right) = (\mu_1 - \mu_2) \left(\frac{AM}{PC} \right)$$

$$\Rightarrow \frac{\mu_1}{PO} - \frac{\mu_2}{PI} = \frac{\mu_1 - \mu_2}{PC}$$

Since $PO = -u$, $PI = -v$, $PC = -R$ so we get

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

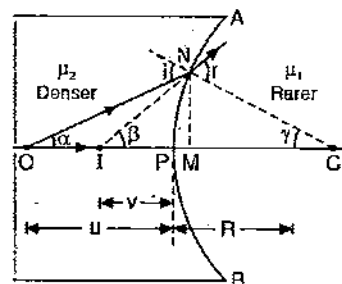
CASE 2 : When the object lies in the denser medium.

Consider a spherical surface of radius R separating the two media 1 and 2 ($\mu_2 > \mu_1$). A point object O is placed on the principal axis to the left of the pole P . The incident ray from O falls on point A and is refracted according to

$$\mu_2 \sin i = \mu_1 \sin r \quad \dots(1)$$

Since the rays are paraxial, so the angle α is small and hence the angles i and r will also be small. Thus, applying such paraxial approximation, then $\sin i \cong i$ and $\sin r \cong r$, so from (1), we have

$$\mu_2 i = \mu_1 r \quad \dots(2)$$



Using the geometrical property that an exterior angle of a triangle is equal to the sum of the two internal opposite angles, we get from triangles AOC and AIC

$$i = \alpha + \gamma \quad \dots(3)$$

$$\text{and } r = \beta + \gamma \quad \dots(4)$$

Substituting the value of i and r from Equations (3) and (4) in Equation (2), we get

$$\begin{aligned} \mu_2 (\alpha + \gamma) &= \mu_1 (\beta + \gamma) \\ \Rightarrow \mu_2 \alpha - \mu_1 \beta &= (\mu_1 - \mu_2) \gamma \quad \dots(5) \end{aligned}$$

Now, since the aperture of the refracting surface is small, so M and P are very close to each other and hence we have

$$\alpha \cong \tan \alpha = \frac{AM}{MO} \cong \frac{AM}{PO},$$

$$\beta \cong \tan \beta = \frac{AM}{MI} \cong \frac{AM}{PI} \text{ and}$$

$$\gamma \cong \tan \gamma = \frac{AM}{MC} \cong \frac{AM}{PC}$$

$$\Rightarrow \mu_2 \left(\frac{AM}{PO} \right) - \mu_1 \left(\frac{AM}{PI} \right) = (\mu_1 - \mu_2) \left(\frac{AM}{PC} \right)$$

$$\Rightarrow \frac{\mu_2}{PO} - \frac{\mu_1}{PI} = \frac{\mu_2 - \mu_1}{PC}$$

Since $PO = -u$, $PI = -v$, $PC = +R$ so we get

$$\frac{\mu_2}{-u} + \frac{\mu_1}{v} = \frac{\mu_2 - \mu_1}{R}$$

CONCEPTUAL NOTE(S)

- For both convex and concave spherical surfaces, the refraction formulae are same, only proper signs of u , v and R are to be used.
- For refraction from rarer to denser medium, the refraction formula is

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

- For refraction from denser to rarer medium, we interchange μ_1 and μ_2 and obtain the refraction formula,

$$\frac{\mu_2}{-u} + \frac{\mu_1}{v} = \frac{\mu_1 - \mu_2}{R}$$

- If the rarer medium is air ($\mu_1 = 1$) and the denser medium has refractive index μ (i.e., $\mu_2 = \mu$), then

- for refraction from air to medium, we have

$$\frac{1}{-u} + \frac{\mu}{v} = \frac{\mu - 1}{R}$$

- for refraction from medium to air, we have

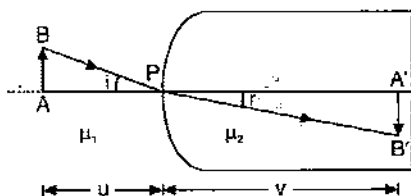
$$\frac{1}{-v} + \frac{\mu}{u} = \frac{\mu - 1}{R}$$

- The factor $\frac{\mu_2 - \mu_1}{R}$ is called power of the spherical refracting surface. It gives a measure of the degree to which the refracting surface can converge or diverge the rays of light passing through it. For air-medium interface, the power is

$$P = \frac{\mu - 1}{R}$$

TRANSVERSE MAGNIFICATION

Instead of a point object O let us now, keep an extended object AB at point O such that its image $A'B'$ will be formed at point I . The distance $x(= -u)$ and $y(= v)$ are related by the above formula.



A ray from point B of the object is incident at point P and is refracted, in accordance with Snell's Law, such that

$$\mu_1 \sin i = \mu_2 \sin r$$

$$\Rightarrow \mu_1 i = \mu_2 r \quad \text{(applying paraxial approximation)}$$

$$\Rightarrow \frac{r}{i} = \frac{\mu_1}{\mu_2} \quad \dots(1)$$

Now, in $\triangle ABP$ and $\triangle A'B'P$, we have

$$AB = u \tan i \approx ui \quad \dots(2)$$

$$\text{and } A'B' = v \tan r \approx vr \quad \dots(3)$$

The magnification is defined as

$$m = \frac{\text{height of image}}{\text{height of object}} = \frac{A'B'}{AB}$$

Using Equations (1), (2) & (3), we get

$$m = \frac{A'B'}{AB} = \frac{vr}{ur} = \left(\frac{v}{u}\right) \left(\frac{\mu_1}{\mu_2}\right)$$

$$\Rightarrow m = \frac{v}{u} \frac{\mu_1}{\mu_2}$$

If m is positive, the image is erect and virtual.

If m is negative, the image is inverted and real.

CONCEPTUAL NOTE(S)

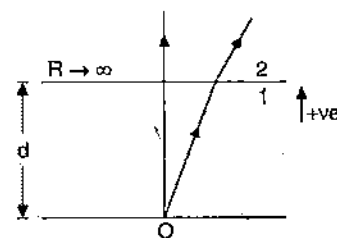
The refraction formula $\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$ is equally applicable to plane refracting surfaces i.e., surfaces for which $R \rightarrow \infty$. Let us derive $\mu = \frac{\text{Real Depth}}{\text{Apparent Depth}}$ using this.

$$\text{Applying } \frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

with proper sign conventions and values, we get

$$\frac{\mu}{-(-d)} + \frac{1}{v} = \frac{1 - \mu}{\infty} = 0$$

$$\Rightarrow v = -\frac{d}{\mu}$$



i.e., image of object O is formed at a distance $\frac{d}{\mu}$ on same side.

$$\text{So, } d_{\text{app}} = \frac{d_{\text{actual}}}{\mu}$$

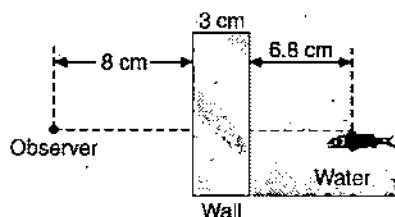
$$\Rightarrow \mu = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

Illustration 41

In figure, a fish watcher watches a fish through a 3 cm thick glass wall of a fish tank. The watcher is in level

with the fish; the index of refraction of the glass is $\frac{8}{5}$ and that of the water is $\frac{4}{3}$.

- (a) To the fish, how far away does the watcher appear to be?
(b) To the watcher, how far away does the fish appear to be?

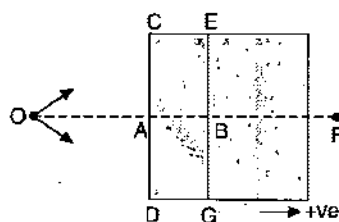


Solution

- (a) $OA = 3$ cm

So, $AI_1 = (n_g)(OA)$

$$\Rightarrow AI_1 = \left(\frac{8}{5}\right)(3) = 4.8 \text{ cm}$$



For refraction at EG ($R \rightarrow \infty$), using

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\Rightarrow \frac{\frac{4}{3}}{BI_2} - \frac{\frac{8}{5}}{-(4.8 + 3)} = 0$$

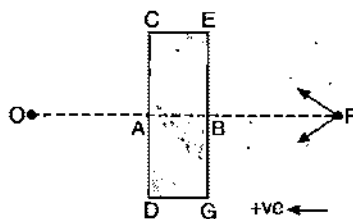
$$\Rightarrow BI_2 = -(7.8) \left(\frac{4}{3}\right) \left(\frac{5}{8}\right) = -6.5 \text{ cm}$$

So, $FI_2 = 6.5 + 6.8 = 13.3$ cm

- (b) For face EF , we have

$$\frac{\frac{8}{5}}{BI_1} - \frac{\frac{4}{3}}{-6.8} = 0 \quad \left\{ \because R \rightarrow \infty \right\}$$

$$\Rightarrow BI_1 = -(6.8) \left(\frac{8}{5}\right) \left(\frac{3}{4}\right) = -8.16 \text{ cm}$$



For face CD , we have

$$\frac{1}{AI_2} - \frac{\frac{8}{5}}{-11.16} = 0 \quad \left\{ \because R \rightarrow \infty \right\}$$

$$\Rightarrow AI_2 = -(11.16) \left(\frac{5}{8}\right) = -6.975 \text{ cm}$$

$$\Rightarrow FI_2 = 8 + 6.975$$

$$\Rightarrow FI_2 = 14.975 \text{ cm}$$

Illustration 42

There are two objects O_1 and O_2 at an identical distance of 20 cm on the two sides of the pole of a spherical concave refracting boundary of radius 60 cm. The indices of refraction of the media on two sides of the boundary are 1 and $\left(\frac{4}{3}\right)$ respectively. Find the location of the object

- (a) O_1 when seen from O_2 .

- (b) O_2 when seen from O_1 .

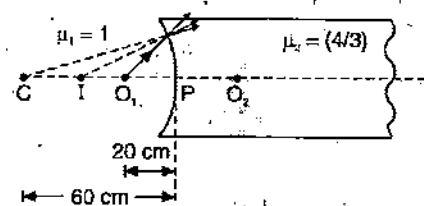
Solution

The formula for refraction from a curved boundary is

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

- (a) From the ray diagram drawn, we get

$$u_1 = -20 \text{ cm}, R = -60 \text{ cm}, \mu_1 = 1, \mu_2 = \frac{4}{3}$$



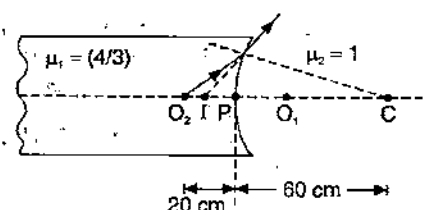
$$\Rightarrow \frac{\left(\frac{4}{3}\right)}{v} - \frac{1}{(-20)} = \frac{\left(\frac{4}{3}\right) - 1}{(-60)}$$

$$\Rightarrow v = -24 \text{ cm}$$

Thus, the object O_1 , will appear at a distance of 24 cm from P towards C .

- (b) Keeping the object O_2 on the left of the pole P as shown, here, we get

$$u_1 = -20 \text{ cm}, R = +60 \text{ cm}, \mu_1 = \frac{4}{3}, \mu_2 = 1$$



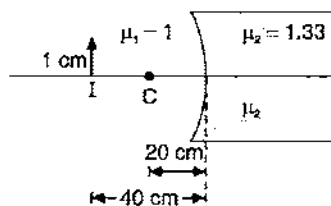
$$\Rightarrow \frac{1}{v} - \frac{\left(\frac{4}{3}\right)}{(-20)} = \frac{1 - \left(\frac{4}{3}\right)}{60}$$

$$\Rightarrow v = -16.36 \text{ cm}$$

Thus, the object O_2 will appear at a distance of 16.36 cm from P towards O_2 .

Illustration 43

An object of height 1 cm is kept at a distance of 40 cm from a concave spherical surface having radius of curvature $R = 20$ cm, separating air and glass having refractive index $\mu = 1.33$. Find the location, height and the nature of the image formed.



Solution

According to Cartesian sign convention,

$$u = -40 \text{ cm}, R = -20 \text{ cm}$$

Applying the formula $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

$$\frac{1.33}{v} - \frac{1}{(-40)} = \frac{1.33 - 1}{(-20)}$$

$$\Rightarrow v = -32 \text{ cm}$$

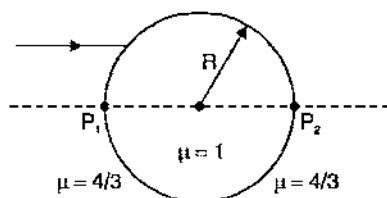
$$\text{Magnification, } m = \frac{h_2}{h_1} = \frac{\mu_1 v}{\mu_2 u} = \frac{1 \times (-32)}{1.33 \times (-40)} = 0.60$$

$$\text{So, height of image, } h_2 = m h_1 = 0.6 \times 1 = 0.6 \text{ cm}$$

The positive sign of magnification indicates that the image is virtual and erect.

Illustration 44

A parallel beam of light travelling in water ($\mu = \frac{4}{3}$) is refracted by a spherical air bubble of radius R situated in water.



- Find the position of the image due to refraction at the first surface and the position of the final image.
- Draw the ray diagram showing the position of the two images.

Solution

- Applying the formula for the refraction at the curved boundary i.e.,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

For, refraction at the first surface, the pole is P_1 and we observe that

$$u \rightarrow -\infty, R = +R, \mu_1 = \frac{4}{3} \text{ and } \mu_2 = 1$$

$$\Rightarrow \frac{1}{v} - \frac{\left(\frac{4}{3}\right)}{(-\infty)} = \frac{1 - \left(\frac{4}{3}\right)}{(+R)}$$

$$\Rightarrow v = -3R$$

Thus, the first image I_1 is formed at a distance of $3R$ to the left of pole P_1 .

This image acts as an object for the refraction at the second surface, with pole P_2 . For this refraction, we have

$$u = -(3R + 2R) = -5R, R = -R, \mu_1 = 1, \mu_2 = \frac{4}{3}$$

$$\Rightarrow \frac{\frac{4}{3}}{v} - \frac{1}{-5R} = \frac{\frac{4}{3} - 1}{-R}$$

$$\Rightarrow v = -\frac{5}{2}R$$

Thus, the final image I_2 is at a distance $\frac{5R}{2}$ from P_2 towards left.

- The ray diagram is shown in figure.

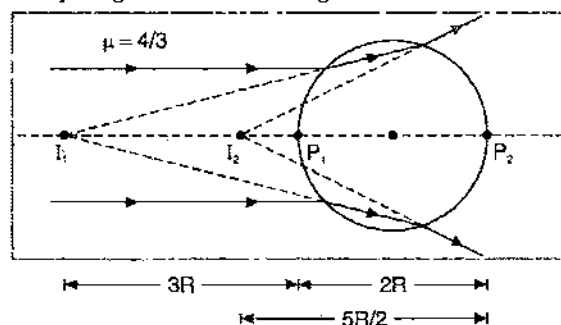


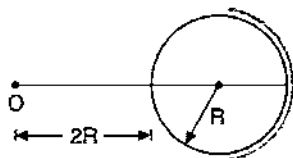
Illustration 45

A solid glass with radius R and an index of refraction 1.5 is silvered over one hemisphere. A small object is located on the axis of the sphere at a distance $2R$ to the left of the vertex of the unsilvered hemisphere. Find the position of final image after all refractions and reflections have taken place.

Solution

The ray of light first gets refracted then reflected and then again refracted. For first refraction and then reflection

the ray of light travels from left to right while for the last refraction it travels from right to left. Hence, the sign convention will change accordingly.



First Refraction :

Using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ with appropriate sign conventions, we get

$$\frac{1.5}{v_1} - \frac{1}{(-2R)} = \frac{1.5 - 1}{+R}$$

$$\Rightarrow v_1 \rightarrow \infty$$

Second Reflection :

Using $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$ with appropriate sign conventions, we get

$$\frac{1}{v_2} + \frac{1}{\infty} = \frac{2}{R}$$

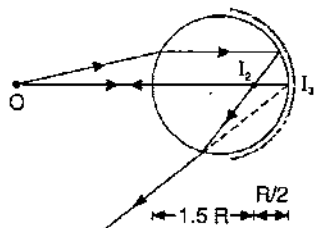
$$\Rightarrow v_2 = \frac{R}{2}$$

Third Reflection :

Again using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ with reversed sign convention, we get

$$\frac{1}{v_3} - \frac{1.5}{-1.5R} = \frac{1 - 1.5}{-R}$$

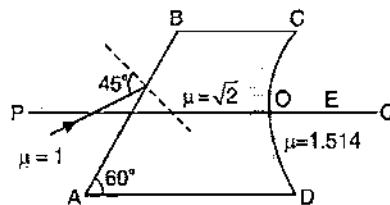
$$\Rightarrow v_3 = -2R$$



i.e., final image is formed at the vertex of the silvered face i.e., at the pole of silvered/curved surface.

Illustration 46

Figure shows an irregular block of material of refractive index $\sqrt{2}$. A ray of light strikes the face AB as shown in the figure. After refraction it is incident on a spherical surface CD of radius of curvature 0.4 m and enters a medium of refractive index 1.514 to meet PQ at E. Find the distance OE upto two places of decimal.



Solution

Applying Snell's Law at face AB, we get

$$(1) \sin 45^\circ = (\sqrt{2}) \sin r$$

$$\Rightarrow \sin r = \frac{1}{2}$$

$$\Rightarrow r = 30^\circ$$

i.e., ray becomes parallel to AD inside the block.

Now applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ on face CD, we get

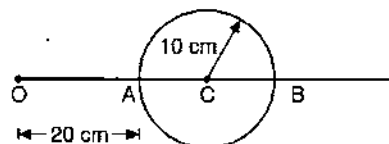
$$\frac{1.514}{OE} - \frac{\sqrt{2}}{\infty} = \frac{1.514 - \sqrt{2}}{0.4}$$

Solving this equation, we get

$$OE \approx 6 \text{ m}$$

Illustration 47

A glass sphere of radius $R = 10 \text{ cm}$ having refractive index $\mu_g = \frac{3}{2}$ is kept inside water. A point object O is placed at 20 cm from A as shown in figure. Find the position and nature of the image when seen from other side of the sphere. Also draw the ray diagram. Given refractive index of water is $\mu_w = \frac{4}{3}$.

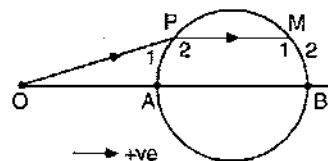


Solution

A ray of light starting from O gets refracted twice. The ray of light is travelling in a direction from left to right. Hence, the distances measured in this direction are taken positive. Applying

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}, \text{ twice with appropriate signs at the}$$

two refracting surfaces, we get



$$\frac{\left(\frac{4}{3}\right)}{-(-20)} + \frac{\left(\frac{3}{2}\right)}{AI_1} = \frac{\left(\frac{3-4}{2-3}\right)}{10}$$

$$\Rightarrow AI_1 = -30 \text{ cm}$$

Now, the first image I_1 , acts as an object for the second surface, so, we have

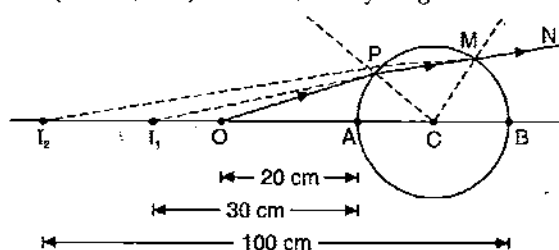
$$BI_1 = u = -(30 + 20) = -50 \text{ cm}$$

Again applying $\frac{\mu_2}{-u} + \frac{\mu_1}{v} = \frac{\mu_1 - \mu_2}{R}$, we get

$$\frac{\left(\frac{3}{2}\right)}{-(-50)} + \frac{\left(\frac{4}{3}\right)}{BI_2} = \frac{\frac{4}{3} - \frac{3}{2}}{-10}$$

$$\Rightarrow BI_2 = -100 \text{ cm}$$

i.e., the final image I_2 is virtual and is formed at a distance 100 cm (towards left) from B. The ray diagram is as shown.

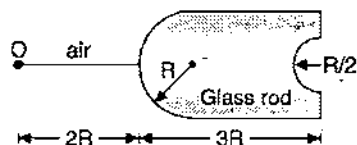


Following points should be kept in mind while drawing the ray diagram.

- At P the ray travels from rarer to a denser medium. Hence, it will bend towards normal PC. At M, it travels from a denser to a rarer medium, hence, moves away from the normal MC.
- The ray PM when extended backwards meets the principal axis at I_1 and the ray MN when extended backwards meets the principal axis at I_2 .

Illustration 48

A glass rod has ends as shown in figure. The refractive index of glass is μ . The object O is at a distance $2R$ from the surface of larger radius of curvature. The distance between apexes of ends is $3R$.



- Find the distance of image formed of the point object from right hand vertex.
- What is the condition to be satisfied if the image is to be real?

Solution

For refraction at curved surface S_1 ,

$$\frac{\mu}{v_1} - \frac{1}{(-2R)} = \frac{(\mu - 1)}{R}$$

$$\Rightarrow \frac{\mu}{v_1} = \frac{2\mu - 3}{2R}$$

$$\Rightarrow v_1 = \frac{2\mu R}{2\mu - 3} \quad \dots(1)$$

The first image acts as object for refraction at second surface S_2 . The origin of our Cartesian coordinate system is now at vertex/pole of surface S_2 . Object distance for second refraction is

$$u_2 = -(3R - v_1)$$

$$\Rightarrow u_2 = -\left(3R - \frac{2\mu R}{2\mu - 3}\right)$$

$$\Rightarrow u_2 = -\left(\frac{4\mu - 9}{2\mu - 3}\right)R$$

For refraction at curved surface S_2 , we have

$$\frac{1}{v_2} - \frac{\mu}{-\left(\frac{4\mu - 9}{2\mu - 3}\right)R} = \frac{(1 - \mu)}{\left(\frac{R}{2}\right)} \quad \dots(2)$$

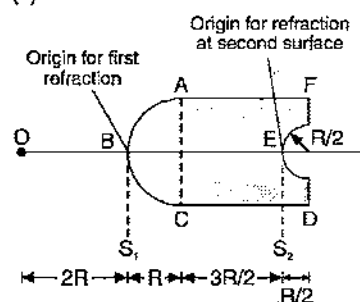
On solving the above expression for v_2 , we get

$$v_2 = \frac{(4\mu - 9)}{(10\mu - 9)(\mu - 2)} \quad \dots(3)$$

The image will be real if v_2 is positive, i.e.,

$$\frac{(4\mu - 9)}{(10\mu - 9)(\mu - 2)} > 0 \quad \dots(4)$$

The equation (4) is satisfied when



Case I :

$$(4\mu - 9) > 0$$

$$\Rightarrow \mu > \frac{9}{4}$$

$$\text{and } (10\mu - 9)(\mu - 2) > 0$$

$$\Rightarrow 0.9 < \mu < 2$$

there is no common solution for this condition and hence this is rejected.

Case II :

$$4\mu - 9 < 0$$

$$\Rightarrow \mu < \frac{9}{4}$$

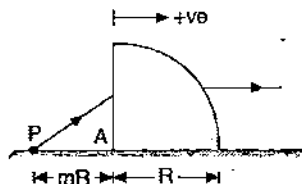
$$(10\mu - 9)(\mu - 2) < 0$$

$$\Rightarrow \mu > 2 \text{ OR } \mu < 0.9$$

Hence the common result is $2 < \mu < 2.25$

Illustration 49

A quarter cylinder of radius R and refractive index 1.5 is placed on a table. A point object P is kept at a distance of mR from it. Find the value of m for which a ray from P will emerge parallel to the table as shown in figure.



Solution

Applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, firstly at the plane surface and then at the curved surface.

For the plane surface, we get

$$\frac{1.5}{AI_1} - \frac{1}{(-mR)} = \frac{1.5 - 1}{\infty} = 0 \quad \left\{ \because R \rightarrow \infty \right\}$$

$$\Rightarrow AI_1 = -(1.5mR)$$

For the curved surface, since the final image is formed at infinity, so we get

$$\frac{1}{\infty} - \frac{1.5}{-(1.5mR + R)} = \frac{1 - 1.5}{-R}$$

$$\Rightarrow \frac{1.5}{(1.5m + 1)R} = \frac{0.5}{R}$$

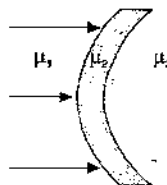
$$\Rightarrow 3 = 1.5m + 1$$

$$\Rightarrow \frac{3}{2}m = 2$$

$$\Rightarrow m = \frac{4}{3}$$

Illustration 50

In the figure, light is incident on the thin lens as shown. The radius of curvature for both the surface is R . Determine the focal length of this system.

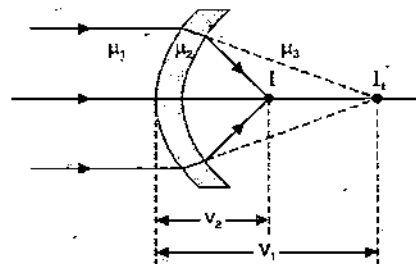


Solution

Since parallel rays after passing through a lens must converge (or appear to converge) at the point. So this point is the place where focus is located and the final image is also formed at the focus.

For refraction at first surface, we get

$$\frac{\mu_2}{v_1} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{+R} \quad \dots(1)$$



For refraction at 2nd surface, we get

$$\frac{\mu_3}{v_2} - \frac{\mu_2}{v_1} = \frac{\mu_3 - \mu_2}{+R} \quad \dots(2)$$

Adding equations (1) and (2), we get

$$\frac{\mu_3}{v_2} = \frac{\mu_3 - \mu_1}{R}$$

$$\Rightarrow v_2 = \frac{\mu_3 R}{\mu_3 - \mu_1}$$

Hence, focal length of the given lens system is

$$f = \frac{\mu_3 R}{\mu_3 - \mu_1}$$

Illustration 51

A parallel beam of light travelling in water having refractive index $\frac{4}{3}$ is refracted by a spherical air bubble of radius 2 mm situated in water. Assuming the light rays to be paraxial.

- Find the position of the image due to refraction at the first surface and the position of the final image.
- Draw a ray diagram showing the positions of both the images.

Solution

- To get the desired result(s), we shall be applying

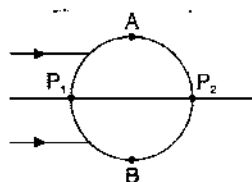
$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}, \text{ one by one on two spherical surfaces.}$$

For first refraction at AP_1B , we have

$$\frac{1}{v_1} - \frac{\frac{4}{3}}{\infty} = \frac{1 - \frac{4}{3}}{+2}$$

$$\Rightarrow \frac{1}{v_1} = -\frac{1}{6}$$

$$\Rightarrow v_1 = -6 \text{ mm}$$



So, the first image will be formed at 6 mm towards left of P_1

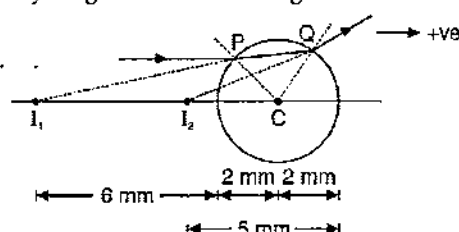
For second refraction at AP_2B , the distance of first image I_1 from P_2 will be $6 \text{ mm} + 4 \text{ mm} = 10 \text{ mm}$ (towards left). So, we get

$$\frac{\frac{4}{3}}{v_2} - \frac{1}{-10} = \frac{\frac{4}{3} - 1}{-2}$$

$$\Rightarrow \frac{4}{3v_2} = -\frac{1}{6} - \frac{1}{10} = -\frac{4}{15}$$

$$\Rightarrow v_2 = -5 \text{ mm}$$

(ii) The ray diagram is shown in figure



CONCEPTUAL NOTE(S)

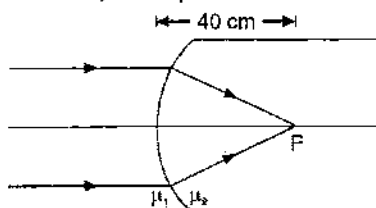
- At P and Q both normal will pass through C.
- At P ray of light is travelling from a denser medium (water) to rarer medium (air) therefore, ray of light will bend away from the normal and on extending meet at I_1 . Similarly at Q, ray of light bends towards the normal.
- Both the images I_1 and I_2 are virtual.

ICE VI

BASED ON REFRACTION AT CURVED SURFACES

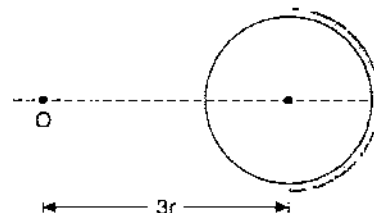
(Solutions on page 1.187)

- A spherical convex surface separates object and image space of refractive index 1 and $\frac{4}{3}$. If radius of curvature of the surface is 10 cm, find its power.

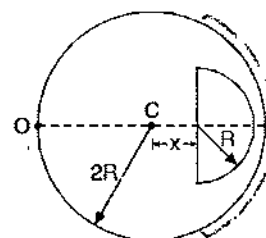


- A hemispherical portion of the surface of a solid glass

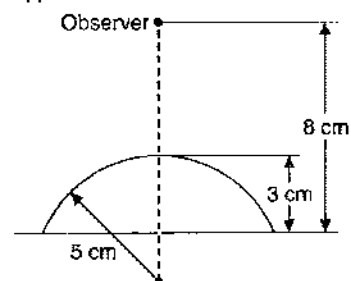
sphere of refractive index 1.5 and of radius r is silvered to make the inner side reflecting. An object is placed at the axis of the sphere at a distance $3r$ from the centre of the sphere. The light from the object is refracted at the unsilvered part, then reflected from the silvered part and again refracted at the unsilvered part. Locate the final image formed.



- Consider the figure shown. A hemispherical cavity of radius R is carved out from a sphere ($\mu = 1.5$) of radius $2R$ such that principal axis of cavity coincides with that of sphere. One side of sphere is silvered as shown. Find the value of x for which the image of an object at O is formed at O itself.



- A glass sphere has a radius of 5 cm and a refractive index of 1.6. A paperweight is constructed by slicing through the sphere on a plane that is 2 cm from the centre of the sphere and perpendicular to a radius of the sphere that passes through the center of the circle formed by the intersection of the plane and the sphere. The paperweight is placed on a table and viewed from directly above by an observer who is 8 cm from the tabletop, as shown in figure. When viewed through the paperweight, how far away does the tabletop appear to the observer?



- One end of a long glass rod having refractive index $\mu = 1.5$ is formed into the shape of a convex surface of radius 6 cm. An object is located in air along the axis of the rod, at a distance of 10 cm from the end of the rod.
 - How far apart are the object and the image formed by the glass rod?
 - For what range of distances from the end of the rod must the object be located in order to produce a virtual image?

6. An object is at a distance of $d = 2.5$ cm from the surface of a glass sphere with a radius $R = 10$ cm. Find the position of the image produced by the sphere. The refractive index of the glass is $\mu = 1.5$.
7. A glass hemisphere of radius 10 cm and refractive index $\mu = 1.5$ is silvered over its curved surface. There is an air bubble in the glass 5 cm from the plane surface along the axis. Find the position of the images of this bubble seen by observer looking along the axis into the flat surface of the hemisphere.
8. A hollow sphere of glass of refractive index μ has a small mark on its interior surface which is observed from a point outside the sphere on the side opposite the center. The inner cavity is concentric with external surface and the thickness of the glass is everywhere equal to the radius of the inner surface. prove that the mark will appear nearer than it really is, by a distance $\frac{(\mu-1)R}{(3\mu-1)}$, where R is the radius of the inner surface.

THIN SPHERICAL LENSES

A lens is a piece of transparent material with two refracting surfaces, at least one of them being curved. It may have one surface plane.

A spherical lens has spherical surfaces as bounds. If the thickness of the lens is small (compared to the radius of curvature of spherical surfaces, the object distance, the image distance, etc.), it is said to be thin.

There are two types of lenses :

- convex or converging lenses,
- concave or diverging lenses.

NAMING CONVENTION FOR LENSES

While naming a lens, the surface with larger radius of curvature is named first. The lens has a nature of the surface that has the smaller radius of curvature.

EXAMPLE : A lens with one surface plane and the other surface convex will be named as Plano-Convex irrespective of its placement and this lens will have converging nature (the same as the nature of the surface having smaller radius of curvature).

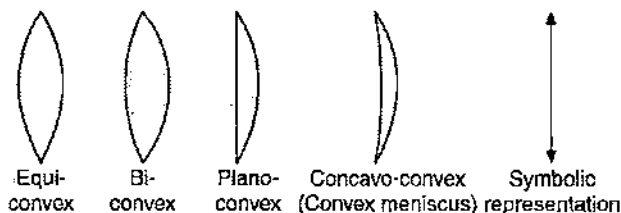
Similarly a Convexo-Concave lens will have a diverging nature and Concavo-Convex lens will have a converging nature.

To summarise, we can say that the first name of a bifocal lens is derived from the name of the surface with bigger radius of curvature and the last name of the lens is derived from the nature of the lens.

CONVEX OR CONVERGING LENSES

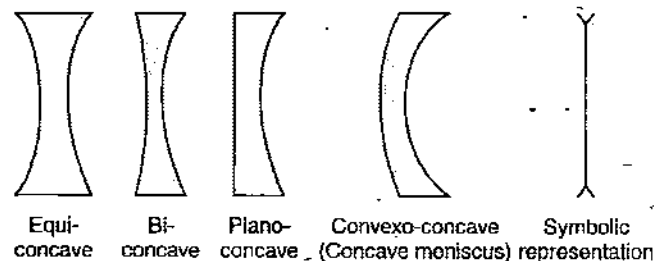
Converging lenses convert a parallel beam of incident rays into a convergent beam. Converging lenses are convex, i.e. such that the thickness at the middle is larger than the thickness of edges. A convex lens is thicker in the centre than at its edges. They include convexo-convex, plano-convex,

and concavo-convex lenses. Sometimes, a converging lens is represented symbolically by a double headed arrow as shown.



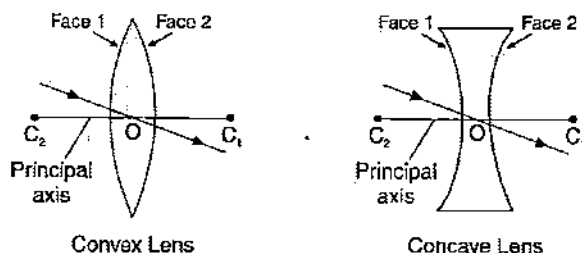
CONCAVE OR DIVERGING LENSES

Diverging lenses convert a parallel beam of rays into a divergent beam. Diverging lenses are concave, i.e. such that the thickness at their edges is larger than the thickness at the middle. A concave lens is thinner at the centre. They include concavo-concave, plano-concave and convexo-concave lenses. Sometimes, a diverging lens is represented symbolically by a line with inverted arrows at its two ends.



OPTICAL CENTRE OF LENS

The central portion of a lens (both convex and concave) behaves as a flat slab. Optical centre O is a point through which any ray passes undeviated.

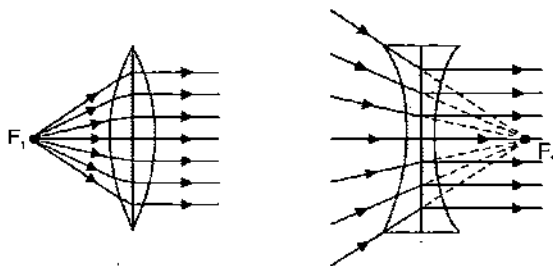


PRINCIPAL AXIS OF A LENS

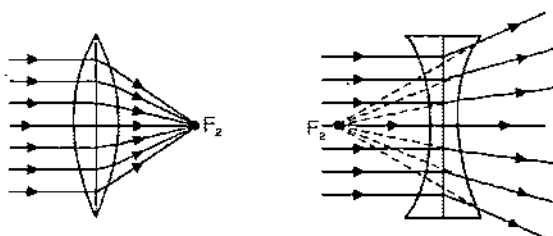
The line joining the centres of curvature C_1 , C_2 is called the principal axis of the lens.

PRINCIPAL FOCUS

A lens has two focal points. The first focal point F_1 is a point object on the principal axis for which the image is at infinity.



The second focal point F_2 is a point image on the principal axis for which the object is at infinity.

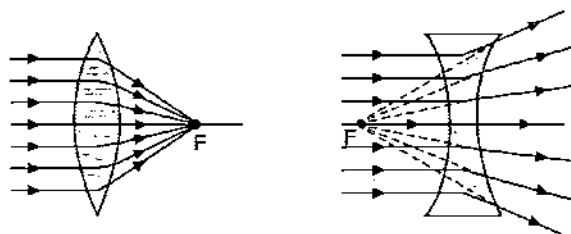


Focal length (f) is the distance between O and the second focus F_2 .

Aperture is the effective diameter of the light transmitting area of the lens. The intensity of the image formed by the lens,

$$I \propto (\text{Aperture})^2$$

Converging and diverging action of a lens is due to the fact that a lens may be thought of a combination of small prisms, as shown in figure. A parallel beam of light, when incident on a convex lens, converges to a point called focus F .



Converging action

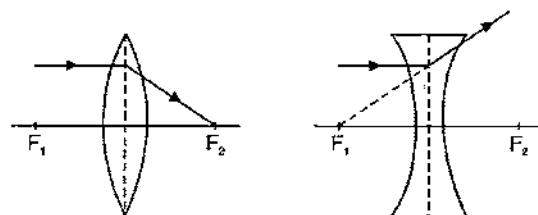
Diverging action

A concave lens diverges a parallel beam of light. It appears to be diverging from a point F , called focus.

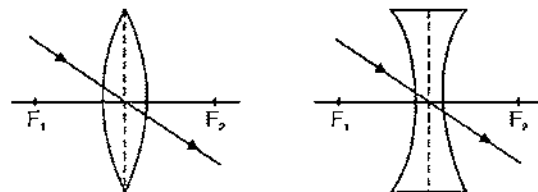
For thin lenses, we need not consider refraction of light at the two surfaces separately. Instead, we say that the light-ray is bent (towards the principal axis in case of convex lens, and away from the principal axis in case of concave lens) when it passes through a thin lens.

RULES FOR OBTAINING IMAGES IN LENSES

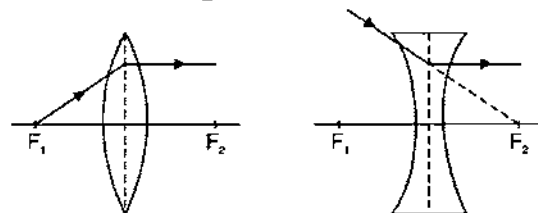
1. A ray parallel to the principal axis, after refraction through the lens, converges to the focus (in case of a convex lens) or appears to diverge from the focus (in case of a concave lens).



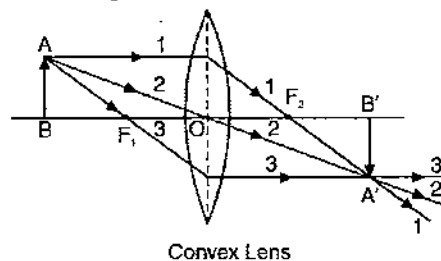
2. A ray passing through the optical centre goes through the lens undeviated.



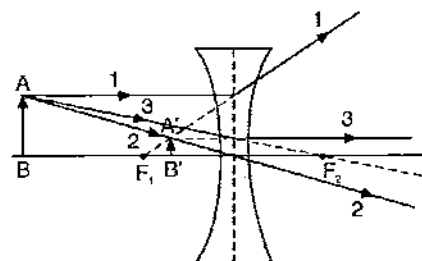
3. A ray passing through the focus (in case of a convex lens) or appearing to pass through the focus (in case of a concave lens) is rendered parallel to the principal axis after refraction through the lens.



Any two of the above three rays can be used to obtain the location of the image.



Convex Lens



Concave Lens

THIN LENS FORMULA FOR A CONVEX LENS

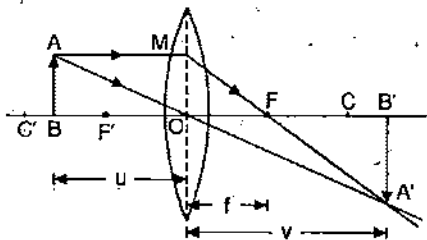
Assumptions used in the derivation of lens formula

- a) The lens used is thin.
- b) The aperture of the lens is small.
- c) The incident and refracted rays make small angles with the principal axis.

d) The object is a small object placed on the principal axis.

CASE 1 : When a real image is formed

Consider an object AB placed perpendicular to the principal axis of a thin convex lens between its F' and C' . A real, inverted and magnified image $A'B'$ is formed beyond C on the other side of the lens, as shown in figure.



Since, $\Delta A'B'O$ and ΔABO are similar,

$$\Rightarrow \frac{A'B'}{AB} = \frac{OB'}{OB} \quad \dots(1)$$

Also $\Delta A'B'F$ and ΔMOF are similar,

$$\Rightarrow \frac{A'B'}{MO} = \frac{FB'}{OF}$$

Since $MO = AB$,

$$\Rightarrow \frac{A'B'}{AB} = \frac{FB'}{OF} \quad \dots(2)$$

From (1) and (2), we get

$$\frac{OB'}{OB} = \frac{FB'}{OF} = \frac{OB' - OF}{OF}$$

Using new Cartesian sign convention, we get

Object distance, $OB = -u$

Image distance, $OB' = +v$

Focal length, $OF = +f$

$$\Rightarrow \frac{v}{-u} = \frac{v-f}{f}$$

$$\Rightarrow vf = -uv + uf$$

$$\Rightarrow uv = uf - vf$$

Dividing both sides by uvf , we get

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Case 2 : When a virtual image is formed

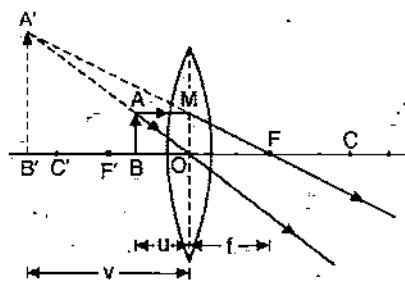
When an object AB is placed between the optical centre O and the focus F of a convex lens, the image $A'B'$ formed by the convex lens is virtual, erect and magnified as shown in figure.

Since, triangles $A'B'O$ and ABO are similar, so we have

$$\frac{A'B'}{AB} = \frac{OB'}{OB} \quad \dots(1)$$

Also, triangles $A'B'F$ and MOF are similar, so we have

$$\frac{A'B'}{OM} = \frac{B'F}{OF}$$



Since $OM = AB$

$$\Rightarrow \frac{A'B'}{AB} = \frac{B'F}{OF} \quad \dots(2)$$

From (1) and (2), we get

$$\frac{OB'}{OB} = \frac{B'F}{OF} = \frac{OB' + OF}{OF}$$

Using new Cartesian sign convention,

Object distance $BO = -u$

Image distance $OB' = -v$

Focal length $OF = +f$

$$\Rightarrow \frac{-v}{-u} = \frac{-v+f}{f}$$

$$\Rightarrow -vf = uv - uf$$

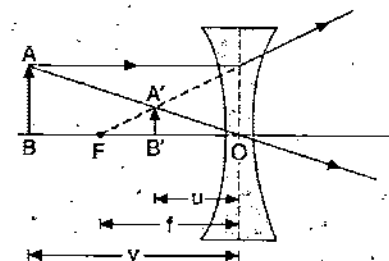
$$\Rightarrow uv = uf - vf$$

Dividing both sides by uvf , we again get

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Thin lens formula for a Concave Lens

Let O be the optical centre and F be the principal focus of concave lens of focal length f . AB is an object placed perpendicular to its principal axis. A virtual, erect and diminished image $A'B'$ is formed due to refraction through the lens.



Since, $\Delta A'B'O$ and ΔABO are similar

$$\text{So, } \frac{A'B'}{AB} = \frac{OB'}{OB} \quad \dots(1)$$

Also, $\Delta A'B'F$ and ΔMOF are similar

$$\text{So, } \frac{A'B'}{OM} = \frac{B'F}{OF}$$

Since $OM = AB$, therefore

$$\frac{A'B'}{AB} = \frac{FB'}{OF}$$

...(2)

$$\Rightarrow \frac{-v}{-u} = \frac{-f+v}{-f}$$

From (1) and (2), we get

$$\frac{OB'}{OB} = \frac{FB'}{OF} = \frac{OF - OB'}{OF}$$

$$\Rightarrow vf = uf - uv$$

$$\Rightarrow uv = uf - vf$$

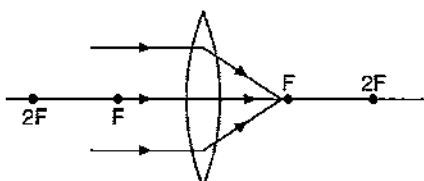
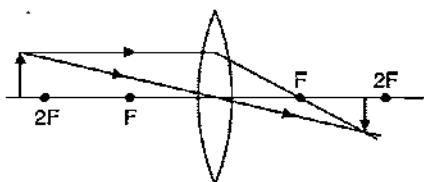
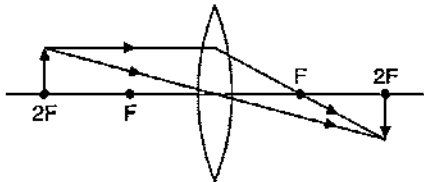
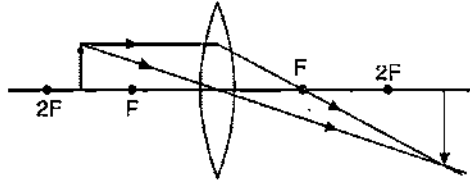
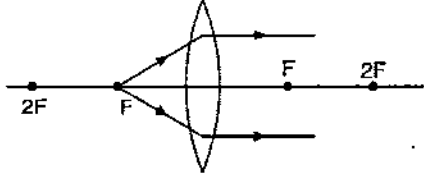
Using new Cartesian sign convention, we get

$$OB = -u, OB' = -v, OF = -f$$

Dividing both sides by uvf , we again get

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

IMAGE FORMATION BY CONVEX LENS

OBJECT POSITION	DIAGRAM	POSITION OF IMAGE	NATURE AND SIZE OF IMAGE
At infinity		At the principal Focus (F) or in the focal plane	Real, inverted and extremely diminished
Beyond 2F		Between F and 2F	Real, inverted and diminished
At 2F		At 2F	Real, inverted and of same size as the object
Between F and 2F		Beyond 2F	Real, inverted and magnified
At F		At infinity	Real, inverted and highly magnified

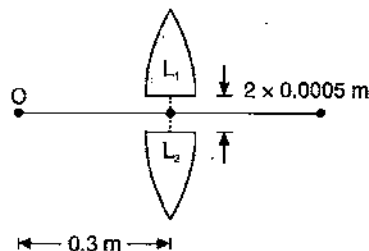
OBJECT POSITION	DIAGRAM	POSITION OF IMAGE	NATURE AND SIZE OF IMAGE
Between F and optical centre		On the same side as the object	Virtual, erect and magnified

IMAGE FORMATION BY A CONCAVE LENS

OBJECT POSITION	DIAGRAM	POSITION OF IMAGE	NATURE AND SIZE OF IMAGE
For all positions of object		Images formed between the optical centre of the lens and the focus (F)	Always forms a Virtual, Erect and Diminished Image

Illustration 52

A point object O is placed at a distance of 0.3 m from a convex lens of focal length 0.2 m . It then cut into two halves each of which is displaced by 0.0005 m as shown in figure. Find the position of the image. If more than one image is formed, find their number and the distance between them.



Solution

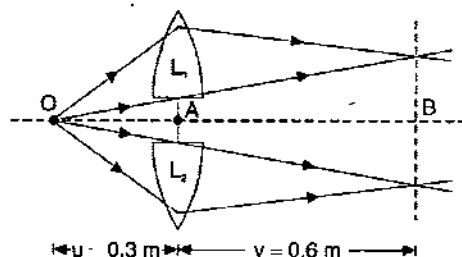
Each part will work as a separate lens and will form its own image. For any part, we have $u = -0.3 \text{ m}$, $f = +0.2 \text{ m}$.

Therefore, from lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we have

$$\frac{1}{v} - \frac{1}{-0.3} = \frac{1}{0.2}$$

$$\Rightarrow v = 0.6 \text{ m}$$

So, each part forms a real image of the point object O at 0.6 m from the lens, as shown in figure.



Since the triangles OL_1I_1 and OL_2I_2 are similar. So, we have

$$\frac{I_1I_2}{L_1L_2} = \frac{OB}{OA} = \frac{u+v}{u}$$

$$\Rightarrow \frac{I_1I_2}{L_1L_2} = \frac{0.3+0.6}{0.3} = \frac{0.9}{0.3} = 3$$

$$\Rightarrow I_1I_2 = 3(L_1L_2) = 3(2 \times 0.0005) = 0.003 \text{ m}$$

Illustration 53

An object is placed 45 cm from a converging lens of focal length 30 cm . A mirror of radius 40 cm is to be placed on the other side of lens so that the object coincides with its image. Find the position of the mirror if it is

(a) convex?

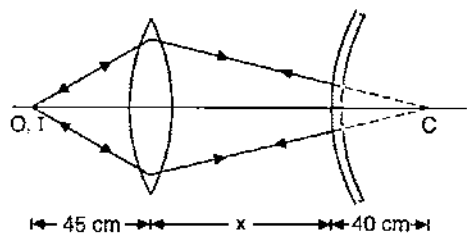
(b) concave?

Solution

The object and image will coincide only if the light ray retraces its path and it will happen only when the ray strikes the mirror normally. In other words the centre of the curvature of the mirror and the rays incident on the mirror are collinear.

- (a) The rays after refraction from lens must be directed towards the centre of curvature of mirror at C . If x is the separation, then for the lens

$$u = -45 \text{ cm}, v = x + 40, f = 30 \text{ cm}$$

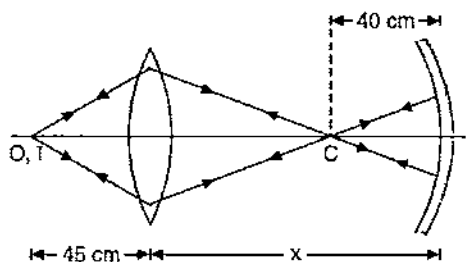


Using lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{x+40} - \frac{1}{-45} = \frac{1}{30}$$

$$\Rightarrow x = \frac{45(30)}{45-30} - 40 = 50 \text{ cm}$$

- (b) In case of concave mirror, the refracted rays from lens meet at C , the centre of curvature (C) of the mirror.



Using lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ where $u = -45 \text{ cm}$,

$$v = x - 40, f = 30 \text{ cm}, \text{ we get}$$

$$\frac{1}{x-40} - \frac{1}{-45} = \frac{1}{30}$$

$$\Rightarrow x - 40 = \frac{45 \times 30}{45-30}$$

$$\Rightarrow x = 90 + 40 = 130 \text{ cm}$$

Illustration 54

A lens with a focal length $f = 30 \text{ cm}$ placed at a distance of $a = 40 \text{ cm}$ from the object produces a sharp image of an object on the screen. A plane parallel plate with thickness of $d = 9 \text{ cm}$ is placed between the lens and the object perpendicular to the optical axis of the lens. Through what distance should the screen be shifted for the image of the object to remain distinct? The refractive index of the glass of the plate is $\mu = 1.8$.

Solution

In the first case,

$$\frac{1}{v_1} + \frac{1}{40} = \frac{1}{30}$$

$$\Rightarrow v_1 = 120 \text{ cm}$$

In the second case, shift due to the glass slab is given by

$$\Delta x = \left(1 - \frac{1}{\mu}\right)t = \left(1 - \frac{1}{1.8}\right)9 = 4 \text{ cm}$$

i.e., u will now become $(40 - 4) = 36 \text{ cm}$, so now we have

$$\frac{1}{v_2} + \frac{1}{36} = \frac{1}{30}$$

$$\Rightarrow v_2 = 180 \text{ cm}$$

Therefore, the screen will have to be shifted 60 cm away from the lens.

Illustration 55

Find the distance of an object from a convex lens of focal length 10 cm if the image formed is two times the size of object. Focal length of the lens is 10 cm .

Solution

A convex lens forms both type of images, real as well as virtual. Since, nature of the image is not mentioned here, so we will have to consider both the cases.

CASE I : When image is real

In this case v is positive and u is negative with $|v| = 2|u|$, so if

$$u = -x \text{ then } v = 2x \text{ and } f = 10 \text{ cm}$$

Substituting in $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{2x} + \frac{1}{x} = \frac{1}{10}$$

$$\Rightarrow \frac{3}{2x} = \frac{1}{10}$$

$$\Rightarrow x = 15 \text{ cm}$$

$x = 15 \text{ cm}$, means object lies between F and $2F$.

CASE II : When image is virtual

In this case v and u both are negative. So if

$$u = -y \text{ then } v = -2y \text{ and } f = 10 \text{ cm}$$

Substituting in, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\Rightarrow \frac{1}{-2y} + \frac{1}{y} = \frac{1}{10}$$

$$\Rightarrow \frac{1}{2y} = \frac{1}{10}$$

$$\Rightarrow y = 5 \text{ cm}$$

$y = 5 \text{ cm}$, means object lies between F and P .

LINEAR MAGNIFICATION (m)

The linear magnification (also called lateral or transverse magnification) m produced by a lens is defined as the ratio of the height of image to the height of the object. So,

$$m = \frac{\text{Height of Image}}{\text{Height of Object}} = \frac{I}{O}$$

Since triangles ABC and $A'B'C$ are similar, so

$$\frac{A'B'}{AB} = \frac{CA'}{CA}$$

Using Conventions, we get

$$A'B' = -I$$

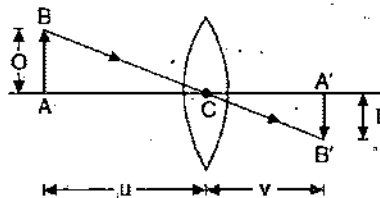
$$AB = O$$

$$CA = -u$$

$$CA' = +v$$

$$\Rightarrow \frac{I}{O} = \frac{v}{-u}$$

$$\Rightarrow m = \frac{I}{O} = \frac{v}{u}$$



Please note that for both the lens and mirror we have

$$m_{\text{real}} = \text{NEGATIVE i.e. } m_{\text{real}} < 0$$

$$m_{\text{virtual}} = \text{POSITIVE i.e. } m_{\text{virtual}} > 0$$

CONCEPTUAL NOTE(S)

- a) **Linear/Transverse/Lateral Magnification** produced by a lens is

$$m = \frac{I}{O} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$

where I is size of image perpendicular to Principal Axis and O is size of object perpendicular to principal Axis.

- b) **Axial Magnification** : Axial magnification is the ratio of the size of image along the principal axis to the size of the object along the principal axis. So

$$m_{\text{axial}} = \frac{\text{Size of Image along Principal Axis}}{\text{Size of Object along Principal Axis}} = \frac{dv}{du}$$

$$\Rightarrow m_{\text{axial}} = \frac{dv}{du} = \frac{v^2}{u^2} = m^2$$

- c) **Areal Magnification** : Areal magnification is the ratio of the area of image to the area of object.

$$m_{\text{areal}} = \frac{\text{Area of Image}}{\text{Area of Object}} = \frac{A_i}{A_o}$$

$$m_{\text{areal}} = \frac{A_i}{A_o} = \frac{v^2}{u^2} = m^2$$

Illustration 56

An object is approaching a thin convex lens of focal length 0.3 m with a speed of 0.01 ms^{-1} . Find the magnitudes of the rates of change of position and lateral magnification of image when the object is at a distance of 0.4 m from the lens.

Solution

Differentiating the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ with respect to time, we get

$$-\frac{1}{v^2} \frac{dv}{dt} + \frac{1}{u^2} \frac{du}{dt} = 0 \quad \{\because f = \text{constant}\}$$

$$\Rightarrow \left(\frac{dv}{dt}\right) = \left(\frac{v^2}{u^2}\right) \frac{du}{dt} \quad \dots(1)$$

Further, substituting proper values in lens formula, we get

$$\frac{1}{v} + \frac{1}{0.4} = \frac{1}{0.3} \quad \{\because u = -0.4 \text{ m}, f = 0.3 \text{ m}\}$$

$$\Rightarrow v = 1.2 \text{ m}$$

Substituting the values in equation (1), we get

Magnitude of rate of change of position of image is

$$\frac{dv}{dt} = 0.09 \text{ ms}^{-1}$$

Lateral magnification, $m = \frac{v}{u}$

Rate of change of lateral magnification is given by

$$\frac{dm}{dt} = \frac{u \frac{dv}{dt} - v \frac{du}{dt}}{u^2} = \frac{(-0.4)(0.09) - (1.2)(0.01)}{(0.4)^2}$$

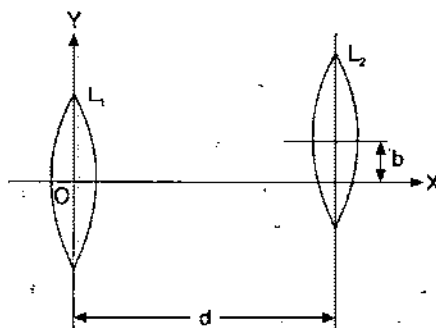
$$\Rightarrow \frac{dm}{dt} = -0.3 \text{ per second}$$

$\Rightarrow$ Magnitude of rate of change of lateral magnification is

$$\frac{dm}{dt} = 0.3 \text{ per second}$$

Illustration 57

Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance d (where $d < f_1$, $d < f_2$) and their principal axes are separated by a vertical distance b as shown in the figure. Taking the centre of the first lens (O) as the origin of co-ordinate system and considering a parallel beam of light coming from the left, find the x and y -coordinates of the focal point of this lens system.



Solution

For the refraction through the first lens, we have $u \rightarrow \infty$, so

$$v_1 = f_1$$

Since, $d < f_2$, the first image (formed by L_1) lies to the right of second lens L_2 , so

$$u_2 = +(f_1 - d)$$

Applying Lens Formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v_2} - \frac{1}{(f_1 - d)} = \frac{1}{f_2}$$

$$\Rightarrow v_2 = \frac{f_2(f_1 - d)}{f_1 + f_2 - d}$$

$$\Rightarrow x = v_2 + d = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}$$

Magnification for second lens is given by

$$m = \frac{v_2}{u_2} = \frac{f_2}{f_1 + f_2 - d}$$

The image due to the second lens is formed below its principal axis and is of the size mb . So, the y coordinate of the focal point system is given by

$$y = b - mb$$

$$\Rightarrow y = b - \frac{f_2 b}{f_1 + f_2 - d}$$

$$\Rightarrow y = \frac{(f_1 - d)b}{f_1 + f_2 - d}$$

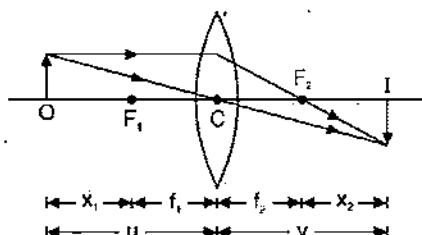
So, the coordinates of the focal point of this system are

$$(x, y) = \left[\frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}, \frac{(f_1 - d)b}{f_1 + f_2 - d} \right]$$

NEWTON'S FORMULA

If the distance of object and image are not measured from optical centre (C), but from first and second principal foci respectively, and if x_1 is the distance of the object from the first focus x_2 is the distance of the image from the second focus and if f is the focal length of the lens, then we have

$$u = -(f + x_1), v = f + x_2$$



According to the lens formula, we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f + x_2} - \frac{1}{-(f + x_1)} = \frac{1}{f}$$

$$\Rightarrow x_1 x_2 = f^2$$

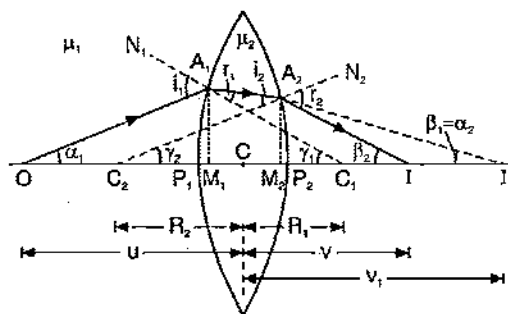
This is called Newton's formula.

where $x_1 = F_1 O$ = distance of object from first principal focus F_1

$x_2 = F_2 I$ = distance of image from second principal focus F_2

LENS MAKER'S FORMULA FOR THIN LENS

Consider a thin lens having its optical centre at C and let O be the point object situated on its principal axis as shown in figure. Light starting from O strikes the first surface of the lens at A_1 and heads towards I_1 , however, refraction takes place at the second surface, thereby giving a final real image at I.



Consider refraction at the first surface only. Let α_1 , β_1 and γ_1 be the angles which the incident ray (OA_1), refracted ray ($A_1 I_1$) and normal ($A_1 C_1$) make with the principal axis.

According to the general law of refraction, applied at A_1 , we get

$$\mu_1 \sin i_1 = \mu_2 \sin r_1$$

Since the angles are small, so $\sin i_1 \approx i_1$ and $\sin r_1 \approx r_1$

$$\Rightarrow \mu_1 i_1 = \mu_2 r_1 \quad \dots (1)$$

In $\Delta A_1 C_1 O$, we have $i_1 = \gamma_1 + \alpha_1$

In $\Delta A_1 C_1 I_1$, we have $\gamma_1 = r_1 + \beta_1$

$$\Rightarrow r_1 = \gamma_1 - \beta_1$$

Substituting for i_1 and r_1 in equation (1), we get

$$\mu_1 (\gamma_1 + \alpha_1) = \mu_2 (\gamma_1 - \beta_1)$$

$$\Rightarrow \mu_2 \beta_1 + \mu_1 \alpha_1 = (\mu_2 - \mu_1) \gamma_1$$

Since the angles are small, so they can be replaced by their tangents.

$$\Rightarrow \mu_2 \tan \beta_1 + \mu_1 \tan \alpha_1 = (\mu_2 - \mu_1) \tan \gamma_1$$

$$\Rightarrow \mu_2 \left(\frac{A_1 M_1}{M_1 I_1} \right) + \mu_1 \left(\frac{A_1 M_1}{M_1 O} \right) = (\mu_2 - \mu_1) \left(\frac{A_1 M_1}{M_1 C_1} \right)$$

For a thin lens, M_1 lies close to C. Therefore, all the distances measured from M_1 can be replaced by those measured from C. Hence we have

$$\frac{\mu_2}{C I_1} + \frac{\mu_1}{C O} = \frac{\mu_2 - \mu_1}{C C_1} \quad \dots (2)$$

Now consider refraction at the second surface. The ray A_1A_2 which is the refracted ray for first surface becomes the incident ray for second surface. Applying general law of refraction at A_2 (light going from denser to rarer medium), we get

$$\mu_2 \sin i_2 = \mu_1 \sin r_2$$

$$\Rightarrow \mu_2 i_2 = \mu_1 r_2 \quad \dots(3)$$

In $\Delta A_2C_2I_1$ we have $i_2 = \gamma_2 + \beta_1$

In ΔA_2C_2I , we have $r_2 = \gamma_2 + \beta_2$

Substituting for i_2 and r_2 in equation (2), we get

$$\mu_2 (\gamma_2 + \beta_1) = \mu_1 (\gamma_2 + \beta_2)$$

$$\Rightarrow \mu_1 \beta_2 - \mu_2 \beta_1 = (\mu_2 - \mu_1) \gamma_2$$

Since angles are small, so replacing the angles by their tangents, we get

$$\mu_1 \tan \beta_2 - \mu_2 \tan \beta_1 = (\mu_2 - \mu_1) \tan \gamma_2$$

$$\Rightarrow \mu_1 \left(\frac{A_2M_2}{M_2I} \right) - \mu_2 \left(\frac{A_2M_2}{M_2I_1} \right) = (\mu_2 - \mu_1) \left(\frac{A_2M_2}{M_2C_2} \right)$$

Since the lens is thin, M_2 lies close to C , so we get

$$\frac{\mu_1}{CI} - \frac{\mu_2}{CI_1} = \frac{\mu_2 - \mu_1}{CC_2} \quad \dots(4)$$

Adding equations (2) and (4), we get

$$\frac{\mu_2}{CI_1} + \frac{\mu_1}{CO} + \frac{\mu_1}{CI} - \frac{\mu_2}{CI_1} = \frac{\mu_2 - \mu_1}{CC_1} + \frac{\mu_2 - \mu_1}{CC_2}$$

$$\Rightarrow \mu_1 \left(\frac{1}{CO} + \frac{1}{CI} \right) = (\mu_2 - \mu_1) \left(\frac{1}{CC_1} + \frac{1}{CC_2} \right)$$

$$\Rightarrow \frac{1}{CO} + \frac{1}{CI} = \left(\frac{\mu_2 - \mu_1}{\mu_1} \right) \left(\frac{1}{CC_1} + \frac{1}{CC_2} \right)$$

$$\Rightarrow \frac{1}{CO} + \frac{1}{CI} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left(\frac{1}{CC_1} + \frac{1}{CC_2} \right)$$

$$\Rightarrow \frac{1}{CO} + \frac{1}{CI} = ({}^1\mu_2 - 1) \left(\frac{1}{CC_1} + \frac{1}{CC_2} \right)$$

Applying sign convention, we have

$$CO = -u, CI = +v, CC_1 = +R_1, CC_2 = -R_2$$

$$\Rightarrow \frac{1}{-u} + \frac{1}{v} = ({}^1\mu_2 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{v} - \frac{1}{u} = ({}^1\mu_2 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(5)$$

Since, focal length of a convex lens is defined as the distance of that point from the centre of lens where a beam coming parallel to principal axis comes to focus after refraction through the lens, so when

$$u \rightarrow \infty \text{ we have } v = f$$

Substituting in equation (5), we get

$$\frac{1}{f} = ({}^1\mu_2 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(6)$$

If the first medium is air, then ${}^1\mu_2 = \mu$, so we have

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

This formula is called Lens Maker's Formula.

PROBLEM SOLVING TRICK(S)

- A concave lens forms virtual, erect and diminished image.
- A convex lens may form real and virtual images. The real image is inverted, it may be diminished or magnified while virtual image formed by convex lens is erect and enlarged.

c) Different Media on either side of Lens

If a lens of refractive index μ_2 has different media on either side, the medium of object space has refractive index μ_1 and that of image space has refractive index μ_3 , then focal length f of lens is

$$\frac{\mu_3}{f} = \frac{\mu_2 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

As a special case if we put $\mu_3 = \mu_1$, we get the Lens Maker's Formula.

Illustration 58

A plano-convex lens has a thickness of 4 cm. When placed on a horizontal table, with the curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of the plane face is found to be $\frac{25}{8}$ cm. Find the focal length of the lens. Assume

thickness to be negligible while finding its focal length.

Solution

When placed on a horizontal table with curved surface in contact with it,

In this case refraction of the rays starting from O takes place from a plane surface as shown in Figure I.

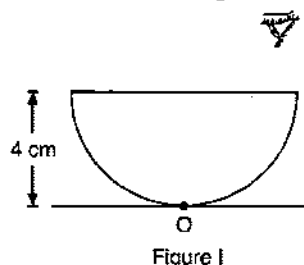


Figure I

So, we can use

$$\text{Apparent Depth} = \frac{\text{Real Depth}}{\mu}$$

$$\Rightarrow 3 = \frac{4}{\mu}$$

$$\Rightarrow \mu = \frac{4}{3}$$

When the plane surface is in contact with the horizontal table.

In this case refraction takes place from a spherical surface as shown in Figure II.

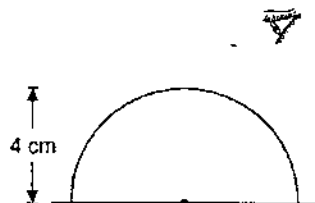


Figure II

Hence, applying

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}, \text{ we get}$$

$$\frac{1}{\frac{25}{8}} - \frac{\frac{4}{3}}{-4} = \frac{1 - \frac{4}{3}}{-R}$$

$$\Rightarrow \frac{1}{3R} = \frac{1}{3} - \frac{8}{25} = \frac{1}{75}$$

$$\Rightarrow R = 25 \text{ cm}$$

Now, to find the focal length we will use the lens maker formula :

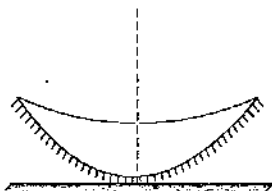
$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f} = \left(\frac{4}{3} - 1 \right) \left(\frac{1}{\infty} - \frac{1}{-25} \right) = \frac{1}{75}$$

$$\Rightarrow f = 75 \text{ cm}$$

Illustration 59

The convex surface of a thin concavo-convex lens of glass of the refractive index 1.5 has a radius of curvature 20 cm. The concave surface has a radius of curvature 60 cm. The convex side is silvered and placed on a horizontal surface.



- Where should a pin be placed on the optic axis such that its image is formed at the same place?
- If the concave part is filled with water of refractive index $\frac{4}{3}$, find the distance through which the pin should be moved, so that the image of the pin again coincides with the pin.

Solution

- Image of object will coincide with it, if the ray of light after refraction from the concave surface falls normally on concave mirror so formed by silvering the convex surface i.e., image after refraction from concave surface should be formed at centre of curvature of concave mirror or at a distance of 20 cm on same side of the combination. Let x be the distance of pin from the given optical system.

Applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get for

$$v = -20 \text{ cm}, u = -x, R = -60 \text{ cm}$$

$$\frac{1.5}{-20} - \frac{1}{-x} = \frac{1.5 - 1}{-60}$$

$$\Rightarrow \frac{1}{x} = \frac{3}{40} - \frac{1}{120} = \frac{8}{120}$$

$$\Rightarrow x = \frac{120}{8} = 15 \text{ cm}$$

- When the concave part is filled with water, then before striking with the concave surface, the ray is first refracted from a plane surface. So, let x be the distance of pin, then the plane surface will form its image at a distance of $h_{app} = \mu t$ i.e., $\frac{4x}{3}$ from it.

Now, using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ with proper signs, we get

$$\frac{1.5}{-20} - \frac{\frac{4}{3}}{\left(\frac{4}{3}x\right)} = \frac{1.5 - \frac{4}{3}}{-60}$$

$$\Rightarrow \frac{1}{x} = \frac{-3}{40} + \frac{1}{360} = \frac{-26}{360}$$

$$\Rightarrow x = \frac{360}{-26} = -13.84 \text{ cm}$$

LENS IMMERSED IN A LIQUID

If a lens (made of glass) of refractive index μ_g is immersed in a liquid of refractive index μ_l , then its focal length in liquid, f_l is given by

$$\frac{1}{f_l} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

If f_a is the focal length of lens of air, then

$$\frac{1}{f_a} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \{ \because \mu_g = \mu_g \}$$

$$\Rightarrow f_l = \left[\frac{\mu_g - 1}{\mu_l - 1} \right] f_a$$

Now three cases arise which are discussed here.

- a) If $\mu_g > \mu_l$, then f_i and f_a are of same sign and $f_i > f_a$.

That is the nature of lens remains unchanged, but its focal length increases and hence power of lens decreases. In other words the convergent lens becomes less convergent and divergent lens becomes less divergent.

- b) If $\mu_g = \mu_l$, then $f_i \rightarrow \infty$ and the lens behaves as a simple glass plate.
c) If $\mu_g < \mu_l$, then f_i and f_a have opposite signs and the nature of lens changes i.e. a convergent lens becomes divergent and vice versa.

Illustration 60

A lens has a power of +5 dioptre in air. Calculate its power if it is completely immersed in water? Given $\mu_g = \frac{3}{2}$

and $\mu_w = \frac{4}{3}$.

Solution

Let f_a and f_w be the focal lengths of the lens in air and water respectively, then

$$P_a = \frac{1}{f_a} \quad \text{and} \quad P_w = \frac{\mu_w}{f_w}$$

Since lens has power +5 D in air, so

$$f_a = \frac{1}{P} = \frac{1}{5} = 0.2 \text{ m} = 20 \text{ cm}$$

Using Lens Maker's formula, we get

$$\frac{1}{f_a} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(1)$$

$$\text{Similarly, } \frac{1}{f_w} = \left(\frac{\mu_g}{\mu_w} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow P_w = \frac{\mu_w}{f_w} = (\mu_g - \mu_w) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(2)$$

Dividing equation (2) by equation (1), we get,

$$\frac{P_w}{P_a} = \frac{(\mu_g - \mu_w)}{(\mu_g - 1)} = \frac{1}{3}$$

$$\Rightarrow P_w = \frac{P_a}{3} = +\frac{5}{3} \text{ D}$$

DISPLACEMENT METHOD

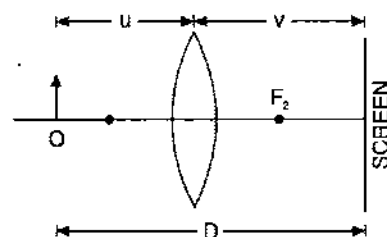
Consider an object and a screen fixed at a distance D apart. Let a lens of focal length f be placed between the object and the screen.

From figure we observe that

$$u + v = D \Rightarrow v = D - u$$

Also from Lens formula

$$\begin{aligned} \frac{1}{v} - \frac{1}{u} &= \frac{1}{f} \\ \Rightarrow \frac{1}{D-u} - \frac{1}{-u} &= \frac{1}{f} \\ \Rightarrow \frac{1}{D-u} + \frac{1}{u} &= \frac{1}{f} \\ \Rightarrow Df &= (D-u)u \\ \Rightarrow u^2 - Du + Df &= 0 \\ \Rightarrow u &= \frac{D \pm \sqrt{D^2 - 4fD}}{2} \end{aligned}$$



For u to be mathematically real,

$$D^2 - 4fD \geq 0$$

$$\Rightarrow D \geq 4f$$

PROBLEM SOLVING TRICK(S)

So, if the object and the screen are placed at a distance less than $4f$, then a virtual image will be formed. Hence, for a real image to be formed $D \geq 4f$.

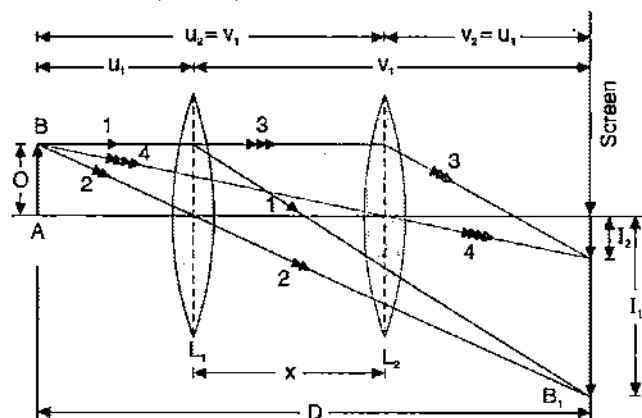
Case I : For $D = 4f$

$$u = v = \frac{D}{2}$$

i.e. the lens is placed exactly between the object and the screen.

Case II : For $D > 4f$

We get two different position of lens (L_1 and L_2) for which the image of object on the screen is distinct and clear.



L_1 First Position of Lens

L_2 Second Position of the Same Lens (shown in grey)

NOTE:

Do not Develop a misconception that there are two lenses, infact the same lens is displaced through x from position L_1 to L_2 .

The object distances for these two positions are given by

$$u_1 = \frac{D - \sqrt{D^2 - 4fD}}{2} \quad \dots(1)$$

$$u_2 = \frac{D + \sqrt{D^2 - 4fD}}{2} \quad \dots(2)$$

Since $u + v = D$, so

$$v_1 = \frac{D + \sqrt{D^2 - 4fD}}{2} \quad \dots(3)$$

$$v_2 = \frac{D - \sqrt{D^2 - 4fD}}{2} \quad \dots(4)$$

We observe that

$$u_1 = v_2 = u \text{ (say)} \quad \dots(5)$$

$$v_1 = u_2 = v \text{ (say)} \quad \dots(6)$$

Let the lens be displaced through x , then we observe from figure that

$$x = v_1 - u_1 = \sqrt{D^2 - 4fD} \quad \dots(7)$$

$$\Rightarrow x^2 = D^2 - 4fD$$

$$\Rightarrow f = \frac{D^2 - x^2}{4D} \quad \dots(8)$$

Using (7) in (1), (2), (3) and (4), we get

$$u_1 = v_2 = \frac{D - x}{2} = u \quad \dots(9)$$

$$v_1 = u_2 = \frac{D + x}{2} = v \quad \dots(10)$$

If m_1 is the magnification for the first position of lens i.e. L_1 , then

$$m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \frac{v}{u} = \frac{D + x}{D - x} \quad \dots(11)$$

If m_2 is the magnification for the second position of the lens i.e. L_2 , then

$$m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \frac{u}{v} = \frac{D - x}{D + x} \quad \dots(12)$$

From (11) and (12), we observe that

$$m_1 m_2 = 1 \quad \dots(13)$$

So, if magnification for position L_1 is m , then magnification for position L_2 is $\frac{1}{m}$.

Also from (13), we get

$$\frac{I_1 I_2}{O O} = 1$$

$$\Rightarrow O^2 = I_1 I_2$$

$$\Rightarrow O = \sqrt{I_1 I_2} \quad \dots(14)$$

i.e. size of the object (O) is the geometric mean of the sizes of the image for two position of lens L_1 and L_2 .

Also,

$$m_1 - m_2 = \frac{D + x}{D - x} - \frac{D - x}{D + x}$$

$$\Rightarrow m_1 - m_2 = \frac{4Dx}{D^2 - x^2}$$

$$\Rightarrow m_1 - m_2 = \frac{x}{\left(\frac{D^2 - x^2}{4D}\right)}$$

$$\Rightarrow m_1 - m_2 = \frac{x}{f}$$

$$\Rightarrow f = \frac{x}{m_1 - m_2} \quad \dots(15)$$

Further if $m_1 = m$, then $m_2 = \frac{1}{m}$

$$\Rightarrow f = \frac{mx}{m^2 - 1} \quad \dots(16)$$

Finally, we observe that

$$\frac{m_1}{m_2} = \left(\frac{D + x}{D - x}\right)^2 \quad \dots(17)$$

PROBLEM SOLVING TRICK(S)

Dear Students, you must keep in mind that actually "Displacement Method" is not in the syllabus, but the Examiner generally asks the problems not in its name but by its concept e.g. an examiner's mind may fabricate a problem not having the name Displacement method but then the problem must be having a clue which may state $D > 4f$ or the lens is displaced to get two real images on screen and stuff like that. So, you are advised not to overlook the topic as this is very important (not by name) but by the concept involved.

Illustration 61

A thin converging lens of focal length f is moved between a candle and a screen. The distance between the candle and the screen is $D (> 4f)$. Show that for two different positions of the lens, two different images can be obtained on the screen. If the ratio of dimensions of the image is β , find the value of $\left(\beta + \frac{1}{\beta}\right)$.

Solution

Let x be the separation between two positions of the lens for which a real image is formed on the screen.

$$\text{Then, } v + u = D \quad \dots(1)$$

$$\text{and } v - u = x \quad \dots(2)$$

$$\text{Solving we get } u = \frac{D - x}{2} \text{ and } v = \frac{D + x}{2}$$

$$\text{Now, } m_1 = \frac{I_1}{O} = \frac{D+x}{D-x}$$

$$\text{and } m_2 = \frac{I_2}{O} = \frac{D-x}{D+x}$$

$$\Rightarrow \frac{I_1}{I_2} = \left(\frac{D+x}{D-x} \right)^2 = \beta$$

$$\Rightarrow \frac{D+x}{D-x} = \sqrt{\beta}$$

$$\Rightarrow x = \left(\frac{\sqrt{\beta}-1}{\sqrt{\beta}+1} \right) D$$

$$\text{Now, } f = \frac{D^2 - x^2}{4D} = \frac{D^2 - \left(\frac{\sqrt{\beta}-1}{\sqrt{\beta}+1} \cdot D \right)^2}{4D} = \frac{D}{2 + \sqrt{\beta} + \frac{1}{\sqrt{\beta}}}$$

$$\Rightarrow \sqrt{\beta} + \frac{1}{\sqrt{\beta}} = \frac{D}{f} - 2$$

$$\Rightarrow \beta + \frac{1}{\beta} = \left(\frac{D}{f} - 2 \right)^2 - 2$$

Illustration 62

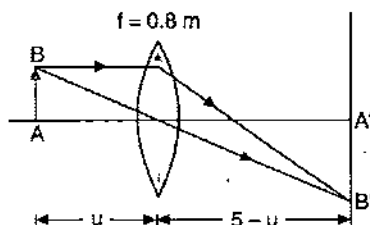
An object is 5 m to the left of a flat screen. A converging lens for which the focal length is $f = 0.8$ m is placed between object and screen.

(a) Show that two lens positions exist that form images on the screen and determine how far are these positions from the object?

(b) How do the two images differ from each other?

Solution

(a) Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get



$$\frac{1}{5-u} - \frac{1}{-u} = \frac{1}{0.8}$$

$$\Rightarrow \frac{1}{5-u} + \frac{1}{u} = \frac{5}{4}$$

$$\Rightarrow \frac{5-u+u}{(5-u)u} = \frac{5}{4}$$

$$\Rightarrow 20 = 25u - 5u^2$$

$$\Rightarrow 5u^2 - 25u + 20 = 0$$

$$\Rightarrow 5u^2 - 20u - 5u + 20 = 0$$

$$\Rightarrow 5u(u-4) - 5(u-4) = 0$$

$$\Rightarrow (5u-5)(u-4) = 0$$

$$\Rightarrow u = 1 \text{ m and } u = 4 \text{ m}$$

Both the values are real, so this means that there exist two positions of lens that form images of object on the screen.

$$(b) \quad m = \frac{v}{u}$$

$$\Rightarrow m_1 = \frac{(5-4)}{(1-4)} = -0.25 \text{ and } m_2 = \frac{(5-1)}{(-1)} = -4$$

Hence, both the images are real and inverted, the first has magnification -0.25 and the second -4 .

Also, we observe that

$$m_1 m_2 = (-0.25)(-4) = 1$$

Illustration 63

For two positions of a converging lens between an object and a screen which are 96 cm apart, two real images are formed. The ratio of the lengths of the two images is 4. Calculate the focal length of the lens.

Solution

$$\text{Since, } \frac{m_1}{m_2} = 4$$

$$\Rightarrow \left(\frac{D+x}{D-x} \right)^2 = 4$$

$$\Rightarrow \frac{D+x}{D-x} = 2$$

Substituting $D = 96$ cm, we get

$$\frac{96+x}{96-x} = 2$$

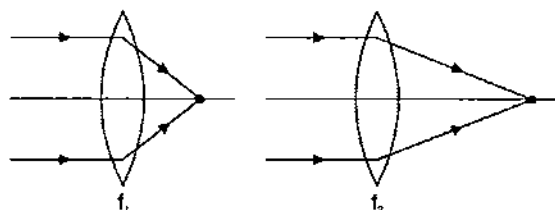
$$\Rightarrow x = 32 \text{ cm}$$

$$\text{Since, } f = \frac{D^2 - x^2}{4D}$$

$$\Rightarrow f = 21.33 \text{ cm}$$

POWER OF A LENS

The power of a lens P is actually the measure of its ability to deviate the incident rays towards axis. The greater the curvature of the two surfaces (i.e., the shorter the focal length f), the greater is the lens action. The shorter the focal length of a lens the more it converges or diverges the light, as shown in figure.



The power of a lens placed in air is actually the reciprocal of the focal length of the lens in metre and is given by

$$P = \frac{1}{f(\text{in metre})} = \frac{100}{f(\text{in cm})}$$

SI unit of power is dioptre (D).

Power of a lens placed in a medium is defined as

$$P_{\text{med}} = \frac{\mu}{f_{\text{med}}}$$

where μ is the refractive index of the medium and f_{med} is the focal length of the lens in that medium.

As a convention, the power of a converging lens (or convex lens) (with focal length positive) is taken to be **positive**. The power of a diverging lens (or concave lens) (with focal length negative) is taken to be **negative**.

Also we must note that for a mirror, power is defined as

$$P = -\frac{1}{f(\text{in metre})} = -\frac{100}{f(\text{in cm})}$$

Thus a convex lens and concave mirror have converging nature and hence they have positive power, whereas the concave lens and convex mirror have diverging nature and hence have negative power.

Nature of Lens / Mirror	Focal Length (f)	Power $P_{\text{lens}} = \frac{1}{f}$, $P_{\text{mirror}} = -\frac{1}{f}$	Converging / Diverging	Ray Diagram
Concave mirror	-ve	+ve	Converging	
Convex lens	+ve	+ve	Converging	
Convex mirror	+ve	-ve	Diverging	
Concave lens	-ve	-ve	Diverging	

Illustration 64

A convergent lens of power 6 D is combined with a diverging lens of -2 D. Find the power and focal length of the combination.

Solution

Here $P_1 = 6 \text{ D}$, $P_2 = -2 \text{ D}$

Power of the combination is given by

$$P = P_1 + P_2 = 6 - 2 = 4 \text{ D}$$

Since $f = \frac{1}{P}$

$$\Rightarrow f = \frac{1}{4} = 0.25 \text{ m} = 25 \text{ cm}$$

Illustration 65

An object 4 cm high is placed at a distance of 10 cm from a convex lens of focal length 20 cm. Find the position, nature and size of the image. Also find the power of the lens.

Solution

Here, $u = -10 \text{ cm}$ (the object assumed to be kept to the left of optical centre)

$f = +20 \text{ cm}$ (positive for a convex lens)

$h_1 = +4 \text{ cm}$ (object kept above the principal axis)

Using the lens formula, we get

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{-10} = \frac{1}{20}$$

$$\Rightarrow v = -20 \text{ cm}$$

That is, the image is 20 cm from the lens, on the same side as the object. Hence, the image is **virtual**. The linear magnification,

$$m = \frac{h_2}{h_1} = \frac{v}{u}$$

So, size of the image is

$$h_2 = h_1 \left(\frac{v}{u} \right) = 4 \times \frac{-20}{-10} = 8 \text{ cm}$$

The positive sign indicates that the image is erect (and **virtual**).

Since, the power of the lens is given by

$$P = \frac{1}{f(\text{in m})} = \frac{1}{+0.2} = +5 \text{ D}$$

Illustration 66

A converging lens forms a five folds magnified image of an object. The screen is moved towards the object by a distance $d = 0.5 \text{ m}$ and the lens is shifted so that the image has the same size as the object. Find the lens power and the initial distance between the object and the screen.

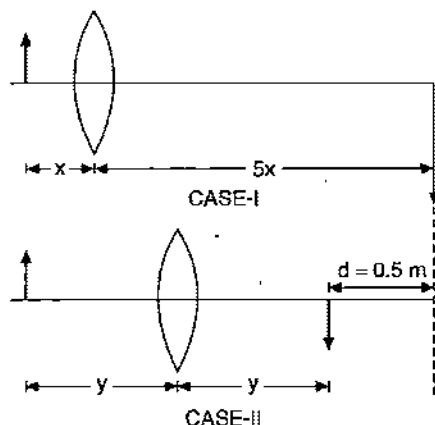
Solution

In the first case image is five times magnified. Hence

$$|v| = 5|u|$$

In the second case image and object are of equal size. Hence

$$|v| = |u|$$



From the two figures, we get

$$6x = 2y + d$$

$$\Rightarrow 6x - 2y = 0.5 \quad \dots(1)$$

Using the lens formula for both the cases, we get for

CASE-I,

$$\frac{1}{5x} - \frac{1}{-x} = \frac{1}{f}$$

$$\Rightarrow \frac{6}{5x} = \frac{1}{f} \quad \dots(2)$$

CASE-II,

$$\frac{1}{y} - \frac{1}{-y} = \frac{1}{f}$$

$$\Rightarrow \frac{2}{y} = \frac{1}{f} \quad \dots(3)$$

Solving these three equations, we get

$$x = 0.1875 \text{ m and } f = 0.15625 \text{ m}$$

Therefore, initial distance between the object and the screen is

$$6x = 1.125 \text{ m}$$

$$\text{Power of the lens, } P = \frac{1}{f} = \frac{1}{0.15625} \text{ D} = 6.4 \text{ D}$$

LENSES IN CONTACT

If two or more lenses of focal lengths $f_1, f_2, \dots$ are placed in contact, then their equivalent focal length f is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \dots = \sum_{i=1}^n \frac{1}{f_i}$$

where $f_1, f_2, \dots$ are to be substituted with proper signs attached.

The power of combination

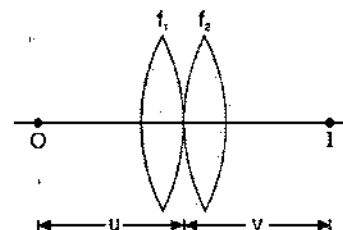
$$P = P_1 + P_2 + \dots = \sum_{i=1}^n P_i$$

Here too, $P_1, P_2, \dots$ are to be substituted with proper signs attached.

The magnification of the combination is

$$M = m_1 \times m_2 \times \dots = \prod_{i=1}^n m_i$$

In many optical instruments, the combination of lenses in contact are used so as to improve the performance of the instrument.



Consider two lenses of focal lengths f_1 and f_2 kept in contact. Let a point object O be placed at a distance u from the combination. The first image (say I_1) after refraction from the first lens is formed at a distance v_1 (whatever may be the sign of v_1) from the combination. This image I_1 acts as an object for the second lens and let v be the distance of the final image from the combination. Applying the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\text{For the first lens, } \frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad \dots(1)$$

$$\text{and for the second lens, } \frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad \dots(2)$$

Adding equations, (1) and (2), we get

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f} \text{ (say)}$$

where, f is the equivalent focal length of the combination. Thus,

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

TWO THIN LENSES SEPARATED BY A DISTANCE

If two thin lenses of focal lengths f_1, f_2 are placed at a distance x apart, then equivalent focal length of combination is

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$$

or Power for the combination is

$$P = P_1 + P_2 - xP_1P_2$$

The net magnification of the combination is still remains

$$m = m_1 \times m_2$$

Illustration 67

Consider a co-axial system of two thin convex lenses of focal length f each separated by a distance d . Draw ray diagrams for image formation corresponding to an object at infinity placed on the principal axis in the following cases: (a) $d < f$ (b) $d = f$ (c) $f < d < 2f$ (d) $d = 2f$ and (e) $d > 2f$. Indicate the nature of the combination (concave, convex or plane) in each case.

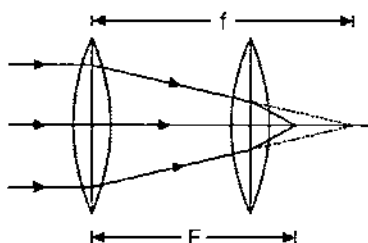
Solution

The formula

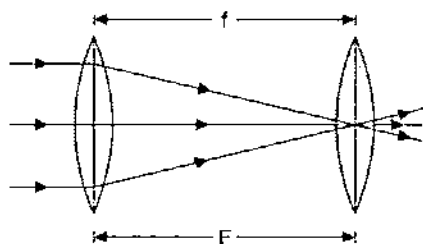
$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

is valid only for small values of d compared to f_1 and f_2 . Therefore, we cannot use this formula in the given cases. However, we can draw the ray diagram to decide the nature of the combination.

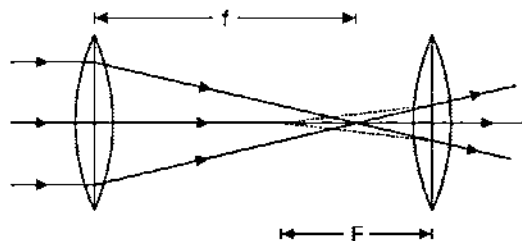
- (a) When $d < f$: The ray diagram is shown in figure. The out-coming rays are convergent. Obviously, the combination is a convex lens with $F < f$.



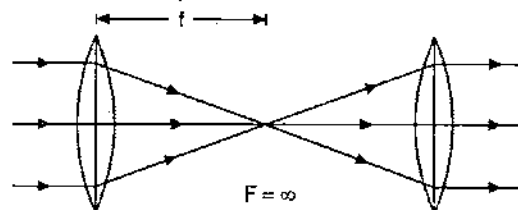
- (b) When $d = f$: The incident parallel beam converges to a point and then passes without any more deviation. The combination behaves like a convex lens of $F = f$.



- (c) When $f < d < 2f$: The incident parallel beam emerges out as a divergent beam. the combination behaves as a divergent or concave lens.



- (d) When $d = 2f$: The incident parallel beam emerges out as a parallel beam but inverted. The combination behaves as a plane glass slab, which inverts the beam.



- (e) When $d > 2f$: The incident parallel beam emerges out as a convergent beam. The combination behaves as a convergent or convex lens.

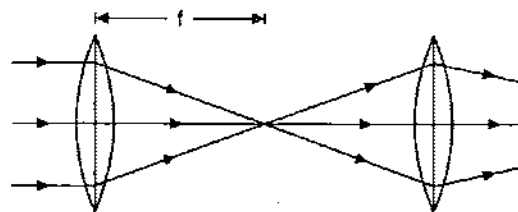


Illustration 68

Two equi-convex lenses of focal lengths 30 cm and 70 cm, made of material of refractive index = 1.5, are held in contact coaxially by a rubber band round their edges. A liquid of refractive index 1.3 is introduced in the space between the lenses filling it completely. Find the position of the image of a luminous point object placed on the axis of the combination lens at a distance of 90 cm from it.

Solution

According to Lens Maker's Formula, we have

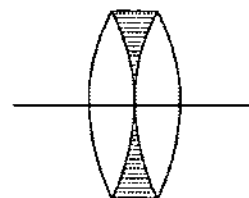
$$\frac{1}{30} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{-R_1} \right)$$

$$\Rightarrow R_1 = 30 \text{ cm}$$

Similarly, radius of curvature of the second lens is 70 cm. Since

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \quad \dots(1)$$

Here, $f_1 = 30 \text{ cm}$, $f_2 = 70 \text{ cm}$



Now f_3 is calculated again using the Lens Maker's Formula, so we get

$$\frac{1}{f_3} = (1.3 - 1) \left(\frac{1}{-30} - \frac{1}{70} \right)$$

$$\Rightarrow f_3 = -70 \text{ cm}$$

$$\Rightarrow F = 30 \text{ cm} \quad \{\text{from equation (1)}\}$$

According to Lens formula, applied on the combination of lenses, we have

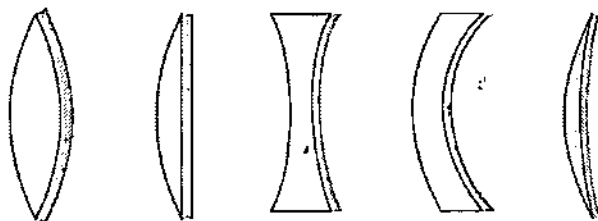
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{(-90)} = \frac{1}{30}$$

$$\Rightarrow v = 45 \text{ cm}$$

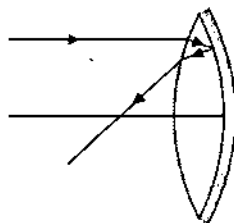
LENSES WITH ONE SILVERED SURFACE

When one face of a lens is silvered as shown in figure it acts like a lens-mirror combination.



Lenses with one face silvered act like lens-mirror combination

It is obvious from the ray diagram as shown in figure that the incident ray of light is refracted through the lens twice (i.e., once when light is incident on the lens and second time when reflected by the mirror) and reflected from the mirror once.



The combination acts like a mirror whose effective power is given by

$$P_{net} = 2P_l + P_m$$

where P_l is the power of the lens and P_m is the power of the mirror.

Since for a mirror we have

$$P_m = -\frac{1}{f_m}$$

and for a lens, we have

$$P_l = \frac{1}{f_l} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

So, the combination acts like a mirror having net focal length given by

$$F_{net} = -\frac{1}{P_{net}}$$

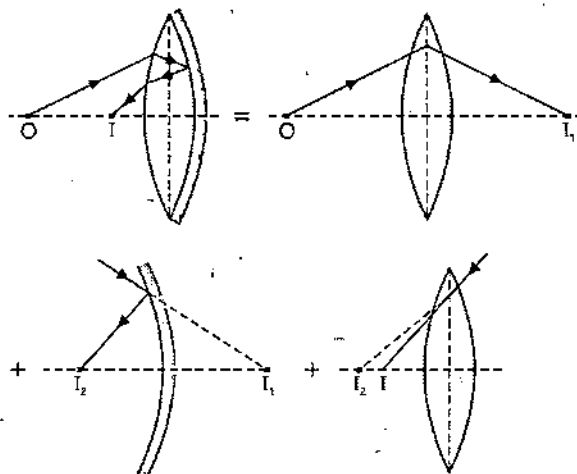
$$\Rightarrow P = -\frac{1}{F} = \frac{2}{f_l} - \frac{1}{f_m}$$

where f_l is focal length of lens and f_m is focal length of spherical mirror formed due to silvering of surface.

To have a fundamental understanding of this we can understand the silvering of lenses using the following arguments.

A ray incident on a lens with its backside silvered will be refracted through the lens twice and will be reflected from the mirror once, as shown.

- Light from object O passes through lens to form image I_1 .
- The image I_1 acts as an object (virtual) for the curved mirror to form image I_2 .
- The image I_2 acts as an object (virtual) for the lens to form the final image I .



The silvered lens acts like a mirror with equivalent focal length F , given by

$$-\frac{1}{F} = \frac{1}{f_l} - \frac{1}{f_m} + \frac{1}{f_l} = \frac{2}{f_l} - \frac{1}{f_m}$$

where f_l is focal length of lens and f_m is focal length of spherical mirror formed due to silvering of surface.

SIGN CONVENTION

While using the above formula, we make use of the following sign conventions:

- f is positive for converging (convex) lens and concave mirror.
- f is negative for diverging (concave) lens and convex mirror.

For example, for a plano-convex lens, from Lens Maker's Formula we get

$$\frac{1}{f_l} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right) = \frac{\mu - 1}{R}$$

$$\Rightarrow f_1 = \frac{R}{\mu - 1}$$

- a) when plane surface is silvered, $f_m \rightarrow \infty$

Since we know that

$$-\frac{1}{F} = \frac{2}{f_1} - \frac{1}{\infty} = \frac{2(\mu - 1)}{R}$$

$$\Rightarrow F = -\frac{R}{2(\mu - 1)}$$

- b) when convex surface is silvered, then in general we know the relation between radius of curvature and the focal length is given by

$$f_m = \frac{R}{2}$$

Since we know that

$$-\frac{1}{F} = \frac{2}{f_1} + \frac{2}{R} = \frac{2(\mu - 1)}{R} + \frac{2}{R} = \frac{2\mu}{R}$$

$$\Rightarrow F = -\frac{R}{2\mu}$$

Illustration 69

The plane surface of a plano-convex lens of focal length 60 cm is silver plated. A point object is placed at a distance 20 cm from the convex face of lens. Find the position and nature of the final image formed.

Solution

$$\text{Since, } P = 2P_1 + P_m$$

$$\Rightarrow -\frac{1}{F} = \frac{2}{f_1} - \frac{1}{f_m}$$

where, $f_1 = +60$ cm and $f_m \rightarrow \infty$

$$\Rightarrow -\frac{1}{F} = \frac{2}{60} - \frac{1}{\infty} = \frac{1}{30}$$

$$\Rightarrow F = -30 \text{ cm}$$

The problem is reduced to a simple case where a point object is placed in front of a concave (converging) mirror of focal length 30 cm.

Using mirror formula i.e.,

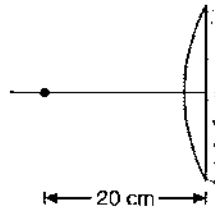
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

where $u = -20$ cm and $f = -30$ cm

$$\Rightarrow \frac{1}{v} + \frac{1}{-20} = \frac{1}{-30}$$

$$\Rightarrow v = 60 \text{ cm}$$

The image is virtual and erect



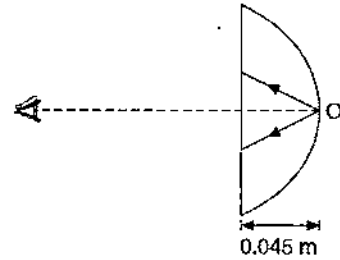
0.03 m and when viewed normally through the curved surface it appears to be 0.036 m. If the actual thickness is 0.045 m, find the

- refractive index of the material of the lens.
- radius of curvature of lens.
- focal length if its plane surface is silvered.
- focal length when the curved surface is silvered.

Solution

- (a) Since, $\mu = \frac{\text{Real Depth}}{\text{Apparent Depth}}$

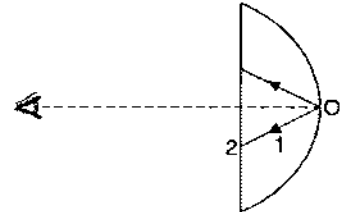
$$\Rightarrow \mu = \frac{d_{\text{actual}}}{d_{\text{app}}} = \frac{0.045}{0.03} = 1.5$$



- (b) Using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

$$\frac{1}{(-0.036)} - \frac{1.5}{(-0.045)} = \frac{1 - 1.5}{(-R)}$$

$$\Rightarrow R = 0.09 \text{ m} = 9 \text{ cm}$$



- (c) If the plane surface is silvered, then

$$\frac{1}{F} = \frac{2}{f_1} + \frac{1}{f_m}$$

But $f_m \rightarrow \infty$

$$\Rightarrow \frac{1}{F} = \frac{2}{f_1}$$

$$\text{where, } \frac{1}{f_1} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{\infty} \right)$$

$$\Rightarrow \frac{1}{F} = \frac{2(\mu - 1)}{R_1}$$

$$\Rightarrow \frac{1}{F} = \frac{2(1.5 - 1)}{+9}$$

$$\Rightarrow F = 9 \text{ cm}$$

The nature is given by applying negative sign to the final result. So, this will behave as a concave mirror.

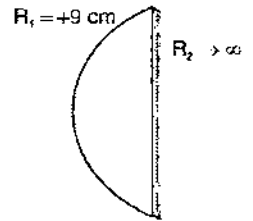


Illustration 70

The greatest thickness of a planoconvex lens when viewed normally through the plane surface appears to be

- (d) When curved surface is silvered
then $R_1 \rightarrow \infty$, $R_2 = -9$ cm

$$\frac{1}{F} = \frac{2}{f_1} - \frac{1}{f_m}$$

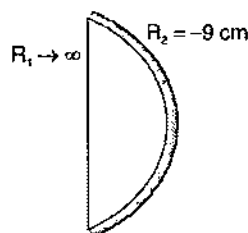
$$\Rightarrow \frac{1}{F} = 2(\mu - 1) \left(\frac{1}{\infty} - \frac{1}{-9} \right) - \frac{2}{-9}$$

$$\Rightarrow \frac{1}{F} = 2 \left(\frac{\mu - 1}{9} \right) + \frac{2}{9}$$

$$\Rightarrow \frac{1}{F} = \frac{2\mu}{9}$$

$$\Rightarrow \frac{1}{F} = \frac{1.5 \times 2}{+9}$$

$$\Rightarrow f = 3 \text{ cm}$$



DEFECTS OF IMAGES : ABERRATIONS

The simple theory of image formation developed for mirrors and lenses suffers from various approximations. As a result, the actual images formed contain several defects. These defects can be broadly divided in two categories.

a) Monochromatic Aberration

The defects, which arise when light of a single colour is used, are called **monochromatic aberrations**.

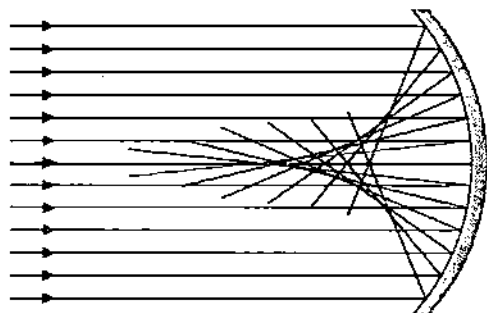
b) Chromatic Aberration

The index of refraction of a transparent medium differs for different wavelengths of the light used. The defects arising from such a variation of the refractive index are termed as **chromatic aberrations**.

MONOCHROMATIC ABERRATIONS

A. SPHERICAL ABERRATION

Throughout the discussion of lenses and mirrors with spherical surfaces, it has been assumed that the aperture of the lens or the mirror is small and the light rays of interest make small angles with the principal axis. Only then, it is possible to have a point image of a point-object.



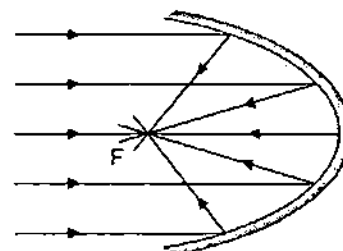
The rays reflect or refract from points at different distances from the principal axis. In general, they meet each other at different points. Thus, the image of a point object is a blurred surface. Such a defect is called **Spherical Aberration**. Figure

shows spherical aberration for a concave mirror for an object at infinity. The rays parallel to the principal axis are incident on the spherical surface of the concave mirror. The rays close to the principal axis (**Paraxial Rays**) are focused at the geometrical focus F of the mirror. The rays farthest from the principal axis (**Marginal Rays**) are focused at a point F' somewhat closer to the mirror. The intermediate rays focus at different points between F and F' . Also, the rays reflected from a small portion away from the pole meet at a point off the axis. Thus, a three-dimensional blurred image is formed.

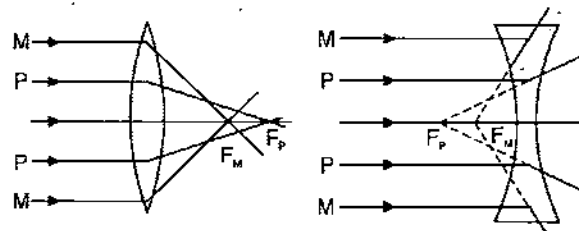
The intersection of this image with the plane of figure is called the **Caustic Curve**.

If a screen is placed perpendicular to the principal axis, a disc image is formed on the screen. As the screen is moved parallel to itself, the disc becomes smallest at one position. This disc is closest to the ideal image and its periphery is called the **Circle of Least Confusion**. The magnitude of spherical aberration may be measured from the distance FF' between the point where the paraxial rays converge and the point where the marginal rays converge.

The parallel rays may be brought to focus at one point if a parabolic mirror is used. Also, if a point source is placed at the focus of a parabolic mirror, the reflected rays will be very nearly parallel. The reflectors used in automobile headlights are made parabolic and the bulb is placed at the focus. The light beam is then nearly parallel and goes up to large distance.



A lens too produces a blurred disc type image of a point object (due to finite aperture of lens). Figure shows the situation for a convex and a concave lens for the rays coming parallel to the principal axis. We see from the figure that the marginal rays deviate a bit strongly and hence, they meet at a point different from that given by geometrical optics formulae. Also, in the situation shown, the spherical aberration is opposite for convex and concave lens. The point F_m , where the marginal rays meet, is to the left of the focus for convex lens and is to the right of the focus for the concave lens.



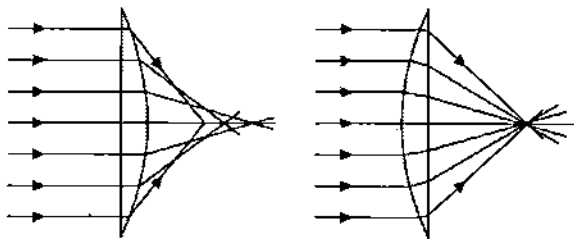
The magnitude of spherical aberration for a lens depends on the radii of curvature and the object distance. For minimum spherical aberration the ratio of radii of curvature of lens is

$$\frac{R_1}{R_2} = \frac{2\mu^2 - \mu - 4}{\mu(2\mu + 1)}$$

However, it cannot be reduced to zero for a single lens which forms a real image of a real object.

A simple method to reduce spherical aberration is to use a stop before and in front of the lens. A stop is an opaque sheet with a small circular opening in it. It only allows a narrow pencil of rays to go through the lens hence reducing the aberration. However, this method reduces the intensity of the image as most of the light is cut off.

Otherwise, the spherical aberration is less if the total deviation of the rays is distributed over the two surfaces of the lens. Example for this is a planoconvex lens forming the image of a distant object. If the plane surface faces the incident rays, the spherical aberration is much larger than that in the case when the curved surface faces the incident rays. In the former case, the total deviation occurs at a single surface whereas it is distributed at both the surfaces in the latter case.



The spherical aberration can also be reduced by using a combination of convex and concave lenses. A suitable combination can reduce the spherical aberration by compensation of positive and negative aberrations.

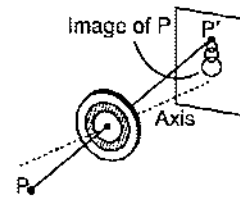
If two thin lenses are separated by a distance d , then condition for minimum spherical aberration is

$$d = f_1 - f_2$$

B. COMA

It has been observed that if a point object is placed on the principal axis of a lens and the image is received on a screen perpendicular to the principal axis, the image has a shape of a disc because of spherical aberration. The basic reason is that the rays passing through different regions of the lens meet the principal axis at different points. If the point object is placed away from the principal axis and the image is received on a screen perpendicular to the axis, the shape of the image is like a comet. This defect is called Coma. The lens fails to converge all the rays passing at different distances from the axis at a single point. The paraxial rays form an image of P at P' . The rays passing through the shaded zone form a circular image on the screen above P' . The rays through outer zones of the lens form bigger circles placed

further above P' . The image seen on the screen thus have a comet-like appearance.



Coma can be reduced by properly designing the radii of curvature of lens surfaces. It can also be reduced by using appropriate stops placed at appropriate distance from the lens.

C. ASTIGMATISM

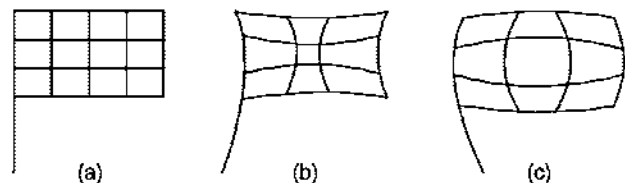
Spherical aberration and coma refer to the spreading of the image of a point object in a plane perpendicular to the principal axis. The image is also spread along the principal axis. Consider a point object placed at a point off the axis of a converging lens. A screen is placed perpendicular to the axis and is moved along the axis. At a certain distance, an approximate line image is focused. If the screen is moved further away, the shape of the image changes but it remains on the screen for quite a distance moved by the screen. The spreading of image along the principal axis is known as Astigmatism (you must not confuse this with a defect of vision having the same name).

D. CURVATURE

So far we have considered the image formed by a lens on a plane. However, it must be kept in mind that the best image may not be formed along a plane. For a point object placed off the axis, the image is spread both along and perpendicular to the principal axis. The best image is, in general, obtained not on a plane but on a curved surface. This defect is known as curvature. It is intrinsically related to astigmatism. The astigmatism or the curvature may be reduced by using proper stops placed at proper locations along the axis.

E. DISTORTION

It is the defect arising when extended objects are imaged. Different portions of the object are, in general, at different distances from the axis. The relation between the object distance and the image distance is not linear and hence, the magnification is not the same for all portions of the extended object. Hence a line object is not imaged into a line but into a curve and shown.



Object (a) and its distorted Images (b) & (c)

CHROMATIC ABERRATION

The inability of a lens to form the white image of a white object is called **chromatic aberration**. In this case the lens forms coloured images of a white object.

The chromatic aberration arises due to the fact that the focal length of a lens depends upon the refractive index of material of the lens. The lens has different refractive indices for different colours or wavelengths in accordance with Cauchy's formula given by

$$\mu = A + \frac{B}{\lambda^2}$$

Accordingly, the refractive index is maximum for violet ($\lambda = 4000\text{\AA}$) and minimum for red ($\lambda = 7800\text{\AA}$). Since

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

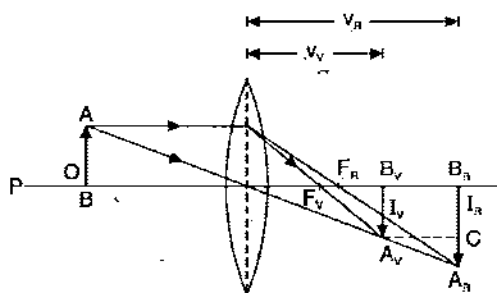
$$\Rightarrow f \propto \frac{1}{\mu - 1}$$

Hence focal length of a lens is maximum for red and minimum for violet

$$\Rightarrow f_{\text{red}} > f_{\text{violet}}$$

Figure represents the chromatic aberration caused by a lens in the image of an object AB of size O .

F_R and F_V are second principal foci for red and violet colours respectively. The images of object AB are of different sizes and of different colours between $A_V B_V$ and $A_R B_R$. The chromatic aberration is of two types.



Chromatic Aberration

A. AXIAL CHROMATIC ABERRATION

This is the spread of images along the principal axis and is given by

$$v_R - v_V = \frac{\omega v^2}{f}$$

where ω is dispersive power, v is distance of image from lens for mean (yellow) colour and f is mean focal length of lens.

If object is at infinity, then axial chromatic aberration,

$$f_R - f_V = \omega f$$

B. LATERAL CHROMATIC ABERRATION

This is the spread of images perpendicular to principal axis and is given by

$$I_R - I_V = \frac{v_R O}{u} - \frac{v_V O}{u} = (v_R - v_V) \frac{O}{u} = \left(\frac{\omega v^2}{f} \right) \frac{O}{u}$$

ACHROMATISM & ACHROMATIC DOUBLET

The lens system free from chromatic aberration is called **achromatic combination**. This is obtained by using two lenses of different materials and different focal lengths and process is called, to **Achromatise** which satisfies the relation

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

$$\Rightarrow \frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2}$$

where ω_1 and ω_2 are dispersive powers of materials of lenses for focal length f_1 and f_2 respectively.

- As ω_1 and ω_2 are always positive, therefore f_1/f_2 must be negative. This means the combination must have one lens convergent and other divergent.
- For the achromatic combination (also called **Achromatic Doublet**) to be convergent, the power of convex lens must be greater or the focal length of convex lens must be smaller than that of concave lens. As dispersive power for crown glass is less than that for flint glass, therefore the convex lens must be made of crown glass while concave lens must be made of flint glass.

Condition for minimum chromatic aberration obtained by two thin lenses of same medium separated by a distance d is

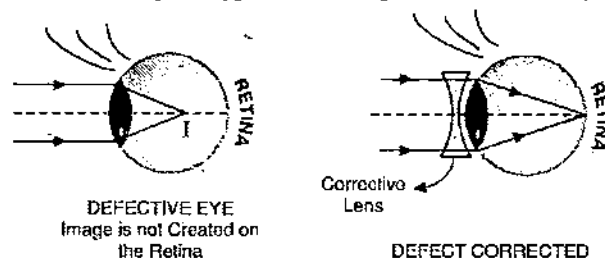
$$d = \frac{f_1 + f_2}{2}$$

DEFECTS OF EYE

A normal eye has nearer point at $D(25\text{ cm})$ called distance of distinct vision and far point at ∞ .

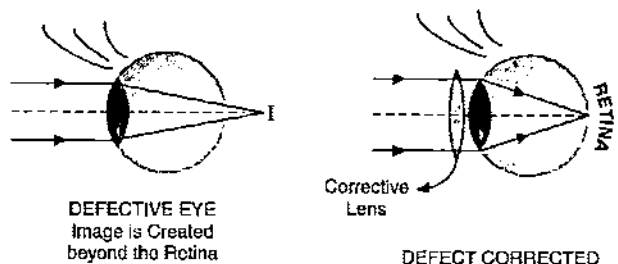
A. SHORT-SIGHTEDNESS OR MYOPIA

A short-sighted eye can see only nearer objects. It is due to elongation of eye-ball. It is corrected by using a concave lens, whose focal length is equal to the far point of defective eye.



B. LONG-SIGHTEDNESS OR HYPERMETROPIA

A long sighted eye can see only farther objects. It is due to contraction of eye-ball. It is corrected by using convex lens. This lens brings the nearer point of defective eye at a distance which equals to the distance of distance of distinct vision $D (= 25 \text{ cm})$.



C. PRESBYOPIA

A presbyopic eye can see objects only within a definite range. This defect is corrected by using bifocal lenses.

D. ASTIGMATISM

It arises due to distortion in spherical shape in cornea. This defect is corrected by using cylindrical lenses.

OPTICAL INSTRUMENTS

An optical instrument is a device which is constructed by a suitable combination of mirrors, prisms and lenses. The principle of working of an optical instrument is based on the laws of reflection and refraction of light.

The common types of optical instrument are

- a) **Projection instruments:** These are used to project on the screen a real, inverted and magnified image of an opaque or transparent object so as to be viewed by a large audience. The object is, however, so fitted that its image is seen in erect form.

An eye, a photographic camera, a projection lantern, an episcopes, an epidiascope, an over-head projector, a film projector, etc., are examples of projection instruments.

- b) **Microscopes:** These are used to see very small objects in magnified form which otherwise cannot be seen distinctly when placed close to the naked eye.

EXAMPLE

A simple microscope and a compound microscope.

- c) **Telescope:** These are used to see astronomical and distant objects in magnified form which, otherwise cannot be seen clearly with the naked eye.

EXAMPLE

An astronomical telescope, a Galilean telescope, a terrestrial telescope, a reflecting telescope, etc.

SIMPLE MICROSCOPE (MAGNIFYING GLASS)

A convex lens of short focal length can be used to see magnified image of a small object and is called a **magnifying glass** or a **simple microscope**.

When a small object is placed between optical centre and focus of a convex lens, its virtual, erect and magnified image is formed on the same side of the lens. The lens is held close to eye and the distance of the object is adjusted, till the image is formed at the least distance of distinct vision from the eye. For a normal eye, the least distance of distinct vision is 25 cm.

MAGNIFYING POWER

Case I : When image is formed at D

It is defined as the ratio of the angle subtended by the image at the eye and the angle subtended by the object seen directly, when both lie at the least distance of distinct vision. It is also called **angular magnification produced by the simple microscope**. It is denoted by M .

By definition, magnifying power of the simple microscope is given by

$$M = \frac{\beta}{\alpha}$$

In particle, angles α and β are small. Therefore, angles α and β can be replaced by their tangents i.e.

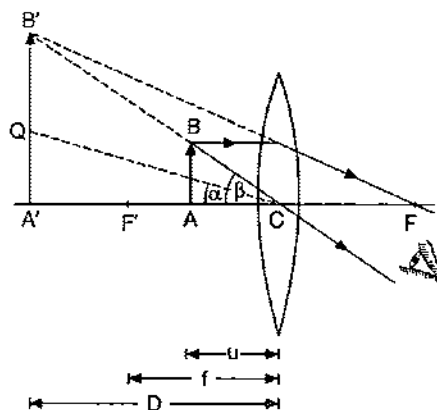
$$M = \frac{\tan \beta}{\tan \alpha} = \frac{CA'}{CA} = \frac{v}{u}$$

If f is focal length of the lens acting as simple microscope, then

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{v}{u} = 1 - \frac{v}{f}$$

$$\Rightarrow M = 1 - \frac{v}{f}$$



Since, image is formed at distance of distinct vision, so according to new Cartesian sign convention.

$$v = -D \text{ and } f = +f$$

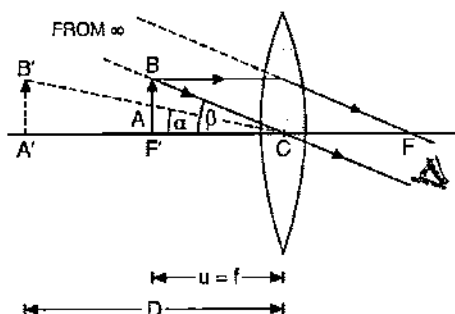
$$\Rightarrow M = 1 + \frac{D}{f}$$

CONCEPTUAL NOTE(S)

- From above it follows that lesser is the focal length of the convex lens used as simple microscope, greater is the value of the magnifying power obtained.
- Further, the positive value of magnifying power of a simple microscope tells that image formed is erect and hence virtual.

Case II : When image is formed at infinity

It is defined as the ratio of the angle formed by the image (situated at infinity) at the eye to the angle formed by object at the eye, when situated at least distance of distinct vision. It is denoted by M .



Draw a line $A'B' = AB$ and perpendicular to principal axis at a distance $CA' = D$ (least distance of distinct vision). Join $B'C$. Then $\angle B'CA' = \alpha$ is the angle formed by object at the eye, when situated at distance D .

The angle formed by the image situated at infinity at the eye is same as the angle formed by the object AB at the eye. Thus, $\angle BCA = \beta$ is the angle formed by the image at the eye,

By definition,

$$M = \frac{\beta}{\alpha}$$

In practice, angles α and β are small. Therefore, angles α and β can be replaced by their tangents i.e.

$$M = \frac{\tan \beta}{\tan \alpha} = \frac{CA'}{CF} = \frac{v}{u} \quad \{\because A'B' = AB\}$$

Applying new cartesian sign conventions

$$CF = -f \text{ and } CA' = -D$$

$$\Rightarrow M = \frac{D}{f}$$

CONCEPTUAL NOTE(S)

It follows that magnifying power of the simple microscope is one less, when the image is formed at infinity. However, the viewing of the image is more comfortable.

USES

- Jewellers and watch makers make use of convex lens of short focal length to obtain a magnified view of the fine jewellery work and the small components of the watches.

- In science laboratories, a magnifying glass is used to see slides and to read the vernier scales attached to the instruments.
- The use of magnifying glass enables us to place the object close to eye, making it appear bright and yet clearly visible. In position AB, object lies close to the eye. In absence of lens, the object will not be clearly visible.
- It is also used by astrologers to read the fate lines of the hand.
- Used by Biology students to see slides.
- Used by detective department to match finger prints.

Illustration 71

A man with normal near point (25 cm) reads a book with small print using a magnifying glass (a thin convex lens) of focal length 5 cm. Find the

- closest and farthest distance at which he can read the book when viewing through the magnifying glass.
- maximum and minimum magnifying power possible using the above simple microscope.

Solution

- For a normal eye, far and near points are ∞ and 25 cm, respectively. So, we have

$$v_{\max} \rightarrow \infty \text{ and } v_{\min} = -25 \text{ cm}$$

Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow u = \frac{f}{\left(\frac{1}{v}\right) - 1}$$

So, u will be minimum, when v is minimum i.e., $v_{\min} = -25 \text{ cm}$

$$\Rightarrow (u)_{\min} = \frac{5}{-\left(\frac{1}{25}\right) - 1} = -\frac{25}{6} = -4.17 \text{ cm}$$

And u will be maximum, when v is maximum i.e., $v_{\max} \rightarrow \infty$

$$\Rightarrow (u)_{\max} = \frac{5}{\left(\frac{1}{\infty}\right) - 1} = -5 \text{ cm}$$

- Since magnifying power for a lens is

$$m = \frac{v}{u}$$

Magnifying power will be minimum, when u is maximum i.e., $u_{\max} = -5 \text{ cm}$

$$\Rightarrow (m)_{\min} = \frac{D}{f} = \frac{-25}{-5} = 5$$

m will be maximum, when u is minimum i.e.,

$$u_{\min} = -\frac{25}{6} = -4.17 \text{ cm}$$

$$\Rightarrow (m)_{\max} = \frac{-25}{\frac{-25}{6}} = 6 \quad \left\{ = 1 + \frac{D}{f} \right\}$$

COMPOUND MICROSCOPE

A compound microscope is used to see extremely small objects.

It consists of two lenses. A lens of **short aperture and short focal length** facing the object is called **object lens** and another lens of **short focal length but large aperture** is called **eye lens**. The two lenses are placed coaxially at the two ends of a tube. To focus over an object, the distance of the object lens from the object is adjusted with the help of rack and pinion arrangement.

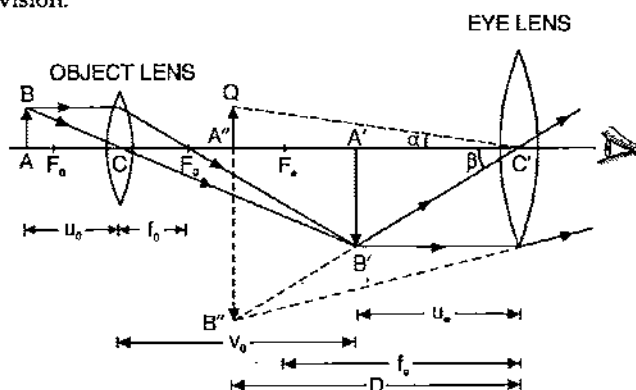
When a small object is placed just outside the focus of the object lens, its real, inverted and magnified image is produced on the other side of the lens between F and $2F$. The image produced by object lens acts as object for the eye lens. The distance of object from the object lens is so adjusted that the final image is formed at the least distance of distinct vision from the eye.

Let AB be an object placed just outside the focus F_0 of the object lens. Its virtual image $A'B'$ is formed on the other side of the lens. The image $A'B'$ lies between focus F_e and optical centre C' of the eye lens and it acts as object for the eye lens. Using the rack and pinion arrangement, the distance between object lens and the object AB is adjusted, till it virtual and magnified image $A''B''$ is formed on the same side at the least distance of distinct vision.

MAGNIFYING POWER

Case I : When image is formed at D

It is defined as the ratio of the angle subtended by the final image at the eye to the angle subtended by the object seen directly, when both are placed at the least distance of distinct vision.



Let $\angle A''C'B'' = \beta$ be the angle subtended by the final image at the eye. Cut $A''Q$ equal to AB and join QC' . Then, $\angle A''C'Q = \alpha$, the angle subtended by the object at the eye, when situated at the least distance of distinct vision. By definition, magnifying power of the compound microscope,

$$M = \frac{\beta}{\alpha}$$

Since the angles α and β are small, they can be replaced by their tangents i.e.

$$M = \frac{\tan \beta}{\tan \alpha} = \frac{A''B''}{AB}$$

$$\left\{ \because \tan \beta = \frac{A''B''}{C'A''} \text{ and } \tan \alpha = \frac{A''Q}{C'A''} = \frac{AB}{C'A''} \text{ (as } A''Q = AB) \right\}$$

Multiplying and dividing by $A'B'$, we have

$$M = \frac{A''B''}{AB} \times \frac{A'B'}{A'B'} = \frac{A'B'}{AB} \times \frac{A''B''}{A'B'}$$

$$\Rightarrow M = m_0 \times m_e$$

where $m_0 = \frac{v_0}{u_0} = \frac{A'B'}{AB}$ = Magnification produced by Object lens.

and $m_e = \frac{v_e}{u_e} = \frac{A''B''}{A'B'}$ = Magnification produced by Eye lens.

Further, for the eye lens,

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$\Rightarrow \frac{v_e}{u_e} = 1 - \frac{v_e}{f_e}$$

$$\Rightarrow m_e = 1 - \frac{v_e}{f_e}$$

Applying new Cartesian sign conventions we get

$$v_e = -D \text{ and } f_e = +f_e$$

$$\Rightarrow m_e = 1 + \frac{D}{f_e}$$

So, finally we get

$$M = m_0 \times m_e = \frac{v_0}{u_0} \left(1 + \frac{D}{f_e} \right)$$

In practice, the focal length of the object lens is very short and the object AB is placed just outside the focus of object lens. Hence

$$u_0 \approx f_0 \text{ (in magnitude)}$$

So that the final image is formed at least distance of distinct vision from the eye, the image $A'B'$ must lie between optical centre and focus of the eye lens. Since, the focal length of the eye lens is also small, the distance of the image $A'B'$ from the object lens is nearly equal to the length (L) of the microscope tube i.e. $v_0 \approx L$

According to new Cartesian sign conventions we have

$$u_0 \approx -f_0 \text{ and } v_0 \approx L$$

Substituting for u_0 and v_0 we get

$$M = -\frac{L}{f_0} \left(1 + \frac{D}{f_e} \right)$$

CONCEPTUAL NOTE(S)

- From the above expression, it follows that a compound microscope will have large magnifying power, if both the object lens and the eye lens are of small focal length. In practice, focal length of object lens is smaller than that of eye lens i.e. $f_o < f_e$. Further the **negative** value of magnifying power of compound microscope tells that final image formed is **inverted**.
- In practice, to eliminate chromatic aberration, a combination of two lenses in contact is used. It is called **objective**.
- In place of an eye lens, a combination of two lenses at certain distance apart satisfying certain conditions (to minimize chromatic and spherical aberrations) is used. It is called **eye piece**.

Case II : When image is formed at infinity

The magnifying power of a compound microscope,

$$M = m_o \times m_e$$

If u_o is distance of object from the object lens and v_o , then distance of image from it, then as obtained earlier,

$$m_o = \frac{v_o}{u_o}$$

Further, as in the case of a simple microscope we have

$$m_e = \frac{D}{f_e}$$

where f_e is focal length of the eye lens.

Therefore, magnifying power of a compound microscope is given by

$$M = \frac{v_o}{u_o} \times \frac{D}{f_e}$$

Illustration 72

A compound microscope has a magnifying power 30. The focal length of its eye-piece is 5 cm. Assuming the final image to be at the least distance of distinct vision (25 cm), find the magnification produced by the objective.

Solution

For a compound microscope, we have

$$M = m_o m_e \quad \dots (1)$$

Since the final image is formed at least distance of distinct vision, the magnification of eye-piece is

$$m_e = \left(1 + \frac{D}{f_e}\right) = 1 + \frac{25}{5} = 6$$

From equation (1), we get

$$-30 = m_o \times 6$$

$$\Rightarrow m_o = -\frac{30}{6} = -5$$

Negative sign implies that image formed by the objective is inverted.

ASTRONOMICAL TELESCOPE (REFRACTING TYPE)

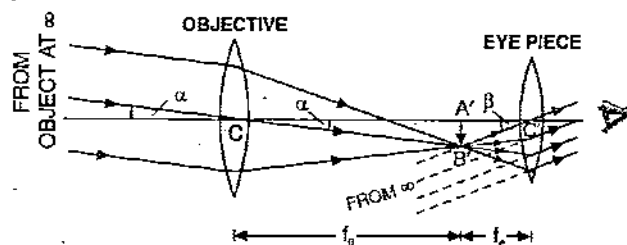
An astronomical telescope is used to see heavenly objects. It produces a virtual and inverted image. As such bodies are round, the inverted image does not affect the observation.

An astronomical refracting telescope consists of two lens systems. The lens system facing the object is called **objective**. It has large aperture and is of **large focal length** (f_o). The other lens system is called **eye-piece**. It has **small aperture** and is of **short focal length** (f_e). The objective and the eye-piece are mounted coaxially in two metallic tubes. The tube holding the eye-piece can be made to slide into the tube holding the objective with the help of rack and pinion arrangement.

The objective forms the real and inverted image of the distant object in its focal plane. The position of the eye-piece is adjusted, till the final image is formed at **least distance of distinct vision**. In case, position of the eye-piece is adjusted such that final image is formed at infinity, the telescope is said to be in **normal adjustment**. The image formation is discussed in the following two cases.

Case I : When final image is formed at infinity (Normal adjustment)

When a parallel beam of light rays from the distant object falls on the objective, its real and inverted image $A'B'$ is formed on the other side of the objective. If the position of eye-piece is adjusted, so that the image $A'B'$ lies at its focus, then the final highly magnified image will be formed at infinity. Under such a situation i.e. in normal adjustment the magnifying power of a telescope is defined as the ratio of the angle subtended by the image at the eye as seen through the telescope to the angle subtended by the object seen directly, when both the object and the image lie at infinity. It is also called **angular magnification** of the telescope and is denoted by M .



As the object is at a very large distance, the angle subtended by it at the eye is practically the same as that subtended by it at the objective.

Thus, $\angle A'CB' = \alpha$ may be considered as the angle subtended by object at the eye. Let $\angle A'C'B' = \beta$.

Then

$$M = \frac{\beta}{\alpha}$$

Since the angles α and β are small, $\alpha \approx \tan \alpha$ and $\beta \approx \tan \beta$. Therefore,

$$M = \frac{\tan \beta}{\tan \alpha} = \frac{CA'}{CA'}$$

Using new Cartesian sign conventions we get

$$CA' = +f_o$$

{ $\therefore$ distance of $A'B'$ from object lens is along the incident light}

$$CA' = -f_e$$

{ $\therefore$ distance of $A'B'$ from eye lens is against incident light}

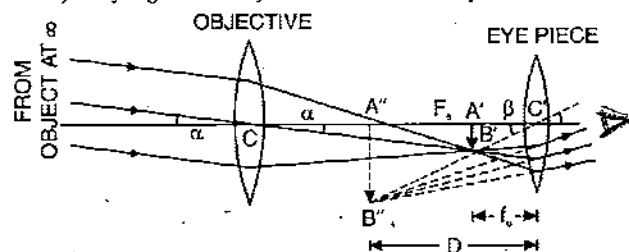
$$\Rightarrow M = \frac{f_o}{f_e}$$

CONCEPTUAL NOTE(S)

- It follows that the magnifying power of a telescope in normal adjustment will be large, if objective is of large focal length and the eye-piece is of short focal length.
- Further, when telescope is in normal adjustment, the distance between the two lenses is equal to sum of their focal lengths ($f_o + f_e$).
- Further, the **negative** value of the magnifying power of the telescope tells that final image formed is inverted and real.

Case II : When final image is formed at least distance of distinct vision

When a parallel beam of light rays from the distant object falls on the objective, its real and inverted image $A'B'$ is formed on the other side of the objective. The position of eye-piece is adjusted so that the final image $A''B''$ is formed at least distance of distinct vision. Under such a situation the magnifying power of a telescope is defined as the ratio of the angle subtended at the eye by the image formed at the least distance of distinct vision to angle subtended at the eye by the object lying at infinity, when seen directly.



Again, as the object is at a very large distance, the angle α subtended by it at the objective is practically the same as that subtended by it at the eye. Therefore, if $\angle A''C'B'' = \beta$, then

$$M = \frac{\beta}{\alpha}$$

Again, as angle α and β are small, they can be replaced by their tangents,

$$\Rightarrow M = \frac{\tan \beta}{\tan \alpha} = \frac{CA'}{CA'}$$

Since, $CA' = f_o$ and $CA' = u_e$,

$$\Rightarrow M = \frac{f_o}{u_e}$$

For eye lens, we have

$$\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$$

$$\Rightarrow \frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e}$$

$$\Rightarrow \frac{1}{u_e} = -\frac{1}{f_e} \left(1 - \frac{f_e}{v_e} \right)$$

$$\Rightarrow M = -\frac{f_o}{f_e} \left(1 - \frac{f_e}{v_e} \right)$$

Applying new Cartesian sign conventions we get

$$f_o = +f_o, v_e = -D, f_e = +f_e$$

$$\Rightarrow M = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

Therefore, a refracting telescope will have large magnifying power, if the object lens is of large focal length and eye lens is of short focal length. Further, the **negative** value of magnifying power of the telescope tells that the final image formed is **inverted** and **real**. Out of the two adjustments discussed, this adjustment gives a higher magnification,

since the factor $\left(1 + \frac{f_e}{D} \right)$ is greater than one.

Also, a telescope does not increase the size of object, but it forms an image nearer so that the angle of vision is increased and hence it appears to us as if the bigger image of object is formed.

Illustration 73

The objective of a telescope is a convex lens of focal length 100 cm. Its eye-piece is also a convex lens of focal length 5 cm. Determine the magnifying power of the telescope for normal adjustment.

Solution

For normal adjustment, the magnifying power of a telescope is given by

$$M = \frac{f_o}{f_e}$$

Here, $f_o = 100$ cm, $f_e = 5$ cm

$$\Rightarrow M = \frac{100}{5} = 20$$

TERRESTRIAL TELESCOPE

A terrestrial telescope is used to observe objects on earth.

An astronomical telescope is used to view heavenly objects since the inversion of their images does not produce any complication. While viewing earthly objects we would prefer to have their images erect and hence, astronomical telescope is not suitable in such cases. By using an additional convex

lens O (of focal length f) in between O_1 and O_2 of an astronomical telescope, we can have the final erect image. The lens O is called **erecting lens**, while the improved version of the telescope is called **Terrestrial Telescope**.

Rays from the distant object get refracted through the objective O_1 , giving a real inverted image A_1B_1 . The erecting lens O is so adjusted that its distance from A_1B_1 is equal to twice its (erecting lens) focal length. An image A_2B_2 having same size as that of A_1B_1 , inverted w.r.t. A_1B_1 and hence erect w.r.t. the object is obtained at a distance $2f$ on other side of O . A_2B_2 acts as an object for lens at O_2 and finally an erect and magnified image is obtained after refraction through O_2 . If the distance O_2B_2 is equal to focal length f_e of the eye lens O_2 , final image is formed at infinity and the telescope is said to be in **normal adjustment** as in Figure 1.

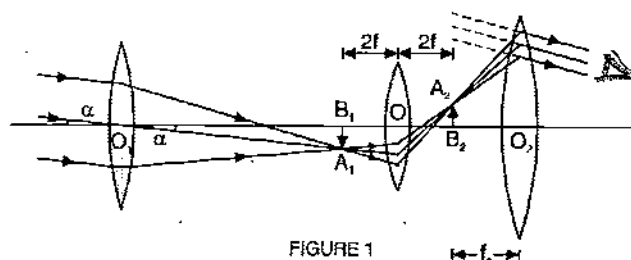


FIGURE 1

If the distance O_2B_2 is less than f_e , then corresponding to a certain value of this distance, a virtual and magnified image is obtained at the distance of distinct vision as shown in Figure 2.

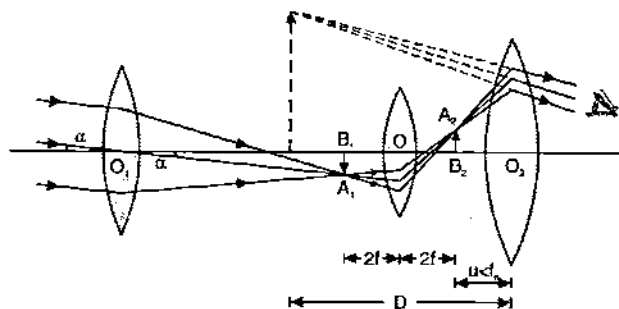
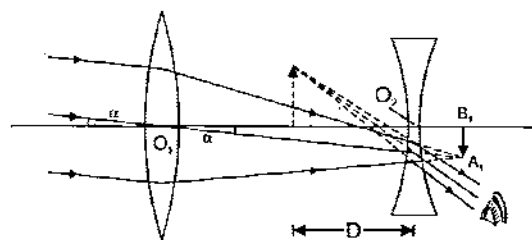


FIGURE 2

Since the sizes of A_2B_2 and A_1B_1 are same, introduction of the erecting lens O has not produced any change in its magnifying power but, has helped in getting the final image erect only. It may also be noted that the use of erecting lens O results in an increase (equal to four times the focal length of erecting lens) in the length of the tube of telescope.

GALILEO'S TELESCOPE

Instead of using a combination of two, lens O_1 and O_2 for getting an erect image, Galileo used only one concave lenses to get the final erect image.



Parallel beam of incident rays from infinity are focussed by the objective O_1 . An inverted image A_1B_1 (shown in grey) would have been formed after refraction through O_1 . Before the rays meet at A_1 , a concave lens (at O_2) intercepts them. The beam diverges and the final erect image A_2B_2 is obtained. The distance O_2B_1 is so adjusted that final image is formed at the distance of distinct vision. If O_2B_1 is equal to the focal length f_e of eye lens at O_2 final image is formed at infinity and the telescope is said to be set in **normal adjustment**. In such a case the length of the tube is equal to the difference between the focal lengths of two lenses. The field of view of this telescope is small because of the use of concave lens.

When set in normal adjustment, its magnifying power M is given by

$$M = \frac{\beta}{\alpha} = \frac{\tan \beta}{\tan \alpha} = \frac{B_1O_2}{A_1B_1} = \frac{B_1O_2}{B_1O_1}$$

$$\Rightarrow M = \frac{F}{f} = \frac{\text{Focal length of objective}}{\text{Focal length of eye lens}}$$

RESOLVING POWER

The resolving power of an optical instrument is defined as the reciprocal of smallest angular separation between two neighbouring objects whose images are just distinctly formed by the instrument. The smallest angular separation is called the **Limit of Resolution**.

A. LIMIT OF RESOLUTION FOR MICROSCOPE

Limit of Resolution is given by

$$S = \frac{\lambda}{2\mu \sin \theta}$$

where $\mu \sin \theta$ is called numerical aperture and μ is refractive index of medium in which it is placed. For high resolution of microscope a beam of electrons is used which has wavelength of the order of $1 \text{ \AA}$.

B. LIMIT OF RESOLUTION FOR TELESCOPE

If a is aperture or diameter of telescope and λ the wavelength, then resolving limit is

$$d\theta \propto \frac{\lambda}{a}$$

For spherical aperture $d\theta = \frac{1.22\lambda}{a}$

Resolving power $\propto \frac{a}{\lambda}$

Resolving power of telescope or microscope has no concern with focal lengths of lenses.

PHOTOMETRY

A. RADIANT FLUX

It is the radiant energy emitted by a body per second in all directions including all wavelengths. Its unit is watt.

B. LUMINOUS FLUX

The radiant energy emitted by a body per second in the visible region (i.e., between wavelengths from 4000Å to 7800Å) is called luminous flux. Its unit is lumen.

C. LUMINOUS INTENSITY OF A LIGHT SOURCE

The luminous intensity of a light source in any direction may be defined as the luminous flux per unit solid angle in that direction.

Its unit is lumen/steradian or candela.

Luminous Intensity

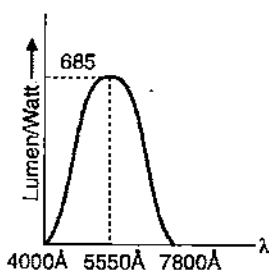
$$I = \frac{\Delta F}{\Delta \Omega} \text{ (in lumen/steradian or candela)}$$

If the light source is isotropic, then luminous flux is uniform in all directions, so that total luminous flux is given by

$$F = 4\pi I \text{ (since total solid angle for all directions is } 4\pi \text{)}$$

If we plot the ratio luminous flux/radiant flux against wavelength of radiation, we get a graph as shown in figure.

The graph indicates that the ratio luminous flux/radiant flux is maximum for 5550Å i.e. for yellow colour. This indicates that our eye is most sensitive for



5550Å i.e. yellow colour. The maximum value of ratio luminous flux/radiant flux is 685 lumen/watt. Thus when the ratio luminous flux/radiant flux is 685 lumen/watt, the luminous efficiency is said to be 100%.

The tungsten filament bulb converts 2-3% electrical energy into visible light energy, while fluorescent tube converts 8-9% electrical energy into visible light energy.

D. ILLUMINANCE OF A SURFACE (E)

It may be defined as the luminous flux falling per unit area on the surface. Its unit is lumenm⁻² or lux.

$$E = \frac{\Delta F}{\Delta S}$$

E. INVERSE SQUARE LAW OBEYED BY ILLUMINANCE

When radiant energy falls normally on a surface, the illuminance E of the surface is inversely proportional to the square of distance of surface point from source i.e.

$$E \propto \frac{1}{r^2} \quad \dots (1)$$

F. LAMBERT'S COSINE LAW OBEYED BY ILLUMINANCE

When radiant energy falls obliquely on a surface, the illuminance E of surface is directly proportional to the cosine of angle made by normal to the surface with the direction of incident radiation. So,

$$E \propto \cos \theta \quad \dots (2)$$

Combining (1) and (2), we get

$$E = \frac{I \cos \theta}{r^2} \quad \dots (3)$$

Total Luminous Energy falling on a surface is given by

$$Q = EA t$$

where E = illuminance of the surface

A = area and

t = exposure time.

The total luminous energy required to be incident on a given type of camera film is constant.

For a box type camera

$$\text{Time of exposure} \propto \left(\frac{f}{d}\right)^2$$

where d = diameter

f = focal length of camera lens.

G. THE LUMINANCE (L) OR BRIGHTNESS

Luminance of a surface is the luminous flux reflected by unit area of the surface normally.

$$\Rightarrow \text{Luminance} = \text{Illuminance} \times \text{Reflection Coefficient.}$$

H. PRINCIPLE OF PHOTOMETRY

If two sources of light of illuminating power I_1 and I_2 are placed at distances r_1 and r_2 from the screen, then the screen will be equally illuminated due to two sources when

$$\frac{I_1}{r_1^2} = \frac{I_2}{r_2^2}$$

ICE VII

BASED ON LENS FORMULA

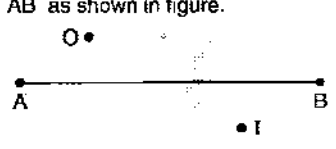
(Solutions on page 1.189)

- The distance between two point sources of light is 24 cm. Find out where would you place a converging lens of focal length 9 cm, so that the images of both the sources are formed at the same point.

2. An object is moved along the principal axis of a convex lens. An image three times the size of the object is obtained when the object is at a distance of 16 cm from the lens and at a distance of 8 cm from the lens. Find the focal length of the lens.
3. The radius of curvature of the convex surface of a plano-convex lens is 10 cm and its focal length is 30 cm. What should be the refractive index of its material?
4. One face of an equi-convex lens ($\mu = 1.5$) of focal length 60 cm is silvered. Does it behave like a concave mirror or convex mirror? Also determine the equivalent focal length of the mirror.
5. A biconvex lens made of glass with a refractive index of $\mu = 1.6$ has a focal length of $f = 10$ cm in air. Calculate the focal length of this lens if it is placed into a transparent medium
 - (a) with a refractive index of $\mu_1 = 1.5$
 - (b) with a refractive index of $\mu_2 = 1.7$
6. A biconvex thin lens is prepared from glass of refractive index $\frac{3}{2}$. The two bounding surfaces have equal radii of 25 cm each. One of the surfaces is silvered from outside to make it reflecting. Where should an object be placed before this lens so that the image coincides with the object.
7. A converging lens of focal length 5 cm is placed in contact with a diverging lens of focal length 10 cm. Find the combined focal length of the system.
8. A biconvex lens of refractive index 1.5 has a focal length of $f = 10$ cm. One of the lens surfaces having a radius of curvature of $R = 10$ cm is coated with silver. Determine the position of the image if the object is at a distance of $u = 15$ cm from the lens.
9. A convex lens is held 45 cm above the bottom of an empty tank. The image of a point at the bottom of a tank is formed 36 cm above the lens. Now a liquid is poured into the tank to a depth of 40 cm. It is found that the distance of the image of the same point on the bottom of the tank is 48 cm above the lens. Find the refractive index of the liquid.
10. A concave spherical mirror with a radius of curvature of 0.2 m is filled with water. Calculate the focal length of this system? Given that refractive index of water is $\frac{4}{3}$.
11. A convex lens of focal length f_1 is placed in front of a luminous point P so that the distance of the point P from lens is greater than focal length and the image formed is at the shortest possible distance. If now a concave lens of very large focal length f_2 be placed in contact with first, find the shift in the position of the image.
12. At what distance from a biconvex lens of focal length $f = 1$ m should a concave spherical mirror with a radius of curvature of $R = 1$ m be placed for a beam incident on the

lens parallel to the major optical axis of the system to leave the lens, remaining parallel to the optical axis, after being reflected from the mirror? Find the image of the object produced by the given optical system.

13. A convex lens of focal length f_1 is placed in front of a luminous point object. The separation between the object and the lens is $3f_1$. A glass slab of thickness t is placed between object and the lens. A real image of the object is formed at the shortest possible distance from the object.
 - (a) Find the refractive index of the slab.
 - (b) If a concave lens of very large focal length f_2 is placed in contact with the convex lens, find the shifting of the image.
14. An optical system consists of two convergent lenses with focal lengths $f_1 = 20$ cm and $f_2 = 10$ cm. The distance between the lenses is $d = 30$ cm. An object is placed at a distance of $u_1 = 30$ cm from the first lens. At what distance from the second lens will the image be obtained?
15. If r be the radius of curvature of each face of thin converging lens whose one face is silvered and μ is the refractive index of lens material, prove that the lens is equivalent to a concave mirror of focal length $\frac{r}{4\mu - 2}$.
16. Three convergent thin lenses of focal lengths $4a$, a and $4a$ respectively are placed in order along the axis so that the distance between consecutive lenses is $4a$. Prove that this combination simply inverts every small object on the axis without change of magnitude or position.
17. The distance between an object and a divergent lens is m times greater than the focal length of the lens. How many times will the image be smaller than the object?
18. An image I is formed of point object O by a lens whose optic axis is AB as shown in figure.



 - (a) State whether it is a convex lens or concave?
 - (b) Draw a ray diagram to locate the lens and its focus.
19. Two thin lenses having focal lengths $f_1 = 7$ cm and $f_2 = 6$ cm are placed at a distance $d = 3$ cm apart. What is the distance of the focus of the system from the second lens? Assume the system to be a centred one.
20. Two glasses with refractive indices of $\mu_1 = 1.5$ and $\mu_2 = 1.7$ are used to make two identical double convex lenses.
 - (a) Find the ratio between their focal lengths.
 - (b) How will each of these lenses act on a ray parallel to its optical axis if the lenses are submerged into a transparent liquid with a refractive index of 1.6?
21. A parallel beam of light is incident on a system consisting of three thin lenses with a common optical axis. The focal

lengths of the lenses are equal to $f_1 = +10$ cm, $f_2 = -20$ cm and $f_3 = +9$ cm, respectively. The distance between the first and the second lenses is 15 cm and between the second and the third 5 cm. Find the position of the point at which the beam converges when it leaves the system of lenses.

22. Consider a plano-concave lens with one of the radii of curvature r made up of a transparent material whose refractive index varies with intensity (I) of incident light as

$$\mu = \mu_0 + aI, \text{ where } a > 0 \text{ and } 0 < \mu_0 < \frac{3}{2}.$$

Calculate the intensity when the focal length is equal to two times the radius of curvature r .

23. Paraxial rays are incident on surfaces of a thin equiconvex glass lens of refractive index μ and having radius of curvature R . If the final image is formed after n internal reflections, calculate distance of this image from pole of the lens.

24. When the plane surface of a plano-convex lens is silvered it is found that the image of the object pin is formed at the position of the object pin placed at a distance of x_1 from the silvered lens. When the same lens is silvered on the curved

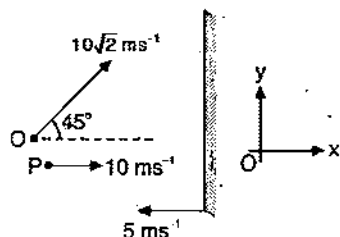
surface the image of the object pin is formed at the position of the object pin placed at a distance of x_2 from the silvered lens. Find, in terms of x_1 and x_2 , the

- focal length of lens
- radius of curvature of the curved surface and
- index of refraction of the medium of lens.

Solved Problems

Problem 1

A plane mirror is moving with a uniform speed of 5 ms^{-1} along negative x -direction and an observer P is moving with a velocity of 10 ms^{-1} along $+x$ direction. Calculate the velocity of image of an object O , moving with a velocity of $10\sqrt{2} \text{ ms}^{-1}$ as shown in the figure, as observed by the observer. Also find its magnitude and direction.



Solution

Let $\vec{v}_O$ be the velocity of the object O , $\vec{v}_P$ be the velocity of the observer P , $\vec{v}_M$ be the velocity of the mirror and $\vec{v}_I$ be the velocity of image (Assume all these velocities w.r.t. ground), then

$$\vec{v}_O = \frac{10\sqrt{2}}{\sqrt{2}}(\hat{i} + \hat{j})$$

$$\vec{v}_O = 10(\hat{i} + \hat{j})$$

$$\vec{v}_P = 10\hat{i}$$

$$\vec{v}_M = -5\hat{i}$$

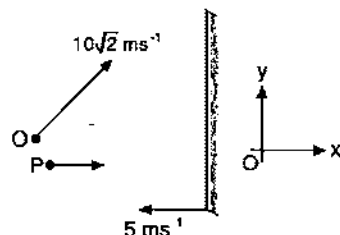
$(\vec{v}_{IM})_{\perp} = -(\vec{v}_{OM})_{\perp}$, where the axis perpendicular to the mirror is the x -axis.

$$\Rightarrow (\vec{v}_I)_x - (\vec{v}_M)_x = -(\vec{v}_O)_x + (\vec{v}_M)_x$$

$$\Rightarrow (\vec{v}_I)_x = 2(\vec{v}_M)_x - (\vec{v}_O)_x$$

$$\Rightarrow (\vec{v}_I)_x = 2(-5\hat{i}) - 10\hat{i}$$

$$\Rightarrow (\vec{v}_I)_x = -20\hat{i}$$



Further, parallel to the mirror, i.e., along y -axis, we have

$$(\vec{v}_I)_y = (\vec{v}_O)_y = 10\hat{j}$$

$$\text{Since } \vec{v}_I = (\vec{v}_I)_x + (\vec{v}_I)_y$$

So, absolute velocity of the image is

$$\vec{v}_I = -20\hat{i} + 10\hat{j}$$

$$\text{Now } \vec{v}_{IP} = \vec{v}_I - \vec{v}_P$$

$$\Rightarrow \vec{v}_{IP} = -20\hat{i} + 10\hat{j} - 10\hat{i}$$

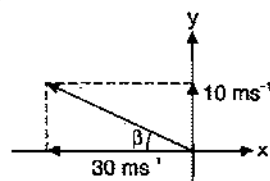
$$\Rightarrow \vec{v}_{IP} = -30\hat{i} + 10\hat{j}$$

$$\Rightarrow |\vec{v}_{IP}| = \sqrt{900 + 100} = 10\sqrt{10} \text{ ms}^{-1}$$

If β is the angle made by $\vec{v}_{IP}$ with $-x$ axis, then

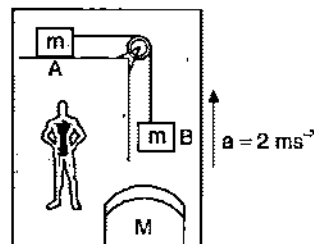
$$\tan \beta = \frac{10}{30}$$

$$\Rightarrow \beta = \tan^{-1}\left(\frac{1}{3}\right), \text{ with } -x \text{ axis}$$



Problem 2

Consider the situation shown in figure. The elevator is going up with an acceleration of 2 ms^{-2} and the focal length of the mirror is 12 cm . All the surfaces are smooth and the pulley is light. The mass pulley system is released from rest (w.r.t. the elevator) at $t=0$ when the distance of B from the mirror is 42 cm . Find the distance between the image of the block B and the mirror at $t=0.2 \text{ s}$. Take $g = 10 \text{ ms}^{-2}$.



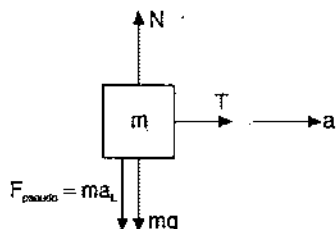
Solution

Let us assume that the acceleration of blocks A and B to be a w.r.t. lift and a_L be the acceleration of lift.

Consider block A, as seen from the reference frame attached to the lift (a non-inertial frame), we get

$$N = mg + ma_L \quad \dots(1)$$

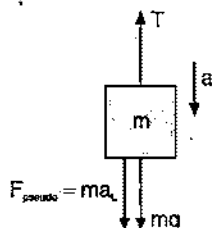
$$T = ma \quad \dots(2)$$



Free Body Diagram of A

Now, consider block B, as seen from the reference frame attached to the lift, we have

$$mg + ma_L - T = ma \quad \dots(3)$$



Free Body Diagram of B

On adding equations (2) and (3), we get

$$a = \frac{g + a_L}{2} = \frac{10 + 2}{2} = 6 \text{ ms}^{-2}$$

So, distance fallen by block (B) is $x = \frac{1}{2}at^2$

$$\Rightarrow x = \frac{1}{2} \times 6 \times (0.2)^2$$

$$\Rightarrow x = 0.12 \text{ m} = 12 \text{ cm}$$

Now, consider reflection at convex mirror, we have

$$u = -(42 - 12) = -40 \text{ cm}$$

$$f = +12 \text{ cm}$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{(-40)} = \frac{1}{12}$$

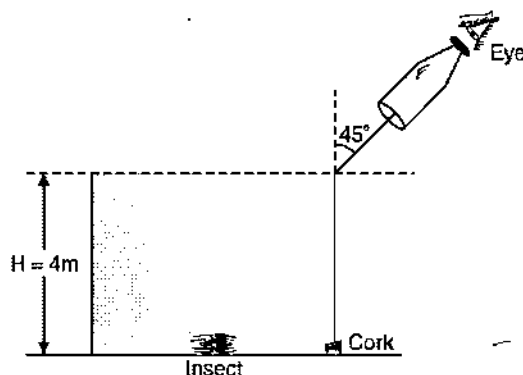
$$\Rightarrow v = \frac{(40)(12)}{40 + 12} = \frac{480}{52}$$

$$\Rightarrow v = 9.23 \text{ cm}$$

Therefore, the distance between the image of block (B) and mirror is 9.23 cm

Problem 3

A fixed cylindrical tank of height $H = 4 \text{ m}$ and radius $R = 3 \text{ m}$ is filled up with a liquid. An observer observes through a telescope fitted at the top of the wall of the tank and inclined at $\theta = 45^\circ$ with the vertical. When the tank is completely filled with liquid, he notices an insect, which is at the center of the bottom of the tank. At $t = 0$, he opens a cork of radius $r = 3 \text{ cm}$ at the bottom of tank. The insect moves in such a way that it is visible for a certain time. Determine



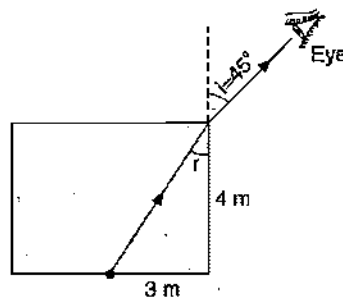
- the refractive index of the liquid
- the velocity of insect as a function of time.

Solution

- At $t = 0$

$$\sin r = \frac{3}{5}$$

$$\mu = \frac{\sin i}{\sin r} = \frac{5}{3\sqrt{2}}$$



- Let at time t , insect be at a distance x from centre of the tank. Since,

$$\frac{x_1}{h} = \tan r = \frac{3}{4}$$

$$\Rightarrow x_1 = \frac{3}{4}h$$

$$\text{So, } x = (H - h) + x_1 - 3$$

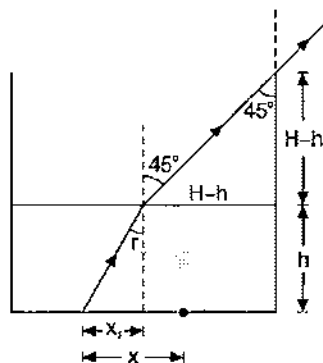
$$\Rightarrow x = 4 - h + \frac{3}{4}h - 3$$

$$\Rightarrow x = 1 - \frac{h}{4}$$

$$\Rightarrow \left(\frac{dx}{dt}\right) = \frac{1}{4} \left(-\frac{dh}{dt}\right) \quad \dots(1)$$

From Equation of Continuity, we have

$$\begin{aligned} -A_1 \left(\frac{dh}{dt}\right) &= A_2 \sqrt{2gh} \\ \Rightarrow \left(-\frac{dh}{\sqrt{h}}\right) &= \frac{A_2 \sqrt{2g}}{A_1} dt \\ \Rightarrow -\int_H^h \frac{dh}{\sqrt{h}} &= \frac{\pi(3 \times 10^{-2})^2}{\pi(3)^2} \sqrt{2 \times 9.8} \int_0^t dt \end{aligned}$$



Substituting $H = 4$ m, we get

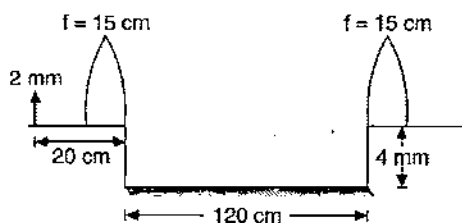
$$\begin{aligned} h &= (2 - 2.21 \times 10^{-4} t)^2 \\ \Rightarrow -\frac{dh}{dt} &= 4.42 \times 10^{-4} (2 - 2.21 \times 10^{-4} t) \end{aligned}$$

So, speed of insect is

$$\begin{aligned} v &= \frac{dx}{dt} = \frac{1}{4} \left(-\frac{dh}{dt}\right) \\ \Rightarrow v &= 1.1 \times 10^{-4} (2 - 2.21 \times 10^{-4} t) \text{ ms}^{-1} \end{aligned}$$

Problem 4

A convex lens of focal length 15 cm is split into two halves and the two halves are placed at a separation of 120 cm. Between these two halves of the convex lens, a plane mirror is placed horizontally and at a distance of 4 mm below the principal axis of the lens halves. An object of length 2 mm is placed at a distance of 20 cm from one half lens as shown in figure.



- Find the position and size of the final image.
- Trace the path of rays forming the image.

Solution

- (a) For refraction at first half lens, using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\begin{aligned} \frac{1}{v} - \frac{1}{-20} &= \frac{1}{15} \\ \Rightarrow v &= 60 \text{ cm} \end{aligned}$$

$$\text{Magnification, } m = \frac{v}{u} = \frac{60}{-20} = -3$$

The image formed by first half lens is shown in Figure-1

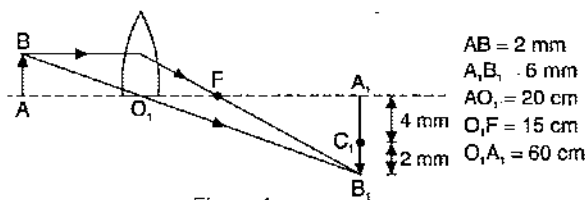


Figure-1

Now, the point B_1 is 6 mm below the principal axis of the lenses. Plane mirror is 4 mm below it. Hence, 4 mm length of A_1B_1 (i.e., A_1C_1) acts as real object for mirror. Mirror forms its virtual image A_2C_2 . So, 2 mm length of A_1B_1 (i.e., C_1B_1) acts as virtual object for mirror. Real image C_2B_2 is formed of this part. Image formed by plane mirror is shown in Figure-2.

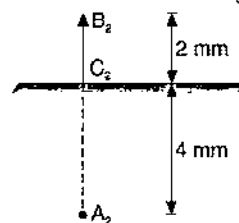


Figure-2

For the second half of the lens, using lens formula

$$\begin{aligned} \frac{1}{v} - \frac{1}{u} &= \frac{1}{f}, \text{ we get} \\ \frac{1}{v} - \frac{1}{-60} &= \frac{1}{15} \\ \Rightarrow v &= +20 \\ m &= \frac{v}{u} = \frac{20}{-60} = -\frac{1}{3} \end{aligned}$$

So, length of final image $A_3B_3 = \frac{1}{3} A_2B_2 = 2$ mm.

However, point B_2 is 2 mm below the optic axis of second half lens. Hence, its image B_3 is formed $\frac{2}{3}$ mm above the principal axis.

Similarly, point A_2 is 8 mm below the principal axis.

Hence, its image is $\frac{8}{3}$ mm above it.

Therefore, image is at a distance of 20 cm behind the second half lens and at a distance of $\frac{2}{3}$ mm above the principal axis.

The size of image is 2 mm and is inverted as compared to the given object. Image formed by second half lens is shown in Figure-3.

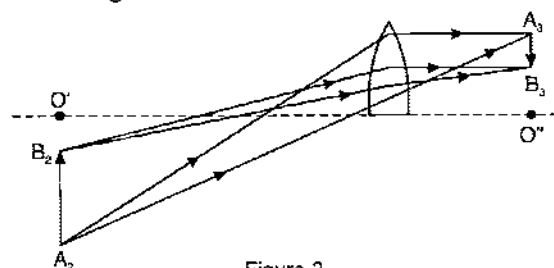


Figure-3

(b) The ray diagram for the final image is shown in Figure-4.

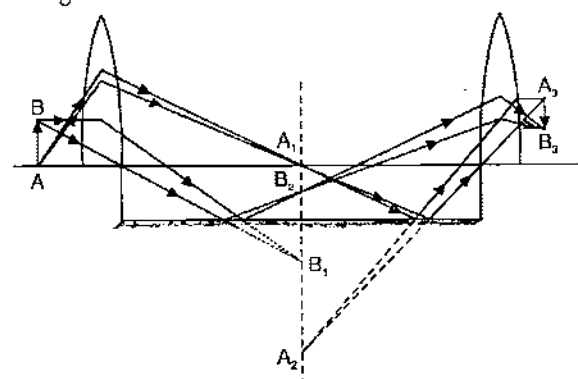
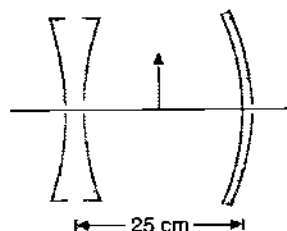


Figure-4

Problem 5

An object lies midway between the lens and a mirror. The mirror's radius of curvature is 20 cm and the lens has a focal length of -16.7 cm. Considering that the rays that leave the object travel first towards the mirror, locate the final image formed by this system. Is this image real or virtual. Is it upright or inverted? What is the overall magnification?



Solution

Step 1 : Image formed by mirror

Using mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$, we get

$$\frac{1}{v_1} + \frac{1}{-12.5} = \frac{2}{-20}$$

$$\Rightarrow v_1 = -50 \text{ cm}$$

$$m_1 = -\frac{v}{u} = -\frac{(-50)}{(-12.5)} = -4$$

So, the image formed by the mirror is at a distance of 50 cm from the mirror to the left of it. It is inverted and four times larger.

Step 2 : Image formed by lens

The image formed by mirror acts as an object for the lens and the image formed by the mirror is at a distance of 25 cm to the left of lens. Using the lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \text{ we get}$$

$$\frac{1}{v_2} - \frac{1}{25} = \frac{1}{-16.7}$$

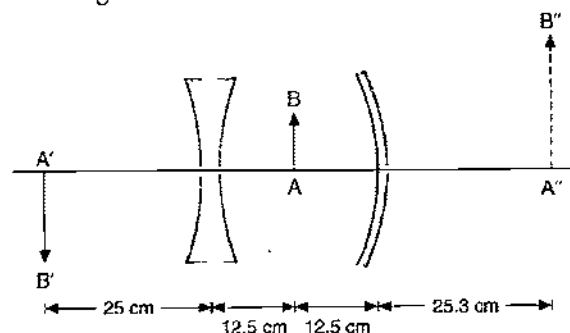
$$\Rightarrow v_2 = -50.3 \text{ cm}$$

$$\text{and } m_2 = \frac{v}{u} = \frac{-50.3}{25} = -2.012$$

Net magnification is given by

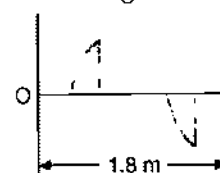
$$m = m_1 \times m_2 \approx 8$$

Hence, the final image is at a distance 25.3 cm to the right of the mirror, virtual, upright enlarged and approximately 8 times. Positions of the two images are shown in figure.



Problem 6

A thin plano-convex lens of focal length f is split into two halves. One of the halves is shifted along the optical axis as shown in figure. The separation between object and image planes is 1.8 m. The magnification of the image, formed by one of the half lens is 2. Find the focal length of the lens and separation between the two halves. Draw the ray diagram for image formation.



Solution

For both the halves, position of object and image is same, however the only difference is of magnification. Magnification for one of the halves is given as $2(>1)$. This can be for the first one, because for this, $|v| > |u|$. Therefore, magnification, $|m| = \left| \frac{v}{u} \right| > 1$. So, for the first half, we have

$$\left| \frac{v}{u} \right| = 2$$

$$\Rightarrow |v| = 2|u|$$

Let $u = -x$, then $v = +2x$

and $|u| + |v| = 1.8 \text{ m}$

$$\Rightarrow 3x = 1.8 \text{ m}$$

$$\Rightarrow x = 0.6 \text{ m}$$

Hence, $u = -0.6 \text{ m}$ and $v = +1.2 \text{ m}$

$$\text{Using } \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{1.2} - \frac{1}{-0.6} = \frac{1}{0.4}$$

$$\Rightarrow f = 0.4 \text{ m}$$

For the second half, we have

$$\frac{1}{f} = \frac{1}{1.2-d} + \frac{1}{-(0.6+d)}$$

$$\Rightarrow \frac{1}{0.4} = \frac{1}{1.2-d} + \frac{1}{(0.6+d)}$$

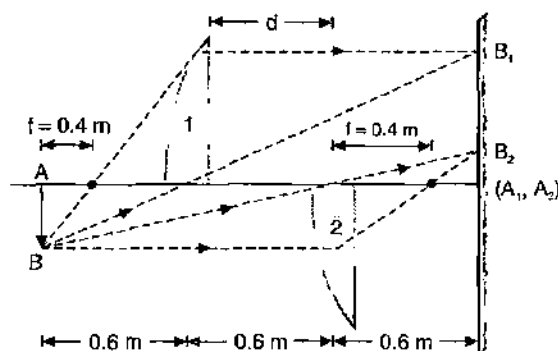
Solving this, we get $d = 0.6 \text{ m}$

Magnification for the second half will be

$$m_2 = \frac{v}{u} = \frac{0.6}{-(1.2)} = -\frac{1}{2}$$

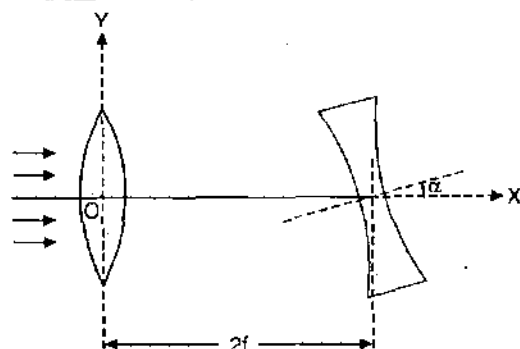
and magnification for the first half is $m_1 = \frac{v}{u} = \frac{1.2}{-(0.6)} = -2$

The ray diagram is as follows:



Problem 7

Two thin lenses of same focal length f are arranged with their principal axes inclined at an angle α as shown in figure. The separation between the optical centers of the lenses is $2f$. A point object lies on the principal axis of the convex lens at a large distance to the left of convex lens.



- (a) Find the co-ordinates of the final image formed by the system of lenses taking O as the origin of the co-ordinate axes.

- (b) Draw the ray diagram.

Solution

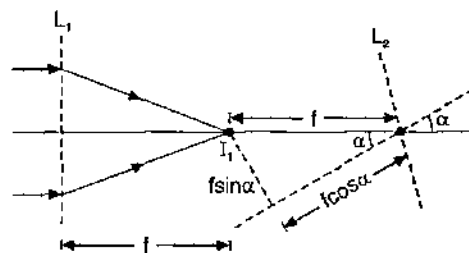
- (a) For concave lens (L_2)

$$\frac{1}{v} - \frac{1}{-f \cos \alpha} = \frac{-1}{f}$$

$$\Rightarrow v = -\left(\frac{f \cos \alpha}{1 + \cos \alpha} \right) \quad \dots(1)$$

The magnification is given by

$$m = \frac{v}{u} = \frac{1}{1 + \cos \alpha}$$



So, height of I_2 from the principal axis of L_2 is

$$h = (f \sin \alpha) m = \frac{f \sin \alpha}{1 + \cos \alpha} \quad \dots(2)$$

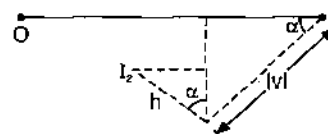
Hence the x-co-ordinate of image I_2 is given by

$$x = 2f - |v| \cos \alpha - h \sin \alpha$$

$$\Rightarrow x = 2f - \frac{f \cos^2 \alpha}{1 + \cos \alpha} - \frac{f \sin^2 \alpha}{1 + \cos \alpha}$$

$$\Rightarrow x = 2f - \frac{f}{1 + \cos \alpha}$$

$$\Rightarrow x = f \left(\frac{2 \cos \alpha + 1}{\cos \alpha + 1} \right)$$



Similarly, y co-ordinate of image I_2

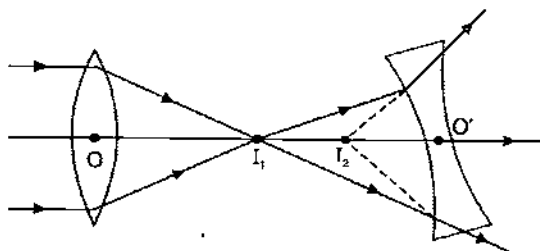
$$y = -(v \sin \alpha - h \cos \alpha)$$

On substituting the values of $|v|$ and h from (1) and (2), we get $y = 0$.

So, the coordinates of the final image are

$$\left[f \left(\frac{2 \cos \alpha + 1}{\cos \alpha + 1} \right), 0 \right]$$

(b) Ray diagram is shown in figure

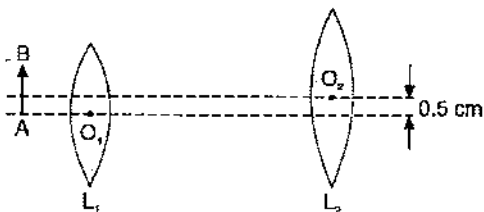


CONCEPTUAL NOTE(S)

The y -co-ordinate of I_2 is zero is very obvious because a ray of light starting from I_1 and passing through O' will suffer no deviation. Hence, I_2 must be formed on this line itself i.e., $y = 0$.

Problem 8

Two thin lenses $f_1 = 10$ cm and $f_2 = 20$ cm are separated by a distance $d = 5$ cm. Their optical centres are displaced a distance $\Delta = 0.5$ cm. A linear object of size 3 cm placed at 30 cm from the optical centre of left lens. Find the nature position and size of final image.



Solution

Step 1: Refraction from the first lens L_1

Using the lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v_1} - \frac{1}{-30} = \frac{1}{10}$$

$$\Rightarrow v_1 = 15 \text{ cm}$$

$$\text{and } m_1 = \frac{v}{u} = \frac{15}{-30} = -\frac{1}{2}$$

Step 2: Refraction from the second lens L_2

Again using the lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v_2} - \frac{1}{(15-5)} = \frac{1}{20}$$

$$\Rightarrow v_2 = \frac{20}{3} \text{ cm} \approx 6.67 \text{ cm}$$

$$\text{and } m_2 = \frac{v}{u} = \frac{20/3}{10} = \frac{2}{3}$$

About Final Image

Net magnification is given by

$$m = m_1 m_2 = -\frac{1}{3}$$

i.e., height of the image is $3 \times \frac{1}{3} = 1$ cm

Since, the net magnification is negative, so the final image is inverted.

Further y -coordinate of a point of the image will be,

$$y_1 = m y_0 - m_2 \Delta \quad \dots(1)$$

with respect to the principal axis of L_1

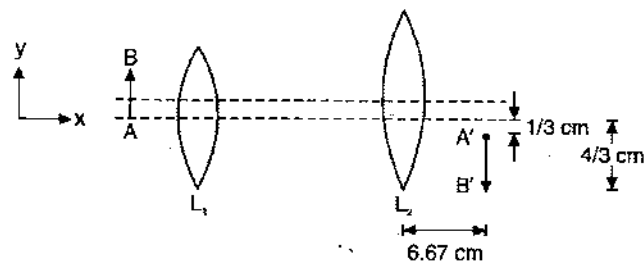
So, y -coordinate of image of A is

$$y_{A'} = \left(-\frac{1}{3}\right)(0) - \left(\frac{2}{3}\right)\left(\frac{1}{2}\right) = -\frac{1}{3} \text{ cm and}$$

y -coordinate of image of B is

$$y_{B'} = \left(-\frac{1}{3}\right)(3) - \left(\frac{2}{3}\right)\left(\frac{1}{2}\right) = -\frac{4}{3} \text{ cm}$$

Thus final image is as shown in figure.

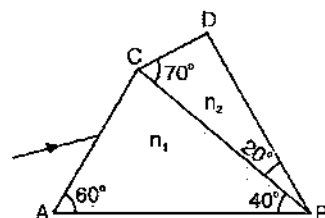


Problem 9

A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together with a gap as shown in the figure. The angles of the prism are as shown, n_1 and n_2 depend on λ , the wavelength of light according to the relations given by

$$n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2} \quad \text{and}$$

$$n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2} \quad \text{where } \lambda \text{ is in nm.}$$



- (a) Calculate the wavelength λ_0 for which rays incident at any angle on the interface BC pass through without bending at that interface.
- (b) For light of wavelength λ_0 , find the angle of incidence i on the face AC such that the deviation produced by the combination of prisms is minimum.

Solution

$$\text{Since, } n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2} \text{ and } n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$$

where, λ is in nm.

- (a) The incident ray will not deviate at BC only if $n_1 = n_2$

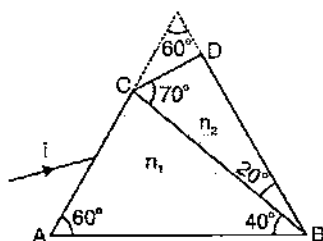
$$\Rightarrow 1.2 + \frac{10.8 \times 10^4}{\lambda_0^2} = 1.45 + \frac{1.80 \times 10^4}{\lambda_0^2} \quad (\lambda = \lambda_0)$$

$$\Rightarrow \frac{9 \times 10^4}{\lambda_0^2} = 0.25$$

$$\Rightarrow \lambda_0 = \frac{3 \times 10^2}{0.5}$$

$$\Rightarrow \lambda_0 = 600 \text{ nm}$$

- (b) The given system happens to be a part of an equilateral prism of prism angle 60° as shown in figure.



At minimum deviation, we have

$$r_1 = r_2 = \frac{60^\circ}{2} = 30^\circ = r \quad \text{[say]}$$

Since according to Snell's Law, we have

$$n_1 = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sin i = n_1 \sin(30^\circ)$$

$$\text{Since, } n_1 = 1.2 + \frac{10.8 \times 10^4}{\lambda_0^2}, \text{ where } \lambda_0 = 600 \text{ nm}$$

$$\Rightarrow \sin i = \left[1.2 + \frac{10.8 \times 10^4}{(600)^2} \right] \left(\frac{1}{2} \right) = \frac{1.5}{2} = \frac{3}{4}$$

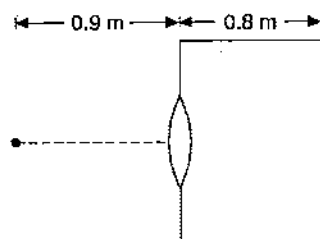
$$\Rightarrow i = \sin^{-1} \left(\frac{3}{4} \right)$$

Problem 10

A thin equiconvex lens of glass of refractive index $\mu = \frac{3}{2}$ and of focal length 0.3 m in air is sealed into an

opening at one end of a tank filled with water $\mu = \frac{4}{3}$. On

the opposite side of the lens, a mirror is placed inside the tank on the tank wall perpendicular to the lens axis, as shown in figure. The separation between the lens and the mirror is 0.8 m. A small object is placed outside the tank in front of. Find the position (relative to the lens) of the image of the object formed by the system.



Solution

Applying Lens Maker's Formula, $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$,

we get

$$\frac{1}{0.3} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} - \frac{1}{-R} \right) \quad \{ \because R_1 = R \text{ and } R_2 = -R \}$$

$$\Rightarrow R = 0.3$$

Now applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ at air glass surface, we get

$$\frac{3}{v_1} - \frac{1}{-(0.9)} = \frac{\left(\frac{3}{2} \right) - 1}{0.3}$$

$$\Rightarrow v_1 = 2.7 \text{ m}$$

So, the first image I_1 will be formed at 2.7 m from the lens.

This image I_1 will act as the virtual object for glass water surface.

Therefore, applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ at glass water surface,

we get

$$\frac{4}{v_2} - \frac{3}{2.7} = \frac{\left(\frac{4}{3} \right) - \left(\frac{3}{2} \right)}{-0.3}$$

$$\Rightarrow v_2 = 1.2 \text{ m}$$

So, the second image I_2 is formed at 1.2 m from the lens or 0.4 m from the plane mirror.

This image I_2 will act as a virtual object for mirror. Therefore, third real image I_3 will be formed at a distance of 0.4 m in front of the mirror after reflection from it. Now this image acts as a real object for water-glass interface. Hence

applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

$$\frac{\frac{3}{2}}{v_4} - \frac{\frac{4}{3}}{-(0.8-0.4)} = \frac{\left(\frac{3}{2}\right) - \left(\frac{4}{3}\right)}{0.3}$$

$$\Rightarrow v_4 = -0.54 \text{ m}$$

So, the fourth image is formed to the right of the lens at a distance of 0.54 m from it.

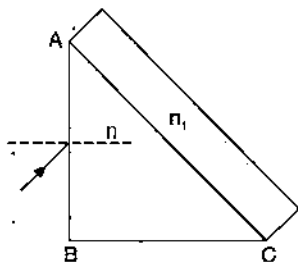
Now finally applying the same formula for glass-air surface, we get

$$\frac{1}{v_5} - \frac{\frac{3}{2}}{-0.54} = \frac{1 - \left(\frac{3}{2}\right)}{-0.3} = -0.9 \text{ m}$$

Hence, the position of final image is 0.9 m relative to the lens (rightwards) i.e., the image is formed 0.1 m behind the mirror.

Problem 11

A right angle prism ($45^\circ-90^\circ-45^\circ$) of refractive index n has a plane of refractive index n_1 ($n_1 < n$) cemented to its diagonal face. The assembly is in air. The ray is incident on AB .



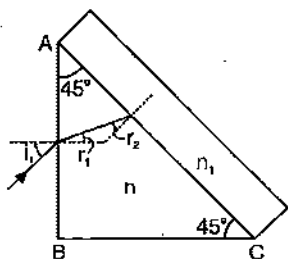
- Calculate the angle of incidence at AB for which the ray strikes the diagonal face at the critical angle.
- Assuming $n = 1.352$, calculate the angle of incidence at AB for which the refracted ray passes through the diagonal face undeviated.

Solution

- Critical angle C at face AC will be given by

$$C = \sin^{-1}\left(\frac{n_1}{n}\right)$$

$$\Rightarrow \sin C = \frac{n_1}{n}$$



Now, it is given that $r_2 = C$

$$\Rightarrow r_1 = A - r_2 = (45^\circ - C)$$

Applying Snell's Law at face AB , we get

$$n = \frac{\sin i_1}{\sin r_1}$$

$$\Rightarrow \sin i_1 = n \sin r_1$$

$$\Rightarrow i_1 = \sin^{-1}(n \sin r_1)$$

Substituting value of r_1 , we get

$$i_1 = \sin^{-1}\{n \sin(45^\circ - C)\}$$

$$\Rightarrow i_1 = \sin^{-1}[n(\sin 45^\circ \cos C - \cos 45^\circ \sin C)]$$

$$\Rightarrow i_1 = \sin^{-1}\left(\frac{n}{\sqrt{2}}\sqrt{1 - \sin^2 C} - \sin C\right) \quad \left\{ \because \sin C = \frac{n_1}{n} \right\}$$

$$\Rightarrow i_1 = \sin^{-1}\left[\frac{n}{\sqrt{2}}\left(\sqrt{1 - \frac{n_1^2}{n^2}} - \frac{n_1}{n}\right)\right]$$

$$\Rightarrow i_1 = \sin^{-1}\left[\frac{1}{\sqrt{2}}(\sqrt{n^2 - n_1^2} - n_1)\right]$$

Therefore, required angles of incidence (i_1) at face AB for which the ray strikes at AC at critical angle is given by

$$i_1 = \sin^{-1}\left[\frac{1}{\sqrt{2}}(\sqrt{n^2 - n_1^2} - n_1)\right]$$

- The ray will pass undeviated through face AC when

$$n_1 = n \text{ or } r_2 = 0^\circ$$

i.e., ray falls normally on face AC

Since it is given that $n_1 < n$, so the option $n_1 = n$ is ruled out, hence

$$r_2 = 0^\circ$$

$$\Rightarrow r_1 = A - r_2 = 45^\circ - 0^\circ = 45^\circ$$

Now applying Snell's Law at face AB , we get

$$n = \frac{\sin i_1}{\sin r_1}$$

$$\Rightarrow 1.352 = \frac{\sin i_1}{\sin(45^\circ)}$$

$$\Rightarrow \sin i_1 = (1.352)\left(\frac{1}{\sqrt{2}}\right)$$

$$\Rightarrow \sin i_1 = 0.956$$

$$\Rightarrow i_1 = \sin^{-1}(0.956) \approx 73^\circ$$

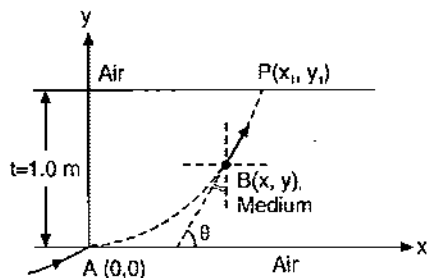
Therefore, required angle of incidence is $i_1 = 73^\circ$.

Problem 12

A ray of light travelling in air is incident at grazing angle (incident angle = 90°) on a long rectangular slab of a transparent medium of thickness $t = 1.0 \text{ m}$. The point of incidence is the origin $A(0, 0)$. The medium has a variable index of refraction $n(y)$ given by

$$n(y) = [ky^{3/2} + 1]^{1/2} \text{ where } k = 1.0(\text{meter})^{-3/2}$$

The refractive index of air is 1.0



- Obtain a relation between the slope of the trajectory of the ray at a point $B(x, y)$ in the medium and the incident angle at the point.
- Obtain an equation for the trajectory $y(x)$ of the ray in the medium.
- Determine the co-ordinates (x_1, y_1) of the point P , where the ray intersects the upper surface of the slab-air boundary.
- Indicate the path of the ray subsequently.

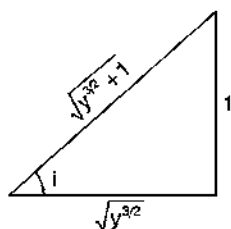
Solution

- $i + \theta = 90^\circ$, $\theta = 90^\circ - i$,
Slope of tangent $= \tan \theta = \tan(90^\circ - i) = \cot i$
- $\tan \theta = \frac{dy}{dx}$
 $\therefore \frac{dy}{dx} = \cot i$... (1)

Applying Snell's Law at A and B

$$n_A \sin i_A = n_B \sin i_B$$

$$n_A = 1 \text{ because } y = 0$$



$$\sin i_A = 1 \text{ because } i_A = 90^\circ \text{ Grazing incidence}$$

$$n_B = \sqrt{ky^{3/2} + 1} = \sqrt{y^{3/2} + 1}$$

$$\text{because } K = 1.0(\text{m})^{-3/2}$$

$$\therefore (1)(1) = \sqrt{(y^{3/2} + 1)} \sin i$$

$$\Rightarrow \sin i = \frac{1}{\sqrt{y^{3/2} + 1}}$$

$$\Rightarrow \cot i = \sqrt{y^{3/2}} \text{ or } y^{3/4} \quad \dots (2)$$

Equating equations (1) and (2), we get

$$\frac{dy}{dx} = y^{3/4} \text{ or } y^{-3/4} dy = dx$$

$$\Rightarrow \int_0^y y^{-3/4} dy = \int_0^x dx \text{ or } 4y^{1/4} = x \quad \dots (3)$$

The required equation of trajectory is $4y^{1/4} = x$

$$\Rightarrow y = \frac{x^4}{4^4} = \frac{x^4}{256}$$

- At the point of intersection on the upper surface,
 $y = 1 \text{ m}$

$$\Rightarrow x = (256)^{1/4} = 4 \text{ m}$$

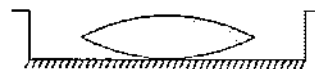
So the co-ordinates are $(4 \text{ m}, 1 \text{ m})$

- As $n_A \sin i_A = n_P \sin i_P$ and as $n_A = n_P = 1$

Therefore, $i_P = i_A = 90^\circ$ i.e., the ray will emerge parallel to the boundary at P i.e., at grazing emergence.

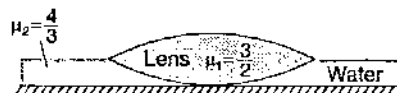
Problem 13

A thin biconvex lens of refractive index $\frac{3}{2}$ is placed on a horizontal plane mirror as shown in the figure. The space between the lens and the mirror is then filled with water of refractive index $\frac{4}{3}$. It is found that when a point object is placed 15 cm above the lens on its principal axis, the object coincides with its own image. On repeating with another liquid, the object and the image again coincide at a distance 25 cm from the lens. Calculate the refractive index of the liquid.



Solution

Let R be the radius of curvature of both the surfaces of the equi-convex lens, then in the first case, the situation is shown in figure.



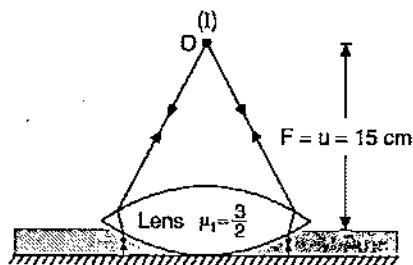
Let f_1 be the focal length of equi-convex lens of refractive index μ_1 and f_2 be the focal length of plano-concave lens (made of water) of refractive index μ_2 . The focal length of the combined lens system is given by

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} = (\mu_1 - 1) \left(\frac{1}{R} - \frac{1}{-R} \right) + (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right)$$

$$\Rightarrow \frac{1}{F} = \left(\frac{3}{2} - 1 \right) \left(\frac{2}{R} \right) + \left(\frac{4}{3} - 1 \right) \left(-\frac{1}{R} \right) = \frac{1}{R} - \frac{1}{3R} = \frac{2}{3R}$$

$$\Rightarrow F = \frac{3R}{2}$$

Now, image coincides with the object when ray of light retraces its path i.e., it falls normally on the plane mirror. This is possible only when object lies at centre of curvature of the lens system.



$$\Rightarrow F = 15 \text{ cm} \quad (\because \text{Distance of object is } 15 \text{ cm})$$

$$\Rightarrow \frac{3R}{2} = 15 \text{ cm}$$

$$\Rightarrow R = 10 \text{ cm}$$

In the second case, let μ be the refractive index of the liquid filled between lens and mirror and let F' be the focal length of new lens system. Then,

$$\frac{1}{F'} = (\mu_l - 1) \left(\frac{1}{R} - \frac{1}{-R} \right) + (\mu - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right)$$

$$\Rightarrow \frac{1}{F'} = \left(\frac{3}{2} - 1 \right) \left(\frac{2}{R} \right) - \frac{(\mu - 1)}{R} = \frac{1}{R} - \frac{\mu - 1}{R} = \frac{(2 - \mu)}{R}$$

$$\Rightarrow F' = \frac{R}{2 - \mu} = \frac{10}{2 - \mu} \quad (\because R = 10 \text{ cm})$$

Now, the image coincides with the object when it is placed at 25 cm distance.

$$\Rightarrow F' = 25$$

$$\Rightarrow \frac{10}{2 - \mu} = 25$$

$$\Rightarrow 50 - 25\mu = 10$$

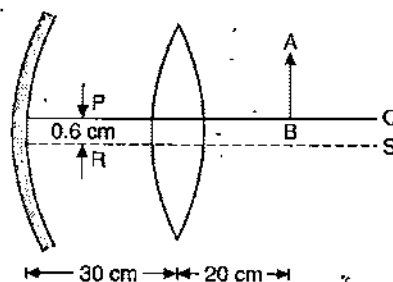
$$\Rightarrow 25\mu = 40$$

$$\Rightarrow \mu = \frac{40}{25} = 1.6$$

$$\Rightarrow \mu = 1.6$$

Problem 14

A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axis PQ and RS parallel but separated in vertical direction by 0.6 cm as shown. The distance between the lens and mirror is 30 cm. An upright object AB of height 1.2 cm is placed on the optic axis PQ of the lens at a distance of 20 cm from the lens. If A'B' is the image after refraction from the lens and the reflection from the mirror, find the distance of A'B' from the pole of the mirror and obtain its magnification. Also locate positions of A' and B' with respect to the optic axis RS.

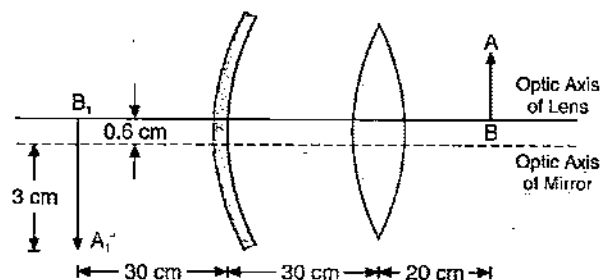


Solution

Rays coming from object AB first refract from the lens and then reflect from the mirror.

For refraction from the lens, we have

$$u = -20 \text{ cm}, f = +15 \text{ cm}$$



Applying lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} - \frac{1}{(-20)} = \frac{1}{15}$$

$$\Rightarrow v = +60 \text{ cm}$$

and linear magnification is given by

$$m_1 = \frac{v}{u} = \frac{+60}{(-20)} = -3$$

So, the first image formed by the lens will be 60 cm from it (or 30 cm from the mirror) towards left and 3 times magnified but inverted. Length of first image A₁B₁ would be A₁B₁ = 1.2 × 3 = 3.6 cm (inverted).

For reflection from mirror, we have

Image formed by lens (A₁B₁) will behave like a virtual object for the mirror at a distance of 30 cm from it as shown. Therefore $u = +30 \text{ cm}$, $f = -30 \text{ cm}$.

Applying mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} + \frac{1}{30} = -\frac{1}{30}$$

$$\Rightarrow v = -15 \text{ cm}$$

and linear magnification is given by

$$m_2 = \frac{v}{u} = \frac{(-15)}{+30} = -\frac{1}{2}$$

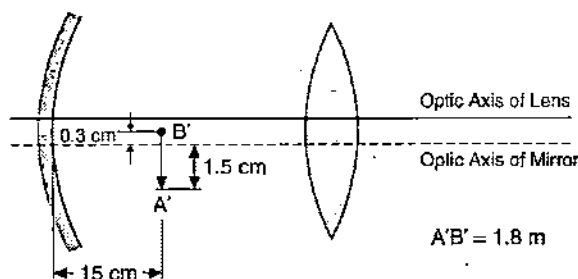
So, the final image $A'B'$ will be located at a distance of 15 cm from the mirror (towards right) and since magnification is $+\frac{1}{2}$, length of final image would be

$$A'B' = 3.6 \times \frac{1}{2} = 1.8 \text{ cm}$$

$$\Rightarrow A'B' = 1.8 \text{ cm}$$

Since the point B_1 is 0.6 cm above the optic axis of mirror, therefore, its image B' would be $(0.6)\left(\frac{1}{2}\right) = 0.3 \text{ cm}$ above optic axis.

Similarly, point A_1 is 3 cm below the optic axis, therefore, its image A' will be $3 \times \frac{1}{2} = 1.5 \text{ cm}$ below the optic axis as shown.



Net magnification of the image is given by

$$m = m_1 \times m_2 = (-3) \left(+\frac{1}{2} \right) = -\frac{3}{2}$$

$$\Rightarrow A'B' = (m)(AB) = \left(-\frac{3}{2} \right) (1.2) = -1.8 \text{ cm}$$

CONCEPTUAL NOTE(S)

If the co-ordinates of the object (X_o, Y_o) are generally known to us with reference to the pole of an optical instrument (whether it is a lens or a mirror), the corresponding co-ordinates of image (X_i, Y_i) are found as follows.

$$X_i \text{ is obtained using } \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \text{ (for a mirror)}$$

$$\text{OR } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \text{ (for a lens)}$$

Here, v is actually X_i and u is X_o i.e., the above formula can be written as $\frac{1}{X_i} \pm \frac{1}{X_o} = \frac{1}{f}$

$$\text{Similarly, } Y_i \text{ is obtained from } m = \frac{Y_i}{Y_o}$$

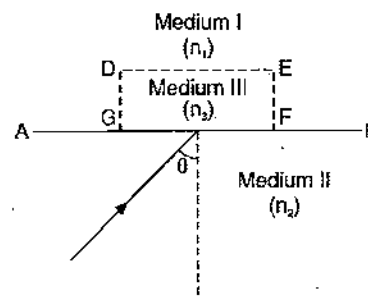
Here, Y_i is Y_i and O is Y_o i.e., the above formula can be written as $m = \frac{Y_i}{Y_o}$ or $Y_i = mY_o$.

Problem 15

Monochromatic light is incident on a plane interface AB between two media of refractive indices n_1 and n_2 ($n_2 > n_1$) at an angle of incidence θ as shown in the figure. The angle θ is infinitesimally greater than the critical angle for the two media so that total internal reflection takes place. Now if a transparent slab $DEFG$ of uniform thickness and of refractive index n_3 is introduced on the interface (as shown in the figure), show that for any value of n_3 all light will ultimately be reflected back again into medium II. Consider separately the cases

(a) $n_3 < n_1$ and

(b) $n_3 > n_1$



Solution

At interface AB , θ is infinitesimally greater (slightly greater) than the critical angle for interface, so

$$\theta > \sin^{-1} \left(\frac{n_1}{n_2} \right)$$

(a) When $n_3 < n_1$

$$\Rightarrow n_3 < n_1 < n_2$$

$$\Rightarrow \frac{n_3}{n_2} < \frac{n_1}{n_2}$$

$$\Rightarrow \sin^{-1} \left(\frac{n_3}{n_2} \right) < \sin^{-1} \left(\frac{n_1}{n_2} \right)$$

Hence, critical angle for Medium III and Medium II will be less than the critical angle for Medium II and Medium I. So, if TIR is taking place between Medium I and Medium II, then TIR will definitely take place between Medium I and Medium III.

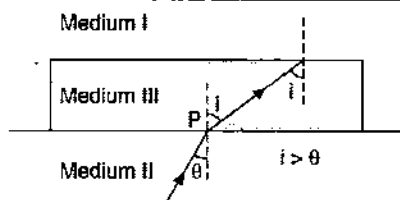
(b) When $n_3 > n_1$, then two further cases may arise.

Case I: $n_1 < n_3 < n_2$

In this case there will be no TIR between Medium I and Medium III but TIR will take place between Medium III and Medium II. This is because:

Ray of light first enters from Medium II to Medium III i.e., from denser to rarer. So,

$$i > \theta$$



Applying Snell's Law at P , we get

$$n_2 \sin \theta = n_3 \sin i$$

$$\Rightarrow \sin i = \left(\frac{n_2}{n_3} \right) \sin \theta$$

Since, $\sin \theta$ is slightly greater than $\frac{n_1}{n_2}$, so

$$\sin i \text{ is slightly greater than } \frac{n_2}{n_3} \times \frac{n_1}{n_2} = \frac{n_1}{n_3}$$

However, $\frac{n_1}{n_3}$ is nothing but $\sin C$ for Medium I, Medium III interface, so

$\sin i$ is slightly greater than $\sin C$ for Medium I, Medium III interface.

$$\Rightarrow i > (C)_{I, III}$$

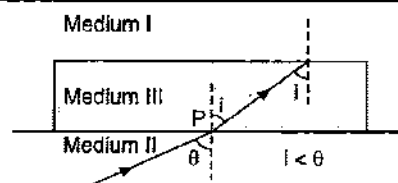
Hence, TIR will now take place on Medium I and Medium III interface and the ray will be reflected back to Medium III.

Case II: $n_1 < n_2 < n_3$

This time while moving from Medium II to Medium III, ray of light will bend towards normal. Again applying Snell's Law at P , we get

$$n_2 \sin \theta = n_3 \sin i$$

$$\Rightarrow \sin i = \frac{n_2}{n_3} \sin \theta$$



Since, $\sin \theta$ slightly greater than $\frac{n_1}{n_2}$

So, $\sin i$ will be slightly greater than $\frac{n_2}{n_3} \times \frac{n_1}{n_2} = \frac{n_1}{n_3}$

However $\frac{n_1}{n_3}$ is $\sin C$ for Medium I and Medium III interface, so

$\sin i > \sin C$ for Medium I and Medium III interface.

$$\Rightarrow i > (C)_{I, III}$$

Therefore, TIR will again take place between Medium I and Medium III and the ray will be reflected back.

CONCEPTUAL NOTE(S)

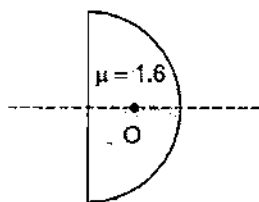
The Cases I and II for $n_3 > n_1$ can be explained by single equation only. But two cases are deliberately formed for better understanding of refraction, Snell's Law and total internal reflection (TIR).

Practice Exercise

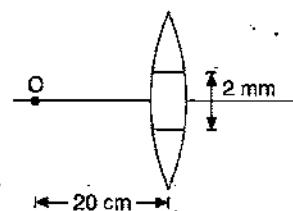
Single Correct Choice Type Questions

This section contains Single Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONLY ONE is correct.

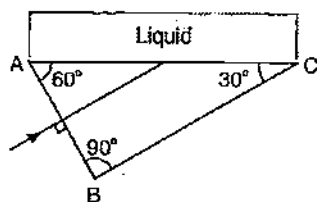
1. A transparent hemisphere has a radius of curvature 8 cm and an index of refraction of 1.6. A small object O is placed on the axis halfway between the plane surface and the spherical surface i.e., 4 cm from each. The distance between the two images when viewed along the axis from the two sides of the hemisphere is approximately



- (A) 7.5 cm (B) 8.5 cm
(C) 2.5 cm (D) 13.5 cm
2. A square wire of side 3.0 cm is placed 25 cm in front of a concave mirror of focal length 10 cm with its centre on the axis of the mirror and its plane normal to the axis. The area enclosed by the image of the wire is
(A) 7.5 cm^2 (B) 6.0 cm^2
(C) 4.0 cm^2 (D) 3.0 cm^2
3. An object is placed at a distance $2f$ from the pole of a convex mirror of focal length f . The linear magnification is
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{3}{4}$ (D) 1
4. A convex lens of focal length 10 cm is painted black at the middle portion as shown in figure. An object is placed at a distance of 20 cm from the lens. Then



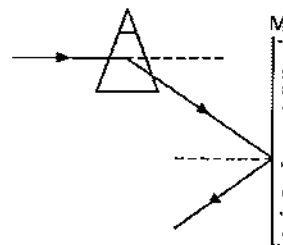
- (A) the distance between the images is 2 mm
(B) the distance between the images is 4 mm
(C) the distance between the two images formed by such a lens is 6 mm
(D) only one image will be formed by the lens
5. An object is placed at 20 cm from a convex mirror of focal length 20 cm. The distance of the image from the pole of the mirror is
(A) infinity (B) 10 cm
(C) 15 cm (D) 40 cm
6. A point object is placed at a distance of 25 cm from a convex lens of focal length 20 cm. When a glass slab of thickness t and refractive index 1.5 is inserted between the lens and the object, the image is formed at infinity. The thickness t of the slab is
(A) 5 cm (B) 10 cm
(C) 15 cm (D) 20 cm
7. Light is incident normally on face AB of a prism as shown in figure. A liquid of refractive index μ is placed on face AC of the prism. The prism is made of glass of refractive index $\frac{3}{2}$. The limits of μ for which total internal reflection takes place at the face AC is



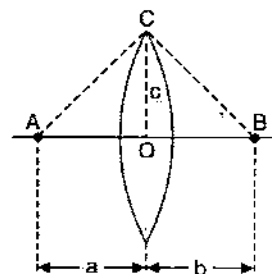
- (A) $\mu < \frac{\sqrt{3}}{2}$ (B) $\mu > \sqrt{3}$
 (C) $\mu < \frac{3\sqrt{3}}{4}$ (D) $\mu > \frac{\sqrt{3}}{2}$
8. An object is placed in front of a convex mirror at a distance of 50 cm. A plane mirror is introduced covering the lower half of the convex mirror. If the distance between the object and the plane mirror is 30 cm, there is no parallax between the images formed by the two mirrors. The radius of curvature of the convex mirror is
 (A) 60 cm (B) 50 cm
 (C) 30 cm (D) 25 cm
9. A concave lens forms the image of an object such that the distance between the object and image is 10 cm and the magnification produced is $\frac{1}{4}$. The focal length of the lens will be
 (A) 10 cm (B) 8.6 cm
 (C) 6.2 cm (D) 4.4 cm
10. For a concave mirror, the magnification of a real image was found to be twice as great when the object was 15 cm from the mirror as it was when the object was 20 cm from the mirror. The focal length of the mirror is
 (A) 5.0 cm (B) 7.5 cm
 (C) 10 cm (D) 12.5 cm
11. The image formed by a convex mirror of focal length 20 cm is half the size of the object. The distance of the object from the mirror is
 (A) 10 cm (B) 20 cm
 (C) 30 cm (D) 40 cm
12. A concave mirror of focal length f in vacuum is placed in a medium of refractive index 2. Its focal length in the medium is
 (A) $\frac{f}{2}$ (B) f
 (C) $2f$ (D) $4f$
13. A spherical mirror forms an erect image three times the size of the object. If the distance between the object and the image is 80 cm, the nature and the focal length of the mirror are
 (A) concave, 30 cm (B) convex, 30 cm

- (C) concave, 15 cm (D) convex, 15 cm
14. A boy of height 1.5 m with his eye level at 1.4 m stands before a plane mirror of length 0.75 m fixed on the wall. The height of the lower edge of the mirror above the floor is 0.8 m. Then
 (A) the boy will see his full image.
 (B) the boy cannot see his hair.
 (C) the boy cannot see his feet.
 (D) the boy cannot see both his hair and feet.

15. A horizontal ray of light passes through a prism of $\mu = 1.5$ whose apex angle is 4° and then strikes a vertical mirror M as shown. For the ray, after reflection to become horizontal, the mirror must be rotated through an angle of



- (A) 1° (B) 2°
 (C) 3° (D) 4°
16. A man of height 1.6 m wishes to see his full image in a plane mirror placed at a distance of 2 m. The minimum length of the mirror should be
 (A) 0.4 m (B) 0.8 m
 (C) 1.6 m (D) 2.4 m
17. A plane mirror reflects a beam of light to form a real image. The incident beam is
 (A) parallel (B) convergent
 (C) divergent (D) any one of the above
18. A plane mirror is approaching you at 10 cm s^{-1} . You can see your image in it. The image will approach you with a speed
 (A) 5 cm s^{-1} (B) 10 cm s^{-1}
 (C) 15 cm s^{-1} (D) 20 cm s^{-1}
19. An object is placed at A ($OA > f$), where, f is the focal length of the lens. The image is formed at B. A perpendicular is erected at O and C is chosen such that $\angle BCA = 90^\circ$. Then the value of f (in terms of a , b and c) is



- (A) $\frac{(a+b)^3}{c^2}$ (B) $\frac{(a+b)c}{(a+c)}$
(C) $\frac{c^2}{a+b}$ (D) $\frac{a^2}{a+b+c}$

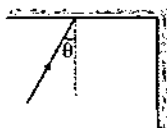
20. An observer moves towards a plane mirror with a speed of 2 ms^{-1} . The speed of the image with respect to the observer is

- (A) 1 ms^{-1} (B) 2 ms^{-1}
(C) 4 ms^{-1} (D) 8 ms^{-1}

21. A concave mirror of focal length f produces a real image n times the size of the object. The distance of the object from the mirror is

- (A) $(n-1)f$ (B) $(n+1)f$
(C) $\left(\frac{n+1}{n}\right)f$ (D) $\left(\frac{n-1}{n}\right)f$

22. Two plane mirrors are arranged at right angles to each other as shown in figure. A ray of light is incident on the horizontal mirror at an angle θ . The value of θ for which the ray emerges parallel to the incoming ray after reflection from the vertical mirror is



- (A) 30° (B) 45°
(C) 60° (D) all of the above

23. A convex mirror of focal length f produces an image $\left(\frac{1}{n}\right)^{\text{th}}$ of the size of the object. The distance of the object from the mirror is

- (A) nf (B) $\frac{f}{n}$
(C) $(n+1)f$ (D) $(n-1)f$

24. A real image formed by a concave mirror is 4.5 times the size of the object. If the mirror is 20 cm from the object, its focal length is

- (A) $\frac{90}{11} \text{ cm}$ (B) $\frac{120}{11} \text{ cm}$
(C) $\frac{150}{11} \text{ cm}$ (D) $\frac{180}{11} \text{ cm}$

25. An object is placed 10 cm in front of a convex mirror of focal length 20 cm. The distance of the image from the mirror is

- (A) $\frac{10}{3} \text{ cm}$ (B) $\frac{20}{3} \text{ cm}$
(C) 10 cm (D) $\frac{40}{3} \text{ cm}$

26. A concave mirror forms the image of an object on a screen. If the lower half of the mirror is covered with an opaque card, the effect would be
(A) to make the image less bright.
(B) to make the lower half of the image disappear.
(C) to make the upper half of the image disappear.
(D) to make the image blurred.

27. Two plane mirrors are inclined at 70° . A ray incident on one mirror at angle θ , after reflection falls on the second mirror and is reflected from there parallel to the first mirror. θ is

- (A) 45° (B) 50°
(C) 55° (D) 60°

28. A man stands in a room with his eyes at the centre of the room. The height of the ceiling is H . The length of the shortest plane mirror, fixed on the wall in front of the man, so that the man can see the full image of the wall behind him is

- (A) $\frac{2H}{3}$ (B) $\frac{H}{2}$
(C) $\frac{H}{3}$ (D) $\frac{H}{4}$

29. An object is placed between two parallel mirrors. The number of images formed is

- (A) 2 (B) 4
(C) 8 (D) infinite

30. A bulb is placed between two plane mirrors inclined at an angle of 60° . The number of images formed is

- (A) 5 (B) 6
(C) 4 (D) 3

31. Two plane mirrors are placed perpendicular to each other. A ray strikes one mirror and after reflection falls on the second mirror. The ray after reflection from the second mirror will be

- (A) perpendicular to the original ray.
(B) parallel to the original ray.
(C) at 45° to the original ray.
(D) can be at any angle to the original ray.

32. A real image is formed by a convex lens, then it is brought in contact with a concave lens such that again a real image is formed. This image will

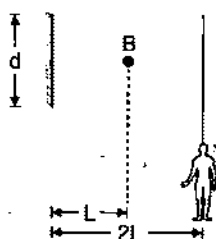
- (A) remain in its original position
(B) shift towards the lens system
(C) shift away from the lens system
(D) shift to infinity

33. Plane mirrors A and B are kept at an angle θ with respect to each other. Light falls on A, is reflected, then falls on B and is reflected. The emergent ray is opposite to the incident direction. Then the angle θ is equal to

- (A) 30° (B) 45°
(C) 60° (D) 90°

- (C) $\frac{3}{2}$ (D) $\sqrt{3}$

34. A point source of light B is placed at a distance L in front of the centre of a mirror of width d hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance $2L$ from it as shown. The greatest distance over which he can see the image of the light source in the mirror is



- (A) $\frac{d}{2}$ (B) d
(C) $2d$ (D) $3d$

35. A thin rod of length $\frac{f}{3}$ lies along the axis of a concave mirror of focal length f . One end of its image touches an end of the rod. The length of the image is

- (A) f (B) $\frac{f}{2}$
(C) $2f$ (D) $\frac{f}{4}$

36. How many images will be formed if two mirrors are fitted on adjacent walls and one mirror on ceiling?

- (A) 5 (B) 7
(C) 11 (D) 2

37. The wavefront that represents the light waves travelling in vacuum along the y -axis is

- (A) $x + y + z = \text{constant}$ (B) $x = \text{constant}$
(C) $y = \text{constant}$ (D) $z = \text{constant}$

38. A boy stands straight in front of a mirror at a distance of 30 cm from it. He sees his erect image whose height is $\frac{1}{5}$ of his real height. The mirror he is using is

- (A) plane (B) convex
(C) concave (D) plano-concave

39. The image of an object placed in front of a concave mirror of focal length 12 cm is formed at a point which is 10 cm more distant from the mirror than the object. The magnification of the image is

- (A) 1.5 (B) 2
(C) 2.5 (D) 3

40. The minimum value of the refractive index for a $90^\circ - 45^\circ - 45^\circ$ prism which is used to deviate a beam through 90° by total internal reflection is

- (A) $\frac{5}{\sqrt{3}}$ (B) $\sqrt{2}$

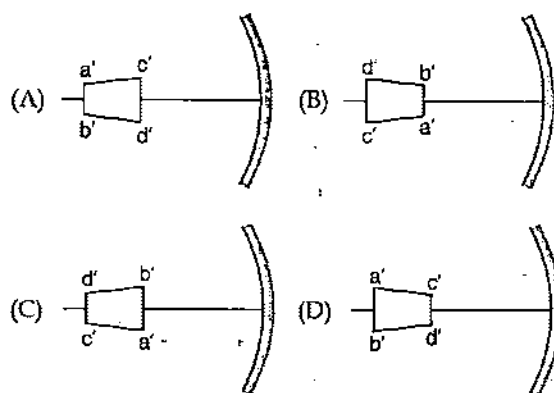
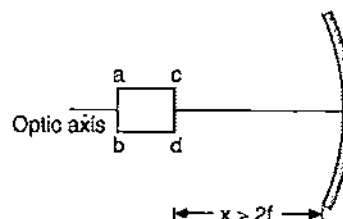
41. An object is moving towards a concave mirror of focal length 24 cm. When it is at a distance of 60 cm from the mirror its speed is 9 cms^{-1} . The speed of its image at that instant, is

- (A) 4 cms^{-1} towards the mirror
(B) 9 cms^{-1} towards the mirror
(C) 4 cms^{-1} away from the mirror
(D) 9 cms^{-1} away from the mirror

42. A ray of light passes through an equilateral prism such that the angle of emergence is equal to the angle of incidence and each is equal to $\left(\frac{3}{4}\right)^{\text{th}}$ of the angle of incidence. The angle of deviation is

- (A) 45° (B) 39°
(C) 20° (D) 30°

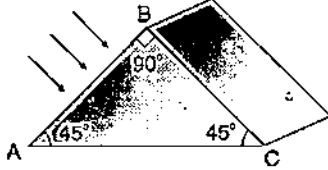
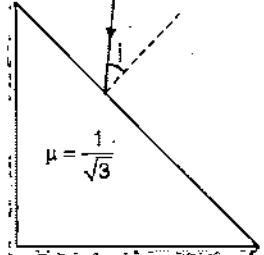
43. An object is placed in front of a concave mirror of focal length f as shown in figure. The correct shape of the image is represented by



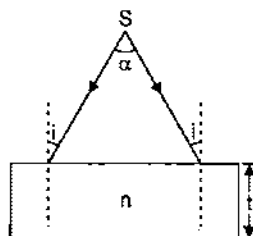
44. The index of refraction of diamond is 2.0. Velocity of light in diamond in cms^{-1} is approximately

- (A) 6×10^{10} (B) 3×10^{10}
(C) 2×10^{10} (D) 1.5×10^{10}

45. A plane mirror is placed at origin parallel of y -axis, facing the positive x -axis. An object starts from

- (2, 0, 0) m with a velocity of $(2\hat{i} + 2\hat{j}) \text{ ms}^{-1}$. The relative velocity of image with respect to object is along
 (A) positive x-axis (B) positive y-axis
 (C) negative x-axis (D) negative y-axis
46. A ray of light passes from vacuum into a medium of refractive index n . If the angle of incidence is twice the angle of refraction, then the angle of incidence is
 (A) $\cos^{-1}\left(\frac{n}{2}\right)$ (B) $\sin^{-1}\left(\frac{n}{2}\right)$
 (C) $2\cos^{-1}\left(\frac{n}{2}\right)$ (D) $2\sin^{-1}\left(\frac{n}{2}\right)$
47. A point of source of light is placed at the bottom of a vessel containing a liquid of refractive index $\frac{5}{3}$. A person is viewing the source from above the surface. There is an opaque disc of radius 1 cm floating on the surface. The centre of the disc lies vertically above the source. The liquid from the vessel is gradually drained out through a tap. The maximum height of the liquid for which the source cannot be seen at all from above is
 (A) $\frac{3}{2}$ cm (B) $\frac{4}{3}$ cm
 (C) $\frac{2}{3}$ cm (D) $\frac{3}{4}$ cm
48. A beam of light consisting of red, green and blue colours is incident on a right-angled prism as shown. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will

 (A) separate part of the red colour from the green and blue colours.
 (B) separate part of the blue colour from the red and green colours.
 (C) separate all the three colours from one another.
 (D) not separate even partially any colour from the other two colours.
49. A thin prism P_1 with angle 4° and made from glass of refractive index is 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of the prism P_2 is
 (A) 5.33° (B) 4°
 (C) 3° (D) 2.6°
50. Total internal reflection of a ray of light is possible when the ray goes from
 (A) denser to rarer medium and the angle of incidence is greater than the critical angle.
 (B) denser to rarer medium and the angle of incidence is less than the critical angle.
 (C) rarer to denser medium and the angle of incidence is greater than the critical angle.
 (D) rarer to denser medium and the angle of incidence is less than the critical angle.
51. The critical angle of light going from medium A into medium B is θ . The speed of light in medium A is v . The speed of light in medium B is
 (A) $\frac{v}{\sin \theta}$ (B) $v \sin \theta$
 (C) $\frac{v}{\tan \theta}$ (D) $v \tan \theta$
52. Glass has refractive index $\frac{3}{2}$ and water has refractive index $\frac{4}{3}$. If the speed of light in glass is $2.00 \times 10^8 \text{ ms}^{-1}$, the speed of light in water in ms^{-1} is
 (A) 1.50×10^8 (B) 1.78×10^8
 (C) 2.25×10^8 (D) 2.67×10^8
53. Two sides of an isosceles right prism are coated with a reflecting coating. A ray of light falls on the hypotenuse at an arbitrary angle i . The value of i for which the ray leaving the prism is parallel to the incident ray is

 (A) 30°
 (B) 60°
 (C) 45°
 (D) any arbitrary angle from $0 < i < \frac{\pi}{2}$
54. A diver in a lake wants to signal his distress to a person sitting on the edge of the lake flashing his water proof torch. He should direct the beam
 (A) vertically upwards.
 (B) horizontally.
 (C) at an angle to the vertical which is slightly less than the critical angle.
 (D) at an angle to the vertical which is slightly more than the critical angle.

55. Critical angle of light passing from a glass to water is minimum for
 (A) red colour (B) green colour
 (C) yellow colour (D) violet colour
56. Mirage is observed in a desert due to the phenomenon of
 (A) interference (B) total internal reflection
 (C) scattering (D) double refraction
57. The distances of an object and its virtual image from the focus of a convex lens of focal length f are 1 cm each, then f is
 (A) $(2 + \sqrt{2})$ cm (B) $(\sqrt{2} + 1)$ cm
 (C) $2\sqrt{2}$ cm (D) 4 cm
58. Total internal reflection can occur when light tends to pass from
 (A) a denser to a rarer medium.
 (B) a rarer to a denser medium.
 (C) one medium to another of different refractive index irrespective of which medium has greater refractive index.
 (D) one medium to another of equal refractive index.
59. A diverging beam of light from a point source S having divergence angle α falls symmetrically on a glass slab as shown. The angles of incidence of the two extreme rays are equal. If the thickness of the glass slab is t and the refractive index n , then the divergence angle of the emergent beam is



- (A) ZERO (B) α
 (C) $\sin^{-1}\left(\frac{1}{n}\right)$ (D) $2\sin^{-1}\left(\frac{1}{n}\right)$
60. A ray incident at an angle of incidence 60° enters a glass sphere of refractive index $\mu = \sqrt{3}$. This ray is reflected and refracted at the farther surface of the sphere. The angle between reflected and refracted rays at this surface is
 (A) 40° (B) 60°
 (C) 70° (D) 90°
61. A water film is formed on a glass block. A light ray is incident on water film from air at an angle 60° . What is the angle of incidence on glass block?

(Refractive Index of Glass = 1.5, Refractive Index of Water = $4/3$)

- (A) $\sin^{-1}\left(\frac{3\sqrt{3}}{8}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 (C) $\sin^{-1}\left(\frac{4\sqrt{3}}{9}\right)$ (D) $\sin^{-1}\left(\frac{9\sqrt{3}}{16}\right)$
62. A stone lies at the bottom of a stream. A boy wants to hit it with a stick. Taking aim the boy holds the stick in the air at an angle of 45° . At what distance from the stone will the stick hit the bottom, if the depth is 32 cm (given $\mu_w = 4/3$)
 (A) 8 cm (B) 12 cm
 (C) 16 cm (D) $12\sqrt{2}$ cm
63. When the surface of the lake is calm, a fish submerged in water will see the entire out-side world within inverted cone whose apex is situated at the eye of the fish and the cone subtends an angle of
 (A) 10° (B) 60°
 (C) 98° (D) 30°
64. A ray of light strikes a glass slab of thickness t . It emerges on the opposite face, parallel to the incident ray but laterally displaced. The lateral displacement is Δx .
 (A) $\Delta x = 0$ (B) $\Delta x = t \sin(i - r) \cos r$
 (C) $\Delta x = \frac{t \sin i}{\cos r}$ (D) $\Delta x = \frac{t \sin(i - r)}{\cos r}$
65. In cold countries the phenomenon of looming (i.e. ship appears in the sky) takes place because
 (A) refractive index of air decreases with height.
 (B) refractive index of air increases with height.
 (C) refractive index does not change with height.
 (D) refractive index becomes infinity at the surface.
66. If D is the deviation of a normally falling light beam on a thin prism of angle A and δ is the dispersive power of the same prism then
 (A) D is independent of A .
 (B) D is independent of refractive index.
 (C) δ is independent of refractive index.
 (D) δ is independent of A .
67. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Its refractive index will be
 (A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{\frac{3}{2}}$
 (C) 2 (D) $\sqrt{2}$
68. A rectangular block of glass (refractive index $3/2$) is kept in water (refractive index $4/3$). The critical angle for total internal reflection is

- (A) $\sin^{-1}\left(\frac{8}{9}\right)$ for a ray of light passing from glass to water.
 (B) $\sin^{-1}\left(\frac{8}{9}\right)$ for a ray of light passing from water to glass.
 (C) $\sin^{-1}\left(\frac{2}{3}\right)$ for a ray of light passing from water to glass.
 (D) $\sin^{-1}\left(\frac{8}{9}\right)$ for a ray of light passing from glass to air.

69. The refractive index of a given piece of transparent quartz is greatest for

- (A) red light (B) violet light
 (C) green light (D) yellow light

70. A well cut diamond appears bright because

- (A) it emits light
 (B) it is radioactive
 (C) of total internal reflection
 (D) of dispersion

71. The maximum refracting angle of a prism of refractive index 2 is

- (A) 30° (B) 45°
 (C) 60° (D) 90°

72. When light passes from one medium to another, the physical quantity that remains unchanged is

- (A) velocity (B) wavelength
 (C) frequency (D) None of these

73. A monochromatic beam of light passes from a denser to a rarer medium. As a result its

- (A) velocity increases (B) velocity decreases
 (C) frequency decreases (D) frequency increases

74. When a ray of light enters a glass slab from air

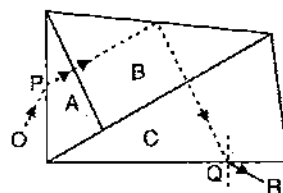
- (A) its wavelength decreases.
 (B) its wavelength increases.
 (C) its frequency increases.
 (D) neither its wavelength nor its frequency changes.

75. A number of images of a candle flame are seen in a thick mirror

- (A) the first image is the brightest.
 (B) the second image is the brightest.
 (C) the last image is the brightest.
 (D) all images are equally bright.

76. Three glass prisms A, B and C of same refractive index are placed in contact with each other as shown in figure with no air gap between the prisms. Monochromatic ray of light OP passes through the prism assembly and

emerges as QR. The condition of minimum deviation is satisfied in the prisms



- (A) A and C (B) B and C
 (C) A and B (D) in all prisms A, B and C

77. A beam of white light is incident on a hollow prism of glass. Then

- (A) the light emerging from prism gives no spectrum.
 (B) the light emerging from prism gives spectrum but the bending of all colours is away from base.
 (C) the light emerging from prism gives spectrum, all the colours bend towards base, the violet most and red the least.
 (D) the light emerging from prism gives spectrum, all the colours bend towards base, the violet the least and red the most.



78. If the critical angle for the medium of a prism is C and the angle of prism is A , then there will be no emergent ray when

- (A) $A < 2C$ (B) $A = 2C$
 (C) $A > 2C$ (D) $A \leq 2C$

79. The angle of a prism is 60° . What is the angle of incidence for minimum deviation? The refractive index of the material of the prism is $\sqrt{2}$.

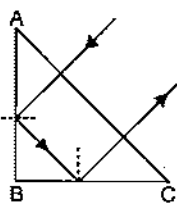
- (A) 45° (B) 60°
 (C) 30° (D) $\sin^{-1}\left(\frac{2}{3}\right)$

80. A ray of light is incident at angle i on one surface of a prism of small angle A and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ , the angle of incidence i is nearly equal to

- (A) $\frac{A}{\mu}$ (B) $\frac{A}{2\mu}$
 (C) μA (D) $\frac{\mu A}{2}$

81. If ${}^i\mu_j$ represents the refractive index when a ray of light goes from medium i to medium j , then the product ${}^2\mu_1 \times {}^3\mu_2 \times {}^4\mu_3$ is equal to

- (A) ${}^3\mu_1$ (B) ${}^3\mu_2$

- (C) $\frac{1}{\mu_4}$ (D) μ_2
82. An air bubble inside a glass slab ($\mu = \frac{3}{2}$) appears to be 6 cm deep when viewed from one side and 4 cm deep when viewed from the opposite side. The thickness of the slab is
(A) 10 cm (B) 6.67 cm
(C) 15 cm (D) None of the above
83. The refracting angle of a prism is A and the refractive index of the material of the prism is $\cot\left(\frac{A}{2}\right)$. The angle of minimum deviation is
(A) $180^\circ - 3A$ (B) $180^\circ + 2A$
(C) $90^\circ - A$ (D) $180^\circ - 2A$
84. The angle of a prism is 30° . The rays incident at 60° at one refracting face suffer a deviation of 30° . The angle of emergence is
(A) 0° (B) 30°
(C) 60° (D) 90°
85. A ray falls on a prism ABC ($AB = BC$) and travels as shown in the figure. The minimum refractive index of the prism material should be
(A) $\frac{4}{3}$ (B) $\sqrt{2}$
(C) $\frac{3}{2}$ (D) $\sqrt{3}$
- 
86. Critical angle is minimum when a light ray passes from
(A) air to glass (B) glass to air
(C) glass to water (D) water to glass
87. A point source of light is placed 4 m below the surface of a liquid of refractive index $\frac{5}{3}$. The minimum diameter of a disc, which should be placed over the source, on the surface of the liquid to cut off all light coming out of water, is
(A) ∞ (B) 6 m
(C) 4 m (D) 3 m
88. A man standing in a swimming pool looks at a stone lying at the bottom. The depth of the swimming pool is h . At what distance from the surface of water is the image of the stone formed? Line of vision is normal. Refractive index of water is n .
(A) $\frac{h}{n}$ (B) $\frac{n}{h}$
(C) h (D) hn
89. The path of a refracted ray of light in a prism is parallel to the base of the prism only when the
(A) light is of a particular wavelength.
(B) ray is incident normally at one face.
(C) ray undergoes minimum deviation.
(D) prism is made of a particular type of glass.
90. A convex lens forms a real image three times larger than the object on a screen. The object and screen are moved until the image becomes twice the size of the object. If the shift of the object is 6 cm then the screen has to be shifted by
(A) 9 cm (B) 18 cm
(C) 36 cm (D) 72 cm
91. A vessel of depth d is half filled with a liquid of refractive index μ_1 and the other half is filled with a liquid of refractive index μ_2 . The apparent depth of the vessel, when looked at normally, is
(A) $d(\mu_1 + \mu_2)$ (B) $d\left(\frac{1}{\mu_1} + \frac{1}{\mu_2}\right)$
(C) $\frac{d}{2}(\mu_1 + \mu_2)$ (D) $\frac{d}{2}\left(\frac{1}{\mu_1} + \frac{1}{\mu_2}\right)$
92. Two point sources S_1 and S_2 are 24 cm apart. Where should a convex lens of focal length 9 cm be placed in between them so that the images of both sources are formed at the same place?
(A) 6 cm from S_1 (B) 10 cm from S_1
(C) 12 cm from S_1 (D) 15 cm from S_1
93. Light travels through a glass plate of thickness t and having refractive index n . If c is the velocity of light in vacuum, the time taken by light to travel this thickness of glass is
(A) $\frac{t}{nc}$ (B) $\frac{t}{n^2c}$
(C) $\frac{nt}{c}$ (D) $\frac{n^2t}{c}$
94. A fish looking up through the water sees the outside world contained in a circular horizon. If the refractive index of water is $\frac{4}{3}$ and the fish is 12 cm below the surface of water, the radius of the circle in cm is
(A) $36\sqrt{5}$ (B) $36\sqrt{7}$
(C) $\frac{36}{\sqrt{7}}$ (D) $4\sqrt{5}$
95. A diver inside water sees the setting sun at
(A) 41° to the horizon (B) 49° to the horizon
(C) 0° to the horizon (D) 45° to the horizon

96. If the refractive index of water is $\frac{4}{3}$ and that of glass is $\frac{5}{3}$, then the critical angle of incidence for light tending to go from glass to water is

(A) $\sin^{-1}\left(\frac{3}{4}\right)$ (B) $\sin^{-1}\left(\frac{3}{5}\right)$
(C) $\sin^{-1}\left(\frac{4}{5}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$

97. Two media A and B of refractive indices $\mu_1 = 1.5$ and $\mu_2 = 2$ are separated by x - z plane. A ray of light travels from A to B. The incident ray and the reflected ray are represented by unit vectors $\vec{u}_1 = a\hat{i} + b\hat{j}$ and $\vec{u}_2 = c\hat{i} + d\hat{j}$. Then

(A) $\frac{a}{c} = \frac{3}{4}$ (B) $\frac{a}{c} = \frac{4}{3}$
(C) $\frac{b}{d} = \frac{3}{4}$ (D) $\frac{b}{d} = \frac{4}{3}$

98. The speed of light in medium A is $2.0 \times 10^8 \text{ ms}^{-1}$ and that in medium B is $2.4 \times 10^8 \text{ ms}^{-1}$. The critical angle of incidence for light tending to go from medium A to medium B is

(A) $\sin^{-1}\left(\frac{5}{12}\right)$ (B) $\sin^{-1}\left(\frac{5}{6}\right)$
(C) $\sin^{-1}\left(\frac{2}{3}\right)$ (D) $\sin^{-1}\left(\frac{3}{4}\right)$

99. The speed of light in glass of refractive index 1.5 is $2 \times 10^8 \text{ ms}^{-1}$. In a certain liquid the speed of light is $2.5 \times 10^8 \text{ ms}^{-1}$. The refractive index of the liquid is

(A) 0.64 (B) 0.80
(C) 1.20 (D) 1.44

100. A ray of light travelling inside a rectangular glass block of refractive index $\sqrt{2}$ is incident on the glass-air surface at an angle of incidence of 45° . The refractive index of air is 1. The ray will

(A) emerge into air without any deviation.
(B) be reflected back into glass.
(C) be absorbed.
(D) emerge into air with an angle of refraction equal to 90° .

101. A fish in water sees an object which is 24 cm above the surface of water. The height of the object above the surface of water that will appear to the fish is

(A) 24 cm (B) 32 cm
(C) 18 cm (D) 48 cm

102. The angle of minimum deviation equals the angle of prism A of an equilateral glass prism. The angle of

incidence at which minimum deviation will be obtained is

(A) $\sin^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (B) 30°
(C) 60° (D) 45°

103. Light is incident at an angle α on one planar end of a transparent cylindrical rod of refractive index n . The least value of n for which the light entering the rod will not emerge from the curved surface of rod, irrespective of value of α is

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
(C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

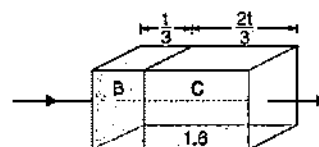
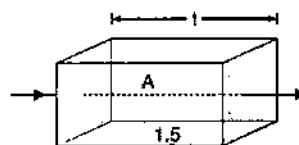
104. For a prism the refractive index (μ) is related to wavelength (λ) as $\mu = A + \frac{B}{\lambda^2}$. The dispersive power is large if

(A) A is large (B) B is large
(C) μ is large (D) A and μ are large

105. An infinitely long rod lies along the axis of a concave mirror of focal length f . The near end of the rod is at a distance $u > f$ from the mirror. The length of the image of the rod is

(A) $\frac{uf}{u+f}$ (B) $\frac{f^2}{u+f}$
(C) $\frac{f^2}{u-f}$ (D) $\frac{uf}{u-f}$

106. Two transparent slabs have the same thickness as shown in figure. One is made of material X of refractive index 1.5. The other is made of two materials Y and Z having thicknesses in the ratio 1 : 2. The refractive index of Z is 1.6. If a monochromatic parallel beam passing through the slabs has the same number of wavelengths inside both, the refractive index of Y is



(A) 1.1 (B) 1.2
(C) 1.3 (D) 1.4

107. A curved mirror of focal length f (in vacuum) is placed in a medium of refractive index 2. Its new focal length in the medium is f' .

- (A) $f' < f$ (B) $f' > f$
(C) $f' = f$ (D) $f' \approx f$

108. If ϵ_0 is the absolute permittivity of free space, μ_0 is absolute permeability of free space, ϵ is the permittivity of medium, μ is permeability of medium and n is the refractive index of medium then,

- (A) $n = \sqrt{\frac{\mu_0 \mu}{\epsilon_0 \epsilon}}$ (B) $n = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}}$
(C) $n = \sqrt{\frac{\mu_0 \epsilon_0}{\mu \epsilon}}$ (D) $n = \frac{\mu \epsilon}{\mu_0 \epsilon_0}$

109. The critical angle of glass ($\mu_g = \frac{3}{2}$) is θ_1 and that of water ($\mu_w = \frac{4}{3}$) is θ_2 . The critical angle for water-glass interface is

- (A) less than θ_1 (B) less than θ_2
(C) between θ_1 and θ_2 (D) greater than θ_2

110. Two plane mirrors M_1 and M_2 are inclined to each other at 70° . A ray incident on the mirror M_1 at an angle θ falls on M_2 and is then reflected parallel to M_1 for

- (A) $\theta = 45^\circ$ (B) $\theta = 50^\circ$
(C) $\theta = 55^\circ$ (D) $\theta = 60^\circ$

111. An object is placed at 20 cm from a convex mirror of focal length 20 cm. The distance of the image from the pole of the mirror is

- (A) infinite (B) 10 cm
(C) 15 cm (D) 40 cm

112. The sun (diameter D) subtends an angle θ radian at the pole of a concave mirror of focal length f . The diameter of the image of the sun formed by the mirror is

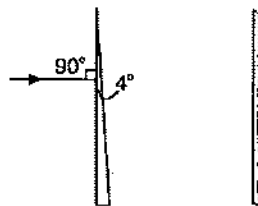
- (A) $f\theta$ (B) $2f\theta$
(C) $\frac{2f\theta}{D}$ (D) $D\theta$

113. An object is placed in front of a convex mirror at a distance of 50 cm. A plane mirror is introduced covering the lower half of the convex mirror. If the distance between the object and the plane mirror is 30 cm, there is no parallax between the images formed by the two mirrors. The radius of curvature of the convex mirror (in cm) is

- (A) 60 (B) 50

- (C) 30 (D) 25

114. A prism having an apex angle 4° and refractive index 1.5 is located in front of a vertical plane mirror as shown in figure. The total angle through which the ray is deviated after reflection from the mirror is given by

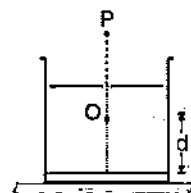


- (A) 176° (B) 4°
(C) 178° (D) 2°

115. A slab of glass of thickness 3 cm and refractive index $\frac{3}{2}$ is placed with its face perpendicular to the principal axis of the concave mirror. If the radius of mirror is 10 cm, the distance at which an object must be placed from the mirror so that the image coincides with the object is

- (A) 9 cm (B) 10 cm
(C) 11 cm (D) 12 cm

116. A tank contains a transparent liquid of refractive index n the bottom of which is made of a mirror as shown. An object O lies at a height d above the mirror. A person P vertically above the object sees O and its image in the mirror and finds the apparent separation to be



- (A) $2nd$ (B) $\frac{2d}{n-1}$
(C) $\frac{2d}{n}$ (D) $\frac{d}{n}(1+n)$

117. A ray of light enters an anisotropic medium from vacuum at grazing incidence. If θ is the angle made by the reflected ray inside the medium with the interface and $n(\theta)$ is the refractive index of the medium then,

- (A) $n(\theta)\sin\theta = 1$ (B) $n(\theta)\cos\theta = 1$
(C) $\frac{n(\theta)}{\sin\theta} = 1$ (D) $\frac{n(\theta)}{\cos\theta} = 1$

118. A person runs with a speed u towards a bicycle moving away from him with speed v . The person approaches his image in the mirror fixed at the rear of bicycle with a speed of

- (A) $u - v$ (B) $u - 2v$
(C) $2u - v$ (D) $2(u - v)$

119. Light travels through a glass plate of thickness t having refractive index μ . If c is the velocity of light in

vacuum, the time taken by the light to travel this thickness of glass is

- (A) $\frac{t}{\mu c}$ (B) $\frac{t}{\mu^2 c}$
(C) $\frac{\mu t}{c}$ (D) $\frac{\mu^2 t}{c}$

120. A real image I is formed by a converging lens L on its optic axis. On introduction of a rectangular glass slab of thickness d and refractive index μ between the image and lens the image displaces it by

- (A) $d(\mu - 1)$ away from L
(B) $d(\mu - 1)$ towards L
(C) $d\left(1 - \frac{1}{\mu}\right)$ away from L
(D) $d\left(1 - \frac{1}{\mu}\right)$ towards L

121. When a ray is refracted from one medium to another, the wavelength changes from $6000 \text{ \AA}$ to $4000 \text{ \AA}$. The critical angle for the interface will be

- (A) $\cos^{-1}\left(\frac{2}{3}\right)$ (B) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$
(C) $\sin^{-1}\left(\frac{2}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

122. A boy stands straight in front of a mirror at a distance of 30 cm away from it. He sees his erect image whose height is one fifth of the original height. The mirror used by him is

- (A) plane (B) convex
(C) concave (D) plano concave

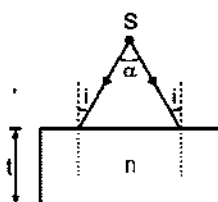
123. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is $\frac{2}{3}$ times the wavelength in free space.

The radius of the curved surface of the lens is

- (A) 1 m (B) 2 m
(C) 3 m (D) 6 m

124. A diverging beam of light from a point source S having divergence angle α , falls symmetrically on a glass slab as shown. The angles of incidence of two extreme rays are equal. If the thickness of the glass slab is t and the refractive index n , then the divergence angle of the emergent beam is

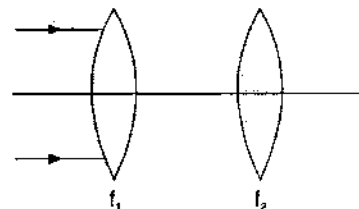
- (A) ZERO (B) α
(C) $\sin^{-1}\left(\frac{1}{n}\right)$ (D) $2\sin^{-1}\left(\frac{1}{n}\right)$



125. The light on reflection from a plane mirror can give a real image when

- (A) the convergent rays are incident on the mirror.
(B) the divergent rays are incident on the mirror.
(C) an object is placed very close to the mirror.
(D) an object is placed very far away from the mirror.

126. Parallel beam of light is incident on the system of two convex lenses of focal length $f_1 = 20 \text{ cm}$ and $f_2 = 10 \text{ cm}$. The distance between the two lenses, so that rays after refraction from both the lenses pass undeviated is



- (A) 30 cm (B) 40 cm
(C) 60 cm (D) 90 cm

127. The plane faces of two identical plano convex lenses, each with focal length f are pressed against each other using an optical glue to form a usual convex lens. The distance from the optical centre at which an object must be placed to obtain the image same as the size of object is

- (A) $\frac{f}{4}$ (B) $\frac{f}{2}$
(C) f (D) $2f$

128. A parallel beam of light incident on a concave lens of focal length 10 cm emerges as a parallel beam from a convex lens placed coaxially, the separation between the lenses being 10 cm. The focal length of the convex lens in cm is

- (A) 10 (B) 20
(C) 15 (D) 30

129. A ray of light is incident on a glass sphere of refractive index $\frac{3}{2}$. The angle of incidence for which a ray that enters the sphere does not come out of the sphere is

- (A) $\tan^{-1}\left(\frac{2}{3}\right)$ (B) $\sin^{-1}\left(\frac{2}{3}\right)$
(C) 45° (D) 90°

130. A thin prism P_1 of angle 4° and made from glass of refractive index 1.54, is combined with another thin prism P_2 made from a glass with refractive index 1.72, to produce dispersion without deviation. The angle of P_2 is

- (A) 5.33° (B) 4°
(C) 3° (D) 2.6°

131. A transparent sphere of radius R made of material of refractive index $\frac{3}{2}$ is kept in air. The distance from the centre of the sphere must a point object be placed so as to form a real image at the same distance from the sphere is

(A) R (B) $2R$
(C) $3R$ (D) $4R$

132. An air bubble in water is to be placed in a way such that a real image is obtained at the same distance from bubble. Taking $\mu_{\text{water}} = \frac{4}{3}$ we have the distance of object from the air bubble as

(A) R
(B) $2R$
(C) $3R$
(D) An air bubble is incapable to form a real image.

133. An object is placed at a distance of 12 cm from a convex lens on its principal axis and a virtual image of certain size is formed. On moving the object 8 cm away from the lens, a real image of the same size as that of virtual image is formed. The focal length of the lens in cm is

(A) 15 (B) 16
(C) 17 (D) 18

134. An air bubble inside a glass slab appears to be 6 cm deep when viewed from one side and 4 cm deep when viewed from the other side. Assuming $\mu_{\text{glass}} = \frac{3}{2}$, the thickness of slab is

(A) 10 cm (B) $\frac{20}{3}$ cm
(C) 15 cm (D) 20 cm

135. On two sides of an oily paper screen, two bulbs A and B are placed at a distance of 20 cm and 30 cm, so that equal intensity is obtained on both sides of screen. If P_A and P_B be the powers of the bulbs A and B respectively then $\frac{P_A}{P_B}$ is

(A) 0.44 (B) 2.25
(C) 1.5 (D) 0.67

136. An achromatic combination pair of a telescope objective will be

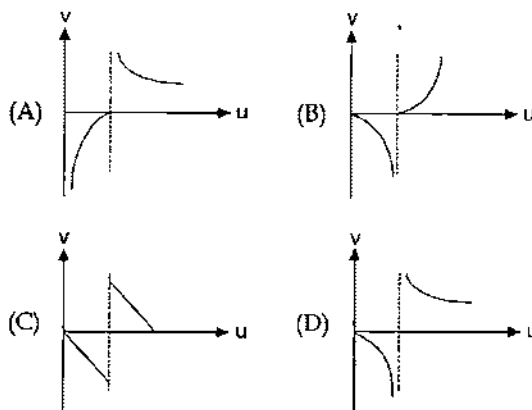
(A) lenses of $f = -50$ cm and power +2 D
(B) lenses of powers 3 D and -5 D respectively
(C) lenses of $f = +20$ cm and power -4.5 D
(D) lenses of $f = +40$ cm and power +2 D

137. A ray of light enters the face of a glass prism of refracting angle A , refractive index μ at an angle of

incidence i . It is observed that no ray emerges from the other face. For this the minimum value of i should be

(A) $\mu \sin A - \cos A$
(B) $\sin^{-1}(\sin A - \mu \cos A)$
(C) $\sin^{-1}[\sqrt{\mu^2 - 1} \sin A - \cos A]$
(D) $\sqrt{\mu^2 - 1} \sin A - \cos A$

138. As the position of an object (u) reflected from a concave mirror is varied, the position of the image (v) also varies. By allowing the u to change from 0 to $+\infty$, the graph between v versus u will be



139. A parallel beam of light emerges from the opposite surface of the sphere when a point source of light lies at the surface of the sphere. The refractive index of the sphere is

(A) $\frac{3}{2}$ (B) $\frac{5}{3}$
(C) 2 (D) $\frac{5}{2}$

140. Two spherical mirrors M_1 and M_2 , one convex and other concave having same radius of curvature R are arranged coaxially at a distance $2R$ (consider their pole separation to be $2R$). A bead of radius a is placed at the pole of the convex mirror as shown. The ratio of the sizes of the first three images of the bead is

(A) 1 : 2 : 3 (B) $1 : \frac{1}{2} : \frac{1}{3}$
(C) $\frac{1}{3} : \frac{1}{11} : \frac{1}{41}$ (D) 3 : 11 : 41

141. Light is incident at an angle α on one planar end of a transparent cylindrical rod of refractive index n . The least value of n for which the light entering the rod will

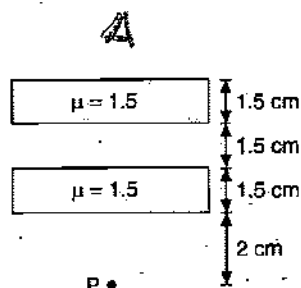
not emerge from the curved surface of rod, irrespective of value of α is

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
(C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

142. A ray of light is incident on one face of prism with refracting angle A ($< 90^\circ$). The incident ray is normal to the other face of the prism. If C is the critical angle for prism-air interface, then the ray will emerge from this face only if

- (A) $\cot C < \cot A + 1$ (B) $\cot C > \cot A + 1$
(C) $\cot A < \cot C + 1$ (D) $\cot A > \cot C + 1$

143. The image of point P when viewed from top of the glass slabs is

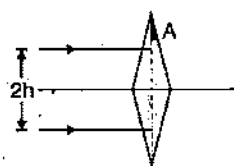


- (A) 2 cm above P (B) 0.5 cm above P
(C) 0.5 cm below P (D) 1 cm above P

144. An isosceles prism has refracting angle A . Its one face is silvered (other than the base). A ray of light falling normally on the face not silvered emerges through the base of the prism normal to it.

- (A) $A = 45^\circ$ (B) $A = 90^\circ$
(C) $A = 36^\circ$ (D) $A = 72^\circ$

145. Two identical thin isosceles prisms of refracting angle A and refractive index μ are placed with their bases touching each other and this system can collectively act as a crude converging lens. A parallel beam of light is incident on this system as shown. The focal length of this so called converging lens is

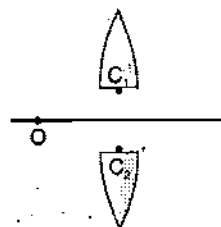


- (A) $f = \frac{h}{\mu A}$ (B) $f = \frac{h}{A}$
(C) $f = \frac{h}{(\mu - 1)A}$ (D) $f = \frac{\mu h}{\mu - 1}$

146. Rays of light from a luminous object are brought to focus at a point A . The rays are intercepted, before meeting at A by a convex lens of focal length 20 cm placed at 24 cm from A and are forced to meet at B . Then AB equals (in cm)

- (A) 12 (B) 24
(C) 6 (D) 48

147. A point object is placed at a distance of 0.3 m from a convex lens of focal length 0.2 m cut into two equal halves, each of which is displaced by 0.0005 m, as shown in figure. If C_1 and C_2 be their optical centres then;



- (A) an image is formed at a distance of 0.6 m from C_1 or C_2 along principal axis.
(B) two images are formed, one at a distance of 0.6 m and other at a distance of 1.2 m from C_1 or C_2 along principal axis.
(C) an image is formed at a distance of 0.12 m from C_1 or C_2 along principal axis.
(D) two images are formed at a distance of 0.6 m from C_1 or C_2 along principal axis at a separation of 0.003 m.

148. All of the following statements are correct except that
(A) the magnification produced by a convex mirror is always less than one.
(B) a virtual, erect, same sized image can be obtained by using the plane mirror.
(C) a virtual, erect, magnified image can be formed by using the concave mirror.
(D) a real, inverted, same sized image can be formed by using a convex mirror.

149. A ray of light undergoes deviation of 30° when incident on an equilateral prism of refractive index $\sqrt{2}$. The angle made by the ray inside the prism with the base of the prism is

- (A) 0° (B) 15°
(C) 30° (D) 45°

150. A convex lens of focal length f forms an image of a heavenly body. The area of the image formed is proportional to

- (A) f^0 (B) f^1
(C) f^2 (D) f^3

151. A plano convex lens has a thickness of 4 cm. When placed on a horizontal table with curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of plane face is found to be $\frac{25}{8}$ cm. The focal length of the lens is

- (A) 50 cm (B) 75 cm
(C) 100 cm (D) 150 cm

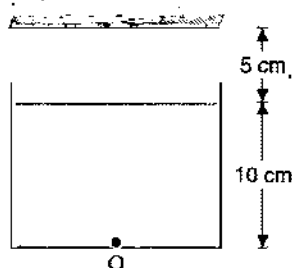
152. If an object is placed between two parallel mirrors, an infinite number of images are formed. If the mirrors are at a distance $2b$ and an object is placed at the middle of the two mirrors, the distance of the n th image from the object is

- (A) nb (B) $\frac{1}{2}nb$
(C) $2nb$ (D) $\frac{1}{4}nb$

153. A ray of light is incident on the plane mirror at rest. The mirror starts turning at a uniform acceleration of $2\pi \text{ rads}^{-2}$. The reflected ray, at the end of $\frac{1}{4}$ s must have turned through

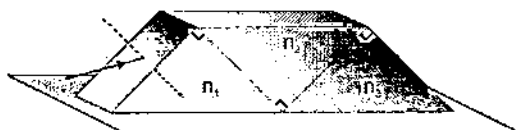
- (A) 90° (B) 45°
(C) 22.5° (D) 11.25°

154. In the situation shown in figure, water ($\mu_w = \frac{4}{3}$) is filled in a beaker upto a height of 10 cm. A plane mirror is fixed at a height of 5 cm from the surface of water. Distance of image from the mirror after reflection from it of an object O at the bottom of the beaker is



- (A) 7.5 cm (B) 10 cm
(C) 12.5 cm (D) 15 cm

155. Three right angled prisms of refractive indices n_1, n_2 and n_3 are fixed together using an optical glue as shown in figure. If a ray passes through the prisms without suffering any deviation, then

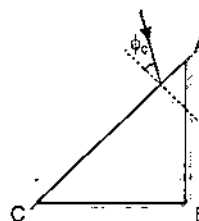


- (A) $n_1 = n_2 = n_3$ (B) $n_1 = n_2 \neq n_3$
(C) $1 + n_1 = n_2 + n_3$ (D) $1 + n_2^2 = n_1^2 + n_3^2$

156. Four lenses are made from same type of glass. The radius of curvature of each face is given. Out of these, the lens having the greatest positive power is

- (A) 10 cm convex and 15 cm convex.
(B) 20 cm convex and 30 cm concave.
(C) 15 cm convex and plane.
(D) 5 cm convex and 10 cm concave.

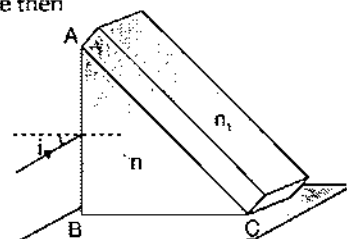
157. The sides of an isosceles right angled prism are silvered. A ray of light falls on the hypotenuse of the prism at an angle ϕ_0 as shown. The ray leaving the prism will



- (A) graze the face AC.
(B) emerge normally to the face AC.
(C) be parallel to the incident ray.
(D) make an angle of 30° with incident ray.

158. A right angled prism ($45^\circ-90^\circ-45^\circ$) of refractive index n has a plate of refractive index n_1 ($n_1 < n$) cemented to its diagonal face. The assembly is in air. A ray is incident on AB as shown. If the ray strikes the diagonal face AC at critical angle then

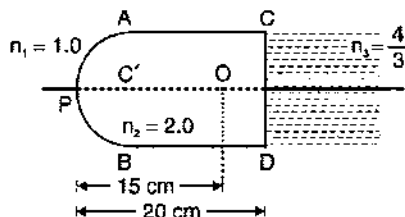
- (A) $\sin i = \left(\frac{n_1}{n}\right)$
(B) $\sin i = \left(\frac{n}{n_1}\right)$
(C) $\sin i = \sqrt{\frac{n^2 - n_1^2}{2}}$
(D) $\sin i = \frac{\sqrt{n^2 - n_1^2} - n_1}{\sqrt{2}}$



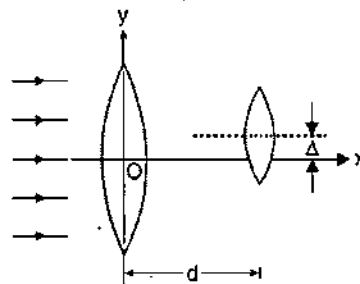
159. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eye piece is 36 cm and the final image is formed at infinity. The focal length f_o of the objective and the focal length f_e of the eye piece are

- (A) $f_o = 45 \text{ cm}$ and $f_e = -9 \text{ cm}$
(B) $f_o = 50 \text{ cm}$ and $f_e = 10 \text{ cm}$
(C) $f_o = 7.2 \text{ cm}$ and $f_e = 5 \text{ cm}$
(D) $f_o = 30 \text{ cm}$ and $f_e = 6 \text{ cm}$

160. The slab of a material of refractive index 2 shown in figure has a curved surface APB of radius of curvature 10 cm and a plane surface CD. On the left of APB is air and on the right of CD is water with refractive indices as given in figure. An object O is placed at a distance of 15 cm from pole P as shown. The distance of the final image of O from P, as viewed from the left is



- (A) 20 cm (B) 30 cm
(C) 40 cm (D) 50 cm
161. A thin rod of length $\frac{1}{3}f$ is placed along the optic axis of a concave mirror of focal length f such that its image which is real and elongated just touches the rod. The magnification is
- (A) $\frac{4}{3}$ (B) $\frac{5}{3}$
(C) $\frac{3}{2}$ (D) None of above
162. A 2 cm diameter coin lies flat at the bottom of a bowl in which the water ($\mu_w = \frac{4}{3}$), is 20 cm deep. If the coin is viewed directly from above, the apparent diameter of the coin is
- (A) 1.67 cm (B) 1.5 cm
(C) 2.67 cm (D) 2 cm
163. A ray of light undergoes deviation of 30° when incident on an equilateral prism of refractive index $\sqrt{2}$. The angle made by the ray inside the prism with the base of the prism is
- (A) 30° (B) 45°
(C) 60° (D) 0°
164. Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance d (where $d < f_1$ and $d < f_2$) and their centres are displaced by a vertical separation Δ as shown. Taking the origin of coordinates O, at the centre of the first lens, the x and y coordinates of the focal point of this lens system, for a parallel beam of rays coming from the left, are given by

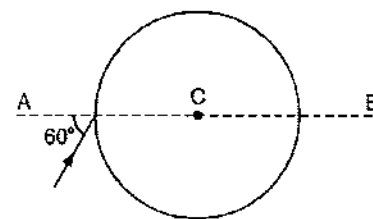


- (A) $x = \frac{f_1 f_2}{f_1 + f_2}$, $y = \Delta$
(B) $x = \frac{f_1 (f_2 + d)}{f_1 + f_2 - d}$, $y = -\frac{\Delta^2}{f_1 + f_2}$
(C) $x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}$, $y = \frac{\Delta(f_1 - d)}{f_1 + f_2 - d}$
(D) $x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}$, $y = 0$

165. Spherical aberration in a thin lens can be reduced by
- (A) using a monochromatic light.
(B) using a doublet combination.
(C) using a circular annular mask over the lens.
(D) increasing the size of the lens.
166. A circular beam of light of diameter $d = 2$ cm falls on a plane surface of a glass slab. The angle of incidence is 60° and refractive index of glass is $\mu = \frac{3}{2}$. The diameter of the refracted beam is
- (A) 2.52 cm (B) 3 cm
(C) 3.26 cm (D) 4 cm
167. Two thin lenses, when in contact, produce a combination of power $+10$ D. When they are 0.25 m apart, the power reduces to $+6$ D. The focal lengths of the lenses (in m) are
- (A) 0.125 and 0.5 (B) 0.125 and 0.125
(C) 0.5 and 0.75 (D) 0.125 and 0.75
168. A real image of a distinct object is formed by a plano convex lens on its principal axis. Spherical aberration
- (A) is absent.
(B) is smaller if the curved surface of the lens faces the object.
(C) is smaller if the plane surface of the lens faces the object.
(D) is the same whichever side of the lens faces the object.
169. A concave mirror is placed on a horizontal table, with its axis directed vertically upwards. Let O be the pole of the mirror and C its centre of curvature. A point object is placed at C. It has a real image, also located at

C. If the mirror is now filled with water, the image will be

- (A) real and will remain at C.
(B) real and located at a point between C and ∞ .
(C) virtual and located at a point between C and O.
(D) real and located at a point between C and O.



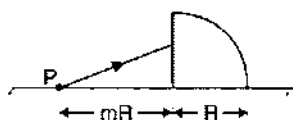
170. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in the glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at the point O and $PO = OQ$. The distance PO is equal to

- (A) $5R$ (B) $3R$
(C) $2R$ (D) $1.5R$

171. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature R . On immersion in a medium of refractive index 1.75, it will behave as a

- (A) convergent lens of focal length $3.5R$.
(B) convergent lens of focal length $3.0R$.
(C) divergent lens of focal length $3.5R$.
(D) divergent lens of focal length $3.0R$.

172. A quarter cylinder of radius R and refractive index 1.5 is placed on a table. A point object P is kept at a distance of mR from it. The value of m for which a ray from P will emerge parallel to the table is



- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) 1 (D) $\frac{4}{3}$

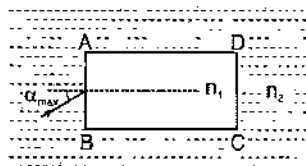
173. A hollow double concave lens is made of very thin transparent material. It can be filled with air or either of two liquids L_1 or L_2 having refractive indices n_1 and n_2 respectively ($n_2 > n_1 > 1$). The lens will diverge a parallel beam of light if it is filled with

- (A) air and placed in air.
(B) air and immersed in L_1 .
(C) L_1 and immersed in L_2 .
(D) L_2 and immersed in L_1 .

174. A ray of light falls on a transparent sphere of refractive index μ , having centre at C as shown in figure. The ray emerges from the sphere parallel to line AB , then

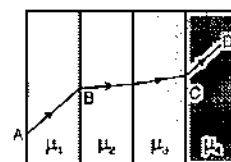
- (A) $\mu = \sqrt{2}$ (B) $\mu = \frac{3}{2}$
(C) $\mu = \sqrt{3}$ (D) $\mu = 2$

175. A rectangular glass slab ABCD of refractive index n_1 , is immersed in water of refractive index n_2 ($n_1 > n_2$). A ray of light is incident at the surface AB of the slab as shown. The maximum value of the angle of incidence α_{max} such that the ray comes out only from the other surface CD is given by



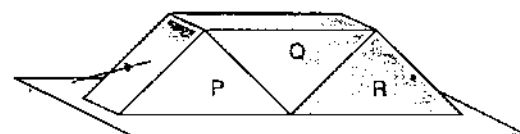
- (A) $\sin^{-1} \left\{ \frac{n_1}{n_2} \cos \left(\sin^{-1} \left(\frac{n_2}{n_1} \right) \right) \right\}$
(B) $\sin^{-1} \left\{ n_1 \cos \left(\sin^{-1} \left(\frac{1}{n_2} \right) \right) \right\}$
(C) $\sin^{-1} \left(\frac{n_1}{n_2} \right)$
(D) $\sin^{-1} \left(\frac{n_2}{n_1} \right)$

176. A ray of light passes through four transparent media with refractive indices μ_1, μ_2, μ_3 and μ_4 as shown in figure. The surface of all media are parallel. If the emergent ray CD is parallel to the incident ray AB, we must have



- (A) $\mu_1 = \mu_2$ (B) $\mu_2 = \mu_3$
(C) $\mu_3 = \mu_4$ (D) $\mu_4 = \mu_1$

177. A given ray of light suffers minimum deviation in an equilateral prism P . Additional prisms Q and R of identical shape and of same material as P are now added as shown in figure. The ray will now suffer



- (A) greater deviation.
(B) no deviation.
(C) same deviation as before.

(D) total internal reflection.

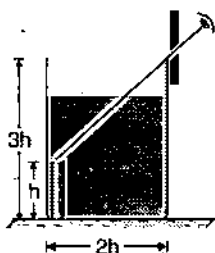
178. Two identical equiconvex lenses of focal length f , made of glass ($\mu_g = \frac{3}{2}$) are kept in contact. The space

between the two lenses is filled with water ($\mu_w = \frac{4}{3}$).

The focal length of the combination is

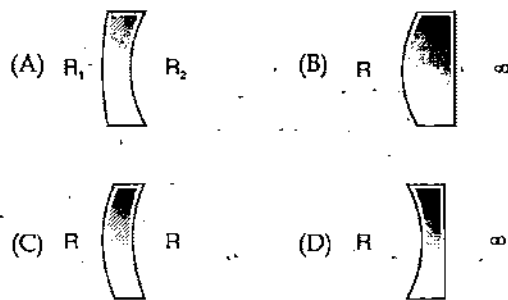
- (A) $\frac{f}{2}$ (B) $\frac{3f}{4}$
(C) f (D) $\frac{4f}{3}$

179. An observer can see through a pin hole, the top of a thin rod of height h , placed as shown in figure. The beaker's height is $3h$ and its radius is h . When the beaker is filled with a liquid upto a height $2h$, he can see the lower end of the rod. Then the refractive index of liquid must be

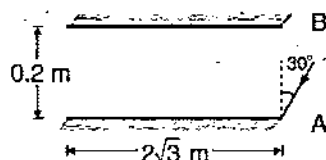


- (A) $\frac{5}{2}$ (B) $\sqrt{\frac{5}{2}}$
(C) $\sqrt{\frac{3}{2}}$ (D) $\frac{3}{2}$

180. Which one of the following spherical lenses does not exhibit dispersion?



181. Two plane mirrors A and B are aligned parallel to each other, as shown in figure. A light ray is incident at an angle of 30° at a point just inside one end of A. The plane of incidence coincides with the plane of figure. The maximum number of times the ray undergoes reflections (including the first one) before it emerges out is



- (A) 28 (B) 30
(C) 32 (D) 34

182. Critical angle for a prism is 36° . The maximum angle of prism for which the emergent ray is possible is

- (A) 18° (B) 36°
(C) 72° (D) 144°

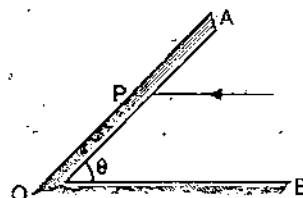
183. A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different positions separated from each other by 20 cm. The lens has a focal length of

- (A) 32.1 cm (B) 15.8 cm
(C) 21.4 cm (D) 10.7 cm

184. A prism, having refractive index $\sqrt{2}$ and refracting angle 30° , has one of the refracting surfaces polished. A beam of light incident on the other refracting surface will retrace its path if the angle of incidence is

- (A) 0° (B) 30°
(C) 45° (D) 60°

185. Two plane mirrors are inclined at angle θ as shown in figure. If a ray parallel to OB strikes the other mirror at P and finally emerges parallel to OA after two reflections, then θ equals



- (A) 30° (B) 45°
(C) 60° (D) 90°

186. An object is placed 20 cm in front of a block of glass 10 cm thick having its farther side silvered. The image is formed 23.2 cm behind the silvered face. The refractive index of glass is

- (A) 1.41 (B) 1.46
(C) 1.51 (D) 1.61

187. Two parallel rays are travelling in a medium of refractive index $\mu_1 = \frac{4}{3}$. However, one of the rays passes through a parallel glass slab of thickness t and refractive index $\mu_2 = \frac{3}{2}$. The path difference between the two rays due to the glass slab is

- (A) $\frac{t}{8}$ (B) $\frac{t}{6}$
(C) $\frac{4t}{3}$ (D) $\frac{3t}{2}$

188. A ray of light entering from air to glass (refractive index 1.5) is partly reflected and partly refracted. If the incident and the reflected rays are at right angles to each other, the angle of refraction is

- (A) $\sin^{-1}\left(\frac{\sqrt{2}}{\sqrt{3}}\right)$ (B) $\sin^{-1}\left(\frac{\sqrt{2}}{3}\right)$
 (C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

189. A beam of light is converging towards a point on a screen. A plane parallel sided plate of glass of thickness t and refractive index μ is introduced in the path of the beam. The convergence point is shifted by

- (A) $t\left(1 - \frac{1}{\mu}\right)$ away (B) $t\left(1 + \frac{1}{\mu}\right)$ away
 (C) $t\left(1 - \frac{1}{\mu}\right)$ nearer (D) $t\left(1 + \frac{1}{\mu}\right)$ nearer

190. The distance of an object from the first focus of an equiconvex lens is 10 cm and the distance of its real image from the second focus is 40 cm. The focal length of the lens is

- (A) 10 cm (B) 20 cm
 (C) 25 cm (D) 40 cm

191. A beam of monochromatic light is incident on one face of an equilateral prism, the angle of incidence being 55° . If the angle of emergence is 46° then the angle of minimum deviation is

- (A) 41° (B) $< 41^\circ$
 (C) $> 41^\circ$ (D) $\geq 41^\circ$

192. When a ray of light is refracted by a prism such that the angle of deviation is minimum, then

- (A) the angle of emergence is equal to the angle of incidence.
 (B) the angle of emergence is greater than the angle of incidence.
 (C) the angle of emergence is smaller than the angle of incidence.
 (D) the sum of the angle of incidence and the angle of emergence is equal to 90° .

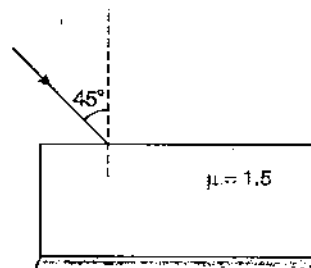
193. The image of a square hole in a screen illuminated by light is obtained on another screen with the help of a converging lens. The distance of the hole from the lens is 40 cm. If the area of the image is nine times that of the hole, the focal length of the lens is

- (A) 30 cm (B) 50 cm
 (C) 60 cm (D) 75 cm

194. A short linear object of length b lies along the axis of a concave mirror of focal length f at a distance u from the pole of the mirror. The size of the image is approximately equal to

- (A) $b\sqrt{\frac{u-f}{f}}$ (B) $b\left(\frac{u-f}{f}\right)^2$
 (C) $b\sqrt{\frac{f}{u-f}}$ (D) $b\left(\frac{f}{u-f}\right)^2$

195. One side of a glass slab of refractive index 1.5 is silvered as shown. A ray of light is incident on the other side at angle of incidence $i = 45^\circ$. The deviation of the ray of light from its initial path when it comes out of the slab is



- (A) 45° (B) 90°
 (C) 120° (D) 180°

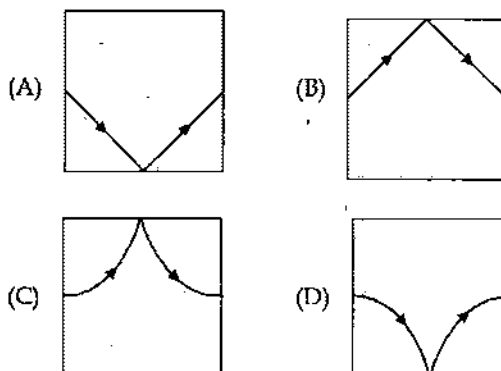
196. A plano-convex lens has a thickness of 4 cm. When placed on a horizontal table with the curved surface in contact with it, the apparent depth of the bottom-most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of the plane face is found to be $\frac{25}{8}$ cm. The focal length of the lens is

- (A) 50 cm (B) 75 cm
 (C) 100 cm (D) 150 cm

197. If f_B and f_R are the focal lengths of a convex lens for blue and red lights respectively and F_B and F_R are the respective values for a concave lens, then

- (A) $f_B > f_R$ and $F_B > F_R$ (B) $f_B < f_R$ and $F_B > F_R$
 (C) $f_B > f_R$ and $F_B < F_R$ (D) $f_B < f_R$ and $F_B < F_R$

198. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. The path of a ray of light inside the liquid is best represented by



199. The focal length of a convex lens is f and the distance of an object from the principal focus is x . The ratio of the size of the real image to the size of the object is

- (A) $\frac{f}{x}$ (B) $\frac{x}{f}$
(C) $\frac{f+x}{f}$ (D) $\frac{f}{f+x}$

200. Focal length of a convex mirror is 10 cm

- (A) image of an object placed at 20 cm is also at 20 cm
(B) image of an object placed at 10 cm is at infinity
(C) both (A) and (B) are correct
(D) both (A) and (B) are incorrect

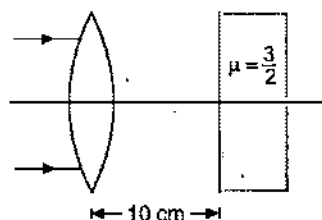
201. An object is placed at a distance x_1 from the principal focus of a lens and its real image is formed at a distance x_2 from the principal focus. The focal length of the lens is

- (A) $x_1 x_2$ (B) $\frac{x_1 x_2}{2}$
(C) $\frac{x_1 + x_2}{2}$ (D) $\sqrt{x_1 x_2}$

202. The plane faces of two identical planoconvex lenses, each having focal length of 40 cm, are pressed against each other to form a usual convex lens. The distance in cm from this lens, at which an object must be placed to obtain a real image with magnification unity is

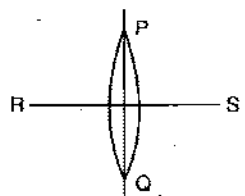
- (A) 10 (B) 20
(C) 40 (D) 80

203. A plane refracting surface of refractive index $\frac{3}{2}$ is placed at a distance of 10 cm from a thin convex lens of focal length 30 cm. The parallel rays incident on lens will converge at a distance of



- (A) 30 cm from the lens. (B) 25 cm from the lens.
(C) 20 cm from the lens. (D) 40 cm from the lens.

204. The figure shows an equiconvex lens of focal length f . If the lens is cut along PQ, the focal length of each half will be



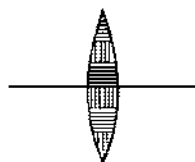
- (A) $\frac{f}{2}$ (B) f

- (C) $2f$ (D) $4f$

205. In PROBLEM 204, if the lens is cut along PQ and RS simultaneously, the focal length of each part will be

- (A) $\frac{f}{2}$ (B) f
(C) $2f$ (D) $4f$

206. The layered lens as shown is made of two types of transparent materials—one indicated by horizontal lines and the other by vertical lines. The number of images formed of an object will be



- (A) 1 (B) 2
(C) 3 (D) 6

207. The distance between an object and its real image formed by a convex lens cannot be

- (A) greater than $2f$ (B) less than $2f$
(C) greater than $4f$ (D) less than $4f$

208. Two thin symmetrical lenses of different nature and of different material have equal radii of curvature $R = 15$ cm are placed close together and immersed in water ($\mu_w = \frac{4}{3}$). The focal length of the system in water is 30 cm. The difference between refractive indices of the two lenses is

- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{1}{4}$ (D) $\frac{3}{4}$

209. A needle of length 5 cm, placed 45 cm from a lens forms an image on a screen placed 90 cm on the other side of the lens. The type of lens and its focal length are

- (A) convex, 30 cm (B) concave, 30 cm
(C) convex, 60 cm (D) concave, 60 cm

210. In PROBLEM 209, the nature and size of the image are

- (A) real, 20 cm (B) real, 10 cm
(C) virtual, 20 cm (D) virtual, 10 cm

211. An object is placed 50 cm in front of a convex surface of radius 20 cm. If the surface separates air from glass of refractive index 1.5, the distance of the image from the lens and its nature are

- (A) 30 cm, real (B) 30 cm, virtual
(C) 300 cm, real (D) 300 cm, virtual

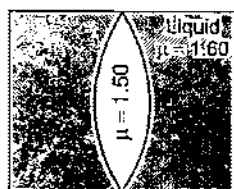
212. One of the refracting surfaces of a prism of angle 30° is silvered. A ray of light incident at an angle of 60° retraces its path. The refractive index of the material of prism is

- (A) $\sqrt{2}$ (B) $\frac{3}{2}$
(C) $\sqrt{3}$ (D) 2
- (B) 25 cm
(C) 20 cm
(D) cannot be estimated with given data

213. A slab of glass of refractive index 1.5 and thickness 3 cm is placed with the faces perpendicular to the principal axis of a concave mirror. If the radius of curvature of the mirror is 10 cm, the distance at which an object must be placed from the mirror so that the image coincides with the object is

- (A) 9 cm (B) 10 cm
(C) 11 cm (D) 12 cm

214. Figure represents a convergent lens placed inside a cell filled with a liquid. The lens has a focal length +20 cm when in air and its material has refractive index 1.50. If the liquid has a refractive index 1.60, the focal length of the lens in the new system is



Lens

- (A) -80 cm (B) +80 cm
(C) -160 cm (D) -24 cm

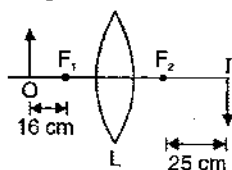
215. A point object O is placed on the principal axis of a convex lens of focal length 20 cm at a distance of 40 cm to the left of it. The diameter of the lens is 10 cm. If the eye is placed 60 cm to the right of the lens at a distance h below the principal axis, then the maximum value of h to see the image will be

- (A) 0 cm (B) 5 cm
(C) 2.5 cm (D) 10 cm

216. For two positions of a lens, the images are obtained on a fixed screen. If the size of object is 2 cm and the size of diminished image is 0.5 cm, the size of the other image will be

- (A) 1 cm (B) 4 cm
(C) 8 cm (D) 16 cm

217. The medium on both sides of lens is air. The distances of object O , image I from first and second foci F_1 and F_2 are shown in figure. The focal length of lens is

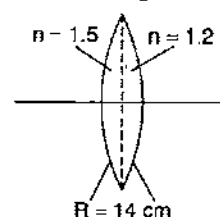


- (A) 16 cm

218. A ray of light travelling in the direction $\frac{1}{2}(\hat{i} + \sqrt{3}\hat{j})$ is incident on a plane mirror. After reflection, it travels along the direction $\frac{1}{2}(\hat{i} - \sqrt{3}\hat{j})$. The angle of incidence is

- (A) 30° (B) 45°
(C) 60° (D) 75°

219. A bi-convex lens is formed with two thin plano convex lenses as shown in the figure. Refractive index n of the first lens is 1.5 and that of the second lens is 1.2. Both the curved surfaces are of the same radius of curvature $R = 14$ cm. For this bi-convex lens, for the object distance of 40 cm, the image distance will be



- (A) -280 cm (B) 40 cm
(C) 21.5 cm (D) 13.3 cm

220. An equiconvex lens, having radius of curvature 33 cm, is placed on a horizontal plane mirror and a pin held 20 cm above the lens coincides with its image. Now the space between the lens and the mirror is filled with a liquid. In order to coincide with the image the pin has to be raised by 5 cm. The refractive index of the liquid is

- (A) 1.33 (B) 1.53
(C) 2.33 (D) 2.66

221. A real image is formed by a convex lens. If we put a concave lens in contact with it, the combination again forms a real image. The new image

- (A) is closer to the lens system.
(B) is farther from the lens system.
(C) is at the original position.
(D) may be anywhere depending on the focal length of the concave lens.

222. A concave mirror has a focal length 20 cm. The distance between the two positions of the object for which the image size is double of the object size is

- (A) 60 cm (B) 40 cm
(C) 30 cm (D) 20 cm

223. A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of the combination in dioptre is

- (A) -1.5 (B) -6.5

(C) +6.5

(D) +6.67

224. Two lenses of powers $+12\text{ D}$ and -2 D are in contact.

The focal length of the combination is

- (A) 10 cm (B) 12.5 cm
(C) 16.6 cm (D) 8.33 cm

225. A point object O is placed

at a distance of 20 cm

from a convex lens of focal

length 10 cm as shown in

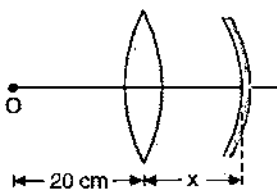
figure. The distance x

from the lens where a

concave mirror of focal

length 60 cm has to be placed so that final image coincides with the object is

- (A) 10 cm
(B) 20 cm
(C) 40 cm
(D) final image can never coincide with the object under the conditions provided.



226. A lens forms a sharp image on a screen. On inserting a parallel sided glass slab between the lens and the screen, it is found necessary to move the screen a distance d away from the lens in order for the image to be sharp again. If the refractive index of the material of the slab is n , the thickness of the slab is

- (A) nd (B) $\frac{d}{n}$
(C) $\frac{(n-1)d}{n}$ (D) $\frac{nd}{n-1}$

227. A plano convex glass

lens ($\mu_g = \frac{3}{2}$) of

radius of curvature

$R = 10\text{ cm}$ is placed at

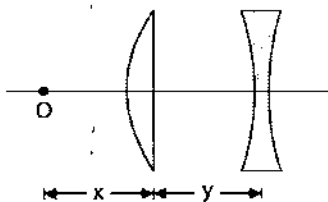
a distance of y from a

concave lens of focal

length 20 cm. The

distance x of a point object O from the plano convex lens so that the position of final image is independent of y is

- (A) 20 cm (B) 30 cm
(C) 40 cm (D) 60 cm



228. A thin lens has focal length f , and its aperture has diameter D . It forms an image of intensity I . If the central part of the aperture, of diameter $\frac{D}{2}$, is blocked by an opaque paper, the focal length of the lens and the intensity of image will become

- (A) $\frac{f}{2}, \frac{I}{2}$ (B) $f, \frac{I}{4}$

(C) $\frac{3f}{4}, \frac{I}{2}$

(D) $f, \frac{3I}{4}$

229. When a ray of light goes from air to a glass slab, then

- (A) its wavelength increases
(B) its wavelength decreases
(C) its frequency increases
(D) neither its wavelength nor its frequency changes.

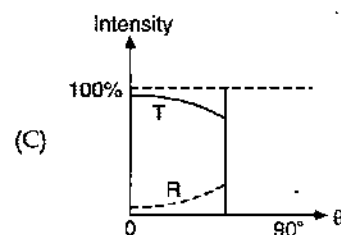
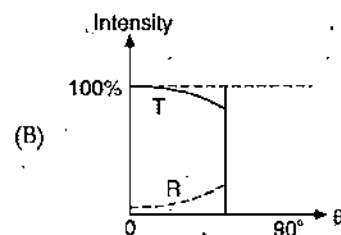
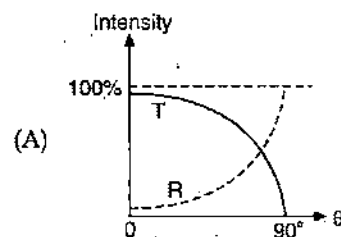
230. In the displacement method, a convex lens is placed in between an object and a screen. If the magnification in the two positions are m_1 and m_2 ($m_1 > m_2$), and the distance between the two positions of the lens is x , the focal length of the lens is

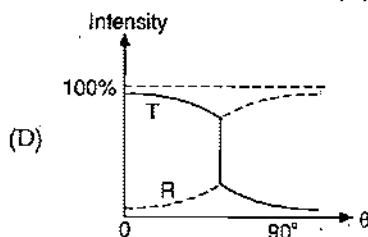
- (A) $\frac{x}{m_1 + m_2}$ (B) $\frac{x}{m_1 - m_2}$
(C) $\frac{x}{(m_1 + m_2)^2}$ (D) $\frac{x}{(m_1 - m_2)^2}$

231. A screen is placed 90 cm from an object. The image of the object on the screen is formed by a convex lens at two different locations separated by 20 cm. The focal length of the lens is

- (A) 10.7 cm (B) 21.4 cm
(C) 15.8 cm (D) 32.1 cm

232. A light ray travelling in glass medium is incident on glass-air interface at an angle of incidence θ . The reflected (R) and transmitted (T) intensities, both as function of θ , are plotted. The correct sketch is





233. A lens is placed between the source of light and a wall. It forms images of area A_1 and A_2 on the wall for its two different positions. The area of the source of light is

(A) $\sqrt{A_1 A_2}$ (B) $\frac{A_1 + A_2}{2}$
 (C) $\sqrt{\left[\frac{\sqrt{A_1} + \sqrt{A_2}}{2}\right]^2}$ (D) $\left(\frac{1}{A_1} + \frac{1}{A_2}\right)^{-1}$

234. The plane face of a plano-convex lens is silvered. If μ be the refractive index and r the radius of curvature of the curved surface, then the system behaves like a concave mirror of radius

(A) $\frac{r}{\mu}$ (B) $\frac{r}{\mu - 1}$
 (C) $r\mu$ (D) $r(\mu - 1)$

235. A ray of light falls on the surface of a spherical paper weight making an angle α with the normal and is refracted in the medium at an angle β . The angle of deviation of the emergent ray from the direction of the incident ray is

(A) $(\alpha - \beta)$ (B) $2(\alpha - \beta)$
 (C) $\frac{(\alpha - \beta)}{2}$ (D) $\beta - \alpha$

236. The magnification of an object placed in front of a convex lens of focal length 20 cm is +2. To obtain a magnification of -2, the object has to be moved by a distance equal to

(A) 40 cm (B) 30 cm
 (C) 20 cm (D) 10 cm

237. A concave mirror is placed on a horizontal table with its axis directed vertically upward. Let (O) be the pole of the mirror and C its centre of curvature. A point object is placed at C. It has a real image also located at C. If the mirror is now filled with water, the image will be
 (A) real and will remain at C.
 (B) real and located at a point between C and ∞ .
 (C) virtual and located at a point between C and O.
 (D) real and located at a point between C and O.

238. All of the following statements are correct except
 (A) the magnification produced by a convex mirror is always less than one.

- (B) a virtual, erect, same-sized image can be obtained using a plane mirror.
 (C) a virtual, erect, magnified image can be formed using a concave mirror.
 (D) a real, inverted, same-sized image can be formed using a convex mirror.

239. When an object is at distances x and y from a lens, a real image and a virtual image is formed respectively having same magnification. The focal length of the lens is given by

(A) $x - y$ (B) $x + y$
 (C) $\sqrt{xy}$ (D) $\frac{x + y}{2}$

240. If the central portion of a convex lens is wrapped in black paper as shown in the figure,
 (A) no image will be formed by the remaining portion of the lens.



- (B) full image will be formed, but it will be less bright.
 (C) the central portion of the image will be missing.
 (D) there will be two images, each produced by one of the exposed portions of the lens.

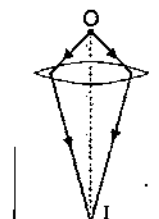
241. A plane mirror made of glass slab ($\mu_g = 1.5$) is 2.5 cm thick and silvered at the back. A point object is placed 5 cm in front of the unsilvered face of the mirror. The position of the final image is

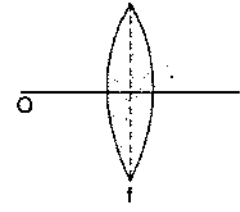
(A) $\frac{16}{3}$ cm from unsilvered face
 (B) $\frac{25}{3}$ cm from unsilvered face
 (C) 12 cm from unsilvered face
 (D) 14 cm from unsilvered face

242. The distance between an object and the screen is 100 cm. A lens produces an image on the screen when placed at either of two positions 40 cm apart. The power of the lens is approximately

(A) 4.25 D (B) 4.50 D
 (C) 4.75 D (D) 5.0 D

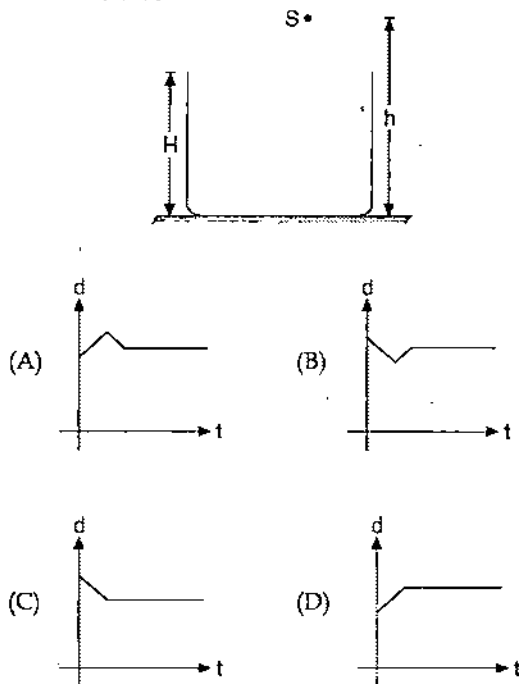
243. A real image of an object is formed by a convex lens at the bottom of an empty beaker. The beaker is now filled with a liquid of refractive index 1.4 to a depth of 7 cm. In order to get the image again at the bottom, the beaker should be moved



- (A) downward by 2 cm
(B) upward by 2 cm
(C) downward by 3 cm
(D) upward by 3 cm
244. The convex surface of a thin concavo-convex lens (refractive index 1.5) has a radius of curvature 20 cm. The concave surface has a radius of curvature 60 cm. The convex side is silvered and placed on a horizontal surface. At what distance from the lens should a pin be placed on the optic axis such that its image is formed at the same place?
(A) 15 cm (B) 7.5 cm
(C) 22.5 cm (D) 30 cm
245. In PROBLEM 244, if the concave part is filled with water (refractive index $\frac{4}{3}$), the distance from the lens at which the pin should be placed to form the image at the same place is
(A) $\frac{90}{13}$ cm (B) $\frac{45}{13}$ cm
(C) $\frac{135}{13}$ cm (D) $\frac{180}{13}$ cm
246. An equiconvex lens of glass ($\mu_g = 1.5$) of focal length 10 cm, silvered on one side behaves like a
(A) convex mirror of focal length 5 cm
(B) convex mirror of focal length 20 cm
(C) concave mirror of focal length 2.5 cm
(D) concave mirror of focal length 10 cm
247. A plano-convex lens of focal length 30 cm has its plane surface silvered. An object is placed 40 cm from the lens on the convex side. The distance of the image from the lens is
(A) 18 cm (B) 24 cm
(C) 30 cm (D) 40 cm
248. Refraction takes place at a concave spherical boundary separating glass air medium. If $\mu_g = \frac{3}{2}$, then for the image to be real, the object distance
(A) is independent of the radius of curvature of the refracting surface
(B) should be greater than the radius of curvature of the refracting surface
(C) should be greater than two times the radius of curvature of the refracting surface
(D) should be greater than three times the radius of curvature of the refracting surface
249. A parallel beam of light incident on a concave lens of focal length 10 cm emerges as a parallel beam from a convex lens placed coaxially, the distance between the lenses being 10 cm. The focal length of the convex lens in cm is
(A) 10 (B) 20
(C) 30 (D) 40
250. An object is placed at a distance of 12 cm from a convex lens on its principal axis and a virtual image of certain size is formed. If the object is moved 8 cm away from the lens, a real image of the same size as that of the virtual image is formed. The focal length of the lens in cm is
(A) 15 (B) 16
(C) 18 (D) 19
251. A thin converging lens of refractive index 1.5 has a power of $+0.5 D$. When this lens is immersed in a liquid, it acts as a diverging lens of focal length 100 cm. The refractive index of the liquid is
(A) $\frac{4}{3}$ (B) $\frac{3}{2}$
(C) $\frac{5}{3}$ (D) 2
252. The distance between an object and its real image formed by a lens is D . If the magnification is m , the focal length of the lens is
(A) $\frac{(m-1)D}{m}$ (B) $\frac{mD}{m+1}$
(C) $\frac{(m-1)D}{m^2}$ (D) $\frac{mD}{(m+1)^2}$
253. A plano convex lens of focal length 16 cm, is to be made of glass of refractive index 1.5. The radius of curvature of the curved surface should be
(A) 8 cm (B) 12 cm
(C) 16 cm (D) 24 cm
254. A real image of a point object O was formed by an equi-convex lens of focal length f and the magnification was found to be unity. Now the lens is cut into two symmetrical pieces as shown by the dotted line and the right part is removed. The position of the image formed by the remaining part is at

(A) f (B) $2f$
(C) $\frac{f}{2}$ (D) Infinity
255. A convex lens, made of a material of refractive index 1.5 and having a focal length of 10 cm is immersed in a liquid of refractive index 3.0. The lens will behave as a

- (A) converging lens of focal length 10 cm.
 (B) diverging lens of focal length 10 cm.
 (C) converging lens of focal length $\frac{10}{3}$ cm.
 (D) diverging lens of focal length 30 cm.

256. A point source S is placed at a height h from the bottom of a vessel of height $H (< h)$. The vessel is polished at the base. If the water is gradually filled in the vessel at a constant rate $\alpha \text{ m}^3 \text{ s}^{-1}$, the distance d of image of the source from the bottom of the vessel varies with time t as



257. A convex lens of glass has power P in air. If it is immersed in water its power will be
 (A) more than P
 (B) less than P
 (C) P
 (D) more than P for some colours and less than P for others

258. A biconvex lens, made of a material of refractive index 1.5, has radius of curvature of each side equal to 0.5 m. The power of the lens is
 (A) 0.5 D (B) 1.0 D
 (C) 1.5 D (D) 2.0 D

259. A convex lens forms a real image 4 cm long on a screen. When the lens is shifted to a new position without disturbing the object or the screen, again real image is formed on the screen which is 16 cm long. The length of the object is
 (A) 8 cm (B) 10 cm

- (C) 12 cm (D) 6 cm

260. The maximum and minimum distances between a convex lens and an object, for the magnification of a real image to be greater than one are
 (A) $2f$ and f (B) f and zero
 (C) ∞ and $2f$ (D) $4f$ and $2f$

261. A point object is placed on the optic axis of a convex lens of focal length f at a distance of $2f$ to the left of it. The diameter of the lens is d . An observer has his eye at a distance of $3f$ to the right of the lens and a distance h below the optic axis. The maximum value of h to see the image is
 (A) $\frac{d}{4}$ (B) $\frac{d}{3}$
 (C) $\frac{d}{2}$ (D) d

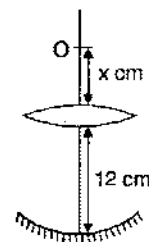
262. A convex lens is immersed in a liquid of refractive index greater than that of glass. It will behave as a
 (A) convergent lens (B) divergent lens
 (C) plane glass (D) homogeneous liquid

263. If the top half of a convex lens is covered with black paper,
 (A) the bottom half of the image will disappear.
 (B) the top half of the image will disappear.
 (C) the magnification will be reduced to half.
 (D) the intensity will be reduced to half.

264. In displacement method, the lengths of images in the two positions of the lens between the object and the screen are 9 cm and 4 cm respectively. The length of the object must be
 (A) 6.25 cm (B) 1.5 cm
 (C) 6 cm (D) 36 cm

265. A convex lens of focal length 15 cm is placed on a plane mirror. An object is placed 20 cm from the lens. The image is formed
 (A) 12 cm in front of the mirror
 (B) 60 cm behind the mirror
 (C) 60 cm in front of the mirror
 (D) 30 cm in front of the mirror

266. A convex lens of focal length 40 cm is held coaxially 12 cm above a concave mirror of focal length 18 cm. An object held x cm above the lens gives rise to an image coincident with it. The x is equal to
 (A) 12 cm (B) 15 cm
 (C) 18 cm (D) 30 cm



267. Two thin lenses of powers 2 D and 3 D are placed in contact. An object is placed at a distance of 30 cm from the combination. The distance in cm of the image from the combination is

- (A) 30 (B) 40
(C) 50 (D) 60

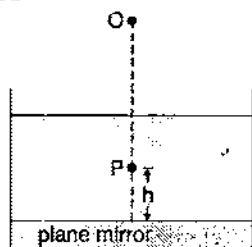
268. Two convex lenses of focal lengths f_1 and f_2 are mounted coaxially separated by a distance. If the power of the combination is zero, the distance between the lenses is

- (A) $|f_1 - f_2|$ (B) $f_1 + f_2$
(C) $\frac{f_1 f_2}{|f_1 - f_2|}$ (D) $\frac{f_1 f_2}{f_1 + f_2}$

269. Chromatic aberration in a lens is caused by

- (A) reflection (B) interference
(C) diffraction (D) dispersion

270. A plane mirror is placed at the bottom of a tank containing a liquid of refractive index μ . A small object P lies at a height h above the mirror. An observer O , vertically above P , outside the liquid, observe P and its image in the mirror. The apparent distance between these two will be



- (A) $h\left(1 + \frac{1}{\mu}\right)$ (B) $\frac{2h}{\mu - 1}$
(C) $2\mu h$ (D) $\frac{2h}{\mu}$

271. A person can see clearly between 1 m and 2 m. His corrective lenses should be

- (A) bifocals with power -0.5 D and additional +3.5 D
(B) bifocals with power -1.0 D and additional +3.0 D
(C) concave with power 1.0 D
(D) convex with power 0.5 D

272. Spherical aberration in a thin lens can be reduced by

- (A) using a monochromatic light.
(B) using a doublet combination.
(C) using a circular annular mask over the lens.
(D) increasing the size of the lens.

273. A virtual image of an object is formed with a magnification of 2, when the object is placed in front of a

concave mirror of focal length f . To obtain a real image of same magnification, the object has to moved by a distance

- (A) $\frac{f}{2}$ (B) $\frac{2f}{3}$
(C) f (D) $\frac{3f}{2}$

274. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and the eyepiece is 36 cm and the final image is formed at infinity. The focal length f_o of the objective and f_e of the eyepiece are

- (A) $f_o = 45$ cm and $f_e = -9$ cm
(B) $f_o = 50$ cm and $f_e = 10$ cm
(C) $f_o = 7.2$ cm and $f_e = 5$ cm
(D) $f_o = 30$ cm and $f_e = 6$ cm

275. A plano convex lens of radius of curvature R fits exactly into a plano concave lens such that their plane surfaces are parallel to each other. If the lenses are made of different materials of refractive indices μ_1 and μ_2 , then focal length of the combination is given by

- (A) $\frac{R}{2 - (\mu_1 + \mu_2)}$ (B) $\frac{R}{2(\mu_1 - \mu_2)}$
(C) $\frac{2R}{\mu_2 - \mu_1}$ (D) $\frac{R}{\mu_1 - \mu_2}$

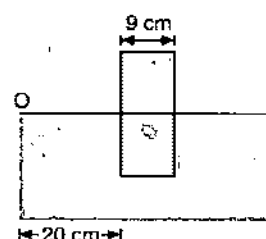
276. A compound microscope has an objective of focal length 2.0 cm and an eye piece of focal length 6.25 cm separated by 15 cm. If the final image is formed at the least distance of distinct vision (25 cm), the distance of the object from the objective is

- (A) 1.5 cm (B) 2.5 cm
(C) 3.0 cm (D) 4.0 cm

277. In PROBLEM 276, the magnifying power of the microscope is

- (A) 10 (B) 15
(C) 20 (D) 30

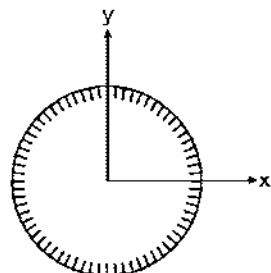
278. A point object is placed at a distance of 20 cm from a glass slab ($\mu_g = \frac{3}{2}$) half immersed in water ($\mu_w = \frac{4}{3}$) as shown in figure. The distance between two images when seen from the other side of the slab is



- (A) 1 cm (B) 2 cm
(C) 4 cm (D) 6 cm
279. A compound microscope has a magnification of 30. The focal length of the eye-piece is 5 cm. If the final image is formed at the least distance of distinct vision (25 cm), the magnification produced by the objective is
(A) 5 (B) 7.5
(C) 10 (D) 15
280. The least distance of distinct vision is 25 cm. The focal length of a convex lens is 5 cm. It can act as a simple microscope of magnifying power
(A) 4 (B) 5
(C) 6 (D) None of these
281. An astronomical telescope has an eye piece of focal length 5 cm. If the angular magnification of normal adjustment is 10, the distance between the objective and the eye piece is
(A) 45 cm (B) 50 cm
(C) 55 cm (D) 110 cm
282. The focal lengths of the objective and the eyepiece of an astronomical telescope are 100 cm and 20 cm respectively. Its magnifying power in normal adjustment is
(A) 5 (B) 2
(C) 25 (D) 4
283. Two convex lenses of focal lengths 0.3 m and 5 cm are used to make a telescope. The distance kept between them is equal to
(A) 0.35 m (B) 5.3 cm
(C) 5.3 m (D) 0.15 m
284. To have larger magnification by a telescope
(A) the objective should be of large focal length and the eyepiece should be of small focal length
(B) both the objective and the eyepiece should be of large focal lengths
(C) both the objective and the eyepiece should be of small focal lengths
(D) the objective should be of small focal length and the eyepiece should be of large focal length
285. The angle of incidence for an equilateral prism is 60° . The refractive index of prism so that the ray inside the prism is parallel to the base of the prism is
(A) $\frac{9}{8}$ (B) $\sqrt{2}$
(C) $\frac{4}{3}$ (D) $\sqrt{3}$
286. Four convergent lenses have focal lengths 100 cm, 10 cm, 4 cm and 0.3 cm. For a telescope with maximum possible magnification, we choose the lenses of focal lengths
(A) 100 cm, 0.3 cm (B) 10 cm, 0.3 cm
(C) 10 cm, 4 cm (D) 100 cm, 4 cm
287. The angular magnification of a telescope which contains an objective of focal length f_1 and eyepiece of focal length f_2 is
(A) $\frac{f_2}{f_1}$ (B) $\frac{f_1 + f_2}{f_2}$
(C) $\frac{f_1}{f_2}$ (D) $\frac{f_1 f_2}{f_1 + f_2}$
288. An achromatic combination is to be made using a convex and a concave lens. The two lenses should have
(A) their power equal.
(B) their refractive indices equal.
(C) their dispersive powers equal.
(D) the product of their powers and dispersive powers equal.
289. For a thin equiconvex lens, the optics axis coincides with the x-axis and the optical centre coincides with the origin. The co-ordinates of a point object and its image are $(-40, 1)$ cm and $(50, -2)$ cm respectively. Lens is located at
(A) $x = 0$ (B) $x = -10$ cm
(C) $x = +20$ cm (D) $x = -30$ cm
290. The near point of a person is 50 cm and the far point is 1.5 m. The spectacles required for reading purpose and for seeing distant objects are respectively
(A) $+2$ D, $-\left(\frac{2}{3}\right)$ D (B) $+\left(\frac{2}{3}\right)$ D, -2 D
(C) -2 D, $+\left(\frac{2}{3}\right)$ D (D) $-\left(\frac{2}{3}\right)$ D, $+2$ D
291. Astigmatism for a human eye can be removed by using
(A) concave lens (B) convex lens
(C) cylindrical lens (D) prismatic lens
292. A hollow convex lens of glass behaves like a
(A) plane mirror (B) concave lens
(C) convex lens (D) glass plate
293. The far point of a myopic eye is 250 cm. The correcting lens should be a
(A) diverging lens of focal length 250 cm.
(B) converging lens of focal length 250 cm.
(C) diverging lens of focal length 125 cm.
(D) converging lens of focal length 125 cm.
294. A person cannot see clearly beyond 50 cm. The power of the lens required to correct his vision is
(A) -0.5 D (B) $+0.5$ D

(C) -2 D (D) $+2\text{ D}$

295. A ray travelling in negative x -direction is directed towards positive y -direction after being reflected from a surface at point P . The reflecting surface is represented by the equation $x^2 + y^2 = a^2$. Then co-ordinates of point P are

(A) $(a, 0)$ (B) $(0.6a, 0.8a)$ (C) $(0.8a, 0.6a)$ (D) $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$

296. A person cannot see clearly objects at a distance less than 100 cm . The power of the spectacles required to see clearly objects at 25 cm is

(A) $+1\text{ D}$ (B) $+3\text{ D}$ (C) $+4\text{ D}$ (D) $+2\text{ D}$

297. An object is kept at a distance of 16 cm from a thin lens and the image formed is real. If the object is kept at a distance of 6 cm from the same lens the image formed is virtual. If the size of the images formed are equal, the focal length of the lens will be

(A) 8 cm (B) 5 cm (C) 11 cm (D) $\sqrt{96}\text{ cm}$

298. A person can see clearly objects lying between 25 cm and 2 m from his eye. His vision can be corrected by using spectacles of power

(A) $+0.25\text{ D}$ (B) $+0.5\text{ D}$ (C) -0.25 D (D) -0.5 D

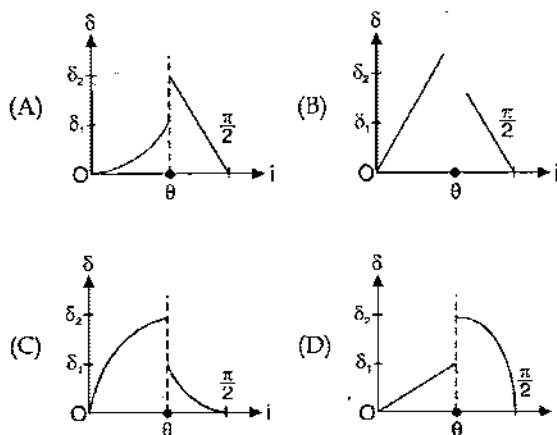
Practice Exercise

Multiple Correct Choice Type Questions

This section contains Multiple Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONE OR MORE is/are correct.

1. The x - y plane is the boundary between two transparent media. Medium 1 with $z \geq 0$ has a refractive index $\sqrt{2}$ and medium 2 with $z < 0$ has a refractive index $\sqrt{3}$. A ray of light in medium 1 given by the vector $\vec{A} = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. The refracted ray makes angle r with $+z$ axis and incident ray makes an angle i with $-z$ axis. Then,
- (A) $i = 120^\circ$ (B) $i = 60^\circ$
(C) $r = 45^\circ$ (D) $r = 135^\circ$

2. A ray of light travels from a medium of refractive index μ to air. Its angle of incidence in the medium is i , measured from the normal to the boundary, and its angle of deviation is δ . The curve that best represents the plot of deviation δ (along y -axis) with angle of incidence i (along x -axis) is



3. In PROBLEM 2,

(A) $\theta = \sin^{-1}\left(\frac{1}{\mu}\right)$ (B) $\theta = \frac{\pi}{2} - \sin^{-1}\left(\frac{1}{\mu}\right)$

(C) $\frac{\delta_2}{\delta_1} = \mu$ (D) $\frac{\delta_2}{\delta_1} = 2$

4. A ray of light from a denser medium strikes a rarer medium at angle of incidence i . The reflected and the refracted rays make an angle of 90° with each other. The angles of reflection and refraction are r and r' respectively. The critical angle is

(A) $\sin^{-1}(\tan r)$ (B) $\sin^{-1}(\tan i)$
(C) $\sin^{-1}(\tan r')$ (D) $\tan^{-1}(\sin i)$

5. A single converging lens is used as a simple microscope. In the position of maximum magnification. Select the correct statement(s).

- (A) the object is placed at the focus of the lens.
(B) the object is placed between the lens and its focus.
(C) the image is formed at infinity.
(D) the object and the image subtend the same angle at the eye.

6. A light of wavelength $6000 \text{ \AA}$ in air enters a medium of refractive index 1.5. Inside the medium, its frequency is ν and its wavelength is λ .

(A) $\nu = 5 \times 10^{14} \text{ Hz}$ (B) $\nu = 7.5 \times 10^{14} \text{ Hz}$
(C) $\lambda = 4000 \text{ \AA}$ (D) $\lambda = 9000 \text{ \AA}$

7. If a converging beam of light is incident on a concave mirror, the reflected light

- (A) may form a real image
(B) must form a real image
(C) may form a virtual image
(D) may be a parallel beam

8. Two points P and Q lie on either side of an axis XY as shown. It is desired to produce an image of P at Q

using a spherical mirror, with XY as the optic axis. The mirror must be



- (A) converging
(B) diverging
(C) positioned to the left of P
(D) positioned to the right of Q

9. An object and a screen are fixed at a distance d apart. When a lens of focal length f is moved between the object and the screen, sharp images of the object are formed on the screen for two positions of the lens. The magnifications produced at these two positions are M_1 and M_2 .

- (A) $d > 2f$ (B) $d > 4f$
(C) $M_1 M_2 = 1$ (D) $|M_1| - |M_2| = \frac{d}{f}$

10. Resolving power of an electron microscope is R_e and that of optical microscope is R_o .

- (A) $R_e > R_o$ (B) $R_e < R_o$
(C) $R_e = R_o$ (D) Data Insufficient

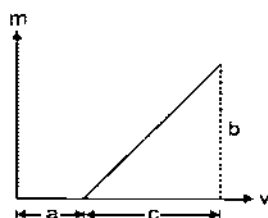
11. In PROBLEM 10, the correct argument for the correct selected option is that

- (A) electrons have greater wavelength than visible light.
(B) electrons have lesser wavelength than visible light.
(C) resolving power is inversely proportional to the wavelength of the wave used for detecting an object by the microscope.
(D) resolving power is inversely proportional to the square of the wavelength of the wave used for detecting an object by the microscope.

12. The distance between two point objects P and Q is 32 cm. A convex lens of focal length 15 cm is placed between them so that the images of both the objects are formed at the same place. The distance of P from the lens could be

- (A) 20 cm (B) 18 cm
(C) 16 cm (D) 12 cm

13. The graph shows the variation of magnification m produced by a convex lens with the image distance v . The focal length of the lens is



(A) $\frac{b}{c}$

(B) $\frac{c}{b}$

(C) a

(D) $\frac{ab}{c}$

14. A lens of focal length f is placed in between an object and screen fixed at a distance D . The lens forms two real images of object on the screen for two of its different positions, a distance x apart. The two real images have magnifications m_1 and m_2 respectively ($m_1 > m_2$).

(A) $f = \frac{x}{m_1 - m_2}$

(B) $m_1 m_2 = 1$

(C) $f = \frac{D^2 - x^2}{4D}$

(D) $D \geq 4f$

15. A planet is observed by an astronomical refracting telescope having an objective of focal length 16 m and an eye piece of focal length 2 cm.

(A) The distance between objective and eye piece is 16.02 m.

(B) The angular magnification of the planet is -800.

(C) The image of the planet is inverted.

(D) The objective is larger than the eye piece.

16. A parallel beam of white light falls on a combination of a concave and a convex lens, both of same material. Their focal lengths are 15 cm and 30 cm respectively for the mean wavelength in white light. On the other side of the lens system, one sees

(A) a coloured pattern with violet at the outer edge.

(B) a coloured pattern with red at the outer edge.

(C) white light again.

(D) that it is unable for the lens to converge the rays at a point.

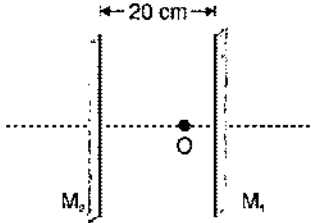
17. Consider a ray of light going from A to B. Let the ray traverse, in going from A to B, distances $s_1, s_2, s_3, \dots, s_m$ in media of indices $n_1, n_2, n_3, \dots, n_m$ respectively.

(A) Total time of flight $t = \frac{1}{c} \sum_{i=1}^m n_i s_i$

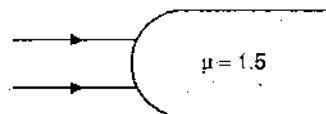
(B) Total time of flight $t = \frac{1}{c} \sum_{i=1}^m s_i$

(C) Optical path length is (O.P.L.) = $\sum_{i=1}^m n_i s_i$

(D) For inhomogeneous media the O.P.L. = $\int_A^B n(s) ds$ and the ray travels along 'Stationary Pathways'.

18. A point object is placed at 30 cm from a convex glass lens ($\mu_g = \frac{3}{2}$) of focal length 20 cm. For the final image of object to be formed at infinity, which of the following is/are correct?
- (A) A concave lens of focal length 60 cm is placed in contact with the convex lens
- (B) A convex lens of focal length 60 cm is placed at a distance of 30 cm from the convex lens.
- (C) The entire convex lens system is immersed in a liquid of refractive index $\frac{4}{3}$
- (D) The entire convex lens system is immersed in a liquid of refractive index $\frac{9}{8}$
19. A converging lens is used to form an image on a screen. When the upper half of the lens is covered by an opaque screen
- (A) half of the image will disappear
- (B) complete image will be formed
- (C) intensity of the image will increase
- (D) intensity of the image will decrease
20. For a mirror, the linear magnification is +2. The conclusion(s) that can be drawn from this information is/are
- (A) The mirror is concave
- (B) The mirror can be convex or concave but it cannot be plane
- (C) The object lies between pole and focus
- (D) The object lies beyond focus
21. A ray of light has speed v_0 , frequency f_0 and wavelength λ_0 in vacuum. When this ray of light enters in a medium of refractive index μ , corresponding values are v , f and λ . Then
- (A) $f = \frac{f_0}{\mu}$ (B) $\lambda = \frac{\lambda_0}{\mu}$
- (C) $v = \frac{v_0}{\mu}$ (D) $f = f_0$
22. For which of the pairs of u and f for curved mirror(s), the image formed is smaller in size.
- (A) $u = -45$ cm, $f = -10$ cm
- (B) $u = -10$ cm, $f = 20$ cm
- (C) $u = -60$ cm, $f = 30$ cm
- (D) $u = -20$ cm, $f = -30$ cm
23. A diverging lens of focal length f_1 is placed in front of and coaxially with a concave mirror of focal length f_2 . Their separation is d . A parallel beam of light incident on the lens returns as a parallel beam from the arrangement. Select the correct statement(s).
- (A) The beam diameters of the incident and reflected beams must be the same.
- (B) $d = 2|f_2| - |f_1|$
- (C) $d = |f_2| - |f_1|$
- (D) If the entire arrangement is immersed in water, the conditions will remain unaltered.
24. An astronomical telescope and a Galilean telescope use identical objective lenses. They have the same magnification, when both are in normal adjustment. The eyepiece of the astronomical telescope has a focal length f . Select the correct statement(s).
- (A) The tube lengths of the two telescopes differ by f .
- (B) The tube lengths of the two telescopes differ by $2f$.
- (C) The Galilean telescope has shorter tube length.
- (D) The Galilean telescope has longer tube length.
25. Two plane mirrors M_1 and M_2 are placed parallel to each other 20 cm apart. A luminous point object 'O' is placed between them at 5 cm from M_1 as shown.
- 
- (A) The distances (in cm) of three nearest images from mirror M_1 are 5, 35 and 45 respectively.
- (B) The distances (in cm) of three nearest images from mirror M_2 are 5, 35 and 45 respectively.
- (C) The distances (in cm) of three nearest images from mirror M_1 are 15, 25 and 55 respectively.
- (D) The distances (in cm) of three nearest images from mirror M_2 are 15, 25 and 55 respectively.
26. In the case of hypermetropia
- (A) the image of a near object is formed behind the retina.
- (B) the image of a distant object is formed in front of the retina.
- (C) a concave lens should be used for correction.
- (D) a convex lens should be used for correction.
27. Which of the following produce a virtual image longer in size than the object?
- (A) Concave lens (B) Convex lens
- (C) Concave mirror (D) Convex mirror

28. A concave mirror has focal length 15 cm. Where should an object be placed in front of the mirror so that the image formed is three times the size of the object?
 (A) 7.5 cm (B) 10 cm
 (C) 17.5 cm (D) 20 cm
29. A concave mirror of focal length f forms an image 2 times the size of object. The object distance from the mirror is
 (A) $\frac{f}{4}$ (B) $\frac{4f}{3}$
 (C) $\frac{3f}{2}$ (D) $\frac{f}{2}$
30. A point object P moves towards a convex mirror with a constant speed V , along its optic axis. The speed of the image
 (A) is always less than V .
 (B) may be less than, equal to or greater than V , depending on the position of P .
 (C) increases as P comes closer to the mirror.
 (D) decrease as P comes closer to the mirror.
31. A bird flies down vertically towards a water surface. To a fish inside the water, vertically below the bird, the bird will appear to
 (A) be closer than its actual distance.
 (B) be farther away than its actual distance.
 (C) move slower than its actual speed.
 (D) move faster than its actual speed.
32. There are three optical media 1, 2 and 3 with their refractive indices $\mu_1 > \mu_2 > \mu_3$. Select the correct statement(s)
 (A) When a ray of light travels from 3 to 1 no TIR will take place.
 (B) Critical angle between 1 and 2 is less than the critical angle between 1 and 3.
 (C) Critical angle between 1 and 2 is more than the critical angle between 1 and 3.
 (D) Chances of TIR are more when ray of light travels from 1 to 3 as compare to the case when it travel from 1 to 2.
33. An equilateral prism has a refractive index $\sqrt{2}$. Select the correct alternative(s).
 (A) Minimum deviation from this prism can be 30°
 (B) Minimum deviation from this prism can be 45°
 (C) At angle of incidence 45° , deviation is minimum
 (D) At angle of incidence 60° , deviation is minimum
34. Parallel rays of light are falling on a convex spherical surface of radius of curvature $R=20$ cm and refractive index $\mu=1.5$ as shown. After refraction from the spherical surface, the parallel rays



- (A) appear to meet after extending the refracted rays backwards.
 (B) actually meet at some point.
 (C) meet (or appear to meet) at a distance of 60 cm from the spherical surface.
 (D) meet (or appear to meet) at a distance of 30 cm from the spherical surface.
35. The focal length of a lens in air and refractive index are f and μ respectively. The focal length changes to f_1 when the lens is immersed in a liquid of refractive index $\frac{\mu}{2}$ and it becomes f_2 when the lens is immersed in a liquid of refractive index 2μ . Then
 (A) $f_1 = -\frac{2(\mu-1)}{f}$ (B) $f_2 = -\frac{2(\mu-1)}{f}$
 (C) $f_1 = \frac{\mu-1}{f}$ (D) $f_2 = \frac{\mu-1}{f}$
36. Two thin lenses, when in contact, produce a combination of power +10 dioptre. When they are 0.25 m apart, the power is reduced to +6 dioptre. The respective powers of the lenses in dioptre, are
 (A) 1 and 9 (B) 2 and 8
 (C) 4 and 6 (D) 5 each
37. A solid, transparent sphere has a small, opaque dot at its centre. When observed from outside, the apparent position of the dot will be
 (A) independent of the refractive index of the sphere.
 (B) closer to the eye than its actual position.
 (C) farther away from the eye than its actual position.
 (D) the same as its actual position.
38. For a concave mirror
 (A) virtual image is always larger in size
 (B) real image is always smaller in size
 (C) real image is always larger in size
 (D) real image may be smaller or larger in size
39. During refraction, ray of light passes undeviated, then
 (A) medium on both sides is same
 (B) angle of incidence is 90°
 (C) angle of incidence is 0°
 (D) medium on other side is rarer
40. A ray of light travelling in a transparent medium falls on a surface separating the medium from air at an angle of incidence 45° . The ray undergoes total internal reflection. If n is the refractive index of the medium

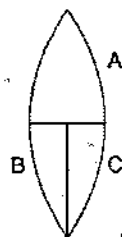
with respect to air, select the possible value(s) of n from the following

- (A) 1.3 (B) 1.4
(C) 1.5 (D) 1.6

41. A convex lens made of glass ($\mu_g = \frac{3}{2}$) has focal length f in air. The image of an object placed in front of it is real, inverted and magnified. Now the whole arrangement is immersed in water ($\mu_w = \frac{4}{3}$) without changing the distance between object and lens, then

- (A) the new focal length becomes $4f$
(B) the new focal length becomes $\frac{f}{4}$
(C) the new image formed will be virtual and magnified.
(D) the new image formed will be real and diminished.

42. A thin, symmetric double-convex lens of power P is cut into three parts A, B and C as shown. The power of



- (A) A is P (B) A is $2P$
(C) B is $\frac{P}{2}$ (D) B is $\frac{P}{4}$

43. A watch glass having uniform thickness and having average radius of curvature of its two surfaces much larger than its thickness is placed in the path of a beam of parallel light. The beam will

- (A) be completely unaffected.
(B) converge slightly.
(C) diverge slightly.
(D) converge or diverge slightly depending on whether the beam is incident from the concave or the convex side.

44. A converging lens of focal length f_1 is placed in front of and coaxially with a convex mirror of focal length f_2 . Their separation is d . A parallel beam of light incident on the lens returns as a parallel beam from the arrangement. Select the correct statement(s).

- (A) The beam diameters of the incident and reflected beams must be the same.
(B) $d = f_1 - 2|f_2|$
(C) $d = f_1 - |f_2|$

- (D) If the entire arrangement is immersed in water, the conditions will remain unaltered.

45. Check the wrong statement(s)

- (A) A concave mirror can give a virtual image.
(B) A concave mirror can give a diminished virtual image.
(C) A convex mirror can give a real image.
(D) A convex mirror can give a diminished virtual image.

46. When lights of different colours move through water, they must have different

- (A) wavelengths (B) frequencies
(C) velocities (D) amplitudes

47. A thin concavo-convex lens has two surfaces of radii of curvature R and $2R$. The material of the lens has a refractive index μ . When kept in air, the focal length of the lens

- (A) will depend on the direction from which light is incident on it.
(B) will be the same, irrespective of the direction from which light is incident on it.
(C) will be equal to $\frac{2R}{\mu - 1}$
(D) will be equal to $\frac{R}{\mu - 1}$

48. A convex mirror is used to form an image of a real object. The image

- (A) always lies between the pole and the focus.
(B) is diminished in size.
(C) is erect.
(D) is real.

49. Which of the following form(s) the virtual and erect image for all positions of object?

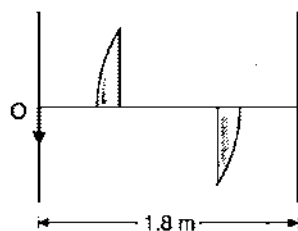
- (A) concave mirror (B) convex lens
(C) convex mirror (D) concave lens

50. A ray of light is incident on a prism of refracting angle A . C is the critical angle for the material of the prism with respect to the surrounding material (say air/vacuum).

- (A) An emergent ray will be there for all values of C .
(B) An emergent ray will be there only for $A < 2C$.
(C) A ray incident at an angle i can pass through the prism if $\sin i > \frac{\sin(A - C)}{\sin C}$ for $C < A < 2C$.
(D) None of above is correct.

51. A thin plane-convex lens of focal length f is split into two equal halves. One of the halves is shifted along the optical axis as shown. The separation between object and image planes is 1.8 m and the magnification of

image formed by one of the half lens is 2. The separation between two halves is d .



- (A) $f = 0.4$ m (B) $f = 0.6$ m
(C) $d = 0.6$ m (D) $d = 0.4$ m

52. A point source of light is placed at a distance h below the surface of a large and deep lake. If f is the fraction of light energy that escapes directly from water surface and μ is refractive index of water then,
(A) f varies as a function of h
(B) f is independent of value of h

(C) $f = \frac{1}{2\sqrt{\mu^2 - 1}}$

(D) $f = \frac{1}{2} \left[1 - \sqrt{1 - \frac{1}{\mu^2}} \right]$

53. An object is placed at a distance $2f$ from the pole of a curved mirror of focal length f .
(A) The linear magnification is 1 for both types of curved mirror.
(B) The linear magnification is 1 for a concave mirror.
(C) The linear magnification is $\frac{1}{3}$ for a convex mirror.
(D) Data Insufficient.
54. If a convergent beam of light passes through a diverging lens, the result
(A) may be a convergent beam.
(B) may be a divergent beam.
(C) may be a parallel beam.
(D) must be a parallel beam.

Practice Exercise

Reasoning Based Questions

This section contains Reasoning type questions, each having four choices (A), (B), (C) and (D) out of which ONLY ONE is correct. Each question contains STATEMENT 1 and STATEMENT 2. You have to mark your answer as

Bubble (A) If both statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1.

Bubble (B) If both statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1.

Bubble (C) If STATEMENT 1 is TRUE and STATEMENT 2 is FALSE.

Bubble (D) If STATEMENT 1 is FALSE but STATEMENT 2 is TRUE.

1. *Statement-1:*

A parallel beam of light traveling in air can be displaced laterally by a parallel transparent slab by distance more than the thickness of the plate.

Statement-2:

The lateral displacement of light traveling in air increases with rise in value of refractive index of slab.

2. *Statement-1:*

Even in absolutely clear water, a diver cannot see very clearly.

Statement-2:

Velocity of light is reduced in water.

3. *Statement-1:*

Spherical aberration of a lens can be reduced by blocking the central portion or peripheral portion of the lens.

Statement-2:

Spherical aberration arises on account of inability of the lens to focus central and peripheral rays at the same point.

4. *Statement-1:*

For total internal reflection, angle of incident in denser medium must be greater than critical angle for the pair of media in contact.

Statement-2:

$\mu = \frac{1}{\sin C}$, where the symbols have their standard meaning.

5. *Statement-1:*

The images formed due to total internal reflections are much brighter than those formed by mirrors or lenses.

Statement-2:

There is no loss of intensity in total internal reflection.

6. *Statement-1:*

A bird in air is diving vertically with speed v_0 over a tank filled with water and having flat silvered bottom serving as plane mirror, it observes velocity of its image in silvered bottom of tank as $2v_0$ upward relative to itself.

Statement-2:

Bird and its image in bottom mirror are always equidistant from bottom mirror.

7. *Statement-1:*

We cannot produce a real image by plane or convex mirrors under any circumstances.

Statement-2:

The focal length of a convex mirror is always taken as positive.

8. *Statement-1:*

If a light ray is incident on any one of the two mirrors inclined at 90° with each other, then finally the emergent ray is antiparallel with incident ray.

Statement-2:

Finally, the reflected and initially incident rays are in same phase when successively reflected from two perpendicularly inclined mirrors.

9. **Statement-1:**
The formula connecting u , v and f for a spherical mirror is valid only for mirrors whose sizes are very small compared to their radii of curvature.
Statement-2:
Laws of reflection are strictly valid for plane surfaces, but not for large spherical surfaces.
10. **Statement-1:**
The focal length of the mirror is f and distance of the object from the focus is u , the magnification of the mirror is $\frac{f}{u}$.
Statement-2:
Magnification = $\frac{\text{size of image}}{\text{size of object}}$.
11. **Statement-1:**
Although the surfaces of the lens used in goggles are curved, it does not have any power.
Statement-2:
In case of goggles, the lenses are concavo-convex and both the surface of lens have equal radii of curvature.
12. **Statement-1:**
A convex lens behaves as a concave lens when placed in a medium of refractive index greater than the refractive index of its material.
Statement-2:
Light in that case will travel through the convex lens from denser to rarer medium. It will bend away from normal, i.e., the convex lens would diverge the rays and behave as concave.
13. **Statement-1:**
The minimum distance between an object and its real image formed by a convex lens is $2f$.
Statement-2:
The distance between an object and its real image is minimum when its magnification is one.
14. **Statement-1:**
The lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ indicates that focal length of a lens depends on distances of object and image from the lens.
Statement-2:
The formula does indicate but when u is changed v also changes, so that f of a particle lens remains constant.
15. **Statement-1:**
For observing traffic at our back, we prefer to use a convex mirror.
Statement-2:
A convex mirror has a much larger field of view than a plane mirror or a concave mirror.
16. **Statement-1:**
When a ray of light enters glass from air, its frequency decreases.
Statement-2:
The velocity of light in glass is less than that in air.
17. **Statement-1:**
A ray incident along normal to the mirror retraces its path.
Statement-2:
In reflection, angle of incidence is always equal to angle of reflection.
18. **Statement-1:**
A concave mirror of focal length in air is used in a medium of refractive index 2. Then the focal length of mirror in medium becomes double.
Statement-2:
The radius of curvature of a mirror is double of the focal length.
19. **Statement-1:**
Light from an object falls on a concave mirror forming a real image of the object. If both the object and mirror are immersed in water, there is no change in position of the image.
Statement-2:
The formation of image by reflection does not depend on surrounding medium, so there is no change in position of image.
20. **Statement-1:**
A convex lens can be convergent in one medium and divergent in other medium.
Statement-2:
In denser medium, convex lens is convergent and in rarer medium, convex lens is divergent.
21. **Statement-1:**
For a prism of refracting angle 60° and refractive index $\sqrt{2}$ minimum deviation is 30° .
Statement-2:
At minimum deviation, $r_1 = r_2 = \frac{A}{2} = 30^\circ$.

22. Statement-1:

There exist two angles of incidence for the same magnitude of deviation (except minimum deviation) by a prism kept in air.

Statement-2:

For a prism kept in air, a ray is incident on first surface and emerges out of second surface (of prism) along the previous emergent ray, then this ray emerges out of first surface along the previous incident ray. This principle is called the Principle of Reversibility of Light.

23. Statement-1:

A plane convex lens is silvered from plane surface. It can act as a diverging mirror.

Statement-2:

Focal length of concave mirror is independent of medium.

24. Statement-1:

Maximum distance of image formed by convex mirror from pole of mirror equals ' f ' for all the objects (real/virtual).

Statement-2:

Convex mirrors forms virtual images for objects placed in front of mirror.

25. Statement-1:

We cannot produce a real image by plane or convex mirror under any circumstances.

Statement-2:

Reflection Law is valid for plane mirror as well as convex mirror.

26. Statement-1:

There is no dispersion of light refracted through a rectangular glass slab.

Statement-2:

Dispersion of light is the phenomenon of splitting of a beam of white light into its constituent colours.

27. Statement-1:

Convex mirror always form a virtual image.

Statement-2:

Focal length of a mirror is half of the radius of curvature.

28. Statement-1:

A fish inside a pond will see a person standing outside taller than he is actually.

Statement-2:

Light rays from person converges into eyes of fish on entering water from air.

29. Statement-1:

Optical fibre has thin glass core coated by glass of small refractive index and is used to send light signals.

Statement-2:

All the rays of light entering the fibre are totally reflected even at very small angles of incidence.

30. Statement-1:

The mirror used in search light are parabolic and not concave spherical.

Statement-2:

In a concave spherical mirror the image formed is always virtual.

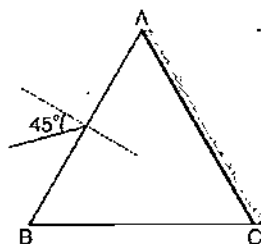
Practice Exercise

Linked Comprehension Type Questions

This section contains Linked Comprehension Type Questions or Paragraph based Questions. Each set consists of a Paragraph followed by questions. Each question has four choices (A), (B), (C) and (D), out of which only one is correct. (For the sake of competitiveness there may be a few questions that may have more than one correct options)

Comprehension 1

A ray of light is incident at 45° on the face AB of an equilateral prism ABC which has the face AC silvered. Based on the information provided answer the following questions.



- The refractive index μ of the material of the prism so that when the ray falls on face BC (after reflecting from AC) it makes an angle 60° with it is
 (A) $\sqrt{3}$ (B) $\sqrt{2}$
 (C) 2 (D) 1.5
- The total deviation, when the ray of light finally emerges from BC is
 (A) 120° (B) 180°
 (C) 150° (D) 90°

Comprehension 2

A convex lens of focal length 20 cm and a concave lens of focal length 10 cm are placed 20 cm apart. In between them an object is placed at distance x from the convex lens. Based on the information provided answer the following questions.

- The value of x (in cm) so that images formed by both the lenses coincide is

- (A) $20(\sqrt{3}-1)$ (B) $\frac{20\sqrt{3}-1}{\sqrt{3}}$
 (C) $\frac{20}{\sqrt{3}}$ (D) $10(\sqrt{3}-1)$
- The linear magnification produced by convex lens and concave lens individually is
 (A) $(\sqrt{3}+1)$ and $\frac{1}{\sqrt{3}}$
 (B) $\sqrt{3}$ and $\frac{1}{\sqrt{3}}$
 (C) $(\sqrt{3}+1)$ and $(\sqrt{3}-1)$
 (D) $\sqrt{3}$ and $(2\sqrt{3}-3)$

Comprehension 3

A telescope is an optical instrument used to increase the visual angle of distant objects such as stars, planets etc. An astronomical telescope consists of two converging lenses. The one facing the object is called objective and the lens close to the eye is called an eyepiece. It can be adjusted by displacing relative to the objective. The angular magnification is defined as the ratio of focal length of objective and eyepiece. One can see the image with unstrained eye if it forms at infinity. An astronomical telescope has an objective of focal length 50 cm and a magnification of 20. Based on above information, answer the following questions.

- Focal length of the eyepiece is
 (A) 2.5 cm (B) 5 cm
 (C) 7.5 cm (D) None of these
- To view remote object by an unstrained normal eye, separation between two lenses will be

- (A) 55 cm (B) 57.5 cm
(C) 60 cm (D) 52.5 cm

7. If object is at a distance 600 m from the telescope to see the image with unstrained eye separation between two lenses should be (in cm)

- (A) 46.65 (B) 47.65
(C) 49.96 (D) 49.65

Comprehension 4

Speed of light in a medium of refractive index n is given by $\frac{c}{n}$ where c is speed of light in vacuum refractive index of a medium depends on wavelength (λ). As wavelength increases refractive index decreases. It is also given

$$\lambda_{\text{red}} > \lambda_{\text{orange}} > \lambda_{\text{yellow}}$$

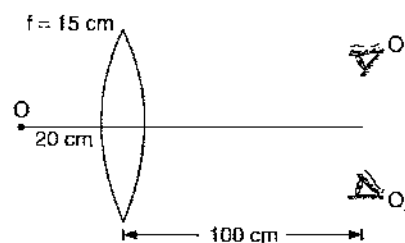
Based on above information, answer the following questions.

8. In glass
(A) orange light travels faster than yellow light
(B) yellow light travels faster than orange light
(C) yellow light travels faster than red light
(D) orange light travels faster than red light
9. The quantity that remains unchanged if light enters from water to glass is
(A) Wavelength and colour
(B) Refractive index and frequency
(C) Frequency and velocity
(D) Colour and frequency
10. The phenomenon that happens because of variation of wavelength is
(A) Aberration
(B) Dispersion
(C) Total internal reflection
(D) Bending of light
11. Which of the following statement is true?
(A) Time taken (t) by yellow light to travel distance x_0 in refractive index n can be $t \leq \frac{nx_0}{c_0}$
(B) Time taken (t) by yellow light to travel distance x_0 in refractive index n can be $t \geq \frac{nx_0}{c}$
(C) Since wavelength of yellow light increases in refractive index n its frequency must decreases.
(D) None of the above
12. Which of the following statement is false?
(A) Light is a electromagnetic wave
(B) Speed of light of each colour is same in vacuum
(C) Time to cover distance x_0 in a medium is same for each colour

- (D) As frequency of light increases then refractive index of glass increases

Comprehension 5

The figure shows a convex lens of focal length 15 cm. A point object is placed on the principle axis of lens at a distance 20 cm from it as shown. On the other side of the lens two observer eyes O_1 and O_2 are situated at a distance 100 cm from the lens at some distance above and below the principle axis.



Now half position of lens below principle axis is painted black. Based on above information, answer the following questions.

13. In initial setup (before painting the lens) which of the following statement is correct.
(A) Observer O_1 will see a real image at 60 cm from the lens but observer O_2 will not be able to see it
(B) Observer O_2 will see a real image at 60 cm from the lens but observer O_1 will not be able to see it
(C) Both the observers will see a real image at 60 cm from lens irrespective the positions of O_1 and O_2
(D) Both the observers may or may not be able to see the image at 60 cm from lens depending on the positions of O_1 and O_2
14. After painting the lens, which of the following observer will not be able to see the image of object, if before this activity both were seeing the image
(A) O_1 (B) O_2
(C) Both O_1 and O_2 (D) Neither O_1 nor O_2
15. After painting the lens, for which observer the intensity of image will be reduced to half?
(A) For O_1
(B) For O_2
(C) Both for O_1 and O_2
(D) Neither for O_1 nor for O_2

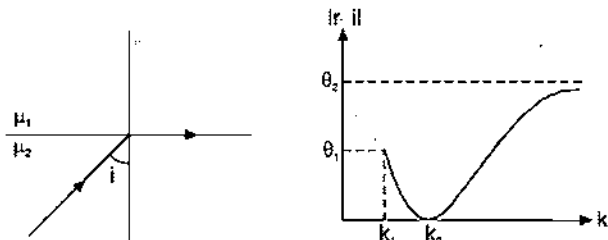
Comprehension 6

The figure shows a ray incident at an angle $i = \frac{\pi}{3}$. If the plot drawn shows the variation of $|r-i|$ versus $\frac{\mu_1}{\mu_2} = k$,

Advanced JEE Physics

Optics & Modern Physics

where r is the angle of refraction, then based on above information, answer the following questions.



16. The value of k_1 is

- (A) $\frac{2}{\sqrt{3}}$ (B) 1
(C) $\frac{1}{\sqrt{3}}$ (D) $\frac{\sqrt{3}}{2}$

17. The value of θ_1 is

- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$
(C) $\frac{\pi}{6}$ (D) ZERO

18. The value of k_2 is

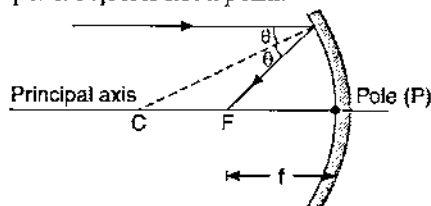
- (A) 1 (B) 2
(C) $\frac{1}{2}$ (D) None of these

Comprehension 7

Spherical aberration in spherical mirrors is a defect which is due to dependence of focal length f on angle of incidence θ as shown in figure is given by

$$f = R - \frac{R}{2} \sec \theta$$

where R is radius of curvature of mirror and θ is the angle of incidence. The rays which are closed to principal axis are called paraxial rays and the rays far away from principal axis are called marginal rays. As a result of above dependence different rays are brought to focus at different points and the image of a point object is not a point.



Based on above information, answer the following questions.

19. If f_p and f_m represent the focal length of paraxial and marginal rays respectively, then correct relationship is

- (A) $f_p = f_m$ (B) $f_p > f_m$

- (C) $f_p < f_m$ (D) None of these

20. If angle of incidence is 60° , then focal length of this marginal ray is

- (A) R (B) $\frac{R}{2}$
(C) $2R$ (D) 0

21. The total deviation suffered by the ray falling on mirror at an angle of incidence equal to 60° is

- (A) 60° (B) 90°
(C) 30° (D) Cannot be determined

Comprehension 8

The XY plane is the boundary between two transparent media. Medium-I with $z \geq 0$ has a refractive index $\sqrt{2}$ and medium-II with $z \leq 0$ has a refractive index $\sqrt{3}$. A ray of light in medium-I given by $\vec{A} = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. Based on the above facts, answer the following questions.

22. The vector representing the incident ray has a magnitude of

- (A) 5 units (B) 10 units
(C) 15 units (D) 20 units

23. The angle of incidence is

- (A) 30° (B) 45°
(C) 60° (D) 90°

24. The angle of refraction is

- (A) 30° (B) 45°
(C) 60° (D) 90°

25. The refracted ray is represented by the vector given by

- (A) $6\sqrt{3}\hat{i} - 8\sqrt{3}\hat{j} + 10\sqrt{3}\hat{k}$
(B) $8\sqrt{3}\hat{j}$
(C) $-10\sqrt{3}\hat{k}$
(D) $6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\sqrt{3}\hat{k}$

26. The vector representing the refracted ray has a magnitude of

- (A) $\sqrt{6}$ units (B) 10 units
(C) $10\sqrt{6}$ units (D) $20\sqrt{6}$ units

27. The unit vector along refracted ray is

- (A) $\frac{3}{5\sqrt{2}}\hat{i} + \frac{4}{5\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$
(B) $3\hat{i} + 4\hat{j} - 5\hat{k}$
(C) $\frac{3}{5\sqrt{2}}\hat{i} - \frac{4}{5\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$
(D) $2\hat{i} - 3\hat{k}$

Comprehension 9

The lens governing the behaviour of the rays namely rectilinear propagation, laws of reflection and refraction can be summarised in one fundamental law known as Fermat's Principle. According to this principle a ray of light travels from one point to another such that the time taken is at a stationary value (maximum or minimum). If c is the velocity of light in a vacuum, the velocity in a medium of refractive index n is $\frac{c}{n}$, hence time taken to travel a

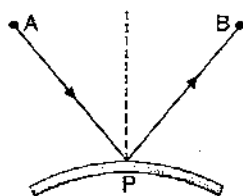
distance l is $\frac{nl}{c}$. If the light passes through a number of

media, the total time taken is $\left(\frac{1}{c}\right)\sum nl$ or $\frac{1}{c}\int n dl$ if refractive index varies continuously. Now, $\sum nl$ is the total optical path, so that Fermat's Principle states then the path of a ray is such that the optical path is at a stationary value. This principle is obviously in agreement with the fact that the ray are straight lines in a homogeneous isotropic medium. It is found that it also agrees with the classical laws of reflection and refraction. Based on above information, answer the following questions.

28. If refractive index of a slab varies as $m = 1 + x^2$ where x is measured from one end, then optical path length of a slab of thickness 1 m is

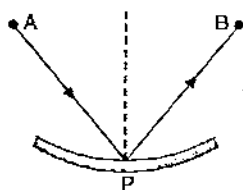
- (A) $\frac{4}{3}$ m (B) $\frac{3}{4}$ m
(C) 1 m (D) None of these

29. The optical path length followed by ray from point A to B, given that laws of reflection are obeyed as shown in figure is



- (A) Maximum (B) Minimum
(C) Constant (D) None of these

30. The optical path length followed by ray from point A to B, given that laws of reflection are obeyed as shown in figure is



- (A) Maximum (B) Minimum
(C) Constant (D) None of these

Comprehension 10

Consider an equiconvex lens of radius R , made of a material of refractive index μ . Its focal length is f_1 when any one face is silvered. Now consider another plano-convex lens of radius R , made of same material having focal length, f_2 when no face is silvered, f_3 when plane face is silvered and f_4 when curved surface is silvered. Based on above information, answer the following questions.

31. f_1 equals

- (A) $\frac{R}{(2\mu-1)}$ (B) $\frac{2R}{(2\mu-1)}$
(C) $\frac{R}{2(2\mu-1)}$ (D) $\frac{2R}{2(2\mu+1)}$

32. f_2 equals

- (A) R (B) $\frac{R}{2\mu-1}$
(C) $\frac{R}{\mu-1}$ (D) $\frac{R}{\mu+1}$

33. f_3 equals

- (A) $\frac{R}{2\mu}$ (B) $\frac{R}{2(\mu-1)}$
(C) $\frac{2R}{\mu}$ (D) $2R\mu$

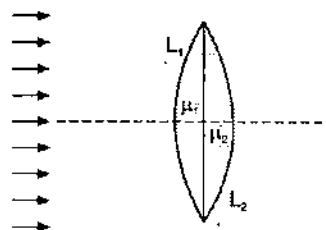
34. f_4 equals

- (A) $\frac{R}{\mu}$ (B) $\frac{2R}{\mu}$
(C) $2R\mu$ (D) $\frac{R}{2\mu}$

Comprehension 11

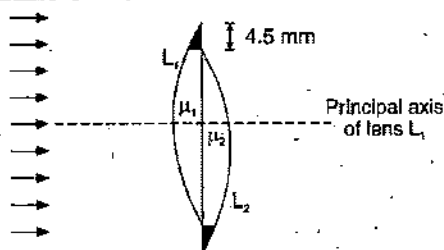
SITUATION-I

Two identical plano-convex lenses L_1 and L_2 having radii of curvature $R = 20$ cm and refractive indices $\mu_1 = 1.4$ and $\mu_2 = 1.5$ are placed as shown in the figure.



SITUATION-II

Now, the second plano-convex lens is shifted vertically downward by a small distance of 4.5 mm and the extended parts of L_1 and L_2 are blackened as shown in figure.



Based on above information, answer the following questions.

35. In SITUATION-I, the position of the image of the parallel beam of light relative to the common principal axis is

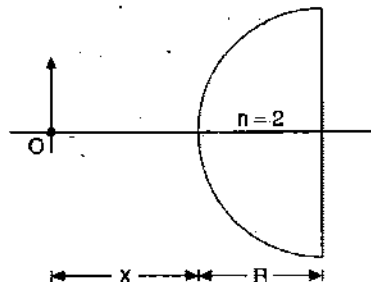
- (A) $\frac{100}{9}$ cm (B) $\frac{100}{3}$ cm
(C) $\frac{200}{3}$ cm (D) $\frac{200}{9}$ cm

36. In SITUATION-II, the new position of the image of the parallel beam is

- (A) $\frac{200}{9}$ cm in front of the lens 2 mm below the principal axis of L_1 .
(B) $\frac{100}{9}$ cm behind the lens 2 mm below the principal axis of L_1 .
(C) $\frac{200}{9}$ cm behind the lens 2.5 mm below the principal axis of L_1 .
(D) $\frac{200}{9}$ cm in front of the lens 2.5 mm below the principal axis of L_1 .

Comprehension 12

A small object O is placed in air at the principal axis at a distance x from the pole of the curved surface of a transparent hemisphere having refractive index 2 and radius R as shown. Based on above information, answer the following questions.



37. The value of x , for which the final image of the object at O will be virtual is

- (A) $2R$ (B) $3R$
(C) $\frac{R}{3}$ (D) $1.5R$

38. The nature of final image of the object when $x = 2R$ is

- (A) Erect and magnified
(B) Inverted and magnified
(C) Erect and same size
(D) Inverted and same size

39. It is observed that for $x = R$, a ray starting from O strikes the spherical surface at grazing incidence. The angle with the normal at which the ray emerges from the plane surface is

- (A) 90° (B) 0°
(C) 30° (D) 60°

Comprehension 13

The refractive indices of the crown glass for blue and red lights are 1.51 and 1.49 respectively and those of the flint glass are 1.77 and 1.73 respectively. An isosceles prism of angle 6° is made of crown glass. A beam of white light is incident at a small angle on this prism. The other flint glass isosceles prism is combined with the crown glass prism such that there is no deviation of the incident light. Based on the above facts, answer the following questions.

40. The refractive index of crown glass for yellow colour is

- (A) 1.51 (B) 1.49
(C) 1.50 (D) 1.59

41. The refractive index of flint glass for yellow colour is

- (A) 1.70 (B) 1.72
(C) 1.73 (D) 1.75

42. The refracting angle of flint glass prism is

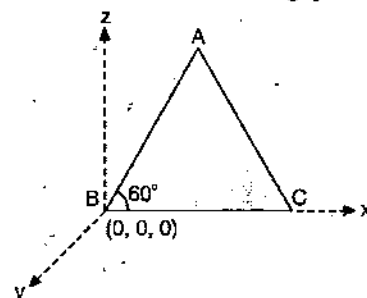
- (A) $+2^\circ$ (B) $+4^\circ$
(C) -2° (D) -4°

43. The net dispersion produced by the combined system is

- (A) 0.02° (B) -0.02°
(C) $+0.04^\circ$ (D) -0.04°

Comprehension 14

An equilateral prism ABC is placed in air with its base side BC lying horizontally along x -axis as shown in figure. A ray of light represented by equation $\sqrt{3}z + x = 10$ is incident at a point P on the face AB of prism. Based on above information, answer the following questions.



44. The value of μ for which the ray grazes the face AC is

- (A) $\frac{3}{2}$ (B) $\frac{4}{3}$
(C) $\frac{2}{\sqrt{3}}$ (D) $\frac{\sqrt{5}}{2}$

45. The direction of the finally refracted ray for $\mu = \frac{3}{2}$ is

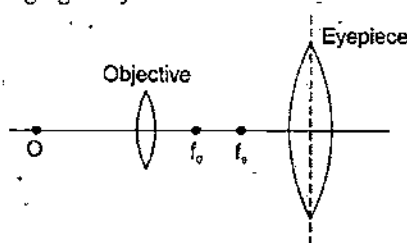
- (A) parallel to x-axis (B) parallel to z-axis
(C) parallel to y-axis (D) parallel to face AB

46. The equation of ray emerging out of prism, if the bottom BC is silvered is

- (A) $z + \sqrt{3}x = 10$ (B) $\sqrt{3}z + x = 10$
(C) $z + \sqrt{3}x = 20$ (D) $x + z = 10\sqrt{3}$

Comprehension 15

The schematic diagram of a compound microscope is shown in the adjacent figure. Its main components are two convex lenses: one acts as the main magnifying lens and is referred to as the objective, and another lens called the eyepiece. The two lenses act independently of each other when bending light rays.



f_o = focal point of objective

f_e = focal point of eyepiece

Light from the object (O) first passes through the objective and enlarged, inverted first image is formed. The eyepiece then magnifies this image. Usually the magnification of the eyepiece is fixed (either $\times 10$ or $\times 15$) and three rotating objective lenses are used: $\times 10$, $\times 40$ and $\times 60$. Angular magnification is defined as the angle subtended by the final image at the eye to the angle subtended by the object placed at least distance of distinct vision (≈ 25 cm) when viewed by the naked eye. Based on above information, answer the following questions.

47. The type of image that would have to be produced by the objective is

- (A) Either virtual or real
(B) Virtual
(C) Real
(D) It depends on the focal length of the lens.

48. Where would the first image have to be produced by the objective relative to the eyepiece such that a second, enlarged image would be generated on the same side of

the eyepiece as the first image? Assume that the first image distance is d_i from the eyepiece.

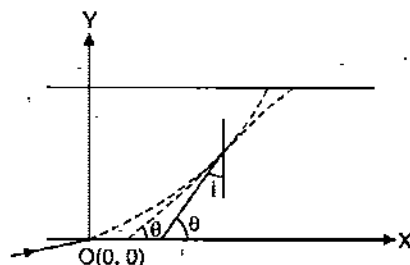
- (A) $d_i < f_e$ (B) $d_i = f_e$
(C) $f_e < d_i < 2f_e$ (D) $d_i > 2f_e$

49. Two compound microscopes A and B were compared. Both had objectives and eyepieces with the same magnification but A gave an overall magnification that was greater than that of B. Which of the following is a possible explanation?

- (A) The distance between object and eyepiece in A is greater than the corresponding distance in B.
(B) The distance between object and eyepiece in A is less than the corresponding distance in B.
(C) The eyepiece and objective positions were reversed in A.
(D) The eyepiece and objective positions were reversed in B.

Comprehension 16

A ray of light travelling in air is incident at grazing angle on a long rectangular slab of a transparent medium of thickness $t = 1.0$ m. The point of incidence O is the origin (0, 0). The medium has a variable index of refraction $\mu(y)$ given by $\mu(y) = [Ky^{3/2} + 1]^{1/2}$ where $K = 1.0(m)^{-3/2}$. The refractive index of air is 1.0.



Based on the above facts, answer the following questions.

50. The relation between the slope of the trajectory of the ray at the point B(x, y) in the medium and the angle of incidence (i) at that point is given by

- (A) $\tan \theta = \sin i$ (B) $\tan \theta = 2 \sin i$
(C) $\tan \theta = \cot i$ (D) $\tan \theta = 2 \cot i$

51. The equation for the trajectory $y(x)$ of the ray in the medium is

- (A) $y = \frac{x^2}{16}$ (B) $y = \frac{x^3}{16}$
(C) $y = \frac{x^4}{16}$ (D) $y = \frac{x^4}{256}$

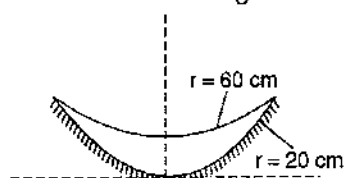
52. The co-ordinates (x, y) of the point where the ray intersects the upper surface of the slab-air boundary are

- (A) (1, 1) m (B) (2, 1) m
(C) (3, 1) m (D) (4, 1) m

53. The ray finally emerges
(A) parallel to the incident ray
(B) perpendicular to the incident ray
(C) at an angle of 30° to the incident ray
(D) at an angle of 45° to the incident ray

Comprehension 17

The convex surface of a thin concavo-convex lens of glass of refractive index 1.5 has a radius of curvature of 20 cm. The concave surface has a radius of curvature of 60 cm. The convex side is silvered and placed on a horizontal surface as shown in the figure.

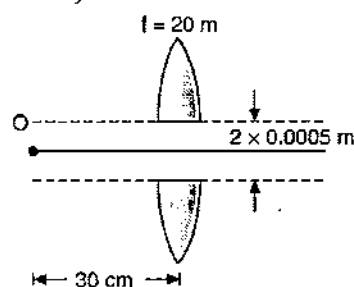


Based on above information, answer the following questions.

54. The focal length of the combination has the magnitude
(A) 1.5 cm (B) 15 cm
(C) 7.5 cm (D) 8.6 cm
55. The combination behaves like
(A) a convex mirror (B) a concave mirror
(C) a convex lens (D) a concave lens
56. A small object is placed on the principal axis of the combination, at a distance of 30 cm in front of the mirror. The magnification of the image is
(A) $-\frac{1}{3}$ (B) $\frac{3}{4}$
(C) 5 (D) $-\frac{1}{4}$

Comprehension 18

A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves each of which is displaced by 0.0005 m as shown in the figure.



Based on above information, answer the following questions.

57. The position at which the image is formed is
(A) 30 cm, right of lens (B) 40 cm, left of lens
(C) 60 cm, right of lens (D) 70 cm, left of lens
58. The total number of images generated by the arrangement is/are
(A) 1 (B) 2
(C) 4 (D) 6
59. The spacing between the images so formed is
(A) 0.1 cm (B) 0.3 cm
(C) 0.5 cm (D) 1 cm

Comprehension 19

A thin biconvex lens of refractive index $\frac{3}{2}$ is placed on a horizontal plane mirror as shown in figure.

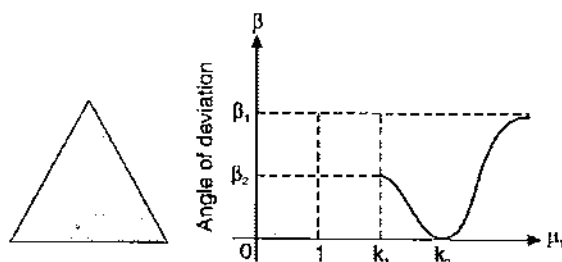


The space between the lens and the mirror is then filled with water of refractive index $\frac{4}{3}$. It is found that when a point object is placed 15 cm above the lens on its principal axis, the object coincides with its own image. On repeating with another liquid, the object and the image again coincide at a distance 25 cm from the lens. Based on the above facts, answer the following questions.

60. The radius of curvature of both the surfaces of the convex lens is R , then the focal length of the convex lens is
(A) R (B) $\frac{R}{2}$
(C) $\frac{R}{4}$ (D) $\frac{R}{8}$
61. When the space between the lens and mirror is filled with water, the focal length of water concave lens is
(A) R (B) $2R$
(C) $3R$ (D) $4R$
62. The radius of curvature R of each surface of convex lens is
(A) 2 cm (B) 5 cm
(C) 10 cm (D) 15 cm
63. The focal length of the liquid concave lens is
(A) $\frac{20}{3}$ cm (B) $\frac{40}{3}$ cm
(C) $\frac{50}{3}$ cm (D) $\frac{70}{3}$ cm
64. The refractive index of the liquid is
(A) 1.1 (B) 1.2
(C) 1.4 (D) 1.6

Comprehension 20

The diagram shows an equilateral prism. The medium on one side of the prism has refractive index μ_1 . The refractive index of the prism is $\mu = \frac{4}{\sqrt{3}}$. The diagram shows variation of magnitude of angle of deviation with respect to μ_1 . Consider the light ray to be incident normally on the first face.



Based on above information, answer the following questions:

65. Value of k_2 is

- (A) $\frac{6}{\sqrt{3}}$ (B) $\frac{4}{\sqrt{3}}$
(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{8}{\sqrt{3}}$

66. Value of k_1 is

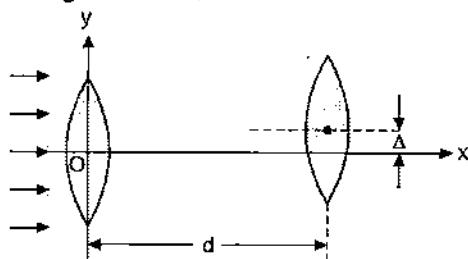
- (A) $\frac{4}{3}$ (B) $\frac{5}{3}$
(C) $\frac{8}{3}$ (D) $\frac{10}{3}$

67. Value of $\beta_1 - \beta_2$ is

- (A) 20° (B) 30°
(C) 60° (D) 90°

Comprehension 21

Two thin convex lenses of focal lengths f_1 and f_2 are separated by a horizontal distance d ($d < f_1$ and $d < f_2$) and their centers are displaced by a vertical separation as shown in the figure. Take the origin of coordinates O at the center of first lens. For a parallel beam of light coming from the left, as shown in figure.



Based on above information, answer the following questions.

68. The x -coordinate of the focal point of this lens system is

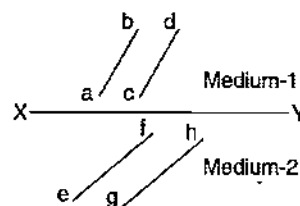
- (A) $\frac{d(f_1 - d) + f_1 f_2}{f_1 + f_2 - d}$ (B) $\frac{f_1 f_2}{f_1 + f_2 - d}$
(C) $\frac{d(f_1 - d)}{f_1 + f_2 - d}$ (D) $\frac{2d(f_1 + d) - f_1 f_2}{f_1 + f_2 - d}$

69. The y -coordinate of the focal point of this lens system is

- (A) $\frac{(f_1 + d)\Delta}{(f_1 + f_2 - d)}$ (B) $\frac{2(f_1 + d)}{(f_1 + f_2 - d)}$
(C) $\frac{2(f_1 - d)\Delta}{(f_1 + f_2 - d)}$ (D) $\frac{(f_1 - d)\Delta}{(f_1 + f_2 - d)}$

Comprehension 22

The figure shows a surface XY separating two transparent media, Medium-1 and Medium-2. The lines ab and cd represent wavefronts of a light wave travelling in Medium-1 and incident on XY . The lines ef and gh represent wavefronts of the light wave in Medium-2 after refraction.



Based on the above facts, answer the following questions.

70. Light travels as a

- (A) parallel beam in each medium
(B) convergent beam in each medium
(C) divergent beam in each medium
(D) divergent beam in one medium and convergent beam in the other medium

71. The phases of the light wave at c , d , e and f are ϕ_c , ϕ_d , ϕ_e and ϕ_f respectively. It is given that $\phi_c \neq \phi_f$

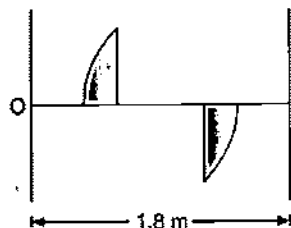
- (A) ϕ_c cannot be equal to ϕ_d
(B) ϕ_d can be equal to ϕ_e
(C) $(\phi_d - \phi_f)$ is equal to $(\phi_e - \phi_c)$
(D) $(\phi_d - \phi_c)$ is not equal to $(\phi_f - \phi_e)$

72. Speed of light is:

- (A) the same in medium-1 and medium-2
(B) larger in medium-1 than in medium-2
(C) larger in medium-2 than in medium-1
(D) different at b and d

Comprehension 23

A thin plano-convex lens of focal length f is split into two halves. One of the halves is shifted along the optical axis. The separation between the object and image planes is 1.8 m. The magnification of the image formed by one of the half lenses is 2.

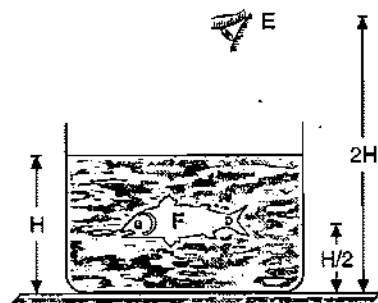


Based on above information, answer the following questions.

73. The focal length of the lens used is,
 (A) 0.4 m (B) 0.6 m
 (C) 1 m (D) 2 m
74. The separation between the two halves of the thin plano-convex lens is
 (A) 0.2 m (B) 0.4 m
 (C) 0.6 m (D) 0.8 m
75. The magnification for the second half lens is
 (A) 0.5 (B) -0.5
 (C) 0.4 (D) -2

Comprehension 24

Consider a beaker filled with water (of refractive index μ) to a height H . A fish F is at a height $\frac{H}{2}$ from the transparent base of the beaker which lies on a surface that happens to be a mirror. An observer whose eye E is at a height $2H$ from the base of beaker is also there. Based on above information, answer the following questions.



76. The distance from itself at which the fish will see the image of the eye by direct observation is
 (A) $H\left(\mu + \frac{1}{2}\right)$ (B) $\frac{H}{2}\left(1 + \frac{\mu}{2}\right)$
 (C) $H\left(\frac{\mu}{2} + 1\right)$ (D) $\frac{H}{2}\left(\mu + \frac{1}{2}\right)$

77. The distance from itself at which the fish sees the image of eye by viewing in the mirror is
 (A) $H\left(\mu + \frac{1}{2}\right)$ (B) $H\left(\mu + \frac{3}{2}\right)$
 (C) $2H\left(\mu + \frac{1}{2}\right)$ (D) $2H\left(\mu + \frac{3}{2}\right)$

78. The distance from itself at which the eye sees the image of the fish by directly observing the fish is
 (A) $2H\left(1 + \frac{1}{2\mu}\right)$ (B) $2H\left(\frac{1}{2} + \frac{1}{\mu}\right)$
 (C) $H\left(1 + \frac{1}{2\mu}\right)$ (D) $H\left(\frac{1}{2} + \frac{1}{\mu}\right)$

79. The distance from itself at which the eye sees the image of the fish by viewing in the mirror is
 (A) $2\left(H + \frac{3}{2\mu}\right)$ (B) $H + \frac{3}{2\mu}$
 (C) $H\left(\frac{1}{2} + \frac{3}{\mu}\right)$ (D) $H\left(1 + \frac{3}{2\mu}\right)$

Comprehension 25

Most materials have the refractive index, $n > 1$. So, when a light ray from air enters a naturally occurring material, then by Snell's law, $\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1}$, it is understood

that the refracted ray bends towards the normal. But it never emerges on the same side of the normal as the incident ray. According to electromagnetism, the refractive index of the medium is given by the relation, $n = \left(\frac{c}{v}\right) = \pm \sqrt{\epsilon_r \mu_r}$, where c

is the speed of electromagnetic waves in vacuum, v its speed in the medium, ϵ_r and μ_r are the relative permittivity and permeability of the medium respectively.

In normal materials, both ϵ_r and μ_r are positive, implying positive n for the medium. When both ϵ_r and μ_r are negative, one must choose the negative root of n . Such negative refractive index materials can now be artificially prepared and are called meta-materials. They exhibit significantly different optical behaviour, without violating any physical laws. Since n is negative, it results in a change in the direction of propagation of the refracted light. However, similar to normal materials, the frequency of light remains unchanged upon refraction even in meta-materials.

80. Choose the correct statement.
 (A) The speed of light in the meta-material is $v = c|n|$
 (B) The speed of light in the meta-material is $v = \frac{c}{|n|}$
 (C) The speed of light in the meta-materials is $v = c$
 (D) The wavelength of the light in the meta-material (λ_m) is given by $\lambda_m = \lambda_{\text{air}} |n|$, where λ_{air}

Practice Exercise

Matrix Match Type Questions

Each question in this section contains statements given in two columns, which have to be matched. The statements in COLUMN-I are labelled A, B, C and D, while the statements in COLUMN-II are labelled p, q, r, s (and t). Any given statement in COLUMN-I can have correct matching with ONE OR MORE statement(s) in COLUMN-II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following examples:

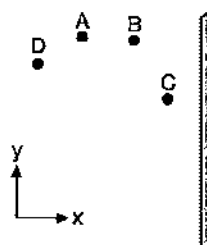
If the correct matches are $A \rightarrow p, s$ and t ; $B \rightarrow q$ and r ; $C \rightarrow p$ and q ; and $D \rightarrow s$ and t ; then the correct darkening of bubbles will look like the following :

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

1. For a real object, match the magnification situations in COLUMN-I, with their respective matches in COLUMN-II.

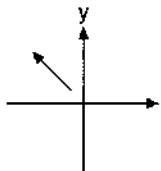
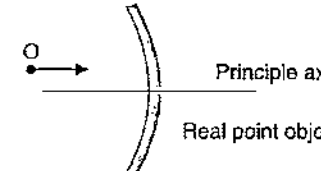
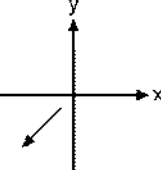
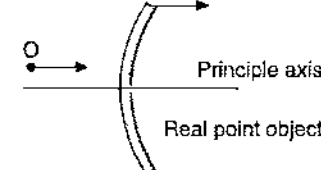
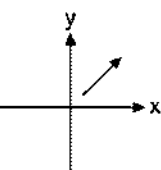
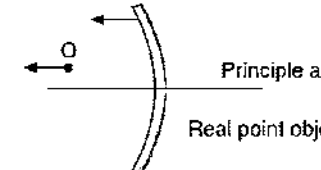
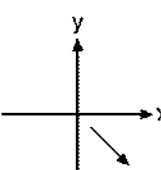
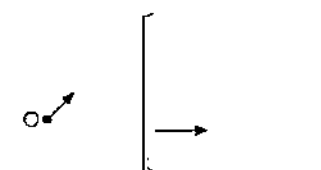
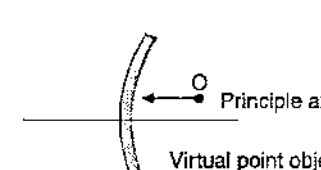
COLUMN-I	COLUMN-II
(A) $m < 0$	(p) Plane mirror
(B) $m > 0$	(q) Convex mirror
(C) $ m < 1$	(r) Concave mirror
(D) $ m \geq 1$	(s) Convex lens
	(t) Concave lens

2. Four particles are moving with different velocities in front of stationary plane mirror that lies in the y - z plane. At $t = 0$, velocity of A is $\vec{v}_A = \hat{i}$, velocity of B is $\vec{v}_B = -\hat{i} + 3\hat{j}$, velocity of C is $\vec{v}_C = 5\hat{i} + 6\hat{j}$, velocity of D is $\vec{v}_D = 3\hat{i} - \hat{j}$. The acceleration of particle A is $\vec{a}_A = 2\hat{i} + \hat{j}$ and acceleration of particle C is $\vec{a}_C = 2\hat{i} + \hat{j}$, whereas the particle B and D move with uniform velocity. Assume no collision to take place till $t = 2$ s, all quantities to be in SI units, the relative velocity of image of object A with respect to object A to be denoted by $\vec{v}_{A'A}$. If velocity of images relative to corresponding objects are given in COLUMN-I and their values at $t = 2$ s are given in COLUMN-II, then match the quantities in COLUMN-I with the corresponding values in COLUMN-II.

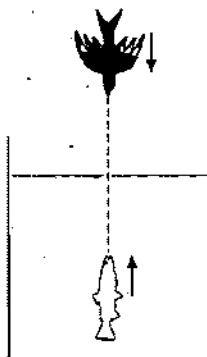


COLUMN-I	COLUMN-II
(A) $\vec{v}_{A'A}$	(p) $2\hat{i}$
(B) $\vec{v}_{B'B}$	(q) $-6\hat{i}$
(C) $\vec{v}_{C'C}$	(r) $-12\hat{i} + 4\hat{j}$
(D) $\vec{v}_{D'D}$	(s) $-10\hat{i}$
	(t) Perpendicular to the plane of mirror

3. The COLUMN-I shows some probable directions of velocity of images formed due to system shown in COLUMN-II. Match the quantities of COLUMN-I with the respective possibilities shown in COLUMN-II.

COLUMN-I	COLUMN-II
(A) 	(p) 
(B) 	(q) 
(C) 	(r) 
(D) 	(s) 
	(t) 

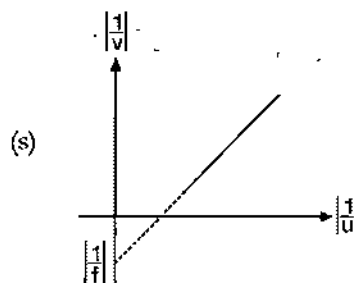
4. A bird in air is diving vertically over a tank with a speed of 6 cm s^{-1} . The base of the tank is silvered. A fish in the tank is rising upward along the same line with a speed of 8 cm s^{-1} . Taking $\mu_{\text{water}} = \frac{4}{3}$, match the quantities in COLUMN-I with their respective values in COLUMN-II.



COLUMN-I	COLUMN-II
(A) Speed of the image of fish, in cms^{-1} as seen by the bird directly	(p) 16
(B) Speed of the image of fish, in cms^{-1} formed after reflection from the mirror as seen by the bird	(q) 0
(C) Speed of image of bird, in cms^{-1} relative to the fish looking upwards	(r) 12
(D) Speed of image of bird, in cms^{-1} relative to the fish looking downwards in the mirror	(s) 8

5. Match the descriptions in COLUMN-I with corresponding plot(s) in COLUMN-II.

COLUMN-I	COLUMN-II
(A) In convex mirror, when object is real and image is virtual	(p)
(B) In convex mirror, when object is virtual and image is real.	(q)
(C) In concave mirror, when object is real and image is virtual.	(r)
(D) In concave mirror, when object is real or virtual and image is real.	



6. If (μ_1, λ_1, v_1) and (μ_2, λ_2, v_2) are the refractive indices, wavelengths and speeds of two light waves respectively, then match the entries of COLUMN-I with the entries of COLUMN-II.

COLUMN-I	COLUMN-II
(A) $\mu_1 > \mu_2$	(p) $v_1 < v_2$
(B) $\mu_1 < \mu_2$	(q) $v_1 > v_2$
(C) $\mu_1 \neq \mu_2$	(r) $\lambda_1 = \lambda_2$
(D) $\mu_1 = \mu_2$	(s) $\lambda_1 < \lambda_2$

7. Match the descriptions in COLUMN-I to corresponding details in COLUMN-II.

COLUMN-I	COLUMN-II
(A) In refraction from a rarer to a denser medium.	(p) Speed of wave does not change.
(B) In refraction.	(q) Wavelength must be decreased.
(C) In reflection from a denser medium.	(r) Frequency does not change.
(D) In reflection.	(s) The reflected ray suffers an additional path change of $\frac{\lambda}{2}$.

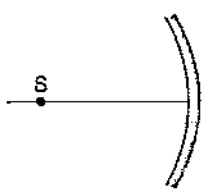
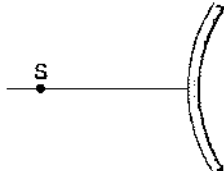
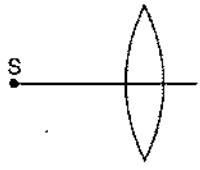
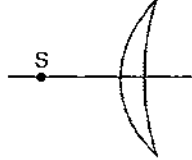
8. Light rays are incident on devices which may cause either reflection or refraction or both. The nature of the incident light and the devices are described in COLUMN-I. Some possible results of this on the rays are, given in COLUMN-II.

COLUMN-I	COLUMN-II
(A) A ray of white light passes from an optically denser medium to an optically rarer medium.	(p) Divergent beam
(B) A parallel beam of monochromatic light passes symmetrically through a glass lens.	(q) Total internal reflection
(C) A ray of white light is incident at an angle on a thick glass sheet.	(r) Lateral shift
(D) A ray of white light is incident on one face of an equivalent glass prism.	(s) Dispersion

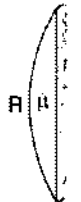
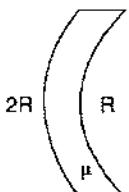
9. For a real object, match the descriptions in COLUMN-I to the corresponding details in COLUMN-II.

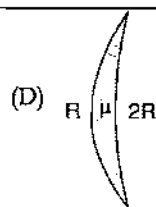
COLUMN-I	COLUMN-II
(A) Convex mirror	(p) Virtual image
(B) Concave mirror	(q) Real image
(C) Convex lens	(r) Enlarged image
(D) Concave lens	(s) Diminished image

10. An optical component and an object S placed along its optic axis are given in COLUMN-I. The distance between the object and the component can be varied. The properties of images are given in COLUMN-II. Match all the properties of images from COLUMN-II with the appropriate components given in COLUMN-I.

COLUMN-I	COLUMN-II
(A) 	(p) Real image (q) Virtual image (r) Magnified image (s) Image at infinity
(B) 	
(C) 	
(D) 	

11. Match the details of COLUMN-I with the respective name and nature described in COLUMN-II.

COLUMN-I	COLUMN-II
(A) 	(p) Converging (q) Concavo-convex (r) Convexo-concave (s) Diverging
(B) 	
(C) 	



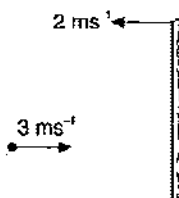
12. Match the following

COLUMN-I	COLUMN-II
(A) Concave mirror, virtual object	(p) Real image
(B) Convex mirror, virtual object	(q) Virtual image
(C) Convex lens, real object	(r) Magnified image
(D) Concave lens, real object	(s) Diminished image

13. For a concave mirror of focal length 20 cm, match the object distances in COLUMN-I to the corresponding details of images formed in COLUMN-II.

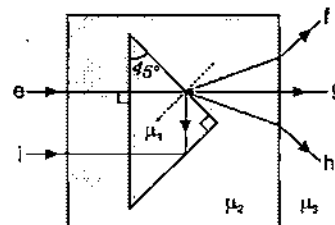
COLUMN-I	COLUMN-II
(A) 10 cm	(p) Magnified, inverted and real
(B) 30 cm	(q) Equal size, inverted and real
(C) 40 cm	(r) Smaller, inverted and real
(D) 50 cm	(s) Magnified, erect and virtual

14. A point object is placed in front of a plane mirror as shown and moving with velocity 3 ms^{-1} towards mirror. Mirror is moving with speed 2 ms^{-1} towards object, then



COLUMN-I	COLUMN-II
(A) Speed of image w.r.t. ground	(p) 10 ms^{-1}
(B) Speed of image w.r.t. mirror	(q) 5 ms^{-1}
(C) Speed of image w.r.t. object	(r) 14 ms^{-1}
(D) Speed of mirror w.r.t. object	(s) 7 ms^{-1}

15. A right angled prism of refractive index μ_1 is placed in a rectangular block of refractive index μ_2 , which is surrounded by a medium of refractive index μ_3 , as shown in the figure. A ray of light e enters the rectangular block at normal incidence. Depending upon the relationships between μ_1 , μ_2 and μ_3 , it takes one of the four possible paths ef , eg , eh , or ei . Match the paths in COLUMN-I with conditions of refractive indices in COLUMN-II and select the correct answer using the codes given below.



COLUMN-I	COLUMN-II
(A) $e \rightarrow f$	(p) $\mu_1 > \sqrt{2}\mu_2$
(B) $e \rightarrow g$	(q) $\mu_2 > \mu_1$ and $\mu_2 > \mu_3$
(C) $e \rightarrow h$	(r) $\mu_1 = \mu_2$
(D) $e \rightarrow i$	(s) $\mu_2 < \mu_1 < \sqrt{2}\mu_2$ and $\mu_2 > \mu_3$

Practice Exercise

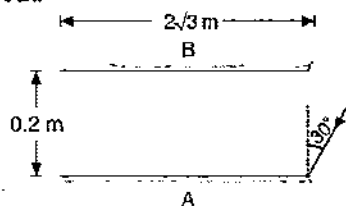
Integer Answer Type Questions

In this section the answer to each of the question is a four digit integer, ranging from 0 to 9999. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answer to question number X (say) is 6092, then the correct darkening of bubbles will look like the following :

X.

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9
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0	1	2	3	4	5	6	7	8	9

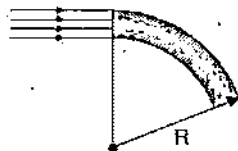
1. Two plane mirrors A and B are aligned parallel to each other, as shown in the figure. A light ray is incident at an angle 30° at a point just inside one end of A. The plane of incidence coincides with the plane of the figure. Find the maximum number of times the ray undergoes reflections (including the first one) before it emerges out.



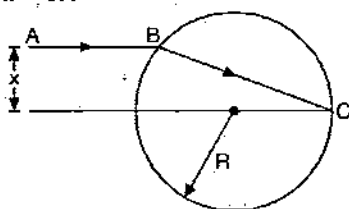
2. Where should an object be placed, in cm, in front of a concave mirror of focal length 30 cm so the image size is 5 times the object size?
3. A rod of length 20 cm is placed along the optical axis of a concave mirror of focal length 30 cm. One end of the rod is at the centre of curvature and the other end lies between F and C. Find the magnitude of the linear magnification of the rod
4. A point source of light S is placed on the major optical axis of concave mirror at a distance of 60 cm. At what

distance, in cm, from the concave mirror should a flat mirror be placed for the rays to converge again at the point S having been reflected from the concave mirror and then from the flat one? Will the position of the point where the rays meet change if they are first reflected from the flat mirror? The radius of the concave mirror is 80 cm.

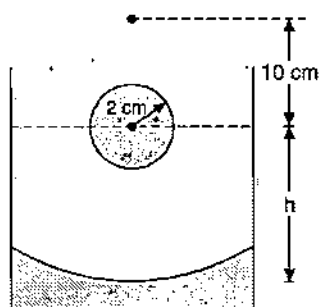
5. A concave mirror forms on a screen a real image of thrice the linear dimensions of the object. Object and screen are moved until the image is twice the size of the object. If the shift of the object is 6 cm, find the shift of the screen and the focal length of the mirror (both in cm).
6. A fish is rising up vertically inside a pond with velocity 4 cm s^{-1} and notices a bird, which is diving vertically downward and its velocity appears to be 16 cm s^{-1} (to the fish). What is the actual velocity of the diving bird, in cm s^{-1} , if refractive index of water is $4/3$.
7. A portion of a straight glass rod of diameter 4 cm and refractive index 1.5 is bent into an arc of radius R cm and a parallel beam of light is incident on it as shown in figure. Find the smallest R, in cm, which permits all the light to pass around the arc.



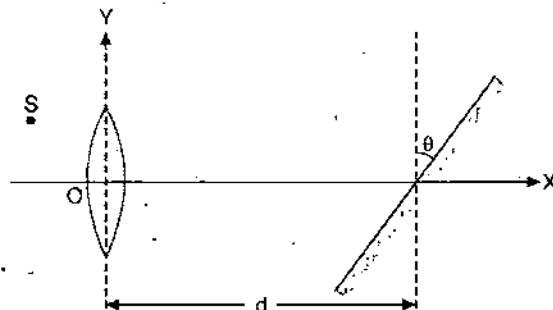
8. A man of height 2 m is standing on level road where because of temperature variation the refractive index of air is varying as $\mu = \sqrt{1 + ay}$, where y is height from road. If $a = 2 \times 10^{-6} \text{ m}^{-1}$. Then find the maximum distance, in km, till which he can see on the road.
9. A ray of light falls on a glass sphere of refractive index $\sqrt{3}$ such that the directions of the incident ray and emergent ray, when produced meet the surface at the same point on the surface. Draw the ray diagram and find the value of angle of incidence, in degree.
10. A spherical ball of transparent material has index of refraction μ . A narrow beam of light AB is aimed as shown. What must the index of refraction be in order that the light is focussed at the point C on the opposite end of the diameter from where the light entered? Given that $x \ll R$.



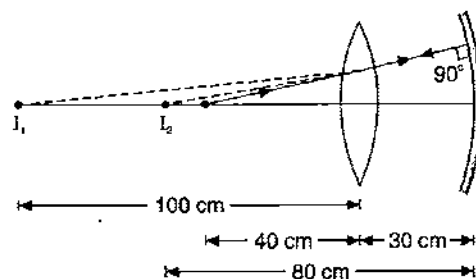
11. The perpendicular faces of a right isosceles prism are coated with silver. The rays incident at an arbitrary angle on the hypotenuse face emerge from the prism after suffering a deviation of $2x$ degree. Find x .
12. A transparent solid sphere of radius 2 cm and density ρ floats in a transparent liquid of density 2ρ kept in a beaker. The bottom of the beaker is spherical in shape with radius of curvature 8 cm and is silvered to make it concave mirror as shown in the figure. When an object is placed at a distance of 10 cm directly above the centre of the sphere its final image coincides with it. Find h (as shown in figure) the height of the liquid surface in the beaker, in cm, from the apex of the bottom. Consider the paraxial rays only. The refractive index of the sphere is $\frac{3}{2}$ and that of the liquid is $\frac{4}{3}$.



13. A thin converging lens of focal length $f = 1.5 \text{ m}$ is placed along y -axis such that its optical centre coincides with the origin. A small light source S is placed at $(-2, 0.1) \text{ m}$. A plane mirror inclined at an angle θ , (where $\tan \theta = 0.3$) is placed as shown in figure, such that y co-ordinate of final image is 0.3 m. Find the distance d , in metre. Also find the x co-ordinate of final image, in metre.



14. An object is placed 12 cm to the left of a diverging lens of focal length -6 cm . A converging lens with a focal length of 12 cm is placed at a distance d to the right of the diverging lens. Find the distance d , in cm, that corresponds to a final image at infinity.
15. Determine the position of the image, in cm, produced by an optical system consisting of a concave mirror with a focal length of 10 cm and a convergent lens with a focal length of 20 cm. The distance from the mirror to the lens is 30 cm and from the lens to the object 40 is cm. Consider only two steps.
16. The figure shows an arrangement of an equiconvex lens of focal length 20 cm and a concave mirror of radius of curvature 80 cm. A point object O is placed on the principal axis at a distance 40 cm from the lens such that the final image is also formed at the position of the object. Find the distance d , in cm. Also draw the ray diagram.
17. A converging beam of rays is incident on a diverging lens. After passing through the lens the rays intersect at a point 15 cm from the lens. If the lens is removed, the point where the rays meet, move 5 cm closer to the



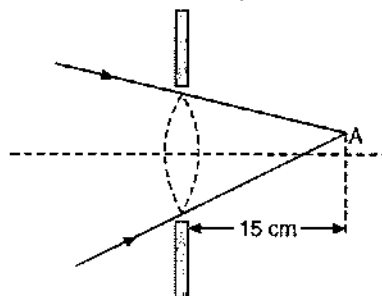
mounting that holds the lens. Find the focal length of the lens, in cm.

18. A lens with a focal length of $f = 30$ cm produces on a screen a sharp image of an object that is at a distance of $a = 40$ cm from the lens. A plane-parallel glass plate having $\mu = 1.8$ and a thickness of $d = 9$ cm is placed between the lens and the object perpendicular to the optical axis of the lens. Through what distance, in cm, should the screen be shifted for the image of the object to remain distinct?

19. The height of a candle flame is 5 cm. A lens produces an image of this flame 15 cm high on a screen. Without touching the lens, the candle is moved over a distance of $\ell = 1.5$ cm away from the lens and a sharp image of the flame 10 cm high is obtained again after shifting the screen. Calculate the focal length of the lens, in cm.

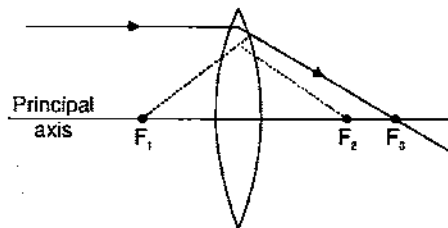
20. The focal length of a convex lens in air is 10 cm. Find its focal length, in cm, in water. Given that $\mu_g = \frac{3}{2}$ and $\mu_w = \frac{4}{3}$.

21. A converging beam of rays passes through a round aperture in a screen as shown in figure. The apex of the beam A is at a distance of 15 cm from the screen. How will the distance from the focus of the rays to the screen change, in cm, if a convergent lens is inserted in the aperture with a focal length of 30 cm? Plot the path of the rays after the lens is fitted.



22. A lens with a focal length of 16 cm produces a sharp image of an object in two positions which are 60 cm apart. Find the distance, in cm, from the object to the screen.

23. An intense beam parallel to the principal axis is incident on a convex lens. Multiple extra images $F_1, F_2, \dots$ are formed due to feeble internal reflections, called flare spots as shown in the figure. The radii of curvature of the lens are 30 cm and 60 cm and the refractive index is 1.5. Find the position of the first flare spot, in cm.



24. A parallel beam of rays is incident on a convergent lens with a focal length of 40 cm. Where should a divergent lens with a focal length of 15 cm be placed for the beam of rays to remain parallel after passing through the two lenses. Give your answer in cm.

25. One side of radius of curvature $R_2 = 120$ cm of a convex lens of material of refractive index $\mu = 1.5$ and focal length $f_1 = 40$ cm is silvered. It is placed on a horizontal surface with silvered surface in contact with it. Another convex lens of focal length $f_2 = 20$ cm is fixed coaxial $d = 10$ cm above the first lens. A luminous point object O on the axis gives rise to an image coincident with it. Find its height, in cm, above the upper lens.

26. A source of light is located from a convergent lens of focal length $f = 30$ cm at a distance double the focal length of the convergent lens. At what distance from the lens should a flat mirror be placed so that the rays reflected from the mirror are parallel after passing through the lens for the second time? Give your answer in cm.

Answers

In Chapter Exercise (ICE) & Practice Exercise Sets

ICE I

BASED ON REFLECTION AT PLANE SURFACES

- | | |
|--|--|
| 1. 30° | 9. $3d$ |
| 2. 4 cm | 10. (a) $12 \text{ cm} \times 8 \text{ cm}$ |
| 3. $\frac{u \cos \alpha (\tan \alpha - \tan \theta)}{g}$ | (b) $12 \text{ cm} \times 4 \text{ cm}$ |
| 4. (a) 60° | 12. $-5(1 + \sqrt{3})\hat{i} + 5\hat{j}$ |
| (b) 240° CCW | 13. $\alpha = 2\theta$ |
| 5. 60° | 14. 30° |
| 6. 100° (CW) | 15. 50° |
| 7. 20 cm, 60 cm, 80 cm, 100 cm and 140 cm | 16. $-\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$ |
| 8. 12 | |

ICE II

BASED ON REFLECTION AT CURVED SURFACES

- | | |
|--|--|
| 1. 5 cm inverted | 8. (a) Concave |
| 2. Concave, 6.67 m | 9. $\left(\frac{\sqrt{3}+1}{2}\right)R$ from the convex mirror |
| 4. 7.5 cm, 12.5 cm | 11. 10 cm |
| 5. (a) 8 cm | 12. $\frac{3}{2}, -\frac{3}{4}$ |
| (b) 16 cm | |
| (c) 48 cm | |
| 6. 15 cm | |
| 7. (a) $\left(\frac{2 + \cos \omega t}{1 + \cos \omega t}\right)f$ | |
| (b) $x = 0$ | |
| (c) $m \rightarrow \infty$ | |

ICE III

BASED ON GENERAL REFRACTION

- 75 cm
- 7.5 cm
- $\Delta v = 0.55 \text{ cm}, \frac{m_2}{m_1} \approx 1.1$
- $2 \times 10^6 \text{ ms}^{-1}, 4000 \text{ \AA} \text{ yellow}$
- 3.5 cm
- $\frac{4}{3}h + d$
- 6.6 cm
- 2.88 m
- $\frac{1}{\sqrt{3}} \text{ cm}$
- 0.7 r
- $\sin^{-1} \left(\frac{L}{R^2} (\sqrt{\mu^2 R^2 - L^2} - \sqrt{R^2 - L^2}) \right)$

ICE IV

BASED ON TOTAL INTERNAL REFLECTION (TIR)

- (a) 54.34°
(b) Yes
- $\frac{1}{B} \left(A - \frac{\sqrt{3}}{2} \right)$
- (a) 2.81 m
(b) 0.23 m
- (b) $\sqrt{2}$
- (a) 26.8°
(b) Yes
- $\frac{h}{\sqrt{\mu^2 - 1}}$
- (a) 40.54°
(b) 26.6°
- 67.3°
- $\frac{4}{3} \text{ cm}$
- $\sqrt{2}$
- $OP \neq \frac{\sqrt{2}}{3} R$
- $\frac{1}{2}$

ICE V

BASED ON PRISM

- (a) 30°
(b) $\frac{\sqrt{7}}{\sqrt{3}}$
- $A = \alpha + \beta - \delta$
$$\mu = \sin \beta \sqrt{1 + \left(\frac{\sin \alpha}{\sin \beta \sin(\alpha + \beta - \delta)} + \cot(\alpha + \beta - \delta) \right)^2}$$
- (a) 157.2°
(b) 128.4°
- $A_{\max} = 83.62^\circ$
- (a) $\sqrt{5}$
(b) 58.8°
- (a) 34.2°
(b) 8.4°
- $\sin^{-1}(\mu \sin \alpha) - \alpha$
- $\sqrt{2}$
- $0^\circ, \sqrt{3}$
- $\delta_v - \delta_r = 4.5^\circ$
- $22^\circ, 56^\circ$
- 0°
- 19°
- 10.1°
- $\delta_{\text{red}} = 30.6^\circ, \delta_{\text{violet}} = 33.4^\circ$
- 2, 10

ICE VI

BASED ON REFRACTION AT CURVED SURFACES

1. 2.5 D
2. Final image is formed at pole of the mirror
3. $x \approx 0.75R$
4. 7.42 cm
5. (a) 80 cm
(b) $u < 12$ cm
6. Final image is formed at 65 cm from first face on the same side of the object.
7. 3.33 cm, infinity

ICE VII

BASED ON LENS FORMULA

1. 6 cm from either of the object
2. 12 cm
3. $\frac{4}{3}$
4. Concave mirror of focal length 15 cm
5. (a) 90 cm
(b) 102 cm
6. 12.5 cm in front of the silvered lens
7. 10 cm
8. 2.14 cm
9. 1.37
10. 7.5 cm
11. $\frac{4t_1^2}{t_2}$
12. 2 m, 1 m
13. (a) $\frac{t}{t-t_1}$
(b) No Shift
14. 7.5 cm
17. $(m+1)$ times smaller
18. (a) Convex
19. 2.4 cm
20. (a) 1.4
21. Rays will become parallel to the optic axis.
22. $i = \frac{3-2\mu_0}{2a}$
23. $\frac{R}{2(\mu n + \mu - 1)}$
24. (a) x_1
(b) $\frac{x_1 x_2}{x_1 - x_2}$
(c) $\frac{x_1}{x_1 - x_2}$

ANSWERS

SINGLE CORRECT CHOICE TYPE QUESTIONS

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. A | 4. D | 5. B |
| 6. C | 7. C | 8. D | 9. D | 10. C |
| 11. B | 12. B | 13. A | 14. C | 15. B |
| 16. B | 17. B | 18. D | 19. C | 20. C |
| 21. C | 22. D | 23. D | 24. D | 25. B |
| 26. A | 27. B | 28. C | 29. D | 30. A |
| 31. B | 32. C | 33. D | 34. D | 35. B |
| 36. B | 37. C | 38. B | 39. A | 40. B |
| 41. C | 42. D | 43. B | 44. D | 45. C |
| 46. C | 47. B | 48. A | 49. C | 50. A |
| 51. A | 52. C | 53. D | 54. C | 55. D |
| 56. B | 57. B | 58. A | 59. B | 60. D |

Advanced JEE Physics

Optics & Modern Physics

61. A	62. B	63. C	64. D	65. A
66. D	67. C	68. A	69. B	70. C
71. C	72. C	73. A	74. A	75. B
76. C	77. A	78. C	79. A	80. C
81. C	82. C	83. D	84. A	85. B
86. B	87. B	88. A	89. C	90. C
91. D	92. A	93. C	94. C	95. A
96. C	97. B	98. B	99. C	100. D
101. C	102. C	103. B	104. B	105. C
106. C	107. C	108. B	109. D	110. B
111. B	112. A	113. D	114. C	115. C
116. C	117. B	118. D	119. C	120. C
121. C	122. B	123. C	124. B	125. A
126. A	127. C	128. B	129. D	130. C
131. C	132. D	133. B	134. C	135. A
136. C	137. C	138. D	139. C	140. C
141. B	142. A	143. D	144. C	145. C
146. A	147. D	148. D	149. A	150. C
151. B	152. C	153. C	154. C	155. D
156. A	157. C	158. D	159. D	160. B
161. C	162. D	163. D	164. C	165. B
166. C	167. A	168. B	169. D	170. A
171. A	172. D	173. D	174. C	175. A
176. D	177. C	178. B	179. B	180. C
181. B	182. C	183. C	184. C	185. C
186. C	187. A	188. B	189. A	190. B
191. B	192. A	193. A	194. D	195. B
196. B	197. D	198. D	199. A	200. D
201. D	202. C	203. D	204. C	205. C
206. B	207. D	208. B	209. A	210. B
211. C	212. C	213. C	214. C	215. C
216. C	217. C	218. A	219. B	220. A
221. B	222. D	223. A	224. A	225. B
226. D	227. A	228. D	229. B	230. B
231. B	232. C	233. B	234. B	235. B
236. C	237. D	238. D	239. D	240. B
241. B	242. C	243. A	244. A	245. D
246. C	247. B	248. D	249. B	250. B
251. C	252. D	253. A	254. D	255. B
256. C	257. B	258. D	259. A	260. A
261. A	262. B	263. D	264. C	265. A
266. B	267. D	268. B	269. D	270. D
271. A	272. C	273. C	274. A	275. D
276. B	277. C	278. B	279. A	280. C
281. C	282. A	283. A	284. A	285. D

286. A	287. C	288. D	289. B	290. A
291. C	292. D	293. A	294. C	295. D
296. B	297. C	298. D		

ANSWERS**MULTIPLE CORRECT CHOICE TYPE QUESTIONS**

1. BC	2. A	3. AD	4. AB	5. BD
6. AC	7. ACD	8. AC	9. BCD	10. A
11. BC	12. AD	13. BC	14. ABCD	15. ABCD
16. BD	17. ACD	18. AD	19. BD	20. AC
21. BCD	22. ABC	23. AB	24. BC	25. AD
26. AD	27. BC	28. BD	29. CD	30. AC
31. BD	32. ACD	33. AC	34. BC	35. B
36. B	37. AD	38. AD	39. AC	40. CD
41. AC	42. AC	43. C	44. AB	45. BC
46. ABC	47. BC	48. ABC	49. CD	50. BC
51. AC	52. BD	53. BC	54. ABC	

ANSWERS**REASONING BASED QUESTIONS**

1. D	2. B	3. A	4. B	5. A
6. C	7. D	8. B	9. C	10. D
11. A	12. A	13. D	14. D	15. A
16. B	17. A	18. D	19. A	20. C
21. B	22. A	23. B	24. D	25. D
26. B	27. D	28. C	29. C	30. C

ANSWERS**LINKED COMPREHENSION TYPE QUESTIONS**

1. B	2. C	3. B	4. D	5. A
6. D	7. C	8. A	9. D	10. B
11. D	12. C	13. D	14. A	15. D
16. D	17. C	18. A	19. C	20. D
21. A	22. D	23. C	24. B	25. D
26. C	27. A	28. A	29. B	30. A
31. C	32. C	33. B	34. D	35. D
36. C	37. C	38. D	39. C	40. C
41. D	42. D	43. D	44. C	45. B
46. B	47. C	48. A	49. A	50. C
51. D	52. D	53. A	54. C	55. B
56. A	57. C	58. B	59. B	60. A
61. C	62. C	63. C	64. D	65. B
66. C	67. B	68. A	69. D	70. A

Advanced JEE Physics

Optics & Modern Physics

71. C	72. B	73. A	74. C	75. D
76. A	77. B	78. C	79. D	80. B
81. C				

ANSWERS

MATRIX MATCH TYPE QUESTIONS

1. A $\rightarrow$ (r, s) B $\rightarrow$ (p, q, r, s, t) C $\rightarrow$ (q, r, s, t) D $\rightarrow$ (p, r, s)	2. A $\rightarrow$ (s, t) B $\rightarrow$ (p, t) C $\rightarrow$ (s, t) D $\rightarrow$ (q, t)	3. A $\rightarrow$ (s) B $\rightarrow$ (p, q, r) C $\rightarrow$ (q, r, s) D $\rightarrow$ (t)	4. A $\rightarrow$ (r) B $\rightarrow$ (q) C $\rightarrow$ (p) D $\rightarrow$ (p)	5. A $\rightarrow$ (p, q) B $\rightarrow$ (r) C $\rightarrow$ (s) D $\rightarrow$ (p, q)
6. A $\rightarrow$ (p, s) B $\rightarrow$ (q) C $\rightarrow$ (p, q, s) D $\rightarrow$ (r)	7. A $\rightarrow$ (q, r) B $\rightarrow$ (r) C $\rightarrow$ (p, r, s) D $\rightarrow$ (p, r)	8. A $\rightarrow$ (p, q, s) B $\rightarrow$ (p, q) C $\rightarrow$ (r) D $\rightarrow$ (p, q, s)	9. A $\rightarrow$ (p, s) B $\rightarrow$ (p, q, r, s) C $\rightarrow$ (p, q, r, s) D $\rightarrow$ (p, s)	10. A $\rightarrow$ (p, q, r, s) B $\rightarrow$ (q) C $\rightarrow$ (p, q, r, s) D $\rightarrow$ (p, q, r, s)
11. A $\rightarrow$ (p) B $\rightarrow$ (p) C $\rightarrow$ (r, s) D $\rightarrow$ (q, p)	12. A $\rightarrow$ (p, s) B $\rightarrow$ (p, q, r, s) C $\rightarrow$ (p, q, r, s) D $\rightarrow$ (q, s)	13. A $\rightarrow$ (s) B $\rightarrow$ (p) C $\rightarrow$ (q) D $\rightarrow$ (r)	14. A $\rightarrow$ (s) B $\rightarrow$ (q) C $\rightarrow$ (p) D $\rightarrow$ (q)	15. A $\rightarrow$ (q) B $\rightarrow$ (r) C $\rightarrow$ (s) D $\rightarrow$ (p)

ANSWERS

INTEGER ANSWER TYPE QUESTIONS

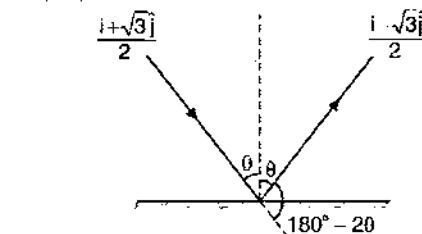
1. 30	2. 24, 36	3. 3	4. 90
5. 36	6. 9	7. 12	8. 2
9. 60	10. 2	11. 180	12. 15
13. 5, 4	14. 8	15. 5	16. 30
17. 30	18. 60	19. 9	20. 40
21. 5	22. 100	23. 12	24. 25
25. 10	26. 45		

Solutions

In Chapter Exercise (ICE)-I

(based on reflection at plane surfaces)

1. The angle between the incident ray and the reflected ray is $180^\circ - 2\theta$, so, we have



$$\cos(180^\circ - 2\theta) = \frac{\left(\frac{1+\sqrt{3}}{2}\right) \cdot \left(\frac{1-\sqrt{3}}{2}\right)}{\left|\frac{1+\sqrt{3}}{2}\right| \left|\frac{1-\sqrt{3}}{2}\right|}$$

$$\Rightarrow -\cos(2\theta) = \frac{(1-3)}{1}$$

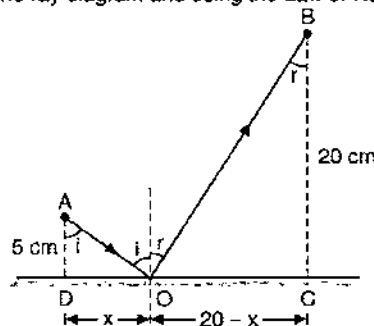
$$\Rightarrow -\cos(2\theta) = -\frac{1}{2}$$

$$\Rightarrow \cos(2\theta) = \frac{1}{2}$$

$$\Rightarrow 2\theta = 60^\circ$$

$$\Rightarrow \theta = 30^\circ$$

2. Drawing the ray diagram and using the Law of Reflection, we get



$$l = r$$

$$\Rightarrow \sin i = \sin r$$

So we can say that $\triangle ADO$ and $\triangle OBC$ are similar

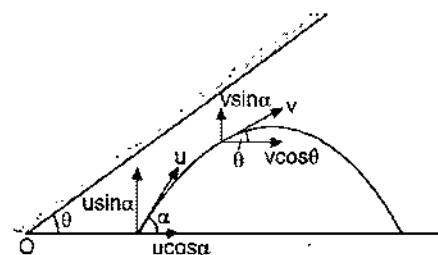
...(1)

$$\Rightarrow \frac{x}{5} = \frac{20-x}{20}$$

$$\Rightarrow x = 4 \text{ cm}$$

So point of incidence of light from A should be at 4 cm from D on mirror.

3. The image will be momentarily at rest when the particle moves parallel to the mirror. Let at the time t the particle has a velocity v parallel to the mirror.



$$v \sin \theta = u \sin \alpha - gt \quad \dots(1)$$

$$\text{and } v \cos \theta = u \cos \alpha$$

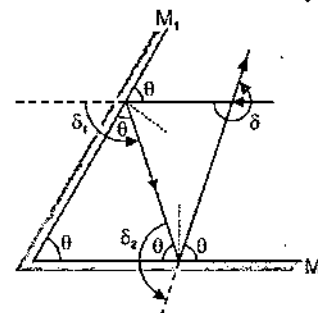
$$\Rightarrow v = \frac{u \cos \alpha}{\cos \theta} \quad \dots(2)$$

From (1) and (2)

$$\left(\frac{u \cos \alpha}{\cos \theta}\right) \sin \theta = u \sin \alpha - gt$$

$$\Rightarrow t = \frac{u \cos \alpha (\tan \alpha - \tan \theta)}{g}$$

- 4.



From the figure, we observe that $3\theta = 180^\circ$

$$\Rightarrow \theta = 60^\circ$$

$$\text{So, } \delta_1 = 180^\circ - 2(30^\circ) = 120^\circ \text{ (CCW)}$$

$$\text{and } \delta_2 = 180^\circ - 2(30^\circ) = 120^\circ \text{ (CCW)}$$

$$\text{So, total deviation } \delta = \delta_1 + \delta_2$$

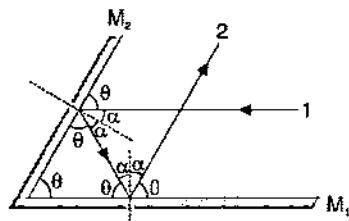
$$\Rightarrow \delta = 240^\circ \text{ (CCW)}$$

Alternatively from the figure, we observe that
 $\delta = 180^\circ + \theta = 240^\circ \text{ (CCW) or } 120^\circ \text{ (CW)}$

5. Various angles made are as shown in figure. In triangle ABC, we observe that

$$\theta + \theta + \theta = 180^\circ$$

$$\Rightarrow \theta = 60^\circ$$

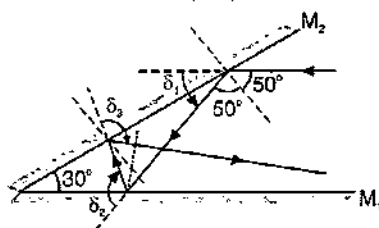


6. From the figure, we observe that

$$\delta_1 = 180^\circ - 2(50^\circ) = 100^\circ \text{ (CCW)}$$

$$\delta_2 = 180^\circ - 2(20^\circ) = 140^\circ \text{ (CW)}$$

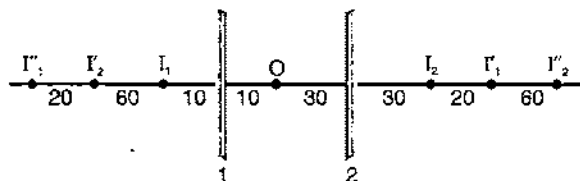
$$\delta_3 = 180^\circ - 2(10^\circ) = 160^\circ \text{ (CW)}$$



$$\text{So, total deviation } \delta = 100^\circ \text{ (CCW)} + 140^\circ \text{ (CW)} + 160^\circ \text{ (CW)}$$

$$\Rightarrow \delta = 100^\circ \text{ (CW) or } 260^\circ \text{ (CCW)}$$

7. The image is formed as far behind the mirror as the object is in front of it. Also image formed by mirror 1 i.e., I_1 , acts as object for mirror 2, so I_1' is formed 50 cm behind the mirror 2 as shown.



Taking all distances to be in cm and plotting them as shown (but not to scale), we get

$$OI_1 = 20 \text{ cm} \quad OI_2 = 60 \text{ cm}$$

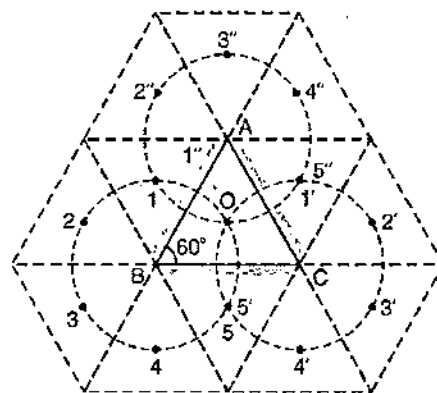
$$OI_1' = 80 \text{ cm} \quad OI_2' = 80 \text{ cm}$$

$$OI_1'' = 100 \text{ cm} \quad OI_2'' = 140 \text{ cm}$$

So, the respective distances are 20 cm, 60 cm, 80 cm, 100 cm and 140 cm

8. Let us first consider the mirrors AB and BC, for which we have

$$\frac{360^\circ}{60^\circ} = 6$$



So, the number of images formed by the combination is given by

$$N = 6 - 1 = 5$$

Combination of Mirrors	Images Formed
AB & BC	1, 2, 3, 4, 5
AC & BC	1', 2', 3', 4', 5'
AB & AC	1'', 2'', 3'', 4'', 5''

These images along with the object must lie on a circle as shown in figure with an angular separation of

$$\frac{360^\circ}{N} = \frac{360^\circ}{5} = 72^\circ$$

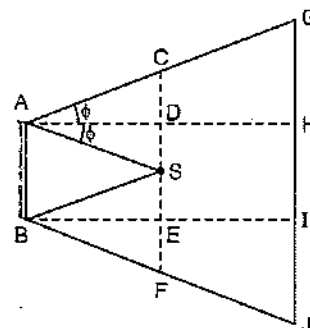
Similarly, the other two combination of mirrors also form 5 images each but we find from symmetry that 5 and 5', 1 and 5'', 1' and 1'' coincide. So the total number of images formed by three mirrors AB, BC and AC is

$$N' = (5)(3) - 3 = 12$$

9. The ray diagram is shown in figure. We observe that

$$HI = AB = d$$

$$DS = CD = \frac{d}{2}$$



$$\text{Also, } AH = 2AD$$

$$\Rightarrow GH = 2CD = 2\left(\frac{d}{2}\right) = d$$

$$\text{Similarly, } IJ = d$$

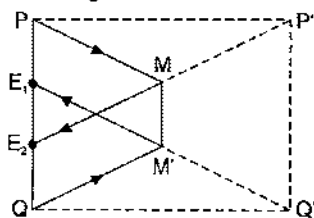
$$\Rightarrow GJ = GH + HI + IJ = d + d + d = 3d$$

10. (a) For a one eyed man, the required size will be half the each dimension of the face i.e., 12 cm x 8 cm

(b) For a two eyed man, the Smallest length of the mirror = Half the length of face

$$\Rightarrow \left(\text{Smallest length of the Mirror}\right) = \frac{1}{2} \times 24 = 12 \text{ cm}$$

The smallest breadth of the mirror is calculated by using the fact that the rays from extreme part of face should reach one of the eyes after reflection from the mirror. The common overlapping portion is then the required breadth of the mirror. The ray diagram is shown in figure.



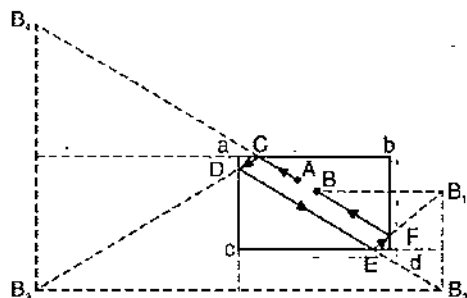
From figure, we get

$$MM' = \frac{1}{2}PQ - \frac{1}{2}E_1E_2$$

$$\Rightarrow \left(\text{Smallest Breadth of the Mirror} \right) = \frac{1}{2}(16 - 8) = 4 \text{ cm}$$

So, the shortest size of mirror is 12 cm by 4 cm.

11. Let us first find the image of point B in mirror bd (shown in figure). Let us then construct image B_1 in mirror cd. Also, B_2 is the image of B_1 in mirror ac and B_3 is the image of B_2 in mirror ab. Let us connect points A and B_3 . Point C is the point of intersection of ab with line AB_3 . Let us now draw line B_3C from B_3 and connect point D at which this line intersects ac with B_2 , E with B_1 and F with B.



It can be stated that the line ACDEFB is the sought path of the beam. Further, we observe that since, B_3CB_1 is an isosceles triangle, CD is the reflection of beam AC.

Similarly, we can show that DE is the reflection of CD and so on. This solution of the problem is not unique, as the beam will not necessarily always be sent initially to mirror ab.

12. Along x-direction i.e., perpendicular to the mirror, we have

$$\left(\text{Relative Velocity of Image w.r.t. mirror} \right) = - \left(\text{Relative Velocity of Object w.r.t. mirror} \right)$$

$$\Rightarrow v_i - v_m = - (v_o - v_m)$$

$$\Rightarrow v_i - (-5 \cos 30^\circ) = - (10 \cos 60^\circ - (-5 \cos 30^\circ))$$

$$\Rightarrow v_i = -5(1 + \sqrt{3}) \text{ ms}^{-1}$$

In the direction parallel to the surface of mirror, i.e., along y-direction we have

$$v_i = v_o$$

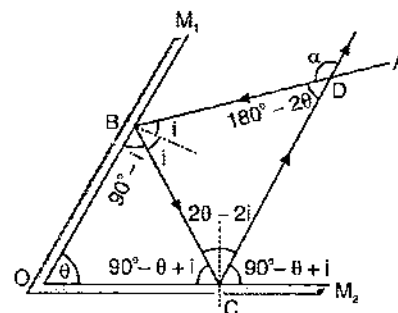
$$\Rightarrow v_i = 10 \sin(60^\circ) = 5 \text{ ms}^{-1}$$

So, velocity of the image

$$\vec{v}_i = (v_i)_x \hat{i} + (v_i)_y \hat{j}$$

$$\Rightarrow \vec{v}_i = -5(1 + \sqrt{3}) \hat{i} + 5 \hat{j}$$

13. Let AB be the incident ray and angle of incidence at the mirror M_1 be i , then



$$\angle CBO = 90^\circ - i$$

$$\Rightarrow \angle BCO = 180^\circ - \theta - (90^\circ - i)$$

$$\Rightarrow \angle BCO = 90^\circ - \theta + i$$

Using the Laws of Reflection, we get

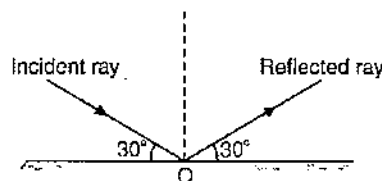
$$\angle DCB = 2\theta - 2i$$

$$\Rightarrow \angle CDB = 180^\circ - 2\theta$$

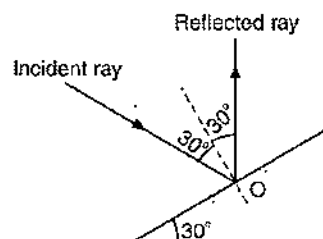
$$\Rightarrow \alpha = 2\theta$$

The angle between incident and emergent ray is 2θ and it is independent of the angle of incidence i .

14. Suppose that a plane mirror is kept horizontal as shown in figure. The reflected ray will make an angle of 30° with horizontal, or an angle of 60° with the vertical.



To make the reflected ray to go vertically upwards, the mirror is required to be rotated about O counterclockwise by 60° . To achieve this, therefore, the plane mirror is required to rotate about O by half the angle, i.e., by 30° , as shown in figure.

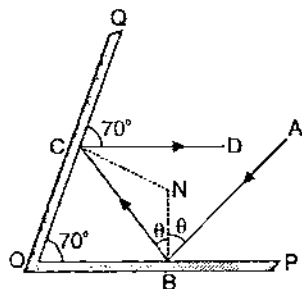


15. Ray AB is incident on mirror OP at an angle θ . The reflected ray BC is incident on second mirror OQ. Finally, the reflected ray CD is parallel to OP. Since CD and OP are parallel, and CO cuts them,

$$\angle QCD = \angle COP = 70^\circ$$

$$\Rightarrow \angle DCN = 90^\circ - \angle QCD = 90^\circ - 70^\circ = 20^\circ$$

$$\Rightarrow \angle NCB = \angle DCN = 20^\circ$$



Further, $\angle OCB = 90^\circ - \angle NCB = 90^\circ - 20^\circ = 70^\circ$

Now, in $\triangle OCB$, we have

$$\angle CBO = 180^\circ - (\angle COB + \angle OCB) = 180^\circ - (70^\circ + 70^\circ) = 40^\circ$$

$$\Rightarrow \theta = \angle NBC = 90^\circ - \angle CBO = 90^\circ - 40^\circ = 50^\circ$$

16. Reflection of a ray of light is just like an elastic collision of a ball with a horizontal ground. The component of the incident ray along the inside normal gets reversed while the component perpendicular to it remains unchanged. So, the component of incident ray vector $\vec{A} = \hat{i} + \hat{j} - \hat{k}$ parallel to normal, i.e., $\hat{i} + \hat{j}$ gets reversed while perpendicular to it, i.e., $-\hat{k}$ remains unchanged. So, the reflected ray is written as,

$$\vec{R} = -\hat{i} - \hat{j} - \hat{k}$$

A unit vector along the reflected ray will be,

$$\hat{r} = \frac{\vec{R}}{R} = \frac{-\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$$

$$\Rightarrow \hat{r} = -\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$$

Solutions

In Chapter Exercise (ICE)-II

(based on reflection at curved surfaces)

- Since, $m = \frac{f}{f-u}$
Now, $f = -f$ and $u = -1.5f$, so
$$m = \frac{-f}{-f+1.5f} \Rightarrow m = \frac{-f}{0.5f}$$

$$\Rightarrow m = -2$$

Since $m = \frac{h_i}{h_o} = -2$
$$\Rightarrow h_i = -2h_o = -5 \text{ cm}$$

The image is 5 cm long. The minus sign shows that it is inverted.
- Given : $u = -25 \text{ m}$, $m = +4$ (since the image is erect).
Now, the magnification is given by
$$m = -\frac{v}{u}$$

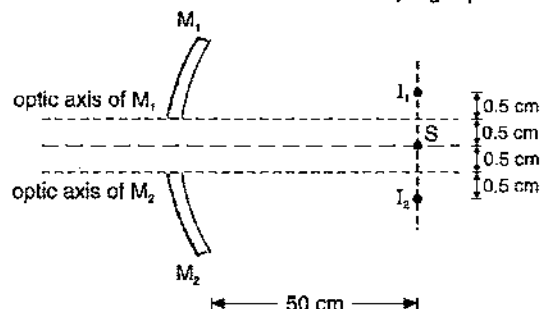
$$\Rightarrow v = -mu = -(+4) \times (-2.5) = 10 \text{ m}$$

Using the mirror formula, we get
$$\frac{1}{10} + \frac{1}{-2.5} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f} = 0.1 - 0.4 = -0.3$$

$$\Rightarrow f = -\frac{1}{0.3} = -\frac{10}{3} \text{ m}$$

Since f is negative so, the mirror is concave.
The radius of curvature of the mirror is given by
$$R = 2f = 2\left(-\frac{10}{3}\right) = -\frac{20}{3} \text{ m} = -6.67 \text{ m}$$
- According to the mirror formula, we have $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$



$$\Rightarrow \frac{1}{v} - \frac{1}{50} = \frac{1}{-25}$$

$$\Rightarrow v = -50 \text{ cm}$$

$$\text{So, } m = -\frac{v}{u} = -1$$

- Given, $f = -10 \text{ cm}$. Since a concave mirror can form real as well as virtual image and since the nature of image is not given in the question. So we will consider two possible cases.

Case 1 (when image is real) :

$$\text{So, } m = -4$$

$$\text{Since } m = -\frac{f}{f-u}$$

$$\Rightarrow -4 = \frac{-10}{-10-u}$$

$$\Rightarrow u = -12.5 \text{ cm}$$

Please note that here,

$|u| > |f|$ and we know that in case of a concave mirror, image is real when object lies beyond F.

Case 2 (When image is virtual) :

$$\text{So, } m = +4$$

$$\text{Since } m = \frac{f}{f-u}$$

$$\Rightarrow 4 = \frac{-10}{-10-u}$$

$$\Rightarrow u = -7.5 \text{ cm}$$

Please, note that here, $|u| < |f|$, as we know that image is virtual when the object lies between F and P.

- Since $f = \frac{R}{2} = -12 \text{ cm}$

Let the object be placed at a distance u from the pole.

Since, we know that magnification m is given by

$$m = \frac{f}{f-u}$$

So, now we apply this formula to these situations one by one.

a) Here, we have $m = +3$

$$\Rightarrow 3 = \frac{-12}{-12-u}$$

$$\Rightarrow -4 = -12-u$$

$$\Rightarrow 12+u = 4$$

$$\Rightarrow u = -8 \text{ cm}$$

b) Here, $m = -3$

$$\Rightarrow -3 = \frac{-12}{-12 - u}$$

$$\Rightarrow 4 = -12 - u \Rightarrow u = -16 \text{ cm}$$

c) Here, $m = -\frac{1}{3}$

$$\Rightarrow -\frac{1}{3} = \frac{-12}{-12 - u}$$

$$\Rightarrow 36 = -12 - u \Rightarrow u = -48 \text{ cm}$$

6. Distance of image formed by the plane mirror is $(b-a)$ i.e., $(b-5)$ cm and distance of object from mirror is $(b+a)$ i.e., $(b+5)$ cm. Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ we get

$$\frac{1}{(b-5)} - \frac{1}{(b+5)} = \frac{1}{20}$$

Solving this equation, we get

$$b = 15 \text{ cm}$$

The coincidence of the images can be established by observing the changes in the relative position of the images when the eye is moved away from the optical axis of the mirror.

When the images are at various distance from the eye the images will be displaced with respect to each other.

When the images are at the same distance, they will coincide irrespective of the placement of the eye.

7. (a) At any instant t , we have

$$u = -(2f + x) = -(2f + f \cos \omega t)$$

Using, the mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} - \frac{1}{2f + f \cos \omega t} = \frac{1}{f}$$

$$\Rightarrow v = -\left(\frac{2 + \cos \omega t}{1 + \cos \omega t}\right)f$$

i.e., distance of image from mirror at time any instant t is

$$\left(\frac{2 + \cos \omega t}{1 + \cos \omega t}\right)f$$

- (b) Ball coincides with its image at centre of curvature, i.e., at $x = 0$

- (c) At $t = \frac{T}{2}$, we have

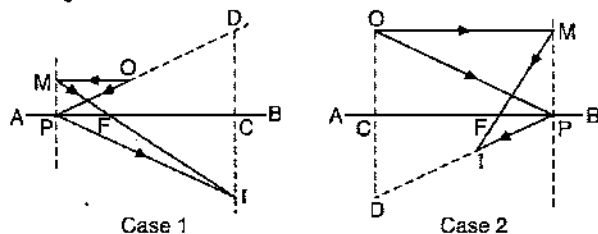
$$\omega t = \pi$$

$$\Rightarrow x = f \cos(\pi) = -f$$

So, $u = -f$ i.e., ball is at focus. So, its image is formed at infinity, so $m \rightarrow \infty$

8. (a) Since the image is on the opposite side of the principal axis, the mirror is concave. Because convex mirror always forms a virtual and erect image.

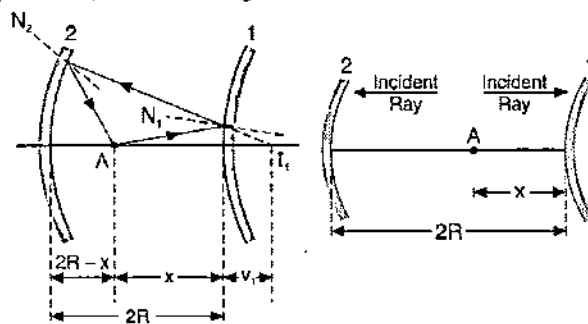
- (b) The ray diagrams for two different cases are shown in figure.



The following steps of construction for drawing the ray diagrams are used.

- From I or O drop a perpendicular on principal axis, such that $CI = CD$ or $OC = CD$.
- Draw a line joining D and O or D and I so that it meets the principal axis at P. The point P will be the pole of the mirror as a ray reflected from the pole is always symmetrical about principal axis.
- From O draw a line parallel to principal axis towards the mirror so that it meets the mirror at M. Join M to I, so that it intersects the principal axis at F. F is the focus of the mirror as any ray parallel to principal axis after reflection from the mirror intersects the principal axis at the focus.

9. Let the point A be at a distance x from the convex mirror as shown in mirror, then assuming the origin to be placed at the pole of convex mirror, we get



For convex mirror, we have

$$\frac{1}{v_1} - \frac{1}{x} = \frac{2}{R}$$

$$\Rightarrow v_1 = \frac{xR}{2x + R}$$

For concave mirror, we have

$$\frac{1}{-(2R - x)} - \frac{1}{\left(2R + \frac{xR}{2x + R}\right)} = -\frac{2}{R}$$

Solving this equation, we get

$$x = \left(\frac{1 + \sqrt{3}}{2}\right)R \text{ and } x = -\left(\frac{\sqrt{3} - 1}{2}\right)R$$

Ignoring the negative value, as we have already used a negative sign with x , so the object should be placed at a distance

$$x = \left(\frac{\sqrt{3} + 1}{2}\right)R \text{ from the convex mirror.}$$

10. Object is placed beyond C. Hence, the image will be real and it will lie between C and F. Further u , v and f all are negative, hence the mirror formula becomes

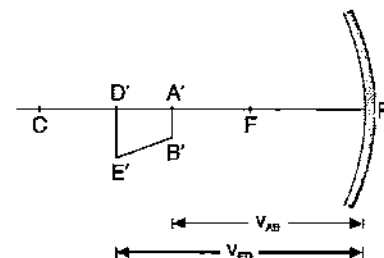
$$-\frac{1}{v} - \frac{1}{u} = -\frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{u - f}{uf}$$

$$\Rightarrow v = \frac{uf}{u - f}$$

Now since, $u_{AB} > u_{ED}$

$$\Rightarrow v_{AB} < v_{ED}$$



Hence, $|m_{AB}| < |m_{ED}|$ $\left\{ \because m = -\frac{v}{u} \right\}$

Therefore, shape of the image will be as shown in figure.

Also note that $v_{AB} < u_{AB}$ and $v_{ED} < u_{ED}$. $|m_{AB}| < 1$ and $|m_{ED}| < 1$

11. Since the image is inverted, so the mirror is concave.

Now, $u = -30$ cm

$$m = -\frac{1}{2} = -\frac{v}{u}$$

$$\Rightarrow v = \frac{u}{2} = -15 \text{ cm}$$

Since $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{(-15)} + \frac{1}{(-30)} = \frac{1}{f}$$

$$\Rightarrow f = -10 \text{ cm}$$

12. Since image touches the rod, the rod must be placed with one end at centre of curvature. However, two cases arise here.

Case I : When the other end lies between C and F

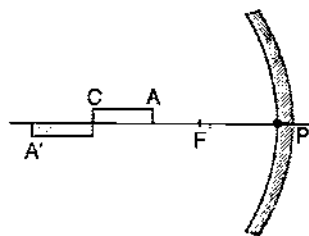
For A, we have

$$u = -\left(2f - \frac{f}{3}\right) = -\frac{5f}{3}$$

$$f = -f$$

Since, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, so we get

$$\frac{1}{v} + \frac{1}{\left(-\frac{5f}{3}\right)} = \frac{1}{(-f)}$$



$$\Rightarrow v = \frac{5f}{2}$$

So, magnification, $m = \frac{\text{Length of Image}}{\text{Length of Object}}$

$$\Rightarrow m = \frac{v_A - v_C}{u_A - u_C} = \frac{-\frac{5f}{2} - (-2f)}{-\frac{5f}{3} - (-2f)} = -\frac{3}{2}$$

Case II : When the other end lies beyond C

For A, we have

$$u = -\left(2f + \frac{f}{3}\right) = -\frac{7f}{3}$$

$$f = -f$$

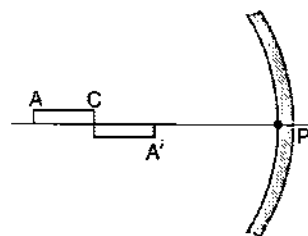
Since $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, so we get

$$\frac{1}{v} + \frac{1}{\left(-\frac{7f}{3}\right)} = \frac{1}{(-f)}$$

$$\Rightarrow v = -\frac{7f}{4}$$

So, magnification, $m = \frac{v_A - v_C}{u_A - u_C}$

$$\Rightarrow m = \frac{-\frac{7f}{4} - (-2f)}{-\frac{7f}{3} - (-2f)} = -\frac{3}{4}$$



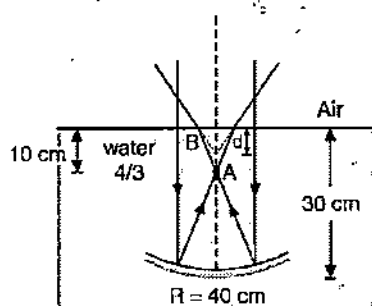
Solutions

In Chapter Exercise (ICE)-III

(based on general refraction)

$$1. \quad d' = \frac{d}{n_{\text{relative}}} = \frac{100}{\frac{4}{3}} = 75 \text{ cm}$$

2. The incident rays will pass undeviated through the water surface and strike the mirror parallel to its principal axis. Therefore for the mirror, object is at ∞ . Its image A (in figure) will be formed at focus which is 20 cm from the mirror. Now for the interface between water and air, $d = 10 \text{ cm}$.



$$\Rightarrow d' = \frac{d}{\left(\frac{n_w}{n_a}\right)} = \frac{10}{\left(\frac{4/3}{1}\right)} = 7.5 \text{ cm}$$

3. Case I : When No Slab is Inserted

According to mirror formula, we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v_1} + \frac{1}{-30} = \frac{1}{-10}$$

$$\Rightarrow v_1 = -15 \text{ cm}$$

$$\text{Magnification, } m_1 = -\frac{v}{u} = -\frac{1}{2} = -0.5$$

Case II : When Slab is Inserted

$$\text{Shift} = \left(1 - \frac{1}{\mu}\right)t = \left(1 - \frac{1}{1.5}\right)6 = 2 \text{ cm}$$

Again, applying the mirror formula, we get

$$\frac{1}{v_2} + \frac{1}{-(30-2)} = \frac{1}{-10}$$

$$\Rightarrow v_2 = -15.55 \text{ cm}$$

$$\text{Magnification, } m_2 = -\frac{v}{u} = -0.55$$

$$\text{So, } \Delta v = 0.55 \text{ cm}$$

$$\frac{m_2}{m_1} \approx 1.1$$

4. The refractive index of glass,

$$\mu = \frac{v_{\text{air}}}{v_{\text{glass}}} = \frac{c}{v}$$

$$\Rightarrow v = \frac{c}{\mu} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ ms}^{-1}$$

Since the frequency of light remains the same when it passes from one medium to another, so we have

$$c = f\lambda_0 \text{ and } v = f\lambda$$

$$\text{Since, } \mu = \frac{c}{v} = \frac{\lambda_0}{\lambda}$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\mu} = \frac{6000}{1.5} = 4000 \text{ \AA}$$

The colour remains yellow, as the colour depends on the frequency and not on the wavelength.

5. Using equation, the total apparent shift is

$$\Delta x = h_1 \left(1 - \frac{1}{\mu_1}\right) + h_2 \left(1 - \frac{1}{\mu_2}\right)$$

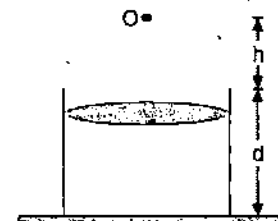
$$\Rightarrow \Delta x = 2 \left(1 - \frac{1}{4/3}\right) + 3 \left(1 - \frac{1}{3/2}\right) = 1.5 \text{ cm}$$

$$\text{Thus, } h = h_1 + h_2 - \Delta x = 2 + 3 - 1.5 = 3.5 \text{ cm}$$

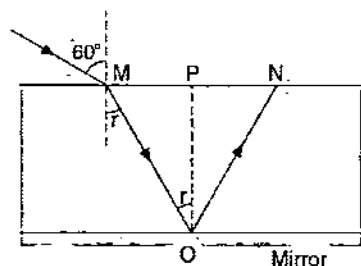
6. When we consider only two steps, then the ray of light starting from object O first gets refracted and then reflected. Distance of image I_1 formed after refraction from the plane surface is given by

$$x = nh + d = \frac{4}{3}h + d$$

Therefore, distance of image I_2 formed by plane mirror will be $\frac{4}{3}h + d$



7. $1.8 = \frac{\sin(60^\circ)}{\sin r}$
 $\Rightarrow r = 28.76^\circ$



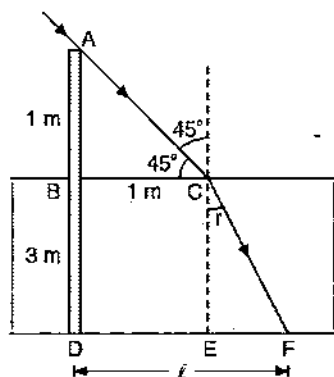
Since, $MP = PO \tan r$
 $\Rightarrow MP = (6) \tan(28.76^\circ)$
 $\Rightarrow MP = 3.3 \text{ cm}$
 So, $MN = 2MP = 6.6 \text{ cm}$

8. From Snell's Law, we have

$$\frac{4}{3} = \frac{\sin(45^\circ)}{\sin r}$$

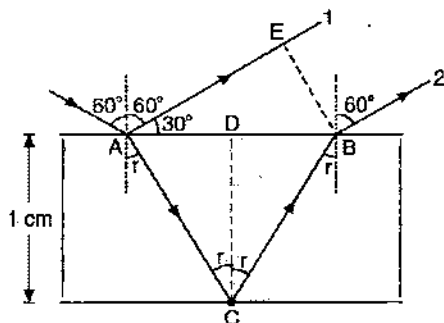
$$\Rightarrow \sin r = \frac{3}{4} \sin(45^\circ)$$

$$\Rightarrow r = 32^\circ$$



Since, $EF = EC \tan r$
 $\Rightarrow EF = (3) \tan 32^\circ = 1.88 \text{ m}$
 Length of shadow at the bottom of the lake is
 $l = DF = DE + EF = 2.88 \text{ m}$

9. From Snell's Law, we have $\sqrt{3} = \frac{\sin(60^\circ)}{\sin r}$
 $\Rightarrow r = 30^\circ$



Since, $AB = 2(AD) = 2(DC \tan r)$

$$\Rightarrow AB = (2)(1) \left(\frac{1}{\sqrt{3}} \right) = \frac{2}{\sqrt{3}} \text{ cm}$$

So, the distance between rays 1 and 2 is given by

$$BE = AB \sin(30^\circ) = \frac{1}{\sqrt{3}} \text{ cm}$$

10. Total deviation suffered by the ray is given by

$$\delta_{\text{Total}} = \delta_p + \delta_o$$

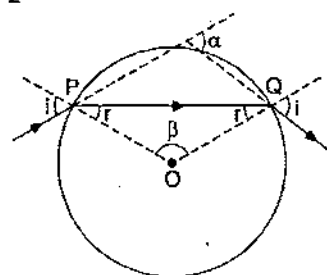
$$\Rightarrow \alpha = (i - r) + (i - r)$$

$$\Rightarrow i - r = \frac{\alpha}{2} \quad \dots(1)$$

Further, in $\triangle OPQ$, we have

$$r + r + \beta = 180^\circ$$

$$\Rightarrow r = 90^\circ - \frac{\beta}{2} \quad \dots(2)$$



From equation (1), we get

$$i = r + \frac{\alpha}{2} = 90^\circ + \left(\frac{\alpha - \beta}{2} \right) \quad \dots(3)$$

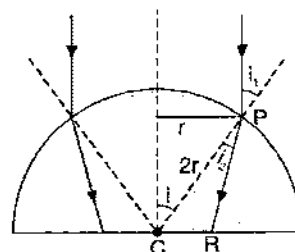
According to Snell's Law, we have

$$\mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow \mu = \frac{\sin \left[90^\circ + \left(\frac{\alpha - \beta}{2} \right) \right]}{\sin \left(90^\circ - \frac{\beta}{2} \right)} = \frac{\cos \left(\frac{\beta - \alpha}{2} \right)}{\cos \left(\frac{\beta}{2} \right)}$$

$$\Rightarrow \cos \left(\frac{\beta - \alpha}{2} \right) = \mu \cos \frac{\beta}{2}$$

11. $\sin i_1 = \frac{r}{2r} = \frac{1}{2}$
 $\Rightarrow i_1 = 30^\circ$



Applying Snell's Law at P, we get

$$\frac{3}{2} = \frac{\sin i_1}{\sin i_2}$$

$$\Rightarrow i_2 = 19.5^\circ$$

Now, Applying Sine Law (Lami's Theorem), on $\triangle CPR$, we get

$$\frac{2r}{\sin(180^\circ - 60^\circ - 19.5^\circ)} = \frac{CR}{\sin(19.5^\circ)} \quad \{\because \angle PCR = 60^\circ\}$$

$$\Rightarrow CR = 0.7r$$

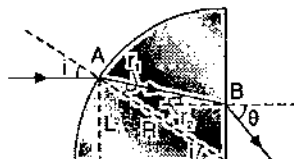
12. Since, $\sin i = \frac{L}{R}$

$$\Rightarrow i = \sin^{-1}\left(\frac{L}{R}\right)$$

According to Snell's Law applied at A, we have

$$\mu = \frac{\sin i}{\sin r_1}$$

$$\Rightarrow r_1 = \sin^{-1}\left(\frac{\sin i}{\mu}\right) = \sin^{-1}\left(\frac{L}{\mu R}\right)$$



Deviation suffered by the ray is

$$\delta = i - r_1 = \sin^{-1}\left(\frac{L}{R}\right) - \sin^{-1}\left(\frac{L}{\mu R}\right)$$

This is also the angle r_2 , so we have

$$r_2 = \sin^{-1}\left(\frac{L}{R}\right) - \sin^{-1}\left(\frac{L}{\mu R}\right)$$

From the knowledge of inverse trigonometry, we have

$$\sin^{-1}(C) - \sin^{-1}(D) = \sin^{-1}(C\sqrt{1-D^2} - D\sqrt{1-C^2})$$

$$\Rightarrow r_2 = \sin^{-1}\left[\frac{L}{R}\sqrt{1-\frac{L^2}{\mu^2 R^2}} - \frac{L}{\mu R}\sqrt{1-\frac{L^2}{R^2}}\right]$$

Now, again applying Snell's Law at B, we get

$$\mu = \frac{\sin \theta}{\sin r_2}$$

$$\Rightarrow \theta = \sin^{-1}(\mu \sin r_2)$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{\mu L}{R}\sqrt{1-\frac{L^2}{\mu^2 R^2}} - \frac{L}{R}\sqrt{1-\frac{L^2}{R^2}}\right)$$

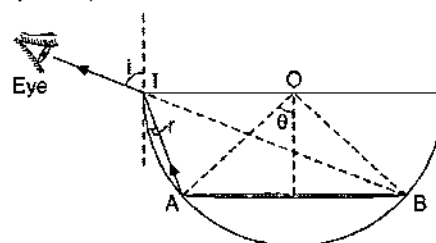
$$\Rightarrow \theta = \sin^{-1}\left(\frac{L}{R^2}\sqrt{\mu^2 R^2 - L^2} - \frac{L}{R^2}\sqrt{R^2 - L^2}\right)$$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{L}{R^2}(\sqrt{\mu^2 R^2 - L^2} - \sqrt{R^2 - L^2})\right)$$

13. In the figure, let AB be the disc and O be the centre of the bowl.

Let $\angle AOM = \theta$

Then by symmetry, $\angle AIB = \theta$



Now from geometry of figure, we have

$$\angle AOI = 2\angle IBA$$

$$\Rightarrow \angle IBA = \frac{1}{2}(90^\circ - \theta)$$

$$\Rightarrow \angle r = \angle IBA = \frac{90^\circ - \theta}{2} \quad \dots(1)$$

$$\text{Also } \angle i = \frac{90^\circ - \theta}{2} + \theta = \frac{90^\circ + \theta}{2} \quad \dots(2)$$

Now, Snell's Law gives

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin\left(\frac{90^\circ + \theta}{2}\right)}{\sin\left(\frac{90^\circ - \theta}{2}\right)}$$

$$\Rightarrow \left(\frac{\cos \frac{\theta}{2} + \sin \frac{\theta}{2}}{\cos \frac{\theta}{2} - \sin \frac{\theta}{2}}\right)^2 = \mu^2$$

$$\Rightarrow \frac{1 + \sin \theta}{1 - \sin \theta} = \mu^2$$

$$\Rightarrow \sin \theta = \frac{\mu^2 - 1}{\mu^2 + 1}$$

$$\text{Since, } \sin \theta = \frac{d}{2a}$$

$$\Rightarrow d = 2a\left(\frac{\mu^2 - 1}{\mu^2 + 1}\right)$$

Solutions

In Chapter Exercise (ICE)-IV

(based on total internal reflection (TIR))

1. (a) Critical angle between 2 and 3, is given by

$$\sin C = \frac{1.3}{1.8}$$

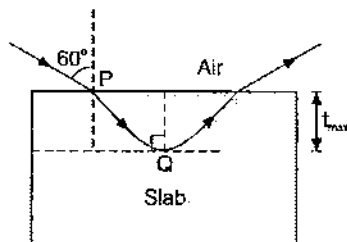
Now, applying Snell's Law, we get

$$1.6 \sin \theta = 1.8 \sin C = (1.8) \left(\frac{1.3}{1.8} \right) = 1.3$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{13}{16} \right) \approx 54.34^\circ$$

- (b) If θ is decreased, the angle of incidence at the interface between 2 and 3 gets decreased or $i < C$, so the light will refract into medium 3.

2. The path of ray is curved as shown in figure. As it travels successively into denser layers, it bends away from normal and TIR takes place at depth where angle of incidence approaches $\frac{\pi}{2}$.



Applying Snell's Law at interfaces P and Q, we get

$$1 \sin(60^\circ) = \mu_g \sin\left(\frac{\pi}{2}\right)$$

$$\Rightarrow \frac{\sqrt{3}}{2} = (A - B t_{\max})$$

$$\Rightarrow t_{\max} = \frac{1}{B} \left(A - \frac{\sqrt{3}}{2} \right)$$

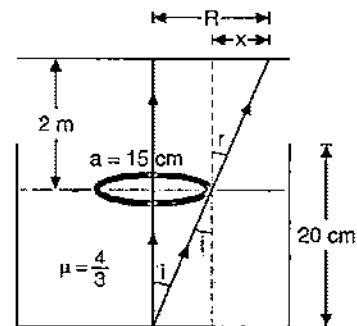
3. (a) Using Snell's Law, we get

$$\mu \sin i = \sin r$$

$$\Rightarrow \frac{4}{3} \left(\frac{15}{\sqrt{15^2 + 20^2}} \right) = 1 \left(\frac{x}{\sqrt{x^2 + 200^2}} \right)$$

Solving, we get

$$x = \frac{800}{3} \text{ cm}$$



So, the radius of shadow is $R = \left(15 + \frac{800}{3} \right) \text{ cm}$

$$\Rightarrow R = \frac{845}{3} \text{ cm} = 2.81 \text{ m}$$

- (b) For shadow to be formed, angle of incidence must be less than critical angle.

Using Snell's Law, we get

$$\frac{4}{3} \left(\frac{a_{\max}}{\sqrt{a_{\max}^2 + 20^2}} \right) = 1 \sin(90^\circ)$$

$$\Rightarrow 16a_{\max}^2 = 9a_{\max}^2 + 9(20^2)$$

$$\Rightarrow 7a_{\max}^2 = 9(20^2)$$

$$\Rightarrow a_{\max} = \left(\frac{9}{7} \right) (20 \text{ cm}) = 0.23 \text{ m}$$

4. (a) At interface AB, applying Snell's Law, we get

$$1 \sin\left(\frac{\pi}{2}\right) = \mu \sin \alpha$$

$$\Rightarrow \sin \alpha = \frac{1}{\mu} \quad \dots(1)$$

At interface BC, applying Snell's Law again, we get

$$\mu \sin(90^\circ - \alpha) = 1(\sin \theta)$$

$$\Rightarrow \sin \theta = \mu \cos \alpha \quad \dots(2)$$

From equation (1) and (2), we get $\sin \theta = \cot \alpha$

- (b) For emergence from BC, we must have

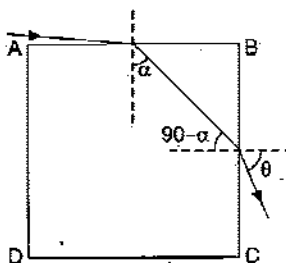
$$90^\circ - \alpha \leq C$$

At grazing incidence we have

$$\begin{aligned} \Rightarrow \alpha &= C \\ \Rightarrow 90 - \alpha &\leq C \\ \Rightarrow 2C &\geq 90^\circ \\ \Rightarrow C &\geq 45^\circ \\ \Rightarrow \sin^{-1}\left(\frac{1}{\mu}\right) &\geq 45^\circ \\ \Rightarrow \frac{1}{\mu} &\geq \frac{1}{\sqrt{2}} \\ \Rightarrow \mu &\leq \sqrt{2} \end{aligned}$$

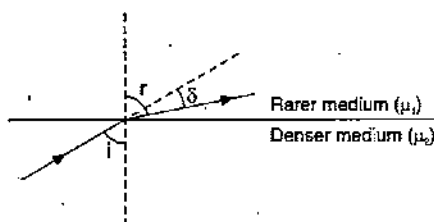
So, the greatest value of refractive index is

$$\mu_{\max} = \sqrt{2}$$



5. Case I:

When angle of incidence (i) is less than critical angle C i.e., $i < C$.



Since, $\delta = \text{Deviation} = r - i$

From Snell's Law, we get

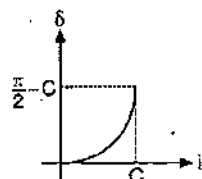
$$\mu_1 \sin i = \mu_2 \sin r$$

$$\Rightarrow r = \sin^{-1}\left(\frac{\mu_1}{\mu_2} \sin i\right)$$

Substituting in equation (1), we get

$$\delta = \sin^{-1}\left(\frac{\mu_1}{\mu_2} \sin i\right) - i$$

This is a non-linear function and graph is given below



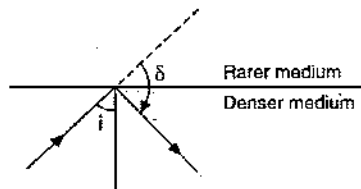
Deviation versus angle of incidence graph when TIR is not taking place.

Case II:

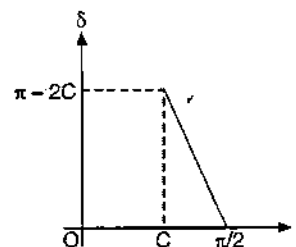
When the angle of incidence i is greater than the critical angle C , i.e., $i > C$

In this case TIR will take place as shown, so deviation is

$$\delta = \pi - 2i$$



This is a linear function and so the graph is given below

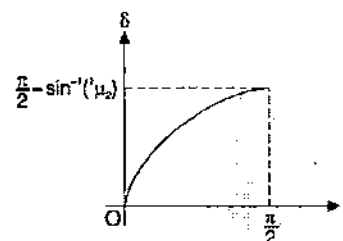


Deviation versus angle of incidence graph when TIR is taking place

CONCEPTUAL NOTE(S)

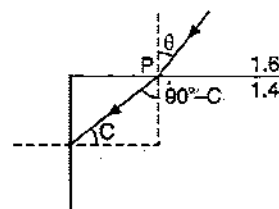
When ray is travelling from rarer to denser medium then deviation is given by

$$\delta = i - \sin^{-1}(\mu_2 \sin i)$$



6. (a) Critical angle between 2 and 3

$$\sin C = \frac{1.2}{1.4} = \frac{6}{7}$$



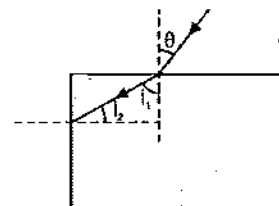
Applying, Snell's Law, at P, we get

$$1.6 \sin \theta = 1.4 \sin (90^\circ - C) = 1.4 \cos C$$

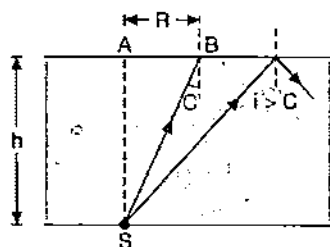
$$\Rightarrow \sin \theta = \frac{1.4 \sqrt{1 - \frac{36}{49}}}{1.6} = 0.45$$

$$\Rightarrow \theta \approx 26.8^\circ$$

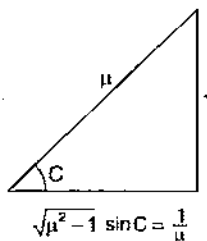
(b) As θ is increased, i_1 will increase or i_2 will decrease or $i_2 < C$ and hence the light will refract in medium 3.



7. As shown in figure, the light from the source will not emerge out of water if $i = C$.



Therefore, minimum radius R corresponds to the situation when $i = C$



In $\triangle SAB$,

$$\begin{aligned} \frac{R}{h} &= \tan C \\ \Rightarrow R &= h \tan C \\ \Rightarrow R &= \frac{h}{\sqrt{\mu^2 - 1}} \end{aligned}$$

8. (a) Applying Snell's Law, we get

$$\begin{aligned} 1.3 \sin \theta_1 &= (1) \sin \theta_2 \\ \Rightarrow \sin \theta_2 &= (1.3) \sin (30^\circ) = \frac{1.3}{2} = 0.65 \\ \Rightarrow \theta_2 &\approx 40.54^\circ \end{aligned}$$

- (b) Applying Snell's Law, we get

$$\begin{aligned} 1.3 \sin \theta_1 &= 1.45 \sin \theta_2 \\ \Rightarrow \sin \theta_2 &= \frac{1.3 \left(\frac{1}{2} \right)}{1.45} = 0.49 \\ \Rightarrow \theta_2 &\approx 26.6^\circ \end{aligned}$$

9. Critical angle, $C = \sin^{-1} \left(\frac{1}{\mu} \right) = \sin^{-1} \left(\frac{1}{1.36} \right) = 47.3^\circ$

For TIR to take place at B, we have

$$\begin{aligned} \beta &> C \\ \Rightarrow \beta &> 47.3^\circ \\ \text{For this to happen, we have} \\ \alpha &< 90^\circ - 47.3^\circ \\ \Rightarrow \alpha &< 42.7^\circ \end{aligned}$$

Applying Snell's Law at A, we get

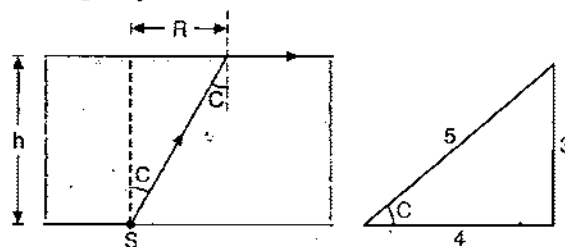
$$\begin{aligned} \theta &< \sin^{-1} (\mu \sin 42.7^\circ) \\ \Rightarrow \theta &< \sin^{-1} [1.36 \sin (42.7^\circ)] \\ \Rightarrow \theta &< 67.3^\circ \end{aligned}$$

So, the maximum value of θ for TIR to take place at B is 67.3° .

10. $\sin C = \frac{1}{\mu} = \frac{3}{5}$

$$\frac{R}{h} = \tan C = \frac{3}{4}$$

$$\therefore h = \frac{4}{3} R = \frac{4}{3} \text{ cm}$$



11. At the maximum, the ray can enter the glass at the grazing angle, so $(i)_{\max} = 90^\circ$

According to Snell's Law

$$\mu_g \sin r_1 = \mu \sin i$$

$$\Rightarrow \sin r_1 = \frac{\mu}{1.5} \sin (90^\circ) = \frac{2\mu}{3}$$

$$\Rightarrow (r_1)_{\max} = \sin^{-1} \left(\frac{2\mu}{3} \right)$$

Since $r_1 + r_2 = 90^\circ$, so

$$\therefore (r_2)_{\min} = \frac{\pi}{2} - \sin^{-1} \left(\frac{2\mu}{3} \right)$$

Again, using Snell's Law, we get $\mu_g = \frac{\sin (30^\circ)}{\sin r_2}$

$$\Rightarrow \frac{3}{2} = \frac{\frac{1}{2}}{\sin \left(\frac{\pi}{2} - r_1 \right)} = \frac{1}{2 \cos r_1}$$

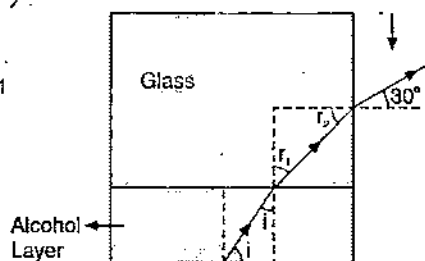
$$\Rightarrow 3 \cos r_1 = 1$$

$$\Rightarrow 3 \sqrt{1 - \frac{4\mu^2}{9}} = 1$$

$$\Rightarrow 1 - \frac{4\mu^2}{9} = \frac{1}{9}$$

$$\Rightarrow \frac{4\mu^2}{9} = \frac{8}{9}$$

$$\Rightarrow \mu = \sqrt{2}$$



Now, when the paper is dry then $\mu = 1$

$$\Rightarrow r_1 = \sin^{-1} \left(\frac{2}{3} \right) \approx 42^\circ$$

Since $r_1 + r_2 = 90^\circ$, so we get

$$r_2 = 48^\circ$$

Critical angle at glass air interface, is given by

$$C = \sin^{-1} \left(\frac{1}{1.5} \right) \approx 42^\circ$$

So, we observe that as $r_2 > C$, so it can't be seen.

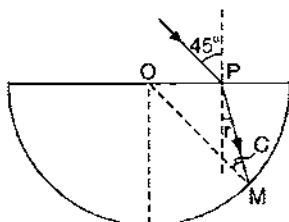
12. According to Snell's Law, we have

$$\sqrt{2} = \frac{\sin (45^\circ)}{\sin r}$$

$$\Rightarrow r = 30^\circ$$

The critical angle C is given by

$$C = \sin^{-1} \left(\frac{1}{\mu} \right) = 45^\circ$$



Applying, Sine Law (i.e., Snell's Law) in $\triangle OPM$, we get

$$\frac{OP}{\sin C} = \frac{OM}{\sin(90^\circ + r)}$$

$$\Rightarrow \frac{OP}{\left(\frac{1}{\sqrt{2}}\right)} = \frac{R}{\cos r} \quad \{R = \text{radius}\}$$

$$\Rightarrow OP = \sqrt{\frac{2}{3}} R$$

As we move away from O, angle PMO will increase.

Therefore, $OP \propto \sqrt{\frac{2}{3}} R$. Same is the case on left side of O.

13. Incident angle i is least for ray AP and this angle should be greater than the critical angle C

$$\text{i.e., } i > C$$

$$\Rightarrow \sin i > \sin C$$

$$\Rightarrow \frac{R}{R+r} > \frac{1}{\mu}$$

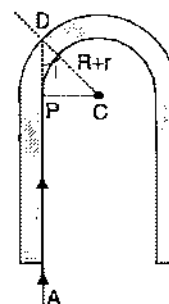
$$\Rightarrow \frac{R}{R+r} > \frac{2}{3}$$

$$3R > 2R + 2r$$

$$\Rightarrow R > 2r$$

$$\Rightarrow \frac{r}{R} < \frac{1}{2}$$

$$\text{Hence, } \left(\frac{r}{R}\right)_{\max} = \frac{1}{2}$$

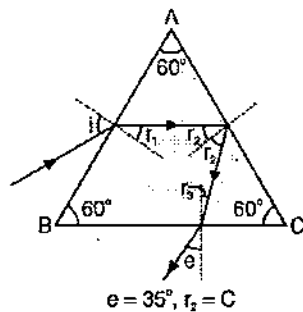


Solutions

In Chapter Exercise (ICE)-V

(based on prism)

1. The ray diagram for the situation discussed is shown in figure.



- (a) From the figure, we observe that

$$r_1 + r_2 = r_2 + r_3 = 60^\circ$$

$$\Rightarrow r_1 = r_3$$

Applying Snell's Law at the faces AB and BC, we get

$$\mu = \frac{\sin i}{\sin r_1} = \frac{\sin(30^\circ)}{\sin r_3}$$

$$\text{Since } r_1 = r_3$$

$$\Rightarrow i = 30^\circ$$

- (b) Since $r_1 + r_2 = A = 60^\circ$

$$\Rightarrow r_1 = 60^\circ - r_2 \equiv 60^\circ - C$$

$$(\because r_2 \approx C)$$

$$\text{Further, } \mu = \frac{\sin i}{\sin r_1} = \frac{\sin(30^\circ)}{\sin(60^\circ - C)}$$

$$\mu = \frac{0.5}{\sin(60^\circ)\cos C - \cos(60^\circ)\sin C}$$

$$\text{Since } \sin C = \frac{1}{\mu}$$

$$\Rightarrow \cos C = \sqrt{1 - \frac{1}{\mu^2}}$$

$$\Rightarrow \mu \left(\frac{\sqrt{3}}{2} \right) \sqrt{1 - \frac{1}{\mu^2}} - \frac{1}{2} = 0.5$$

$$\Rightarrow \left(\frac{\sqrt{3}}{2} \right) \sqrt{\mu^2 - 1} = 1$$

$$\Rightarrow \mu = \sqrt{\frac{7}{3}}$$

2. $r_1 + r_2 = A$

$$\text{Since } i + e = A + \delta$$

$$\Rightarrow \delta = \alpha + \beta - A$$

Further applying Snell's Law at incident surface and emergent surface, we get

$$\mu = \frac{\sin \alpha}{\sin r_1} \text{ and } \frac{\sin r_2}{\sin \beta} = \frac{1}{\mu}$$

$$\Rightarrow \frac{\sin \alpha}{\sin r_1} = \frac{\sin \beta}{\sin r_2}$$

$$\Rightarrow \frac{\sin \alpha}{\sin(A - r_2)} = \frac{\sin \beta}{\sin r_2}$$

$$\frac{\sin(A - r_2)}{\sin r_2} = \frac{\sin \alpha}{\sin \beta}$$

$$\Rightarrow \frac{\sin A \cos r_2}{\sin r_2} - \frac{\cos A \sin r_2}{\sin r_2} = \frac{\sin \alpha}{\sin \beta}$$

$$\Rightarrow \sin A \cot r_2 = \frac{\sin \alpha}{\sin \beta} + \cos A$$

$$\Rightarrow \cot r_2 = \frac{\sin \alpha}{\sin \beta \sin A} + \cot A$$

$$\text{Since } \mu = \frac{\sin \beta}{\sin r_2} = \sin \beta \operatorname{cosec} r_2$$

$$\Rightarrow \mu = \sin \beta \sqrt{1 + \cot^2 r_2}$$

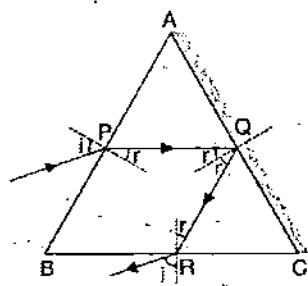
$$\Rightarrow \mu = \sin \beta \sqrt{1 + \left(\frac{\sin \alpha}{\sin \beta \sin A} + \cot A \right)^2}$$

$$\text{Since } A = \alpha + \beta - \delta$$

$$\Rightarrow \mu = \sin \beta \sqrt{1 + \left(\frac{\sin \alpha}{\sin \beta \sin(\alpha + \beta - \delta)} + \cot(\alpha + \beta - \delta) \right)^2}$$

3. At minimum deviation, we have

$$r = \frac{A}{2} = 30^\circ$$



- (a) Applying Snell's Law at AB, we get

$$1.5 = \frac{\sin i}{\sin(30^\circ)}$$

$$\Rightarrow i = 48.6^\circ$$

Since, $\delta_{\text{Total}} = \delta_P + \delta_Q + \delta_R$

$$\Rightarrow \delta_{\text{Total}} = (i - r) + (180^\circ - 2r) + (i - r)$$

$$\Rightarrow \delta_{\text{Total}} = 180^\circ + 2i - 4r$$

$$\Rightarrow \delta_{\text{Total}} = 157.2^\circ$$

- (b) Again applying Snell's Law for water-glass interface, we get

$$\frac{4}{3} \sin i' = \frac{3}{2} \sin(30^\circ)$$

$$\Rightarrow i' = 34.2^\circ$$

$$\Rightarrow \delta_{\text{Total}} = 180^\circ + 2i' - 4r$$

$$\Rightarrow \delta_{\text{Total}} = 128.4^\circ$$

4. Since, the condition for no emergence is

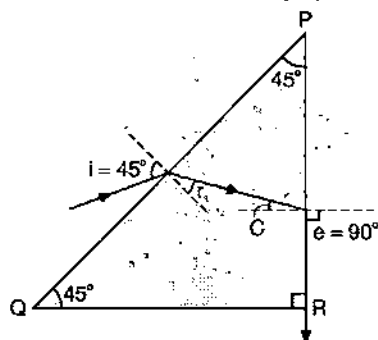
$$A > 2C$$

$$\Rightarrow A > 2 \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$\Rightarrow A > 2 \sin^{-1}\left(\frac{1}{1.5}\right) > 83.62^\circ$$

Therefore, $A_{\text{max}} = 83.62^\circ$, for escaping of the ray through the adjacent face.

5. The situation is shown in figure



- (a) Since, $e = 90^\circ$

$$\text{Also, } r_2 = C = \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$\text{Now, } i = A = 45^\circ \text{ and } r_1 = A - r_2 = 45^\circ - C$$

Applying Snell's Law at AB, we get

$$\mu = \frac{\sin i}{\sin r_1} = \frac{\sin(45^\circ)}{\sin(45^\circ - C)}$$

$$\Rightarrow \mu = \frac{\sin(45^\circ)}{\sin(45^\circ) \cos C - \cos(45^\circ) \sin C}$$

$$\text{Since } \sin C = \frac{1}{\mu}$$

$$\Rightarrow \cos C = \sqrt{1 - \frac{1}{\mu^2}}$$

$$\Rightarrow \mu \sqrt{1 - \frac{1}{\mu^2}} - 1 = 1$$

$$(\sin 45^\circ = \cos 45^\circ)$$

$$\Rightarrow \mu^2 - 1 = 4$$

$$\Rightarrow \mu = \sqrt{5}$$

- (b) At minimum deviation, we have

$$r_1 = r_2 = \frac{A}{2} = 22.5^\circ$$

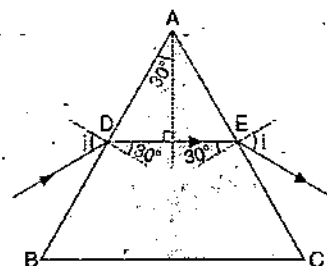
$$\text{Since, } \sqrt{5} = \frac{\sin i}{\sin(22.5^\circ)}$$

$$\Rightarrow i = 58.8^\circ$$

6. (a) Applying Snell's law at D, we get

$$\left(\frac{4}{3}\right) \sin i = \left(\frac{3}{2}\right) \sin 30^\circ$$

$$\Rightarrow i = 34.2^\circ$$



- (b) Total deviation suffered by the ray is

$$\delta = \delta_D + \delta_E = 2\delta_D$$

$$\Rightarrow \delta = 2(i - 30^\circ) = 8.4^\circ$$

7. At near normal incidence, $i \approx r_1 = 0^\circ$

$$\text{Since } r_1 + r_2 = A$$

$$\Rightarrow r_2 = A$$

From Snell's Law applied at the face from where the refracted ray emerges, we get

$$\mu = \frac{\sin e}{\sin r_2}$$

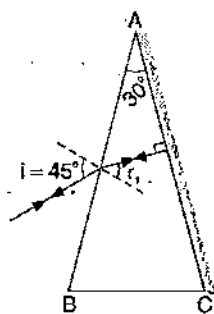
$$\Rightarrow e = \sin^{-1}(\mu \sin A)$$

$$\text{Now, deviation } \delta = i + e - A = \sin^{-1}(\mu \sin A) - A$$

$$\Rightarrow \delta = \sin^{-1}(\mu \sin A) - A \quad \{\because i = 0^\circ\}$$

8. For the ray to retrace its path, it must be incident normally to the face AC. So, we have

$$r_2 = 0^\circ$$



Since $r_1 + r_2 = A$

$$\Rightarrow r_1 = A = 30^\circ$$

From Snell's Law, we have

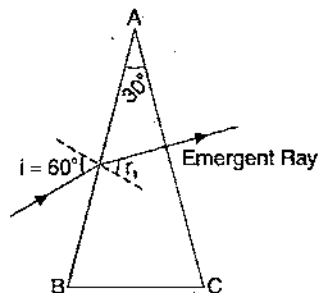
$$\mu = \frac{\sin i}{\sin r_1} = \sqrt{2}$$

9. From the statement of the problem, we gather the information that

$$i = 60^\circ, A = 30^\circ, \delta = 30^\circ$$

$$\text{Since, } \delta = i + e - A$$

$$\Rightarrow e = \delta - i + A = 30^\circ - 60^\circ + 30^\circ = 0^\circ$$



i.e., the emergent ray is perpendicular to the face through which it emerges.

Further, $r_2 = 0$ and $r_1 + r_2 = A$

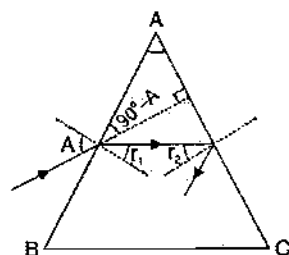
$$\Rightarrow r_1 = A = 30^\circ \quad \text{(as } e = 0)$$

From Snell's Law applied at face AC, we get

$$\mu = \frac{\sin i}{\sin r_1} = \sqrt{3}$$

10. From geometry, we observe that the angle of incidence at the face AB is A. Applying Snell's Law at face AB, we get

$$\mu = \frac{\sin A}{\sin r_1} \quad \dots(1)$$



If C is the critical angle of the prism, then

$$\mu = \frac{1}{\sin C} \quad \dots(2)$$

From (1) and (2), we get

$$\frac{\sin A}{\sin r_1} = \frac{1}{\sin C}$$

$$\Rightarrow \sin r_1 = \sin A \sin C \quad \dots(3)$$

The ray does not emerge from the other face AC, when

$$r_2 > C$$

Since, $r_1 + r_2 = A$

$$\Rightarrow A - r_1 > C$$

$$\Rightarrow r_1 < A - C$$

$$\Rightarrow \sin r_1 < \sin(A - C)$$

$$\Rightarrow \sin A \sin C < \sin A \cos C - \sin C \cos A$$

$$\Rightarrow 1 < \cot C - \cot A$$

$$\Rightarrow \cot A < \cot C - 1$$

11. For Violet Light : According to Snell's Law, applied at the plane of incidence, we get

$$\mu = \frac{\sin i}{\sin r_1}$$

$$\Rightarrow 1.66 = \frac{\sin(50^\circ)}{\sin r_1}$$

$$\Rightarrow r_1 = 27.5^\circ$$

Since, $r_1 + r_2 = A$

$$\Rightarrow r_2 = A - r_1 = 32.5^\circ$$

Applying Snell's Law at the plane of emergence, we get

$$\mu = \frac{\sin e}{\sin r_2}$$

$$\Rightarrow 1.66 = \frac{\sin e}{\sin(32.5^\circ)}$$

$$\Rightarrow e = 63.1^\circ$$

Since, $\delta_v = (i + e) - A$

$$\Rightarrow \delta_v = 53.1^\circ$$

For Red Light : According to Snell's Law, applied at the plane of incidence, we get

$$1.62 = \frac{\sin(50^\circ)}{\sin r_1}$$

$$\Rightarrow r_1 = 28.2^\circ$$

Since, $r_1 + r_2 = A$

$$\Rightarrow r_2 = A - r_1 = 31.8^\circ$$

Applying Snell's Law at the plane of emergence, we get

$$1.62 = \frac{\sin e}{\sin(31.8^\circ)}$$

$$\Rightarrow e = 58.6^\circ$$

Since, $\delta_R = (i + e) - A = 48.6^\circ$

So, angular dispersion is given by

$$\delta_v - \delta_R = 4.5^\circ$$

12. For minimum deviation, we have

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow 1.3 \sin(45^\circ) = \sin(45^\circ + \delta_m)$$

Solving, this we get

$$\delta_m = 22^\circ$$

For maximum deviation, we have the emergent ray to be grazing on the surface of emergence. So,

$$e = 90^\circ, r_2 = C \text{ and } r_1 = 90^\circ - r_2 = 90^\circ - C$$

$$\Rightarrow \delta_{\max} = i + e - A$$

We can find i by using $\mu = \frac{\sin i}{\sin r_1}$ and $\sin C = \frac{1}{\mu}$. Substituting

the values, we get

$$\delta_{\max} = 56^\circ$$

13. Given that, $i = 60^\circ$, $A = 30^\circ$ and $\delta = 30^\circ$

Since, $\delta = i + e - A$

Substituting the values we get, $e = 0^\circ$

Now, $e = 0^\circ$, means that the emergent ray is normal to the face through which it emerges.

14. Given that $A = 30^\circ$ and $i = 0^\circ$, so $r_1 = 0^\circ$

Since, $r_1 + r_2 = A$

$$\Rightarrow r_2 = A = 30^\circ$$

Further applying Snell's Law at the plane of emergence, we get

$$1.5 = \frac{\sin e}{\sin r_2}$$

Substituting the values, we get

$$e = 49^\circ$$

$$\Rightarrow \delta = i + e - A = 19^\circ$$

15. For no total internal reflection, when the ray leaves the prism,

$$r_2 = C$$

$$\text{But } \sin C = \frac{1}{\mu} = \frac{1}{1.6}$$

$$r_2 = C = \sin^{-1}\left(\frac{1}{1.6}\right) = 38.7^\circ$$

Further $r_1 + r_2 = A = 45^\circ$

$$\Rightarrow r_1 = 45^\circ - 38.7^\circ = 6.3^\circ$$

$$\text{Now, } i = \sin^{-1}(\mu \sin r_1) = \sin^{-1}[1.6 \times \sin(6.3^\circ)] = 10.1^\circ$$

16. For red ray, we have

$$e = 0^\circ = r_2$$

Since, $r_1 + r_2 = A$

$$\Rightarrow r_1 = A = 45^\circ$$

Now, according to Snell's Law, we have

$$1.37 = \frac{\sin i}{\sin(45^\circ)}$$

$$\Rightarrow i = 75.6^\circ$$

$$\Rightarrow \delta_{\text{red}} = i + e - A = 30.6^\circ$$

For violet ray, we have

$$1.42 = \frac{\sin(75.6^\circ)}{\sin r_1}$$

$$\Rightarrow r_1 = 43^\circ$$

$$\Rightarrow r_2 = A - r_1 = 2^\circ$$

$$\{\because r_1 + r_2 = A\}$$

Again, applying Snell's Law at the emerging face, for violet rays, we get

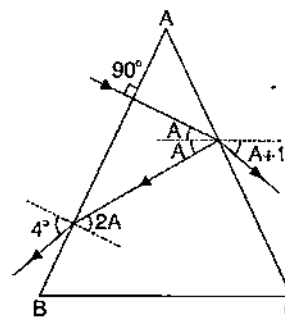
$$1.42 = \frac{\sin e}{\sin r_2}$$

$$\Rightarrow e = 2.84^\circ$$

$$\Rightarrow \delta_{\text{violet}} = i + e - A = 33.4^\circ$$

17. As the angles are small we can take,

$$\sin \theta \approx \theta$$



Applying Snell's Law for the two emerging rays at AC and AB, we get

$$\mu = \frac{\sin(A + 1^\circ)}{\sin A} = \frac{\sin(4^\circ)}{\sin(2A)}$$

$$\Rightarrow \mu \approx \frac{A + 1^\circ}{A} = \frac{4^\circ}{2A}$$

$$\Rightarrow A = 1^\circ \text{ and } \mu = 2$$

Solutions

In Chapter Exercise (ICE)-VI

(based on refraction at curved surfaces)

1. Let us see where do the parallel rays converge (or diverge) on the principal axis. Let us call it the focus and the corresponding length the focal length f . Using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ with appropriate values and signs, we get

$$\frac{4}{f} - \frac{1}{\infty} = \frac{4-1}{10}$$

$$\Rightarrow f = 40 \text{ cm} = 0.4 \text{ m}$$

Since, the rays are converging, its power should be positive. Hence,

$$P \text{ (in dioptre)} = \frac{1}{f \text{ (metre)}} = \frac{1}{0.4}$$

$$\Rightarrow P = 2.5 \text{ dioptre} = 2.5 \text{ D}$$

2. For first refraction at the unsilvered surface, we have

$$\frac{1.5}{v_1} - \frac{1}{(-2r)} = \frac{1.5-1}{r}$$

$$\Rightarrow v_1 \rightarrow \infty$$

i.e., rays become parallel to the principal axis.

Hence the image formed by the curved mirror will lie at the focus of the mirror i.e., a distance $\frac{r}{2}$ from pole of mirror.

$$\text{So, } v_2 = \frac{r}{2} \quad \text{(from pole of the mirror)}$$

For second refraction at the unsilvered surface, we have

$$\frac{1}{v_3} - \frac{1.5}{\left(-\frac{3r}{2}\right)} = \frac{1-1.5}{(-r)}$$

$$\Rightarrow v_3 = -2r$$

i.e., final image is formed at pole of the mirror.

3. For the image of object O to be formed at O , the light should fall normally on mirror. First image I_1 (after refraction from the plane surface) will be formed at a distance of $\frac{2R+x}{\mu}$ from plane

surface, because $d_{\text{app}} = \frac{d_{\text{actual}}}{\mu}$.

Now applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ with the idea that I_2 is formed at C , because light falls normally on the mirror.

$$\Rightarrow \frac{\mu}{-(R+x)} - \frac{1}{-\left(\frac{2R+x}{\mu} + R\right)} = \frac{\mu-1}{-R}$$

Solving this equation for $\mu = 1.5$, we get
 $x \approx 0.75R$

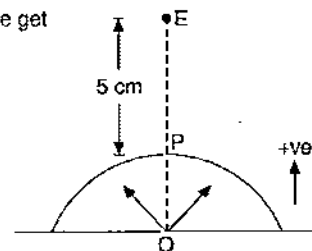
4. Applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

$$\frac{1}{PI} - \frac{1.6}{(-3)} = \frac{1-1.6}{-5}$$

$$\Rightarrow PI = -2.42 \text{ cm}$$

$$\Rightarrow EI = (5 + 2.42) \text{ cm}$$

$$\Rightarrow EI \approx 7.42 \text{ cm}$$



5. (a) Applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

$$\frac{1.5}{v} - \frac{1}{-1} = \frac{1.5-1}{6}$$

$$\Rightarrow v = -90 \text{ cm}$$

So, the distance between object and its image is 80 cm

- (b) Again applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\Rightarrow \frac{1.5}{v} - \frac{1}{(-u)} = \frac{1.5-1}{6}$$

$$\Rightarrow v = \frac{18}{\left(1 - \frac{12}{u}\right)}$$

v is negative when $\frac{12}{u} > 1$

$$\Rightarrow u < 12 \text{ cm}$$

6. Applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ twice, we get

$$\frac{1.5}{v_1} - \frac{1}{(-2.5)} = \frac{1.5-1}{+10}$$

$$\Rightarrow v_1 = -\frac{30}{7} \text{ cm}$$

$$\text{Further, } \frac{1}{v_2} - \frac{1.5}{\left(20 + \frac{30}{7}\right)} = \frac{1-1.5}{-10}$$

$$\Rightarrow v_2 = -85 \text{ cm}$$

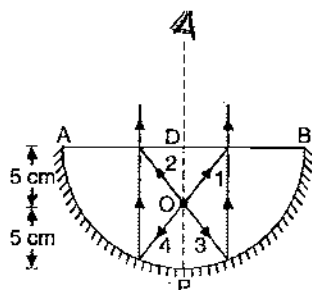
i.e., final image is formed at 85 cm from first face on the same side of the object.

7. First image will be formed by direct rays 1 and 2, etc.

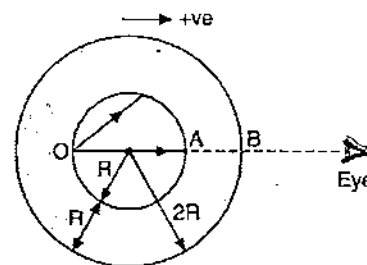
$$DI_1 = \frac{DO}{\mu} = \frac{5}{1.5} = 3.33 \text{ cm}$$

Second image will be formed by reflected rays 3 and 4, etc.

Object is placed at the focus of the mirror. Hence, I_2 is formed at infinity.



8. We have to see the image of O from the other side



Applying, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ twice, we get

$$\frac{\mu}{AI_1} - \frac{1}{(-2R)} = \frac{\mu-1}{-R}$$

$$\Rightarrow AI_1 = \frac{2\mu R}{1-2\mu}$$

$$\text{Further, } \frac{1}{BI_2} - \frac{\mu}{(AI_1 - R)} = \frac{1-\mu}{-R}$$

Solving this equation, we get

$$BI_2 = -\frac{2R(4\mu-1)}{3\mu-1}$$

So, the distance between the final image and the object is

$$d = 3R - \frac{2R(4\mu-1)}{3\mu-1} = \frac{(\mu-1)R}{(3\mu-1)}$$

Solutions

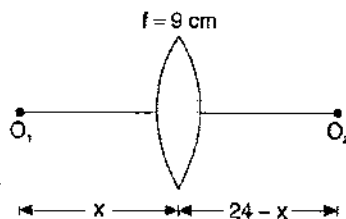
In Chapter Exercise (ICE)-VII

(based on lens formula)

1. When the images of both the sources are formed at the same point, then v will be same for both (in value). However for one case the image will be real and for the other case it will be virtual. So,

For O_1 , we have $\frac{1}{v} + \frac{1}{x} = \frac{1}{9}$... (1)

For O_2 , we have $-\frac{1}{v} + \frac{1}{24-x} = \frac{1}{9}$... (2)



Adding equations (1) and (2), we get

$$\frac{1}{x} + \frac{1}{24-x} = \frac{2}{9}$$

Solving we get $x = 6$ cm

Hence, the lens should be kept at a distance of 6 cm from either of the object.

2. Here the image formed can be virtual as well as real, so the value of m should be $+3$ in one case (virtual image) and -3 in the other (real image). Magnification of $+3$ can be obtained only when the object is placed within F (i.e., for smaller value of the object distance). The magnification of -3 is obtained when the object is kept between F and $2F$ (i.e., for greater value of object distance). So,

$$m = +3, \text{ for } u = -8 \text{ cm}$$

Therefore, from the definition of magnification, we have

$$m = \frac{v}{u}$$

$$\Rightarrow +3 = \frac{v}{-8}$$

$$\Rightarrow v = 3 \times (-8) = -24 \text{ cm}$$

Using lens formula, we get

$$\frac{1}{-24} - \frac{1}{-8} = \frac{1}{f}$$

$$\Rightarrow \frac{-1+3}{24} = \frac{1}{f}$$

$$\Rightarrow f = 12 \text{ cm}$$

Note that you would get the same answer by considering the other case ($m = -3$, for $u = -16$ cm)

3. The Lens Maker's Formula is given by

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Here, $f = 30$ cm, $R_1 = 10$ cm, $R_2 = \infty$

$$\Rightarrow \frac{1}{30} = (\mu - 1) \left(\frac{1}{10} - \frac{1}{\infty} \right) = \frac{\mu - 1}{10}$$

$$\Rightarrow 30\mu - 30 = 10$$

$$\Rightarrow \mu = \frac{4}{3}$$

4. Applying Lens Maker's Formula,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Here, $\mu = 1.5$, $f = 60$ cm, $R_1 = +R$, $R_2 = -R$

$$\Rightarrow \frac{1}{60} = (1.5 - 1) \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$\Rightarrow R = 60 \text{ cm}$$

Therefore, the focal length of the spherical silvered surface, is given by

$$f_m = \frac{R}{2} = \frac{60}{2} = +30 \text{ cm}$$

(Positive, because it is a converging mirror)

The equivalent focal length of the lens-mirror combination is then given by

$$-\frac{1}{F} = \frac{2}{f} - \frac{1}{f_m} = \frac{2}{60} - \frac{1}{(-30)}$$

$$\Rightarrow F = -15 \text{ cm}$$

The negative sign indicates that the combination behaves as a concave mirror.

5. According to the Lens Maker's Formula, we have

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \text{ where } f = 10 \text{ cm} \quad \dots (1)$$

When placed in medium 1, then

$$\frac{1}{f_1} = \left(\frac{\mu_1}{\mu_2} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(2)$$

When placed in medium 2, then

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(3)$$

From equations (1) and (2), we get

$$\frac{f_1}{f_2} = \left(\frac{\mu_1}{\mu_2} - 1 \right) \left(\frac{1.6 - 1}{1.5 - 1} \right) = 9$$

$$\Rightarrow f_1 = 9f_2 = 90 \text{ cm}$$

i.e., still it behaves as a converging lens

From equations (1) and (3), we get

$$\frac{f_2}{f_1} = \left(\frac{\mu_2}{\mu_1} - 1 \right) \left(\frac{1.6 - 1}{1.7 - 1} \right) = -10.2$$

$$\Rightarrow f_2 = -10.2f_1 = -102 \text{ cm}$$

i.e., it now behaves like a diverging lens.

6. Here $R_1 = +25 \text{ cm}$, $R_2 = -25 \text{ cm}$, $\mu_1 = 1$ and $\mu_2 = \frac{3}{2}$

Image coincides with object, hence, $u = v = -x$ (say) $\rightarrow +ve$

$$\frac{1}{-x} - \frac{1}{x} = \frac{2(3/2)}{-25} - \frac{2(3/2-1)}{25}$$

$$\Rightarrow \frac{2}{x} = \frac{3}{25} + \frac{1}{25} = \frac{4}{25}$$

$$\Rightarrow x = 12.5 \text{ cm}$$

Hence, the object should be placed at a distance 12.5 cm in front of the silvered lens.



7. Given, $f_1 = +5 \text{ cm}$ and $f_2 = -10 \text{ cm}$

The combined focal length F is given by

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{5} - \frac{1}{10} = +\frac{1}{10}$$

$$\Rightarrow F = +10 \text{ cm}$$

i.e., the combination behaves as a converging lens of focal length 10 cm.

8. According to Lens Maker's Formula, we have

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{10} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{-10} \right)$$

$$\Rightarrow R_1 = +10 \text{ cm}$$

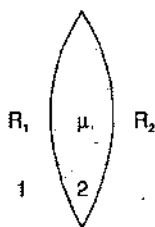
Now, using, $\frac{1}{v} + \frac{1}{u} = \frac{2(\mu_2/\mu_1)}{R_2} - \frac{2(\mu_2/\mu_1 - 1)}{R_1}$

Substituting the values, we get

$$\frac{1}{v} - \frac{1}{15} = \frac{2(1.5)}{-10} - \frac{2(1.5 - 1)}{+10}$$

$$\Rightarrow v = -2.14 \text{ cm}$$

9. Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get



$$\frac{1}{36} - \frac{1}{-45} = \frac{1}{f}$$

$$\Rightarrow f = 20 \text{ cm}$$

In the second case, let μ be the refractive index of the liquid, then

$$\frac{1}{48} - \frac{1}{\left(5 + \frac{40}{\mu} \right)} = \frac{1}{20}$$

Solving, we get

$$\mu = 1.37$$

10. The system behaves like a mirror of focal length given by,

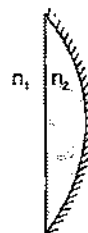
$$\frac{1}{F} = \frac{2(\mu_2/\mu_1)}{R_2} - \frac{2(\mu_2/\mu_1 - 1)}{R_1}$$

Substituting the values with appropriate signs, we get

$$\frac{1}{F} = \frac{2\left(\frac{4}{3}\right)}{-20} \quad \{\because R_1 \rightarrow \infty\}$$

$$\Rightarrow F = -7.5 \text{ cm}$$

So, the system behaves as a concave mirror of focal length 7.5 cm.



11. For a convex lens, the distance between an object and its real image is minimum when $u = 2f_1$ and $v = 2f_1$

When concave lens is placed in contact, then we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_{\text{comb}}}$$

$$\frac{1}{v'} + \frac{1}{2f_1} = \frac{1}{f_1} - \frac{1}{f_2} = \frac{f_2 - f_1}{f_1 f_2}$$

$$\Rightarrow v' = \frac{2f_1 f_2}{f_2 - 2f_1}$$

Shift of image is

$$\Delta v = v' - v = \frac{2f_1 f_2}{f_2 - 2f_1} - 2f_1 = \frac{4f_1^2}{f_2 - 2f_1}$$

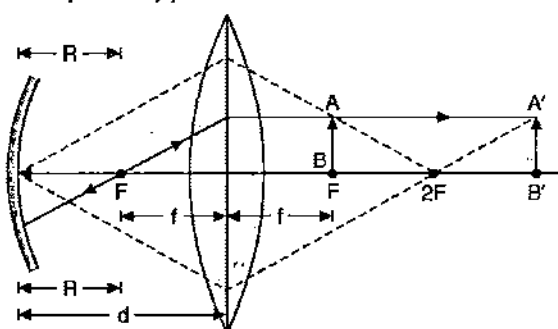
Since, $f_2 \gg f_1$, so the shift of image is

$$\Delta v \approx \frac{4f_1^2}{f_2}$$

12. The following two cases are possible.

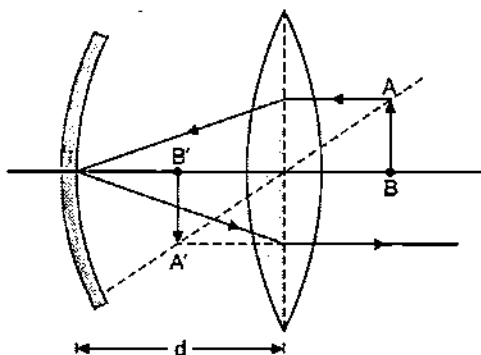
CASE - I :

The mirror is at a distance of $d = f + R = 2 \text{ m}$ from the lens. The path of the beam parallel to the optical axis of the system and the image of object AB are shown in figure. Image A'B' (direct and real) is obtained to full scale with the object in any position.



CASE - II :

The mirror is at a distance of $d = f = R = 1$ m from the lens. The image of object $A'B'$, also full scale, will be inverted and virtual with the object in any position.



13. For a convex lens, distance between an object and its real image is minimum when $u = 2f$. Hence,

$$\begin{aligned} \text{(a)} \quad 3f_1 - \left(1 - \frac{1}{\mu}\right)t &= 2f_1 \\ \Rightarrow \left(1 - \frac{1}{\mu}\right)t &= f_1 \\ \Rightarrow \frac{1}{\mu} &= 1 - \frac{f_1}{t} = \frac{t - f_1}{t} \\ \Rightarrow \mu &= \frac{t}{t - f_1} \end{aligned}$$

- (b) If a concave lens of very large focal length f_2 (i.e., very small power) is placed in contact with the convex lens, then its power and hence the focal length are almost unaltered. Therefore, there will be no shifting of the image.

14. Using the lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

for the first lens

$$\begin{aligned} \frac{1}{v_1} + \frac{1}{30} &= \frac{1}{20} \\ \Rightarrow v_1 &= 60 \text{ cm} \end{aligned}$$

For the second lens

$$\begin{aligned} \frac{1}{v_2} - \frac{1}{30} &= \frac{1}{10} \\ \Rightarrow v_2 &= 7.5 \text{ cm} \end{aligned}$$

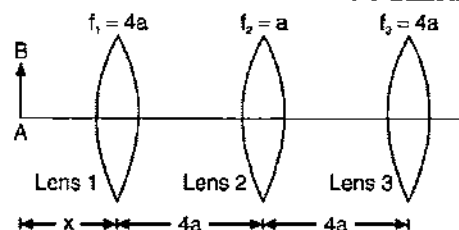
15. Since, $\frac{1}{F} = \frac{2\left(\frac{\mu_2}{\mu_1}\right)}{R_2} - \frac{2\left(\frac{\mu_2}{\mu_1} - 1\right)}{R_1} = \frac{2\mu}{-r} - \frac{2(\mu - 1)}{r}$

$$\begin{aligned} \Rightarrow \frac{1}{F} &= -\left(\frac{4\mu - 2}{r}\right) \\ \Rightarrow F &= -\left(\frac{r}{4\mu - 2}\right) \end{aligned}$$

i.e., the lens is equivalent to a concave mirror of focal length $\frac{r}{4\mu - 2}$

16. Let x be the distance between first lens and the object AB.

Applying the lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ thrice, we get for lens 1,



$$\frac{1}{v_1} + \frac{1}{x} = \frac{1}{4a} \quad \dots(1)$$

$$\Rightarrow v_1 = \frac{4ax}{x - 4a}$$

$$\Rightarrow m_1 = \frac{v}{u} = \frac{v_1}{-x} = \frac{4a}{4a - x} \quad \dots(2)$$

Similarly for Lens 2, we get

$$\frac{1}{v_2} - \frac{1}{v_1 - 4a} = \frac{1}{a}$$

$$\Rightarrow \frac{1}{v_2} - \frac{x - 4a}{16a^2} = \frac{1}{a} \quad \dots(3)$$

$$\Rightarrow v_2 = \frac{16a^2}{x + 12a}$$

$$\Rightarrow m_2 = \frac{\left(\frac{16a^2}{x + 12a}\right)}{\left(\frac{16a^2}{x - 4a}\right)} = \frac{x - 4a}{x + 12a} \quad \dots(4)$$

Similarly for lens 3, we get

$$\frac{1}{v_3} - \frac{1}{v_2 - 4a} = \frac{1}{4a} \quad \dots(5)$$

$$\Rightarrow v_3 = -(x + 8a)$$

$$\Rightarrow m_3 = \frac{x + 12a}{4a} \quad \dots(6)$$

So, we observe from (2), (4) and (6) that

$$m_1 m_2 m_3 = -1$$

Also, $v_3 = -(x + 8a)$ means that the object and its image lie at the same place and $m_1 m_2 m_3 = -1$ means final image is of the same size but inverted.

17. According to the lens formula, we have, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Since $u = -mf$, so we get

$$\begin{aligned} \frac{1}{v} + \frac{1}{mf} &= \frac{1}{(-f)} \\ \Rightarrow v &= -\left(\frac{m}{m + 1}\right)f \end{aligned}$$

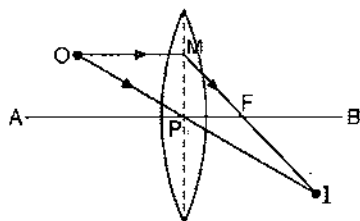
So, lateral magnification is given by

$$m_L = \frac{v}{u} = \frac{1}{m + 1}$$

Hence, the image will be $(m + 1)$ times smaller than the object.

18. (a) Since a concave lens always forms an erect image whereas the given image I is on the other side of the optic axis, so the lens is convex.

- (b) Join O with I. Line OI cuts the optic axis AB at pole (P) of the lens. The dotted line shows the position of lens.



From point O, draw a line parallel to AB which after refraction must pass through the focus F of the lens.

19. Let parallel rays be incident on first lens, then

$$v_1 = f_1 = 7 \text{ cm}$$

For the second lens, we have

$$\frac{1}{v_2} - \frac{1}{(7-3)} = \frac{1}{6}$$

$$\Rightarrow \frac{1}{v_2} - \frac{1}{4} = \frac{1}{6}$$

$$\Rightarrow v_2 = 2.4 \text{ cm}$$

20. (a) From Lens Maker's Formula, we get

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f} = \frac{\mu_2 - 1}{\mu_1 - 1} \cdot \frac{1.7 - 1}{1.5 - 1} = \frac{7}{5} = 1.4$$

- (b) In this liquid the first lens will be a diverging (as refractive index of liquid > 1.5) and the second a converging one (as refractive index of liquid < 1.7).

21. Since the incident beam is parallel, so we have

$$v_1 = f_1 = 10 \text{ cm}$$

For the second lens, we have

$$\frac{1}{v_2} - \frac{1}{-5} = \frac{1}{-20}$$

$$\Rightarrow v_2 = -4 \text{ cm}$$

For the third lens, we have

$$\frac{1}{v_3} - \frac{1}{-9} = \frac{1}{9}$$

$$\Rightarrow v_3 \rightarrow \infty$$

i.e., rays will become parallel to the optic axis.

22. Since the focal length is equal to two times the radius of curvature, so $f = -2r$

$$\frac{1}{(-2r)} = (\mu_0 + a1 - 1) \left(\frac{1}{\infty} - \frac{1}{r} \right)$$

$$\Rightarrow I = \frac{3 - 2\mu_0}{2a}$$

23. The rays will first get refracted, then n -times reflected and finally again refracted. So, using $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ for first refraction, we get

$$\frac{\mu}{v_1} - \frac{1}{\infty} = \frac{\mu - 1}{R}$$

$$\Rightarrow v_1 = \left(\frac{\mu}{\mu - 1} \right) R$$

For first reflection, let us use the mirror formula, i.e.,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$$

$$\Rightarrow \frac{1}{v_1} + \left(\frac{\mu - 1}{\mu R} \right) = \frac{-2}{R}$$

$$\Rightarrow \frac{1}{v_1} = - \left(\frac{3\mu - 1}{\mu R} \right)$$

For second reflection, similarly we get

$$\frac{1}{v_2} + \frac{3\mu - 1}{\mu R} = \frac{-2}{R}$$

$$\Rightarrow \frac{1}{v_2} = - \left(\frac{5\mu - 1}{\mu R} \right)$$

Similarly after n th reflections, we get

$$\frac{1}{v_n} = - \left[\frac{(2n + 1)\mu - 1}{\mu R} \right]$$

Finally again using the refraction formula, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, applied at the second curved surface, we get

$$\frac{1}{v_1} - \left[\frac{(2n + 1)\mu - 1}{R} \right] = \frac{1 - \mu}{-R}$$

$$\Rightarrow v_1 = \frac{R}{2(\mu n + \mu - 1)}$$

$$24. \frac{1}{f_1} = \frac{2 \left(\frac{\mu_2}{\mu_1} \right)}{R_2} - \frac{2 \left(\frac{\mu_2}{\mu_1} - 1 \right)}{R_1}$$

$$\Rightarrow \frac{1}{f_1} = \frac{2\mu}{\infty} - \frac{2(\mu - 1)}{R}$$

$$\Rightarrow f_1 = \frac{-R}{2(\mu - 1)}$$

So, the system behaves as a concave mirror of focal length $\frac{R}{2(\mu - 1)}$. The object will coincide with image when the object is placed at centre of curvature. So, we get

$$x_1 = 2|f_1| = \frac{R}{\mu - 1} \quad \dots(1)$$

In the second case, $\frac{1}{f_2} = \frac{-2\mu}{R}$

$$\Rightarrow f_2 = -\frac{R}{2\mu}$$

$$\Rightarrow x_2 = 2|f_2| = \frac{R}{\mu} \quad \dots(2)$$

Solving equations (1) and (2), we get

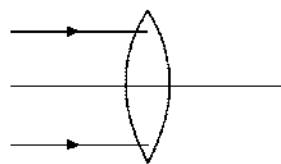
$$\mu = \frac{x_1}{x_1 - x_2}$$

$$\text{and } R = \frac{x_1 x_2}{x_1 - x_2}$$

Now, according to Lens Maker's Formula, we have

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} \right) = \frac{\mu - 1}{R} = \frac{1}{x_1} \quad \{\text{from equation (1)}\}$$

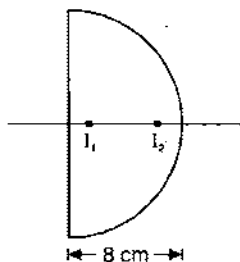
$$\Rightarrow f = x_1$$



Solutions

Single Correct Choice Type Questions

1. [C]



Distance of image from the plane surface is

$$x_1 = \frac{4}{1.6} = 2.5 \text{ cm}$$

$$\left\{ \because d_{\text{app}} = \frac{d_{\text{actual}}}{\mu} \right\}$$

For the curved surface, we have

$$\frac{1.6}{4} + \frac{1}{x_2} = \frac{1 - 1.6}{-8}$$

$$\Rightarrow x_2 \approx -3 \text{ cm}$$

The minus sign means the image is on the side where the object lies. So,

$$I_1 I_2 = (8 - 2.5 - 3) \text{ cm} = 2.5 \text{ cm}$$

2. [C]

Area of object = 9 cm²

Also, we know that

$$\text{Areal Magnification} = m_{\text{ar}} = \frac{A_t}{A_o} = \frac{v^2}{u^2} = \left(\frac{f}{f-u} \right)^2$$

$$\Rightarrow \frac{A_t}{9} = \left[\frac{-10}{-25 - (-10)} \right]^2$$

$$\Rightarrow A_t = \left(\frac{2}{3} \right)^2 \times 9$$

$$\Rightarrow A_t = 4 \text{ cm}^2$$

3. [A]

The similar thing is extended and applied here too. Here the answer fabricated by the MISCONCEPTION is 1 (but we must know this is the answer only for a Concave Mirror (or Convex Lens). For Convex Mirror we have

$$m = \frac{f}{f-u}$$

$$\Rightarrow m = \frac{f}{f - (-2f)}$$

$$\Rightarrow m = \frac{1}{3}$$

4. [D]

Only one image will be formed by this lens system, because the optic axis of both the parts coincide. Two images would have been formed if their optic axis would have been different.

5. [B]

MISCONCEPTION

Maximum students are gripped by the misconception that when object lies at focus then image is formed at infinity. This is true but only for a Concave Mirror (Convex Lens) and is absolutely wrong when applied to Convex Mirror (Concave Lens)

How To Proceed Then?

If object placed at focus then just check out the MIRROR. If CONCAVE then image is formed at infinity and if Convex then

$$\text{apply } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{20} = \frac{1}{+20}$$

$$\Rightarrow v = 10 \text{ cm}$$

6. [C]

Image will be formed at infinity, when the object is placed at focus of the lens i.e., at 20 cm from the lens. So, we have

$$\text{Shift } \Delta x = 25 - 20 = \left(1 - \frac{1}{\mu} \right) t$$

$$\Rightarrow 5 = \left(1 - \frac{1}{1.5} \right) t$$

$$\Rightarrow t = \frac{5 \times 1.5}{0.5} = 15 \text{ cm}$$

7. [C]

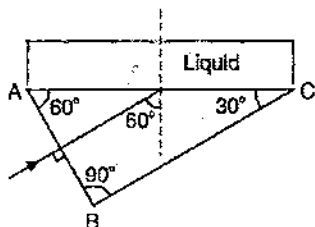
Critical angle between glass and liquid interface is

$$\sin C = \frac{\mu}{3/2} = \frac{2\mu}{3}$$

Angle of incidence at face AC is 60°

For TIR to take place, we have

$$i > C$$



$$\Rightarrow \sin(60^\circ) > \frac{2\mu}{3}$$

$$\Rightarrow \mu < \frac{3\sqrt{3}}{4}$$

8. [D]

Since no parallax exists between the images formed by two mirrors (convex and plane) hence the images for both coincide. But for a plane mirror an image is as far behind the mirror as the object is in front of it. Hence the image for plane mirror should be 30 cm behind it or 10 cm behind the convex mirror. So for convex mirror,

$$u = -50 \text{ cm}, v = +10 \text{ cm}$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{10} + \frac{1}{-50} = \frac{1}{f}$$

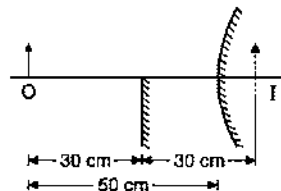
$$\Rightarrow \frac{1}{f} = \frac{5-1}{50}$$

$$\Rightarrow f = \frac{50}{4}$$

$$\Rightarrow f = 12.5 \text{ cm}$$

$$\text{Since } R = 2f$$

$$\Rightarrow R = 25 \text{ cm}$$



9. [D]

Concave lens forms the virtual image of a real object. So, we have

$$m = -\frac{v}{u} = \frac{1}{4}. \text{ Now, if}$$

$$u = -4x, \text{ then } v = -x \text{ then } 3x = 10 \text{ cm}$$

$$\Rightarrow x = \frac{10}{3} \text{ cm}$$

$$\Rightarrow u = -\frac{40}{3} \text{ cm}$$

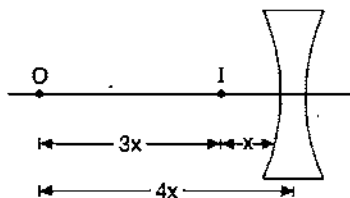
$$\text{and } v = -\frac{10}{3} \text{ cm}$$

$$\text{Substituting in } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \text{ we get}$$

$$\frac{1}{f} = \frac{-3}{10} + \frac{3}{40}$$

$$\Rightarrow f = -\frac{40}{9}$$

$$\Rightarrow f = -4.4 \text{ cm}$$



10. [C]

$$|m_{\text{real}}| = 2|m_{\text{virtual}}|$$

$$\Rightarrow m_{\text{real}} = -2m_{\text{virtual}}$$

$$\Rightarrow \frac{f}{f - (-15)} = -2 \frac{f}{f - (-20)}$$

$$\Rightarrow f + 20 = 2f + 30$$

$$\Rightarrow f = -10 \text{ cm}$$

11. [B]

$$m = +\frac{1}{2}$$

$$\Rightarrow \frac{1}{2} = \frac{20}{20 - u}$$

$$\Rightarrow 40 = 20 - u$$

$$\Rightarrow u = -20 \text{ cm}$$

12. [B]

Focal length of mirror is independent of the refractive index of medium in which it is placed.

13. [A]

Since image formed is erect, hence it must be virtual. So,

$$m = \frac{v}{u} = 3$$

$$\text{Also } |u| + |v| = D$$

$$|u| + 3|u| = 80$$

$$|u| = 20$$

Since object always lies on negative side. So,

$$u = -20 \text{ cm}$$

$$\Rightarrow 3 = \frac{f}{f - (-20)}$$

$$\Rightarrow 3f + 60 = f$$

$$\Rightarrow f = -30 \text{ cm}$$

Negative sign indicates the mirror is concave.

An Advice

I would always advice you to write $|u| + |v| = D$ wherever you are given the distance between object and image as no error will creep in these because the MOD signs prevent the errors.

15. [B]

$$\text{Since, } \delta = (\mu - 1)A = (1.5 - 1)(4) = 2^\circ$$

$$\Rightarrow i = \delta = 2^\circ$$

Let the mirror be rotated by an angle θ , then

$$i' = (2^\circ + \theta)$$

$$\text{Since, } \delta_{\text{total}} = 180^\circ$$

$$\Rightarrow \delta + 180^\circ - 2i' = 180^\circ$$

$$\Rightarrow \delta = 2i'$$

$$\Rightarrow 2^\circ = 2(2^\circ + \theta)$$

$$\theta = -2^\circ$$

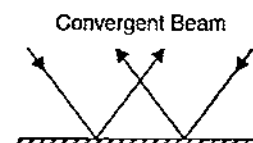
Here, negative sign implies that i gets decreased or $i' = 0$. i.e., light should fall normally on mirror.

16. [B]

The minimum length of the mirror required for the purpose is half the height of the person.

17. [B]

A divergent beam appears to converge behind the mirror thus giving a virtual image. So, a convergent beam will give a real image.



19. [C]

Applying $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{b} + \frac{1}{a} = \frac{1}{f}$$

$$\Rightarrow f = \frac{ab}{a+b}$$

...(1)

Further in right triangle ACB, we have

$$AC^2 + BC^2 = AB^2$$

$$\Rightarrow (a^2 + c^2) + (b^2 + c^2) = (a+b)^2$$

$$\Rightarrow a^2 + b^2 + 2c^2 = a^2 + b^2 + 2ab$$

$$\Rightarrow ab = c^2$$

Substituting this in equation (1), we get

$$f = \frac{c^2}{a+b}$$

20. [C]

If the mirror approaches the object or the object approaches the stationary mirror with speed v then image approaches object with speed $2v$.

21. [C]

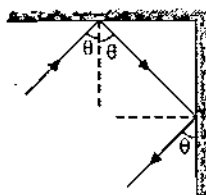
$$m_{\text{real}} = -n = \frac{f}{f-u}$$

$$\Rightarrow -nf + nu = f$$

$$\Rightarrow u = \left(\frac{n+1}{n} \right) f$$

22. [D]

The incident and the second reflected ray make the same angle θ with vertical. Hence, they are parallel for any value of θ .



23. [D]

$$\frac{1}{n} = \frac{f}{f-u}$$

$$f-u = nf$$

$$\Rightarrow u = (1-n)f = -(n-1)f$$

According to the sign convention used u must always be negative.

24. [D]

$$m_{\text{real}} = -4.5$$

$$m = \frac{f}{f-u}$$

$$-4.5 = \frac{f}{f-(-20)}$$

$$\Rightarrow -4.5f - 90 = f$$

$$\Rightarrow 90 = -5.5f$$

$$\Rightarrow f = -\frac{90}{5.5}$$

$$\Rightarrow f = \frac{900}{55} = -\frac{180}{11} \text{ cm}$$

Negative sign with focal length implies that the mirror is a concave mirror.

25. [B]

$$f = +20 \text{ cm}$$

$$u = -10 \text{ cm}$$

$$\text{Since, } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

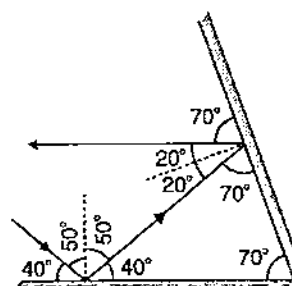
$$\Rightarrow \frac{1}{v} - \frac{1}{10} = \frac{1}{20}$$

$$\Rightarrow v = \frac{20}{3} \text{ cm}$$

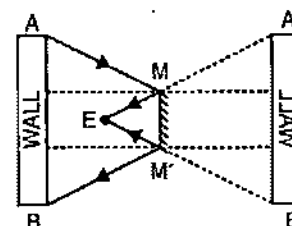
26. [A]

Covering the lower half will just make the image less bright (not blurred) as less number of rays will be reflected as compared to the previous case.

27. [B]



28. [C]



From symmetry we observe that length of mirror is one third of height of wall.

$$\Rightarrow l = \frac{H}{3}$$

29. [D]

$$n = \frac{360}{\theta} \rightarrow \infty$$

30. [A]

$$\frac{360}{\theta} = 6 \text{ which is Even}$$

$$\Rightarrow n = \frac{360}{\theta} - 1 = 5$$

31. [B]

Ray's after reflections from two perpendicular mirrors are always parallel to incident ray irrespective of angle of incidence.

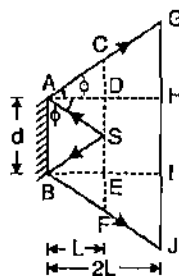
32. [C]

Power of concave lens must be less than that of the convex lens to form a real image. So, net power will decrease or focal length will increase. For real image v is positive, u is negative and f is positive. Applying lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ (substituting all values with sign), we get

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

u is constant, f is increasing. So v will also increase.

34. [D]



35. [B]

Since an elongated image is formed and it touches one end of the rod, so the rod must lie with one end at $2F$ and other end between $2F$ and F (shown in figure).

$$\text{For end A, } u = -\frac{5f}{3}$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{3}{5f} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{3}{5f} - \frac{1}{f} = \frac{3-5}{5f} = -\frac{2}{5f}$$

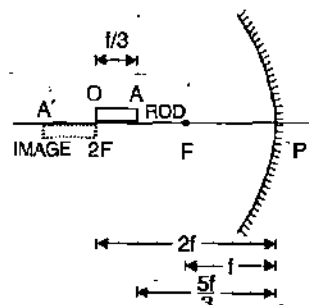
$$\Rightarrow \frac{1}{v} = -\frac{2}{5f}$$

$$\Rightarrow v = -\frac{5f}{2}$$

$$\Rightarrow PA' = \frac{5f}{2}$$

$$\Rightarrow OA' = \frac{5f}{2} - 2f$$

$$\Rightarrow OA' = \frac{f}{2}$$



36. [B]

First consider two adjacent walls (not the ceiling). If n_1 is the number of images formed due to these perpendicular walls, then

$$n_1 = \frac{360}{90} - 1$$

$$\Rightarrow n_1 = 3$$

Now, when we consider the mirror on the ceiling then it will make a total of 4 images (one of the original object and three images of the previous arrangement) so, total images formed equals 7. However, if the object were the observer himself then total number of images is $7 - 1 = 6$

37. [C]

Velocity of light is always normal to the wavefront.

38. [B]

Both concave and convex mirror give virtual image but a concave mirror gives a magnified virtual image (when object placed between F and P). Since the boy sees his image of diminished size, so the mirror must be convex.

39. [A]

Let object be placed at a distance x from mirror.

$$u = -x, v = -(x+10), f = -12 \text{ cm}$$

$$\text{Since, } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{-(x+10)} + \frac{1}{-x} = \frac{1}{-12}$$

$$\Rightarrow x = 20 \text{ cm}$$

$$\Rightarrow u = -20 \text{ cm,}$$

$$v = -30 \text{ cm}$$

$$\text{So, } m = \frac{v}{u} = \frac{-30}{-20} = -1.5$$

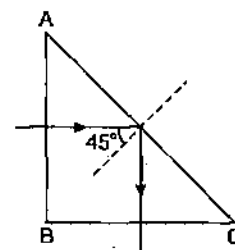
Negative sign with magnification indicates image is real.

40. [B]

At face AB , the ray of light suffers no deviation, so applying Snell's Law at face AC , we get

$$\mu = \frac{1}{\sin C} = \frac{1}{\sin(45^\circ)}$$

$$\Rightarrow \mu_{\min} = \sqrt{2}$$



41. [C]

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow v^{-1} + u^{-1} = f^{-1} = \text{constant}$$

Take derivative w.r.t. time on both sides

$$\frac{d}{dt}(v^{-1}) + \frac{d}{dt}(u^{-1}) = 0$$

$$\Rightarrow (-1)v^{-2} \frac{dv}{dt} + (-1)u^{-2} \frac{du}{dt} = 0$$

$$\Rightarrow \frac{dv}{dt} = -\frac{v^2}{u^2} \left(\frac{du}{dt} \right)$$

$$\Rightarrow \text{Image speed} = -\frac{v^2}{u^2} (\text{object speed})$$

when object moves towards mirror u decreases with passage of time and hence $\frac{du}{dt} = -9 \text{ cms}^{-1}$

$$\Rightarrow \frac{dv}{dt} = -\left(\frac{f}{f-u} \right)^2 \frac{du}{dt}$$

$$\Rightarrow \frac{dv}{dt} = -\left(\frac{-24}{-24+60} \right)^2 \frac{du}{dt}$$

$$\Rightarrow \frac{dv}{dt} = -\left(\frac{2}{3} \right)^2 \frac{du}{dt}$$

$$\Rightarrow \frac{dv}{dt} = -\frac{4}{9}(-g)$$

$$\Rightarrow \frac{dv}{dt} = +4 \text{ cms}^{-1}$$

Positive value of $\frac{dv}{dt}$ indicates that v increases with the passage of time i.e. image must be going away from mirror.

42. [D]

$A = 60^\circ$ for equilateral prism.

$$i = e = \frac{3}{4}A$$

Since, $i + e = A + D$

$$\Rightarrow 2i = A + D$$

$$\Rightarrow 2\left(\frac{3}{4}A\right) = A + D$$

$$\Rightarrow D = \frac{A}{2} = 30^\circ$$

43. [B]

Since, $u_{ab} > u_{cd}$

$$\Rightarrow v_{ab} < v_{cd} \quad \left(\because \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \right)$$

Since, $m = -\frac{v}{u}$

$$\Rightarrow |m_{ab}| < |m_{cd}|$$

44. [D]

$$\mu = \frac{c}{v}$$

$$\Rightarrow 2 = \frac{3 \times 10^8}{v}$$

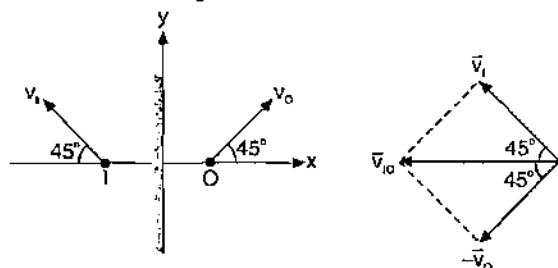
$$\Rightarrow v = 1.5 \times 10^8 \text{ ms}^{-1}$$

$$\Rightarrow v = 1.5 \times 10^{10} \text{ cms}^{-1}$$

45. [C]

$$|\vec{v}_O| = |\vec{v}_I| = \sqrt{(2)^2 + (2)^2} = 2\sqrt{2} \text{ ms}^{-1}$$

Relative velocity of image with respect to object is in negative x-direction as shown in figure.



46. [C]

$$i = 2r$$

Since, $n = \frac{\sin i}{\sin r}$

$$\Rightarrow n = \frac{\sin i}{\sin\left(\frac{i}{2}\right)} = \frac{2\sin\left(\frac{i}{2}\right)\cos\left(\frac{i}{2}\right)}{\sin\left(\frac{i}{2}\right)}$$

$$\left\{ \because 2\sin\left(\frac{i}{2}\right)\cos\left(\frac{i}{2}\right) = \sin i \right\}$$

$$\Rightarrow n = 2\cos\left(\frac{i}{2}\right)$$

$$\Rightarrow \cos\left(\frac{i}{2}\right) = \frac{n}{2}$$

$$\Rightarrow i = 2\cos^{-1}\left(\frac{n}{2}\right)$$

47. [B]

The source cannot be seen if angle of incidence in the denser medium is greater than the critical angle.

Since,

$$r = \frac{d}{\sqrt{\mu^2 - 1}} = \frac{h}{\sqrt{\mu^2 - 1}}$$

$$\Rightarrow h = (1) \sqrt{\left(\frac{5}{3}\right)^2 - 1}$$

$$\Rightarrow h = \frac{4}{3} \text{ cm}$$

48. [A]

$$\sin C = \frac{1}{\mu}$$

TIR will take place at AC if $i > C$ i.e. $45^\circ > C$ for

Red Colour

$$\mu_R = 1.39$$

$$\Rightarrow \sin C_R = \frac{1}{1.39} > \frac{1}{1.41} = \sin 45^\circ$$

$$\Rightarrow \sin C_R > \sin 45^\circ$$

$$\Rightarrow C_R > 45^\circ$$

(No TIR will take place)

Green Colour

$$\mu_G = 1.44$$

$$\Rightarrow \sin C_G = \frac{1}{1.44} < \frac{1}{1.41} = \sin 45^\circ$$

$$\Rightarrow \sin C_G < \sin 45^\circ$$

$$\Rightarrow C_G < 45^\circ$$

(TIR will take place for Green Colour)

Blue Colour

$$\mu_B = 1.47$$

$$\Rightarrow \sin C_B = \frac{1}{1.47} < \frac{1}{1.41} = \sin 45^\circ$$

$$\Rightarrow \sin C_B < \sin 45^\circ$$

$$\Rightarrow C_B < 45^\circ$$

(TIR will take place for Blue Colour)

So, Red is separated from Green and Blue.

49. [C]

$$A_2 = \left(\frac{n_1 - 1}{n_2 - 1} \right) A_1$$

$$\Rightarrow A_2 = \left(\frac{1.54 - 1}{1.72 - 1} \right) 4$$

$$\Rightarrow A_2 = 3^\circ$$

51. [A]

$$\sin C = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} = \frac{V_{\text{denser}}}{V_{\text{rarer}}}$$

$$\Rightarrow \sin 0 = \frac{V_{\text{denser}}}{V_{\text{rarer}}} = \frac{V_A}{V_B}$$

$$\Rightarrow \sin \theta = \frac{V}{V_B}$$

$$\Rightarrow V_B = \frac{V}{\sin \theta}$$

52. [C]

Since, $\mu v = \text{constant}$

$$\Rightarrow \mu_a v_a = \mu_w v_w$$

$$\Rightarrow \frac{3}{2}(2 \times 10^8) = \frac{4}{3}v_w$$

$$\Rightarrow v_w = 2.25 \times 10^8 \text{ ms}^{-1}$$

53. [D]

After two reflections from two mirrors placed at right angles, the emergent ray will always be parallel to the incident ray for any value of i .

54. [C]

Since light has to travel from denser to rarer medium so, it must be made incident in the denser medium at an angle less than critical angle.

55. [D]

$C \propto \lambda$

Since $\lambda_{\text{Red}} > \lambda_{\text{Violet}}$

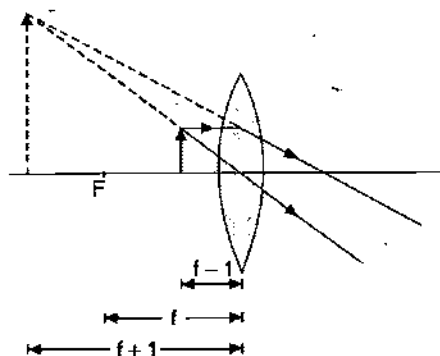
$$\Rightarrow C_{\text{Red}} > C_{\text{Violet}}$$

57. [B]

According to the problem, we have, $u = -(f-1)$

$$v = -(f+1)$$

$$f = +f$$



Applying $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{-(f+1)} + \frac{1}{f-1} = \frac{1}{f}$$

$$\Rightarrow f^2 - 2f - 1 = 0$$

$$\Rightarrow f = (\sqrt{2} + 1) \text{ cm}$$

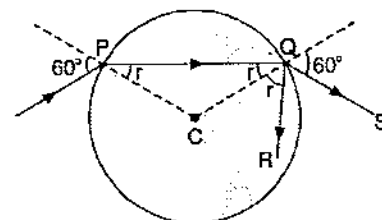
59. [B]

Since rays after passing through the glass slab just suffer lateral displacement hence we have angle between the emergent rays as α .

60. [D]

Applying Snell's Law $\mu = \frac{\sin i}{\sin r}$, we get

$$\Rightarrow \sin r = \frac{\sin i}{\mu}$$



$$\Rightarrow \sin r = \frac{\sin 60^\circ}{\sqrt{3}} = \frac{1}{2}$$

$$\Rightarrow r = 30^\circ$$

Since, $PC = QC$

$$\Rightarrow \angle CPQ = \angle PQC = \angle r = 30^\circ$$

Angle between reflected ray QR and refracted ray QS at the other face is $180^\circ - r - 60^\circ = 90^\circ$ $\{\therefore r = 30^\circ\}$

61. [A]

According to Snell's Law

$$\frac{\sin i}{\sin r} = \frac{\mu_w}{\mu_a}$$

$$\Rightarrow \sin r = \frac{1}{\mu_w} \sin i$$

$$\Rightarrow \sin r = \frac{1}{4/3} \sin 60$$

$$\Rightarrow r = \sin^{-1} \left(\frac{3\sqrt{3}}{8} \right)$$

62. [B]

$$\angle DBM = 45^\circ$$

$$\Rightarrow DM = h \tan 45^\circ = h = 32 \text{ cm}$$

$$\mu = \frac{\sin i}{\sin r}$$

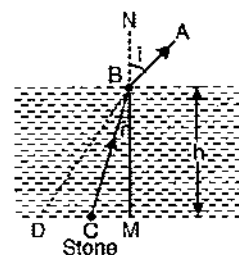
$$\sin r = \frac{\sin i}{\mu} = \frac{\sin 45^\circ}{4/3} = \frac{3}{4\sqrt{2}}$$

$$\Rightarrow \tan r = \frac{\sin r}{\cos r} = \frac{\sin r}{\sqrt{1 - \sin^2 r}}$$

$$\Rightarrow \tan r = \frac{3}{\sqrt{1 - \frac{9}{32}}} = \frac{3}{\sqrt{\frac{23}{32}}} = \frac{3}{\sqrt{23}}$$

$$\Rightarrow CM = h \tan r = 32 \times \frac{3}{\sqrt{23}} = 20 \text{ cm}$$

$$\Rightarrow CD = DM - CM = 32 - 20 = 12 \text{ cm}$$

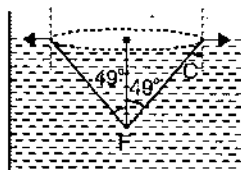


63. [C]

$$\sin C = \frac{1}{\mu_w} = \frac{3}{4}$$

$$\Rightarrow C = 49^\circ$$

Apex angle of cone is $2C = 98^\circ$



64. [D]

In $\triangle OAN'$

$$\sin(i-r) = \frac{\Delta x}{OA}$$

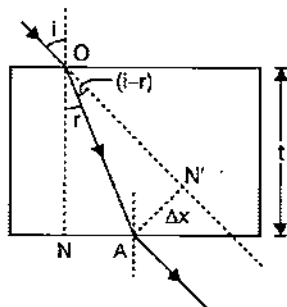
$$\Rightarrow \Delta x = OA \sin(i-r)$$

In $\triangle OAN$

$$\cos r = \frac{ON}{OA}$$

$$\Rightarrow OA = \frac{t}{\cos r}$$

$$\Rightarrow \Delta x = \frac{t \sin(i-r)}{\cos r}$$



67. [C]

For grazing incidence and emergence, we have

$$i = e = 90^\circ, r_1 = r_2 = \frac{A}{2} = 30^\circ$$

According to Snell's Law, we get

$$\mu = \frac{\sin i}{\sin r_1} = 2$$

68. [A]

$$\sin C = \frac{\mu_r}{\mu_d} = \frac{4/3}{3/2}$$

$$\Rightarrow C = \sin^{-1}\left(\frac{8}{9}\right)$$

and light must go from denser to rarer medium.

69. [B]

According to Snell's Law

$$\mu\lambda = \text{constant}$$

Since $\lambda_{\text{Red}} > \lambda_{\text{Violet}}$

$$\Rightarrow \mu_{\text{Red}} < \mu_{\text{Violet}}$$

71. [C]

$$\text{Critical angle } C = \sin^{-1}\left(\frac{1}{\mu}\right)$$

$$\Rightarrow C = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$

When $A > 2C$, the ray does not emerge from the prism. So, maximum refracting angle can be 60° .

73. [A]

According to Snell's Law

$$\mu v = \text{constant}$$

74. [A]

According to Snell's Law

$$\mu\lambda = \text{constant}$$

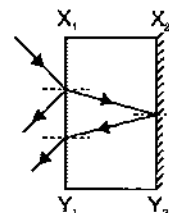
$$\Rightarrow \mu_a \lambda_a = \mu_g \lambda_g$$

$$\Rightarrow \lambda_g = \frac{\mu_a \lambda_a}{\mu_g}$$

$$\Rightarrow \lambda_g < \lambda_a$$

75. [B]

This happens due to multiple refractions and reflections. The first image is formed due to reflection at X_1Y_1 and is fainter. The second image is formed due to reflection at X_2Y_2 and is brightest all other images formed further are faint.



76. [C]

In both A and B, the refracted ray is parallel to the base of prism.

79. [A]

$$\mu = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \sqrt{2} \sin 30 = \sin\left(\frac{A+D_m}{2}\right)$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \sin\left(\frac{A+D_m}{2}\right)$$

$$\Rightarrow \frac{60+D_m}{2} = 45$$

$$\Rightarrow \frac{D_m}{2} = 15$$

$$\Rightarrow D_m = 30^\circ$$

$$\text{Further we know that at minimum deviation } i = \frac{A+D_m}{2}$$

$$\Rightarrow i = 45^\circ$$

80. [C]

Normal Emergence implies

$$e = 0$$

Since $i + e = A + D$

For prism with small A,

$$D = (\mu - 1)A$$

$$\Rightarrow i = A + (\mu - 1)A$$

$$\Rightarrow i \cong \mu A$$

81. [C]

$$\text{Since, } {}^1\mu_2 \times {}^2\mu_3 \times {}^3\mu_4 \times {}^4\mu_1 = 1$$

$$\Rightarrow {}^4\mu_3 \times {}^3\mu_2 \times {}^2\mu_1 \times {}^1\mu_4 = 1$$

$$\Rightarrow {}^4\mu_3 \times {}^3\mu_2 \times {}^2\mu_1 = \frac{1}{{}^1\mu_4}$$

82. [C]
Let thickness of slab be t and real depth from first side be x .
Then

$$\mu = \frac{x}{6} \quad (\text{when viewed from first side})$$

$$\mu = \frac{t-x}{4} \quad (\text{when viewed from second side})$$

$$\Rightarrow x = 6\left(\frac{3}{2}\right) \quad \left\{ \because \mu = \frac{3}{2} \right\}$$

$$\Rightarrow x = 9 \text{ cm}$$

$$\Rightarrow \frac{3}{2} = \frac{t-9}{4}$$

$$\Rightarrow t = 15 \text{ cm}$$

83. [D]

$$\text{Since, } \mu = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \cot\left(\frac{A}{2}\right) = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \cos\left(\frac{A}{2}\right) = \sin\left(\frac{A+D_m}{2}\right) \quad \left\{ \because \cot\left(\frac{A}{2}\right) = \frac{\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)} \right\}$$

$$\Rightarrow \sin\left(\frac{\pi}{2} - \frac{A}{2}\right) = \sin\left(\frac{A}{2} + \frac{D_m}{2}\right)$$

$$\Rightarrow D_m = \pi - 2A = 180^\circ - 2A$$

84. [A]

$$i + e = A + D$$

$$\Rightarrow 60^\circ + e = 30^\circ + 30^\circ$$

$$\Rightarrow e = 0^\circ$$

85. [B]

$$\sin(45^\circ) = \frac{1}{\mu}$$

$$\Rightarrow \mu = \sqrt{2}$$

86. [B]

$$\sin C_1 = \frac{\mu_1}{\mu_2} = \frac{\mu_a}{\mu_g} = \frac{1}{\mu_g}$$

$$\sin C_2 = \frac{\mu_w}{\mu_g} > \sin C_1$$

$$C_1 < C_2$$

87. [B]

$$\text{Since } r = \frac{d}{\sqrt{\mu^2 - 1}}$$

$$\Rightarrow r = \frac{4}{\sqrt{\left(\frac{5}{3}\right)^2 - 1}}$$

$$\Rightarrow r = \frac{4 \times 3}{4} = 3 \text{ m}$$

$$\text{Diameter} = 2r = 6 \text{ m}$$

88. [A]

$$\text{Apparent Depth} = \frac{h}{n}$$

90. [C]

From mirror formula, we get

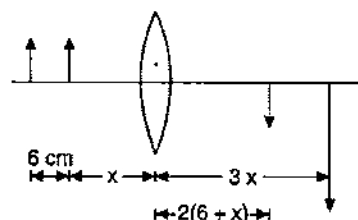
$$\frac{1}{3x} + \frac{1}{x} = \frac{1}{f}$$

$$\Rightarrow \frac{4}{3x} = \frac{1}{f} \quad \dots(1)$$

From mirror formula, we get

$$\frac{1}{2(6+x)} + \frac{1}{6+x} = \frac{1}{f}$$

$$\Rightarrow \frac{3}{2(6+x)} = \frac{1}{f} \quad \dots(2)$$



Equations (1) and (2) give

$$\frac{4}{3x} = \frac{3}{2(6+x)}$$

$$\Rightarrow 9x = 48 + 8x$$

$$\Rightarrow x = 48 \text{ cm}$$

$$\text{Shift of screen} = 3x - 2(6+x)$$

$$\text{Shift of screen} = 3 \times 48 - 2(6+48) = 36 \text{ cm}$$

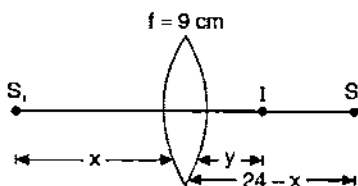
91. [D]

$$\text{Apparent Depth} = \frac{d}{2\mu_1} + \frac{d}{2\mu_2}$$

92. [A]

In this case, one of the image will be real and the other will be virtual. Let us assume that image of S_1 is real and that of S_2 is

virtual. Then, applying $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$



for S_1 , we get

$$\frac{1}{y} + \frac{1}{x} = \frac{1}{9} \quad \dots(1)$$

for S_2 , we get

$$\frac{1}{y} + \frac{1}{24-x} = \frac{1}{9} \quad \dots(2)$$

Solving equations (1) and (2), we get

$$x = 6 \text{ cm}$$

CONCEPTUAL NOTE(S)

This question may have following answer 6 cm from S_1 or 18 cm from S_2 and 18 cm from S_1 or 6 cm from S_2 .

93. [C]

Optical path length = nt

So time taken = $\frac{nt}{c}$

94. [C]

To a fish the outer world is seen in a circle of radius $\frac{d}{\sqrt{\mu^2 - 1}}$, d

is the depth at which the fish swims.

$$\Rightarrow r = \frac{12}{\sqrt{\frac{16}{9} - 1}}$$

$$\Rightarrow r = \frac{36}{\sqrt{7}} \text{ cm}$$

95. [A]

For the diver

$$\sin C = \frac{1}{\mu_w} = \frac{1}{(4/3)} \quad \left(\text{Remember that } \mu_{\text{water}} = \frac{4}{3} \right)$$

$$\Rightarrow C = 49^\circ \text{ with vertical}$$

$$\Rightarrow \theta = 90 - 49^\circ = 41^\circ \text{ with the horizon.}$$

96. [C]

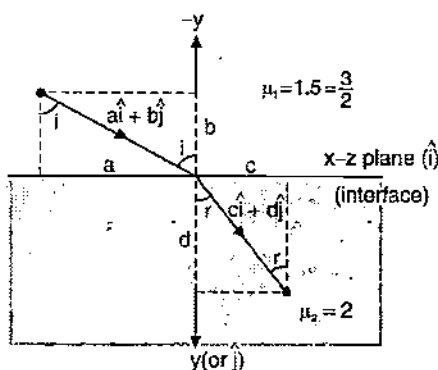
$$\sin C = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} = \frac{\mu_w}{\mu_g}$$

$$\Rightarrow \sin C = \frac{4/3}{5/3} = \frac{4}{5}$$

97. [B]

Applying Snell's Law at the interface separating two media, we get

$$\mu_1 \sin i = \mu_2 \sin r$$



From the figure, we get

$$\left(\frac{3}{2} \right) \left(\frac{a}{\sqrt{a^2 + b^2}} \right) = 2 \left(\frac{c}{\sqrt{c^2 + d^2}} \right) \quad \dots (1)$$

Since $a\hat{i} + b\hat{j}$ and $c\hat{i} + d\hat{j}$ are unit vectors, so we get

$$\sqrt{a^2 + b^2} = \sqrt{c^2 + d^2} = 1$$

Substituting in equation (1), we get

$$\frac{a}{c} = \frac{4}{3}$$

98. [B]

$$\sin C = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} = \frac{\mu_a}{\mu_A}$$

$$\text{Since, } \mu_A v_A = \mu_a v_a$$

$$\Rightarrow \sin C = \frac{v_A}{v_a} = \frac{2}{2.4}$$

$$\Rightarrow \sin C = \frac{5}{6}$$

99. [C]

According to Snell's Law

$$\mu v = \text{constant}$$

$$\Rightarrow \mu_g v_g = \mu_r v_r$$

$$\Rightarrow (1.5) \times (2 \times 10^8) = \mu_r (2.5 \times 10^8)$$

$$\Rightarrow \mu_r = \frac{3}{2.5} = \frac{30}{25} = 1.2$$

100. [D]

$$\sin C = \frac{1}{\sqrt{2}}$$

$$\Rightarrow C = 45^\circ$$

For $i = C$ in denser medium angle of refraction in rarer medium is 90° .

101. [C]

$$\text{Apparent Depth} = \frac{\text{Real Depth}}{\mu}$$

$$\Rightarrow \text{Apparent Depth} = \frac{24}{4/3}$$

$$\Rightarrow \text{Apparent Depth} = 18 \text{ cm}$$

102. [C]

$$\text{Given that } A = \delta_m = 60^\circ$$

At minimum deviation, we have

$$i = \left(\frac{A + \delta_m}{2} \right) = 60^\circ$$

103. [B]

$$r + i = 90$$

$$i = 90 - r$$

For ray not to emerge from curved surface

$$i > C$$

$$\Rightarrow \sin i > \sin C$$

$$\Rightarrow \sin(90 - r) > \sin C$$

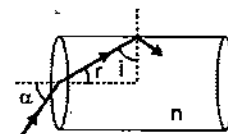
$$\Rightarrow \cos r > \sin C$$

$$\Rightarrow \sqrt{1 - \sin^2 r} > \frac{1}{n}$$

$$\Rightarrow 1 - \frac{\sin^2 i}{n^2} > \frac{1}{n^2}$$

$$\Rightarrow 1 > \frac{1}{n^2} (1 + \sin^2 i)$$

$$\Rightarrow n^2 > 1 + \sin^2 i$$



$$\left(\because \sin C = \frac{1}{n} \right)$$

$$-\infty < \frac{1}{\sqrt{2}} < \infty$$

$$\Rightarrow n > \sqrt{2} \quad \{\sin i \rightarrow 1\}$$

$$\Rightarrow \text{least value} = \sqrt{2}$$

104. [B]

$$\text{Dispersive Power} = \frac{d\mu}{d\lambda}$$

$$\Rightarrow \frac{d\mu}{d\lambda} = -\frac{2B}{\lambda^3}$$

So, as B increases, dispersive power increases.

105. [C]

For the near end of the rod applying $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Since, here u and f are negative, so

$$|v| = \frac{uf}{u-f}$$

The farther end of the rod is at infinity, so its image will be formed at focus. Hence,

Length of the image is $\ell = |v| - f$

$$\Rightarrow \ell = \frac{uf}{u-f} - f = \frac{f^2}{u-f}$$

106. [C]

For A

$$\text{Total number of waves} = \frac{(1.5)t}{\lambda} \quad \dots(1)$$

$$\therefore \left(\text{Total number of waves} \right) = \left(\frac{\text{Optical path length}}{\text{wavelength}} \right)$$

For B and C

Total number of waves

$$= \frac{n_B \left(\frac{t}{3} \right)}{\lambda} + \frac{(1.6) \left(\frac{2t}{3} \right)}{\lambda} \quad \dots(2)$$

Equating (1) and (2)

$$\Rightarrow n_B = 1.3$$

107. [C]

Focal length of curved mirrors is independent of the refractive index of the medium in which the mirror is placed.

108. [B]

$$n = \frac{c}{v}$$

$$\Rightarrow n = \frac{\sqrt{\mu_0 \epsilon_0}}{\sqrt{\mu \epsilon}}$$

$$\Rightarrow n = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}}$$

109. [D]

$$\sin \theta_1 = \frac{1}{\mu_g} \text{ and } \sin \theta_2 = \frac{1}{\mu_w}$$

Since, $\mu_g > \mu_w$, $\theta_1 < \theta_2$

The critical angle θ between glass-water interface is given by

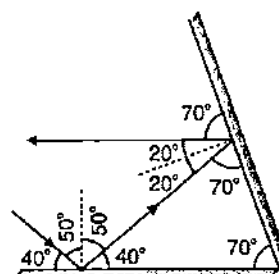
$$\sin \theta = \frac{\mu_w}{\mu_g}$$

$$\Rightarrow \theta > \theta_2$$

CONCEPTUAL NOTE(S)

Critical angle increases as the relative refractive index is decreased.

110. [B]



111. [B]

$$u = -20 \text{ cm}, f = +20 \text{ cm}, v = ?$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{-20} = \frac{1}{20}$$

$$\Rightarrow v = 10 \text{ cm}$$

Students generally give (A) as the answer to this problem because they have in their mind that if an object is placed at centre of curvature C, then image is also formed at C, but this is true only for the case of a convex lens or a concave mirror. So correct answer is (B).

113. [D]

Since no parallax exists between the images formed by two mirrors (convex and plane) hence the images for both coincide. But for a plane mirror an image is as far behind the mirror as the object is in front of it. Hence the image for plane mirror should be 30 cm behind it or 10 cm behind the convex mirror. So for convex mirror,

$$u = -50 \text{ cm}, v = +10 \text{ cm}$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{10} + \frac{1}{-50} = \frac{1}{f}$$

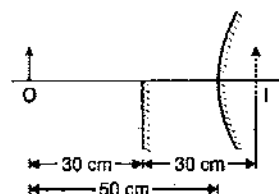
$$\Rightarrow \frac{1}{f} = \frac{5-1}{50}$$

$$\Rightarrow f = \frac{50}{4}$$

$$\Rightarrow f = 12.5 \text{ cm}$$

$$\text{Since } R = 2f$$

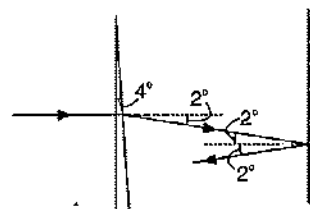
$$\Rightarrow R = 25 \text{ cm}$$



114. [C]

$$\text{Since, } \delta_{\text{prism}} = (\mu - 1)A = (1.5 - 1)4^\circ = 2^\circ$$

$$\Rightarrow \delta_{\text{total}} = \delta_{\text{prism}} + \delta_{\text{mirror}}$$

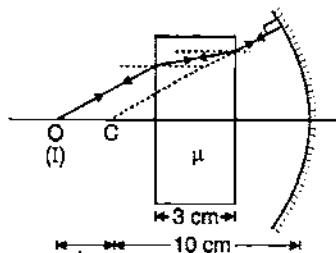


$$\Rightarrow \delta_{\text{total}} = (\mu - 1)A + (180 - 2i)$$

$$\Rightarrow \delta_{\text{total}} = (1.5 - 1)4^\circ + (180 - 2 \times 2^\circ)$$

$$\Rightarrow \delta_{\text{total}} = 2^\circ + 176^\circ = 178^\circ$$

115. [C]



$$\Delta x = 3 \left[1 - \frac{1}{\left(\frac{3}{2} \right)} \right] \Delta x$$

$$\Rightarrow \Delta x = 1 \text{ cm}$$

$$\Rightarrow \text{Object distance} = 10 + 1 = 11 \text{ cm}$$

116. [C]

Apparent depth = $\frac{d}{n}$

Hence apparent separation = $\frac{2d}{n}$

117. [B]

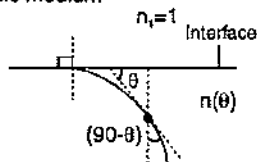
According to Snell's Law for anisotropic medium

$$n(\theta) \sin(90 - \theta) = \text{constant}$$

or $n_1 \sin i = n(\theta) \cos \theta$

$$\Rightarrow (1) \sin 90 = n(\theta) \cos \theta$$

$$\Rightarrow n(\theta) \cos \theta = 1$$



118. [D]

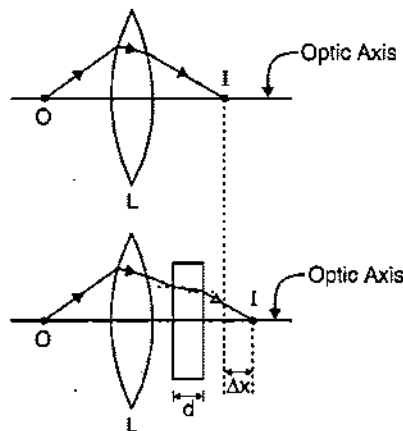
Velocity of approach of man towards the bicycle = $(u - v)$
Hence velocity of approach of image towards bicycle is $2(u - v)$.

119. [C]

Optical path length = μt

Time taken = $\frac{\mu t}{c}$

120. [C]



$$\Delta x = d \left(1 - \frac{1}{\mu} \right)$$

121. [C]

$$\sin C = \frac{4000}{6000}$$

$$\Rightarrow C = \sin^{-1} \left(\frac{2}{3} \right)$$

122. [B]

A virtual, erect image is obtained by using the mirror. The mirror can be both concave and convex. But a virtual image obtained by a concave mirror is always magnified and hence the mirror must be convex as we are getting diminished image.

123. [C]

$$\mu = \frac{c}{v_{\text{med}}} = \frac{f \lambda_{\text{air}}}{f \lambda_{\text{med}}} = \frac{3}{2}$$

Now, $v = +8 \text{ m}$, $m = -\frac{1}{3} = \frac{v}{u}$

$$\Rightarrow u = -24 \text{ m}$$

Since, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\Rightarrow \frac{1}{f} = \frac{1}{8} + \frac{1}{24} = \frac{4}{24}$$

$$\Rightarrow f = 6 \text{ m}$$

For a plano-convex lens

$$f = \frac{R}{\mu - 1}$$

$$6 = \frac{R}{0.5}$$

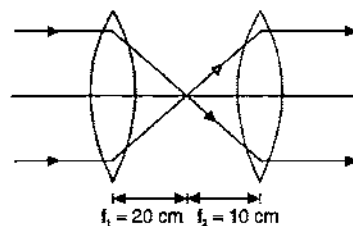
$$\Rightarrow R = 3 \text{ m}$$

124. [B]

Since rays after passing through the glass slab just suffer lateral displacement hence we have angle between the emergent rays as α .

126. [A]

As shown in figure the distance between the lenses should be 30 cm.



127. [C]

Two plano convex lens of focal length f , when combined using optical glue will give rise to a convex lens of focal length $\left(\frac{f}{2} \right)$.

i.e. size of image = size of object.

$\Rightarrow$ Object at $2F$ i.e. at a distance f from optical centre.

128. [B]

$$f_{\text{combination}} \rightarrow \infty$$

Since $\frac{1}{f_{\text{combination}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$

$$\Rightarrow 0 = \frac{1}{-10} + \frac{1}{f_2} - \frac{10}{(-10)f_2}$$

$$\Rightarrow 0 = \frac{1}{-10} + \frac{1}{f_2} + \frac{1}{f_2}$$

$$\Rightarrow \frac{1}{10} = \frac{2}{f_2}$$

$$\Rightarrow f_2 = 20 \text{ cm}$$

129. [D]

Since, $\angle ABO = \angle OAB = C$

Further, by definition, we have

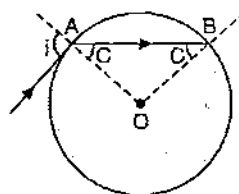
$$\sin C = \frac{1}{\mu} = \frac{2}{3}$$

Applying Snell's Law at A, we get

$$\frac{\sin i}{\sin C} = \frac{3}{2}$$

$$\Rightarrow \sin i = \left(\frac{3}{2}\right) \sin C = \left(\frac{3}{2}\right) \left(\frac{2}{3}\right) = 1$$

$$\Rightarrow i = 90^\circ$$



130. [C]

For combination to produce dispersion without deviation.

$$(n_1 - 1)A_1 = (n_2 - 1)A_2$$

$$\Rightarrow A_2 = \left(\frac{n_1 - 1}{n_2 - 1}\right)A_1$$

$$\Rightarrow A_2 = \left(\frac{1.54 - 1}{1.72 - 1}\right)4^\circ$$

$$\Rightarrow A_2 = \frac{3}{4}(4^\circ)$$

$$\Rightarrow A_2 = 3^\circ$$

131. [C]

$$\frac{\mu}{\infty} + \frac{1}{-u} = \frac{\mu - 1}{R}$$

$$\Rightarrow u = -\frac{R}{\mu - 1} \text{ from the surface}$$

$\Rightarrow$ Total distance from centre

$$\Rightarrow x = |u| + R = R + \frac{R}{\mu - 1}$$

$$x = R + 2R$$

$$x = 3R$$

132. [D]

An air bubble in water always behaves as a concave lens and hence is always incapable to form a real image.

133. [B]

$$m_{\text{virtual}} = m_{\text{real}}$$

$$\Rightarrow \frac{f}{f + (-12)} = \frac{f}{f + (-20)}$$

$$\Rightarrow \frac{f}{f - 12} = \frac{f}{f - 20}$$

$$\Rightarrow f - 20 = -f + 12$$

$$\Rightarrow 2f = 32$$

$$\Rightarrow f = 16 \text{ cm}$$

134. [C]

$$4 = \frac{t}{\mu}$$

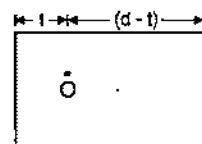
$$\text{and } 6 = \frac{d - t}{\mu}$$

$$\Rightarrow 6\mu = d - 4\mu$$

$$\Rightarrow d = 10\mu$$

$$\Rightarrow d = 15 \text{ cm}$$

$$\left\{ \begin{array}{l} \mu = \frac{\text{Real Depth}}{\text{Apparent Depth}} \end{array} \right\}$$



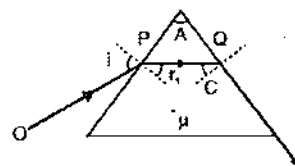
135. [A]

$$\frac{P_A}{P_B} = \left(\frac{f_B}{f_A}\right)^2 = \frac{4}{9} = 0.44$$

136. [C]

An 'achromatic combination' is made from a concave lens and a convex lens with greater power of convex lens so as to make the rays converge at a point.

137. [C]



$$\frac{\sin i}{\sin r_1} = \mu \quad (\text{at } P) \quad \dots(1)$$

$$\text{and } r_1 + C = A \quad \dots(2)$$

$$\Rightarrow \sin i = \mu \sin(A - C)$$

$$\Rightarrow \sin i = \mu (\sin A \cos C - \cos A \sin C)$$

$$\Rightarrow \sin i = \mu \left(\sin A \sqrt{1 - \frac{1}{\mu^2}} - \cos A \frac{1}{\mu} \right) \quad \left\{ \because \sin C = \frac{1}{\mu} \right\}$$

$$\Rightarrow i = \sin^{-1} \left[\left(\sqrt{\mu^2 - 1} \right) \sin A - \cos A \right]$$

138. [D]

At $u = f$ (focal length)

$$v \rightarrow \infty$$

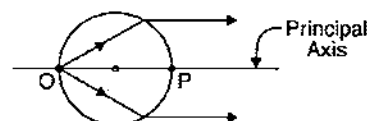
At $u = 0$ (i.e., object is at pole)

$$v = 0$$

(image is also at pole)

Satisfying these two conditions, only OPTION (D) is correct.

139. [C]



Considering pole at P, we have

$$\frac{\mu}{-2R} + \frac{1}{\infty} = \frac{\mu - 1}{-R} \quad \left\{ \because \frac{\mu_1}{\mu} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R} \right\}$$

$$\Rightarrow \frac{\mu}{2} = \mu - 1$$

$$\Rightarrow \frac{\mu}{2} = 1$$

$$\Rightarrow \mu = 2$$

140. [C]

The first image is formed due to the reflection from concave mirror M_2 .

$$\Rightarrow \frac{1}{v_1} + \frac{1}{(-2R)} = \frac{2}{-R}$$

$$\Rightarrow \frac{1}{v_1} = \frac{1}{2R} - \frac{4}{2R}$$

$$\Rightarrow \frac{1}{v_1} = \frac{-3}{2R}$$

$$\Rightarrow v_1 = \frac{-2R}{3}$$

$$\Rightarrow m_1 = -\left(\frac{-2R}{\frac{-3}{2R}}\right) = \frac{1}{3}$$

$$\Rightarrow \text{Radius of image circle} = \frac{1}{3}a, \text{ (in magnitude).}$$

Now the second image is formed by convex mirror M_1 .

The second image will be formed because the image formed by the first acts as object for it.

$$\Rightarrow \text{Object distance} = 2R - \frac{2R}{3} = \frac{4R}{3}$$

$$\Rightarrow \frac{1}{v_2} + \frac{1}{\left(\frac{4R}{3}\right)} = \frac{1}{\left(\frac{R}{2}\right)}$$

$$\Rightarrow \frac{1}{v_2} = \frac{2}{R} - \frac{3}{4R}$$

$$\Rightarrow v_2 = \frac{4R}{11}$$

$$\Rightarrow m_2 = \frac{-v_2}{u_2} = \frac{-\frac{4R}{11}}{-\frac{4R}{3}} = \frac{3}{11}$$

$$\Rightarrow m_2 = \frac{3}{11}$$

$$\text{So radius of second image} = a_2 = \frac{3}{11} \cdot \frac{a}{3} = \frac{a}{11}$$

$$\text{Similarly radius of third image is } a_3 = \frac{a}{41}$$

141. [B]

$$r + i = 90$$

$$i = 90 - r$$

For ray not to emerge from curved surface

$$i > C$$

$$\Rightarrow \sin i > \sin C$$

$$\Rightarrow \sin(90 - r) > \sin C$$

$$\Rightarrow \cos r > \sin C$$

$$\Rightarrow \sqrt{1 - \sin^2 r} > \frac{1}{n}$$

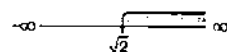
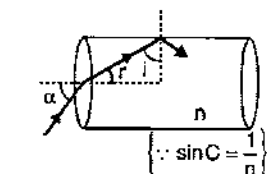
$$\Rightarrow 1 - \frac{\sin^2 i}{n^2} > \frac{1}{n^2}$$

$$\Rightarrow 1 > \frac{1}{n^2}(1 + \sin^2 i)$$

$$\Rightarrow n^2 > 1 + \sin^2 i$$

$$\Rightarrow n > \sqrt{2}$$

$$\Rightarrow \text{least value} = \sqrt{2}$$



($\sin i \rightarrow 1$)

143. [D]

The two slabs will shift the image by a distance

$$\Delta x = 2\left(1 - \frac{1}{\mu}\right)t$$

$$\Rightarrow \Delta x = 2\left(1 - \frac{1}{1.5}\right)(1.5) = 1 \text{ cm}$$

Therefore, final image will be 1 cm above point P.

144. [C]

Since

$$PR \parallel N_2$$

$$\Rightarrow \alpha = 2i$$

$$\text{Also } A + 2\alpha = 180^\circ$$

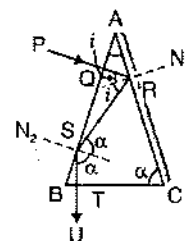
$$\Rightarrow A + 2(2i) = 180$$

$$\Rightarrow i + 4i = 180$$

$$\Rightarrow 5i = 180$$

$$\Rightarrow i = 36^\circ$$

$$\Rightarrow A = 36^\circ$$



145. [C]

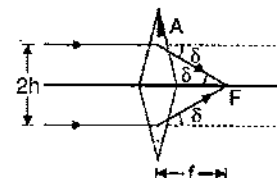
$$\text{Since } \tan \delta = \frac{h}{f}$$

$$\Rightarrow f = \frac{h}{\tan \delta}$$

Further

$$\Rightarrow \delta = (\mu - 1)A$$

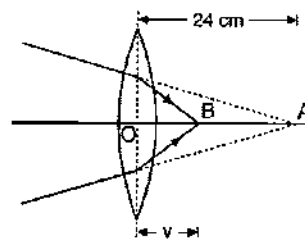
$$\Rightarrow f = \frac{h}{(\mu - 1)A}$$



146. [A]

If no lens had been there the rays would have met at A. On inserting the lens the rays meet at B. Thus A acts as an object for the lens.

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$



$$u = +24 \text{ cm}, v = ?, f = +24 \text{ cm}$$

$$\Rightarrow v = 12 \text{ cm}$$

$$\Rightarrow \text{Distance } AB = OA - OB = 24 - 12 = 12 \text{ cm}$$

147. [D]

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

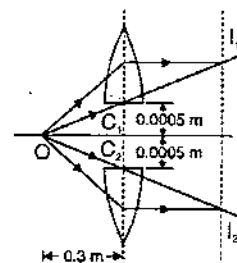
$$\Rightarrow \frac{1}{v} - \frac{1}{(-0.3)} = \frac{1}{0.2}$$

$$\Rightarrow v = 0.6 \text{ m}$$

Δ 's OC_1C_2 and OI_1I_2 are similar

Hence

$$\frac{0.001}{0.3} = \frac{1 \cdot I_2}{0.3 + 0.6}$$



$$\Rightarrow I_1 I_2 = 0.003 \text{ m}$$

148. [D]

A convex mirror can never form a real image.

149. [A]

$$\text{Given, } \mu = \sqrt{2}, A = 60^\circ, \delta = 30^\circ$$

For the minimum deviation, we have

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

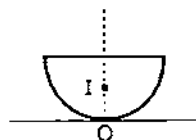
Substituting the values of A and μ we get

$$\delta_m = 30^\circ$$

i.e., the given deviation $\delta = 30^\circ$ is actually the minimum deviation δ_m . At minimum deviation the ray inside the prism is parallel to the base of the prism (in case of an equilateral prism).

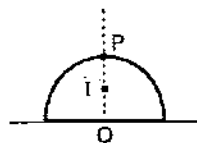
151. [B]

CASE I



$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}} = \frac{4}{3}$$

CASE II



Consider origin to be at the pole P.

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

$$\Rightarrow \frac{\frac{4}{3}}{-(-4)} + \frac{1}{\left(-\frac{25}{8}\right)} = \frac{1 - \left(\frac{4}{3}\right)}{R}$$

$$\Rightarrow R = -25 \text{ cm}$$

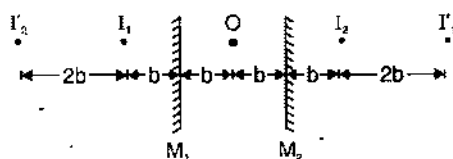
According to Lens Maker's Formula

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\Rightarrow \frac{1}{f} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{\infty} - \frac{1}{-25}\right)$$

$$\Rightarrow f = 75 \text{ cm}$$

152. [C]



I_1 is the direct image of O by mirror M_1

I_2 is the direct image of O by mirror M_2

I_1 acts as object for mirror M_2 and gives an image I_1' and similarly I_2 acts as object for mirror M_1 and gives image I_2' .

Since I_1 is located at a distance $3b$ from M_2 .

So, I_1' must be at a distance $3b$ from M_2 .

Similarly I_2' must be at a distance $3b$ from M_1 .

From above we observe that the first image is at a distance $2b$ from O.

Second image is at a distance $4b$ from O and so on. So, n th image is at a distance $2nb$ from O.

153. [C]

$$\theta = \frac{1}{2}(2\pi)\left(\frac{1}{4}\right)^n \quad \left\{ \because \theta = \frac{1}{2} \alpha t^2 \right\}$$

$$\Rightarrow \theta = \frac{\pi}{16}$$

Reflected ray turns through an angle $2\theta = 22.5^\circ$

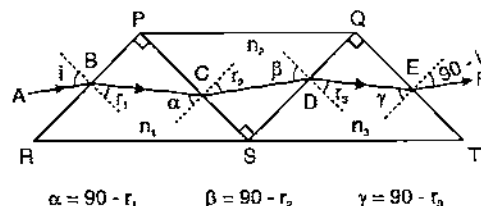
154. [C]

Distance of first image (I_1) formed due to refraction from the plane surface of water is

$$d_{\text{app}} = \frac{d_{\text{real}}}{\mu} = \frac{10}{4/3} = 7.5 \text{ cm}$$

Now distance of this image from the plane mirror is $x = 5 + 7.5 = 12.5 \text{ cm}$. Therefore, distance of second image (I_2) from the mirror will also be equal to 12.5 cm from it.

155. [D]



$$\alpha = 90^\circ - r_1, \quad \beta = 90^\circ - r_2, \quad \gamma = 90^\circ - r_3$$

At B

$$\sin i = n_1 \sin r_1$$

$$\Rightarrow \sin^2 i = n_1^2 \sin^2 r_1 \quad \dots(1)$$

At C

$$n_1 \sin(90^\circ - r_1) = n_2 \sin r_2$$

$$\Rightarrow n_1^2 \sin^2 r_2 = n_1^2 \cos^2 r_1 \quad \dots(2)$$

At D

$$n_2 \sin(90^\circ - r_2) = n_3 \sin r_3$$

$$\Rightarrow n_2^2 \cos^2 r_2 = n_3^2 \sin^2 r_3 \quad \dots(3)$$

At E

$$n_3 \sin(90^\circ - r_3) = (1) \sin(90^\circ - i)$$

$$\Rightarrow \cos^2 i = n_3^2 \cos^2 r_3 \quad \dots(4)$$

Adding (1), (2), (3) & (4)

$$\Rightarrow 1 + n_2^2 = n_1^2 + n_3^2$$

156. [A]

$$P = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

158. [D]

$$(90-r) + (90-C) + 45 = 180$$

$$\Rightarrow r + C = 45^\circ$$

$$\Rightarrow r = 45 - C$$

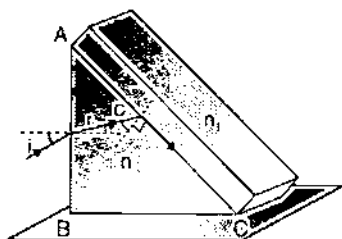
$$\text{Since } \frac{\sin i}{\sin r} = n$$

$$\Rightarrow \sin i = n \sin r$$

$$\Rightarrow \sin i = n \sin(45 - C)$$

$$\Rightarrow \sin i = n(\sin 45 \cos C - \cos 45 \sin C)$$

$$\Rightarrow \sin i = \frac{n}{\sqrt{2}}(\cos C - \sin C)$$



$$\text{Further, } \sin C = \frac{n_1}{n}$$

$$\Rightarrow \sin i = \frac{n}{\sqrt{2}} \left[\sqrt{1 - \frac{n_1^2}{n^2}} - \frac{n_1}{n} \right]$$

$$\Rightarrow \sin i = \frac{1}{\sqrt{2}} \left[\sqrt{n^2 - n_1^2} - n_1 \right]$$

$$\Rightarrow i = \sin^{-1} \left[\frac{\sqrt{n^2 - n_1^2} - n_1}{\sqrt{2}} \right]$$

159. [D]

$$f_o + f_e = 36 \text{ cm}$$

$$\frac{f_o}{f_e} = 5$$

$$\Rightarrow f_o = 30 \text{ cm and } f_e = 6 \text{ cm.}$$

160. [B]

$$\frac{n_1}{v} + \frac{n_2}{-u} = \frac{n_1 - n_2}{R}$$

$$\Rightarrow \frac{1}{v} - \frac{2}{(-15)} = \frac{1-2}{-10}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{2}{15} = -\frac{1}{30}$$

$$\Rightarrow v = -30 \text{ cm}$$

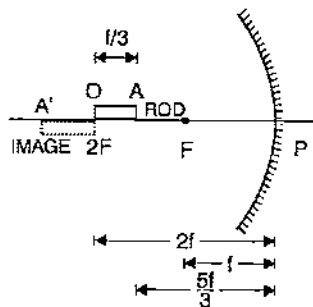
161. [C]

Since an elongated image is formed and it touches one end of the rod, so the rod must lie with one end at $2F$ and other end between $2F$ and F (shown in figure).

$$\text{For end A, } u = -\frac{5f}{3}$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{3}{5f} = \frac{1}{-f}$$



$$\Rightarrow \frac{1}{v} = \frac{3}{5f} - \frac{5}{5f}$$

$$\Rightarrow \frac{1}{v} = -\frac{2}{5f}$$

$$\Rightarrow v = -\frac{5f}{2}$$

$$\Rightarrow PA' = \frac{5f}{2}$$

$$\Rightarrow OA' = \frac{5f}{2} - 2f$$

$$\Rightarrow OA' = \frac{f}{2}$$

$$\Rightarrow m = \frac{OA'}{OA} = \frac{\frac{f}{2}}{\frac{f}{3}} = \frac{3}{2}$$

162. [D]

For a plane refracting surface, the lateral magnification is 1. So, the image of the coin will be of the same size as the coin itself.

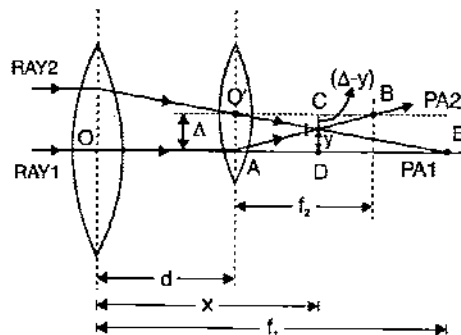
163. [D]

$$\delta = \frac{A}{2} = \frac{60}{2} = 30$$

$\Rightarrow$ The ray passes symmetrically through the prism parallel to its base.

$$\text{Hence } \theta = 0^\circ$$

164. [C]



For drawing the above ray diagram, we must consider two rays RAY 1 and RAY 2 in such a manner that RAY 1 passes through optical centre of first convex lens and is parallel to the Principal Axis for the second convex lens (PA2).

So wherever the RAY 1 cuts PA2 is the focal length f_2 of second convex lens. Similarly we have RAY 2 parallel to PA1 and let this pass through O' so as to cut PA1 at a distance equal to focal length (f_1) for the first convex lens.

$\triangle IDE$ and $\triangle O'AE$ are similar

$$\Rightarrow \frac{y}{\Delta} = \frac{f_1 - x}{f_1 - d} \quad \dots(1)$$

$\triangle ICB$ and $\triangle O'B$ are similar

$$\Rightarrow \frac{\Delta - y}{\Delta} = \frac{(f_2 + d) - x}{f_2}$$

$$\Rightarrow \frac{y}{\Delta} = \frac{d}{f_2} - \frac{x}{f_2}$$

$$\Rightarrow \frac{y}{\Delta} = \frac{x}{f_2} - \frac{d}{f_1}$$

From (1) and (2)

$$\frac{x}{f_2} - \frac{d}{f_1} = \frac{f_1 - x}{f_1 - d}$$

$$\Rightarrow \frac{x}{f_2} - \frac{d}{f_1} = \frac{f_1}{f_1 - d} - \frac{x}{f_1 - d}$$

$$\Rightarrow x \left(\frac{1}{f_2} + \frac{1}{f_1 - d} \right) = \frac{f_1}{f_1 - d} + \frac{d}{f_1}$$

$$\Rightarrow x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}$$

Similarly we can calculate value of y .

OBJECTIVE TRICK

If $\Delta = 0$, y must be zero (satisfied by A, B, C)

If $d \rightarrow f_1 + f_2$, then $x \rightarrow \infty$ (satisfied by B, C, D)

If $d \rightarrow f_1$, then $x \rightarrow f_1$ (satisfied by C, D)

So, we observe that **OPTION (C)** is the only option satisfying both the x and y in special situations mentioned above.

166. [C]

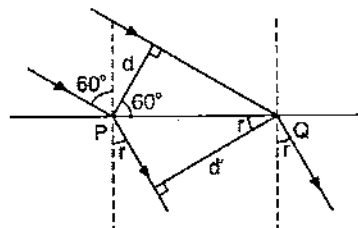
Let d' be the diameter of refracted beam. Then

$$d = PQ \cos(60^\circ)$$

and $d' = PQ \cos r$

$$\Rightarrow \frac{d'}{d} = \frac{\cos r}{\cos 60^\circ} = 2 \cos r$$

$$\Rightarrow d' = 2d \cos r$$



$$\text{Since, } \sin r = \frac{\sin i}{\mu} = \frac{\frac{\sqrt{3}}{2}}{2} = \frac{1}{\sqrt{3}}$$

$$\text{Also, } \cos r = \sqrt{1 - \sin^2 r}$$

$$\Rightarrow \cos r = \sqrt{\frac{2}{3}}$$

$$\Rightarrow d' = (2)(2) \sqrt{\frac{2}{3}} = 4 \sqrt{\frac{2}{3}} \text{ cm}$$

$$\Rightarrow d' \approx 3.26 \text{ cm}$$

167. [A]

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$$

170. [A]

$$\frac{1}{-(-x)} + \frac{1.5}{x} = \frac{1.5 - 1}{R}$$

$$\Rightarrow \frac{2.5}{x} = \frac{0.5}{R}$$

$$\Rightarrow x = 5R$$

171. [A]

$$\frac{1}{f'} = \left(\frac{1.5}{1.75} - 1 \right) \left(\frac{-2}{R} \right)$$

$$\Rightarrow \frac{1}{f'} = \frac{0.25}{1.75} \times \frac{2}{R}$$

$$\Rightarrow f' = \frac{R}{2} \left(\frac{175}{25} \right)$$

$$\Rightarrow f' = 3.5R$$

172. [D]

For refraction at plane surface,

$$\frac{1}{-(-mR)} + \frac{1.5}{v} = \frac{1.5 - 1}{\infty}$$

$$\Rightarrow v = -1.5 mR$$

For refraction at the curved surface

$$u' = -(1.5 mR + R), v' \rightarrow \infty$$

$$\Rightarrow \frac{1}{\infty} - \frac{1.5}{-(1.5 mR + R)} = \frac{1 - 1.5}{-R}$$

$$\Rightarrow \frac{3}{(1.5 m + 1)} = 1$$

$$\Rightarrow 3 = 1.5 m + 1$$

$$\Rightarrow 1.5 m = 2$$

$$\Rightarrow m = \frac{4}{3}$$

174. [C]

Deviation by a sphere is $2(i - r)$

Here, deviation $\delta = 60^\circ = 2(i - r)$

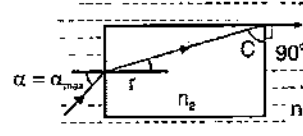
$$\Rightarrow i - r = 30^\circ$$

$$\Rightarrow r = i - 30^\circ = 60^\circ - 30^\circ = 30^\circ$$

According to Snell's Law, we have $\mu = \frac{\sin i}{\sin r}$

$$\Rightarrow \mu = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$

175. [A]



$$\frac{\sin \alpha}{\sin r} = \frac{n_1}{n_2}$$

$$\Rightarrow \sin \alpha = \frac{n_1}{n_2} \sin r$$

...(1)

For TIR at other end

$$\Rightarrow \sin C = \frac{n_2}{n_1}$$

Also $C = 90^\circ - r$

$$\Rightarrow \sin \alpha = \frac{n_1}{n_2} \sin(90^\circ - C) \quad [\text{Put } r = 90^\circ - C \text{ in (1)}]$$

$$\Rightarrow \sin \alpha = \frac{n_1}{n_2} \cos C$$

$$\Rightarrow \sin \alpha = \frac{n_1}{n_2} \left[\cos \left[\sin^{-1} \left(\frac{n_2}{n_1} \right) \right] \right]$$

176. [D]

CD is parallel to AB, so both media must have equal refractive indices.

177. [C]

Since all the prisms P, Q and R are made of same material, so the deviation suffered by the combination remains the same as the combination forms a part of the bigger sphere.

178. [B]

Let R be the radius of curvature of each surface. Then applying Lens Maker's Formula, we get

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$\Rightarrow R = f$$

For the water lens, again applying the Lens Maker's Formula, we get

$$\frac{1}{f'} = \left(\frac{4}{3} - 1 \right) \left(-\frac{1}{R} - \frac{1}{R} \right) = \frac{1}{3} \left(-\frac{2}{f} \right)$$

$$\Rightarrow \frac{1}{f'} = -\frac{2}{3f}$$

Since, we know that for lenses placed in contact, we have

$$\frac{1}{F} = \frac{1}{f} + \frac{1}{f'} + \frac{1}{f''}$$

$$\Rightarrow \frac{1}{F} = \frac{1}{f} + \frac{1}{f'} + \frac{1}{f''}$$

$$\Rightarrow \frac{1}{F} = \frac{2}{f} - \frac{2}{3f} = \frac{4}{3f}$$

$$\Rightarrow F = \frac{3f}{4}$$

179. [B]

$$\frac{\sin i}{\sin r} = \frac{1}{n}$$

Since

$$\tan r = \frac{2h}{2h} = 1$$

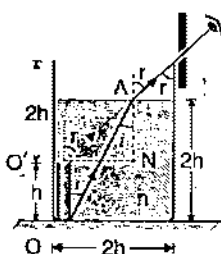
$$\Rightarrow r = 45^\circ$$

$$\Rightarrow \sin i = \frac{h}{h\sqrt{5}}$$

$$\Rightarrow \sin i = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \frac{1}{n} = \frac{\sqrt{5}}{1}$$

$$\Rightarrow n = \sqrt{\frac{5}{2}}$$



180. [C]

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For no dispersion

$$d \left(\frac{1}{f} \right) = 0$$

$$\Rightarrow d\mu \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = 0$$

$$\Rightarrow R_1 = R_2$$

181. [B]

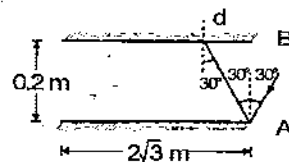
$$\frac{d}{0.2} = \tan 30^\circ$$

$$\Rightarrow d = \frac{0.2}{\sqrt{3}} = \frac{2}{10\sqrt{3}}$$

Total number of reflections

$$= n = \frac{\ell}{d} = \frac{2\sqrt{3}}{\left(\frac{2}{10\sqrt{3}} \right)}$$

$$\Rightarrow n = 30$$



182. [C]

$$A_{\max} = 2C$$

$$\Rightarrow A_{\max} = 2(36^\circ)$$

$$\Rightarrow A_{\max} = 72^\circ$$

183. [C]

$$f = \frac{D^2 - x^2}{4D}$$

$$\Rightarrow f = \frac{(90)^2 - (20)^2}{4(90)}$$

$$\Rightarrow f = 21.4 \text{ cm}$$

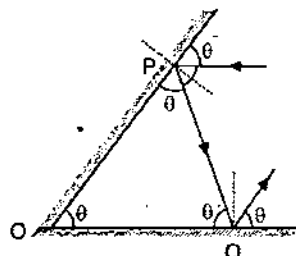
184. [C]

$$\text{Since } \mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sqrt{2} = \frac{\sin i}{\sin 30^\circ}$$

$$\Rightarrow i = 45^\circ$$

185. [C]

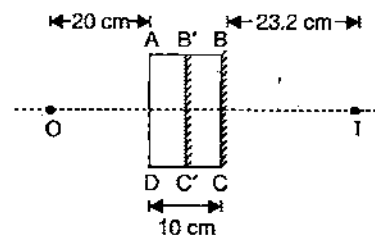


$$\angle OPQ = \angle OQP = \theta$$

$$\Rightarrow 3\theta = 180^\circ$$

$$\Rightarrow \theta = 60^\circ$$

186. [C]



AB' = Apparent Depth = x

AB = 10 cm

The face BC appears to be shifted to B'C'. If x be the apparent depth, then AB' = x and B'B = (10 - x).

Also, according to Laws of Reflection an image is formed as far behind the mirror as the object is in front of it

$$\Rightarrow 20 + x = 23.2 + (10 - x)$$

$$\Rightarrow 2x = 13.2$$

$$\Rightarrow x = 6.6 \text{ cm}$$

$$\text{Since } \mu = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$

$$\Rightarrow \mu = \frac{10}{6.6} = 1.51$$

187. [A]

$$\text{Since } \mu_2 t = \mu_1 t'$$

$$\Rightarrow t' = \frac{\mu_2}{\mu_1} t$$

$$\Rightarrow \Delta x = t' - t$$

$$\Rightarrow \Delta x = \left(\frac{\mu_2}{\mu_1} - 1 \right) t = \left(\frac{3}{2} - 1 \right) t = \frac{t}{2}$$

188. [B]

By Laws of Reflection

$$i = r$$

Also

$$i + r = 90$$

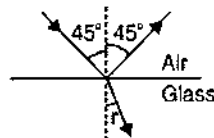
$$\Rightarrow i = 45$$

According to Snell's Law

$$\mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow 1.5 = \frac{3}{2} = \frac{\sin 45}{\sin r}$$

$$\Rightarrow r = \sin^{-1} \left(\frac{\sqrt{2}}{3} \right)$$



189. [A]

$$\text{Shift} = t \left(1 - \frac{1}{\mu} \right) \text{ away}$$

190. [B]

According to Newton's Formula, we have

$$x_1 x_2 = f^2$$

$$\Rightarrow (10)(40) = f^2$$

$$\Rightarrow f^2 = 400$$

$$\Rightarrow f = 20 \text{ cm}$$

191. [B]

$$A = 60^\circ$$

$$i = 55^\circ, e = 46^\circ$$

$$\text{Since } i + e = A + D$$

$$\Rightarrow 55 + 46 = 60 + D$$

$$\Rightarrow D = 41^\circ$$

$$\text{So, } D_m < D$$

192. [A]

$$\text{At } D = D_m$$

$$i = e$$

193. [A]

$$\text{Areal Magnification} = 9$$

$$\Rightarrow m_{ar} = 9 = \left(\frac{f}{f+u} \right)^2$$

$$\Rightarrow \frac{f}{f+u} = -3$$

(Since we get a real image)

$$\Rightarrow \frac{f}{f-40} = -3$$

$$\Rightarrow -3f + 120 = f$$

$$\Rightarrow 4f = 120$$

$$\Rightarrow f = 30 \text{ cm}$$

194. [D]

Axial Magnification is given by

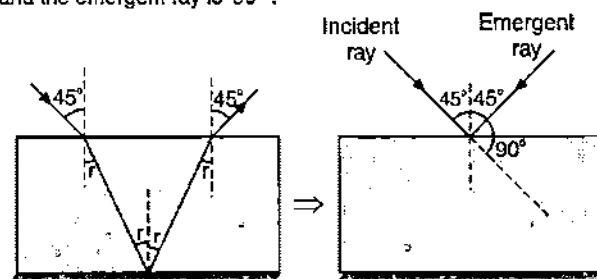
$$m_{ax} = \frac{I \text{ along PA}}{O \text{ along PA}} = \frac{v^2}{u^2}$$

$$\Rightarrow m_{ax} = \frac{I}{O} = \left(\frac{f}{f-u} \right)^2$$

$$\Rightarrow I = O \left(\frac{f}{f-u} \right)^2$$

195. [B]

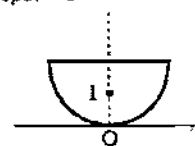
From the figure it is clear that the angle between incident ray and the emergent ray is 90° .



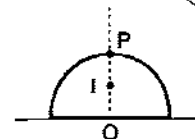
196. [B]

Case I

$$n = \frac{\text{Real Depth}}{\text{Apparent Depth}} = \frac{4}{3}$$



Case II



Consider origin to be at the pole P.

$$\frac{n_1}{-u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$$

$$\Rightarrow \frac{4}{-(-4)} + \frac{1}{\left(-\frac{25}{8} \right)} = \frac{1 - \left(\frac{4}{3} \right)}{R}$$

$$\Rightarrow R = -25 \text{ cm}$$

According to Lens Maker's Formula

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f} = \left(\frac{4}{3} - 1\right) \left(\frac{1}{\infty} - \frac{1}{-25}\right)$$

$$\Rightarrow f = 75 \text{ cm}$$

197. [D]

According to Lens Maker's Formula

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow f \propto \frac{1}{(\mu - 1)}$$

Since $\mu_{\text{Red}} < \mu_{\text{Violet}}$

$$\Rightarrow f_{\text{Red}} > f_{\text{Violet}}$$

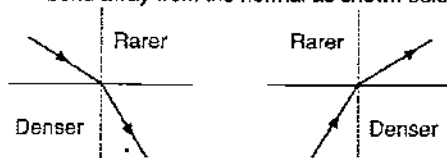
$$\Rightarrow f_{\text{Red}} > f_{\text{Blue}}$$

Always keep in mind that whenever you are asked to compare (greater than or less than) u , v or f you must not apply sign conventions for comparison.

198. [D]

Since, the refractive index is increasing linearly from top to the bottom, so the light cannot travel in a straight line in the liquid as shown in options (A) and (B).

Initially it will bend towards normal and after reflecting from the bottom it will bend away from the normal as shown below.



199. [A]

$$u = -(f + x)$$

$$m = \frac{f}{f + u}$$

$$\Rightarrow m = \frac{f}{f - f - x}$$

$$\Rightarrow m = -\frac{f}{x}$$

Negative sign confirms that real magnification is negative.

200. [D]

For a convex mirror, both (A) and (B) are incorrect.

201. [D]

$$u = -(f + x_1)$$

$$v = (f + x_2)$$

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f + x_2} + \frac{1}{f + x_1} = \frac{1}{f}$$

$$\Rightarrow f(2f + x_1 + x_2) = f^2 + (x_1 + x_2)f + x_1x_2$$

$$\Rightarrow f^2 = x_1x_2$$

$$\Rightarrow f = \sqrt{x_1x_2}$$

202. [C]

Two plano-convex lenses of focal length f on combining give a convex lens of focal length $\frac{f}{2}$. To obtain a real image of

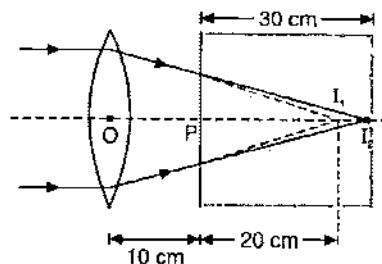
magnification unity the object must be at $2F$ (or C) and so image is also formed at $2F$ (or C). Hence $u = -2f = -40 \text{ cm}$

203. [D]

The lens will converge the rays at its focus i.e., 30 cm from the lens or 20 cm from the refracting surface, so we have

$$PI_1 = 20 \text{ cm}$$

$$\text{Now, } PI_2 = \mu(PI_1) = \frac{3}{2} \times 20 = 30 \text{ cm}$$



Hence, the rays will converge at a distance of 40 cm from the lens.

204. [C]

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{-R} \right)$$

$$\Rightarrow \frac{1}{f} = (\mu - 1) \left(\frac{2}{R} \right)$$

...(1)

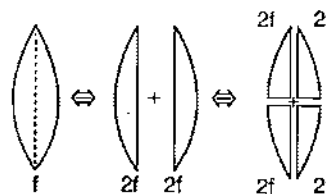
On cutting

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$\Rightarrow f = 2f$$

205. [C]

When a lens is cut parallel to principal axis its focal length remains the same. Hence each part will have a focal length $2f$.



206. [B]

According to Lens Maker's Formula

$$\frac{1}{v} - \frac{1}{u} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

The lens is made of two materials, so for a single object distance, two different image distances are obtained i.e. two images are formed.

208. [B]

Let f_1 and f_2 be the focal lengths of the lenses of refractive indices μ_1 and μ_2 in water, then

$$\frac{1}{f_1} = \left(\frac{\mu_1}{\mu_w} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_1} = \left(\frac{\mu_1}{\mu_w} - 1 \right) \left(\frac{2}{R} \right)$$

...(1)

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_w} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{f_2} = \left(\frac{\mu_2}{\mu_w} - 1 \right) \left(\frac{-2}{R} \right) \quad \dots(2)$$

Adding equations (1) and (2), we get

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{2(\mu_1 - \mu_2)}{\mu_w R}$$

$$\Rightarrow \frac{1}{30} = \frac{2(\mu_1 - \mu_2)}{\mu_w R}$$

$$\Rightarrow (\mu_1 - \mu_2) = \frac{\mu_w R}{60}$$

Substituting the values, we get

$$(\mu_1 - \mu_2) = \frac{4 \times 15}{3 \times 60} = \frac{1}{3}$$

209. [A]

Since image is formed on the screen, hence it must be real. So, Lens used must be convex (and cannot be concave)

$$u = -45 \text{ cm}, v = +90 \text{ cm}$$

Since, for a lens

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{90} + \frac{1}{45} = \frac{1}{f}$$

$$\Rightarrow f = +30 \text{ cm}$$

Further again the positive sign with focal length indicates the lens is convex.

210. [B]

$$\text{Size of Image} = \frac{v}{u} (\text{Size of Object})$$

$$\Rightarrow I = \frac{90}{-45} (5) = -10 \text{ cm}$$

Here again the negative sign with 1 indicates that it is formed below origin i.e. inverted i.e. real.

211. [C]

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

$$\Rightarrow \frac{1}{-(-50)} + \frac{1.5}{v} = \frac{1.5 - 1}{20}$$

$$\Rightarrow \frac{1}{50} + \frac{1.5}{v} = \frac{0.5}{20}$$

$$\Rightarrow \frac{1.5}{v} = \frac{1}{40} - \frac{1}{50}$$

$$\Rightarrow \frac{1.5}{v} = \frac{10}{2000}$$

$$\Rightarrow v = \frac{15 \times 200}{10} = +300 \text{ cm}$$

Positive side indicates that image is real.

212. [C]

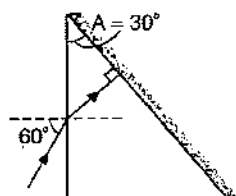
$$r_2 = 0^\circ$$

$$\Rightarrow r_1 = A = 30^\circ$$

$$\text{and } i_1 = 60^\circ$$

So, from Snell's Law, we get

$$\mu = \frac{\sin i_1}{\sin r_1} = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$

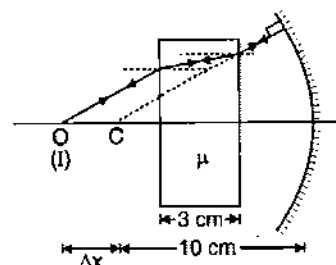


213. [C]

$$\Delta x = 3 \left[1 - \frac{1}{\left(\frac{3}{2} \right)} \right]$$

$$\Rightarrow \Delta x = 1 \text{ cm}$$

$$\Rightarrow \text{Object distance} = 10 + 1 = 11 \text{ cm}$$

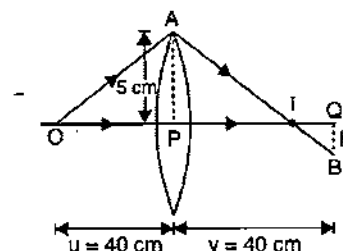


214. [C]

$$f_t = \frac{\mu_2 - 1}{\mu_2 - 1} \times f_a = \frac{1.5 - 1}{1.5 - 1} \times 20$$

$$\Rightarrow f_t = -\frac{0.5 \times 1.6}{0.1} \times 20 = -160 \text{ cm}$$

215. [C]



$$f = 20 \text{ cm}, u = -40 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \text{ gives}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} - \frac{1}{40}$$

$$\Rightarrow v = 40 \text{ cm}$$

The situation is shown in figure.

From similar triangles PAI and IQB

$$\frac{5}{h} = \frac{40}{20}$$

$$\Rightarrow h = \frac{5}{2} = 2.5 \text{ cm}$$

216. [C]

$$\text{Since } O^2 = I_1 I_2$$

$$\Rightarrow 4 = (0.5) I_2$$

$$\Rightarrow I_2 = 8 \text{ cm}$$

217. [C]

According to Newton's Formula

$$x_1 x_2 = f^2$$

$$\Rightarrow f = \sqrt{16 \times 25} = 20 \text{ cm}$$

218. [A]

$$\cos(180^\circ - 2\theta) = \frac{\left(\frac{i + \sqrt{3}j}{2}\right) \cdot \left(\frac{i - \sqrt{3}j}{2}\right)}{\left|\frac{i + \sqrt{3}j}{2}\right| \left|\frac{i - \sqrt{3}j}{2}\right|}$$

$$\Rightarrow -\cos 2\theta = \frac{(1-3)}{4}$$

$$\Rightarrow -\cos 2\theta = \frac{-1}{2}$$

$$\Rightarrow \cos 2\theta = \frac{1}{2}$$

$$\Rightarrow 2\theta = 60^\circ$$

$$\Rightarrow \theta = 30^\circ$$

219. [B]

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{14} - \frac{1}{\infty} \right) + (1.2 - 1) \left(\frac{-1}{\infty} - \frac{1}{-14} \right)$$

$$\Rightarrow f = 20 \text{ cm}$$

$$\Rightarrow \text{As object distance} = 40 \text{ cm}$$

$$\Rightarrow \text{Image distance} = 40 \text{ cm}$$

220. [A]

Focal Length of Lens is $f_l = 20 \text{ cm}$.

Focal length of combination is

$$f_c = 20 + 5 = 25 \text{ cm}$$

when the space between lens and mirror is filled with a liquid then

$$\frac{1}{f_c} = \frac{1}{f_l} + \frac{1}{f}$$

$$\frac{1}{25} = \frac{1}{f_l} + \frac{1}{20}$$

$$\Rightarrow \frac{1}{f_l} = \frac{1}{25} - \frac{1}{20}$$

$$\Rightarrow f_l = -100 \text{ cm}$$

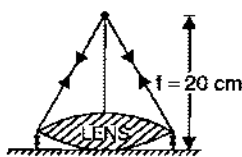
For this liquid lens

$$\frac{1}{f_l} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow -\frac{1}{100} = (\mu - 1) \left(\frac{1}{-33} - \frac{1}{\infty} \right)$$

$$\Rightarrow \mu - 1 = 0.33$$

$$\Rightarrow \mu = 1.33 = \frac{4}{3}$$



222. [D]

For real image

$$u = -u_1, v = -2u_1, f = -20 \text{ cm}$$

Substituting in $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{-2u_1} - \frac{1}{u_1} = -\frac{1}{20}$$

$$\Rightarrow u_1 = 30 \text{ cm}$$

For virtual image $u = -u_2, v = 2u_2, f = -20 \text{ cm}$

Again applying $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{2u_2} - \frac{1}{u_2} = -\frac{1}{20}$$

$$\Rightarrow u_2 = 10 \text{ cm}$$

So, the distance between two positions of the object is $u_1 - u_2$

$$\Rightarrow u_1 - u_2 = 30 \text{ cm} - 10 \text{ cm} = 20 \text{ cm}$$

223. [A]

$$\frac{1}{f} = \frac{1}{40} + \frac{1}{-25}$$

$$\Rightarrow f = \frac{-40 \times 25}{40 - 25}$$

$$\Rightarrow f = -66.67 \text{ cm}$$

$$P = \frac{10}{f(\text{in cm})} = -1.5 \text{ D}$$

224. [A]

$$P = P_1 + P_2$$

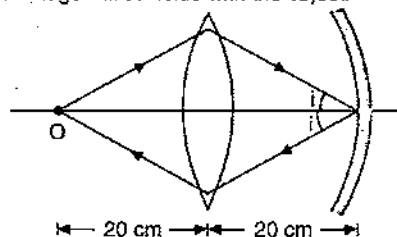
$$\Rightarrow P = 10 \text{ D}$$

$$f = \frac{1}{P} = 0.1 \text{ m}$$

$$\Rightarrow f = 10 \text{ cm}$$

225. [B]

Object is placed at a distance of $2f$ from the lens of focal length f i.e., the image formed by the lens will be at a distance of $2f$ or 20 cm from the lens. So, if the concave mirror is placed in this position, the first image will be formed at its pole and it will reflect all the rays symmetrically to other side as shown below and the final image will coincide with the object.



226. [D]

$$\text{Shift} = d = t \left(1 - \frac{1}{n} \right)$$

$$\Rightarrow t = \frac{nd}{n-1}$$

227. [A]

Focal length of plano convex lens using Lens Maker's Formula is

$$\frac{1}{f} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{10} - \frac{1}{\infty}\right)$$

$$\Rightarrow f = 20 \text{ cm}$$

If point object O is placed at a distance of 20 cm from the plano convex lens rays become parallel and final image is formed at second focus or 20 cm from concave lens which is independent of y .

228. [D]

Focal length remains unchanged.

$I \propto \text{Area of Aperture}$

$$I = kD^2$$

$$I' = k \left[D^2 - \left(\frac{D}{2}\right)^2 \right]$$

$$\Rightarrow I' = \frac{3I}{4}$$

229. [B]

$$\lambda = \frac{v}{f}$$

In moving from air to glass, f remains unchanged while v decreases. Hence, λ should decrease.

230. [B]

$$f = \frac{x}{m_1 - m_2} \quad (\text{See 'Displacement Method' in SYNOPSIS})$$

231. [B]

$$f = \frac{D^2 - x^2}{4D}$$

$$\Rightarrow f = \frac{(0.9)^2 - (0.2)^2}{4(0.9)}$$

$$\Rightarrow f = 0.214 \text{ m}$$

$$\Rightarrow f = 21.4 \text{ cm}$$

232. [C]

When $\theta < C$ partial transmission and reflection will occur. When $\theta > C$, only reflection takes place.

234. [B]

$$\frac{1}{F} = \frac{2}{f_t} + \frac{1}{f_m}$$

$$\Rightarrow \frac{1}{F} = \frac{2}{f_t} + \frac{1}{\infty}$$

$$\Rightarrow F = \frac{f_t}{2}$$

Also

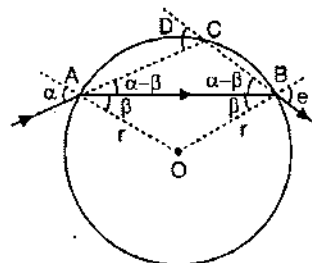
$$\frac{1}{f_t} = (\mu - 1) \left(\frac{1}{-r} - \frac{1}{\infty} \right)$$

$$\Rightarrow f_t = \frac{-r}{\mu - 1}$$

$$\Rightarrow F = -\frac{r}{2(\mu - 1)}$$

Negative sign indicates that mirror is concave.

235. [B]



Since angles opposite to equal sides are equal. So, in $\triangle AOB$ we have $\angle ABO = \beta$. In $\triangle CAB$, external angle equals the sum of internal opposite angles. So,

$$D = (\alpha - \beta) + (\alpha - \beta)$$

$$D = 2(\alpha - \beta)$$

236. [C]

When magnification is $m = +2$, then we have

$$u = -x$$

$$v = -2x$$

$$f = +20$$

Applying Mirror Formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{-2x} + \frac{1}{x} = \frac{1}{20}$$

$$\Rightarrow x = 10 \text{ cm}$$

To have a magnification of $m' = -2$, we have

$$u = -y$$

$$v = +2y$$

and $f = +20$

$$\Rightarrow \frac{1}{2y} + \frac{1}{y} = \frac{1}{20}$$

$$\Rightarrow y = 30 \text{ cm}$$

So, the object has to be moved by

$$y - x = 20 \text{ cm}$$

237. [D]

When water is filled in the mirror, a plano convex lens is formed, so now combination contains a plano-convex lens and a mirror. The effective focal length of combination is less than the focal length of above mirror, so image is shifted downwards.

239. [D]

The given lens is a convex lens. Let the magnification be m , then for real image, we have

$$\frac{1}{mx} + \frac{1}{x} = \frac{1}{f} \quad \dots(1)$$

and for virtual image, we have

$$\frac{1}{-my} + \frac{1}{y} = \frac{1}{f} \quad \dots(2)$$

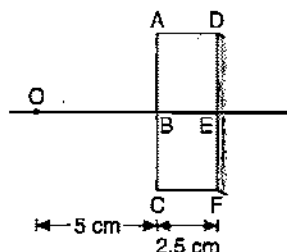
From equation (1) and (2), we get

$$f = \frac{x+y}{2}$$

241. [B]

Let I_1 , I_2 and I_3 be the images formed due to

- refraction from ABC
- reflection from DEF and
- again refraction from ABC



Then $BI_1 = (5)\mu_g = (5)(1.5) = 7.5$ cm

Now $EI_1 = (7.5 + 2.5) = 10$ cm

$\Rightarrow EI_2 = 10$ cm behind the mirror

Now $BI_2 = (10 + 2.5) = 12.5$ cm

$\Rightarrow BI_3 = \frac{12.5}{\mu_g} = \frac{12.5}{1.5} = \frac{25}{3}$ cm

242. [C]

$f = \frac{D^2 - x^2}{4D}$ (By Displacement Method)

$\Rightarrow P = \frac{4D}{D^2 - x^2}$

$\Rightarrow P = \frac{4(1)}{1 - 0.16}$

$\Rightarrow P = \frac{4}{0.84}$

$\Rightarrow P = 4.76$ D

243. [A]

Shift $= d \left(1 - \frac{1}{\mu} \right) = 7 \left(1 - \frac{1}{1.4} \right)$

$\Rightarrow$ Shift = 2 cm (downwards)

244. [A]

$\frac{1}{F} = \frac{2}{f} + \frac{1}{f_m}$

$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

$\Rightarrow \frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{60} \right)$

$\Rightarrow f = 60$ cm and

$f_m = \frac{20}{2} = 10$ cm

$\Rightarrow \frac{1}{F} = \frac{2}{60} + \frac{1}{10}$

$\Rightarrow F = 7.5$ cm

For image to be formed at the same place where object is situated we have $u = 2F = 15$ cm

245. [D]

$\frac{1}{F} = \frac{2}{f} + \frac{2}{f_w} + \frac{1}{f_m}$

(Because here we have two refractions at the concave surface of lens and two refraction for water lens). Since

$\frac{1}{f_w} = \left(\frac{4}{3} - 1 \right) \left(\frac{1}{60} - \frac{1}{\infty} \right)$



$\Rightarrow f_w = 180$ cm

$\Rightarrow \frac{1}{F} = \frac{2}{60} + \frac{2}{180} + \frac{1}{10}$

$\Rightarrow \frac{1}{F} = \frac{13}{90}$ cm

$\Rightarrow F = \frac{90}{13}$ cm

So, required answer to get asked condition is $2F$. Hence

$u = \frac{180}{13}$ cm

246. [C]

For equiconvex lens, we have

$|R_1| = |R_2| = f = 10$ cm

Now $P = 2P_L + P_M$

$\Rightarrow \frac{1}{F} = 2(\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) - \frac{2}{R_2}$

$\Rightarrow \frac{1}{F} = 2(1.5 - 1) \left(\frac{1}{10} - \frac{1}{-10} \right) - \frac{2}{-10}$

$\Rightarrow F = -2.5$ cm

Therefore, the system will behave like a concave mirror of focal length 2.5 cm.



247. [B]

$\frac{1}{F} = \frac{2}{f} + \frac{1}{f_m}$

Since $f_m \rightarrow \infty$

$\Rightarrow F = \frac{f}{2} = 15$ cm

Further

$\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$

$\Rightarrow \frac{1}{v} - \frac{1}{(-40)} = \frac{1}{15}$

$\Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{40}$

$\Rightarrow v = 24$ cm

248. [D]

Applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

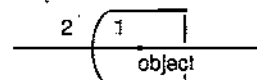
$\frac{1}{v} - \frac{1.5}{(-u)} = \frac{1 - (1.5)}{-R}$

$\Rightarrow \frac{1}{v} + \frac{3}{2u} = \frac{1}{2R}$

For v to be positive, we have

$\frac{1}{2R} > \frac{3}{2u}$

$\Rightarrow u > 3R$



249. [B]

$f_{\text{combination}} \rightarrow \infty$

Since $\frac{1}{f_{\text{combination}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_2}$

$\Rightarrow 0 = \frac{1}{-10} + \frac{1}{f_2} - \frac{10}{(-10)f_2}$

$$\Rightarrow 0 = \frac{1}{-10} + \frac{1}{f_2} + \frac{1}{f_2}$$

$$\Rightarrow \frac{1}{10} = \frac{2}{f_2}$$

$$\Rightarrow f_2 = 20 \text{ cm}$$

250. [B]

$$\frac{f}{f+(-12)} = -\frac{f}{f+(-20)}$$

$$f-12 = -(f-20)$$

$$f-12 = -f+20$$

$$2f = 32$$

$$f = 16 \text{ cm}$$

251. [C]

$$5 = (0.5) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{R_1} - \frac{1}{R_2} = 10$$

$$-1 = (\mu_g - 1)10$$

$$\Rightarrow \left(\frac{\mu_g}{\mu_r} - 1 \right) = -0.1$$

$$\Rightarrow \frac{1.5}{\mu_r} = 0.9$$

$$\Rightarrow \mu_r = \frac{1.5}{0.9} = \frac{5}{3}$$

252. [D]

$$|u| + |v| = D$$

$$m = \frac{v}{u}$$

$$\Rightarrow v = mu$$

$$|u| + m|u| = D$$

$$\Rightarrow |u| = \frac{D}{m+1}$$

$$\Rightarrow u = -\left(\frac{D}{m+1} \right)$$

Since image formed is real, so it must be on positive side. Hence

$$|v| = +v = \frac{mD}{m+1}$$

For a lens

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{m+1}{mD} + \frac{m+1}{D} = \frac{1}{f}$$

$$\Rightarrow f = \frac{mD}{(m+1)^2}$$

253. [A]

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$\Rightarrow \frac{1}{16} = 0.5 \frac{1}{R}$$

$$\Rightarrow R = 8 \text{ cm}$$

254. [D]

$$\text{Since } m = \frac{f}{f+u}$$

$$\Rightarrow -1 = \frac{f}{f+u}$$

$$\Rightarrow -f - u = f$$

$$\Rightarrow u = -2f$$

When the lens is cut, the focal length of the remaining portion is $2f$.

Applying Lens formula, we get

$$\frac{1}{v} - \frac{1}{(-2f)} = \frac{1}{2f}$$

$$\Rightarrow v \rightarrow \infty$$

255. [B]

$$\frac{1}{10} = (1.5 - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{5} \quad \dots(1)$$

On immersing it in a liquid of refractive index 3

$$\frac{1}{f} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = \left(\frac{1.5}{3} - 1 \right) \left(\frac{1}{5} \right)$$

$$\Rightarrow \frac{1}{f} = -\frac{1}{10}$$

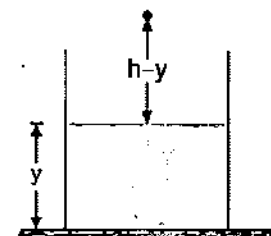
$$\Rightarrow f = -10 \text{ cm}$$

Negative sign indicates diverging nature of lens in the liquid.

256. [C]

Since, we have, at time t

$$d = y + \mu(h - y) = \mu h - (\mu - 1)y$$



If A is the area of the tank, then we have

$$y = \frac{at}{A} \quad (\because Ay = at)$$

$$\Rightarrow d = \mu h - \frac{(\mu - 1)at}{A}$$

i.e., $d-t$ graph is a straight line with negative slope and positive intercept. But d becomes constant once $y = H$.

257. [B]

On immersing in water f increases and hence P decreases.

258. [D]

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{0.5} - \frac{1}{-0.5} \right)$$

$$\Rightarrow \frac{1}{f} = 0.5 \left(\frac{2}{0.5} \right)$$

$$\Rightarrow P = 2D$$

259. [A]

$$O = \sqrt{I_1 I_2}$$

$$O = \sqrt{4 \times 16} = 8 \text{ cm}$$

260. [A]

For a lens making real image

$$-m = \frac{f}{f+u}$$

$$\Rightarrow -mf - mu = f$$

$$\Rightarrow -mu = f(1+m)$$

$$\Rightarrow u = -f \left(1 + \frac{1}{m} \right)$$

$$\Rightarrow |u| = f \left(1 + \frac{1}{m} \right)$$

For $|u|$ to be MINIMUM m must be MAXIMUM i.e. $m \rightarrow \infty$

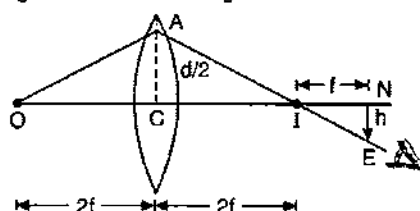
$$\Rightarrow |u|_{\min} = f$$

For $|u|$ to be MAXIMUM m must be MINIMUM i.e. $m = 1$

$$\Rightarrow |u|_{\max} = 2f$$

261. [A]

The ray diagram is as shown in figure



Since triangles CAI and NEI are similar, so we have

$$\frac{h}{d/2} = \frac{f}{2f}$$

$$\Rightarrow \frac{2h}{d} = \frac{1}{2}$$

$$\Rightarrow h = \frac{d}{4}$$

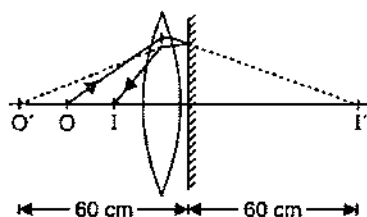
265. [A]

If mirror would have been absent then image is formed on the other side of lens (at I').

So

$$\frac{1}{v} - \frac{1}{-20} = \frac{1}{15}$$

$$\Rightarrow v = 60 \text{ cm}$$



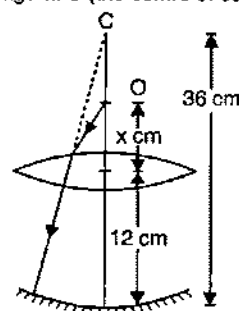
Since the mirror reflects the ray back, so O' serves as a virtual object and forms a real image I in front of mirror.

$$\frac{1}{v} + \frac{1}{-60} = \frac{1}{15}$$

$$\Rightarrow v = +12 \text{ cm}$$

266. [B]

Rays from O must fall normally on the mirror, only then the lens forms a virtual image at C (the centre of curvature).



$$\Rightarrow \frac{1}{x} + \frac{1}{-(36-12)} = \frac{1}{40}$$

$$\Rightarrow x = 15 \text{ cm}$$

267. [D]

$$\frac{1}{F} = \frac{1}{(1/2)} + \frac{1}{(1/3)} = 5$$

$$\Rightarrow F = 0.2 \text{ m} = 20 \text{ cm}$$

Since

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{(-30)} = \frac{1}{20}$$

$$\Rightarrow v = 60 \text{ cm}$$

268. [B]

$$P_{\text{comb}} = P_1 + P_2 - xPP_2$$

$$0 = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$$

$$\Rightarrow x = f_1 + f_2$$

270. [D]

Apparent Separation = 2(Apparent Depth)

$$\Rightarrow \text{Apparent Separation} = \frac{2h}{\mu}$$

271. [A]

For distant vision

$$u = -\infty$$

$$\Rightarrow \frac{1}{-2} - \frac{1}{-\infty} = \frac{1}{f}$$

$$\Rightarrow f = -2 \text{ m}$$

$$\Rightarrow P = -0.5 \text{ D}$$

For near vision

$$u = -D = -25 \text{ cm}$$

$$\Rightarrow \frac{1}{-1} - \frac{1}{-0.25} = \frac{1}{f}$$

$$\Rightarrow P = +3 \text{ D}$$

Hence he must use bifocal lenses with

$$P = -0.5 \text{ D and additional } +3.5 \text{ D}$$

(to give +3 D net)

273. [C]

In the first case. Let x be the distance of object from the mirror. Then

$$u = -x, v = +2x \text{ and } f = -f$$

Using $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{2x} - \frac{1}{x} = -\frac{1}{f}$$

$$\Rightarrow x = \frac{f}{2}$$

In the second case, let y be the distance of object from the mirror. Then

$$u = -y, v = -2y \text{ and } f = -f$$

$$\Rightarrow \frac{1}{-2y} - \frac{1}{y} = -\frac{1}{f}$$

$$\Rightarrow y = \frac{3}{2}f$$

So, object will have to be moved by a distance of $y - x = f$.

274. [A]

$$\frac{f_0}{f_0} = -5$$

$$\text{Separation} = f_0 + f_0$$

$$36 = -4f_0$$

$$\Rightarrow f_0 = -9 \text{ cm}$$

$$\Rightarrow f_0 = 45 \text{ cm}$$

275. [D]

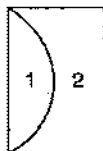
For lenses placed in contact, we have

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\Rightarrow \frac{1}{F} = (\mu_1 - 1) \left(\frac{1}{\infty} + \frac{1}{R} \right) + (\mu_2 - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right)$$

$$\Rightarrow \frac{1}{F} = \frac{\mu_1 - \mu_2}{R}$$

$$\Rightarrow F = \frac{R}{\mu_1 - \mu_2}$$



276. [B]

$$v_0 = -D = -25 \text{ cm}, f_0 = 6.25 \text{ cm}$$

Since

$$\frac{1}{f_0} = \frac{1}{v_0} + \frac{1}{u_0}$$

$$\Rightarrow \frac{1}{u_0} = \frac{1}{f_0} - \frac{1}{v_0}$$

$$\Rightarrow \frac{1}{u_0} = \frac{1}{6.25} - \frac{1}{(-25)}$$

$$\Rightarrow u_0 = 5 \text{ cm}$$

$$\Rightarrow v_0 = 15 - 5 = 10 \text{ cm}$$

Also, $f_0 = 2 \text{ cm}$ and

$$\frac{1}{f_0} = \frac{1}{v_0} + \frac{1}{u_0}$$

$$\Rightarrow \frac{1}{2} = \frac{1}{10} + \frac{1}{u_0}$$

$$\Rightarrow u_0 = -2.5 \text{ cm}$$

277. [C]

$$m = m_0 \times m_e = \frac{v_0}{u_0} \times \frac{v_e}{u_e} = \frac{10}{2.5} \times \frac{25}{5}$$

$$\Rightarrow m = 20$$

278. [B]

$$(\text{Shift})_1 = t \left(1 - \frac{1}{\mu_g} \right) = 9 \left(1 - \frac{1}{3/2} \right) = 3 \text{ cm}$$

$$(\text{Shift})_2 = t \left(1 - \frac{\mu_w}{\mu_g} \right) = 9 \left(1 - \frac{8}{9} \right) = 1 \text{ cm}$$

So, distance between two images is given by

$$(\text{Shift})_1 - (\text{Shift})_2 = 2 \text{ cm}$$

279. [A]

$$m = m_0 \times m_e = m_0 \left(1 + \frac{D}{f_0} \right)$$

$$\Rightarrow m_0 = \frac{m}{\left(1 + \frac{D}{f_0} \right)}$$

$$\Rightarrow m_0 = \frac{30}{1+5} = 5$$

280. [C]

$$m = 1 + \frac{D}{f}$$

$$\Rightarrow m = 1 + \frac{25}{5} = 6$$

281. [C]

$$m = \frac{f_0}{f_0} = 10$$

$$\Rightarrow f_0 = 50 \text{ cm}$$

$$\text{Separation} = f_0 + f_0 = 55 \text{ cm}$$

282. [A]

$$m = \frac{f_0}{f_0} = 5$$

283. [A]

$$\text{Separation} = f_0 + f_0$$

$$\text{Separation} = 0.3 + 0.05 = 0.35 \text{ m}$$

285. [D]

For an equilateral prism, we have

$$A = 60^\circ$$

Since, the ray inside the prism is parallel to its base, so we have the condition of minimum deviation. So, $i = e = 60^\circ$ and

$$\delta_m = i + e - A = (60^\circ + 60^\circ) - 60^\circ = 60^\circ$$

$$\text{Since, } \mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$$

$$\Rightarrow \mu = \frac{\sin(60^\circ)}{\sin(30^\circ)} = \sqrt{3}$$

286. [A]

$\frac{f_o}{f_e}$ must be maximum

289. [B]

Linear magnification $m = \frac{-2 \text{ cm}}{1 \text{ cm}} = -2$

So, the image is real and inverted

$$|v| = 2|u|$$

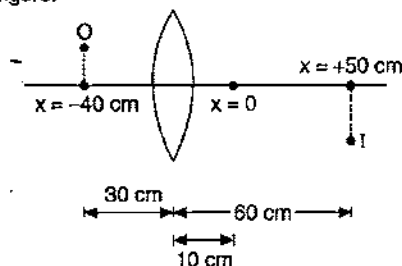
Let $|u| = x$ then $|v| = 2x$

Now $|u| + |v| = 50 - (-40) = 90$

$$\Rightarrow x + 2x = 90$$

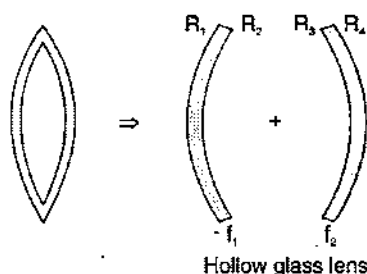
$$\Rightarrow x = 30 \text{ cm}$$

So, the distance of object from the lens is 30 cm and of object is 60 cm, i.e., the lens must be located at $x = -10 \text{ cm}$ as shown in figure.



292. [D]

Hollow convex lens is as shown in figure. Applying Lens Maker's Formula, we get



$$\frac{1}{f} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = 0 \quad \{\because R_1 = R_2\}$$

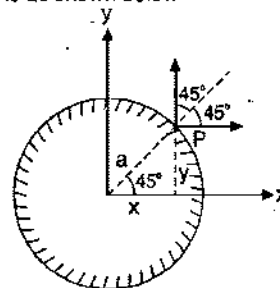
$$\Rightarrow f \rightarrow \infty$$

Similarly, $f_2 \rightarrow \infty$

So, a hollow, convex lens of any material will behave like a glass plate.

295. [D]

The ray diagram is as shown below



$$\text{Since, } x = \frac{a}{\sqrt{2}}$$

$$\text{and } y = \frac{a}{\sqrt{2}}$$

$$\Rightarrow P = \left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}} \right)$$

297. [C]

$$\frac{-f}{f-16} = \frac{f}{f-6}$$

$$\Rightarrow -f + 16 = f - 6$$

$$\Rightarrow 2f = 22$$

$$\Rightarrow f = 11 \text{ cm}$$

Solutions

Multiple Correct Choice Type Questions

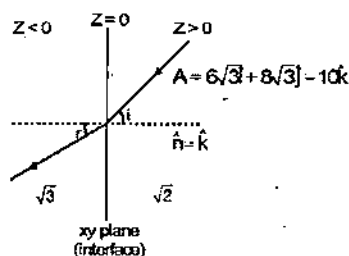
1. [B, C]

The normal to the interface is along $\hat{k}$.

$$\therefore \cos i = \frac{\vec{A} \cdot (-\hat{k})}{|\vec{A}| |\hat{k}|} = \frac{+10}{20} = \frac{1}{2}$$

$$\Rightarrow i = 60^\circ$$

(i is the angle which incident ray makes with $-Z$ -axis)



But according to Snell's Law

$$\sqrt{2} \sin 60 = \sqrt{3} \sin r$$

$$\Rightarrow \sin r = \frac{\sqrt{2} \sqrt{3}}{\sqrt{3} \cdot 2}$$

$$\Rightarrow \sin r = \frac{1}{\sqrt{2}}$$

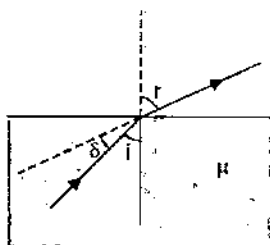
$$\Rightarrow r = 45^\circ$$

3. [A, D]

Combined Solution to 2 & 3

For $i < C$, no TIR will take place, so we have deviation (δ) given by

$$\delta = r - i$$



Now, according to Snell's Law, we have

$$\mu \sin i = \sin r$$

$$\Rightarrow r = \sin^{-1}(\mu \sin i)$$

$$\Rightarrow \delta = \sin^{-1}(\mu \sin i) - i$$

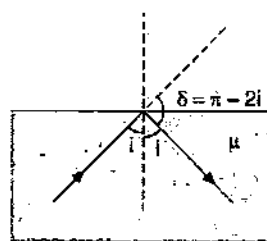
...(1)

This is a non-linear variation of δ with i . Also, we have δ to be maximum when $i = C$ and hence $r = \frac{\pi}{2}$. So

$$\delta_{\max} = \delta_1 = \frac{\pi}{2} - C$$

...(2)

Further when $i > C$, then TIR takes place and the incident ray is reflected back in the denser medium as shown in the figure.



$$\text{So, } \delta = \pi - 2i$$

i.e., δ decreases linearly with i . So,

$$\delta_{\max} = \delta_2 = \pi - 2C$$

...(3)

From (2) and (3), we get

$$\delta_2 = 2\delta_1$$

4. [A, B]

$$\sin C = \frac{\mu_{\text{rarer}}}{\mu_{\text{denser}}} = \frac{\mu_r}{\mu_d}$$

$$i + r' = 90^\circ$$

According to Snell's Law

$$\mu_d \sin i = \mu_r \sin(r')$$

$$\Rightarrow \frac{\sin i}{\sin(r')} = \frac{\mu_r}{\mu_d} = \sin C$$

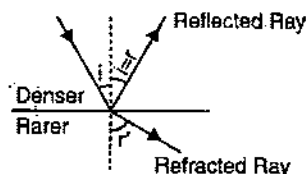
$$\Rightarrow \frac{\sin i}{\sin(90 - i)} = \sin C$$

$$\Rightarrow \sin C = \tan i$$

$$\Rightarrow C = \sin^{-1}(\tan i)$$

$$\Rightarrow C = \sin^{-1}(\tan r)$$

(Because $i = r$)



6. [A, C]

According to Snell's Law, we have

$$\mu_1 \lambda_1 = \mu_2 \lambda_2$$

$$\Rightarrow (1)(6000) = (1.5)\lambda_2$$

$$\Rightarrow \lambda_2 = 4000 \text{ \AA}$$

Since frequency does not change when light goes from one medium to another, so

$$v = \frac{c}{\lambda_1}$$

$$\Rightarrow v = \frac{3 \times 10^8}{6000 \times 10^{-10}} = 5 \times 10^{14} \text{ Hz}$$

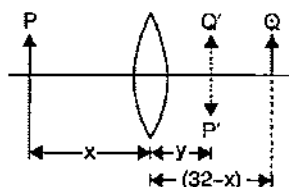
9. [B, C, D]

See Displacement Method.

12. [A, D]

Since images are formed at the same place, So one image must be real and other must be virtual.

$$|x| + |y| = 32$$



For P

$$\frac{1}{y} - \frac{1}{-x} = \frac{1}{15}$$

$$\Rightarrow \frac{1}{y} + \frac{1}{x} = \frac{1}{15} \quad \dots(1)$$

For Q

$$\frac{1}{-y} - \frac{1}{-(32-x)} = \frac{1}{15}$$

$$\Rightarrow \frac{1}{-y} + \frac{1}{32-x} = \frac{1}{15} \quad \dots(2)$$

Adding (1) and (2), we get

$$\frac{1}{x} + \frac{1}{32-x} = \frac{2}{15}$$

$$\frac{32}{x(32-x)} = \frac{2}{15}$$

$$\Rightarrow 32x - x^2 = 240$$

$$\Rightarrow x^2 - 32x + 240 = 0$$

$$\Rightarrow x^2 - 20x - 12x + 240 = 0$$

$$\Rightarrow x(x-20) - 12(x-20) = 0$$

$$\Rightarrow (x-12)(x-20) = 0$$

$$\Rightarrow x = 12 \text{ cm}, 20 \text{ cm}$$

13. [B, C]

$$m = \frac{f-v}{f}$$

$$\Rightarrow m = 1 - \frac{v}{f}$$

$$\Rightarrow \text{Slope} = -\frac{1}{f} = \frac{b}{c}$$

$$\Rightarrow f = \frac{c}{b}$$

Also at $m = 0$, $v = a$

$$\Rightarrow 0 = f - a$$

$$\Rightarrow f = a$$

14. [A, B, C, D]

All are the consequences of the

"DISPLACEMENT METHOD TO FIND FOCAL LENGTH OF A CONVEX LENS"

15. [A, B, C, D]

Objective and eye piece are separated by a distance $(f_o + f_e) = 16.02 \text{ m}$ (OPTION (A))

$$\text{Angular Magnification} = -\frac{f_o}{f_e} = -\frac{16}{0.02} = -800 \quad \text{(OPTION (B))}$$

A telescope produces an image which is always inverted.

(OPTION (C))

In a telescope an objective is larger than the eye piece.

(OPTION (D))

16. [B, D]

$$\frac{1}{f} = \frac{1}{-15} + \frac{1}{30}$$

$$\Rightarrow f = -30 \text{ cm (Diverging in nature)}$$

(OPTION (D))

Since red deviates the least and violet deviates the maximum. So a coloured pattern with red on the outside is observed.

17. [A, C, D]

Optical path length in passing from 1st medium is $n_1 s_1$.

Optical path length in passing from 2nd medium is $n_2 s_2$ and so on.

$$\text{So total optical path length} = \sum_{i=1}^n n_i s_i \quad \text{(OPTION (C))}$$

$$\Rightarrow \text{Total time of flight } t = \frac{1}{c} \sum_{i=1}^n n_i s_i \quad \text{(OPTION (A))}$$

For inhomogeneous media optical path length is

$$\text{O.P.L.} = \int_A^B n(s) ds \text{ and the ray must travel along a path in which}$$

time taken to go from A to B is minimum. Such paths are called stationary pathways and this is the statement of Fermat's Least Action Principle or Fermat's Principle of Least Time.

18. [A, D]

The final image is formed at infinity when the combined focal length of the two lenses (in contact) is 30 cm i.e.,

$$\frac{1}{30} = \frac{1}{20} + \frac{1}{f}$$

$$f = -60 \text{ cm}$$

So, when another concave lens of focal length 60 cm is kept in contact with the first lens.

Similarly, if μ be the refractive index of a liquid in which focal length of the given convex lens becomes 30 cm. Then from Lens Maker's Formula, we have

$$\frac{1}{20} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \dots(1)$$

$$\frac{1}{30} = \left(\frac{3}{\mu} - 1\right) \left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \dots(2)$$

From equations (1) and (2), we get

$$\mu = \frac{9}{8}$$

19. [B, D]

When upper half of the lens is covered, image is formed by the rays coming from lower half of the lens i.e., the image will be formed by lesser number of rays. Therefore, intensity of image will decrease. However, complete image will be formed.

20. [A, C]

$m = +2$, means image is virtual, erect and magnified. A virtual and magnified image can be formed only by a concave mirror and that too when object lies between pole and focus.

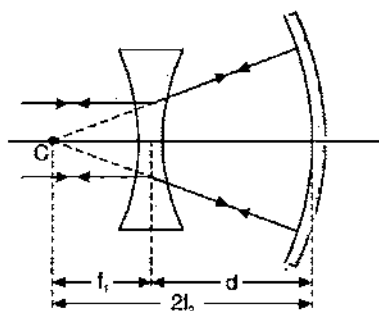
21. [B, C, D]

When passing from vacuum to a medium, frequency remains unchanged while speed and wavelength decreases μ times.

22. [A, B, C]

For convex mirror (having positive focal length) the image is always smaller in size. For concave mirror (having negative focal length) the image is smaller when the object lies beyond $2f$.

23. [A, B]



24. [B, C]

The tube length of an astronomical telescope, in normal adjustment, is $(f_o + f_e)$ and that of Galilean telescope, in normal adjustment is $(f_o - f_e)$ where f_o and f_e are focal lengths of objective and eye piece respectively. In this case, $f_e = f$, so difference in tube lengths is

$$\ell_1 - \ell_2 = (f_o + f_e) - (f_o - f_e) = 2f_e = 2f$$

27. [B, C]

A concave mirror and a convex lens give virtual magnified image for a particular object position (i.e. when object lies between F and P (or C))

28. [B, D]

A concave mirror can give both real and virtual magnified images. Since nothing is specified, so

$$m = \pm 3$$

$$\Rightarrow \pm 3 = \frac{-15}{-15 - u}$$

$$+u = -10 \text{ cm}$$

$$-u = -20 \text{ cm}$$

29. [C, D]

When the image is virtual, then

$$u = -x, \text{ so } v = +2x$$

From the mirror formula, we get

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{2x} - \frac{1}{x} = \frac{1}{-f}$$

$$\Rightarrow x = \frac{f}{2}$$

When the image is real, then

$$u = -y, \text{ so } v = -2y$$

Again applying the mirror formula, we get

$$\frac{1}{-2y} - \frac{1}{y} = \frac{1}{-f}$$

$$\Rightarrow \frac{3}{2y} = \frac{1}{f}$$

$$\Rightarrow y = \frac{3f}{2}$$

30. [A, C]

When the object moves from infinity to the pole of the mirror, the virtual image moves from focus to the pole.

31. [B, D]

Since, for refraction at a plane surface, we have

$$\frac{\mu_1}{u} = \frac{\mu_2}{v} \quad \dots (1)$$

If x be the height of the bird above the water surface, then for the light travelling from the bird to the fish, we have

$$\mu_1 = 1, \mu_2 = \mu \text{ and } u = -x$$

So, from (1), we get

$$\frac{1}{(-x)} = \frac{\mu}{v}$$

$$\Rightarrow v = -\mu x$$

$$\Rightarrow |v| = \mu x$$

Now speed of the bird is $\frac{dx}{dt}$

So, apparent speed of the bird is

$$\left| \frac{dv}{dt} \right| = \mu \frac{dx}{dt}$$

$$\Rightarrow \left| \frac{dv}{dt} \right| > \frac{dx}{dt} \quad (\because \mu > 1)$$

32. [A, C, D]

TIR takes place when ray of light travels from denser to rarer medium.

$$\text{Further, } \sin C_{12} = \frac{\mu_2}{\mu_1} \text{ and } \sin C_{13} = \frac{\mu_3}{\mu_1}$$

$$\text{Since, } \frac{\mu_2}{\mu_1} > \frac{\mu_3}{\mu_1}$$

$$C_{12} > C_{13}$$

Smaller the value of critical angle, more are the chances of TIR.

33. [A, C]

$$\text{Since, } \mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin \frac{A}{2}}$$

For $\mu = \sqrt{2}$ and $A = 60^\circ$, we get

$$\delta_m = 30^\circ$$

Further, at minimum deviation, we have

$$r_1 = r_2 = \frac{A}{2} = 30^\circ$$

Applying Snell's Law, we get

$$\sin i_1 = \mu \sin r_1$$

$$\Rightarrow \sin i_1 = (\sqrt{2}) \sin(30^\circ) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow i_1 = 45^\circ$$

34. [B, C]

Using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$, we get

$$\frac{1.5}{v} - \frac{1}{\infty} = \frac{1.5 - 1}{20}$$

$$\Rightarrow v = +60 \text{ cm}$$

Since v is positive, the rays actually meet.

35. [B]

$$\text{In air, } \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(1)$$

When immersed in a liquid of refractive index $\frac{\mu}{2}$,

$$\frac{1}{f_1} = \left(\frac{\mu}{2} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(2)$$

From (1) and (2), we get

$$\frac{1}{f} = \frac{\mu - 1}{f_1}$$

$$\Rightarrow f_1 = \frac{\mu - 1}{f}$$

When immersed in a liquid of refractive index 2μ ,

$$\frac{1}{f_2} = \left(2\mu - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(3)$$

From (1) and (3), we get

$$\frac{1}{f} = (\mu - 1) \left(-\frac{2}{f_2} \right)$$

$$\Rightarrow f_2 = -\frac{2(\mu - 1)}{f}$$

36. [B]

When in contact, we have

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\Rightarrow 10 = \frac{1}{f_1} + \frac{1}{f_2} \quad \left\{ \because P = \frac{1}{F} = 10 \right\}$$

$$\Rightarrow f_1 + f_2 = 10f_1f_2 \quad \dots(1)$$

When at a separation of 0.25 m, we have

$$\frac{1}{F'} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1f_2}$$

$$\Rightarrow 6 = \frac{1}{f_1} + \frac{1}{f_2} - \frac{0.25}{f_1f_2}$$

$$\Rightarrow 6 = 10 - \frac{0.25}{f_1f_2}$$

$$\Rightarrow f_1f_2 = \frac{1}{16} \quad \dots(2)$$

From (1), we get

$$f_1 + f_2 = \frac{10}{16} = \frac{5}{8}$$

Now, from (2) we get

$$f_1 \left(\frac{5}{8} - f_1 \right) = \frac{1}{16}$$

$$\Rightarrow 16f_1^2 - 10f_1 + 1 = 0$$

$$\Rightarrow 16f_1^2 - 8f_1 - 2f_1 + 1 = 0$$

$$\Rightarrow 8f_1(2f_1 - 1) - 1(2f_1 - 1) = 0$$

$$\Rightarrow (8f_1 - 1)(2f_1 - 1) = 0$$

$$\Rightarrow f_1 = \frac{1}{8} \text{ or } f_1 = \frac{1}{2}$$

$$\Rightarrow P_1 = 8D \text{ or } P_1 = 2D$$

37. [A, D]

$$\text{Since, } \frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

where $\mu_1 = \mu$, $\mu_2 = 1$, $u = -R$, $R = -R$

$$\Rightarrow \frac{1}{v} - \frac{\mu}{(-R)} = \frac{1 - \mu}{-R}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{R}$$

$$\Rightarrow v = R$$

38. [A, D]

Real image is smaller in size if object lies beyond $2f$ and it is larger if object lies between f and $2f$.

39. [A, C]

A ray can pass undeviated when $\mu_1 = \mu_2$ or the ray is incident normally i.e., angle of incidence is 0° .

40. [C, D]

For TIR, we have

$$i > C$$

$$\Rightarrow \sin i > \sin C$$

$$\Rightarrow \sin(45^\circ) > \sin C$$

$$\text{Since, } \sin C = \frac{1}{n}$$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{1}{n}$$

$$\Rightarrow n > \sqrt{2}$$

$$\Rightarrow n > 1.414$$

41. [A, C]

Applying Lens Maker's Formula, we get

$$\frac{1}{f_{\text{air}}} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f_{\text{water}}} = \left(\frac{3/2}{4/3} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

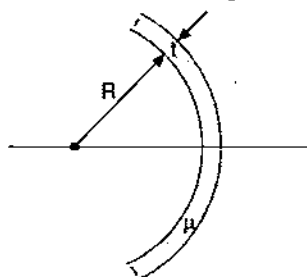
From these two equations, we get

$$f_{\text{water}} = 4f_{\text{air}} = 4f$$

In air, the image was inverted, real and magnified i.e., the object must be lying between f and $2f$. Now the focal length has changed to $4f$. Therefore, the object now lies between pole and focus and so the new image formed will be virtual and magnified.

43. [C]

Let t be the thickness of the watch glass, R be the radius of curvature of inner surface, so $(R+t)$ is radius of curvature of outer surface. If μ be the refractive index of this thin lens (watch glass), from Lens Maker's Formula, we get



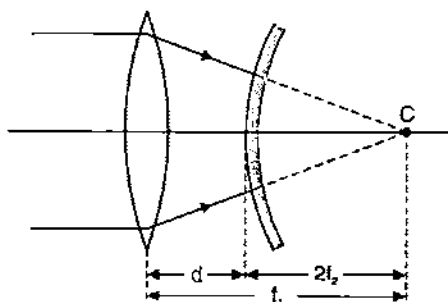
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} = (\mu - 1) \left[\frac{1}{-R} - \frac{1}{-(R+t)} \right]$$

$$\Rightarrow \frac{1}{f} = (\mu - 1) \left(\frac{1}{R+t} - \frac{1}{R} \right) = -\frac{(\mu - 1)t}{R(R+t)} < 0$$

So, the watch glass will have a diverging nature.

44. [A, B]



45. [B, C]

A concave mirror can give a virtual image (object lies between F and P) but the image is always magnified and a convex mirror can never give a real image.

46. [A, B, C]

Since, the lights used are of different colours, so they have different frequencies and hence (B) is also correct.

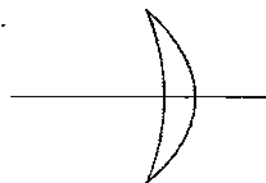
47. [B, C]

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{-2R} - \frac{1}{-R} \right)$$

$$\Rightarrow \frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{2R} \right)$$

$$\Rightarrow \frac{1}{f} = \frac{\mu - 1}{2R}$$

$$\Rightarrow f = \frac{2R}{\mu - 1}$$



The focal length of lens has nothing to do with the direction from which the light is incident on it.

49. [C, D]

For all object positions a convex mirror (or a concave lens) forms a virtual and erect image.

53. [B, C]

For a concave mirror, when object is placed at $2F$ (or C) real image of the same size as that object is formed at $2F$ (or C).

$$\Rightarrow m_{\text{concave}} = \frac{-f}{-f - (-2f)} = -1$$

For convex mirror

$$\Rightarrow m_{\text{convex}} = \frac{f}{f - (-2f)} = \frac{1}{3}$$

Solutions

Reasoning Based Questions

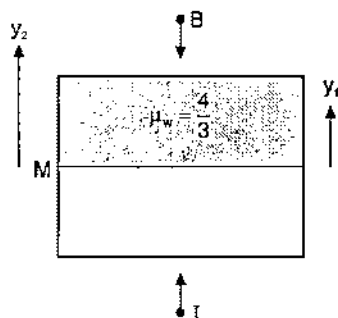
1. [D]

Lateral displacement, $\Delta x = t \sin i \left(1 - \frac{1}{\mu} \right) < t$.

2. [B]

Both Statements are true but Statement-2 is not the correct explanation to Statement-1.

6. [C]



$$\frac{y_{IM}}{4/3} = \frac{y_2 - y_1}{1} + \frac{y_1}{4/3}$$

$$\Rightarrow y_{IM} = y_1 + \frac{4}{3}(y_2 - y_1) \neq y_2$$

$$\text{But } y_{IB} = 2 \left[\frac{3y_1}{4} + (y_2 - y_1) \right]$$

$$\Rightarrow \frac{dy_{IB}}{dt} = 2 \frac{dy_2}{dt} = 2v_0$$

7. [D]

When the object is virtual, a real image can be formed by a plane or convex mirror.

8. [B]

Both Statements are correct but Statement-2 is not correct explanation of Statement-1.

10. [D]

For a mirror, $m = \frac{f}{f - u}$.

11. [A]

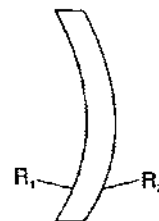
According to Lens Maker's Formula,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Since, $R_1 = R_2$

$$\Rightarrow \frac{1}{f} = 0$$

So, power = $\frac{1}{f} = 0$



16. [B]

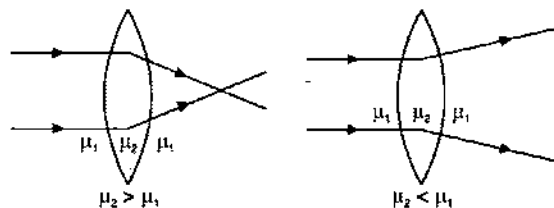
When light goes from one medium to another, its frequency remains unchanged.

19. [A]

Both the Statements are true, and Statement-2 is the correct explanation to Statement-1.

20. [C]

Statement-1 is true & Statement-2 is false.



21. [B]

$A = 60^\circ$

$$\mu = \sqrt{2}$$

$$i = \frac{A + D_m}{2} = \frac{60 + 30}{2} = 45^\circ$$

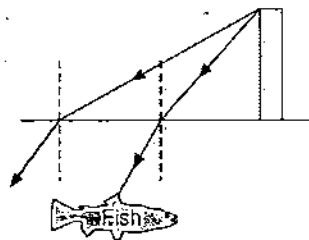
$$\text{Since, } r_2 = \frac{A}{2} = 30^\circ$$

So, According to Snell's Law, we have

$$\frac{\sin i}{\sin r} = \frac{\sin(45^\circ)}{\sin(30^\circ)} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2}$$

Both Statement-1 and Statement-2 are true but Statement-2 is not the correct explanation of Statement-1.

22. [A]
Both Statements are true and Statement-2 is correct explanation of Statement-1.
23. [B]
Both Statements are true and Statement-2 is correct explanation of Statement-1.
24. [D]
Statement-1 false and Statement-2 is true.
25. [D]
When the object is virtual, the convex mirror will give a real image.
26. [B]
After refraction at two parallel faces of a glass slab, a ray of light emerges in a direction parallel to the direction of incidence of white light on the slab. As rays of all colours emerge in the same direction (of incidence of white light), hence there is no dispersion, but only lateral displacement.
28. [C]



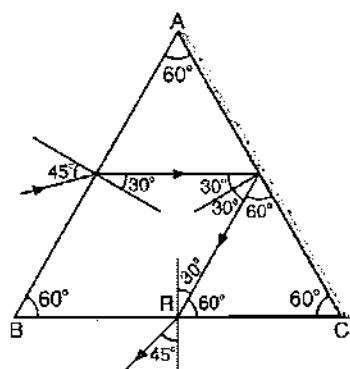
29. [C]
Angle of incidence at any location should be greater than critical angle.
30. [C]
In search light, we need intense parallel beam of light. When source of light is placed at focus of concave mirror, only paraxial rays are rendered parallel due to large aperture of mirror. Marginal rays give a divergent beam but in case of parabolic mirror, when source is at focus, beam of light produced over the entire cross-section of mirror is a parallel beam.

Solutions

Linked Comprehension Type Questions

1. [B]

Different angles by geometry and the given conditions are shown in figure.



$$\mu = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2}$$

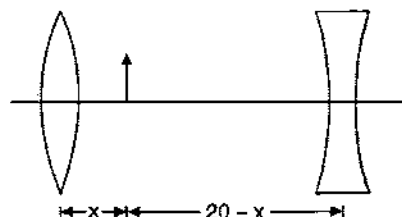
2. [C]

$$S_{\text{Total}} = S_p + S_o + S_R$$

$$S_{\text{Total}} = (45^\circ - 30^\circ) + (180^\circ - 2 \times 30^\circ) + (45^\circ - 30^\circ) = 150^\circ$$

3. [B]

Image formed by concave lens is virtual for all positions of object i.e., image by concave lens lies between the two lenses. For both the images to coincide, image by convex lens should also lie in between the two lenses the two lens or image by convex lens should also be virtual.



For convex lens :

$$\frac{1}{-v} - \frac{1}{-x} = \frac{1}{20}$$

$$\Rightarrow \frac{1}{x} - \frac{1}{v} = \frac{1}{20} \quad \dots(1)$$

For concave lens :

$$\frac{1}{-(20-v)} - \frac{1}{-(20-x)} = \frac{1}{-10}$$

$$\Rightarrow \frac{1}{20-x} - \frac{1}{20-v} = \frac{1}{-10}$$

Solving this equation, we get

$$x = \frac{20(\sqrt{3}-1)}{\sqrt{3}} \text{ cm} \quad \dots(2)$$

$$\Rightarrow v = 20(\sqrt{3}-1) \text{ cm}$$

4. [D]

Magnification by convex lens is given by

$$m_1 = \frac{v_1}{u_1} = \frac{-v}{-x} = \frac{v}{x} = \frac{20(\sqrt{3}-1)}{20(\sqrt{3}-1)/\sqrt{3}} = \sqrt{3}$$

Magnification by concave lens is given by

$$m_2 = \frac{v_2}{u_2} = \frac{-(20-v)}{-(20-x)} = \frac{20-v}{20-x}$$

$$m_2 = \frac{40-20\sqrt{3}}{20-\frac{20(\sqrt{3}-1)}{\sqrt{3}}} = (2\sqrt{3}-3)$$

8. [A]

Since refractive index decreases with increase of λ and velocity decreases with increase of refractive index, so we have

$$v_{\text{red}} > v_{\text{orange}} > v_{\text{yellow}}$$

9. [D]

Colour and frequency remains unchanged as it is property of source.

10. [B]

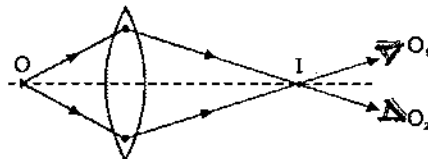
Dispersion depends on wavelength

11. [D]
Since light takes shortest path, so

$$t = \frac{nx_0}{c}$$

12. [C]
Speed of light changes with refractive index.

13. [D]
Since, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$



$$\Rightarrow \frac{1}{v} + \frac{1}{20} = \frac{1}{15}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{60}$$

$$\Rightarrow v = 60 \text{ cm}$$

14. [A]
Observer O_1 cannot see the image because light will be absorbed by the blackened portion of the lens.

16. [D]
According to Snell's Law, we have

$$\frac{\sin i}{\sin r} = \frac{\mu_1}{\mu_2} \Rightarrow K \sin i = 60^\circ$$

$$\Rightarrow \sin(60^\circ) = K \sin r \Rightarrow r = 90^\circ$$

$$\Rightarrow K = \frac{\sqrt{3}}{2}$$

17. [C]
 $\theta_1 = \frac{\pi}{6}$

18. [A]
Since, $\theta = 0^\circ$
 $\Rightarrow i = r$
 $\Rightarrow K_2 = 1$

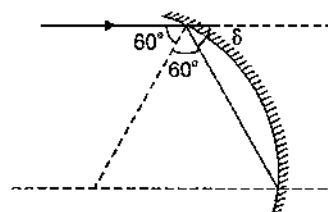
19. [C]
Since, $f = R - \frac{R}{2} \sec \theta$

$$\text{So, for } 0 > \theta, f < \frac{R}{2}$$

$$\Rightarrow f_b < f_m$$

20. [D]
 $f_m = R - \frac{R}{2} \sec(60^\circ) = 0$

21. [A]



$$\delta = 180 - 120 = 60^\circ$$

Laws of reflection is universally true

22. [D]
 $A = \sqrt{(6\sqrt{3})^2 + (8\sqrt{3})^2 + (-10)^2}$
 $A = 20 \text{ units}$

23. [C]
 $\cos i = \frac{\vec{A} \cdot (-\hat{k})}{|\vec{A}|} = \frac{(6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}) \cdot (-\hat{k})}{20}$
 $\cos i = \frac{10}{20} = \frac{1}{2}; i = 60^\circ$

24. [B]
 $\frac{\sin i}{\sin r} = \frac{\sqrt{3}}{\sqrt{2}}, \sin r = \frac{\sqrt{2}}{\sqrt{3}} \sin i$
 $\frac{\sin i}{\sin r} = \frac{\sqrt{2}}{\sqrt{3}} \sin 60^\circ = \frac{\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{2}} = 1$
 $\Rightarrow r = 45^\circ$

25. [D]
Let the vector representing the refracted ray be
 $\vec{A}' = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} + C\hat{k}$

$$\text{Since, } \cos r = \frac{\vec{A}' \cdot (-\hat{k})}{|\vec{A}'|}$$

$$\Rightarrow \cos r = \frac{(6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} + C\hat{k}) \cdot (-\hat{k})}{\sqrt{(6\sqrt{3})^2 + (8\sqrt{3})^2 + C^2}}$$

$$\Rightarrow \cos r = \frac{-C}{\sqrt{108 + 192 + C^2}}$$

$$\Rightarrow \cos 45^\circ = \frac{-C}{\sqrt{300 + C^2}}; \frac{1}{\sqrt{2}} = \frac{-C}{\sqrt{300 + C^2}}$$

$$\Rightarrow C = \pm 10\sqrt{3}$$

Since the refracted ray travels downwards,

$$\Rightarrow C = -10\sqrt{3}$$

$$\Rightarrow \vec{A}' = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\sqrt{3}\hat{k}$$

26. [C]
 $|\vec{A}'| = \sqrt{(6\sqrt{3})^2 + (8\sqrt{3})^2 + (-10\sqrt{3})^2} = 10\sqrt{6}$

27. [A]
 $\hat{n} = \frac{\vec{A}'}{|\vec{A}'|} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{4}{5\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$

28. [A]
Optical path length (OPL) = $\int n dx$

$$\Rightarrow \text{OPL} = \int_0^1 (1+x^2) dx$$

$$\Rightarrow \text{OPL} = \left(x + \frac{x^3}{3} \right) \Big|_0^1 = \frac{4}{3} \text{ m}$$

31. [C]

$$P_{\text{eq}} = 2P_{\text{lens}} + P_{\text{mirror}}$$

$$\text{where } P_{\text{lens}} = \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\text{and } P_{\text{mirror}} = -\frac{1}{f} = -\frac{2}{R}$$

$$\frac{1}{f} = 2(\mu - 1) \left(\frac{1}{R} - \frac{1}{(-R)} \right) - \frac{2}{(-R)}$$

$$\Rightarrow \frac{1}{f} = \frac{4(\mu - 1)}{R} + \frac{2}{R}$$

$$\Rightarrow \frac{1}{f} = \frac{4\mu}{R} - \frac{4}{R} + \frac{2}{R}$$

$$\Rightarrow \frac{1}{f} = \frac{4\mu}{R} - \frac{2}{R}$$

$$\Rightarrow \frac{1}{f} = \frac{2}{R} (2\mu - 1)$$

$$\Rightarrow f = \frac{R}{2(2\mu - 1)}$$

32. [C]

For a plano-convex lens, we have

$$\frac{1}{f_2} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$\Rightarrow \frac{1}{f_2} = \frac{\mu - 1}{R}$$

$$\Rightarrow f_2 = \frac{R}{\mu - 1}$$

33. [B]

When plane surface is silvered, then

$$\frac{1}{f_3} = \frac{2}{f} - \frac{1}{\infty}$$

$$(\because f_m \rightarrow \infty)$$

$$\frac{1}{f_3} = \frac{2(\mu - 1)}{R}$$

$$\Rightarrow f_3 = \frac{R}{2(\mu - 1)}$$

34. [D]

When curved surface is silvered, then

$$\frac{1}{f_4} = \frac{2}{f} - \frac{1}{f_m}$$

$$\Rightarrow \frac{1}{f_4} = 2(\mu - 1) \left(\frac{1}{\infty} - \frac{1}{(-R)} \right) - \frac{2}{(-R)}$$

$$\Rightarrow \frac{1}{f_4} = \frac{2(\mu - 1)}{R} + \frac{2}{R}$$

$$\Rightarrow \frac{1}{f_4} = \frac{2\mu}{R}$$

$$\Rightarrow f_4 = \frac{R}{2\mu}$$

35. [D]

Focal length of lens L_1 and L_2 is given by using the Lens Maker's Formula, so

$$\frac{1}{f_1} = (\mu_1 - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$\Rightarrow f_1 = 50 \text{ cm}$$

$$\text{Similarly, } \frac{1}{f_2} = (\mu_2 - 1) \left(\frac{1}{\infty} - \frac{1}{(-R)} \right)$$

$$\Rightarrow f_2 = 40 \text{ cm}$$

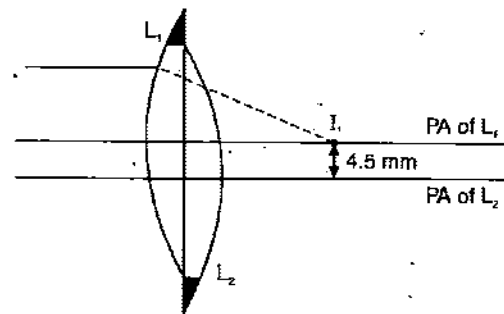
So, equivalent focal length is given by

$$\frac{1}{f_{\text{eq}}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\Rightarrow f_{\text{eq}} = \frac{200}{9} \text{ cm}$$

36. [C]

Image formed by L_1 lies 50 cm behind it and on principal axis of L_1 . This will act as an object for L_2 . So, for L_2 ,



we have

$$u = +50 \text{ cm}, f_2 = +40 \text{ cm}$$

Applying Len's Formula, we get

$$\frac{1}{v} - \frac{1}{50} = \frac{1}{40}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{50} + \frac{1}{40}$$

$$\Rightarrow v = \frac{200}{9} \text{ cm}$$

Magnification produced by L_2 is $m_2 = \frac{v}{u} = \frac{4}{9}$

However, for L_2 , the image I_1 is at a distance of 4.5 mm above its principal axis i.e., (PA of L_2). So, distance of image I_2 from PA of L_2 is

$$y = m_2 (4.5 \text{ mm})$$

$$\Rightarrow y = \frac{4}{9} (4.5 \text{ mm}) = 2 \text{ mm}$$

Hence I_2 is at a distance of $(4.5 - 2) \text{ mm} = 2.5 \text{ mm}$ from PA of L_1 .

37. [C]

Applying refraction at curved surface formula, i.e.,

$$\frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}, \text{ we get}$$

$$\frac{1}{x} + \frac{2}{v_1} = \frac{1}{R}$$

$$\Rightarrow v_1 = \frac{2xR}{x-R} \quad (\text{from pole of curved surface})$$

This image formed will be at a distance $v' = (v_1 - R)$ from the plane surface. Now again applying the above formula, but at the plane surface, we get

$$\frac{1}{v} - \frac{2}{v'} = 0 \quad (\because \text{for plane surface } R \rightarrow \infty)$$

$$\Rightarrow \frac{1}{v} = \frac{2}{v_1 - R}$$

Substituting $v_1 = \frac{2xR}{x-R}$, we get

$$\frac{1}{v} = \frac{2(x-R)}{R(x+R)}$$

For the image to be virtual, we have

$$\frac{1}{v} < 0$$

$$\Rightarrow \frac{2(x-R)}{R(x+R)} < 0$$

$$\Rightarrow x < R$$

This condition is only satisfied by $x = \frac{R}{3}$.

38. [D]

$$\text{Since, } v_1 = \frac{(2x)R}{x-R}$$

So, for $x = 2R$, we get

$$v_1 = 4R$$

Also, $u = -2R$

$$\text{Since, } m_1 = \left(\frac{\mu_1}{\mu_2} \right) \left(\frac{v}{u} \right)$$

$$\Rightarrow m_1 = \left(\frac{1}{2} \right) \left(\frac{4R}{-2R} \right)$$

$$\Rightarrow m_1 = -1$$

However for plane mirror, we have

$$m_2 = 1$$

$$m_{\text{net}} = m_1 m_2 = -1$$

Since, $m_{\text{net}} = -1$, so the final image formed is real, inverted and of same size.

39. [C]

$$\frac{\sin(90^\circ)}{\sin r} = 2$$

$$\Rightarrow \sin r = \frac{1}{2}$$

$$\Rightarrow r = 30^\circ$$

In triangle OAC, we have

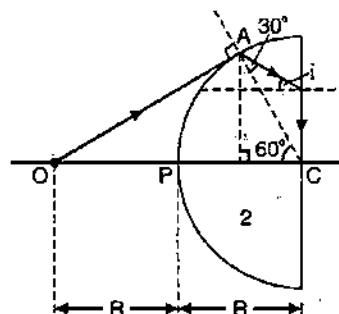
OC = 2R, AC = R, so by Pythagora's Theorem, we get

$$OA = \sqrt{3}R$$

$$\Rightarrow \angle ACP = 60^\circ$$

Using geometry, we get

$$i = 30^\circ$$



Since, refractive index of this is 2, so the critical angle C is given by

$$C = \sin^{-1} \left(\frac{1}{2} \right) = 30^\circ$$

Since, $i = C$, so the ray grazes the plane surface.

40. [C]

Since yellow is the mean colour, so

$$\mu_y = \frac{\mu_b + \mu_r}{2} = \frac{1.51 + 1.49}{2} = 1.50$$

41. [D]

Again yellow is the mean colour, so

$$\mu'_y = \frac{\mu'_b + \mu'_r}{2} = \frac{1.77 + 1.73}{2} = 1.75$$

42. [D]

For no deviation to take place, deviation (given by $\delta = (\mu - 1)A$) by one must be cancelling the deviation due to the other. For this (a) the prisms must be arranged upside down

$$(b) \frac{A'}{A} = - \left[\frac{\mu_y - 1}{\mu'_y - 1} \right]$$

$$\Rightarrow A' = - \left(\frac{1.50 - 1}{1.75 - 1} \right) 6^\circ = -4^\circ$$

43. [D]

$$\text{Net dispersion} = (\delta_b - \delta_r) + (\delta'_r - \delta'_b)$$

$$\begin{aligned} &= (\mu_b - \mu_r)A + (\mu'_r - \mu'_b)A' \\ &= (1.51 - 1.49)6^\circ - (1.77 - 1.73)4^\circ \\ &= 0.02 \times 6^\circ - 0.04 \times 4^\circ \\ &= 0.12^\circ - 0.16^\circ = -0.04^\circ \end{aligned}$$

44. [C]

$$\text{Since, } \sqrt{3}z + x = 10$$

$$\Rightarrow z = \frac{-x}{\sqrt{3}} + \frac{10}{\sqrt{3}}$$

$$\Rightarrow \text{Slope} = \frac{-1}{\sqrt{3}}$$

$$\Rightarrow \tan \theta = \frac{-1}{\sqrt{3}}$$

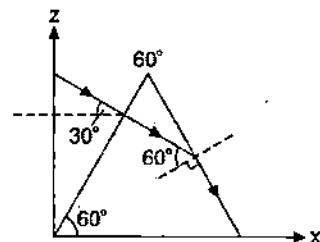
$$\Rightarrow \theta = -30^\circ$$

So, the ray is incident normally on the face AB. Thus angle of incidence on face AC is 60° .

For grazing the face AC, we have

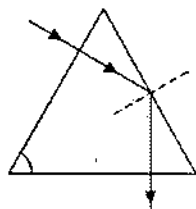
$$\mu \sin(60^\circ) = 1 \times \sin(90^\circ)$$

$$\Rightarrow \mu = \frac{2}{\sqrt{3}}$$



45. [B]

For $\mu = \frac{3}{2}$ the ray will be internally reflected, so, the ray is normal at face BC. Hence finally, the refracted ray is parallel to z-axis.



46. [B]

If BC is silvered, the ray will retrace its path. Hence equation of ray coming out of the prism is $\sqrt{3}z + x = 10$.

47. [C]

The objective lens must form a real image for eyepiece to magnify it.

48. [A]

The image formed by the objective must lie within the focus of the eyepiece.

49. [A]

To obtain best magnification, the object must be placed just beyond the focus of the objective lens. In this case, the first image distance from the objective is very large.

50. [C]

$$i + \theta = 90^\circ, \theta = 90^\circ - i,$$

$$\text{Slope of tangent} = \tan \theta = \tan(90^\circ - i) = \cot i$$

51. [D]

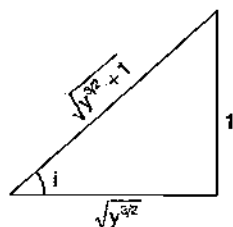
$$\text{but } \tan \theta = \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \cot i \quad \dots(1)$$

Applying Snell's Law at A and B

$$n_A \sin i_A = n_B \sin i_B$$

$$n_A = 1 \text{ because } y = 0$$



$$\sin i_A = 1 \text{ because } i_A = 90^\circ \text{ (Grazing incidence)}$$

$$n_B = \sqrt{Ky^{3/2} + 1} = \sqrt{y^{3/2} + 1}$$

$$\text{because } K = 1.0(\text{m})^{-3/2}$$

$$\therefore (1)(1) = \sqrt{y^{3/2} + 1} \sin i$$

$$\Rightarrow \sin i = \frac{1}{\sqrt{y^{3/2} + 1}}$$

$$\therefore \cot i = \sqrt{y^{3/2}} \text{ or } y^{3/4} \quad \dots(2)$$

Equating equations (1) and (2), we get

$$\frac{dy}{dx} = y^{3/4}$$

$$\Rightarrow y^{-3/4} dy = dx$$

$$\Rightarrow \int_0^y y^{-3/4} dy = \int_0^x dx$$

$$\Rightarrow 4y^{1/4} = x$$

...(3)

The required equation of trajectory is $4y^{1/4} = x$

$$\Rightarrow y = \frac{x^4}{4^4} = \frac{x^4}{256}$$

52. [D]

At the point of intersection on the upper surface, $y = 1 \text{ m}$

$$\Rightarrow x = (256)^{1/4} = 4 \text{ m}$$

So the co-ordinates are (4 m, 1 m)

53. [A]

As $n_A \sin i_A = n_B \sin i_B$ and as $n_A = n_B = 1$

Therefore, $i_B = i_A = 90^\circ$ i.e., the ray will emerge parallel to the boundary at P i.e., at grazing emergence.

54. [C]

$$\frac{1}{F} = \frac{2}{f_1} - \frac{1}{f_2}$$

$$\Rightarrow \frac{1}{F} = 2(1.5 - 1) \left(\frac{1}{20} - \frac{1}{60} \right) - \frac{2}{(-20)}$$

$$\Rightarrow \frac{1}{F} = \frac{1}{20} - \frac{1}{60} + \frac{2}{20} = \frac{8}{60}$$

$$\Rightarrow F = -\frac{60}{8} = -7.5 \text{ cm}$$

55. [B]

Since $F = \infty$, so combination behaves like a concave mirror.

56. [A]

$$m = \frac{f}{f-u} = \frac{-7.5}{-7.5+30} = \frac{-7.5}{22.5} = -\frac{1}{3}$$

57. [C]

58. [B]

59. [B]

Combine solution of 57, 58, & 59

Each part will work as a separate lens and will form its own image. For any part, we have $u = -0.3 \text{ m}$, $f = +0.2 \text{ m}$.

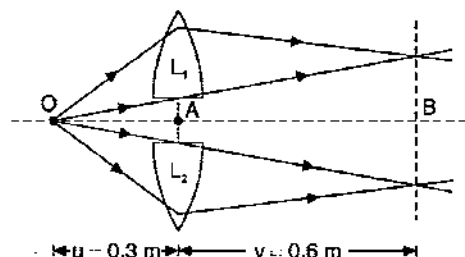
Therefore, from lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{-0.3} = \frac{1}{0.2}$$

$$\Rightarrow v = 0.6 \text{ m}$$

So, each part forms a real image of the point object O at 0.6 m from the lens, as shown in figure.



Since the triangles OL_1I_2 and OI_1L_2 are similar. So, we have

$$\frac{I_1I_2}{L_1L_2} = \frac{OB}{OA} = \frac{u+v}{u}$$

$$\Rightarrow \frac{I_1I_2}{L_1L_2} = \frac{0.3+0.6}{0.3} = \frac{0.9}{0.3} = 3$$

$$\Rightarrow I_1I_2 = 3(L_1L_2) = 3(2 \times 0.0005) = 0.003 \text{ m}$$

60. [A]

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} - \frac{1}{-R} \right) = \frac{1}{2} \times \frac{2}{R} = \frac{1}{R}$$

$$\Rightarrow f = R$$

61. [C]

$$\frac{1}{f} = \left(\frac{4}{3} - 1 \right) \left(\frac{1}{-R} - \frac{1}{\infty} \right) = -\frac{1}{3R}$$

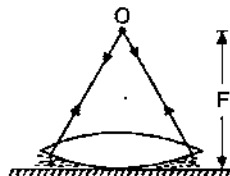
$$\Rightarrow f = -3R$$

62. [C]

$$\text{Now, } \frac{1}{F} = \frac{1}{f} + \frac{1}{f} = \frac{1}{R} - \frac{1}{3R} = \frac{2}{3R}$$

$$\Rightarrow F = \frac{3R}{2}$$

Since, image coincides with the object so, clearly rays of light must have retraced their path after reflection. This is possible only when rays of light must have fallen normally on the plane mirror. For this, the object is at the focus of the lens system.



$$\Rightarrow F = \text{distance of object} = 15 \text{ cm}$$

$$\Rightarrow \frac{3R}{2} = 15 \text{ cm}$$

$$\Rightarrow R = 10 \text{ cm}$$

63. [C]

If f is the focal length of the liquid concave lens, then

$$\frac{1}{25} = \frac{1}{10} + \frac{1}{f} \text{ or } \frac{1}{f} = \frac{-3}{50}$$

$$\Rightarrow f = -\frac{50}{3} \text{ cm}$$

64. [D]

Using Lens-Maker's Formula for the liquid concave lens,

$$\frac{1}{50} = (\mu - 1) \left(\frac{1}{-R} - \frac{1}{\infty} \right) = -\frac{\mu - 1}{R} = -\frac{\mu - 1}{10}$$

$$\Rightarrow \frac{3}{50} = \frac{\mu - 1}{10}$$

$$\Rightarrow \mu = 1 + 0.6 = 1.6$$

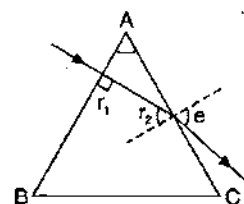
65. [B]

Since the ray is incident normally, so

$$r_1 = 0$$

Further, $r_1 + r_2 = A = 60^\circ$

$$\Rightarrow r_2 = A = 60^\circ$$



Applying Snell's Law at AC, we get

$$\mu \sin r_2 = \mu_1 \sin e$$

$$\Rightarrow \mu_1 \sin e = \frac{4}{\sqrt{3}} \times \sin 60^\circ$$

$$\Rightarrow \sin e = \frac{4}{\sqrt{3}\mu_1} \times \frac{\sqrt{3}}{2} = \frac{2\sqrt{3}}{\sqrt{3}} \times \frac{1}{\mu_1} = \frac{2}{\mu_1} \quad \dots(1)$$

Further, deviation is given by

$$\beta = e - r_2$$

From the graph, we observe that when $\mu_1 = k_2$, $\beta = 0$

$$\text{So, } \beta = e - r_2 = 0$$

$$\Rightarrow e = r_2 = 60^\circ$$

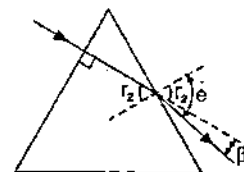
So, equation (1) becomes

$$\sin(60^\circ) = \frac{2}{\mu_1}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{2}{\mu_1}$$

$$\Rightarrow \mu_1 = \frac{4}{\sqrt{3}}$$

$$\Rightarrow k_2 = \frac{4}{\sqrt{3}}$$



66. [C]

When $\mu_1 < k_2$,

light will not emerge, it can be seen from graph, so we have

$$r_2 = C \text{ when } \mu_1 = k_2$$

$$\Rightarrow \sin r_2 = \sin C = \frac{\mu}{\mu_1}$$

$$\Rightarrow \sin(60^\circ) = \frac{\mu}{\mu_1} \quad \{\because r_2 = 60^\circ\}$$

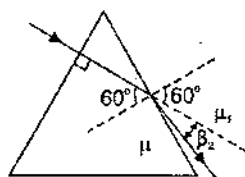
$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{4}{\sqrt{3}} \times \frac{1}{\mu_1}$$

$$\Rightarrow \mu_1 = \frac{8}{3}$$

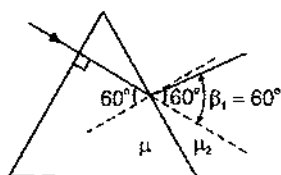
$$\Rightarrow k_1 = \frac{8}{3}$$

67. [B]

From the graph, we observe that β_1 is the maximum deviation and β_2 is the minimum deviation



$$\text{So, } \beta_2 = \frac{\pi}{2} - 60^\circ = 30^\circ \text{ and } \beta_1 = 60^\circ$$



$$\Rightarrow \beta_1 - \beta_2 = 60^\circ - 30^\circ = 30^\circ$$

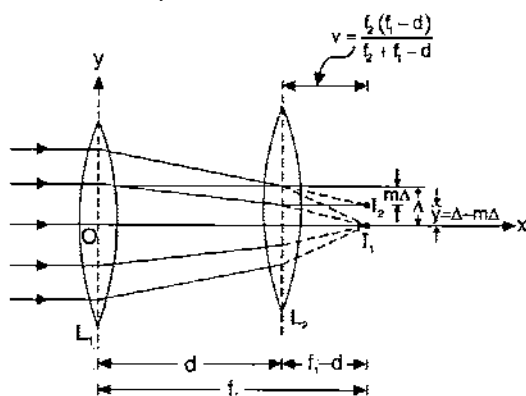
68. [A]

From the first lens parallel beam of light is focussed at its focus i.e., at a distance f_1 from it. This image I_1 acts as virtual object for second lens L_2 . Therefore, for L_2

$$u = +(f_1 - d), f = +f_2$$

$$\Rightarrow \frac{1}{v} - \frac{1}{f} = \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{f_2} + \frac{1}{f_1 - d}$$

$$\Rightarrow v = \frac{f_2(f_1 - d)}{f_2 + f_1 - d}$$



Therefore, x-coordinate of its focal point will be

$$x = d + v = d + \frac{f_2(f_1 - d)}{f_2 + f_1 - d}$$

$$\Rightarrow x = \frac{f_1 f_2 + d(f_1 - d)}{f_1 + f_2 - d}$$

69. [D]

Linear magnification for L_2 , is given by

$$m = \frac{v}{u} = \left(\frac{f_2(f_1 - d)}{f_2 + f_1 - d} \right) \left(\frac{1}{f_1 - d} \right) = \frac{f_2}{f_1 + f_2 - d}$$

Therefore, second image will be formed at a distance of $m\Delta$ or

$$\left(\frac{f_2}{f_1 + f_2 - d} \right) (\Delta) \text{ below its optic axis.}$$

Therefore, y-coordinate of the focus of system is given by

$$y = \Delta - \left(\frac{f_2 \Delta}{f_1 + f_2 - d} \right)$$

$$\Rightarrow y = \frac{(f_1 - d)\Delta}{f_1 + f_2 - d}$$

70. [A]

Wavefronts are parallel in both media. Therefore, light which propagates perpendicular to wavefront travels as a parallel beam in each medium.

71. [C]

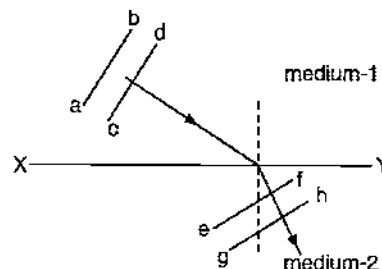
All points on a wavefront are at the same phase

$$\therefore \phi_c = \phi_e \text{ and } \phi_i = \phi_o$$

$$\therefore \phi_e - \phi_i = \phi_c - \phi_o$$

72. [B]

In medium-2 wavefront bends away from the normal after refraction. Therefore, ray of light which is perpendicular to wavefront bends towards the normal in medium-2 during refraction. So, medium-2 is denser or its speed in medium-1 is more.



75. [D]

For both the halves, position of object and image is same, however the only difference is of magnification. Magnification for one of the halves is given as $2(>1)$. This can be for the first one, because for this, $|v| > |u|$. Therefore, magnification,

$$|m| = \left| \frac{v}{u} \right| > 1. \text{ So, for the first half, we have}$$

$$\left| \frac{v}{u} \right| = 2$$

$$\Rightarrow |v| = 2|u|$$

$$\text{Let } u = -x, \text{ then } v = +2x$$

$$\text{and } |u| + |v| = 1.8 \text{ m}$$

$$\Rightarrow 3x = 1.8 \text{ m}$$

$$\Rightarrow x = 0.6 \text{ m}$$

$$\text{Hence, } u = -0.6 \text{ m and } v = +1.2 \text{ m}$$

$$\text{Using } \frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{1.2} - \frac{1}{-0.6} = \frac{1}{0.4}$$

$$\Rightarrow f = 0.4 \text{ m}$$

For the second half, we have

$$\frac{1}{f} = \frac{1}{1.2 - d} + \frac{1}{-(0.6 + d)}$$

$$\Rightarrow \frac{1}{0.4} = \frac{1}{1.2 - d} + \frac{1}{(0.6 + d)}$$

Solving this, we get $d = 0.6 \text{ m}$

Magnification for the second half will be

$$m_2 = \frac{v}{u} = \frac{0.6}{-(1.2)} = -\frac{1}{2}$$

and magnification for the first half is $m_1 = \frac{v}{u} = \frac{1.2}{-(0.6)} = -2$

76. [A]

For direct observation, the eye E appears to be farther than it actually is. So,

$$x = \mu H + \frac{H}{2} = H\left(\mu + \frac{1}{2}\right)$$

77. [B]

The distance of eye from mirror is

$$\mu H + H$$

So, distance of eye as seen by the fish in the mirror is

$$x = (\mu H + H) + \frac{H}{2}$$

$$\Rightarrow x = \frac{3H}{2} + \mu H$$

$$\Rightarrow x = H\left(\mu + \frac{3}{2}\right)$$

78. [C]

For direct observation by the eye E, the fish appears to be closer than it actually is. So, we have

$$x_1 = H + \frac{H}{2\mu}$$

$$\Rightarrow x = H\left(1 + \frac{1}{2\mu}\right)$$

79. [D]

The distance of the mirror as observed by the eye is $H + \frac{H}{\mu}$. So,

the distance of the fish as seen by the eye in the mirror is

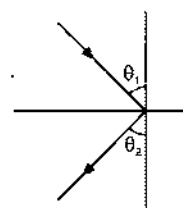
$$x = H + \frac{H}{\mu} + \frac{H}{2\mu}$$

$$\Rightarrow x = H\left(1 + \frac{3}{2\mu}\right)$$

81. [C]

$$1 \sin \theta_1 = -n \sin \theta_2$$

$$\sin \theta_2 = -\frac{1}{n} \sin \theta_1$$



Solutions

Matrix/Match Type Questions

- A $\rightarrow$ (r, s)
 B $\rightarrow$ (p, q, r, s, t)
 C $\rightarrow$ (q, r, s, t)
 D $\rightarrow$ (p, r, s)

(A) $m < 0$, means real image, possible for concave mirror and convex lens. So, (A) $\rightarrow$ (r, s).
 (B) $m > 0$, means virtual image, possible for all i.e., plane mirror, convex mirror (always), concave mirror (when object lies between focus and pole), concave lens (always) and convex lens (when object lies between optical centre and focus). So, (B) $\rightarrow$ (p, q, r, s, t).
 (C) $|m| < 1$, means diminished image, possible for concave mirror, convex lens, convex mirror and concave lens. So, (C) $\rightarrow$ (q, r, s, t).
 (D) $|m| \geq 1$, means magnified (> 1) and same sized ($= 1$) image, possible for concave mirror/convex lens (both $|m| > 1$) and plane mirror ($|m| = 1$). So, (D) $\rightarrow$ (p, r, s).

- A $\rightarrow$ (s, t)
 B $\rightarrow$ (p, t)
 C $\rightarrow$ (s, t)
 D $\rightarrow$ (q, t)

$$\vec{v}_A = \vec{i} + \vec{a}t = \vec{i} + (2\vec{i} + \vec{j})(2) = 5\vec{i} + 2\vec{j}$$

$$\Rightarrow \vec{v}_{A'} = -5\vec{i} + 2\vec{j}$$

$$\Rightarrow \vec{v}_{A'A} = \vec{v}_{A'} - \vec{v}_A = -10\vec{i}$$

Similarly, $\vec{v}_B = (-\vec{i} + 3\vec{j})$

$$\Rightarrow \vec{v}_{B'} = (\vec{i} + 3\vec{j})$$

$$\Rightarrow \vec{v}_{B'B} = \vec{v}_{B'} - \vec{v}_B = 2\vec{i}$$

- A $\rightarrow$ (s)
 B $\rightarrow$ (p, q, r)
 C $\rightarrow$ (q, r, s)
 D $\rightarrow$ (t)
- A $\rightarrow$ (r)
 B $\rightarrow$ (q)
 C $\rightarrow$ (p)
 D $\rightarrow$ (p)

- (A) Velocity of fish in air $= 8 \times \frac{3}{4} = 6 \uparrow$
 Velocity of fish w.r.t. bird $= 6 + 6 = 12 \uparrow$
 (B) Velocity of image of fish after reflection from mirror in air $= 8 \times \frac{3}{4} = 6 \downarrow$
 w.r.t. bird $= -6 + 6 = 0$
 (C) Velocity of bird as seen from water $= 6 \times \frac{4}{3} = 8 \downarrow$
 Velocity of bird w.r.t. fish $= 8 + 8 = 16 \downarrow$
 (D) Velocity of bird in water after reflection from mirror $= 8 \downarrow$
 w.r.t. fish $= 8 - 8 = 0$

- A $\rightarrow$ (p, q)
 B $\rightarrow$ (r)
 C $\rightarrow$ (s)
 D $\rightarrow$ (p, q)
- A $\rightarrow$ (p, s)
 B $\rightarrow$ (q)
 C $\rightarrow$ (p, q, s)
 D $\rightarrow$ (r)
- A $\rightarrow$ (q, r)
 B $\rightarrow$ (r)
 C $\rightarrow$ (p, r, s)
 D $\rightarrow$ (p, r)
- A $\rightarrow$ (p, q, s)
 B $\rightarrow$ (p, q)
 C $\rightarrow$ (r)
 D $\rightarrow$ (p, q, s)
- A $\rightarrow$ (p, s)
 B $\rightarrow$ (p, q, r, s)
 C $\rightarrow$ (p, q, r, s)
 D $\rightarrow$ (p, s)
- A $\rightarrow$ (p, q, r, s)
 B $\rightarrow$ (q)
 C $\rightarrow$ (p, q, r, s)
 D $\rightarrow$ (p, q, r, s)

11. A → (p)
B → (p)
C → (r, s)
D → (q, p)

12. A → (p, s)
B → (p, q, r, s)
C → (p, q, r, s)
D → (q, s)

(A) For convex mirror :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\Rightarrow v = \left(\frac{uf}{u-f} \right)$$

Now, v may be positive and negative, depending on values of u and f .

$$\text{Since, } |m| = \left| \frac{v}{u} \right| = \left(\frac{f}{u-f} \right) = \left(\frac{1}{\frac{u}{f} - 1} \right)$$

Again $|m|$ can be greater than or less than 1.

(B) For concave mirror :

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{-f}$$

$$\Rightarrow v = - \left(\frac{fu}{f+u} \right)$$

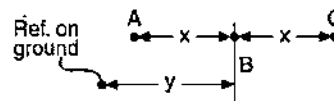
So, v is always negative i.e., image is always real.

$$\text{Further, } |m| = \left| \frac{v}{u} \right| = \frac{1}{\frac{u}{f} + 1}$$

i.e., m is always less than 1 or image is always diminished.

13. A → (s)
B → (p)
C → (q)
D → (r)
A → when object lies between pole and focus image is virtual, magnified and erect.
B → when object lies between focus and centre of curvature, image is real, inverted and magnified.
C → when object lies at centre of curvature, image is real, inverted and of equal size.
D → when object lies beyond centre of curvature, image is real, inverted and smaller in size.

14. A → (s)
B → (q)
C → (p)
D → (q)



$$(A) \frac{d}{dt}(x+y) = \frac{dx}{dt} + \frac{dy}{dt} = 5 + 2 = 7 \text{ ms}^{-1}$$

$$(B) \frac{dx}{dt} = 3 + 2 = 5 \text{ ms}^{-1}$$

$$(C) \frac{d(2x)}{dt} = 2 \frac{dx}{dt} = 10 \text{ ms}^{-1}$$

$$(D) \frac{dx}{dt} = 5 \text{ ms}^{-1}$$

15. A → (q)
B → (r)
C → (s)
D → (p)

A $\mu_2 > \mu_1$ (towards normal)

$\mu_2 > \mu_3$ (away from normal)

B $\mu_1 = \mu_2$ (no change in path)

$$\angle i = 0$$

$$\Rightarrow \angle r = 0 \text{ on the block}$$

C $\mu_1 > \mu_2$ (away from the normal)

$\mu_2 > \mu_3$ (away from the normal)

$$\mu_1 \times \frac{1}{\sqrt{2}} = \mu_2 \sin r$$

$$\Rightarrow \sin r = \frac{\mu_1}{\sqrt{2}\mu_2}$$

Since $\sin r < 1$

$$\Rightarrow \mu_1 < \sqrt{2}\mu_2$$

D For TIR : $45^\circ > C$

$$\Rightarrow \sin 45^\circ > \sin C$$

$$\Rightarrow \frac{1}{\sqrt{2}} > \frac{\mu_2}{\mu_1}$$

$$\Rightarrow \mu_1 > \sqrt{2}\mu_2$$

Solutions

Integer Answer Type Questions

1. Forward shift of point of incidence due to a single reflection

$$x = 0.2 \tan(30^\circ) = \frac{0.2}{\sqrt{3}}$$

Hence required number of reflections required is

$$N = \frac{\text{Mirror Length}}{\text{Forward Shift}} = \frac{2\sqrt{3}}{\left(\frac{0.2}{\sqrt{3}}\right)} = 30$$

2. Since, for a concave mirror, we can have both virtual and real image, so

for real image, $m = -5$ and

for virtual image, $m = +5$

$$\text{Since we know that } m = \frac{f}{f - u}$$

where $f = -30$ cm and $m = \pm 5$ because the image can be real or virtual.

For real image, $m = -5$

$$\Rightarrow -5 = \frac{-30}{-30 - u}$$

$$\Rightarrow u = -36 \text{ cm}$$

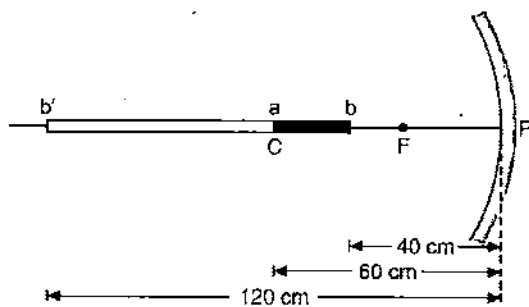
For virtual image, $m = +5$

$$\Rightarrow 5 = \frac{-30}{-30 - u}$$

$$\Rightarrow u = -24 \text{ cm}$$

Hence, the object must be placed at 24 cm or 36 cm in front of the concave mirror

3. The image of the end a of the rod which lies at the centre of curvature C is formed at C.



The position of the image of end b can be obtained using mirror formula, according to which we have

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

where $u = -40$ cm, $f = -30$ cm

$$\Rightarrow \frac{1}{v} + \frac{1}{-40} = \frac{1}{-30}$$

$$\Rightarrow v = -120 \text{ cm}$$

Length of the image (rod) is given by

$$\ell_1 = 120 - 60 = 60 \text{ cm}$$

$$\text{So, magnification } m = -\frac{dv}{du} = -\left(\frac{v_2 - v_1}{u_2 - u_1}\right)$$

$$\Rightarrow m = -\left(\frac{120 - 60}{60 - 40}\right)$$

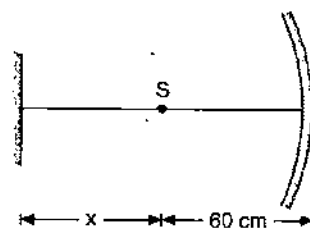
$$\Rightarrow m = -3$$

$$\Rightarrow |m| = 3$$

4. Applying mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for concave mirror, we get

$$\frac{1}{v} + \frac{1}{60} = \frac{1}{(-40)}$$

$$\Rightarrow v = -120 \text{ cm}$$



Now, for the rays to again converge at S, (after reflection from the plane mirror)

$$\left(\text{Distance of S from Plane Mirror}\right) = \left(\text{Distance of Image formed by Concave Mirror}\right)$$

$$\Rightarrow x = 120 - (x + 60)$$

$$\Rightarrow x = 30 \text{ cm}$$

So, the desired distance is 90 cm

From reversibility principle, it hardly matters whether the ray of light is first reflected from the concave mirror or plane mirror.

5. Since the image formed is real, so for the first case, we have

$$m = -3$$

$$\Rightarrow \frac{v_1}{u_1} = -3$$

$$\Rightarrow v_1 = 3u_1$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{(-3u_1)} + \frac{1}{(-u_1)} = \frac{1}{-f}$$

$$\Rightarrow f = \frac{3u_1}{4} \quad \dots(1)$$

Similarly, for the second case, we have

$$m = -2$$

$$\Rightarrow \frac{v_2}{u_2} = -2$$

$$\Rightarrow v_2 = 2u_2$$

$$\text{Since } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{(-2u_2)} + \frac{1}{(-u_2)} = \frac{1}{-f}$$

$$\Rightarrow f = \frac{2u_2}{3} \quad \dots(2)$$

From (1) and (2), we get

$$\frac{3u_1}{4} = \frac{2u_2}{3}$$

$$\Rightarrow u_2 = \frac{9u_1}{8} \quad \dots(3)$$

But according to the problem, we are given that the shift of the object is 6 cm, so we get

$$u_2 - u_1 = 6 \text{ cm}$$

$$\Rightarrow \frac{9}{8}u_1 - u_1 = 6$$

$$\Rightarrow u_1 = 48 \text{ cm}$$

$$\text{Since, } v_1 = 3u_1$$

$$\Rightarrow v_1 = 144 \text{ cm}$$

$$\text{So, } f = \frac{3u_1}{4} = \frac{3}{4}(48) = 36 \text{ cm}$$

$$\text{Now } u_2 = \frac{9u_1}{8} = \frac{9}{8}(48) = 54 \text{ cm}$$

$$\text{and } v_2 = 2u_2 = 2(54) = 108 \text{ cm}$$

$$\text{So, shift of the screen is } (v_1 - v_2)$$

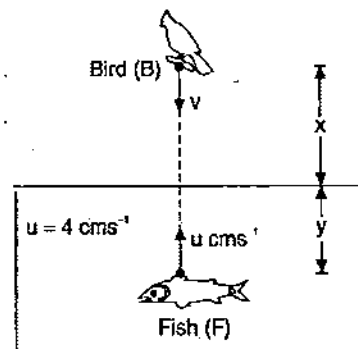
$$\Rightarrow (v_1 - v_2) = 144 - 108 = 36 \text{ cm}$$

6. Let at some instant bird is at a height of x from the water surface and it is diving downwards with v cm s^{-1}

At this instant fish is at a depth y below water surface. Then the distance between fish and image of bird at this instance will be,

$$s = \mu x + y$$

$$\Rightarrow s = \frac{4}{3}x + y$$



Differentiating w.r.t. time, we get

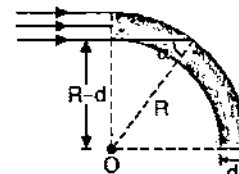
$$\left(\frac{ds}{dt} \right) = \frac{4}{3} \left(\frac{dx}{dt} \right) + \left(\frac{dy}{dt} \right)$$

$$\Rightarrow 16 = \frac{4}{3}(v) + u, \text{ where } u = 4 \text{ cm s}^{-1}$$

$$\Rightarrow v = 9 \text{ cm s}^{-1}$$

7. The necessary and sufficient condition for all the rays to pass around the arc is that the ray with least angle of incidence should get internally reflected i.e., should suffer TIR.

From the figure, it becomes obvious that the ray with least angle of incidence is the one which is incident almost grazingly with the inner wall.



For this ray, let α be the angle of incidence, then we observe that

$$\sin \alpha = \frac{R-d}{R} \text{ where } d \text{ is the diameter of tube}$$

For TIR, $\alpha \geq C$

$$\Rightarrow \sin \alpha \geq \sin C$$

$$\Rightarrow \frac{R-d}{R} \geq \frac{1}{\mu}$$

$$\Rightarrow R \geq \frac{\mu d}{\mu - 1}$$

Since, $\mu = 1.5$ and $d = 4 \text{ cm}$, so we get

$$R = 12 \text{ cm}$$

Hence, the least radius required is 12 cm.

8. Since, $\theta = 90^\circ - i$

$$\Rightarrow \tan \theta = \cot i$$

$$\Rightarrow \frac{dy}{dx} = \cot i \quad \dots(1)$$

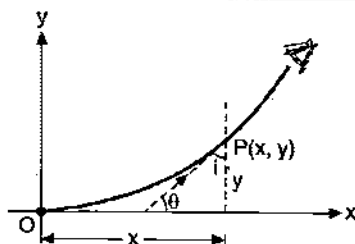
According to Snell's Law at O and P, we have

$$\mu_o \sin i_o = \mu_p \sin i_p$$

$$\text{Since } \mu = \sqrt{1+ay}$$

$$\Rightarrow \text{At } y = 0, \mu = 1$$

$$\Rightarrow \sin(90^\circ) = (\sqrt{1+ay}) \sin i$$



$$\Rightarrow \sin i = \frac{1}{\sqrt{1+ay}}$$

$$\Rightarrow \cot i = \sqrt{ay} = \frac{dy}{dx}$$

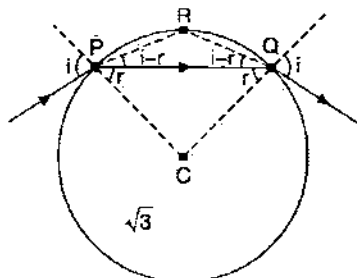
$$\Rightarrow \int_0^y \frac{dy}{\sqrt{ay}} = \int_0^x \frac{dx}{\sqrt{ay}}$$

$$\Rightarrow x = 2\sqrt{\frac{y}{a}}$$

Substituting $y = 2 \text{ m}$ and $a = 2 \times 10^{-6} \text{ m}^{-1}$, we get

$$x_{\text{max}} = 2000 \text{ m} = 2 \text{ km}$$

9. The ray diagram for the situation is shown in figure.



Since, $\angle PCQ = \pi - 2r$, $\angle PRQ = \pi - 2(i - r)$

From the property of a circle, we get

$$2(\angle PRQ) + \angle PCQ = 2\pi$$

$$\Rightarrow 2\pi - 4(i - r) + \pi - 2r = 2\pi$$

$$\Rightarrow r = 2i - \frac{\pi}{2}$$

According to Snell's Law, we have

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin i}{\sin(2i - \frac{\pi}{2})}$$

$$\Rightarrow \sqrt{3} = \frac{\sin i}{-\cos 2i} = \frac{\sin i}{2\sin^2 i - 1}$$

$$\Rightarrow 2\sqrt{3}\sin^2 i - \sin i - \sqrt{3} = 0$$

$$\Rightarrow \sin i = \frac{1 \pm \sqrt{1+24}}{4\sqrt{3}} = \frac{1 \pm 5}{4\sqrt{3}}$$

Rejecting the negative value, we get

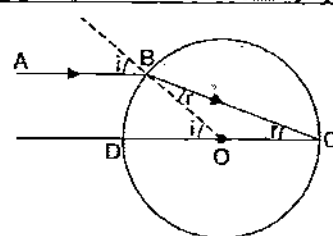
$$\sin i = \frac{\sqrt{3}}{2}$$

$$\Rightarrow i = 60^\circ$$

10. From the figure, we observe that

$$BO = OC$$

So, if $\angle OBC = \angle BCO = r$ (say) and i be the angle of incidence, then



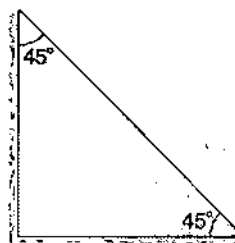
$$i = r + r = 2r$$

Since, we know that

$$\mu = \frac{\sin i}{\sin r}$$

$$\Rightarrow \mu = \frac{\sin 2r}{\sin r} = \frac{2r}{r} = 2$$

11. After reflecting twice from two plane mirrors at right angles, a ray of light gets deviated by 180° , irrespective the angle of incidence, so the emergent ray is antiparallel to incident ray and hence angle of deviation is 180° .



12. From Archimedes Principle, we know that

$$\frac{V_{\text{immersed}}}{V_{\text{total}}} = \frac{\rho_{\text{body}}}{\rho_{\text{liquid}}}$$

$$\Rightarrow \frac{V_{\text{immersed}}}{V} = \frac{\rho_{\text{sphere}}}{\rho_{\text{liquid}}} = \frac{p}{2p}$$

$$\Rightarrow \frac{V_{\text{immersed}}}{V} = \frac{1}{2}$$

i.e., half the sphere is inside the liquid. For the image to coincide with the object, light should fall normally on the sphere. Using,

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \text{ twice, we have}$$

$$\frac{\frac{3}{2}}{v_1} - \frac{1}{(-8)} = \frac{(\frac{3}{2}) - 1}{+2}$$

$$\Rightarrow v_1 = 12 \text{ cm}$$

$$\text{Further, } \frac{\frac{4}{h-10}}{3} - \frac{\frac{3}{8}}{2} = \frac{\frac{4}{h-10} - \frac{3}{8}}{-2}$$

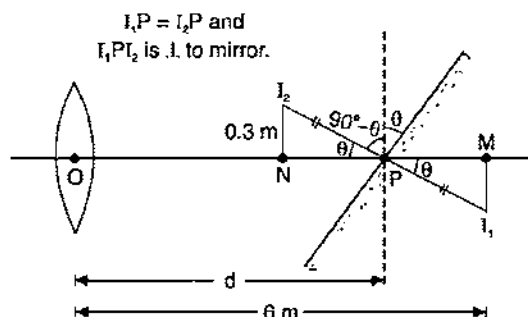
Solving this equation, we get
 $h = 15 \text{ cm}$

13. For the lens, $u = -2 \text{ m}$, $f = +1.5 \text{ m}$

$$\Rightarrow \frac{1}{v} - \frac{1}{-2} = \frac{1}{1.5}$$

$$\Rightarrow v = 6 \text{ m}$$

$$\text{Since, } m = \frac{6}{(-2)} = -3$$



Therefore, y co-ordinate of image formed by the lens is given by

$$y = m(0.1) = -0.3 \text{ m}$$

In triangle $PN I_2$, we have

$$\tan \theta = \frac{I_2 N}{NP}$$

$$\Rightarrow \tan \theta = \frac{0.3}{NP} = 0.3$$

$$\{\because \tan \theta = 0.3\}$$

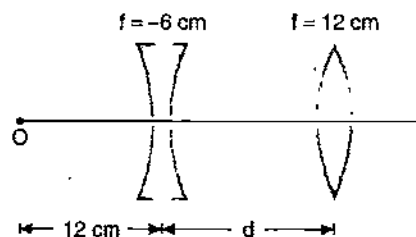
$$\Rightarrow NP = MP = 1 \text{ m}$$

$$\Rightarrow d = 6 - 1 = 5 \text{ m}$$

and x co-ordinate of final image I_2 is,

$$x = d - 1 = 4 \text{ m}$$

14. Applying lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ twice we get



$$\frac{1}{v_1} - \frac{1}{-12} = \frac{1}{-6}$$

...(1)

$$\frac{1}{\infty} - \frac{1}{v_1 - d} = \frac{1}{12}$$

...(2)

Solving equations (1) and (2), we get

$$v_1 = -4 \text{ cm}$$

and $d = 8 \text{ cm}$

15. Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v_1} + \frac{1}{40} = \frac{1}{20}$$

$$\Rightarrow v_1 = 40 \text{ cm}$$

Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v_2} + \frac{1}{10} = \frac{1}{-10}$$

$$\Rightarrow v_2 = -5 \text{ cm}$$

so, the final image is formed at a distance of 5 cm from the mirror towards lens.

16. From Lens Maker's Formula, we get

$$\frac{1}{20} = (1.5 - 1) \left(\frac{1}{R} - \frac{1}{-R} \right)$$

$$\Rightarrow R = 20 \text{ cm}$$

Applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ twice with the condition that rays must fall normally on the concave mirror, we get

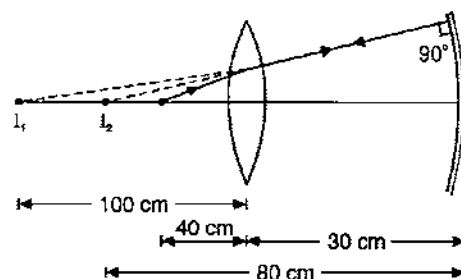
$$\frac{1.5}{v_1} - \frac{1.2}{-40} = \frac{1.5 - 1.2}{+20} \quad \dots(1)$$

$$\frac{2}{d - 80} - \frac{1.5}{v_1} = \frac{2 - 1.5}{-20} \quad \dots(2)$$

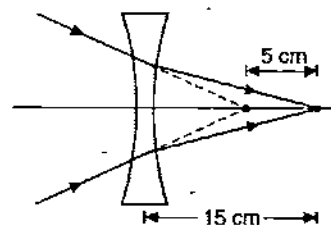
Solving equations (1) and (2), we get

$$d = 30 \text{ cm} \text{ and } v_1 = -100 \text{ cm}$$

The ray diagram is as shown in figure.



17. Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get



$$\frac{1}{f} = \frac{1}{15} - \frac{1}{10} = -\frac{1}{30}$$

$$\Rightarrow f = -30 \text{ cm}$$

18. Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v_1} - \frac{1}{(-40)} = \frac{1}{30}$$

$$\Rightarrow v_1 = 120 \text{ cm}$$

$$\text{Shift due to the slab } \Delta x = \left(1 - \frac{1}{\mu} \right) d = \left(1 - \frac{1}{1.8} \right) 9 = 4 \text{ cm}$$

$$\text{So, } u' = -(40 - \Delta x) = -36 \text{ cm}$$

$$\Rightarrow \frac{1}{v_2} - \frac{1}{(-36)} = \frac{1}{30}$$

$$\Rightarrow v_2 = 180 \text{ cm}$$

Therefore, the screen has to be shifted away from the lens by a distance

$$x = v_2 - v_1 = 60 \text{ cm}$$

19. Lateral magnification in first case is -3 , so if $u = -x$ then $v = +3x$

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{+3x} + \frac{1}{x} = \frac{1}{f}$$

$$\Rightarrow \frac{4}{3x} = \frac{1}{f}$$

$$\Rightarrow x = \frac{4f}{3} \quad \dots(1)$$

In the second case magnification is -2 , so now we have

$$u = -(x+1.5), v = 2(x+1.5)$$

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{2(x+1.5)} + \frac{1}{(x+1.5)} = \frac{1}{f}$$

$$\Rightarrow \frac{3}{2(x+1.5)} = \frac{1}{f}$$

$$\Rightarrow f = \frac{2}{3}(x+1.5) \quad \dots(2)$$

Solving equations (1) and (2), we get

$$f = 9 \text{ cm}$$

20. Using Lens Maker's Formula for both the cases, we get

$$\frac{1}{f_{\text{air}}} = (\mu_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(1)$$

$$\text{and } \frac{1}{f_{\text{water}}} = \left(\frac{\mu_g}{\mu_w} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \dots(2)$$

Dividing equation (1) by (2), we get

$$\frac{f_{\text{water}}}{f_{\text{air}}} = \frac{(\mu_g - 1)}{\left(\frac{\mu_g}{\mu_w} - 1 \right)}$$

Substituting the values, we get

$$f_{\text{water}} = \left(\frac{3-1}{\frac{3}{2}-1} \right) f_{\text{air}} = \left(\frac{2}{\frac{1}{2}} \right) f_{\text{air}} = 4 f_{\text{air}}$$

$$\Rightarrow f_{\text{water}} = 4 f_{\text{air}} = 4 \times 10 = 40 \text{ cm}$$

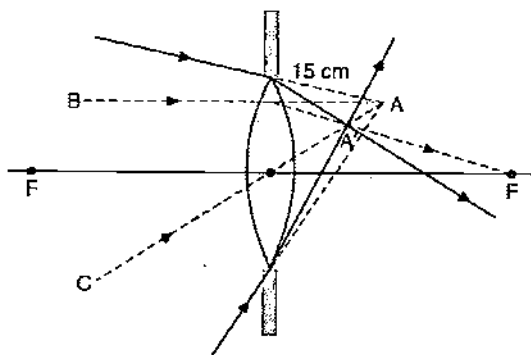
21. Using the lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$u = +15 \text{ cm}, f = +30 \text{ cm}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{15} = \frac{1}{30}$$

$$\Rightarrow v = +10 \text{ cm}$$

Therefore, the focus of the rays will move 5 cm closer to the screen. The ray diagram is as shown in figure.



22. Since we know that

$$f = \frac{D^2 - x^2}{4D}$$

Substituting $f = 16 \text{ cm}$ and $x = 60 \text{ cm}$, we get

$$16 = \frac{D^2 - (60)^2}{4D}$$

$$\Rightarrow D^2 - 3600 = 64D$$

$$\Rightarrow D^2 - 64D - 3600 = 0$$

$$\Rightarrow D^2 - 100D + 36D - 3600 = 0$$

$$\Rightarrow D(D - 100) + 36(D - 100) = 0$$

$$\Rightarrow (D - 100)(D + 36) = 0$$

But $D \neq 36 \text{ cm}$, because it happens to be less than 60 cm . So, $D = 100 \text{ cm}$

23. For the first flare spot, the lens acts as if its right face is silvered. The equivalent focal length is given by

$$\frac{1}{F} = \frac{2 \left(\frac{\mu_2}{\mu_1} \right)}{R_2} - \frac{2 \left(\frac{\mu_2}{\mu_1} - 1 \right)}{R_1}$$

$$\Rightarrow \frac{1}{F} = \frac{2 \times 1.5}{-60} - \frac{2(1.5 - 1)}{+30}$$

$$\Rightarrow F = -12 \text{ cm}$$

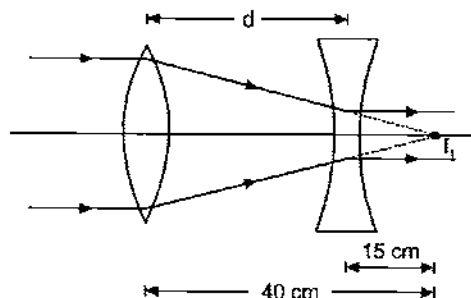
Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{F}$ with $u \rightarrow \infty$, we get

$$\frac{1}{v} + \frac{1}{\infty} = \frac{1}{-12}$$

$$\Rightarrow v = -12 \text{ cm}$$

24. Method 1 :

Ray diagram is as shown



$$d = 25 \text{ cm}$$

Method 2 :

Since this combination just behaves as a plane glass plate, so the power of the combination is zero. Since

$$P_{\text{comb}} = \frac{1}{f_{\text{comb}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{x}{f_1 f_2}$$

$$\Rightarrow 0 = \frac{1}{40} + \frac{1}{(-15)} - \frac{x}{(40)(-15)}$$

$$\Rightarrow \frac{x}{(40)(15)} = \frac{1}{15} - \frac{1}{40}$$

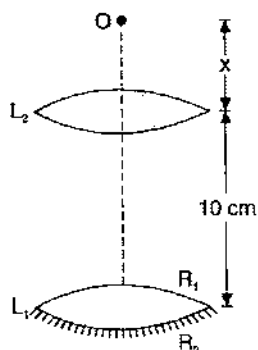
$$\Rightarrow \frac{x}{(40)(15)} = \frac{40 - 15}{(40)(15)}$$

$$\Rightarrow x = 25 \text{ cm}$$

25. Applying Lens Maker's Formula, we get

$$\frac{1}{40} = (1.5 - 1) \left(\frac{1}{120} + \frac{1}{R_1} \right)$$

$\Rightarrow R_1 = 24 \text{ cm}$



Applying lens formula, for L_2 , we get

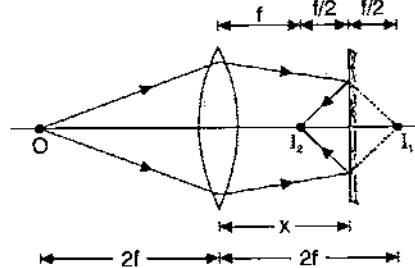
$$\frac{1}{v_1} + \frac{1}{x} = \frac{1}{20} \quad \dots(1)$$

Using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ for unsilvered side of L_1 , we get

$$\frac{1.5}{(-120)} - \frac{1}{(v_1 - 10)} = \frac{1.5 - 1}{24} \quad \dots(2)$$

Solving equations (1) and (2), we get
 $x = 10 \text{ cm}$

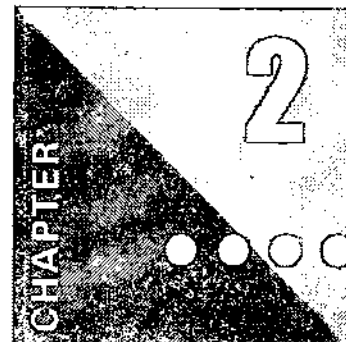
26. The ray diagram is as shown in figure for first two steps. If the rays reflected from the mirror are parallel after passing through the lens for the second time, then I_2 must lie at first focus of lens. So, the desired distance is given by



$$x = (2f + 2f) - \frac{f}{2}$$

$$\Rightarrow x = \frac{3f}{2} = \frac{3(30)}{2} = 45 \text{ cm}$$

Wave Optics



Contents	
WAVE OPTICS	2.1
Solved Practice Problems	2.41
Practice Exercise Sets	
➤ Single Correct Choice Type Questions.....	2.52
➤ Multiple Correct Choice Type Questions	2.66
➤ Reasoning Based Questions (Assertion Reason Type)	2.69
➤ Linked Comprehension Type Questions (Paragraph Type)	2.71
➤ Matrix Match Type Questions (Column Matching Type)	2.77
➤ Integer Answer Type Questions	2.80
Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	2.83
Solutions to In Chapter Exercises (ICE).....	2.86
Solutions to Practice Exercise Sets	2.92

Wave Optics

INTRODUCTION

The phenomenon of interference, diffraction and polarisation exhibited by light could not be explained on the basis of Newton's Corpuscular Theory. In 1678, Huygen suggested that light propagates in the form of waves. The first historic experiment in favour of wave theory was done by Foucault, who in 1850 found experimentally that velocity of light in denser medium is less than that in the rarer medium which was contrary to Newton's Corpuscular Theory.

NEWTON'S CORPUSCULAR THEORY

Newton proposed that light is made up of tiny, light and elastic particles called **corpuscles** which are emitted by a luminous body. These corpuscles travel with speed equal to the speed of light in all directions in straight lines and carry energy with them. When the corpuscles strike the retina of the eye, they produce the sensation of vision. The corpuscles of different colour are of different sizes (red corpuscles larger than blue corpuscles).

The corpuscular theory explains that light carry energy and momentum, light travels in a straight line, Propagation of light in vacuum, Laws of reflection and refraction. However, it fails to explain the phenomenon of interference, diffraction and polarization.

WAVE OPTICS

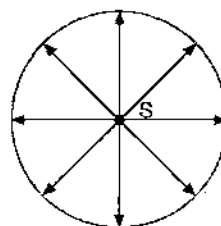
Wave optics is the study of the wave nature of light. Interference and diffraction are two main phenomena giving convincing evidence that light is a wave.

WAVEFRONTS AND RAYS

The locus of all the points vibrating in same phase of oscillation is called a **wavefront (WF)** i.e. a wavefront is defined as a surface joining the points vibrating in the same phase. The direction of propagation of light (ray of light) is along the normal to the Wavefront. The speed with which the wavefront moves onwards from the source is called the

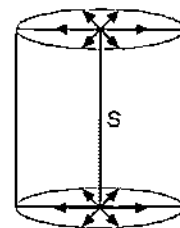
phase velocity or wave velocity. The energy travels outwards along straight lines emerging from the source, normally to the wavefront, that is, along the radii of the spherical wavefront. These lines are called the **rays**.

For a point source in a homogeneous medium the wavefront is spherical.



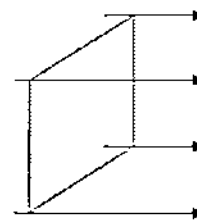
Spherical wavefront

For a linear source of light the wavefront is cylindrical.



Cylindrical wavefront

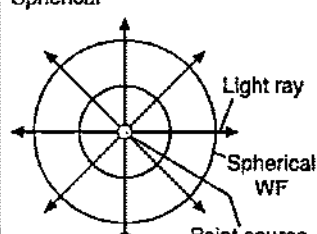
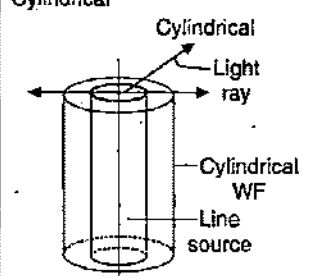
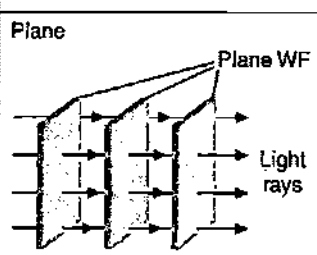
A small part of a spherical or cylindrical wavefront from a distant source will appear plane and is, therefore, called a **plane wavefront**.



Plane wavefront

CONCEPTUAL NOTE(S)

Different types of wavefront

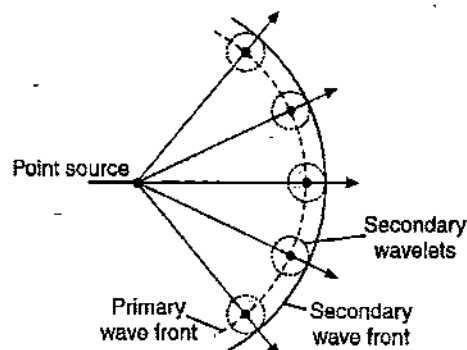
Type of wavefront	Intensity	Amplitude
Spherical 	$I \propto \frac{1}{r^2}$	$A \propto \frac{1}{r}$
Cylindrical 	$I \propto \frac{1}{r}$	$A \propto \frac{1}{\sqrt{r}}$
Plane 	$I \propto r^0$	$A \propto r^0$

HUYGEN'S PRINCIPLE

This principle is useful for determining the position of a given wavefront at any further time if its present position is known. The principle may be stated in three parts.

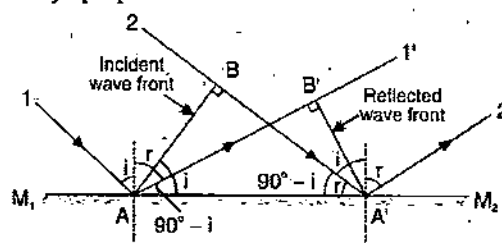
- Every point on the given wavefront may be regarded as the source of the new disturbance.
- The new disturbances from each point spread out in all directions with the velocity of light in the same manner as the original source of light does and these new disturbances are called **secondary wavelets**.
- The surface of tangency to the secondary wavelets in forward direction at any time gives the position of the new wavefront at that time. This new wavefront is called the **Secondary Wavefront**.

This principle explained successfully, the reflection, refraction, total internal reflection, interference and diffraction but failed to explain the rectilinear propagation of light.



LAWS OF REFLECTION ON THE BASIS OF HUYGEN'S THEORY

Let AB be the plane wave front incident on a plane mirror M_1M_2 at $\angle BAA' = i$, where 1, 2 are the corresponding incident rays perpendicular to AB .



According to Huygen's principle every point on AB is a source of secondary wavelets, so

$$BA' = ct, \text{ where } c \text{ is speed of light}$$

The secondary wavelets from A will travel the same distance ct in the same time. So,

$$AA' = ct$$

Now, $\angle AA'B = 90^\circ - i$, so that $\angle A'AB = i$, ($0 < i < 90^\circ$)

Also, $\angle A'AB' = 90^\circ - r$, so that $\angle AA'B' = r$, ($0 < r < 90^\circ$)

From $\triangle AB'A'$, we have

$$\sin r = \frac{AB'}{AA'} = \frac{ct}{AA'} \quad \dots(1)$$

From $\triangle A'BA$, we have

$$\sin i = \frac{AB}{AA'} = \frac{ct}{AA'} \quad \dots(2)$$

From equation (1) and (2), we get

$$\sin i = \sin r$$

$$\angle i = \angle r \text{ which is the Law of Reflection}$$

LAW OF REFRACTION ON THE BASIS OF HUYGEN'S THEORY

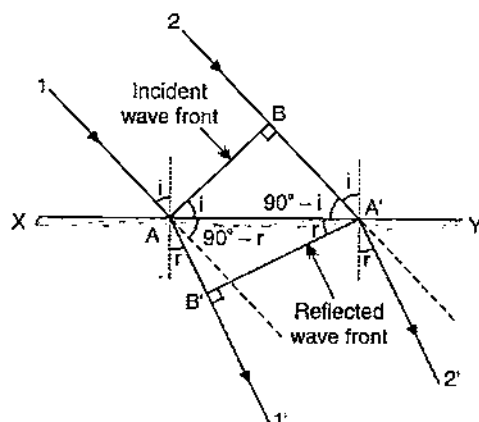
XY is a plane surface that separates a denser medium of refractive index μ from a rarer medium. If v_1 is velocity of light in rarer medium and v_2 is velocity of light in denser medium, then by definition

$$\mu = \frac{v_1}{v_2} \quad \dots(1)$$

AB is a plane wave front incident on XY at $\angle BAA' = \angle i$, where 1, 2 are the corresponding incident rays normal to AB . According to Huygen's principle

$$BA' = v_1 t \quad \dots(2)$$

The secondary wavelets from A travel in the denser medium with a velocity v_2 and would cover a distance $AB' = v_2 t$ in the same time.



So, from $\triangle ABA'$ and $\triangle AB'A'$

$$\sin i = \frac{BA'}{AA'} \text{ and } \sin r = \frac{AB'}{AA'}$$

$$\Rightarrow \frac{\sin i}{\sin r} = \frac{BA'}{AA'} \times \frac{AA'}{AB'} = \frac{BA'}{AB'} = \frac{v_1 t}{v_2 t} = \frac{v_1}{v_2}$$

So, from equation (1), we get

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \mu \text{ which is the Snell's Law of Refraction.}$$

INTERFERENCE

When two waves of same frequency, nearly same amplitude and constant initial phase difference travel in the same direction along same straight line, they superimpose in such a way that in the region of superposition, the intensity is maximum at some points and minimum at some other points. This modification in intensity in the region of superposition is called **Interference**. The sources having the same frequency and constant initial phase difference are called **coherent sources**. The phenomenon of interference is based on the Law of Conservation of Energy.

COHERENT SOURCES

Two sources which emit light of the same wavelength with zero or a constant phase difference are called **coherent sources**.

Unlike sound waves, two independent sources of light cannot be coherent. Sound is a bulk property of matter. So, two independent sources of sound can produce coherent waves. However, two independent sources of light cannot

be coherent. The emission of light from any source is from a very large number of atoms and the emission from each atom is random and independent of each other. Therefore, there is no stable phase relationship between radiations from two independent sources. So, for two sources to be coherent, they must be derived from the same parent source.

In practice, coherent sources are obtained either by dividing the wavefront (as in the case of Young's Double Slit Experiment, Fresnel's biprism, Lloyd mirror, etc.) or by dividing the amplitude (as in the case of thin films, Newton rings, etc.) of the incoming waves from a single source.

A laser discovered in 1960, is different from common light sources. Its atoms act in a cooperative manner so as to produce intense, monochromatic, unidirectional and coherent light. Thus, two independent laser beams can produce observable interference on a screen.

METHODS OF PRODUCING COHERENT SOURCES

A. DIVISION OF WAVEFRONT

In this method the wavefront is divided into two parts by the use of mirrors, or lenses or prisms. Well known methods are Young's double slit arrangement, Fresnel's biprism and Lloyd's single mirror.

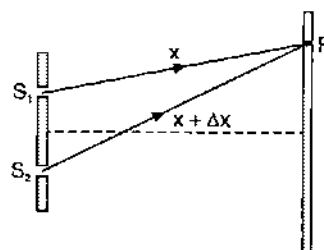
B. DIVISION OF AMPLITUDE

In this method the amplitude of the incoming beam is divided into two parts by means of partial reflection of refraction. These divided parts travel different paths and are finally brought together to produce interference. This class of interference requires broad sources of light. The common examples of such interference of light are the brilliant colours seen when a thin film of transparent material like soap bubble or thin film of kerosene oil spread on the surface of water is exposed to an extended source of light. This kind of interference exists in two types.

- Interference due to waves reflected from both the front and back surfaces of the film.
- Interference due to transmitted waves.

INTERFERENCE : MATHEMATICAL TREATMENT

Two waves (whether sound or light) of equal frequencies travelling almost in the same direction show interference. Consider two waves coming from sources S_1 and S_2 . These reach point P with a path difference Δx , having amplitude A_1 and A_2 .



$$y_1 = A_1 \sin(\omega t - kx) \quad \dots(1)$$

$$\text{and } y_2 = A_2 \sin[\omega t - k(x + \Delta x)]$$

$$\Rightarrow y_2 = A_2 \sin(\omega t - kx - \phi) \quad \dots(2)$$

where $\phi = k\Delta x = \left(\frac{2\pi}{\lambda}\right)\Delta x$ and Δx is the path difference.

By Principle of Superposition, the resultant wave at P is

$$y = y_1 + y_2 = A_1 \sin(\omega t - kx) + A_2 \sin(\omega t - kx - \phi)$$

$$\Rightarrow y = (A_1 + A_2 \cos \phi) \sin(\omega t - kx) - (A_2 \sin \phi) \cos(\omega t - kx) \quad \dots(3)$$

Substituting $A_1 + A_2 \cos \phi = A \cos \theta$ and $A_2 \sin \phi = A \sin \theta$, we get

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \phi \quad \dots(4)$$

Equation (3), becomes

$$y = A \sin(\omega t - kx - \theta) \quad \dots(5)$$

where, $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$ and $\tan \theta = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$

The intensity of the resultant wave is

$$I = \frac{1}{2} \rho v \omega^2 A^2 = KA^2 = K[A_1^2 + A_2^2 + 2A_1A_2 \cos \phi]$$

$$\Rightarrow I = I_1 + I_2 + \underbrace{2\sqrt{I_1 I_2} \cos \phi}_{\text{Interference term}} \quad \dots(6)$$

Thus, when interference of two waves of equal intensities occur, the intensity of maxima becomes 4 times that of single wave and that of minima becomes zero.

CONDITION FOR MAXIMA : Constructive Interference

From equation (6), I is maximum, when

$$\cos \phi = +1$$

$$\Rightarrow \phi = 0, 2\pi, 4\pi, \dots$$

$$\Rightarrow \phi = 2n\pi$$

$$\Rightarrow \phi = \left(\frac{2\pi}{\lambda}\right)x = 2n\pi, \text{ where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \Delta x = (2n)\frac{\lambda}{2}, \text{ where } n = 0, 1, 2, 3, \dots$$

So, intensity will be maximum when phase difference ϕ is an even multiple of π or path difference Δx is an even multiple of $\frac{\lambda}{2}$.

$$\Rightarrow I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} = (\sqrt{I_1} + \sqrt{I_2})^2 = (A_1 + A_2)^2$$

CONDITION FOR MINIMA : Destructive Interference

Intensity I will be minimum, when

$$\cos \phi = -1$$

$$\Rightarrow \phi = \pi, 3\pi, 5\pi, \dots$$

$$\Rightarrow \phi = (2n+1)\pi$$

$$\Rightarrow \phi = \left(\frac{2\pi}{\lambda}\right)x = (2n+1)\pi, \text{ where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \Delta x = (2n+1)\frac{\lambda}{2}, \text{ where } n = 0, 1, 2, 3, \dots$$

So, intensity will be maximum when phase difference ϕ is an odd multiple of π or path difference Δx is an odd multiple of $\frac{\lambda}{2}$.

$$\Rightarrow I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2} = (\sqrt{I_1} - \sqrt{I_2})^2 = K(A_1 - A_2)^2$$

$$\text{The ratios } \frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2}$$

If $I_1 = I_2 = I_0$ (i.e., $A_1 = A_2$), we have

$$I_{\max} = 4I_0 \text{ and } I_{\min} = 0$$

Thus, when interference of two waves of equal intensities occur, the intensity of maxima becomes four times that of single wave and that of minima becomes zero.

In Young's Double Slit Experiment popularly known as YDSE, usually the intensities I_1 and I_2 are equal, so $I_1 = I_2 = I_0$

Since, $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$, so we get

$$I = 2I_0(1 + \cos \phi)$$

$$\Rightarrow I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

PHASE DIFFERENCE AND PATH DIFFERENCE

If two waves travel different lengths of path to reach a point, they may not be in phase with each other. The phase difference depends on the path difference as $\phi = \frac{2\pi}{\lambda}\Delta x$, where Δx is the difference in length of path traversed by the waves.

The phase difference between two light waves can change if the waves travel through different materials having different refractive indices:

Suppose, we have two waves having identical wavelengths λ , initially in phase, in air. One of the waves travel through medium 1 of refractive index μ_1 and length L and other wave travels through same length L in another medium of refractive index μ_2 . As wavelength differs in a medium, the two waves may not remain in phase.

The path difference after crossing through the medium is given by

$$\Delta x = (n_1 - n_2)L$$

where n_1 is number of wavelengths in medium 1 and

n_2 is number of wavelengths in medium 2

$$\Rightarrow \Delta x = \left(\frac{L}{\lambda_1} - \frac{L}{\lambda_2} \right) \lambda = \left(\frac{\lambda}{\lambda_1} - \frac{\lambda}{\lambda_2} \right) L$$

Since $\frac{\lambda}{\lambda_1} = \mu_1$ and $\frac{\lambda}{\lambda_2} = \mu_2$, so

$$\Delta x = (\mu_1 - \mu_2)L$$

Illustration 1

Two light rays, initially in phase and having wavelength 6×10^{-7} m, go through different plastic layers of the same thickness, 7×10^{-6} m. The indices of refraction are 1.65 for one layer and 1.49 for the other.

- What is the equivalent phase difference between the rays when they emerge?
- If those two rays then reach a common point, does the interference result in complete darkness, maximum brightness, intermediate illumination but closer to complete darkness, or intermediate illumination but closer to maximum brightness?

Solution

$$(a) \Delta x = (\mu_1 - \mu_2)t = (1.65 - 1.49)(7 \times 10^{-6})$$

$$\Delta x = 1.12 \times 10^{-6} \text{ m}$$

Since, Phase difference $\phi = \left(\frac{2\pi}{\lambda} \right) (\Delta x)$

$$\Rightarrow \phi = \left(\frac{2\pi}{6 \times 10^{-7}} \right) (1.12 \times 10^{-6})$$

$$\Rightarrow \phi = 11.72 \text{ radian}$$

- To discuss this case, two options arise

Option 1 : Waves are in phase, then using

$$I = I_{\max} \cos^2 \left(\frac{\phi}{2} \right), \text{ we get}$$

$$I = I_{\max} \cos^2 \left(\frac{11.72}{2} \right) = 0.8 I_{\max}$$

This value is intermediate illumination closer to maximum brightness.

Option 2 : Waves are out of phase, then

$$\phi_{\text{net}} = 11.72 \pm \pi$$

$$\Rightarrow \phi_{\text{net}} = 14.86 \text{ rad}$$

$$\Rightarrow I = I_{\max} \cos^2 \left(\frac{14.86}{2} \right)$$

$$\Rightarrow I = 0.17 I_0$$

This value is intermediate illumination closer to darkness.

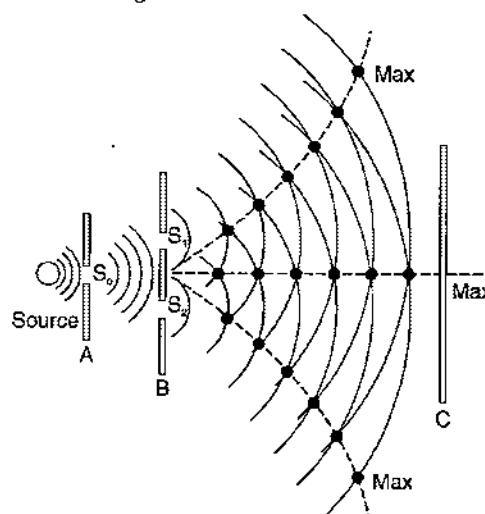
THEORY OF INTERFERENCE : MAXIMA & MINIMA

THEORY OF DIVISION OF WAVEFRONT: YOUNG'S DOUBLE SLIT EXPERIMENT

The phenomenon of interference of light waves arising from

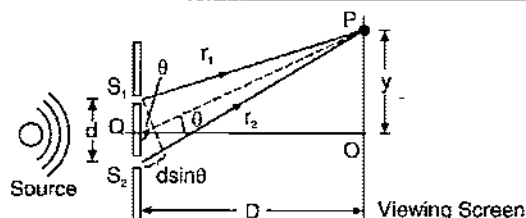
two sources was first demonstrated by Thomas Young in 1801.

Light is incident on screen A, which is provided with a narrow slit, S_0 . The cylindrical waves emerging from this slit arrive at screen B, which contains two narrow, parallel slits, S_1 and S_2 . Light emerges from these two slits as cylindrical waves. In effect, slits S_1 and S_2 act as individual light sources that are in phase as they originate from the same cylindrical wavefront. The light from the two slits produces a visible pattern on screen C. The pattern consists of a series of bright and dark parallel bands called fringes. The overall light amplitude at a given point on the screen is the result of the superposition of the two wave amplitudes from S_1 and S_2 . Two waves that add constructively give a bright fringe, and any two waves that add destructively produce a dark fringe.



Schematic diagram of Young's double-slit experiment. The narrow slits act as sources of cylindrical waves. Slits S_1 and S_2 behave as coherent sources which produce an interference pattern on screen C.

YDSE (QUANTITATIVE TREATMENT) : METHOD 1



Geometric construction for describing Young's double-slit experiment. Note that the path difference between the two rays is $r_2 - r_1 = d \sin \theta$

Consider a point P on the viewing screen located a perpendicular distance D from the two identical slits S_1 and S_2 , which are separated by a distance d. Let us assume that the source is equidistant from the two slits and is

monochromatic, that is, emitting light of a single wavelength λ . Under these assumptions, the waves emerging from slits S_1 and S_2 have the same frequency and amplitude and are in phase. The light intensity on the screen at P is the resultant of light coming from both slits. Note that a wave from the lower slit travels farther than a wave from the upper slit by an amount equal to $d \sin \theta$. This distance is called the **path difference**, x , where

$$x = r_2 - r_1 = d \sin \theta \quad \dots(1)$$

The value of this path difference will determine whether or not the two waves are in phase when they arrive at P . If the path difference is either zero or some integral multiple of the wavelength, the two waves are in phase at P and constructive interference results. Therefore, the condition for bright fringes, or constructive interference, at P is given by

$$x = d \sin \theta = n\lambda \quad (n = 0, \pm 1, \pm 2, \pm 3 \dots) \quad \dots(2)$$

The index number n is called the **order number** of the fringe. The central bright fringe at $\theta = 0$ ($n = 0$) is called the **zeroth order maximum**. The first maximum on either side, when $n = \pm 1$, is called the **first order maximum**, etc.

Similarly, when the path difference is an odd multiple of $\frac{\lambda}{2}$,

the two waves arriving at P will be opposite in phase and will give rise to destructive interference. Therefore, the condition for dark fringes, or destructive interference, at P is given by

$$x = d \sin \theta = (2n + 1) \frac{\lambda}{2} \quad (n = 0, \pm 1, \pm 2 \dots) \quad \dots(3)$$

It is useful to obtain expressions for the positions of the bright and dark fringes measured vertically from O to P . We shall assume that $D > d$ and consider only points P that are close to O . In this case, θ is small, and so we can use the approximation $\sin \theta \approx \tan \theta$. From the large triangle OPQ in Figure, we see that

$$\sin \theta \approx \tan \theta = \frac{y}{D} \quad \dots(4)$$

Using this result together with equation (2), we see that the positions of the bright fringes measured from O are given by

$$y_{\text{bright}} = n \left(\frac{\lambda D}{d} \right) \quad \dots(5)$$

From this expression, we find that the separation between any two adjacent bright fringes called **Fringe Width** is equal to $\frac{\lambda D}{d}$, that is,

$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d} (n + 1) - \frac{\lambda D}{d} n = \frac{\lambda D}{d} \quad \dots(6)$$

Similarly, using equation (3) and (4), we find that the dark fringes are located at

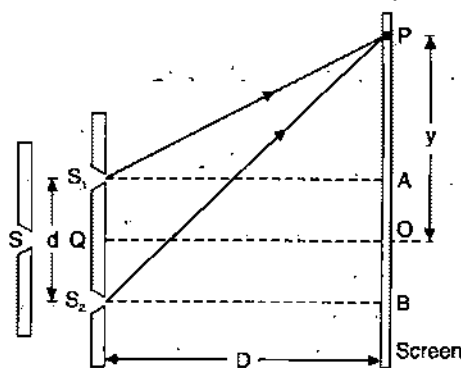
$$y_{\text{dark}} = (2n + 1) \frac{\lambda D}{2d} \quad \dots(7)$$

This result shows that the separation between adjacent dark fringes is also equal to $\beta = \frac{\lambda D}{d}$. Since the quantities D and d are both measurable, we see that the double-slit interference pattern, together with equation (6), provides a direct determination of the wavelength λ . Young used this technique to make the first measurement of the wavelength of light.

YDSE (QUANTITATIVE TREATMENT) : METHOD 2

Consider that two coherent sources of light S_1 and S_2 are placed at a distance d apart and a screen is placed at a distance D from the plane of the two sources. Let P be a point on the screen at a distance y from a point O exactly opposite to the centre of the two sources S_1 and S_2 . If x is path difference between the light waves reaching point P from the sources S_1 and S_2 , then

$$x = S_2P - S_1P$$



In right angled ΔS_2BP , we have

$$S_2P^2 = S_2B^2 + BP^2 = D^2 + \left(y + \frac{d}{2}\right)^2$$

Also, in right angled ΔS_1AP , we have

$$S_1P^2 = S_1A^2 + AP^2 = D^2 + \left(y - \frac{d}{2}\right)^2$$

$$\Rightarrow S_2P^2 - S_1P^2 = \left(D^2 + \left(y + \frac{d}{2}\right)^2\right) - \left(D^2 + \left(y - \frac{d}{2}\right)^2\right)$$

$$\Rightarrow (S_2P + S_1P)(S_2P - S_1P) = \left(y + \frac{d}{2}\right)^2 - \left(y - \frac{d}{2}\right)^2$$

Since $S_2P - S_1P = x$ (the path difference between the two light waves), the above equation becomes

$$(S_2P + S_1P)x = 4y \left(\frac{d}{2}\right) = 2yd$$

$$\Rightarrow x = \frac{2yd}{S_2P + S_1P}$$

In practice, the point P lies very close to the centre of screen, so we have

$$S_2P = S_1P = D$$

$$\Rightarrow x = \frac{2yd}{D+D} = \frac{2yd}{2D}$$

$$\Rightarrow x = \frac{yd}{D} \quad \dots(1)$$

For Maxima, we know that path difference x must be an even multiple of $\frac{\lambda}{2}$, so

$$x = (2n)\frac{\lambda}{2}, \text{ where } n = 0, 1, 2, \dots$$

$$\Rightarrow \frac{yd}{D} = n\lambda, \text{ where } n = 0, 1, 2, \dots$$

$$\Rightarrow y_n = n\left(\frac{\lambda D}{d}\right), \text{ where } n = 0, 1, 2, \dots$$

At $y=0$ (i.e., for $n=0$) we get a Central Bright Fringe.

For Minima, we know that path difference x must be an odd multiple of $\frac{\lambda}{2}$, so

$$x = (2n+1)\frac{\lambda}{2}, \text{ where } n = 0, 1, 2, \dots$$

$$\Rightarrow \frac{yd}{D} = (2n+1)\frac{\lambda}{2}, \text{ where } n = 0, 1, 2, \dots$$

$$\Rightarrow y_n = (2n+1)\frac{\lambda D}{2d}, \text{ where } n = 0, 1, 2, \dots$$

$$\Rightarrow y_n = \left(n + \frac{1}{2}\right)\frac{\lambda D}{d}, \text{ where } n = 0, 1, 2, \dots$$

Illustration 2

If the maximum intensity in YDSE is I_0 , find the intensity at a point on the screen where

(a) the phase difference between the two interfering beams is $\frac{\pi}{3}$.

(b) the path difference between the two interfering beams is $\frac{\lambda}{4}$.

Solution

$$(a) \text{ Since, } I = I_{\max} \cos^2\left(\frac{\phi}{2}\right)$$

where $I_{\max}$ is I_0 i.e., intensity due to independent sources is $\frac{I_0}{4}$. Therefore, at

$$\phi = \frac{\pi}{3}$$

$$\Rightarrow \frac{\phi}{2} = \frac{\pi}{6}$$

$$\Rightarrow I = I_0 \cos^2\left(\frac{\pi}{6}\right) = \frac{3}{4}I_0$$

(b) Phase difference corresponding to the given path difference Δx is given by

$$\phi = \left(\frac{2\pi}{\lambda}\right)\Delta x$$

$$\Rightarrow \phi = \left(\frac{2\pi}{\lambda}\right)\left(\frac{\lambda}{4}\right)$$

$$\Rightarrow \phi = \frac{\pi}{2}$$

$$\Rightarrow \frac{\phi}{2} = \frac{\pi}{4}$$

$$\Rightarrow I = I_0 \cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$

Illustration 3

In YDSE, the interference pattern is found to have an intensity ratio between the bright and dark fringes as 9. Find the ratio of

(a) intensities.

(b) amplitudes of the two interfering waves.

Solution

In case of interference, we have

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$(a) \quad I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$\text{and } I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\text{Since, } \frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \frac{9}{1}$$

$$\Rightarrow \frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} = \frac{3}{1}$$

Solving, we get

$$\frac{I_1}{I_2} = \frac{4}{1} = 4$$

(b) Since, $I \propto A^2$,

$$\Rightarrow \frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2$$

$$\Rightarrow \left(\frac{A_1}{A_2}\right)^2 = 4$$

$$\Rightarrow \frac{A_1}{A_2} = 2$$

Illustration 4

The intensity of the light coming from one of the slits in YDSE is double the intensity from the other slit. Find the ratio of the maximum intensity to the minimum intensity in the interference fringe pattern observed.

Solution

Since, we know that

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2$$

According to the problem, we have

$$I_1 = 2I_0 \text{ and } I_2 = I_0$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{2} + 1}{\sqrt{2} - 1} \right)^2 = 34$$

Illustration 5

In a Young's double-slit experiment the distance between the slits is 1 mm and the distance of the screen from the slits is 1 m. If light of wavelength $6000 \text{ \AA}$ is used, find the distance between the second dark fringe and the fourth bright fringe.

Solution

The position of the second dark fringe is given by

$$y_2(\text{dark}) = (2n-1) \frac{\lambda D}{2d} = (4-1) \frac{\lambda D}{2d} = \frac{3}{2} \left(\frac{\lambda D}{d} \right)$$

The position of the 4th bright fringe is given by

$$y_4(\text{bright}) = \frac{n\lambda D}{d} = \frac{4\lambda D}{d}$$

Therefore, the separation is given by

$$\Delta y = y_4(\text{bright}) - y_2(\text{dark}) = \left(4 - \frac{3}{2} \right) \frac{\lambda D}{d}$$

$$\Rightarrow \Delta y = \frac{5}{2} \times \frac{6000 \times 10^{-10} \times 1}{10^{-3}} = 1.5 \times 10^{-3} = 1.5 \text{ mm}$$

Illustration 6

In a YDSE, the separation between the coherent sources is 6 mm, the separation between coherent sources and the screen is 2 m. If light of wavelength $6000 \text{ \AA}$ is used, then

- find the fringe width.
- find the position of the third maxima.
- find the position of the second minima.

Solution

- Since fringe width is given by $\beta = \frac{\lambda D}{d}$, so we have

$$\beta = \frac{\lambda D}{d} = \frac{(6000 \times 10^{-10})(2)}{6 \times 10^{-3}} = 0.2 \text{ mm}$$

- Position of third maxima is obtained by substituting $n=3$ in the equation $y_n = n \left(\frac{\lambda D}{d} \right)$, so we get

$$y_3 = \frac{3\lambda D}{d} = 3\beta = 3(0.2) = 0.6 \text{ mm}$$

- Position of second minima is obtained by putting $n=2$

in the equation $y_n = (2n-1) \frac{\lambda D}{2d}$, so we get

$$y_2 = \left(2 - \frac{1}{2} \right) \frac{\lambda D}{d} = \frac{3}{2} \beta = \frac{3}{2} (0.2) = 0.3 \text{ mm}$$

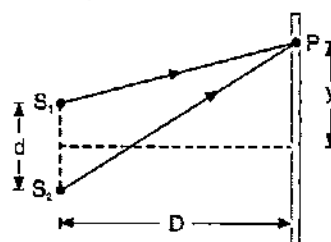
Illustration 7

Young's double slit experiment is carried out using microwaves of wavelength $\lambda = 3 \text{ cm}$. Distance between the slits is $d = 5 \text{ cm}$ and the distance between the plane of slits and the screen is $D = 100 \text{ cm}$. Find the number of maximas and their positions on the screen.

Solution

$$\left(\begin{array}{c} \text{The maximum path} \\ \text{difference that} \\ \text{can be produced} \end{array} \right) = \left(\begin{array}{c} \text{Distance between the} \\ \text{coherent sources} \\ \text{i.e., } 5 \text{ cm} \end{array} \right)$$

Thus, in this case we can have only three maximas, one central maxima and two on its either side (for a path difference of $\lambda = 3 \text{ cm}$)



For maximum intensity at P, we have

$$S_2P - S_1P = \lambda$$

$$\Rightarrow \sqrt{\left(\frac{y+d}{2} \right)^2 + D^2} - \sqrt{\left(\frac{y-d}{2} \right)^2 + D^2} = \lambda$$

Substituting $d = 5 \text{ cm}$, $D = 100 \text{ cm}$ and $\lambda = 3 \text{ cm}$ and solving the equation, we get

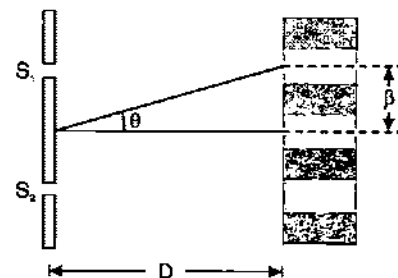
$$y = \pm 75 \text{ cm}$$

Thus, the three maximas will be at

$$y = 0 \text{ and } y = \pm 75 \text{ cm}$$

FRINGE WIDTH & ANGULAR FRINGE WIDTH

The separation between two consecutive bright (or dark) fringes is called the **fringe width** (β), given by



$$\beta = y_{n+1} - y_n = \frac{\lambda D}{d}$$

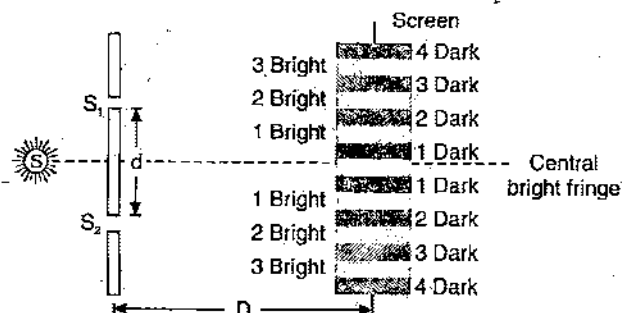
Note that fringe width β is independent of n . That is, the interference fringes have same width throughout.

The angular fringe width is given by

$$\Delta\phi = \frac{\beta}{D} = \frac{\lambda}{d}$$

CONCEPTUAL NOTE(S)

In YDSE alternate bright and dark bands obtained on the screen. These bands are called Fringes.



d = Distance between slits

D = Distance between slits and screen

λ = Wavelength of monochromatic light emitted from source

- Central fringe is always bright, because at central position the path difference, $x = 0$ and hence the phase difference, $\phi = 0^\circ$ and hence the Central Bright Fringe is also called the zeroth maxima.
- The n th minima comes before the n th maxima.
- The fringe pattern obtained due to a slit is more bright than that due to a point.
- If the slit widths are unequal, the minima will not be complete dark. For very large width uniform illumination occurs.
- If one slit is illuminated with red light and the other slit is illuminated with blue light, no interference pattern is observed on the screen.
- If the two coherent sources consist of object and its reflected image, the central fringe is dark instead of bright one.

Illustration 8

In a YDSE bi-chromatic light of wavelengths 400 nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1 m. Find the minimum distance between two successive regions of complete darkness.

Solution

Let n_1 minima of 400 nm coincides with n_2 minima of 560 nm, then

$$(2n_1 - 1)400 = (2n_2 - 1)560$$

$$\Rightarrow \frac{2n_1 - 1}{2n_2 - 1} = \frac{7}{5} = \frac{14}{10} = \frac{21}{15}$$

i.e., 4th minima of 400 nm coincides with 3rd minima of 560 nm

The location of this minima is

$$y_4 = \frac{7(1000)(400 \times 10^{-6})}{2 \times 0.1} = 14 \text{ mm}$$

Next, 11th minima of 400 nm will coincide with 8th minima of 560 nm

Location of this minima is

$$y_{11} = \frac{21(1000)(400 \times 10^{-6})}{2 \times 0.1} = 42 \text{ mm}$$

So, required separation is

$$\Delta y = 42 - 14 = 28 \text{ mm}$$

Illustration 9

Light from a source consists of two wavelengths $\lambda_1 = 6500 \text{ \AA}$ and $\lambda_2 = 5200 \text{ \AA}$. If the separation between the sources from each other is 6.5 mm and that from the screen is 2 m, find the minimum value of $y (\neq 0)$ where the maxima of both the wavelengths coincide.

Solution

Let n_1 maxima of λ_1 coincides with n_2 maxima of λ_2 .

Then, $y_{n_1} = y_{n_2}$

$$\Rightarrow \frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{5200}{6500} = \frac{4}{5}$$

Thus, fourth maxima of λ_1 coincides with fifth maxima of λ_2 . The minimum value of $y (\neq 0)$ is given by

$$y = \frac{4\lambda_1 D}{d} = \frac{4(0.65 \times 10^{-6})(2)}{6.5 \times 10^{-3}} = 0.8 \text{ mm}$$

Illustration 10

Two coherent sources are 0.3 mm apart. They are 0.9 m away from the screen. The second dark fringe is at a distance of 0.3 cm from the centre. Find the distance of fourth bright fringe from the centre. Also find the wavelength of light used.

Solution

Second dark fringe $n = 2$ will be obtained at

$$y = (2n - 1) \frac{\lambda D}{2d} = \frac{3\lambda D}{2d} \quad \dots (1)$$

$$\Rightarrow \frac{\lambda D}{d} = \frac{2}{3} y$$

$$\Rightarrow \beta = \frac{\lambda D}{d} = \frac{2}{3} (0.3) = 0.2 \text{ m}$$

Fourth bright fringe from the centre will be obtained at

$$y_4 = 4\beta = 0.8 \text{ cm}$$

From equation (1), we get

$$\lambda = \frac{2yd}{3D} = \frac{2 \times 0.3 \times 10^{-3} \times 0.3 \times 10^{-2}}{3 \times 0.9}$$

$$\Rightarrow \lambda = 6.67 \times 10^{-7} \text{ m}$$

Illustration 11

In a Young's double slit experiment, two narrow slits 0.8 mm apart are illuminated by a source of light of wavelength 4000 Å. How far apart are the adjacent bright bands in the interference pattern observed on a screen 2 m away?

Solution

Since, $d = 0.8 \times 10^{-3} \text{ m}$, $\lambda = 4000 \text{ Å} = 4000 \times 10^{-10} \text{ m}$, $D = 2 \text{ m}$

The distance between the adjacent bright bands or the fringe width is given by

$$\beta = \frac{\lambda D}{d} = \frac{4000 \times 10^{-10} \times 2}{0.8 \times 10^{-3}} \text{ m} = 1 \text{ mm}$$

Illustration 12

A beam of light consisting of two wavelengths, 6500 Å and 5200 Å, is used to obtain interference fringes in a Young's double slit experiment. The distance between the slits is 2 mm and the distance between the plane of the slits and the screen is 120 cm.

- Find the distance of the third bright fringe on the screen from the central maximum for the wavelength 6500 Å
- What is the least distance from the central maximum where the bright fringes due to both the wavelengths coincide?

Solution

According to the problem, we have

$$\lambda_1 = 6500 \text{ Å} = 6500 \times 10^{-10} \text{ m}$$

$$\lambda_2 = 5200 \text{ Å} = 5200 \times 10^{-10} \text{ m}$$

$$d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$$

$$D = 120 \text{ cm} = 1.2 \text{ m}$$

$$(a) \quad y_n = \frac{n\lambda D}{d}$$

$$\Rightarrow y_3 = \frac{3 \times 6500 \times 10^{-10} \times 1.2}{2 \times 10^{-3}} = 1.17 \times 10^{-3} \text{ m} = 1.17 \text{ mm}$$

- The least distance from the central maximum where the bright fringes due to both the wavelength coincide corresponds to that value of n for which

$$\frac{n\lambda_1 D}{d} = \frac{(n+1)\lambda_2 D}{d}$$

$$\Rightarrow n\lambda_1 = (n+1)\lambda_2$$

$$\Rightarrow \frac{n+1}{n} = \frac{6500}{5200} = \frac{5}{4}$$

$$\Rightarrow n = 4$$

$$\Rightarrow y_4 = \frac{n\lambda D}{d} = \frac{4 \times 6500 \times 10^{-10} \times 1.2}{2 \times 10^{-3}}$$

$$\Rightarrow y_4 = 1.56 \times 10^{-3} \text{ m} = 1.56 \text{ mm}$$

Illustration 13

A convergent lens with a focal length of $f = 10 \text{ cm}$ is cut into two halves that are then moved apart to a distance of $d = 0.5 \text{ mm}$ (a double lens). Find the fringe width on screen at a distance of 60 cm behind the lens if a point source of monochromatic light ($\lambda = 5000 \text{ Å}$) is placed in front of the lens at a distance of $a = 15 \text{ cm}$ from it.

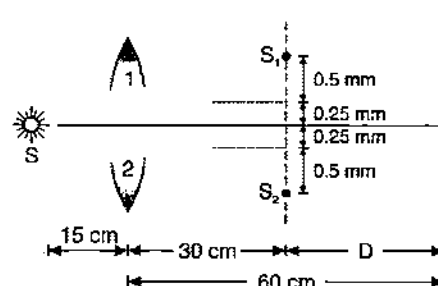
Solution

Applying lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} + \frac{1}{15} = \frac{1}{10}$$

$$\Rightarrow v = 30 \text{ cm}$$

$$\text{Since, } m = \frac{v}{u} = \frac{30}{-15} = -2$$



Distance between two slits is $d = 1.5 \text{ mm}$

Distance between slits and screen is $D = 30 \text{ cm}$

$$\text{Fringe width } \beta = \frac{\lambda D}{d} = \frac{(5 \times 10^{-7})(0.3)}{(1.5 \times 10^{-3})} = 10^{-4} \text{ m}$$

$$\beta = 0.1 \text{ mm}$$

INTERFERENCE EXPERIMENT IN WATER

In water (liquid), of refractive index μ the wavelength decreases from λ to $\lambda' = \frac{\lambda}{\mu}$. Therefore, if interference experiment is performed in water the fringe width decreases from β to β' , such that

$$\beta = \frac{\lambda D}{d} \text{ and } \beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d}$$

$$\Rightarrow \beta' = \frac{\beta}{\mu}$$

Illustration 14

A Young's double slit arrangement produces interference fringes for sodium light ($\lambda = 5890 \text{ \AA}$) that are 0.20° apart. What is the angular fringe separation if the entire arrangement is immersed in water? Refractive index of water is $\frac{4}{3}$.

Solution

The wavelength of light in water is $\lambda_w = \frac{\lambda}{\mu}$

Angular fringe-width in air, $\theta_a = \frac{\lambda}{d}$

Angular fringe-width in water, $\theta_w = \frac{\lambda_w}{d}$

$$\text{So, } \theta_w = \frac{\lambda_w}{d} = \frac{\lambda}{\mu d} = \frac{\theta_a}{\mu} = \frac{0.20^\circ}{\frac{4}{3}} = 0.15^\circ$$

FRINGE VISIBILITY (V)

With the help of the concept of visibility, the knowledge about coherence, fringe contrast and interference pattern is obtained. Fringe visibility V is defined as

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$$

If $I_{\min} = 0$, $V = 1$ (maximum) i.e., fringe visibility will be the best.

Also if $I_{\max} = 0$ then $V = -1$

and if $I_{\max} = I_{\min}$, then $V = 0$

INTENSITY DISTRIBUTION

When two coherent light waves of intensity I_1 and I_2 with a constant phase difference ϕ superimpose, then the resultant intensity is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

In YDSE, usually the intensities I_1 and I_2 are equal, so $I_1 = I_2 = I_0$

Since, $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$, so we get

$$I = 2I_0(1 + \cos \phi)$$

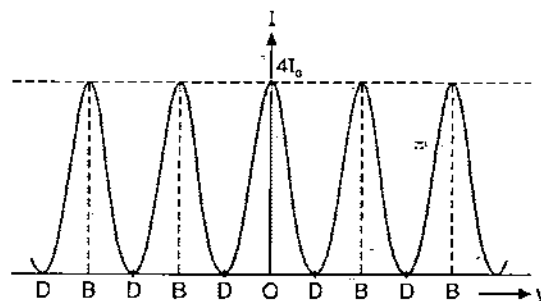
$$\Rightarrow I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

For maxima, $\phi = 2n\pi$

$$\Rightarrow I_{\max} = 4I_0$$

For minima, $\phi = (2n+1)\pi$

$$\Rightarrow I_{\min} = 0$$



Intensity distribution on the screen as a function of y in YDSE
 $I_{\max} = 4I_0$ for bright fringe and $I_{\min} = 0$ for dark fringe.

PROBLEM SOLVING TRICK(S)

a) Interference occurs due to Law of Conservation of Energy. Actually, redistribution of energy takes place.

b) If w_1 and w_2 are the widths of the slits and I_1 and I_2 is the intensity of light (with respective amplitudes a_1 and a_2) passing through slits, then

$$\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \frac{w_1}{w_2}$$

$$c) \frac{I_{\min}}{I_{\max}} = \frac{(a_1 - a_2)^2}{(a_1 + a_2)^2} = \left(\frac{\sqrt{w_1} - \sqrt{w_2}}{\sqrt{w_1} + \sqrt{w_2}} \right)^2 = \left(\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}} \right)^2$$

d) If point source is used to illuminate the two slits, the intensity emerging from the slit is proportional to area of exposed part of slit. In case of identical slits,

$$I_1 = I_2$$

$$\Rightarrow a_1 = a_2$$

e) When white light is used to illuminate the slit, we obtain an interference pattern consisting of a central white fringe having on both sides symmetrically a few coloured fringes and then a uniform illumination.

f) If ϕ is the phase difference between two waves of intensities I_1 and I_2 , then

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

g) If x is the path difference, then

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\left(\frac{2\pi x}{\lambda}\right)$$

h) In YDSE, if n_1 fringes are visible in a field of view with light of wavelength λ_1 , while n_2 with light of wavelength λ_2 in the same field, then $n_1 \lambda_1 = n_2 \lambda_2$.

i) Separation (Δx) between fringes

(a) Between n^{th} bright and m^{th} bright fringes ($n > m$)
 $\Delta x = (n - m)\beta$

(b) Between n^{th} bright and m^{th} dark fringe

$$(i) \text{ If } n > m \text{ then } \Delta x = \left(n - m + \frac{1}{2}\right)\beta$$

$$(ii) \text{ If } n < m \text{ then } \Delta x = \left(m - n - \frac{1}{2}\right)\beta$$

j) Identification of central bright fringe : To identify central bright fringe, monochromatic light is replaced by white light.

Due to overlapping central maxima will be white with red edges. On the other side of it we shall get a few coloured band and then uniform illumination.

Illustration 15

In YDSE, light of wavelength 60 nm is used. The separation between the sources is 6 mm and between the sources and the screen is 2 m. Find the positions of a point lying between third maxima and third minima where the intensity is three-fourth of the maximum intensity on the screen.

Solution

$$\text{Since, } I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

$$\text{where } I = \frac{3}{4}(4I_0) = 3I_0$$

$$\Rightarrow \cos\left(\frac{\phi}{2}\right) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{\phi}{2} = n\pi \pm \frac{\pi}{6}$$

$$\Rightarrow \phi = 2n\pi \pm \frac{\pi}{3}$$

$$\text{Since, } \phi = \frac{2\pi}{\lambda} \Delta x \text{ where } \Delta x = \frac{y_n d}{D}$$

$$\Rightarrow \frac{2\pi y_n d}{\lambda D} = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow y_n = \left(n \pm \frac{1}{6}\right) \frac{\lambda D}{d}$$

For the point lying between third minima and third maxima, we have

$$n = 3$$

$$\Rightarrow y_3 = \left(3 - \frac{1}{6}\right) \frac{\lambda D}{d}$$

$$\Rightarrow y_3 = \frac{17}{6} \left(\frac{\lambda D}{d}\right)$$

Substituting $\lambda = 0.6 \times 10^{-6} \text{ m}$, $D = 2 \text{ m}$, $d = 6 \text{ mm}$, we get

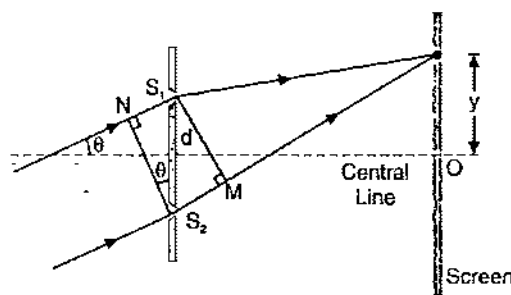
$$y_3 = \frac{17}{6} \frac{(0.6 \times 10^{-6})(2)}{6 \times 10^{-3}} = 5.67 \text{ mm}$$

YDSE WHEN INCIDENT RAYS ARE NOT PARALLEL TO CENTRAL LINE

In this case, the rays reaching S_1 and S_2 already have an initial path difference. So, net path difference between the rays reaching the point P is given by

$$\Delta x = (NS_1 + S_1P) - S_2P$$

$$\Rightarrow \Delta x = |NS_1 - (S_2P - S_1P)| = |NS_1 - MS_2|$$



Now, $NS_1 = d \sin \theta$ and

$$MS_2 = S_2P - S_1P = \frac{yd}{D} \text{ (as done earlier)}$$

$$\Rightarrow \Delta x = \left| d \sin \theta - \frac{yd}{D} \right|$$

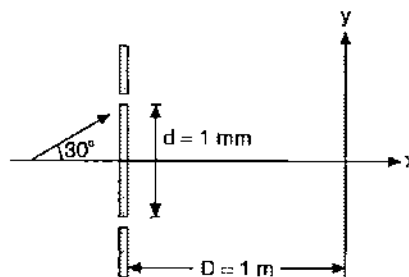
Once the path difference Δx is known, then

$$\text{FOR MAXIMA, } \Delta x = (2n) \frac{\lambda}{2}, n = 0, 1, 2, 3, \dots$$

$$\text{FOR MINIMA, } \Delta x = (2n+1) \frac{\lambda}{2}, n = 0, 1, 2, 3, \dots$$

Illustration 16

A coherent parallel beam of microwaves of wavelength $\lambda = 0.5 \text{ mm}$ falls on a Young's double slit apparatus. The separation between the slits is 1.0 mm. The intensity of microwaves is measured on a screen placed parallel to the plane of the slits at a distance of 1.0 m from it as shown in the figure.



- If the incident beam falls normally on the double slit apparatus, find the y-coordinates of all the interference minima on the screen.
- If the incident beam makes an angle of 30° with the x-axis (as in the dotted arrow shown in figure), find the y-coordinates of the first minima on either side of the central maximum.

Solution

- Given $\lambda = 0.5 \text{ mm}$, $d = 1 \text{ mm}$, $D = 1 \text{ m}$

When the incident beam falls normally

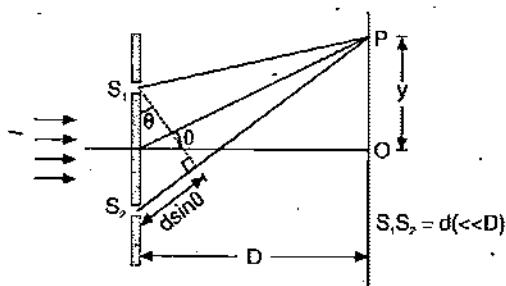
Path difference between the two rays S_2P and S_1P is

$$\Delta x = S_2P - S_1P \approx d \sin \theta$$

For minimum intensity,

$$d \sin \theta = (2n-1) \frac{\lambda}{2}, n = 1, 2, 3, \dots$$

$$\Rightarrow \sin \theta = \frac{(2n-1)\lambda}{2d} = \frac{(2n-1)0.5}{2 \times 1} = \frac{2n-1}{4}$$



Since, $\sin \theta \leq 1$

$$\Rightarrow \frac{(2n-1)}{4} \leq 1$$

$$\Rightarrow n \leq 2.5$$

So, n can be either 1 or 2

When $n=1$, we have $\sin \theta_1 = \frac{1}{4}$

$$\Rightarrow \tan \theta_1 = \frac{1}{\sqrt{15}}$$

When, $n=2$, we have $\sin \theta_2 = \frac{3}{4}$

$$\Rightarrow \tan \theta_2 = \frac{3}{\sqrt{7}}$$

Since, $y = D \tan \theta = \tan \theta (D=1 \text{ m})$

So, the position of minima will be

$$y_1 = D \tan \theta_1 = \frac{1}{\sqrt{15}} \text{ m}$$

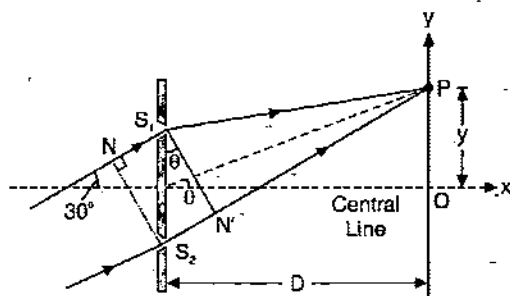
$$y_2 = D \tan \theta_2 = \frac{3}{\sqrt{7}} \text{ m}$$

Since, minima can be on either side of centre O , so there will be four minima at positions $\pm \frac{1}{\sqrt{15}} \text{ m}$ and

$\pm \frac{3}{\sqrt{7}} \text{ m}$ on the screen.

- (b) Path difference between the rays before entering the slits S_1 and S_2 is

$$NS_1 = \Delta x_1 = d \sin(30^\circ) = \frac{d}{2}$$



Path difference between the rays after passing through the slits S_1 and S_2 is

$$S_2P - S_1P = \Delta x_2 = d \sin \theta$$

So, net path difference is given by

$$\Delta x = |NS_1 + S_1P - S_2P| = |NS_1 - (S_2P - S_1P)|$$

$$\Rightarrow \Delta x = |\Delta x_1 - \Delta x_2| = \left| \frac{d}{2} - d \sin \theta \right|$$

For first minima, we have

$$\Delta x = \frac{\lambda}{2}$$

$$\Rightarrow \left| \frac{d}{2} - d \sin \theta \right| = \frac{\lambda}{2}$$

$$\Rightarrow \frac{d}{2} - d \sin \theta = \pm \frac{\lambda}{2}$$

$$\Rightarrow \frac{1}{2} - \sin \theta = \pm \frac{\lambda}{2d}$$

$$\Rightarrow \sin \theta = \frac{1}{2} \mp \frac{\lambda}{2d}$$

Since $d = 1 \text{ mm}$, $\lambda = 0.5 \text{ mm}$

$$\Rightarrow \sin \theta = \frac{1}{2} \mp \frac{0.5}{2}$$

$$\Rightarrow \sin \theta = \frac{1}{2} \mp \frac{1}{4}$$

$$\Rightarrow \sin \theta = \frac{1}{4} \text{ and } \sin \theta = \frac{3}{4}$$

Since the position of first minima on either side of central maxima is

$$\tan \theta = \frac{y}{D}$$

$$\Rightarrow y = D \tan \theta$$

For $\sin \theta = \frac{1}{4}$, we have $\tan \theta = \frac{1}{\sqrt{15}}$

$$\Rightarrow y = \frac{1}{\sqrt{15}} \text{ m}$$

For $\sin \theta = \frac{3}{4}$, we have $\tan \theta = \frac{3}{\sqrt{7}}$

$$\dots(1) \Rightarrow y = \frac{3}{\sqrt{7}} \text{ m}$$

YDSE FOR SOURCE NOT PLACED AT THE CENTRAL LINE

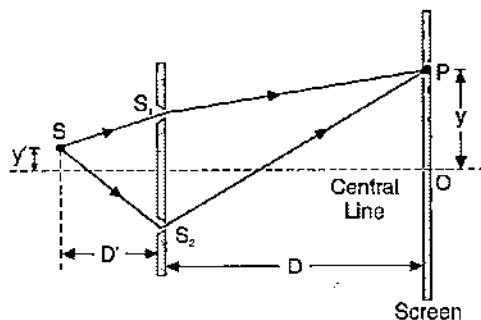
If the source of light is not placed at the central line but a little beyond (slightly up or down) the central line, then the waves reaching S_1 and S_2 will already have an initial path difference.

Path difference of waves meeting at the point P is

$$\Delta x = (SS_2 + S_2P) - (SS_1 + S_1P)$$

$$\Rightarrow \Delta x = (SS_2 - SS_1) + (S_2P - S_1P)$$

$$\Rightarrow \Delta x = \frac{y'd}{D'} + \frac{y'd}{D}$$



where, y' is the distance of the source S above or below the central line and D' is the distance of S from S_1 and S_2 .

Similarly, here once we calculated the path difference, then

FOR MAXIMA, $\Delta x = (2n)\frac{\lambda}{2}$, $n = 0, 1, 2, 3, \dots$

FOR MINIMA, $\Delta x = (2n+1)\frac{\lambda}{2}$, $n = 0, 1, 2, 3, \dots$

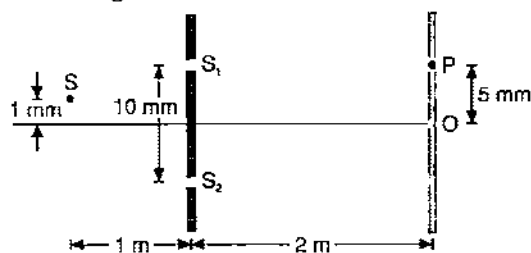
PROBLEM SOLVING TRICK(S)

At the position of central maxima, we have

$$\Delta x = 0$$

Illustration 17

In the Young's Double Slit experiment a point source of $\lambda = 5000 \text{ \AA}$ is placed slightly off the central axis as shown in the figure.



- Find the nature and order of the interference at the point P .
- Find the nature and order of the interference at O .
- Where should we place a film of refractive index $\mu = 1.5$ and what should be its thickness so that a maxima of zero order is placed at O .

Solution

- The optical path difference between the two waves arriving at P is

$$\Delta x = \frac{y_1 d}{D_1} + \frac{y_2 d}{D_2} = \frac{(1)(10)}{10^3} + \frac{(5)(10)}{2 \times 10^3}$$

$$\Rightarrow \Delta x = 3.5 \times 10^{-2} \text{ mm} = 0.035 \text{ mm}$$

To calculate the order of interference, we shall calculate

$$n = \frac{\Delta x}{\lambda}$$

$$\Rightarrow n = \frac{0.035 \times 10^{-3} \text{ m}}{5000 \times 10^{-10} \text{ m}}$$

$$\Rightarrow n = 70$$

$$\Rightarrow \Delta x = 70\lambda$$

So, 70th order maxima is obtained at P .

$$(b) \text{ At } O, \Delta x = \frac{y_1 d}{D_1} = 10^{-2} \text{ mm} = 0.01 \text{ mm}$$

Now, we observe that $\Delta x = 20\lambda$

So, 20th order maxima is obtained at O .

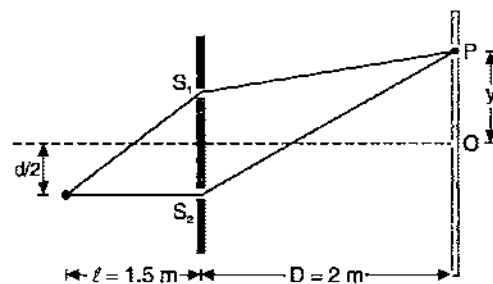
$$(c) (\mu - 1)t = 0.01 \text{ mm}$$

$$\Rightarrow t = \frac{0.01}{1.5 - 1} = 0.02 \text{ mm} = 20 \mu\text{m}$$

Since the pattern has to be shifted upwards, therefore, the film must be placed in front of S_1 .

Illustration 18

In the YDSE the monochromatic source of wavelength λ is placed at a distance $\frac{d}{2}$ from the central axis (as shown in the figure), where d is the separation between the two slits S_1 and S_2 .



- Find the position of the central maxima.
- Find the order of interference formed at O .
- Find the minimum thickness of the film of refractive index $\mu = 1.5$ to be placed in front of S_2 so that intensity at O becomes $\frac{3}{4}$ th of the maximum intensity.

Given $\lambda = 6000 \text{ \AA}$ and $d = 6 \text{ mm}$.

Solution

$$(a) (\Delta x)_{\text{net}} = 0$$

$$\Rightarrow \frac{y_1 d}{D_1} = \frac{y_2 d}{D_2}$$

$$\Rightarrow \frac{d/2}{1.5} = \frac{y}{2}$$

$$\Rightarrow y = \frac{d}{1.5} = \frac{6}{1.5} = 4 \text{ mm}$$

(b) At O , net path difference is given by

$$\Delta x = \frac{y_1 d}{D_1} = \frac{\left(\frac{d}{2}\right)(d)}{D_1} = \frac{(6 \times 10^{-3})^2}{2 \times 1.5}$$

$$\Rightarrow \Delta x = 12 \times 10^{-6} \text{ m}$$

$$\Rightarrow \Delta x = 120 \times 10^{-7} \text{ m}$$

$$\text{Since, } \lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$$

$$\Rightarrow \Delta x = 20\lambda$$

So at O , the bright fringe of order 20 will be obtained.

$$(c) I = I_{\max} \cos^2\left(\frac{\phi}{2}\right)$$

$$\Rightarrow \frac{3}{4} I_{\max} = I_{\max} \cos^2\left(\frac{\phi}{2}\right)$$

$$\Rightarrow \frac{\phi}{2} = \frac{\pi}{6}$$

$$\Rightarrow \phi = \frac{\pi}{3} = \left(\frac{2\pi}{\lambda}\right)(\mu - 1)t$$

$$\Rightarrow t = \frac{\lambda}{6(\mu - 1)} = \frac{6000}{6(1.5 - 1)} = 2000 \text{ \AA}$$

MISSING WAVELENGTH IN FRONT OF ONE SLIT IN YDSE

Suppose P is a point of observation in front of slit S_1 as shown. The wavelengths missing are the ones obtained by using the condition of destructive interference, i.e.,

$$\Delta x = (2n + 1) \frac{\lambda}{2}, \text{ where } n = 0, 1, 2, 3, \dots$$

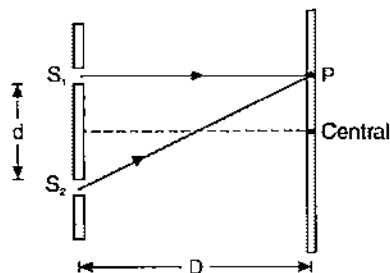
$$\text{Now } \Delta x = \sqrt{D^2 + d^2} - D$$

$$\Rightarrow \Delta x = D \left[\left(1 + \frac{d^2}{D^2} \right)^{\frac{1}{2}} - 1 \right] \quad \dots(1)$$

$$\text{Since, } D \gg d, \text{ so } \left(1 + \frac{d^2}{D^2} \right)^{\frac{1}{2}} \cong 1 + \frac{d^2}{2D^2}$$

So, (1) becomes

$$\Delta x = \frac{d^2}{2D} = (2n + 1) \frac{\lambda}{2}$$



$$\Rightarrow \lambda_{\text{missing}} = \frac{d^2}{(2n + 1)D}, \text{ where } n = 0, 1, 2, 3, \dots$$

By putting $n = 0, 1, 2, 3, \dots$ Missing wavelengths are

$$\lambda = \frac{d^2}{D}, \frac{d^2}{3D}, \frac{d^2}{5D}, \dots$$

Illustration 19

White light is used in a YDSE with separation between the sources to be 0.9 m and separation between the sources and the screen to be 1 m. Light reaching the screen at position $y = 1 \text{ mm}$ is passed through a prism and its spectrum is obtained. Find the missing lines in the visible region of this spectrum.

Solution

Path difference is given by

$$\Delta x = \frac{yd}{D} = (9 \times 10^{-4})(1 \times 10^{-3}) = 900 \text{ nm}$$

$$\text{For minima } \Delta x = (2n + 1) \frac{\lambda}{2}, \text{ where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \lambda = \frac{2\Delta x}{(2n + 1)} = \frac{1800}{(2n + 1)}$$

$$\Rightarrow \lambda_{\text{missing}} = \frac{1800}{1}, \frac{1800}{3}, \frac{1800}{5}, \frac{1800}{7}, \dots$$

Of these, 600 nm and 360 nm lie in the visible range, so these will be missing lines in the visible spectrum.

SUSTAINED INTERFERENCE

The interference pattern in which the positions of maxima and minima remain fixed is called a **sustained interference**.

CONDITIONS FOR SUSTAINED INTERFERENCE

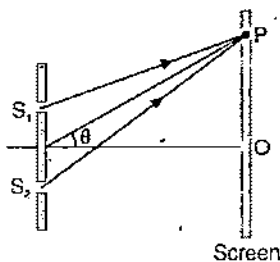
- The fundamental condition for sustained interference is that the two sources should be coherent i.e., the initial phase difference between the two interfering waves must remain constant with time.
- The amplitudes of the two waves should be equal or nearly equal. This will give good contrast between bright and dark fringes.
- The two sources should be very closely spaced, otherwise the fringes will be too close for the eye to resolve.
- The sources should be monochromatic, otherwise there will be overlapping of interference patterns due to different wavelengths, which will reduce the contrast.
- The frequencies of the two interfering waves must be equal.
- The sources should be narrow.

Since two independent sources cannot be coherent, a sustained interference pattern can be obtained only if the two sources simultaneously and, therefore, the phase difference between them remains constant.

ORDER OF FRINGES

If the slits are vertical, as shown in figure, path difference is,

$$\Delta x = d \sin \theta$$



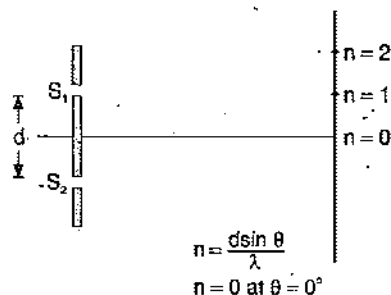
This path difference increases as θ increases.

The order of fringe n is given by

$$d \sin \theta = n\lambda$$

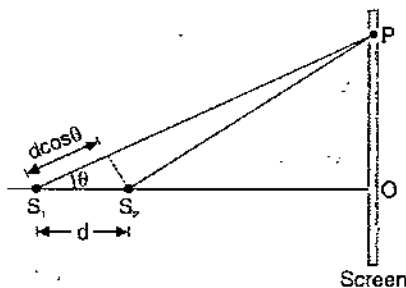
$$\Rightarrow n = \frac{d \sin \theta}{\lambda}$$

The order of fringe increases as we move away from point O on the screen.



When the slits are horizontal, as shown in figure, then the path difference is

$$\Delta x = d \cos \theta$$



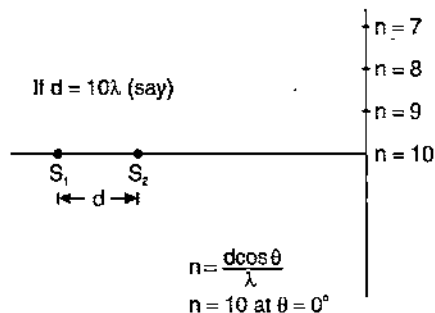
This path difference decreases as θ increases

The order of fringe n is given by

$$d \cos \theta = n\lambda$$

$$\Rightarrow n = \frac{d \cos \theta}{\lambda}$$

The order of fringe decreases as we move away from point O .



CONCEPTUAL NOTE(S)

To calculate the number of maxima or minima that can be obtained on the screen, we use the fact that value of $\sin \theta$ (or $\cos \theta$) can never be greater than 1. For example in the first case when the slits are vertical.

$$\sin \theta = \frac{n\lambda}{d} \quad \text{(for maximum intensity)}$$

Since, $\sin \theta \leq 1$

$$\Rightarrow \frac{n\lambda}{d} \leq 1 \Rightarrow n \leq \frac{d}{\lambda}$$

Suppose in some question $\frac{d}{\lambda}$ comes out say 3.6 then total number of maxima on the screen will be 7. Corresponding to $n = 0, \pm 1, \pm 2, \pm 3$.

So, highest order of Interference maxima is

$$n_{\max} = \left[\frac{d}{\lambda} \right]$$

where $[]$ represents the greatest integer function. So, total number of maxima obtained are

$$N = 2n_{\max} + 1$$

Similarly, highest order of interference minima is

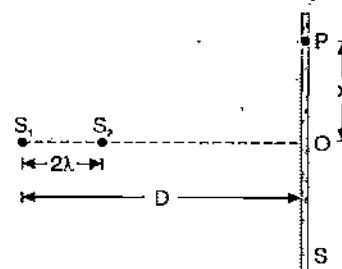
$$n_{\min} = \left[\frac{d}{\lambda} + \frac{1}{2} \right]$$

where $[]$ represents the greatest integer function. So, total number of minima obtained are

$$N = 2n_{\min}$$

Illustration 20

Two coherent narrow slits emitting light of wavelength λ in the same phase are placed parallel to each other at a small separation of 2λ . The light is collected on a screen S which is placed at a distance $D (\gg \lambda)$ from the slit S_1 as shown in figure. Find the finite distance x such that the intensity at P is equal to intensity at O .



Solution

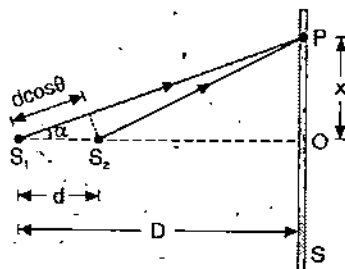
Path difference for waves reaching at O is

$$S_1O - S_2O = 2\lambda$$

i.e., maximum intensity is obtained at O.

Next maxima will be obtained at point P where,

$$S_1P - S_2P = \lambda$$



$$\Rightarrow d \cos \theta = \lambda$$

$$\Rightarrow (2\lambda) \cos \theta = \lambda$$

$$\Rightarrow \cos \theta = \frac{1}{2}$$

$$\Rightarrow \theta = 60^\circ$$

Now in ΔS_1PO

$$\frac{PO}{S_1O} = \tan \theta$$

$$\Rightarrow \frac{x}{D} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow x = \sqrt{3}D$$

CONCEPTUAL NOTE(S)

At point O, path difference is 2λ i.e., we obtain second order maxima. At point P, where path difference is λ (so, $x = \sqrt{3}D$), we get first order maxima. The next, i.e., zero order maxima will be obtained where path difference, $d \cos \theta = 0$

$$\Rightarrow \theta = 90^\circ$$

$$\Rightarrow x \rightarrow \infty$$

So, our answer, i.e., finite distance of x should be $x = \sqrt{3}D$, corresponding to first order maxima.

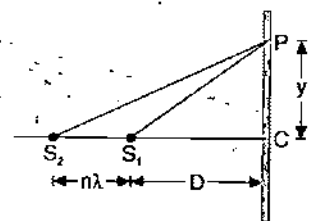
Illustration 21

Two point sources are $d = n\lambda$ apart. A screen is held at right angles to the line joining the two sources at a distance D from the nearest source. Calculate the distance of the point on the screen, where the first bright fringe (excluding the centre one) is observed. Assume $D \gg d$.

Solution

At point C path difference is $n\lambda$. Therefore, nth bright fringe will be observed. Next bright fringe is observed where path difference is $(n-1)\lambda$, so

$$S_2P - S_1P = (n-1)\lambda \quad \dots(1)$$



$$\text{Now, } S_1P = (D^2 + y^2)^{1/2} = D \left(1 + \frac{y^2}{D^2} \right)^{1/2} \approx D + \frac{y^2}{2D} \quad \dots(2)$$

$$\text{Similarly, } S_2P = (D + d) + \frac{y^2}{2(D + d)} \quad \dots(3)$$

Substituting (2) and (3) in (1) i.e., $S_2P - S_1P = (n-1)\lambda$, we get

$$d - \frac{y^2 d}{2D(D + d)} = (n-1)\lambda$$

Since, $d = n\lambda$

$$\Rightarrow n\lambda - \frac{y^2(n\lambda)}{2D(D + n\lambda)} = n\lambda - \lambda$$

$$\Rightarrow \frac{ny^2}{2D(D + n\lambda)} = 1$$

$$\Rightarrow y = \sqrt{\frac{2D(D + n\lambda)}{n}}$$

OPTICAL PATH

It is defined as distance travelled by light in vacuum in the same time in which it travels a given path length in a medium. If light travels a path length d in a medium at speed v , the time taken by it will be $t = \frac{d}{v}$. So, the optical path length is

$$OPL = ct = c \left(\frac{d}{v} \right) = \left(\frac{c}{v} \right) d = \mu d \quad \left\{ \because \mu = \frac{c}{v} \right\}$$

Since, for all media $\mu > 1$, optical path length is always greater than geometrical path length.

When two light waves arrive at a point by travelling different distances in different media, the phase difference between the two is related by their optical path difference, instead of simple path difference. So,

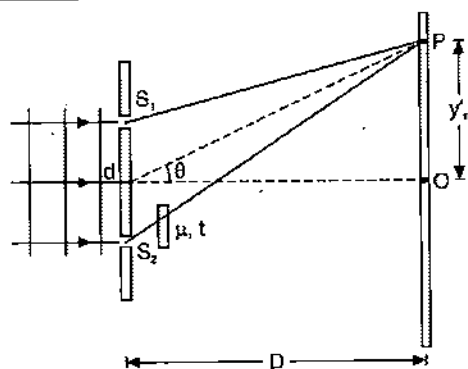
$$\text{Phase Difference} = \frac{2\pi}{\lambda} (OPL)$$

DISPLACEMENT OR SHIFTING OF FRINGE PATTERN IN YDSE

When a transparent film of thickness t and refractive index μ is introduced in front of one of the slits, the fringe pattern shifts in the direction where the film is placed.

How much is the fringe shift?

Consider the YDSE arrangement shown in the figure.



A film of thickness t and refractive index μ is placed in front of the lower slit.

The optical path difference is given by

$$x = [(S_2P - t) + \mu t] - S_1P$$

$$\Rightarrow x = (S_2P - S_1P) + t(\mu - 1)$$

$$\text{Since } S_2P - S_1P = d \sin \theta$$

$$\Rightarrow x = d \sin \theta + t(\mu - 1)$$

$$\text{Since } \sin \theta \approx \tan \theta = \frac{y'_n}{D}$$

$$\Rightarrow x = \frac{dy'_n}{D} + t(\mu - 1)$$

The maxima will be obtained when the path difference is an even multiple of $\frac{\lambda}{2}$ i.e.,

$$x = (2n) \frac{\lambda}{2}$$

$$\Rightarrow (2n) \frac{\lambda}{2} = \frac{dy'_n}{D} + t(\mu - 1)$$

$$\Rightarrow y'_n = \frac{n\lambda D}{d} - (\mu - 1) \frac{tD}{d}$$

In the absence of film, the position of the n th maxima is given by equation

$$y_n = \frac{n\lambda D}{d}$$

Therefore, the fringe shift (FS) is given by

$$FS = y_n - y'_n = \frac{D}{d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t \quad \left\{ \because \beta = \frac{\lambda D}{d} \right\}$$

Note that the shift is in the direction where the film is introduced.

CONCEPTUAL NOTE(S)

- The entire pattern shifts towards the side where the plate is introduced and there is no other change in the pattern.
- To measure this shift white light must be used because with monochromatic light all the fringes will exactly be similar and hence no shift can be observed.
- The effective path in air is increased by an amount $(\mu - 1)t$ due to introduction of the plate i.e., the additional path

difference is $(\mu - 1)t$.

d) If shift is equivalent to n fringes then $n = \frac{(\mu - 1)t}{\lambda}$ or

$$t = \frac{n\lambda}{(\mu - 1)}$$

e) The shift Δx is independent of the order of fringe n , i.e. Shift of zero order maxima = Shift of n th order maxima.

f) Shift is independent of wavelength.

Illustration 22

Interference fringes are produced by a double slit arrangement and a piece of plane parallel glass of refractive index 1.5 is interposed in one of the interfering beam. If the fringes are displaced through 30 fringe widths for light of wavelength 6×10^{-5} cm, find the thickness of the plate.

Solution

Path difference due to the introduction of glass slab is

$$\Delta x = (\mu - 1)t$$

Thirty fringes are displaced due to the introduction of slab. So,

$$\Delta x = 30\lambda$$

$$\Rightarrow (\mu - 1)t = 30\lambda$$

$$\Rightarrow t = \frac{30\lambda}{\mu - 1} = \frac{30 \times 6 \times 10^{-5}}{1.5 - 1}$$

$$\Rightarrow t = 3.6 \times 10^{-3} \text{ cm}$$

Illustration 23

In Young's double slit experiment using monochromatic light the fringe pattern shifts by a certain distance on the screen when a mica sheet of refractive index 1.6 and thickness 1.964 microns is introduced in the path of one of the interfering waves. The mica sheet is then removed and the distance between the slits and the screen is doubled. It is found that the distance between successive maxima (or minima) now is the same as the observed fringe shift upon the introduction of the mica sheet. Calculate the wavelength of the monochromatic light used in the experiment.

Solution

Shifting of fringes due to introduction of slab in the path of one of the slits is given by

$$\Delta y = \frac{(\mu - 1)tD}{d} \quad \dots(1)$$

Now, the distance between the screen and slits is doubled.

Hence, the new fringe width will become

$$\beta' = \frac{\lambda(2D)}{d} \quad \dots(2)$$

Given, $\Delta y = \beta'$

$$\Rightarrow \frac{(\mu-1)tD}{d} = \frac{\lambda(2D)}{d}$$

$$\Rightarrow \lambda = \frac{(\mu-1)t}{2} = \frac{(1.6-1)(1.964 \times 10^{-6})}{2}$$

$$\Rightarrow \lambda = 0.5892 \times 10^{-6} \text{ m} = 5892 \text{ \AA}$$

Illustration 24

In a YDSE, the two coherent sources are separated from each other by 6 mm and from the screen by 2 m. A light of wavelength 6000 Å is used. A film of refractive index 1.5 is introduced in front of the lower slit such that the third maxima shifts to the origin.

- Find the thickness of the film.
- Find the positions of the fourth maxima.

Solution

- Since third minima shifts to the origin, therefore, the fringe shift (FS) is equal to three fringe widths i.e., 3β , so we have

$$FS = y_3 = 3 \left(\frac{\lambda D}{d} \right)$$

Since we know that the fringe shift (FS) is given by

$$FS = (\mu-1) \frac{tD}{d}$$

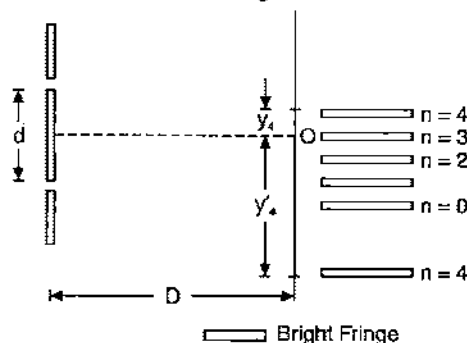
$$\Rightarrow (\mu-1) \frac{tD}{d} = 3 \frac{\lambda D}{d}$$

$$\Rightarrow t = \frac{3\lambda}{\mu-1}$$

Since, $\lambda = 0.6 \times 10^{-6} \text{ m}$, $\mu = 1.5$

$$\Rightarrow t = \frac{(3)(0.6 \times 10^{-6})}{1.5-1} = 3.6 \text{ } \mu\text{m}$$

- There are two positions of fourth maxima, one above and the other below the origin. So, we have



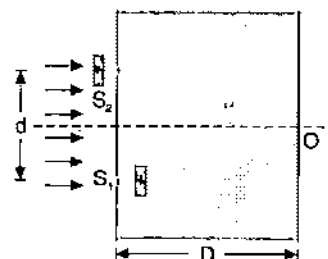
$$y_4 = 1\beta = \frac{\lambda D}{d} = 0.2 \text{ mm and}$$

$$y_4' = -7\beta = -7 \left(\frac{\lambda D}{d} \right) = -1.4 \text{ mm}$$

Illustration 25

A Young double slit apparatus is immersed in a liquid of refractive index μ_1 . The slit plane touches the liquid surface. A parallel beam of monochromatic light of wavelength λ (in air) is incident normally on the slits.

- Find the fringe width
- If one of the slits (say S_2) is covered by a transparent slab of refractive index μ_2 and thickness t as shown, find the new position of central maxima.
- Now the other slit S_1 is also covered by a slab of same thickness and refractive index μ_3 as shown in figure due to which the central maxima recovers its position find the value of μ_3 .



- Find the ratio of intensities at O in the three conditions (a), (b) and (c).

Solution

- Fringe width is given by

$$\beta = \frac{\lambda'D}{d} = \frac{\lambda D}{\mu_1 d} \quad \left\{ \because \lambda' = \frac{\lambda}{\mu} \right\}$$

- Position of central maximum is shifted upwards by a distance

$$\Delta y = \frac{(\mu_2-1)tD}{d} \quad \dots(1)$$

- Downward shift is now given by

$$\Delta y' = \frac{\left(\frac{\mu_3}{\mu_1} - 1 \right) tD}{d} \quad \dots(2)$$

Since the central maxima recovers its position, so

$$\Delta y = \Delta y'$$

So, from (1) and (2), we get

$$\frac{(\mu_2-1)tD}{d} = \frac{\left(\frac{\mu_3}{\mu_1} - 1 \right) tD}{d}$$

$$\Rightarrow \frac{\mu_2}{\mu_1} = \mu_3$$

$$\Rightarrow \mu_3 = \mu_1 \mu_2$$

- Since, $I = I_{\max} \cos^2 \left(\frac{\phi}{2} \right)$

$$\text{where } \phi = \left(\frac{2\pi}{\lambda} \right) \Delta x$$

$$\Rightarrow \frac{\phi}{2} = \left(\frac{\pi}{\lambda}\right) \Delta x$$

$$\Rightarrow I \propto \cos^2\left(\frac{\phi}{2}\right)$$

In the first and third case, $\Delta x = 0$ while in second case, $\Delta x = (\mu_2 - 1)t$. Therefore, the desired ratio is,

$$I_1 : I_2 : I_3 = 1 : \cos^2\left(\frac{\pi(\mu_2 - 1)t}{\lambda}\right) : 1$$

PROBLEM SOLVING TECHNIQUE(S)

If two thin plates are also inserted just after S_1 and S_2 , then our first task is to find the path difference. In the figure shown, path of ray 1 is more than path of ray 2 by a distance,

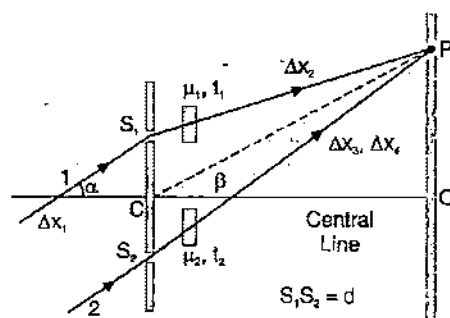
$$\Delta x_1 = d \sin \alpha \text{ and } \Delta x_2 = (\mu_1 - 1)t_1$$

and path of ray 2 is greater than path of ray 1 by a distance,

$$\Delta x_3 = d \sin \beta \text{ and } \Delta x_4 = (\mu_2 - 1)t_2$$

Therefore, net path difference is,

$$\Delta x = (\Delta x_1 + \Delta x_2) - (\Delta x_3 + \Delta x_4)$$



Once, we know the path difference Δx , then

$$\text{FOR MAXIMA, } \Delta x = (2n) \frac{\lambda}{2}, n = 0, 1, 2, \dots$$

$$\text{FOR MINIMA, } \Delta x = (2n + 1) \frac{\lambda}{2}, n = 0, 1, 2, \dots$$

Illustration 26

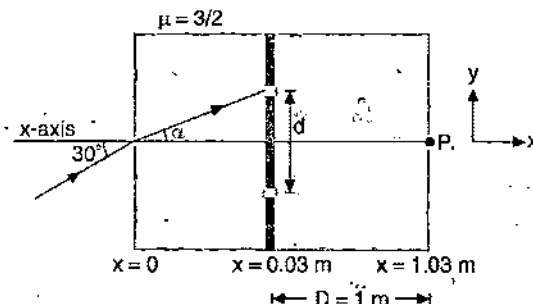
A large opaque sheet placed parallel to the yz plane at $x = 0.03 \text{ m}$. The region $x \geq 0$ is filled with a transparent liquid of refractive index $\frac{3}{2}$. A wide monochromatic beam of light of wavelength 900 nm falls on the yz -plane at $x = 0$ making an angle of 30° with the x -axis. The sheet has two slits parallel to z -axis at $y = \pm 0.9 \text{ mm}$. The intensity of the wave is measured on a screen placed at $x = 1.03 \text{ m}$ parallel to the sheet.

- Find the intensity at a point P on the screen where $y = z = 0$.
- The lower slit is covered by a transparent strip of refractive index 1.4 and thickness 4.2 mm . Now find the intensity at point P .

$$\tan(20^\circ) = \frac{1}{2\sqrt{2}}$$

Solution

- The situation in the problem is shown in the figure. So, we observe that separation between the sources is



$$d = 2 \times 0.9 = 1.8 \text{ mm}$$

$$\text{Since, we know that } \lambda' = \frac{\lambda}{\mu} = \frac{900}{\frac{3}{2}} = 600 \text{ nm}$$

Applying Snell's Law at the $x = 0$ interface, we get

$$\frac{3}{2} = \frac{\sin(30^\circ)}{\sin \alpha}$$

$$\Rightarrow \sin \alpha = \frac{1}{3}$$

$$\Rightarrow \alpha = 20^\circ$$

{Using Trigonometry}

Initial path difference is

$$\Delta x = d \sin \alpha = (1.8) \left(\frac{1}{3}\right) = 0.6 \text{ mm}$$

$$\Rightarrow \phi = \frac{2\pi}{\lambda'} \Delta x = \frac{2\pi}{600 \times 10^{-9}} \times 0.6 \times 10^{-3} = 2000\pi$$

$$\Rightarrow I = I_{\max} \cos^2\left(\frac{\phi}{2}\right) = I_{\max}$$

- Net path difference at P is now

$$\Delta x = (0.6 \text{ mm}) + \left(\frac{1.5}{1.4} - 1\right)(4.2 \text{ mm}) = 0.3 \text{ mm}$$

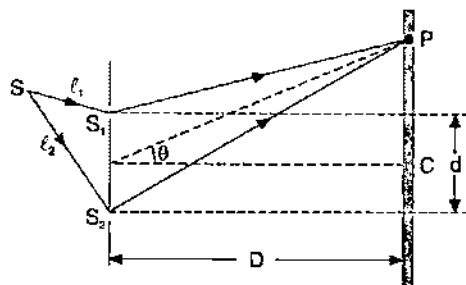
$$\Rightarrow \phi = 1000\pi$$

$$\Rightarrow I = I_{\max}$$

Illustration 27

In a Young's Double slit Experiment, the light source is at distance $l_1 = 20 \mu\text{m}$ and $l_2 = 40 \mu\text{m}$ from the slits. The light of wavelength $\lambda = 500 \text{ nm}$ is incident on slits separated at a distance $10 \mu\text{m}$. A screen is placed at a distance $D = 2 \text{ m}$ away from the slits as shown in figure.

- Find the values of θ relative to the central line where maxima appear on the screen?
- How many maxima will appear on the screen?
- What should be minimum thickness of a slab of refractive index 1.5 be placed on the path of one of the ray so that minima occurs at C ?



Solution

- (a) The optical path difference between the beams arriving at P is given by

$$\Delta x = (l_2 - l_1) + d \sin \theta$$

The condition for maximum intensity is,

$$\Delta x = n\lambda, \text{ where } n = 0, \pm 1, \pm 2, \dots$$

$$\Rightarrow \sin \theta = \frac{1}{d} (\Delta x - (l_2 - l_1)) = \frac{1}{d} (n\lambda - (l_2 - l_1))$$

$$\Rightarrow \sin \theta = \frac{1}{10 \times 10^{-6}} (n \times 500 \times 10^{-9} - 20 \times 10^{-6})$$

$$\Rightarrow \sin \theta = 2 \left(\frac{n}{40} - 1 \right)$$

$$\Rightarrow \sin \theta = \frac{n}{20} - 2$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{n}{20} - 2 \right)$$

- (b) Since we know that

$$|\sin \theta| \leq 1$$

$$\Rightarrow -1 \leq \left(\frac{n}{20} - 2 \right) \leq 1$$

$$\Rightarrow -20 \leq (n - 40) \leq 20$$

$$\Rightarrow 20 \leq n \leq 60$$

Hence, number of maxima obtained is

$$N = 60 - 20 = 40$$

- (c) At C, phase difference, $\phi = \frac{2\pi}{\lambda} \Delta x = \left(\frac{2\pi}{\lambda} \right) (l_2 - l_1)$

$$\Rightarrow \phi = \left(\frac{2\pi}{500 \times 10^{-9}} \right) (20 \times 10^{-6}) = 80\pi$$

Hence, maximum intensity will appear at C.

For minimum intensity at C, we have

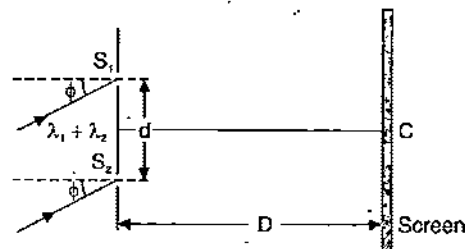
$$(\mu - 1)t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{\lambda}{2(\mu - 1)} = \frac{500 \times 10^{-9}}{2 \times 0.5} = 500 \text{ nm}$$

Illustration 28

In a Young's double slit experiment a parallel beam containing wavelengths $\lambda_1 = 4000 \text{ \AA}$ and $\lambda_2 = 5600 \text{ \AA}$

incident at an angle $\phi = 30^\circ$ on a diaphragm having narrow slits at a separation $d = 2 \text{ mm}$. The screen is placed at a distance $D = 40 \text{ cm}$ from slits. A mica slab of thickness $t = 5 \text{ mm}$ is placed in front of one of the slits and whole the apparatus is submerged in water. If the central bright fringe is observed at C, calculate



- (a) the refractive index of the slab.
(b) the distance of the first black line from C. Both wavelengths are in air. Take $\mu_w = \frac{4}{3}$.

Solution

- (a) To observe bright fringe at C, the mica slab should be placed in front of S_2 . In that case, net path difference at C is,

$$\Delta x = d \sin \phi - (\mu_r - 1)t \quad \left\{ \because \mu_r = \frac{\mu_{\text{slab}}}{\mu_w} \right\}$$

For central bright at C we have

$$\Delta x = 0$$

$$\Rightarrow d \sin \phi = (\mu_r - 1)t$$

$$\Rightarrow (\mu_r - 1) = \frac{d \sin \phi}{t} = \frac{(2 \times 10^{-3}) \sin(30^\circ)}{5 \times 10^{-3}} = 0.2$$

$$\Rightarrow \mu_r = 1.2$$

$$\Rightarrow \frac{\mu_{\text{slab}}}{\mu_w} = 1.2$$

$$\Rightarrow \mu_{\text{slab}} = 1.2 \mu_w = 1.2 \left(\frac{4}{3} \right) = 1.6$$

- (b) A black line is formed at the position where both the wavelengths interfere destructively. Distance of n th dark fringe from C is given by

$$y = \frac{(2n - 1)\lambda D}{2d}$$

$$\text{For black line, } \frac{(2n_1 - 1)\lambda'_1 D}{2d} = \frac{(2n_2 - 1)\lambda'_2 D}{2d}$$

where λ'_1 and λ'_2 are wavelengths in water.

$$\Rightarrow \frac{\lambda'_1}{\lambda'_2} = \frac{\mu_w}{\mu_w} = \frac{\lambda_1}{\lambda_2} = \frac{4000}{5600}$$

Substituting these values in equation (1), we get

$$\frac{2n_1 - 1}{2n_2 - 1} = \frac{7}{5}$$

For minimum value $n_1 = 4$ and $n_2 = 3$

Hence, distance of first black line is given by

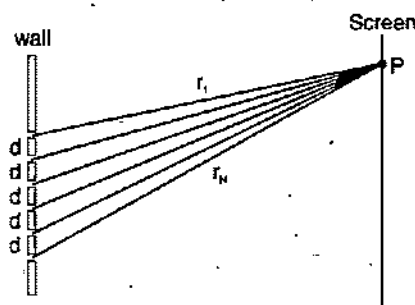
$$y = \frac{(2 \times 4 - 1)(4000 \times 10^{-10})40 \times 10^{-2} \times 3}{2 \times 2 \times 10^{-3} \times 4}$$

$$\Rightarrow y = 2.1 \times 10^{-1} \text{ m}$$

$$\Rightarrow y = 0.21 \text{ mm}$$

MULTIPLE SLIT INTERFERENCE PATTERN

Let us now look at the case where we have a general number N of equally spaced slits, instead of two equally spaced slits. As an assumption, we have shown in figure a set-up of six equally spaced slits.



Set up for 6 Equally Spaced Slits

Similar to the $N=2$ case discussed already, we will make the far-field assumption that the distance of the sources from the screen is much larger than the total span of the slits, which is $(N-1)d$. We can then say, as we did in the $N=2$ case, that all the paths to a given point P on the screen have approximately the same length in a multiplicative (but not additive) sense, which implies that the amplitudes of the interfering waves are all essentially equal and we can also say that all the paths are essentially parallel (because of far-field assumption). A close-up version near the slits is shown in figure. Also, each path length is $d \sin \theta$ longer than the one just above it. So the lengths take the form of $r_n = r_1 + (n-1)d \sin \theta$.

To find the total wave at a given point at an angle θ on the screen, we need to add up the N individual waves. The procedure is the same as in the $N=2$ case, except that now we simply have more terms in the sum. If a be the amplitude due to an individual source, then the equations of waves interfering at the point P' are given by

$$y_1 = a \sin(\omega t)$$

$$y_2 = a \sin(\omega t + \phi)$$

$$y_3 = a \sin(\omega t + 2\phi)$$

$$y_4 = a \sin(\omega t + 3\phi)$$

$\vdots$

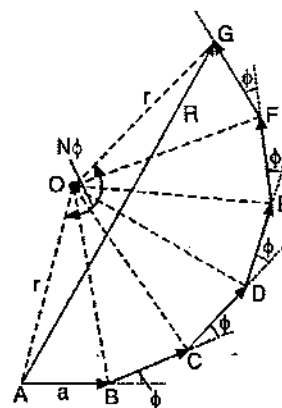
$$y_n = a \sin[\omega t + (N-1)\phi]$$

At angle θ , the path difference between any two successive slits is $\Delta x = d \sin \theta$. So, the corresponding phase difference ϕ is given by

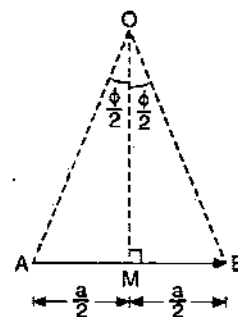
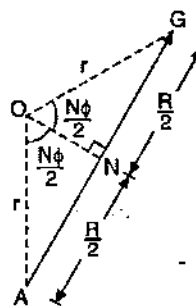
$$\phi = \left(\frac{2\pi}{\lambda} \right) \Delta x = \frac{2\pi}{\lambda} (d \sin \theta)$$

RESULTANT WAVE AMPLITUDE (USING PHASORS)

The above set of equations can be represented by phasor diagram shown in figure (for a set of six sources generalised to N sources).



If R be the amplitude of the resultant of N interfering waves, then from above phasor diagram, on extracting triangles OAG and OAB , we get following figures to be used for evaluation of R .



OAG and OAB are isosceles triangles so for triangle OAG , we have

$$\frac{R}{2} = a \sin\left(\frac{N\phi}{2}\right) \quad \dots(1)$$

Triangle OAB , we have

$$\frac{a}{2} = a \sin\left(\frac{\phi}{2}\right) \quad \dots(2)$$

Dividing (1) and (2), we get

$$\frac{R}{a} = \frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)}$$

So, the resultant amplitude R is given by

$$R = a \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right] \quad \dots(3)$$

If I_R be the resultant intensity, then

$$I_R = R^2$$

$$\Rightarrow I_R = a^2 \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right]^2$$

If I_0 be the intensity due to an individual source, then

$$I_R = I_0 \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right]^2 \quad \dots(4)$$

CHECK POINT

For $N=2$, we get

$$I_R = I_0 \left[\frac{\sin\phi}{\sin\left(\frac{\phi}{2}\right)} \right]^2$$

$$\text{Since } \sin\phi = 2\sin\left(\frac{\phi}{2}\right)\cos\left(\frac{\phi}{2}\right)$$

$$\Rightarrow I_R = 4I_0 \cos^2\left(\frac{\phi}{2}\right) = 4a^2 \cos^2\left(\frac{\phi}{2}\right)$$

RESULTANT WAVE EQUATION

To find the resultant wave equation, we shall be using the concept of complex numbers. From our knowledge of complex numbers, we know that

$$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$$

$$\Rightarrow \text{Im}(e^{i\omega t}) = \sin(\omega t) \text{ and } \text{Re}(e^{i\omega t}) = \cos(\omega t)$$

Since the resultant wave equation is obtained by adding individual waves, so we get

$$y = a[\sin(\omega t) + \sin(\omega t + \phi) + \dots + \sin(\omega t + (N-1)\phi)]$$

$$\Rightarrow y = a\text{Im}[e^{i\omega t} + e^{i(\omega t + \phi)} + \dots + e^{i(\omega t + (N-1)\phi)}]$$

$$\Rightarrow y = a\text{Im}[e^{i\omega t}(1 + e^{i\phi} + \dots + e^{i(N-1)\phi})]$$

$$\text{Since, } 1 + r + r^2 + \dots + r^{N-1} = 1 \left(\frac{1-r^N}{1-r} \right)$$

$$\Rightarrow y = a\text{Im}\left[e^{i\omega t} \left(\frac{1 - e^{iN\phi}}{1 - e^{i\phi}} \right)\right]$$

$$\Rightarrow y = a\text{Im}\left[e^{i\omega t} \left(\frac{1 - e^{i\left(\frac{N\phi}{2}\right)} e^{i\left(\frac{N\phi}{2}\right)}}{1 - e^{i\left(\frac{\phi}{2}\right)} e^{i\left(\frac{\phi}{2}\right)}} \right)\right]$$

$$\Rightarrow y = a\text{Im}\left[e^{i\omega t} \left(\frac{e^{i\left(\frac{N\phi}{2}\right)}}{e^{i\left(\frac{\phi}{2}\right)}} \left(\frac{e^{i\left(\frac{N\phi}{2}\right)} - e^{-i\left(\frac{N\phi}{2}\right)}}{e^{i\left(\frac{\phi}{2}\right)} - e^{-i\left(\frac{\phi}{2}\right)}} \right) \right]$$

$$\Rightarrow y = a\text{Im}\left[e^{i\left(\omega t + (N-1)\frac{\phi}{2}\right)} \left(\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right)\right]$$

$$\text{But } \text{Im}\left[e^{i\left(\omega t + (N-1)\frac{\phi}{2}\right)}\right] = \sin\left(\omega t + (N-1)\frac{\phi}{2}\right)$$

$$\Rightarrow y = a \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right] \sin\left(\omega t + (N-1)\frac{\phi}{2}\right)$$

Again here, we have directly come across the amplitude of the resultant wave given by

$$R = a \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right]$$

$$\Rightarrow I_R = I_0 \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right]^2, \text{ where } I_0 = a^2$$

We observe that at the centre of screen, I_R is indeterminate.

So, the maximum intensity at the midpoint of the screen i.e., at $\theta=0^\circ$ is obtained by taking the limit when $\phi \rightarrow 0^\circ$.

(Please note that since $\phi = \frac{2\pi}{\lambda}(d\sin\theta)$, so $\phi \rightarrow 0^\circ$ when $\theta \rightarrow 0^\circ$). So,

$$I_{\max} = \lim_{\phi \rightarrow 0} I_0 \frac{\sin^2\left(\frac{N\phi}{2}\right)}{\sin^2\left(\frac{\phi}{2}\right)}$$

For $\phi \rightarrow 0$, we have $\sin\phi \approx \phi$

$$\Rightarrow \sin^2\left(\frac{N\phi}{2}\right) \approx \frac{N^2\phi^2}{4} \text{ and } \sin^2\left(\frac{\phi}{2}\right) = \frac{\phi^2}{4}$$

$$\Rightarrow I_{\max} = I_0 \left[\frac{\left(\frac{N^2\phi^2}{4}\right)}{\left(\frac{\phi^2}{4}\right)} \right]$$

$$\Rightarrow I_{\max} = N^2 I_0$$

LOCATION OF SECONDARY MINIMA(S)

I_R has zero values, when

$$\sin^2\left(\frac{N\phi}{2}\right) = 0$$

$$\Rightarrow \frac{N\phi}{2} = \text{Integral Multiple of } \pi$$

$$\Rightarrow \frac{N\phi}{2} = m\pi, \text{ where } m = 0, 1, 2, 3, \dots$$

$$\Rightarrow \phi = (2m) \frac{\pi}{N}$$

$$\Rightarrow \phi = \text{Even Multiple of } \frac{\pi}{N} \quad \dots(5)$$

However, one exception to this is when $\frac{\phi}{2}$ is also an integral multiple of π , because the denominator in equation (4) is also zero.

$$\text{So, } \frac{\phi}{2} = m'\pi, \text{ where } m' = 0, 1, 2, 3, \dots$$

$$\Rightarrow \phi = (2m')\pi$$

$$\Rightarrow \phi = \text{Even multiple of } \pi \quad \dots(6)$$

So, from (5) and (6), we conclude that $I_R = 0$, when

$$\phi = (2m) \frac{\pi}{N} \text{ excluding } \phi = 0, 2\pi, 4\pi, 6\pi, \dots$$

Positions of Primary Maxima

i.e., $\phi = (\text{Even Multiple}) \frac{\pi}{N}$, excluding Positions of Primary Maxima (located at $0, 2\pi, 4\pi, 6\pi, \dots$)

LOCATION OF SECONDARY MAXIMA(S)

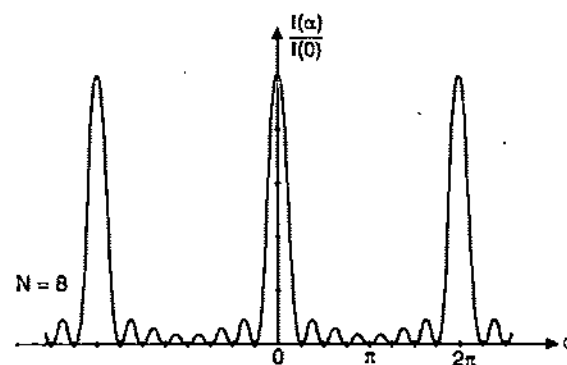
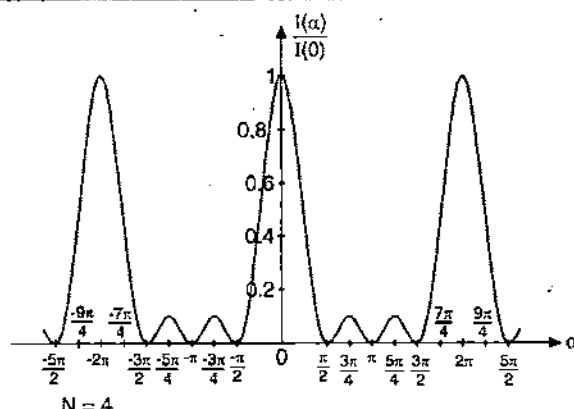
To find the locations of the secondary maxima (i.e., small bumps) we have to find the local maxima of I_R by taking the derivative I_R w.r.t. ϕ (i.e., $\frac{dI_R}{d\phi}$) and then equating it to zero.

So,

$$\frac{dI_R}{d\phi} = 0$$

$$\Rightarrow N \tan\left(\frac{\phi}{2}\right) = \tan\left(\frac{N\phi}{2}\right)$$

This equation has to be solved numerically. However, for large N , the solutions of ϕ are generally very close to odd multiples of $\frac{\pi}{N}$ excluding the values of $\phi = 2\pi \pm \frac{\pi}{N}$, because these values will be lying well within the primary maxima region. Just to make you understand, we are plotting $\frac{I_R}{I_0}$ (with ϕ for $N=4$ and $N=8$).



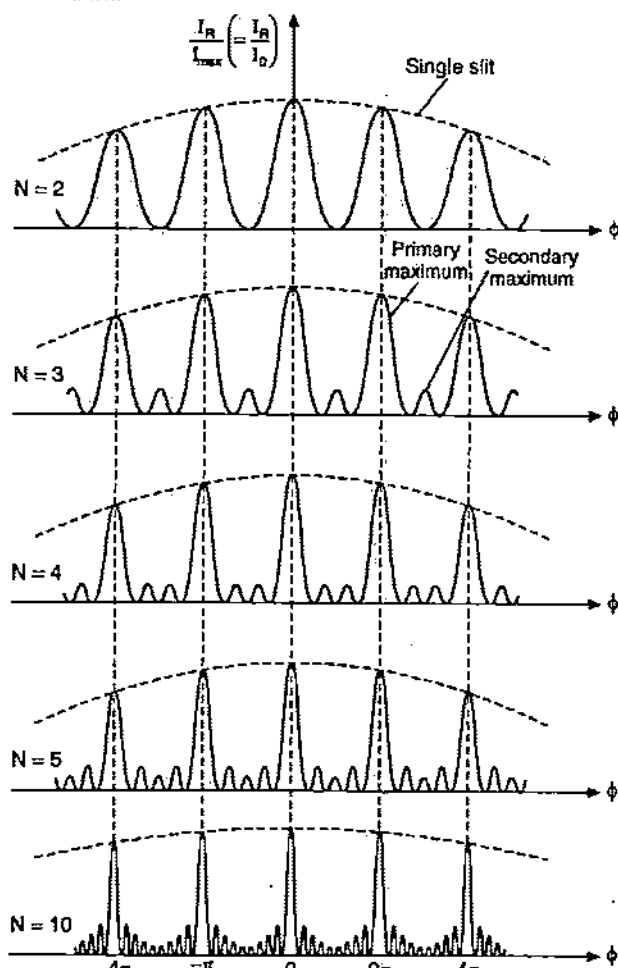
CONCEPTUAL NOTE(S)

- It is customary not to deal with the resultant intensity alone, but rather to deal with the resultant intensity relative to the maximum intensity i.e., $\frac{I_R}{I_{\max}}$

$$\Rightarrow \frac{I_R}{I_{\max}} = \frac{I_R}{I_0} = \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{N \sin\left(\frac{\phi}{2}\right)} \right]^2 \quad \dots(1)$$
- $\lim_{\phi \rightarrow 0} \left(\frac{I_R}{I_{\max}} \right) = \lim_{\phi \rightarrow 0} \left(\frac{I_R}{I_0} \right) = 1$
- $\frac{I_R}{I_{\max}}$ has a periodicity of 2π in ϕ i.e., repeats itself for integral multiples of 2π .
- The number of zeros between the main peaks is $(N-1)$, where N is the number of slits used.
- The number of secondary maxima (little bumps) between the main peaks is $(N-2)$, where N is the number of slits used.
 For $N=2$ (for two slits), we get ZERO Secondary Maxima.
 For $N=3$ (for 3 slits), we get One Secondary Maxima.
 For $N=4$ (for 4 slits), we get Two Secondary Maxima and so on.
- A point worth noting here is that the height of the secondary maxima (little bumps) i.e., the bump sizes are symmetric around $\phi = \pi$ (or in general any multiple of π). Also, we know that since

$$I_R = I_0 \left[\frac{\sin\left(\frac{N\phi}{2}\right)}{\sin\left(\frac{\phi}{2}\right)} \right]^2$$

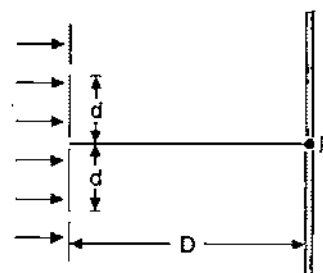
Hence, the bump size is shortest at $\phi = \pi$, because then the denominator in the equation (1), will be having a maximum value at $\phi = \pi$, due to which I_R becomes the least at $\phi = \pi$. Furthermore, the bump size grows as they get closer to the main peaks, as shown for various slits taken.



g) As N , the number of slits, is increased, the primary maxima (the tallest peaks in each graph) become narrower but remain fixed in position and the number of secondary maxima increases. For any value of N , the decrease in intensity in maxima to the left and right of the central maximum, indicated by the blue dashed arcs, is due to diffraction patterns from the individual slits, which can be neglected here.

Illustration 29

A light wave of wavelength 500 nm falls upon three slits a distance 0.5 mm each from one another. A screen is placed at a distance 2 m from slits. Find



- the distances from P where intensity reduces to zero.
- the distances from P where next bright fringe are observed.
- the ratio of intensities of bright fringes observed on the screen.

Solution

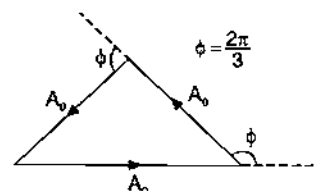
- In case of three slits, intensity becomes zero, when phase difference between any two waves is,

$$\phi = 2n\pi + \frac{2\pi}{3}, \text{ where } n = 0, 1, 2, \dots$$

The corresponding path difference, $\Delta x = \left(\frac{\lambda}{2\pi}\right)\phi$

$$\Rightarrow d \sin \theta = \left(\frac{\lambda}{2\pi}\right)\left(\frac{2\pi}{3} + 2n\pi\right) = n\lambda + \frac{\lambda}{3}$$

$$\Rightarrow \sin \theta = \frac{\lambda}{d}\left(n + \frac{1}{3}\right)$$



For small angles, $\sin \theta \approx \tan \theta = \frac{y}{D}$

$$\Rightarrow \frac{y}{D} = \frac{\lambda}{d}\left(n + \frac{1}{3}\right)$$

$$\Rightarrow y = \frac{\lambda D}{d}\left(n + \frac{1}{3}\right)$$

Substituting the values, we have

$$y = \frac{500 \times 10^{-9} \times 2}{0.5 \times 10^{-3}}\left(n + \frac{1}{3}\right) = 2 \times 10^{-3}\left(n + \frac{1}{3}\right) \text{ m}$$

$$\Rightarrow y = 2\left(n + \frac{1}{3}\right) \text{ mm, where } n = 0, 1, 2, \dots$$

$$\Rightarrow y = \frac{2}{3} \text{ mm for } n = 0, y = \frac{8}{3} \text{ mm for } n = 1 \text{ etc.}$$

- Bright fringes are obtained on the screen where

$$(i) \quad \phi = 2n\pi, n = 1, 2, 3, \dots$$

$$\Delta x = \left(\frac{\lambda}{2\pi}\right)(\phi) = n\lambda$$

$$\Rightarrow d \sin \theta = n\lambda$$

$$\Rightarrow \sin \theta = \frac{n\lambda}{d}$$

For small angles,

$$\sin \theta \approx \tan \theta = \frac{y}{D}$$

$$\Rightarrow \frac{y}{D} = \frac{n\lambda}{d}$$

$$\Rightarrow y = \frac{n\lambda D}{d} = \frac{n(500 \times 10^{-9})(2)}{(0.5 \times 10^{-3})}$$

$$\Rightarrow y = 2n \times 10^{-3} \text{ m}$$

$$\Rightarrow y = (2n) \text{ mm} \quad (\text{where } n = 1, 2, 3, \dots \text{ etc.})$$

These are called **primary** maximas.

$$(ii) \phi = (2n+1)\pi, \quad n = 1, 2, \dots$$

Proceeding in the similar manner, we get

$$y = 2\left(n + \frac{1}{2}\right) \text{ mm} \quad (\text{where } n = 1, 2, 3, \dots)$$

These are called **secondary** maximas.

Note that $y=0$ is also a secondary maxima because at P , $\phi = \pi$.

(c) At principal maximas, we have $\phi = 2\pi, 4\pi, \dots$, etc.

$$\begin{array}{l} \longrightarrow A_0 \\ \longrightarrow A_0 \Rightarrow \longrightarrow A = 3A_0 \\ \longrightarrow A_0 \end{array}$$

Resultant amplitude $R = 3A_0$

$$\Rightarrow I_R = 9I_0 \quad (\because I \propto A^2)$$

While at secondary maximas, ($\phi = \pi, 3\pi, 5\pi, \dots$)

$$\begin{array}{l} A_0 \longleftarrow \longrightarrow A_0 \\ \longrightarrow A_0 \Rightarrow \longrightarrow A = A_0 \end{array}$$

Resultant amplitude, $R' = A_0$

$$\Rightarrow I'_R = I_0$$

So, the desired ratio is therefore, 9:1

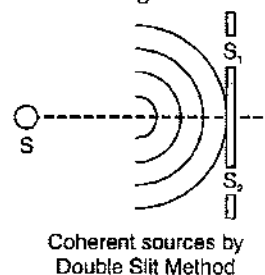
COHERENT SOURCES BY DIVISION OF WAVEFRONT

When two or more waves travel through a medium simultaneously, the resultant intensity at any point, in the medium depends on whether they interfere constructively or destructively which, in turn, depends upon the phase difference between them. Resultant intensity, at any point, remains constant with time if the phase difference between them does not change. Two independent sources can never have same phase or a constant phase difference, because if we try to have interference with two independent sources, then net intensity at any point undergoes a continuous change due to a change in the phase difference between them. As a result of this no fixed interference pattern can be observed. The interference pattern of such sources is so short-lived that its photograph with the fastest available

camera cannot be obtained. To obtain a fixed interference pattern we must have two sources which either have no phase difference or have a constant difference of phase. These sources are called **coherent sources**. It has been generally observed that coherent sources are obtained when they are derived from the same parent source. The methods for obtaining coherent sources (derived from the same parent source) are given below.

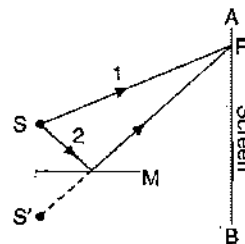
A. DOUBLE SLIT METHOD

Light from a source S is limited to a narrow beam with the help of a slit. The emergent light is made to fall upon a screen containing two slits S_1 and S_2 placed symmetrically with respect to the slit. Here, both S_1 and S_2 are illuminated by the same wavefront. Therefore the beams of light coming out from S_1 and S_2 have no phase difference. Thus S_1 and S_2 can be treated as the coherent sources. Young used this technique in his famous Young's double slit experiment.



B. A SOURCE AND ITS OWN VIRTUAL IMAGE

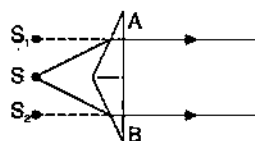
Light from a source S is made to fall on a plane mirror M . Point of observation P on a screen AB receives direct light as well as light reflected from M . To an observer, reflected light appears to come from a source S' (virtual image of S). So interference at P takes place between waves coming from S and S' . Since S' is not an independent source, being the virtual image of S , it will have the same phase as S . Hence the two are taken to be coherent sources. Lloyd made use of this arrangement in Lloyd single mirror experiment.



C. BIPRISM METHOD

Light from a source S is made to fall on an assembly of two right angled prisms A and B joined base to base as shown in Figure. S_1 and S_2 are the virtual image of S produced by refraction through prisms A and B respectively. Being

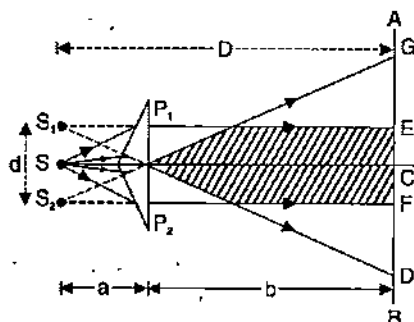
virtual images of the same source, S_1 and S_2 have same phase and hence can be treated as the coherent sources. This type of arrangement is made use of in Fresnel's biprism experiment.



FRESNEL'S BIPRISM

It is one of the convenient laboratory arrangements for producing interference fringes. It consists of a combination of two right angled prisms with their bases joined together so that their faces are inclined to each other at angle of $179^\circ 20'$. Source of light is taken in the form of a narrow slit S , illuminated by the monochromatic light and is held symmetrically at a distance of about 5 cm from the biprism.

Light from S gets refracted by prism P_1 and P_2 , thereby, producing virtual images S_1 and S_2 , which can be taken as two coherent sources producing interference. Light beams from S_1 and S_2 strike the screen in the regions ED and FG respectively. EF is the common region where both the beams can be found. Therefore interference pattern can be observed in the region EF .



The separation between these sources may be found by using the formula for deviation caused by a thin prism. If α is the small angle of biprism, μ refractive index of material of biprism and a the separation of source S from biprism, then deviation caused by prism.

$$\delta = (\mu - 1)\alpha$$

From Figure, $d = 2a\delta = 2a(\mu - 1)\alpha$

DETERMINATION OF λ

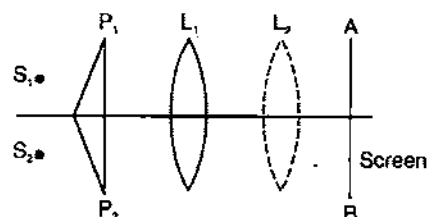
Biprism method can be used to determine the wave length of light. The fringe width β for the interference pattern obtained is given by,

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \lambda = \frac{\beta d}{D} \quad \dots(1)$$

- Determination of D** : It is the distance between source and screen. It can be measured with an ordinary metre rod.
- Determination of β** : A low power travelling microscope is used to find the total separation x between a number of fringes, say 20 and hence $\beta = \frac{x}{20}$.
- Determination of d** : d can be calculated by using displacement method. A convex lens is placed in between the biprism and the screen. It is observed that for two positions L_1 and L_2 of the lens, the images of S_1 and S_2 can be focussed on the screen AB . Let x and y be the distances between these images when the lens is at L_1 and L_2 respectively. Then,

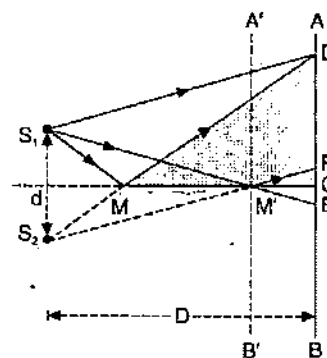
$$d = \sqrt{xy}$$



Substituting for β , D and d in equation (1), λ can be calculated.

LLOYD'S SINGLE MIRROR

This experimental set-up for producing interference fringes, was devised by Dr. Lloyd in 1834. Light from a source S_1 in the form of narrow slit is held in such a way that the light is incident, at almost grazing incidence, upon a mirror MM' which is blackened at the back to avoid internal reflections. S_2 is the virtual image of source S_1 obtained after reflection from MM' . Screen AB is placed to receive light coming directly from S_1 as well as that reflected from the mirror. Reflected light can be supposed to be coming from source S_2 . DF is the common region on the screen where both the beams are received and hence interference is obtained in region DF .



Experimental set up for Lloyd's single mirror

The point C lies symmetrically w.r.t. S_1 and S_2 and also lies outside the interference region, zero order fringe is not visible. It can be seen by moving the screen to position $A'B'$ so that it just touches the mirror. It will be observed that the zero order fringe at M' is dark instead of being bright as demanded by the theory of interference fringes since at M' path difference is zero. This indicates that the beam which suffers reflection from MM' undergoes in phase of π -radian.

Lloyd's single mirror can be used to determine the wave length of light.

If, a is the height of source S_1 above MM' , then

$$d = 2a$$

If, D is the distance of source S_1 from screen AB , then fringe width β is given by

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \lambda = \frac{\beta d}{D} = \frac{\beta(2a)}{D}$$

β can be determined, experimentally, by using a low power microscope. Knowing β , a and D value of λ can be calculated.

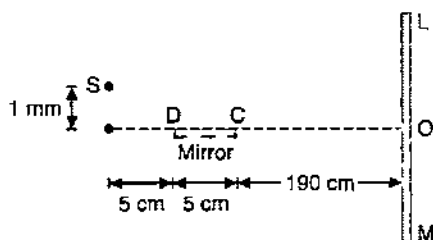
CONCEPTUAL NOTE(S)

Central spot, in case of Lloyd's single mirror is a dark one instead of being bright. This proves that there is a phase change of π -radian, when a transverse wave (light) is reflected from a denser medium.

Illustration 30

The arrangement for a mirror experiment is shown in the figure. S is a point source of frequency 6×10^{14} Hz. D and C represent the two ends of a mirror placed horizontally and LOM represents the screen.

Determine the position of the region where the fringes will be visible and calculate the number of fringes.

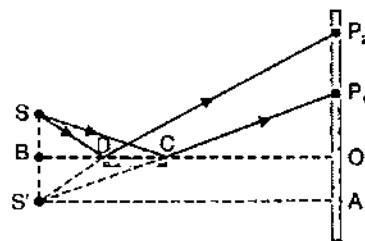


Solution

Fringes will be observed in the region between P_1 and P_2 because the reflected rays lie only in this region.

From similar triangles BDS' and $S'P_2A$, $\frac{AP_2}{BS'} = \frac{AS'}{BD}$

$$\Rightarrow AP_2 = \frac{(AS')(BS')}{BD} = \frac{(190+5+5)(0.1)}{5} = 4 \text{ cm}$$



Similarly in triangles BCS' and $S'P_1A$, we have

$$\frac{AP_1}{BS'} = \frac{AS'}{BC}$$

$$\Rightarrow AP_1 = \frac{(AS')(BS')}{BC} = \frac{(190+5+5)(0.1)}{10} = 2 \text{ cm}$$

$$\Rightarrow P_1P_2 = AP_2 - AP_1 = 2 \text{ cm}$$

Wavelength of the light $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^{14}} = 5 \times 10^{-7} \text{ m}$

$$\text{Fringe width } \beta = \frac{\lambda D}{d}$$

Since, $D = S'A = (190+5+5) = 200 \text{ cm} = 2 \text{ m}$,

$$d = SS' = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$$

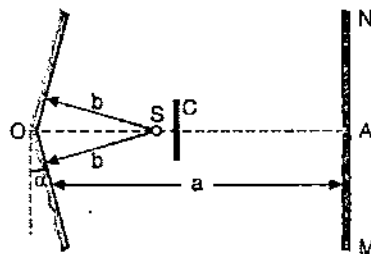
$$\Rightarrow \beta = \frac{(5 \times 10^{-7})(2)}{2 \times 10^{-3}} = 5 \times 10^{-4} \text{ m} = 0.05 \text{ cm}$$

Number of fringes is

$$N = \frac{P_1P_2}{\beta} = 40$$

Illustration 31

Two flat mirrors form an angle close to 180° . A source of light S is placed at equal distances b from the mirrors. Find the interval between adjacent interference bands on screen MN at a distance $OA = a$ from the point of intersection of the mirror. The wavelength of the light wave is known and equal to λ . Shield C does not allow the light to pass directly from the source to the screen.

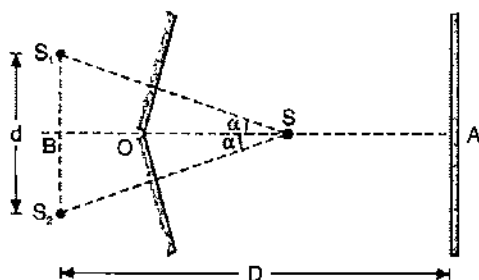


Solution

Fringe width is given by

$$\beta = \frac{\lambda D}{d}$$

where $D = AB \approx a + b$ and $d = S_1S_2$



In ΔS_1SB , we have

$$\frac{d}{2} = 2b \frac{\alpha}{2}$$

$$\Rightarrow d = 2b\alpha$$

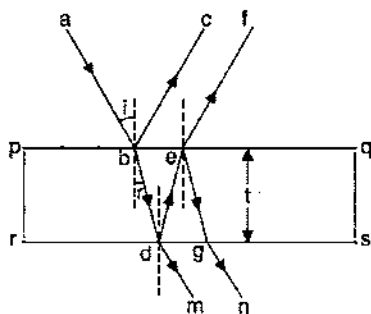
$$\Rightarrow \beta = \frac{\lambda(a+b)}{2b\alpha}$$

THEORY OF DIVISION OF AMPLITUDE

A. REFLECTED LIGHT

If μ is the refractive index of material of film of thickness t , then path difference between the waves abc and $abdef$ is $2\mu t \cos r$

Additional path difference due to reflection at denser medium (at b) is $\frac{\lambda}{2}$



So, effective path difference is

$$x = 2\mu t \cos r + \frac{\lambda}{2}$$

For **maxima** or constructive interference to take place, we have

$$2\mu t \cos r + \frac{\lambda}{2} = (2n) \frac{\lambda}{2}$$

$$\Rightarrow 2\mu t \cos r = (2n-1) \frac{\lambda}{2}, n = 1, 2, 3, \dots$$

For **minima** or destructive interference to take place

$$2\mu t \cos r + \frac{\lambda}{2} = (2n+1) \frac{\lambda}{2}$$

$$\Rightarrow 2\mu t \cos r = n\lambda, n = 1, 2, 3, \dots$$

B. TRANSMITTED LIGHT

In transmitted light system there is no phase difference or path difference due to reflection or transmission as all reflections take place from rarer medium.

So, the effective path difference is

$$x = 2\mu t \cos r$$

For maxima

$$2\mu t \cos r = n\lambda$$

and for minima

$$2\mu t \cos r = (2n-1) \frac{\lambda}{2}, n = 1, 2, 3, \dots$$

Obviously, the conditions of interference in reflected and transmitted lights are opposite to each other, therefore if the film appears dark in reflected light, it will appear bright in transmitted light and vice versa. With the use of white light, the colours visible in reflected light will be complementary to those visible in transmitted light, i.e., the colours absent in one system will be present in the other system; the sum of two constituting the white light.

Illustration 32

A thick glass slab ($\mu = 1.5$) is to be viewed in reflected white light. It is proposed to coat the slab with a thin layer of a material having refractive index 1.3 so that the wavelength $6000 \text{ \AA}$ is suppressed. Find the minimum thickness of the coating required.

Solution

Optical path difference for the reflected light from coating and slab is

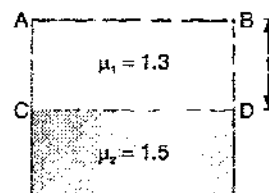
$$\Delta x = 2\mu t$$

For minimum intensity,

$$2\mu_1 t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{\lambda}{4\mu_1} = \frac{6000}{4 \times 1.3}$$

$$\Rightarrow t = 1154 \text{ \AA}$$



CONCEPTUAL NOTE(S)

Both reflected rays (one from AB and the another from CD) get a phase change of π .

Illustration 33

A parallel beam of white light falls on a thin film whose refractive index is equal to $\frac{4}{3}$. The angle of incidence $i = 53^\circ$. What must be the minimum film thickness if the reflected light is to be coloured yellow (λ of yellow = $0.6 \mu\text{m}$) most intensively? Given $\tan 53^\circ = \frac{4}{3}$.

Solution

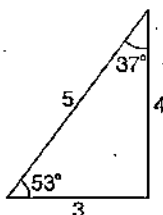
According to Snell's Law, we have

$$\mu = \frac{\sin i}{\sin r}$$

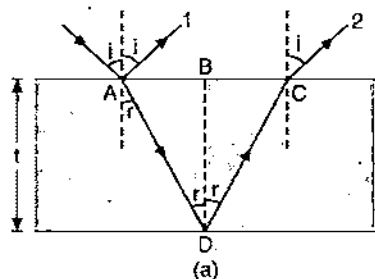
$$\Rightarrow \frac{4}{3} = \frac{\sin(53^\circ)}{\sin r} = \frac{4}{5}$$

$$\Rightarrow \sin r = \frac{3}{5}$$

$$\Rightarrow r = 37^\circ$$



From Figure (a) :

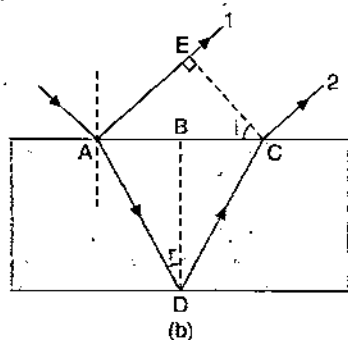


Path difference between 2 and 1 is $\Delta x_1 = 2(AD)$

$$\Rightarrow \Delta x_1 = 2BD \sec r = 2t \sec r$$

Their optical path corresponding to Δx_1 is $2\mu t \sec r$

From Figure (b) :



Path difference between 1 and 2 is given by

$$\Delta x_2 = AC \sin i = (2t \tan r) \sin i$$

$$\Rightarrow (\Delta x)_{\text{net}} = \Delta x_1 - \Delta x_2 = 2\mu t \sec r - 2t(\tan r)(\sin i)$$

$$\Rightarrow \Delta x_{\text{net}} = 2 \times \frac{4}{3} \times t \times \frac{5}{4} - 2 \times t \times \frac{3}{4} \times \frac{4}{5}$$

$$\Rightarrow \Delta x_{\text{net}} = \frac{32}{15} t$$

Since reflection takes place at the surface of denser medium, so phase difference between 1 and 2 is π .

So, for constructive interference, we have

$$\frac{32}{15} t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{15\lambda}{64} = \frac{15 \times 0.6}{64} = 0.14 \mu\text{m}$$

ICE-1

BASED ON INTERFERENCE

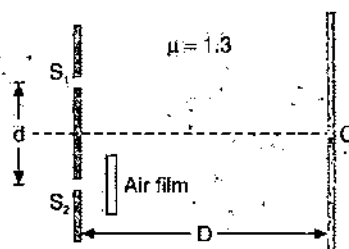
(Solutions on page 2.86)

- In a Young's Double Slit Experiment carried out in a liquid of refractive index $\mu = 1.3$, a thin film of air is formed in front of the lower slit as shown in the figure. If a maxima of third order is formed at the origin O, find the

- thickness of the air film,
- positions of the fourth maxima.

The wavelength of light in air is $\lambda_0 = 0.78 \mu\text{m}$ and

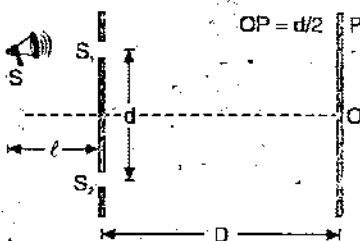
$$\frac{D}{d} = 1000.$$



- In YDSE, if light of wavelength $5000 \text{ \AA}$ is used, find the thickness of a glass slab ($\mu = 1.5$) which should be placed before the upper the upper slit S_1 so that the central maximum now lies at a point where 5^{th} bright fringe was lying earlier (before inserting the slab).

- A source S of wavelength λ is kept directly behind the slit S_1 in a double slit apparatus. Find the phase difference at a point O which is equidistant from S_1 and S_2 . If $D \gg d$, what will be the phase difference at P if a liquid of refractive index μ is filled

- between the screen and the slits?
- between the slits and the source S?



- In solar cells, a silicon solar cell ($\mu = 3.5$) is coated with a thin film of silicon monoxide SiO ($\mu = 1.45$) to minimize reflective losses from the surface. Determine the minimum thickness of SiO that produces the least reflection at a wavelength of 550 nm , near the centre of the visible spectrum.

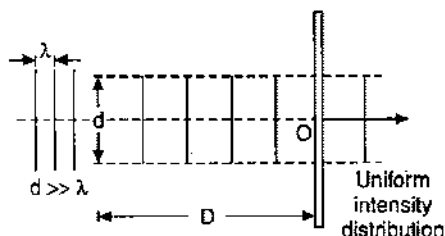
- A parallel beam of green light of wavelength 546 nm passes through a slit of width 0.4 mm . The transmitted light is collected on a screen 40 cm away. Find the distance between the two first order minima.

- Calculate the minimum thickness of a soap bubble film ($\mu = 1.33$) that results in constructive interference in the

- reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600 \text{ nm}$.
- Monochromatic light of wavelength $5000 \text{ \AA}$ is used in YDSE, with slit separation 1 mm , distance between screen and slits 1 m . If intensity at the two slits are, $I_1 = 4I_0$, $I_2 = I_0$, find
 - fringe width β .
 - distance of 5^{th} minima from the central maxima on the screen.
 - intensity at $y = \frac{1}{3} \text{ mm}$.
 - distance of the 1000^{th} maxima.
 - distance of the 5000^{th} maxima.
 - S_1 and S_2 are two point sources of radiation that are radiating waves in phase with each other of wavelength 400 nm . The sources are located on x -axis at $x = 6.5 \text{ }\mu\text{m}$ and $x = -6 \text{ }\mu\text{m}$, respectively.
 - Determine the phase difference (in radian) at the origin between the radiation from S_1 and the radiation from S_2 .
 - Suppose a slab of transparent material with thickness $1.5 \text{ }\mu\text{m}$ and index of refraction $\mu = 1.5$ is placed between $x = 0$ and $x = 1.5 \text{ }\mu\text{m}$. What then is the phase difference (in radian) at the origin between the radiation from S_1 and the radiation from S_2 ?
 - In a Young's double slit experiment using monochromatic light the fringe pattern shifts by a certain distance on the screen when a mica sheet of refractive index 1.6 and thickness 1.964 micron is introduced in the path of one of the interfering waves. The mica sheet is then removed and the distance between the slits and the screen is doubled. It is found that the distance between successive maxima (or minima) now is the same as the observed fringe shift upon the introduction of the mica sheet. Calculate the wavelength of the monochromatic light used in the experiment.
 - In Young's experiment a thin glass plate is placed in the path of one of the interfering rays. This causes the central light band to shift into a position which was initially occupied by the fifth bright band (not counting the central one). The ray falls onto the plate perpendicularly. The refractive index of the plate is 1.5 . The wavelength is $6 \times 10^{-7} \text{ m}$. What is the thickness of the plate?
 - In a double slit pattern ($\lambda = 6000 \text{ \AA}$), the first order and tenth order maxima fall at 12.50 mm and 14.75 mm from a particular reference point. If λ is changed to $5500 \text{ \AA}$, find the position of zero order and tenth order fringes, other arrangements remaining the same.
 - In YDSE, the two slits are separated by 0.1 mm and they are 0.5 m from the screen. The wavelength of light used is $5000 \text{ \AA}$. Find the distance between 7^{th} maxima and 11^{th} minima on the screen.
 - What is the effect on the interference fringes in a YDSE due to each of the following operations?
 - The screen is moved away from the plane of the slits
 - The (monochromatic) source is replaced by another (monochromatic) source of shorter wavelength
 - The separation between the two slits is increased
 - The monochromatic source is replaced by source of white light
 - The whole experiment is carried out in a medium of refractive index μ
 - In a Young's double slit experiment, the slits are 2 mm apart and are illuminated with a mixture of two wavelengths $\lambda_1 = 750 \text{ nm}$ and $\lambda_2 = 900 \text{ nm}$. At what minimum distance from the common central bright fringe on a screen 2 m from the slits will a bright fringe from one interference pattern coincide with a bright fringe from the other?
 - Bichromatic light is used in YDSE having wavelengths $\lambda_1 = 400 \text{ nm}$ and $\lambda = 700 \text{ nm}$. Find minimum order of λ_1 which overlaps with λ_2 .
 - In Young's double slit experiment set-up with light of wavelength $\lambda = 6000 \text{ \AA}$, distance between the two slits is 2 mm and distance between the plane of slits and the screen is 2 m . The slits are of equal intensity. When a sheet of glass of refractive index 1.5 (which permits only a fraction η of the incident light to pass through) and thickness $8000 \text{ \AA}$ is placed in front of the lower slit, it is observed that the intensity at a point P , 0.15 mm above the central maxima does not change. Find the value of η .
 - In a Young's double slit experiment set up, the wavelength of light used is 546 nm . The distance of screen from slits is 1 metre . The slit separation is 0.3 mm .
 - Compare the intensity at a point P distant 10 mm from the central fringe where the intensity is I_0 .
 - Find the number of bright fringes between P and the central fringe.
 - In a Young's double slit experiment, two wavelengths of 500 nm and 700 nm were used. What is the minimum distance from the central maximum where their maxima coincide again? Take $\frac{D}{d} = 10^3$. Symbols have their usual meanings.
 - When a thin sheet of a transparent material of thickness $7.2 \times 10^{-4} \text{ cm}$ is introduced in the path of one of the interfering beams, the central fringe shift to a position occupied by the sixth bright fringe. If $\lambda = 6 \times 10^{-6} \text{ cm}$, find the refractive index of the sheet.

DIFFRACTION : INTRODUCTION & CLASSIFICATION

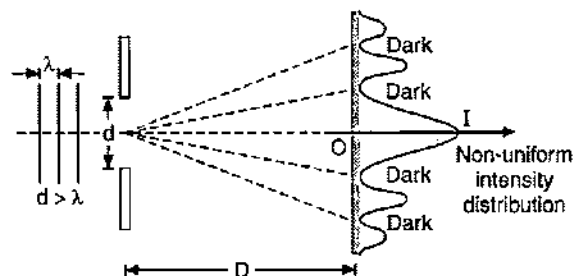
When light waves pass through a small aperture, an interference pattern is observed rather than a sharp spot of light cast by the aperture. This shows that light spreads in various directions beyond the aperture into regions where a shadow would be expected if light travelled in straight lines.



Light passing through two slits does not produce two distinct bright areas on a screen. Instead, an interference pattern is observed on the screen which shows that the light has deviated from a straight-line path and has entered the otherwise shadowed region. Other waves, such as sound waves and water waves, also have this property of being able to bend around corners.

This deviation of light from a straight-line path is called **diffraction**. Diffraction results from the interference of light from many coherent sources. In principle, the intensity of a diffraction pattern at a given point in space can be computed using Huygens' principle, where each point on the wavefront at the source of the pattern is taken to be a point source.

The phenomenon of bending of light around the corners of an obstacle/aperture of the size of the wave length of light is called diffraction. The phenomenon resulting from the superposition of secondary wavelets originating from different parts of the same wave front is define as diffraction of light. Diffraction is the characteristic of all types of waves. Greater the wave length of wave higher will be it's degree of diffraction.



CONCEPTUAL NOTE(S)

Diffraction, can be regarded as a consequence of interference from many coherent wave sources. In other words, the phenomena of diffraction and interference are basically equivalent.

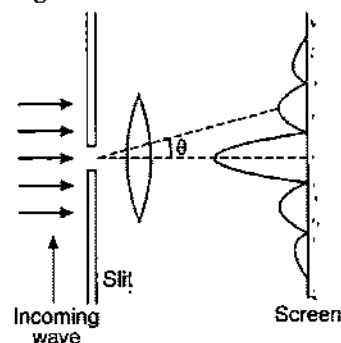
Diffraction phenomena are usually classified as being one of two types, which are named after the men who first explained them. The first type is called **Fraunhofer Diffraction** and the second is called **Fresnel's Diffraction**.

A. FRAUNHOFER DIFFRACTION

This occurs when the rays reaching a point are approximately parallel i.e. when both the source and screen are effectively at infinite distance from the diffracting device. In this case, the incident light is a plane wave so that the

phase of the light at each point in the aperture is the same. This can be achieved experimentally either by placing the observing screen at a large distance from the aperture or by using a converging lens to focus parallel rays on the screen, as in Figure. Note that a bright fringe is observed along the axis at $\theta=0$, with alternating bright and dark fringes on either side of the central bright fringe.

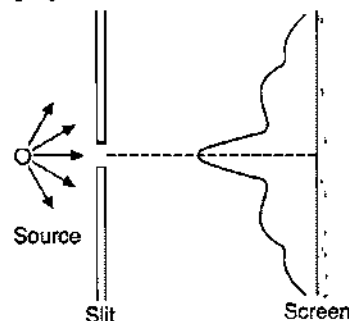
Common examples : Diffraction at single slit, double slit and diffraction grating.



B. FRESNEL'S DIFFRACTION

When the observing screen is placed at a finite distance from the slit and no lens is used to focus parallel rays, the observed pattern is called a **Fresnel Diffraction Pattern**. Fresnel diffraction is rather complex to treat quantitatively.

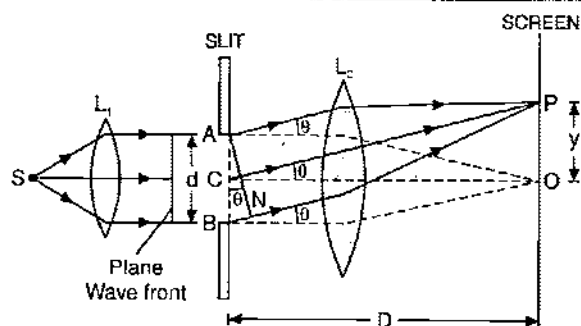
Common examples : Diffraction at a straight edge, narrow wire or small opaque disc etc.



A Fresnel diffraction pattern of a single slit is observed when the incident rays are not parallel and the observing screen is at a finite distance from the slit.

FRAUNHOFER DIFFRACTION AT A SINGLE SLIT

Consider that a monochromatic source of light S , emitting light waves of wavelength λ , is placed at the principal focus of the convex lens L_1 . A parallel beam of light i.e., a plane wavefront gets incident on a narrow slit AB of width d as shown in figure. The diffraction pattern is obtained on a screen lying at a distance D from the slit and at the focal plane of the convex lens L_2 .



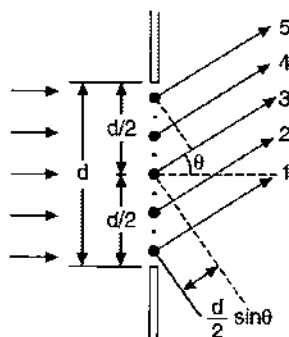
According to rectilinear propagation of light, a bright image of the slit is expected at the centre O of the screen. But in practice, we get a diffraction pattern i.e., a central maximum at the centre O flanked by a number of dark and bright fringes called secondary maxima and minima on either side of the point O .

The diffraction pattern is obtained on the screen, which lies at the focal plane of the convex lens L_2 . It is found that

- the width of the central maximum is twice as that of a secondary maximum and
- the intensity of the secondary maxima goes on decreasing with the order of maxima. These observations are explained on the basis of the phenomenon of diffraction using the following mathematical treatment.

EXPLANATION & MATHEMATICAL TREATMENT

Consider Fraunhofer diffraction by a single slit as shown in Figure. Important features of this problem can be deduced by examining waves coming from various portions of the slit. According to Huygens' principle, each portion of the slit acts as a source of waves. Hence, light from one portion of the slit can interfere with light from another portion, and the resultant intensity on the screen will depend on the direction θ .



Diffraction of light by a narrow slit of width d . Each portion of the slit acts as a point source of waves. The path difference between rays 1 and 3 or between rays 2 and 4 is equal to $(d/2) \sin \theta$

To analyze the resultant diffraction pattern, it is convenient to divide the slit in two halves as in Figure. All the waves that originate from the slit are in phase. Consider waves 1 and 3, which originate from the bottom and center of the slit, respectively. Wave 1 travels farther than wave 3 by an

amount equal to the path difference $\left(\frac{d}{2}\right) \sin \theta$, where d is the width of the slit. Similarly, the path difference between waves 2 and 4 is also equal to $\left(\frac{d}{2}\right) \sin \theta$. If this path difference is exactly one half of a wavelength (corresponding to a phase difference of 180°), the two waves cancel each other and destructive interference results. This is true, in fact, for any two waves that originate at points separated by half the slit width, since the phase difference between two such points is 180° . Therefore, waves from the upper half of the slit interfere destructively with waves from the lower half of the slit, when

$$\frac{d}{2} \sin \theta = \frac{\lambda}{2}$$

$$\Rightarrow \sin \theta = \frac{\lambda}{d}$$

Similarly, destructive interference (minima) occurs when the path difference $\left(\frac{d}{2}\right) \sin \theta$ equals $\lambda, \frac{3\lambda}{2}, 2\lambda, \frac{5\lambda}{2}$, etc. These points occur at progressively larger values of θ . Therefore, the general condition for destructive interference is

$$\sin \theta = m \frac{\lambda}{d} \quad (m = \pm 1, \pm 2, \pm 3, \dots) \quad \dots(1)$$

where $|m| \leq \frac{d}{\lambda}$

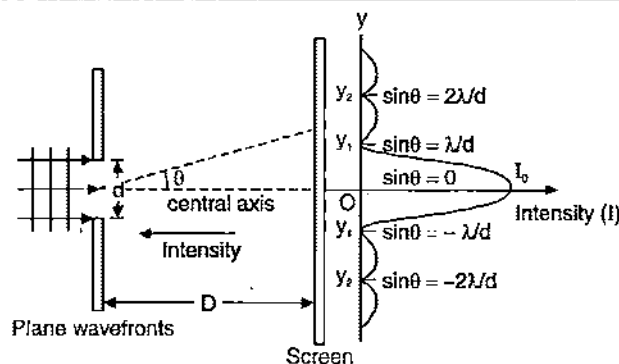
Equation (1) gives the values of θ for which the diffraction pattern has zero intensity. However, it tells us nothing about the variation in intensity along the screen. The general features of the intensity distribution along the screen are shown in Figure. A broad central bright fringe is observed, flanked by much weaker alternating maxima. The central bright fringe corresponds to those points opposite the slit for which the path difference is zero, or $\theta = 0$. All waves originating from the slit reach this region in phase, hence constructive interference results. The various dark fringes (points of zero intensity) occur at the values of θ that satisfy equation (1). The positions of the weaker maxima lie approximately halfway between the dark fringes. Note that the central bright fringe is twice as wide as the weaker maxima.

Angular width of central maxima is $\frac{2\lambda}{d}$ and width of central

maxima is $\frac{2\lambda D}{d}$, where D is the distance of the screen from the slit.

The intensity distribution of the diffraction pattern is quite different from the interference pattern produced due to superposition of light from two coherent sources. The point O on the central axis is the brightest. The angular position (θ) of n^{th} diffraction minima is given by

$$d \sin \theta = n\lambda \quad n = 1, 2, 3, 4, \dots$$



Position of the various minima for the Fraunhofer diffraction pattern of a single slit of width d .

For small angle θ , we have $\sin \theta \approx \theta$. Thus, as shown in the figure, the angular position of the 1st, 2nd, 3rd, ... minima are $\frac{\lambda}{d}$, $\frac{2\lambda}{d}$, $\frac{3\lambda}{d}$, respectively on either side of the central axis. A maximum is approximately halfway between two adjacent minima.

The central bright fringe is the widest, with width

$$\beta = \frac{\lambda}{d} + \frac{\lambda}{d} = \frac{2\lambda}{d}$$

The secondary bright fringes are narrower.

Note that as the slit width d increases, the width of the central diffraction maximum decreases. That is, there is less spreading out of the light by the slit. The secondary maxima also decrease in width and become weaker. When d becomes much greater than λ , the secondary maxima disappear.

The intensity I of the diffraction pattern as a function of θ is given as

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

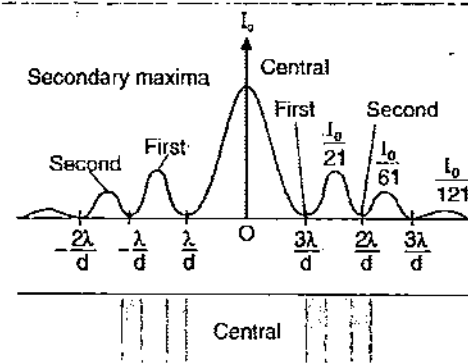
Since $\sin \theta \approx \frac{y}{D}$, so we get

$$\text{where } \alpha = \frac{\pi d \sin \theta}{\lambda} \approx \pi \left(\frac{y d}{\lambda D} \right)$$

The intensity of secondary maxima is much less. Compared to the intensity of central maximum (I_0), the intensity of the first of the secondary maxima is only 4.5%, of the second is only 1.6%, of the third is merely 0.83%..... The successive secondary maxima decrease rapidly in intensity.

CONCEPTUAL NOTE(S)

- a) If the intensity of the central maxima is I_0 , then the intensity of the first and second secondary maxima are found to be $\frac{I_0}{21}$ and $\frac{I_0}{61}$. Thus diffraction fringes are of unequal width and unequal intensities. Hence the ratio of the intensities of secondary maxima to central maxima are $1: \frac{1}{21}: \frac{1}{61}: \frac{1}{121}$



- As the slit width increases (relative to wavelength) the width of the central diffraction maxima decreases, that is, the light undergoes less flaring by the slit. The secondary maxima also decrease in width (and become weaker).
- If $d \gg \lambda$, the secondary maxima due to the slit disappear, we then no longer have single slit diffraction.
- When the slit width is reduced by a factor of 2, the amplitude of the wave at the centre of the screen is reduced by a factor of 2, so the intensity at the centre is reduced by a factor of 4.

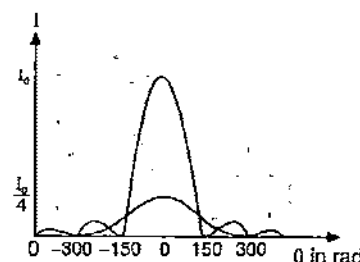


Illustration 34

A slit of width d is illuminated by white light.

- For what value of d , will the first minimum for red light of $\lambda = 6500 \text{ \AA}$ be at $\theta = 15^\circ$?
- What is the wavelength λ' of the light whose first side-maximum is at $\theta = 15^\circ$, thus coinciding with the first minimum for the red light?

Solution

- The angular position θ_n of n^{th} minimum is given by

$$d \sin \theta_n = n\lambda$$

Here, $n = 1$, $\lambda = 6500 \times 10^{-10} \text{ m}$, $\theta = 15^\circ$

$$\Rightarrow d = \frac{n\lambda}{\sin \theta_n} = \frac{1 \times 6500 \times 10^{-10}}{\sin 15^\circ} \approx 2.5 \mu\text{m}$$

- This maximum is approximately halfway between the first and second minima produced with light of wavelength λ' . Thus, by putting $n = 1.5$, we get

$$d \sin \theta = 1.5\lambda'$$

$$\Rightarrow \lambda' = \frac{d \sin \theta}{1.5} = \frac{2.5 \times 10^{-6} \times \sin 15^\circ}{1.5} = 430 \text{ nm} = 4300 \text{ \AA}$$

This is the wavelength of violet light. Note that the first side-maximum for light of $\lambda' = 4300 \text{ \AA}$ will always coincide with the first minimum for light of $\lambda = 6500 \text{ \AA}$, no matter what the slit-width is.

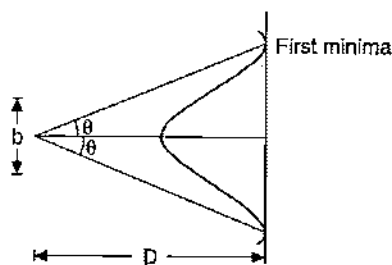
Illustration 35

Angular width of central maximum in the Fraunhofer diffraction pattern of a slit is measured. The slit is illuminated by light of wavelength $6000 \text{ \AA}$. When the slit is illuminated by light of another wavelength, the angular width decreases by 30%. Calculate the wavelength of this light. The same decrease in the angular width of central maximum is obtained when the original apparatus is immersed in a liquid. Find refractive index of the liquid.

Solution

- (a) Given $\lambda = 6000 \text{ \AA}$

Let b be the width of slit and D the distance between screen and slit.



First minima is obtained at $b \sin \theta = \lambda$

$$\Rightarrow b \theta = \lambda \sin \theta \approx \theta$$

$$\Rightarrow \theta = \frac{\lambda}{b}$$

$$\text{Angular width of first maxima} = 2\theta = \frac{2\lambda}{b} \propto \lambda$$

Angular width will decrease by 30% when λ is also decreased by 30%.

Therefore, new wavelength

$$\lambda' = \left\{ (6000) - \left(\frac{30}{100} \right) 6000 \right\} = 4200 \text{ \AA}$$

- (b) When the apparatus is immersed in a liquid of refractive index μ , the wavelength is decreased μ times. Therefore,

$$4200 \text{ \AA} = \frac{6000 \text{ \AA}}{\mu}$$

$$\Rightarrow \mu = \frac{6000}{4200}$$

$$\Rightarrow \mu = 1.429 \approx 1.43$$

VALIDITY OF GEOMETRICAL OPTICS

When a slit of width d is illuminated by a parallel beam of light, the angular spread of diffracted light is approximately

$\frac{\lambda}{d}$. Therefore, after travelling a distance D , the diffracted beam acquires a width $\frac{D\lambda}{d}$.

Geometrical optics is based on rectilinear propagation of light, which is just an approximation. We can say that geometrical optics is valid, if the width $\frac{D\lambda}{d}$ of the diffracted beam is less than the size of the slit, that is

$$\frac{D\lambda}{d} < d$$

$$\Rightarrow D < \frac{d^2}{\lambda}$$

Illustration 36

For what distance is the ray optics a good approximation, if the slit is 3 mm wide and the wavelength of light is $5000 \text{ \AA}$?

Solution

$$D < \frac{d^2}{\lambda} = \frac{(3 \times 10^{-3})^2}{5000 \times 10^{-10}} = 18 \text{ m}$$

Thus, upto a distance of 18 m, we can assume rectilinear propagation of light to a good approximation.

INTERFERENCE & DIFFRACTION : A COMPARISON

INTERFERENCE	DIFFRACTION
(a) It results from interaction of light coming from two different wave fronts originating from two coherent sources.	(a) It results from interaction of light coming from different parts of the same wavefront.
(b) Here, the fringes may or may not be of the same width.	(b) Here the fringes are always of varying width.
(c) The fringes of minimum intensity are dark (or perfectly dark when waves are of same amplitude).	(c) The fringes of minimum intensity are not perfectly dark.
(d) All bright fringes possess the same intensity.	(d) The intensity of all the bright fringes is not same. It is maximum for central fringe and decreases sharply for first, second etc. bright fringes.
(e) An interference pattern consists a good contrast between the dark and bright fringes.	(e) In diffraction pattern the contrast between the bright and dark fringes is comparatively poor.

POLARIZATION OF LIGHT

According to Maxwell, light possesses electromagnetic nature. An electromagnetic wave consists of varying electric and magnetic fields, such that the two fields are mutually perpendicular to each other and to the direction of propagation of waves. The optical phenomena i.e., phenomena concerning light may primarily be attributed to the vibrations of electric field vector in a direction perpendicular to the direction of propagation of light. In ordinary or unpolarised light, the vibrations of electric field vector are regularly or symmetrically distributed in a plane perpendicular to the direction of the propagation of the light.

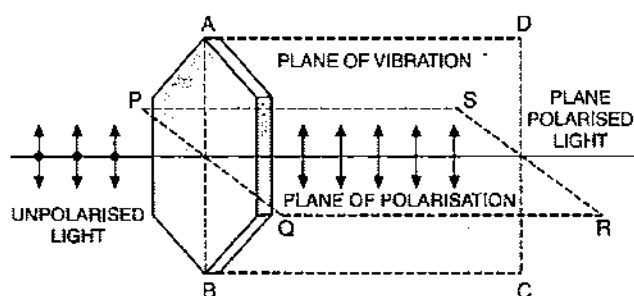


UNPOLARISED LIGHT (REPRESENTATION)

In an ordinary ray of light, the electric vibrations are in all the directions but perpendicular to the direction of propagation of the light. Such a ray of light is called a ray of ordinary or unpolarised light. It is schematically represented as shown. The arrows represent vibrations in the plane of the paper, while the dots represent vibrations in a direction perpendicular to the plane of the paper.

The phenomenon, due to which the vibrations of light are restricted to a particular plane, is called the polarisation of light.

When ordinary light i.e. unpolarised light passes through a tourmaline crystal, out of all the vibrations which are symmetrical about the direction of propagation, only those pass through it, which are parallel to its crystallographic axis AB. Therefore, on emerging through the crystal, the vibrations no longer remain symmetrical about the direction of propagation but are confined to a single plane (see Figure).

**PLANE OF VIBRATION**

The plane (ABCD), which contains the vibrations of plane polarised light, is called the plane of vibration.

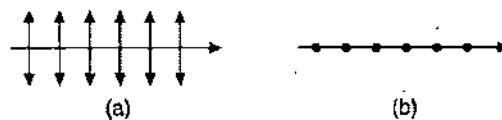
PLANE OF POLARISATION

The plane (PQRS) perpendicular to the plane of vibrations is called the plane of polarisation.

PLANE POLARISED LIGHT

It may be defined as the light, in which the vibrations of the light (vibrations of the electric vector) are restricted to a particular plane.

In a plane polarised light, the vibrations are restricted to a fixed plane, so that vibrations are perpendicular to direction of propagation of light. Figure (a) represents plane polarised light having vibrations in the plane of the paper and Figure (b) represents the plane polarised light having vibrations in a plane perpendicular to the plane of the paper.



POLARISED LIGHT (REPRESENTATION)

PROBLEM SOLVING TRICK(S)

The vibrations in plane polarised light are perpendicular to the plane of polarisation.

POLARIZATION BY REFLECTION

Polarized light may also be obtained by the process of reflection. When an unpolarized light beam is reflected, light is completely polarized, partially polarized, or unpolarized, depending on the angle of incidence. If the angle of incidence is either 0 or 90° (normal or grazing angles), the reflected beam is unpolarized. However, for intermediate angles of incidence, the reflected light is polarized to some extent.

Suppose an unpolarized light beam is incident on a surface as in figure. The beam can be described by two electric field components, one parallel to the surface (the dots) and the other perpendicular to the first and to the direction of propagation (the arrows). It is found that the parallel component reflects more strongly than the other component, and this results in a partially polarized beam. Furthermore, the refracted ray is also partially polarized.

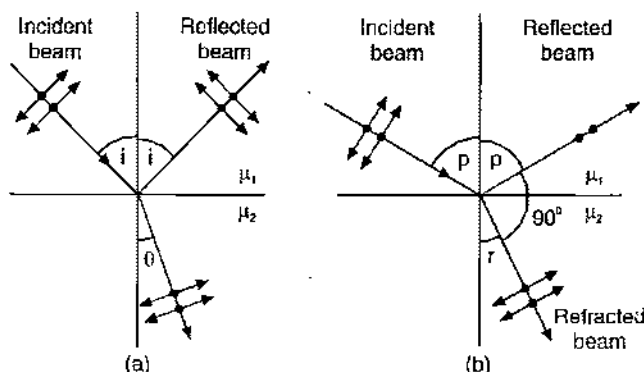
Now suppose the angle of incidence, i , is varied until the angle between the reflected and refracted beams is 90° . At this particular angle of incidence, the reflected beam is completely polarized with its electric field vector parallel to the surface, while the refracted beam is partially polarized. The angle of incidence at which this occurs is called the polarizing angle, p .

From figure, we see that at the polarizing angle, $p + 90^\circ + r = 180^\circ$, so that $r = 90^\circ - p$. Using Snell's Law, we have

$$\mu = \frac{\sin p}{\sin r}$$

Since $\sin r = \sin(90^\circ - p) = \cos p$, the expression for μ can be written

$$\mu = \frac{\sin p}{\cos p} = \tan p$$



(a) When unpolarized light is incident on a reflecting surface, the reflected and refracted beams are partially polarized. (b) The reflected beam is completely polarized when the angle of incidence equals the polarizing angle θ_p .

This expression is called **Brewster's Law**, and the polarizing angle p is sometimes called **Brewster's Angle**, after its discoverer, Sir David Brewster (1781-1868). For example, the Brewster's angle for crown glass ($\mu = 1.52$) is $p = \tan^{-1}(1.52) = 56.7^\circ$. Since μ varies with wavelength for a given substance, the Brewster's angle is also a function of the wavelength.

Polarization by reflection is a common phenomenon. Sunlight reflected from water, glass, snow and metallic surfaces is partially polarized. If the surface is horizontal, the electric field vector of the reflected light will have a strong horizontal component. Sunglasses made of polarizing material reduce the glare of reflected light. The transmission axes of the lenses are oriented vertically so as to absorb the strong horizontal component of the reflected light.

Illustration 37

A ray of light strikes a glass plate at an angle of 60° . If the reflected and refracted rays are perpendicular to each other, find the refractive index of glass.

Solution

Reflected and refracted rays are mutually perpendicular only when the angle of incidence is equal to polarising angle, hence

$$i_p = 60^\circ$$

So, refractive index is

$$\mu = \tan i_p = \tan 60^\circ = \sqrt{3} = 1.732$$

LAW OF MALUS

When a plane polarised light is seen through an analyser, the intensity of transmitted light varies as the analyser is rotated in its own plane about the incident direction. In 1809, E.N. Malus discovered that when a beam of completely plane

polarised light is passed through analyser, the intensity I of transmitted light varies directly as the square of the cosine of the angle θ between the transmission directions of polariser and analyser. This statement is known as the **Law of Malus**.

Mathematically, according to Malus Law, we have

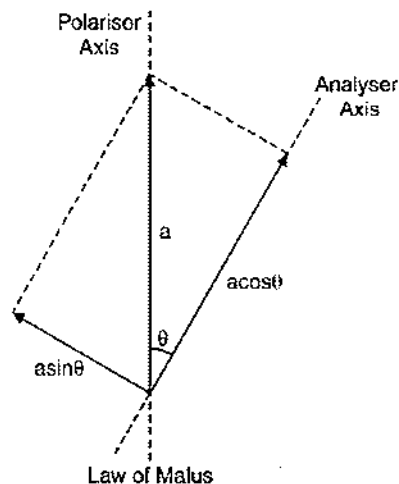
$$I \propto \cos^2 \theta$$

$$\Rightarrow I = I_0 \cos^2 \theta$$

where I_0 is the maximum intensity of transmitted light. It may be noted that I_0 is equal to half the intensity of unpolarised light incident on the polariser.

EXPLANATION OF THE LAW

Let the planes of polariser and analyser are inclined to each other at an angle θ as shown in figure. Let I_0 be the intensity and a the amplitude of the plane polarised light transmitted by the polariser.



The amplitude a of the light incident on the analyser has two rectangular components,

- $a \cos \theta$, parallel to the plane of transmission of the analyser, and
- $a \sin \theta$, perpendicular to the plane of transmission of the analyser.

So only the component $a \cos \theta$ is transmitted by the analyser. The intensity of light transmitted by the analyser is

$$I = k(a \cos \theta)^2$$

$$\Rightarrow I = k a^2 \cos^2 \theta$$

$$\Rightarrow I = I_0 \cos^2 \theta$$

where $I_0 = k a^2$, is the maximum intensity of light transmitted by the analyser (when $\theta = 0^\circ$). The above equation is the **Law of Malus** or **Malus Law**.

CONCEPTUAL NOTE(S)

- a) When $\theta = 0^\circ$ or 180° , $\cos \theta = \pm 1$

$$\Rightarrow I = I_0$$

So when the transmission directions of polariser and

analyser are parallel or antiparallel to each other, the maximum intensity of plane polarised light is transmitted by the analyser and is equal to the intensity emerging from the polariser.

b) When $\theta = 90^\circ$, $\cos\theta = 0$

$$\Rightarrow I = 0$$

So when the transmission directions of polariser and analyser are perpendicular to each other, the intensity of light transmitted through the analyser is zero.

c) When a beam of unpolarised light is incident on the polariser, then

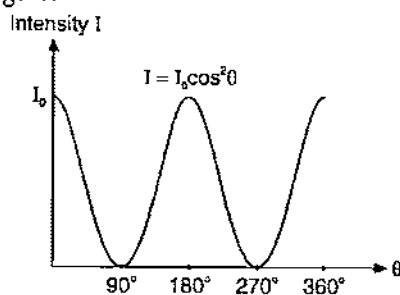
$$I = I_0 (\cos^2 \theta)$$

$$\text{Since, } \langle \cos^2 \theta \rangle = \frac{1}{2}$$

$$\Rightarrow I = \frac{I_0}{2}$$

INTENSITY CURVE

As the angle θ between the transmission directions of polariser and analyser is varied, the intensity I of the light transmitted by the analyser varies as a function of $\cos^2 \theta$, as shown in figure.



POLARISATION BY SCATTERING

When we look at the blue portion of the sky through a polaroid and rotate the polaroid, the transmitted light shows rise and fall of intensity. This shows that the light from the blue portion of the sky is plane polarised. This is because sunlight gets scattered (i.e., its direction is changed) when it encounters the molecules of the earth's atmosphere. The scattered light seen in a direction perpendicular to the direction of incidence is found to be plane polarised.

Explanation. Figure shows the unpolarised light incident on a molecule. The dots show vibrations perpendicular to the plane of paper and double arrows show vibrations in the plane of paper. The electrons in the molecule begin to vibrate in both of these directions. The electrons vibrating parallel to the double arrows cannot send energy towards an observer looking at 90° to the direction of the sun because their acceleration has no transverse component. The light scattered by the molecules in this direction has only dots. It is polarised perpendicular to the plane of paper. This explains the polarisation of light scattered from the sky.

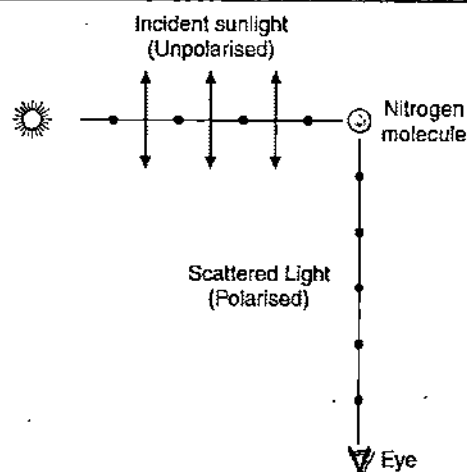


Illustration 38

Two 'crossed' polaroids A and B are placed in the path of a light-beam. In between these, a third polaroid C is placed whose polarisation axis makes an angle θ with the polarisation axis of the polaroid A . If the intensity of light emerging from the polaroid A is I_0 , then show that the intensity of light emerging from polaroid B will be $\frac{1}{4} I_0 \sin^2(2\theta)$.

Solution

By Malus Law, the intensity of light emerging from the middle polaroid C will be

$$I_1 = I_0 \cos^2 \theta$$

This intensity I_1 falls on the polaroid B whose polarisation axis makes an angle of $(90^\circ - \theta)$ with the polarisation axis of the polaroid C . Therefore, the intensity of light emerging from B will be

$$I_2 = I_1 \cos^2(90^\circ - \theta) = (I_0 \cos^2 \theta) \cos^2(90^\circ - \theta)$$

$$\Rightarrow I_2 = I_0 \cos^2 \theta \sin^2 \theta = \frac{I_0}{4} (2 \sin \theta \cos \theta)^2$$

$$\Rightarrow I_2 = \frac{1}{4} I_0 \sin^2(2\theta)$$

Illustration 39

Two polaroids are placed 90° to each other. What happens when $N-1$ more polaroids are inserted between two crossed polaroids (at 90° to each other). Their axes are equally spaced. How does the transmitted intensity behave for large N ?

Solution

Transmitted intensity through first polaroid is

$$I_1 = I_0 \cos^2 \theta$$

where I_0 is the original intensity. Similarly, the transmitted intensity through second polaroid will be

$$I_2 = I_1 \cos^2 \theta = I_0 \cos^4 \theta$$

If N polaroids are used, then

$$I_N = I_0 (\cos \theta)^{2N}$$

As the optic axes of the polaroids are equally inclined, so angle of rotation θ is same for each polaroid. Thus

$$\frac{I_N}{I_0} = (\cos \theta)^{2N}$$

Since, angle between successive polaroids is given by

$$\theta = \frac{90^\circ}{N} = \frac{\pi}{2N} \text{ radian}$$

For large N , θ becomes small, so we get

$$\left(\cos \frac{\pi}{2N} \right)^{2N} = \left(1 - \frac{\pi^2}{8N^2} + \dots \right)^{2N} \approx \left(1 - \frac{2N\pi^2}{8N^2} + \dots \right)$$

which approaches 1 for large N . So, fractional intensity, is

$$\frac{I_N}{I_0} \approx 1$$

$$\Rightarrow I_N \approx I_0$$

Illustration 40

A beam of plane-polarised falls normally on a polariser (cross-sectional area $3 \times 10^{-4} \text{ m}^2$) which rotates about the axis of the ray with an angular velocity of 31.4 rads^{-1} . Find the energy of light passing through the polariser per revolution and the intensity of the emergent beam if the flux of energy of the incident ray is 10^{-3} W .

Solution

Cross-sectional area of polaroid, $A = 3 \times 10^{-4} \text{ m}^2$

Angular velocity, $\omega = 3.14 \text{ rads}^{-1}$

Time taken to complete one revolution,

$$T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{31.4} = 0.2 \text{ s}$$

$$(\text{Energy incident/sec}) = 10^{-3} \text{ W}$$

So, intensity of incident polarised beam is given by

$$I_0 = \frac{\text{Energy incident/sec}}{\text{Area}} = \frac{10^{-3}}{3 \times 10^{-4}} = \frac{10}{3} \text{ Wm}^{-2}$$

Since, $I = I_0 \cos^2 \theta$ where $\langle \cos^2 \theta \rangle = \frac{1}{2}$

So, average intensity transmitted is

$$I_{av} = \frac{I_0}{2} = \frac{10}{3 \times 2} = 1.67 \text{ Wm}^{-2}$$

Light energy passing through polariser per revolution is given by

$$E = I_{av} AT = \frac{10}{6} (3 \times 10^{-4}) (0.2) = 10^{-4} \text{ J}$$

ICE II

BASED ON DIFFRACTION AND POLARISATION

(Solutions on page 2.89)

1. A slit of width 0.025 m is placed in front of a lens of focal length 50 cm . The slit is illuminated with light of wavelength $5900 \text{ \AA}$. Calculate the distance between the centre and first dark band of diffraction pattern obtained on a screen placed at the focal plane of the lens.
2. Two spectral lines of sodium D_1 and D_2 have wavelengths of approximately $5890 \text{ \AA}$ and $5896 \text{ \AA}$. A sodium lamp sends incident plane wave onto a slit of width 2 micrometre . A screen is located 2 m from the slit. Find the spacing between the first maxima of two sodium lines as measured on the screen.
3. In young's double slit experiment, the distance d between the slits S_1 and S_2 is 1 mm . What should the width of each slit be so as to obtain 10 maxima of the double slit pattern within the central maximum of the single slit pattern?
4. Estimate the distance for which ray optics is a good approximation for an aperture of 4 mm and wavelength 400 nm .
5. Two towers on the top of two hills are 40 km apart. The line joining them passes 50 m above a hill half way between the towers. What is the longest wavelength of radiowaves which can be sent between the towers without appreciable diffraction effects?
6. A slit of width d is illuminated by white light. For what value of d will the first minimum for red light ($\lambda = 6500 \text{ \AA}$) fall at an angle $\theta = 30^\circ$?
7. A screen is placed 2 m away from a single narrow slit. Calculate the slit width if the first minimum lies 5 mm on either side of central maximum. Incident plane waves have a wavelength of $5000 \text{ \AA}$.
8. Determine the angular separation between central maximum and first order maximum of the diffraction pattern due to a single slit of width 0.25 mm when light of wavelength $5890 \text{ \AA}$ is incident on it normally.
9. Two polarising sheets have their polarising directions parallel so that the intensity of the transmitted light is maximum. Through what angle must the either sheet be turned if the intensity is to drop by one-half?
10. A polariser and an analyser are oriented so that the maximum light is transmitted. What is the fraction of maximum light transmitted when analyser is rotated through (a) 30° (b) 60° ?
11. Two polaroids are crossed to each other. If one of them is rotated through 60° , then what percentage of the incident unpolarised light will be transmitted by the polaroids?
12. Two polaroids are placed at 90° to each other and the transmitted intensity is zero. What happens when one more polaroid is placed between these two bisecting the angle

between them?

13. A polaroid examines two adjacent plane-polarised light beams A and B whose planes of polarisation are mutually at right angles. In one position of the polaroid, the beam B shows zero intensity. From this position a rotation of 30° shows the two beams of equal intensities. Find the intensity ratio $\frac{I_A}{I_B}$ of the two beams.
14. Show that when a ray of light is incident on the surface of a transparent medium at the polarising angle, the reflected and transmitted rays are perpendicular to each other.
15. Unpolarised light of intensity 32 Wm^{-2} passes through three polarisers such that the transmission axis of the last polariser is crossed with the first, if the intensity of the

emerging light is 3 Wm^{-2} , what is the angle between the transmission axes of the first two polarisers? At what angle will the transmitted intensity be maximum?

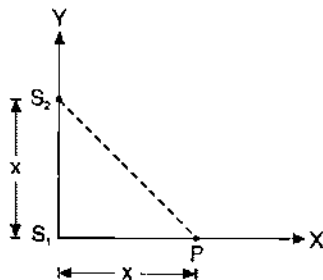
Solved Problems

Problem 1

An interference pattern is observed due to two coherent sources S_1 placed at origin and S_2 placed at $(0, 3\lambda, 0)$, where λ is the wavelength of the sources. A detector D is moved along the positive x -axis. Find the coordinates on the x -axis (excluding $x=0$ and ∞) where maximum intensity is observed.

Solution

At $x=0$, path difference is 3λ . Hence, third order maxima will be obtained. At $x \rightarrow \infty$, path difference is zero. Hence, zero order maxima is obtained. In between first and second order maxima will be obtained.



For First order maxima, we have

$$S_2P - S_1P = \lambda$$

$$\Rightarrow \sqrt{x^2 + 9\lambda^2} - x = \lambda$$

$$\Rightarrow \sqrt{x^2 + 9\lambda^2} = x + \lambda$$

Squaring both sides, we get $x^2 + 9\lambda^2 = x^2 + \lambda^2 + 2x\lambda$

Solving this, we get

$$x = 4\lambda$$

For Second order maxima, we have

$$S_2P - S_1P = 2\lambda$$

$$\Rightarrow \sqrt{x^2 + 9\lambda^2} - x = 2\lambda$$

$$\Rightarrow \sqrt{x^2 + 9\lambda^2} = (x + 2\lambda)$$

Squaring both sides, we get

$$x^2 + 9\lambda^2 = x^2 + 4\lambda^2 + 4x\lambda$$

Solving, we get

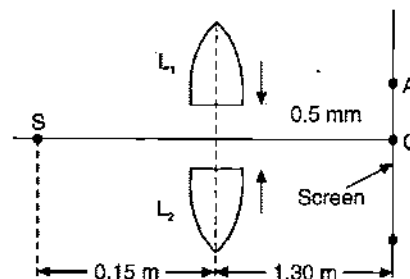
$$x = \frac{5}{4}\lambda = 1.25\lambda$$

Hence, the desired x coordinates are

$$x = 1.25\lambda \text{ and } x = 4\lambda$$

Problem 2

In given figure, S is a monochromatic point source emitting light of wavelength $\lambda = 500 \text{ nm}$. A thin lens of circular shape and focal length 0.10 m is cut into two identical halves L_1 and L_2 by a plane passing through a diameter. The two halves are placed symmetrically about the central axis SO with a gap of 0.5 mm . The distance along the axis from S to L_1 and L_2 is 0.15 m while that from L_1 and L_2 to O is 1.30 m . The screen at O is normal to SO .



- If the third intensity maximum excluding central maximum, occurs at the point A on the screen, find the distance OA .
- If the gap between L_1 and L_2 is reduced from its original value of 0.5 mm , will the distance OA increase, decrease, or remain the same.

Solution

- For the lens, $u = -0.15 \text{ m}$, $f = +0.10 \text{ m}$

Therefore, using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ we get

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{(-0.15)} + \frac{1}{(0.10)}$$

$$\Rightarrow v = 0.3 \text{ m}$$

$$\text{Linear magnification, } m = \frac{v}{u} = \frac{0.3}{-0.15} = -2$$

Hence, two images S_1 and S_2 of S will be formed at 0.3 m from the lens as shown in figure. Image S_1 due to part 1 will be formed at 0.5 mm above its optics axis ($m = -2$). Similarly, S_2 due to part 2 is formed 0.5 mm below the optic axis of this part as shown.

Hence, distance between S_1 and S_2 is $d = 1.5 \text{ mm}$

$$\text{Also, } D = 1.30 - 0.30 = 1.0 \text{ m} = 10^3 \text{ mm}$$

$$\text{and } \lambda = 500 \text{ nm} = 5 \times 10^{-4} \text{ mm}$$

So, fringe width is given by

$$\beta = \frac{\lambda D}{d} = \frac{(5 \times 10^{-4})(10^3)}{(1.5)} \text{ mm} = \frac{1}{3} \text{ mm}$$

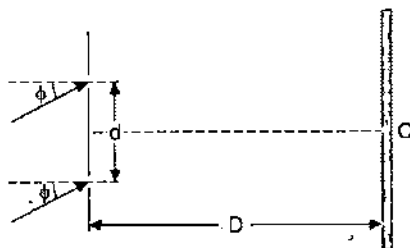
Now, as the point A is at the third maxima

$$\Rightarrow OA = 3\beta = 3\left(\frac{1}{3}\right) = 1 \text{ mm}$$

- (ii) If the gap between L_1 and L_2 is reduced, d will decrease. Hence, the fringe width β will increase or the distance OA will increase.

Problem 3

Light of wavelength $\lambda = 500 \text{ nm}$ falls on two narrow slits placed a distance $d = 50 \times 10^{-4} \text{ cm}$ apart, at an angle $\phi = 30^\circ$ relative to the slits shown in figure. ON the lower slit a transparent slab of thickness 0.1 nm and refractive index $\frac{3}{2}$ is placed. The interference pattern is observed on a screen at a distance $D = 2 \text{ m}$ from the slits.

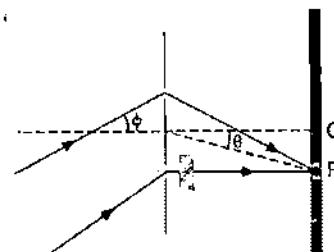


- Locate the position of the central maxima.
- Find the order of minima closest to centre C of screen.
- How many fringes will pass over C , if we remove the transparent slab from the lower slit?

Solution

- (a) Path difference is given by

$$\Delta x = d \sin \phi + d \sin \theta - (\mu - 1)t$$



For central maxima, $\Delta x = 0$

$$\Rightarrow \sin \theta = \frac{(\mu - 1)t}{d} - \sin \phi$$

$$\Rightarrow \sin \theta = \frac{\left(\frac{3}{2} - 1\right)(0.1)}{50 \times 10^{-3}} - \sin(30^\circ) = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

- (b) At C , $\theta = 0^\circ$, so we get

$$\Delta x = d \sin \phi - (\mu - 1)t$$

$$\Rightarrow \Delta x = (50 \times 10^{-3})\left(\frac{1}{2}\right) - \left(\frac{3}{2} - 1\right)(0.1)$$

$$\Rightarrow \Delta x = 0.025 - 0.05 = -0.025 \text{ mm}$$

Substituting, $\Delta x = n\lambda$, we get

$$n = \frac{\Delta x}{\lambda} = \frac{-0.025}{500 \times 10^{-6}} = -50$$

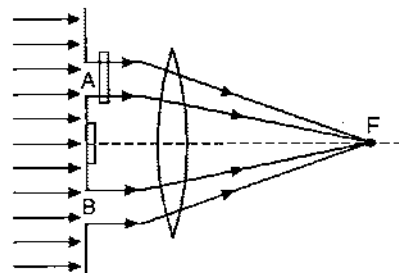
Hence, at C there will be maxima. Therefore the order of minima closest to the C are -49 .

- (c) Number of fringes shifted upwards is

$$N = \frac{(\mu - 1)t}{\lambda} = \frac{\left(\frac{3}{2} - 1\right)(0.1)}{500 \times 10^{-6}} = 100$$

Problem 4

In a modified Young's double slit experiment, a monochromatic uniform and parallel beam of light of wavelength $6000 \text{ \AA}$ and intensity $\left(\frac{10}{\pi}\right) \text{ Wm}^{-2}$ is incident normally on two apertures A and B of radii 0.001 m and 0.002 m respectively. A perfectly transparent film of thickness $2000 \text{ \AA}$ and refractive index 1.5 for the wavelength of $6000 \text{ \AA}$ is placed in front of aperture A (shown in figure).



Calculate the power (in W) received at the focal spot F of the lens. The lens is symmetrically placed with respect to the apertures. Assume that 10% of the power received by each aperture goes in the original direction and is brought to the focal spot.

Solution

Power received by aperture A is given by

$$P_A = I(\pi r_A^2) = \frac{10}{\pi}(\pi)(0.001)^2 = 10^{-5} \text{ W} \quad \left\{ \because I = \frac{P}{A} \right\}$$

Power received by aperture B is given by

$$P_B = I(\pi r_B^2) = \frac{10}{\pi}(\pi)(0.002)^2 = 4 \times 10^{-5} \text{ W}$$

Only 10% of P_A and P_B goes to the original direction, so

Portion of P_A going to original direction is $P_1 = 10^{-6} \text{ W}$

Portion of P_B going to original direction is $P_2 = 4 \times 10^{-6} \text{ W}$

Path difference created by slab is given by

$$\Delta x = (\mu - 1)t = (1.5 - 1)(2000) = 1000 \text{ Å}$$

Corresponding phase difference is given by

$$\phi = \left(\frac{2\pi}{\lambda} \right) \Delta x = \frac{2\pi}{6000} \times 1000 = \frac{\pi}{3}$$

Now, resultant power at the focal point is given by

$$P = P_1 + P_2 + 2\sqrt{P_1 P_2} \cos \phi$$

$$\Rightarrow P = 10^{-6} + 4 \times 10^{-6} + 2\sqrt{(10^{-6})(4 \times 10^{-6})} \cos \left(\frac{\pi}{3} \right)$$

$$\Rightarrow P = 7 \times 10^{-6} \text{ W}$$

Problem 5

A central portion with a width of $d = 0.5 \text{ mm}$ is cut out of a convergent lens having a focal length of $f = 10 \text{ cm}$, as shown in figure. Both halves are tightly fitted against each other. The lens receives monochromatic light ($\lambda = 5000 \text{ Å}$) from a point source at a distance of 5 cm from it.



- At what distance should a screen be fixed on the opposite side of the lens to observe three interference bands on it?
- What is the maximum possible number of interference bands that can be observed in this installation?

Solution

Applying the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

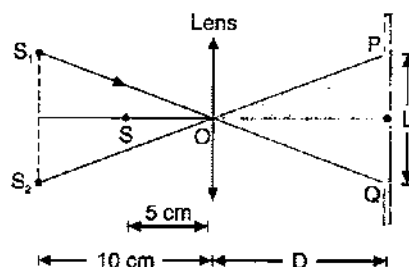
$$\frac{1}{v} + \frac{1}{5} = \frac{1}{10}$$

$$\Rightarrow v = -10 \text{ cm}$$

$$\Rightarrow m = \frac{v}{u} = \frac{-10}{-5} = 2$$

i.e., two virtual sources are formed with distance between them

$$d = 0.5 \text{ mm}$$



$$\text{Fringe width } \beta = \frac{\lambda(D+10)}{d}$$

Fringes are observed between the region P and Q (waves interfere in this region only), where

$$\frac{L}{D} = \frac{d}{10}$$

$$\Rightarrow L = \frac{Dd}{10}$$

Number of fringes that can be observed on the screen is given by

$$N = \frac{L}{\beta} = \frac{d^2 D}{10\lambda(D+10)} \quad \dots(1)$$

Substituting the values, we get

$$3 = \frac{(0.05)^2 D}{10 \times 5 \times 10^{-5} (D+10)}$$

Solving this equation, we get

$$D = 15 \text{ cm}$$

From equation (1), we have

$$N = \frac{d^2}{10\lambda \left(1 + \frac{10}{D} \right)}$$

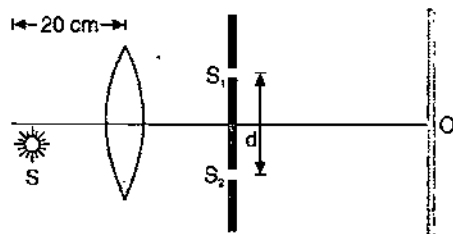
N will be maximum when $D \rightarrow \infty$

$$\Rightarrow \frac{10}{D} \rightarrow 0$$

$$\Rightarrow N_{\max} = \frac{d^2}{10\lambda} = \frac{(0.05)^2}{10 \times 5 \times 10^{-5}} = 5$$

Problem 6

A point source is placed at a distance $\frac{d}{2}$ below the principal axis of an equiconvex lens of refractive index $\frac{3}{2}$ and radius 20 cm. The emergent light from lens having wavelength $\lambda = 5000 \text{ \AA}$ falls on the slits S_1 and S_2 separated by $d = 1 \text{ mm}$ which are placed symmetrically along the principal axis. The resulting interference pattern is observed on the screen kept at a distance $D = 1 \text{ m}$ from the slit plane.



- (a) Find the position of central maxima and its width
(b) Find the intensity at point O.

Solution

Using Lens Maker's Formula, we get

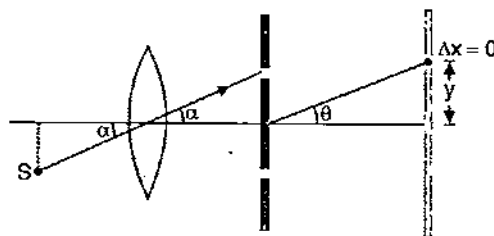
$$\frac{1}{f} = \left(\frac{3}{2} - 1\right) \left(\frac{1}{20} - \frac{1}{-20}\right)$$

$$\Rightarrow f = 20 \text{ cm}$$

Since source lies in focal plane of lens. So, all the emergent rays will be parallel. So,

$$\tan \alpha = \frac{d/2}{20} = \frac{d}{40} = \frac{1}{400} \approx \sin \alpha$$

Initial path difference $(\Delta x)_{\text{initial}} = d \sin \alpha$



Let the central maxima is obtained at angle θ . Then

$$d \sin \theta = d \sin \alpha$$

$$\Rightarrow \sin \theta = \sin \alpha$$

$$\Rightarrow \tan \theta = \tan \alpha$$

$$\Rightarrow \frac{y}{D} = \frac{d}{40}$$

$$\Rightarrow y = \frac{Dd}{40}$$

$$\Rightarrow y = \frac{100}{40} d = 2.5d = 2.5 \text{ mm}$$

At O, net path difference is given by

$$\Delta x = d \sin \alpha = \frac{1}{400} \text{ mm}$$

$$\text{Since, } \phi = \frac{2\pi}{\lambda} \Delta x$$

$$\Rightarrow \phi = \frac{2\pi}{5000 \times 10^{-10}} \times \frac{1}{400} \times 10^{-3}$$

$$\Rightarrow \phi = 10\pi$$

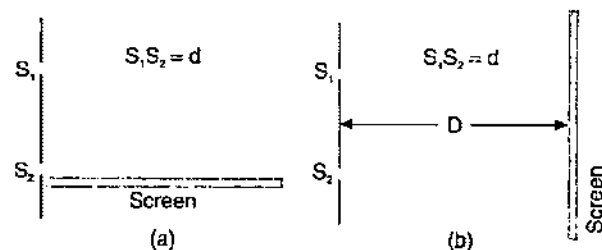
$$\text{Since, } I = I_{\text{max}} \cos^2 \left(\frac{\phi}{2} \right)$$

$$\Rightarrow I = I_{\text{max}}$$

Problem 7

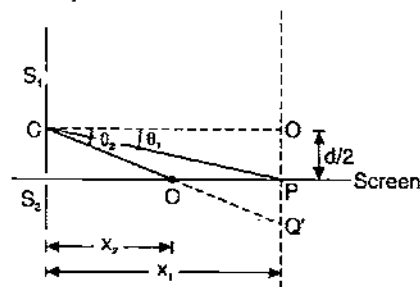
The interference pattern of a Young's double slit experiment is observed in two ways by placing the screen in two possible ways as shown in figure (a) and (b). The distance between two consecutive right most minima on the screen of figure (a) using light of wavelength $\lambda_1 = 4000 \text{ \AA}$ is observed to be 600 times the fringe width in the screen of figure (b) using the wavelength $\lambda_2 = 6000 \text{ \AA}$. If D (as shown in figure) is 1 m then find the separation between the coherent sources S_1 and S_2 . Given that

$$d > \frac{3\lambda_1}{2}$$



Solution

Had the screen been perpendicular to S_2P , then P and Q' would have been the positions of first and second minima (last two).



Since, the angular positions of minima do not depend on the position of the screen, so the second minima is formed at Q on the screen.

For right most minima at P, we have

$$d \sin \theta_1 = \frac{\lambda_1}{2} \quad \dots (1)$$

For small angles, we have

$$\sin \theta_1 \approx \tan \theta_2 = \frac{d/2}{x_2}$$

Substituting in equation (1), we get

$$x_2 = \frac{d^2}{\lambda_1} \quad \dots(2)$$

For next minima at Q , we have

$$d \sin \theta_2 = \frac{3}{2} \lambda_1 \quad \dots(3)$$

For small angles, we have

$$\sin \theta_2 \approx \tan \theta_2 = \frac{d}{x_2}$$

Substituting in equation (3), we get

$$x_2 = \frac{d^2}{3\lambda_1} \quad \dots(4)$$

$$\Rightarrow PQ = x_1 - x_2 = \frac{2d^2}{3\lambda_1} \quad \dots(5)$$

In the second case, fringe width is given by

$$\beta = \frac{\lambda_2 D}{d} \quad \dots(6)$$

Since it is given that $PQ = 600\beta$

$$\Rightarrow \frac{2d^2}{3\lambda_1} = 600 \frac{\lambda_2 D}{d}$$

$$\Rightarrow d^3 = 900 \lambda_1 \lambda_2 D$$

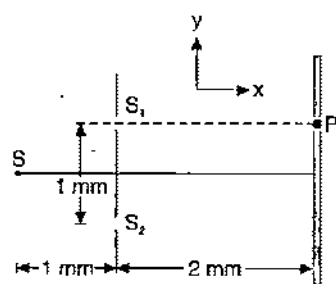
$$\Rightarrow d^3 = 900 \times 4000 \times 6000 \times 10^{-20} \times 1$$

$$\Rightarrow d^3 = 216 \times 10^{-12}$$

$$\Rightarrow d = 6 \times 10^{-4} \text{ m} = 0.6 \text{ mm}$$

Problem 8

In a Young's double slit experiment set-up source S of wavelength $5000 \text{ \AA}$ illuminates two slits S_1 and S_2 , which act as two coherent sources. The source S oscillates about its shown position according to the equation $y = 0.5 \sin(\pi t)$, where y is in millimetres and t in seconds.



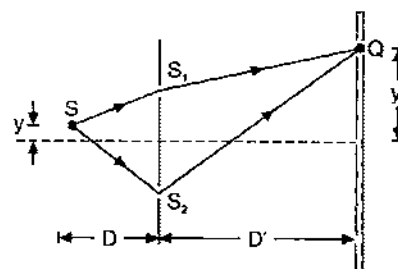
- (a) Locate the position of the central maxima as a function of time.

- (b) Calculate the minimum value of t for which the intensity at point P on the screen exactly in front of the upper slit becomes maximum.

Solution

- (a) Net path difference of the waves reaching at Q , is

$$\Delta x = \frac{yd}{D} + \frac{y'd}{D'}$$



For central maximum, $\Delta x = 0$

$$\Rightarrow y' = -\frac{D'}{D} y$$

$$\Rightarrow y' = -\left(\frac{2}{1}\right)(0.5 \sin(\pi t))$$

$$\Rightarrow y' = -\sin(\pi t) \text{ mm}$$

- (b) $y' = \frac{d}{2}$, at point P exactly in front of S_1 , so we have

$$\Delta x = \left(\frac{yd}{D}\right) + \left(\frac{d^2}{2D'}\right)$$

For maximum intensity, we have path difference to be an even multiple of $\frac{\lambda}{2}$, so

$$\Delta x = (2n) \frac{\lambda}{2} = n\lambda$$

Substituting the values, we get

$$0.5 \sin(\pi t) + 0.25 = 0.5n$$

$$\Rightarrow \sin(\pi t) = \frac{0.5n - 0.25}{0.5}$$

For minimum value of t , we have $n = 1$

$$\Rightarrow \sin(\pi t) = 0.5$$

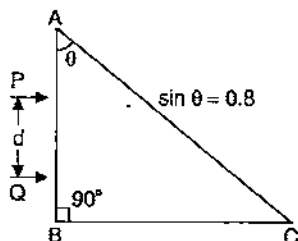
$$\Rightarrow \pi t = \frac{\pi}{6}$$

$$\Rightarrow t = \frac{1}{6} = 0.167 \text{ s}$$

Problem 9

Two parallel beams of light P and Q (separation d) containing radiations of wavelengths $4000 \text{ \AA}$ and $5000 \text{ \AA}$ (which are mutually coherent in each wavelength separately) are incident normally on a prism as shown in figure. The refractive index of the prism as a function of

wavelength is given by the relation, $\mu(\lambda) = 1.20 + \frac{b}{\lambda^2}$ where λ is in $\text{\AA}$ and b is positive constant. The value of b is such that the condition for total reflection at the face AC is just satisfied for one wavelength and is not satisfied for the other.

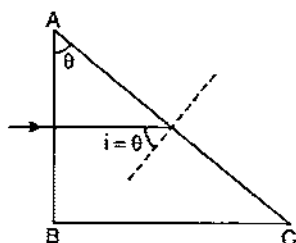


- Find the value of b
- Find the deviation of the beams transmitted through the face AC
- A convergent lens is used to bring these transmitted beams into focus. If the intensities of the upper and the lower beams immediately after transmission from the face AC , are $4I$ and I respectively, find the resultant intensity at the focus.

Solution

- Total internal reflection (TIR) will take place first for those wavelength for which critical angle is small or μ is large.

From the given expression of μ , it is more for the wavelength for which value of λ is less.



Thus, condition of TIR is just satisfied for $4000 \text{ \AA}$

$$\Rightarrow i = C \text{ for } 4000 \text{ \AA}$$

$$\Rightarrow \theta = C$$

$$\Rightarrow \sin \theta = \sin C$$

Since $\sin \theta = 0.8$ and $\sin C = \frac{1}{\mu}$, so we get

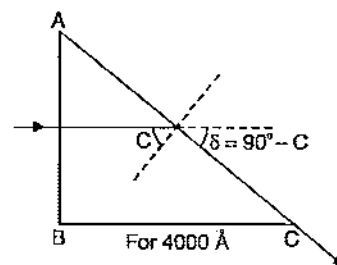
$$\Rightarrow 0.8 = \frac{1}{\mu} \quad (\text{for } 4000 \text{ \AA})$$

$$\Rightarrow 0.8 = \frac{1}{1.20 + \frac{b}{(4000)^2}}$$

Solving this equation, we get

$$b = 8.0 \times 10^5 (\text{\AA})^2$$

- For, $4000 \text{ \AA}$ condition of TIR is just satisfied. Hence, it will emerge from AC , just grazingly.

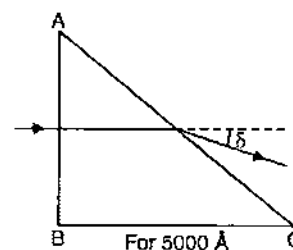


So, deviation for $4000 \text{ \AA}$ is given by

$$\Rightarrow \delta_{4000 \text{ \AA}} = 90 - i = 90 - \sin^{-1}(0.8) \approx 37^\circ$$

For $5000 \text{ \AA}$, we have

$$\mu = 1.2 + \frac{b}{\lambda^2} = 1.2 + \frac{8 \times 10^5}{(5000)^2} = 1.232$$



Applying, $\mu = \frac{\sin i_{\text{air}}}{\sin i_{\text{medium}}}$

$$\Rightarrow 1.232 = \frac{\sin i_{\text{air}}}{\sin \theta} = \frac{\sin i_{\text{air}}}{0.8}$$

$$\Rightarrow i_{\text{air}} = 80.26^\circ$$

So, deviation for $5000 \text{ \AA}$ is given by

$$\delta_{5000 \text{ \AA}} = i_{\text{air}} - i_{\text{medium}} = 80.26^\circ - \sin^{-1}(0.8) = 27.13^\circ$$

- The intensity of the upper beam ($4000 \text{ \AA}$) and lower beam ($5000 \text{ \AA}$) after transmission are $4I$ and I respectively, then

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

Since no phase change takes place for the waves refracted from the lens, so $\phi = 0^\circ$.

$$\Rightarrow I_R = 4I + I + 2\sqrt{(4I)I} \cos(0^\circ)$$

$$\Rightarrow I_R = 9I$$

Problem 10

A glass plate of refractive index 1.5 is coated with a thin layer of thickness t and refractive index 1.8 . Light of wavelength λ travelling in air is incident normally on the layer. It is partly reflected at the upper and the lower surfaces of the layer and the two reflected rays interfere. Write the condition for their constructive interference. If $\lambda = 648 \text{ nm}$, obtain the least value of t for which the rays interfere constructively.

Solution

Incident ray AB is partly reflected as ray 1 from the upper surface and partly reflected as ray 2 from the lower surface of the layer of thickness t and refractive index $\mu_1 = 1.8$ as shown in figure. Path difference between the two rays would be

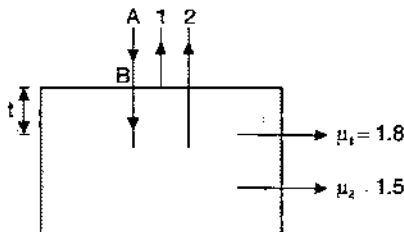
$$\Delta x = 2\mu_1 t = 2(1.8)t = 3.6t$$

Ray 1 is reflected from a denser medium, therefore, it undergoes a phase change of π , whereas the ray 2 gets reflected from a rarer medium, therefore, there is no change in phase of ray 2.

Hence, phase difference between rays 1 and 2 would be $\Delta\phi = \pi$. Therefore, condition of constructive interference will be

$$\Delta x = \left(n - \frac{1}{2}\right)\lambda \quad \text{where } n = 1, 2, 3, \dots$$

$$\Rightarrow 3.6t = \left(n - \frac{1}{2}\right)\lambda$$



Least values of t is corresponding to $n = 1$ or

$$t_{\min} = \frac{\lambda}{2 \times 3.6}$$

$$\Rightarrow t_{\min} = \frac{648}{7.2} \text{ nm}$$

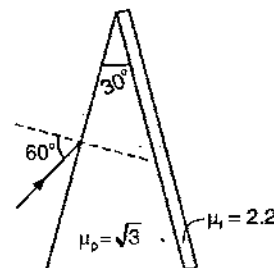
$$\Rightarrow t_{\min} = 90 \text{ nm}$$

CONCEPTUAL NOTE(S)

- For a wave (whether it is sound or electromagnetic), a medium is denser or rarer is decided from the speed of wave in that medium. In denser medium speed of wave is less. For example, water is rarer for sound, while denser for light compared to air because speed of sound in water is more than in air, while speed of light is less.
- In transmission/refraction, no phase change takes place. In reflection, there is a change of phase of π when it is reflected by a denser medium and phase change is zero if it is reflected by a rarer medium.
- If two waves in phase interfere having a path difference of Δx ; then condition of maximum intensity would be $\Delta x = n\lambda$, $n = 0, 1, 2, \dots$
But if two waves, which are already out of phase (a phase difference of π) interfere with path difference Δx , then condition of maximum intensity will be $\Delta x = \left(n - \frac{1}{2}\right)\lambda$, $n = 1, 2, \dots$

Problem 11

Shown in the figure is a prism of refracting angle 30° and refractive index $\mu_p (= \sqrt{3})$. The face AC of the prism is covered with a thin film of refractive index $\mu_f (= 2.2)$. A monochromatic light of wavelength $\lambda = 550 \text{ nm}$ falls on the face AB at an angle of incidence of 60° . Calculate



- the angle of emergence.
- the minimum value of thickness t of the coated film so that the intensity of the emergent ray is maximum.

Solution

- Applying Snell's Law, we get

$$\sin i_1 = \mu \sin r_1$$

$$\Rightarrow \sin(60^\circ) = \sqrt{3} \sin r_1$$

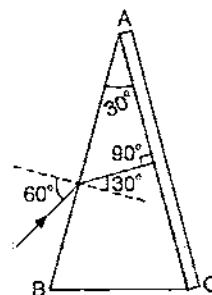
$$\Rightarrow \sin r_1 = \frac{1}{2}$$

$$\Rightarrow r_1 = 30^\circ$$

$$\text{Since, } r_1 + r_2 = A$$

$$\Rightarrow r_2 = A - r_1 = 30^\circ - 30^\circ = 0^\circ$$

Therefore, ray of light falls normally on the face AC and angle of emergence $e = i_2 = 0^\circ$.



- Multiple reflection occurs between surfaces of film. Intensity will be maximum if interference takes place in the transmitted wave.

For maximum thickness, we have

$$\Delta x = 2\mu t = \lambda$$

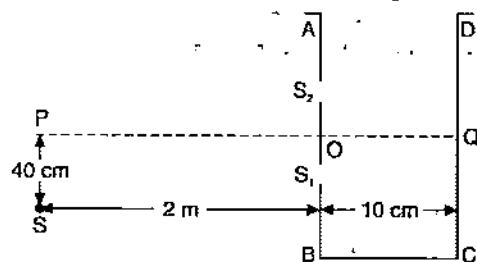
where t is the thickness of coated film

$$\Rightarrow t = \frac{\lambda}{2\mu} = \frac{550}{2 \times 2.2} = 125 \text{ nm}$$

Problem 12

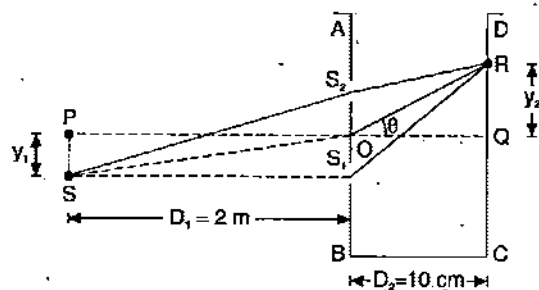
A vessel $ABCD$ of 10 cm width has two small slits S_1 and S_2 sealed with identical glass plates of equal thickness. The distance between the slits is 0.8 mm . POQ is the line perpendicular to the plane AB and passing through O , the middle point of S_1 and S_2 . A monochromatic light source is kept at S , 40 cm below P and 2 m from the vessel, to illuminate the slits as shown in the figure alongside.

- (a) Calculate the position of the central bright fringe on the other wall CD with respect to the line OQ.
- (b) Now, a liquid is poured into the vessel and filled upto OQ. The central bright fringe is found to be at Q. Calculate the refractive index of the liquid.



Solution

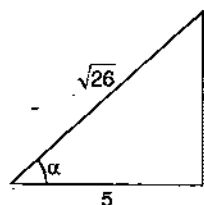
- (a) Given $y_1 = 40 \text{ cm}$, $D_1 = 2 \text{ m} = 200 \text{ cm}$, $D_2 = 10 \text{ cm}$



$$\tan \alpha = \frac{y_1}{D_1} = \frac{40}{200} = \frac{1}{5}$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{1}{5}\right)$$

$$\Rightarrow \sin \alpha = \frac{1}{\sqrt{26}} \approx \frac{1}{5} = \tan \alpha$$



Path difference between SS_1 and SS_2 is

$$\Delta x_1 = SS_1 - SS_2$$

$$\Rightarrow \Delta x_1 = d \sin \alpha = (0.8 \text{ mm})\left(\frac{1}{5}\right)$$

$$\Rightarrow \Delta x_1 = 0.16 \text{ mm} \quad \dots(1)$$

Now, let at point R on the screen, central bright fringe is observed (i.e., net path difference = 0).

Path difference between S_2R and S_1R is

$$\Delta x_2 = S_2R - S_1R$$

$$\Rightarrow \Delta x_2 = d \sin \theta \quad \dots(2)$$

Central bright fringe will be observed when net path difference is zero.

$$\Rightarrow \Delta x_2 - \Delta x_1 = 0$$

$$\Rightarrow \Delta x_2 = \Delta x_1$$

$$\Rightarrow d \sin \theta = 0.16$$

$$\Rightarrow (0.8) \sin \theta = 0.16$$

$$\Rightarrow \sin \theta = \frac{0.16}{0.8} = \frac{1}{5}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{24}}$$

$$\Rightarrow \sin \theta \approx \frac{1}{5}$$

$$\Rightarrow \tan \theta \approx \sin \theta = \frac{y_2}{D_2} = \frac{1}{5}$$

$$\Rightarrow y_2 = \frac{D_2}{5} = \frac{10}{5} = 2 \text{ cm}$$

Therefore, central bright fringe is observed at 2 cm above point Q on side CD.

Alternate solution for (a)

Δx at R will be zero if $\Delta x_1 = \Delta x_2$

$$\Rightarrow d \sin \alpha = d \sin \theta$$

$$\Rightarrow \alpha = \theta$$

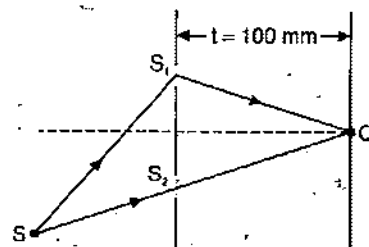
$$\Rightarrow \tan \alpha = \tan \theta$$

$$\Rightarrow \frac{y_1}{D_1} = \frac{y_2}{D_2}$$

$$\Rightarrow y_2 = \left(\frac{D_2}{D_1}\right) y_1 = \left(\frac{10}{200}\right)(40) \text{ cm}$$

$$\Rightarrow y_2 = 2 \text{ cm}$$

- (b) The central bright fringe will be observed at point Q, if the path difference created by the liquid slab of thickness $t = 10 \text{ cm}$ or 100 mm is equal to Δx_1 , so that the net path difference at Q becomes zero.



$$\Rightarrow (\mu - 1)t = \Delta x_1$$

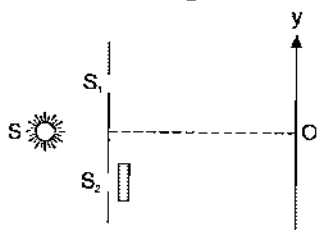
$$\Rightarrow (\mu - 1)(100) = 0.16$$

$$\Rightarrow \mu - 1 = 0.0016$$

$$\Rightarrow \mu = 1.0016$$

Problem 13

The Young's double slit experiment is done in a medium of refractive index $\frac{4}{3}$. A light of 600 nm wavelength is falling on the slits having 0.45 mm separation. The lower slit S_2 is covered by a thin glass sheet of thickness 10.4 μm and refractive index 1.5. The interference pattern is observed on a screen placed 1.5 m from the slits as shown in the figure.



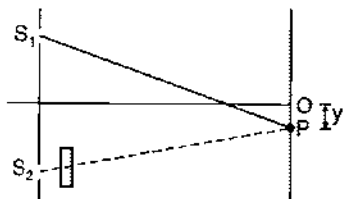
- Find the location of central maximum (bright fringe with zero path difference) on the y-axis.
- Find the light intensity of point O relative to the maximum fringe intensity.
- Now, if 600 nm light is replaced by white light of range 400 to 700 nm, find the wavelengths of the light that form maxima exactly at point O.

[All wavelengths in the problem are for the given medium of refractive index $\frac{4}{3}$. Ignore dispersion]

Solution

Given $\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}$,

$d = 0.45 \text{ mm} = 0.45 \times 10^{-3} \text{ m}$ and $D = 1.5 \text{ m}$



Thickness of glass sheet, $t = 10.4 \mu\text{m} = 10.4 \times 10^{-6} \text{ m}$

Refractive index of the medium, $\mu_m = \frac{4}{3}$

And refractive index of glass sheet, $\mu_g = 1.5$

- Let central maximum is obtained at a distance y below point O.

$$\Rightarrow \Delta x_1 = S_1P - S_2P = \frac{yd}{D}$$

Path difference due to glass sheet is given by

$$\Delta x_2 = \left(\frac{\mu_g}{\mu_m} - 1 \right) t$$

Net path difference will be zero, when we have

$$\Delta x_1 = \Delta x_2$$

$$\Rightarrow \frac{yd}{D} = \left(\frac{\mu_g}{\mu_m} - 1 \right) t$$

$$\Rightarrow y = \left(\frac{\mu_g}{\mu_m} - 1 \right) t \frac{D}{d}$$

Substituting the values, we get

$$y = \left(\frac{1.5}{4/3} - 1 \right) \frac{10.4 \times 10^{-6} (1.5)}{0.45 \times 10^{-3}}$$

$$\Rightarrow y = 4.33 \times 10^{-3} \text{ m}$$

$$\Rightarrow y = 4.33 \text{ mm}$$

- At O, $\Delta x_1 = 0$ and $\Delta x_2 = \left(\frac{\mu_g}{\mu_m} - 1 \right) t$

So, net path difference is

$$\Delta x = \Delta x_2$$

Corresponding phase difference, $\Delta\phi = \left(\frac{2\pi}{\lambda} \right) \Delta x$

Substituting the values, we get

$$\phi = \Delta\phi = \frac{2\pi}{6 \times 10^{-7}} \left(\frac{1.5}{4/3} - 1 \right) (10.4 \times 10^{-6})$$

$$\Rightarrow \phi = \left(\frac{13}{3} \right) \pi$$

Now, $I(\phi) = I_{\max} \cos^2 \left(\frac{\phi}{2} \right)$

$$\Rightarrow I = I_{\max} \cos^2 \left(\frac{13\pi}{6} \right)$$

$$\Rightarrow I = \frac{3}{4} I_{\max}$$

- At O, path difference is $\Delta x = \Delta x_2 = \left(\frac{\mu_g}{\mu_m} - 1 \right) t$

For maximum intensity at O, we have

$$\Delta x = n\lambda, \text{ where } n = 1, 2, 3, \dots$$

$$\Rightarrow \lambda = \frac{\Delta x}{1}, \frac{\Delta x}{2}, \frac{\Delta x}{3}, \dots \text{ and so on}$$

$$\Rightarrow \Delta x = \left(\frac{1.5}{4/3} - 1 \right) (10.4 \times 10^{-6} \text{ m})$$

$$\Rightarrow \Delta x = \left(\frac{1.5}{4/3} - 1 \right) (10.4 \times 10^3 \text{ nm}) = 1300 \text{ nm}$$

So, maximum intensity will be corresponding to

$$\lambda = 1300 \text{ nm}, \frac{1300}{2} \text{ nm}, \frac{1300}{3} \text{ nm}, \frac{1300}{4} \text{ nm}, \dots$$

$$\Rightarrow \lambda = 1300 \text{ nm}, 650 \text{ nm}, 433.33 \text{ nm}, 325 \text{ nm}, \dots$$

The wavelength in the range 400 nm to 700 nm are 650 nm and 433.33 nm.

Problem 14

In Young's experiment, the source is red light of wavelength 7×10^{-7} m. When a thin glass plate of refractive index 1.5 at this wavelength is put in the path of one of the interfering beams, the central bright fringe shifts by 10^{-3} m to the position previously occupied by the 5th bright fringe. Find the thickness of the plate. When the source is now changed to green light of wavelength 5×10^{-7} m, the central fringe shifts to a position initially occupied by the 6th bright fringe due to red light. Find the refractive index of glass for green light. Also estimate the change in fringe width due to the change in wavelength.

Solution

Path difference due to the glass slab,

$$\Delta x = (\mu - 1)t = (1.5 - 1)t = 0.5t$$

Due to this slab, 5 red fringes have been shifted upwards. So, we have

$$\Delta x = 5\lambda_{\text{red}}$$

$$\Rightarrow 0.5t = (5)(7 \times 10^{-7} \text{ m})$$

$$\Rightarrow t = \text{thickness of glass slab} = 7 \times 10^{-6} \text{ m}$$

Let μ' be the refractive index for green light, then

$$\Delta x' = (\mu' - 1)t$$

Now the shifting is of 6 fringes of red light. So, we have

$$\Delta x' = 6\lambda_{\text{red}}$$

$$\Rightarrow (\mu' - 1)t = 6\lambda_{\text{red}}$$

$$\Rightarrow (\mu' - 1) = \frac{(6)(7 \times 10^{-7})}{7 \times 10^{-6}} = 0.6$$

$$\Rightarrow \mu' = 1.6$$

Since the shifting of 5 bright fringes was equal to 10^{-3} m

$$\Rightarrow 5\beta_{\text{red}} = 10^{-3} \text{ m, where } \beta \text{ is the Fringe width}$$

$$\Rightarrow \beta_{\text{red}} = \frac{10^{-3}}{5} \text{ m} = 0.2 \times 10^{-3} \text{ m}$$

Now since $\beta = \frac{\lambda D}{d}$

$$\Rightarrow \beta \propto \lambda$$

$$\Rightarrow \frac{\beta_{\text{green}}}{\beta_{\text{red}}} = \frac{\lambda_{\text{green}}}{\lambda_{\text{red}}}$$

$$\Rightarrow \beta_{\text{green}} = \beta_{\text{red}} \frac{\lambda_{\text{green}}}{\lambda_{\text{red}}} = (0.2 \times 10^{-3}) \left(\frac{5 \times 10^{-7}}{7 \times 10^{-7}} \right)$$

$$\Rightarrow \beta_{\text{green}} = 0.143 \times 10^{-3} \text{ m}$$

$$\Rightarrow \Delta\beta = \beta_{\text{green}} - \beta_{\text{red}} = (0.143 - 0.2) \times 10^{-3} \text{ m}$$

$$\Rightarrow \Delta\beta = -5.71 \times 10^{-5} \text{ m}$$

Problem 15

A double slit apparatus is immersed in a liquid of refractive index 1.33. It has slit separation of 1 mm and distance between the plane of slits and screen is 1.33 m. The slits are illuminated by a parallel beam of light whose wavelength in air is 6300 Å.

(i) Calculate the fringe width.

(ii) One of the slits of the apparatus is covered by a thin glass sheet of refractive index 1.53. Find the smallest thickness of the sheet to bring the adjacent minimum at the axis.

Solution

Given $\mu = 1.33$, $d = 1 \text{ mm}$, $D = 1.33 \text{ m}$ and $\lambda = 6300 \text{ Å}$

(i) Wavelength of light in the given liquid is

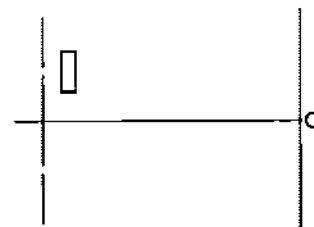
$$\lambda' = \frac{\lambda}{\mu} = \frac{6300}{1.33} \text{ Å} \approx 4737 \text{ Å} = 4737 \times 10^{-10} \text{ m}$$

$$\Rightarrow \text{Fringe width, } \beta = \frac{\lambda' D}{d}$$

$$\Rightarrow \beta = \frac{(4737 \times 10^{-10} \text{ m})(1.33 \text{ m})}{(1 \times 10^{-3} \text{ m})}$$

$$\Rightarrow \beta = 6.3 \times 10^{-4} \text{ m} = 0.63 \text{ mm}$$

(ii) Let t be the thickness of the glass slab.



Path difference due to glass slab at centre O is given by

$$\Delta x = \left(\frac{\mu_{\text{glass}}}{\mu_{\text{liquid}}} - 1 \right) t = \left(\frac{1.53}{1.33} - 1 \right) t$$

$$\Rightarrow \Delta x = 0.15t$$

Now, for the intensity to be minimum at O , this path difference should be equal to $\frac{\lambda'}{2}$

$$\Rightarrow \Delta x = \frac{\lambda'}{2}$$

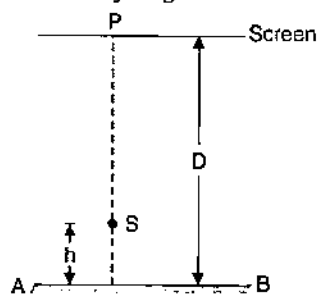
$$\Rightarrow 0.15t = \frac{4737}{2} \text{ Å}$$

$$\Rightarrow t = 15790 \text{ Å} = 1.579 \text{ μm}$$

Problem 16

A point source S emitting light of wavelength 600 nm is placed at a very small height h above a flat reflecting surface AB (shown in figure). The intensity of the reflected light is 36% of the incident intensity. Interference

fringes are observed on a screen placed parallel to the reflecting surface at a very large distance D from it.



- What is the shape of the interference fringes on the screen?
- Calculate the ratio of the minimum to the maximum intensities in the interference fringes formed near the point P (shown in the figure).
- If the intensity at point P corresponds to a maximum, calculate the minimum distance through which the reflecting surface AB should be shifted so that the intensity at P again becomes maximum.

Solution

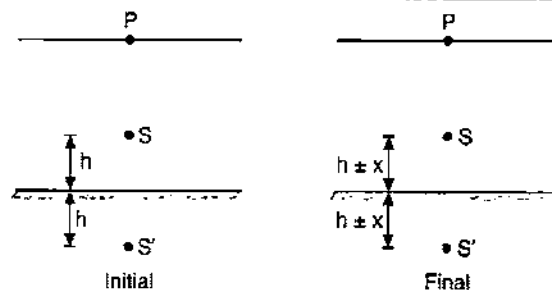
- Since there is symmetry about the line SP , so the shape of the interference fringes will be circular.
- Intensity of light reaching on the screen directly from the source $I_1 = I_0$ (say) and intensity of light reaching on the screen after reflecting from the mirror is $I_2 = 36\%$ of $I_0 = 0.36I_0$.

$$\Rightarrow \frac{I_1}{I_2} = \frac{I_0}{0.36I_0} = \frac{1}{0.36}$$

$$\Rightarrow \sqrt{\frac{I_1}{I_2}} = \frac{1}{0.6}$$

$$\Rightarrow \frac{I_{\min}}{I_{\max}} = \frac{\left(\sqrt{\frac{I_1}{I_2}} - 1\right)^2}{\left(\sqrt{\frac{I_1}{I_2}} + 1\right)^2} = \frac{\left(\frac{1}{0.6} - 1\right)^2}{\left(\frac{1}{0.6} + 1\right)^2} = \frac{1}{16}$$

- Initially path difference at P between two waves reaching from S and S' is as shown.



Since the ray is reflected from the surface of a denser medium, so it suffers an additional path change of $\frac{\lambda}{2}$ or a phase change of π .

For maximum at P , path difference equals $n\lambda$.

If AB is shifted by x , then this will cause an additional path difference of $2\left(x - \frac{\lambda}{2}\right)$ (for object and its image taken as coherent sources). Since reflection takes place at surface of denser medium, so this will produce an additional phase change of π or a path change of $\frac{\lambda}{2}$.

So, we get

$$2\left(x - \frac{\lambda}{2}\right) = n\lambda$$

$$2x - \lambda = n\lambda$$

$$\Rightarrow 2x = (n+1)\lambda$$

$$\Rightarrow x = (n+1)\frac{\lambda}{2} \text{ where } n = 0, 1, 2, 3, \dots$$

Now, to get minimum value of x , n must be minimum i.e., $n = 0$

$$\Rightarrow x = \frac{\lambda}{2}$$

$$\Rightarrow x = \frac{600}{2} = 300 \text{ nm}$$

Practice Exercise

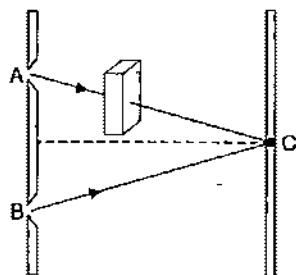
Single Correct Choice Type Questions

This section contains Single Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONLY ONE is correct.

- In an interference pattern produced by two identical slits, the intensity at the site of the central maximum is I . The intensity at the same spot when either of two slits is closed is
 - $\frac{I}{2}$
 - $\frac{I}{4}$
 - $\frac{I}{2\sqrt{2}}$
 - $\frac{I}{\sqrt{2}}$
- In YDSE bi-chromatic light of wavelengths 400 nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1 m. The minimum distance between two successive regions of complete darkness is
 - 4 mm
 - 5.6 mm
 - 14 mm
 - 28 mm
- In the ideal double-slit experiment, when a glass-plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wavelength λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is
 - 2λ
 - $\frac{2\lambda}{3}$
 - $\frac{\lambda}{3}$
 - λ
- Young's double slit experiment is carried out by using green, red and blue light, one color at a time. The fringe widths recorded are β_G , β_R and β_B respectively. Then,
 - $\beta_G > \beta_B > \beta_R$
 - $\beta_B > \beta_R > \beta_G$
 - $\beta_R > \beta_B > \beta_G$
 - $\beta_R > \beta_G > \beta_B$
- Huygens' conception of secondary waves
 - helps us to find the focal length of a thick lens
 - is a geometrical method to find the position of a wave-front at a later or an earlier instant
 - is used to determine the velocity of light
 - is used to explain polarization of light
- Longitudinal waves do not exhibit
 - refraction
 - reflection
 - diffraction
 - polarization
- The idea of the quantum nature of light has emerged in an attempt to explain
 - interference
 - diffraction
 - polarization
 - radiation spectrum of a black body
- In the spectrum of light of a luminous heavenly body the wavelength of a spectral line is measured to be 4747 Å while actual wavelength of the line is 4700 Å. The relative velocity of the heavenly body with respect to earth will be (velocity of light is $3 \times 10^8 \text{ ms}^{-1}$)
 - $3 \times 10^5 \text{ ms}^{-1}$ moving towards the earth
 - $3 \times 10^5 \text{ ms}^{-1}$ moving away from the earth
 - $3 \times 10^6 \text{ ms}^{-1}$ moving towards the earth
 - $3 \times 10^6 \text{ ms}^{-1}$ moving away from the earth
- A grating has $5000 \text{ lines cm}^{-1}$. The maximum order visible with wavelength 6000 Å
 - 2
 - 3
 - 4
 - 0
- A beam of monochromatic light enters from vacuum into a medium of refractive index n . The ratio of the wavelengths of the incident and refracted waves is
 - $n:1$
 - $1:n$
 - $n^2:1$
 - $1:n^2$

11. In Young's double slit experiment, 62 fringes are seen in visible region for sodium light of wavelength $5893 \text{ \AA}$. If violet light of wavelength $4358 \text{ \AA}$ is used in place of sodium light, then number of fringes seen will be
(A) 54 (B) 64
(C) 74 (D) 84
12. Monochromatic light of wavelength $5000 \text{ \AA}$ illuminates a pair of slits 1 mm apart. The separation of bright fringes in the interference pattern formed on a screen 2 m away is
(A) 0.25 mm (B) 0.1 mm
(C) 0.01 mm (D) 1.0 mm
13. Air has refractive index 1.0003 . The thickness of an air column, which will have one more wavelength of yellow light ($6000 \text{ \AA}$) than in the same thickness of vacuum is
(A) 2 mm (B) 2 cm
(C) 2 m (D) 2 km
14. Two coherent monochromatic light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are
(A) $4I$ and I (B) $5I$ and $3I$
(C) $9I$ and I (D) $9I$ and $3I$
15. There is a wavelength corresponding to each colour. How many colours are possible, then
(A) 3 (B) 1
(C) 7 (D) None of these
16. Though quantum theory of light can explain a number of phenomena observed with light, it is necessary to retain the wave nature of light to explain the phenomenon of
(A) photo-electric effect
(B) diffraction
(C) Compton effect
(D) black body radiation
17. Which of the following cannot be polarized?
(A) Radio wave (B) X-rays
(C) Infrared radiation (D) Sound waves in air
18. A blue object on a white background when seen through a blue filter will appear
(A) blue on a white background
(B) black on a blue background
(C) blue on red background
(D) invisible
19. Illumination of the sun at noon is maximum because
(A) the sun is nearer to the earth at noon
(B) rays are incident almost normally
(C) refraction of light is minimum at noon
(D) scattering is reduced at noon
20. In Young's double slit experiment, carried out with light of wavelength $\lambda = 5000 \text{ \AA}$, the distance between the slits is 0.2 mm and the screen is at 200 cm from the slits. The central maximum is at $x=0$. The third maximum (taking the central maximum as zeroth maximum) will be at x equal to
(A) 1.67 cm (B) 1.5 cm
(C) 0.5 cm (D) 5.0 cm
21. Ray optics is valid when characteristic dimensions are
(A) of the same order as the wavelength of light
(B) much smaller than the wavelength of light
(C) much larger than the wavelength of light
(D) of the order of 1 mm
22. In order that a thin film of oil floating on the surface of water shows colours due to interference, the thickness of the oil film should be of the order of
(A) 1 cm (B) $10 \text{ \AA}$
(C) $5000 \text{ \AA}$ (D) $10000 \text{ \AA}$
23. The blue cross on a white background illuminated with white light is observed through a red filter. The pattern seen is
(A) a red cross on a black background
(B) a blue cross on a red background
(C) a red cross on a blue background
(D) a black cross on a blue background
24. The deflection of light in a gravitational field was predicted first by
(A) Einstein (B) Newton
(C) Max Planck (D) Maxwell
25. Both the particle and wave aspects of the wave aspects of light appear to be used in
(A) photoelectric effect
(B) gamma emission
(C) interference
(D) classical mechanics
26. At sunset, the sun seems to be
(A) higher than it really is
(B) lower than it really is
(C) exactly where it really is
(D) lower than it would be at sunrise
27. In Huygens' wave theory, the locus of all the points in the same state of vibration is called a
(A) half period zone (B) vibrator
(C) wavefront (D) ray
28. In Young's experiment, monochromatic light is used to illuminate the two slits A and B . Interference fringes are observed on a screen placed in front of the slits.

Now if a thin glass plate is placed normally in the path of the beam coming from the slit

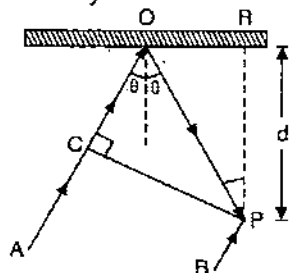


- (A) The fringes will disappear
(B) The fringe width will increase
(C) The fringe width will increase
(D) There will be no change in the fringe width but the pattern shifts

29. When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is

- (A) ZERO (B) I_0
(C) $\frac{1}{2}I_0$ (D) $\frac{1}{4}I_0$

30. In the diagram, CP represent a wavefront and AO and BP, the corresponding two rays. Find the condition on θ for constructive interference at P between the ray BP and reflected ray OP



- (A) $\cos\theta = \frac{3\lambda}{2d}$ (B) $\cos\theta = \frac{\lambda}{4d}$
(C) $\sec\theta - \cos\theta = \frac{\lambda}{d}$ (D) $\sec\theta - \cos\theta = \frac{4\lambda}{d}$

31. In Young's double-slit experiment the separation between the slits is doubled and the distance between the slit and the screen is halved. The fringe-width becomes

- (A) one-fourth (B) half
(C) double (D) quadruple

32. In Young's double slit experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width is

- (A) unchanged (B) halved
(C) doubled (D) quadrupled

33. Two coherent point sources s_1 and s_2 vibrating in phase emit light of wavelength λ . The separation between the sources is 2λ . The smallest distance from s_2 on a line passing through s_2 and perpendicular to s_1s_2 , where a minimum of intensity occurs is

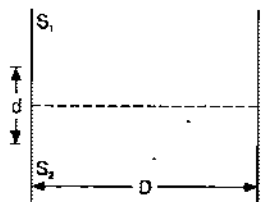
- (A) $\frac{7\lambda}{12}$ (B) $\frac{15\lambda}{4}$
(C) $\frac{\lambda}{2}$ (D) $\frac{3\lambda}{4}$

34. In a Young's double slit experiment, the fringes are displaced by a distance x when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of same thickness, the shift of fringes is $\left(\frac{3}{2}\right)x$. The

refractive index of second plate is

- (A) 1.75 (B) 1.50
(C) 1.25 (D) 1.00

35. Consider a usual set-up of Young's double slit experiment with slits of equal intensity as shown in the figure. Take O as origin and the Y axis as indicated. If average intensity between $y_1 = -\frac{\lambda D}{4d}$ and $y_2 = +\frac{\lambda D}{4d}$ equals n times the intensity of maxima, then n equals (take average over phase difference)



- (A) $\frac{1}{2}\left(1 + \frac{2}{\pi}\right)$ (B) $2\left(1 + \frac{2}{\pi}\right)$
(C) $\left(1 + \frac{2}{\pi}\right)$ (D) $\frac{1}{2}\left(1 - \frac{2}{\pi}\right)$

36. A plate of thickness t made of a material of refractive index μ is placed in front of one of the slits in a double slit experiment. What should be the minimum thickness t which will make the intensity at the centre of the fringe pattern zero?

- (A) $(\mu - 1)\frac{\lambda}{2}$ (B) $(\mu - 1)\lambda$
(C) $\frac{\lambda}{2(\mu - 1)}$ (D) $\frac{\lambda}{(\mu - 1)}$

37. Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between

- these polaroids then the intensity of light emerging from the last polaroid will be
- (A) $\left(\frac{I_0}{8}\right)\sin^2 2\theta$ (B) $\left(\frac{I_0}{4}\right)\sin^2 2\theta$
 (C) $\left(\frac{I_0}{2}\right)\cos^4 \theta$ (D) $I_0 \cos^4 \theta$
38. In Young's double slit experiment intensity at a point is $\left(\frac{1}{4}\right)$ of the maximum intensity. Angular position of this point is
- (A) $\sin^{-1}\left(\frac{\lambda}{d}\right)$ (B) $\sin^{-1}\left(\frac{\lambda}{2d}\right)$
 (C) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$ (D) $\sin^{-1}\left(\frac{\lambda}{4d}\right)$
39. In the Young's double slit experiment using a monochromatic light of wavelength λ , the path difference (in terms of an integer n) corresponding to any point having half the peak intensity is
- (A) $(2n+1)\frac{\lambda}{2}$ (B) $(2n+1)\frac{\lambda}{4}$
 (C) $(2n+1)\frac{\lambda}{8}$ (D) $(2n+1)\frac{\lambda}{16}$
40. If two slightly different wavelengths are present in the light used in Young's double-slit experiment, then
- (A) the sharpness of fringes will be more than the case when only one wavelength is present
 (B) the sharpness of fringes will decrease as we move away from the central fringe
 (C) the central fringe will be white
 (D) the central fringe will be dark
41. Two identical coherent sources placed on a diameter of a circle of radius R at separation x ($x \ll R$) symmetrically about the centre of the circle. The sources emit identical wavelength λ each. The number of points on the circle with maximum intensity is ($x = 5\lambda$)
- (A) 20 (B) 22
 (C) 24 (D) 26
42. Laser is
- (A) intense, coherent and monochromatic
 (B) only intense and coherent
 (C) only coherent and monochromatic
 (D) only intense and monochromatic
43. Imagine a hypothetical convex lens material which can transmit all the following radiation. This lens will have minimum focal length for
- (A) ultraviolet rays (B) infrared rays
 (C) radio waves (D) X-rays
44. A star emitting yellow light starts accelerating towards earth, its colour as seen from the earth will
- (A) turn gradually red
 (B) turn suddenly red
 (C) remains same
 (D) turn gradually blue
45. The transverse nature of light is shown by
- (A) interference of light
 (B) refraction of light
 (C) polarization of light
 (D) dispersion of light
46. Laser light is considered to be coherent, because it consists of
- (A) many wavelengths
 (B) uncoordinated wavelengths
 (C) coordinated waves of exactly the same wavelength
 (D) divergent beams
47. The wavelength of light observed on the earth, from a moving star is found to decrease by 0.05%. Relative to the earth the star is
- (A) Moving away with a velocity of $1.5 \times 10^5 \text{ ms}^{-1}$
 (B) Coming closer with a velocity of $1.5 \times 10^5 \text{ ms}^{-1}$
 (C) Moving away with a velocity of $1.5 \times 10^4 \text{ ms}^{-1}$
 (D) Coming closer with a velocity of $1.5 \times 10^4 \text{ ms}^{-1}$
48. A beam of electron is used in an YDSE experiment. The slit width is d . When the velocity of electron is increased, then
- (A) No interference is observed
 (B) Fringe width increases
 (C) Fringe width decreases
 (D) Fringe width remains same
49. The ratio of the intensity at the centre of a bright fringe to the intensity at a point one-quarter of the distance between two fringe from the centre is
- (A) 2 (B) $\frac{1}{2}$
 (C) 4 (D) 16
50. The ratio of intensities of consecutive maxima in the diffraction pattern due to a single slit is
- (A) 1 : 4 : 9 (B) 1 : 2 : 3
 (C) $1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$ (D) $1 : \frac{1}{\pi^2} : \frac{9}{\pi^2}$
51. A beam of natural light falls on a system of 6 polaroids, which are arranged in succession such that each polaroid is turned through 30° with respect to the preceding one. The percentage of incident intensity that passes through the system will be
- (A) 100% (B) 50%
 (C) 30% (D) 12%

52. The maximum number of possible interference maxima for slit-separation equal to twice the wavelength in Young's double-slit experiment is
(A) Infinite (B) Five
(C) Three (D) Zero

53. A rocket is going towards moon with a speed v . The astronaut in the rocket sends signals of frequency ν towards the moon and receives them back on reflection from the moon. What will be the frequency of the signal received by the astronaut (Take $v \ll c$)

- (A) $\frac{c}{c-v} \nu$ (B) $\frac{c}{c-2v} \nu$
(C) $\frac{2v}{c} \nu$ (D) $\frac{2c}{v} \nu$

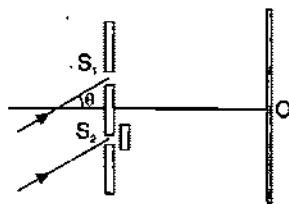
54. In Young's double slit experiment the y -coordinates of central maxima and 10th maxima are 2 cm and 5 cm respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5 the corresponding y -coordinates will be

- (A) 2 cm, 7.5 cm (B) 3 cm, 6 cm
(C) 2 cm, 4 cm (D) $\frac{4}{3}$ cm, $\frac{10}{3}$ cm

55. In Young's double slit experiment how many maxima can be obtained on a screen (including the central maximum) on both sides of the central fringe if $\lambda = 2000 \text{ \AA}$ and $d = 7000 \text{ \AA}$

- (A) 12 (B) 7
(C) 18 (D) 4

56. A monochromatic beam of light falls on YDSE apparatus at some angle (say θ) as shown in figure. A thin sheet of glass is inserted in front of the lower slit S_2 . The central bright fringe (path difference = 0) will be obtained



- (A) At O
(B) Above O
(C) Below O
(D) Anywhere depending on angle θ , thickness of plate t and refractive index of glass μ

57. In Young's double slit experiment, the two slits act as coherent sources of equal amplitude A and wavelength λ . In another experiment with the same set up the two slits are of equal amplitude A and wavelength λ but are incoherent. The ratio of the

intensity of light at the mid-point of the screen in the first case to that in the second case is

- (A) 1:2 (B) 2:1
(C) 4:1 (D) 1:1

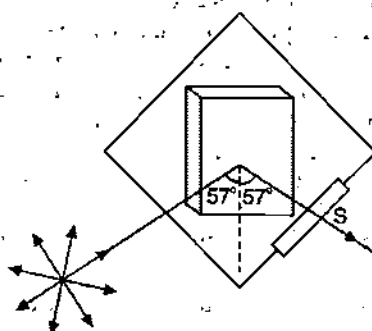
58. In the ideal double-slit experiment, when a glass-plate (refractive index 1.5) of thickness t is introduced in the path of one of the interfering beams (wavelength λ), the intensity at the position where the central maximum occurred previously remains unchanged. The minimum thickness of the glass-plate is

- (A) 2λ (B) $\frac{2\lambda}{3}$
(C) $\frac{\lambda}{3}$ (D) λ

59. In a Young's double-slit experiment, the intensity ratio of maxima and minima is infinite. The ratio of the amplitudes of two sources

- (A) is infinity (B) is unity
(C) is two (D) cannot be predicted

60. Figure represents a glass plate placed vertically on a horizontal table with a beam of unpolarised light falling on its surface at the polarising angle of 57° with the normal. The electric vector in the reflected light on screen S will vibrate with respect to the plane of incidence in a



- (A) Vertical plane
(B) Horizontal plane
(C) Plane making an angle of 45° with the vertical
(D) Plane making an angle of 57° with the horizontal

61. A clear sheet of polaroid is placed on the top of similar sheet so that their axes make an angle $\sin^{-1}\left(\frac{3}{5}\right)$ with each other. The ratio of intensity of the emergent light to that of unpolarised incident light is

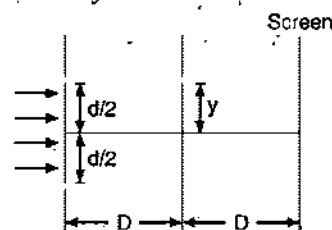
- (A) 16:25 (B) 9:25
(C) 4:5 (D) 8:25

62. Optically active substances are those substances which

- (A) produce polarized light
(B) rotate the plane of polarization of polarized light
(C) produce double refraction

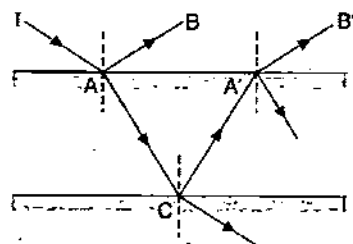
- (D) convert a plane polarized light into circularly polarized light
63. If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled
- (A) I_0 (B) $\frac{I_0}{2}$
(C) $2I_0$ (D) $4I_0$
64. The critical angle of a certain medium is $\sin^{-1}\left(\frac{3}{5}\right)$. The polarising angle of the medium is
- (A) $\sin^{-1}\left(\frac{4}{5}\right)$ (B) $\tan^{-1}\left(\frac{5}{3}\right)$
(C) $\tan^{-1}\left(\frac{3}{4}\right)$ (D) $\tan^{-1}\left(\frac{4}{3}\right)$
65. The ray of light is incident on glass of refractive index 1.5 at polarising angle. The angle of deviation of the incident ray in glass is
- (A) 57° (B) 33°
(C) 24° (D) 114°
66. Double refraction of light is shown by
- (A) quartz and calcite only
(B) calcite only
(C) calcite and ice only
(D) calcite, ice and quartz
67. A slit of width a is illuminated by red light of wavelength $6500 \text{ \AA}$. The first minimum will fall at $\theta = 30^\circ$ if a is
- (A) $3250 \text{ \AA}$ (B) $6.5 \times 10^{-4} \text{ mm}$
(C) $1.3 \text{ \mu m}$ (D) $2.6 \times 10^{-4} \text{ cm}$
68. The resolution of the human eye is $1'$. The resolving power of the human eye is nearly
- (A) 360 (B) 3600
(C) 36000 (D) 360000
69. Colours of thin films are due to
- (A) dispersion of light (B) interference of light
(C) absorption of light (D) scattering of light
70. A person standing at a distance of 3.6 km can just resolve two poles. The distance between the poles is
- (A) 0.1 m (B) 100 m
(C) 1 m (D) 10 m
71. A heavenly body is receding from earth such that the fractional change in λ is 1, then its velocity is
- (A) c (B) $\frac{3c}{5}$
(C) $\frac{c}{5}$ (D) $\frac{2c}{5}$
72. A star is moving towards the earth with a speed of $4.5 \times 10^6 \text{ ms}^{-1}$. If the true wavelength of a certain line in the spectrum received from the star is $5890 \text{ \AA}$, its apparent wavelength will be about ($c = 3 \times 10^8 \text{ ms}^{-1}$)
- (A) $5890 \text{ \AA}$ (B) $5978 \text{ \AA}$
(C) $5802 \text{ \AA}$ (D) $5896 \text{ \AA}$
73. Lights of wavelength $\lambda_1 = 4500 \text{ \AA}$, $\lambda_2 = 6000 \text{ \AA}$ are sent through a double-slit arrangement simultaneously. Then
- (A) no interference pattern will be formed
(B) the third bright fringe of λ_1 will coincide with the fourth bright fringe of λ_2
(C) the third bright fringe of λ_2 will coincide with fourth bright fringe of λ_1
(D) the fringes of wavelength λ_1 will be wider than the fringes of wavelength λ_2
74. Two slits separated by a distance of 1 mm are illuminated with light of wavelength $6 \times 10^{-7} \text{ m}$. The interference fringes are observed on a screen placed 1 m from the slits. The distance between the second dark fringe and the fourth bright fringe is equal to
- (A) 0.5 mm (B) 1.0 mm
(C) 1.5 mm (D) 2.0 mm
75. Interference fringes were produced in Young's double-slit experiment using light of wavelength $5000 \text{ \AA}$. When a film of thickness $2.5 \times 10^{-3} \text{ cm}$ was placed in front of one of the slits, the fringe pattern shifted by a distance equal to 20 fringe-width. The refractive index of the material of the film is
- (A) 1.25 (B) 1.35
(C) 1.4 (D) 1.5
76. In Young's double-slit experiment, an interference pattern is obtained on a screen by a light of wavelength $6000 \text{ \AA}$, coming from the coherent sources S_1 and S_2 . At certain point P on the screen third dark fringe is formed. Then the path difference $S_1P - S_2P$ in microns is
- (A) 0.75 (B) 1.5
(C) 3.0 (D) 4.5
77. Young's double slit experiment is made in a liquid. The 10th bright fringe in liquid lies where 6th dark fringe lies in vacuum. The refractive index of the liquid is approximately
- (A) 1.8 (B) 1.54
(C) 1.67 (D) 1.2

78. A point source emits light equally in all directions. Two points P and Q are at distances 9 m and 25 m respectively from the source. The ratio of the amplitudes of the waves P and Q is
(A) 9:25 (B) 25:9
(C) $9^2:25^2$ (D) $25^2:9^2$
79. In Young's double slit experiment the y-co-ordinates of central maxima and 10th maxima are 2 cm and 5 cm respectively. When the YDSE apparatus is immersed in a liquid of refractive index 1.5 the corresponding y-co-ordinates will be
(A) 2 cm, 7.5 cm (B) 3 cm, 6 cm
(C) 2 cm, 4 cm (D) $\frac{4}{3}$ cm, $\frac{10}{3}$ cm
80. In Young's double slit experiment $\frac{d}{D} = 10^{-4}$ (d = distance between slits, D = distance of screen from the slits). At a point P on the screen resulting intensity is equal to the intensity due to individual slit I_0 . Then the distance of point P from the central maximum is ($\lambda = 6000 \text{ \AA}$)
(A) 2 mm (B) 1 mm
(C) 0.5 mm (D) 4 mm
81. The Young's double-slit experiment is carried out with light of wavelength $5000 \text{ \AA}$. The distance between the slits is 0.2 mm and the screen is at 200 cm from the slits. The central maximum is at $x = 0$. The third maximum will be at x equal to
(A) 1.67 cm (B) 1.5 cm
(C) 0.5 cm (D) 5.0 cm
82. In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm , number of fringes observed in the same segment of the screen is given by
(A) 12 (B) 18
(C) 24 (D) 30
83. The blue colour of the sky is explained by
(A) refraction (B) reflection
(C) polarisation (D) scattering
84. A Young's double slit experiment uses a monochromatic source. The shape of the interference fringes formed on a screen is
(A) Straight line (B) Parabola
(C) Hyperbola (D) Circle
85. Among the two interfering monochromatic sources A and B; A is ahead of B in phase by 66° . If the observation be taken from point P, such that $PB - PA = \frac{\lambda}{4}$. Then the phase difference between the waves from A and B reaching P is
(A) 156° (B) 140°
(C) 136° (D) 126°
86. Two coherent sources S_1 and S_2 are separated by a distance four times the wavelength λ of the source. The sources lie along y-axis whereas a detector moves along +x-axis. Leaving the origin and far off points the number of points where maxima are observed is
(A) 2 (B) 3
(C) 4 (D) 5
87. The first minimum due to a Fraunhofer diffraction using light of wavelength 500 nm and a slit of width 0.5 mm will be formed at an angle (in minutes)
(A) 2.42 (B) 3.43
(C) 4.84 (D) 1.71
88. Aperture of the human eye is 2 mm. Assuming the mean wavelength of light to be $5000 \text{ \AA}$, the angular resolution limit of the eye is nearly
(A) 2 minute (B) 1 minute
(C) 0.5 minute (D) 1.5 minute
89. In the Young's double slit experiment apparatus shown in figure, the ratio of maximum to minimum intensity on the screen is 9. The wavelength of light used is λ , then the value of y is



- (A) $\frac{\lambda D}{d}$ (B) $\frac{\lambda D}{2d}$
(C) $\frac{\lambda D}{3d}$ (D) $\frac{\lambda D}{4d}$
90. In Young's experiment, using red and blue lights of wavelengths $7800 \text{ \AA}$ and $5200 \text{ \AA}$ respectively, the value of n for which n th red fringe coincides with $(n+1)$ th blue fringe is
(A) 2 (B) 3
(C) 4 (D) 5
91. In Young's experiment for interference of light with two slits, maxima occur at angles for which $\sin \theta = \frac{m\lambda}{d}$. Here d is

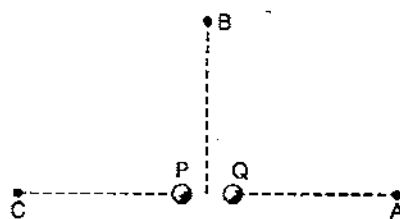
- (A) distance of slits from the screen
(B) distance between dark and bright fringes
(C) distance between slits
(D) width of m th fringe
92. Interference is observed in a chamber with air present inside the chamber. The chamber is then evacuated and the same light is again used to produce interference. A careful observer will see
(A) no change in the pattern
(B) that the fringewidth slightly increases
(C) that the fringewidth slightly decreases
(D) no interference pattern
93. Finger prints of a piece of paper may be detected by sprinkling fluorescent powder on the paper and then looking into it under
(A) yellow light (B) brightness
(C) infrared light (D) ultraviolet light
94. Two nicol prisms (polariser and analyser) have their axes at angles of 30° in between. If I is the intensity of light falling on first nicol, then that of emerging light is
(A) $0.125I$ (B) $0.25I$
(C) $0.375I$ (D) $0.5I$
95. The Young's double-slit experiment is performed with blue light and green light of wavelengths $4360 \text{ \AA}$ and $5460 \text{ \AA}$ respectively. If X is the distance of 4th maximum from the central one, then
(A) $X(\text{Blue}) = X(\text{Green})$
(B) $X(\text{Blue}) < X(\text{Green})$
(C) $X(\text{Blue}) > X(\text{Green})$
(D) $\frac{X(\text{Blue})}{X(\text{Green})} = \frac{5460}{4360}$
96. A ray of light of intensity I is incident on a parallel glass-slab at a point A as shown in figure. It undergoes partial reflection and refraction. At each reflection 25% of incident energy is reflected. The rays AB and $A'B'$ undergo interference. The ratio $I_{\max}/I_{\min}$ is



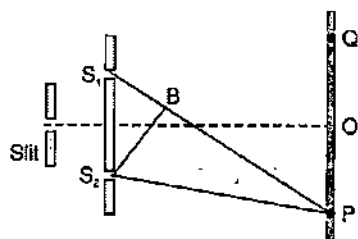
- (A) 4:1 (B) 8:1
(C) 7:1 (D) 49:1
97. Plane polarised light is passed through a polaroid. On viewing through the polaroid we find that when the

polaroid is given one complete rotation about the direction of light

- (A) the intensity of light gradually decreases to zero and remains at zero
(B) the intensity of light gradually increases to a maximum and remains maximum
(C) there is no change in the intensity of light
(D) the intensity of light varies such that it is twice maximum and twice zero
98. Figure here shows P and Q as two equally intense coherent sources emitting radiations of wavelength 20 m . The separation PQ is 5 m and phase of P is ahead of the phase of Q by 90° . A , B and C are three distant points of observation equidistant from the mid-point of PQ . The intensity of radiations at A , B , C will bear the ratio



- (A) 0:1:4 (B) 4:1:0
(C) 0:1:2 (D) 2:1:0
99. In the figure is shown Young's double slit experiment. Q is the position of the first bright fringe on the right side of O . P is the 11th fringe on the other side, as measured from Q . If the wavelength of the light used is $6000 \times 10^{-10} \text{ m}$, then S_1B will be equal to

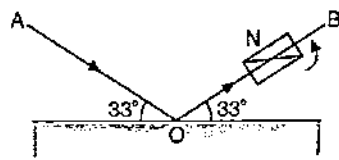


- (A) $6 \times 10^{-6} \text{ m}$ (B) $6.6 \times 10^{-6} \text{ m}$
(C) $3.138 \times 10^{-7} \text{ m}$ (D) $3.144 \times 10^{-7} \text{ m}$
100. The maximum intensity in Young's double slit experiment is I_0 . Distance between the slits is $d = 5\lambda$, where λ is the wavelength of monochromatic light used in the experiment. What will be the intensity of light in front of one of the slits on a screen at a distance $D = 10d$
- (A) $\frac{I_0}{2}$ (B) $\frac{3}{4}I_0$
(C) I_0 (D) $\frac{I_0}{4}$

101. The polarising angle of diamond is 67° . The critical angle of diamond is nearest to
 (A) 22° (B) 34°
 (C) 45° (D) 60°

102. No longitudinal wave will show
 (A) interference (B) diffraction
 (C) TIR (D) polarisation

103. A beam of light AO is incident on a glass slab ($\mu = 1.54$) in a direction as shown in figure. The reflected ray OB is passed through a Nicol prism on viewing through a Nicol prism, we find on rotating the prism that

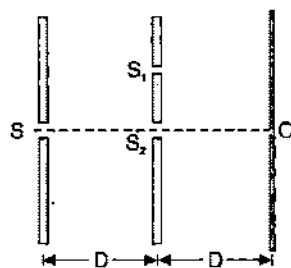


- (A) The intensity is reduced down to zero and remains zero
 (B) The intensity reduces down some what and rises again
 (C) There is no change in intensity
 (D) The intensity gradually reduces to zero and then again increases

104. When one of the slits of Young's experiment is covered with a transparent sheet of thickness 4.8 mm , the central fringe shifts to a position originally occupied by the 30^{th} bright fringe. What should be the thickness of the sheet if the central fringe has to shift to the position occupied by 20^{th} bright fringe

- (A) 3.8 mm (B) 1.6 mm
 (C) 7.6 mm (D) 3.2 mm

105. Two ideal slits S_1 and S_2 are at a distance d apart, and illuminated by light of wavelength λ passing through an ideal source slit S placed on the line through S_2 as shown. The distance between the planes of slits and the source slit is D . A screen is held at a distance D from the plane of the slits. The minimum value of d for which there is darkness at O is



- (A) $\sqrt{\frac{3\lambda D}{2}}$ (B) $\sqrt{\lambda D}$

- (C) $\sqrt{\frac{\lambda D}{2}}$ (D) $\sqrt{3\lambda D}$

106. Two waves originating from sources S_1 and S_2 having zero phase difference and common wavelength λ will show completely destructive interference at a point P if $S_1P - S_2P$ is

- (A) 5λ (B) $\frac{3\lambda}{4}$
 (C) 2λ (D) $\frac{11\lambda}{2}$

107. A thin air film between a plane glass plate and a convex lens is irradiated with parallel beam of monochromatic light and is observed under a microscope. We see

- (A) uniform brightness
 (B) complete darkness
 (C) field crossed over by concentric bright and dark rings
 (D) field crossed over by straight bright and dark fringes

108. In Young's double-slit experiment, the intensity of light at a point on the screen where the path difference is λ is I , λ being the wavelength of light used. The intensity at a point where the path difference is $\frac{\lambda}{4}$ will be

- (A) $\frac{I}{4}$ (B) $\frac{I}{2}$
 (C) I (D) ZERO

109. In a two slit experiment with monochromatic light fringes are obtained on a screen placed at some distance from the slits. If the screen is moved by $5 \times 10^{-2} \text{ m}$ towards the slits, the change in fringe width is $3 \times 10^{-5} \text{ m}$. If separation between the slits is 10^{-3} m , the wavelength of light used is

- (A) $6000 \text{ \AA}$ (B) $5000 \text{ \AA}$
 (C) $3000 \text{ \AA}$ (D) $4500 \text{ \AA}$

110. In Young's experiment using monochromatic light, the fringe pattern shifts by a certain distance on the screen when a mica sheet of refractive index 1.6 and thickness 1.964 micron is introduced in the path of one of the interfering waves. The mica sheet is then removed and the distance between the slits and the screen is doubled. It is found that the distance between successive maxima now is the same as the observed fringe shift upon the introduction of the mica sheet. The wavelength of light is

- (A) $5762 \text{ \AA}$ (B) $5825 \text{ \AA}$
 (C) $5892 \text{ \AA}$ (D) $6500 \text{ \AA}$

111. The phenomenon of interference is shown by

- (A) longitudinal mechanical waves only
(B) transverse mechanical waves only
(C) non-mechanical transverse waves only
(D) all the above types of waves

112. In Young's double-slit experiment, if L is the distance between the slits and the screen upon which the interference pattern is observed, x is the average distance between the adjacent fringes and d is the slit separation, then the wavelength of light is

- (A) $\frac{xd}{L}$ (B) $\frac{xL}{d}$
(C) $\frac{Ld}{x}$ (D) $\frac{1}{Ldx}$

113. Interference can take place between

- (A) transverse waves only, but not in longitudinal waves
(B) longitudinal waves only, but not in transverse waves
(C) both longitudinal and transverse waves
(D) light waves only, but not sound waves

114. Young's double-slit experiment is performed with light of wavelength $\lambda = 6000 \text{ \AA}$. A glass plate of thickness 0.01 mm and $\mu = 1.5$ is introduced. The number of fringes shifting in the system is

- (A) 2000 (B) 8
(C) 120 (D) 4910

115. The contrast in the fringes in an interference pattern depends on

- (A) fringe width
(B) wavelength
(C) intensity ratio of the sources
(D) distance between the slits

116. In a Young's double slit experiment the source S and the two slits A and B are vertical with slit A above slit B . The fringes are observed on a vertical screen K . The optical path length from S to B is increased very slightly (by introducing a transparent material of higher refractive index) and the optical path length from S to A is not changed, as a result the fringe system on K moves

- (A) Vertically downwards slightly
(B) Vertically upwards slightly
(C) Horizontally, slightly to the left
(D) Horizontally, slightly to the right

117. In the Young's double slit experiment, if the phase difference between the two waves interfering at a point is ϕ , the intensity at that point can be expressed by the expression

(A) $I = \sqrt{A^2 + B^2 \cos^2 \phi}$ (B) $I = \frac{A}{B} \cos \phi$

(C) $I = A + B \cos \frac{\phi}{2}$ (D) $I = A + B \cos \phi$

Where A and B depend upon the amplitudes of the two waves.

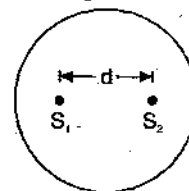
118. The time period of rotation of the sun is 25 days and its radius is $7 \times 10^8 \text{ m}$. The Doppler shift for the light of wavelength $6000 \text{ \AA}$ emitted from the surface of the sun will be

- (A) $0.04 \text{ \AA}$ (B) $0.40 \text{ \AA}$
(C) $4.00 \text{ \AA}$ (D) $40.0 \text{ \AA}$

119. A flake of glass (refractive index 1.5) is placed over one of the openings of a double slit apparatus. The interference pattern displaces itself through seven successive maxima towards the side where the flake is placed. If wavelength of the diffracted light is $\lambda = 600 \text{ nm}$, then the thickness of the flake is

- (A) 2100 nm (B) 4200 nm
(C) 8400 nm (D) None of these

120. Two coherent sources separated by distance d are radiating in phase having wavelength λ . A detector moves in a big circle around the two sources in the plane of the two sources. The angular position of $n = 4$ interference maxima is given as



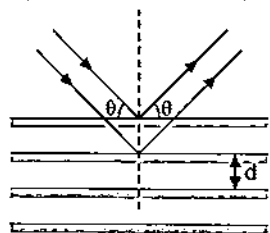
(A) $\sin^{-1} \left(\frac{n\lambda}{d} \right)$ (B) $\cos^{-1} \left(\frac{4\lambda}{d} \right)$

(C) $\tan^{-1} \left(\frac{d}{4\lambda} \right)$ (D) $\cos^{-1} \left(\frac{\lambda}{4d} \right)$

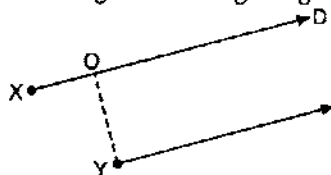
121. White light may be considered to be a mixture of waves with λ ranging between $3900 \text{ \AA}$ and $7800 \text{ \AA}$. An oil film of thickness $10,000 \text{ \AA}$ is examined normally by reflected light. If $\mu = 1.4$, then the film appears bright for

- (A) $4308 \text{ \AA}, 5091 \text{ \AA}, 6222 \text{ \AA}$
(B) $4000 \text{ \AA}, 5091 \text{ \AA}, 5600 \text{ \AA}$
(C) $4667 \text{ \AA}, 6222 \text{ \AA}, 7000 \text{ \AA}$
(D) $4000 \text{ \AA}, 4667 \text{ \AA}, 5600 \text{ \AA}, 7000 \text{ \AA}$

122. The k line of singly ionised calcium has a wavelength of 393.3 nm as measured on earth. In the spectrum of one of the observed galaxies, this spectral line is located at 401.8 nm . The speed with which the galaxy is moving away from us, will be

- (A) 6480 kms^{-1} (B) 3240 kms^{-1}
(C) 4240 kms^{-1} (D) None of these
123. In a YDSE bi-chromatic light of wavelengths 400 nm and 560 nm are used. The distance between the slits is 0.1 mm and the distance between the plane of the slits and the screen is 1 m . The minimum distance between two successive regions of complete darkness is
(A) 4 mm (B) 5.6 mm
(C) 14 mm (D) 28 mm
124. In a double slit arrangement fringes are produced using light of wavelength $4800 \text{ \AA}$. One slit is covered by a thin plate of glass of refractive index 1.4 and the other with another glass plate of same thickness but of refractive index 1.7 . By doing so the central bright shifts to original fifth bright fringe from centre. Thickness of glass plate is
(A) $8 \text{ }\mu\text{m}$ (B) $6 \text{ }\mu\text{m}$
(C) $4 \text{ }\mu\text{m}$ (D) $10 \text{ }\mu\text{m}$
125. Polarisation of light establishes
(A) corpuscular theory of light
(B) quantum nature of light
(C) transverse nature of light
(D) all the three
126. A ray of unpolarised light is incident on glass plate at the polarising angle, then
(A) the reflected and transmitted rays will be completely plane polarised
(B) the reflected ray is completely polarised and the transmitted ray is partially polarised
(C) the reflected ray is partially polarised and the transmitted ray is completely polarised
(D) the reflected ray and the transmitted ray will be partially polarised
127. To observe diffraction, the size of the obstacle
(A) should be of the same order as the wavelength
(B) should be much larger than the wavelength
(C) has no relation to wavelength
(D) should be exactly half the wavelength
128. A slit is illuminated by red light of wavelength $6500 \text{ \AA}$. The first minimum is obtained at $\theta = 30^\circ$. The width of the slit is
(A) $3200 \text{ \AA}$ (B) 1.24 micron
(C) $6.5 \times 10^{-4} \text{ mm}$ (D) $2.6 \times 10^{-4} \text{ cm}$
129. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ . If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is
- (A) $90^\circ + \phi$ (B) $\sin^{-1}(\mu \cos \phi)$
(C) 90° (D) $90^\circ - \sin^{-1}\left(\frac{\sin \phi}{\mu}\right)$
130. A beam of plane polarized light falls normally on a polarizer of cross sectional area $3 \times 10^{-4} \text{ m}^2$. Flux of energy of incident ray is 10^{-3} W . The polarizer rotates with an angular frequency of 31.4 rads^{-1} . The energy of light passing through the polarizer per revolution will be
(A) 10^{-4} Joule (B) 10^{-3} Joule
(C) 10^{-2} Joule (D) 10^{-1} Joule
131. A beam with wavelength λ falls on a stack of partially reflecting planes with separation d . The angle θ that the beam should make with the planes so that the beams reflected from successive planes may interfere constructively is (where $n = 1, 2, \dots$)
- 
- (A) $\sin^{-1}\left(\frac{n\lambda}{d}\right)$ (B) $\tan^{-1}\left(\frac{n\lambda}{d}\right)$
(C) $\sin^{-1}\left(\frac{n\lambda}{2d}\right)$ (D) $\cos^{-1}\left(\frac{n\lambda}{2d}\right)$
132. A diffraction pattern is obtained using a beam of red light. If the red light is replaced by blue light, then
(A) the diffraction pattern remains unchanged
(B) diffraction bands become narrower and crowded together
(C) bands become broader and farther apart
(D) bands disappear
133. At sunrise or at sunset the sun appears to be reddish while at mid-day the sun looks white. The reason is that
(A) the sun is less hot at sunrise or at sunset than at noon
(B) diffraction sends red rays to the earth at these time
(C) refraction is responsible for this effect
(D) scattering due to dust particles and air molecules
134. Light of wavelength $6328 \text{ \AA}$ is incident normally on a slit having a width of 0.2 mm . The distance of the screen from the slit is 0.9 m . The angular width of the central maximum is
(A) 0.09 degree (B) 0.72 degree
(C) 0.18 degree (D) 0.36 degree

135. Two point sources X and Y emit waves of same frequency and speed but Y lags in phase behind X by $2\pi l$ radian. If there is a maximum in direction D the distance XO using n as an integer is given by



- (A) $\frac{\lambda}{2}(n-l)$ (B) $\lambda(n+l)$
(C) $\frac{\lambda}{2}(n+l)$ (D) $\lambda(n-l)$
136. In a Young's double slit experiment, the slits are 2 mm apart and are illuminated with a mixture of two wavelength $\lambda_0 = 750 \text{ nm}$ and $\lambda = 900 \text{ nm}$. The minimum distance from the common central bright fringe on a screen 2m from the slits where a bright fringe from one interference pattern coincides with a bright fringe from the other is
(A) 1.5 mm (B) 3 mm
(C) 4.5 mm (D) 6 mm
137. If sound waves can be assumed to be diffracted, which of the following objects will diffract sound waves in air from a 384 Hz tuning fork
(A) A sphere of radius 1 cm
(B) A sphere of radius 1 mm
(C) A sphere of radius 1 m
(D) A sphere of radius 10 m
138. A beam of light of wavelength 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between the first dark fringes on either side of the central bright fringe is
(A) 1.2 cm (B) 1.2 mm
(C) 2.4 cm (D) 2.4 mm
139. A nicol prism is based on the action of
(A) refraction (B) double refraction
(C) dichroism (D) both (B) and (C)
140. A thin sheet of glass (refractive index 1.5) of thickness 6 micron, introduced in the path of one of the interfering beams in a double-slit experiment, shifts the central fringe to a position earlier occupied by the fifth bright fringe. The wavelength of light used is
(A) 3000 Å (B) 6000 Å
(C) 4500 Å (D) 7000 Å
141. Which of the following cannot be polarised?
(A) Radio waves (B) β rays
(C) Infrared rays (D) γ rays
142. When light is incident on a transparent surface at the polarizing angle, which of the following is completely polarized?
(A) Reflected light
(B) Refracted light
(C) Both reflected as well as refracted light
(D) Neither reflected nor refracted light
143. An optically active substance
(A) produces polarized light
(B) rotates the plane of polarization of polarized light
(C) converts a plane polarized light into circularly polarized light
(D) converts a circularly polarized light into plane polarized light
144. Diffraction pattern of a single slit consists of a central bright band which is
(A) wide, and is flanked by alternate dark and bright bands of decreasing intensity
(B) narrow, and is flanked by alternate dark and bright bands of equal intensity
(C) wide, and is flanked by alternate dark and bright bands of equal intensity
(D) narrow, and is flanked by alternate dark and bright bands of decreasing intensity
145. In Young's experiment with one source and two slits, one of the slits is covered with black paper. Then
(A) the fringes will be darker
(B) the fringes will be narrower
(C) the fringes will be broader
(D) no fringes will be obtained and the screen will have uniform illumination
146. The distance between two coherent sources is 0.1 mm. The fringe width on a screen 1.2 m away from the sources is 6.0 mm. The wavelength of light used is
(A) 4000 Å (B) 5000 Å
(C) 6000 Å (D) 7200 Å
147. If three slits are used in Young's experiment instead of two, we get
(A) no fringe pattern
(B) the same fringe pattern as that with two slits
(C) a pattern with fringe width reduced to half of that in the two slit pattern
(D) alternate bright and dim fringes
148. When a transparent parallel plate of uniform thickness t and refractive index n is interposed normally in the path of a beam of light, the optical path is
(A) increased by nt
(B) decreased by nt
(C) decreased by $(n-1)t$
(D) increased by $(n-1)t$

149. In Young's experiment, monochromatic light is used to illuminate the two slits and interference fringes are observed on a screen placed in front of the slits. Now if a thin glass plate is placed normally in the path of the beam coming from one of the slits, then
- the fringes will disappear
 - the fringe-width will decrease
 - the fringe-width will increase
 - there will be no change in the fringe-width
150. If the Young's double slit-experiment is performed with white light, then the colour which will have maximum fringe width is
- Blue
 - Green
 - Yellow
 - Violet
151. In the interference pattern, energy is
- created at the positions of maxima
 - destroyed at the positions of minima
 - conserved but is redistributed
 - not conserved
152. Fluorescent tubes give more light than a filament bulb of same power because
- the tube contains gas at low temperature
 - ultraviolet light is converted into visible light by fluorescence
 - light is diffused through the walls of the tube
 - it produces more heat than bulb
153. Energies of photons of four different electromagnetic radiations are given below. The energy value corresponds to a visible photon is equal to
- 1 eV
 - 2 eV
 - 5 eV
 - 1000 eV
154. Atomic spectrum should be
- pure emission line spectrum
 - emission band spectrum
 - absorption line spectrum
 - absorption band spectrum
155. If a torch is used in place of monochromatic light in Young's experiment
- fringes will appear as for monochromatic light
 - fringes will appear for a moment and then they will disappear
 - no fringes will appear
 - only bright fringes will appear
156. A beam of unpolarized light of intensity I is passed first through a tourmaline crystal A and then through another tourmaline crystal B oriented so that its principal plane is parallel to that of A. If A is now rotated by 45° in a plane perpendicular to the direction of the incident ray, the intensity of the emergent light will be
- $\frac{I}{2}$
 - $\frac{I}{\sqrt{2}}$
 - I
 - $\frac{I}{4}$
157. Interference pattern is obtained on a screen due to two identical coherent sources of monochromatic light. The intensity of the central bright fringe is I . When one of the sources is blocked, the intensity become I_0 . The intensity in the two situations are related as
- $I = I_0$
 - $I = 2I_0$
 - $I = 3I_0$
 - $I = 4I_0$
158. The phase difference between two wave trains giving rise to a dark fringe in Young's double-slit experiment is (where n is an integer)
- ZERO
 - $2\pi n + \frac{\pi}{2}$
 - $2\pi n + \pi$
 - $2\pi n + \frac{\pi}{4}$
159. A Young's double-slit set-up for interference is shifted from air to within water. Then the
- fringe pattern disappears
 - fringewidth decreases
 - fringewidth increases
 - fringewidth remains unchanged
160. Two interfering beams have intensities in the ratio of 9:4. Then the ratio of maximum to minimum intensity in the interference pattern is
- 25:1
 - 13:5
 - 5:1
 - 3:2
161. In interference with two coherent beams of light, the fringe width is proportional
- to wavelength
 - to inverse wavelength
 - to square of wavelength
 - to inverse square of wavelength
162. The fringe pattern observed in Young's double-slit experiment is
- a diffraction pattern
 - an interference pattern
 - a combination of diffraction and interference patterns
 - neither a diffraction nor an interference pattern
163. In Young's interference experiment with one source and two slits, one slit is covered with a cellophane sheet which absorbs half the intensity. Then
- no fringes are obtained
 - bright fringes will be brighter and dark fringes will be darker
 - all fringes will be dark

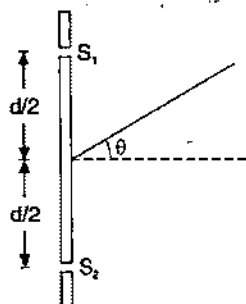
- (D) bright fringes will be less bright and dark fringes will be less dark
164. The distance between sources in a biprism of angle α and refractive index μ , if the source is placed at a distance a from it is
 (A) $2(\mu-1)\alpha$ (B) $2(\mu-1)\alpha a$
 (C) $(\mu-1)\alpha$ (D) $(\mu-1)\alpha a$
165. To obtain a sustained interference pattern, we require two sources which emit radiation of
 (A) the same frequency.
 (B) nearly the same frequency.
 (C) the same frequency having a definite phase relationship.
 (D) different wavelengths.
166. A thin mica sheet of thickness 2×10^{-6} m and refractive index ($\mu=1.5$) is introduced in the path of the first wave. The wavelength of the wave used is $5000 \text{ \AA}$. The central bright maximum will shift
 (A) 2 fringes upward (B) 2 fringes downward
 (C) 10 fringes upward (D) None of these
167. In Young's double slit experiment, the slits are 0.5 mm apart and interference pattern is observed on a screen placed at a distance of 1 m from the plane containing the slits. If wavelength of the incident light is $6000 \text{ \AA}$, then the separation between the third bright fringe and the central maxima is
 (A) 4 mm (B) 3.5 mm
 (C) 3 mm (D) 2.5 mm
168. Interference fringes are obtained due to the interference of waves from two coherent sources of light with amplitudes a_1 and a_2 ($a_1 = 2a_2$). The ratio of the maximum and minimum intensities of light in the interference pattern is
 (A) 2 (B) 4
 (C) 9 (D) ∞
169. In the double-slit experiment, the distance of the second dark fringe from the central line is 3 mm. The distance of the fourth bright fringe from the central line is
 (A) 6 mm (B) 8 mm
 (C) 12 mm (D) 16 mm
170. In Young's double-slit experiment, we get 60 fringes in the field of view if we use light of wavelength $4000 \text{ \AA}$. The number of fringes we will get in the same field of view if we use light of wavelength $6000 \text{ \AA}$ is
 (A) 40 (B) 90
 (C) 60 (D) 50
171. With a monochromatic light, the fringe-width obtained in a double-slit experiment is 1.33 mm. If the whole set-up is immersed in water of refractive index 1.33, the new fringe-width will be
 (A) 1.33 mm (B) 1 mm
 (C) $1.33 \times 1.33 \text{ mm}$ (D) $\frac{1.33}{2} \text{ mm}$
172. Two waves having amplitudes in the ratio 5:1 produce interference. The ratio of the maximum to the minimum intensity is
 (A) 25:1 (B) 6:4
 (C) 9:4 (D) 3:2
173. If the intensities of the two interfering beams in Young's double-slit experiment are I_1 and I_2 , then the contrast between the maximum and minimum intensities is good when
 (A) $|I_1 - I_2|$ is large
 (B) $|I_1 - I_2|$ is small
 (C) either I_1 or I_2 is zero
 (D) $I_1 = I_2$

Practice Exercise

Multiple Correct Choice Type Questions

This section contains Multiple Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONE OR MORE is/are correct.

1. In an interference arrangement similar to Young's double-slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by a distance $d = 150.0$ m. The intensity $I(\theta)$ is measured as a function of θ , where θ is defined as shown. If I_0 is the maximum intensity, then $I(\theta)$ for $0 \leq \theta \leq 90^\circ$ is given by



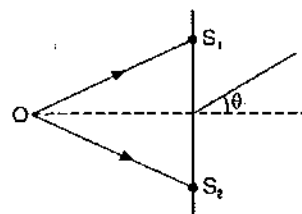
- (A) $I(\theta) = \frac{I_0}{2}$ for $\theta = 30^\circ$
 (B) $I(\theta) = \frac{I_0}{4}$ for $\theta = 90^\circ$
 (C) $I(\theta) = I_0$ for $\theta = 0^\circ$
 (D) $I(\theta)$ is constant for all values of θ

2. In a Young's double slit experiment, the separation between the two slits is d and the wavelength of the light is λ . The intensity of light falling on slit 1 is four times the intensity of light falling on slit 2. Choose the correct choice(s).

- (A) If $d = \lambda$, the screen will contain only one maximum

- (B) If $\lambda < d < 2\lambda$, at least one more maximum (besides the central maximum) will be observed on the screen
 (C) If the intensity of light falling on slit 1 is reduced so that it becomes equal to that of slit 2, the intensities of the observed dark and bright fringes will increase
 (D) If the intensity of light falling on slit 2 is increased so that it becomes equal to that of slit 1, the intensities of the observed dark and bright fringes will increase

3. In a modified YDSE experiment if point source of monochromatic light O is placed in such a manner that $OS_1 - OS_2 = \frac{\lambda}{4}$, where λ is the wavelength of light and S_1, S_2 are the slits separated by a distance 2λ . Then value(s) of θ for which a maxima is obtained is/are



- (A) $\sin^{-1}\left(\frac{1}{8}\right)$ (B) $\sin^{-1}\left(-\frac{1}{4}\right)$
 (C) $\sin^{-1}\left(\frac{5}{6}\right)$ (D) $\sin^{-1}\left(-\frac{7}{8}\right)$

4. Two coherent waves represented by

$$y_1 = A \sin\left(\frac{2\pi x_1}{\lambda} - \omega t + \frac{\pi}{4}\right) \text{ and}$$

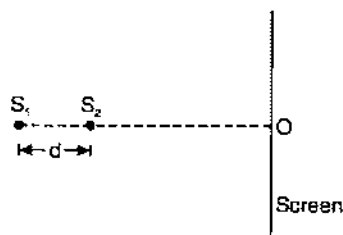
$$y_2 = A \sin\left(\frac{2\pi x_2}{\lambda} - \omega t + \frac{\pi}{6}\right)$$

are superposed. The two waves will produce

- (A) constructive interference at $x_1 - x_2 = \frac{11\lambda}{24}$
- (B) constructive interference at $x_1 - x_2 = \frac{23\lambda}{24}$
- (C) destructive interference at $x_1 - x_2 = \frac{23\lambda}{24}$
- (D) destructive interference at $x_1 - x_2 = \frac{11\lambda}{24}$

5. To observe a stationary interference pattern formed by two light waves, it is not necessary that they must have
- (A) the same frequency
 - (B) the same amplitude
 - (C) a constant phase difference
 - (D) the same intensity

6. Two point monochromatic and coherent sources of light of wavelength λ are placed on the dotted line in front of a large screen. The source emit waves in phase with each other. The distance between S_1 and S_2 is d while their distance from the screen is much larger. Then for



- (A) $d = \frac{3\lambda}{2}$, O will be a minima
- (B) $d = 3\lambda$, there will be a total of 6 minima on screen
- (C) $d = \lambda$, there will be one maxima on the screen
- (D) $d = 2\lambda$, there will be two maxima on the screen

7. If white light is used in a Young's double-slit experiment, then
- (A) bright white fringe is formed at the centre of the screen
 - (B) fringes of different colours are observed clearly only in the first order
 - (C) the first-order violet fringes are closer to the centre of the screen than the first order red fringes
 - (D) the first-order red fringes are closer to the centre of the screen than the first order violet fringes

8. In the Young's double slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes as 9. This implies that -
- (A) the intensities at the screen due to the two slits are 5 units and 4 units respectively
 - (B) the intensities at the screen due to the two slits can be 4 units and 1 unit respectively
 - (C) the amplitude ratio is 3

- (D) the amplitude ratio is 2

9. Four coherent light waves are represented by

- (i) $y_1 = a_1 \sin(\omega t)$
- (ii) $y_2 = a_2 \sin(\omega t + \phi)$
- (iii) $y_3 = a_1 \sin(2\omega t)$
- (iv) $y_4 = a_2 \sin(2\omega t + \phi)$

Interference fringes may be observed due to superposition of

- (A) (i) and (ii)
- (B) (i) and (iii)
- (C) (ii) and (iv)
- (D) (iii) and (iv)

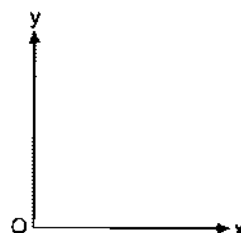
10. If one of the slits of a standard Young's double slit experiment is covered by a thin parallel slit glass so that it transmits only one half the light intensity of the other, then

- (A) The fringe pattern will get shifted towards the covered slit
- (B) The fringe pattern will get shifted away from the covered slit
- (C) The bright fringes will become less bright and the dark ones will become more bright
- (D) The fringe width will remain unchanged

11. If the first minima in a Young's slit experiment occurs directly in front of one of the slits, (distance between slits and screen $d = 5$ cm) then the wavelength of the radiation used can be

- (A) 2 cm
- (B) 4 cm
- (C) $\frac{2}{3}$ cm
- (D) $\frac{4}{3}$ cm

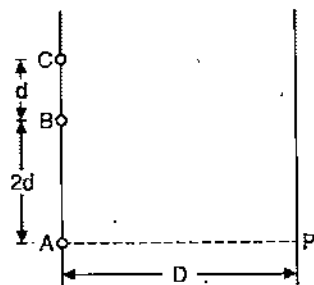
12. Two coherent sources A and B emitting light of wavelength λ are placed at positions $(-D, 0)$ and $(-D, 3\lambda)$ respectively $D \gg \lambda$



- (A) number of minima on y-axis is 6
- (B) number of minima is more than number of maxima on y-axis
- (C) number of maxima on x-axis is 3
- (D) number of maxima on x-axis is more than number of minima on x-axis

13. In the figure A, B and C are three slits each of them individually producing the same intensity I_0 at P when they are illuminated by parallel beam of light of

wavelength λ . It is given that $BP - AP = \frac{\lambda}{2}$. Also given that $d \ll D$, then wavelength λ and resultant intensity I at P will be



(A) $\lambda = \frac{2d^2}{D}$

(B) $\lambda = \frac{4d^2}{D}$

(C) $I = 2I_0$

(D) $I = I_0$

14. White light is used to illuminate the two slits in a Young's double-slit experiment. The separation between the slits is b and the screen is at a distance $d (\gg b)$ from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are

(A) $\lambda = \frac{b^2}{d}$

(B) $\lambda = \frac{2b^2}{d}$

(C) $\lambda = \frac{b^2}{3d}$

(D) $\lambda = \frac{2b^2}{3d}$

15. The fringe width in Young's double-slit experiment can be increased by decreasing

(A) separation of the slits

(B) frequency of the source of light

(C) distance between slit and screen

(D) wavelength of the source of light

Practice Exercise

Reasoning Based Questions

This section contains Reasoning type questions, each having four choices (A), (B), (C) and (D) out of which ONLY ONE is correct. Each question contains STATEMENT 1 and STATEMENT 2. You have to mark your answer as

Bubble (A) If both statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1.

Bubble (B) If both statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1.

Bubble (C) If STATEMENT 1 is TRUE and STATEMENT 2 is FALSE.

Bubble (D) If STATEMENT 1 is FALSE but STATEMENT 2 is TRUE.

1. *Statement-1:*

If the phase difference between the light waves emerging from the slits of the Young's experiment is π -radian, the central fringe will be dark.

Statement-2:

Phase difference is equal to $\frac{2\pi}{\lambda}$ times the path difference.

2. *Statement-1:*

When a thin transparent sheet is placed in front of both the slits of Young's experiment, the fringe width will increase.

Statement-2:

In Young's experiment the fringe width is proportional to wavelength of the source used.

3. *Statement-1:*

In Young's double slit experiment, we observe an interference pattern on the screen if both the slits are illuminated by two bulbs of same power.

Statement-2:

The interference pattern is observed when source is monochromatic and coherent.

4. *Statement-1:*

No interference pattern is detected when two coherent sources are infinitely close to each other in simple YDSE.

Statement-2:

The fringe width is inversely proportional to the distance between the two slits in simple YDSE.

5. *Statement-1:*

The minimum slit separation d for interference to produce at least one maxima other than central maxima is 3λ .

Statement-2:

For a maxima, path difference equals $n\lambda$. The maximum value of path difference is d .

6. *Statement-1:*

Two slits in YDSE are illuminated by two different sodium lamps emitting light of same wavelength. No interference pattern is observed.

Statement-2:

To obtain interference pattern, source must be coherent. Two different light sources can never be coherent.

7. *Statement-1:*

In Young's double slit experiment interference pattern disappears when one of the slits is covered by transparent slab.

Statement-2:

Interference occurs due to superimposition of light wave from two coherent sources.

8. *Statement-1:*

When a thin transparent sheet is placed in front of both the slits of Young's experiment, the fringe width will increase.

Statement-2:

In Young's experiment the fringe width is proportional to wavelength of the source used.

9. *Statement-1:*

Total number of maxima obtained over screen remains same whether Young's Double slit experiment is performed in air or in water with same setup.

Statement-2:

$$\beta_{\text{water}} = \frac{\beta_{\text{air}}}{\mu_{\text{water}}} \quad (\text{in Young's double slit experiment}).$$

10. *Statement-1:*

Interference obeys the Law of Conservation of Energy.

Statement-2:

The energy is redistributed in case of interference.

11. *Statement-1:*

Geometrical optics can be regarded as the limiting case of wave optics.

Statement-2:

When size of obstacle or opening is very large compared to the wavelength of light then wave nature can be ignored and light can be assumed to be travelling in straight line.

12. *Statement-1:*

Light from two coherent sources is reaching the screen. If the path difference at a point on the screen for yellow light is $\frac{3\lambda}{2}$, then the fringe at the point will be coloured.

Statement-2:

Two coherent sources always have constant phase relationship.

13. *Statement-1:*

The maximum intensity in interference pattern is four times the intensity due to each slit.

Statement-2:

Intensity is directly proportional to square of amplitude.

14. *Statement-1:*

Interference can be obtained by using two different lamps.

Statement-2:

Two different lamps are incoherent sources as constant phase difference cannot be maintained between them.

15. *Statement-1:*

Interference pattern is made by using blue light instead of red light, the fringes become narrower.

Statement-2:

In Young's double slit experiment, fringe width is given by relation $\beta = \frac{\lambda D}{d}$.

16. *Statement-1:*

Interference pattern is obtained on a screen due to two identical coherent sources of monochromatic light. The intensity at the central part of the screen becomes one-fourth if one of the source is blocked.

Statement-2:

The resultant intensity is the sum of the intensities due to two sources.

17. *Statement-1:*

Thin films such as a soap bubble or a thin layer of oil on water show beautiful colours when illuminated by monochromatic light.

Statement-2:

Colour in film are obtained due to interference between reflected light from the upper & lower layer of film.

18. *Statement-1:*

The fringe obtained at the centre of the screen is known as zeroth order fringe, or the central fringe.

Statement-2:

Path difference between the wave from S_1 and S_2 , reaching the central fringe (or zero order fringe) is zero.

19. *Statement-1:*

The phase difference between any two points on a wavefront is zero.

Statement-2:

Light from the source reaches every point of the wavefront at the same time.

20. *Statement-1:*

In Young's double slit experiment interference pattern disappears when one of the slits is closed.

Statement-2:

Interference occurs due to superimposition of light wave from two coherent sources.

21. *Statement-1:*

If a clean glass slide is observed under white light, one does not observe any colours. However, if this slide is touched with oily hands, coloured fringes appear on the slide.

Statement-2:

These fringes are due to interference of reflected light, reflected from the upper and lower surfaces of the thin oil film.

Practice Exercise

Linked Comprehension Type Questions

This section contains Linked Comprehension Type Questions or Paragraph based Questions. Each set consists of a Paragraph followed by questions. Each question has four choices (A), (B), (C) and (D), out of which only one is correct. (For the sake of competitiveness there may be a few questions that may have more than one correct options)

Comprehension 1

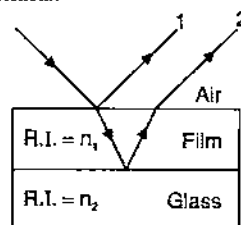
In a Young's experiment the upper slit is covered by a thin glass plate of refractive index $\mu = 1.4$. The interference pattern is observed using light of interference $5000 \text{ \AA}$. Based on above information, answer the following questions.

- It is observed that
 - the central maxima shifts upwards.
 - the central maxima shifts downwards.
 - fringe pattern will change after introduction of thin plate.
 - none of the above phenomenon is observed.
- Now a thin plate of refractive index 1.7 is placed in front of lower slit then
 - central maxima will be obtained above centre.
 - central maxima will be obtain at centre.
 - there will not be any change in central maxima after introduction of thin plate.
 - no conclusion can be withdrawn without knowing the thickness of the plate.
- Assume the thickness of both thin glass plate is t , the path difference between waves incident at center is
 - $0.5t$
 - $0.3t$
 - $0.8t$
 - $\frac{1.7}{1.5}t$
- If minima is to be obtained at center then minimum value of t is (source wavelength is λ)
 - $t = \frac{\lambda}{0.6}$
 - $t = \frac{\lambda}{0.3}$
 - $t = \lambda$
 - $t = \frac{\lambda}{0.15}$

- If intensity at center is $\frac{3}{4}$ of maximum intensity then minimum value of t is
 - $t = 2777.7 \text{ \AA}$
 - $t = 3188 \text{ \AA}$
 - $t = 4188.8 \text{ \AA}$
 - $t = 2122.9 \text{ \AA}$

Comprehension 2

A thin film of a specific material can be used to decrease the intensity of reflected light. There is destructive interference of waves reflected from upper and lower surfaces of the film. These films are called non-reflecting or antireflection coatings. The process of coating the lens or surface with non-reflecting film is called blooming as shown in the figure. The refracting index of coating (n_1) is less than that of the glass (n_2). Based on above information, answer the following questions.

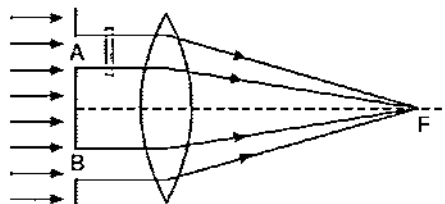


- If the light of wavelength λ is incident normally and the thickness of film is t then optical path difference between waves reflected from upper and lower surface of film is
 - $2n_1t$
 - $2n_1t - \frac{\lambda}{2}$
 - $2n_1t + \frac{\lambda}{2}$
 - $2t$

7. Magnesium fluoride (MgF_2) is generally used as anti-reflection coating. If refractive index of MgF_2 is 1.38 then minimum thickness of film required for $\lambda = 550 \text{ nm}$ is
 (A) 112.4 nm (B) 78.2 nm
 (C) 99.64 nm (D) 225 nm.
8. Assuming that the thickness of film in above problem is not technically possible to manufacture, then next thickness of film required is (approximately)
 (A) 298.9 nm (B) 271.7 nm
 (C) 304.7 nm (D) 550 nm

Comprehension 3

In a modified Young's double slit experiment, a monochromatic uniform and parallel beam of light of wavelength $6000 \text{ \AA}$ and intensity $\frac{10}{\pi} \text{ Wm}^{-2}$ is incident normally on two circular apertures A and B of radii 0.001 m and 0.002 m respectively. A perfect transparent film of thickness $2000 \text{ \AA}$ and refractive index 1.5 for the wavelength of $6000 \text{ \AA}$ is placed in front of aperture A. The lens is symmetrically placed with respect to the apertures. Assume that 10% of the power received by each aperture goes in the original direction and is brought to the focal spot.

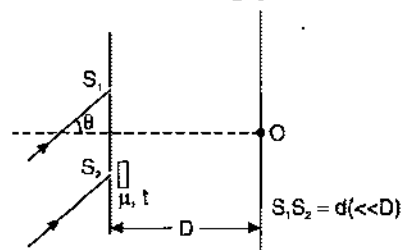


Based on the above facts, answer the following questions.

9. The power received at A is
 (A) 10^{-5} W (B) $4 \times 10^{-5} \text{ W}$
 (C) 10^{-6} W (D) $4 \times 10^{-6} \text{ W}$
10. The power received at B is
 (A) 10^{-5} W (B) $4 \times 10^{-5} \text{ W}$
 (C) 10^{-6} W (D) $4 \times 10^{-6} \text{ W}$
11. The power transmitted through A is
 (A) 10^{-5} W (B) $4 \times 10^{-5} \text{ W}$
 (C) 10^{-6} W (D) $4 \times 10^{-6} \text{ W}$
12. The power transmitted through B is
 (A) 10^{-5} W (B) $4 \times 10^{-5} \text{ W}$
 (C) 10^{-6} W (D) $4 \times 10^{-6} \text{ W}$
13. The path difference introduced by the film is
 (A) 10^{-3} m (B) 10^{-5} m
 (C) 10^{-7} m (D) 10^{-9} m
14. The phase difference introduced by the film is
 (A) π radian (B) $\frac{\pi}{2}$ radian
 (C) $\frac{\pi}{3}$ radian (D) $\frac{\pi}{4}$ radian
15. The power in watt received at the focal point F of the lens is
 (A) $2 \mu\text{W}$ (B) $5 \mu\text{W}$
 (C) $6 \mu\text{W}$ (D) $7 \mu\text{W}$

Comprehension 4

A monochromatic beam of light of wavelength $\lambda = 600 \text{ nm}$ falls on Young's double slit experiment apparatus as shown in figure. A thin sheet of glass is inserted in front of lower slit S_2 . Based on above information, answer the following questions.

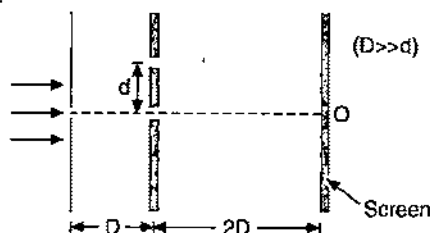


16. The central bright fringe can be obtained
 (A) at O
 (B) at O or below O
 (C) at O or above O
 (D) anywhere on the screen
17. If central bright fringe is obtained on screen at O, then we have
 (A) $(\mu - 1)t = d \sin \theta$ (B) $(\mu - 1)t = d \cos \theta$
 (C) $\mu t = d \theta$ (D) $\frac{t}{\mu - 1} = \frac{d}{\sin \theta}$
18. The phase difference between central maxima and fifth minima is
 (A) $\frac{\pi}{6}$ (B) 9π
 (C) $\frac{3\pi}{2}$ (D) $8\pi \pm \frac{\pi}{6}$
19. Fringe width for the pattern obtained on screen, if $\lambda = 600 \text{ nm}$, $\mu = 1.5$, $d = 3 \text{ mm}$, $D = 2 \text{ m}$ and $\theta = 30^\circ$ is
 (A) $2 \times 10^5 \text{ nm}$ (B) $4 \times 10^5 \text{ nm}$
 (C) 10^4 nm (D) $3 \times 10^6 \text{ nm}$

20. Assume if $\theta \left(< \frac{\pi}{2} \right)$ is increased then for a given value of μ
- central maxima will move downwards.
 - central maxima will move upward.
 - fringe width will increase.
 - fringe width will decrease.

Comprehension 5

In the arrangement shown in the figure, the distance D is large compared to the separation d between the slits. Monochromatic light of wavelength λ is incident on the slit, based on the information provided answer the following questions.



21. The minimum value of d for which there is a dark fringe at the point O is
- $\sqrt{\frac{\lambda D}{2}}$
 - $\sqrt{\frac{D\lambda}{3}}$
 - $\sqrt{\lambda D}$
 - not possible to be calculated
22. The position of first bright fringe for the minimum value of d is
- $\frac{d}{2}$ below
 - d above
 - $\frac{3d}{2}$ below
 - $\frac{3d}{2}$ above
23. The fringe width is
- $\frac{3D\lambda}{4d}$
 - $\frac{3D\lambda}{2d}$
 - $\frac{D\lambda}{d}$
 - $\frac{2D\lambda}{d}$

Comprehension 6

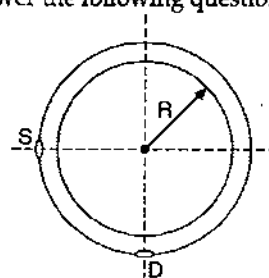
If light incident on a thin film has wavelength as 900 nm and refractive index of film is 1.5. Based on above information, answer the following questions.

24. Minimum thickness of film needed for constructive interference in reflected light system is
- 100 nm
 - 150 nm
 - 200 nm
 - 250 nm

25. Minimum thickness of film for destructive interference in transmitted light system is
- 150 nm
 - 200 nm
 - 250 nm
 - 100 nm

Comprehension 7

A narrow tube is bent in the form of circle of radius R as shown. Two small holes S and D are made in the tube at the positions right angles to each other. A source placed at S generates a wave of intensity I_0 which is equally divided into two parts. One part travels along the longer path, while the other travels along the shorter path. Both the part waves meet at point D where a detector is placed. Based on above information, answer the following questions.



26. Maximum intensity produced at D is given by
- $4I_0$
 - $3I_0$
 - $2I_0$
 - I_0
27. The maximum value of wavelength λ to produce a maximum at D is given by
- πR
 - $2\pi R$
 - $\frac{\pi R}{2}$
 - $\frac{3\pi R}{2}$
28. The maximum value of wavelength λ to produce a minimum at D is given by
- πR
 - $2\pi R$
 - $\frac{\pi R}{2}$
 - $\frac{3\pi R}{2}$

Comprehension 8

When Fresnel's biprism experiment is performed in air then distance between coherent sources is 0.5 mm and distance between source and screen is 1 m. Fringe width obtained in air is 1 mm. Refractive index of biprism is 1.5. Now the experiment is performed in water having refractive index $\mu_w = \frac{4}{3}$. If the refractive index of the biprism is $\mu = \frac{3}{2}$. Based on above information, answer the following questions.

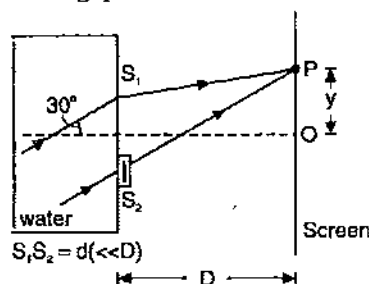
29. The distance between coherent sources in water is
- 1 mm
 - $\frac{1}{2}$ mm
 - $\frac{1}{4}$ mm
 - $\frac{1}{8}$ mm

30. The fringe width in water is
 (A) 1 mm (B) 2 mm
 (C) 3 mm (D) 4 mm
31. The wavelength of light in air is
 (A) 4000 Å (B) 4500 Å
 (C) 5000 Å (D) 6000 Å

Comprehension 9

In a YDSE experiment, the two slits are covered with transparent membranes of negligible thickness which allow light to pass through it but does not allow water. A glass slab of thickness $t = 0.41 \text{ mm}$ and refractive index $\mu_g = 1.5$ is placed in front of one of the slits as shown in figure. The separation between the slits is $d = 0.30 \text{ mm}$. The entire space to the left of the slits is filled with water of refractive index $\mu_w = \frac{4}{3}$. A coherent light of intensity I and absolute

wavelength $\lambda = 5000 \text{ Å}$ is being incident on the slits making an angle of 30° with horizontal. Screen is placed at a distance $D = 1 \text{ m}$ from the slits. Based on above information, answer the following questions.

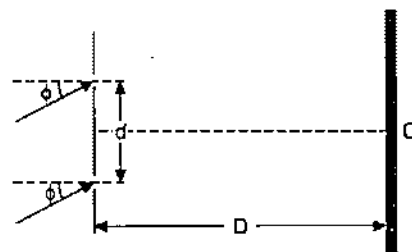


32. At point O, equidistant from slits we get
 (A) 9th dark fringe (B) 10th dark fringe
 (C) 11th bright fringe (D) 10th bright fringe
33. Central maxima is located at
 (A) $y = +\frac{5}{6} \text{ cm}$ (B) $y = -\frac{5}{6} \text{ cm}$
 (C) $y = +\frac{5}{3} \text{ cm}$ (D) $y = -\frac{5}{3} \text{ cm}$
34. The ratio of intensity at point P at $y = \frac{1}{8} \text{ cm}$ on screen and maximum intensity is
 (A) 0 (B) 1
 (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Comprehension 10

Light of wavelength $\lambda = 500 \text{ nm}$ falls on two narrow slits placed a distance $d = 50 \times 10^{-4} \text{ cm}$ apart, at an angle $\phi = 30^\circ$ relative to the slits shown in figure. On the lower slit a transparent slab of thickness 0.1 nm and refractive index

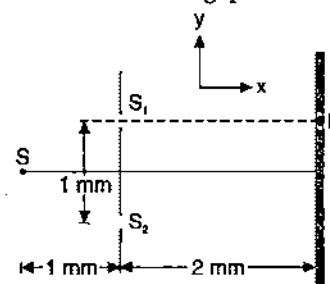
$\frac{3}{2}$ is placed. The interference pattern is observed on a screen at a distance $D = 2 \text{ m}$ from the slits. Based on above information, answer the following questions.



35. The angular position of the central maxima w.r.t. central line is
 (A) 60° (B) 45°
 (C) 30° (D) 15°
36. The order of minima closest to centre C of screen is
 (A) 50 (B) 49
 (C) 48 (D) 47
37. The number of fringes that will pass over C, when the transparent slab from the lower slit is removed is
 (A) 100 (B) 98
 (C) 96 (D) 94

Comprehension 11

In a Young's double slit experiment set-up source S of wavelength 5000 Å illuminates two slits S_1 and S_2 , which act as two coherent sources. The source S oscillates about its shown position according to the equation $y = 0.5 \sin(\pi t)$, where y is in millimetres and t in seconds. Based on above information, answer the following questions.



38. The position of the central maxima as a function of time is best represented by the equation
 (A) $-\cos(\pi t)$ (B) $\cos(\pi t)$
 (C) $-\sin[(\pi + 1)t]$ (D) $-\sin[\pi(1 + t)]$
39. The minimum time t at which the intensity at point P on the screen exactly in front of the upper slit becomes maximum is
 (A) $\frac{1}{2} \text{ s}$ (B) $\frac{1}{3} \text{ s}$

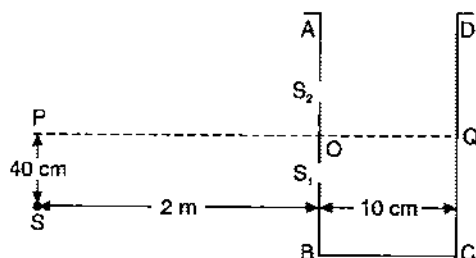
- (C) $\frac{1}{6}$ s (D) 1 s

40. The minimum time t at which the intensity at point P on the screen exactly in front of the upper slit becomes minimum is

- (A) 1 s (B) $\frac{1}{2}$ s
(C) $\frac{1}{3}$ s (D) $\frac{1}{6}$ s

Comprehension 12

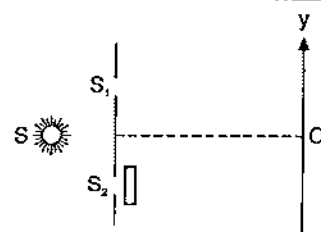
A vessel $ABCD$ of 10 cm width has two small slits S_1 and S_2 sealed with identical glass plates of equal thickness. The distance between the slits is 0.8 mm. POQ is the line perpendicular to the plane AB and passing through O , the middle point of S_1 and S_2 . A monochromatic light source is kept at S , 40 cm below P and 2 m from the vessel, to illuminate the slits as shown in the figure. Based on above information, answer the following questions.



41. The position of the central bright fringe on the other wall CD with respect to the line OQ is
(A) 2 cm below Q (B) 2 cm above Q
(C) 4 cm below Q (D) 4 cm above Q
42. It is observed that when a liquid is poured into the vessel and filled upto OQ , the central bright fringe is found to be at Q . The refractive index of the liquid is
(A) 1.0008 (B) 1.0004
(C) 1.0016 (D) 1.0012

Comprehension 13

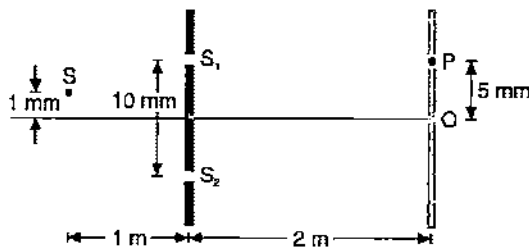
The Young's double slit experiment is done in a medium of refractive index $\frac{4}{3}$. A light of 600 nm wavelength is falling on the slits having 0.45 mm separation. The lower slit S_2 is covered by a thin glass sheet of thickness 10.4 μ m and refractive index 1.5. The interference pattern is observed on a screen placed 1.5 m from the slits as shown in the figure. Assume that all wavelengths in the problem are for the given medium of refractive index $\frac{4}{3}$ and ignore dispersion. Based on above information, answer the following questions.



43. The central maximum formed on the y -axis is located at
(A) $y = \frac{11}{3}$ mm above O
(B) $y = \frac{11}{3}$ mm below O
(C) $y = \frac{14}{3}$ mm above O
(D) $y = \frac{13}{3}$ mm below O
44. The ratio of light intensity of point O to the maximum fringe intensity is
(A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) $\frac{3}{4}$ (D) 1
45. Assuming the 600 nm light to be replaced by white light of range 400 to 700 nm, the wavelengths of the light that form maxima exactly at point O are
(A) $\frac{1300}{3}$ nm, 500 nm (B) $\frac{1400}{3}$ nm, 600 nm
(C) $\frac{1300}{3}$ nm, 650 nm (D) $\frac{1400}{3}$ nm, 650 nm

Comprehension 14

In the Young's double slit experiment a point source of $\lambda = 5000 \text{ \AA}$ is placed slightly off the central axis as shown in the figure. Based on the information provided, answer the following questions.



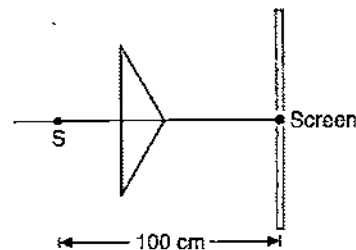
46. The order and nature of the interference at the point P is
(A) 50th order, maxima (B) 50th order, minima
(C) 70th order, maxima (D) 70th order, minima

47. The order and the nature of the interference at O is
 (A) 50th order, minima (B) 50th order, maxima
 (C) 20th order, minima (D) 20th order, maxima
48. If the zero order maxima is formed at O , then
 (A) we should place a film of refractive index $\mu = 1.5$, thickness $10 \mu\text{m}$ in front of S_2 .
 (B) we should place a film of refractive index $\mu = 1.5$, thickness $20 \mu\text{m}$ in front of S_2 .
 (C) we should place a film of refractive index $\mu = 1.5$, thickness $10 \mu\text{m}$ in front of S_1 .
 (D) we should place a film of refractive index $\mu = 1.5$, thickness $20 \mu\text{m}$ in front of S_1 .

Comprehension 15

A narrow slit S allows monochromatic light of wavelength $\lambda = 6000 \text{ \AA}$ to fall on a prism of very small angle as shown in figure. A screen is placed at a distance $l = 100 \text{ cm}$ from the source to obtain an interference pattern. To determine the distance between the virtual images formed by the prism an experiment is done. The prism and screen are kept fixed and a convex lens is moved between the prism and the screen. For two positions of the lens (between the prism and the screen) we get two sharp point images on the screen in each case. The images are separated

from each other by a distance 6 mm and 1.5 mm in the other. Now lens is removed and interference pattern is obtained on the screen. Based on the information provided, answer the following questions.



49. Focal length of the lens is
 (A) 16 cm (B) 20 cm
 (C) 36 cm (D) 40 cm
50. Fringe width of the pattern on the screen is
 (A) 0.1 mm (B) 0.2 mm
 (C) 0.3 mm (D) 0.4 mm
51. If screen is displaced slightly away from prism, then
 (A) No interference pattern is observed
 (B) fringe width remains same.
 (C) fringe width decreases.
 (D) fringe width increases.

Practice Exercise

Matrix Match Type Questions

Each question in this section contains statements given in two columns, which have to be matched. The statements in COLUMN-I are labelled A, B, C and D, while the statements in COLUMN-II are labelled p, q, r, s (and t). Any given statement in COLUMN-I can have correct matching with **ONE OR MORE** statement(s) in COLUMN-II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following examples:

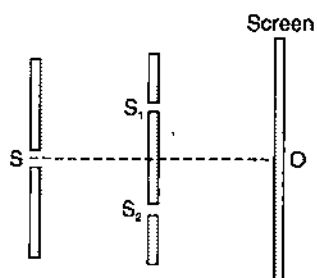
If the correct matches are $A \rightarrow p, s$ and t ; $B \rightarrow q$ and r ; $C \rightarrow p$ and q ; and $D \rightarrow s$ and t ; then the correct darkening of bubbles will look like the following :

	p	q	r	s	t
A	☒	☐	☐	☐	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

1. COLUMN-I shows the effect on the fringe pattern in YDSE corresponding to the changes mentioned in COLUMN-II. Match the effects in COLUMN-I with the corresponding causes in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Angular fringe width remains same	(p) Screen is moved away from the plane of the slits
(B) Angular fringe width changes	(q) Wavelength of light used is decreased
(C) Fringe width (linear separation between two consecutive fringes) changes	(r) The separation between the slits is increased
(D) The fringe pattern may disappear	(s) The width of the source slit is increased

2. Figure shows a set-up to perform Young's double slit experiment. A monochromatic source of light is placed at S. S_1 and S_2 act as coherent sources and interference pattern is obtained on the screen. Match COLUMN-I with COLUMN-II keeping in mind the Young's double slit experiment.



COLUMN-I	COLUMN-II
(A) S is removed and two real sources emitting light of same wavelength are placed at S_1 and S_2 .	(p) Interference fringes disappear.
(B) Width of S_1 is two times the width of S_2 .	(q) There is uniform illumination on a large part of the screen.
(C) S_1 is closed.	(r) The zero order fringe will not form at O .
(D) A thin transparent plate is placed in front of S_1 . Assuming negligible absorption by the plate.	(s) Intensity of a dark fringe will be non-zero, but less than the intensity of bright fringe.

3. Match the contents of COLUMN-I with the respective phenomenon in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Shining of diamonds.	(p) interference
(B) Light waves projected on oil surface shows seven colours.	(q) total internal reflection
(C) Huygen's wave theory of light cannot explain.	(r) origin of spectra
(D) Phenomena which is not Explained by Huygen's construction of wavelength.	(s) photoelectric effect

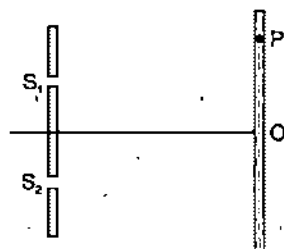
4. In Young's Double Slit Experiment, if distance between slits is d , distance between slit and screen is D , wavelength of light used is λ . Then match COLUMN-I with COLUMN-II.

COLUMN-I	COLUMN-II
(A) For bright fringe, path difference.	(p) $\frac{D\lambda}{2d}$
(B) For dark fringe, path difference.	(q) $\frac{D}{d}(\mu - 1)t$
(C) Displacement of fringe when glass plate of thickness t is placed.	(r) $n\lambda$
(D) Distance between central maxima and first dark fringe when glass plate of thickness t is used.	(s) $(2n - 1)\frac{\lambda}{2}$

5. Match the contents of COLUMN-I with the respective contents of COLUMN-II.

COLUMN-I	COLUMN-II
(A) Sources of variable phase difference.	(p) Incoherent sources
(B) Point on a wavefront behaves as a light source.	(q) Coherent sources
(C) Net displacement is the vector sum of individual displacement.	(r) Superposition principle
(D) Young's double slit experiment uses.	(s) Huygen's principle

6. In the YDSE apparatus shown in figure, Δx is the path difference between S_2P and S_1P . If, now a glass slab is introduced in front of S_2 , then match the contents of COLUMN-I with the respective matches in COLUMN-II.



COLUMN-I	COLUMN-II
(A) Fringe width will	(p) increase
(B) Fringe pattern will	(q) decrease
(C) Number of fringes between O and P will	(r) remain same
(D) Δx at P will	(s) shift upward
	(t) shift downward

7. Match the quantities in COLUMN-I with their respective matches in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Point source of light	(p) Spherical wavefront
(B) Limit of resolution of telescope	(q) Amplitude division
(C) Interference	(r) Superposition of waves
(D) Coherent sources	(s) Radius of lens

8. In the light of possibility of occurrence of phenomena listed in COLUMN-I match the listings in COLUMN-I to the corresponding waves in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Reflection	(p) Non-mechanical waves
(B) Interference	(q) Electromagnetic waves
(C) Diffraction	(r) Visible light waves
(D) Polarisation	(s) Sound waves

Practice Exercise

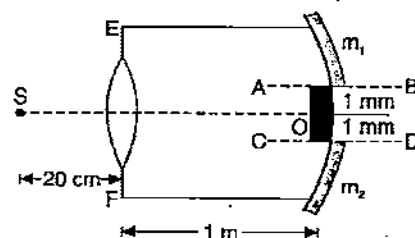
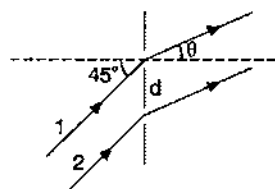
Integer Answer Type Questions

In this section the answer to each of the question is a four digit integer, ranging from 0 to 9999. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answer to question number X (say) is 6092, then the correct darkening of bubbles will look like the following :

X.

0	●	0	0
1	1	1	1
2	2	2	●
3	3	3	3
4	4	4	4
5	5	5	5
●	6	6	6
7	7	7	7
8	8	8	8
9	9	●	9

- Interference pattern with Young's double slit 1.5 mm apart are formed on a screen at a distance 1.5 m from the plane of slits. In the path of the beam from one of the slits, a transparent film of 10 micron thickness and the refractive index 1.6 is interposed while in the path of the beam from the other slit a transparent film of 15 micron thickness and a refractive index 1.2 is interposed. Find the displacement of the fringe pattern, in mm.
- Two coherent radio point sources that are separated by 2 m are radiating in phase with a wavelength of 0.25 m. If a detector moves in a large circle around their midpoint, at how many points will the detector show a maximum signal?
- In the figure shown the distance between the slits is $d = 20\lambda$, where λ is the wavelength of light used. Find the angle θ , in degree, where
 - central maxima (where path difference is zero) is obtained.
 - third order maxima is obtained.
- An equiconvex lens of focal length 10 cm (in air) and refractive index $\frac{3}{2}$ is put at a small opening on a tube of length 1 m fully filled with liquid of refractive index $\frac{4}{3}$. A concave mirror of radius of curvature 20 cm is cut into two halves m_1 and m_2 and placed at the end of the tube. m_1 and m_2 are placed such that their principal axes AB and CD respectively are separated by 1 mm each from the principle axes of the lens.

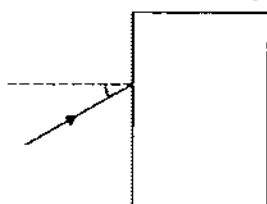


A slit S placed in air illuminates the lens with light of frequency 7.5×10^{14} Hz. The light reflected from m_1

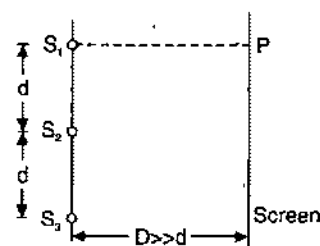
and m_2 forms interference pattern on the left and EF of the tube. O is an opaque substance to cover the hole created by the placement of m_1 and m_2 . Find :

- The position of the image, in cm, formed by lens-water combination.
- The distance, in mm, between the images formed by m_1 and m_2 .
- Width of the fringes on EF , in μm .

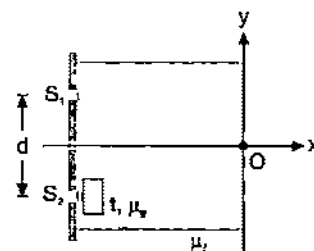
- A ray of light is incident on the left vertical face of the glass slab. If the incident light has an intensity I and on each reflection the intensity decreases by 90% and on each refraction the intensity decreases by 10%, find the ratio of the intensities of maximum to minimum in the reflected pattern.



- Two slits are separated by 0.32 mm. A beam of 500 nm light strikes the slits producing an interference pattern. Determine the number of maxima observed in the angular range $-30^\circ < \theta < 30^\circ$.
- Interference fringes were produced by Young's double slit method, the wavelength of light used being 6000 Å. The separation between the two slits is 2 mm. The distance between the slits and screen is 10 cm. When a transparent plate of thickness 0.5 mm is placed over one of the slits, the fringe pattern is displaced by 5 mm. If μ be the refractive index of the material of the plate, then find 5μ .
- In Young's double slit experiment mixture of two light wave having wavelengths $\lambda_1 = 500 \text{ nm}$ and $\lambda_2 = 700 \text{ nm}$ are being used. Find the position next to central maxima, where maxima due to both waves coincides. (Given $\frac{D}{d} = 1000$)
- Consider the interference at P between waves emanating from three coherent sources in same phase located at S_1 , S_2 and S_3 . If intensity due to each source is $I_0 = 12 \text{ Wm}^{-2}$ at P and $\frac{d^2}{2D} = \frac{\lambda}{3}$ then find the resultant intensity at P , in Wm^{-2} .



- In a YDSE (young double slit experiment) screen is placed 1 m from the slits wavelength of light used is 6000 Å. The fringes formed on the screen are observed by a student sitting close to the slits. The student's eye can distinguish two neighboring fringes, if they subtend an angle more than 1 minute of the arc. Calculate the maximum distance between the slits, in mm, so that fringes are clearly visible. Give your answer to the nearest integer.
- A parallel beam of white light falls from air on a thin film in air whose refractive index is $\sqrt{3}$. The angle of incidence is $i = 60^\circ$. Find the minimum film thickness (in nanometer), if the reflected light is most intense for $\lambda = 6000 \text{ Å}$.
- In a modified YDSE the region between screen and slits is immersed in a liquid whose refractive index varies with time as $\mu_t = \frac{5}{2} - \frac{t}{4}$, until it reaches a steady state value $\frac{5}{4}$. A glass plate of thickness $T = 36 \mu\text{m}$ and refractive index $\mu = \frac{3}{2}$ is introduced in front of one of the slits. If the separation between the sources and the screen is 1 m and the separation between the sources is 2 mm, then calculate the speed of the central maxima, in mms^{-1} , when it is at O .



- In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is now changed to 400 nm, what will be the new number of fringes observed in the same segment of screen.

14. A glass wedge of angle 0.01 radian is illuminated by monochromatic light of wavelength $6000 \text{ \AA}$ falling normally on it. Find the distance from the edge of the wedge, in mm where the 10^{th} fringe will be observed due to the reflected light.
15. In Young's double slit experiment the two slits act as coherent sources of equal amplitude A and wavelength λ . In another experiment with the same set up the two slits are source of equal amplitude A and wavelength λ but are incoherent. Find the ratio of intensity of light at the mid-point of the screen in the first case to that in second case.

Answers

In Chapter Exercise (ICE) & Practice Exercise Sets

ICE I

BASED ON INTERFERENCE

1. (a) $7.8 \mu\text{m}$
(b) 0.6 mm
2. $5 \mu\text{m}$
3. (a) $\frac{2\pi}{\lambda} \left[\left(\sqrt{d^2 + \ell^2} - d \right) + \frac{\mu d^2}{2D} \right]$
(b) $\frac{2\pi}{\lambda} \left(\mu \left(\sqrt{d^2 + \ell^2} - \ell \right) + \frac{d^2}{2D} \right)$
4. 94.8 nm
5. 0.546 mm
6. 112.78 nm
7. (a) 0.5 mm
(b) 2.25 mm
(c) $3I_0$
(d) $\frac{1}{\sqrt{3}} \text{ meter}$
(e) $n = 5000$ is not possible
8. (a) $\frac{5\pi}{2}$
(b) $\frac{25\pi}{4}$
9. $5890 \text{ \AA}$
10. $1.2 \mu\text{m}$
11. Zero order maxima will remain unchanged.
Tenth order will now be at 4.55 mm .
12. 8.75 mm
14. 4.5 mm
15. 7
16. 0.21
17. (a) $3 \times 10^{-4} I_0$
(b) 5.49
18. 3.5 mm
19. 1.5

ICE II

BASED ON DIFFRACTION & POLARISATION

1. 11.8 mm
2. $9 \times 10^{-4} \text{ m}$
3. 0.2 mm
4. 40 m
5. 12.5 cm
6. $1.3 \mu\text{m}$
7. 0.2 mm
8. $3.534 \times 10^{-3} \text{ rad}$
9. $\pm 45^\circ, \pm 135^\circ$
10. (a) 0.75
(b) 0.25
11. 37.5%
12. $\frac{I_0}{4}$
13. 1:3
15. $30^\circ, 45^\circ$

ANSWERS**SINGLE CORRECT CHOICE TYPE QUESTIONS**

1. B	2. D	3. A	4. D	5. B
6. D	7. D	8. D	9. B	10. A
11. D	12. D	13. A	14. C	15. D
16. B	17. D	18. D	19. B	20. B
21. C	22. C	23. D	24. B	25. D
26. A	27. C	28. D	29. C	30. B
31. A	32. D	33. A	34. A	35. A
36. C	37. A	38. C	39. B	40. B
41. A	42. A	43. D	44. D	45. C
46. C	47. B	48. B	49. A	50. C
51. D	52. B	53. B	54. C	55. B
56. D	57. B	58. A	59. B	60. A
61. D	62. B	63. D	64. B	65. C
66. D	67. C	68. B	69. B	70. C
71. A	72. C	73. C	74. C	75. C
76. B	77. A	78. B	79. C	80. A
81. B	82. B	83. D	84. C	85. A
86. B	87. B	88. B	89. C	90. A
91. C	92. B	93. D	94. C	95. B
96. D	97. D	98. D	99. A	100. A
101. A	102. D	103. D	104. D	105. C
106. D	107. C	108. B	109. A	110. C
111. D	112. A	113. C	114. B	115. C
116. A	117. D	118. A	119. C	120. B
121. A	122. A	123. D	124. A	125. C
126. B	127. A	128. B	129. C	130. A
131. C	132. B	133. D	134. D	135. A
136. C	137. C	138. D	139. D	140. B
141. B	142. A	143. B	144. A	145. D
146. B	147. D	148. D	149. D	150. C
151. C	152. B	153. B	154. A	155. C
156. D	157. D	158. C	159. B	160. A
161. A	162. C	163. D	164. B	165. C
166. A	167. B	168. C	169. B	170. A
171. B	172. C	173. D		

ANSWERS**MULTIPLE CORRECT CHOICE TYPE QUESTIONS**

1. AC	2. AB	3. AD	4. BD	5. BD
6. ABC	7. ABC	8. BD	9. AB	10. ACD
11. AC	12. ABC	13. BD	14. AC	15. AB

ANSWERS

REASONING BASED QUESTIONS

1. B	2. D	3. D	4. B	5. D
6. A	7. D	8. D	9. D	10. A
11. A	12. D	13. B	14. D	15. A
16. C	17. D	18. D	19. A	20. A
21. A				

ANSWERS

LINKED COMPREHENSION TYPE QUESTIONS

1. A	2. D	3. B	4. A	5. A
6. A	7. C	8. A	9. A	10. B
11. C	12. D	13. C	14. C	15. D
16. D	17. A	18. B	19. B	20. B
21. B	22. C	23. D	24. B	25. A
26. C	27. A	28. B	29. D	30. C
31. C	32. D	33. D	34. C	35. C
36. B	37. A	38. D	39. C	40. A
41. B	42. C	43. D	44. C	45. C
46. C	47. D	48. D	49. A	50. B
51. D				

ANSWERS

MATRIX MATCH TYPE QUESTIONS

1. $A \rightarrow (p, s)$ $B \rightarrow (q, r)$ $C \rightarrow (p, q, r)$ $D \rightarrow (q, r, s)$	2. $A \rightarrow (p, q)$ $B \rightarrow (s)$ $C \rightarrow (p, q)$ $D \rightarrow (r)$	3. $A \rightarrow (q)$ $B \rightarrow (p)$ $C \rightarrow (s)$ $D \rightarrow (r)$	4. $A \rightarrow (r)$ $B \rightarrow (s)$ $C \rightarrow (q)$ $D \rightarrow (p)$	5. $A \rightarrow (p)$ $B \rightarrow (s)$ $C \rightarrow (r)$ $D \rightarrow (q, r)$
6. $A \rightarrow (r)$ $B \rightarrow (t)$ $C \rightarrow (r)$ $D \rightarrow (p)$	7. $A \rightarrow (p)$ $B \rightarrow (s)$ $C \rightarrow (r)$ $D \rightarrow (q)$	8. $A \rightarrow (p, q, r, s)$ $B \rightarrow (p, q, r, s)$ $C \rightarrow (p, q, r, s)$ $D \rightarrow (p, q, r)$		

ANSWERS

INTEGER ANSWER TYPE QUESTIONS

1. 3	2. 32	3. (a) 45, (b) 59	4. (a) 80, (b) 4, (c) 60
5. 361	6. 739	7. 6	8. 3
9. 36	10. 2	11. 100	12. 3
13. 18	14. 3	15. 2	

Solutions

In Chapter Exercise (ICE)-I

(based on Interference)

$$1. (a) \left(1 - \frac{1}{\mu}\right)t = \frac{3\lambda}{\mu}$$

$$\Rightarrow t = \frac{3\lambda}{(\mu-1)} = \frac{3 \times 0.78}{1.3-1} = 7.8 \mu\text{m}$$

(b) Upwards

$$\frac{yD}{D} - \left(1 - \frac{1}{\mu}\right)t = \frac{4\lambda}{\mu}$$

$$\Rightarrow y = 4.2 \text{ mm}$$

Downwards

$$t\left(1 - \frac{1}{\mu}\right) + \frac{yD}{D} = \frac{4\lambda}{\mu}$$

$$\Rightarrow y = 0.6 \text{ mm}$$

2. Since we are given that
Shift = 5 (fringe width)

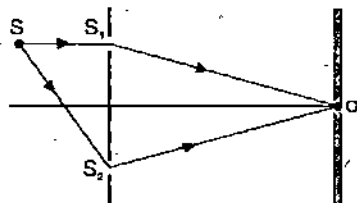
$$\Rightarrow \frac{(\mu-1)D}{d} = \frac{5\lambda D}{d}$$

$$\Rightarrow t = \frac{5\lambda}{\mu-1} = \frac{25000}{1.5-1} = 50,000 \text{ \AA} = 5 \mu\text{m}$$

3. $\Delta x = SS_2O - SS_1O = \sqrt{d^2 + \ell^2} - \ell$

$$\Rightarrow \phi = \frac{2\pi}{\lambda} \Delta x$$

$$\Rightarrow \phi = \frac{2\pi}{\lambda} (\sqrt{d^2 + \ell^2} - \ell)$$



(a) Geometric path difference between S_2P and S_1P is

$$y = \left(\frac{d}{2}\right) \frac{D}{D} = \frac{d^2}{2D}$$

Their optical path difference = $\frac{\mu d^2}{2D}$

Therefore, net path difference $\Delta x = (\sqrt{d^2 + \ell^2} - \ell) + \frac{\mu d^2}{2D}$

So, phase difference is given by

$$\phi = \frac{2\pi}{\lambda} \left[(\sqrt{d^2 + \ell^2} - \ell) + \frac{\mu d^2}{2D} \right]$$

(b) When liquid is filled between slits and source S, then

$$\Delta x = \mu \left((\sqrt{d^2 + \ell^2} - \ell) + \frac{d^2}{2D} \right)$$

$$\Rightarrow \phi = \frac{2\pi}{\lambda} \left(\mu (\sqrt{d^2 + \ell^2} - \ell) + \frac{d^2}{2D} \right)$$

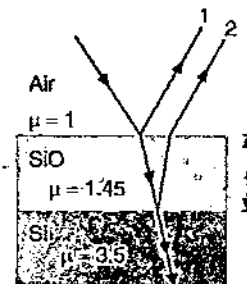
4. The reflected light will be minimum when rays 1 and 2 (shown in figure) meet the condition of destructive interference. Rays 1 and 2 both suffer an additional phase change of π (or 180°) after being reflected at surface of a denser medium. The net change in phase due to reflection is therefore zero and the condition for a reflection minimum

requires a path difference of $\frac{\lambda}{2}$.

Hence,

$$2\mu t = \frac{\lambda}{2}$$

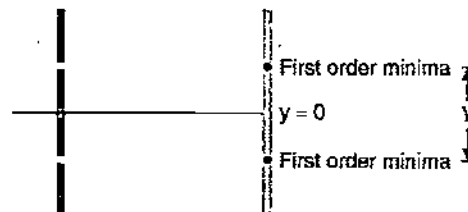
$$\Rightarrow t = \frac{\lambda}{4\mu} = \frac{550}{4(1.45)} = 94.8 \text{ nm}$$



5. The desired distance is

$$y = \text{fringe width} = \frac{\lambda D}{d} = \frac{546 \times 10^{-9} \times 0.4}{0.4 \times 10^{-3}}$$

$$\Rightarrow y = 0.546 \times 10^{-3} \text{ m} = 0.546 \text{ mm}$$



6. For constructive interference in case of soap film,

$$2\mu t = \left(n - \frac{1}{2}\right)\lambda$$

$$n = 1, 2, 3, \dots$$

For minimum thickness t , $n = 1$

$$\Rightarrow 2\mu t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{\lambda}{4\mu} = \frac{500}{4 \times 1.33} = 112.78 \text{ nm}$$

$$7. (a) \beta = \frac{\lambda D}{d} = \frac{(5000 \times 10^{-10})(1)}{1 \times 10^{-3}} = 0.5 \text{ mm}$$

$$(b) y = (2n - 1) \frac{\lambda D}{d}$$

For $n = 5$

$$y = 2.25 \text{ mm}$$

$$(c) \text{ At } y = \frac{1}{3} \text{ mm, } y \ll D$$

$$\Rightarrow \Delta x = \frac{y d}{D}$$

$$\Rightarrow \Delta \phi = \frac{2\pi}{\lambda} \Delta x = 2\pi \frac{y d}{\lambda D} = \frac{4\pi}{3}$$

Now resultant intensity, is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta \phi$$

$$\Rightarrow I = 4I_0 + I_0 + 2\sqrt{4I_0^2} \cos \Delta \phi = 5I_0 + 4I_0 \cos\left(\frac{4\pi}{3}\right) = 3I_0$$

$$(d) \frac{d}{\lambda} = \frac{10^{-3}}{0.5 \times 10^{-6}} = 2000$$

$n = 1000$ is not very much less than 2000

Hence now $\Delta x = d \sin \theta$ must be used, so we get

$$d \sin \theta = n\lambda = 1000\lambda$$

$$\Rightarrow \sin \theta = 1000 \frac{\lambda}{d} = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

Since, $y = D \tan \theta$

$$\Rightarrow y = \frac{1}{\sqrt{3}} \text{ meter}$$

(e) Highest order maxima

$$n_{\max} = \left[\frac{d}{\lambda}\right] = 2000$$

Hence, $n = 5000$ is not possible.

$$8. (a) \Delta x = 0.5 \mu\text{m} = 5 \times 10^{-7} \text{ m}$$

So, phase difference is given by

$$\phi = \left(\frac{2\pi}{\lambda}\right)(\Delta x) = \left(\frac{2\pi}{400 \times 10^{-9}}\right)(5 \times 10^{-7})$$

$$\Rightarrow \phi = \frac{5\pi}{2}$$

$$(b) \Delta x' = (\mu - 1)t = (1.5 - 1)(1.5) = 0.75 \mu\text{m}$$

$$\Rightarrow (\Delta x)_{\text{net}} = \Delta x + \Delta x' = 1.25 \mu\text{m} = 1.25 \times 10^{-6} \text{ m}$$

$$\Rightarrow \phi' = \left(\frac{2\pi}{400 \times 10^{-9}}\right)(1.25 \times 10^{-6})$$

$$\Rightarrow \phi' = \frac{25\pi}{4}$$

9. Since, shift = 2(Fringe Width)

$$\Rightarrow \frac{(\mu - 1)tD}{d} = \frac{\lambda(2D)}{d}$$

$$\Rightarrow \lambda = \frac{(\mu - 1)t}{2} = \frac{(1.6 - 1)(1.964 \times 10^{-6})}{2}$$

$$\Rightarrow \lambda = 0.589 \times 10^{-6} \text{ m}$$

$$\Rightarrow \lambda = 5890 \text{ \AA}$$

10. Shift = 5 (fringe width)

$$\Rightarrow (\mu - 1) \frac{tD}{d} = \frac{\lambda D}{d}$$

$$\Rightarrow t = \frac{\lambda}{\mu - 1} = \frac{6 \times 10^{-7}}{(1.5 - 1)} = 1.2 \times 10^{-6} \text{ m}$$

$$\Rightarrow t = 1.2 \mu\text{m}$$

$$11. \beta = \frac{(14.75 - 12.50)}{10 - 1} \text{ mm} = 0.25 \text{ mm}$$

$$\beta' = \frac{\lambda'}{\lambda} \beta = \frac{5500}{6000} \times 0.25 \text{ mm} = 0.23 \text{ mm}$$

Zero order maxima will remain unchanged. Earlier it was at 12.5 mm – 12.25 mm.

Tenth order will now be at 12.25 mm + 10 β' = 14.55 mm.

$$12. \text{ Given, } d = 0.1 \text{ mm} = 10^{-4} \text{ m, } D = 0.5 \text{ m and } \lambda = 5000 \text{ \AA} = 5 \times 10^{-7} \text{ m}$$

$$\Delta y = (y_{11})_{\text{dark}} - (y_7)_{\text{bright}} = \frac{(2 \times 11 - 1)\lambda D}{2d} - \frac{7\lambda D}{d}$$

$$\Rightarrow \Delta y = \frac{7\lambda D}{2d} = \frac{7 \times 5 \times 10^{-7} \times 0.5}{2 \times 10^{-4}} = 8.75 \times 10^{-3} \text{ m} = 8.75 \text{ mm}$$

13. (a) Angular separation $\left(= \frac{\lambda}{d}\right)$ of the fringes remains constant.

But the linear separation or fringe width increases in proportion to the distance (D) from the screen.

(b) As λ decreases, fringe width ($\beta \propto \lambda$) decreases.

(c) As d increases, fringe width $\left(\beta \propto \frac{1}{d}\right)$ decreases.

(d) The interference pattern due to different component colours of white light overlap (in-coherently). The central bright fringes of different colours are at the same position. Therefore, the central fringe is white. Since blue colour has the lower λ , the fringe closest on either side of the central white fringe is blue; the farthest is red.

(e) Since in a medium the wavelength of light is $\lambda' = \frac{\lambda}{\mu}$, therefore the fringe width is given by $\beta = \frac{\lambda' D}{d} = \frac{\lambda D}{\mu d}$.

Thus, fringe width decreases by μ .

14. For bright fringes to coincide, we have

$$\frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{900}{750} = \frac{6}{5}$$

i.e., 5th order maxima of larger wavelength λ_2 will overlap with 6th order maxima of smaller wavelength λ_1 .

$$\text{So, } y_{\text{min}} = \frac{6\lambda D}{d} = \frac{6 \times 750 \times 10^{-9} \times 2}{2 \times 10^{-3}}$$

$$\Rightarrow y_{\min} = 4.5 \times 10^{-3} \text{ m}$$

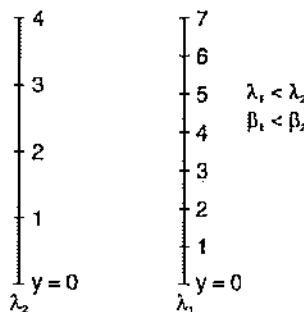
$$\Rightarrow y_{\min} = 4.5 \text{ mm}$$

15. Let n_1 bright fringe of λ_1 overlaps with n_2 bright fringe of λ_2 . Then,

$$\frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$$

$$\Rightarrow n_1 \lambda_1 = n_2 \lambda_2$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{700}{400} = \frac{7}{4}$$



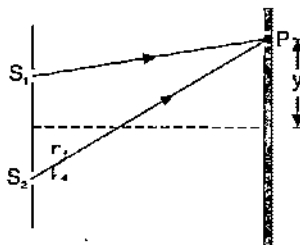
The ratio $\frac{n_1}{n_2} = \frac{7}{4}$ implies that 7th bright fringe of λ_1 will overlap with 4th bright fringe of λ_2 . Similarly 14th of λ_1 will overlap with 8th of λ_2 and so on.

So the minimum order of λ_1 which overlaps with λ_2 is 7.

16. Without inserting the slab, path difference at P is given by

$$\Delta x = \frac{y d}{D} = \frac{0.15 \times 10^{-3} \times 2 \times 10^{-3}}{2}$$

$$\Rightarrow \Delta x = 1.5 \times 10^{-7} \text{ m}$$



Corresponding phase difference at P is

$$\phi = \left(\frac{2\pi}{\lambda} \right) (\Delta x)$$

$$\Rightarrow \phi = \left(\frac{2\pi}{6000 \times 10^{-10}} \right) (1.5 \times 10^{-7}) = \frac{\pi}{2}$$

$$\Rightarrow \frac{\phi}{2} = \frac{\pi}{4}$$

Since, intensity at P is given by $I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$

$$\Rightarrow I = 4I_0 \cos^2 \left(\frac{\pi}{4} \right) = 2I_0$$

Phase difference after placing the glass sheet is

$$\phi' = \phi + \frac{2\pi}{\lambda} (\mu - 1)t$$

$$\Rightarrow \phi' = \frac{\pi}{2} + \frac{2\pi}{6000 \times 10^{-10}} (1.5 - 1)(8000 \times 10^{-10})$$

$$\Rightarrow \phi' = \frac{11\pi}{6}$$

The intensity at P is now given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\Rightarrow I' = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \left(\frac{11\pi}{6} \right) = 2I_0 \quad \{ \because I' = 2I_0 \}$$

Solving this equation, we get,

$$\eta = 0.21$$

17. (a) Since, $\Delta x = \frac{y d}{D} = \frac{(0.3 \times 10^{-3})(10 \times 10^{-3})}{1} = 3 \times 10^{-6} \text{ m}$

$$\Rightarrow \phi = \left(\frac{2\pi}{\lambda} \right) (\Delta x) = \left(\frac{2\pi}{546 \times 10^{-9}} \right) (3 \times 10^{-6})$$

$$\Rightarrow \phi = 34.52 \text{ radian} = 1978^\circ$$

$$\Rightarrow \frac{\phi}{2} = 989^\circ$$

$$\text{Since, } I_P = I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\Rightarrow I_P = 3 \times 10^{-4} I_0$$

- (b) Fringe width $\beta = \frac{\lambda D}{d} = \frac{(546 \times 10^{-9})(1)}{(0.3 \times 10^{-3})} = 1.82 \times 10^{-3} \text{ m}$

$$\Rightarrow \beta = 1.82 \text{ mm}$$

Therefore, number of fringes between point P and the central fringe are given by

$$N = \frac{10}{1.82} = 5.49$$

18. Let n_1 bright fringe corresponding to wavelength $\lambda_1 = 500 \text{ nm}$ coincides with n_2 bright fringe corresponding to wavelength $\lambda_2 = 700 \text{ nm}$.

$$\Rightarrow \frac{n_1 \lambda_1 D}{d} = \frac{n_2 \lambda_2 D}{d}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{7}{5}$$

This implies that 7th maxima of λ_1 coincides with 5th maxima of λ_2 . Similarly 14th maxima of λ_1 will coincide with 10th maxima of λ_2 and so on.

So, minimum distance is

$$y_{\min} = \frac{n_1 \lambda_1 D}{d} = 7 \times 5 \times 10^{-7} \times 10^3$$

$$\Rightarrow y_{\min} = 3.5 \times 10^{-3} \text{ m} = 3.5 \text{ mm}$$

19. Here $\lambda = 6 \times 10^{-5} \text{ cm}$, $t = 7.2 \times 10^{-4} \text{ cm}$

Displacement of the central bright fringe, $\Delta x = 6\beta$

$$\text{Since } \Delta x = \frac{\beta}{\lambda} (\mu - 1)t$$

$$\Rightarrow \frac{\beta}{\lambda} (\mu - 1)t = 6\beta$$

$$\Rightarrow \mu - 1 = \frac{6\lambda}{t} = \frac{6 \times 6 \times 10^{-5}}{7.2 \times 10^{-4}} = 0.5$$

$$\Rightarrow \mu = 0.5 + 1 = 1.5$$

Solutions

In Chapter Exercise (ICE)-II

(based on diffraction & polarisation)

1. For first dark band, $\sin \theta = \frac{\lambda}{d}$

As the diffraction pattern is obtained in the focal plane of lens, therefore

$$\tan \theta = \frac{x}{f}$$

For small θ , $\tan \theta \approx \sin \theta$

$$\Rightarrow \frac{x}{f} = \frac{\lambda}{d}$$

$$\Rightarrow x = \left(\frac{\lambda}{d}\right) f$$

Since, $\lambda = 5900 \text{ \AA} = 59 \times 10^{-8} \text{ m}$, $f = 50 \text{ cm} = 0.50 \text{ m}$

$$d = 0.025 \text{ m} = 2.5 \times 10^{-5} \text{ m}$$

$$\Rightarrow x = \frac{59 \times 10^{-8} \times 0.50}{2.5 \times 10^{-5}} = 11.8 \times 10^{-3} \text{ m} = 11.8 \text{ mm}$$

2. Given, $\lambda_1 = 5890 \times 10^{-10} \text{ m}$, $\lambda_2 = 5896 \times 10^{-10} \text{ m}$,

$$d = 2 \mu\text{m} = 10^{-6} \text{ m}, D = 2 \text{ m}$$

Distance of first secondary maximum from the centre of the screen is

$$x = \frac{3D\lambda}{2d}$$

For the two wavelengths, we have

$$x_1 = \frac{3D\lambda_1}{2d} \text{ and } x_2 = \frac{3D\lambda_2}{2d}$$

Spacing between the first two maximum of sodium lines is given by

$$\Delta x = x_2 - x_1 = \frac{3D}{2d}(\lambda_2 - \lambda_1)$$

$$\Rightarrow \Delta x = \frac{3 \times 2 \times (5896 - 5890) \times 10^{-10}}{2 \times 2 \times 10^{-6}} = 9 \times 10^{-4} \text{ m}$$

3. The linear separation between n bright fringes in an interference pattern on the screen is given by

$$x_n = \frac{n\lambda D}{d}$$

Since, $x_n \ll D$, the angular separation between n bright fringes should be

$$\theta_n = \frac{x_n}{D} = \frac{n\lambda}{d}$$

For 10 bright fringes, we get, $\theta_{10} = \frac{10\lambda}{D}$

The angular width of the central maximum in the diffraction pattern due to slit of width a is

$$2\theta_1 = \frac{2\lambda}{a}$$

Since, 10 maxima of the double slit pattern within the central maximum of the single slit pattern

$$\Rightarrow 10 \frac{\lambda}{d} < 2 \frac{\lambda}{a}$$

$$\Rightarrow a \leq \frac{d}{5}$$

$$\Rightarrow a = \frac{1}{5} \text{ mm} = 0.2 \text{ mm}$$

4. Since, $D_F = \frac{d^2}{\lambda}$

where $d = 4 \text{ mm} = 4 \times 10^{-3} \text{ m}$, $\lambda = 400 \text{ nm} = 4 \times 10^{-7} \text{ m}$

$$\Rightarrow D_F = \frac{(4 \times 10^{-3})^2}{4 \times 10^{-7}} = 40 \text{ m}$$

Thus ray optics is valid upto a distance of 40 m from the aperture.

5. Here size of Fresnel zone D_F at the middle hill must be less than 50 m.

Distance of either of the two hills from the middle hill is

$$D = \frac{40}{2} \text{ km} = 20,000 \text{ m}$$

Since size of Fresnel's zone, $D_F = \sqrt{\lambda D}$

$$\Rightarrow \sqrt{\lambda D} \ll 50$$

$$\Rightarrow \lambda D \ll 2500$$

$$\Rightarrow \lambda \ll \frac{2500}{D} = \frac{2500}{20,000} = 0.125 \text{ m} = 12.5 \text{ cm}$$

Thus wavelengths longer than 12.5 cm will undergo serious diffraction effects.

6. Given, $\lambda = 6500 \text{ \AA} = 6500 \times 10^{-10} \text{ m}$, $\theta = 30^\circ$

For first minimum, $d \sin \theta = \lambda$

$$\Rightarrow d = \frac{\lambda}{\sin \theta} = \frac{6500 \times 10^{-10}}{\sin 30^\circ} = \frac{6500 \times 10^{-10}}{0.5}$$

$$\Rightarrow d = 1.3 \times 10^{-6} \text{ m} = 1.3 \mu\text{m}$$

7. Given, $D = 2 \text{ m}$, $x_1 = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$,

$$\lambda = 5000 \text{ \AA} = 5 \times 10^{-7} \text{ m}$$

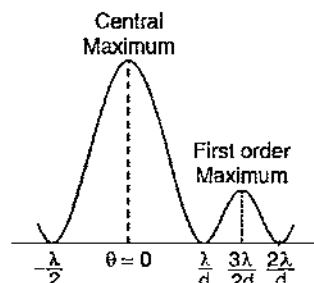
Width of central maximum is

$$\beta_0 = 2x_1 = 2 \times 5 \times 10^{-3} \text{ m} = 10^{-2} \text{ m}$$

So, slit width is

$$d = \frac{2D\lambda}{\beta_0} = \frac{2 \times 2 \times 5 \times 10^{-7}}{10^{-2}} = 2 \times 10^{-4} \text{ m} = 0.2 \text{ mm}$$

8. From figure, it is clear that the angular separation between central maximum and first order minimum is



$$\theta = \frac{3\lambda}{2d} - 0 = \frac{3\lambda}{2d}$$

$$\Rightarrow \theta = \frac{3 \times 5890 \times 10^{-10}}{2 \times 0.25 \times 10^{-3}}$$

$$\Rightarrow \theta = 3.534 \times 10^{-3} \text{ rad}$$

9. Since, $I = \frac{I_0}{2}$

Using Malus Law, $I = I_0 \cos^2 \theta$

$$\Rightarrow \frac{I_0}{2} = I_0 \cos^2 \theta$$

$$\Rightarrow \cos \theta = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \pm 45^\circ, \pm 135^\circ$$

The same effect occurs no matter which sheet is rotated or in which direction it is rotated.

10. According to Malus Law, $I = I_0 \cos^2 \theta$

(a) Here $\theta = 30^\circ$

$$\Rightarrow I = I_0 \cos^2(30^\circ) = I_0 \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4} I_0$$

$$\Rightarrow \frac{I}{I_0} = 0.75$$

(b) Here $\theta = 60^\circ$

$$\Rightarrow I = I_0 \cos^2(60^\circ) = I_0 \left(\frac{1}{2}\right)^2 = \frac{I_0}{4}$$

$$\Rightarrow \frac{I}{I_0} = 0.25$$

11. Let I_0 be intensity of incident unpolarised light. Then the intensity of light emerging from the first polaroid will be

$$I_1 = \frac{I_0}{2}$$

If θ is the angle between the transmission directions of the two polaroids, then the intensity of light emerging from second polaroid is

$$I_2 = I_1 \cos^2 \theta = \frac{I_0}{2} \cos^2 \theta$$

Initially the two polaroids are crossed to each other i.e., $\theta = 90^\circ$. When polaroid is rotated through 60° , the angle between their polarising directions will become

$$\theta = 90^\circ - 60^\circ = 30^\circ$$

$$\Rightarrow I_2 = \frac{I_0}{2} \cos^2(30^\circ) = \frac{I_0}{2} \times \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{8} I_0$$

$$\Rightarrow \frac{I_2}{I_0} = 0.375$$

$$\text{So, transmitted percentage is } \frac{I_2}{I_0} \times 100 = 0.375 \times 100 = 37.5\%$$

12. Obviously, the axis of P_3 is inclined at 45° to the axes of P_1 and P_2

Let amplitude of linearly polarised light emerging from P_1 be E_0

The amplitude of light emerging from P_3 is $E_0 \cos 45^\circ = \frac{E_0}{\sqrt{2}}$

Next the amplitude of light emerging from P_2 is

$$\frac{E_0}{\sqrt{2}} \cos 45^\circ = \frac{E_0}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{E_0}{2}$$

Let intensity transmitted by $P_1 = I_0 \propto E_0^2$

So, intensity after passing through P_2 and P_3 is

$$I \propto \left(\frac{E_0}{2}\right)^2$$

$$\Rightarrow I = \frac{I_0}{4}$$

Thus the intensity becomes one-fourth of the maximum transmitted intensity.

13. The planes of polarisation of light beams A and B are mutually at right angles. Initially, the beam B shows zero intensity. Therefore, $\theta = 90^\circ$ for beam B and $\theta = 0^\circ$ for beam A. When the polaroid is rotated through 30° , we have

$$\theta = 60^\circ \text{ for beam B and}$$

$$\theta = 30^\circ \text{ for beam A}$$

In this position, we have

$$\left(\text{Intensity of emerging beam A}\right) = \left(\text{Intensity of emerging beam B}\right)$$

$$\Rightarrow I_A \cos^2(30^\circ) = I_B \cos^2(60^\circ)$$

$$\Rightarrow \frac{I_A}{I_B} = \frac{\cos^2(60^\circ)}{\cos^2(30^\circ)} = \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{1}{3} = 1:3$$

14. From Snell's Law, $\frac{\sin i_p}{\sin r_p} = \mu$

From Brewster Law, $\tan i_p = \frac{\sin i_p}{\cos i_p} = \mu$

$$\Rightarrow \frac{\sin i_p}{\sin r_p} = \frac{\sin i_p}{\cos i_p}$$

$$\Rightarrow \sin r_p = \cos i_p$$

$$\Rightarrow \sin r_p = \sin(90^\circ - i_p)$$

$$\Rightarrow r_p = 90^\circ - i_p$$

$$\Rightarrow i_p + r_p = 90^\circ$$

Hence the reflected and transmitted rays are perpendicular to each other.

15. Let P_1 , P_2 , P_3 be the three polarisers and θ be the angle between the transmission axes of P_1 and P_2 . As P_1 and P_3 are crossed, the angle between P_2 and P_3 is $(90^\circ - \theta)$.

Let I_0 be the intensity of the unpolarised light falling on P_1 .

Then the intensity of light emerging from P_1 will be

$$I_1 = \frac{I_0}{2}$$

By Malus Law, the intensity of light emerging from P_2 is given by

$$I_2 = I_1 \cos^2 \theta = \frac{I_0}{2} \cos^2 \theta$$

The intensity of light emerging from P_3 is given by

$$I_3 = I_2 \cos^2 (90^\circ - \theta) = \frac{1}{2} I_0 \cos^2 \theta \sin^2 \theta = \frac{1}{8} I_0 \sin^2 (2\theta)$$

$$\Rightarrow \sin^2 (2\theta) = \frac{8I_3}{I_0} = \frac{8 \times 3}{32} = \frac{3}{4}$$

$$\Rightarrow \sin(2\theta) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow 2\theta = 60^\circ$$

$$\Rightarrow \theta = 30^\circ$$

$$\text{So, } I_3 = \frac{I_0}{8} \sin^2 (2\theta)$$

I_3 will be maximum when $\sin^2 (2\theta) = 1$ (maximum), i.e.,

$$\sin(2\theta) = 1 = \sin(90^\circ)$$

$$\Rightarrow \theta = 45^\circ$$

Solutions

Single Correct Choice Type Questions

1. [B]

$$I_{\max} = I = (a + a)^2$$

$$\Rightarrow I = 4a^2 = 4I_0$$

When either of the two slits is covered then

$$I' = (a + 0)^2 = a^2 = \frac{I}{4}$$

2. [D]

Let n^{th} minima of 400 nm coincides with m^{th} minima of 560 nm, then

$$(2n-1)\left(\frac{400}{2}\right) = (2m-1)\left(\frac{560}{2}\right)$$

$$\Rightarrow \frac{2n-1}{2m-1} = \frac{7}{5} = \frac{14}{10} = \dots$$

i.e., 4th minima of 400 nm coincides with 3rd minima of 560 nm.

Location of this minima is,

$$Y_1 = \frac{(2 \times 4 - 1)(1000)(400 \times 10^{-9})}{2 \times 0.4} = 14 \text{ mm}$$

Next 11th minima of 400 nm will coincide with 8th minima of 560 nm

Location of this minima is,

$$Y_2 = \frac{(2 \times 11 - 1)(1000)(400 \times 10^{-9})}{2 \times 0.1} = 42 \text{ mm}$$

 $\therefore$ Required distance = $Y_2 - Y_1 = 28 \text{ mm}$

3. [A]

Path difference due to slab should be integral multiple of λ or $\Delta x = n\lambda$

$$\Rightarrow (\mu - 1)t = n\lambda \quad n = 1, 2, 3, \dots$$

$$\Rightarrow t = \frac{n\lambda}{\mu - 1}$$

For minimum value of t , $n = 1$

$$\Rightarrow t = \frac{\lambda}{\mu - 1} = \frac{\lambda}{1.5 - 1} = 2\lambda$$

4. [D]

As $\lambda_R > \lambda_G > \lambda_B$

$$\Rightarrow \beta_R > \beta_G > \beta_B \text{ as } \beta \propto \lambda$$

8. [D]

$$\Delta\lambda = \frac{v_A}{c} \lambda$$

$$\Rightarrow v_A = \frac{\Delta\lambda \cdot c}{\lambda} = \frac{47 \times 3 \times 10^9}{4700}$$

$$\Rightarrow v_A = 3 \times 10^6 \text{ ms}^{-1} \text{ away from earth}$$

9. [B]

The relation among angle of diffraction θ , order n and number of lines per cm of the grating N is $\sin\theta = Nn\lambda$. The maximum value of $\sin\theta = 1$. Hence maximum value for $n = \frac{1}{N\lambda}$. $\lambda = 6000 \times 10^{-8} \text{ cm}$, $N = 5000 \text{ lines/cm}$, we get $n = 3.33$. The order of spectrum is an integer. Thus we cannot see the fourth order, but can see the third order.

10. [A]

Since $\mu\lambda = \text{constant}$

$$\Rightarrow \mu_1\lambda_1 = \mu_2\lambda_2$$

$$\Rightarrow (1)\lambda_1 = n\lambda_2$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = n$$

11. [D]

$$n_1\lambda_1 = n_2\lambda_2$$

$$\Rightarrow 62 \times 5893 = n_2 \times 4358$$

$$\Rightarrow n_2 = 84$$

12. [D]

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \beta = \frac{(5000 \times 10^{-10})(2)}{(10^{-3})}$$

$$\Rightarrow \beta = 10^{-3} \text{ m} = 1 \text{ mm}$$

13. [A]

Let the required thickness be $t \text{ \AA}$. So, number of wavelengths in vacuum is $\frac{t}{6000}$.

$$\Rightarrow \text{Number of wavelengths in air is} \\ \frac{1.003t}{6000}$$

According to the problem

$$\frac{t}{6000} = \frac{1.003t}{6000} + 1$$

$$\Rightarrow t = 2 \text{ mm}$$

14. [C]

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = 9I$$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2 = I$$

15. [D]

Theoretically infinite colours are possible, hence infinite wavelengths.

17. [D]

Because sound waves are longitudinal in nature.

18. [D]

The blue filter will allow only blue light to pass through. So light from the object passes through filter. Similarly, the white background will also look blue through the filter. Thus we have a blue object under a blue background, which makes it indistinguishable.

19. [B]

Intensity of illumination at a point is decided by three factors; (i) power of the source P (ii) distance from the source r and (iii) angle of incidence of rays $\cos \theta$

$$\Rightarrow I = P \cos \frac{\theta}{r^2}$$

At noon the sun's rays are normally incident making $\theta = 0$. Hence I is maximum at noon.

20. [B]

Distance of third maxima from central maxima is

$$x = \frac{3\lambda D}{d} = \frac{3 \times 5000 \times 10^{-10} \times (200 \times 10^{-2})}{0.2 \times 10^{-3}} = 1.5 \text{ cm}$$

22. [C]

For observation of colours, the thickness of the film should be of the order of the wavelength of visible light.

23. [D]

The red filter will pass only red light. Hence a blue cross will appear as black. The white background will appear as red when seen through a red filter. Thus the observer will see a black cross under a red background.

24. [B]

Newton's concept of light is that it is made of corpuscles or particles. All particles are deflected by earth's gravitational field. Hence they are also deflected by gravitational field. In the general theory of relativity, Einstein predicts deflection of light by the gravitational field. The angle of deflection predicted by both are however not the same. Einstein's prediction agrees with experimental results.

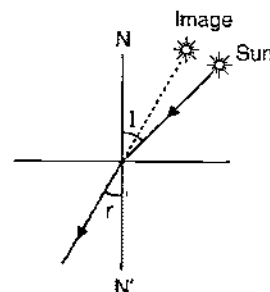
25. [D]

In photoelectric effect, the photon is treated as a particle having a quantum of energy $h\nu$. It emits one electron. Here the particle and wave aspect of light appear in two sides of equation.

Einstein's photoelectric equation appears to be interlinking of the two aspects.

26. [A]

This is the effect of refraction. Rays from the sun are passing from vacuum to air when they enter into the earth's atmosphere. They bend towards the normal making the sun to appear at a higher altitude as shown in figure.



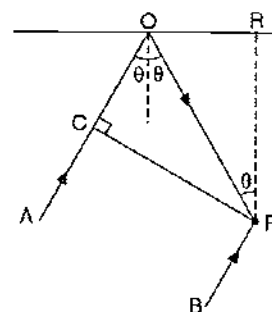
28. [D]

In the presence of thin glass plate, the fringe pattern shifts, but no change in fringe width.

29. [C]

If an unpolarised light is converted into plane polarised light by passing through a polaroid, its intensity becomes half.

30. [B]



$$PR = d$$

$$\therefore PO = d \sec \theta$$

$$\text{and } CO = PO \cot 2\theta = d \sec \theta \cos 2\theta$$

path difference between the two rays is,

$$\Delta x = CO + PO = (d \sec \theta + d \sec \theta \cos 2\theta)$$

phase difference between the two rays is

$$\Delta \phi = \pi \text{ (one is reflected, while another is direct)}$$

Therefore, condition for constructive interference should be

$$\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \dots$$

$$\Rightarrow d \sec \theta (1 + \cos 2\theta) = \frac{\lambda}{2}$$

$$\Rightarrow \left(\frac{d}{\cos \theta} \right) (2 \cos^2 \theta) = \frac{\lambda}{2}$$

$$\Rightarrow \cos \theta = \frac{\lambda}{4d}$$

31. [A]

$$\beta = \frac{\lambda D}{d} \text{ and}$$

$$\beta' = \frac{\lambda (D/2)}{2d} = \frac{\beta}{4}$$

32. [D]

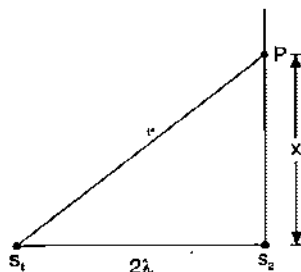
$$\beta = \frac{\lambda D}{d}$$

d is halved and D is doubled

So, Fringe width β will become four times.

33. [A]

Path difference at s_2 is 2λ . Therefore for minimum intensity at P



Let x be the minimum distance from s_2 . Then

$$s_1P - s_2P = \frac{3\lambda}{2} \times \frac{\lambda}{2} \quad \dots(1)$$

$$\Rightarrow \sqrt{4\lambda^2 + x^2} - x = \frac{3\lambda}{2}$$

Solving this equation we get

$$x = \frac{7\lambda}{12}$$

NOTE : If we substitute $s_1P - s_2P = \frac{\lambda}{2}$ in equation (1) we get

$$x = \frac{15\lambda}{4} \text{ which is greater than } \frac{7\lambda}{12}$$

34. [A]

$$\text{Shift} = \frac{(\mu - 1)tD}{d}$$

$$x = \frac{(1.5 - 1)tD}{d} \quad \dots(1)$$

$$\text{and } \frac{3}{2}x = \frac{(\mu - 1)tD}{d} \quad \dots(2)$$

Dividing equation (1) by (2)

$$\frac{2}{3} = \frac{0.5}{\mu - 1}$$

$$2\mu - 2 = 1.5$$

$$2\mu = 3.5$$

$$\mu = 1.75$$

35. [A]

Phase difference corresponding to $y_1 = -\frac{\pi}{2}$ and that for

$$y_2 = +\frac{\pi}{2}$$

$\Rightarrow$ Average intensity between y_1 and y_2

$$\Rightarrow I_{av} = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} I_{max} \cos^2\left(\frac{\phi}{2}\right) d\phi = I_{max} \frac{(\pi + 2)}{2\pi}$$

So, the required ratio is $\frac{1}{2} \left(1 + \frac{2}{\pi}\right)$

36. [C]

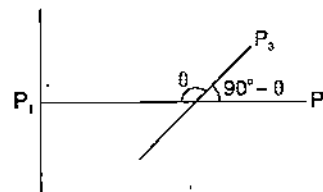
Intensity at the centre will be zero if path difference $= \frac{\lambda}{2}$

$$\Rightarrow (\mu - 1)t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{\lambda}{2(\mu - 1)}$$

37. [A]

No light is emitted from the second polaroid, so P_1 and P_2 are perpendicular to each other



Let the initial intensity of light is I_0 . So Intensity of light after

transmission from first polaroid $= \frac{I_0}{2}$

Intensity of light emitted from P_3

$$I_1 = \frac{I_0}{2} \cos^2 \theta$$

Intensity of light transmitted from last polaroid i.e., from

$$P_2 = I_1 \cos^2 (90^\circ - \theta) = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta$$

$$P_2 = \frac{I_0}{8} (2 \sin \theta \cos \theta)^2 = \frac{I_0}{8} \sin^2 2\theta$$

38. [C]

$$I = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

$$\Rightarrow \phi = \frac{2\pi}{3}$$

$$\Rightarrow \Delta x \times \left(\frac{2\pi}{\lambda}\right) = \frac{2\pi}{3} = \frac{\lambda}{3}$$

$$\Rightarrow \sin \theta = \frac{\Delta x}{d}$$

$$\Rightarrow \sin \theta = \frac{\lambda}{3d}$$

39. [B]

$$I = I_{max} \cos^2\left(\frac{\phi}{2}\right)$$

$$\frac{1}{2} = \cos^2\left(\frac{\phi}{2}\right)$$

$$\Rightarrow \cos \phi = 0$$

$$\Rightarrow \phi = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$\Rightarrow \Delta x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$$

$$\Rightarrow \Delta x = (2n + 1) \frac{\lambda}{4}$$

41. [A]

Path difference at P is

$$\Delta x = 2 \left(\frac{x}{2} \cos \theta \right) = x \cos \theta$$

For intensity to be maximum

$$\Delta x = n\lambda$$

$$\Rightarrow x \cos \theta = n\lambda$$

$$\Rightarrow \cos \theta = \frac{n\lambda}{x}$$

Since, $\cos \theta \neq 1$

$$\Rightarrow \frac{n\lambda}{x} \neq 1$$

$$\Rightarrow n \neq \frac{x}{\lambda}$$

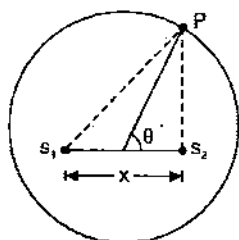
Substituting $x = 5\lambda$, we get

$$n \neq 5$$

$$\Rightarrow n = 1, 2, 3, 4, 5, \dots$$

Therefore, in all four quadrants there can be 20 maxims. There are more maxims at $\theta = 0^\circ$ and $\theta = 180^\circ$.

But $n = 5$ corresponds to $\theta = 90^\circ$ and $\theta = 270^\circ$ which are coming only twice while we have multiplied it four times. Therefore, total number of maxims are still 20 i.e., $n = 1$ to 4 in four quadrants (total 16) plus four more at $\theta = 0^\circ, 90^\circ, 180^\circ$ and 270° .



$$\{n = 0, 1, 2, 3, \dots\}$$

$$\Rightarrow I_2 = 4I_0 \cos^2 \left(\frac{\pi}{2} \right) = 2I_0 \quad \dots(2)$$

$$\text{Using (1) and (2), } \frac{I_1}{I_2} = \frac{4I_0}{2I_0} = 2$$

50. [C]

$$I = I_0 \left[\frac{\sin \alpha}{\alpha} \right]^2, \text{ where } \alpha = \frac{\phi}{2}$$

$$\text{For } n^{\text{th}} \text{ secondary maxima } d \sin \theta = \left(\frac{2n+1}{2} \right) \lambda$$

$$\Rightarrow \alpha = \frac{\phi}{2} = \frac{\pi}{\lambda} [d \sin \theta] = \left(\frac{2n+1}{2} \right) \pi$$

$$\therefore I = I_0 \left[\frac{\sin \left(\frac{(2n+1)\pi}{2} \right)}{\left(\frac{(2n+1)\pi}{2} \right)} \right]^2 = \frac{I_0}{\left\{ \frac{(2n+1)}{2} \pi \right\}^2}$$

$$\text{So } I_0 : I_1 : I_2 = I_0 : \frac{4}{9\pi^2} I_0 : \frac{4}{25\pi^2} I_0 = 1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$$

42. [A]

Laser is intense, coherent (all photons in phase) and monochromatic (only single wavelength).

43. [D]

The focal length increases with increases of wavelength.

$\frac{1}{f} \propto (n-1)$ nearly $\propto \frac{1}{\lambda^2}$ where λ is wavelength. In the given set of radiations, X-rays have the minimum wavelength. Hence focal length is minimum for X-rays.

44. [D]

As the star accelerates towards the earth, its velocity will increase. By Doppler effect in light, a source approaching an observer shows violet shift. So the colour of the star will shift towards the shorter wavelength region, that is blue.

47. [B]

$$\frac{\Delta \lambda}{\lambda} = \frac{v}{c}$$

$$\Rightarrow \frac{0.05}{100} = \frac{v}{3 \times 10^8}$$

$$\Rightarrow v = 1.5 \times 10^5$$

(Since wavelength is decreasing, so star coming closer)

48. [B]

Momentum of the electron will increase. So the wavelength $\left(\lambda = \frac{h}{p} \right)$ of electrons will decrease and fringe width decreases as $\beta \propto \lambda$.

49. [A]

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\text{At central position } I_1 = 4I_0 \quad \dots(1)$$

Since the phase difference between two successive fringes is 2π , the phase difference between two points separated by a distance equal to one quarter of the distance between the two,

$$\text{successive fringes is equal to } \delta = (2\pi) \left(\frac{1}{4} \right) = \frac{\pi}{2} \text{ radian}$$

51. [D]

If I is the final intensity and I_0 is the initial intensity then

$$I = \frac{I_0}{2} (\cos^2 30^\circ)^5$$

$$\Rightarrow \frac{I}{I_0} = \frac{1}{2} \times \left(\frac{\sqrt{3}}{2} \right)^{10} = 0.12$$

52. [B]

For maxima $\Delta = d \sin \theta = n\lambda$

$$\Rightarrow 2\lambda \sin \theta = n\lambda$$

$$\Rightarrow \sin \theta = \frac{n}{2}$$

since value of $\sin \theta$ cannot be greater 1.

$$\therefore n = 0, 1, 2$$

Therefore only five maxims can be obtained on both side of the screen.

53. [B]

In this case, we can assume as if both the source and the observer are moving towards each other with speed v . Hence

$$v' = \frac{c - u_o}{c - u_s} v = \frac{c - (-v)}{c - v} v = \frac{c + v}{c - v} v$$

$$v' = \frac{(c + v)(c - v)}{(c - v)^2} v = \frac{c^2 - v^2}{c^2 + v^2 - 2vc} v$$

$$\text{Since } v \ll c, \text{ therefore } v' = \frac{c^2}{c^2 - 2vc} = \frac{c}{c - 2v} v$$

54. [C]

Fringe width $\beta \propto \lambda$. Therefore, λ and hence β decreases 1.5 times when immersed in liquid. The distance between central maxima and 10th maxima is 3 cm in vacuum. When immersed in liquid it will reduce to 2 cm. Position of central maxima will not change while 10th maxima will be obtained at $y = 4$ cm.

55. [B]

For maximum intensity on the screen $d \sin \theta = n\lambda$

$$\Rightarrow \sin \theta = \frac{n\lambda}{d} = \frac{n(2000)}{7000} = \frac{n}{3.5}$$

Since maximum value of $\sin \theta$ is 1

So $n = 0, 1, 2, 3$ only. Thus only seven maxims can be obtained on both sides of the screen.

56. [D]

If $d \sin \theta = (\mu - 1)t$, central fringe is obtained at O

If $d \sin \theta > (\mu - 1)t$, central fringe is obtained above O and

If $d \sin \theta < (\mu - 1)t$, central fringe is obtained below O

57. [B]

Resultant intensity $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$

At central position with coherent source and $I_1 = I_2 = I_0$

$$I_{\text{coh}} = 4I_0 \quad \dots(1)$$

In case of incoherent at a given point, ϕ varies randomly with time so $(\cos \phi)_{\text{av}} = 0$

$$\therefore I_{\text{incoh}} = I_1 + I_2 = 2I_0 \quad \dots(2)$$

$$\text{Hence } \frac{I_{\text{coh}}}{I_{\text{incoh}}} = \frac{2}{1}$$

58. [A]

According to given condition

$$(\mu - 1)t = n\lambda \text{ for minimum } t, n = 1$$

So, $(\mu - 1)t_{\text{min}} = \lambda$

$$t_{\text{min}} = \frac{\lambda}{\mu - 1} = \frac{\lambda}{1.5 - 1} = 2\lambda$$

59. [B]

$$\left(\frac{I_1 + I_2}{I_1 - I_2} \right)^2 \rightarrow \infty \Rightarrow I_1 - I_2 \rightarrow 0$$

$$\Rightarrow I_1 = I_2$$

60. [A]

When unpolarised light is made incident at polarising angle, the reflected light is plane polarised in a direction perpendicular to the plane of incidence.

Therefore $\vec{E}$ in reflected light will vibrate in vertical plane with respect to plane of incidence.

61. [D]

If I is the intensity of the incident unpolarised light, the intensity transmitted by the first is $\frac{I}{2}$. This is the intensity of incident light

on the second polaroid. Intensity transmitted by the second polaroid is $\left(\frac{I}{2} \right) \cos^2 \theta$, where θ is the angle between the axes.

Here $\sin \theta = \frac{3}{5}$, $\cos \theta$ is therefore $\frac{4}{5}$.

$$\frac{I}{2} \cos^2 \theta = \frac{I}{2} \times \left(\frac{4}{5} \right)^2 = \frac{8}{25} I$$

$$\frac{8}{25} \text{ is the required ratio.}$$

63. [D]

If you divide the original slit into N strips and represents the light from each strip, when it reaches the screen, by a phasor, then at

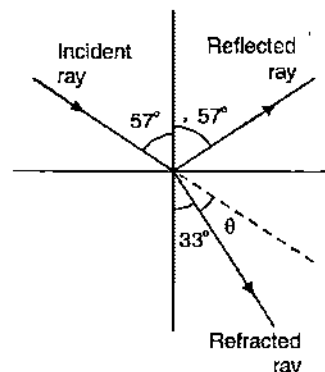
the central maximum in the diffraction pattern you add N phasors, all in the same direction and each with the same amplitude. The intensity is therefore N^2 . If you double the slit width, you need $2N$ phasors, if they are each to have the amplitude of the each to have the amplitude of the phasors you used for the narrow slit. The intensity at the central maximum is proportional to $(2N)^2$ and is, therefore, four times the intensity for the narrow slit.

64. [B]

$\sin C = \frac{3}{5}$. Hence $n = \frac{5}{3}$. If i is the polarising angle $\tan i = n$.

$$i = \tan^{-1}(n) = \tan^{-1}\left(\frac{5}{3}\right).$$

65. [C]



For glass of refractive index 1.5, polarising angle is 57° , $\tan i = 1.5$. At this stage the angle between the refracted ray and the reflected ray is 90° . The required angle from figure is, $\theta = 57 - 33 = 24^\circ$.

66. [D]

The property of double refraction is shown by quartz, calcite and ice also.

67. [C]

For first minimum

$$\lambda = a \sin \theta$$

$$\Rightarrow 6.5 \times 10^{-7} = a \sin 30$$

$$\Rightarrow a = 13 \times 10^{-7} \text{ m}$$

$$\Rightarrow a = 1.3 \mu\text{m}$$

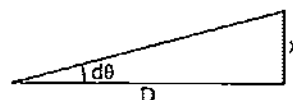
68. [B]

The resolving power of the human eye is $\frac{1}{d\theta}$ is the smallest angle subtended by two objects just distinguishable,

$d\theta = r = \frac{1}{60^\circ}$. Converting this into radian and taking 1 radian as

nearly 60° (Actually $57^\circ 18'$) we get the resolving power as 3600.

70. [C]



If x is the distance between poles and D is the distance of the eye from the poles as shown in figure we have $d\theta = \tan\theta = \frac{x}{D}$.

Here $D = 3.6 \text{ km} = 3600 \text{ m}$ and $d\theta = \frac{1}{3600} \text{ radian}$. Hence $x = 1 \text{ m}$.

71. [A]

$$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$

$$\Rightarrow 1 = \frac{v}{c}$$

$$\Rightarrow v = c$$

72. [C]

$$\lambda' = \lambda \left(1 - \frac{v}{c} \right) = 5890 \left(1 - \frac{4.5 \times 10^6}{3 \times 10^8} \right) \approx 5802 \text{ Å}$$

73. [C]

$$\frac{\lambda_1}{\lambda_2} = \frac{4500}{6000} = \frac{3}{4}$$

$$\Rightarrow 4\lambda_1 = 3\lambda_2$$

i.e. the fourth bright fringe of λ_1 coincides with the third bright fringe of λ_2 .

74. [C]

From here, we must take a note that central maxima ($n=0$) is flanked by first minima ($n=1$) and then first maxima ($n=1$).

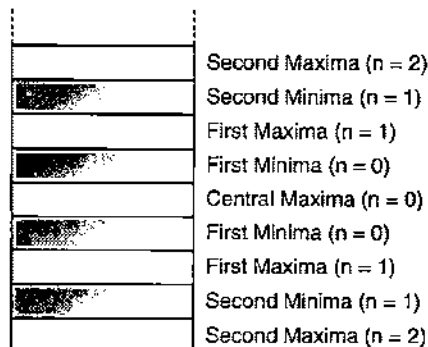
Position of the $(n+1)$ th dark fringe is

$$x_{n+1} = (2n+1) \frac{\lambda D}{2d}$$

So, position of second dark fringe is

$$x_2 = [2(1)+1] \frac{\lambda D}{2d}$$

$$\Rightarrow x_2 = \frac{3\lambda D}{2d}$$



Similarly, position of n th bright fringe is

$$x_n = n \left(\frac{\lambda D}{d} \right)$$

$$\Rightarrow x_4 = \frac{4\lambda D}{d}$$

So, separation is

$$\Delta x = x_4 - x_2 = \frac{5}{2} \left(\frac{\lambda D}{d} \right)$$

$$\Rightarrow \Delta x = 1.5 \text{ mm}$$

75. [C]

$$\text{Shift, } \Delta x = [(\mu - 1)t] \frac{D}{d}$$

$$\Rightarrow 20\beta = (\mu - 1) \frac{tD}{d}$$

$$\Rightarrow 20 \left(\frac{\lambda D}{d} \right) = (\mu - 1) \frac{tD}{d}$$

$$\Rightarrow 20\lambda = (\mu - 1)t$$

$$\Rightarrow \mu = \frac{20\lambda}{t} + 1$$

$$\Rightarrow \mu = \frac{20 \times 5000 \times 10^{-10}}{2.5 \times 10^{-5}} + 1$$

$$\Rightarrow \mu = \frac{4}{10} + 1$$

$$\Rightarrow \mu = 1.4$$

76. [B]

For dark fringe at P

$$S_1P - S_2P = \Delta = \frac{(2n-1)\lambda}{2}$$

Here $n = 3$ and $\lambda = 6000$

$$\text{So, } \Delta = \frac{5\lambda}{2} = 5 \times \frac{6000}{2} = 15000 \text{ Å} = 1.5 \text{ micron}$$

77. [A]

Fringe width $w = \frac{\lambda D}{d}$. When the apparatus is immersed in a liquid, λ and hence w is reduced μ (refractive index) times.

$$10w' = (5.5)w$$

$$\Rightarrow 10\lambda' \left(\frac{D}{d} \right) = (5.5) \frac{\lambda D}{d}$$

$$\Rightarrow \frac{\lambda}{\lambda'} = \frac{10}{5.5} = \mu$$

$$\Rightarrow \mu = 1.8$$

78. [B]

The intensity of light received is inversely proportional to the square of distance. Hence the amplitude of the light wave received is inversely proportional to the distance.

79. [C]

Fringe width $\beta \propto \lambda$. Therefore, λ and hence w will decrease 1.5 times when immersed in liquid. The distance between central maxima and 10^{th} maxima is 3 cm in vacuum. When immersed in liquid it will reduce to 2 cm. Position of central maxima will not change while 10^{th} maxima will be obtained at $y = 4 \text{ cm}$.

80. [A]

$$I = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$I_0 = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\Rightarrow \cos \left(\frac{\phi}{2} \right) = \frac{1}{2}$$

$$\Rightarrow \frac{\phi}{2} = \frac{\pi}{3}$$

$$\Rightarrow \phi = \frac{2\pi}{3} = \left(\frac{2\pi}{\lambda}\right) \Delta x$$

$$\Rightarrow \frac{1}{3} = \left(\frac{1}{\lambda}\right) y \frac{d}{D} \quad \left\{ \because \Delta x = \frac{yd}{D} \right\}$$

$$\Rightarrow y = \frac{\lambda}{3 \times \frac{d}{D}} = \frac{6 \times 10^{-7}}{3 \times 10^{-4}}$$

$$\Rightarrow y = 2 \times 10^{-3} \text{ m} = 2 \text{ mm}$$

81. [B]
For third maximum

$$x = n \left(\frac{\lambda D}{d} \right)$$

$$\Rightarrow x = (3) \frac{5000 \times 10^{-10} \times 2}{0.2 \times 10^{-3}}$$

$$\Rightarrow x = 1.5 \text{ cm}$$

82. [B]
 $n_1 \lambda_1 = n_2 \lambda_2$

$$\Rightarrow n_2 = n_1 \times \frac{\lambda_1}{\lambda_2} = 12 \times \frac{600}{400} = 18$$

83. [D]
Blue of the sky is explained by scattering. Scattered light obeys Rayleigh's law. $I \propto \frac{1}{\lambda^4}$. Hence shorter wavelengths are scattered with more intensity. In visible light, violet and blue have the minimum wavelength. Of the two, blue is more intense.

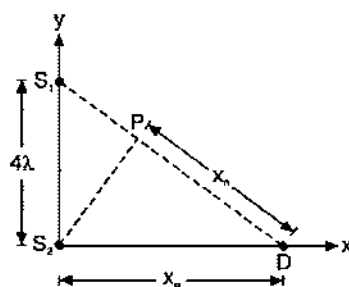
84. [C]
The interference fringes for two slits are hyperbolic.

85. [A]
Total phase difference
= Initial phase difference + Phase difference due to path
 $\Rightarrow \phi = 66^\circ + \frac{360^\circ}{\lambda} \times \Delta x = 66^\circ + \frac{360^\circ}{\lambda} \times \frac{\lambda}{4} = 66^\circ + 90^\circ = 156^\circ$

86. [B]
From $\Delta S_1 S_2 D$,

$$(S_1 D)^2 = (S_1 S_2)^2 + (S_2 D)^2$$

$$(S_1 P + PD)^2 = (S_1 S_2)^2 + (S_2 D)^2$$



Here $S_1 P$ is the path difference $= n\lambda$ for maximum intensity.

$$\therefore (n\lambda + x_n)^2 = (4\lambda)^2 + (x_n)^2$$

$$\Rightarrow x_n = \frac{16\lambda^2 - n^2\lambda^2}{2n\lambda}$$

$$\text{Then } x_1 = \frac{16\lambda^2 - \lambda^2}{2\lambda} = 7.5\lambda$$

$$x_2 = \frac{16\lambda^2 - 4\lambda^2}{4\lambda} = 3\lambda$$

$$x_3 = \frac{16\lambda^2 - 9\lambda^2}{6\lambda} = \frac{7}{6}\lambda$$

$$x_4 = 0$$

So, Number of points for maxima becomes 3.

87. [B]
If a is the width of the slit, the minimum of Fraunhofer diffraction pattern appears for $a \sin \theta = n\lambda$, where n is an integer. For the first minimum $n=1$. Then $\sin \theta = \frac{\lambda}{a} = 10^{-3} \text{ rad}$; $1 \text{ rad} = 57^\circ$.
Hence $\theta = 0.057^\circ = 0.057 \times 60' = 3.42'$.

88. [B]
Angular limit of resolution of human eye is

$$\frac{1.22\lambda}{a} = R (\text{say})$$

$$\Rightarrow R = \frac{1.22 \times 5 \times 10^{-7}}{2 \times 10^{-3}} \text{ rad}$$

$$\Rightarrow R = \frac{1.22 \times 5 \times 10^{-7}}{2 \times 10^{-3}} \times \frac{180}{\pi} \text{ degree}$$

$$\Rightarrow R = 0.0175 \text{ degree}$$

$$\Rightarrow R = 0.0175 \times 60 \text{ minute}$$

$$\Rightarrow R = 1 \text{ minute}$$

89. [C]
 $\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1/I_2} + 1}{\sqrt{I_1/I_2} - 1} \right)^2 = \frac{9}{1}$
 $\Rightarrow \frac{x+1}{x-1} = 3$
 $\Rightarrow x = 2$
 $\Rightarrow \frac{I_1}{I_2} = 4$
 $\Rightarrow I_1 = 4I_2$

i.e., if $I_2 = I_0$, then $I_1 = 4I_0$

$$I_0 = 4I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\Rightarrow \phi = \frac{2\pi}{3}$$

$$\Rightarrow \left(\frac{2\pi}{\lambda} \right) \left(\frac{yd}{D} \right) = \frac{2\pi}{3}$$

$$\Rightarrow y = \frac{\lambda D}{3d}$$

90. [A]
 $(n+1)5200 = n(7800)$
 $\Rightarrow n = 2$

92. [B]
Air has a refractive index slightly greater than one. So, when the chamber is evacuated, the refractive index decreases and hence wavelength increases as a result of which fringe width increases slightly.

93. [D]
Ultraviolet radiations, when fall on a fluorescent material, are converted into visible radiations. Hence we should use ultraviolet radiation.

Δ_2 = Path difference between the waves after emerging from slits.

$$\Delta_1 = SS_1 - SS_2 = \sqrt{D^2 + d^2} - D$$

$$\text{and } \Delta_2 = S_1O - S_2O = \sqrt{D^2 + d^2} - D$$

$$\Rightarrow \Delta = 2\left\{\left(D^2 + d^2\right)^{\frac{1}{2}} - D\right\} = 2\left\{\left(D^2 + \frac{d^2}{2D}\right) - D\right\}$$

$$\Rightarrow \Delta = \frac{d^2}{D} \quad (\text{From Binomial expansion})$$

For obtaining dark at O, Δ must be equals to $(2n-1)\frac{\lambda}{2}$ i.e.,

$$\frac{d^2}{D} = (2n-1)\frac{\lambda}{2}$$

$$\Rightarrow d = \sqrt{\frac{(2n-1)\lambda D}{2}}$$

For minimum distance $n=1$ so $d = \sqrt{\frac{\lambda D}{2}}$

106. [D]

For destructive interference, path difference must be an odd multiple of $\frac{\lambda}{2}$.

107. [C]

These concentric bright and dark fringes are called Newton's Rings.

108. [B]

$$I = I_0 + I_0 + 2I_0 \cos(2\pi)$$

$$\Rightarrow I = 4I_0$$

$$\text{For } x = \frac{\lambda}{4}, \phi = \frac{2\pi}{\lambda}x = \frac{\pi}{2}$$

$$\Rightarrow I' = I_0 + I_0 + 0$$

$$\Rightarrow I' = 2I_0 = \frac{I}{2}$$

109. [A]

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \beta \propto D$$

$$\Rightarrow \frac{\beta_1}{\beta_2} = \frac{D_1}{D_2}$$

$$\Rightarrow \frac{\beta_1 - \beta_2}{\beta_2} = \frac{D_1 - D_2}{D_2}$$

$$\Rightarrow \frac{\Delta\beta}{\Delta D} = \frac{\beta_2}{D_2} = \frac{\lambda_2}{d_2}$$

$$= \lambda_2 = \frac{3 \times 10^{-5}}{5 \times 10^{-2}} \times 10^{-3} = 6 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

110. [C]

$$[(n-1)t] \frac{D}{d} = \frac{\lambda(2D)}{d}$$

$$\Rightarrow \lambda = \frac{(n-1)t}{2} = 5892 \text{ \AA}$$

112. [A]

$$x = \frac{\lambda L}{d}$$

$$\Rightarrow \lambda = \frac{xd}{L}$$

113. [C]

Interference is a physical effect of superposition of wave motion. This can happen for any kind of wave motion, i.e., light waves, sound waves, matter waves, etc.

114. [B]

The extra path difference produced by the glass plate of thickness t and refractive index μ is $(\mu-1)t$. Due to this if n is the number of fringes which shift, $(\mu-1)t = n\lambda$. $t = 0.01 \times 10^{-3} \text{ m}$, $\lambda = 6000 \times 10^{-10} \text{ m}$, $\mu = 1.5$. This gives a value for $n = 8$.

117. [D]

$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos \phi$$

$$\text{Put } a_1^2 + a_2^2 = A \text{ and } a_1a_2 = B$$

$$\Rightarrow I = A + B \cos \phi$$

118. [A]

$$\Delta\lambda = \lambda \frac{v}{c} \text{ and } v = r\omega$$

$$v = 7 \times 10^8 \times \frac{2\pi}{25 \times 24 \times 3600}, c = 3 \times 10^8 \text{ ms}^{-1}$$

$$\Rightarrow \Delta\lambda = 0.04 \text{ \AA}$$

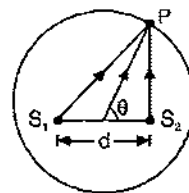
119. [C]

$$\text{Shift} = \frac{\beta}{\lambda}(\mu-1)t$$

$$\Rightarrow 7\beta = \frac{\beta}{\lambda}(\mu-1)t$$

$$\Rightarrow t = \frac{7\lambda}{(\mu-1)} = \frac{7 \times 600}{(1.5-1)} = 8400 \text{ nm}$$

120. [B]



Here path difference at a point P on the circle is given by

$$\Delta x = d \cos \theta \quad \dots(1)$$

For maxima at P

$$\Delta x = n\lambda \quad \dots(2)$$

From equation (1) and (2)

$$n\lambda = d \cos \theta$$

$$\Rightarrow \theta \cos^{-1} \left(\frac{n\lambda}{d} \right) = \cos^{-1} \left(\frac{4\lambda}{d} \right)$$

121. [A]

The film appears bright when the path difference

$$(2\mu t \cos r) \text{ is equal to odd multiple of } \frac{\lambda}{2}$$

$$\text{i.e., } 2\mu t \cos r = \frac{(2n-1)\lambda}{2} \text{ where } n = 1, 2, 3, \dots$$

$$\Rightarrow \lambda = \frac{4\mu t \cos r}{(2n-1)}$$

$$\Rightarrow \lambda = \frac{4 \times 1.4 \times 10,000 \times 10^{-10} \times \cos 0}{(2n-1)} = \frac{56000}{(2n-1)} \text{ \AA}$$

$$\Rightarrow \lambda = 56000 \text{ \AA}, 18666 \text{ \AA}, 8000 \text{ \AA}, 6222 \text{ \AA}, 5091 \text{ \AA}, 4308 \text{ \AA}, 3733 \text{ \AA}.$$

The wavelength which are not within specified range are to be refracted.

122. [A]

$$\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$

$$\Rightarrow \frac{(401.8 - 393.3)}{393.3} = \frac{v}{3 \times 10^8}$$

$$\Rightarrow v = 6.48 \times 10^6 \text{ ms}^{-1} = 6480 \text{ kms}^{-1}$$

123. [D]

Let n th minima of 400 nm coincides with m th minima of 560 nm then

$$(2n-1)400 = (2m-1)560$$

$$\Rightarrow \frac{2n-1}{2m-1} = \frac{7}{5} = \frac{14}{10} = \frac{21}{15}$$

i.e., 4th minima of 400 nm coincides with 3rd minima of 560 nm

The location of this minima is

$$= \frac{7(1000)(400 \times 10^{-6})}{2 \times 0.1} = 14 \text{ mm}$$

Next, 11th minima of 400 nm will coincide with 8th minima of 560 nm

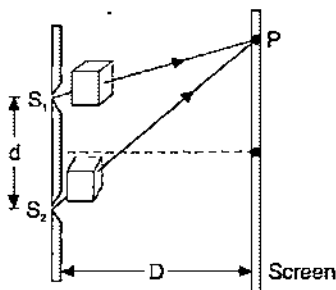
Location of this minima is

$$= \frac{21(1000)(400 \times 10^{-6})}{2 \times 0.1} = 42 \text{ mm}$$

$\therefore$ Required distance = 28 mm

124. [A]

$$\text{Shift } \Delta x = \frac{\beta}{\lambda}(\mu-1)t$$



$$\text{Shift due to one plate } \Delta x_1 = \frac{\beta}{\lambda}(\mu_1-1)t$$

$$\text{Shift due to another path } \Delta x_2 = \frac{\beta}{\lambda}(\mu_2-1)t$$

$$\text{Net shift } \Delta x = \Delta x_2 - \Delta x_1 = \frac{\beta}{\lambda}(\mu_2 - \mu_1)t \quad \dots(1)$$

$$\text{Also it is given that } \Delta x = 5\beta \quad \dots(2)$$

$$\text{Hence } 5\beta = \frac{\beta}{\lambda}(\mu_1 - \mu_2)t$$

$$\Rightarrow t = \frac{5\lambda}{(\mu_2 - \mu_1)} = \frac{5 \times 4800 \times 10^{-10}}{(1.7 - 1.4)} = 8 \times 10^{-6} \text{ m} = 8 \text{ }\mu\text{m}$$

126. [B]

At the polarising angle, the reflected ray is fully polarised while the transmitted ray is partially polarised. In fact a method to produce plane polarised light is by reflection at the polarising angle.

128. [B]

$$a = \frac{\lambda}{\theta}$$

$$\Rightarrow a = \frac{6500 \times 10^{-10}}{\left(\frac{\pi}{6}\right)}$$

$$\Rightarrow a = 1.24 \times 10^{-6} \text{ m}$$

$$\Rightarrow a = 1.24 \text{ }\mu\text{m}$$

130. [A]

Using Malus Law, $I = I_0 \cos^2 \theta$

As here polariser is rotating i.e., all the values of θ are possible.

$$I_{av} = \frac{1}{2\pi} \int_0^{2\pi} I d\theta = \frac{1}{2\pi} \int_0^{2\pi} I_0 \cos^2 \theta d\theta$$

On integration we get $I_{av} = \frac{I_0}{2}$

$$\text{where } I_0 = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{P}{A} = \frac{10^{-3}}{3 \times 10^{-4}} = \frac{10 \text{ Watt}}{3 \text{ m}^2}$$

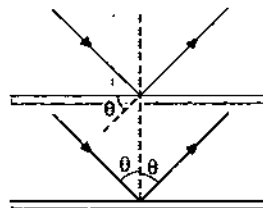
$$\Rightarrow I_{av} = \frac{1}{2} \times \frac{10}{3} = \frac{5}{3} \text{ Watt}$$

$$\text{and Time period } T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{31.4} = \frac{1}{5} \text{ sec}$$

So, energy of light passing through the polariser per revolution

$$= I_{av} \times \text{Area} \times T = \frac{5}{3} \times 3 \times 10^{-4} \times \frac{1}{5} = 10^{-4} \text{ J}$$

131. [C]



$$\text{Path difference} = 2d \sin \theta$$

For constructive interference

$$2d \sin \theta = n\lambda$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{n\lambda}{2d} \right)$$

133. [D]

During sunset and sunrise we see the unscattered part of the sunlight. Sun's rays have to travel long distance and hence are incident obliquely. All shorter wavelengths are scattered by

Rayleigh's law for scattered intensity $I \propto \frac{1}{\lambda^4}$. Only the red-orange rays reach us. During noon the sun's rays are normally

incident on the earth. Hence a portion of both the shorter

wavelengths and longer wavelengths reach us. The sun looks white.

134. [D]

$$\beta = \frac{\lambda D}{d}$$

So, angular width, 2θ is

$$2\theta = \frac{2\beta}{D} = \frac{2\lambda}{d}$$

$$\theta = \frac{6328 \times 10^{-10}}{0.2 \times 10^{-3}}$$

$$\Rightarrow \theta = 3164 \times 10^{-6} \text{ radian}$$

$$\Rightarrow \theta = 3164 \times 10^{-6} \times \frac{180}{\pi} \text{ degree}$$

$$\Rightarrow \theta = 0.18 \text{ degree}$$

$$\Rightarrow 2\theta = 0.36 \text{ degree}$$

135. [A]

$$\text{For maxima } 2\pi n = \frac{2\pi}{\lambda}(XO) - 2\pi\ell$$

$$\Rightarrow \frac{2\pi}{\lambda}(XO) = 2\pi(n + \ell)$$

$$\Rightarrow (XO) = \lambda(n + \ell)$$

136. [C]

From the given data, note that the fringe width (β_1) for $\lambda_1 = 900 \text{ nm}$ is greater than fringe width (β_2) for $\lambda_2 = 750 \text{ nm}$. This means that at though the central maxima of the two coincide, but first maximum for $\lambda_1 = 900 \text{ nm}$ will be further away from the first maxima for $\lambda_2 = 750 \text{ nm}$ and so on. A stage may come when this mismatch equals β_2 , then again maxima of $\lambda_1 = 900 \text{ nm}$, will coincide with a maxima of $\lambda_2 = 750 \text{ nm}$, let this correspond to n^{th} order fringe for λ_1 . Then it will correspond to $(n+1)^{\text{th}}$ order fringe for λ_2 .

$$\text{Therefore } \frac{n\lambda_1 D}{d} = \frac{(n+1)\lambda_2 D}{d}$$

$$\Rightarrow n \times 900 \times 10^{-9} = (n+1)750 \times 10^{-9}$$

$$\Rightarrow n = 5$$

Minimum distance from

$$\text{Central maxima} = \frac{n\lambda_1 D}{d} = \frac{5 \times 900 \times 10^{-9} \times 2}{2 \times 10^{-3}}$$

$$= 45 \times 10^{-4} \text{ m} = 4.5 \text{ mm}$$

137. [C]

One of the essential condition to observe diffraction is that the dimension of the object should be of the order of wavelength of the wave. The velocity of sound in air is nearly 350 ms^{-1} . Hence the wavelength of sound waves produced by a tuning fork of frequency 384 Hz is $\lambda = \frac{v}{f} = 1 \text{ m}$ nearly. Hence the diameter of the sphere should be of the same order.

138. [D]

Distance between the first dark fringes is

$$\Delta x = \frac{2\lambda D}{d} = 2.4 \text{ mm}$$

139. [D]

The action of a Nicol prism is based on double refraction and dichroism. When a ray of light enters into a calcite crystal, it is split into two rays O and E. Both are plane polarised in mutually perpendicular planes. One of them suffers total reflection and absorption while the other comes out as plane polarised. Selective absorption is called dichroism.

140. [B]

$$\text{Shift, } \Delta x = [(\mu - 1)t] \frac{D}{d}$$

$$\text{Here } \Delta x = 5\beta$$

$$\Rightarrow 5\left(\frac{\lambda D}{d}\right) = [(\mu - 1)t] \frac{D}{d}$$

$$\Rightarrow 5\lambda = (\mu - 1)t$$

$$\Rightarrow \lambda = (\mu - 1) \frac{t}{5}$$

$$\Rightarrow \lambda = \frac{(0.5)(6 \times 10^{-8})}{5}$$

$$\Rightarrow \lambda = 6 \times 10^{-7} \text{ m}$$

$$\Rightarrow \lambda = 6000 \text{ Å}$$

141. [B]

Only transverse waves can be polarised. (A) and (C) and (D) are electromagnetic radiations, which are transverse waves while (B), that is, β rays are electrons.

146. [B]

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \frac{6}{1000} = \frac{\lambda(1.2)}{\left(\frac{0.1}{1000}\right)}$$

$$\Rightarrow \lambda = 5000 \text{ Å}$$

150. [C]

Using the equation $\beta = \frac{D\lambda}{d}$, $\beta \propto \lambda$. Of the given colours, yellow has the maximum wavelength and hence the maximum fringe width.

152. [B]

The tubelight has coating of fluorescent material which converts ultraviolet light into visible light. Hence the light is more intense.

153. [B]

We take one wavelength in the visible region and calculate the energy just to know its order. Let us take 6000 Å (red) for convenience of a rough calculation. Energy $E = \frac{hc}{\lambda}$. Taking

$$h = 6.6 \times 10^{-34}, c = 3 \times 10^8, \lambda = 6000 \times 10^{-10} \text{ m we get } \frac{hc}{\lambda} \text{ in}$$

joules. Dividing this by 1.6×10^{-19} to convert to eV, we get $E = 2 \text{ eV}$. For energy 1 eV , wavelength will be 12000 Å .

Energy 5 eV means wavelength $\frac{2}{5}$ of 6000 , that is 2400 Å .

Both these are invisible, recalling that the visible range is 4000 to 7000 Å .

154. [A]

The energy levels of atoms are well defined. Hence transition between them should give a well defined frequency, which means a single frequency, and hence a single line.

155. [C]

The torch produces light waves which are non-coherent in nature.

156. [D]

For unpolarised light

$$I' = \frac{I_0}{2} \cos^2 \phi$$

$$\Rightarrow I' = \frac{I}{2} \cos^2 \left(\frac{\pi}{4} \right)$$

$$\Rightarrow I' = \frac{I}{4}$$

158. [C]

The path difference of two waves producing destructive interference or darkness is $(2n+1)\left(\frac{\lambda}{2}\right)$ where n is an integer 0, 1, 2, 3, etc. A path difference of λ corresponds to a phase difference of 2π rad and a path difference of $\frac{\lambda}{2}$ corresponds to a phase difference of π . Hence the corresponding phase difference is $(2n+1)\pi$.

159. [B]

$$\beta' = \frac{\beta}{\mu} < \beta$$

160. [A]

$$\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \frac{9}{4}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{3}{2}$$

$$\text{Since, } \frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2 = \left(\frac{3+2}{3-2} \right)^2$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{25}{1}$$

161. [A]

$$\text{Using the equation } \beta = \frac{D\lambda}{d}; \beta \propto \lambda$$

164. [B]

In a biprism, the distance between sources is due to deviation produced by two small angled prisms. The deviation produced by each prism is $(\mu-1)\alpha$. The total deviation is $2(\mu-1)\alpha$. Separation of sources $2(\mu-1)\alpha a$.

166. [A]

$$\text{Shift} = \frac{\beta}{\lambda} (\mu-1)t = \frac{\beta}{(5000 \times 10^{-10})} (1.5) \times 2 \times 10^{-6} = 2\beta$$

i.e., 2 fringes upwards.

167. [B]

$$\text{Separation } n^{\text{th}} \text{ bright fringe and central maxima is } x_n = \frac{n\lambda D}{d}$$

$$\text{So, } x_3 = \frac{3 \times 6000 \times 10^{-10} \times 1}{0.5 \times 10^{-3}} = 3.5 \text{ mm.}$$

168. [C]

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2 = 9$$

169. [B]

For second dark fringe, $n=1$

$$\text{Since, } x = (2n+1) \frac{\lambda D}{2d}$$

$$\Rightarrow 3 = \frac{3\lambda D}{2d} \quad \dots(1)$$

For fourth bright fringe, $n=4$. Since

$$x = n \left(\frac{\lambda D}{d} \right)$$

$$\Rightarrow x = 4 \left(\frac{\lambda D}{d} \right) \quad \dots(2)$$

$$\Rightarrow \frac{x}{3} = \frac{4}{(3/2)}$$

$$\Rightarrow x = 8 \text{ mm}$$

170. [A]

For same field of view

$$n_1 \beta_1 = n_2 \beta_2$$

$$\Rightarrow n_1 \left(\frac{\lambda_1 D}{d} \right) = n_2 \left(\frac{\lambda_2 D}{d} \right)$$

$$\Rightarrow n_1 \lambda_1 = n_2 \lambda_2$$

$$\Rightarrow 60(4000) = n_2(6000)$$

$$\Rightarrow n_2 = 40$$

171. [B]

$$\beta' = \frac{\beta}{\mu} = 1 \text{ mm}$$

172. [C]

$$\frac{a_1}{a_2} = \frac{5}{1}$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{36}{16} = \frac{9}{4}$$

173. [D]

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

For contrast to be good $I_1 = I_2$ and hence

$$I_{\max} = 4I_1$$

$$I_{\min} = 0$$

Solutions

Multiple Correct Choice Type Questions

1. [A, C]

The intensity of light is $I(\theta) = I_0 \cos^2\left(\frac{\delta}{2}\right)$

where $\delta = \frac{2\pi}{\lambda}(\Delta x) = \left(\frac{2\pi}{\lambda}\right)(d \sin \theta)$

(i) For $\theta = 30^\circ$

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{10^{15}} = 300 \text{ m and } d = 150 \text{ m, we get}$$

$$\delta = \left(\frac{2\pi}{300}\right)(150)\left(\frac{1}{2}\right) = \frac{\pi}{2}$$

$$\Rightarrow \frac{\delta}{2} = \frac{\pi}{4}$$

$$\Rightarrow I(\theta) = I_0 \cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2} \quad \text{(option (A))}$$

(ii) For $\theta = 90^\circ$

$$\delta = \left(\frac{2\pi}{300}\right)(150)(1) = \pi$$

$$\Rightarrow \frac{\delta}{2} = \frac{\pi}{2} \text{ and } I(\theta) = 0$$

(iii) For $\theta = 0^\circ$, $\delta = 0$ or $\frac{\delta}{2} = 0$

$$I(\theta) = I_0 \quad \text{(option (C))}$$

3. [A, D]

The phase difference corresponding to $OS_1 - OS_2 = \frac{\lambda}{4}$ is $\frac{\pi}{2}$.

Hence, the net phase difference is given by

$$\phi = \frac{\pi}{2} - \frac{2\pi}{\lambda}(2\lambda \sin \theta)$$

$$\Rightarrow \phi = \frac{\pi}{2} - 4\pi \sin \theta$$

For maxima, we have $\phi = n\pi$, where $n = 0, \pm 1, \pm 2, \dots$

$$\Rightarrow \frac{\pi}{2} - 4\pi \sin \theta = n\pi$$

$$\Rightarrow \sin \theta = \frac{1-n}{4}$$

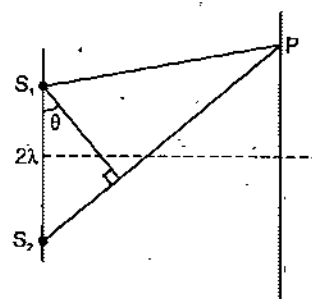
For, $n = 0$, $\sin \theta = \frac{1}{8}$

$$n = \pm 1, \sin \theta = -\frac{1}{8}, \frac{3}{8}$$

$$n = \pm 2, \sin \theta = -\frac{3}{8}, \frac{5}{8}$$

$$n = \pm 3, \sin \theta = -\frac{5}{8}, \frac{7}{8}$$

$$n = 4, \sin \theta = -\frac{7}{8}$$



6. [A, B, C]

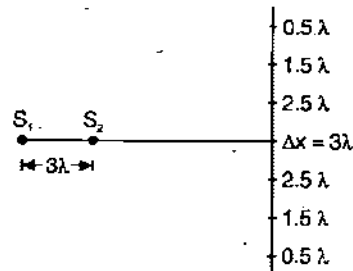
Path difference at O is d

if $d = \lambda, 2\lambda, 3\lambda, \dots$, then O is a maxima.

if, $d = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \dots$ then O is a minima

The path difference on screen is > 0

For $d = 3\lambda$, we have



8. [B, D]

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = 9$$

$$\Rightarrow \sqrt{I_1} + \sqrt{I_2} = 3(\sqrt{I_1} - \sqrt{I_2})$$

$$\Rightarrow 2\sqrt{I_1} = 4\sqrt{I_2}$$

$$\Rightarrow I_1 = 4I_2$$

$$\text{Since, } \frac{I_1}{I_2} = \left(\frac{A_1}{A_2}\right)^2$$

$$\Rightarrow \frac{A_1}{A_2} = \sqrt{\frac{I_1}{I_2}} = 2$$

10. [A, C, D]

$$I_1 = I_2 = I$$

$$I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2 = \left(\sqrt{I} + \sqrt{\frac{I}{2}}\right)^2 < 4I$$

$$I_{\min} = \left(\sqrt{I} - \sqrt{\frac{I}{2}}\right)^2 > 0$$

11. [A, C]

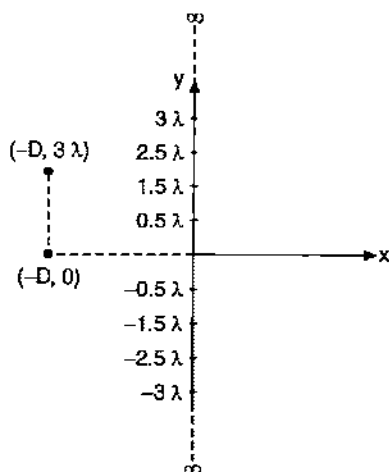
$$\text{Path difference} = \sqrt{D^2 + d^2} - D = 1 \text{ cm}$$

$$\text{Also, } (\sqrt{D^2 + d^2} - D) = (2n-1) \frac{\lambda}{2}$$

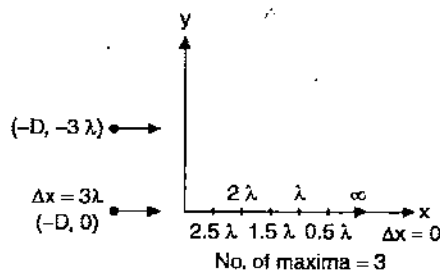
$$\Rightarrow \lambda = \frac{2(1)}{2n-1}$$

$$\Rightarrow \lambda = 2 \text{ cm}, \frac{2}{3} \text{ cm}, \frac{2}{5} \text{ cm}, \dots$$

12. [A, B, C]



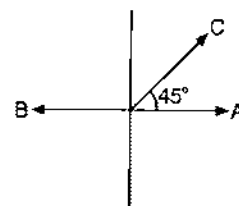
Δx at ∞ on y -axis is 3λ and -3λ . there will be 6 minima and 5 maxima



13. [B, D]

$$\sqrt{D^2 + 4d^2} - D = \frac{\lambda}{2}$$

$$\text{gives, } \lambda = \frac{4d^2}{D}$$



Intensity is due to C only
 $= I_0$

14. [A, C]

$$x = \text{Path Difference} = \sqrt{b^2 + d^2} - d$$

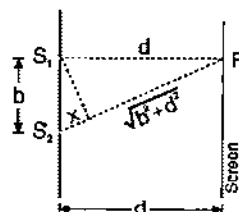
$$\Rightarrow x = d \left(\sqrt{1 + \frac{b^2}{d^2}} - 1 \right)$$

$$\Rightarrow x = d \left(1 + \frac{b^2}{2d^2} - 1 \right)$$

$$\left\{ \text{Since } \frac{b^2}{d^2} \ll 1, \text{ Hence } \sqrt{1 + \frac{b^2}{d^2}} = 1 + \frac{b^2}{2d^2} \right\}$$

$$\Rightarrow x = \frac{b^2}{2d}$$

For wavelengths to be missing we must find positions of minima.



At minima path difference is an odd multiple of $\frac{\lambda}{2}$

$$\Rightarrow x = \frac{b^2}{2d} = (2n+1) \frac{\lambda}{2}; n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \lambda = \frac{b^2}{(2n+1)d}; n = 0, 1, 2, \dots$$

$$\Rightarrow \lambda = \frac{b^2}{d}, \frac{b^2}{3d}, \frac{b^2}{5d}, \dots$$

15. [A, B]

$$\beta = \frac{\lambda D}{d}$$

For increasing β , d must be decreased and λ must be increased (i.e. frequency must be decreased).

Solutions

Reasoning Based Questions

2. [D]

Fringe width $\beta = \frac{\lambda D}{d}$ shall remain the same as the waves travel in air only, after passing through the thin transparent sheet. Due to introduction of thin sheet, only path difference of the wave is changed due to which there is shift of position of fringes only, which is given as $\Delta x = \frac{D(n-1)t}{d}$, where n is refractive index of thin sheet and t is its thickness.

3. [D]

If both the slits are illuminated by two bulbs of same power, no interference pattern will be observed on the screen. This is because waves reaching at any point on the screen do not have a constant phase difference, as phase difference from two non coherent sources changes randomly. Therefore, maxima and minima would also change their positions randomly and in quick succession. This will result in general illumination of the screen.

5. [D]

$\Delta x = d = n\lambda$, for $n=1$, $d=\lambda$ and here we will have three maxima.

9. [D]

Statement-1 is false but Statement-2 is true.

14. [D]

Statement-1 is false, Statement-2 is true.

15. [A]

Statement-1 is true & Statement-2 is true and Statement-2 is correct explanation of the Statement-1.

Solutions

Linked Comprehension Type Questions

1. [A]
It will be shifted upward
2. [D]
Cannot be predicted without knowing the thickness of thin plate.
3. [B]
 $(\mu_1 - \mu_2)t = \Delta x = 0.3t$
4. [A]
 $(\mu_1 - \mu_2)t = \Delta x = \frac{\lambda}{2}$
 $\Rightarrow t = \frac{\lambda}{2(0.3)} = \frac{\lambda}{0.6}$
5. [A]
 $\frac{3}{4}I_0 = I_0 \cos^2 \frac{\phi}{2} = \frac{\sqrt{3}}{2}$
 $\phi = \frac{\pi}{3}$
 $\Rightarrow \Delta x = \frac{\Delta \phi}{2\pi} \lambda = \frac{\lambda}{6}$
 $(\mu_1 - \mu_2)t = \frac{\lambda}{6} \Rightarrow t = \frac{\lambda}{(6)(0.3)} = \frac{5000 \text{ Å}}{1.8} = 2777.7 \text{ Å}$
6. [A]
Optical Path Difference (OPD) = $2n_1t$
7. [C]
For destructive interference, we have
 $2n_1t_{\min} = \frac{\lambda}{2}$
 $\Rightarrow t_{\min} = \frac{550}{4 \times 1.38} = 99.64 \text{ nm}$
8. [A]
 $2n_1t = \frac{3\lambda}{2}$
 $\Rightarrow t = 3t_{\min} \approx 298.9 \text{ nm}$
9. [A]
Power received at A,
 $P_A = \frac{10}{\pi} \pi (0.001)^2 = 10^{-5} \text{ W}$
10. [B]
Power received at B,
 $P_B = \frac{10}{\pi} \pi (0.002)^2 = 4 \times 10^{-5} \text{ W}$
11. [C]
Power transmitted through A,
 $P'_A = \frac{10}{100} \times 10^{-5} \text{ W} = 10^{-6} \text{ W}$
12. [D]
Power transmitted through B,
 $P'_B = \frac{10}{100} \times 4 \times 10^{-5} \text{ W} = 4 \times 10^{-6} \text{ W}$
13. [C]
Path difference,
 $\Delta = (\mu - 1)t = (1.5 - 1) \times 2000 \times 10^{-10} \text{ m}$
 $\Delta = 10^{-7} \text{ m}$
14. [C]
Phase difference, $\phi = \frac{2\pi}{\lambda} \Delta = \frac{2\pi \times 10^{-7}}{6000 \times 10^{-10}}$
 $\Rightarrow \phi = \frac{\pi}{3} \text{ radian}$
15. [D]
We know that, $P \propto a^2$ or $P = ka^2$, where k is any arbitrary positive constant.
Now, $a_A = \sqrt{\frac{10^{-6}}{k}}$, $a_B = \sqrt{\frac{4 \times 10^{-6}}{k}}$
Resultant amplitude,
 $a_r = \sqrt{a_A^2 + a_B^2 + 2a_A a_B \cos \phi}$
Substituting values and simplifying,
 $a_r^2 = \frac{7 \times 10^{-6}}{k}$
Resultant power, $P_r = ka_r^2 = 7 \times 10^{-6} \text{ W} = 7 \text{ } \mu\text{W}$

16. [D]
Anywhere on screen because there is no relation between θ , μ .

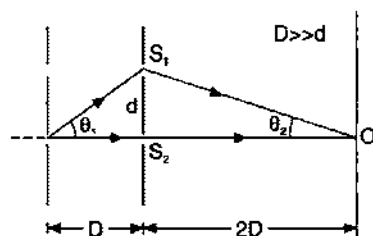
17. [A]
Total path difference is given by
 $\Delta x = (\mu - 1)t - d \sin \theta$
For central bright fringe to be obtained at O, we have
 $\Delta x = 0$
 $\Rightarrow (\mu - 1)t = d \sin \theta$

18. [B]
 $\Delta \phi = (2n - 1)\pi$, where $n = 5$

19. [B]
Fringe width $\beta = \frac{\lambda D}{d} = 4 \times 10^5 \text{ nm}$

20. [B]
Since, $\Delta x = d \sin \theta - (\mu - 1)t$, so if θ increases then $d \sin \theta$ will increase and hence Δx increases, so that the central maxima will shift upward.

21. [B]
Net path difference is given by
 $d \sin \theta_1 + d \sin \theta_2 = \frac{\lambda}{2}$ (for minimum d)



$$\Rightarrow d \left(\frac{d}{D} \right) + d \left(\frac{d}{2D} \right) = \frac{\lambda}{2} \quad (\sin \theta \approx \tan \theta)$$

$$\Rightarrow \frac{3d^2}{2D} = \frac{\lambda}{2}$$

$$\Rightarrow d = \sqrt{\frac{\lambda D}{3}}$$

22. [C]

For central maxima, we have
 $d \sin \theta_1 + d \sin \theta_2 = 0$

$$\Rightarrow \frac{d}{D} + d \frac{y}{2D} = 0$$

$$\Rightarrow y = -2d$$

So, the position of first bright fringe is given by

$$y = -2d + \frac{d}{2} = -\frac{3}{2}d$$

So, the first bright fringe is located $\frac{3}{2}d$ below O

23. [D]
 $d \sin \theta_1 + d \sin \theta_2 = n\lambda$

$$\Rightarrow \frac{d^2}{D} + \frac{nyd}{2D} = n\lambda \quad \dots(1)$$

$$\text{Also, } \frac{d^2}{D} + \frac{nyd + 1}{2D} = (n + 1)\lambda \quad \dots(2)$$

$$(2) - (1) \text{ we get } \frac{d}{2D} \beta = \lambda$$

$$\Rightarrow \beta = \frac{2D}{d} \lambda$$

24. [B]
In reflected light system, for constructive interference, we have

$$2\mu t = (2n - 1) \frac{\lambda}{2}$$

For minimum thickness, we have

$$n = 1$$

$$\Rightarrow 2\mu t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{900}{4 \times 1.5}$$

$$\Rightarrow t = 150 \text{ nm}$$

25. [A]
In transmitted light system, for destructive interference, we have

$$2\mu t = (2n - 1) \frac{\lambda}{2}$$

Again, for minimum thickness, we have

$$n = 1$$

$$\Rightarrow t = \frac{900}{4 \times 1.5} = 150 \text{ nm}$$

26. [C]
Path difference between the waves reaching D is

$$\Delta x = \frac{3\pi R}{2} - \frac{\pi R}{2} = \frac{2\pi R}{2} = \pi R$$

Since, $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$

$$\Rightarrow I_R = (\sqrt{I_1} + \sqrt{I_2})^2 \text{ when } \phi = 2n\pi, n = 0, 1, 2, 3, \dots$$

Since the wave generated is divided equally in two parts, so we have

$$I_1 = I_2 = \frac{I_0}{2}$$

$$\Rightarrow I_R = \left(\sqrt{\frac{I_0}{2}} + \sqrt{\frac{I_0}{2}} \right)^2$$

$$\Rightarrow I_R = 2I_0$$

27. [A]

Since $\Delta x = \pi R$

For maxima, we have

$$\Delta \phi = (2n)\pi, \text{ where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \left(\frac{2\pi}{\lambda} \right) (\pi R) = (2n)\pi$$

$$\Rightarrow \lambda_{\max} = \frac{\pi R}{n_{\min}} = \frac{\pi R}{1}$$

$$\Rightarrow \lambda_{\max} = \pi R$$

28. [B]

Similarly, for minima, we have

$$\Delta \phi = (2n + 1)\pi, \text{ where } n = 0, 1, 2, 3, \dots$$

$$\Rightarrow \left(\frac{2\pi}{\lambda}\right)(\pi R) = (2n+1)\pi$$

$$\Rightarrow \lambda_{\max} = \frac{2\pi R}{2n+1}$$

$$\Rightarrow \lambda_{\max} = 2\pi R$$

29. [D]

$$\text{Since, } d_{\text{air}} = 2a(\mu - 1)\alpha$$

$$\text{and } d_{\text{water}} = 2a\left(\frac{\mu}{\mu_w} - 1\right)\alpha$$

$$\Rightarrow \frac{d_{\text{water}}}{d_{\text{air}}} = \frac{\mu - 1}{\mu_w - 1}$$

$$\Rightarrow \frac{d_{\text{water}}}{0.5} = \frac{\frac{3/2}{4/3} - 1}{\frac{3}{2} - 1} = \frac{1}{4}$$

$$\Rightarrow d_{\text{water}} = \frac{1}{4} \times 0.5 = \frac{1}{8} \text{ mm}$$

30. [C]

$$\text{Since, } \beta_{\text{air}} = \frac{D\lambda_{\text{air}}}{2a(\mu - 1)\alpha}$$

$$\text{and } \beta_{\text{water}} = \frac{D\lambda_{\text{water}}}{2a(\mu - \mu_w)\alpha}$$

$$\Rightarrow \frac{\beta_{\text{water}}}{\beta_{\text{air}}} = \frac{\mu - 1}{\mu - \mu_w} = \frac{\frac{3}{2} - 1}{\frac{3}{2} - \frac{4}{3}}$$

$$\Rightarrow \beta_{\text{water}} = 3 \times 1 = 3 \text{ mm}$$

31. [C]

$$\text{Since, } \beta = \frac{\lambda D}{d}$$

$$\Rightarrow 1 \times 10^{-3} = \frac{1 \times \lambda}{0.5 \times 10^{-3}}$$

$$\Rightarrow \lambda = 5 \times 10^{-7} \text{ m} = 5000 \text{ \AA}$$

32. [D]

At O, optical path difference is

$$\Delta x = (\mu_g - 1)t - \mu_w d \sin \theta$$

$$\Rightarrow \Delta x = 0.5 \times 0.41 \times 10^{-3} - \frac{4}{3} \times 3 \times 10^{-4} \times \frac{1}{2}$$

$$\Rightarrow \Delta x = 5 \times 10^{-6} \text{ m}$$

Since, $\lambda = 5000 \text{ \AA}$, so we observe that

$$\Delta x = 10\lambda$$

So, 10^{th} bright fringe, will be formed at O.

33. [D]

For central maxima, we have

$$\Delta x = 0$$

$$\text{where, } \Delta x = \frac{yd}{D} + (\mu_g - 1)t - \mu_w d \sin \theta$$

$$\Rightarrow \frac{yd}{D} + (\mu_g - 1)t - \mu_w d \sin \theta = 0$$

Substituting values, we get

$$\frac{yd}{D} = -10\lambda$$

$$\Rightarrow y = -10 \left(\frac{\lambda D}{d} \right) = -\frac{5}{3} \times 10^{-2} \text{ m}$$

34. [C]

At P, we have

$$\Delta x = \frac{yd}{D} + (\mu_g - 1)t - \mu_w d \sin \theta$$

$$\Rightarrow \Delta x = \frac{10^{-2} \times 3 \times 10^{-4}}{8 \times 1} + 10\lambda = 10\lambda + \frac{3\lambda}{4}$$

$$\Delta \phi = \frac{2\pi}{\lambda} \left(10\lambda + \frac{3\lambda}{4} \right) = 20\pi + \frac{3\pi}{2}$$

$$\text{So, } I_p = \sqrt{I^2 + I^2 + 2I^2 \cos \left(20\pi + \frac{3\pi}{2} \right)}$$

$$\text{Since } \cos \left(20\pi + \frac{3\pi}{2} \right) = \cos \left(\frac{3\pi}{2} \right) = 0$$

$$\Rightarrow I_p = \sqrt{2}I$$

$$\text{Further } I_{\max} = \frac{I}{2} + \frac{I}{2} + 2 \sqrt{\left(\frac{I}{2} \right) \left(\frac{I}{2} \right)} \cos(0^\circ)$$

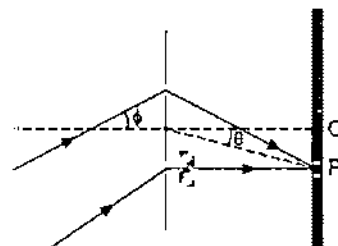
$$\Rightarrow I_{\max} = I + I = 2I$$

$$\Rightarrow \frac{I_p}{I_{\max}} = \frac{\sqrt{2}I}{2I} = \frac{1}{\sqrt{2}}$$

35. [C]

Path difference is given by

$$\Delta x = d \sin \phi + d \sin \theta - (\mu - 1)t$$



For central maxima, $\Delta x = 0$

$$\Rightarrow \sin \theta = \frac{(\mu - 1)t}{d} - \sin \phi$$

$$\Rightarrow \sin \theta = \frac{\left(\frac{3}{2} - 1 \right) (0.1)}{50 \times 10^{-3}} - \sin(30^\circ) = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

36. [B]

At C, $\theta = 0^\circ$, so we get

$$\Delta x = d \sin \phi - (\mu - 1)t$$

$$\Rightarrow \Delta x = (50 \times 10^{-3}) \left(\frac{1}{2} \right) - \left(\frac{3}{2} - 1 \right) (0.1)$$

$$\Rightarrow \Delta x = 0.025 - 0.05 = -0.025 \text{ mm}$$

Substituting, $\Delta x = n\lambda$, we get

$$n = \frac{\Delta x}{\lambda} = \frac{-0.025}{500 \times 10^{-6}} = -50$$

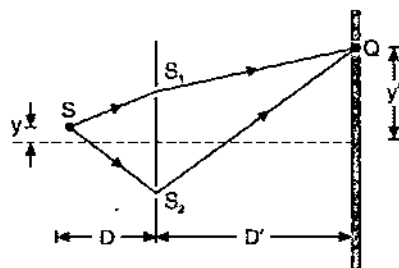
Hence, at C there will be maxima. Therefore the order of minima closest to the C are -49 .

37. [A]
Number of fringes shifted upwards is

$$N = \frac{(\mu - 1)t}{\lambda} = \frac{\left(\frac{3}{2} - 1\right)(0.1)}{500 \times 10^{-9}} = 100$$

38. [D]
Net path difference of the waves reacting at Q, is

$$\Delta x = \frac{yd}{D} + \frac{y'd}{D'}$$



For central maximum, $\Delta x = 0$

$$\Rightarrow y' = -\frac{D'}{D}y$$

$$\Rightarrow y' = -\left(\frac{2}{1}\right)(0.5 \sin(\pi t))$$

$$\Rightarrow y' = -\sin(\pi t) \text{ mm}$$

$$\Rightarrow y' = \sin(\pi + \pi t) = -\sin(\pi t)$$

39. [C]
 $y' = \frac{d}{2}$, at point P exactly in front of S_1 , so we have

$$\Delta x = \left(\frac{yd}{D}\right) + \left(\frac{d^2}{2D'}\right)$$

For maximum intensity, we have path difference to be an even multiple of $\frac{\lambda}{2}$, so

$$\Delta x = (2n)\frac{\lambda}{2} = n\lambda$$

Substituting the values, we get

$$0.5 \sin(\pi t) + 0.25 = 0.5n$$

$$\Rightarrow \sin(\pi t) = \frac{0.5n - 0.25}{0.5}$$

For minimum value of t , we have $n = 1$

$$\Rightarrow \sin(\pi t) = 0.5$$

$$\Rightarrow \pi t = \frac{\pi}{6}$$

$$\Rightarrow t = \frac{1}{6}$$

40. [A]

For minimum intensity, path difference is an odd multiple of $\frac{\lambda}{2}$, so

$$\Delta x = (2n + 1)\frac{\lambda}{2}$$

$$\Rightarrow 0.5 \sin(\pi t) + 0.25 = \frac{0.5}{2}$$

$$\Rightarrow \sin(\pi t) = 0$$

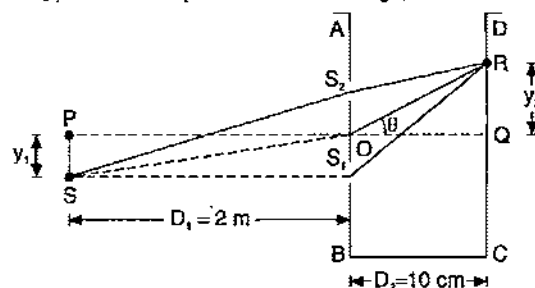
$$(\because n = 1)$$

$$\Rightarrow \pi t = \pi$$

$$\Rightarrow t = 1 \text{ s}$$

41. [B]

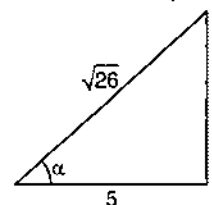
Given $y_1 = 40 \text{ cm}$, $D_1 = 2 \text{ m} = 200 \text{ cm}$, $D_2 = 10 \text{ cm}$



$$\tan \alpha = \frac{y_1}{D_1} = \frac{40}{200} = \frac{1}{5}$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{1}{5}\right)$$

$$\Rightarrow \sin \alpha = \frac{1}{\sqrt{26}} \approx \frac{1}{5} = \tan \alpha$$



Path difference between SS_1 and SS_2 is

$$\Delta x_1 = SS_1 - SS_2$$

$$\Rightarrow \Delta x_1 = d \sin \alpha = (0.8 \text{ mm})\left(\frac{1}{5}\right)$$

$$\Rightarrow \Delta x_1 = 0.16 \text{ mm} \quad \dots(1)$$

Now, let at point R on the screen, central bright fringe is observed (i.e., net path difference = 0).

Path difference between S_2R and S_1R is

$$\Delta x_2 = S_2R - S_1R$$

$$\Rightarrow \Delta x_2 = d \sin \theta \quad \dots(2)$$

Central bright fringe will be observed when net path difference is zero.

$$\Rightarrow \Delta x_2 - \Delta x_1 = 0$$

$$\Rightarrow \Delta x_2 = \Delta x_1$$

$$\Rightarrow d \sin \theta = 0.16$$

$$\Rightarrow (0.8) \sin \theta = 0.16$$

$$\Rightarrow \sin \theta = \frac{0.16}{0.8} = \frac{1}{5}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{24}}$$

$$\Rightarrow \sin \theta \approx \frac{1}{5}$$

$$\Rightarrow \tan \theta \approx \sin \theta = \frac{y_2}{D_2} = \frac{1}{5}$$

$$\Rightarrow y_2 = \frac{D_2}{5} = \frac{10}{5} = 2 \text{ cm}$$

Therefore, central bright fringe is observed at 2 cm above point Q on side CD.

Alternate solution for (a)

Δx at R will be zero if $\Delta x_1 = \Delta x_2$

$$\Rightarrow d \sin \alpha = d \sin \theta$$

$$\Rightarrow \alpha = \theta$$

$$\Rightarrow \tan \alpha = \tan \theta$$

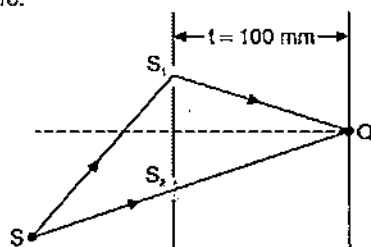
$$\Rightarrow \frac{y_1}{D_1} = \frac{y_2}{D_2}$$

$$\Rightarrow y_2 = \frac{D_2}{D_1} \cdot y_1 = \left(\frac{10}{200}\right)(40) \text{ cm}$$

$$\Rightarrow y_2 = 2 \text{ cm}$$

42. [C]

The central bright fringe will be observed at point Q, if the path difference created by the liquid slab of thickness $t = 10 \text{ cm}$ or 100 mm is equal to Δx_1 , so that the net path difference at Q becomes zero.



$$\Rightarrow (\mu - 1)t = \Delta x_1$$

$$\Rightarrow (\mu - 1)(100) = 0.16$$

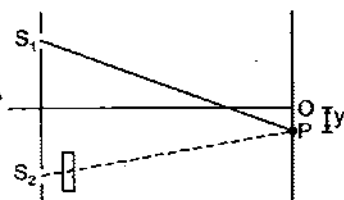
$$\Rightarrow \mu - 1 = 0.0016$$

$$\Rightarrow \mu = 1.0016$$

43. [D]

Given $\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}$,

$d = 0.45 \text{ mm} = 0.45 \times 10^{-3} \text{ m}$ and $D = 1.5 \text{ m}$



Thickness of glass sheet, $t = 10.4 \mu\text{m} = 10.4 \times 10^{-6} \text{ m}$

Refractive index of the medium, $\mu_m = \frac{4}{3}$

And refractive index of glass sheet, $\mu_g = 1.5$

Let central maximum is obtained at a distance y below point O.

$$\Rightarrow \Delta x_1 = S_1P - S_2P = \frac{yd}{D}$$

Path difference due to glass sheet is given by

$$\Delta x_2 = \left(\frac{\mu_g}{\mu_m} - 1\right)t$$

Net path difference will be zero, when we have

$$\Delta x_1 = \Delta x_2$$

$$\Rightarrow \frac{yd}{D} = \left(\frac{\mu_g}{\mu_m} - 1\right)t$$

$$\Rightarrow y = \left(\frac{\mu_g}{\mu_m} - 1\right)t \frac{D}{d}$$

Substituting the values, we get

$$y = \left(\frac{1.5}{4/3} - 1\right) \frac{10.4 \times 10^{-6} (1.5)}{0.45 \times 10^{-3}}$$

$$\Rightarrow y = 4.33 \times 10^{-3} \text{ m}$$

$$\Rightarrow y = 4.33 \text{ mm} = \frac{13}{3} \text{ mm}$$

44. [C]

$$\text{At O, } \Delta x_1 = 0 \text{ and } \Delta x_2 = \left(\frac{\mu_g}{\mu_m} - 1\right)t$$

So, net path difference

$$\Delta x = \Delta x_2$$

Corresponding phase difference $\Delta \phi = \left(\frac{2\pi}{\lambda}\right) \Delta x$

Substituting the values, we get

$$\phi = \Delta \phi = \frac{2\pi}{6 \times 10^{-7}} \left(\frac{1.5}{4/3} - 1\right) (10.4 \times 10^{-6})$$

$$\Rightarrow \phi = \left(\frac{13}{3}\right)\pi$$

$$\text{Now, } I(\phi) = I_{\max} \cos^2\left(\frac{\phi}{2}\right)$$

$$\Rightarrow I = I_{\max} \cos^2\left(\frac{13\pi}{6}\right)$$

$$\Rightarrow I = \frac{3}{4} I_{\max}$$

45. [C]

$$\text{At O, path difference is } \Delta x = \Delta x_2 = \left(\frac{\mu_g}{\mu_m} - 1\right)t$$

For maximum intensity at O, we have

$$\Delta x = n\lambda, \text{ where } n = 1, 2, 3, \dots$$

$$\Rightarrow \lambda = \frac{\Delta x}{1}, \frac{\Delta x}{2}, \frac{\Delta x}{3}, \dots \text{ and so on}$$

$$\Rightarrow \Delta x = \left(\frac{1.5}{4/3} - 1\right) (10.4 \times 10^{-6} \text{ m})$$

$$\Rightarrow \Delta x = \left(\frac{1.5}{4/3} - 1\right) (10.4 \times 10^3 \text{ nm}) = 1300 \text{ nm}$$

So, maximum intensity will be corresponding to

$$\lambda = 1300 \text{ nm}, \frac{1300}{2} \text{ nm}, \frac{1300}{3} \text{ nm}, \frac{1300}{4} \text{ nm}, \dots$$

$$\Rightarrow \lambda = 1300 \text{ nm}, 650 \text{ nm}, 433.33 \text{ nm}, 325 \text{ nm}, \dots$$

The wavelength in the range 400 nm to 700 nm are 650 nm

$$\text{and } 433.33 \text{ nm} \left(= \frac{1300}{3} \text{ nm} \right)$$

46. [C]

The optical path difference between the two waves arriving at P is

$$\Delta x = \frac{y_1 d}{D_1} + \frac{y_2 d}{D_2} = \frac{(1)(10)}{10^3} + \frac{(5)(10)}{2 \times 10^3}$$

$$\Rightarrow \Delta x = 3.5 \times 10^{-2} \text{ mm} = 0.035 \text{ mm}$$

To calculate the order of interference, we shall calculate

$$n = \frac{\Delta x}{\lambda}$$

$$\Rightarrow n = \frac{0.035 \times 10^{-3} \text{ m}}{5000 \times 10^{-10} \text{ m}}$$

$$\Rightarrow n = 70$$

$$\Rightarrow \Delta x = 70\lambda$$

So, 70th order maxima is obtained at P.

47. [D]

$$\text{At } O, \Delta x = \frac{y_1 d}{D_1} = 10^{-2} \text{ mm} = 0.01 \text{ mm}$$

Now, we observe that $\Delta x = 20\lambda$

So, 20th order maxima is obtained at O.

48. [D]

$$(\mu - 1)t = 0.01 \text{ mm}$$

$$\Rightarrow t = \frac{0.01}{1.5 - 1} = 0.02 \text{ mm} = 20 \mu\text{m}$$

Since the pattern has to be shifted upwards, therefore, the film must be placed in front of S_1 .

49. [A]

Using the concept of Displacement Method, we get

$$O = \sqrt{I_1 I_2} = 3 \text{ mm}$$

$$\text{Since, } m = \frac{f}{f + u}$$

$$\Rightarrow \frac{100 - x}{x} = 4$$

$$\Rightarrow x = 20 \text{ cm}$$

So, we get $u = -20$, $v = +80$

$$\text{Since } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow f = 16 \text{ cm}$$

50. [B]

$$\beta = \frac{\lambda D}{d} = \frac{6000 \times 10^{-10} \times 1}{3 \times 10^{-3}} = 0.2 \text{ mm}$$

51. [D]

$$\beta = \frac{\lambda D}{d}$$

When D is increased, then β is increased.

Solutions

Matrix Match Type Questions

1. $A \rightarrow (p, s)$
 $B \rightarrow (q, r)$
 $C \rightarrow (p, q, r)$
 $D \rightarrow (q, r, s)$

2. $A \rightarrow (p, q)$
 $B \rightarrow (s)$
 $C \rightarrow (p, q)$
 $D \rightarrow (r)$

3. $A \rightarrow (q)$
 $B \rightarrow (p)$
 $C \rightarrow (s)$
 $D \rightarrow (r)$

4. $A \rightarrow (r)$
 $B \rightarrow (s)$
 $C \rightarrow (q)$
 $D \rightarrow (p)$

5. $A \rightarrow (p)$
 $B \rightarrow (s)$
 $C \rightarrow (r)$
 $D \rightarrow (q, r)$

6. $A \rightarrow (r)$
 $B \rightarrow (l)$
 $C \rightarrow (r)$
 $D \rightarrow (p)$

If glass slab is introduced across S_2 this effective path increases so central maxima will be shifted downward but fringe width remains same.

7. $A \rightarrow (p)$
 $B \rightarrow (s)$
 $C \rightarrow (r)$
 $D \rightarrow (q)$

8. $A \rightarrow (p, q, r, s)$
 $B \rightarrow (p, q, r, s)$
 $C \rightarrow (p, q, r, s)$
 $D \rightarrow (p, q, r)$

Solutions

Integer Answer Type Questions

1. Shift $\Delta y = (\mu_2 - 1)t_2 - (\mu_1 - 1)t_1 \left| \frac{D}{d} \right|$

$$\Rightarrow \Delta y = (1.2 - 1) \times 15 \times 10^{-6} - (1.6 - 1) \times 10 \times 10^{-6} \left(\frac{1.5}{1.5 \times 10^{-3}} \right)$$

$$\Rightarrow \Delta y = 3 \times 10^{-3} \text{ m}$$

$$\Rightarrow \Delta y = 3 \text{ mm}$$

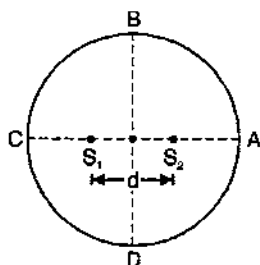
2. Since, $\lambda = 0.25 \text{ m}$ and $d = 2 \text{ m} = 8\lambda$

At A and C, $\Delta x = d = 8\lambda$

Since this is an even multiple of λ , so maximum intensity is obtained.

At B and D, $\Delta x = 0$

Since this is an even multiple of λ , so again maximum intensity will be obtained.



Further, between A and B seven maxims corresponding to $\Delta x = 7\lambda, 6\lambda, 5\lambda, 4\lambda, 3\lambda, 2\lambda$ and λ are obtained.

Similarly between B and C, C and D, and D and A.

So, total number of maxims is $N = 4 \times 7 + 4 = 32$

3. Ray 1 has a longer path than that of ray 2 by a distance $d \sin(45^\circ)$, before reaching the slits. Afterwards ray 2 has a path longer than ray 1 by a distance $d \sin \theta$. The net path difference is therefore, $d \sin \theta - d \sin 45^\circ$

(a) Central maximum is obtained where, net path difference is zero, so we have

$$d \sin \theta - d \sin(45^\circ) = 0$$

$$\Rightarrow \theta = 45^\circ$$

(b) Third order maxima is obtained where net path difference is 3λ , so we have

$$d \sin \theta - d \sin(45^\circ) = 3\lambda$$

$$\Rightarrow \sin \theta = \sin(45^\circ) + \frac{3\lambda}{d}$$

Substituting $d = 20\lambda$, we get

$$\sin \theta = \sin(45^\circ) + \frac{3\lambda}{20\lambda}$$

$$\Rightarrow \theta \approx 59^\circ$$

4. (a) $\frac{1}{10} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} + \frac{1}{R} \right)$

$$\Rightarrow R = 10 \text{ cm}$$

Now applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$ twice, we get

$$\frac{3/2}{v_1} - \frac{1}{-20} = \frac{3/2 - 1}{10} \text{ and} \quad \dots(1)$$

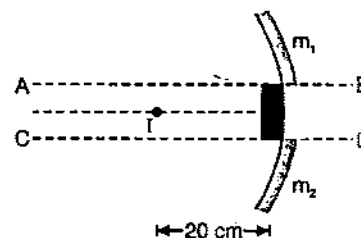
$$\frac{4/3}{v_2} - \frac{3/2}{v_1} = \frac{4/3 - 3/2}{-10} \quad \dots(2)$$

Adding equation (1) and (2), we get

$$v_2 = 80 \text{ cm}$$

(b) The image formed by (lens + water) system will act as an object for the mirror.

This is below the axis of m_1 and at the same distance as the centre is, therefore, its image will be formed vertically above at 1 mm from AB. Similarly m_2 will form an image of I_1 , 1 mm below CD.



$$\Rightarrow I_1 I_2 = 1 + 1 + 1 = 4 \text{ mm}$$

(c) $d = I_1 I_2 = 4 \text{ mm}$, $D = 80 \text{ cm}$

$$\text{Since, } \lambda = \frac{v}{f}$$

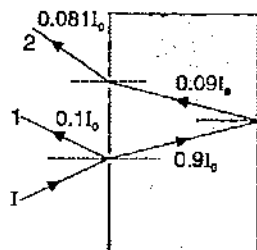
$$\Rightarrow \lambda = \frac{c}{\mu_w f} = \left(\frac{3 \times 10^8}{7.5 \times 10^{14} \times 4} \right)$$

$$\Rightarrow \beta = \frac{\lambda D}{d} = \frac{3 \left(\frac{3 \times 10^8}{7.5 \times 10^{14}} \right) \left(\frac{80 \times 10^{-2}}{4 \times 10^{-3}} \right)}{m} = 6 \times 10^{-6} \text{ m}$$

$$\Rightarrow \beta = 60 \mu\text{m}$$

5. The intensities of the rays due to successive reflections and refractions are shown in figure. So,

$$I_1 = 0.1I_0, I_2 = 0.081I_0$$



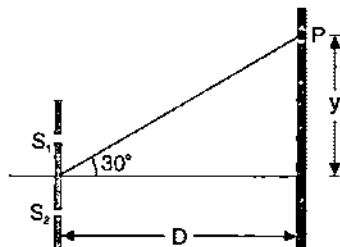
$$\Rightarrow \sqrt{\frac{I_1}{I_2}} = \frac{10}{9}$$

$$\Rightarrow \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1} \right)^2 = (19)^2 = 361$$

6. Fringe width $\beta = \frac{\lambda D}{d}$ and $y = \frac{D}{\sqrt{3}}$

Therefore, number of fringe widths in a distance y are given by

$$n = \frac{y}{\beta} = \frac{d}{\sqrt{3}\lambda} = \frac{0.32 \times 10^{-3}}{(\sqrt{3})(500 \times 10^{-9})} = 369.5$$



Therefore, total number of maxima obtained in the angular range $-30^\circ < \theta < 30^\circ$ (including the central one) is

$$N = 2 \times 369 + 1 = 739$$

7. Here, $d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$, $D = 10 \text{ cm} = 0.10 \text{ m}$,
 $t = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$, $\Delta x = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$,
 $\lambda = 6 \times 10^{-7} \text{ m}$

$$\text{Since, } \Delta x = \frac{D}{d}(\mu - 1)t$$

$$\Rightarrow \mu - 1 = \frac{\Delta x d}{D t} = \frac{5 \times 10^{-3} \times 2 \times 10^{-3}}{0.10 \times 0.5 \times 10^{-3}} = 0.2$$

$$\mu = 1 + 0.2 = 1.2$$

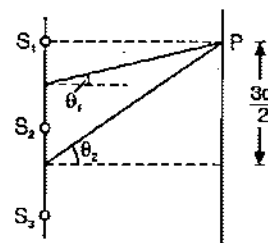
$$\Rightarrow 5\mu = 6$$

9. $S_2P - S_1P = d \sin \theta_1 = d \left(\frac{d}{2D} \right) = \frac{d^2}{2D} = \frac{\lambda}{3}$

$$\Rightarrow \Delta \phi = \frac{2\pi}{3}$$

$$S_3P - S_2P = d \sin \theta_2 = d \left(\frac{3d}{2D} \right) = \frac{3d^2}{2D} = \lambda$$

$$\Rightarrow \Delta \phi = 2\pi$$



If I_R be the resultant intensity of the wave at the point P , then

$$I_R = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

$$\Rightarrow I_R = I_0 + 4I_0 + 2\sqrt{(I_0)(4I_0)} \cos(120^\circ)$$

$$\Rightarrow I_R = 3I_0 = 3(12 \text{ Wm}^{-2}) = 36 \text{ Wm}^{-2}$$

10. Angular fringe width $\theta = \frac{\beta}{D} = \frac{\lambda}{d}$

According to the given condition

$$\frac{\lambda}{d} \geq \frac{\pi}{180 \times 60}$$

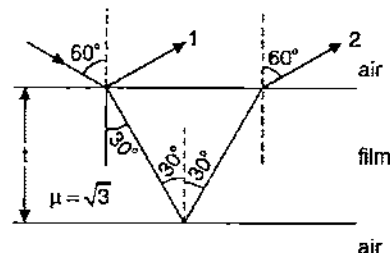
$$\Rightarrow d < \frac{6 \times 10^{-7} \times 180 \times 60}{\pi}$$

$$d_{\max} = 2.06 \times 10^{-3} \text{ m} = 2.06 \text{ mm}$$

11. According to Snell's Law, we have

$$t \sin(30^\circ) = \sqrt{3} \sin r$$

$$\Rightarrow r = 30^\circ$$



The Optical path difference is given by

$$\Delta x = 2\mu t \sec r - 2t \tan r \sin i$$

$$\Rightarrow \Delta x = 2\sqrt{3}(t \sec 30^\circ) - 2t \tan(30^\circ) \sin(60^\circ)$$

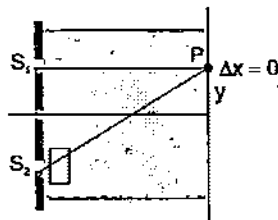
$$\Delta x = 4t - 2t \left(\frac{1}{\sqrt{3}} \right) \left(\frac{\sqrt{3}}{2} \right) = 3t$$

Since the Ray 1 is reflected at the surface of the denser medium so it suffers an additional phase change of π or a path change of $\frac{\lambda}{2}$. So, for constructive interference, we have

$$3t = \frac{\lambda}{2}$$

$$\Rightarrow t = \frac{\lambda}{6} = 1000 \text{ \AA} = 100 \text{ nm}$$

12.



The path difference at O is

$$\Delta x = (S_2P - T)_{\text{liquid}} + T_{\text{glass}} - (S_1P)_{\text{liquid}}$$

$$\Delta x = \frac{yd}{D} + \left(\frac{\mu T}{\mu_e} - T \right)_{\text{liquid}}$$

At the central maxima, we have path difference

$$\Delta x = 0$$

$$\Rightarrow y = -\frac{DT}{d} \left(\frac{\mu - \mu_e}{\mu_e} \right) = -\frac{TD}{d} \left(\frac{4 - t}{10 - t} \right)$$

Central maxima will be at O when $y = 0$

$$\Rightarrow t = 4 \text{ s}$$

Now, the speed of the central maxima is given by

$$v = \left| \frac{dy}{dt} \right| = \frac{6DT}{(10 - t)^2 d}$$

$$\text{At } t = 4 \text{ s, we have } v = \frac{6DT}{36d} = \frac{DT}{6d}$$

$$\Rightarrow v = \frac{(1)(36 \times 10^{-6})}{(6)(2 \times 10^{-3})} = 3 \times 10^{-3} \text{ ms}^{-1} = 3 \text{ mms}^{-1}$$

13. Fringe width, $\beta = \frac{\lambda D}{d} \propto \lambda$

Since the wavelength is decreasing from 600 nm to 400 nm, so fringe width also decreases by a factor of $\frac{4}{6} = \frac{2}{3}$. So the

number of fringes will increase by a factor of $\frac{3}{2}$. Hence the new number of fringes formed is

$$N = \frac{3}{2}(12) = 18$$

14. $\theta = 0.01$ radian

$$n = 10$$

$$\lambda = 6000 \times 10^{-8} \text{ cm}$$

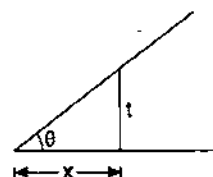
$$\Delta x = 2t = n\lambda$$

$$\text{Since, } \theta = \frac{t}{x}$$

$$\Rightarrow t = \theta x$$

$$\Rightarrow 2\theta x = n\lambda$$

$$\Rightarrow x = \frac{n\lambda}{2\theta} = 3 \times 10^{-1} \text{ m} = 3 \text{ mm}$$



15. When coherent, then Δx at centre is zero, so, we have the resultant intensity to be

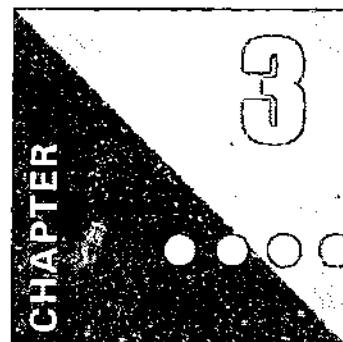
$$I_1 = 4I$$

When incoherent, then the sources will not interfere and it will be a general illumination at the point, so we have the resultant intensity to be,

$$I_2 = 2I$$

$$\Rightarrow \frac{I_1}{I_2} = 2$$

Dual Nature of Radiation & Matter



Contents

DUAL NATURE OF RADIATION & MATTER.....	3.1
Solved Practice Problems	3.15
Practice Exercise Sets	
➤ Single Correct Choice Type Questions.....	3.21
➤ Multiple Correct Choice Type Questions	3.33
➤ Reasoning Based Questions (Assertion Reason Type)	3.36
➤ Linked Comprehension Type Questions (Paragraph Type)	3.38
➤ Matrix Match Type Questions (Column Matching Type)	3.40
➤ Integer Answer Type Questions	3.43
Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	3.45
Solutions to In Chapter Exercises (ICE).....	3.48
Solutions to Practice Exercise Sets	3.52

Dual Nature of Radiation & Matter

MATTER WAVES

Light possesses dual nature i.e. it behaves both as a wave and as a particle. In some phenomena e.g., interference, diffraction and polarisation, it behaves as waves because they are explained on the basis of Wave theory while in some other phenomena e.g. photoelectric effect, Compton effect, it behaves as particles (photons).

Since nature demands symmetry, therefore de-Broglie thought that matter must have dual nature. The particle nature of matter is well known and hence de-Broglie thought that material particles must possess wave-nature.

de-Broglie's POSTULATE

According to de-Broglie a material particle in motion must have a wave like character and the wavelength associated with it is given by

$$\lambda = \frac{h}{p} \quad \dots(1)$$

where, h is the Planck's constant whose value is given by $h = 6.63 \times 10^{-34}$ Js and

p = momentum of the particle

de-Broglie assumed this expression in analogy with photon because momentum of photon is

$$p = \frac{h}{\lambda}$$

$$\Rightarrow \lambda = \frac{h}{p}$$

If m is the mass of particle and v the velocity, then momentum of particle is $p = mv$.

$$\text{So, de-Broglie wavelength } \lambda = \frac{h}{mv} \quad \dots(2)$$

If E_k is kinetic energy of particle, then

$$p = \sqrt{2mE_k} \quad \left\{ \because E_k = \frac{p^2}{2m} \right\}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mE_k}} \quad \dots(3)$$

For charged particles accelerated through a potential difference of V volts,

Kinetic energy i.e. $E_k = qV$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mqV}} \quad \dots(4)$$

For electrons accelerated through a potential difference of V volts

$$\text{de-Broglie wavelength } \lambda = \frac{h}{\sqrt{2meV}}$$

Substituting $m = 9.1 \times 10^{-31}$ kg, $h = 6.62 \times 10^{-34}$ Js, $e = 1.6 \times 10^{-19}$ C, we get

$$\lambda = \sqrt{\frac{150}{V}} \times 10^{-10} \quad \dots(5)$$

$$\Rightarrow \lambda = \sqrt{\frac{150}{V}} \text{ \AA} = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

This is expression for de-Broglie wavelength associated with electron accelerated through a potential difference of V .

For neutral particles (like neutrons, atoms) at temperature T , kinetic energy of most of particles,

$E_k = k_B T$, where k_B is the Boltzmann's constant.

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mk_B T}} \quad \dots(6)$$

The wave nature is possessed by all particles neutral or charged. The wave nature was first verified by Davisson and Germer for slow electrons.

Illustration 1

Obtain the de-Broglie wavelength associated with thermal neutrons at room temperature (27°C). Hence explain why a fast neutron beam needs to be thermalised

with the environment before it can be used for neutron diffraction experiments.

Solution

Average kinetic energy of a neutron at absolute temperature T is

$$\frac{1}{2}mv^2 = \frac{3}{2}k_B T$$

$$\Rightarrow \frac{p^2}{2m} = \frac{3}{2}k_B T \quad \left\{ \because p = mv \right\}$$

$$\Rightarrow p = \sqrt{3mk_B T}$$

de-Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{3mk_B T}}$$

Given $m_n = 1.675 \times 10^{-27}$ kg, $k_B = 1.38 \times 10^{-23}$ J mol⁻¹ K⁻¹

$$T = 27 + 273 = 300 \text{ K}, h = 6.63 \times 10^{-34} \text{ Js}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 1.675 \times 10^{-27} \times 1.38 \times 10^{-23} \times 300}}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-10}}{4.56} \text{ m} = 1.45 \times 10^{-10} \text{ m}$$

$$\Rightarrow \lambda = 1.45 \text{ Å}$$

As this wavelength is comparable to interatomic spacing ($\sim 1 \text{ Å}$) in a crystal, so thermal neutrons can be used for diffraction experiments. A high energy neutron beam should be first thermalised before using it for diffraction.

CHARACTERISTICS OF MATTER WAVES

- Matter wave represents the probability of finding a particle in space.
- Matter waves are not electromagnetic in nature.
- de-Broglie or matter wave is independent of the charge on the material particle. It means, matter wave of de-Broglie wave is associated with every moving particle (whether charged or uncharged).
- Practical observation of matter waves is possible only when the de-Broglie wavelength is of the order of the size of the particles.
- Electron microscope works on the basis of de-Broglie waves.
- The phase velocity of the matter waves can be greater than the speed of the light.
- Matter waves can propagate in vacuum, hence they are not mechanical waves.
- The number of de-Broglie waves associated with n^{th} orbital electron is n .
- Only those circular orbits around the nucleus are stable whose circumference is integral multiple of de-Broglie wavelength associated with the orbital electron.

de-BROGLIE WAVELENGTH ASSOCIATED WITH THE CHARGED PARTICLES

The energy of a charged particle accelerated through potential difference V is $E = \frac{1}{2}mv^2 = qV$

Hence de-Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$$

Using the above formula, we get

$$\lambda_{\text{Electron}} = \frac{12.27}{\sqrt{V}} \text{ Å}$$

$$\lambda_{\text{Proton}} = \frac{0.286}{\sqrt{V}} \text{ Å}$$

$$\lambda_{\text{Deuteron}} = \frac{0.202}{\sqrt{V}} \text{ Å}$$

$$\lambda_{\alpha\text{-particle}} = \frac{0.101}{\sqrt{V}} \text{ Å}$$

de-BROGLIE WAVELENGTH ASSOCIATED WITH UNCHARGED PARTICLES

For Neutron de-Broglie wavelength is given as

$$\lambda_{\text{Neutron}} = \frac{0.286 \times 10^{-10}}{\sqrt{E(\text{in eV})}} \text{ m} = \frac{0.286}{\sqrt{E(\text{in eV})}} \text{ Å}$$

Energy of thermal neutrons at ordinary temperature is given by

$$E = k_B T$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mk_B T}}$$

where T is the Absolute temperature, k_B is the Boltzmann's constant given by $k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$

$$\text{So, } \lambda_{\text{Thermal neutron}} = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 1.38 \times 10^{-23} \times T}} = \frac{30.83}{\sqrt{T}} \text{ Å}$$

HEISENBERG'S UNCERTAINTY PRINCIPLE

According to Heisenberg it is impossible to measure the position and momentum of a particle simultaneously with 100% accuracy. This is called Heisenberg's uncertainty principle. Uncertainty principle successfully explains the

- Non-existence of electrons in the nucleus
- Finite size of spectral lines.

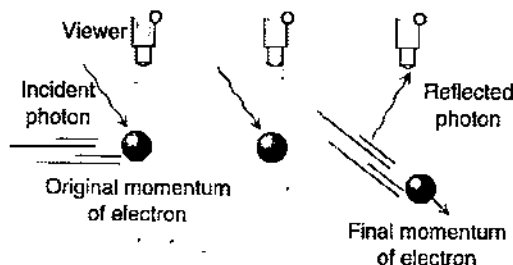
If Δx and Δp are uncertainties in determining the position and momentum of the particle simultaneously, then

$$\Delta x \Delta p \geq \frac{h}{4\pi}, \text{ where } h = 6.63 \times 10^{-34} \text{ Js}$$

$$\Rightarrow m \Delta x \Delta v \geq \frac{h}{4\pi}$$

This principle is universal and holds for all microscopic and macroscopic particles. The principle is also unaffected by experimental techniques.

If $\Delta x = 0$, then $\Delta p \rightarrow \infty$ and if $\Delta p = 0$ then $\Delta x \rightarrow \infty$, i.e. if we are able to measure the exact position of the particle (say an electron) then the uncertainty in the measurement of the linear momentum of the particle is infinite. Similarly, if we are able to measure the exact linear momentum of the particle i.e., $\Delta p = 0$, then we cannot measure the exact position of the particle at that time.



An electron cannot be observed without changing its momentum

This principle is also applicable to energy and time, angular momentum and angular displacement. Hence,

$$\Delta E \Delta t \geq \frac{h}{4\pi}$$

$$\Delta L \Delta \theta \geq \frac{h}{4\pi}$$

PROBLEM SOLVING TRICK(S)

a) For numerical problems, we shall use

$$\Delta x \Delta p \approx h \quad \Delta E \Delta t \approx h \quad \Delta L \Delta \theta \approx h$$

b) If the radius of the nucleus is r then the probability of finding the electron inside the nucleus is $\Delta x = 2r$ and uncertainty in momentum is $\Delta p = \frac{h}{4\pi r}$.

Illustration 2

Find the de Broglie wavelength corresponding to the root-mean-square velocity of hydrogen molecules at room temperature (20°C).

Solution

$$\text{Since, } v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$v_{\text{rms}} = \sqrt{\frac{3 \times 8.31 \times 293}{2 \times 10^{-3}}} = 1911 \text{ ms}^{-1}$$

$$\text{Now, } \lambda = \frac{h}{p} = \frac{h}{mv_{\text{rms}}}$$

Mass of one hydrogen molecule is given by

$$m = \frac{2}{6.02 \times 10^{26}} \text{ kg} = 3.32 \times 10^{-27} \text{ kg}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-34}}{3.32 \times 10^{-27} \times 1911} \text{ m}$$

$$\Rightarrow \lambda = 1.04 \times 10^{-10} \text{ m} = 1.04 \text{ \AA}$$

QUANTUM NATURE OF LIGHT

Some phenomena like photoelectric effect, Compton effect, Raman effect could not be explained by Wave theory of light. Therefore quantum theory of light was proposed by Einstein who extended the Planck's hypothesis to explain Black Body radiation. According to quantum theory of light or radiation, the energy of an electromagnetic wave is not continuously distributed over the wave front (just like the energy possessed by water waves). Instead Plank proposed that an electro-magnetic wave travels in the form of discrete packets or bundles of energy called *Quanta*.

So, according to Plank, "light is propagated in bundles of small energy, each bundle being called a photon and possessing energy", given by

$$E = h\nu = \frac{hc}{\lambda}$$

where, ν is frequency, λ is wavelength of light, h is Planck's constant whose value is $6.63 \times 10^{-34} \text{ Js}$ and $c = 3 \times 10^8 \text{ ms}^{-1}$

PROPERTIES OF PHOTONS

1. In the interaction of radiation with matter, the radiation behaves as if it is made up of particles called Photons. This fundamental is also called as Corpuscular Theory of Light. So, Light behaves both as a particle and a wave.
2. All photons emitted by any source travel through free space with a speed equal to the speed of light i.e. $c = 3 \times 10^8 \text{ ms}^{-1}$.
3. Each photon has a definite energy depending upon the frequency ν of the radiation and this energy is independent of the intensity. So,

$$E = h\nu = \frac{hc}{\lambda} \text{ (in joule)}$$

4. If λ is in $\text{\AA}$, then $E = \frac{12375}{\lambda} \text{ eV}$.
5. If the intensity of the light of given wavelength is increased, then there is an increase in the number of photons incident per second per unit area on a surface. However, energy of the photon remains the same as long as the frequency or the wavelength is unchanged.
6. The speed of the photon changes as it travels through different media due to the change in its wavelength.
7. The frequency of the photon does not change when it goes from one medium to the other.
8. In the situations when a photon collides with a material particle, the total energy and momentum remains conserved. However, the number of photons may not be conserved in a collision because during the

collision photon(s) may be absorbed or new photon(s) may be created.

9. A photon is an electrically neutral particle which is not deflected by electric and magnetic field.
10. Rest mass of photon is zero.

Since the mass of a particle m moving with a speed v is given by

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where m_0 is the rest mass of the particle.

$$\Rightarrow m_0 = m \sqrt{1 - \frac{v^2}{c^2}}$$

Now, for a photon, $v = c$, hence

$$m_0 = 0$$

So from here we can conclude that the rest mass of a photon is always zero, i.e. we cannot have a frame of reference where the photon is at rest.

11. Dynamic or kinetic mass of a photon, is determined by using the Einstein's Mass-Energy Equivalence, i.e.,

$$E = h\nu = mc^2$$

$$\Rightarrow m = \frac{h\nu}{c^2} = \frac{h}{c\lambda}$$

12. The Linear Momentum of a photon is found by using the de-Broglie relation according to which, we have

$$\lambda = \frac{h}{p}, \text{ where } \lambda = \frac{c}{\nu}$$

$$\Rightarrow p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

However, this result is also obtained by using the fact that the total energy of a subatomic particle of rest mass m_0 , moving with a velocity v , having momentum p is given by

$$E^2 = p^2c^2 + m_0^2c^4$$

Now, for a photon, $m_0 = 0$, so we have from above expression that

$$E = pc$$

$$\text{Since, } E = h\nu = \frac{hc}{\lambda}$$

$$\Rightarrow pc = \frac{hc}{\lambda}$$

$$\Rightarrow p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

13. The number of photons N , each of energy E , emitted from a source of monochromatic radiation of wavelength λ and energy W and power P

$$N = \frac{W}{E} = \frac{W}{h\nu} = \frac{Pt}{h\nu}$$

14. Intensity of light (I)

Energy crossing per unit area normally per second is called intensity or energy flux, i.e.

$$I = \frac{E}{At} = \frac{P}{A} \quad \left(\frac{E}{t} = P = \text{radiation power} \right)$$

At a distance r from a point source of power P intensity is given by

$$I = \frac{P}{4\pi r^2}$$

$$\Rightarrow I \propto \frac{1}{r^2}$$

15. Number of photons falling per second (n)

If P is the power of radiation and E is the energy of a photon then $n = \frac{P}{E}$.

MOMENTUM AND RADIATION PRESSURE

An electromagnetic wave consists of photons capable of transporting linear momentum. The linear momentum P possessed by an electromagnetic wave is related to the energy E it transports according to the relation

$$P = \frac{E}{c} \quad \dots(1)$$

If the wave is incident in the direction perpendicular to a surface and gets completely absorbed by the surface, then equation (1), tells us the linear momentum imparted to the surface.

If surface is perfectly reflecting, the momentum change of the wave is doubled. Consequently the momentum imparted to the surface is also doubled.

According to Newton's Second Law, the force exerted by an electromagnetic wave on a surface is given by the equation,

$$F = \frac{\Delta p}{\Delta t}$$

From equation (1), $\frac{\Delta p}{\Delta t} = \frac{1}{c} \left(\frac{\Delta E}{\Delta t} \right)$

$$\Rightarrow F = \frac{1}{c} \left(\frac{\Delta E}{\Delta t} \right) \quad \dots(2)$$

Intensity (I) of a wave is the energy transported per unit area per unit time.

$$\Rightarrow I = \left(\frac{1}{A} \right) \frac{\Delta E}{\Delta t}$$

$$\Rightarrow \frac{\Delta E}{\Delta t} = IA$$

Substituting in equation (2), $F = \frac{IA}{c}$

$$\Rightarrow \frac{F}{A} = \text{pressure} = \frac{I}{c}$$

$$\Rightarrow P_{\text{rad}} = \frac{I}{c}$$

$\frac{I}{c}$ is also equal to the energy density (energy per unit volume) u .

Hence, $P_{\text{rad}} = u$... (3)

The radiation pressure is thus equal to the energy density ($\text{Nm}^{-2} = \text{Jm}^{-3}$). At a perfectly reflecting surface the pressure on the surface is doubled. Thus, we can write,

$$P_{\text{rad}} = \frac{I}{c} = u \quad \{\text{wave totally absorbed}\}$$

$$\text{and } P_{\text{rad}} = \frac{2I}{c} = 2u \quad \{\text{wave totally reflected}\}$$

Illustration 3

Find the energy, the mass and the momentum of a photon of ultraviolet radiation of 280 nm wavelength.

Solution

$$\text{Given, } \lambda = 280 \times 10^{-9} \text{ m}$$

$$\text{Since, } E = \frac{hc}{\lambda}$$

$$\Rightarrow E = \frac{(4.316 \times 10^{-15} \text{ eV sec})(3 \times 10^8 \text{ ms}^{-1})}{(280 \times 10^{-9} \text{ m})} = 4.6 \text{ eV}$$

$$\text{Mass of photon is } m = \frac{E}{c^2}$$

$$\Rightarrow m = \frac{4.6 \times 1.6 \times 10^{-19}}{(3 \times 10^8)^2} = 8.2 \times 10^{-36} \text{ kg}$$

Momentum of a photon is

$$P = \frac{E}{c}$$

$$\Rightarrow P = \frac{4.6 \times 1.6 \times 10^{-19}}{3 \times 10^8} = 2.45 \times 10^{-27} \text{ kg ms}^{-1}$$

Illustration 4

The intensity of sunlight on the surface of earth is 1400 Wm^{-2} . Assuming the mean wavelength of sunlight to be $6000 \text{ \AA}$, calculate

- the photon flux arriving at 1 m^2 area on earth perpendicular to light radiations, and
- the number of photons emitted from the sun per second assuming the average radius of Earth's orbit is $1.49 \times 10^{11} \text{ m}$.

Solution

(a) Energy of a photon

$$E = \frac{hc}{\lambda} = \frac{12400}{6000} = 2.06 \text{ eV} = 3.3 \times 10^{-19} \text{ J}$$

$$\text{Photon flux} = \frac{IA}{E} = \frac{(1400)(1)}{3.3 \times 10^{-19}} = 4.22 \times 10^{21}$$

$$(b) \quad n = \frac{P}{E} = \frac{I(4\pi R^2)}{E}$$

$$\Rightarrow n = \frac{(1400)(4\pi)(1.49 \times 10^{11})^2}{3.3 \times 10^{-19}} = 1.18 \times 10^{45}$$

Illustration 5

A small plate of a metal is placed at a distance of 2 m from a monochromatic light source of wavelength $4.8 \times 10^{-7} \text{ m}$ and power 1 watt. The light falls normally on the plate. Find the number of photons striking the metal plate per square meter per second.

Solution

$$E = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34})(3 \times 10^8)}{(4.8 \times 10^{-7})} = 4.125 \times 10^{-19} \text{ J}$$

Number of photons striking the metal plate per square meter per second is

$$n = \left(\frac{P}{E} \right) \left(\frac{1}{4\pi r^2} \right)$$

$$\Rightarrow n = \left(\frac{1}{4.125 \times 10^{-19}} \right) \frac{1}{(4\pi)(2)^2} = 4.82 \times 10^{16} \text{ m}^{-2} \text{ s}^{-1}$$

Illustration 6

Find the number of photons entering the pupil of our eye per second corresponding to the minimum intensity of white light that we humans can perceive ($\sim 10^{-10} \text{ Wm}^{-2}$). Take the area of the pupil to be about 0.4 cm^2 and the average frequency of white light to be about $6 \times 10^{14} \text{ Hz}$.

Solution

$$\text{Minimum intensity, } I = 10^{-10} \text{ Wm}^{-2}$$

$$\text{Area of pupil, } A = 0.4 \text{ cm}^2 = 0.4 \times 10^{-4} \text{ m}^2$$

$$\text{Average frequency, } \nu = 6 \times 10^{14} \text{ Hz}$$

Energy of one photon

$$E = h\nu = 6.63 \times 10^{-34} \times 6 \times 10^{14} \text{ J} = 4 \times 10^{-19} \text{ J}$$

Let n be the number of photons crossing per square metre area per second.

Now Intensity = Energy incident per square metre area per second

$$\Rightarrow I = \text{Total Energy of } n \text{ photons}$$

$$\Rightarrow I = n \times \text{Energy of one photon}$$

$$\Rightarrow n = \frac{\text{Intensity}}{\text{Energy of one photon}}$$

$$\Rightarrow n = \frac{10^{-10} \text{ Wm}^{-2}}{4 \times 10^{-19} \text{ J}} = 2.5 \times 10^8 \text{ m}^{-2} \text{ s}^{-1}$$

So, the number of photons entering the pupil of our eye per second is

$$N = n(\text{Area of the pupil})$$

$$\Rightarrow N = 2.5 \times 10^8 \times 0.4 \times 10^{-4} \text{ s}^{-1} = 10^4 \text{ s}^{-1}$$

Illustration 7

Find the number of photons emitted per second by a 25 W source of monochromatic light of wavelength $6600 \text{ \AA}$. What is the photoelectric current assuming 3% efficiency for photoelectric effect? Given $h = 6.6 \times 10^{-34} \text{ Js}$.

Solution

Energy of each photon is

$$E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6600 \times 10^{-10}} = 3 \times 10^{-19} \text{ J}$$

Total energy emitted per second by 25 W source is

$$E = 25 \text{ J}$$

Number of photons emitted per second is

$$n = \frac{25}{3 \times 10^{-19}} = 8.33 \times 10^{19}$$

Photoelectric current (I) is

$$I = \left(\frac{\text{3\% of photons}}{\text{emitted per second}} \right) \times \left(\frac{\text{charge on}}{\text{electron}} \right)$$

$$\Rightarrow I = \frac{3}{100} \times \frac{25}{3 \times 10^{-19}} \times 1.6 \times 10^{-19} = 0.4 \text{ A}$$

Illustration 8

A source emits monochromatic light of frequency $5.5 \times 10^{14} \text{ Hz}$ at a rate of 0.1 W. Of the photons given out, 0.15% fall on the cathode of a photocell which gives a current of $6 \text{ \mu A}$ in an external circuit.

- Find the energy of a photon.
- Find the number of photons leaving the source per second.
- Find the percentage of the photons falling on the cathode which produce photoelectrons.

Solution

$$(a) \text{ Since, } E = h\nu = (6.6 \times 10^{-34})(5.5 \times 10^{14}) = 36.3 \times 10^{-20} \text{ J}$$

$$\Rightarrow E = 2.27 \text{ eV}$$

(b) Number of photons leaving the source per second is

$$n = \frac{P}{E} = \frac{0.1}{36.3 \times 10^{-20}} = 2.75 \times 10^{17}$$

(c) Number of photons falling on cathode per second is

$$n_1 = \frac{0.15}{100} \times 2.75 \times 10^{17} = 4.125 \times 10^{14}$$

Number of photoelectrons emitting per second is

$$n_2 = \frac{6 \times 10^{-6}}{1.6 \times 10^{-19}} = 3.75 \times 10^{13}$$

$$\text{So, } \left(\frac{\text{\%age of Photons}}{\text{falling on Cathode}} \right) = \frac{n_2}{n_1} \times 100$$

$$\Rightarrow \left(\frac{\text{\%age of Photons}}{\text{falling on Cathode}} \right) = \frac{3.75 \times 10^{13}}{4.125 \times 10^{14}} \times 100 = 9\%$$

Illustration 9

A cylindrical rod of some laser material $5 \times 10^{-2} \text{ m}$ long and 10^{-2} m in diameter contains 2×10^{25} ions per m^3 . If on excitation all the ions are in the upper energy level and de-excite simultaneously emitting photons in the same direction, calculate the maximum energy contained in a pulse of radiation of wavelength $6.6 \times 10^{-7} \text{ m}$. If the pulse lasts for 10^{-7} second, calculate the average power of the laser during the pulse.

Solution

Total number of ions in the rod is

$$N = \left(\frac{\text{Number of ions}}{\text{per unit volume}} \right) \times \left(\frac{\text{Volume of}}{\text{the rod}} \right)$$

$$\Rightarrow N = (2 \times 10^{25} \text{ m}^{-3}) \times (3.14 \times (0.005)^2 \times 5 \times 10^{-2} \text{ m}^3)$$

$$\Rightarrow N = 7.85 \times 10^{19}$$

As all the ions de-excite simultaneously, the number of photons emitted in the same direction is also 7.85×10^{19} .

So, the energy contained in a pulse of radiation of wavelength $6.6 \times 10^{-7} \text{ m}$ is

$$E = n \left(\frac{hc}{\lambda} \right)$$

$$\Rightarrow E = \frac{hc}{\lambda} \times 7.85 \times 10^{19}$$

$$\Rightarrow E = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6.6 \times 10^{-7}} = 23.55 \text{ J}$$

$$\text{Average power } P = \frac{\text{Energy}}{\text{Time}}$$

$$P = \frac{23.55 \text{ J}}{10^{-7} \text{ s}} = 23.55 \times 10^7 \text{ W} = 235.5 \text{ MW}$$

ICE 1

BASED ON PHOTON PROPERTIES

(Solutions on page 3.48)

- The intensity of direct sunlight before it passes through the earth's atmosphere is 1.4 kWm^{-2} . If it is completely absorbed find the corresponding radiation pressure.
- According to the maxwell theory of electrodynamics an electron going in a circle should emit radiations of frequency equal to its frequency of revolution. What would be the wavelength of the radiation emitted by a hydrogen atom in ground state if this rule is followed?
- An electron is accelerated by a potential difference of 25 volt. Find the de-Broglie wavelength associated with it.
- Find the number of photons emitted per second by a MW transmitter of 10 kW power emitting radiowaves of wavelength 500 m.
- If 5% of the energy supplied to an incandescent light bulb is radiated as visible light, how many visible light photons are emitted by 100 watt bulb. Assume wavelength of all visible

photons to be $5600 \text{ \AA}$. Given $h = 6.625 \times 10^{-34} \text{ Js}$.

6. Calculate the number of photons in 6.62 J of radiation energy of frequency 10^{12} Hz . Given $h = 6.62 \times 10^{-34} \text{ Js}$.
7. Monochromatic light of frequency $6 \times 10^{14} \text{ Hz}$ is produced by a laser. The power emitted is $2 \times 10^{-3} \text{ W}$.
 - (a) What is the energy of each photon in the light?
 - (b) How many photons per second, on the average, are emitted by the source?
8. Show that a free electron at rest cannot absorb a photon and thereby acquire kinetic energy equal to the energy of the photon. Would the conclusion change if the free electron was moving with a constant velocity?
9. An electron microscope uses electrons accelerated by a voltage of 50 kV . Determine the de-Broglie wavelength associated with the electrons. If other factors (such as numerical aperture etc.) are taken to be roughly the same, how does the resolving power of an electron microscope compare with that of an optical microscope which uses yellow light?
10. An electron and proton are possessing the same amount of kinetic energy. Which of the two have greater wavelength?
11. An electron and a photon have same de Broglie wavelength (say $1 \text{ \AA}$). Which one possesses more kinetic energy?
12. An electron and a proton have same wavelength. Which one possesses more energy?

EMISSION OF ELECTRONS

As we are aware of the fact that metals have free electrons (negatively charged particles) which are responsible for their conductivity. However, the free electrons cannot normally escape out of the metal surface. If an electron attempts to come out of the metal, the metal surface acquires a positive charge and pulls the electron back to the metal. The free electron is thus held inside the metal surface by the attractive forces of the ions. Consequently, the electron can come out of the metal surface only if it is supplied some minimum energy to overcome the attractive pull of the metal.

This minimum energy required by an electron to escape from the metal surface is called the *work function* of the metal. It is generally denoted by W or ϕ_0 and is measured in eV (electron volt). One electron volt is the energy gained by an electron when it has been accelerated by a potential difference of 1 volt, so

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

This unit of energy is commonly used in atomic and nuclear physics.

The work function, W depends on the properties of the metal and the nature of its surface. The work function for platinum is the highest $W_{\text{Pt}} = 5.65 \text{ eV}$, whereas it is the lowest for caesium i.e., $W_{\text{Cs}} = 2.14 \text{ eV}$.

The minimum energy required for the electron emission from the metal surface can be supplied to the free electrons by any one of the following physical processes.

THERMIONIC EMISSION

By suitably heating a metal, sufficient thermal energy can be imparted to the free electrons to enable them to come out of the metal. The free electrons so emitted are called as *Thermions*.

FIELD EMISSION

When the metal surface is subjected to very strong electric field of the order ranging from 10^3 Vm^{-1} to 10^8 Vm^{-1} , the electrons (beyond a certain limit) start coming out of the metal surface. This method of emission is dangerous and less efficient. This method of emission is also called the *Cold Cathode Emission*.

PHOTO-ELECTRIC EMISSION

When light of certain minimum energy (or minimum frequency or maximum wavelength) illuminates or falls on a metal surface, electrons are emitted from the metal surface. The emitted electrons are called *photoelectrons*. In case of Photoelectric emission, the rate of emission of photoelectrons is very low.

SECONDARY EMISSION

When fast moving electrons strike a metal surface, then some of their energy is transferred to the free electrons of the metal. Due to this, when free electrons gain energy more than the work function, then they are emitted from the metal surface. These emitted electrons are called the *secondary electrons*.

PHOTOELECTRIC EFFECT

The phenomenon of emission of electrons from a metallic surface by the use of light (or radiant) energy of certain minimum frequency (or maximum wavelength) is called *photoelectric effect*. The emitted electrons are called as *photoelectrons*. The phenomenon was discovered by Hallwach in 1888. For photoelectric emission the metal used must have low work function e.g., alkali metals. Cesium is assumed to be the best metal for photoelectric effect. To escape from the surface, the electron must absorb enough energy from the incident radiation to overcome the attraction of nucleus of the atom of the metal surface. The explanation to the photoelectric effect given by Einstein is based on the Law of Conservation of Energy. Before discussing the effect further, we must understand the following terms.

A. WORK FUNCTION (OR THRESHOLD ENERGY) (W_0)

The minimum energy of incident radiation, required to eject the electrons from metallic surface is defined as work

function of that surface. It is the characteristic of a metal surface.

$$W_0 = h\nu_0 = \frac{hc}{\lambda_0} \text{ (in joule), where}$$

ν_0 = Threshold frequency and

λ_0 = Threshold wavelength

Work function in electron volt is given by

$$W_0 \text{ (eV)} = \frac{hc}{e\lambda_0} = \frac{12375}{\lambda_0 \text{ (in } \text{\AA})}$$

B. THRESHOLD FREQUENCY (ν_0)

The minimum frequency of incident radiations required to eject the electron from metal surface is defined as threshold frequency.

If incident frequency $\nu < \nu_0 \Rightarrow$ No photoelectron emission.

For most metals the threshold frequency is in the ultraviolet (corresponding to wavelengths between 200 and 300 nm), but for potassium and cesium oxides it is in the visible spectrum (λ between 400 and 700 nm)

C. THRESHOLD WAVELENGTH (λ_0)

The maximum wavelength of incident radiations required to eject the electrons from a metallic surface is defined as threshold wavelength.

If incident wavelength $\lambda > \lambda_0$, then No photoelectron emission will take place.

Illustration 10

In an experiment on photoelectric effect light of wavelength 400 nm is incident on a caesium plate at the rate of 5 W. The potential of the collector plate is made sufficiently positive with respect to emitter so that the current reaches the saturation value. Assuming that on the average one out of every 10^6 photons is able to eject a photoelectron, find the photocurrent in the circuit.

Solution

$$E = \frac{12375}{4000} = 3.1 \text{ eV}$$

Number of photoelectrons emitted per second

$$n = \left(\frac{1}{10^6} \right) \left(\frac{5}{3.1 \times 1.6 \times 10^{-19}} \right) = 1 \times 10^{13} \text{ per second}$$

$$\text{Since, } i = \frac{q}{t} = \frac{Ne}{t} = \left(\frac{N}{t} \right) e = ne$$

$$\Rightarrow i = (ne) = 1 \times 10^{13} \times 1.6 \times 10^{-19} = 1.6 \times 10^{-6} \text{ A} = 1.6 \text{ } \mu\text{A}$$

LAWS OF PHOTOELECTRIC EMISSION

We thus have the following laws of photoelectric emission, derived from the experimental observations.

- For each emitting metal, there is a certain minimum frequency ν_0 (or maximum wavelength λ_0), called the threshold frequency of the incident radiation, below (above which) which no emission of photoelectron takes place, no matter how great is the intensity. The value of ν_0 (or λ_0) is different for different emitting surfaces.
- The process of emission of photoelectrons is an instantaneous process. There is no time lag ($< 10^{-8}$ s) between the incidence of radiation and the emission of photoelectrons.
- Photoelectric effect is a one photon-one electron phenomenon i.e. even if photon has an energy sufficient to strike off 3 electrons (say) it can only strike off one electron with the excess energy being imparted to the struck off electron as kinetic energy.
- The number of photoelectrons emitted per second, that is, photoelectric current is directly proportional to the intensity of the incident radiation but is independent of the frequency (or wavelength) of light.
- The velocities (or the energies) of the emitted photoelectrons vary between zero and a definite maximum ($v_{\max}$). The proportion of photoelectrons having a particular velocity is independent of the light intensity.
- The maximum velocity, $v_{\max}$, and hence the maximum kinetic energy is independent of the intensity of the incident light, but depends on its frequency, increasing linearly with the increase of the frequency of the incident light.

EINSTEIN'S EXPLANATION OF PHOTO-ELECTRIC EFFECT

The wave theory of light could not explain the observed characteristics of photoelectric effect. Einstein extended Planck's quantum idea for light to explain photo-electric effect.

According to his idea, the energy of electromagnetic radiation is not continuously distributed over the wave front like the energy of water waves but remains concentrated in packets of energy content $h\nu$, where ν is frequency of radiations and h is universal Planck's constant ($= 6.625 \times 10^{-34}$ Js). Each packet of energy moves with the speed of light. The assumptions of Einstein's theory are

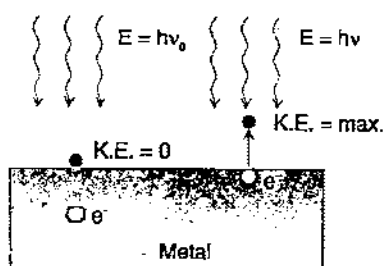
- The photoelectric effect is the result of collision of two particles, one of which is a photon of incident light and the other is an electron of photo-metal.
- The electron of photo-metal is bound with the nucleus by Coulomb attractive forces. The minimum energy required to free an electron from its bondage is called work function, $W = \phi_0 = h\nu_0$.
- The incident photon interacts with a single electron and loses its energy in two parts

- (i) Firstly, in getting the electron released from the bondage of the nucleus.
- (ii) Secondly, to impart kinetic energy to emitted electron.
- d) The efficiency of photoelectric effect is less than 1%, i.e. less than 1% of photons are capable of ejecting photoelectrons.

Accordingly, if $h\nu$ is the energy of incident photon, then

$$h\nu = W + K_{\max}$$

$$\Rightarrow K_{\max} = h\nu - W \quad \dots(1)$$



This is Einstein's photoelectric equation, where W is work function and

$$K_{\max} = \frac{1}{2}mv_{\max}^2 = eV_s$$

is the maximum kinetic energy of photo-electrons emitted.

Equation (1) is referred as Einstein's photo electric equation that explains all experimental results of photo-electric effect and is based on the Law of Conservation of Energy.

Illustration 11

Ultraviolet light of wavelength $2000 \text{ \AA}$ causes photoemission from a surface. The stopping potential is 2 V .

- (a) Find the work function in eV
- (b) Find the maximum speed of the photoelectrons.

Solution

- (a) Using Einstein relation

$$W = \frac{hc}{\lambda} - eV_s$$

$$\Rightarrow W = \frac{12400}{2000} - 2 = 4.2 \text{ eV}$$

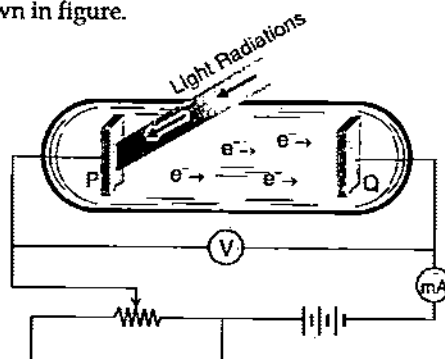
- (b) Since $\frac{1}{2}mv_{\max}^2 = eV_s$

$$\Rightarrow v_{\max} = \sqrt{\frac{2eV_s}{m}} = \sqrt{\frac{2(1.6 \times 10^{-19})(2)}{9.1 \times 10^{-31}}}$$

$$\Rightarrow v_{\max} = 8.4 \times 10^5 \text{ ms}^{-1}$$

EXPERIMENTAL SETUP FOR PHOTOELECTRIC EFFECT

It consists of two conducting electrodes, the anode (Q) and cathode (P) which are enclosed in an evacuated glass tube as shown in figure.



The battery or some other source of potential difference creates an electric field in the direction from anode to cathode. Light of certain wavelength or frequency falling on the surface of cathode causes a current to flow in the external circuit. This current is called the **photoelectric current**.

When the potential difference increases, the photo electric current also increases till saturation is reached. As the polarity of battery is reversed (i.e. plate Q is at negative potential w.r.t. plate P) the electrons start moving back towards the cathode. It is observed that at a particular negative potential of plate Q, no electron reaches the plate Q i.e. the current becomes zero. This negative potential for which the photo-electric current is zero is called the **stopping potential** denoted by V_0 . Maximum kinetic energy (in eV) of photo electrons in terms of stopping potential will therefore be $K_{\max} = |V_0| \text{ eV}$

Illustration 12

When a beam of 10.6 eV photons of intensity 2 Wm^{-2} falls on a platinum surface of area $1 \times 10^{-4} \text{ m}^2$ and work function 5.6 eV , 0.53% of the incident photons eject photo electrons. Find the number of photoelectrons emitted per second and their minimum and maximum energies (in eV). Given that $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

Solution

Number of photoelectrons emitted per second

$$\frac{N}{t} = \frac{(\text{Intensity})(\text{Area})}{(\text{Energy of each photon})} \times \frac{0.53}{100}$$

$$\Rightarrow \frac{N}{t} = \frac{(2)(1 \times 10^{-4})}{(10.6 \times 1.6 \times 10^{-19})} \times \frac{0.53}{100} = 6.25 \times 10^{11} \text{ s}^{-1}$$

Minimum kinetic energy of photoelectrons is

$$K_{\min} = 0$$

and maximum kinetic energy is,

$$K_{\max} = E - W = (10.6 - 5.6) \text{ eV} = 5 \text{ eV}$$

Illustration 13

A monochromatic light source of frequency illuminates a metallic surface and ejects photoelectrons. The photo electrons having maximum energy are just able to ionize the hydrogen atoms in ground state. When the entire experiment is repeated with an incident radiation of frequency $\frac{5}{6}f$, the photoelectrons so emitted are able to excite the hydrogen atom beam which then emits a radiation of wavelength 1215 Å.

- (a) What is the frequency of radiation.
(b) Find the work function of the metal.

Solution

- (a) Using Einstein's equation of photoelectric effect i.e.

$$K_{\max} = hf - W$$

$$\text{where } K_{\max} = 13.6 \text{ eV}$$

$$\Rightarrow hf - W = 13.6 \text{ eV} \quad \dots(1)$$

So, when the experiment is repeated, then

$$h\left(\frac{5}{6}f\right) - W = \frac{12375}{1215} = 10.2 \text{ eV} \quad \dots(2)$$

Solving equations (1) and (2), we get

$$\frac{hf}{6} = 3.4 \text{ eV}$$

$$\Rightarrow f = \frac{(6)(3.4)(1.6 \times 10^{-19})}{(6.63 \times 10^{-34})} = 4.92 \times 10^{15} \text{ Hz}$$

- (b) From equation (1), we have

$$W = hf - 13.6$$

$$\Rightarrow W = 6(3.4) - 13.6$$

$$\Rightarrow W = 6.8 \text{ eV}$$

Illustration 14

A photon with an energy of 4.9 eV ejects photoelectrons from tungsten. When the ejected electron enters a constant magnetic field of strength $B = 2.5 \text{ mT}$ at an angle of 60° with the field direction, the maximum pitch of the helix described by the electron is found to be 2.7 mm. Find the work function of the metal in electron-volts. Given that specific charge of electron is $1.76 \times 10^{11} \text{ C kg}^{-1}$.

Solution

Pitch of helical path is

$$p = (v \cos \theta) T = \frac{vT}{2} \quad \left\{ \because \theta = 60^\circ \right\}$$

$$\text{where, } T = \frac{2\pi m}{qB} = \frac{2\pi}{B\alpha} \quad \left\{ \because \alpha = \frac{q}{m} \right\}$$

$$\Rightarrow p = \frac{\pi v}{B\alpha}$$

$$\Rightarrow v = \frac{B\alpha p}{\pi} \quad \dots(1)$$

$$\Rightarrow v = \frac{(2.5 \times 10^{-3})(1.76 \times 10^{11})(2.7 \times 10^{-3})}{3.14}$$

$$\Rightarrow v = 0.38 \times 10^6 \text{ ms}^{-1}$$

$$\text{Since, } KE = \frac{1}{2}mv^2 = E - W$$

$$\Rightarrow W = E - \frac{1}{2}mv^2 \quad \dots(2)$$

Substituting value of v from equation (1) in equation (2), we get

$$W = 4.9 - \frac{1}{2} \frac{(9.1 \times 10^{-31})(0.38 \times 10^6)^2}{1.6 \times 10^{-19}}$$

$$\Rightarrow W = (4.9 - 0.4) \text{ eV}$$

$$\Rightarrow W = 4.5 \text{ eV}$$

Illustration 15

If the wavelength of the incident radiation is increased from 3000 Å to 3010 Å, find the corresponding change in the stopping potential V .

Solution

According to Einstein's Photo-electric equation, we have

$$eV_1 = E_1 - W \quad \dots(1)$$

$$eV_2 = E_2 - W \quad \dots(2)$$

Subtracting (2) from (1), we get

$$e(V_1 - V_2) = (E_1 - E_2)$$

$$\Rightarrow V_1 - V_2 = \frac{hc}{e} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

$$\Rightarrow V_1 - V_2 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19} \times 10^{-10}} \left(\frac{1}{3000} - \frac{1}{3010} \right) = 0.012 \text{ V}$$

Illustration 16

A light beam of wavelength 400 nm is incident on a metal of work function 2.2 eV. A particular electron absorbs a photon and makes 2 collisions before coming out of the metal

- (a) Assuming that 10% of extra energy is lost to the metal in each collision find the final kinetic energy of this electron as it comes out of the metal.
(b) Under the same assumptions find the maximum number of collisions the electron should suffer before it becomes unable to come out of the metal.

Solution

$$(a) \text{ Since, } E(\text{in eV}) = \frac{12375}{\lambda(\text{in Å})}$$

$$\Rightarrow E = \frac{12375}{4000} = 3.1 \text{ eV}$$

Energy of electron after first collision is

$$E_1 = (90\% \text{ of } E) = 2.79 \text{ eV} \quad (\because 10\% \text{ is lost})$$

Energy of electron after second collision

$$E_2 = (90\% \text{ of } E_1) = 2.51 \text{ eV}$$

Hence, KE of this electron after emitting from the metal surface = $(2.51 - 2.2) \text{ eV} = 0.31 \text{ eV}$

(b) Energy after third collision,

$$E_3 = (90\% \text{ of } E_2) = 2.26 \text{ eV}$$

Similarly, $E_4 = (90\% \text{ of } E_3) = 2.03 \text{ eV} < W$

So, after four collisions the electron will not be able to come out of the metal.

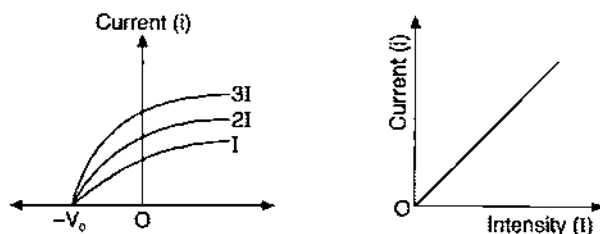
CHARACTERISTICS OF PHOTO ELECTRIC EFFECT

The following observations were made to study the effect of changes in various factors while studying the Photo Electric Effect.

A. EFFECT OF INTENSITY

Intensity of light means the energy incident per second per unit area. For a given frequency, if intensity of incident light is increased, the photoelectric current increases and with decrease of intensity, the photoelectric current decreases, but the stopping potential remains the same.

In photoelectric effect current (i) is directly proportional to intensity (I) of incident light.

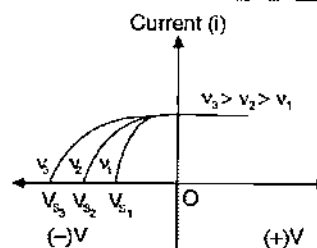


This means that the intensity of incident light affects the photoelectric current but leaves the maximum kinetic energy of photoelectrons unchanged.

B. EFFECT OF FREQUENCY

When the intensity of incident light is kept fixed and frequency is increased, the photoelectric current remains the same but the stopping potential increases.

If the frequency is decreased, the stopping potential decreases and at a particular frequency of incident light, the stopping potential becomes zero. This value of frequency of incident light for which the stopping potential is zero is called **threshold frequency** ν_0 . If the frequency of incident light (ν) is less than the threshold frequency (ν_0), no photo-electric emission takes place.



Thus, the increase of frequency increases maximum kinetic energy of photoelectrons but leaves the photoelectric current unchanged.

C. EFFECT OF PHOTO-METAL

When frequency and intensity of incident light are kept fixed and photo-metal is changed, we observe that stopping potential (V_s) versus frequency (ν) graphs are parallel straight lines, cutting frequency axis at different points. This shows that threshold frequency are different for different metals, the slope $\left(\frac{V_s}{\nu}\right)$ for all the metals is same and hence universal constant.

Since we know that

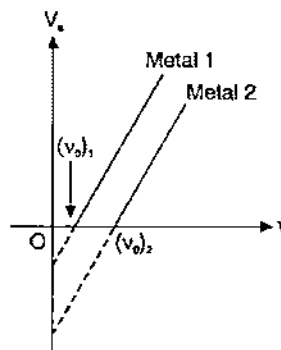
$$eV_s = h\nu - W$$

$$\Rightarrow V_s = \left(\frac{h}{e}\right)\nu - \left(\frac{W}{e}\right)$$

Comparing with the line $y = mx + c$, where m is the slope and c is the intercept on the y axis.

Then, we observe that the graph is a straight line with slope $\frac{h}{e}$ (a universal constant) and negative intercept $\frac{W}{e}$ (depending on the nature of the metal).

In figure threshold frequency and work function are greater for Metal 2 as compared to Metal 1.



D. EFFECT OF TIME

There is no time lag between incidence of light and the emission of photo-electrons.

PROBLEM SOLVING TRICK(S)**FORMULAE FOR WORKING THE PROBLEMS ON PHOTO-ELECTRIC EFFECT**

Maximum Kinetic Energy of photo-electrons

$$E_k = eV_0 = \frac{1}{2}mv_{\max}^2$$

If λ_0 is the threshold wavelength and ν_0 the threshold frequency,

Work function of photo-metal,

$$\phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

Threshold frequency is minimum frequency and Threshold wavelength is maximum wavelength of incident light to cause photoelectric effect.

Einstein's Photo-electric Equation may be expressed as or $E_k = h\nu - \phi_0$

$$E_k = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

condition for photoelectric emission is $h\nu \geq \phi_0$ or equivalently $\nu \geq \nu_0$ or equivalently $\lambda \leq \lambda_0$.

GRAPH BETWEEN $K_{\max}$ and ν

Let us plot a graph between the maximum kinetic energy $K_{\max}$ and the frequency of the falling photon ν or the incident light. According to Einstein's Photo-Electric equation, we have

$$h\nu = W + E_k$$

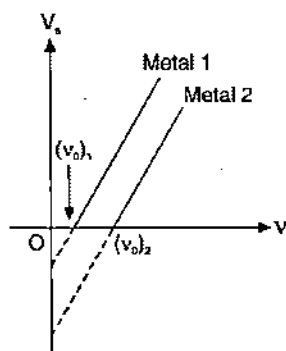
$$\Rightarrow K_{\max} = h\nu - W$$

Comparing with the line $y = mx + c$, where m is the slope and c is the intercept on the y axis.

Then, we observe that the graph is a straight line with slope h (a universal constant) and negative intercept W (depending on the nature of the metal).

For Metal 2, we observe that

$$W_2 > W_1 \text{ and hence } (\nu_0)_2 > (\nu_0)_1$$



Also, we observe that when $\nu = \nu_0$, the threshold frequency, then, $K_{\max} = 0$

DETERMINATION OF PHOTOELECTRIC CURRENT

Let P be the power of a point source of electromagnetic radiations, then intensity I at a distance r from the source is given by

$$I = \frac{P}{4\pi r^2} \text{ Wm}^{-2}$$

If A is the area of a metal surface on which radiations are incident, then the power received by the plate is

$$P' = IA = \left(\frac{P}{4\pi r^2} \right) A$$

If ν is the frequency of radiation, then the energy of photon is given by

$$E = h\nu$$

The number of photons incident on the plate per second (called photon flux) is given by

$$\phi = \frac{P'}{E} = \left[\frac{P}{4\pi r^2} \times A \right] \frac{1}{h\nu}$$

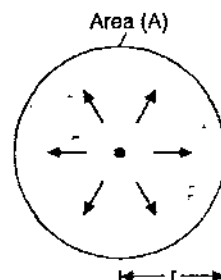
If $\nu > \nu_0$ (threshold frequency) and photon efficiency of the metal plate is $\eta\%$, then the number of photoelectrons emitted per second is given by

$$n = \frac{\phi\eta}{100} = \left[\frac{P}{4\pi r^2} \times A \right] \frac{\eta}{100 h\nu}$$

Finally, the photocurrent i is given by

$$i = ne$$

where e is the charge of an electron ($e = 1.6 \times 10^{-19} \text{ C}$)

**Illustration 17**

Light of wavelength 180 nm ejects photoelectrons from a plate of a metal whose work function is 2 eV . If a uniform magnetic field of $50 \mu\text{T}$ is applied parallel to the plate, what would be the radius of the path followed by electrons ejected normally from the plate with maximum energy.

Solution

$$\lambda = 180 \text{ nm} = 1800 \text{ \AA}$$

$$\Rightarrow E = \frac{12375}{1800} = 6.875 \text{ eV}$$

$$\text{Since, } K_{\max} = E - W = 4.875 \text{ eV}$$

$$\text{Since, } r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$$

Substituting the values, we get

$$r = \frac{\sqrt{2 \times 4.875 \times 1.6 \times 10^{-19} \times 9.1 \times 10^{-31}}}{5 \times 10^{-5} \times 1.6 \times 10^{-19}} = 0.15 \text{ m} = 15 \text{ cm}$$

Illustration 18

In a photoelectric effect set-up, a point source of light of power $3.2 \times 10^{-3} \text{ W}$ emits mono energetic photons of energy 5 eV . The source is located at a distance of 0.8 m from the centre of a stationary metallic sphere of work function 3 eV and of radius $8 \times 10^{-3} \text{ m}$. The efficiency of photoelectron emission is one for every 10^6 incident photons. Assume that the sphere is isolated and electrons are instantly swept away after emission.

- Calculate the number of photoelectrons emitted per second.
- Find the ratio of the wavelength of incident light to the de Broglie wavelength of the fastest photoelectrons emitted.
- It is observed that the photoelectron emission stops at a certain time t after the light source is switched on. Why?
- Evaluate the time t .

Solution

- Number of photoelectrons emitted per second is

$$n = \left(\frac{1}{10^6} \right) \left(\frac{P}{E} \right) \left(\frac{\pi r^2}{4\pi R^2} \right)$$

$$\Rightarrow n = \left(\frac{1}{10^6} \right) \left(\frac{3.2 \times 10^{-3}}{5 \times 1.6 \times 10^{-19}} \right) \left(\frac{1}{4\pi \times 0.8 \times 0.8} \right)$$

$$(\pi \times 8 \times 10^{-3} \times 8 \times 10^{-3})$$

$$\Rightarrow n = 10^5 \text{ sec}^{-1}$$

- $K_{\max} = E - W = 2 \text{ eV}$

Since, for an electron, we have

$$\lambda_2 = \sqrt{\frac{150}{KE(\text{in eV})}} \text{ \AA}$$

$$\Rightarrow \lambda_2 = \sqrt{\frac{150}{2}} = 8.66 \text{ \AA}$$

Further, Wavelength of incident photon is

$$\lambda_1 = \frac{12375}{5} = 2475 \text{ \AA}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} \approx 286$$

- Photoemission will stop when potential on the sphere becomes equal to the stopping potential.
- $K_{\max} = 2 \text{ eV}_0$. Therefore, the stopping potential V_0 is 2 volt. Let t be the desired time. Then

$$V_0 = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{Ne}{4\pi\epsilon_0 r} = \frac{(nt)e}{4\pi\epsilon_0 r}$$

$$\Rightarrow t = \frac{V_0 r}{\left(\frac{1}{4\pi\epsilon_0} \right) (ne)} = \frac{2 \times 8 \times 10^{-3}}{9 \times 10^9 \times 10^5 \times 1.6 \times 10^{-19}} = 111 \text{ s}$$

ICE II

BASED ON PHOTOELECTRIC EFFECT

(Solutions on page 3.50)

- What will be the maximum kinetic energy of the photoelectrons ejected from magnesium (for which the work function $W = 3.7 \text{ eV}$) when irradiated by ultraviolet light of frequency $1.5 \times 10^{16} \text{ sec}^{-1}$.
- A 40 W ultraviolet light source of wavelength $2480 \text{ \AA}$ illuminates a magnesium (Mg) surface placed 2 m away. Determine the number of photons emitted from the source per second and the number incident on unit area of the magnesium surface per second. The photoelectric work function for magnesium is 3.68 eV . Calculate the kinetic energy of the fastest electrons ejected from the surface. Determine the maximum wavelength for which the photoelectric effect can be observed with a Mg surface.
- The hydrogen atom in its ground state is excited by means of monochromatic radiation. Its resulting spectrum has six different lines. These radiations are incident on a metal plate. It is observed that only two of them are responsible for photoelectric effect. If the ratio of maximum kinetic energy of photoelectrons in the two cases is 5 then find the work function of the metal.
[Take ground state energy of H-atom = -13.6 eV].
- A metallic surface is irradiated with monochromatic light of variable wavelength. Above a wavelength of $5000 \text{ \AA}$, no photoelectrons are emitted from the surface. With an unknown wavelength, stopping potential of 3 V is necessary to eliminate the photocurrent. Find the unknown wavelength.
- A light source, emitting three wavelengths $5000 \text{ \AA}$, $6000 \text{ \AA}$ and $7000 \text{ \AA}$, has a total power of 10^{-3} W and a beam diameter of 2 mm . The power density is distributed equally amongst the three wavelengths. The beam shines normally on a metallic surface of area on 10^{-4} m^2 and having a work function of 1.9 eV . Assuming that each photon liberates an electron, calculate the charge emitted per second from the metal surface.
- A beam of light consists of four wavelength $4000 \text{ \AA}$, $4800 \text{ \AA}$, $6000 \text{ \AA}$ and $7000 \text{ \AA}$, each of intensity $1.5 \times 10^{-3} \text{ Wm}^{-2}$. The beam falls normally on an area 10^{-4} m^2 of a clean metallic surface of work function 1.9 eV . Assuming no loss of light energy calculate the number of photoelectrons liberated per second.
- In an experiment on photo electric emission, following observations were made :
(i) Wavelength of the incident light = $1.98 \times 10^{-7} \text{ m}$
(ii) Stopping potential = 2.5 V .
Find the
(a) threshold frequency.
(b) work function and
(c) energy of photo electrons with maximum speed.
- Radiation of wavelength $5461 \text{ \AA}$ falls on a photo cathode and electrons with a maximum kinetic energy of 0.18 eV

are emitted. When radiation of wavelength $1849 \text{ \AA}$ falls on the same surface a (negative) potential of 4.6 V has to be applied to the collector electrode to reduce the photoelectric current is zero. Find the value of h and cutoff wavelength.

9. Illuminating the surface of a certain metal alternately with light of wavelengths $\lambda_1 = 0.35 \mu\text{m}$ and $\lambda_2 = 0.54 \mu\text{m}$, it was found that the corresponding maximum velocities of photo electrons differ by a factor $\eta = 2$. Find the work function of that metal.
10. When a surface is irradiated with light of $\lambda = 4950 \text{ \AA}$ a photocurrent appears which vanishes if a retarding potential 0.6 V is applied. When a different source of light is used it is found that critical retarding potential is changed to 1.1 V . Find the work function of emitting surface and wavelength of second source. If photoelectrons after emission from surface are subjected to a magnetic field of 10 T , what changes will be observed in the above two retarding potentials?

11. The photoelectric work function of potassium is 2.3 eV . If light having a wavelength of $2800 \text{ \AA}$ falls on potassium, find
- (a) the kinetic energy in electron volt of the most energetic electrons ejected.
 - (b) the stopping potential in volt.

Solved Problems

Problem 1

A mercury arc lamp provides 0.2 watt of ultraviolet radiation at a wavelength of $\lambda = 2537 \text{ \AA}$.

Assume no other wavelength to be present. The cathode of photoelectric device consists of potassium and has an effective area of 4 cm^2 . The arc lamp is at a distance of 1 m from the cathode. Given that work function for potassium is $W = 2.22 \text{ eV}$.

- According to classical theory, what time of exposure to the radiation should be required for a potassium atom (radius $2 \text{ \AA}$) to accumulate sufficient energy to eject a photoelectron?
- What is the energy of a single photon from the source?
- What is the flux of photons (number per second) at the cathode? To what saturation current does this flux correspond if the photo conversion efficiency is 10%. Photo conversion efficiency is the probability of a photon being successful in knocking out an electron.
- Find the cut-off potential.

Solution

- Energy required to just eject a photoelectron from the potassium surface is equal to the work function, i.e., $2.22 \text{ eV} = 3.552 \times 10^{-19} \text{ J}$

As per the classical theory, energy flow is a continuous process and a photoelectron will be ejected if a potassium atom receives this amount of energy over a length of time.

The potassium atom is at a distance of 1 m from the source. Hence, intensity of ultraviolet radiation on the potassium surface is,

$$I = \frac{P}{4\pi r^2} = \frac{0.2}{4\pi(1)^2} = \frac{0.2}{4\pi} \text{ watt m}^{-2}$$

Cross-sectional area of the potassium atom is,

$$A = \pi(2 \times 10^{-10})^2 = 4\pi \times 10^{-20} \text{ m}^2$$

Hence, the exposure time is given by,

$$t = \frac{3.552 \times 10^{-19}}{\left(\frac{0.2}{4\pi}\right)(4\pi \times 10^{-20})}$$

- Since, $E = \frac{12375}{\lambda(\text{in \AA})} \text{ eV}$

$$\Rightarrow E = \frac{12375}{2537} = 4.87 \text{ eV}$$

- Let N be the number of photons reaching the cathode per second. Then intensity at cathode is,

$$I' = N \times E$$

But energy falling per second on the cathode is

$$\left(\frac{0.2}{4\pi}\right)(4\pi \times 10^{-4}), \text{ so}$$

$$N = \frac{0.2 \times 4 \times 10^{-4}}{4\pi E} = \frac{0.2 \times 4 \times 10^{-4}}{4\pi \times 4.87 \times 1.6 \times 10^{-19}}$$

$$\Rightarrow N = 8.12 \times 10^{12} \text{ photon sec}^{-1}$$

Since, 10% of these photons are able to eject electrons.

$$\Rightarrow i = 0.1 \times N \times 1.6 \times 10^{-19} \text{ A} = 65 \text{ nA}$$

- $K_{\text{max}} = E - W = (4.87 - 2.22) \text{ eV} = 2.65 \text{ eV}$

$$\Rightarrow \text{Stopping potential} = 2.65 \text{ V}$$

Problem 2

In a photocell the plates P and Q have a separation of 10 cm , which are connected through a galvanometer without any cell. Bi-chromatic light of wavelengths $4000 \text{ \AA}$ and $6000 \text{ \AA}$ are incident on plate Q whose work function is 2.39 eV . If a uniform magnetic field B exists parallel to the plates, find the minimum value of B for which the galvanometer shows zero deflection.

Solution

Energy of photons corresponding to light of wavelength $\lambda_1 = 4000 \text{ \AA}$ is

$$E_1 = \frac{12375}{4000} = 3.1 \text{ eV}$$

and that corresponding to $\lambda_2 = 6000 \text{ \AA}$ is,

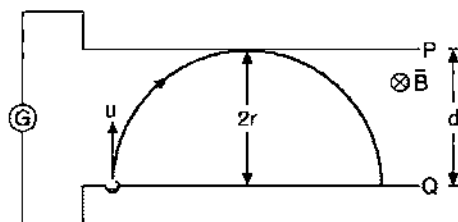
$$E_2 = \frac{12375}{6000} = 2.06 \text{ eV}$$

Given that the work function for the metal is $W = 2.39 \text{ eV}$, so we observe that

$$E_2 < W \text{ and } E_1 > W$$

Hence photoelectric emission is possible with λ_1 only.

Photoelectrons experience magnetic force and move along a circular path.



The galvanometer will indicate zero deflection when the photoelectrons complete semi-circular path before reaching the plate P.

Thus, $d = 2r = 10 \text{ cm}$

$$\Rightarrow r = 5 \text{ cm} = 0.05 \text{ m}$$

$$\text{Further } r = \frac{mv}{Bq} = \frac{\sqrt{2Km}}{Bq}$$

$$\Rightarrow B_{\min} = \frac{\sqrt{2Km}}{rq}$$

where $K = E_1 - W = (3.1 - 2.39) = 0.71 \text{ eV}$

Substituting the values, we have

$$B_{\min} = \frac{\sqrt{2 \times 0.71 \times 1.6 \times 10^{-19} \times 9.109 \times 10^{-31}}}{(0.05)(1.6 \times 10^{-19})}$$

$$\Rightarrow B_{\min} = 5.68 \times 10^{-5} \text{ Tesla}$$

Problem 3

In a photoelectric effect setup, a point source of light of power $3.2 \times 10^{-3} \text{ W}$ emits mono energetic photons of energy 5 eV . The source is located at a distance of 0.8 m from the centre of a stationary metallic sphere of work function 3 eV and of radius 8×10^{-3} . The efficiency of photoelectron emission is one for every 10^6 incident photons. Sphere is initially neutral and that the photoelectrons are instantly swept away after emission.

- Calculate the number of photoelectrons emitted per second.
- It is observed that the photoelectrons emission stops at a certain time t after the light source is switched on. Evaluate time t .

Solution

- Intensity of light at a distance 0.8 m from the source

$$I = \frac{(3.2 \times 10^{-3} \text{ Js}^{-1})}{4\pi(0.8)^2 \text{ m}^2} \approx 4 \times 10^{-4} \frac{\text{watt}}{\text{m}^2}$$

Energy incident on the metallic sphere in unit time is

$$E_1 = \pi(8 \times 10^{-3})^2 (4 \times 10^{-4}) = 8.04 \times 10^{-8} \text{ W}$$

Energy of one single photon is

$$E_2 = 5 \times 1.6 \times 10^{-19} \text{ J} = 8 \times 10^{-19} \text{ J}$$

Therefore, total number of photons incident on the sphere per second is

$$n_1 = \frac{E_1}{E_2} = \frac{8.04 \times 10^{-8}}{8 \times 10^{-19}} \approx 10^{11}$$

Since, the efficiency of photoelectric emission is one for every 10^6 . Hence, total number of photoelectrons per second is

$$n_2 = \frac{n_1}{10^6} = \frac{10^{11}}{10^6} = 10^5$$

- Maximum kinetic energy of photoelectrons

$$K_{\max} = E - W = 2 \text{ eV}$$

So, stopping potential $V_0 = 2 \text{ V}$

Total positive charge on the sphere after time t is,

$$q = (n_2 e)t = (10^5)(1.6 \times 10^{-19})t$$

$$\Rightarrow q = (1.6 \times 10^{-14})t$$

$$\text{Potential on the sphere } V = \frac{q}{4\pi\epsilon_0 r}$$

Photoemission will stop when this potential becomes the stopping potential, so

$$\frac{1}{4\pi\epsilon_0} \frac{(1.6 \times 10^{-14})t}{r} = 2$$

$$\Rightarrow \frac{(9 \times 10^9)(1.6 \times 10^{-14})t}{(8 \times 10^{-3})} = 2$$

$$\Rightarrow t = 111 \text{ s}$$

Problem 4

When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A expressed in eV and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A - 1.50 \text{ eV})$. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then find

- the work function W_A of metal A and the work function W_B of metal B.
- the maximum kinetic energy T_A of the electrons ejected from metal A.

Solution

$$K_{\max} = E - W$$

$$\text{Therefore, } T_A = 4.25 - W_A \quad \dots(1)$$

$$T_B = (T_A - 1.50) = 4.70 - W_B \quad \dots(2)$$

Equations (1) and (2) gives,

$$W_B - W_A = 1.95 \text{ eV} \quad \dots(3)$$

de-Broglie wavelength is given by

$$\lambda = \frac{h}{\sqrt{2Km}}$$

$$\Rightarrow \lambda \propto \frac{1}{\sqrt{K}} \quad K = \text{KE of electron}$$

$$\Rightarrow \frac{\lambda_B}{\lambda_A} = \sqrt{\frac{K_A}{K_B}}$$

$$\Rightarrow 2 = \sqrt{\frac{T_A}{T_A - 1.5}}$$

$$\Rightarrow T_A = 2 \text{ eV}$$

From equation (1), we get

$$W_A = 4.25 - T_A = 2.25 \text{ eV}$$

From equation (3), we get

$$W_B = W_A + 1.95 \text{ eV} = (2.25 + 1.95)$$

$$\Rightarrow W_B = 4.20 \text{ eV}$$

$$\Rightarrow T_B = 4.70 - W_B = 4.70 - 4.20 = 0.50 \text{ eV}$$

Problem 5

Ultraviolet light of wavelengths $800 \text{ \AA}$ and $700 \text{ \AA}$ when allowed to fall on hydrogen atoms in their ground state is found to liberate electrons with kinetic energy 1.8 eV and 4.0 eV respectively. Find the value of Planck's constant.

Solution

When $800 \text{ \AA}$ wavelength falls on hydrogen atom (in ground state) 13.6 eV energy is used in liberating the electron. The rest is given as kinetic energy to the electron.

Hence, $K = E - 13.6 (\text{in eV})$

$$\Rightarrow (1.8 \times 1.6 \times 10^{-19}) = \frac{hc}{800 \times 10^{-10}} - 13.6 \times 1.6 \times 10^{-19} \quad \dots(1)$$

Similarly for the second wavelength, we have

$$(4 \times 1.6 \times 10^{-19}) = \frac{hc}{700 \times 10^{-10}} - 13.6 \times 1.6 \times 10^{-19} \quad \dots(2)$$

Solving these two equations, we get

$$h = 6.6 \times 10^{-34} \text{ Js}$$

Problem 6

A beam of light has three wavelengths $4144 \text{ \AA}$, $4972 \text{ \AA}$ and $6216 \text{ \AA}$ with a total intensity of

$3.6 \times 10^{-3} \text{ Wm}^{-2}$ equally distributed amongst the three wavelengths. The beam falls normally on an area 1 cm^2 of a clean metallic surface of work function 2.3 eV . Assume that there is no loss of light by reflection and that each energetically capable photon ejects one electron. Calculate the number of photoelectrons liberated in two seconds.

Solution

$$\text{Since, } E (\text{in eV}) = \frac{12375}{\lambda (\text{in \AA})}$$

Energy of photon having wavelength $4144 \text{ \AA}$

$$E_1 = \frac{12375}{4144} \text{ eV} = 2.99 \text{ eV}$$

Similarly,

$$E_2 = \frac{12375}{4972} \text{ eV} = 2.49 \text{ eV and}$$

$$E_3 = \frac{12375}{6216} \text{ eV} = 1.99 \text{ eV}$$

Since, only E_1 and E_2 are greater than the work function $W = 2.3 \text{ eV}$, only first two wavelengths are capable for ejecting photoelectrons. Given intensity is equally distributed in all wavelengths. Therefore, intensity corresponding to each wavelength is

$$\frac{3.6 \times 10^{-3}}{3} = 1.2 \times 10^{-3} \text{ Wm}^{-2}$$

or energy incident, per second in the given area

$$(A = 1 \text{ cm}^2 = 10^{-4} \text{ m}^2) \text{ is}$$

$$I = 1.2 \times 10^{-3} \times 10^{-4} = 1.2 \times 10^{-7} \text{ Js}^{-1}$$

Let n_1 be the number of photons incident per unit time in the given area corresponding to first wavelength. Then,

$$n_1 = \frac{I}{E_1} = \frac{1.2 \times 10^{-7}}{2.99 \times 1.6 \times 10^{-19}} = 2.5 \times 10^{11}$$

Similarly,

$$n_2 = \frac{I}{E_2} = \frac{1.2 \times 10^{-7}}{2.49 \times 1.6 \times 10^{-19}} = 3 \times 10^{11}$$

Since, each energetically capable photon ejects electron, hence total number of photoelectrons liberated in 2 s is

$$N = t(n_1 + n_2)$$

$$\Rightarrow N = 2(n_1 + n_2)$$

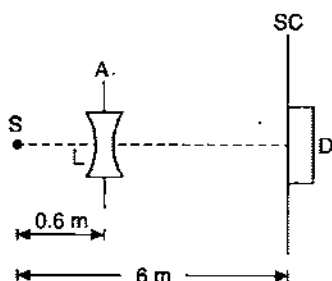
$$\Rightarrow N = 2(2.5 + 3) \times 10^{11}$$

$$\Rightarrow N = 1.1 \times 10^{12}$$

Problem 7

A monochromatic point source S radiating wavelength $6000 \text{ \AA}$, with power 2 watt , an aperture A of diameter 0.1 m and a large screen SC are placed as shown in figure. A photoemissive detector D of surface area 0.5 cm^2 is placed at the centre of the screen. The efficiency

of the detector for the photoelectron generation per incident photon is 0.9.



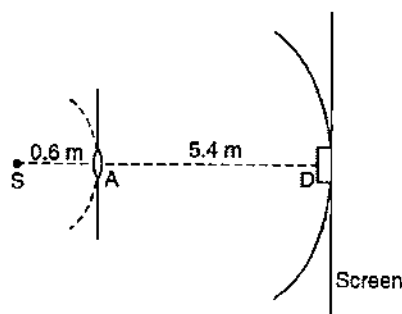
- Calculate the photon flux at the centre of the screen and the photocurrent in the detector.
- If the concave lens L of focal length 0.6 m is inserted in the aperture as shown, find the new values of photon flux and photocurrent. Assume a uniform average transmission of 80% from the lens.
- If the work function of the photoemissive surface is 1 eV , calculate the values of the stopping potential in the two cases (without and with the lens in the aperture).

Solution

- Energy of one photon,

$$E = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34})(3 \times 10^8)}{6000 \times 10^{-10}}$$

$$\Rightarrow E = 3.3 \times 10^{-19}\text{ J}$$



Power of the source is $2\text{ W} = 2\text{ J s}^{-1}$. Therefore, number of photons emitting per second is

$$n_1 = \frac{2}{3.3 \times 10^{-19}} = 6.06 \times 10^{18}\text{ s}^{-1}$$

At distance 0.6 m , number of photons incident per unit area per unit time is

$$n_2 = \frac{n_1}{4\pi(0.6)^2} = 1.34 \times 10^{18}\text{ m}^{-2}\text{ s}^{-1}$$

Area of aperture is,

$$S_1 = \frac{\pi}{4}d^2 = \frac{\pi}{4}(0.1)^2 = 7.85 \times 10^{-3}\text{ m}^2$$

So, total number of photons incident per unit time on the aperture,

$$n_3 = n_2 S_1 = (1.34 \times 10^{18})(7.85 \times 10^{-3})\text{ s}^{-1}$$

$$\Rightarrow n_3 = 1.052 \times 10^{16}\text{ s}^{-1}$$

This aperture will become new source of light.

Now these photons are further distributed in all directions. Hence, at the location of the detector, photons incident per unit area per unit time is

$$n_4 = \frac{n_3}{4\pi(6-0.6)^2} = \frac{1.052 \times 10^{16}}{4\pi(5.4)^2}$$

$$\Rightarrow n_4 = 2.87 \times 10^{13}\text{ m}^{-2}\text{ s}^{-1}$$

This is the photon flux at the centre of the screen.

Area of detector is 0.5 cm^2 or $0.5 \times 10^{-4}\text{ m}^2$. Therefore, total number of photons incident on the detector per unit time is

$$n_5 = (0.5 \times 10^{-4})(2.87 \times 10^{13}) = 1.435 \times 10^9\text{ s}^{-1}$$

The efficiency of photoelectron generation is 0.9 . Hence, total photoelectrons generated per unit time is

$$n_6 = 0.9n_5 = 1.2915 \times 10^9\text{ s}^{-1}$$

Hence, photocurrent in the detector is

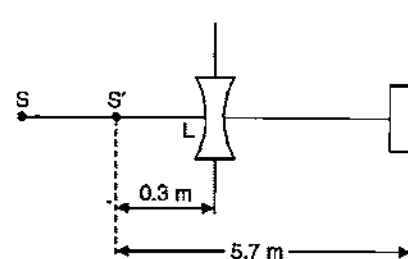
$$i = (e)n_6 = (1.6 \times 10^{-19})(1.2915 \times 10^9) = 2.07 \times 10^{-10}\text{ A}$$

- Using the lens formula, we get

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow v = -0.3\text{ m}$$

i.e., image of source (say S' , is formed at 0.3 m) from the lens.



Total number of photons incident per unit time on the lens are still n_3 or $1.052 \times 10^{16}\text{ s}^{-1}$. Since, 80% of it transmits to second medium, therefore, at a distance of 5.7 m from S' number of photons incident per unit area per unit time will be

$$n_7 = \frac{\left(\frac{80}{100}\right)(1.052 \times 10^{16})}{(4\pi)(5.7)^2}$$

$$\Rightarrow n_7 = 2.06 \times 10^{13}\text{ m}^{-2}\text{ s}^{-1}$$

This is the photon flux at the detector.

New value of photocurrent is given by

$$i' = (2.06 \times 10^{13})(0.5 \times 10^{-4})(0.9)(1.6 \times 10^{-19})$$

$$\Rightarrow i' = 1.483 \times 10^{-10}\text{ A}$$

- Energy of incident photons (in both the cases) is

$$E(\text{in eV}) = \frac{12375}{\lambda(\text{in Å})}$$

$$\Rightarrow E = \frac{12375}{6000 \text{ Å}} = 2.06 \text{ eV}$$

Work function $W = 1 \text{ eV}$

Maximum kinetic energy of photoelectrons in both cases,

$$K_{\text{max}} = E - W = 1.06 \text{ eV}$$

or the stopping potential will be 1.06 V .

Problem 8

Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one dimensional array with nodes at each of the atomic sites. It is found that one such standing wave is formed if the distance d between the atoms of the array is 2 Å . A similar standing wave is again formed if d is increased to 2.5 Å but not for any intermediate value of d . Find the energy of the electron in eV and the least value of d for which the standing wave of the type described above can form.

Solution

From the figure it is clear that

$$\text{So, } p \cdot \left(\frac{\lambda}{2}\right) = 2 \text{ Å and}$$

$$(p+1) \cdot \frac{\lambda}{2} = 2.5 \text{ Å}$$

$$\Rightarrow \frac{\lambda}{2} = (2.5 - 2) \text{ Å} = 0.5 \text{ Å}$$

$$\Rightarrow \lambda = 1 \text{ Å} = 10^{-10} \text{ m}$$

de-Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

where K is the kinetic energy of electron

$$\Rightarrow K = \frac{h^2}{2m\lambda^2}$$

$$\Rightarrow K = \frac{(6.63 \times 10^{-34})^2}{2(9.1 \times 10^{-31})(10^{-10})^2}$$

$$\Rightarrow K = 2.415 \times 10^{-17} \text{ J}$$

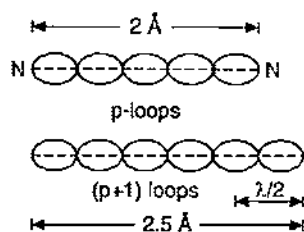
$$\Rightarrow K = \left(\frac{2.415 \times 10^{-17}}{1.6 \times 10^{-19}}\right) \text{ eV}$$

$$\Rightarrow K = 150.8 \text{ eV}$$

The least value of d will be, when only one loop is formed. So, we have

$$d_{\text{min}} = \frac{\lambda}{2}$$

$$\Rightarrow d_{\text{min}} = 0.5 \text{ Å}$$



Problem 9

When a beam of 10.6 eV photons of intensity 2 Wm^{-2} falls on a platinum surface of area $1 \times 10^{-4} \text{ m}^2$ and work function 5.6 eV . 0.53% of the incident photons eject photoelectrons. Find the number of photoelectrons emitted per second and their minimum and maximum energies (in eV). Take $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

Solution

Energy of incident photons,

$$E_i = 10.6 \text{ eV}$$

$$\Rightarrow E_i = 10.6 \times 1.6 \times 10^{-19} \text{ J}$$

$$\Rightarrow E_i = 16.96 \times 10^{-19} \text{ J}$$

Energy incident per unit area per unit time (intensity) = 2 J

So, number of photons incident on unit area in unit time is

$$\frac{n}{A} = \frac{2}{16.96 \times 10^{-19}} = 1.18 \times 10^{18}$$

Therefore, number of photons incident per unit time on given area ($1 \times 10^{-4} \text{ m}^2$) is

$$n = (1.18 \times 10^{18})(1 \times 10^{-4})$$

$$\Rightarrow n = 1.18 \times 10^{14}$$

But only 0.53% of incident photons emit photoelectrons, so number of photoelectrons emitted per second (n) is

$$n = \left(\frac{0.53}{100}\right)(1.18 \times 10^{14})$$

$$\Rightarrow n = 6.25 \times 10^{11}$$

$$K_{\text{min}} = 0$$

$$\text{and } K_{\text{max}} = E_i - W$$

$$\Rightarrow K_{\text{max}} = (10.6 - 5.6) \text{ eV} = 5 \text{ eV}$$

$$\Rightarrow K_{\text{max}} = 5 \text{ eV}$$

$$\text{and } K_{\text{min}} = 0$$

Problem 10

A small plate of a metal having work function of 1.17 eV is placed at a distance of 2 m from a monochromatic light source of wave length $4.8 \times 10^{-7} \text{ m}$ and power 1 W . The light falls normally on the plate. Find the number of photons striking the metal plate per m^2 per sec. If a constant uniform magnetic field of strength 10^{-4} T is applied parallel to the metal surface, find the radius of the largest circular path followed by the emitted photoelectrons. Given $h = 6.6 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ ms}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$ and electron mass $m = 9.1 \times 10^{-31} \text{ kg}$.

Solution

Energy of photons of wavelength $4.8 \times 10^{-7} \text{ m}$ is

$$E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4.8 \times 10^{-7}} = 4.125 \times 10^{-19} \text{ J}$$

Power of source = $1 \text{ W} = 1 \text{ Js}^{-1}$

So, rate of emission of photons from the source is

$$n = \frac{1 \text{ Js}^{-1}}{4.125 \times 10^{-19} \text{ J}} = 2.424 \times 10^{18} \text{ s}^{-1}$$

These photons move in all directions randomly. At a distance r from the source, the photons fall normally over a spherical surface of area $4\pi r^2$. The plate is at a distance $r = 2 \text{ m}$. Hence the number of photons striking the surface per m^2 per second is

$$n = \frac{2.424 \times 10^{18}}{4 \times 3.14 \times (2)^2} = 4.82 \times 10^{16}$$

The maximum KE of a photoelectron emitted from the plate is

$$K_{\max} = \frac{hc}{\lambda} - W_0$$

$$\Rightarrow K_{\max} = 4.125 \times 10^{-19} - 1.17 \times 1.6 \times 10^{-19} = 2.253 \times 10^{-19} \text{ J}$$

Hence the maximum velocity of the photoelectron is

$$v_{\max} = \sqrt{\frac{2K_{\max}}{m}}$$

$$\Rightarrow v_{\max} = \sqrt{\frac{2 \times 2.253 \times 10^{-19}}{9.1 \times 10^{-31}}} = 7.03 \times 10^5 \text{ ms}^{-1}$$

Radius of the largest circular path of the photoelectrons in the magnetic field is

$$r = \frac{mv_{\max}}{eB} = \frac{9.1 \times 10^{-31} \times 7.03 \times 10^5}{1.6 \times 10^{-19} \times 10^{-4}} = 4 \times 10^{-2} \text{ m} = 4 \text{ cm}$$

Problem 11

Two metallic plates A and B each of area $5 \times 10^{-4} \text{ m}^2$, are placed parallel to each other at separation of 1 cm . Plate B carries a positive charge of $33.7 \times 10^{-12} \text{ C}$. A monochromatic beam of light, with photons of energy 5 eV each, starts falling on plate A at $t = 0$ so that 10^6 photons fall on it per square meter per second. Assume that the photoelectron is emitted for every 10^6 incident photons. Also assume that all the emitted photoelectrons are collected by plate B and the work function of plate A remains constant at the value 2 eV .

(a) the number of photoelectrons emitted up to $t = 10 \text{ s}$

(b) the magnitude of the electric field between the plates A and B at $t = 10 \text{ s}$ and

(c) The kinetic energy of the most energetic photoelectrons emitted at $t = 10 \text{ s}$ when it reaches plate B.

Neglect the time taken by the photoelectron to reach plate B.

(Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

Solution

Area of plates $A = 5 \times 10^{-4} \text{ m}^2$

Distance between the plates $d = 1 \text{ cm} = 10^{-2} \text{ m}$

(a) Number of photoelectrons emitted upto $t = 10 \text{ s}$ are

$$n = \frac{\left(\text{number of photons falling in unit area in unit time} \right) \times (\text{area} \times \text{time})}{10^6}$$

$$\Rightarrow n = \frac{1}{10^6} [(10)^6 \times (5 \times 10^{-4}) \times (10)]$$

$$\Rightarrow n = 5 \times 10^7$$

(b) At time $t = 10 \text{ s}$, charge on plate A

$$q_A = +ne = (5 \times 10^7)(1.6 \times 10^{-19})$$

$$\Rightarrow q_A = 8 \times 10^{-12} \text{ C}$$

and charge on plate B,

$$q_B = (33.7 \times 10^{-12} - 8 \times 10^{-12})$$

$$\Rightarrow q_B = 25.7 \times 10^{-12} \text{ C}$$

Since, electric field between the plates is

$$E = \frac{(q_B - q_A)}{2A\epsilon_0}$$

$$\Rightarrow E = \frac{(25.7 - 8) \times 10^{-12}}{2 \times (5 \times 10^{-4})(8.85 \times 10^{-12})} = 2 \times 10^3 \text{ NC}^{-1}$$

(c) Energy of photoelectrons at plate A is

$$K = E - W = (5 - 2) \text{ eV} = 3 \text{ eV}$$

Increase in energy of photoelectrons is

$$(eEd) \text{ joule} = (Ed) \text{ eV}$$

$$\Rightarrow (\text{Increase in Energy}) = (2 \times 10^3)(10^{-2}) \text{ eV} = 20 \text{ eV}$$

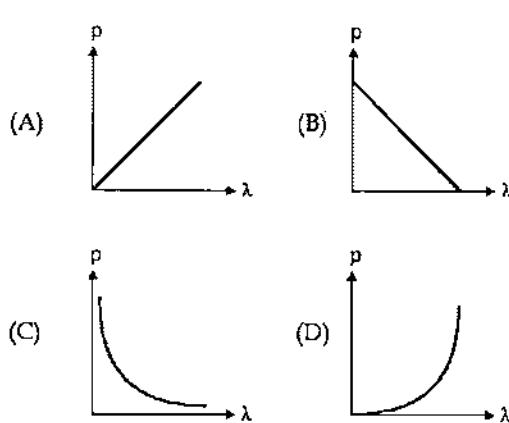
Energy of photoelectrons at plate B is

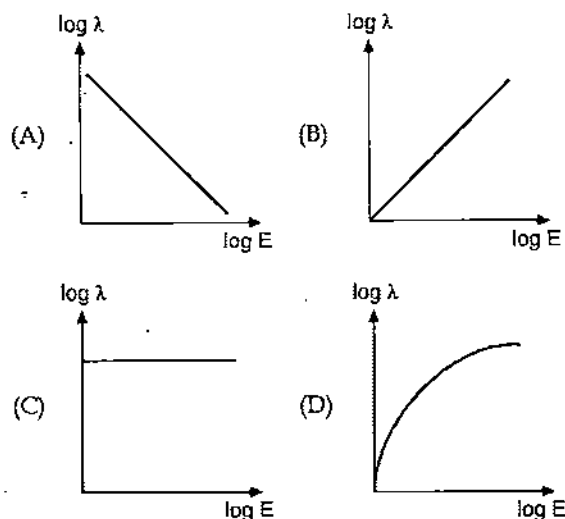
$$(20 + 3) \text{ eV} = 23 \text{ eV}$$

Practice Exercise

Single Correct Choice Type Questions

This section contains Single Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONLY ONE is correct.

- Consider a source emitting 100 W of green light at a wavelength of 500 nm. The number of photons emerging from source per second is
 (A) 2.5×10^{19} photon per second
 (B) 25×10^{20} photon per second
 (C) 25×10^{19} photon per second
 (D) 25×10^{27} photon per second
- The distance d of a 100 W lamp is continuously increased from a photocell. The photoelectric current I varies with distance d as
 (A) $I \propto d^2$ (B) $I \propto d$
 (C) $I \propto \frac{1}{d^2}$ (D) $I \propto \frac{1}{d}$
- When a metallic surface is illuminated with monochromatic light of wavelength λ , the stopping potential is $5V_0$. When the same surface is illuminated with light of wavelength 3λ , the stopping potential is V_0 . Then the work function of the metallic surface is
 (A) $\frac{hc}{6\lambda}$ (B) $\frac{hc}{5\lambda}$
 (C) $\frac{hc}{4\lambda}$ (D) $\frac{2hc}{4\lambda}$
- An electron of mass m , when accelerated through a potential difference V has de-Broglie wavelength λ . The de-Broglie wavelength associated with a proton of mass M accelerated through the same potential difference will be
 (A) $\lambda \left(\frac{m}{M} \right)$ (B) $\lambda \sqrt{\frac{m}{M}}$
 (C) $\lambda \left(\frac{M}{m} \right)$ (D) $\lambda \sqrt{\frac{M}{m}}$
- Which of the following graphs represents the variation of particle momentum and the associated de Broglie wavelength?

- If Planck's constant is denoted by h and the electronic charge by e , experiments on photoelectric effect allow the determination of
 (A) only h (B) only e
 (C) both h and e (D) only $\frac{h}{e}$
- The maximum energy of the electrons released in a photocell is independent of
 (A) frequency of incident light
 (B) intensity of incident light
 (C) nature of cathode rays
 (D) None of these
- If the energy and wavelength of electron are E and λ , then the graph between $\log E$ and $\log \lambda$ will be



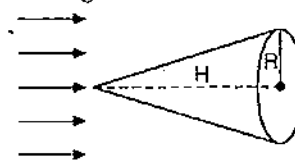
9. The total energy E of a sub-atomic particle of rest mass m_0 moving at non-relativistic speed v is
 (A) $E = m_0 c^2$ (B) $E = \frac{1}{2} m_0 v^2$
 (C) $E = m_0 c^2 + \frac{1}{2} m_0 v^2$ (D) $E = m_0 c^2 - \frac{1}{2} m_0 v^2$
10. An electron is 2000 times lighter than a proton. Both are moving such that their matter waves have a length of $1 \text{ \AA}$. The ratio of their kinetic energy in approximation is
 (A) 1:1 (B) 1:2000
 (C) 2000:1 (D) 1:200
11. The de Broglie wavelength of a particle is approximately the same as that of a photon with the same energy.
 (A) The energy of the particle is much greater than its rest energy.
 (B) The energy of the particle is much less than its rest energy.
 (C) The energy of the particle equals its rest energy.
 (D) Data insufficient to arrive at a conclusion.
12. Two lumps of clay each of rest mass m_0 , collide with a speed of $\frac{4}{5}c$ head on and stick together. The mass of the composite lump thus formed is
 (A) $\frac{10}{3}m_0$ (B) $\frac{5}{3}m_0$
 (C) $\frac{5}{6}m_0$ (D) $\frac{5}{12}m_0$
13. An electron moving with velocity $2 \times 10^7 \text{ ms}^{-1}$ describes a circle in a magnetic field of strength

$2 \times 10^{-2} \text{ T}$. If $\left(\frac{e}{m}\right)$ of electron is $1.76 \times 10^{11} \text{ C kg}^{-1}$, then the diameter of the circle is nearly

- (A) 1.1 m (B) 1.1 mm
 (C) 1.1 cm (D) 11 cm

14. The radiation emitted, when an electron jumps from $n=3$ to $n=2$ orbit in a hydrogen atom, falls on a metal to produce photoelectron. The electrons from the metal surface with maximum kinetic energy are made to move perpendicular to a magnetic field of $\frac{1}{320} \text{ T}$ in a radius of 10^{-3} m . The work function of metal is
 (A) 1.03 eV (B) 1.89 eV
 (C) 0.86 eV (D) 2.03 eV

15. The radiation force experienced by a body exposed to radiation of intensity I assuming surface of body to be perfectly absorbing is



- (A) $\frac{\pi R^2 I}{c}$ (B) $\frac{2\pi R^2 I}{c}$
 (C) $\frac{4\pi R^2 I}{c}$ (D) None of these

16. A radiation of energy E falls normally on a perfectly reflecting surface. The change in momentum of radiation is

- (A) $\frac{E}{c}$ (B) $\frac{2E}{c}$
 (C) Ec (D) $\frac{E}{c^2}$

17. An electron, accelerated by a potential difference V , has de Broglie wavelength λ . If the electron is accelerated by a potential difference $4V$, its de Broglie wavelength will be

- (A) 2λ (B) 4λ
 (C) $\frac{\lambda}{2}$ (D) $\frac{\lambda}{4}$

18. The photoelectric work function of a metal is 1 eV. Light of wavelength $\lambda = 3000 \text{ \AA}$ falls on it. The photoelectrons will come out with approximate speed equal to

- (A) 10 ms^{-1} (B) 10^2 ms^{-1}
 (C) 10^4 ms^{-1} (D) 10^6 ms^{-1}

19. A proton and an α -particle are accelerated through the same potential difference. The ratio of their de Broglie wavelengths is

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$
(C) $2\sqrt{2}$ (D) 2

20. In an electron microscope if the potential is increased from 20 kV to 80 kV, the resolving power R of the microscope will become

(A) R (B) $2R$
(C) $4R$ (D) $\frac{R}{2}$

21. Boron has two isotopes ${}_5B^{10}$ and ${}_5B^{11}$. If the atomic weight of boron is 10.81, the ratio of ${}_5B^{10}$ to ${}_5B^{11}$ in nature is

(A) $\frac{19}{81}$ (B) $\frac{20}{53}$
(C) $\frac{15}{10}$ (D) $\frac{10}{11}$

22. In photoelectric emission the number of electrons ejected per second is proportional to the

(A) intensity of light
(B) wavelength of light
(C) frequency of light
(D) work function of the material

23. For a photon, the de Broglie relation is given by

(A) $\lambda = \frac{h}{mc}$ (B) $\lambda = \frac{h}{p}$
(C) $\lambda \rightarrow \text{infinity}$ (D) Data Insufficient

24. A desk lamp illuminates a desk top with light of wavelength λ . The amplitude of this electromagnetic wave is E_0 . Assuming illumination to be normally on the surface, the number of photons striking the desk per second per unit area N is

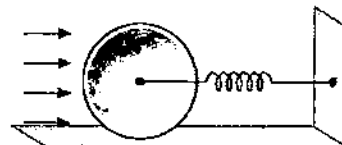
(A) $N = \frac{\lambda \epsilon_0 E_0^2}{h}$ (B) $N = \frac{2\lambda \epsilon_0 E_0^2}{h}$
(C) $N = \frac{\lambda \epsilon_0 E_0^2}{2h}$ (D) Data Insufficient

25. A small metal plate (work-function W_0) is kept at a distance d from a singly ionised fixed ion. A monochromatic light beam is incident on the metal plate and photoelectrons are emitted. The maximum wavelength of light so that the photo-electrons may go round the ions along a circle is

(A) $\frac{8\pi\epsilon_0 W_0 d + e^2}{8\pi h c \epsilon_0 d}$ (B) $\frac{8\pi h c \epsilon_0 d}{8\pi\epsilon_0 W_0 d + e^2}$

(C) $\frac{2\left(\frac{hc}{W_0} - e^2\right)}{n}$ (D) $\frac{8\pi\epsilon_0 d}{hc e W_0}$

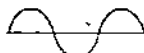
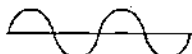
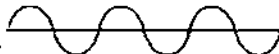
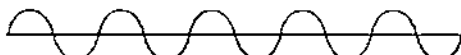
26. A perfectly reflecting body sphere in shape of radius R is placed on a path of parallel light beam of intensity I shown in figure. One end of a spring is attached to centre of sphere and the other end to a rigid wall as shown in figure. Assuming the sphere to be in equilibrium, the spring constant of spring to be K , then compression in spring is



(Neglect any thermal effect and friction is absent)

(A) $\frac{2I}{KC} \pi R^2$ (B) $\frac{I}{KC} \pi R^2$
(C) $\frac{I\pi R^2}{C}$ (D) None of these

27. The de-Broglie wave present in the fifth Bohr orbit is

(A) 
(B) 
(C) 
(D) 

28. The interatomic distance between atoms in a crystal is $2.8 \text{ \AA}$. Then if such a crystal is used in Davisson-Germer experiment, the maximum order of diffraction that can be observed for a beam of electrons accelerated by 100 V shall be

(A) $n=1$ (B) $n=2$
(C) $n=10$ (D) $n \rightarrow \infty$

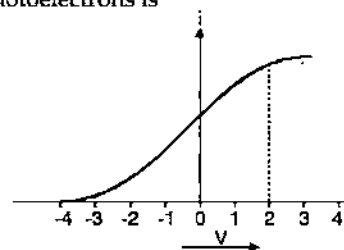
29. The energy of a photon of wavelength λ is

(A) $hc\lambda$ (B) $\frac{hc}{\lambda}$
(C) $\frac{\lambda}{hc}$ (D) $\frac{h\lambda}{c}$

30. The work function of a metal is 4 eV. For the emission of photoelectrons of zero velocity from the metal surface, the wavelength of the incident radiation should be

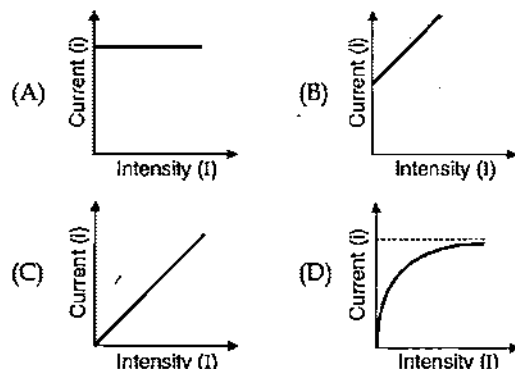
- (A) 1700 Å (B) 2700 Å
(C) 3100 Å (D) 5900 Å
31. Photoelectric effect is the phenomenon in which
(A) photons come out of a metal when it is hit by a beam of electrons
(B) photons come out of the nucleus of an atom under the action of an electric field.
(C) electrons come out of a metal with a constant velocity which depends on the frequency and intensity of incident radiation.
(D) electrons come out of a metal with different velocities not greater than a certain value which depends only on the frequency of the incident light and not on its intensity.
32. The photoelectric effect is the ejection of electrons from the surface of a metal when
(A) it is heated to a high temperature.
(B) electrons of suitable velocity strike it.
(C) radiation of suitable wavelength falls on it.
(D) it is placed in a strong electric field.
33. Matter waves are
(A) electromagnetic waves.
(B) transverse mechanical waves.
(C) longitudinal mechanical waves.
(D) neither electromagnetic nor mechanical waves.
34. A photon of wavelength 1000 Å has energy 12.3 eV. If light of wavelength 5000 Å, having intensity I , falls on a metal surface, the saturation current is 0.40 μA and the stopping potential is 1.36 V. The work function of the metal is
(A) 2.47 eV (B) 1.36 eV
(C) 1.10 eV (D) 0.43 eV
35. In PROBLEM 34, if the intensity of light is made $4I$, the stopping potential will become
(A) 1.36 V (B) 2.72 V
(C) 5.44 V (D) 21.76 V
36. In PROBLEM 34, if the intensity of light is made $4I$, the saturation current will become
(A) 0.4 μA (B) 0.8 μA
(C) 1.6 μA (D) 6.4 μA
37. In a photoemissive cell with exciting wavelength λ , the fastest electron has a speed v . If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electrons will be
(A) $v\sqrt{\frac{3}{4}}$ (B) $v\sqrt{\frac{4}{3}}$
(C) less than $v\sqrt{\frac{4}{3}}$ (D) greater than $v\sqrt{\frac{4}{3}}$
38. The de-Broglie wavelength of a molecule of thermal energy $k_B T$ (k_B = Boltzmann constant and T = absolute temperature), is
(A) $\lambda = \sqrt{\frac{h}{2mk_B T}}$ (B) $\frac{h}{\sqrt{2mk_B T}}$
(C) $h\sqrt{2mk_B T}$ (D) $\frac{h}{4m^2 k_B^2 T^2}$
39. A proton is accelerated through a potential V . The de Broglie wavelength associated with it is
(A) $\frac{12.27}{\sqrt{V}}$ Å (B) $\frac{0.287}{\sqrt{V}}$ Å
(C) $\frac{12.27}{\sqrt{V}}$ fm (D) $\frac{0.287}{\sqrt{V}}$ fm
40. In Davisson-Germer experiment Ni crystal acts as
(A) an ideal reflector
(B) three dimensional diffraction grating
(C) an ideal absorber
(D) two dimensional diffraction grating
41. The angle between the incident and the diffracted electron in the Davisson-Germer experiment is called as
(A) angle of incidence
(B) angle of diffraction
(C) angle of scattering
(D) none of the above
42. In Davisson-Germer experiment maximum intensity is observed at
(A) 50° and 54 volt (B) 54° and 50 volt
(C) 50° and 50 volt (D) 65° and 50 volt
43. In Davisson-Germer experiment, an electron beam of 60 eV energy falls normally to the surface of the crystal and maximum intensity is obtained at an angle of 60° to the direction of incident beam. The inter-atomic distance in the lattice plane of the crystal is
(A) 18 Å (B) 3.6 Å
(C) 1.8 Å (D) 0.18 Å
44. The incorrect statement in connection with Davisson and Germer experiment is
(A) The inter-atomic distance in nickel crystal is of the order of the de-Broglie wavelength.
(B) Electrons of constant energy are obtained by the electron gun.
(C) Nickel crystal acts as a three dimensional diffracting grating.
(D) Davisson-Germer experiment is an interference experiment.

45. In Davisson-Germer experiment the relation between Bragg's angle ϕ and diffraction angle θ is
 (A) $\theta = 90^\circ - \phi$ (B) $\theta = \frac{90^\circ - \phi}{2}$
 (C) $\theta = 180^\circ - \phi$ (D) $\phi = \left(\frac{180^\circ - \theta}{2}\right)$
46. The ionization chamber used in Davisson-Germer experiment, acts as
 (A) emitter (B) collector
 (C) source (D) radiator
47. The distance between two consecutive atoms of the crystal lattice is $1.227 \text{ \AA}$. The maximum order of diffraction of electrons accelerated through 10^4 V will be
 (A) 10 (B) $\frac{1}{10}$
 (C) 100 (D) $\frac{1}{100}$
48. The human eye can barely detect a yellow light ($6000 \text{ \AA}$) that delivers $1.7 \times 10^{-18} \text{ watt}$ to the retina. Nearly how many photons per second does the retina receive?
 (A) 50 (B) 5
 (C) 500 (D) More than 5 million
49. The ratio of the specific charge of a proton to that of an α -particle is
 (A) 1:4 (B) 1:2
 (C) 4:1 (D) 2:1
50. A photosensitive surface is receiving light of wavelength $5000 \text{ \AA}$ at the rate of 10^{-7} Js^{-1} . The number of photons received per second is
 (A) 2.5×10^{12} (B) 2.5×10^{11}
 (C) 2.5×10^{10} (D) 2.5×10^9
51. Photons of frequency ν fall on a metal surface for which the threshold frequency is ν_0 . Then,
 (A) all ejected electrons have the same kinetic energy $h(\nu - \nu_0)$.
 (B) the ejected electrons have a distribution of kinetic energy from zero to $h(\nu - \nu_0)$.
 (C) the most energetic electrons have kinetic energy $h\nu$
 (D) the average kinetic energy of ejected electrons is $h\nu_0$.
52. When radiation of wavelength $3000 \text{ \AA}$ is incident on a photosensitive surface, the kinetic energy of electrons is 2.5 eV . The stopping potential for radiation of wavelength $1500 \text{ \AA}$ will be
 (A) 2.5 V
 (B) 5.0 V
 (C) less than 5.0 V but more than 2.5 V
 (D) more than 5.0 V
53. The de Broglie wavelength of a particle of mass m moving with a kinetic energy E is
 (A) $\sqrt{\frac{h}{2mE}}$ (B) $\frac{h}{\sqrt{2mE}}$
 (C) $\frac{h}{2mE}$ (D) $\frac{\sqrt{h}}{2mE}$
54. Two photons of energy 2.5 eV each are incident on a metal plate whose work function is 4.0 eV , then the number of electrons emitted from the metal surface will be
 (A) one (B) two
 (C) None of these (D) more than two
55. Of the following moving with same momentum, the one which has largest wavelength is
 (A) an electron.
 (B) a proton.
 (C) an α -particle.
 (D) all have same de-Broglie wavelength.
56. The maximum velocity of an electron emitted by light of wavelength λ incident on the surface of a metal of work-function ϕ is
 (A) $\sqrt{\frac{2(hc + \lambda\phi)}{m\lambda}}$ (B) $\frac{2(hc - \lambda\phi)}{m}$
 (C) $\sqrt{\frac{2(hc - \lambda\phi)}{m\lambda}}$ (D) $\sqrt{\frac{2(h\lambda - \phi)}{m}}$
 where h = Planck's constant, m = mass of electron and c = speed of light
57. An α -particle is accelerated through a potential difference of 200 V . The increase in its kinetic energy in electron volt will be
 (A) 100 eV (B) 200 eV
 (C) 400 eV (D) 800 eV
58. Figure represents the graph of photo current i versus applied voltage (V). The maximum energy of the emitted photoelectrons is



- (A) 2 eV
(C) 0 eV
- (B) 4 eV
(D) 4 J

59. Which one of the following graphs represents correctly the variation of photoelectric current (i) with intensity (I) of incident radiations



60. The potential difference between the cathode and anode in a cathode ray tube is V . Then the speed acquired by the electrons is proportional to

- (A) V
(C) V^2
- (B) $\sqrt{V}$
(D) $V^{3/2}$

61. The duration of a laser pulse is 10^{-8} s. The uncertainty in its energy will be

- (A) 6.6×10^{-26} J
(C) 6.6×10^{-12} J
- (B) 6.6×10^{-34} J
(D) $\frac{1}{6.6} \times 10^{26}$ J

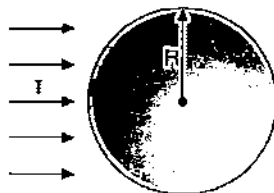
62. Which one of the following statements about photons is incorrect?

- (A) Rest mass of a photon is zero
(B) Momentum of a photon of frequency ν is $\frac{h\nu}{c}$
(C) Energy of a photon of frequency ν is $h\nu$
(D) Photons exert no pressure

63. A parallel beam of uniform, monochromatic light of wavelength $2640 \text{ \AA}$ has an intensity of 100 W m^{-2} . The number of photons in 1 mm^3 of this radiation are

- (A) 222
(C) 442
- (B) 335
(D) 555

64. Consider a sphere of radius R exposed to radiation of intensity I as shown in figure. If surface of sphere is partially reflection and reflection coefficient is 0.3, then radiation force experienced is



- (A) $\frac{\pi R^2 I}{c}$
(C) $\frac{0.3\pi R^2 I}{c}$
- (B) $\frac{1.7\pi R^2 I}{c}$
(D) None of these

65. Two electrons are moving with the same speed v . One electron enters a region of uniform electric field while the other enters a region of uniform magnetic field, then after sometime if the de-Broglie wavelengths of the two are λ_1 and λ_2 , then

- (A) $\lambda_1 = \lambda_2$
(C) $\lambda_1 < \lambda_2$
- (B) $\lambda_1 > \lambda_2$
(D) $\lambda_1 > \lambda_2$ or $\lambda_1 < \lambda_2$

66. An electromagnetic radiation of wavelength λ has the same momentum as an electron moving with a speed $2 \times 10^5 \text{ ms}^{-1}$.

- (A) $\lambda = 2.64 \text{ nm}$
(C) $\lambda = 3.64 \text{ nm}$
- (B) $\lambda = 1.64 \text{ nm}$
(D) $\lambda = 4.64 \text{ nm}$

67. Let K_1 be the maximum kinetic energy of photoelectrons emitted by light of wavelength λ_1 and K_2 corresponding to wavelength λ_2 . If $\lambda_1 = 2\lambda_2$ then

- (A) $2K_1 = K_2$
(C) $K_1 < \frac{K_2}{2}$
- (B) $K_1 = 2K_2$
(D) $K_1 > 2K_2$

68. Solar constant of the sun is $\sigma = 8.106 \times 10^4 \text{ J min}^{-1} \text{ m}^{-2}$ and average sun earth distance is $1.5 \times 10^8 \text{ km}$. The yearly loss in the mass of the sun is

- (A) $13.8 \times 10^{17} \text{ kg}$
(C) $1.38 \times 10^{17} \text{ kg}$
- (B) $1.38 \times 10^{19} \text{ kg}$
(D) $13.8 \times 10^{20} \text{ kg}$

69. A photon strikes a free electron at rest and is scattered straight backward. If the speed of electron after collision is αc , where $\alpha \ll 1$ then,

- (A) electron's kinetic energy is a fraction α of photon's initial energy.
(B) electron's kinetic energy is a fraction $\frac{1}{\alpha}$ of photon's initial energy.
(C) electron's kinetic energy is a fraction α^2 of photon's initial energy.
(D) electron's kinetic energy is a fraction $\frac{1}{\alpha^2}$ of photon's initial energy.

70. Which one of the following does not fit into the group?

- (A) Photon
(C) Proton
- (B) Graviton
(D) Meson

71. Two photons approach each other. The relative velocity of approach is

- (A) $\frac{c}{2}$ (B) c
(C) $2c$ (D) $4c$

72. A particle of mass $3m$ at rest decays into two particles of masses m and $2m$ having non-zero velocities. The ratio of the de-Broglie wavelengths of the particles $\left(\frac{\lambda_1}{\lambda_2}\right)$ is

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) 1 (D) 2

73. A small ball is projected with initial speed u and at an angle θ with horizontal from ground. The de-Broglie wavelength of ball at the moment its velocity vector becomes perpendicular to initial velocity vector is

- (A) $\frac{h}{mu}$ (B) $\frac{h}{mu \sin \theta}$
(C) $\left(\frac{h}{mu}\right) \tan \theta$ (D) $\frac{h}{mu \cos \theta}$

74. The force exerted by a photon of intensity 1.4 kWm^{-2} if it falls on a perfect absorber of radius 2 m is

- (A) $2.35 \times 10^{-4} \text{ N}$ (B) 10^8 N
(C) $8.35 \times 10^4 \text{ N}$ (D) $8.8 \times 10^{-8} \text{ N}$

75. Photoelectrons are emitted with maximum kinetic energy E from a metal surface when light of frequency ν falls on it when light of frequency ν' falls on the same metal, the maximum kinetic energy of emitted photoelectrons is found to be $2E$, then ν' is

- (A) $\nu' = \nu$ (B) $\nu' = 2\nu$
(C) $\nu' > 2\nu$ (D) $\nu' < 2\nu$

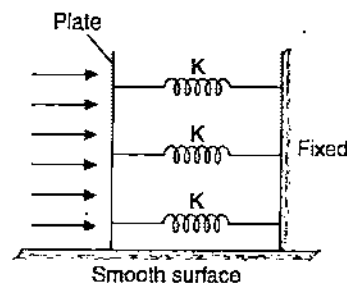
76. A material particle with a rest mass m_0 is moving with speed of light c . The de-Broglie wavelength associated is given by

- (A) $\frac{h}{m_0 c}$ (B) $\frac{m_0 c}{h}$
(C) ZERO (D) ∞

77. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV successively illuminate a metal of work function 0.5 eV . The ratio of the maximum speeds of the emitted electrons will be

- (A) $1:5$ (B) $1:4$
(C) $1:2$ (D) $1:1$

78. Light of intensity I is incident normally on a perfectly reflecting plate of area A kept in a gravity free space. If the photons strike the plate symmetrically and initially the springs are in their natural lengths, then the maximum compression in the springs is



- (A) $\frac{IA}{KC}$ (B) $\frac{2IA}{3KC}$
(C) $\frac{3IA}{KC}$ (D) $\frac{4IA}{3KC}$

79. The work function of a metallic surface is 5.01 eV . Photoelectrons are emitted when light of wavelength $2000 \text{ \AA}$ falls on it. The potential difference required to stop the fastest photoelectrons is

- (A) 1.2 V (B) 2.4 V
(C) 3.6 V (D) 4.8 V

80. A proton, accelerated through a potential difference V has a certain de Broglie wavelength. In order to have the same de Broglie wavelength, an α -particle must be accelerated through a potential difference

- (A) $4V$ (B) $8V$
(C) $\frac{V}{4}$ (D) $\frac{V}{8}$

81. Ultraviolet light wavelength 300 nm and intensity 1.0 Wm^{-2} falls on the surface of a photoelectric material. If one percent of the incident photons produce photoelectrons, then the number of photoelectrons emitted per second from an area of 1.0 cm^2 of the surface is nearly

- (A) 9.61×10^{14} (B) 4.12×10^{13}
(C) 1.51×10^{12} (D) 2.13×10^{11}

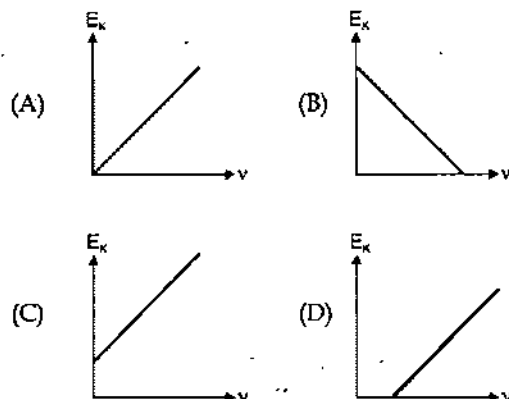
82. A proton and an α -particle are injected into a uniform electric field at right angles to the direction of field with equal kinetic energy. Then

- (A) the proton trajectory will be less curved than α -particle trajectory.
(B) the α -particle trajectory will be less curved than proton trajectory.
(C) both the trajectories will be equally curved.
(D) both trajectories will be straight.

83. de Broglie waves are associated with

- (A) moving charged particles only.
(B) moving neutral particles only.
(C) all moving particles.
(D) all particles whether in motion or at rest.

84. The maximum kinetic energy (E_k) of photoelectrons varies with the frequency (ν) of the incident radiation as

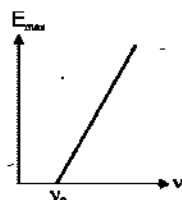


85. Stopping potential for photoelectrons
 (A) does not depend on the frequency of the incident light.
 (B) does not depend on the nature of cathode material.
 (C) depends on both the frequency of the incident light and the nature of the cathode material.
 (D) depends on the intensity of the incident light.

86. A particle of mass 10^{-31} kg is moving with a speed of 10^5 ms $^{-1}$. The de Broglie wavelength of the particle is
 (A) 6.63×10^{-8} m (B) $6.63 \text{ \AA}$
 (C) $66.3 \text{ \AA}$ (D) 6.63×10^{-7} m

87. A moving particle is associated with a wave packet or group of waves. The group velocity is equal to
 (A) velocity of light
 (B) velocity of sound
 (C) velocity of particle
 (D) $\frac{1}{\text{particle velocity}}$

88. The maximum kinetic energy ($E_{\max}$) of photoelectrons emitted in a photoelectric cell varies with frequency (ν) as shown in the graph. The slope of the graph is equal to



- (A) charge of the electron
 (B) $\frac{e}{m}$ of the electron
 (C) work function of the emitter

- (D) Planck's constant

89. A photon of frequency ν is incident on a metal surface whose threshold frequency is ν_0 . The maximum kinetic energy of the emitted electron will be

- (A) $h(\nu - \nu_0)$ (B) $h(\nu + \nu_0)$
 (C) $\frac{1}{2}h(\nu - \nu_0)$ (D) $\frac{1}{2}h(\nu + \nu_0)$

90. A stream of photons impinging normally on a completely absorbing screen in vacuum exerts a pressure $\mathcal{P}$. If I is the irradiance then,

- (A) $\mathcal{P} = \frac{2I}{c}$ (B) $\mathcal{P} = Ic$
 (C) $\mathcal{P} = \frac{I}{c}$ (D) $\mathcal{P} = 2Ic$

91. An electron and a photon have same wavelength. If p is the momentum of electron and E the energy of photon. The magnitude of $\frac{p}{E}$ in SI unit is

- (A) $\frac{1}{2c}$ (B) $\frac{1}{c}$
 (C) $\frac{2}{c}$ (D) None of these

92. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV. The stopping potential in volt is

- (A) 2 (B) 4
 (C) 6 (D) 10

93. A sensor is exposed for time t to a lamp of power P placed at a distance l . The sensor has an opening that is $4d$ in diameter. Assuming all energy of the lamp is given off as light, the number of photons entering the sensor if the wavelength of light is λ is

- (A) $N = \frac{P\lambda d^2 t}{hcl^2}$ (B) $N = \frac{4P\lambda d^2 t}{hcl^2}$
 (C) $N = \frac{P\lambda d^2 t}{4hcl^2}$ (D) $N = \frac{P\lambda d^2 t}{16hcl^2}$

94. When a metallic surface is illuminated by a monochromatic light of wavelength λ , the stopping potential for photoelectric current is $3V_0$. When the same surface is illuminated by light of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength for this surface for photoelectric effect is

- (A) 6λ (B) $\frac{4\lambda}{3}$
 (C) 4λ (D) 8λ

95. The momentum of a photon of an electromagnetic radiation is $3.3 \times 10^{-29} \text{ kgms}^{-1}$. The frequency of the associated waves is ($h = 6.6 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ ms}^{-1}$)

(A) $3.0 \times 10^3 \text{ Hz}$ (B) $6.0 \times 10^3 \text{ Hz}$
(C) $7.5 \times 10^{12} \text{ Hz}$ (D) $1.5 \times 10^{13} \text{ Hz}$

96. If the wavelength of incident radiation in a photoelectric experiment is decreased then

(A) the photoelectric current will decrease.
(B) the photoelectric current will increase.
(C) the stopping potential will decrease.
(D) the stopping potential will increase.

97. The threshold wavelength for a photosensitive surface is $6000 \text{ \AA}$ and the wavelength of incident light is $5000 \text{ \AA}$. Then the maximum energy of emitted electrons would be

(A) 0.041 eV (B) 0.41 eV
(C) 4.1 eV (D) 41 eV

98. An electron of mass m and charge e initially at rest gets accelerated by a constant electric field E . The rate of change of de-Broglie wavelength of this electron at time t , ignoring relativistic effects

(A) $-\frac{h}{eEt^2}$ (B) $-\frac{eht}{E}$
(C) $-\frac{mht}{eEt^2}$ (D) $-\frac{h}{eE}$

99. Of the following, the one which has the largest de Broglie wavelength for the same speed is

(A) electron (B) proton
(C) α -particle (D) oxygen atom

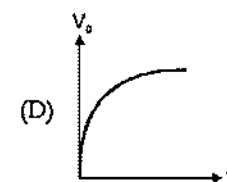
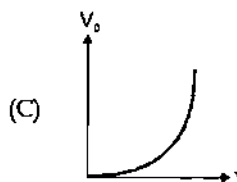
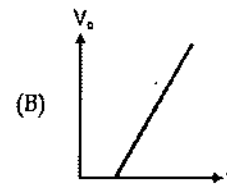
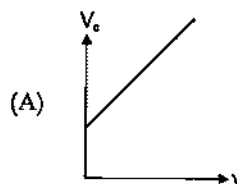
100. A small photocell is placed at a distance of 4 m from a photosensitive surface. When light falls on the surface the current is 5 mA . If the distance of cell is decreased to 1 m , the current will become

(A) 1.25 mA (B) $\left(\frac{5}{16}\right) \text{ mA}$
(C) 20 mA (D) 80 mA

101. Einstein's photoelectric equation is $E_k = h\nu - \phi$. In this equation E_k refers to

(A) kinetic energy of all the emitted electrons.
(B) mean kinetic energy of emitted electrons.
(C) maximum kinetic energy of emitted electrons.
(D) minimum kinetic energy of emitted electrons.

102. In photoelectric effect, the graph showing the variation of cut-off voltage (V_0) with the frequency of incident radiation (ν) is



103. An electron is accelerated through a potential difference of 100 V . Its kinetic energy will be

(A) 100 J (B) 100 erg
(C) 100 eV (D) 100 MeV

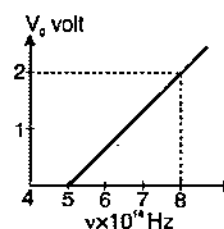
104. If 5% of the energy supplied to a bulb is radiated as visible light, the number of visible quanta emitted per second by a 100 W bulb, assuming the wavelength of visible light to be $5.6 \times 10^{-5} \text{ cm}$, is

(A) 1.4×10^{19} (B) 1.4×10^{20}
(C) 2×10^{19} (D) 2×10^{20}

105. A neutron is confined to a nucleus of size 10^{-14} m . The minimum momentum of the electron may be

(A) $6.6 \times 10^{-20} \text{ kgms}^{-1}$ (B) $3.3 \times 10^{-20} \text{ kgms}^{-1}$
(C) $3.3 \times 10^{-48} \text{ kgms}^{-1}$ (D) $6.6 \times 10^{-48} \text{ kgms}^{-1}$

106. The stopping potential (V_0) versus frequency plot of a substance is shown in figure. The threshold wavelength is

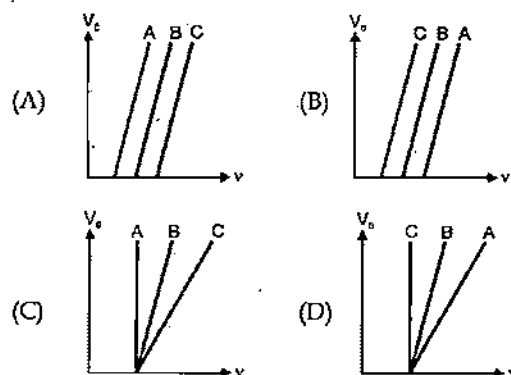


(A) $5 \times 10^{14} \text{ m}$
(B) $6000 \text{ \AA}$
(C) $5000 \text{ \AA}$
(D) cannot be estimated from given data

107. The energy of incident photon is 12.375 eV while the energy of scattered photon is 9.375 eV . Then the kinetic energy of recoil electron is

(A) 3 eV (B) less than 3 eV
(C) more than 3 eV (D) 21.75 eV

108. When electrons are accelerated through potential difference of V volt, the de-Broglie wavelength associated is given by
- (A) $\lambda = \sqrt{\frac{150}{V}} \text{ \AA}$ (B) $\lambda = \sqrt{\frac{150}{V}} \text{ m}$
 (C) $\lambda = \frac{150}{\sqrt{V}} \text{ \AA}$ (D) $\lambda = \frac{\sqrt{150}}{V} \text{ \AA}$
109. An electron and a proton are accelerated through the same potential. If their masses are m_e and m_p respectively, then the ratio of their de Broglie wavelength is
- (A) 1 (B) $\sqrt{\frac{m_e}{m_p}}$
 (C) $\frac{m_p}{m_e}$ (D) $\sqrt{\frac{m_p}{m_e}}$
110. A particle of mass 1 g is located in a box of size 2 cm. The uncertainty in the momentum of the electron will be
- (A) $3.3 \times 10^{-32} \text{ kgms}^{-1}$ (B) $6.6 \times 10^{-32} \text{ kgms}^{-1}$
 (C) $3.3 \times 10^{-33} \text{ kgms}^{-1}$ (D) $6.6 \times 10^{-34} \text{ kgms}^{-1}$
111. If E_1 , E_2 , E_3 are the respective kinetic energies of an electron, an alpha-particle and a proton, each having the same de-Broglie wavelength, then
- (A) $E_1 > E_3 > E_2$ (B) $E_2 > E_3 > E_1$
 (C) $E_1 > E_2 > E_3$ (D) $E_1 = E_2 = E_3$
112. The kinetic energy of electron is E , when the incident light has wavelength λ . To increase the K.E. to $2E$, the incident light must have wavelength
- (A) $\frac{hc}{E\lambda - hc}$ (B) $\frac{hc\lambda}{E\lambda + hc}$
 (C) $\frac{h\lambda}{E\lambda + hc}$ (D) $\frac{hc\lambda}{E\lambda - hc}$
113. We wish to observe an object which is $2.5 \text{ \AA}$ in size. The minimum energy photon that can be used is
- (A) 5 keV (B) 8 keV
 (C) 10 keV (D) 12 keV
114. A certain mass of ice at 0°C melts into water at 0°C and there by gains 1 kg mass. If initial mass of ice is m_0 then,
- (A) $m_0 = 2.69 \times 10^{11} \text{ kg}$ (B) $m_0 = 1 \text{ kg}$
 (C) $m_0 = 2.69 \times 10^{10} \text{ kg}$ (D) $m_0 = 9 \times 10^{16} \text{ kg}$
115. A star of mass M_0 , radius R_0 contracts to radius R . Energy radiated by the star assuming uniform density in each case while temperature remains unchanged is
- (A) $M_0 c^2$ (B) $M_0 c^2 \left[1 - \left(\frac{R}{R_0} \right)^2 \right]$
 (C) $M_0 c^2 \left(1 - \frac{R}{R_0} \right)$ (D) $M_0 c^2 \left[1 - \left(\frac{R}{R_0} \right)^3 \right]$
116. The number of red photons ($\lambda = 663 \text{ nm}$) that must strike a totally reflecting screen per second at normal incidence so that a force of 1 N is exerted on the screen is
- (A) $n = 5 \times 10^{23}$ (B) $n = 5 \times 10^{24}$
 (C) $n = 5 \times 10^{25}$ (D) $n = 5 \times 10^{26}$
117. A 20 amu atom emits photon of $6.6 \text{ \AA}$ while making a transition from excited state to ground state. The recoil energy of the atom will be
- (A) $1.5 \times 10^{-23} \text{ J}$ (B) $3.5 \times 10^{-23} \text{ J}$
 (C) $5.1 \times 10^{-23} \text{ J}$ (D) $7.5 \times 10^{-23} \text{ J}$
118. How many red photon (wavelength λ) must strike a totally reflecting screen per second at normal incidence, if the exerted force is to be 1 N ?
- (A) $\frac{\lambda}{h}$ (B) $\frac{2\lambda}{h}$
 (C) $\frac{\lambda}{2h}$ (D) infinity
119. The surface of a metal is illuminated with the light of 400 nm. The kinetic energy of the ejected photoelectrons was found to be 1.68 eV. The work function of the metal for $hc = 1240 \text{ eV-nm}$ is
- (A) 3.09 eV (B) 1.41 eV
 (C) 1.51 eV (D) 1.68 eV
120. A cathode of a photoelectric cell is changed such that the work function changes from W_1 to W_2 ($W_1 < W_2$). If the current before and after changes are I_1 and I_2 all other conditions remaining unchanged, then (assuming $h\nu > W_2$)
- (A) $I_1 = I_2$ (B) $I_1 < I_2$
 (C) $I_1 > I_2$ (D) $I_1 < I_2 < 2I_1$
121. The momentum of a photon of frequency ν is
- (A) $\frac{h\nu}{c^2}$ (B) $\frac{h\nu}{c}$
 (C) $h\nu c$ (D) $h\nu c^2$
122. The work functions for three different metals A, B and C are ϕ_A , ϕ_B and ϕ_C respectively with $\phi_A > \phi_B > \phi_C$. The graphs between stopping potential (V_0) and frequency ν of incident radiation for them would look like



123. In order to increase the kinetic energy of ejected photoelectrons, there should be an increase in
 (A) intensity of radiation.
 (B) wavelength of radiation.
 (C) frequency of radiation.
 (D) both wavelength and intensity of radiation.

124. Which of the following arrangements corresponds to decreasing order of specific charge?
 (A) Electron, proton, α -particle
 (B) Proton, α -particle, electron
 (C) α -particle, electron, proton
 (D) Electron, α -particle, proton

125. A photoelectric cell is illuminated by a small bright source of light placed at 1 m. If the same source of light is placed 2 m away, the electrons emitted by the cathode
 (A) each carries one quarter of its previous momentum.
 (B) each carries one quarter of its previous energy.
 (C) are half the previous number.
 (D) are one quarter of the previous number.

126. Light of frequency 1.5 times the threshold frequency is incident on a photo-sensitive material. If the frequency is halved and the intensity is doubled, the photoelectric current becomes
 (A) four times (B) double
 (C) half (D) ZERO

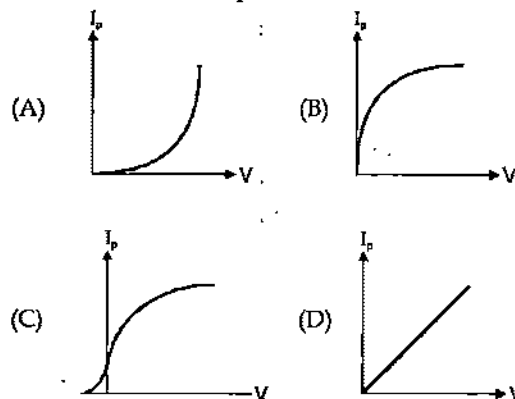
127. The ratio of the de Broglie wavelengths of a proton and an α -particle will be 1:2 if their
 (A) kinetic energies are in the ratio 1:8
 (B) kinetic energies are in the ratio 8:1
 (C) velocities are in the ratio 1:8
 (D) velocities are in the ratio 8:1

128. Cathode rays moving with same velocity v describe an approximate circular path of radius r metre in an electric field of strength x volt metre⁻¹. If the speed of the cathode rays is doubled to $2v$, the value of electric

field needed so that the rays describe the same approximate circular path (volt metre⁻¹) is

- (A) $2x$ (B) $3x$
 (C) $4x$ (D) $6x$

129. Which of the following graphs gives the variation of photoelectric current (I_p) with the voltage (V) applied to the electrodes of a photo cell?



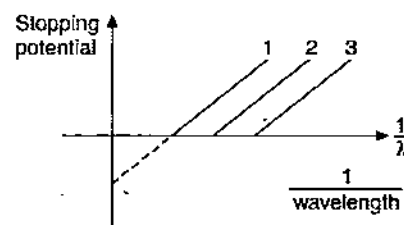
130. An electron with speed v and a photon with a speed c have the same de-Broglie wavelength. If the K.E. and momentum of electrons is E_e and P_e and that of photon is E_{ph} and P_{ph} respectively, then the correct statement is

- (A) $\frac{E_e}{E_{ph}} = \frac{2c}{v}$ (B) $\frac{E_e}{E_{ph}} = \frac{v}{2c}$
 (C) $\frac{P_e}{P_{ph}} = \frac{2c}{v}$ (D) None of these

131. The energy of a photon corresponding to the visible light of maximum wavelength is approximately
 (A) 1 eV (B) 1.6 eV
 (C) 3.2 eV (D) 7 eV

132. Planck's work was connected with
 (A) wave nature of matter
 (B) photoelectric effect
 (C) structure of atom
 (D) quantum nature of radiation.

133. The graph shows stopping potential versus $\frac{1}{\text{wavelength}}$ for three metals, then

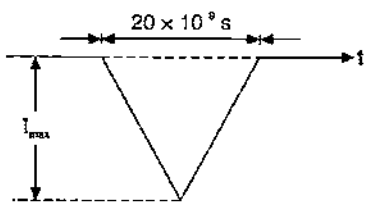


- (A) Planck's constant for metal (1) is greatest
 (B) Work function for metal (3) is greatest
 (C) Threshold frequency for metal (1) is greatest
 (D) Threshold wavelength is maximum for metal (3)
134. The work function of a substance is 4 eV. The longest wavelength of light that can cause photoelectron emission from this substance is approximately
 (A) 540 nm (B) 400 nm
 (C) 310 nm (D) 220 nm
135. A radio transmitter operates at a frequency of 880 kHz and a power of 10 kW. The number of photons emitted per second is
 (A) 1.71×10^{31} (B) 1327×10^{34}
 (C) 13.27×10^{34} (D) 13.27×10^{44}
136. The largest momentum we can expect for a microwave photon is
 (A) 6.6×10^{-27} kgms⁻¹ (B) 6.6×10^{-34} kgms⁻¹
 (C) 6.6×10^{-31} kgms⁻¹ (D) 6.6×10^{-30} kgms⁻¹
137. An electron is moving with a velocity of $\frac{c}{10}$. The de-Broglie wavelength associated with it is
 (A) 0.48×10^{-10} m (B) 0.24×10^{-10} Å
 (C) 0.24×10^{-10} m (D) 1.24×10^{-10} m
138. The kinetic energy of the body is twice the rest mass energy. The ratio of the relativistic mass of the body to its rest mass is
 (A) 1 (B) 2
 (C) 3 (D) infinite
139. The number of complete de-Broglie wavelengths associated with the electron in n th orbit of hydrogen atom is
 (A) n (B) n^2
 (C) $\frac{1}{n}$ (D) n^4
140. In a photoelectric cell, the current stops when the collecting plate is one volt negative with respect to the emitting metal. The maximum kinetic energy of the photoelectrons is
 (A) 1 erg (B) 1 J
 (C) 1.6×10^{-19} J (D) 1.6×10^{-19} eV

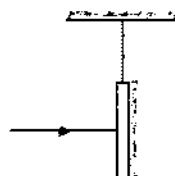
Practice Exercise

Multiple Correct Choice Type Questions

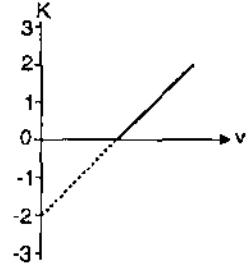
This section contains Multiple Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONE OR MORE is/are correct.

- When photons of energy 4.25 eV strike the surface of a metal, the ejected photoelectrons have maximum kinetic energy T_A eV and de Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A - 1.50)$ eV. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then
 - the work function of A is 2.25 eV.
 - the work function of B is 4.20 eV.
 - $T_A = 2.00$ eV.
 - $T_B = 2.75$ eV.
- A collimated beam of light of flux density 30 kWm^{-2} is incident normally on a 100 mm^2 completely absorbing screen. If $\mathcal{P}$ is the pressure exerted on the screen and Δp is the momentum transferred to the screen during a 1000 s interval then,
 - $\mathcal{P} = 10^{-3} \text{ Nm}^{-2}$
 - $\mathcal{P} = 10^{-4} \text{ Nm}^{-2}$
 - $\Delta p = 10^{-4} \text{ kgms}^{-1}$
 - $\Delta p = 10^{-5} \text{ kgms}^{-1}$
- A photomultiplier tube is to be used to detect light pulses each of which consists of a small but fixed number of photons. The average photoelectric efficiency is 10%. That is photon has 10% probability of causing the emission of a detectable photoelectron. Assume the photomultiplier gain is 10^6 and that the output current as a function of time can be approximated as shown in figure.
 
- i_{max} when averaged over many pulses is $80 \mu\text{A}$. Then which of the following statements is/are true.
 - The charge carried by one pulse is $8 \times 10^{-13} \text{ C}$
 - Number of photoelectrons emitted per light pulse is 5
 - Number of photons in one light pulse is 50
 - Number of electrons carried by one pulse is 5×10^5
- Light from a monochromatic source is incident normally on a small photo sensitive surface S having work function ϕ . If power of the source is W and a is the distance between the source and S , then
 - the number of photons striking the surface per unit time will be $\left(\frac{W\lambda S}{4\pi hca^2} \right)$
 - the maximum energy of the emitted electrons will be $\left(\frac{hc}{\lambda} - \phi \right)$
 - the stopping potential needed to stop the most energetic photons will be $\frac{e}{\lambda}(hc - \lambda\phi)$
 - photo emission occurs only if $0 \leq \lambda \leq \frac{hc}{\phi}$
- When ultraviolet radiation is incident on a surface, no photoelectrons are emitted. If another beam causes photoelectrons to be emitted from the surface, it may consist of
 - radio waves
 - infrared rays
 - X-rays
 - gamma rays
- When the intensity of a light source is increased,
 - the number of photons emitted by the source in unit time increases
 - the total energy of the photons emitted per unit time increases

- (C) more energetic photons are emitted
(D) faster photons are emitted
7. Light rays are incident on an opaque sheet. The correct statement(s) is/are
(A) Light rays exert a force on the sheet
(B) Light rays transfer an energy to the sheet
(C) Light rays transfer momentum to the sheet
(D) Light rays transfer impulse to the sheet
8. According to Heisenberg's Uncertainty Principle,
(A) $\Delta x \Delta p \geq \frac{h}{4\pi}$ (B) $\Delta E \Delta t \geq \frac{h}{4\pi}$
(C) $\Delta \theta \Delta L \geq \frac{h}{4\pi}$ (D) $\Delta x \Delta v \geq \frac{h}{4\pi m}$
where the symbols bear the usual meaning.
9. In which of the following situations, the heavier of the two particles will have a smaller de-Broglie wavelength.
(A) The particles move with the same speed
(B) The particles move with the same linear momentum
(C) The particles move with the same kinetic energy
(D) The particles have fallen through the same height
10. The threshold wavelength for photoelectric emission from a material is $5200 \text{ \AA}$. This material when illuminated with monochromatic radiation emits photoelectrons.
(A) 1 W UV (B) 50 W UV
(C) 1 W IR (D) 50 W IR
11. Light rays are incident on a metallic sheet. Then,
(A) the force exerted is independent of frequency of light incident
(B) the force depends on the direction of light incident
(C) the pressure is independent of frequency of light incident
(D) the pressure is proportional to the area of the plate
12. The momentum of a single photon of red light of frequency $400 \times 10^{12} \text{ Hz}$ moving through free space is
(A) ZERO (B) $8.8 \times 10^{-28} \text{ kgms}^{-2}$
(C) $1.65 \times 10^{-6} \text{ MeV/c}$ (D) Data Insufficient
13. A small mirror is suspended by a thread as shown in figure. A short pulse of monochromatic light rays is incident normally on the mirror and gets reflected. Which of the following statements is/are correct?



- (A) Mirror will start to oscillate.
(B) Wavelength of reflected rays will be greater than that of incident rays.
(C) Wavelength of reflected rays may be less than that of incident rays
(D) None of these
14. For a 75 W point light source assuming all the electric power consumed goes into emitted light of wavelength 600 nm, then
(A) frequency of the emitted light is $5 \times 10^{14} \text{ Hz}$
(B) number of photons emitted per second is 2.3×10^{20}
(C) this emitted light on falling on a metal surface of work function 1.07 eV, will emit photoelectrons having kinetic energy between 0 and 1 eV
(D) on doubling the distance of this metal surface from the point source maximum kinetic energy of photoelectrons emitted becomes 0.25 eV
15. Photoelectric effect supports the quantum nature of light because
(A) there is a minimum frequency of light below which no photoelectrons are emitted.
(B) the maximum kinetic energy of photoelectrons depends only on the frequency of light and not on its intensity.
(C) even when the metal surface is faintly illuminated, the photoelectrons leave the surface immediately.
(D) electric charge of the photoelectrons is quantized.
16. If the wavelength of light in an experiment on photoelectric effect is doubled,
(A) the photoelectric emission will not take place
(B) the photoelectric emission may or may not take place
(C) the stopping potential will increase
(D) the stopping potential will decrease
17. A metallic surface ejects electrons when exposed to green light of intensity I but no photoelectrons are emitted when exposed to yellow light of intensity I . It is possible to eject electrons from the same surface by
(A) yellow light of some intensity which is more than I .
(B) green light of any intensity.
(C) red light of any intensity.
(D) violet light of any intensity.
18. In an experiment of photoelectric effect, light from a point source of monochromatic light of wavelength $3000 \text{ \AA}$ is incident on a metal surface. The kinetic energies of photoelectrons range from zero to $4 \times 10^{-19} \text{ J}$, then
(A) stopping potential for this light is 2.5 V
(B) threshold wavelength for the material is $7590 \text{ \AA}$

- (C) stopping potential will be doubled on reducing the distance and the wavelength of light source to half
(D) saturation current will be doubled on reducing the distance of source to half
19. The threshold wavelength for photoelectric emission from a material is $5200 \text{ \AA}$. Photoelectrons will be emitted when this material is illuminated with monochromatic radiation from a
(A) 50 watt infrared lamp
(B) 1 watt infrared lamp
(C) 50 watt ultraviolet lamp
(D) 1 watt ultraviolet lamp
20. Light of wavelength 496 nm is incident on a metal surface causing ejection of photoelectrons for which stopping potential is 1.5 V , then
(A) the work function of the surface is 1 eV
(B) de-Broglie wavelength of fastest photoelectron is 100 nm
(C) to move the fastest electron in a circle of radius 1 m , perpendicular magnetic field B required is $4 \mu\text{T}$
(D) this fastest electron when strikes zinc target can produce X-rays
21. Which of the following statements about photoelectric effect is/are false?
(A) It exhibits the particle nature of radiation
(B) Electrons are emitted only if the radiation has a frequency above a certain value
(C) All the electrons emitted by radiation of a particular frequency have the same energy
(D) Changing the intensity of radiation changes the maximum energy with which the electrons can be emitted
22. Radiations of monochromatic waves of wavelength 400 nm are made incident on the surface of metals Zn , Fe and Ni of work functions 3.4 eV , 4.8 eV and 5.9 eV respectively (take $hc = 12400 \text{ eV \AA}$)
(A) maximum KE associated with photoelectrons from the surface of any metal is 0.3 eV
(B) no photoelectrons are emitted from the surface of Ni
- (C) if the wavelength of source of radiation is doubled then KE of photoelectrons is also doubled
(D) photoelectrons will be emitted from the surface of all the three metals if the wavelength of incident radiations is less than 200 nm
23. When a monochromatic point source of light is at a distance of 0.2 m from a photo-electric cell, the cut-off voltage and the saturation current are respectively 0.6 V and 18.0 mA . If the same source is placed 0.6 m away from the photoelectric cell, then
(A) the stopping potential will be 0.2 V .
(B) the stopping potential will be 0.6 V .
(C) the saturation current will be 6.0 mA .
(D) the saturation current will be 2.0 mA .
24. Figure represents a graph of kinetic energy (K) of photoelectrons (in eV) and frequency (ν) for a metal used as cathode in photoelectric experiment. The work function of metal is
- 
- (A) 1 eV
(B) 1.5 eV
(C) 2 eV
(D) 3 eV
25. In PROBLEM 24, the threshold frequency is nearly
(A) $5 \times 10^{14} \text{ Hz}$
(B) $10 \times 10^{14} \text{ Hz}$
(C) $2.5 \times 10^{14} \text{ Hz}$
(D) cannot be estimated
26. It is necessary to consider light as a stream of photons to explain
(A) Photoelectric effect
(B) Compton effect
(C) Polarization of light
(D) Diffraction of light

Practice Exercise

Reasoning Based Questions

This section contains Reasoning type questions, each having four choices (A), (B), (C) and (D) out of which ONLY ONE is correct. Each question contains STATEMENT 1 and STATEMENT 2. You have to mark your answer as

Bubble (A) If both statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1.

Bubble (B) If both statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1.

Bubble (C) If STATEMENT 1 is TRUE and STATEMENT 2 is FALSE.

Bubble (D) If STATEMENT 1 is FALSE but STATEMENT 2 is TRUE.

1. *Statement-1:*

Threshold wavelength of certain metal is λ_0 . Light of wavelength slightly less than λ_0 is incident on the plate. It is found that after some time the emission of electrons stops.

Statement-2:

The ejected electrons experience force of attraction due to development of positive charges on plate which after certain time is adequate enough to hold them to plate itself.

2. *Statement-1:*

A photon has no rest mass, yet it carries definite momentum.

Statement-2:

Momentum of photon is due to its energy and hence its equivalent mass.

3. *Statement-1:*

In a photoelectric effect, the current increases when positive potential of collector is increased, before saturation of current.

Statement-2:

The number of emitted photoelectrons increases.

4. *Statement-1:*

The de Broglie wavelength of an electron accelerated through 941 volt is $0.4 \text{ \AA}$.

Statement-2:

Higher the accelerating potential, smaller is the de-Broglie wavelength.

5. *Statement-1:*

Work function of copper is greater than that of sodium. But both will have same value of the threshold frequency and threshold wavelength.

Statement-2:

The frequency is inversely proportional to wavelength.

6. *Statement-1:*

In case of an electron and a photon having same momentum, wavelength associated with electron is smaller.

Statement-2:

Electron cannot move with a speed of photon.

7. *Statement-1:*

The photoelectrons produced by a monochromatic light beam incident on a metal surface, have a spread in their kinetic energies.

Statement-2:

The work function of the metal varies as a function of depth from the surface.

8. *Statement-1:*

A proton, a deuteron and an α -particle are accelerated by the same potential difference. Their velocities will be in the ratio of $1:1:\sqrt{2}$.

Statement-2:

Kinetic energy, $E = qV = \frac{1}{2}mv^2$

9. Statement-1:

Photoelectric effect demonstrates the wave nature of light.

Statement-2:

The number of photoelectrons is proportional to the frequency of light.

10. Statement-1:

A photon has no rest mass, yet it carries momentum.

Statement-2:

Momentum depends more on velocity than that of mass.

11. Statement-1:

The threshold frequency of photoelectric effect supports the particle nature of sunlight.

Statement-2:

If frequency of incident light is less than the threshold frequency, electrons are not emitted from metal surface.

12. Statement-1:

Effective mass of photon varies with wavelength.

Statement-2:

$E = mc^2$ is the relation between mass and energy.

13. Statement-1:

Though light of a single frequency (monochromatic light) is incident on a metal, the energies of emitted photoelectrons are different.

Statement-2:

The energy of electrons just after they absorb photons incident on metal surface may be lost in collision with other atoms in the metal before the electron is ejected out of metal.

14. Statement-1:

A photon and an electron, both of energy 1 MeV has same wavelength.

Statement-2:

$E = 22m_e c^2 = 10^6 \text{ eV}$

15. Statement-1:

The velocity of body of rest mass m_0 is $\frac{\sqrt{3}}{2}c$ (where c is the velocity of light in vacuum) then mass of the body is $2m_0$.

Statement-2:

Moving mass is given as $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$

16. Statement-1:

In the process of photoelectric emission by monochromatic light, all the emitted photo-electrons possess the same kinetic energy.

Statement-2:

In photoelectric effect a single photon interacts with a single electron and electron is emitted only if energy of each of incident photon is greater than the work function.

17. Statement-1:

de-Broglie wavelength of an electron accelerated through a potential difference of V volt is $\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$.

Statement-2:

de-Broglie wavelength of an electron is given by $\lambda = \frac{h}{mv}$

Practice Exercise

Linked Comprehension Type Questions

This section contains Linked Comprehension Type Questions or Paragraph based Questions. Each set consists of a Paragraph followed by questions. Each question has four choices (A), (B), (C) and (D), out of which only one is correct. (For the sake of competitiveness there may be a few questions that may have more than one correct options)

Comprehension 1

A photocell is operating in saturation mode with a photo current $20 \mu\text{A}$ when a monochromatic radiation of wave length $3000 \text{ \AA}$ and power 1-MW is incident. When another monochromatic radiation of wave length $1500 \text{ \AA}$ and power 5 MW is incident, it is observed that the maximum velocity of photo electron is doubled. Assuming efficiency of photoelectron generation per incident photon to be same for both the cases. Based on above information, answer the following questions.

- The threshold wavelength for the cell is
(A) $3500 \text{ \AA}$ (B) $4000 \text{ \AA}$
(C) $4500 \text{ \AA}$ (D) $5000 \text{ \AA}$
- The saturation current in second case is
(A) $50 \mu\text{A}$ (B) $40 \mu\text{A}$
(C) $60 \mu\text{A}$ (D) $45 \mu\text{A}$
- The efficiency of photoelectron generation per incident photon is
(A) 8.5% (B) 8.25%
(C) 8% (D) 8.75%

Comprehension 2

The radiations emitted when an electron jumps from $n=3$ to $n=2$ orbit of hydrogen atom falls on a metal to produce photoelectrons. The electrons emitted from the metal surface with maximum kinetic energy are made to move perpendicular to a magnetic field of strength $\frac{1}{320} \text{ T}$ in a radius of 10^{-3} m . Based on above information, answer the following questions.

- The kinetic energy of the electrons is
(A) 0.56 eV (B) 1.32 eV
(C) 0.86 eV (D) 1.76 eV
- The work function of the metal is
(A) 1.89 eV (B) 1.03 eV
(C) 1.58 eV (D) 2.32 eV
- Wavelength of radiation is nearly
(A) $6565 \text{ \AA}$ (B) $5555 \text{ \AA}$
(C) $4545 \text{ \AA}$ (D) $3535 \text{ \AA}$

Comprehension 3

de-Broglie suggested that every moving particle has a wavelength associated with it, which is given by $\lambda = \frac{h}{p}$ or

$\lambda = \frac{h}{\sqrt{2mK}}$. With the help of these formulae we know that wavelength for a charged particle accelerated through a potential V is $\lambda = \frac{h}{\sqrt{2mqV}}$. Based on above information, answer the following questions.

- Proton and alpha particle are accelerated through same potential difference. Then ratio of their wavelength is
(A) $1:\sqrt{2}$ (B) $2\sqrt{2}:1$
(C) $2:1$ (D) $1:2\sqrt{2}$
- When an electron is accelerated through 150 V potential difference, then the wavelength associated with it is approximately
(A) $1 \text{ \AA}$ (B) $2 \text{ \AA}$
(C) $3 \text{ \AA}$ (D) $4 \text{ \AA}$

9. If electron and alpha particle have same momentum the ratio of their wavelength is
 (A) 1840:1 (B) 1:1840
 (C) 1:1 (D) None of these

Comprehension 4

A large number of identical balls each having a mass of 66.3 g are thrown with speed of 5 ms^{-1} into a house through two tall, narrow, parallel windows spaced 0.6 m apart, the choice of window as target being random at each throw. Fringes are formed on a wall 12 m behind the windows. Assuming the Planck's constant to have a hypothetical value of $h = 6.63 \times 10^{-3} \text{ Js}$. Based on above information, answer the following questions.

10. de-Broglie wavelength of the balls is
 (A) 0.2 m (B) 0.02 m
 (C) 2 m (D) 20 m
11. Fringe width obtained on the wall is
 (A) 0.01 m (B) 0.2 m
 (C) 0.5 m (D) 0.4 m
12. If electrons are used in place of balls, moving with speed 10^7 ms^{-1} in the experiment then
 (A) Fringe width will decrease
 (B) No interference pattern is obtained
 (C) Fringe width will not increase
 (D) Interference pattern is obtained but it is not possible to observe it

Comprehension 5

A surface has light of wavelength $\lambda = 496 \text{ nm}$ incident on it, causing the ejection of photoelectrons for which the stopping potential is found to be 1.5 V. Based on above information, answer the following questions.

13. The de-Broglie wavelength of the fastest photoelectron emitted is
 (A) 75 nm (B) 100 nm
 (C) 135 nm (D) 235 nm
14. To move the fastest electron in a circle of radius 1 m, the value of perpendicular magnetic field B is
 (A) $4.1 \times 10^{-6} \text{ T}$ (B) $2.1 \times 10^{-5} \text{ T}$
 (C) $3.2 \times 10^{-5} \text{ T}$ (D) $5.2 \times 10^{-6} \text{ T}$

15. The threshold wavelength for photoelectric emission to occur is nearly
 (A) 1250 nm (B) 2000 nm
 (C) 2250 nm (D) 3000 nm

Comprehension 6

A parallel beam of monochromatic light ($\lambda = 663 \text{ nm}$) of intensity 30 kWm^{-2} is incident normally on a 100 mm^2 completely absorbing screen for 10 s. Based on above information, answer the following questions.

16. Pressure exerted by beam on the surface is
 (A) 10^{-5} Pa (B) 10^{-4} Pa
 (C) $2 \times 10^{-4} \text{ Pa}$ (D) $5 \times 10^{-5} \text{ Pa}$
17. Momentum transferred to the screen during the interval is
 (A) $10^{-8} \text{ kgms}^{-1}$ (B) $5 \times 10^{-8} \text{ kgms}^{-1}$
 (C) $10^{-7} \text{ kgms}^{-1}$ (D) $2 \times 10^{-7} \text{ kgms}^{-1}$
18. Number of photons striking the screen during the interval is
 (A) 5×10^{19} (B) 10^{20}
 (C) 2×10^{20} (D) 5×10^{20}

Comprehension 7

A metallic surface, when illuminated by light of frequency $8 \times 10^{14} \text{ Hz}$ and $12 \times 10^{14} \text{ Hz}$ emits photoelectrons of maximum kinetic energy 0.5 eV and 2.0 eV. Based on the information provided answer the following questions.

19. The value of Planck's constant is
 (A) $6.0 \times 10^{-34} \text{ Js}$ (B) $6.2 \times 10^{-34} \text{ Js}$
 (C) $6.4 \times 10^{-34} \text{ Js}$ (D) $6.6 \times 10^{-34} \text{ Js}$
20. The work function of metal is
 (A) 0.5 eV (B) 1.5 eV
 (C) 2.5 eV (D) 3.5 eV
21. de-Broglie wavelength of electron when its energy is 0.5 eV
 (A) 8.68 Å (B) 17.35 Å
 (C) 21.25 Å (D) 24.54 Å

Practice Exercise

Matrix Match Type Questions

Each question in this section contains statements given in two columns, which have to be matched. The statements in COLUMN-I are labelled A, B, C and D, while the statements in COLUMN-II are labelled p, q, r, s (and t). Any given statement in COLUMN-I can have correct matching with ONE OR MORE statement(s) in COLUMN-II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following examples:

If the correct matches are $A \rightarrow p, s$ and t ; $B \rightarrow q$ and r ; $C \rightarrow p$ and q ; and $D \rightarrow s$ and t ; then the correct darkening of bubbles will look like the following :

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

1. In a photoelectric effect experiment, if f is the frequency of radiations incident on the metal surface and I is the intensity of incident radiations, then match the quantities in COLUMN-I with their matches in COLUMN-II.

COLUMN-I	COLUMN-II
(A) If distance between cathode and anode is increased.	(p) Stopping potential increases.
(B) If I is increased keeping f and work function constant.	(q) Saturation current increases.
(C) Work function is decreased keeping f and I constant.	(r) Maximum kinetic energy of photo electron increases.
(D) If f is increased keeping I and work function constant.	(s) Stopping potential remain same.

2. In a photoelectric effect experiment, if the following changes are made, then match the COLUMN-I with COLUMN-II.

COLUMN-I	COLUMN-II
(A) If intensity of incident light is increased keeping its frequency constant.	(p) Stopping potential will increase.
(B) If work function of photo sensitive electrode is increased.	(q) Current through circuit may stop.
(C) If frequency of incident light is increased and its intensity is decreased.	(r) Maximum kinetic energy of ejected photoelectrons will increase.
(D) If frequency of incident light is increased keeping its intensity constant.	(s) Saturation current will increase.

3. Match the properties in COLUMN-I with their respective phenomenon in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Photon character of radiation.	(p) Photoelectric effect.
(B) Wave character of radiation.	(q) Compton effect.
(C) Interaction of a photon with an electron, such that photon energy is much greater than the binding energy of electron, is more likely to result in.	(r) Diffraction.
(D) Interaction of a photon with an electron, such that photon energy is equal to or slightly greater than the binding energy of electron, is more likely to result in.	(s) Interference.

4. Match the wavelengths in COLUMN-I to the respective matches in COLUMN-II.

COLUMN-I	COLUMN-II
(A) $0.1 \text{ \AA}$	(p) de-Broglie wavelength of electron in X-ray tube.
(B) $1 \text{ \AA}$	(q) Photoelectric threshold wavelength.
(C) $10 \text{ \AA}$	(r) X-ray wavelength.
(D) $5000 \text{ \AA}$	(s) de-Broglie wavelength of most energetic photoelectron emitted from metal surface in photoelectric effect.

5. Some quantities related to photoelectric effect are mentioned under COLUMN-I and COLUMN-II. Match each quantity on COLUMN-I with the corresponding quantity in COLUMN-II on which it depends.

COLUMN-I	COLUMN-II
(A) de-Broglie wavelength of photoelectron.	(p) Frequency of light.
(B) Force due to radiation falling on metal plate.	(q) Work function.
(C) Stopping potential.	(r) Area of photo sensitive plate.
(D) Saturation current.	(s) Intensity of light (at constant v).

6. In a photoelectric effect experiment, if
- f
- is the frequency of radiations incident on the metal surface and
- I
- is the intensity of incident radiations, then match the following.

COLUMN-I	COLUMN-II
(A) Work function is decreased keeping f and I constant.	(p) Stopping potential increases.
(B) If I is increased keeping f and work function constant.	(q) Saturation current increases.
(C) If distance between cathode and anode is increased.	(r) Maximum kinetic energy of photoelectron increases.
(D) If f is increased keeping I and work function constant.	(s) Stopping potential remain same.

7. With respect to photoelectric effect experiment, match the entries of COLUMN-I with the entries of COLUMN-II.

COLUMN-I	COLUMN-II
(A) If I is increased keeping f and ϕ constant.	(p) Stopping potential increases.
(B) If f (frequency) is increased keeping I (intensity) and ϕ (work function) constant.	(q) Saturation photocurrent increases.
(C) If ϕ is decreased keeping f and I constant.	(r) Maximum K.E. of the photoelectrons increases.
(D) If the distance between anode and cathode increases.	(s) Stopping potential remains the same.

8. If radiation of energy E , intensity I falls on different kinds of surfaces mentioned, then match the quantities in COLUMN-I with their respective answers in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Radiation pressure for a perfectly absorbing surface.	(p) $(1 + \rho) \frac{I}{c}$
(B) Radiation pressure for a perfectly reflecting surface.	(q) $\frac{2E}{c}$
(C) Radiation pressure for a surface of reflection coefficient (ρ) .	(r) $\frac{I}{c}$
(D) Impulse on a perfectly absorbing surface.	(s) $\frac{2I}{c}$
(E) Impulse on a perfectly reflecting surface.	(t) $(1 + \rho) \frac{E}{c}$
(F) Impulse on a surface of reflection coefficient (ρ) .	(u) $\frac{E}{c}$

Practice Exercise

Integer Answer Type Questions

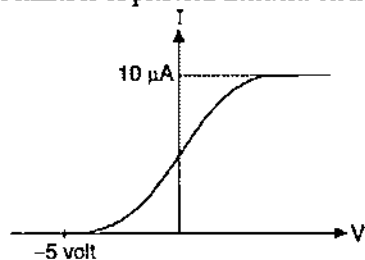
In this section the answer to each of the question is a four digit integer, ranging from 0 to 9999. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answer to question number X (say) is 6092, then the correct darkening of bubbles will look like the following :

X.

0	●	0	0
1	1	1	1
2	2	2	●
3	3	3	3
4	4	4	4
5	5	5	5
●	6	6	6
7	7	7	7
8	8	8	8
9	9	●	9

- A metal plate is placed 5 metre from a monochromatic light source whose power output is 10^{-3} W. Consider that a given ejected photoelectrons may collect its energy from a circular area of the plate as large as ten atomic diameters (10^{-9} m) in radius. The energy required to remove an electron through the metal surface is about 5 eV. Assuming light to be a wave, how long, in hour, would it take for such a 'target' to soak up this much energy from such a light source.
- Energy from the sun is received on the earth at the rate of $2 \text{ cal cm}^{-2} \text{ min}^{-1}$. If average wavelength of solar light be taken as $6600 \text{ \AA}$, then $x \times 10^{18}$ photons are received on earth per cm^2 per minute, find x . Take $1 \text{ cal} = 4.2 \text{ J}$, $c = 3 \times 10^8 \text{ ms}^{-1}$.
- Compute the typical de-Broglie wavelength of an electron in a metal at 27°C and compare it with the mean separation between two electrons in a metal which is given to be about $2 \times 10^{-10} \text{ m}$.
- Light of wavelength 180 nm ejects photoelectrons from a plate of metal whose work-function is 2 eV . If a uniform magnetic field of $5 \times 10^{-5} \text{ T}$ be applied parallel to the plate, what would be radius of the path, in mm, followed by electrons ejected normally from the plate with maximum energy.
- The maximum kinetic energy of photoelectrons emitted from a certain metallic surface is 30 eV when monochromatic radiation of wavelength λ falls on it. When the same surface is illuminated with light of wavelength 2λ , the maximum kinetic energy of photoelectrons is observed to be 10 eV . Calculate the wavelength λ and determine the maximum wavelength of incident radiation (both in $\text{\AA}$) for which photoelectrons can be emitted by this surface. Given $h = 4 \times 10^{-15} \text{ eVs}$ and $c = 3 \times 10^8 \text{ ms}^{-1}$.
- On a certain metal light of frequency $\nu = 5\nu_0$ falls then maximum velocity of electrons emitted is $8 \times 10^6 \text{ ms}^{-1}$, where ν_0 is threshold frequency of metal. If $\nu = 2\nu_0$ then the maximum velocity of photoelectron is $x \times 10^6 \text{ ms}^{-1}$. Find x .
- A metallic sphere of radius 21 cm is kept in the path of a parallel beam of light of intensity $\frac{1}{110} \text{ Wm}^{-2}$. The force exerted by beam on the sphere is $x \times 10^{-13} \text{ N}$. Find x .

8. The wavelength of light incident on a metal surface is reduced from 300 nm to 200 nm (both are less than threshold wavelength). Find the change in the stopping potential, in volt, for photoelectrons emitted from the surface. Take $h = 6.6 \times 10^{-34}$ Js.
9. In the photoelectric experiment, if we use a monochromatic light, the I-V curve is as shown. If work function of the metal is 2 eV, estimate the power of light used (in W). Assume efficiency of photo emission to be $10^{-3}\%$, i.e., number of photoelectrons emitted are $10^{-3}\%$ of number of photons incident on metal.
10. If photons of ultraviolet light of energy 12 eV are incident on a metal surface of work function of 4 eV, then find the stopping potential (in eV).
11. A proton and an α -particle are fired through the same magnetic fields which is perpendicular to their velocity vectors. Both move such that radius of curvature of their path is the same. Find the ratio of their de-Broglie wavelengths.



Answers

In Chapter Exercise (ICE) & Practice Exercise Sets

ICE II

BASED ON PHOTON PROPERTIES

- | | |
|---|---|
| 1. $4.7 \times 10^{-6} \text{ Nm}^{-2}$ | 7. (a) $3.98 \times 10^{-19} \text{ J}$ |
| 2. $453 \text{ \AA}$ | (b) 5×10^{15} |
| 3. $2.5 \text{ \AA}$ | 9. $5.5 \times 10^{-12} \text{ m}$ |
| 4. 2.51×10^{31} | 10. $\lambda_s > \lambda_p$ |
| 5. 1.43×10^{19} | 11. $E_p > E_s$ |
| 6. 10^{22} | 12. $E_p > E_s$ |

ICE II

BASED ON PHOTOELECTRIC EFFECT

- | | |
|-----------------------------|---|
| 1. 2.8 eV | 8. $6.6 \times 10^{-34} \text{ Js}$, $5990.25 \text{ \AA}$ |
| 2. $3363 \text{ \AA}$ | 9. 1.9 eV |
| 3. 11.93 eV | 10. 1.9 eV , $4125 \text{ \AA}$, No change is observed |
| 4. $2260 \text{ \AA}$ | 11. (a) 2.1 eV |
| 5. 9.28 mC | (b) 2.1 V |
| 6. 1.12×10^{12} | |
| 7. (a) 10^{15} Hz | |
| (b) 6.25 eV | |
| (c) 2 eV | |

ANSWERS

SINGLE CORRECT CHOICE TYPE QUESTIONS

1. C	2. C	3. A	4. B	5. C
6. D	7. B	8. A	9. C	10. C
11. A	12. A	13. C	14. A	15. A
16. B	17. C	18. D	19. C	20. B
21. A	22. A	23. B	24. C	25. B
26. B	27. D	28. B	29. B	30. C
31. D	32. C	33. D	34. C	35. A
36. C	37. D	38. B	39. B	40. B
41. B	42. A	43. C	44. D	45. D
46. B	47. A	48. B	49. D	50. B
51. B	52. D	53. B	54. C	55. D
56. C	57. C	58. B	59. C	60. B
61. A	62. D	63. C	64. A	65. D
66. C	67. C	68. C	69. A	70. C
71. B	72. C	73. C	74. A	75. C
76. C	77. C	78. D	79. A	80. D
81. C	82. A	83. C	84. D	85. C
86. A	87. C	88. D	89. A	90. C
91. B	92. B	93. A	94. C	95. D
96. D	97. B	98. A	99. A	100. D
101. C	102. B	103. C	104. A	105. B
106. B	107. A	108. A	109. D	110. A
111. A	112. B	113. A	114. A	115. D
116. D	117. A	118. C	119. B	120. A
121. B	122. B	123. C	124. A	125. D
126. D	127. D	128. C	129. C	130. B
131. B	132. D	133. B	134. C	135. A
136. C	137. C	138. C	139. A	140. C

ANSWERS

MULTIPLE CORRECT CHOICE TYPE QUESTIONS

1. ABC	2. BD	3. ABC	4. ABD	5. CD
6. AB	7. ABCD	8. ABCD	9. ACD	10. AB
11. ABC	12. BC	13. AB	14. ABC	15. ABC
16. BD	17. BD	18. AB	19. CD	20. AC
21. CD	22. BD	23. BD	24. C	25. A
26. AB				

ANSWERS

REASONING BASED QUESTIONS

1. A	2. A	3. C	4. B	5. D
6. D	7. C	8. D	9. D	10. C
11. A	12. A	13. A	14. A	15. A
16. D	17. D			

ANSWERS

LINKED COMPREHENSION TYPE QUESTIONS

1. C	2. A	3. B	4. C	5. B
6. A	7. B	8. A	9. C	10. B
11. D	12. D	13. B	14. A	15. A
16. B	17. C	18. B	19. A	20. C
21. B				

ANSWERS

MATRIX MATCH TYPE QUESTIONS

1. A → (s) B → (q, s) C → (p, r) D → (p, r)	2. A → (s) B → (q) C → (p, r) D → (p, r)	3. A → (p, q) B → (r, s) C → (q) D → (p)	4. A → (p) B → (r) C → (s) D → (q)	5. A → (p, q) B → (p, r, s) C → (p, q) D → (s)
6. A → (p, r) B → (q, s) C → (s) D → (p, r)	7. A → (q, s) B → (p, r) C → (p, r) D → (s)	8. A → (r) B → (s) C → (p) D → (u) E → (q) F → (t)		

ANSWERS

INTEGER ANSWER TYPE QUESTIONS

1. 20	2. 28	3. 31	4. 148
5. 1200	6. 4	7. 42	8. 2
9. 7	10. 8	11. 2	

Solutions

In Chapter Exercise (ICE)-I

(based on photon properties)

1. For completely absorbing surface,

$$P_{rad} = \frac{I}{C} = \frac{1.4 \times 10^3}{3 \times 10^8} = 4.7 \times 10^{-6} \text{ Nm}^{-2}$$

2. $f = \frac{v_1}{2\pi r_1} = \frac{c}{\lambda}$

$$\Rightarrow \lambda = \frac{2\pi r_1}{v_1} = \frac{(2\pi)(3 \times 10^8)(0.529 \times 10^{-10})(10^{19})}{(2.2 \times 10^9)} \text{ Å}$$

$$\Rightarrow \lambda = 453 \text{ Å}$$

3. For an electron, de-Broglie wavelength is given by,

$$\lambda = \sqrt{\frac{150}{V}} = \sqrt{\frac{150}{25}} = \sqrt{6}$$

$$\Rightarrow \lambda \approx 2.5 \text{ Å}$$

4. Power of transmitter = 10 kW = 10^4 W

So, total energy emitted per second is 10^4 J

$$\text{Energy of one photon, } E = h\nu = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{500} \text{ J}$$

Number of photons emitted per second is

$$n = \frac{\text{Total energy emitted per second}}{\text{Energy of one photon}}$$

$$\Rightarrow n = \frac{10^4 \times 500}{6.63 \times 10^{-34} \times 3 \times 10^8} = 2.51 \times 10^{31}$$

5. Energy of one photon is

$$E = \frac{hc}{\lambda}$$

$$\Rightarrow E = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{5600 \times 10^{-10}} = 3.5 \times 10^{-19} \text{ J}$$

Since a 100 W bulb supplies 100 J of energy per second.

So, energy released per second as visible photons is

$$E' = \frac{100 \times 5}{100} = 5 \text{ J}$$

$\therefore$ Number of photons emitted per second as visible light is

$$\frac{E'}{E} = \frac{5}{3.5 \times 10^{-19}} = 1.43 \times 10^{18}$$

6. Energy of each photon is

$$E = h\nu = 6.62 \times 10^{-34} \times 10^{12} = 6.62 \times 10^{-22} \text{ J}$$

Number of photons present in 6.62 J of radiation energy is calculated by using

$$E = N(h\nu)$$

$$\Rightarrow N = \frac{E}{h\nu} = \frac{6.62}{6.62 \times 10^{-22}} = 10^{22}$$

7. (a) Energy of each photon,

$$E = h\nu = 6.63 \times 10^{-34} \times 6 \times 10^{14} = 3.98 \times 10^{-19} \text{ J}$$

(b) If N is the number of photons emitted per second by the source, then

Power transmitted in the beam is

$$P = N(\text{energy of each photon})$$

$$\Rightarrow P = NE$$

$$\Rightarrow N = \frac{P}{E} = \frac{2 \times 10^{-3} \text{ W}}{3.98 \times 10^{-19} \text{ J}}$$

$$\Rightarrow N = 5 \times 10^{15} \text{ photons per second.}$$

8. The total energy of an electron with rest mass m_0 and momentum p is

$$E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

where c is the speed of light in free space. Law of Conservation of Energy requires,

$$\left(\begin{array}{c} \text{Rest mass} \\ \text{of electron} \end{array} \right) + \left(\begin{array}{c} \text{Energy of} \\ \text{photon} \end{array} \right) = \left(\begin{array}{c} \text{Total energy of} \\ \text{moving electron} \end{array} \right)$$

$$\Rightarrow m_0 c^2 + h\nu = \sqrt{m_0^2 c^4 + p^2 c^2} \quad \dots(1)$$

Law of Conservation of Momentum requires.

$$\left(\begin{array}{c} \text{Initial momentum} \\ \text{of electron} \end{array} \right) + \left(\begin{array}{c} \text{Momentum} \\ \text{of photon} \end{array} \right) = \left(\begin{array}{c} \text{Final momentum} \\ \text{of electron} \end{array} \right)$$

$$\Rightarrow 0 + \frac{h}{\lambda} = p$$

$$\Rightarrow \frac{h\nu}{c} = p$$

$$\Rightarrow h\nu = pc$$

$$\therefore \lambda = \frac{c}{\nu} \quad \dots(2)$$

Squaring both sides of equation (1), we get

$$m_0^2 c^4 + h^2 \nu^2 + 2m_0 c^2 h\nu = m_0^2 c^4 + p^2 c^2$$

$$\Rightarrow 2(m_0 c^2)(h\nu) = 0 \quad \text{(using (2))}$$

This is impossible. In a frame, where the initial electron is moving with uniform velocity, the same conclusion must hold

because if a process is forbidden in one inertial frame, it is also forbidden in another inertial frame.

9. KE of an electron is

$$E_k = 50 \text{ eV} = 1.6 \times 10^{-19} \times 5 \times 10^4 \text{ J} = 8 \times 10^{-15} \text{ J}$$

de-Broglie wavelength of electrons is

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 8 \times 10^{-15}}} \text{ m}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-11}}{12.07} \text{ m} = 5.5 \times 10^{-12} \text{ m}$$

For yellow light, wavelength $\lambda = 5.9 \times 10^{-7} \text{ m}$

As resolving power is inversely proportional to wavelength, so the resolving power of an electron microscope is about 10^5 times greater than that of an optical microscope.

10. The kinetic energy of a particle of mass m can be expressed in terms of its momentum p as follows

$$KE = \frac{1}{2}mv^2 = \frac{1}{2} \cdot \frac{(mv)^2}{m} = \frac{p^2}{2m}$$

As de-Broglie wavelength, $\lambda = \frac{h}{p}$

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \frac{p_p}{p_e} = \frac{\sqrt{2m_p \cdot KE}}{\sqrt{2m_e \cdot KE}} = \sqrt{\frac{m_p}{m_e}}$$

As $m_e < m_p$, therefore, $\lambda_e > \lambda_p$

Thus the electron has greater de-Broglie wavelength.

11. Kinetic energy of electron is

$$E_e = \frac{1}{2}mv^2 = \frac{1}{2} \cdot \frac{(mv)^2}{m} = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} \quad \left\{ \because p = \frac{h}{\lambda} \right\}$$

Energy of a photon is totally kinetic and is given by

$$E_p = hv = \frac{hc}{\lambda}$$

$$\Rightarrow \frac{E_p}{E_e} = \frac{hc}{\lambda} \cdot \frac{2m\lambda^2}{h^2} = \frac{2mc\lambda}{h}$$

$$\Rightarrow \frac{E_p}{E_e} = \frac{m_p}{m_e}$$

Thus the KE of a photon is greater than that of an electron of same wavelength.

12. The total relativistic energy of a particle is

$$E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

As wavelength λ is same for both electron and proton,

Since, momentum, $p = \frac{h}{\lambda}$ is same for both particles and hence

$p^2 c^2$ is same for both.

But rest mass m_0 of a proton is greater than that of an electron, therefore, the energy of a proton is more than that of an electron of same wavelength.

Solutions

In Chapter Exercise (ICE)-II

(based on photoelectric effect)

1. Since, $E = h\nu = (4.316 \times 10^{-15} \text{ eV} \cdot \text{sec})(1.5 \times 10^{15} \text{ sec}^{-1}) \approx 6.5 \text{ eV}$
 $\Rightarrow K_{\max} = E - W = (6.5 - 3.7) \text{ eV} = 2.8 \text{ eV}$

2. $E = \frac{12375}{2480} \approx 5 \text{ eV} = 8 \times 10^{-19} \text{ J}$

Number of photons emitted per second from the source is given by

$$n_1 = \frac{P}{E} = \frac{40}{8 \times 10^{-19}} = 5 \times 10^{19}$$

Number of photons incident per unit area per second on magnesium surface is

$$n_2 = \frac{n_1}{4\pi r^2} = \frac{5 \times 10^{19}}{4\pi(2)^2} \approx 10^{18}$$

From Einstein's Photoelectric equation, we have

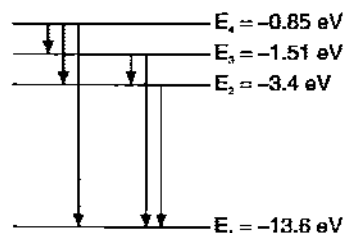
$$K_{\max} = E - W = 5 - 3.68 = 1.32 \text{ eV}$$

Since, $\lambda_0 (\text{in } \text{\AA}) = \frac{12375}{W (\text{in eV})}$

$$\Rightarrow \lambda_0 = \frac{12375}{W} = \frac{12375}{3.68} = 3363 \text{ \AA}$$

3. $\frac{K_1}{K_2} = 5$

$$\Rightarrow \frac{\Delta E_1 - W}{\Delta E_2 - W} = 5 \quad \dots(1)$$



Here, $\Delta E_1 = E_4 - E_1 = 12.75 \text{ eV}$ and $\Delta E_2 = E_3 - E_1 = 12.09 \text{ eV}$

Substituting in equation (1) and solving we get
 $W = 11.93 \text{ eV}$

4. $K_{\max} = E - W$

$$\Rightarrow 3 \text{ eV} = \frac{12375}{\lambda} - \frac{12375}{5000}$$

$$\Rightarrow \lambda = 2260 \text{ \AA}$$

5. Since, $E (\text{in eV}) = \frac{12375}{\lambda (\text{in } \text{\AA})}$

$$\Rightarrow E_1 = \frac{12375}{5000} = 2.475 \text{ eV}$$

$$E_2 = \frac{12375}{6000} = 2.06 \text{ eV} \text{ and}$$

$$E_3 = \frac{12375}{7000} = 1.77 \text{ eV}$$

Since, W is 1.9 eV , photons of energy E_1 and E_2 can only emit photoelectrons. Charge emitted per second is

$$\frac{q}{t} = (1.6 \times 10^{-19}) \left(\frac{1}{3} \right) \frac{(10^{-3})(10^{-4})}{\pi \times (10^{-3})^2 \times 2.475 \times 1.6 \times 10^{-19}} + (1.6 \times 10^{-19}) \left(\frac{1}{3} \right) \frac{(10^{-3})(10^{-4})}{\pi \times (10^{-3})^2 \times 2.06 \times 1.6 \times 10^{-19}}$$

$$\Rightarrow \frac{q}{t} = 9.28 \times 10^{-3} \text{ C} = 9.28 \text{ mC}$$

6. $E_1 = \frac{12375}{4000} = 3.1 \text{ eV}$

$$E_2 = \frac{12375}{4800} = 2.57 \text{ eV}$$

$$E_3 = \frac{12375}{6000} = 2.06 \text{ eV} \text{ and}$$

$$E_4 = \frac{12375}{7000} = 1.77 \text{ eV}$$

Therefore, light of wavelengths $4000 \text{ \AA}$, $4800 \text{ \AA}$ and $6000 \text{ \AA}$ can only emit photoelectrons.

So, number of photoelectrons emitted per second is

$$n = \frac{I_1 A_1}{E_1} + \frac{I_2 A_2}{E_2} + \frac{I_3 A_3}{E_3}$$

$$\Rightarrow n = IA \left(\frac{E_1 E_2 + E_2 E_3 + E_1 E_3}{E_1 E_2 E_3} \right)$$

$$\Rightarrow n = \frac{(1.5 \times 10^{-3})(10^{-4})}{1.6 \times 10^{-19}} \left[\frac{3.1 \times 2.57 + 2.57 \times 2.06 + 3.1 \times 2.06}{3.1 \times 2.57 \times 2.06} \right]$$

$$\Rightarrow n = 1.12 \times 10^{12}$$

7. (a) $h\nu_0 = W = 3.75 \times 1.6 \times 10^{-19} \text{ J}$

$$\Rightarrow \nu_0 = \frac{3.75 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} = 10^{15} \text{ Hz}$$

$$(b) E = \frac{12375}{1980} \text{ eV} = 6.25 \text{ eV}$$

$$\Rightarrow K_{\max} = 2.5 \text{ eV}$$

$$(c) K_{\max} = 2 \text{ eV}$$

$$8. K_{\max} = E - W = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$0.18 \times 1.6 \times 10^{-19} = \frac{h(3 \times 10^8)}{5461 \times 10^{-10}} - \frac{h(3 \times 10^8)}{\lambda_0} \quad \dots(1)$$

$$\text{Further, } eV_0 = E - W = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$\Rightarrow 4.6 \times 1.6 \times 10^{-19} = \frac{h(3 \times 10^8)}{1849 \times 10^{-10}} - \frac{h(3 \times 10^8)}{\lambda_0} \quad \dots(2)$$

Solving equations (1) and (2), for h and λ_0 , we get

$$h = 6.6 \times 10^{-34} \text{ Js}$$

$$\text{and } \lambda_0 = 5990.25 \text{ \AA}$$

9. From Einstein's Photo-Electric Equation, we have

$$\frac{1}{2}mv_1^2 = \frac{hc}{\lambda_1} - W \quad \dots(1)$$

$$\frac{1}{2}mv_2^2 = \frac{hc}{\lambda_2} - W \quad \dots(2)$$

Dividing equation (1) with equation (2), and using the given fact that $v_1 = 2v_2$, we get

$$\frac{\frac{hc}{\lambda_1} - W}{\frac{hc}{\lambda_2} - W} = 4$$

$$\Rightarrow 3W = 4\left(\frac{hc}{\lambda_2}\right) - \left(\frac{hc}{\lambda_1}\right) = \frac{4 \times 12375}{5400} - \frac{12375}{3500} = 5.63 \text{ eV}$$

$$\Rightarrow W = 1.9 \text{ eV}$$

10. From Einstein's Photo-Electric Equation, we have

$$0.6 = \frac{12375}{4950} - W \quad \dots(1)$$

$$1.1 = \frac{12375}{\lambda} - W \quad \dots(2)$$

Solving above two equations, we get

$$W = 1.9 \text{ eV and } \lambda = 4125 \text{ \AA}$$

No change is observed.

11. Since, $W = 2.3 \text{ eV}$ and $\lambda = 2800 \text{ \AA}$

$$E(\text{in eV}) = \frac{12375}{\lambda(\text{in \AA})} = \frac{12375}{2800} = 4.4 \text{ eV}$$

(a) From Einstein's Photo-Electric Equation, we have

$$K_{\max} = E - W$$

$$\Rightarrow K_{\max} = (4.4 - 2.3) \text{ eV}$$

$$\Rightarrow K_{\max} = 2.1 \text{ eV}$$

(b) Since, $K_{\max} = eV_0$

$$\Rightarrow 2.1 \text{ eV} = eV_0$$

$$\Rightarrow V_0 = 2.1 \text{ V}$$

Solutions

Single Correct Choice Type Questions

1. [C]
 $E = nh\nu$
 $\Rightarrow \frac{E}{t} = P = \left(\frac{n}{t}\right)h\nu$

3. [A]
 $5 eV_0 = \frac{hc}{\lambda} - W$... (1)
 $\Rightarrow eV_0 = \frac{hc}{3\lambda} - W$... (2)
 Solving equation (1) & (2), we get
 $\frac{hc}{\lambda} - W = \frac{5hc}{3\lambda} - 5W$
 $\Rightarrow 4W = \frac{2hc}{3\lambda}$
 $\Rightarrow W = \frac{hc}{6\lambda}$

4. [B]
 $\lambda = \frac{h}{\sqrt{2meV}}$
 $\lambda' = \frac{h}{\sqrt{2MeV}}$
 $\Rightarrow \lambda' = \lambda \sqrt{\frac{m}{M}}$

8. [A]
 $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$
 $\Rightarrow \lambda = kE^{-\frac{1}{2}}$
 $\Rightarrow \log_e \lambda = \log k - \frac{1}{2} \log E$

9. [C]
 $E = mc^2$
 $\Rightarrow E = m_0 c^2 \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$
 Since $\frac{v}{c} \ll 1$

$$\Rightarrow E = m_0 c^2 \left(1 + \frac{v^2}{2c^2}\right)$$

$$\Rightarrow E = m_0 c^2 + \frac{1}{2} m_0 v^2$$

10. [C]
 Since both have same de Broglie wavelength, hence both must have equal value of momentum.
 Since, $E = \frac{p^2}{2m}$
 $\Rightarrow \frac{E_e}{E_p} = \frac{m_p}{m_e} = 1840 \approx 2000$ {nearest possible approximation to answer}

11. [A]
 $E^2 = p^2 c^2 + m_0^2 c^4$
 $\Rightarrow E^2 = p^2 c^2 + E_0^2$ {where $E_0 = m_0 c^2$ }
 $\Rightarrow p = \frac{E}{c} \sqrt{1 - \left(\frac{E_0}{E}\right)^2} = \frac{E}{c}$ (for $E \gg E_0$)
 $\Rightarrow \lambda = \frac{h}{p} = \frac{hc}{E}$
 For a photon
 $E = h\nu = \frac{hc}{\lambda_1}$
 $\Rightarrow \lambda_1 = \frac{hc}{E} = \lambda$

12. [A]
 Energy of each lump before collision is
 $E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - \left(\frac{4}{5}\right)^2}} = \frac{5}{3} m_0 c^2$

The energy of composite lump after collision will be Mc^2 .
 By energy conservation principle we get
 Total initial energy = Total final energy
 $\Rightarrow \frac{5}{3} m_0 c^2 + \frac{5}{3} m_0 c^2 = Mc^2$
 $\Rightarrow M = \frac{10}{3} m_0$

13. [C]

$$r = \frac{mv}{qB}$$

$$\Rightarrow r = \frac{2 \times 10^7}{1.76 \times 10^{11} \times 2 \times 10^{-2}}$$

$$\Rightarrow r = 0.0055 \text{ m}$$

$$\Rightarrow D = 2r = 0.011 \text{ m}$$

$$\Rightarrow D = 1.1 \text{ cm}$$

14. [A]

$$E_3 - E_2 = 13.6 \left[\frac{1}{2^2} - \frac{1}{3^2} \right]$$

$$\Rightarrow \Delta E = \frac{13.6 \times 5}{36} = 1.89 \text{ eV}$$

Photoelectrons with $K_{\max}$ are moving on circular path, so

$$r = \frac{mv}{qB}$$

$$\Rightarrow mv = qBr$$

$$\Rightarrow p = qBr = 1.6 \times 10^{-19} \times \frac{1}{3200} \times 10^{-3}$$

$$\Rightarrow p = \frac{1}{2} \times 10^{-24} = 5 \times 10^{-25} \text{ kgms}^{-1}$$

$$\text{Energy of photoelectron is } K_{\max} = \frac{p^2}{2m}$$

$$\Rightarrow K_{\max} = \frac{25 \times 10^{-50}}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$\Rightarrow K_{\max} = 0.86 \text{ eV}$$

Now using Einstein equation, we get

$$h\nu = \phi + K_{\max}$$

$$\Rightarrow 1.89 = 0.56 + \phi$$

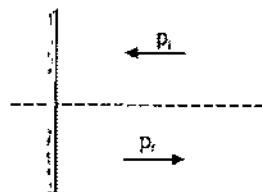
$$\Rightarrow \phi = 1.03 \text{ eV}$$

15. [A]

$$F = \frac{I}{c} (\text{Effective Area})$$

$$\Rightarrow F = \frac{I}{c} (\pi R^2) = \frac{\pi R^2 I}{c}$$

16. [B]



$$\text{Let } p = \frac{E}{c}$$

$$\Rightarrow \Delta p = \frac{E}{c} - \left(-\frac{E}{c} \right) = \frac{2E}{c}$$

17. [C]

$$\lambda = \frac{h}{\sqrt{2meV}}$$

So, when V becomes $4V$, λ becomes $\frac{\lambda}{2}$

18. [D]

Energy of falling light

$$E = \frac{hc}{\lambda} = \frac{(4.14 \times 10^{-15})(3 \times 10^8)}{3 \times 10^{-7}}$$

$$\Rightarrow E = 4.14 \text{ eV}$$

So, electrons are emitted with a kinetic energy $= 4.14 - 1 = 3.14 \text{ eV}$

$$\Rightarrow \frac{1}{2} mv^2 = 3.14 \times 1.6 \times 10^{-19}$$

$$\Rightarrow v = \sqrt{\frac{2 \times 3.14 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}}$$

$$\Rightarrow v = 1.2 \times 10^6 \text{ ms}^{-1}$$

19. [C]

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow \lambda = \frac{h}{m \sqrt{\frac{2qV}{m}}}$$

$$\left\{ \because \frac{1}{2} mv^2 = eV \right\}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mqV}}$$

$$\Rightarrow \frac{\lambda_p}{\lambda_a} = \sqrt{\frac{m_a q_a}{m_p q_p}} = 2\sqrt{2}$$

$$\left\{ \because m_a = 4m_p, q_a = 2q_p \right\}$$

20. [B]

$$R \propto \frac{1}{\lambda} \text{ and } \lambda = \frac{h}{\sqrt{2mqV}} \propto \frac{1}{\sqrt{V}}$$

$$\Rightarrow R \propto \sqrt{V}$$

$$\Rightarrow \frac{R_2}{R_1} = \sqrt{\frac{80 \text{ kV}}{20 \text{ kV}}} = 2$$

$$\Rightarrow R_2 = 2R_1 = 2R$$

21. [A]

$$A = \frac{N_1 A_1 + N_2 A_2}{N_1 + N_2}$$

$$\Rightarrow 10.81 = \frac{N_1(10) + N_2(11)}{N_1 + N_2}$$

Solving, we get

$$\frac{N_1}{N_2} = \frac{0.19}{0.81} = \frac{19}{81}$$

23. [B]

$$\text{For a photon } \lambda = \frac{h}{p} = \frac{hc}{E}$$

For a particle of mass m moving with a velocity v de Broglie relationship is given by

$$\lambda = \frac{h}{mv}$$

24. [C]

$$E = n \left(\frac{hc}{\lambda} \right)$$

$$\text{Also } \frac{1}{2} \epsilon_0 E_0^2 = \frac{\text{Energy}}{\text{Volume}}$$

$$\Rightarrow \text{Energy} = \left(\frac{1}{2} \epsilon_0 E_0^2 \right) (\text{Volume of cavity})$$

$$\Rightarrow \text{Energy} = \left(\frac{1}{2} \epsilon_0 E_0^2 \right) (\text{Area} (c \Delta t))$$

$$\Rightarrow n \frac{hc}{\lambda} = \frac{1}{2} \epsilon_0 E_0^2 (A c \Delta t)$$

$$\frac{n}{A \Delta t} = \frac{(\text{Number of photons striking the desk})}{(\text{Area})(\text{time})}$$

$$\Rightarrow \frac{n}{A \Delta t} = N = \frac{\lambda \epsilon_0 E_0^2}{2 h}$$

26. [B]

$$Kx_0 = \frac{1}{C} (\pi R^2)$$

$$\Rightarrow x_0 = \frac{1}{KC} (\pi R^2)$$

30. [C]

According to Einstein's Photoelectric Equation

$$E_k = h\nu_0 - \phi$$

$$\Rightarrow 0 = h\nu_0 - \phi$$

$$\Rightarrow \phi = h\nu_0 = \frac{hc}{\lambda_0}$$

$$\Rightarrow 4 \times 1.6 \times 10^{-19} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{\lambda_0}$$

$$\Rightarrow \lambda_0 = 3100 \text{ \AA}$$

34. [C]

Stopping potential of 1.36 V implies $E_k = 1.36 \text{ eV}$.

Since $h\nu = \phi_0 + E_k$

$$\Rightarrow \frac{(4.14 \times 10^{-15})(3 \times 10^8)}{5000 \times 10^{-10}} = \phi_0 + 1.36$$

$$\Rightarrow 2.46 = \phi_0 + 1.36$$

$$\Rightarrow \phi_0 = 1.1 \text{ eV}$$

35. [A]

Stopping potential is independent of intensity and depends upon frequency.

36. [C]

Saturation current $\propto$ (Intensity)

$$\Rightarrow \text{Saturation current} = 4(0.4 \mu\text{A})$$

$$\Rightarrow \text{Saturation current} = 1.6 \mu\text{A}$$

37. [D]

$$\frac{1}{2}mv^2 = \frac{hc}{\lambda} - W \quad \dots(1)$$

$$\frac{1}{2}mv'^2 = \frac{hc}{3\lambda/4} - W \quad \dots(2)$$

$$\text{Dividing, } \left(\frac{v'}{v} \right)^2 = \frac{\frac{4hc}{3\lambda} - W}{\frac{hc}{\lambda} - W}$$

$$v' > v \sqrt{\frac{4}{3}}$$

38. [B]

$$\frac{1}{2}mv^2 = k_B T$$

$$v = \sqrt{\frac{2k_B T}{m}}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mk_B T}}$$

39. [B]

$$\frac{1}{2}m_p v^2 = eV$$

$$\Rightarrow v = \sqrt{\frac{2eV}{m_p}}$$

Since, $\lambda = \frac{h}{m_p v}$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2m_p eV}}$$

$$\Rightarrow \lambda = \frac{0.287}{\sqrt{V}} \text{ \AA} \left(\begin{array}{l} \text{is de-Broglie wavelength} \\ \text{for a proton accelerated} \\ \text{through a potential } V \end{array} \right)$$

$$\Rightarrow \lambda = \frac{12.27}{\sqrt{V}} \text{ \AA} \left(\begin{array}{l} \text{is de-Broglie wavelength} \\ \text{for an electron accelerated} \\ \text{through a potential } V \end{array} \right)$$

48. [B]

$$P = \frac{nhc}{\lambda t}$$

$$\Rightarrow 1.7 \times 10^{-18} = \frac{n \times 6.6 \times 10^{-34} \times 3 \times 10^8}{6000 \times 10^{-10} \times 1}$$

$$\Rightarrow n = 5.15 \approx 5$$

49. [D]

$$\frac{(e/m_p)}{(2e/4m_p)} = 2$$

50. [B]

According to Planck's Quantisation Law.

$$E = nh\nu = n \left(\frac{hc}{\lambda} \right)$$

$$\Rightarrow \frac{E}{t} = \left(\frac{n}{t} \right) \left(\frac{hc}{\lambda} \right)$$

$$\Rightarrow 10^{-7} = \left(\frac{n}{t} \right) \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{5000 \times 10^{-10}}$$

$$\Rightarrow \frac{n}{t} = 2.5 \times 10^{11}$$

52. [D]

$$\frac{hc}{\lambda} = \phi_0 + 2.5$$

$$\frac{hc}{(\lambda/2)} = \phi_0 + E'_k$$

$$E'_k - 2.5 = \frac{hc}{\lambda}$$

$$E'_k = 2.5 + 4 \cdot 14$$

$$E'_k = 6.64 \text{ eV}$$

So, stopping potential is 6.64 V > 5 V

53. [B]

$$\lambda = \frac{h}{mv}$$

If E is kinetic energy of electron, then

$$E = \frac{1}{2} mv^2$$

$$\Rightarrow v = \sqrt{\frac{2E}{m}}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mE}}$$

54. [C]

Energy of falling photon < Threshold energy. So, no photoelectric effect takes place.

55. [D]

$$\text{Since } \lambda = \frac{h}{p}$$

For identical p's, λ are identical

56. [C]

$$h\nu = E_k + \phi$$

$$\Rightarrow h\nu - \phi = \frac{1}{2} mv^2$$

$$\Rightarrow v = \sqrt{\frac{2(h\nu - \phi)}{m}}$$

$$\Rightarrow v = \sqrt{\frac{2(hc - \lambda\phi)}{m\lambda}}$$

58. [B]

Stopping potential is 4 V. So, maximum K.E. is 4 eV.

59. [C]

Photoelectric current $\propto$ Intensity

$$\Rightarrow i \propto I$$

$$\Rightarrow i = kI$$

60. [B]

$$\frac{1}{2} mv^2 = eV$$

$$\Rightarrow v = \sqrt{\frac{2eV}{m}}$$

61. [A]

Since $\Delta E \Delta t \geq h$

$$\Rightarrow \Delta E (10^{-8}) = 6.6 \times 10^{-34}$$

$$\Rightarrow \Delta E = 6.6 \times 10^{-26} \text{ J}$$

63. [C]

$$I = \frac{N h \nu}{A \Delta t} = \frac{N \left(\frac{hc}{\lambda} \right)}{\frac{1 \text{ mm}}{c} \times 1 \text{ mm}^2}$$

$$\Rightarrow N = \frac{100 \times 10^{-9} \times 2640 \times 10^{-10}}{(3 \times 10^8)^2 \times 6.63 \times 10^{-34}}$$

$$\Rightarrow N = 442 \text{ photons mm}^{-3}$$

64. [A]

$$F = \frac{I}{c} (\text{Effective Area})$$

$$\Rightarrow F = \frac{I}{c} (\pi R^2) = \frac{\pi R^2 I}{c}$$

65. [D]

Speed of first electron may increase or decrease, depending on the direction of electric field. However in the case of electron entering the magnetic field speed remains constant. Since, from de-Broglie relation

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow \lambda_1 > \lambda_2 \text{ OR } \lambda_2 > \lambda_1$$

are both the possibilities.

66. [C]

$$\frac{h}{\lambda} = m_e v$$

$$\Rightarrow \lambda = 3.64 \text{ nm}$$

67. [C]

$$K_1 = \frac{hc}{\lambda_1} - W \quad \dots(1)$$

$$K_2 = \frac{hc}{\lambda_2} - W \quad \dots(2)$$

Since, $\lambda_1 = 2\lambda_2$, so (1) becomes

$$K_1 = \frac{hc}{2\lambda_2} - W = \frac{1}{2} \left(\frac{hc}{\lambda_2} \right) - W$$

$$\Rightarrow K_1 = \frac{1}{2} (K_2 + W) - W$$

$$\Rightarrow K_1 = \frac{K_2}{2} - \frac{W}{2}$$

$$\Rightarrow K_1 < \frac{K_2}{2}$$

68. [C]

Total energy radiated per minute from sun is

$$E_{\text{radiated}} = \sigma (4\pi R_{\text{sun}}^2)$$

Energy radiated annually is given by

$$E_{\text{total}} = 24 \times 60 \times 365 \times E_{\text{radiated}}$$

$$\Rightarrow \text{Annual loss of mass} = \Delta m = \frac{E_{\text{total}}}{c^2} = 1.38 \times 10^{17} \text{ kg}$$

69. [A]

Since $v = \alpha c$

{ where $\alpha \ll 1$ }

the electron is moving at non-relativistic speed.

By Law of Conservation of Momentum

$$\frac{h}{\lambda} + \frac{h}{\lambda'} = mv$$

$$\Rightarrow \frac{hc}{\lambda} + \frac{hc}{\lambda'} = mc v = \frac{mv^2}{\alpha} \quad \dots(1)$$

By Law of Conservation of Energy

$$\frac{1}{2} mv^2 = \frac{hc}{\lambda} - \frac{hc}{\lambda'} \quad \dots(2)$$

Adding (1) & (2), we get

$$2 \frac{hc}{\lambda} = \frac{1}{2} mv^2 + \frac{1}{\alpha} (mv^2)$$

$$\Rightarrow \frac{hc}{\lambda} = E_f = \frac{mv^2}{4} + \frac{mv^2}{2\alpha}$$

Since, $\alpha \ll 1$

$$\Rightarrow \frac{1}{\alpha} \gg 1$$

$$\Rightarrow \frac{mv^2}{4} \ll \frac{mv^2}{2\alpha}$$

$$\Rightarrow E_f = \frac{1}{\alpha} \left(\frac{mv^2}{2} \right)$$

$$\Rightarrow \frac{1}{2}mv^2 = \alpha E_f$$

70. [C]

Photons are exchange particles for electromagnetic interactions.
Gravitons are exchange particles for gravitational interactions.
Mesons are exchange particles for nuclear interactions.
Whereas protons are not any sort of exchange particles.

71. [B]

Relativistic relative velocity of approach is given by

$$\bar{v}_r = \frac{\bar{v}_1 - \bar{v}_2}{1 - \frac{\bar{v}_1 \cdot \bar{v}_2}{c^2}}$$

$$\Rightarrow v_r = \frac{c - (-c)}{1 - \frac{(c)(-c)}{c^2}}$$

$$\Rightarrow v_r = \frac{2c}{1 + \frac{c^2}{c^2}}$$

$$\Rightarrow v_r = c$$

Never expect the relative velocity to grow beyond the speed of light.

72. [C]

By Law of Conservation of Momentum, we have

$$0 = mv_1 + 2m(-v_2)$$

$$\Rightarrow mv_1 = 2mv_2$$

...(1)

Now, according to de-Broglie relation, we have

$$\lambda = \frac{h}{p}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{p_2}{p_1} = 1$$

73. [C]

The vector is perpendicular to initial velocity vector at $t = \frac{u}{g \sin \theta}$

and at this instant its speed is

$$v = u \cot \theta$$

$$\text{Since, } \lambda = \frac{h}{mv}$$

$$\Rightarrow \lambda = \frac{h}{mu \cot \theta}$$

$$\Rightarrow \lambda = \left(\frac{h}{mu} \right) \tan \theta$$

74. [A]

$$F_{ex} = \frac{P}{c} = \frac{IA}{c}$$

where $I = 1.4 \times 10^3 \text{ W m}^{-2}$

$$\Rightarrow A = 4\pi r^2 = 4 \times 3.14 \times (2)^2 \text{ m}^2$$

$$\Rightarrow F_{ex} = 2.35 \times 10^{-4} \text{ N}$$

75. [C]

$$KE = hv + \phi$$

...(1)

$$2KE = hv' + \phi$$

...(2)

$$\Rightarrow 2(hv + \phi) = hv' + \phi$$

$$\Rightarrow v' = 2v + \frac{\phi}{h}$$

$$\Rightarrow v' > 2v$$

76. [C]

$$\lambda = \frac{h}{mv} \text{ where,}$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

As $v \rightarrow c$

$m \rightarrow \infty$

$$\Rightarrow \lambda \rightarrow 0$$

77. [C]

$$E_k = hv - \phi_0$$

$$\Rightarrow (E_k)_1 = 1 - 0.5 = 0.5 \text{ eV}$$

$$\text{Similarly } (E_k)_2 = 2.5 - 0.5 = 2 \text{ eV}$$

$$\Rightarrow \frac{(E_k)_1}{(E_k)_2} = \frac{1}{4}$$

$$\Rightarrow \frac{v_1^2}{v_2^2} = \frac{1}{4}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{1}{2}$$

78. [D]

$$\text{Force} = \frac{2IA}{C}$$

$$\Rightarrow Fx = \left(\frac{1}{2} Kx^2 \right) \times 3$$

$$\Rightarrow \left(\frac{2IA}{C} \right) x = \frac{3}{2} Kx^2$$

$$\Rightarrow x = \frac{4}{3} \left(\frac{IA}{KC} \right)$$

79. [A]

$$KE = E_k = hv - hv_0 = hv - \phi_0$$

$$\Rightarrow E_k = \frac{hc}{\lambda} - \phi_0$$

$$\Rightarrow E_k = \frac{(4.14 \times 10^{-15})(3 \times 10^8)}{2 \times 10^{-7}} - 5.01$$

$$\Rightarrow E_k = 6.21 - 5.01$$

$$\Rightarrow E_k = 1.2 \text{ eV}$$

So, stopping potential required is 1.2 V

80. [D]
For two particles to have same de-Broglie wavelength

$$\sqrt{mqV} = \text{constant}$$

$$\Rightarrow m_p q_p V = m_a q_a V'$$

$$\Rightarrow V' = \frac{m_p q_p}{m_a q_a} V$$

$$\Rightarrow V' = \frac{V}{8}$$

81. [C]
Energy of each photon,

$$E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{300 \times 10^{-9}}$$

$$E = 6.6 \times 10^{-19} \text{ J}$$

Power of source

$$P = IA = 1.0 \times 1.0 \times 10^{-4} = 10^{-4} \text{ watt}$$

So, number of photons per sec is

$$\frac{N}{t} = \frac{P}{E} = \frac{10^{-4}}{6.6 \times 10^{-19}}$$

Number of electrons emitted is

$$N' = \frac{1}{100} \times \frac{10^{-4}}{6.6 \times 10^{-19}}$$

$$\Rightarrow N' = 1.51 \times 10^{12} \text{ per second}$$

82. [A]
Transverse deflection is

$$y = \frac{1}{2} at^2 = \frac{1}{2} \left(\frac{qE}{m} \right) \left(\frac{x}{v} \right)^2 = \frac{1}{2} \frac{qEx^2}{mv^2}$$

$$\Rightarrow y = \frac{\frac{1}{2} qEx^2}{2 \cdot \left(\frac{1}{2} mv^2 \right)} = \frac{1}{4} \frac{qEx^2}{E_k}$$

For same electric field E , kinetic energy E_k and length x , $y \propto q$, q is smaller for proton, so y is smaller for proton. So proton's trajectory will be less curved than α -particle's trajectory.

84. [D]
 $E_k = h\nu - h\nu_0$
 $E_k = y$, $v = x$, $h\nu_0 = \text{constant}(k)$
 $\Rightarrow y = hx - k$ (Equation of a straight line)

86. [A]
 $\lambda = \frac{h}{mv}$
 $\Rightarrow \lambda = \frac{6.626 \times 10^{-34}}{(10^{-31})(10^5)}$
 $\Rightarrow \lambda = 6.63 \times 10^{-8} \text{ m}$

88. [D]
 $E_k = h\nu - h\nu_0$
is the equation of a straight line with slope h .

89. [A]
 $E_k = h\nu - h\nu_0$

90. [C]
The time rate of change of momentum equals force.

$$\Rightarrow F = \frac{\Delta p}{\Delta t}$$

Force per unit area is pressure

$$\Rightarrow \mathcal{P} = \frac{1}{A} \frac{\Delta p}{\Delta t}$$

But for a photon $E = pc$, where

p = momentum of photon

E = energy of photon

$$\Delta E = c \Delta p$$

$$\Rightarrow \mathcal{P} = \frac{1}{A} \frac{\Delta E}{\Delta t} = \frac{1}{c} \left(\frac{\Delta E}{A \Delta t} \right) = \frac{1}{c} (I)$$

Because irradiance I is defined as energy per unit area per unit time.

91. [B]

$$\lambda_e = \lambda_p$$

$$\Rightarrow \frac{h}{mv} = \frac{hc}{E}$$

$$\Rightarrow \frac{p}{E} = \frac{1}{c}$$

92. [B]

$$\text{Stopping Potential } V_s = \frac{K_{\max}}{e}$$

$$\Rightarrow V_s = 4 \text{ V}$$

93. [A]

$$E = \frac{hc}{\lambda}$$

Number of photons emitted is

$$\frac{Pt}{\left(\frac{hc}{\lambda} \right)} = n_0$$

$$\Rightarrow n_0 = \frac{P\lambda t}{hc}$$

Since the radiation is spherically symmetric, so total number of photons entering the sensor is n_0 times the ratio of aperture area to the area of a sphere of radius ℓ .

$$\Rightarrow N = n_0 \frac{\pi (2d)^2}{4\pi \ell^2} = \frac{P\lambda t}{hc} \frac{d^2}{\ell^2}$$

94. [C]

$$\frac{hc}{\lambda} = \phi_0 + 3V_0 \quad \dots(1)$$

$$\frac{hc}{2\lambda} = \phi_0 + V_0 \quad \dots(2)$$

Subtracting

$$2V_0 = \frac{hc}{2\lambda}$$

$$\Rightarrow V_0 = \frac{hc}{4\lambda}$$

$$\Rightarrow \phi_0 = \frac{hc}{4\lambda}$$

$$\Rightarrow \frac{hc}{\lambda_0} = \frac{hc}{4\lambda}$$

$$\Rightarrow \lambda_0 = 4\lambda$$

95. [D]

$$p = \frac{h\nu}{c} \text{ (for a photon)}$$

$$\Rightarrow 3.3 \times 10^{-29} = \frac{6.6 \times 10^{-34} \nu}{3 \times 10^8}$$

$$\Rightarrow \nu = 1.5 \times 10^{13} \text{ Hz}$$

96. [D]

Decreasing the λ of incident photon means energy of incident light is increased, so E_k increases and hence stopping potential increases.

97. [B]

$$E_k = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$E_k = \frac{4.14 \times 10^{-15} \times 3 \times 10^8}{10^{-7}} \left(\frac{1}{5} - \frac{1}{6} \right)$$

$$\Rightarrow E_k = 0.414 \text{ eV}$$

Wherever necessary we take $h = 6.626 \times 10^{-34} \text{ Js}$

$$\Rightarrow h = 4.14 \times 10^{-15} \text{ eVs}$$

98. [A]

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow \frac{d\lambda}{dt} = \frac{h}{m} \frac{d}{dt} (v^{-1}) = \frac{h}{m} (-1) v^{-2} \frac{dv}{dt}$$

$$\Rightarrow \frac{d\lambda}{dt} = -\frac{h}{mv^2} \cdot a$$

$$\text{Also } v = u + at$$

$$\Rightarrow v = at \quad \{\because u = 0\}$$

$$\Rightarrow \frac{d\lambda}{dt} = -\frac{h}{ma^2 t^2} \cdot a = -\frac{h}{mat^2}$$

$$\text{Also } ma = eE$$

$$\Rightarrow \frac{d\lambda}{dt} = -\frac{h}{eEt^2}$$

99. [A]

$$\lambda = \frac{h}{mv}$$

Since electron is the lightest particle of these four. So λ_e is maximum.

100. [D]

$$I \propto \frac{1}{d^2}$$

when d becomes $\frac{d}{4}$, I becomes 16I

102. [B]

$$h\nu = h\nu_0 + eV_0$$

$$\Rightarrow eV_0 = h\nu - h\nu_0$$

$$\Rightarrow V_0 = \frac{h}{e} \nu - \frac{h}{e} \nu_0$$

which is again equation of a straight line with slope $\left(\frac{h}{e} \right)$.

104. [A]

5% of 100 W is 5 Js^{-1}

$$\Rightarrow 5 = \frac{n \left(\frac{hc}{\lambda} \right)}{t}$$

$$\Rightarrow \frac{n}{t} = \frac{5\lambda}{hc}$$

$$\Rightarrow \frac{n}{t} = \frac{5(5.6 \times 10^{-7})}{6.626 \times 10^{-34} \times 3 \times 10^8}$$

$$\Rightarrow \frac{n}{t} = 1.4 \times 10^{19}$$

105. [B]

Since $\Delta x \Delta p = h$

$$\Rightarrow (2 \times 10^{-14}) \Delta p = 6.6 \times 10^{-34}$$

$$\Rightarrow \Delta p = 3.3 \times 10^{-20} \text{ kgms}^{-1}$$

106. [B]

$$\nu_0 = 5 \times 10^{14} \text{ Hz}$$

Since, $c = \nu_0 \lambda$

$$\Rightarrow \lambda = \frac{3 \times 10^8}{5 \times 10^{14}}$$

$$\Rightarrow \lambda = 6000 \text{ \AA}$$

107. [A]

Kinetic energy of recoil is

$$E_k = h\nu - h\nu'$$

$$\Rightarrow E_k = 12.375 - 9.375$$

$$\Rightarrow E_k = 3 \text{ eV}$$

108. [A]

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Putting values of h, m, e , we get

$$\Rightarrow \lambda = \sqrt{\frac{150}{V}} \text{ \AA}$$

$$\Rightarrow \lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

109. [D]

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p q_p}{m_e q_e}} = \sqrt{\frac{m_p}{m_e}}$$

$$\{\because q_e = q_p\}$$

110. [A]

$$\Delta x = \frac{2}{100} \text{ m}$$

Since

$$\Delta x \Delta p = h$$

$$\Rightarrow \Delta p = \frac{h}{\Delta x} = \frac{6.6 \times 10^{-34}}{(2/100)}$$

$$\Rightarrow \Delta p = 3.3 \times 10^{-32} \text{ kgms}^{-1}$$

111. [A]

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\Rightarrow mE = \text{constant}$$

$$m_e < m_p < m_\alpha$$

$$\Rightarrow E_1 > E_3 > E_2$$

112. [B]

$$E = \frac{hc}{\lambda} - \phi \quad \dots(1)$$

$$2E = \frac{hc}{\lambda'} - \phi \quad \dots(2)$$

$$\Rightarrow 2E - E = hc \left[\frac{1}{\lambda'} - \frac{1}{\lambda} \right] = E$$

$$\Rightarrow E + \frac{hc}{\lambda} = \frac{hc}{\lambda'}$$

$$\Rightarrow \frac{E\lambda + hc}{\lambda} = \frac{hc}{\lambda'}$$

$$\Rightarrow \lambda' = \left(\frac{hc\lambda}{E\lambda + hc} \right)$$

113. [A]

In order for scattering to occur, the wavelength of the waves must be of the same order of magnitude or smaller than the size of the object being observed. Hence the largest possible wavelength we can use in the present problem is $\lambda_{\text{max}} = 2.5 \text{ \AA}$. Hence minimum energy is

$$E_{\text{min}} = \frac{hc}{\lambda_{\text{max}}}$$

$$\Rightarrow E_{\text{min}} = \frac{12.40 \times 10^3}{2.5 \text{ \AA}} \text{ eV \AA}$$

$$\Rightarrow E_{\text{min}} = 4.96 \times 10^3 \text{ eV}$$

$$\Rightarrow E_{\text{min}} = 5 \text{ keV}$$

114. [A]

$$mc^2 = m_0 L$$

$$(1)(3 \times 10^8)^2 = m_0 (80 \times 4200)$$

$$\Rightarrow m_0 = 2.67 \times 10^{11} \text{ kg}$$

115. [D]

Density of star of mass M_0 , radius R_0 is

$$\rho = \frac{M_0}{\frac{4}{3}\pi R_0^3}$$

On contracting the new mass becomes

$$M = \frac{4}{3}\pi R^3 \rho = M_0 \left(\frac{R}{R_0} \right)^3$$

Loss in mass due to contraction = $-\Delta M$

$$\Rightarrow \text{Loss} = M_0 - M$$

$$\Rightarrow \text{Energy radiated} = c^2 \Delta M$$

116. [D]

$$F = \frac{\Delta p}{\Delta t} = \frac{n}{\Delta t} \left[\frac{h}{\lambda} - \left(-\frac{h}{\lambda} \right) \right] = N \left(\frac{2h}{\lambda} \right)$$

{ $\because$ Photons rebound with the same initial value of momentum }

$\Rightarrow N$ = Total number of photons striking the totally reflecting

$$\text{screen} = \frac{F}{\left(\frac{2h}{\lambda} \right)} = \frac{F\lambda}{2h}$$

$$\Rightarrow N = 5 \times 10^{26}$$

117. [A]

$$\sqrt{2mE} = \frac{h}{\lambda}$$

$$\Rightarrow E = \frac{h^2}{2m\lambda^2} = \frac{(6.6 \times 10^{-34})^2}{2 \times 20 \times 1.66 \times 10^{-27} \times (6.6 \times 10^{-10})^2}$$

$$\Rightarrow E = \frac{10^{-48+27}}{40 \times 1.66} = \frac{1}{4 \times 1.66} \times 10^{-22} \text{ J}$$

$$\Rightarrow E = 1.5 \times 10^{-23} \text{ J}$$

118. [C]

$$\frac{2nh}{\lambda} = 1$$

$$\Rightarrow n = \frac{\lambda}{2h}$$

119. [B]

$$\text{Since, } \frac{hc}{\lambda} = \frac{1}{2}mv^2 + \phi$$

$$\Rightarrow \phi = \frac{hc}{\lambda} - \frac{1}{2}mv^2$$

$$\Rightarrow \phi = \frac{1240}{400} - 1.68 = 1.41 \text{ eV}$$

120. [A]

Both are independent of each other.

121. [B]

For a photon

$$E = pc = hv$$

$$\Rightarrow p = \frac{hv}{c}$$

124. [A]

$$S_a = \frac{e}{m_e}$$

$$S_p = \frac{e}{m_p}$$

$$S_a = \frac{2e}{4m_p} = \frac{e}{2m_p} = \frac{1}{2} S_p$$

S_a is maximum, then comes S_p and then S_α

125. [D]

$$I \propto \frac{1}{d^2}$$

On doubling the distance the intensity becomes one fourth i.e. only one fourth of photons now strike the target in comparison to the previous number. Since photoelectric effect is a one photon-one electron phenomena, so only one-fourth photoelectrons are emitted out of the target hence reducing the current to one fourth the previous value.

126. [D]

Let threshold frequency be ν_0 .

$$\nu = 1.5 \nu_0$$

When the new frequency is halved

$$\nu' = 0.75\nu_0 < \nu_0$$

So, no photoelectric effect takes place.

127. [D]

$$\frac{\lambda_p}{\lambda_a} = \frac{m_a \nu_a}{m_p \nu_p}$$

$$\Rightarrow \frac{1}{2} = 4 \frac{\nu_p}{\nu_a}$$

$$\Rightarrow \frac{\nu_p}{\nu_a} = 8$$

128. [C]

$$y = \frac{1}{2} at^2 = \frac{1}{2} \frac{eE}{m} \frac{\ell^2}{\nu^2}$$

For same (nearly) circular path y is same

$$\Rightarrow \frac{E}{\nu^2} = \text{constant}$$

Here $E = x$

$$\Rightarrow \frac{x}{\nu^2} = \frac{x'}{(2\nu)^2}$$

$$\Rightarrow x' = 4x$$

130. [B]

$$\text{Given that } \frac{h}{p} = \frac{hc}{E} = \frac{h}{mv}$$

$$\Rightarrow E = mvc$$

$$\Rightarrow \frac{E_0}{E_p} = \frac{\frac{1}{2}mv^2}{mvc} = \frac{v}{2c}$$

131. [B]

$$\lambda_{\max} = 7500 \text{ \AA}$$

$$\text{So, } E = \frac{hc}{\lambda}$$

$$\Rightarrow E = \frac{(4.14 \times 10^{-15} \text{ eVs})(3 \times 10^8 \text{ ms}^{-1})}{7500 \times 10^{-10} \text{ m}}$$

$$\Rightarrow E = 1.6 \text{ eV}$$

133. [B]

$$eV_s = KE_{\max} = h\nu - W$$

$$V_s = \frac{hc}{e\lambda} - \frac{W}{e}$$

[$\because W = \text{work function}$]

134. [C]

$$\lambda (\text{in \AA}) = \frac{12400}{E (\text{in eV})}$$

$$\Rightarrow \lambda = \frac{12400}{4} = 3100 \text{ \AA} = 310 \text{ nm}$$

135. [A]

$$E = nh\nu$$

$$\Rightarrow \frac{E}{t} = \left(\frac{n}{t}\right)h\nu$$

$$\Rightarrow 10000 = \frac{n}{t} (6.6 \times 10^{-34}) (880 \times 1000)$$

$$\Rightarrow \frac{n}{t} = 1.71 \times 10^{31}$$

136. [C]

Maximum microwave frequency = $3 \times 10^{11} \text{ Hz}$

$$\text{Since, } p = \frac{E}{c} = \frac{h\nu}{c}$$

$$\Rightarrow p_{\max} = \frac{(6.626 \times 10^{-34})(3 \times 10^{11})}{(3 \times 10^8)}$$

$$\Rightarrow p_{\max} = 6.626 \times 10^{-31} \text{ kgms}^{-1}$$

137. [C]

$$\lambda = \frac{h}{mv} = \frac{6.62 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^7}$$

$$\Rightarrow \lambda = 0.24 \times 10^{-10} \text{ m}$$

138. [C]

$$(m - m_0)c^2 = 2m_0c^2$$

$$\Rightarrow m = 3m_0$$

140. [C]

Stopping potential = 1 V

$$\Rightarrow E_k = 1 \text{ eV}$$

$$\Rightarrow \frac{1}{2}mv^2 = 1.6 \times 10^{-19} \text{ J}$$

Solutions

Multiple Correct Choice Type Questions

1. [A, B, C]

$$h\nu = K.E.(T) + \text{Work function } (W)$$

$$\Rightarrow h\nu = T + W$$

$$\Rightarrow 4.25 \text{ eV} = T_A + W_A \text{ (for Metal A)}$$

$$\Rightarrow 4.70 \text{ eV} = T_B + W_B \text{ (for Metal B)}$$

$$\text{Since } T_B = (T_A - 1.5) \text{ eV}$$

$$\text{Also } \lambda = \frac{h}{p}$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2mT}}$$

$$\left\{ \because \frac{p^2}{2m} = T = K.E. \right\}$$

$$\Rightarrow \frac{\lambda_A}{\lambda_B} = \sqrt{\frac{T_B}{T_A}}$$

$$\text{Since } \lambda_A = \frac{1}{2} \lambda_B$$

$$\Rightarrow T_A = 4T_B$$

$$\Rightarrow T_B = T_A - 1.50 \text{ gives}$$

$$\Rightarrow T_B = 4T_B - 1.5$$

$$\Rightarrow T_B = 0.5 \text{ eV}$$

$$\Rightarrow T_A = 2 \text{ eV}$$

$$\Rightarrow W_A = 2.25 \text{ eV}$$

$$\Rightarrow W_B = 4.20 \text{ eV}$$

2. B, D

$$\text{Since } \mathcal{P} = \frac{I}{A} = 10^4 \text{ Nm}^{-2}$$

$$\Rightarrow \mathcal{P} = \frac{F}{A} = \frac{1}{A} \cdot \frac{\Delta p}{\Delta t}$$

$$\Rightarrow \Delta p = \mathcal{P} A \Delta t = 10^{-5} \text{ kgms}^{-1}$$

3. [A, B, C]

The total quantity of charge carried by one pulse of current is

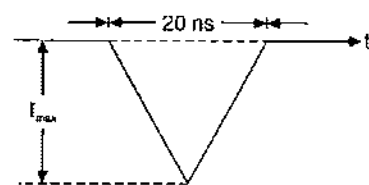
$$Q = \int I dt$$

which is the area of the triangle in figure. Thus

$$Q = \frac{1}{2} (20 \times 10^{-9}) (80 \times 10^{-6}) = 8 \times 10^{-13} \text{ C}$$

and the number of electrons carried by one pulse is

$$n = \frac{Q}{e}$$



$$\Rightarrow n = \frac{8 \times 10^{-13}}{1.6 \times 10^{-19}} = 5 \times 10^6$$

Then the number of photoelectrons emitted per light pulse is

$$n' = \frac{n}{10^9} = 5$$

and hence the number of photons in one light pulse is

$$N = \frac{n'}{0.1} = 50$$

8. [A, B, C, D]

According to Heisenberg's Uncertainty Principle the product of

(a) uncertainty in position and uncertainty in momentum cannot be greater than $\frac{h}{4\pi}$.(b) uncertainty in energy and uncertainty in time cannot be greater than $\frac{h}{4\pi}$.(c) uncertainty in angular position and uncertainty in angular momentum cannot be greater than $\frac{h}{4\pi}$.(d) uncertainty in generalised coordinate and generalised momentum cannot be greater than $\frac{h}{4\pi}$.

10. [A, B]

Wavelength of UV radiation is less than $5200 \text{ \AA}$ where as wavelength for IR radiation is greater than $5200 \text{ \AA}$. Hence photoelectric effect will be shown by UV radiation irrespective of its intensity.

12. B, C

For a photon
 $pc = h\nu$

$$\Rightarrow p = \frac{h\nu}{c}$$

$$\Rightarrow p = 8.8 \times 10^{-28} \text{ kgms}^{-2}$$

$$\Rightarrow p = 1.65 \times 10^{-6} \text{ MeV/c}$$

14. [A, B, C]

(A) $f = \frac{c}{\lambda} = \frac{3 \times 10^8}{600 \times 10^{-9}} = 5 \times 10^{14}$

(B) $N = \frac{P}{hf}$

(C) $KE_{\text{max}} = \frac{1240}{600} - 1.07 = 1$

(D) KE_{max} depends upon frequency of incident photons and not distance of source.

17. [B, D]

$\lambda_{\text{red}} > \lambda_{\text{violet}}$

VIBGYOR pattern shows that VIBG all have λ less than that of yellow colour and hence can initiate photoelectric effect irrespective of intensity.

18. [A, B]

$K_{\text{max}} = 4 \times 10^{-19} \text{ J} = \frac{4 \times 10^{-19}}{1.6 \times 10^{-19}} = 2.5 \text{ eV}$

$\Rightarrow$ Stopping potential = 2.5 eV

Since, $K_{\text{max}} = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$

$\Rightarrow 2.5 = \frac{1240}{300} - \frac{1240}{\lambda_0}$

$\Rightarrow \lambda_0 = 759 \text{ nm}$

19. [C, D]

Infrared light has wavelength greater than 5200 Å and UV has $\lambda < 5200 \text{ Å}$. So UV lamp will be able to get the photoelectrons emitted by a surface irrespective of intensity.

20. [A, C]

$K_{\text{max}} = 1.5 \text{ eV}$

$\lambda = \frac{h}{\sqrt{2mK}}$

$\Rightarrow \lambda = 1 \text{ nm}$

E of incident photon = $\frac{1241}{496} = 2.5 \text{ eV}$

$\Rightarrow \phi = 2.5 - 1.5 = 1 \text{ eV}$

Since, $qvB = \frac{mv^2}{r}$

$\Rightarrow B = \frac{mv}{qr} = \sqrt{\frac{2mK_{\text{max}}}{qr}}$

$\Rightarrow B = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times e \times 1.5}}{e \times 1}$

$\Rightarrow B = \sqrt{\frac{3 \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19}}}$

$\Rightarrow B = 4.13 \times 10^{-6} \text{ T} \approx 4 \mu\text{T}$

X-rays are of order of 0.1 nm = 12.4 KeV

22. [B, D]

Energy of photon incident

$h\nu = \frac{12400}{4000} \text{ eV}$

$\Rightarrow h\nu = 3.1 \text{ eV}$

Since, $h\nu < \text{work function of all metals}$

Hence no electron will come out

If $\lambda = 200 \text{ nm}$

then $h\nu = \frac{hc}{\lambda} = \frac{12400}{2000} = 6.2 \text{ eV}$

Since, 6.2 eV > work function of all metals, hence photoelectron will be emitted.

23. [B, D]

Cut off voltage is independent of intensity and hence remains the same. Since distance becomes 3 times, so I becomes $\frac{1}{9}$.

Hence photocurrent also decreases by this factor i.e. becomes $\frac{18}{9} = 2 \text{ mA}$

24. [C]

Work function is the intercept on K-axis i.e. 2 eV

25. [A]

$2 = 4.14 \times 10^{-15} \nu$

$\Rightarrow \nu = 4.8 \times 10^{14} \text{ Hz}$

26. [A, B]

Photoelectric effect and Compton effect are explained on particle nature of light i.e. light is considered to be made up of a stream of photons.

Solutions

Reasoning Based Questions

6. [D]

$$\lambda = \frac{h}{p}, \text{ Same for both}$$

8. [D]

$$\text{Kinetic energy; } E = \frac{1}{2}mv^2 = qV$$

$$\Rightarrow v \propto \sqrt{\frac{q}{m}} \text{ because } V \text{ is constant}$$

$$\Rightarrow v_p : v_d : v_a = \sqrt{\frac{q_p}{m_p}} : \sqrt{\frac{q_d}{m_d}} : \sqrt{\frac{q_a}{m_a}}$$

$$\Rightarrow v_p : v_d : v_a = \sqrt{\frac{e}{m}} : \sqrt{\frac{e}{2m}} : \sqrt{\frac{2e}{4m}}$$

$$\Rightarrow v_p : v_d : v_a = 1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{2}} = \sqrt{2} : 1 : 1$$

13. [A]

Energy of photoelectron emitted is different because after absorbing the photon electrons within metals collide with other atom before being ejected out of metal.

14. [A]

$$\frac{\lambda_e}{\lambda_{ph}} = \sqrt{\frac{E}{2m_0c^2}} = \sqrt{\frac{10^6 \text{ eV}}{1 \text{ MeV}}} = 1$$

15. [A]

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow 2m_0 = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow v = \frac{\sqrt{3}}{2}c$$

16. [D]

$$K_{\max} = hv = \phi$$

KE of emitted photoelectrons varies from zero to $K_{\max}$

17. [D]

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

Solutions

Linked Comprehension Type Questions

1. [C]

$$\text{Since, } K_{\max} = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$\Rightarrow \frac{1}{2}mv_{\max}^2 = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_0} \quad \dots(1)$$

$$\frac{1}{2}mv_{\max}^2 = \frac{hc}{\lambda_2} - \frac{hc}{\lambda_0} \quad \dots(2)$$

Dividing (2) by (1), we get

$$\left(\frac{v_{\max 2}}{v_{\max 1}}\right)^2 = \frac{\frac{hc}{\lambda_2} - \frac{hc}{\lambda_0}}{\frac{hc}{\lambda_1} - \frac{hc}{\lambda_0}}$$

$$\Rightarrow (2)^2 = \frac{1 - \frac{\lambda_0}{\lambda_2}}{1 - \frac{\lambda_0}{\lambda_1}}$$

$$\Rightarrow \frac{4}{3000} - \frac{4}{\lambda_0} = \frac{1}{1500} - \frac{1}{\lambda_0}$$

$$\Rightarrow \frac{4}{3000} - \frac{1 \times 2}{1500 \times 2} = \frac{3}{\lambda_0}$$

$$\Rightarrow \frac{2}{3000} = \frac{3}{\lambda_0}$$

$$\Rightarrow \lambda_0 = 4500 \text{ \AA}$$

2. [A]

$$I_s \propto P\lambda$$

$$\Rightarrow \frac{I_{s2}}{I_{s1}} = \frac{P_2 \lambda_2}{P_1 \lambda_1}$$

$$\Rightarrow \frac{I_{s2}}{20} = \frac{5 \times 1500}{1 \times 3000}$$

$$\Rightarrow I_{s2} = 50 \text{ } \mu\text{A}$$

3. [B]

For case I, we have

$$n = \frac{P\lambda}{hc}$$

$$\text{Since, } n_0 = \frac{I_s}{e}$$

Efficiency of photoelectron generation per incident photon is

$$\eta = \left(\frac{n_0}{n} \times 100\%\right) = \frac{I_s \times hc}{e \times P\lambda} \times 100$$

$$\Rightarrow \eta = \frac{20 \times 10^{-6} \times 6.6 \times 10^{-34} \times 3 \times 10^8 \times 100}{1.6 \times 10^{-19} \times 1 \times 10^{-3} \times 3000 \times 10^{-10}}$$

$$\Rightarrow \eta = \left(\frac{20}{160}\right)\left(\frac{66}{10}\right) = \frac{66}{8} = 8.25\%$$

4. [C]

$$\frac{mv^2}{r} = qvB$$

$$\Rightarrow \sqrt{2mE_k} = qBr$$

$$\Rightarrow E_k = \frac{(qBr)^2}{2m}$$

$$\Rightarrow E_k = 0.86 \text{ eV}$$

5. [B]

$$E_3 - E_2 = 13.6 \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$\Rightarrow \Delta E = 1.89 \text{ eV}$$

$$\text{Since, } \phi = E - K_{\max}$$

$$\Rightarrow \phi = 1.89 - 0.86 = 1.03 \text{ eV}$$

6. [A]

$$\lambda = \frac{hc}{\Delta E}$$

$$\Rightarrow \lambda = 6565 \text{ \AA}$$

7. [B]

$$\lambda \propto \frac{1}{\sqrt{mq}}$$

8. [A]

$$\lambda_0 = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

9. [C]

$$\text{Since, } \lambda = \frac{h}{p}$$

Further, $p_{\text{electron}} = p_{\alpha}$

$\Rightarrow$ Wavelength will be same

10. [B]

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(66.3 \times 10^{-3})(5)}$$

$$\Rightarrow \lambda = 0.02 \text{ m}$$

11. [D]

Fringe width β is given by

$$\beta = \frac{\lambda D}{d} = \frac{12 \times 0.02}{0.6}$$

$$\Rightarrow \beta = 0.4 \text{ m}$$

12. [D]

de-Broglie wavelength of electron is

$$\lambda = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^7}$$

$$\Rightarrow \lambda \approx 10^{-21} \text{ m}$$

$$\text{Fringe width } \beta = \frac{\lambda D}{d} \approx \left(\frac{12}{0.6} \right) \times 10^{-21} \approx 10^{-22} \text{ m}$$

Fringe width is so large that it is not possible to observe it.

13. [B]

$$\lambda = \frac{h}{\sqrt{2mE}} = \sqrt{\frac{150}{1.5}} = 10 \text{ Å} = 100 \text{ nm}$$

14. [A]

$$qvB = \frac{mv^2}{r}$$

$$\Rightarrow B = \frac{mv}{qr} = 4.1 \times 10^{-6} \text{ T}$$

15. [A]

$$\text{Photon energy} = \frac{hc}{\lambda} = \frac{12375}{4960} = 2.5 \text{ eV}$$

Since, $V_s = 1.5 \text{ V}$

So, work function is given by

$$\phi = 2.5 - 1.5 = 1 \text{ eV}$$

$$\Rightarrow \lambda_{\min} = \frac{12375}{1} = 12375 \text{ Å} = 1237 \text{ nm}$$

$$\Rightarrow \lambda_{\max} \approx 1250 \text{ nm}$$

16. [B]

$$p = \frac{I}{c} = \frac{3 \times 10^4}{3 \times 10^8} = 10^{-4} \text{ Pa}$$

17. [C]

$$\Delta p = P \Delta t = 10^{-4} \times 10^{-4} \times 10 = 10^{-7} \text{ kgms}^{-1}$$

18. [B]

$$\frac{Nh}{\lambda} = \Delta p$$

$$\Rightarrow N = \frac{10^{-7} \times 663 \times 10^{-9}}{6.63 \times 10^{-34}} = 10^{20}$$

19. [A]

Since, $K_{\max} = h\nu - W$

If $K_{\max}$ and W both are in eV, but $h\nu$ is in joule, then we have

$$K_{\max} = \frac{h\nu}{e} - W$$

So for both the frequency values, we have

$$0.5 = \frac{h(8 \times 10^{14})}{e} - W \quad \dots(1)$$

$$2 = \frac{h(12 \times 10^{14})}{e} - W \quad \dots(2)$$

Subtracting equation (1) from (2), we get

$$1.5 = \frac{h(4 \times 10^{14})}{e}$$

$$\Rightarrow h = \frac{1.5 \times 1.6 \times 10^{-19}}{4 \times 10^{14}}$$

$$\Rightarrow h = 6 \times 10^{-34} \text{ Js}$$

20. [C]

From equation (1), we get

$$0.5 = \frac{(6 \times 10^{-34})(8 \times 10^{14})}{1.6 \times 10^{-19}} - W$$

$$\Rightarrow 0.5 = 3 - W$$

$$\Rightarrow W = 3 - 0.5 = 2.5 \text{ eV}$$

21. [B]

$$\lambda = \frac{h}{\sqrt{2mK}} = \frac{12.27}{\sqrt{V}} \text{ Å}$$

$$\Rightarrow \lambda = \frac{12.27}{\sqrt{0.5}} \text{ Å} = 12.27\sqrt{2} \text{ Å}$$

$$\Rightarrow \lambda = 17.35 \text{ Å}$$

Solutions

Matrix Match Type Questions

1. A → (s)
B → (q, s)
C → (p, r)
D → (p, r)

2. A → (s)
B → (q)
C → (p, r)
D → (p, r)

3. A → (p, q)
B → (r, s)
C → (q)
D → (p)

4. A → (p)
B → (r)
C → (s)
D → (q)

de-Broglie wavelength of electron in X-Ray tube

$$\lambda = \frac{h}{\sqrt{2mKE}} = \frac{h}{\sqrt{2meV}}$$

$$\Rightarrow \lambda = \frac{1.227 \times 10^{-9}}{\sqrt{V}} = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

(A) Accelerating potential

$V = 10^4 \text{ eV}$ in X-ray tube

$$\lambda = \frac{1.227}{10^2} \times 10^{-9} = 1.227 \times 10^{-11} \text{ m}$$

$$\lambda = 0.1 \text{ \AA}$$

(B) Wavelength associated with X-rays

$$\lambda = \frac{12.4 \times 10^{-7}}{10^4} = 12.4 \times 10^{-11} \text{ m}$$

$$\lambda = 1.2 \times 10^{-10} \text{ m}$$

$$\lambda = 1 \text{ \AA}$$

(C) de-Broglie wavelength of most energetic photoelectron

$$\lambda = \frac{1.227 \times 10^{-9}}{\sqrt{V}} = 1.227 \times 10^{-9} = 12.2 \times 10^{-10} \text{ m}$$

$$\lambda = 10 \text{ \AA}$$

(D) $E = \frac{hc}{\lambda}$

$$\Rightarrow \lambda = \frac{hc}{E} = \frac{hc}{eV} = \frac{12.4 \times 10^{-7} \text{ m}}{V}$$

$$V = 2.5 \text{ V}$$

$$\lambda = \frac{12.4}{2.5} \times 10^{-7} = 5000 \text{ \AA}$$

5. A → (p, q)
B → (p, r, s)
C → (p, q)
D → (s)

6. A → (p, r)
B → (q, s)
C → (s)
D → (p, r)

7. A → (q, s)
B → (p, r)
C → (p, r)
D → (s)

8. A → (r)
B → (s)
C → (p)
D → (u)
E → (q)
F → (t)

Solutions

Integer Answer Type Questions

1. The target area is $S_1 = \pi(10^{-9})^2 = \pi \times 10^{-18} \text{ m}^2$.

The area of a 5 metre sphere centred on the light source is,
 $S_2 = 4\pi(5)^2 = 100 \pi \text{ m}^2$.

Thus, if the light source radiates uniformly in all directions the rate P at which energy falls on the target is given by,

$$P = (10^{-3} \text{ watt}) \left(\frac{S_1}{S_2} \right) = (10^{-3}) \left(\frac{\pi \times 10^{-18}}{100 \times \pi} \right) = 10^{-23} \text{ Js}^{-1}$$

Assuming that all power is absorbed, the required time is,

$$t = \left(\frac{5 \text{ eV}}{10^{-23} \text{ Js}^{-1}} \right) \left(\frac{1.6 \times 10^{-19} \text{ J}}{1 \text{ eV}} \right) \approx 20 \text{ hr}$$

2. Rate of energy received from the sun

$$Q = 2 \text{ cal cm}^{-2} \text{ min}^{-1} = 2 \times 4.2$$

$$\Rightarrow Q = 8.4 \text{ J cm}^{-2} \text{ min}^{-1}$$

Energy of a photon received from the sun is

$$E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6600 \times 10^{-10}} = 3 \times 10^{-19} \text{ J}$$

If n is the number of photons reaching the earth per cm^2 per minute, then their total energy will be $(3 \times 10^{-19} n)$ joule.

$$\Rightarrow (3 \times 10^{-19}) n = 8.4$$

$$\Rightarrow n = \frac{8.4}{3 \times 10^{-19}} = 28 \times 10^{18} \text{ photons per minute per cm}^2$$

$$\Rightarrow x = 28$$

3. Mass of an electron is

$$m = 9.11 \times 10^{-31} \text{ kg and } T = 27 + 273 = 300 \text{ K}$$

$\therefore$ de-Broglie wavelength of electrons is

$$\lambda = \frac{h}{\sqrt{3mkT}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 9.11 \times 10^{-31} \times 1.38 \times 10^{-23} \times 300}} \text{ m}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-9}}{\sqrt{3 \times 9.11 \times 1.38 \times 3}} = \frac{6.63 \times 10^{-9}}{10.64}$$

$$\Rightarrow \lambda = 6.2 \times 10^{-9} \text{ m}$$

Mean separation between two electrons in a metal is

$$r = 2 \times 10^{-10} \text{ m}$$

$$\Rightarrow \frac{\lambda}{r} = \frac{6.2 \times 10^{-9}}{2 \times 10^{-10}} = 31$$

4. The maximum kinetic energy of the emitted electron is

$$K_{\max} = \frac{1}{2} m v_{\max}^2 = \frac{hc}{\lambda} - W_0$$

$$\Rightarrow K_{\max} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{180 \times 10^{-9}} - 2 \times 1.6 \times 10^{-19} = 7.8 \times 10^{-19} \text{ J}$$

$$\Rightarrow v_{\max} = \sqrt{\frac{K_{\max}}{m}} = \sqrt{\frac{2 \times 7.8 \times 10^{-19}}{9.1 \times 10^{-31}}} = 1.3 \times 10^6 \text{ ms}^{-1}$$

The magnetic field provides the centripetal force to the electron, so

$$Bev_{\max} = \frac{mv_{\max}^2}{r}$$

$$\Rightarrow r = \frac{mv_{\max}}{Be} = \frac{9.1 \times 10^{-31} \times 1.3 \times 10^6}{5 \times 10^{-5} \times 1.6 \times 10^{-19}} = 0.148 \text{ m}$$

$$\Rightarrow r = 148 \text{ mm}$$

5. The Einstein's photoelectric equation for the first case can be written as

$$\frac{hc}{\lambda} = W + K_1 \quad \dots(1)$$

When illuminated with light of wavelength 2λ ,

$$\frac{hc}{2\lambda} = W + K_2 \quad \dots(2)$$

Subtracting (2) from (1), we get

$$\frac{hc}{\lambda} - \frac{hc}{2\lambda} = K_1 - K_2$$

$$\Rightarrow \lambda = \frac{hc}{2(K_1 - K_2)} = \frac{(4 \times 10^{-15})(3 \times 10^8)}{2(30 - 10)}$$

$$\Rightarrow \lambda = 300 \times 10^{-10} \text{ m} = 300 \text{ Å}$$

If $\lambda_{\max}$ is the maximum wavelength of the photons with which photoelectrons can be emitted, then

$$\frac{hc}{\lambda_{\max}} = W_0 = \frac{hc}{\lambda} - K_1 = \frac{(4 \times 10^{-15})(3 \times 10^8)}{300 \times 10^{-10}} - 30 \text{ eV}$$

$$\Rightarrow E = 40 - 30 = 10 \text{ eV}$$

$$\Rightarrow \lambda_{\max} = \frac{hc}{10 \text{ eV}} = \frac{(4 \times 10^{-15})(3 \times 10^8)}{10} = 1200 \times 10^{-10} \text{ m}$$

$$\Rightarrow \lambda_{\max} = 1200 \text{ Å}$$

6. Since $K_{\max} = h\nu - h\nu_0$

$$\Rightarrow \frac{1}{2} m(8 \times 10^5)^2 = h(5\nu_0 - \nu_0) \quad \dots(1)$$

For the second case, we have

$$KE = \frac{1}{2} mv^2 = h(2\nu_0 - \nu_0) \quad \dots(2)$$

$$\Rightarrow \frac{(8 \times 10^5)^2}{v^2} = 4$$

$$\Rightarrow v = 4 \times 10^5 \text{ ms}^{-1}$$

7. $F = (\text{Area}) \frac{1}{c}$ here effective Area = πR^2

$$\Rightarrow F = (\pi R^2) \frac{I}{c}$$

$$\Rightarrow F = \frac{22}{7} \times (21 \times 10^{-2})^2 \times \frac{1}{110} \times \frac{1}{3 \times 10^8}$$

$$\Rightarrow F = \frac{(22)(21 \times 10^{-2})(21 \times 10^{-2})}{7 \times 110 \times 3 \times 10^8}$$

$$\Rightarrow F = 42 \times 10^{-13} \text{ N}$$

$$\Rightarrow x = 42$$

8. Since, $eV_1 = \frac{hc}{\lambda_1} - \phi$ and

$$eV_2 = \frac{hc}{\lambda_2} - \phi$$

Subtracting, we get

$$e(V_2 - V_1) = hc \left(\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} \right)$$

$$\Rightarrow V_2 - V_1 = \frac{hc}{e} \left(\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} \right)$$

$$\Rightarrow V_2 - V_1 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19}} \times \frac{100}{6 \times 10^{-6}}$$

$$\Rightarrow V_2 - V_1 = \frac{66}{32} \times 10^{-34+8+2+19+5}$$

$$\Rightarrow V_2 - V_1 = \frac{33}{16} \approx 2 \text{ V}$$

9. Since, $eV_s = h\nu - W$

$$\Rightarrow 5 \text{ eV} = h\nu - 2 \text{ eV}$$

$$\Rightarrow h\nu = 7 \text{ eV}$$

So, total number of photons incident is

$$n = \frac{P}{h\nu}$$

Since, $\eta = 10^{-3}\%$

$$\Rightarrow n_{\text{emitted}} = n \times \frac{10^{-3}}{100}$$

$$\Rightarrow i = \frac{q}{t} = \left(\frac{n_{\text{emitted}}}{t} \right) e$$

where, from the graph

$$i = 10 \times 10^{-8} \text{ A}$$

$$\Rightarrow 10 \times 10^{-8} = \frac{P}{h\nu} \times \frac{10^{-3}}{100} \times 1.6 \times 10^{-19}$$

$$\Rightarrow 10 \times 10^{-8} = \frac{P \times 10^{-3} \times 1.6 \times 10^{-19}}{7 \times 1.6 \times 10^{-18} \times 100}$$

$$\Rightarrow P = 7 \text{ W}$$

10. According to Einstein's Photo-Electric Equation, we have

$$K_{\max} = eV_s = h\nu - W$$

$$\Rightarrow eV_s = 12 \text{ eV} - 4 \text{ eV}$$

$$\Rightarrow eV_s = 8 \text{ eV}$$

$$\Rightarrow V_s = 8 \text{ V}$$

11. Magnetic force experienced by a charged particle in a magnetic field is given by

$$\text{Since, } r = \frac{mv}{qB}$$

The de Broglie wave/length is given by

$$\lambda = \frac{h}{mv} = \frac{h}{qBr}$$

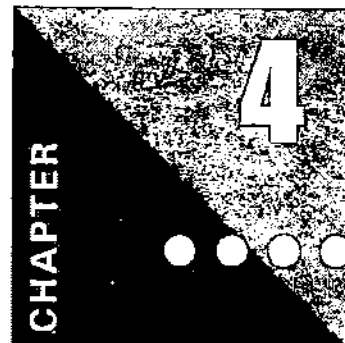
$$\left\{ \because r = \frac{mv}{qB} \right\}$$

$$\Rightarrow \frac{\lambda_p}{\lambda_a} = \frac{q_a r_a}{q_p r_p}$$

$$\text{Given, } \frac{r_a}{r_p} = 1 \text{ and } \frac{q_a}{q_p} = 2$$

$$\Rightarrow \frac{\lambda_p}{\lambda_a} = 2$$

Atomic & Nuclear Physics



Contents

(a) ATOMIC PHYSICS	4.1
(b) NUCLEAR PHYSICS	4.21
Solved Practice Problems	4.48
Practice Exercise Sets	
➤ Single Correct Choice Type Questions.....	4.58
➤ Multiple Correct Choice Type Questions	4.73
➤ Reasoning Based Questions (Assertion Reason Type)	4.78
➤ Linked Comprehension Type Questions (Paragraph Type)	4.81
➤ Matrix Match Type Questions (Column Matching Type)	4.89
➤ Integer Answer Type Questions	4.92
Answers to In Chapter Exercises (ICE) & Practice Exercise Sets	4.94
Solutions to In Chapter Exercises (ICE).....	4.99
Solutions to Practice Exercise Sets	4.106

Atomic & Nuclear Physics

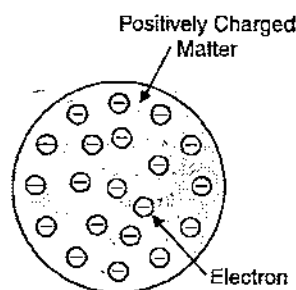
(a) Atomic Physics

STRUCTURE OF AN ATOM : AN INTRODUCTION

All matter is made up of tiny particles known as atoms. There are about 105 different kinds of atoms, and they combine with each other in different ways to form groups called molecules. All matter has been found to be composed of atoms or molecules, and the basic knowledge of atoms and their constitution gives us valuable information about the behaviour of matter.

THOMSON EMPIRICAL MODEL / THOMSON PLUM PUDDING MODEL

J.J. Thomson gave the first idea regarding structure of atom. The model is known after him as Thomson's atom model. According to this, entire positive charge is distributed uniformly in the form of a sphere. Negatively charged electrons are arranged within this sphere lying here and there. The model is popularly known as plum-pudding model. Every electron is attracted towards the centre of uniformly charged sphere while they exert a force of repulsion upon each other. The electrons get themselves arranged in such a way that the force of attraction and that of repulsion balance each other. When disturbed, electrons vibrate to and fro within the atom and cause emission of visible, infra-red and ultra-violet light.



Thomson's atom model

Thomson's atom model satisfied the requirements of the atom and the demands of electro-magnetic theory. According to this model, hydrogen can give rise to a single

spectral line. Experimentally, hydrogen is found to give several series, each series consisting of several lines. This indicated that Thomson's atom model needed modifications which was modified by Rutherford.

RUTHERFORD'S ATOMIC MODEL

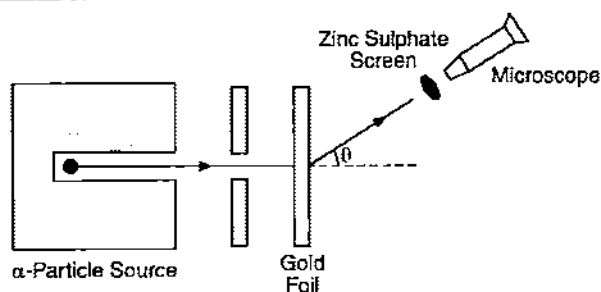
The correct description of the distribution of positive and negative charges within an atom was made in 1911 by a New Zealander while working at Manchester University in England. This was Ernest Rutherford, who was later called as Lord Rutherford for his many scientific achievements. He entered into physics during that crucial period of its development when the phenomenon of natural radioactivity had just been discovered, and he was first to realize that radioactivity represents a spontaneous disintegration of heavy unstable atoms.

Rutherford realized that important information about the inner structure of atoms can be obtained by the study of collisions between the rushing α particles incident on the atoms of various materials that form the target on which the α particle beam is incident.

EXPERIMENTAL ARRANGEMENT

The basic idea of the experimental arrangement used by Rutherford in his studies was explained as follows:

A piece or speck of α -emitting radioactive material is placed in a lead shield with a hole that allows a narrow beam of the α -particles to pass through it. In front of this arrangement is placed a gold thin metal foil to deflect or scatter the α particles. After passing through the gold foil the deflected particles are incident on a pivoted fluorescent screen with a magnifier through which the tiny flashes of light were observed whenever an α -particle struck the screen. The whole apparatus placed in an evacuated chamber, so that the particles would not collide with air molecules.



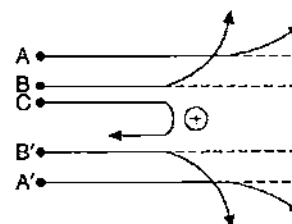
OBSERVATIONS

Rutherford performed experiments on the scattering of alpha particles by extremely thin metals foils and made the following observations.

- Most of the α -particles, either passed straight through the metal foil or suffered only small deflections. This could not be explained by Thomson's atom model.
- A few particles were deflected through angles which were less than or equal to 90° .
- Very few particles were deflected through angles greater than 90° . It was observed that about only 1 in 8000 particles was found to be deflected greater than 90° . Sometimes a particle was found to be deflected through 180° . In other words, it was sent back in the same direction from where it came. The large angle of scattering came as a greater surprise. It could not be explained by Thomson's atom model. It was one of the main reasons for rejecting Thomson's atom model.

CONCLUSIONS

- The fact that most of the α -particles passed undeflected led to the conclusion that an atom has a lot of empty space in it.
- α -particles are heavy particles having high initial speeds. These could be deflected through large angles only by a strong electrical force. This led Rutherford to the conclusion that entire positive charge and nearly the entire mass of the atom were concentrated in a tiny central core. Rutherford named this core as Nucleus.
- The difference in deflection of various particles can be explained as follows:
 α -particles which pass at greater distances away from the nucleus, shown as A and A' in figure, suffer a small deflection due to smaller repulsion exerted by the nucleus upon them. The particles like B and B' which pass close to the nucleus experience a comparatively greater force and hence get deflected through greater angles. A particle C which travels directly towards the nucleus is first slowed down by the repulsive force. Such a particle finally stops and then, is repelled along the direction of its approach. Thus, it gets repelled back after suffering a deviation of 180° .



Different deviations for different α particles

Also during the experiment following conclusions were made

- If ϕ is the angle made by a scattered particle with its original direction of motion and N is the number of particles available in that direction, it was found that,

$$\frac{1}{\sin^4 \frac{\phi}{2}} \propto \log_e N$$

- If t is the thickness of the foil and N is the number of α -particles scattered in a particular direction ($\phi = \text{constant}$) it was observed that

$$\frac{N}{t} = \text{constant}$$

$$\Rightarrow \frac{N_1}{N_2} = \frac{t_1}{t_2}$$

RUTHERFORD'S ATOM-MODEL POSTULATES

On the basis of the conclusions drawn from Rutherford's experiment, a new atom model was proposed. This atom model, known as Rutherford's atom model, had the following characteristics.

- An atom consists of equal amounts of positive and negative charge so, the atom, as a whole is electrically neutral.
- The entire positive charge of the atom and practically its entire mass is concentrated in a small region which forms the core of the atom, called the nucleus.
- The negative charge, which is contained in the atom in the form of electrons, is distributed all around the nucleus, but separated from it.
- In order to explain the stability of electron at a certain distance from the nucleus, it was proposed by Rutherford that the electrons revolve round the nucleus in circular orbits. The electrostatic force of attraction between the nucleus and the electron provides the centripetal force to the electron to revolve in the orbit.
- The nuclear diameter is of the order of 10^{-14} m. This can be calculated as follows.

Let an α -particle having velocity v approach a nucleus (head-on) having a charge $+Ze$. The velocity of α -particle decreases till it comes to rest at a distance r_0 .

from the nucleus. It is, then, repelled back along the direction of approach, r_0 gives the radius of nucleus.

$$\text{Initial K.E. of } \alpha\text{-particle} = \frac{1}{2}mv^2$$

$$\text{Initial P.E. of } \alpha\text{-particle} = 0$$

$$\text{Final K.E. of } \alpha\text{-particle} = 0$$

$$\text{Final P.E. of } \alpha\text{-particle} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_0}$$

By Law of Conservation of Energy

$$(U + K)_{\text{initial}} = (U + K)_{\text{final}}$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_0}$$

$$\Rightarrow r_0 = \frac{2}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{mv^2} \right)$$

For α -particle, $q_1 = 2e$

$$\Rightarrow r_0 = \frac{2}{4\pi\epsilon_0} \left(\frac{2Ze^2}{mv^2} \right)$$

$$\Rightarrow r_0 = \frac{4}{4\pi\epsilon_0} \left(\frac{Ze^2}{mv^2} \right) = 4 \times 9 \times 10^9 \frac{Ze^2}{mv^2}$$

In one of the experiments, α -particles of velocity $2 \times 10^7 \text{ ms}^{-1}$ were bombarded upon gold foil with $Z = 79$

So, for $Z = 79$

$$e = 1.59 \times 10^{-19} \text{ C}$$

$$m = 4 \times 1.67 \times 10^{-27} \text{ kg}$$

$$v = 2 \times 10^7 \text{ ms}^{-1}$$

we get,

$$r_0 = 4 \times 9 \times 10^9 \times \frac{79 \times (1.59 \times 10^{-19})^2}{4 \times 1.67 \times 10^{-27} \times (2 \times 10^7)^2}$$

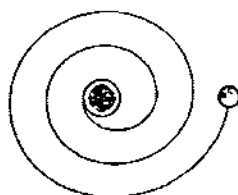
$$\Rightarrow r_0 = 2.69 \times 10^{-14} \text{ m}$$

This gives the order of the radius of nucleus.

FAILURE OF RUTHERFORD MODEL

- a) According to laws of electro-magnetic theory, a charged particle in accelerated motion must radiate energy in the form of electro-magnetic radiation. As the electron revolves in a circular orbit, it is constantly subjected to

centripetal acceleration $\frac{v^2}{r}$.



Electron spiralling inwards

So it must radiate energy continuously. Bohr calculated that this emission of radiation would cause the electrons in an atom to lose all their energy and fall into the nucleus within a hundred - millionth of a second following a spiral path. Thus the whole atomic structure should collapse. Since matter composed of atoms exists permanently, as far as we know, there was obviously something wrong here.

- b) According to Rutherford model, electron can revolve in any orbit. So, it must emit continuous radiations of all frequencies. But elements emit spectral lines of only definite frequencies.

Illustration 1

An isolated hydrogen atom emits a photon of 10.2 eV.

- Determine the momentum of photon emitted
- Calculate the recoil momentum of the atom
- Find the kinetic energy of the recoil atom. [Mass of proton, $m_p = 1.67 \times 10^{-27} \text{ kg}$]

Solution

- (a) Momentum of the photon is

$$p_1 = \frac{E}{c} = \frac{10.2 \times 1.6 \times 10^{-19}}{3 \times 10^8} = 5.44 \times 10^{-27} \text{ kgms}^{-1}$$

- (b) Applying the Law of Conservation of Linear Momentum, we get



$$p_2 = p_1 = 5.44 \times 10^{-27} \text{ kgms}^{-1}$$

- (c) $K = \frac{1}{2}mv^2$ (v = recoil speed of atom, m = mass of hydrogen atom)

$$\Rightarrow K = \frac{1}{2}m \left(\frac{p}{m} \right)^2 = \frac{p^2}{2m}$$

Substituting the value of the momentum of atom, we get

$$K = \frac{(5.44 \times 10^{-27})^2}{2 \times 1.67 \times 10^{-27}} = 8.86 \times 10^{-27} \text{ J}$$

BOHR'S ATOMIC MODEL

BOHR'S THEORY OF THE HYDROGEN LIKE ATOMS

To rectify the drawbacks of Rutherford Model, Bohr proposed a theory which applies to hydrogen atom and species like He^+ , Li^{++} etc. Here a single electron revolves around a stationary nucleus of positive charge Ze where $Z=1$ for hydrogen atom, $Z=2$ for He^+ etc. Bohr in defiance of the well-established laws of classical mechanics

and electrodynamics, proposed the following postulates to support his atomic model.

A. CIRCULAR ORBITS

The atom consists of central nucleus, containing the entire positive charge and almost all mass of the atom. The electrons revolve around the nucleus in certain discrete circular orbits. The necessary centripetal force for circular orbit is provided by Coulomb's attraction between the electron and nucleus. So,

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(e)}{r^2}$$

where, m = mass of electron,
 r = radius of circular orbit,
 v = speed of electron in circular orbit,
 Ze = charge on nucleus,
 Z = atomic number,
 e = charge on electron = -1.6×10^{-19} C

B. STATIONARY ORBITS

The allowed orbits for electron are those in which the electron does not radiate energy. These orbits are also called stationary orbits.

C. QUANTUM CONDITION (BOHR'S QUANTISATION RULE)

The stationary orbits are those in which angular momentum of electron is an integral multiple of $\frac{h}{2\pi}$ ($= \hbar$)

i.e., $mvr = n\left(\frac{h}{2\pi}\right)$, n being integer or the principle quantum number

D. STATIONARY NUCLEUS

The nucleus is so heavy, that its motion may be neglected.

E. CONSTANCY OF MASS

The mass of the electron in motion is assumed to be constant.

F. BOHR'S TRANSITION RULE

When an electron jumps from one stationary orbit to another, a photon is emitted or absorbed having energy equal to the difference of energies between initial and final states and being given by

$$E_i - E_f = h\nu$$

$$\Rightarrow v = \frac{E_i - E_f}{h}$$

G. RADIUS OF ORBIT

Since, we have

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(e)}{r^2} \quad \dots(1)$$

$$\text{and } mvr = \frac{nh}{2\pi} \quad \dots(2)$$

$$\text{From (2), } v = \frac{nh}{2\pi mr}$$

Put in (1), we get

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z}$$

$$\Rightarrow r_n = (0.53) \frac{n^2}{Z} \text{ \AA}$$

So, for H-like atoms

$$r_n \propto \frac{n^2}{Z}$$

H. VELOCITY OF ELECTRON IN NTH ORBIT

$$\text{Since, } v = \frac{nh}{2\pi mr}$$

$$\text{and } r = \frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z}$$

$$\Rightarrow v = \left(\frac{e^2}{2hc_0} \right) \frac{Z}{n}$$

$$\Rightarrow v = \left(\frac{e^2}{2hc_0 c} \right) \left(\frac{cZ}{n} \right)$$

$$\Rightarrow v = \alpha \left(\frac{cZ}{n} \right)$$

where $\alpha = \frac{e^2}{2hc_0 c}$ is the fine structure constant (a pure number) whose value is $\frac{1}{137}$.

$$\Rightarrow v = \left(\frac{1}{137} \right) \frac{cZ}{n}$$

i.e. velocity of electron in Bohr's First Orbit is $\frac{c}{137}$, in

Second Orbit is $\frac{c}{274}$ and so on.

I. KINETIC ENERGY OF ELECTRON (E_k)

Since, we have

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(e)}{r^2}$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$$

$$\Rightarrow E_k = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$$

J. POTENTIAL ENERGY (U) OF ELECTRON IN nth ORBIT

$$U = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(e)}{r}$$

$$\Rightarrow U = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

K. TOTAL ENERGY (E) OF ELECTRON IN nth ORBIT

Total Energy = K.E. + P.E.

$$\Rightarrow E = \frac{Ze^2}{8\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$\Rightarrow E = -\frac{Ze^2}{8\pi\epsilon_0 r}$$

So, we conclude that

$$\text{Total Energy} = -\text{K.E.} = \frac{1}{2}(\text{P.E.})$$

Further, since $r = \frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z}$

$$\Rightarrow E = -\left(\frac{m e^4}{8 h^2 \epsilon_0^2}\right) \frac{Z^2}{n^2}$$

$$\Rightarrow E = -(13.6) \frac{Z^2}{n^2} \text{ eV}$$

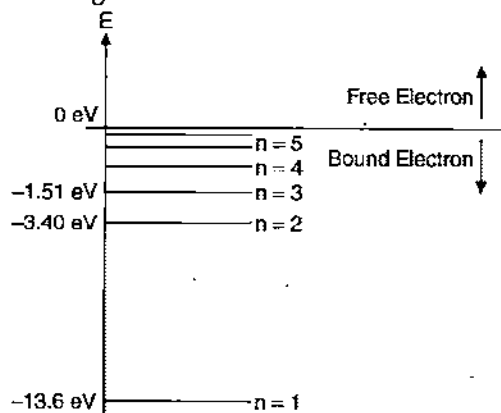
Also $E = -\left(\frac{m e^4}{8 \epsilon_0^2 h^3}\right) c h \frac{Z^2}{n^2}$

$$\Rightarrow E = -(Rch) \frac{Z^2}{n^2}$$

where $R = \text{Rydberg's constant} = \frac{m e^4}{8 \epsilon_0^2 h^3} = 1.097 \times 10^7 \text{ m}^{-1}$ and

$Rch = \text{Rydberg's Energy} = 2.17 \times 10^{-18} \text{ J} = 13.6 \text{ eV}$ is the electron energy in first orbit of H atom.

An energy level diagram for the hydrogen atom ($Z=1$) is shown in the figure.



The vertical axis represents energy. The (arbitrary) zero of energy is taken as the energy of a stationary electron, infinitely far from the positive nucleus. The lowest energy

level ($n=1$) is known as the ground state. The energy level corresponding to $n=2$ is called the first excited state and so on. In this diagram zero energy level corresponds to $n=\infty$ which is the ionized state of the atom.

Illustration 2

Using Bohr's theory show that when n is very large the frequency of radiation emitted by hydrogen atom due to transition of electron from n to $(n-1)$ is equal to frequency of revolution of electron in its orbit.

Solution

Frequency of revolution electron in n th orbit is given by

$$f_{\text{revolution}} = \frac{1}{T} = \frac{v}{2\pi r} = \frac{\left(\frac{e^2}{2h\epsilon_0 c}\right) \frac{Z}{n}}{2\pi \left(\frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z}\right)}$$

$$\Rightarrow f_{\text{revolution}} = \left(\frac{m e^4}{4 \epsilon_0^2 h^3}\right) \frac{Z^2}{n^3} \quad \dots(1)$$

Further, frequency of transition from n to $(n-1)$ is

$$hf = \frac{m e^4 Z^2}{8 h^2 \epsilon_0^2} \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right]$$

$$\Rightarrow hf = \frac{m e^4 Z^2}{8 h^2 \epsilon_0^2} \left[\frac{2n-1}{n^2 (n-1)^2} \right]$$

When n is large then

$$2n-1 \approx 2n \text{ and } n-1 \approx n$$

$$\Rightarrow hf \approx \frac{m e^4 Z^2}{8 h^2 \epsilon_0^2} \frac{2n}{n^4}$$

$$\Rightarrow f_{\text{transition}} \approx \left(\frac{m e^4}{4 \epsilon_0^2 h^3}\right) \frac{Z^2}{n^3} \quad \dots(2)$$

So, from (1) and (2), we observe that for large n ,

$$f_{\text{revolution}} = f_{\text{transition}} \text{ (between adjacent levels)}$$

This Principle is also called "BOHR'S CORRESPONDENCE PRINCIPLE".

Illustration 3

Assume a hypothetical hydrogen atom in which the potential energy between electron and proton at separation

r is given by $U = k \left(\log r - \frac{1}{2} \right)$, where k is a constant. For

such a hypothetical hydrogen atom, calculate the radius of n th Bohr's orbit and energy levels.

Solution

Force of interaction between electron and proton is given by

$$F = -\frac{dU}{dr} = \frac{k}{r}$$

Force is negative. It means there is attraction between the particles and they are bound to each other. This force provides the necessary centripetal force for the electron. So, we have

$$\frac{mv^2}{r} = \frac{k}{r} \quad \dots(1)$$

According to Bohr's assumption, we have

$$mvr = n \left(\frac{h}{2\pi} \right) \quad \dots(2)$$

Solving equations (1) and (2), we get

$$r = \frac{nh}{2\pi\sqrt{mk}} \text{ and } v = \sqrt{\frac{k}{m}}$$

$$\text{Since, } E = U + \frac{1}{2}mv^2$$

$$\Rightarrow E = k \log_e r - \frac{k}{2} + \frac{k}{2} = k \log_e r$$

$$\text{So, } r_n = \frac{nh}{2\pi\sqrt{mk}} \text{ and } E_n = k \log_e \left(\frac{nh}{2\pi\sqrt{mk}} \right)$$

Illustration 4

An electron is orbiting in a circular orbit of radius r under the influence of a constant magnetic field of strength B . Assuming that Bohr's postulate regarding the quantisation of angular momentum holds good for this electron, find

- the allowed values of the radius r of the orbit.
- the kinetic energy of the electron in orbit.
- the potential energy of interaction between the magnetic moment of the orbital current due to the electron moving in its orbit and the magnetic field B .
- the total energy of the allowed energy levels.
- the total magnetic flux due to the magnetic field B passing through the n th orbit.

(Assume that the charge on the electron is $-e$ and the mass of the electron is m).

Solution

$$(a) \text{ Since, } r = \frac{mv}{Be} \quad \dots(1)$$

From Bohr's Quantisation Rule, we have

$$mvr = \frac{nh}{2\pi} \quad \dots(2)$$

Solving these two equations, we get

$$r = \sqrt{\frac{nh}{2\pi Be}} \text{ and } v = \sqrt{\frac{nhBe}{2\pi m^2}}$$

$$(b) K = \frac{1}{2}mv^2 = \frac{nhBe}{4\pi m}$$

$$(c) M = iA = \left(\frac{e}{T} \right) (\pi r^2) = \left(\frac{e}{\frac{2\pi r}{v}} \right) (\pi r^2) = \frac{e\pi r^2}{2}$$

$$\Rightarrow M = \frac{e}{2} \sqrt{\frac{nh}{2\pi Be}} \sqrt{\frac{nhBe}{2\pi m^2}} = \frac{nhe}{4\pi m}$$

$$\text{Since, } U = -MB \cos 180^\circ$$

$$\Rightarrow U = \frac{nheB}{4\pi m}$$

The angle between $\vec{M}$ and $\vec{B}$ will be 180° because instead of taking electronic current, we have to take conventional current which moves opposite to electronic current.

$$(d) E = U + K = \frac{nheB}{2\pi m}$$

$$(e) |\phi| = B\pi r^2 = \frac{nh}{2e}$$

Illustration 5

Light from a discharge tube containing hydrogen atoms falls on the surface of a piece of sodium. The kinetic energy of the fastest photoelectrons emitted from sodium is 0.73 eV. The work function for sodium is 1.82 eV.

Find

- the energy of the photons causing the photoelectrons emission.
- the quantum numbers of the two levels involved in the emission of these photons
- the change in the angular momentum of the electron in the hydrogen atom, in the above transition, and
- the recoil speed of the emitting atom assuming it to be at rest before the transition. (Ionization potential of hydrogen is 13.6 eV).

Solution

- From Einstein's equation of photoelectric effect,

$$\left(\begin{array}{c} \text{Energy of} \\ \text{photons causing} \\ \text{the photoelectric} \\ \text{emission} \end{array} \right) = \left(\begin{array}{c} \text{Maximum} \\ \text{kinetic energy} \\ \text{of emitted} \\ \text{photons} \end{array} \right) + \left(\begin{array}{c} \text{work} \\ \text{function} \end{array} \right)$$

$$\Rightarrow E = K_{\max} + W = (0.73 + 1.82) \text{ eV}$$

$$\Rightarrow E = 2.55 \text{ eV}$$

- In case of a hydrogen atom,

$$E_1 = -13.6 \text{ eV}$$

$$E_2 = -3.4 \text{ eV}$$

$$E_3 = -1.5 \text{ eV}$$

$$E_4 = -0.85 \text{ eV}$$

$$\text{Since, } E_4 - E_2 = 2.55 \text{ eV}$$

Therefore, quantum numbers of the two levels involved in the emission of these photons are 4 and 2 i.e., from $4 \rightarrow 2$.

- Change in angular momentum transition from 4 to 2 will be

$$\Delta L = L_2 - L_1$$

$$\Rightarrow \Delta L = 2\left(\frac{h}{2\pi}\right) - 4\left(\frac{h}{2\pi}\right)$$

$$\Rightarrow \Delta L = -\frac{h}{\pi}$$

(d) By Law of Conservation of Linear Momentum, we have

$$\left(\text{Momentum of hydrogen atom}\right) = \left(\text{Momentum of emitted photon}\right)$$

$$\Rightarrow mv = \frac{E}{c} \quad (m = \text{mass of hydrogen atom})$$

$$\Rightarrow v = \frac{E}{mc}$$

$$\Rightarrow v = \frac{(2.55)(1.6 \times 10^{-19})}{(1.67 \times 10^{-27})(3 \times 10^8)}$$

$$\Rightarrow v = 0.814 \text{ ms}^{-1}$$

FREQUENCY OF EMITTED RADIATION

If electron jumps from initial state n_i to a final state n_f , then frequency of emitted or absorbed radiation ν is given by

$$E_i - E_f = h\nu$$

$$\text{or } \nu = \frac{E_i - E_f}{h} = Z^2 R c \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

If c is the speed of light and λ the wavelength of emitted or absorbed radiation, then

$$\nu = \frac{c}{\lambda} = Z^2 R c \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

So, Wave number ($\bar{\nu}$) is given by

$$\bar{\nu} = \frac{1}{\lambda} = Z^2 R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

This relation holds for radiations emitted by hydrogen-like atoms i.e.

$$H(Z=1), \text{He}^+(Z=2), \text{Li}^{++}(Z=3) \text{ and } \text{Be}^{+++}(Z=4).$$

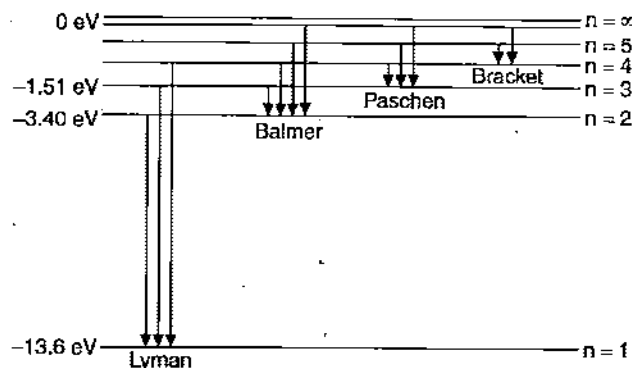
HYDROGEN SPECTRUM

Since, wave number ($\bar{\nu}$) is given by

$$\bar{\nu} = \frac{1}{\lambda} = Z^2 R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

where R is the Rydberg constant.

The various transitions for the hydrogen atom are shown in the following figure. All transitions starting from $n=2$ onwards and ending at $n=1$ belong to the Lyman Series. Likewise, all transitions starting from $n=3$ onwards and ending at $n=2$ belong to the Balmer Series. The other spectral series' names are mentioned in the figure.



	INITIAL STATE	FINAL STATE	WAVELENGTH FORMULA	FIRST MEMBER - SECOND MEMBER	SERIES LIMIT - $n_i \rightarrow \infty$ TO n_f	MAXIMUM WAVELENGTH ($n_i + 1$) TO n_f	LINES FOUND IN
LYMAN	$n_i = 2, 3, 4, 5, 6, \dots$	$n_f = 1$	$\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n_i^2} \right)$	$n_i = 2$ to $n_f = 1$ $n_i = 3$ to $n_f = 1$	From ∞ to 1 $\lambda = \frac{1}{R}$ $\lambda = 911 \text{ \AA}$	From 2 to 1 $\lambda = \frac{4}{3R}$ $\lambda = 1216 \text{ \AA}$	UV Region
BALMER	$n_i = 3, 4, 5, 6, 7, \dots$	$n_f = 2$	$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right)$	$n_i = 3$ to $n_f = 2$ $n_i = 4$ to $n_f = 2$	From ∞ to 2 $\lambda = \frac{4}{R}$ $\lambda = 3646 \text{ \AA}$	From 3 to 2 $\lambda = \frac{36}{5R}$ $\lambda = 6563 \text{ \AA}$	Visible Region
PASCHEN	$n_i = 4, 5, 6, 7, 8, \dots$	$n_f = 3$	$\frac{1}{\lambda} = R \left(\frac{1}{3^2} - \frac{1}{n_i^2} \right)$	$n_i = 4$ to $n_f = 3$ $n_i = 5$ to $n_f = 3$	From ∞ to 3 $\lambda = \frac{9}{R}$ $\lambda = 8204 \text{ \AA}$	From 4 to 3 $\lambda = \frac{144}{7R}$ $\lambda = 18753 \text{ \AA}$	IR Region

BRACKETT	$n_i = 5, 6, 7, 8, 9, \dots$	$n_f = 4$	$\frac{1}{\lambda} = R \left(\frac{1}{4^2} - \frac{1}{n_i^2} \right)$	$n_i = 5 \text{ to } n_f = 4$ $n_i = 6 \text{ to } n_f = 4$	From ∞ to 4 $\lambda = \frac{16}{R}$ $\lambda = 14585 \text{ \AA}$	From 5 to 4 $\lambda = \frac{400}{9R}$ $\lambda = 40515 \text{ \AA}$	IR Region
PFUND	$n_i = 6, 7, 8, 9, 10, \dots$	$n_f = 5$	$\frac{1}{\lambda} = R \left(\frac{1}{5^2} - \frac{1}{n_i^2} \right)$	$n_i = 6 \text{ to } n_f = 5$ $n_i = 7 \text{ to } n_f = 5$	From ∞ to 5 $\lambda = \frac{25}{R}$ $\lambda = 22790 \text{ \AA}$	From 6 to 5 $\lambda = \frac{900}{11R}$ $\lambda = 74583 \text{ \AA}$	Far IR Region

Illustration 6

A single electron orbits around a stationary nucleus of charge $+Ze$, where Z is a constant and e is the magnitude of electronic charge. It requires 47.2 eV to excite the electron from second Bohr orbit to the third Bohr orbit.

- Find the value of Z
- Find the energy required to excite the electron from $n=3$ to $n=4$
- Find the wavelength of radiation required to remove electron from first Bohr's Orbit to infinity.
- Find the kinetic energy, potential energy and angular momentum of the electron in the first Bohr orbit.

Solution

- (a) Given $\Delta E_{23} = 47.2 \text{ eV}$

$$\text{Since, } \Delta E = 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

$$\Rightarrow 47.2 = 13.6Z^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$\Rightarrow Z = 5$$

- (b) To find ΔE_{34} , $n_1 = 3$, $n_2 = 4$.

$$\Delta E = 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

$$\Rightarrow \Delta E = 13.6 \times 5^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = 16.53 \text{ eV}$$

- (c) Ionization energy is the energy required to excite the electron from $n=1$ to $n=\infty$

$$\text{Thus, } \Delta E = 13.6 \times 5^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2} \right) = 340 \text{ eV}$$

The respective wavelength is

$$\lambda = \frac{hc}{\Delta E} = \frac{12400}{\Delta E} = \frac{12400}{340} = 36.47 \text{ \AA}$$

- (d) $K = -E = +340 \text{ eV}$

$$U = 2E = -680 \text{ eV}$$

$$L = \frac{h}{2\pi} \times \frac{6.63 \times 10^{-34}}{2\pi} = 1.056 \times 10^{-34} \text{ Js}$$

Illustration 7

The wavelength of the first line of Lyman series for hydrogen is identical to that of the second line of Balmer series for some hydrogen like ion x . Calculate energies of the first four levels, of x .

Solution

Wavelength of the first line of Lyman series for hydrogen atom will be given by the equation

$$\frac{1}{\lambda_1} = R \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3R}{4} \quad \dots(1)$$

The wavelength of second Balmer line for hydrogen like ion x is

$$\frac{1}{\lambda_2} = Rz^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{3Rz^2}{16} \quad \dots(2)$$

Given that $\lambda_1 = \lambda_2$

$$\Rightarrow \frac{1}{\lambda_1} = \frac{1}{\lambda_2}$$

$$\Rightarrow \frac{3R}{4} = \frac{3Rz^2}{16}$$

$$\Rightarrow z = 2$$

Hence the ion x is actually He^+ .

The energies of first four levels of ion x are,

$$E_1 = -(13.6)z^2 = -54.4 \text{ eV}$$

$$E_2 = \frac{E_1}{(2)^2} = -13.6 \text{ eV}$$

$$E_3 = \frac{E_1}{(3)^2} = -6.04 \text{ eV and}$$

$$E_4 = \frac{E_1}{(4)^2} = -3.4 \text{ eV}$$

Illustration 8

A doubly ionised lithium atom is hydrogen-like with atomic number 3.

- (a) Find the wavelength of the radiation required to excite the electron in Li^{2+} from the first to the third Bohr orbit. (Ionisation energy of the hydrogen atom equals 13.6 eV).

- (b) How many spectral lines are observed in the emission spectrum of the above excited system?

Solution

Given, $Z = 3$

Since, $E_n \propto \frac{Z^2}{n^2}$

- (a) To excite the atom from $n=1$ to $n=3$, energy of photon required is,

$$E_{1 \rightarrow 3} = E_3 - E_1 = \frac{(-13.6)(3)^2}{(3)^2} - \left(\frac{(-13.6)(3)^2}{(1)^2} \right)$$

$$\Rightarrow E_{1 \rightarrow 3} = 108.8 \text{ eV}$$

Corresponding wavelength will be,

$$\lambda (\text{in } \text{\AA}) = \frac{12375}{E (\text{in eV})} = \frac{12375}{108.8} = 113.74 \text{ \AA}$$

- (b) From n^{th} orbit total number of emission lines can be

$\frac{n(n-1)}{2}$. So, number of emission lines is

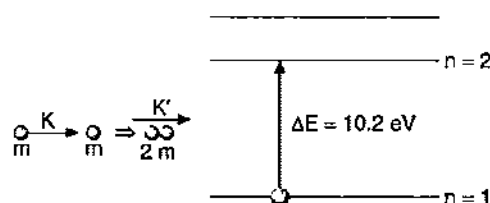
$$N = \frac{3(3-1)}{2} = 3$$

Illustration 9

A moving hydrogen atom makes a head on collision with a stationary hydrogen atom. Before collision both atoms are in ground state and after collision they move together. What is the minimum value of the kinetic energy of the moving hydrogen atom, such that one of the atoms reaches one of the excitation state.

Solution.

Let K be the kinetic energy of the moving hydrogen atom and K' , the kinetic energy of combined mass after collision.



By Law of Conservation of Linear Momentum, we have

$$p = p'$$

$$\Rightarrow \sqrt{2Km} = \sqrt{2K'(2m)}$$

$$\Rightarrow K = 2K' \quad \dots(1)$$

$$\text{From Conservation of Energy, } K = K' + \Delta E \quad \dots(2)$$

Solving equations (1) and (2), we get

$$\Delta E = \frac{K}{2}$$

Now minimum value of ΔE for hydrogen atom is 10.2 eV, so we have

$$\Delta E \geq 10.2 \text{ eV}$$

$$\Rightarrow \frac{K}{2} \geq 10.2$$

$$\Rightarrow K \geq 20.4 \text{ eV}$$

Therefore, the minimum kinetic energy of moving hydrogen is 20.4 eV.

Illustration 10

A moving H-atom makes a head on perfectly inelastic collision with a stationary Li^{++} ion. Before collision H-atom and Li^{++} ion are both in their first excited states. What is the velocity of the moving H atom if after collision H is found in its ground state and Li^{++} ion in its second excited state. Take mass of hydrogen atom, $m_H = 1.66 \times 10^{-27} \text{ kg}$ and mass of Li^{++} to be $m_{\text{Li}^{++}} = 7m_H$.

Solution

For Li^{++} , we have $Z = 3$

$$\Rightarrow E_2 = \frac{-13.6(3)^2}{(2)^2} = -30.6 \text{ eV}$$

$$\Rightarrow E_3 = \frac{-13.6(3)^2}{(3)^2} = -13.6 \text{ eV}$$

Energy required for Li^{++} ion to go from first excited state ($n=2$) to second excited state ($n=3$) is

$$\Delta E = -13.6 - (-30.6) = 17 \text{ eV}$$

Energy released by hydrogen atom to go from first excited state to ground state is

$$\Delta E' = -3.4 - (-13.6) = 10.2 \text{ eV}$$

So, $\Delta E - \Delta E' = 17 - 10.2 = 6.8 \text{ eV}$ is the energy that should come from loss in KE in collision.

From Law of Conservation of Linear Momentum, velocity of combined mass is

$$v = \frac{m_1 u_1}{m_1 + m_2} \quad (m_2 = 7m_1)$$

$$\Rightarrow \Delta KE = \frac{1}{2} m_1 u_1^2 - \frac{1}{2} (m_1 + m_2) v^2$$

Since, $\Delta K = 6.8 \text{ eV}$

Solving these equations, we get

$$u_1 = 3.9 \times 10^4 \text{ ms}^{-1}$$

Illustration 11

Find the largest and shortest wavelengths in the Lyman series for hydrogen. In what region of the electromagnetic spectrum does each series lie?

Solution

The transition equation for Lyman series is given by,

$$\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right), \quad n = 2, 3, \dots$$

The largest wavelength is corresponding to $n=2$, so

$$\frac{1}{\lambda_{\max}} = 1.097 \times 10^7 \left(\frac{1}{1} - \frac{1}{4} \right)$$

$$\Rightarrow \frac{1}{\lambda_{\max}} = 0.823 \times 10^7$$

$$\Rightarrow \lambda_{\max} = 1.2154 \times 10^{-7} \text{ m}$$

$$\Rightarrow \lambda_{\max} = 1215 \text{ Å}$$

The shortest wavelength corresponds to $n \rightarrow \infty$, so

$$\frac{1}{\lambda_{\min}} = 1.097 \times 10^7 \left(\frac{1}{1} - \frac{1}{\infty} \right)$$

$$\Rightarrow \lambda_{\min} = 0.911 \times 10^{-7} \text{ m} = 911 \text{ Å}$$

Both of these wavelengths lie in ultraviolet (UV) region of electromagnetic spectrum.

ABOUT RYDBERG CONSTANT

Rydberg Constant is given by

$$R = \frac{me^4}{8\epsilon_0^2 ch^3}$$

where m is mass of electron.

It appears that Rydberg constant is same for all elements, but it is not so. Actually Rydberg constant is different for different elements. The reason is that in Bohr's Theory nucleus is assumed infinitely heavy as compared to electron. But if the mass of nucleus is taken into account then electron mass m must be replaced by reduced mass (μ), where

$$\mu = \frac{mM}{m+M}$$

where M is mass of nucleus.

$$\Rightarrow \mu = \frac{mM}{m+M} = \frac{m}{1 + \frac{m}{M}}$$

If $M \rightarrow \infty$ (infinity), $\mu \rightarrow m$ (Mass of electron)

Therefore

$$R = \frac{\mu e^4}{8\epsilon_0^2 ch^3} = \frac{me^4}{8 \left(1 + \frac{m}{M} \right) \epsilon_0^2 ch^3}$$

If for a heavy nucleus, Rydberg Constant R is represented by

$$R_\infty = \frac{me^4}{8\epsilon_0^2 ch^3}$$

Then Rydberg constant for an element is given by

$$R = \frac{R_\infty}{1 + \frac{m}{M}}$$

Clearly this depends on mass of nucleus M and so Rydberg constant is different for different elements. Greater is M , larger is the value of Rydberg constant R . Thus, Rydberg constant increases with increase of mass of nucleus.

Illustration 12

The nucleus of a deuterium has a mass of $3.34 \times 10^{-27} \text{ kg}$ as compared to $1.67 \times 10^{-27} \text{ kg}$ for the hydrogen. Calculate the wavelength difference between the first Balmer line emitted by hydrogen and the first Balmer line emitted by deuterium. Given that the mass of electron is $m_e = 9.109 \times 10^{-31} \text{ kg}$.

Solution

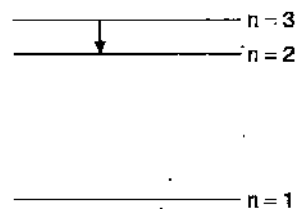
The first Balmer line corresponds to the transition from $n=3$ to $n=2$. In case of hydrogen atom, we have

$$\Delta E = E_3 - E_2$$

$$\Rightarrow \Delta E = \left\{ -\frac{13.6}{3^2} - \left(-\frac{13.6}{2^2} \right) \right\} \text{ eV} = 1.89 \text{ eV}$$

$$\text{So, wavelength } \lambda = \frac{12375}{\Delta E (\text{in eV})}$$

$$\Rightarrow \lambda = \frac{12375}{1.89} = 6547.6 \text{ Å}$$



For ordinary hydrogen reduced mass of proton and electron is,

$$\mu_1 = \frac{(1.67 \times 10^{-27})(9.109 \times 10^{-31})}{(1.67 \times 10^{-27} + 9.109 \times 10^{-31})}$$

$$\Rightarrow \mu_1 = 9.10408 \times 10^{-31} \text{ kg}$$

For deuterium atom reduced mass of nucleus and electron is,

$$\mu_2 = \frac{(3.34 \times 10^{-27})(9.109 \times 10^{-31})}{(3.34 \times 10^{-27} + 9.109 \times 10^{-31})}$$

$$\Rightarrow \mu_2 = 9.10654 \times 10^{-31} \text{ kg}$$

All energies are proportional to μ , whereas the wavelengths are inversely proportional to μ . The wavelength of photon emitted in case of hydrogen is given by

$$\lambda_1 = \frac{(6547.6)(9.109 \times 10^{-31})}{(9.10408 \times 10^{-31})} = 6551 \text{ Å}$$

Similarly in case of deuterium, wavelength of photon emitted is,

$$\lambda_2 = \frac{(6547.6)(9.109 \times 10^{-31})}{(9.10654 \times 10^{-31})} = 6549 \text{ Å}$$

$$\Rightarrow \Delta \lambda = \lambda_2 - \lambda_1 = 2 \text{ Å}$$

Illustration 13

Calculate the separation between the particles of a system in the ground state, the corresponding binding energy and wavelength of first line in Lyman series if such

a system is positronium consisting of an electron and positron revolving round their common centre of mass.

Solution

$$\text{Reduced mass } \mu = \frac{(m)(m)}{(m+m)} = \frac{m}{2}$$

$$\text{Since, } r \propto \frac{1}{m}$$

For, H atom, we have

$$r_1 = 0.53 \text{ \AA}$$

$$\Rightarrow r = (2)(0.53) = 1.06 \text{ \AA}$$

Since, $E \propto m$

{binding energy}

For, H atom, we have

$$E_1 = 13.6 \text{ eV}$$

$$\Rightarrow E = \frac{13.6}{2} = 6.8 \text{ eV}$$

For, H atom, we have

$$E_{2 \rightarrow 1} = 10.2 \text{ eV} \quad \text{(First line of Lyman series)}$$

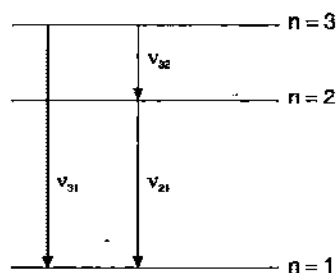
$$\Rightarrow E_{21} = \frac{10.2}{2} = 5.1 \text{ eV}$$

$$\Rightarrow \lambda = \frac{12375}{5.1} \text{ \AA} = 2426 \text{ \AA}$$

RITZ COMBINATION PRINCIPLE

If an electron is initially in an excited state with say $n=3$, then it may transit downward from $n=3$ level to $n=1$ level directly. Alternatively, it may first transit from $n=3 \rightarrow n=2$ and subsequently from $n=2 \rightarrow n=1$. In the first case if ν_{31} be the frequency of the photon emitted

$$h\nu_{31} = E_3 - E_1 \quad \dots(1)$$



In the second case, two different spectral lines (photons) of frequency ν_{32} and ν_{21} respectively would be emitted given by

$$h\nu_{32} = E_3 - E_2 \text{ and } h\nu_{21} = E_2 - E_1 \quad \dots(2)$$

(1) can be rewritten as

$$h\nu_{31} = (E_3 - E_2) + (E_2 - E_1)$$

$$\Rightarrow h\nu_{31} = h\nu_{32} + h\nu_{21}$$

$$\Rightarrow \nu_{31} = \nu_{32} + \nu_{21}$$

Ritz made this discovery empirically (1908) long before Bohr proposed his theory and is known as Ritz combination principle.

Generalising, we may write, labelling the photon frequency by appropriate integers, as follows:

$$h\nu_{sm} = E_s - E_m$$

$$\Rightarrow h\nu_{sm} = (E_s - E_r) + (E_r - E_m)$$

$$\Rightarrow h\nu_{sm} = h\nu_{sr} + h\nu_{rm} \quad (m < n < s) \quad \dots(3)$$

Since all combinations predicted by (3) are not actually observed, there has been an imposition of some rules, the so-called selection rules, to eliminate certain combinations. Bohr's theory provides, as discussed above, a proper explanation of the combination principle.

CONCEPTUAL NOTE(S)

- If an electron in a single atom jumps to a level having principal quantum number n , then the maximum number of photons (emitted) will be $(n-1)$.
- If in a hydrogen like sample the electrons of many atoms jump to a level having principal quantum number n , then the maximum number of photons emitted = $C_2 = \frac{n(n-1)}{2}$.
- For example if many atoms are in third excited state ($n=4$), then the number of photons emitted for spectral lines seen = $\frac{4 \times 3}{2} = 6$.
- If an atom goes to excited state by absorbing certain energy, then it may emit a number of photons in succession; the sum of the energies of all emitted photons will be equal to the amount of energy absorbed. For example if an atom absorbing energy E reaches to an excited level and it returns to ground state by emitted wavelength λ_1, λ_2 and λ_3 , then

$$E = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} + \frac{hc}{\lambda_3}$$

- If an energetic electron strikes an electron (target electron) of an atom in ground state then electron jumps to an excited state by absorbing energy equal to the difference of ground state energy and excited state energy and the remaining energy is still carried by the incident (or striking) electron.
- The de-Broglie quantum condition: According to de-Broglie, only those orbits are allowed as stationary orbits in which circumference of orbit is equal to the integral multiple of de-Broglie wavelength.

$$\text{i.e. } 2\pi r = n\lambda$$

$$\Rightarrow 2\pi r = n \left(\frac{h}{mv} \right)$$

$$\Rightarrow mvr = n \left(\frac{h}{2\pi} \right)$$

This is same as Bohr's quantum condition.

- "An atom possesses discrete levels" was verified by Franck Hertz Experiment.

MERIT AND DEMERITS OF BOHR'S THEORY

The merits of Bohr's theory can hardly be overestimated. It saved physics at a time when it was in the grip of severe crisis. But, like every physical theory, this theory also has drawbacks. We enumerate below some of its merits and demerits.

MERITS

- The determination of the ratio of the mass of an electron to that of a proton in terms of the reduced mass concept of Bohr's theory agrees excellently with the value obtained by other methods.
- The general principle used by Bohr has also been successfully applied to a great number of phenomena such as the excitation and ionisation of atoms, X-ray spectra etc.
- The validity of the theory is further confirmed by the fact that the theory predicts new undiscovered series lines (spectral) which have later been actually observed.
- The theory has been instrumental to the discovery of heavy hydrogen (deuterium) by H.C. Urey.
- It gives a convincing explanation and a very simple and elegant picture of the origin of spectral lines.
- The agreement between the empirically determined value of the Rydberg constant and that evaluated by Bohr in terms of fundamental constants offers an excellent proof of the truth of Bohr's theory.

DEMERITS

- There is an ad hoc nature in the assumptions of Bohr in that the quantum idea of the stationary orbits is mixed up with the classical idea of coulomb force.
- The assumption of only circular orbits is utterly unjustified. In fact, Bohr in his original paper suggested that the orbit might be an ellipse instead of a circle.
- The spectral series, though agree excellently in case of hydrogen, are at variance with the theory for multi-electron atomic systems, e.g. the helium, singly ionised lithium etc. In these cases, it becomes necessary to introduce a magnetic quantum number.
- It cannot suggest any explanation whatsoever for the origin of the fine structure of the spectral lines.
- Bohr's theory is also unable to account for the multiple structure of spectral lines. For example, the doublet of sodium, triplets of magnesium etc. cannot be explained from Bohr's theory.
- It cannot make any calculation about the transitions or the selection rules which apply to them.
- It could not explain the splitting up of spectral lines when an atom is subjected to electric field (phenomenon called Stark Effect) or magnetic field (phenomenon called Zeeman Effect).

CRITICAL POTENTIAL

The resonance potential, the excitation potentials and the ionisation potentials are all included in the wider term critical potentials.

RESONANCE POTENTIAL

A minimum potential V is required to accelerate the bombarding electron to an energy V (in electron-volt) in order that an atom may be excited from its ground state to the next higher state. This potential is called the resonance potential.

EXCITATION POTENTIAL

The various values of the potential required to impart the necessary energy to excite an atom to different higher states are known as excitation potentials.

IONISATION POTENTIAL

The minimum potential necessary to supply the required energy to ionise an atom is called the ionisation potential or the first ionisation potential.

PROBLEM SOLVING TRICK(S)

Let us illustrate the above definitions by taking the case of hydrogen atom. For H atom we have :

$$E_n = -\frac{me^4}{8\epsilon_0^2 n^2 h^2} = -\frac{2.17 \times 10^{-18}}{n^2} \text{ J} = -\frac{13.6}{n^2} \text{ eV}$$

So the energy of the 1st, 2nd, 3rd, ..., ∞ -th orbits are respectively -13.6 eV , -3.4 eV , -1.51 eV , ..., 0 eV . Hence,

$$\text{Resonance potential} = -3.4 - (-13.6) = 10.2 \text{ eV}$$

$$\text{First excitation potential} = \text{resonance potential} = 10.2 \text{ eV}$$

$$\text{Second excitation potential} = -1.51 - (-13.6) = 12.09 \text{ eV}$$

$$\text{Ionisation potential} = 0 - (-13.6) = 13.6 \text{ eV}$$

Illustration 14

Find the quantum number n corresponding to excited state of He^+ ion if on transition to the ground state, the ion emits two photons in succession with wavelengths 108.5 nm and 30.4 nm . The ionization energy of H atom is 13.6 eV .

Solution

The energy transitions for the given wavelengths are

$$\Delta E_1 = \frac{12400}{\lambda_1} = \frac{12400}{1085} = 11.43 \text{ eV}$$

$$\Delta E_2 = \frac{12400}{\lambda_2} = \frac{12400}{304} = 40.79 \text{ eV}$$

$$\text{Total energy emitted } \Delta E = \Delta E_1 + \Delta E_2 = 52.22 \text{ eV}$$

$$\Rightarrow \Delta E = 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

where, ΔE is the energy emitted

$$\Rightarrow 52.34 = 13.6 \times 2^2 \left(\frac{1}{1^2} - \frac{1}{n^2} \right)$$

Thus, $n = 5$

Illustration 15

A hydrogen atom in a state of binding energy 0.85 eV makes a transition to a state of excitation energy of 10.2 eV. Find the energy and wavelength of photon emitted.

Solution

Since the binding energy is always negative, therefore,

$$E_i = -0.85 \text{ eV}$$

Let n_i be the initial binding state of the electron, then

$$E_n = -13.6 \frac{Z^2}{n_i^2}$$

$$\Rightarrow -0.85 = -13.6 \frac{Z^2}{n_i^2}$$

$$\Rightarrow n_i = 4$$

$$\text{Binding energy is } E_n = -\frac{13.6Z^2}{n^2}$$

$$\Rightarrow 0.85 \text{ eV} = \frac{-13.6(1)^2}{n_2^2}$$

$$\Rightarrow n_2 = 4$$

Let the electron now goes to an energy level n whose excitation energy is 10.2 eV. Since the excitation energy ΔE is defined with respect to ground state, therefore

$$E = 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$$

$$\Rightarrow 10.2 = 13.6 \times 1^2 \left(\frac{1}{1^2} - \frac{1}{n_f^2} \right)$$

$$\Rightarrow n_f = 2$$

So the electron makes a transition from energy level $n_i = 4$ to $n_f = 2$

Thus, the energy released is $\Delta E = E_4 - E_2$

$$\Rightarrow \Delta E = 13.6 \left[\frac{1}{2^2} - \frac{1}{4^2} \right] = 2.55 \text{ eV}$$

$$\text{Since, } \lambda = \frac{hc}{\Delta E} = \frac{12400}{2.55 \text{ eV}} = 5511 \text{ \AA}$$

Illustration 16

A particle of charge equal to that of an electron, $-e$ and mass 208 times the mass of electron (called a μ -meson) moves in a circular orbit around a nucleus of charge $+3e$

(take the mass of the nucleus to be infinite). Assuming that the Bohr model of the atom is applicable to this system;

- Calculate the radius of n th Bohr orbit
- Find the value of n , for which the radius of orbit is approximately the same as that of first Bohr orbit for the hydrogen atom;
- Find the wavelength of radiation emitted when the μ -meson jumps from the third orbit to first orbit (Rydberg's constant $= 1.097 \times 10^7 \text{ m}^{-1}$).

Solution

- Radius of the n th Bohr orbit for hydrogen atom is

$$r_n = 0.53 \frac{n^2}{Z}$$

$$\text{Since, } r \propto \frac{1}{m}$$

Radius of n th orbit for μ -meson is

$$r_n = \frac{0.53n^2}{(208)Z}$$

$$\Rightarrow r_n = (8.5 \times 10^{-4})n^2$$

- $(8.5 \times 10^{-4})n^2 = 0.53$

$$\Rightarrow n^2 = 623$$

$$\Rightarrow n \approx 25$$

- In case of hydrogen like atom,

$$\Delta E = E_3 - E_1 = 13.6Z^2 \left(1 - \frac{1}{3^2} \right) = 12.08 \text{ eV}$$

Since, $\Delta E \propto m$

So for μ -meson, we have

$$\Delta E = (12.8)(208) = 22.6 \text{ keV}$$

$$\Rightarrow \lambda = \frac{12400}{\Delta E} = \frac{12400}{22.6 \times 10^3} = 0.548 \text{ \AA}$$

Illustration 17

A hydrogen like atom of atomic number z is in an excited state of quantum number $2n$. It can emit a maximum energy photon of 204 eV. If it makes a transition to quantum state n , a photon of energy 40.8 eV is emitted. Find n , z and the ground state energy (in eV) for this atom. Also, calculate the minimum energy (in eV) that can be emitted by this atom during de-excitation. Ground state energy of hydrogen atom is 13.6 eV.

Solution

Given, $E_{2n} - E_1 = 204 \text{ eV}$

$$\Rightarrow (13.6)z^2 \left(1 - \frac{1}{4n^2} \right) = 204 \quad \dots(1)$$

Also, $E_{2n} - E_n = 40.8 \text{ eV}$

$$\Rightarrow 13.6z^2 \left(\frac{1}{n^2} - \frac{1}{4n^2} \right) = 40.8 \quad \dots(2)$$

Solving equations (1) and (2), we get

$$n=2 \text{ and } z=4$$

Since, $E_n = \frac{(-13.6)z^2}{n^2}$ eV, so we have

$$E_1 = (-13.6)(4)^2 \text{ eV}$$

$$\Rightarrow E_1 = -217.6 \text{ eV}$$

During de-excitation, minimum energy emitted is,

$$E_{\min} = E_{2n} - E_{2n-1} = E_4 - E_3$$

$$\Rightarrow E_{\min} = \frac{-217.6}{4^2} - \left(\frac{-217.6}{3^2} \right) = 10.58 \text{ eV}$$

Illustration 18

Calculate the energy of a H-atom in the first excited state, if the potential energy is assumed to be zero in the ground state.

Solution

Since, we know that, in ground state, $n=1$, we have

$$TE = -KE = \frac{PE}{2}$$

$$\Rightarrow PE = 2(TE) = 2(-13.6 \text{ eV})$$

$$\Rightarrow PE = -27.2 \text{ eV}$$

However, we have assumed this energy to be zero i.e., potential energy is increased by 27.2 eV. Since, kinetic energy in all energy states will remain unchanged whereas potential energy and hence, the total energy in all states will increase by 27.2 eV. Further, first excited state means $n=2$, so

$$E_2 = -3.4 \text{ eV (previously)}$$

$$\Rightarrow E'_2 = -3.4 + 27.2 = 23.8 \text{ eV (now)}$$

Illustration 19

An imaginary particle has a charge equal to that of an electron and mass 100 times the mass of the electron. It moves in a circular orbit around a nucleus of charge $+4e$. Take the mass of the nucleus to be infinite. Assuming that the Bohr's model is applicable to this system.

- Derive an expression for the radius of n th Bohr orbit.
- Find the wavelength of the radiation emitted when the particle jumps from fourth orbit to the second orbit.

Solution

$$(a) \text{ We have } \frac{m_p v^2}{r_n} = \frac{1}{4\pi\epsilon_0} \frac{ze^2}{r_n^2} \quad \dots(1)$$

The quantization of angular momentum gives,

$$m_p v r_n = \frac{nh}{2\pi} \quad \dots(2)$$

Solving equations (1) and (2), we get

$$r = \frac{n^2 h^2 \epsilon_0}{z\pi m_p e^2}$$

Substituting $m_p = 100 m$

where $m = \text{mass of electron and } z=4$

$$\text{we get, } r_n = \frac{n^2 h^2 \epsilon_0}{400\pi m e^2}$$

(b) As we know, $E_1^H = -13.60 \text{ eV}$

$$\text{and } E_n \propto \left(\frac{z^2}{n^2} \right) m$$

For the given particle,

$$E_4 = \frac{(-13.60)(4)^2}{(4)^2} \times 100 = -1360 \text{ eV}$$

$$\text{and } E_2 = \frac{(-13.60)(4)^2}{(2)^2} \times 100 = -5440 \text{ eV}$$

$$\Rightarrow \Delta E = E_4 - E_2 = 4080 \text{ eV}$$

$$\Rightarrow \lambda (\text{in } \text{\AA}) = \frac{12375}{\Delta E (\text{in eV})} = \frac{12375}{4080} = 3 \text{ \AA}$$

Illustration 20

Electrons in hydrogen-like atom ($Z=3$) make transitions from the fifth to the fourth orbit and from the fourth to the third orbit. The resulting radiations are incident normally on a metal plate and eject photoelectrons. The stopping potential for the photoelectrons ejected by the shorter wavelength is 3.95 V. Calculate the work function of the metal, and the stopping potential for the photoelectrons ejected by the longer wavelength (Rydberg constant $= 1.094 \times 10^7 \text{ m}^{-1}$).

Solution

The stopping potential for shorter wavelength is 3.95 V i.e., maximum kinetic energy of photoelectrons corresponding to shorter wavelength will be 3.95 eV. Further energy of incident photons corresponding to shorter wavelength will be in transition from $n=4$ to $n=3$.

$$E_{4 \rightarrow 3} = E_4 - E_3 = \frac{-(13.6)(3)^2}{(4)^2} - \left(\frac{-(13.6)(3)^2}{(3)^2} \right)$$

$$\Rightarrow E_{4 \rightarrow 3} = 5.95 \text{ eV}$$

Now, from the equation,

$$K_{\max} = E - W$$

we have $W = E - K_{\max} = E_{4 \rightarrow 3} - K_{\max}$

$$\Rightarrow W = (5.95 - 3.95) \text{ eV} = 2 \text{ eV}$$

Longer wavelength will correspond to transition from $n=5$ to $n=4$. From the relation, we get

$$\frac{1}{\lambda} = R z^2 \left(\frac{1}{N_1^2} - \frac{1}{N_2^2} \right)$$

The longer wavelength,

$$\frac{1}{\lambda} = (1.094 \times 10^7)(3)^3 \left(\frac{1}{16} - \frac{1}{25} \right)$$

$$\Rightarrow \lambda = 4.514 \times 10^{-7} \text{ m} = 4514 \text{ \AA}$$

Energy corresponding to this wavelength is

$$E = \frac{12375 \text{ eV \AA}}{4514 \text{ \AA}} = 2.74 \text{ eV}$$

So, maximum kinetic energy of photo-electrons is

$$K_{\max} = E - W = (2.74 - 2) \text{ eV} = 0.74 \text{ eV}$$

Hence, the stopping potential is 0.74 V.

Illustration 21

Hydrogen atom in its ground state is excited by means of monochromatic radiation of wavelength 975 Å. How many different lines are possible in the resulting spectrum? Calculate the longest wavelength amongst them. You may assume the ionization energy for hydrogen atom as 13.6 eV.

Solution

Energy corresponding to given wavelength is

$$E(\text{in eV}) = \frac{12375}{\lambda(\text{in \AA})} = \frac{12375}{975} = 12.69 \text{ eV}$$

Now, let the electron excites to n^{th} energy state. Then,

$$E_n - E_1 = 12.69$$

$$\Rightarrow \frac{(-13.6)}{(n^2)} - (-13.6) = 12.69$$

$$\Rightarrow n \approx 4$$

i.e., electron excites to 4th energy state

Total number of lines in emission spectrum would be

$$N = \frac{n(n-1)}{2} = \frac{4 \times 3}{2} = 6$$

Longest wavelength will correspond to the minimum energy and minimum energy is released in transition from $n = 4$ to $n = 3$.

$$E_{4-3} = E_4 - E_3 = \frac{-13.6}{(4^2)} - \left(\frac{-13.6}{(3)^2} \right) = 0.66 \text{ eV}$$

Hence, longest wavelength will be,

$$\lambda_{\max} = \frac{12375}{E(\text{in eV})} = \frac{12375}{0.66} \text{ \AA} = 1.875 \times 10^4 \text{ m}$$

$$\Rightarrow \lambda_{\max} = 1.875 \text{ \mu m}$$

X-RAYS

PRODUCTION

When fast moving electrons strike a target of high melting point and high atomic weight (like tungsten, platinum molybdenum), X-rays are produced. X-rays were discovered by Roentgen, therefore, they are also known as Roentgen rays. Roentgen discovered X-rays in 1895.

Coolidge modified the Roentgen tube. A modern X-ray tube consists of

- An electron source, preferably a filament heated by the passage of an electric current which may be varied.
- A heavy target of high melting point inclined at 45° to the path of electron beam, kept cooled by circulating cold water internally.
- A source of high potential difference applied across the filament and the target, keeping target positive with respect to filament. When the filament is heated, a fine beam of electrons strikes the target to produce X-rays.

CONTROL OF INTENSITY

The intensity of incident electrons determines the intensity of X-rays, i.e., greater is the number of electrons striking the target, more intense are the X-rays produced.

CONTROL OF PENETRATING POWER

The potential difference across the filament and target determines the energy and hence the penetrating power of X-rays.

NEED OF COOLING DEVICE

Only about 1% of incident electron's energy is converted into X-rays and the remaining 99% is converted into heat, therefore cooling device is essential with an X-ray tube.

HARD AND SOFT X-RAYS

X-rays upto 4 Å have high penetrating power and are called hard X-rays while those of $\lambda > 4 \text{ \AA}$ are called soft X-rays.

X-RAY SPECTRA CLASSIFICATION

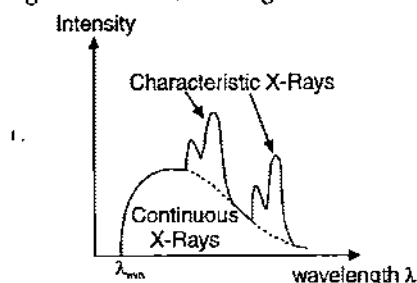
In X-ray tube, when high speed electrons strikes the target, they penetrate the target. They lose their kinetic energy and come to rest inside the metal. The electron before finally being stopped makes several collisions with the atoms in the target. At each collision one of the following two types of X-rays may get formed.

- Continuous X-rays
- Characteristic X-rays

A. CONTINUOUS SPECTRUM

It consists of radiations of all possible wavelengths within a definite wavelength range having a definite short wavelength limit. These are produced due to deceleration of electrons near heavy nucleus. These decelerated electrons emit radiations which lie in X-ray region. As the electrons suffer collisions at all angles, right from the glancing collision to the direct hit, they suffer varying decelerations

and hence radiations of all possible wavelengths within a certain range are emitted, forming the continuous spectrum.



The maximum limiting frequency $\nu_{\max}$ or minimum limiting wavelength $\lambda_{\min}$ is obtained when entire kinetic energy of bombarding electron is converted to X-ray energy. If V_0 is the accelerating potential, then

$$eV_0 = h\nu_{\max}$$

$$\Rightarrow \nu_{\max} = \frac{eV_0}{h}$$

In terms of $\lambda_{\min}$

$$eV_0 = \frac{hc}{\lambda_{\min}}$$

$$\lambda_{\min} = \frac{hc}{eV_0}$$

This relation gives the short wavelength limit of continuous X-ray spectrum.

If f is the fraction of kinetic energy of electrons converted into X-ray, then wavelength of emitted X-ray photon is given by

$$f(eV_0) = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{feV_0}$$

Illustration 22

Find the cut-off wavelength of the X-rays emitted by an X-ray tube operating at 30 kV.

Solution

For minimum wavelength, the total kinetic energy should be converted into an X-ray photon.

$$\text{Thus, } \lambda = \frac{hc}{E} = \frac{12400}{E} = \frac{12400}{30 \times 10^3} = 0.41 \text{ \AA}$$

Illustration 23

If an X-ray tube operates at the voltage of 10 kV, find the ratio of the de-Broglie wavelength of the incident electrons to the shortest wavelength of X-rays produced. The specific charge of electron is $1.8 \times 10^{11} \text{ Ckg}^{-1}$.

Solution

de Broglie wavelength (λ_d) when a charge q is accelerated by a potential difference of V volt is given by

$$\lambda_d = \frac{h}{\sqrt{2mqV}} \quad \dots(1)$$

For cut-off wavelength of X-rays, we have

$$qV = \frac{hc}{\lambda_m}$$

$$\Rightarrow \lambda_m = \frac{hc}{qV} \quad \dots(2)$$

From equations (1) and (2), we get

$$\frac{\lambda_d}{\lambda_m} = \frac{\sqrt{qV}}{c}$$

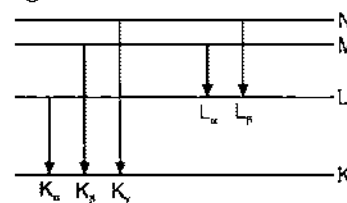
For electron, we have $\frac{q}{m} = 1.8 \times 10^{11} \text{ Ckg}^{-1}$

$$\Rightarrow \frac{\lambda_d}{\lambda_m} = \frac{\sqrt{1.8 \times 10^{11} \times 10 \times 10^3}}{3 \times 10^8} = 0.1$$

B. CHARACTERISTIC SPECTRUM

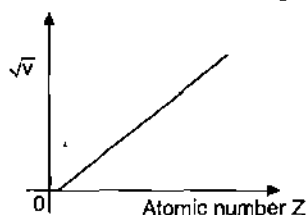
The minimum wavelength depends on the electron energy, but not on the target material. The line spectrum depends on the element used as target. These characteristic X-rays are produced when an electron knocks out an atomic electron from one of the inner levels. The ejected electron leaves a vacancy, which is then filled by an electron falling from a higher level. When an electron jumps from a higher energy orbit E_1 to lower energy orbit E_2 , it radiates energy ($E_1 - E_2$). Thus this energy difference is radiated in the form of X-rays of very small but definite wavelength which depends upon the target material. The X-ray spectrum consists of sharp lines and is called characteristic X-ray spectrum.

If the transitions are to the $n=1$ level, the X-rays are labelled $K_\alpha, K_\beta, \dots$. If they are to the $n=2$ level, they are labelled $L_\alpha, L_\beta, \dots$ etc. In the figure shown, the energy level diagram for an atom is drawn. The arrows indicate the transitions that give rise to the different series of X-rays.



In 1913, Moseley noted that the characteristic lines shifted systematically as the target material was changed. He plotted the square root of the frequency of the K_α line

versus the atomic number Z for many elements. The straight line he obtained is shown in the figure.



Moseley's plot did not pass through the origin. Let us see, why?. Once one of the two electrons in the $n=1$ level is ejected, an electron in the next highest level will drop to the lower state to fill the vacancy and in the process it emits the K_α frequency. For this electron the electric field due to the nucleus is screened by the remaining electron in the $n=1$ level. Moseley estimated that the effective nuclear charge for the K_α transition is $(Z-1)e$. Thus Moseley's Law for the frequency of the K_α line is

$$\sqrt{\nu_{K_\alpha}} = a(Z-1)$$

where $a = \sqrt{\frac{3}{4}} R c$ in which R is the Rydberg constant c is the speed of light.

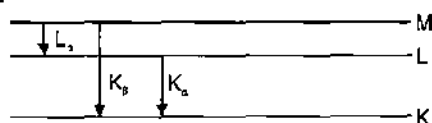
The wavelength of K - lines is given by

$$\frac{1}{\lambda} = (Z-1)^2 \left[1 - \frac{1}{n^2} \right] \text{ where } n = 2, 3, 4, \dots$$

Illustration 24

Show that the frequency of K_β X-ray of a material equals to the sum of frequencies of K_α and L_α X-rays of the same material.

Solution



The energy level diagram of an atom with one electron knocked out is shown above.

Energy of K_α X-ray is $E_{K_\alpha} = E_L - E_K$ of K_β X-ray is $E_{K_\beta} = E_M - E_K$

and, of L_α X-ray is $E_{L_\alpha} = E_M - E_L$

thus, $E_{K_\alpha} = E_{K_\beta} + E_{L_\alpha}$ or $\nu_{K_\beta} = \nu_{K_\alpha} + \nu_{L_\alpha}$

MOSELEY'S LAW : GENERAL TREATMENT

According to Moseley, for a given transition, the frequencies of characteristic lines emitted by different elements obey the relation

$$\nu = a(Z-b)^2$$

also called Moseley's Law.

where a and b are constants for a given transition. The constants a and b do not depend on the nature of target. Different values of b are as follows

$b=1$ for K_α - radiations of all substances.

$b=7.4$ for L_α - radiations of all substances.

$b=19.2$ for M_α - radiations of all substances.

If the bombarding electron knocks off an atomic electron from K shell and if the vacancy is filled in from $L, M, N, \dots$ shells the characteristics X-rays form K -series.

For K_α -line, $E_L - E_K = h\nu_{K_\alpha}$

For K_β -line, $E_M - E_K = h\nu_{K_\beta}$

Both these transitions are highly probable. Similarly, if bombarding electron knocks off atomic electron from L -shell and if the vacancy is filled in from $M, N, \dots$ shells, the characteristics X-rays form L -series and so on.

MOSELEY'S LAW : CONCLUSIONS REVISITED

- Moseley's Law supported Bohr's theory.
- It experimentally determined the atomic number (Z) of elements.
- This law established the importance of ordering of elements in periodic table by atomic number and not by atomic weight.
- Gaps in Moseley's data for $A=43, 61, 72, 75$ suggested existence of new elements which were later discovered.
- The atomic numbers of Cu, Ag and Pt were established to be 29, 47 and 78 respectively.
- When a vacancy occurs in the K -shell, there is still one electron remaining in the K -shell. An electron in the L -shell will feel an effective charge of $(Z-1)e$ due to $+Ze$ from the nucleus and $-e$ from the remaining K -shell electron, because L -shell orbit is well outside the K -shell orbit.
- Wave length of characteristic spectrum is

$$\frac{1}{\lambda} = R(Z-b)^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

- Energy of X-ray radiations is given by

$$\Delta E = h\nu = \frac{hc}{\lambda} = Rhc(Z-b)^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

- If transition takes place from $n_2=2$ to $n_1=1$ we get the K_α line, for which we have

$$(i) \quad a = \sqrt{\frac{3RC}{4}} = 2.47 \times 10^{15} \text{ Hz}$$

- Frequency is given by

$$\nu_{K_\alpha} = RC(Z-1)^2 \left(1 - \frac{1}{2^2} \right) = \frac{3RC}{4} (Z-1)^2$$

$$\Rightarrow \nu_{K_\alpha} = 2.47 \times 10^{15} (Z-1)^2 \text{ Hz}$$

$$(iii) E_{K\alpha} = 10.2(Z-1)^2 \text{ eV}$$

(iv) In general the wavelength of all the K-lines are

$$\text{given by } \frac{1}{\lambda_K} = R(Z-1)^2 \left(1 - \frac{1}{n^2}\right)$$

where $n = 2, 3, 4, \dots$

$$\text{While for } K_{\alpha} \text{ line } \lambda_{K\alpha} = \frac{1216}{(Z-1)^2} \text{ \AA}$$

Illustration 25

In Moseley's equation, we have $\sqrt{f} = a(z-b)$, where a and b are constants. Find their values with the help of the following data.

Element	z	Wavelength of K_{α} X-rays
Mo	42	0.71 \AA
Co	27	1.785 \AA

Solution

Since, According to Moseley's Law we have

$$\sqrt{f} = a(z-b)$$

$$\text{where } f = \frac{c}{\lambda}$$

So, for the first element, we have

$$\sqrt{\frac{c}{\lambda_1}} = a(z_1 - b) \quad \dots(1)$$

and for the second element, we have

$$\sqrt{\frac{c}{\lambda_2}} = a(z_2 - b) \quad \dots(2)$$

From equations (1) and (2), we get

$$\sqrt{c} \left(\frac{1}{\sqrt{\lambda_1}} - \frac{1}{\sqrt{\lambda_2}} \right) = a(z_1 - z_2) \quad \dots(3)$$

Now, we know that $c = 3 \times 10^8 \text{ ms}^{-1}$ and further it is given to us that

$$\lambda_1 = 0.71 \times 10^{-10} \text{ m}$$

$$\lambda_2 = 1.785 \times 10^{-10} \text{ m}$$

$$z_1 = 42 \text{ and}$$

$$z_2 = 27$$

Solving above three equations, we get

$$a = 5 \times 10^7 (\text{Hz})^{1/2} \text{ and } b = 1.37$$

Illustration 26

Determine the energy of the characteristic X-ray (K_{β}) emitted from a tungsten ($z = 74$) target when an electron drops from the M shell ($n = 3$) to a vacancy in the K shell ($n = 1$).

Solution

Energy associated with the electron in the K shell is approximately

$$E_K = -(74-1)^2 (13.6 \text{ eV}) = -72474 \text{ eV}$$

An electron in the M shell is subjected to an effective nuclear charge that depends on the number of electrons in the $n = 1$ and $n = 2$ states because these electrons shield the M electrons from the nucleus.

Since, there are eight electrons in the $n = 2$ state and one remaining in the $n = 1$ state, hence roughly nine electrons shield M electrons from the nucleus, so we have

$$z_{\text{eff}} = z - 9$$

Hence, the energy associated with an electron in the M shell is

$$E_M = \frac{-13.6 z_{\text{eff}}^2}{3^2} \text{ eV} = \frac{-13.6(z-9)^2}{3^2} \text{ eV}$$

$$\Rightarrow E_M = -\frac{(13.6)(74-9)^2}{9} \text{ eV} = -6384 \text{ eV}$$

Therefore, emitted X-ray has an energy equal to

$$E_M - E_K = [-6384 - (-72474)] \text{ eV} = 66090 \text{ eV}$$

Illustration 27

X-rays are incident on a target metal atom having 30 neutrons. The ratio of atomic radius of the target atom and ${}^4_2\text{He}$ is $(14)^{1/3}$.

(a) Find the atomic number of target atom.

(b) Find the frequency of K_{α} line emitted by this metal.

$$R = 1.1 \times 10^7 \text{ m}^{-1}, c = 3 \times 10^8 \text{ ms}^{-1}$$

Solution

(a) From the relation $r \propto A^{1/3}$

$$\Rightarrow \frac{r_2}{r_1} = \left(\frac{A_2}{A_1} \right)^{1/3}$$

$$\Rightarrow \left(\frac{A_2}{4} \right)^{1/3} = (14)^{1/3}$$

$$\Rightarrow A_2 = 56$$

(b) $Z_2 = A_2 - \text{number of neutrons}$

$$\Rightarrow Z_2 = 56 - 30 = 26$$

$$\text{Since, } f_{K\alpha} = Rc(Z-1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3Rc}{4} (Z-1)^2$$

Substituting the given values of R , c and Z , we get

$$f_{K\alpha} = 1.55 \times 10^{18} \text{ Hz}$$

Illustration 28

Characteristic X-rays of frequency 4.2×10^{18} Hz are produced when transitions from L-shell, K-shell take place in a certain target material. Use Mosley's Law to determine the atomic number of the target material. Given Rydberg constant $R = 1.1 \times 10^7 \text{ m}^{-1}$.

Solution

$$\Delta E = h\nu = Rhc(Z-b)^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For K-series, $b = 1$

$$\Rightarrow \nu = Rc(Z-1)^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Substituting the values, we get

$$4.2 \times 10^{18} = (1.1 \times 10^7)(3 \times 10^8)(Z-1)^2 \left(\frac{1}{1} - \frac{1}{4} \right)$$

$$\Rightarrow (Z-1)^2 = 1697$$

$$\Rightarrow Z-1 \approx 41$$

$$\Rightarrow Z = 42$$

PROPERTIES OF X-RAYS

- X-rays are electromagnetic waves of very short wavelength of order of $1 \text{ \AA}$. Therefore, they can exhibit properties of reflection, refraction, interference, diffraction, polarisation like ordinary light. Due to this property they help in the study of crystal structure.
- They travel in vacuum with speed of light i.e. $c = 3 \times 10^8 \text{ ms}^{-1}$
- They are electrically neutral, hence cannot be deflected by electric and magnetic fields.
- They do not possess magnetic moment.
- They have ionising power. Therefore, when they pass through a gas, the gas is ionised.
- They have penetrating power. They can penetrate light substances like wood, flesh, thick paper, thin sheets of metals, but cannot penetrate heavy substances like lead, calcium, barium sulphate etc.
- When incident on certain metals, they liberate electrons. This effect is called photoelectric effect.
- They cause fluorescence in many substances like barium, cadmium, zinc sulphide etc.
- They have destructive effect on living tissues. Therefore the persons working with X-rays often wear lead clothes.

BRAGG'S LAW

When an X-ray beam of wavelength λ is incident on a crystal of inter planar spacing d at grazing angle θ , then the directions of diffraction maxima are given by

$$2d \sin \theta = n\lambda \quad \dots (1)$$

where n is an integer, called order of maxima. Equation (1) is called Bragg's equation

$$\lambda = \frac{2d \sin \theta}{n}$$

For maximum wavelength

$$n_{\min} = 1 \text{ and } (\sin \theta)_{\max} = 1$$

$$\Rightarrow \lambda_{\max} = 2d$$

Hence equation (1) has solution only for $\lambda \leq 2d$.

INTENSITY OF TRANSMITTED X-RAY

The intensity of monochromatic X-ray beam after penetrating a thickness x of a target material is given by $I = I_0 e^{-\mu x}$, where μ is a constant called Absorption Coefficient. Its value depends upon nature of material. μ increases with increase of λ and atomic number Z of absorbing material. μ is maximum for lead.

ICE**BASED ON ATOMIC STRUCTURE & PROPERTIES**

(Solutions on page 4.99)

- A doubly ionised lithium atom is hydrogen like with atomic number 3.
 - Find the wavelength of the radiation require to excite the electron in Li^{++} from the first to the third Bohr orbit (ionisation energy of the hydrogen atom equals 13.6 eV).
 - How many spectral lines are observed in the emission spectrum of the above excited system.
- Find the ionization energy of a doubly ionized lithium atom.
- An electron and a proton are separated by a large distance and the electron approaches the proton with a kinetic energy of 2 eV . If the electron is captured by the proton to form a hydrogen atom in the ground state, what wavelength photon would be given off?
- Electrons in hydrogen like atoms ($Z = 3$) make transitions from the fifth of the fourth orbit and from the fourth to the third orbit. The resulting radiation are incident normally on a metal plate and eject photoelectrons. The stopping potential for the photoelectrons ejected by the shorter wavelength is 3.95 volts . Calculate the work function of the metal and the stopping potential for the photoelectrons ejected by the longer wavelength.
- Use Moseley's Law with $b = 1$ to find the frequency of the K_α X-rays of $\text{La} (Z = 57)$ if the frequency of the K_α X-rays of $\text{Cu} (Z = 29)$ is known to be $1.88 \times 10^{18} \text{ Hz}$.
- When the voltage applied to an X-ray tube is increased $\eta = 1.5$ times, the short wave limit of an X-ray continuous spectrum shifts by $\Delta \lambda = 26 \text{ pm}$. Find the initial voltage applied to the tube.

7. Find the cutoff wavelength for the continuous X-rays coming from an X-ray tube operating at 40 kV.

8. Certain gas of identical hydrogen like atoms has all its atoms in a particular upper energy level. The atoms make transition to a higher energy level when a monochromatic radiation, having wavelength $1654 \text{ \AA}$, is incident upon it. Subsequently, the atoms emit radiation of only three different photon energies.

- Identify the atom
- Obtain the ionization energy for the gas atoms.
- If the atoms of the gas are to be excited to such a level which gives radiation of only six different photon energies, what should be energy of incident radiation.

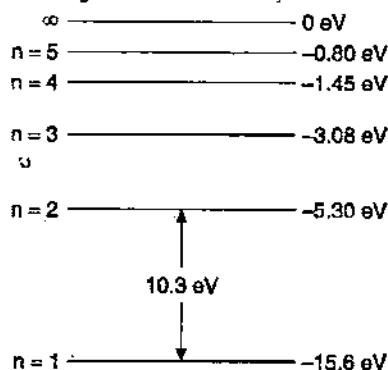
9. Electrons are emitted from an electron gun at almost zero velocity and are accelerated by an electric field E through a distance of 1 m . The electrons are now scattered by an atomic hydrogen sample in ground state. What should be the minimum value of E so that red light of wavelength $6563 \text{ \AA}$ may be emitted in the hydrogen?

10. A hot gas emits radiation of wavelengths $460 \text{ \AA}$, $831 \text{ \AA}$ and $1035 \text{ \AA}$ only. Assume that the atoms have only two excited states and the difference between consecutive energy levels decreases as energy is increased. Taking the energy of the highest energy state to be zero. Find the energies of the ground state and the first excited state.

11. A mixture of hydrogen atoms (in their ground state) and hydrogen like ions (in their first excited state) are being excited by electrons which have been accelerated by same potential difference V volts. After excitation when they come directly into ground state, the wavelengths of emitted light are found in the ratio $5:1$. Calculate the minimum value of V for which both the atoms get excited after collision with electrons. Also find the atomic number of other ion and the energy of emitted light by hydrogen atoms and ions.

12. A stationary He^+ emitted a photon corresponding to the first line of Lyman series. This photon liberated a photoelectron from a stationary hydrogen atom in the ground state. Find the velocity of the photoelectron.

13. The energy levels of a hypothetical one electron atom are shown in the figure.



- Find the ionization potential of this atom.
- Find the short wavelength limit of the series terminating at $n=2$.
- Find the excitation potential for the state $n=3$.

(d) Find wave number of the photon emitted for the transition $n=3$ to $n=1$.

(e) What is the minimum energy that an electron will have after interacting with this atom in the ground state if the initial kinetic energy of the electron is

- 6 eV
- 11 eV

14. Using the known values for hydrogen atom, calculate

- radius of third orbit for Li^{+2}
- speed of electron in fourth orbit for He^+ .

15. Find an expression for the magnetic dipole moment and magnetic field induction at the center of a Bohr's hypothetical hydrogen atom in the n th orbit of the electron in terms of universal constants.

16. A small particle of mass m moves in such a way that the potential energy $U = ar^2$ where a is a constant and r is the distance of the particle from the origin. Assuming Bohr's model of quantization of angular momentum and circular orbits, find the radius of n th allowed orbit.

17. (a) If two times the λ_{min} of continuous X-ray spectra of target atom A at 34.3 kV is same as the wavelength of K_{α} line of target atom B at 40 kV, then determine the atomic number of the atom B.

(b) Stopping potentials of 24, 100, 110 and 115 kV are measured for photoelectrons emitted from a certain element when it is radiated with monochromatic X-ray. If this element is used as a target in an X-ray tube, what will be the wavelength of K_{α} line?

18. If potential energy in first orbit is taken to be zero, then find the kinetic energy, potential energy and total energy in first and second orbit of hydrogen atom.

19. Wavelengths belonging to Balmer series lying in the range of 450 nm to 750 nm were used to eject photoelectrons from a metal surface whose work function is 2 eV. Find (in eV) the maximum kinetic energy of the emitted photoelectrons. Take $hc = 1242 \text{ eV nm}$.

20. The potential energy of a particle varies as :

$$U(x) = \begin{cases} E_0 & 0 \leq x \leq 1 \\ 0 & x > 1 \end{cases}$$

For $0 \leq x \leq 1$, the de-Broglie wavelength is λ_1 and for $x > 1$, the de-Broglie wavelength is λ_2 . Total energy of the particle is $2E_0$. Find $\frac{\lambda_1}{\lambda_2}$.

21. An electron in a hydrogen like atom is in an excited state. It has a total energy of -3.4 eV . Calculate.

- the kinetic energy,
- the de-Broglie wavelength of the electron.

Atomic & Nuclear Physics

(b) Nuclear Physics

NUCLEAR STRUCTURE

The nucleus of an atom consists of two types of particles, protons and neutrons. A proton has a positive charge equal to 1.6×10^{-19} C and a mass equal to 1.6726×10^{-27} kg. A neutron has no charge and its mass is 1.6749×10^{-27} kg. Thus a neutron is slightly heavier than a proton. Protons or neutrons, being the particles present inside nucleus, together are called Nucleons.

The total number of protons in the nucleus is called its atomic number (Z). The total number of nucleons in the nucleus is called its mass number (A). If N is the number of neutrons, then,

$$A = Z + N$$

No electrons are present inside the nucleus.

If X is the chemical symbol for an element then its nucleus is represented as ${}_Z^AX$ or as ${}_ZX^A$

ISOTOPES

Nuclides having the same charge number (Z) but different mass number (A) are called isotopes. All the isotopes are chemically similar and hence they occupy the same position in the periodic table.

ISOBARS

Nuclides having the same mass number (A) but different atomic number (Z) are called isobars.

ISOTONES

Nuclides having the same neutron number ($A - Z$) but different mass number (A) are called isotones.

ATOMIC MASS UNIT (u OR AMU)

Atomic and nuclear masses are generally expressed in terms of atomic mass unit (a.m.u.).

$$1 \text{ u} = \frac{1}{12} (\text{mass of one atom of } {}_6\text{C}^{12}) = 1.66056 \times 10^{-27} \text{ kg}$$

NUCLEAR RADIUS

Assuming that the nuclei are spherical, their radii are well represented by the empirical formula

$$R = R_0 A^{1/3}$$

where $R_0 = 1.1 \times 10^{-15} \text{ m} = 1.1 \text{ fermi (fm)}$

NUCLEAR DENSITY

The density of a nucleus of mass M and mass number A can be written as

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{A \text{ amu}}{\frac{4}{3}\pi (R_0 A^{1/3})^3} = \frac{(A \times 1.67 \times 10^{-27}) \text{ kg}}{\frac{4}{3}\pi (1.1 \times 10^{-15} \text{ m})^3 A}$$

$$\Rightarrow \rho \approx 2.9 \times 10^{17} \text{ kgm}^{-3}$$

This comes out to be $\sim 10^{17} \text{ kgm}^{-3}$, which is extremely large as compared to the density of ordinary matter which is $\sim 10^3 \text{ kgm}^{-3}$.

CONCEPTUAL NOTE(S)

Since density is independent of mass number A , so all nuclei same density. So, whether two nuclei are isobars, or isotopes or isotones. They must possess same density.

THE NUCLEAR FORCE

The force which binds the protons and neutrons inside the nucleus is neither electrical nor gravitational. It is an entirely different kind of force called the strong nuclear force. This force is extremely complex in nature. Some of its main characteristics are mentioned below.

- Nuclear forces are attractive in nature. Their magnitude, which depends upon inter nucleon distance is of very high order.

- b) Nuclear forces are charge independent. Nature of force remains the same whether we consider force between two protons, between two neutrons or between a proton and a neutron.
- c) These are short range forces. Nuclear forces operate between two nucleons situated in close neighbourhood only.
- d) Nuclear forces decrease very quickly with distance between two nucleons. Their rate of decrease is much more rapid than that of inverse square law forces (Coulombic forces). The forces become negligible when the nucleons are more than 10^{-14} cm apart.
- e) Nuclear forces are spin dependent. Nucleons having parallel spin are more strongly bound to each other than those having anti-parallel spin.

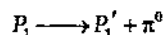
YUKAWA THEORY OR MESON THEORY OF NUCLEAR FORCES

According to this theory, a nucleon consists of a core surrounded by a cloud of mesons, which may be charged or neutral. The mesons constantly get exchanged, back and forth, between two neighbouring nucleons. In this process the two nucleons remain bound to each other.

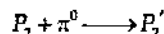
A. PROTON-PROTON INTERACTION

It is the force between two neighbouring protons. It is due to the exchange of π^0 meson between them. It is represented in the form of a reaction as follows.

Proton P_1 emits π^0 and gets converted into a proton P_1' , having different co-ordinates. So,

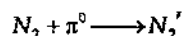
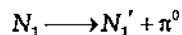


This π^0 is absorbed by P_2 which also gets converted into a new proton P_2' . Hence,



B. NEUTRON-NEUTRON INTERACTION

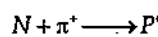
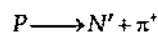
It is the force between two neutrons. It is also due to exchange of π^0 between them



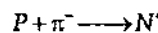
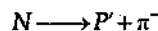
C. PROTON-NEUTRON INTERACTION

It is the force between a proton and a neutron situated close to each other. It can be take place in following two ways.

- a) Due to exchange of π^+ meson.
- Proton emits π^+ meson and gets converted to a neutron. Another neutron absorbs this π^+ meson to get converted to a proton. So,



- b) Due to exchange of π^- meson.
- π^- meson is emitted by the neutron which is absorbed by the proton.



DOG-BONE ANALOGY

The above interactions can be explained with the dog bone analogy according to which we consider the two interacting nucleons to be two dogs having a common bone clenched in between their teeth very firmly. Each one of these dogs wants to take the bone and hence they cannot be separated easily. They seem to be bound to each other with a strong attractive force (which is the bone) though the dogs themselves are strong enemies. The meson plays the same role of the common bone in between two nucleons.

RADIOACTIVITY

The phenomenon of spontaneous emission of radiations (α , β , γ etc.) by certain nuclei is called radioactivity.

The phenomenon of radioactivity was discovered by Becquerel.

LAWS OF RADIOACTIVE DISINTEGRATION

A. RUTHERFORD-SODDY LAWS

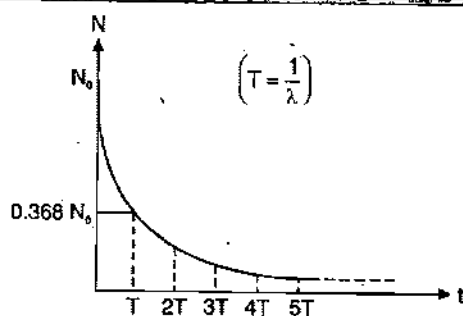
- Radioactivity is nuclear disintegration phenomenon. It is independent of all physical and chemical conditions.
- The disintegration is random and spontaneous statistical process. It is a matter of chance for any atom to disintegrate first.
- The radioactive substances emit α or β particles along with γ -rays. These rays originate from the nuclei of disintegrating atoms and form fresh radioactive products with different physical and chemical properties.

The rate of decay of nuclei $\left(-\frac{dN}{dt}\right)$ is directly proportional to the number of undecayed nuclei (N) in the sample at time t .

$$\Rightarrow -\frac{dN}{dt} \propto N$$

$$\Rightarrow \frac{dN}{dt} = -\lambda N$$

where λ is constant of proportionality called Decay Constant or Disintegration Constant.



$$\Rightarrow \int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt$$

$$\Rightarrow \log_e \left(\frac{N}{N_0} \right) = -\lambda t$$

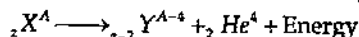
$$\Rightarrow N = N_0 e^{-\lambda t}$$

where, N is the number of un-decayed nuclei in the sample at time t and N_0 is the number of un-decayed nuclei in the sample at time $t=0$ (initially).

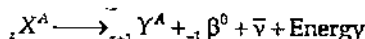
So, we conclude that the number of un-decayed nuclei in the sample decays exponentially with time.

B. DISPLACEMENT LAWS

- a) When a nuclide emits α -particle, its mass number is reduced by four and atomic number by two, i.e.,

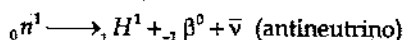


- b) When a nuclide emits a β -particle, its mass number remains unchanged but atomic number increases by one, i.e.,

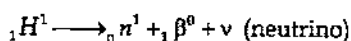


where $\bar{\nu}$ is the antineutrino

The β -particle is not present initially in the nucleus but is produced due to disintegration of neutron into a proton, i.e.,



When a proton is converted into a neutron, positive β -particle or positron is emitted.



- c) When a nuclide emits a gamma photon, neither the atomic number nor the mass number changes.

HALF LIFE ($T_{1/2}$)

The half life period of a radioactive substance is defined as the time in which one-half of the radioactive substance is disintegrated. If N_0 is initial number of radioactive atoms present, then in a half life time $T_{1/2}$, the number of undecayed radioactive atoms will be $\frac{N_0}{2}$ and in next half life

$\frac{N_0}{4}$ and so on.

CONCEPTUAL NOTE(S)

So, at $t = T_{1/2}$, $N = \frac{N_0}{2}$

$$N_0 \xrightarrow{T_{1/2}} \frac{N_0}{2} \xrightarrow{T_{1/2}} \frac{N_0}{2^2} \xrightarrow{T_{1/2}} \frac{N_0}{2^3} \xrightarrow{\dots} \xrightarrow{nT_{1/2}} \frac{N_0}{2^n}$$

$$\text{where } n = \frac{t}{T_{1/2}} = \frac{\text{Time Lapsed}}{\text{Half Life}}$$

Since $N = N_0 e^{-\lambda t}$

$$\Rightarrow \frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

$$\Rightarrow e^{-\lambda T_{1/2}} = \frac{1}{2}$$

$$\Rightarrow \lambda T_{1/2} = \log_e 2 = 0.693$$

$$\Rightarrow T_{1/2} = \frac{0.693}{\lambda}$$

Table : Fraction of active/decayed atom at different time

Time (t)	Remaining fraction of active atoms (N/N_0) probability of survival	Fraction of atoms decayed $\frac{N_0 - N}{N_0}$ probability of decay
$t = 0$	1 (= 100%)	0
$t = T_{1/2}$	$\frac{1}{2}$ (= 50%)	$\frac{1}{2}$ (= 50%)
$t = 2T_{1/2}$	$\frac{1}{4}$ (= 25%)	$\frac{3}{4}$ (= 75%)
$t = 3T_{1/2}$	$\frac{1}{8}$ (= 12.5%)	$\frac{7}{8}$ (= 87.5%)
$t = 10(T_{1/2})$	$\left(\frac{1}{2}\right)^{10} \approx 0.1\%$	$\approx 99.9\%$
$t = nT_{1/2}$	$\left(\frac{1}{2}\right)^n$	$\left\{1 - \left(\frac{1}{2}\right)^n\right\}$

Illustration 29

At a given instant there are 25% undecayed radioactive nuclei in a sample. After 10 seconds the number of undecayed nuclei reduces to 12.5%. Calculate

- mean life of the nuclei
- the time in which the number of undecayed nuclei will further reduce to 6.25% of the reduced number.

Solution

- In 10 s, number of nuclei has been reduced to half (25% to 12.5%).

Therefore, its half-life is

$$t_{1/2} = 10 \text{ s}$$

Relation between half-life and mean life is

$$t_{\text{mean}} = \frac{t_{1/2}}{\log_e(2)} = \frac{0.693}{0.693} \text{ s}$$

$$\Rightarrow t_{\text{mean}} = 14.43 \text{ s}$$

- (b) From initial 100% to reduction till 6.25%, it takes four half lives.

$$100\% \xrightarrow{t_{1/2}} 50\% \xrightarrow{t_{1/2}} 25\% \xrightarrow{t_{1/2}} 12.5\% \xrightarrow{t_{1/2}} 6.25\%$$

$$\Rightarrow t = 4t_{1/2} = 4(10) \text{ s} = 40 \text{ s}$$

$$\Rightarrow t = 40 \text{ s}$$

MEAN LIFE (τ)

The mean life or average life of a radioactive substance is equal to the average time for which the nuclei of atoms of the radioactive substance exist.

The average life of a sample can be calculated by finding the total life of all the nuclei of the substance and then dividing it by the total number of nuclei present in the sample initially. Mathematically

$$\tau = \frac{\int_0^{\infty} t dN}{\int_0^{\infty} dN} = \frac{1}{N_0} \int_0^{\infty} t dN = \frac{N_0}{N_0}$$

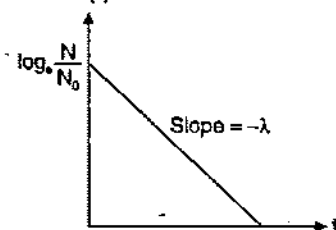
$$\Rightarrow \tau = \frac{1}{\lambda}$$

CONCEPTUAL NOTE(S)

- a) From $N = N_0 e^{-\lambda t}$, we get

$$\log_e \left(\frac{N}{N_0} \right) = -\lambda t$$

So, the slope of the line shown in the graph i.e. the magnitude of inverse of slope of $\log_e \left(\frac{N}{N_0} \right)$ vs t curve is known as mean life (τ).



- b) From $N = N_0 e^{-\lambda t}$, if $t = \frac{1}{\lambda} = \tau$

$$\Rightarrow N = N_0 e^{-1} = N_0 \left(\frac{1}{e} \right) = 0.37 N_0 = 37\% \text{ of } N_0$$

i.e. mean life is the time interval in which number of undecayed atoms (N) becomes $\frac{1}{e}$ times or 0.37 times or 37% of original number of atoms. or

It is the time in which number of decayed atoms ($N_0 - N$) becomes $\left(1 - \frac{1}{e} \right)$ times or 0.63 times or 63% of original

number of atoms.

$$\text{c) Since, } T_{1/2} = \frac{0.693}{\lambda}$$

$$\Rightarrow \frac{1}{\lambda} = \tau = \frac{1}{0.693} (T_{1/2}) = 1.44 (T_{1/2})$$

i.e. mean life is about 44% more than that of half life. Which gives us $\tau > T_{1/2}$

So, we conclude that

$$T_{1/2} = 0.693 \tau$$

$$\Rightarrow T_{1/2} < \tau$$

Illustration 30

The mean lives of a radio-active substances are 1620 years and 405 years for α emission and β emission respectively. Find out the time during which three-fourth of a sample will decay if it is decaying both by α emission and β emission simultaneously.

Solution

Let at some instant of time t , number of atoms of the radioactive substance are N . It may decay either by α emission or by β emission. So, we can write,

$$\left(-\frac{dN}{dt} \right)_{\text{net}} = \left(-\frac{dN}{dt} \right)_{\alpha} + \left(-\frac{dN}{dt} \right)_{\beta}$$

If the effective decay constant is λ , then

$$\lambda N = \lambda_{\alpha} N + \lambda_{\beta} N$$

$$\Rightarrow \lambda = \lambda_{\alpha} + \lambda_{\beta} = \frac{1}{1620} + \frac{1}{405}$$

$$\Rightarrow \lambda = \frac{1}{324} \text{ year}^{-1}$$

$$\text{Since, } N = N_0 e^{-\lambda t}$$

$$\Rightarrow \frac{N_0}{4} = N_0 e^{-\lambda t}$$

$$\Rightarrow -\lambda t = \log_e \left(\frac{1}{4} \right) = -1.386$$

$$\Rightarrow \left(\frac{1}{324} \right) t = 1.386$$

$$\Rightarrow t = 449 \text{ year}$$

Illustration 31

A radioactive nucleus can decay by two different processes, the half life for the first process is t_1 and that for the second process is t_2 . Show that the effective half-

life t of the nucleus is given by $\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}$.

Solution

Let at any instant t , number of nuclei in radioactive sample be N . Then it may decay by either of two different processes. So,

$$-\frac{dN}{dt} = -\left(\frac{dN}{dt}\right)_1 - \left(\frac{dN}{dt}\right)_2$$

$$\Rightarrow \lambda N = \lambda_1 N + \lambda_2 N$$

$$\Rightarrow \lambda = \lambda_1 + \lambda_2$$

$$\Rightarrow \frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \quad \left\{ \because t_{1/2} = t = \frac{0.693}{\lambda} \right\}$$

Illustration 32

Prove mathematically that mean life or average life of a radioactive substance is $t_{av} = \frac{1}{\lambda}$.

Solution

Let N be the number of atoms that exist at time t . Between t and $t + dt$ let dN atoms are decayed, then Mean or average life is

$$t_{av} = \frac{\int_0^{\infty} t dN}{\int_0^{\infty} dN}$$

$$\text{Since, } -\frac{dN}{dt} = \lambda N$$

$$\Rightarrow dN = -\lambda N dt$$

$$\Rightarrow t_{av} = \frac{-\int_0^{\infty} t \lambda N dt}{-N_0}$$

$$\text{Since, } N = N_0 e^{-\lambda t}, \text{ so}$$

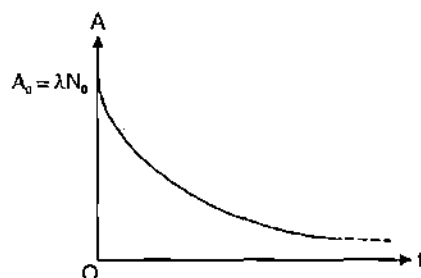
$$t_{av} = \frac{\int_0^{\infty} t \lambda N_0 e^{-\lambda t} dt}{N_0}$$

This integration is done by parts to get the result

$$t_{av} = \frac{1}{\lambda}$$

ACTIVITY OF RADIOACTIVE SUBSTANCE (A)

The activity of a radioactive substance means the rate of decay (or the number of disintegration/sec). This is denoted by



$$A = -\frac{dN}{dt} = -\frac{d}{dt}(N_0 e^{-\lambda t}) = \lambda N_0 e^{-\lambda t} = A_0 e^{-\lambda t} \text{ where}$$

$$A_0 = (\lambda N_0) \text{ is the activity at time } t = 0.$$

So, activity of a radioactive sample decreases exponentially with time.

UNITS OF RADIOACTIVITY

A. BACQUEREL

In S.I. system the unit of radioactivity is bacquerel.

1 bacquerel = 1 disintegration/sec = 1 dps.

B. RUTHERFORD

It is defined as a unit of activity equal to 10^6 dps.

C. CURIE

The traditional unit of activity is curie. It is defined as 3.7×10^{10} dps which is also equal to the radioactivity of 1 g of pure Radium.

Illustration 33

The half-life of Cobalt 60 is 5.25 years. How long after its activity have decreased to about one-eighth of its original value?

Solution

The activity is proportional to the number of undecayed atoms.

In each half-life, half the remaining sample decays.

Since, $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$, therefore, three half-lives or

15.75 yr are required for the sample to decay to $\frac{1}{8}$ th its original strength.

Illustration 34

A count rate-meter is used to measure the activity of a given sample. At one instant the meter shows 4750 counts per minute. Five minutes later it shows 2700 counts per minute.

(a) Find the decay constant

(b) Also, find the half life of the sample

Solution

$$\text{Initial velocity } A_i = \left. \frac{dN}{dt} \right|_{t=0} = \lambda N_0 = 4750 \quad \dots(1)$$

$$\text{Final velocity } A_f = \left. \frac{dN}{dt} \right|_{t=5} = \lambda N = 2700 \quad \dots(2)$$

Dividing (1) by (2), we get

$$\frac{4750}{2700} = \frac{N_0}{N_t}$$

The decay constant is given by

$$\lambda = \frac{2.303}{t} \log_e \left(\frac{N_0}{N_t} \right)$$

$$\Rightarrow \lambda = \frac{2.303}{5} \log_e \left(\frac{4750}{2700} \right) = 0.113 \text{ min}^{-1}$$

Half life of the sample is

$$T = \frac{0.693}{\lambda} = \frac{0.693}{0.113} = 6.14 \text{ min}$$

Illustration 35

The half-life of radium is 1620 years. How many radium atoms decay in 1 s in a 1 g sample of radium. The atomic weight of radium is 226 gmol^{-1} .

Solution

Number of atoms in 1 g sample is

$$N = \left(\frac{0.001}{226} \right) (6.02 \times 10^{26}) = 2.66 \times 10^{21} \text{ atoms}$$

The decay constant is

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{(1620)(3.16 \times 10^7)} = 1.35 \times 10^{-11} \text{ s}^{-1}$$

Taking one year = $3.16 \times 10^7 \text{ s}$

$$\text{Now, } \frac{\Delta N}{\Delta t} = \lambda N = (1.35 \times 10^{-11})(2.66 \times 10^{21}) = 3.6 \times 10^{10} \text{ s}^{-1}$$

Thus, 3.6×10^{10} nuclei decay in one second.

Illustration 36

A F^{32} radio nuclide with half life $T = 14.3$ days is produced in a reactor at a constant rate $q = 2 \times 10^9$ nuclei per second. How soon after the beginning of production of that radio nuclide will its activity be equal to $R = 10^9$ disintegrations per second?

Solution

$$N = \frac{R}{\lambda} = \frac{10^9}{\frac{0.693}{14.3 \times 3600}} = 7.43 \times 10^{13}$$

$$\text{Now, } \frac{dN}{dt} = q - \lambda N$$

$$\Rightarrow \int_0^N \frac{dN}{q - \lambda N} = \int_0^t dt$$

$$\Rightarrow N = \frac{q}{\lambda} (1 - e^{-\lambda t})$$

Substituting the values, we get

$$7.43 \times 10^{13} = \frac{2 \times 10^9}{\left(\frac{0.693}{14.3 \times 3600} \right)} \left(1 - e^{-\left(\frac{0.693}{14.3 \times 3600} \right) t} \right)$$

Solving this equation we get $t = 14.3 \text{ hr}$.

Illustration 37

There is a stream of neutrons with a kinetic energy of 0.0327 eV. If the half life of neutrons be 700 s, what fraction of neutrons will decay before they travel a distance of 10 m? Mass of neutron equal to $1.675 \times 10^{-27} \text{ kg}$.

Solution

$$\text{Since, } K = \frac{1}{2} mv^2$$

$$\Rightarrow v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 0.0327 \times 1.6 \times 10^{-19}}{1.675 \times 10^{-27}}} = 2.5 \times 10^3 \text{ ms}^{-1}$$

$$\Rightarrow t = \frac{s}{v} = \frac{10}{2.5 \times 10^3} = 4 \times 10^{-3} \text{ s}$$

$$\text{Fraction decayed is } \frac{N}{N_0} = \frac{N_0(1 - e^{-\lambda t})}{N_0} = 1 - e^{-\lambda t}$$

$$\Rightarrow \frac{N}{N_0} = 1 - e^{-0.693 \times 4 \times 10^{-3} / 700}$$

$$\Rightarrow \frac{N}{N_0} = 3.96 \times 10^{-6}$$

Illustration 38

A radionuclide X is produced at constant rate α . At time $t = 0$, number of nuclei of X are zero. Find

- the maximum number of nuclei of X .
- the number of nuclei at time t .

Decay constant of X is λ .

Solution

- Let N be the number of nuclei of X at time t .



Rate of formation of $X = \alpha$ [given]

Rate of disintegration = λN

Number of nuclei of X will increase until both the rates will become equal.

Therefore,

$$\alpha = \lambda N_{\max}$$

$$\Rightarrow N_{\max} = \frac{\alpha}{\lambda}$$

- Net rate of formation of X at time t is,

$$\frac{dN}{dt} = \alpha - \lambda N$$

$$\Rightarrow \frac{dN}{\alpha - \lambda N} = dt$$

Integrating with proper limits, we have

$$\int_0^N \frac{dN}{\alpha - \lambda N} = \int_0^t dt$$

$$\Rightarrow N = \frac{\alpha}{\lambda} (1 - e^{-\lambda t})$$

This expression shows that number of nuclei of X are increasing exponentially from 0 to $\frac{\alpha}{\lambda}$.

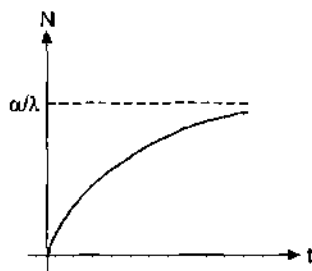


Illustration 39

Natural uranium is a mixture of three isotopes $^{234}_{92}\text{U}$, $^{235}_{92}\text{U}$ and $^{238}_{92}\text{U}$ with mass percentage 0.01%, 0.71% and 99.28% respectively. The half life of three isotopes are 2.5×10^5 years, 7.1×10^8 years and 4.5×10^9 years respectively. Determine the share of radioactivity of each isotope into the total activity of the natural uranium.

Solution

Let R_1 , R_2 and R_3 be the activities of $^{234}_{92}\text{U}$, $^{235}_{92}\text{U}$ and $^{238}_{92}\text{U}$ respectively.

$$\text{Total activity } R = R_1 + R_2 + R_3$$

$$\text{Share of } ^{234}_{92}\text{U}, \frac{R_1}{R} = \frac{\lambda_1 N_1}{\lambda_1 N_1 + \lambda_2 N_2 + \lambda_3 N_3}$$

Let m be the total mass of natural uranium.

$$\text{Then } m_1 = \frac{0.01}{100}m, m_2 = \frac{0.71}{100}m \text{ and } m_3 = \frac{99.28}{100}m$$

$$\text{Now, } N_1 = \frac{m_1}{M_1}, N_2 = \frac{m_2}{M_2} \text{ and } N_3 = \frac{m_3}{M_3}$$

where M_1 , M_2 and M_3 are atomic weights.

$$\begin{aligned} \Rightarrow \frac{R_1}{R} &= \frac{\left(\frac{m_1}{M_1}\right) \frac{1}{T_1}}{\frac{m_1}{M_1} \frac{1}{T_1} + \frac{m_2}{M_2} \frac{1}{T_2} + \frac{m_3}{M_3} \frac{1}{T_3}} \\ &= \frac{\frac{0.01/100}{234} \times \frac{1}{2.5 \times 10^5 \text{ years}}}{\left(\frac{0.01/100}{234}\right) \left(\frac{1}{2.5 \times 10^5}\right) + \left(\frac{0.71/100}{235}\right) \left(\frac{1}{7.1 \times 10^8}\right) + \left(\frac{99.28/100}{238}\right) \left(\frac{1}{4.5 \times 10^9}\right)} \end{aligned}$$

$$\Rightarrow \frac{R_1}{R} \times 100\% = 0.648 \approx 64.8\%$$

Similarly, share of $^{235}_{92}\text{U}$ is 0.016%

and of $^{238}_{92}\text{U}$ is 35.184%

Illustration 40

Uranium ores on the earth at the present time typically have a composition consisting of 99.3% of the isotope $^{238}_{92}\text{U}$ and 0.7% of the isotope $^{235}_{92}\text{U}$. The half lives of these isotopes are 4.47×10^9 y and 7.04×10^8 y respectively. If these isotopes were equally abundant when the earth was formed, estimate the age of the earth.

Solution

Let N_0 be number of atoms of each isotope at the time of formation of the earth ($t=0$) and N_1 and N_2 the number of atoms at present ($t=t$). Then

$$N_1 = N_0 e^{-\lambda_1 t} \quad \dots(1)$$

$$\text{and } N_2 = N_0 e^{-\lambda_2 t} \quad \dots(2)$$

$$\Rightarrow \frac{N_1}{N_2} = e^{(\lambda_2 - \lambda_1)t} \quad \dots(3)$$

Further it is given that

$$\frac{N_1}{N_2} = \frac{99.3}{0.7} \quad \dots(4)$$

Equating (3) and (4) and taking log both sides, we get

$$(\lambda_2 - \lambda_1)t = \log_e \left(\frac{99.3}{0.7} \right)$$

$$\Rightarrow t = \left(\frac{1}{\lambda_2 - \lambda_1} \right) \log_e \left(\frac{99.3}{0.7} \right)$$

Substituting the values, we get

$$t = \frac{1}{\frac{0.693}{7.04 \times 10^8} - \frac{0.693}{4.47 \times 10^9}} \log_e \left(\frac{99.3}{0.7} \right)$$

$$\Rightarrow t = 5.97 \times 10^9 \text{ y}$$

Illustration 41

In the chemical analysis of a rock the mass ratio of two radioactive isotopes is found to be 100:1. The mean lives of the two isotopes are 4×10^9 years and 2×10^9 years respectively. If it is assumed that at the time of formation the atoms of both the isotopes were in equal proportional, calculate the age of the rock. Ratio of the atomic weights of the two isotopes is 1.02:1.

Solution

At the time of observation i.e., at time t , we have

$$\frac{m_1}{m_2} = \frac{100}{1} \quad [\text{given}]$$

Further it is given that

$$\frac{A_1}{A_2} = \frac{1.02}{1}$$

$$\text{Number of atoms } N = \frac{m}{A}$$

$$\Rightarrow \frac{N_1}{N_2} = \frac{m_1}{m_2} \times \frac{A_2}{A_1} = \frac{100}{1.02} \quad \dots(1)$$

Let N_0 be the number of atoms of both the isotopes at the time of formation, then

$$\frac{N_1}{N_2} = \frac{N_0 e^{-\lambda_1 t}}{N_0 e^{-\lambda_2 t}} = e^{(\lambda_2 - \lambda_1)t} \quad \dots(2)$$

Equating (1) and (2), we get

$$e^{(\lambda_2 - \lambda_1)t} = \frac{100}{1.02}$$

$$\Rightarrow (\lambda_2 - \lambda_1)t = \log_e(100) - \log_e(1.02)$$

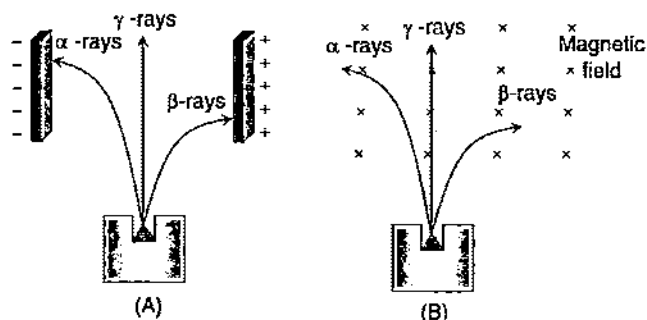
$$\Rightarrow t = \frac{\log_e(100) - \log_e(1.02)}{\left(\frac{1}{2 \times 10^9} - \frac{1}{4 \times 10^9}\right)}$$

Substituting the values, we have

$$t = 1.834 \times 10^{10} \text{ years}$$

NUCLEAR RADIATIONS

According to Rutherford's experiment when a sample of radioactive substance is put in a lead box and allow the emission of radiation through a small hole only. When the radiation enters into the external electric field, they split into three parts (α -rays, β -rays and γ -rays)



a) α -decay : Nearly 90% of the 2500 known nuclides are radioactive; they are not stable but decay into other nuclides

- When unstable nuclides decay into different nuclides, they usually emit alpha (α) or beta (β) particles.
- Alpha emission occurs principally with nuclei that are too large to be stable. When a nucleus emits an alpha particle, its N and Z values each decrease by two and A decreases by four.
- Alpha decay is possible whenever the mass of the original neutral atom is greater than the sum of the

masses of the final neutral atom and the neutral helium-atom.

b) β -decay : There are different simple type of β -decay β^- , β^+ and electron capture.

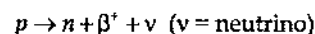
(i) A beta minus particle (β^-) is an electron. Emission of β^- involves transformation of a neutron into a proton, an electron and a third particle called an antineutrino ($\bar{\nu}$).

(ii) β^- decay usually occurs with nuclides for which the neutron to proton ratio $\left(\frac{N}{Z} \text{ ratio}\right)$ is too large for stability.

(iii) In β^- decay, N decreases by one, Z increases by one and A doesn't change.

(iv) β^- decay can occur whenever the neutral atomic mass of the original atom is larger than that of the final atom.

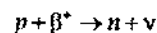
(v) Nuclides for which $\frac{N}{Z}$ is too small for stability can emit a positron, the electron's antiparticle, which is identical to the electron but with positive charge. The basic process called beta plus β^+ decay



(vi) β^+ decay can occur whenever the neutral atomic mass of the original atom is at least two electron masses larger than that of the final atom

(vii) The mass of ν and $\bar{\nu}$ is zero. The spin of both is $\frac{1}{2}$ in units of $\frac{h}{2\pi}$. The charge on both is zero. The spin of neutrino is antiparallel to its momentum while that of antineutrino is parallel to its momentum.

(viii) There are a few nuclides for which β^+ emission is not energetically possible but in which an orbital electron (usually in the k-shell) can combine with a proton in the nucleus to form a neutron and a neutrino. The neutron remains in the nucleus and the neutrino is emitted.



c) γ -decay : The energy of internal motion of a nucleus is quantized. A typical nucleus has a set of allowed energy levels, including a ground state (state of lowest energy) and several excited states. Because of the great strength of nuclear interactions, excitation energies of nuclei are typically of the order of 1 MeV, compared with a few eV for atomic energy levels. In ordinary physical and chemical transformations the nucleus always remains in its ground state. When a nucleus is placed in an excited state, either by bombardment with high-energy particles or by a radioactive transformation, it can decay to the ground state by

emission of one or more photons called gamma rays or gamma-ray photons, with typical energies of 10 keV to 5 MeV. This process is called gamma (γ) decay. All the known conservation laws are obeyed in γ -decay.

The intensity of γ -decay after passing through x thickness of a material is given by $I = I_0 e^{-\mu x}$ (μ = absorption co-efficient)

Table : Properties of α , β and γ -rays

Features	α -particles	β -particles	γ -rays
1. Identity	Helium nucleus or doubly ionised helium atom (${}_2\text{He}^4$)	Fast moving electron ($-\beta^0$ or β^-)	Photons (E.M. waves)
2. Charge	$+2e$	$-e$	ZERO
3. Mass $4m_p$ (m_p = mass of proton = 1.87×10^{-27})	$4m_p$	m_e	Massless
4. Speed	$\approx 10^7 \text{ ms}^{-1}$	1% to 99% of speed of light	Speed of light
5. Range of kinetic energy	4 MeV to 9 MeV	All possible values between a minimum certain value to 1.2 MeV	Between a minimum value to 2.23 MeV
6. Penetration power (γ, β, α)	1 (Stopped by a paper)	100 (100 times of α)	10,000 (100 times of β upto 30 cm of iron (or Pb) sheet)
7. Ionisation power ($\alpha > \beta > \gamma$)	10,000	100	1
8. Effect of electric or magnetic field	Deflected	Deflected	Not deflected
9. Energy spectrum	Line and discrete	Continuous	Line and discrete
10. Mutual interaction with matter	Produces heat	Produces heat	Produces, photo-electric effect, Compton effect, pair production
11. Equation of decay	${}_Z X^A \xrightarrow{\alpha\text{-decay}} {}_{Z-2} Y^{A-4} + {}_2\text{He}^4$ ${}_Z X^A \xrightarrow{n_\alpha} {}_Z Y^{A'}$ $\Rightarrow n_\alpha = \frac{A - A'}{4}$	${}_Z X^A \rightarrow {}_{Z+1} Y^A + {}_{-1}e^0 + \bar{\nu}$ ${}_Z X^A \xrightarrow{n_\beta} {}_Z Y^{A'}$ $\Rightarrow n_\beta = (2n_\alpha - Z + Z')$	${}_Z X^A \rightarrow {}_Z X^A + \gamma$

RADIOACTIVE SERIES

- If the isotope that results from a radioactive decay is itself radioactive then it will also decay and so on.
- The sequence of decays is known as radioactive decay series. Most of the radio-nuclides found in nature are members of four radioactive series. These are as follows

Table : Four radioactive series

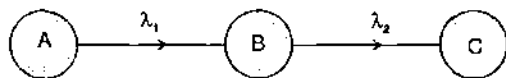
Mass number	Series (Nature)	Parent	Stable end product	Integer n
$4n$	Thorium (natural)	${}_{90}\text{Th}^{232}$	${}_{82}\text{Pb}^{208}$	52
$4n+1$	Neptunium (Artificial)	${}_{93}\text{Np}^{237}$	${}_{83}\text{Bi}^{209}$	52

$4n+2$	Uranium (Natural)	${}_{92}\text{U}^{238}$	${}_{82}\text{Pb}^{206}$	51
$4n+3$	Actinium (Natural)	${}_{89}\text{Ac}^{227}$	${}_{82}\text{Pb}^{207}$	51

- c) The $4n+1$ series starts from ${}_{94}\text{Pu}^{241}$ but commonly known as neptunium series because neptunium is the longest lived member of the series.
- d) The $4n+3$ series actually starts from ${}_{92}\text{U}^{235}$.

SUCCESSIVE DISINTEGRATION AND RADIOACTIVE EQUILIBRIUM

Suppose a radioactive element A disintegrates to form another radioactive element B which in turn disintegrates to still another element C; such decays are called successive disintegration.



Rate of disintegration of $A = \frac{dN_1}{dt} = -\lambda_1 N_1$ (which is also the rate of formation of B)

Rate of disintegration of $B = \frac{dN_2}{dt} = -\lambda_2 N_2$

$$\therefore \left(\begin{array}{c} \text{Net rate of} \\ \text{formation} \\ \text{of B} \end{array} \right) = \left(\begin{array}{c} \text{Rate of} \\ \text{disintegration} \\ \text{of A} \end{array} \right) - \left(\begin{array}{c} \text{Rate of} \\ \text{disintegration} \\ \text{of B} \end{array} \right)$$

$$= \lambda_1 N_1 - \lambda_2 N_2$$

EQUILIBRIUM

In radioactive equilibrium, the rate of decay of any radioactive product is just equal to its rate of production from the previous member.

i.e., $\lambda_1 N_1 = \lambda_2 N_2$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{N_2}{N_1} = \frac{\tau_2}{\tau_1} = \frac{(T_{1/2})_2}{(T_{1/2})_1}$$

Illustration 42

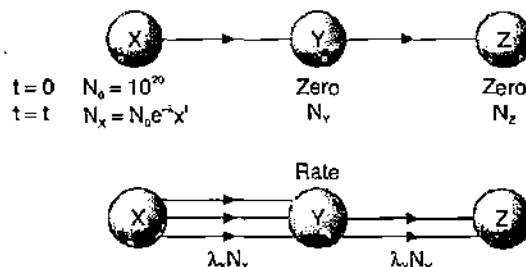
A radioactive nucleus X decays to a nucleus Y with a decay constant $\lambda_X = 0.1 \text{ s}^{-1}$. Y further decays to a stable nucleus Z with a decay constant $\lambda_Y = \frac{1}{30} \text{ s}^{-1}$. Initially there are only X nuclei and their number is $N_0 = 10^{20}$. Setup the rate equations for the populations of X, Y and Z. The population of the Y nucleus as a function of time is given by

$$N_Y(t) = \left\{ \frac{N_0 \lambda_X}{(\lambda_X - \lambda_Y)} \right\} \{ \exp(-\lambda_Y t) - \exp(-\lambda_X t) \}$$

Find the time at which N_Y is maximum and determine the population of X and Z at that instant.

Solution

Let at time $t = t$, number of nuclei of Y and Z are N_Y and N_Z . Then



Rate equation of the populations of X, Y and Z are

$$\left(\frac{dN_X}{dt} \right) = -\lambda_X N_X \quad \dots(1)$$

$$\left(\frac{dN_Y}{dt} \right) = \lambda_X N_X - \lambda_Y N_Y \quad \dots(2)$$

$$\text{and } \left(\frac{dN_Z}{dt} \right) = \lambda_Y N_Y \quad \dots(3)$$

$$\text{Given } N_Y(t) = \frac{N_0 \lambda_X}{\lambda_X - \lambda_Y} [e^{-\lambda_Y t} - e^{-\lambda_X t}]$$

$$\text{For } N_Y \text{ to be maximum } \frac{dN_Y(t)}{dt} = 0$$

$$\text{i.e., } \lambda_X N_X = \lambda_Y N_Y \quad \{\text{from equation (2)}\} \quad \dots(4)$$

$$\Rightarrow \lambda_X (N_0 e^{-\lambda_X t}) = \lambda_Y \frac{N_0 \lambda_X}{\lambda_X - \lambda_Y} [e^{-\lambda_Y t} - e^{-\lambda_X t}]$$

$$\Rightarrow \frac{\lambda_X - \lambda_Y}{\lambda_Y} = \frac{e^{-\lambda_Y t}}{e^{-\lambda_X t}} - 1$$

$$\Rightarrow \frac{\lambda_X}{\lambda_Y} = e^{(\lambda_X - \lambda_Y)t}$$

$$\Rightarrow (\lambda_X - \lambda_Y)t \log_e(e) = \log_e \left(\frac{\lambda_X}{\lambda_Y} \right)$$

$$\Rightarrow t = \frac{1}{(\lambda_X - \lambda_Y)} \log_e \left(\frac{\lambda_X}{\lambda_Y} \right)$$

Substituting the values of λ_X and λ_Y , we have

$$t = \frac{1}{\left(0.1 - \frac{1}{30} \right)} \log_e \left(\frac{0.1}{\frac{1}{30}} \right) = 15 \log_e(3)$$

$$\Rightarrow t = 16.48 \text{ s}$$

The population of X at this moment

$$N_X = N_0 e^{-\lambda_X t} = (10^{20}) e^{-(0.1)(16.48)}$$

$$\Rightarrow N_X = 1.92 \times 10^{19}$$

Since, $N_Y = \frac{N_X \lambda_X}{\lambda_Y}$ [From equation (4)]

$$\Rightarrow N_Y = (1.92 \times 10^{19}) \frac{(0.1)}{\frac{1}{30}} = 5.76 \times 10^{19}$$

$$\Rightarrow N_Z = N_0 - N_X - N_Y$$

$$\Rightarrow N_Z = 10^{20} - 1.92 \times 10^{19} - 5.76 \times 10^{19}$$

$$\Rightarrow N_Z = 2.32 \times 10^{19}$$

Illustration 43

Consider a radioactive disintegration according to the equation $A \rightarrow B \rightarrow C$. Decay constant of A and B is same and equal to λ . Number of nuclei of A , B and C are N_0 , 0, 0 respectively at $t=0$. Find

(a) Number of nuclei of B as function of time t .

(b) Time t at which the activity of B is maximum and the value of maximum activity of B .

Solution

(a) Let the number of nuclei at any instant be shown in the table.

	A	B	C
At $t=0$	N_0	0	0
At t	N_1	N_2	N_3

where, $N_1 = N_0 e^{-\lambda t}$... (1)

Now, $\frac{dN_2}{dt} = \lambda(N_1 - N_2)$

$$\Rightarrow \frac{dN_2}{dt} = \lambda N_0 e^{-\lambda t} - \lambda N_2$$

$$\Rightarrow dN_2 + \lambda N_2 dt = \lambda N_0 e^{-\lambda t} dt$$

$$\Rightarrow e^{\lambda t} dN_2 + \lambda N_2 e^{\lambda t} dt = \lambda N_0 dt$$

$$\Rightarrow d(N_2 e^{\lambda t}) = \lambda N_0 dt$$

Integrating, we get

$$\int_0^{N_2} d(N_2 e^{\lambda t}) = \int_0^t \lambda N_0 dt$$

$$\Rightarrow N_2 = \lambda N_0 t e^{-\lambda t}$$

(b) Activity of B is, $R_2 = \lambda N_2 = \lambda^2 N_0 t e^{-\lambda t}$

For maximum activity, we have

$$\frac{dR_2}{dt} = 0$$

$$\Rightarrow t = \frac{1}{\lambda}$$

$$\Rightarrow R_{\max} = \frac{\lambda N_0}{e}$$

BASED ON NUCLEUS PROPERTIES & RADIOACTIVITY

(Solutions on page 4.102)

- At time $t=0$, number of nuclei of a radioactive substance are 100. At $t=1$ s these numbers become 90. Find the number of undecayed nuclei at $t=2$ s.
- Find the amount of heat generated by 1 mg of Po^{210} preparation during the mean life period of these nuclei if the emitted alpha particles are known to possess kinetic energy 5.3 MeV and practically all daughter nuclei are formed directly in the ground state.
- The radioactivity of a uranium specimen with mass number 238 is $2.5 \times 10^4 \text{ s}^{-1}$, the specimen's mass is 2 g. Find the half-life.
- Ac^{227} has a half life of 21.8 years with respect to radioactive decay. The decay follows two parallel paths, one leading the Th^{227} and the other leading to Fr^{223} . The percentage yields of these two daughter nuclides are 1.2% and 98.8% respectively. What is the rate constant in yrs^{-1} , for each of the separate paths?
- The disintegration rate of a certain radioactive sample at any instant is 4750 disintegrations per minute. Five minutes later the rate becomes 2700 per minute. Calculate
(a) decay constant and
(b) half-life of the sample.
- In an agricultural experiment, a solution containing 1 mole of a radioactive material ($T_{1/2} = 14.3$ days) was injected into the roots of a plant. The plant was allowed 70 hours to settle down and then activity was measured in its fruit. If the activity measured was $1 \mu\text{Ci}$, what percentage of activity is transmitted from the root to the fruit in steady state?
- A sample of 1 g of ^{109}Bi with a half life of 2.7×10^7 year decays into a stable isotope of thallium by emitting α particles.
(a) What is the activity of the sample?
(b) What will be the activity of the sample after 2 years?
(c) After what time does the activity reduces to 25% of the original activity?
- A number N_0 of atoms of a radioactive element are placed inside a closed volume. The radioactive decay constant for the nuclei of this element is λ_1 . The daughter nuclei that form as a result of the decay process are assumed to be radioactive, too, with a radioactive decay constant λ_2 . Determine the time variation of the number of such nuclei. Consider two limiting cases, when $\lambda_1 \gg \lambda_2$ and $\lambda_1 \ll \lambda_2$.
- Calculate the probability that a radioactive atom having a mean life of 10 days decays during the fifth day.
- Old wood from an Egyptian tomb, 4500 years old has C-14 activity of $7.3 \text{ dis. min}^{-1} \text{ g}^{-1}$. Old wood known to be 2500 years old has a C-14 activity $9.3 \text{ dis. min}^{-1} \text{ g}^{-1}$.

- (a) What is half life for C-14?
(b) What is the activity of fresh wood?

11. Determine the amount of ${}_{84}\text{Po}^{210}$ (polonium) necessary to provide a source of α particles of 5 millicurie strength. If half life of polonium is 138 days, given $1 \text{ curie} = 3.7 \times 10^{10}$ disintegrations/sec.
12. The specific activity of a preparation consisting of radioactive Co^{60} and non-radioactive Co^{59} is equal to 2.2×10^{12} dissec $^{-1}$ gm. The half life of Co^{60} is 71.3 days. Find the ratio of the mass of radioactive cobalt in that preparation to the total mass of the preparation.
13. The half lives of radioisotopes P^{32} and P^{33} are 14 days and 25 days respectively. These radio isotopes are mixed in the ratio of 4 : 1 of their atoms. If the initial activity of the mixed sample is 3 mCi, find the activity of the mixed isotopes after 60 days.

MASS ENERGY EQUIVALENCE

According to Einstein the mass and energy are equivalent i.e., mass can be converted into energy and vice-versa. The mass energy equivalence relation is

$$\Delta E = c^2 \Delta m$$

Accordingly, annihilation of 1 kg mass is equivalent to energy given by

$$\Delta E = 1 \times (3 \times 10^8)^2$$

$$\Rightarrow \Delta E = 9 \times 10^{16} \text{ J}$$

Energy corresponding to annihilation of 1 amu of mass is

$$\Delta E = (1.67 \times 10^{-27})(9 \times 10^{16}) \text{ J}$$

$$\Rightarrow \Delta E = \frac{(1.67 \times 10^{-27})(9 \times 10^{16})}{1.6 \times 10^{-19}} \text{ eV}$$

$$\Rightarrow \Delta E = 931.5 \text{ MeV}$$

MASS DEFECT AND BINDING ENERGY

It has been observed that the mass of a nucleus is always less than the mass of its constituent nucleons (i.e., protons + neutrons). The difference between the total mass of the nucleons and the mass of the nucleus is called the mass defect (Δm). This is due to the fact that when nucleons combine to form a nucleus, the binding energy of nucleons is liberated. The binding energy is equal to the work that must be done to split the nucleus into particles constituting it.

Let $m({}_Z X^A)$ be the mass of nucleus,

m_p = the mass of proton and

m_n = mass of neutron

then, the mass defect is given by

$$\Delta m = \text{mass of constituent nucleons} - \text{mass of nucleus}$$

$$\Delta m = [Zm_p + (A-Z)m_n] - m({}_Z X^A)$$

This mass defect exists in the form of binding energy of nucleus, which is responsible for binding the nucleons into a small nucleus. So,

$$\text{Binding energy of nucleus} = (\Delta m)c^2 = (931.5)\Delta m \text{ (in MeV)}$$

$$\text{and binding energy per nucleon} = \frac{(\Delta m)c^2}{A}$$

If the masses are taken in atomic mass unit, the binding energy is given by

$$\text{B.E.} = [Zm_p + (A-Z)m_n] - m({}_Z X^A) \times 931.5 \text{ MeV}$$

Dividing the binding energy by the number of nucleons A , in the nucleus, we obtain the binding energy per nucleon. The stability of a nucleus is measured by the binding energy per nucleon.

The more the binding energy per nucleon of the nucleus the more stable a nucleus is.

Packing fraction (f) is the mass defect per nucleon. So,

$$f = \frac{\Delta m}{A}$$

CONCEPTUAL NOTE(S)

- a) It is not the binding energy which accounts for the stability of nucleus.
b) The stability of nucleus is governed by binding energy per nucleon. The more the binding energy per nucleon, the more stable a nucleus is.

Illustration 44

Find the binding energy of ${}^{12}_6\text{C}$? Also find the binding energy per nucleon. Given mass of ${}^1_1\text{H} = 1.0078 \text{ u}$, ${}^1_0\text{n} = 1.0087 \text{ u}$, ${}^{12}_6\text{C} = 12.00004 \text{ u}$.

Solution

One atom of ${}^{12}_6\text{C}$ consists of 6 protons, 6 electrons and 6 neutrons. The mass of the un-combined protons and electrons is the same as that of six ${}^1_1\text{H}$ atoms (if we ignore the very small binding energy of each proton-electron pair).

$$\text{Mass of six } {}^1_1\text{H atoms} = 6 \times 1.0078 = 6.0468 \text{ u}$$

$$\text{Mass of six neutrons} = 6 \times 1.0087 = 6.0522 \text{ u}$$

$$\text{Total mass of component particles} = 12.0990 \text{ u}$$

$$\text{Mass of } {}^{12}_6\text{C atom} = 12.00004$$

$$\text{Mass defect} = 0.0990 \text{ u}$$

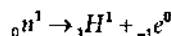
$$\text{Binding energy} = (931)(0.099) = 92 \text{ MeV}$$

$$\text{Binding energy per nucleon} = \frac{92}{12} = 7.66 \text{ MeV}$$

Illustration 45

A neutron breaks into a proton and electron. Calculate the energy produced in this reaction in MeV. Mass of an electron = 9×10^{-31} kg, Mass of proton = 1.6725×10^{-27} kg, Mass of neutron 1.6747×10^{-27} kg. Speed of light = 3×10^8 ms⁻¹.

Solution



$$\text{Mass defect } (\Delta m) = \left(\begin{array}{c} \text{Mass} \\ \text{of} \\ \text{neutron} \end{array} \right) - \left(\begin{array}{c} \text{mass of proton} \\ + \\ \text{mass of electron} \end{array} \right)$$

$$\Rightarrow \Delta m = [(1.6747 \times 10^{-27}) - (1.6725 \times 10^{-27} + 9 \times 10^{-31})]$$

$$\Rightarrow \Delta m = 0.0013 \times 10^{-27} \text{ kg}$$

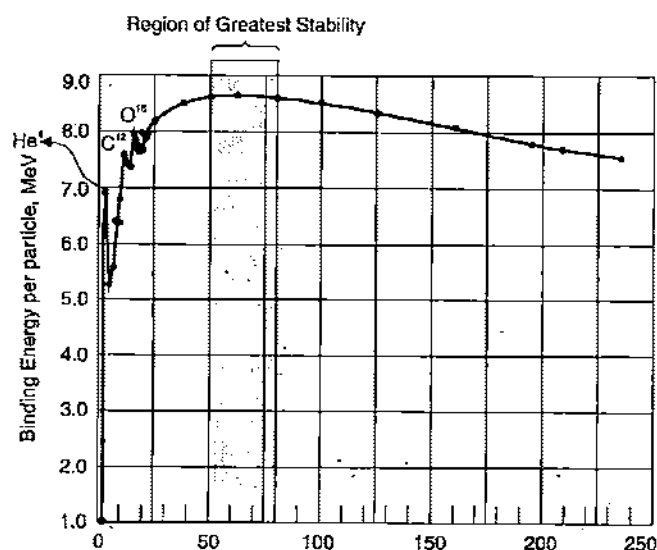
$$\text{Energy released } Q = \Delta mc^2$$

$$\Rightarrow Q = (0.0013 \times 10^{-27}) \times (3 \times 10^8)^2 = 1.17 \times 10^{-13} \text{ J}$$

$$\Rightarrow Q = \frac{1.17 \times 10^{-13}}{1.6 \times 10^{-19}} = 0.73 \times 10^6 \text{ eV} = 0.73 \text{ MeV}$$

VARIATION OF BINDING ENERGY PER NUCLEON WITH MASS NUMBER A

The graph represents the average binding energy per nucleon in MeV against mass number A. It is observed that the binding energy for nuclei (except ${}_2\text{He}^4$, ${}_6\text{C}^{12}$ and ${}_8\text{O}^{16}$) rises first sharply, reaches a maximum value 8.5 MeV at A=50 and then falls slowly, decreasing to 7.6 MeV for elements of higher mass number A=240. Following facts can be concluded from this curve.



- The binding energy per nucleon for light nuclei, such as ${}_1\text{H}^2$, is very small (= 1 MeV).
- The binding energy per nucleon increases rapidly for nuclei up to mass number 20 and the curve possesses

peaks corresponding to nuclei ${}_2\text{He}^4$, ${}_6\text{C}^{12}$ and ${}_8\text{O}^{16}$. The peaks indicate that these nuclei are more stable than those in their neighbourhood. It confirms the reason for extraordinary stability of α -particle.

- After mass number 20, binding energy per nucleon increases gradually and for mass number between 40 and 120, the curve becomes more or less flat. The average value of binding energy per nucleon in this region is about 8.5 MeV. For $A=56$ (${}_{26}\text{Fe}^{56}$), the binding energy per nucleon is maximum and it is equal to 8.8 MeV.
- After mass number 120, binding energy per nucleon starts decreasing and drops to 7.6 MeV for uranium. This low value of binding energy per nucleon in case of heavy nuclei is unable to have control over the repulsion between the large number of protons. Such nuclei are unstable and are found to disintegrate by emitting α -particles. The emission of α -particle not only decreases repulsive force inside the nucleus but also increases the value of B.E./A of the nucleus due to its extraordinary stable structure (α -particle has large binding energy). It is called α -decay.

Sometimes, the heavy nuclei increase the value of their B.E./A by emitting an electron. It is called β -decay. Inside the nucleus, an electron does not exist. It is created at the time of β -decay due to conversion of a neutron into proton. The β -decay leads to increase in Coulomb's repulsive force, but it increases B.E./A and also improves the neutron-proton ratio.

All such nuclei, which undergo α and β -decay are called radioactive nuclei.

- The binding energy per nucleon has a low value for both very light and very heavy nuclei. In order to attain higher value of binding energy per nucleon, the lighter nuclei may unite together to form a heavier nucleus (process of nuclear fusion) or a heavier nucleus may split into lighter nuclei (process of nuclear fission). In both the nuclear processes, the resulting nucleus acquires greater value of binding energy per nucleon along with the liberation of enormous amount of energy.

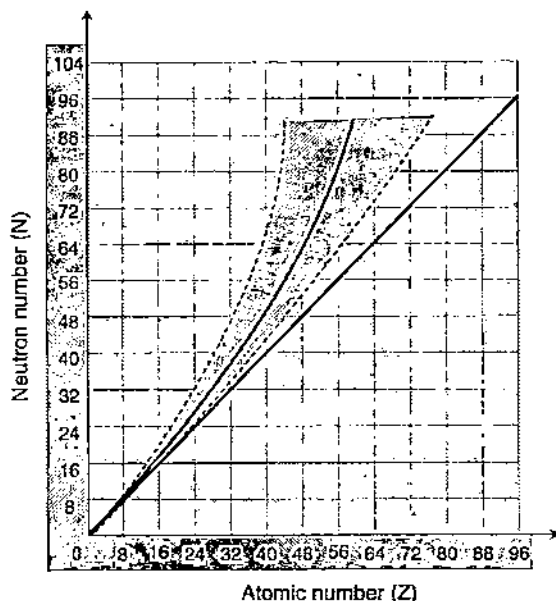
NUCLEAR STABILITY

Among about 1500 known nuclides, less than 260 are stable. The others are unstable that decay to form other nuclides by emitting α , β -particles and γ -EM waves. (This process is called radioactivity). The stability of nucleus is determined by many factors. Few such factors are given below :

- Neutron-proton ratio ($\frac{N}{Z}$ Ratio) :** The chemical properties of an atom are governed entirely by the number of protons (Z) in the nucleus, the stability of

an atom appears to depend on both the number of protons and the number of neutrons.

- (i) For lighter nuclei, the greatest stability is achieved when the number of protons and neutrons are approximately equal ($N \approx Z$) i.e., $\frac{N}{Z} = 1$.
- (ii) Heavy nuclei are stable only when they have more neutrons than protons. Thus heavy nuclei are neutron rich compared to lighter nuclei (for heavy nuclei, more is the number of protons in the nucleus, greater is the electrical repulsive force between them. Therefore more neutrons are added to provide the strong attractive forces necessary to keep the nucleus stable.)
- (iii) Figure shows a plot of N versus Z for the stable nuclei. For mass number upto about $A = 40$. For larger value of Z the nuclear force is unable to hold the nucleus together against the electrical repulsion of the protons unless the number of neutrons exceeds the number of protons.



At $\text{Bi}(Z=83, A=209)$, the neutron excess in $N - Z = 43$. There are no stable nuclides with $Z > 83$.

- b) Even or odd numbers of Z or N : The stability of a nuclide is also determined by the consideration whether it contains an even or odd number of protons and neutrons.
 - (i) It is found that an even-even nucleus (even Z and even N) is more stable (60% of stable nuclide have even Z and even N).
 - (ii) An even-odd nucleus (even Z and odd N) or odd-even nuclide (odd Z and even N) is found to be less stable while the odd-odd nucleus is found to be less stable.

(iii) Only five stable odd-odd nuclides are known : ${}^2_1\text{H}$, ${}^6_3\text{Li}$, ${}^{10}_5\text{B}$, ${}^{14}_7\text{N}$ and ${}^{180}_{75}\text{Ta}$

- c) **Binding energy per nucleon**: The stability of a nucleus is determined by value of its binding energy per nucleon. In general, higher the value of binding energy per nucleon, the more is the stability of the nucleus.

ARTIFICIAL TRANSMUTATION - NUCLEAR REACTIONS

A radio-active substance breaks up by emitting radiations. The daughter nucleus, left behind, has different physical and chemical properties and is assigned a new place in the periodic chart. Thus radio-activity is the phenomenon by which a substance gets converted into another one. This change can be brought about by artificial method, by bombarding a given nucleus with some radiation. The particles constituting the incident radiation must possess sufficient kinetic energy so as to penetrate into the given nucleus. As they enter the given nucleus, a compound nucleus (an intermediate state) is formed which is generally unstable. The compound nucleus then breaks up to produce product nucleus by emitting radiation. The process is, schematically, represented as



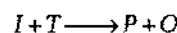
This is a reaction in which only the nuclei take part. Orbital electrons have no contribution to it. Such reactions are known as nuclear reactions.

LAWS GOVERNING NUCLEAR REACTIONS

- a) **Law of Charge**: The electric charge involved in nuclear reactions must be same before and after the reaction. So, charge number is conserved in nuclear reactions.
- b) **Law of Number of Nucleons**: The total number of nucleons involved in a nuclear reaction must be same before and after the reaction. So, mass number is also conserved in nuclear reactions.
- c) **Law of Conservation of Energy**: The total energy (rest mass energy + K.E.) of the reacting particles must be equal to the total energy of the product particles.
- d) **Law of Conservation of Linear Momentum**: The total linear momentum of the reacting particles must be equal to the total linear momentum of the product particles.
- e) **Law of Conservation of Angular Momentum**: Total angular momentum of nuclei before and after reaction must be the same.

Q-VALUE OF A NUCLEAR REACTION

Consider a nuclear reaction, schematically represented by equation



Let E_I , E_P and E_O be the kinetic energies associated with I , P and O respectively while the target T is at rest initially. Q -value of a nuclear reaction is given by,

$$Q = E_p + E_o - E_i$$

Let m_i , m_T , m_P , m_O , respectively, be the masses of I , T , P and O

BEFORE REACTION	AFTER REACTION
Energy of $I = m_i c^2 + E_i$	Energy of $P = m_P c^2 + E_P$
Energy of $T = m_T c^2$	Energy of $O = m_O c^2 + E_O$
Total energy of the system $= m_i c^2 + E_i + m_T c^2$	Total energy of the system $= m_P c^2 + E_P + m_O c^2 + E_O$

By Law of Conservation of Energy,

$$m_i c^2 + E_i + m_T c^2 = m_P c^2 + E_P + m_O c^2 + E_O$$

$$\Rightarrow Q = E_p + E_o - E_i = [(m_i + m_T) - (m_P + m_O)] c^2$$

where Δm is the mass defect between initial and final particles.

CASE I Exo-ergic reactions: A reaction is said to be exo-ergic if Q is positive.

$$\Rightarrow (m_P + m_O) < (m_i + m_T)$$

The part of mass which disappears gets converted into the energy in accordance with Einstein's Mass-Energy equivalence.

CASE II Endo-ergic reaction: A reaction is said to be endo-ergic if Q is negative.

$$\Rightarrow (m_P + m_O) > (m_i + m_T)$$

i.e., the sum of the masses of product particles is greater than that of reactant particles. For this reaction to proceed, the incoming particle must possess an energy equivalent to the mass defect.

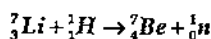
The minimum amount of energy that a bombarding particle must have in order to initiate an endoergic reaction, is called Threshold Energy E_{th} .

$$E_{th} = -Q \left(\frac{m_P}{m_T} + 1 \right)$$

where, m_P is the mass of the projectile i.e. the nucleus used to hit the target and m_T is the mass of the target nucleus

Illustration 46

How much energy must a bombarding proton possess to cause the reaction.



Solution

Since the mass of an atom include the masses of the atomic electrons, the appropriate number of electron masses must be subtracted from the given values.

Reactants	Products
${}^7_3\text{Li}$ 7.01600 - $3m_e$	${}^7_4\text{Be}$ 7.01693 - $4m_e$
${}^1_1\text{H}$ 1.0783 - $1m_e$	${}^1_0\text{n}$ 1.0866
Total 8.02383 - $4m_e$	Total 8.02559 - $4m_e$

The Q -value of the reaction

$$Q = -0.00176 \text{ u} = -1.65 \text{ MeV}$$

The energy is supplied as kinetic energy of the bombarding proton. The incident proton must have more than this energy because the system must possess some kinetic energy even after the reaction, so that momentum is conserved.

With momentum conservation taken into account, the minimum kinetic energy that the incident particle can be found with the formula.

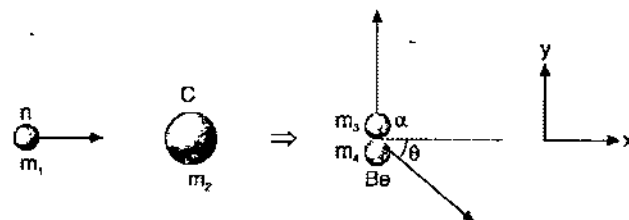
$$E_{th} = -\left(1 + \frac{m}{M}\right)Q = -\left(1 + \frac{1}{7}\right)(-1.65) = 1.89 \text{ MeV}$$

Illustration 47

A neutron with kinetic energy $K = 10 \text{ MeV}$ activates a nuclear reaction $n + {}^{12}_6\text{C} \rightarrow {}^9_4\text{Be} + \alpha$. Find the kinetic energy of the alpha particles outgoing at right angle to the direction of incoming neutrons.

Take $n = 931.5 \text{ MeV}$ and threshold energy of reaction (E_{th}) = 6.17 MeV.

Solution



Since,

$$Q + K_1 = K_3 + K_4 \quad \dots(1)$$

Applying Law of Conservation of Linear Momentum

Along x-axis

$$\sqrt{2m_1 K_1} = \sqrt{2m_3 K_3} \cos \theta \quad \dots(2)$$

Along y-axis

$$\sqrt{2m_3 K_3} = \sqrt{2m_4 K_4} \sin \theta \quad \dots(3)$$

Squaring and adding equations (2) and (3), we get

$$m_1 K_1 + m_3 K_3 = m_4 K_4$$

$$\Rightarrow K_4 = \frac{m_1}{m_4} K_1 + \frac{m_3}{m_4} K_3$$

Substituting value of K_1 in equation (1) and rearranging, we get

$$Q + \left(1 - \frac{m_1}{m_4}\right) K_1 = \left(1 + \frac{m_3}{m_4}\right) K_3$$

$$\text{where, } Q = \frac{-E_{\alpha}}{\left(1 + \frac{m_1}{m_2}\right)} = \frac{-6.17}{1 + \frac{1}{12}} = -5.69 \text{ MeV}$$

$$\Rightarrow \left(1 + \frac{4}{9}\right)K_3 = -5.69 + \left(1 - \frac{1}{9}\right)(10)$$

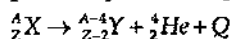
$$\Rightarrow K_3 = 2.21 \text{ MeV}$$

ALPHA DECAY

Alpha decay is a process in which an unstable nucleus transforms itself into a new nucleus by emitting an alpha particle (a helium nucleus, ${}^4_2\text{He}$).

Since an α -particle has two protons and two neutrons, so after an α -decay, the parent nucleus is transformed into a daughter nucleus with mass number smaller by 4 and atomic number smaller by 2.

An alpha decay can be expressed by the equation

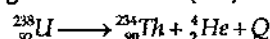


Here Q is the energy released in the process and is determined from Einstein's mass-energy relation which gives

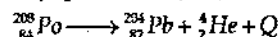
$$Q = (m_X - m_Y - m_{\alpha})c^2$$

where m_X , m_Y and m_{α} are the masses of the parent nucleus X , daughter nucleus Y and the α -particle respectively. The energy Q is shared by the daughter nucleus X and the α -particle. As the parent nucleus is at rest before its α -decay, the α -particles are emitted with fixed energy and hence are mono-energetic. This energy can be determined by Applying the Laws of Conservation of Energy and Momentum.

For example, uranium (238) on emitting an α -particle changes into thorium (234) as



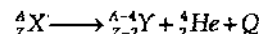
Similarly, polonium (208) is transmuted into lead (204) as



Generally, the nuclei with mass number 210 or more undergo α -decay. In such nuclei, the long range repulsive forces between the protons dominate over the short range nuclear forces which bind the various nucleons together. By emitting α -particles, these nuclei achieve greater stability. An α -particle has a high value of binding energy ($\approx 28 \text{ MeV}$). After the emission of an α -particle, the binding energy per nucleon of the emitting nucleus increases and the residual nucleus becomes more stable.

SPEED OF EMITTED α -PARTICLES

Consider the alpha decay process equation i.e.,



The speed of the emitted α -particles can be calculated by using the Laws of Conservation of Energy and Momentum.

Suppose the parent nucleus ${}^A_Z\text{X}$ be at rest before decay. Let v_{α} and v_Y be the velocities of the α -particle and the daughter nucleus. Applying the Law of Conservation of Momentum, we get

$$m_Y v_Y = m_{\alpha} v_{\alpha} \quad \dots(1)$$

As the energy Q released in the decay process appears in the form of kinetic energy of α -particle and the daughter nucleus, so we have

$$\frac{1}{2} m_{\alpha} v_{\alpha}^2 + \frac{1}{2} m_Y v_Y^2 = Q$$

Substituting the value of v_Y from equation (1), we get

$$\frac{1}{2} m_{\alpha} v_{\alpha}^2 + \frac{1}{2} \frac{m_{\alpha}^2 v_{\alpha}^2}{m_Y^2} = Q$$

$$\Rightarrow \frac{1}{2} m_Y m_{\alpha} v_{\alpha}^2 + \frac{1}{2} m_{\alpha}^2 v_{\alpha}^2 = m_Y Q$$

$$\Rightarrow \frac{1}{2} (m_Y + m_{\alpha}) m_{\alpha} v_{\alpha}^2 = m_Y Q$$

$$\Rightarrow K_{\alpha} = \frac{1}{2} m_{\alpha} v_{\alpha}^2 = \left(\frac{m_Y}{m_Y + m_{\alpha}} \right) Q$$

Since, $m_Y = (A - 4) \text{ amu}$ and $m_{\alpha} = 4 \text{ amu}$, therefore,

$$K_{\alpha} = \frac{1}{2} m_{\alpha} v_{\alpha}^2 = \left(\frac{A - 4}{A} \right) Q$$

$$\Rightarrow v_{\alpha} = \sqrt{\frac{2K_{\alpha}}{m_{\alpha}}} = \sqrt{\frac{2(A - 4)Q}{Am_{\alpha}}}$$

For example, in the α -decay of a radon nucleus ${}^{222}_{86}\text{Rn}$, we have

$$Q = 5.587 \text{ MeV}$$

$$\Rightarrow K_{\alpha} = \left(\frac{A - 4}{A} \right) Q = \left(\frac{222 - 4}{222} \right) \times 5.587 \text{ MeV}$$

$$\Rightarrow K_{\alpha} = 5.486 \text{ MeV} = 5.486 \times 1.6 \times 10^{-19} \text{ J}$$

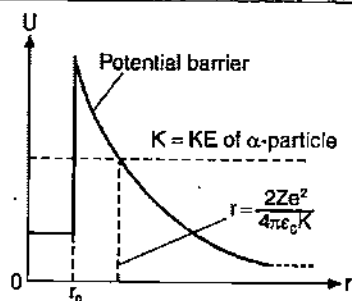
Since, $m_{\alpha} = 4 \text{ amu} = 4 \times 1.66 \times 10^{-27} \text{ kg}$

$$\Rightarrow v_{\alpha} = \sqrt{\frac{2 \times 5.486 \times 1.6 \times 10^{-19}}{4 \times 1.66 \times 10^{-27}}} \text{ ms}^{-1}$$

$$\Rightarrow v_{\alpha} = 1.62 \times 10^7 \text{ ms}^{-1}$$

THEORY OF α -DECAY (TUNNELLING EFFECT)

The α -particles emitted by different radioactive nuclei have kinetic energy ranging from 4 to 9 MeV. The nucleus of an α -emitter poses a barrier of height about 25 MeV. Figure shows a plot of the potential energy U of the system consisting of the α -particle and the residual nucleus. The α -particles are short of about 16 to 25 MeV of energy, needed for the emission. Therefore, classically, we cannot explain the emission of α -particles by radioactive nuclei.



Plot of potential energy U of an α -particle as a function of distance r from the centre of the residual nucleus.

In 1928, Gamow, Congdon and Gurney explained the emission of α -particles in terms of the penetration of the nuclear potential barrier on the basis of Quantum Theory. According to this theory, we have

- An α -particle may exist as an entity (already formed) inside a nucleus before it escapes from the nucleus.
- The α -particle is in a state of constant motion inside the nucleus with a speed of about 10^7 ms^{-1} .
- Quantum mechanically, even an α -particle having insufficient kinetic energy has a small but finite probability p of its crossing the potential barrier.

As the size of the nucleus $\approx 10^{-14} \text{ m}$ and speed of α -particle $\approx 10^7 \text{ ms}^{-1}$, the α -particle takes about 10^{-21} s to move across the nucleus. Thus α -particle presents itself before the potential barrier 10^{21} times in a second. The probability of escape of an α -particle from a nucleus will be

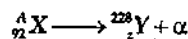
$$P = pv$$

As v is large (10^{21} s^{-1}), so P is sufficiently large and the α -particle can tunnel through the energy barrier which is classically impossible. Hence α -decay occurs as a result of barrier tunnelling.

The barrier tunnelling explains why every ${}^{238}_{92}\text{U}$ nuclide in a sample of ${}^{238}_{92}\text{U}$ atoms does not decay at once, even when its decay process has a positive Q value. Consequently, the half-lives for α -decay of most of the alpha unstable nuclei are very long. For example, the half-life of ${}^{238}_{92}\text{U}$ for α -decay is 4.5×10^9 year.

Illustration 48

A nucleus X initially at rest, undergoes alpha-decay, according to the equation



- Find the value of A and z in the above process.
- The α -particle in the above process is found to move in a circular track of radius $1.1 \times 10^2 \text{ m}$ in a uniform magnetic field of $3 \times 10^3 \text{ T}$. Find the energy (in MeV) released during the process and binding energy of the parent nucleus X .

Given: $m_y = 228.03 \text{ amu}$ $m_\alpha = 4.003 \text{ amu}$
 $m({}^1_0 n) = 1.009 \text{ amu}$ $m({}^1_1 \text{H}) = 1.008 \text{ amu}$
 $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg} = 931.5 \text{ MeVc}^{-2}$

Solution

(a) The given equation is,
 ${}^A_Z X \longrightarrow {}^{228}_{92} Y + {}^4_2 \text{He}$
 $A = 228 + 4 = 232$

and $92 = z + 2$

$\Rightarrow z = 90$

(b) $\frac{m_\alpha v_\alpha^2}{r} = qv_\alpha B$

$\Rightarrow v_\alpha = \sqrt{\frac{rqB}{m_\alpha}}$

$\Rightarrow v_\alpha = \sqrt{\frac{1.1 \times 10^2 \times 2 \times 1.6 \times 10^{-19} \times 3 \times 10^3}{4.003 \times 1.66 \times 10^{-27}}}$

$\Rightarrow v_\alpha = 4 \times 10^6 \text{ ms}^{-1}$

Applying Law of Conservation of Linear Momentum, we get

$m_\alpha v_\alpha = m_y v_y$

$\Rightarrow v_y = \frac{m_\alpha v_\alpha}{m_y} = \frac{(4.003)(4 \times 10^6)}{(228.03)}$

$\Rightarrow v_y = 7 \times 10^4 \text{ ms}^{-1}$

Therefore, energy released during the process is

$E = \frac{1}{2} [m_\alpha v_\alpha^2 + m_y v_y^2]$

$\Rightarrow E = \frac{(1.66 \times 10^{-27})}{(2 \times 1.6 \times 10^{-13})}$

$[(4.003)(4 \times 10^6)^2 + (228.03)(7 \times 10^4)^2] \text{ MeV}$

$\Rightarrow E = 0.34 \text{ MeV} = \frac{0.34}{931.5} \text{ amu} = 0.000365 \text{ amu}$

Therefore, mass of ${}^{232}_{92} X = m_y + m_\alpha + 0.000365 = 232.033365 \text{ u}$

So, mass defect is

$\Delta m = 92(1.008) + (232 - 92)(1.009) - 232.033365$

$\Rightarrow \Delta m = 1.962635 \text{ amu}$

$\Rightarrow \text{Binding energy} = 1.962635 \times 931.5 \text{ MeV} = 1828.2 \text{ MeV}$

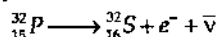
BETA DECAY

The process of spontaneous emission of an electron (e^-) or a positron (e^+) from a nucleus is called beta decay.

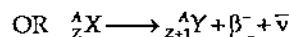
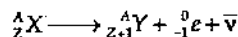
Like α -decay, β -decay is a spontaneous process, with a definite disintegration energy and half-life. It is also a statistical process, obeying the Law of radioactive Decay.

In beta minus (β^-) decay, the mass number of the radioactive nucleus remains unchanged but its atomic

number increases by one. An electron and a new particle antineutrino ($\bar{\nu}$) are emitted from the nucleus, as in the decay.

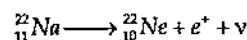


In general, the beta minus decay may be represented as

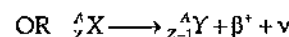
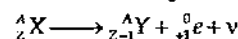


The electron emitted from the nucleus is called a beta particle, denoted by β^{-} .

In beta plus (β^{+}) decay, the mass number of the radioactive nucleus remains unchanged but its atomic number decreases by one. A positron (e^{+}) and a new particle neutrino (ν) are emitted from the nucleus, as in the decay.



In general, the beta plus decay may be represented as

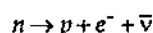


The positron so emitted is called a beta plus particle (β^{+})

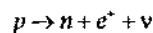
The positron is an antiparticle of electron. It has a positive charge equal in magnitude to the charge on an electron and has a mass equal to the mass of an electron. Similarly, neutrino and antineutrino are antiparticles of each other.

Both are massless, chargeless particles having spins $\pm \frac{1}{2}$.

Although a nucleus contains no electrons, positrons and neutrinos, yet can it eject these particles. It is believed that electrons, positrons and neutrinos are created during the process of beta decay. If the unstable nucleus has excess neutrons than needed for stability, a neutron converts itself into a proton. So in a beta-minus decay, an electron and an antineutrino are created and emitted from the nucleus via the reaction given by



If the unstable nucleus has excess protons than that needed for stability, a proton converts itself into a neutron. So in a beta-plus decay, a positron and a neutrino are created and emitted from the nucleus via the reaction given by

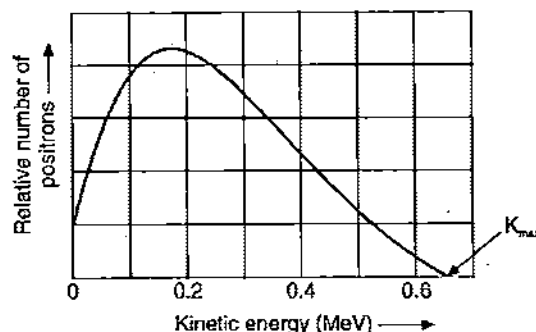


Clearly, a beta decay process involves the conversion of a neutron into a proton or vice versa. These nucleons have nearly equal masses. That is why the mass number A of a nuclide undergoing beta decay does not change.

CONTINUOUS ENERGY SPECTRUM FOR BETA RAYS

In both α - and β -decays, the disintegration energy Q depends on the nature of the radionuclide. In the α -decay of a particular radionuclide, every emitted α -particle has a definite amount of kinetic energy.

However in β -decay, the disintegration energy is shared in all proportions between the three particles i.e., daughter nucleus, electron (or positron) and antineutrino (or neutrino). As a result, the kinetic energy of the electrons (or positrons) is not fixed. Their energy varies from zero to a maximum value K_{max} . Thus β -rays have a continuous energy spectrum, as shown in Figure.

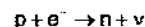


The distribution of the kinetic energies of positrons emitted in the decay of ${}_{29}^{64}\text{Cu}$.

The maximum kinetic energy or end point energy K_{max} must be equal to disintegration energy Q . When the electron (or positron) has maximum energy, the energy carried by the daughter nucleus and neutrino is nearly zero.

CONCEPTUAL NOTE(S)

- Since neutrinos (or antineutrinos) are massless and chargeless, they interact so weakly with matter that it becomes very difficult to detect them. They can penetrate through earth without being absorbed. By ingenious experiments, neutrinos have been detected and their mass and spin or intrinsic angular momentum have been measured.
- Electron capture.** Some proton rich nuclei capture one of the atomic electrons (usually from the K shell). A proton in the nucleus combines this electron forming a neutron. A neutrino created in the process is emitted from the nucleus.



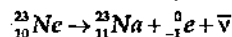
The entire process may be represented as



This process is called **electron capture** or **K-capture**. The vacancy created in the K shell is filled by transition of electrons from the outer shells. This results in the emission of characteristic X-rays.

Illustration 49

Neon - 23 beta decays in the following way :



Find the minimum and maximum kinetic energy that the beta particle ${}_{-1}^0e^{-}$ can have. The atomic masses of ${}^{23}\text{Ne}$ and ${}^{23}\text{Na}$ are $22.9945u$ and $22.9898u$, respectively.

Solution

Reactants	Products
${}^{23}_{10}\text{Ne} \quad 22.9945 - 10m_e$	${}^{23}_{11}\text{Na} \quad 22.9898 - 11m_e$
	${}^0_{-1}e - m_e$
Total $22.9945 - 10m_e$	Total $22.9898 - 10m_e$

$$\text{Mass defect} = 22.9945 - 22.9898 = 0.0047u$$

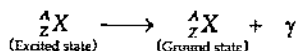
$$Q = (0.0047)(931) = 4.4 \text{ MeV}$$

The β -particle and neutrino share this energy. Hence the energy of the β -particle can range from 0 to 4.4 MeV.

GAMMA DECAY

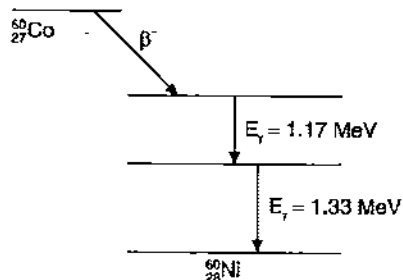
The process of emission of a γ -ray photon during the radioactive disintegration of a nucleus is called **gamma decay**.

As the emitted γ -ray photons have zero rest mass and carry no charge, so in a γ -decay the mass number and atomic number of the nucleus remain unchanged and no new element is formed. A γ -decay can be expressed as



A nucleus does not contain photons, yet it can emit photons. These photons are created during the emission process. We know that a nucleus can exist in different energy states. After an α or a β -decay, the daughter nucleus is usually left in the excited state. It attains the ground state by single or successive transitions by emitting one or more photons. As the nuclear states have energies of the order of MeV, therefore, the photons emitted by the nuclei have energy of the order of several MeV. The wavelength of such high energy photons is a fraction of an angstrom. The short wavelength electromagnetic waves emitted by nuclei are called γ -rays.

An example of γ -decay is shown through an energy level diagram shown in Figure. Here an unstable ${}^{60}_{27}\text{Co}$ nucleus is transformed via a β -decay into an excited ${}^{60}_{28}\text{Ni}$ nucleus, which in turn reaches the stable ground state by emitting photons of energies 1.17 MeV and 1.33 MeV, in two successive γ -decay processes.



Energy-level diagram showing the emission of γ -rays by a ${}^{60}_{27}\text{Co}$ nucleus subsequent to beta decay.

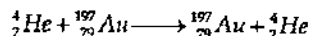
Usually, γ -rays are emitted after α - or β -decay, but there are long lived radioactive nuclei that emit only γ -rays.

CLASSIFICATION OF NUCLEAR REACTIONS

Nuclear reactions can be classified into the following categories.

A. ELASTIC SCATTERING

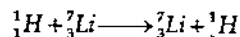
The incident particle gets deflected without any change in its energy, i.e.,



The bombarding particle passes sufficiently at large distance away from the target nucleus so as to get repulsion which changes its direction of motion without any change in its energy.

B. INELASTIC SCATTERING

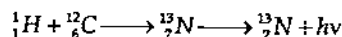
If the bombarding particle passes close to target it gets deflected. Due to strong repulsion, the target particle also acquires some energy. So the energy left with the scattered particle is less than that it had initially.



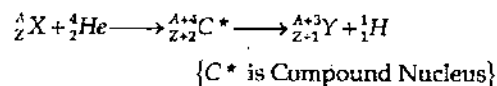
${}^7_3\text{Li}$ means existence of ${}^7_3\text{Li}$ in one of its excited states.

C. SIMPLE CAPTURE

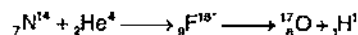
The incoming particle is captured by the target nucleus. The product nucleus which is generally in the form of excited state decays to the ground state by emitting γ -ray of energy $h\nu$.

**D. DISINTEGRATION (NUCLEAR TRANSMUTATIONS)**

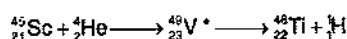
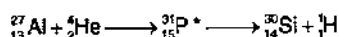
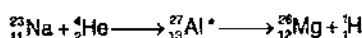
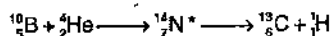
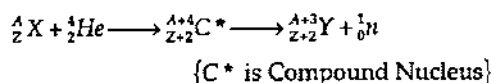
The intermediate compound nucleus breaks up and results in a product nucleus and an outgoing particle. The product nucleus has different chemical properties as compared to the target particle. Majority of nuclear reactions belong to this category. Such nuclear disintegrations are called Nuclear Transmutations.

a) Disintegration by α -particles**(i) (α, p) reactions****EXAMPLES**

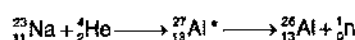
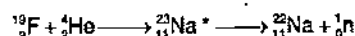
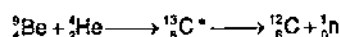
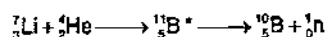
The historical experiment of Rutherford is an α induced transmutation, an (α, p) reaction.



and is exoergic in nature ($Q > 0$). Other useful (α, p) reactions are

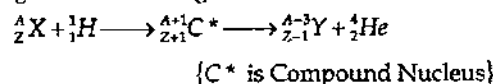
(ii) (α, n) reactions

EXAMPLES

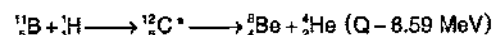
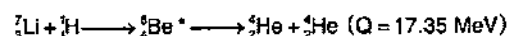
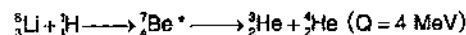


b) Disintegration by protons

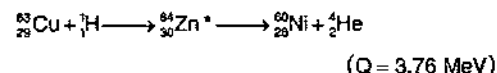
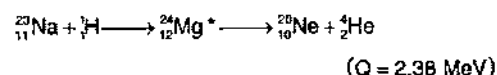
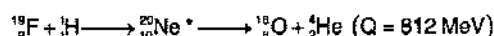
- (i) (p, α) reactions: When the reactions yield α -particles. The (p, α) reactions are usually exoergic and have the general form



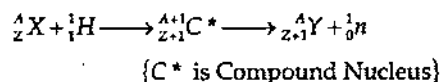
EXAMPLES



{The product nucleus ${}^8_4\text{Be}$ is highly unstable and decays almost immediately as ${}^8_4\text{Be} \longrightarrow {}^4_2\text{He} + {}^4_2\text{He}$, so that the final reaction gives three α -particles}

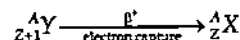


- (ii) (p, n) reactions: When the reactions yield neutrons. The general equation of this type of reactions is

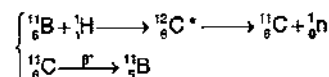


the product nucleus being isobaric with the target nucleus.

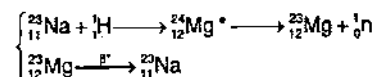
Since two isobars differing in Z by unity cannot both be stable, the product nucleus is β^+ active, decaying by β^+ emission (or electron capture) into ${}_Z^AX$ (the same as target nucleus):



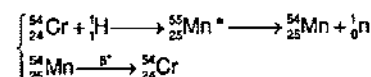
EXAMPLES



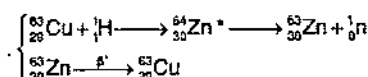
(Q = -1.763 MeV) (Half-life = 2.5 min)



(Q = -4.84 MeV) (Half-life = 12.3 s)



(Q = -2.16 MeV) (Half-life = 310 days)

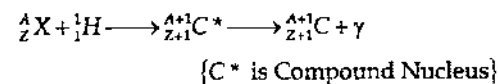


(Q = -4.15 MeV) (Half-life = 38.5 min)

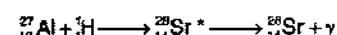
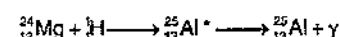
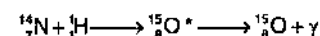
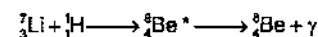
The (p, n) reaction is always endoergic.

- (iii) (p, γ) reactions: When the reactions yield γ -photons. The compound nucleus formed by absorption of proton by the target nucleus does not emit any nuclear particle but goes down to the ground state emitting one or more γ -photons. The (p, γ) reaction is the radiative capture of proton.

The general equation of (p, γ) reaction is



EXAMPLES

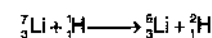


- (iv) (p, d) reactions: When the reactions yield deuterons. The general equation of this type of reaction is



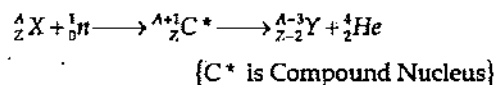
This is an example of direct reaction without any formation of the compound nucleus.

EXAMPLES



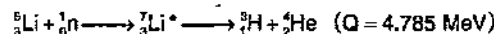
c) Disintegration by neutrons

- (i) (n, α) reactions: When the reactions yield α -particles. The general equation for (n, α) reactions is

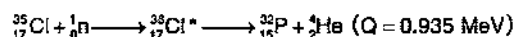
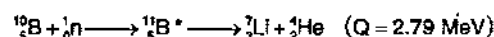


The (n, α) reactions are usually exoergic, i.e. Q is positive, particularly for medium heavy nuclei.

EXAMPLES



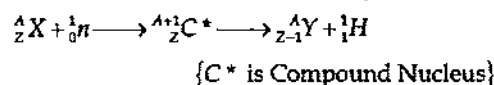
followed by ${}_1^3H \xrightarrow{\beta^-} {}_2^3He + {}_0^{-1}e + \bar{\nu}_e$



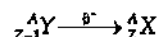
The first two reactions are utilised in the construction of neutron detectors as they have fairly large cross-sections. The first reaction also gives a method of producing tritium which is useful in nuclear fusion.

- (ii) (n, p) reactions: When the reactions yield protons.

The general equation for (n, p) reactions is

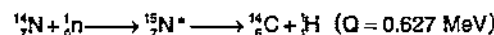


The product nucleus Y is an isobar of the target nucleus X with Z -value one unit lower and is thus β^- active decaying to the target nucleus. So,

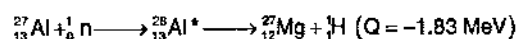


So, the (n, p) reactions are such that the initial and final nuclides are identical. The process therefore appears to be a conversion of neutron into a proton and an electron.

EXAMPLES



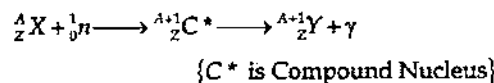
followed by ${}_{6}^{14}C \xrightarrow[5568 \text{ yr}]{\beta^-} {}_7^{14}N + {}_0^{-1}e + \bar{\nu}_e$



followed by ${}_{12}^{27}Mg \xrightarrow[10 \text{ min}]{\beta^-} {}_{13}^{27}Al + {}_0^{-1}e + \bar{\nu}_e$

Only the first reaction is induced by thermal neutrons.

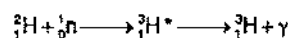
- (iii) (n, γ) reactions: When the reactions yield γ -photons. This is the most important neutron-induced transmutation, known as **radiative capture** of neutrons, and has the following general equation.



The product nucleus is thus the same as the compound nucleus in the ground state. The (n, γ) reaction is always exoergic ($Q > 0$) and can be induced by almost zero energy neutrons.

The radiative capture raises the target nucleus to an excited isomeric state and by releasing the excitation energy as γ photons, the product nucleus becomes an isotope of the target nucleus. The isotopic product nuclei are generally β^- active, as it has a higher neutron-proton ratio compared to the original one. In fact, this method of inducing β^- -activity is used extensively with copious supply of neutrons from reactors.

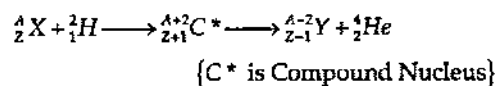
EXAMPLES



followed by ${}_1^3H \xrightarrow[12.4 \text{ yr}]{\beta^-} {}_2^3He + {}_0^{-1}e + \bar{\nu}_e$

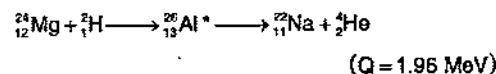
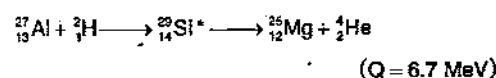
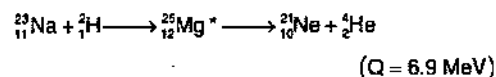
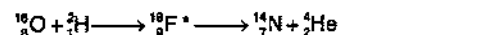
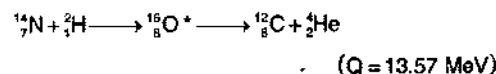
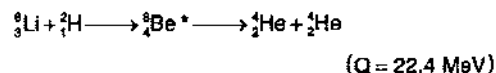
- d) **Disintegration by deuterons**

- (i) (d, α) reactions: When the reactions yield α -particles. The general equation of (d, α) reactions is



The Q -values are usually positive and the reactions exoergic. Some example of (d, α) reactions are given as under.

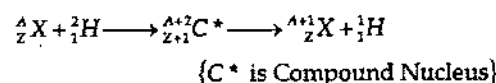
EXAMPLES



Since the α -particles ejected from the compound nucleus are to cross high potential barrier, the (d, α) reactions occur at fairly high energy of deuteron and for low Z target nuclei.

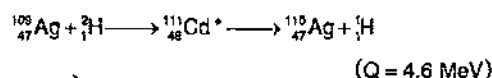
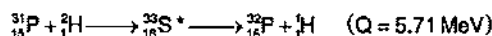
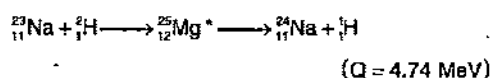
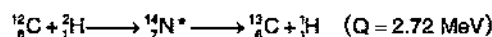
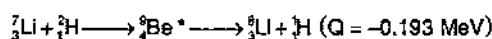
- (ii) (d, p) reactions: When the reactions yield protons.

The general equation of (d, p) reactions is



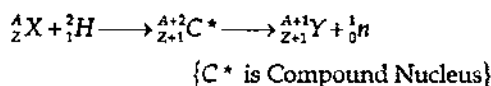
The Q -values are usually positive and the reactions exoergic. For some light nuclei, however, Q may be negative. Some examples of (d, p) reactions are as under.

EXAMPLES



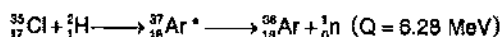
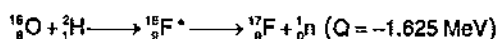
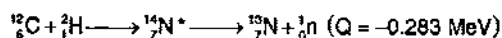
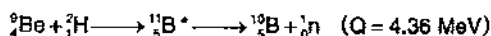
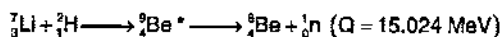
The products of (d, p) reactions are usually radioactive

- (iii) (d, n) reactions: When the reactions yield neutrons. The general equation of (d, n) reaction is

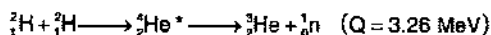
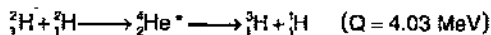


With some exceptions, the (d, n) reactions are exoergic and the Q-values are positive. The product nucleus Y is an isotope of the compound nucleus. Some examples of (d, n) reactions are given below.

EXAMPLES



When deuterons bombard deuterons both (d, p) and (d, n) reactions may be observed because of the two alternative decay schemes for the compound nucleus.

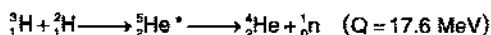


The product ${}^3_1\text{H}$ is tritium, an isotope of hydrogen. Its nucleus is called triton which is β -active.



The other product ${}^3_2\text{He}$ is a stable isotope of helium.

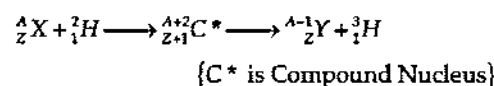
The tritium may be bombarded with deuterons to produce (d, n) reaction



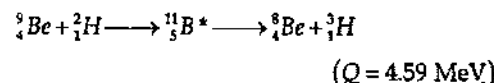
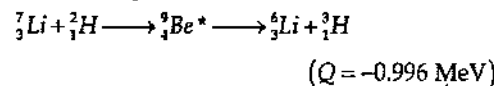
The reactions with beryllium and deuterium serve as sources of neutrons. A thick beryllium target if bombarded with 1 MeV deuterons (accelerated in a cyclotron or Van de Graaff generator) yields about 10^8 neutrons per sec per μA deuteron-current absorbed in the target.

- (iv) (d, t) reactions: When the reactions yield tritium.

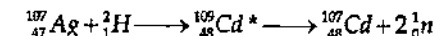
The general equation of (d, t) reactions is



The product nucleus Y is an isotope of the target nucleus X. The cross-section of such reactions is low. Some examples are as under.



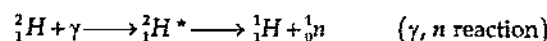
NOTE : At higher energies (> 20 MeV) of deuterons the (d, 2n), (d, 2p), (d, 3n) etc. reactions in which more than one particle (two or more) is emitted from the compound nucleus become important, e.g.



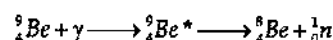
- e) Photo disintegration (γ -induced transmutations)

This type of reactions, called photo disintegrations or photonuclear reactions, occur when sufficiently high-energy photons enter into a nucleus. The energy of the incident photon must be greater than the binding energy of a nuclear particle (separation energy) like neutron, proton, α -particle etc. to produce (γ, n), (γ, p), (γ, α) etc. reactions.

The photo disintegration of deuteron, discovered by Chadwick and Goldhaber, deserves special mention, for it was from this reaction that they evaluated neutron mass.

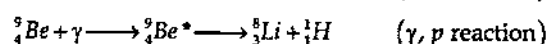
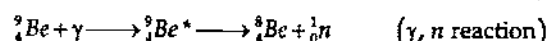
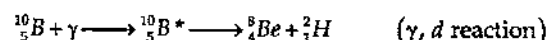
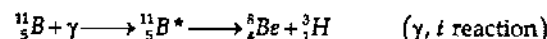
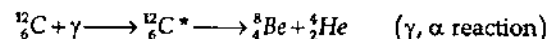


Another example of (γ, n) reaction is



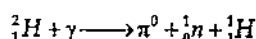
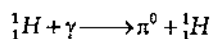
This reaction is used for preparing the source for photo-neutrons.

Other γ -induced reactions are:



The last (γ, n) reaction serves as a convenient source of neutrons.

Neutral π -mesons can be artificially produced by the interaction of γ -rays with hydrogen and deuterium.



The first reaction can occur only if the energy of the γ -photon is not less than the threshold value equal to the mass-energy of the neutral pion mass. Steinberg could produce π^0 -mesons by bombarding light targets such as hydrogen or beryllium with high energy x-radiation from an electron synchrotron.

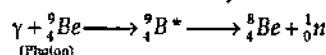


Illustration 50

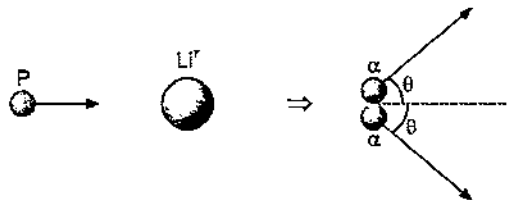
A proton is bombarded on a stationary lithium nucleus. As a result of the collision two α particles are produced. If the direction of motion of the α particles with the initial direction of motion makes an angle $\cos^{-1}\left(\frac{1}{4}\right)$, find the kinetic energy of the striking proton.

Given binding energies per nucleon of Li^7 and He^4 are 5.60 and 7.06 MeV respectively. (Assume mass of proton $\approx$ mass of neutron).

Solution

Q value of the reaction is given by

$$Q = (2 \times 4 \times 7.06 - 7 \times 5.6) \text{ MeV} = 17.28 \text{ MeV}$$



Applying Law of Conservation of Energy for Collision, we get

$$K_p + Q = 2K_\alpha \quad \dots(1)$$

where K_p and K_α are the kinetic energies of proton and α particle respectively.

Applying Law of Conservation of Linear Momentum, we get

$$\sqrt{2m_p K_p} = 2\sqrt{2m_\alpha K_\alpha} \cos \theta \quad \dots(2)$$

$$\Rightarrow K_p = 16K_\alpha \cos^2 \theta = (16K_\alpha) \left(\frac{1}{4}\right)^2 \quad \left\{ \because m_\alpha = 4m_p \right\}$$

$$\Rightarrow K_\alpha = K_p \quad \dots(3)$$

Solving equations (1) and (3) with $Q = 17.28 \text{ MeV}$, we get

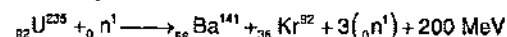
$$K_p = 17.28 \text{ MeV}$$

NUCLEAR FISSION

The splitting of heavy nucleus into two or more fragments of comparable masses, with an enormous release of energy is called nuclear fission.

CONCEPTUAL NOTE(S)

When slow neutrons are bombarded on ${}_{92}\text{U}^{235}$, the fission takes place according to reaction



- In nuclear fission the sum of masses before reaction is greater than the sum of masses after reaction, the difference in mass being released in the form of fission energy.
- The phenomenon of nuclear fission was discovered by Otto Hans and F. Strassmann in 1939 and was explained by N. Bohr and J.A. Wheeler on the basis of liquid drop model of nucleus.
- It may be pointed out that it is not necessary that in each fission of uranium, the two fragments Ba^{141} and Kr^{92} are formed but they may be any stable isotopes of middle weight atoms. The most probable division is into two fragments containing about 40% and 60% of the original nucleus with the emission of 2 or 3 neutrons per fission. So, average number of neutrons produced per fission is 2.5.
- Most of energy released appears in the form of kinetic energy of fission fragments.
- The fission of U^{235} takes place by fast neutrons.

CHAIN REACTION

If on the average more than one of the neutrons produced in each fission are capable of causing further fission, the number of fissions taking place at successive stages goes on increasing at a rapid rate, giving rise to self sustained sequence of fission known as chain reaction. The chain reaction takes place only if the size of the fissionable material is greater than a certain size called the critical size.

There are two types of chain reactions.

A. UNCONTROLLED CHAIN REACTION

In this process the number of fissions in a given interval on the average goes on increasing and the system will have the explosive tendency. This forms the principle of atom bomb. If a nuclear reaction is uncontrolled then in about $1 \mu\text{s}$, energy of order of $2 \times 10^3 \text{ J}$ is released.

B. CONTROLLED CHAIN REACTION (AS IN A NUCLEAR REACTOR)

In this process the number of fissions in a given interval is maintained constant by absorbing a desired number of neutrons. This forms the principle of nuclear reactor, consisting of the following parts:

- Fuel:** The fuel is U^{235} or U^{233} or Pu^{239}
- Moderator:** A moderator is a suitable material to slow down neutrons produced in the fission. The best choice as moderators are heavy water (D_2O) and graphite (C).
- Controller:** To maintain the steady rate of fission, the neutron absorbing material known as controller is used. The control rods are made of Cadmium or Boron-steel.

- d) **Coolant:** To remove the considerable amount of heat produced in the fission process, suitable cooling fluids known, as coolants are used. The usual coolants are water, carbon-dioxide, air etc.
- e) **Reactor shield:** The intense neutrons and gamma radiation produced in nuclear reactors are harmful for human body. To protect the workers from such radiations, the reactor core is surrounded by concrete wall, called the reactor shield.

CRITICAL MASS

If the amount of uranium is too small, then the liberated neutrons have large scope to escape from the surface and the chain reaction may stop before enough energy is released for explosion. Therefore, in order for explosion to occur, the mass uranium has to be greater than some minimum value, called the critical mass.

REPRODUCTION FACTOR

It is the ratio of the rate of neutron production and the rate at which the neutrons disappear.

Whether a mass of active material will sustain a chain reaction or not is determined by the reproduction factor (K). If $K \geq 1$, the chain reaction will be sustained. If $K = 1$, the mass is said to be critical.

Illustration 51

Polonium ($^{210}_{84}\text{Po}$) emits ^4_2He particles and is converted into lead ($^{206}_{82}\text{Pb}$). This reaction is used for producing electric power in a space mission. Po^{210} has half life of 138.6 days. Assuming an efficiency of 10% for the thermoelectric machine, how much ^{210}Po is required to produce $1.2 \times 10^7 \text{ J}$ of electric energy per day at the end of 693 days. Also find the initial activity of the material.

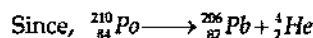
Given: Masses of nuclei

$$^{210}\text{Po} = 209.98264 \text{ amu}, \quad ^{206}\text{Pb} = 205.97440 \text{ amu}, \\ ^4_2\text{He} = 4.00260 \text{ amu}.$$

$$1 \text{ amu} = 931 \text{ MeV} \quad \text{and}$$

$$\text{Avogadro number} = 6 \times 10^{23} \text{ mol}^{-1}$$

Solution



$$\Rightarrow \Delta m = 0.00564 \text{ amu}$$

$$\text{Energy liberated per reaction} = (\Delta m)931 \text{ MeV} = 8.4 \times 10^{-13} \text{ J}$$

$$\text{Electrical energy produced} = 8.4 \times 10^{-14} \text{ J}$$

Let $m \text{ g}$ of ^{210}Po is required to produce the desired energy, then

$$N = \frac{m}{210} \times 6 \times 10^{23}$$

$$\text{Also, } \lambda = \frac{0.693}{t_{1/2}} = 0.005 \text{ per day}$$

$$\Rightarrow \left(-\frac{dN}{dt}\right) = \lambda N = \frac{(0.005)(6 \times 10^{23})m}{210} \text{ per day}$$

So, electrical energy produced per day is

$$E = \frac{(0.005)(6 \times 10^{23})m}{210} \times 8.4 \times 10^{-14} \text{ J}$$

$$\text{Since, } E = 1.2 \times 10^7 \quad \text{(given)}$$

$$\Rightarrow m = 10 \text{ g}$$

Activity at the end of 693 days is

$$R = \frac{0.005 \times 6 \times 10^{23} \times 10}{210} = \frac{10^{21}}{7} \text{ per day} = R_0 \left(\frac{1}{2}\right)^n$$

where, n is the number of half lives

$$\Rightarrow n = \frac{693}{138.6} = 5$$

$$\Rightarrow R_0 = R(2)^5 = 32 \times \frac{10^{21}}{7} = 4.57 \times 10^{21} \text{ per day}$$

Illustration 52

In a nuclear reactor, fission is produced in 1 g for U^{235} (235.0439 u) in 24 hours by a slow neutron (1.0087 u). Assume that $^{92}_{36}\text{Kr}$ (91.8973 u) and $^{141}_{56}\text{Ba}$ (140.9139 amu) are produced in all reactions and no energy is lost.

- (a) Write the complete reaction
(b) Calculate the total energy produced in kilowatt hour.
Given $1 \text{ u} = 931 \text{ MeV}$.

Solution

The nuclear fission reaction is



$$\text{Mass defect } \Delta m = [(m_u + m_n) - (m_{Ba} + m_{Kr} + 3m_n)]$$

$$\Delta m = 256.0526 - 235.8373 = 0.2153 \text{ u}$$

$$\text{Energy released, } \alpha = 0.2153 \times 931 = 200 \text{ MeV}$$

$$\text{Number of atoms in } 1 \text{ g} = \frac{6.02 \times 10^{23}}{235} = 2.56 \times 10^{21}$$

Energy released in fission of 1 g of U^{235} is

$$Q = 200 \times 2.56 \times 10^{21} = 5.12 \times 10^{23} \text{ MeV}$$

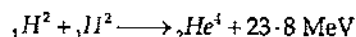
$$\Rightarrow Q = (5.12 \times 10^{23}) \times (1.6 \times 10^{-13}) = 8.2 \times 10^{10} \text{ J}$$

$$\Rightarrow Q = \frac{8.2 \times 10^{10}}{3.6 \times 10^6} \text{ kWh} = 2.28 \times 10^4 \text{ kWh}$$

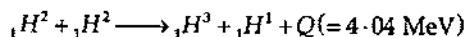
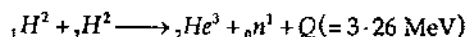
NUCLEAR FUSION

The phenomenon of combination of two or more light nuclei to form a heavy nucleus with release of enormous amount of energy is called the nuclear fusion. The sum of masses before fusion must be greater than the sum of masses after fusion,

the difference in mass appearing as fusion energy. The fusion of two deuterium nuclei into helium is expressed as



It may be pointed out that this fusion reaction does not actually occur. Due to huge quantity of energy release, the helium nucleus ${}_2He^4$ has got such a large value of excitation energy that it breaks up by the emission of a proton or a neutron as soon as it is formed, giving rise to the following reactions.



The fusion process occurs at extremely high temperature and high pressure just as it takes place at sun where temperature is 10^7 K . So, fusion reactions are also called Thermo-nuclear reactions.

Nuclear fusion has the possibility of being a much better source of energy than fission due to the following reasons.

- In fusion there is no radiation hazard as no radioactive material is used.
- The fuel needed for fission (U-235 etc.) is not available easily whereas hydrogen needed for fusion can be obtained in huge quantity.
- The energy released per nucleon is much more in fusion than in fission.

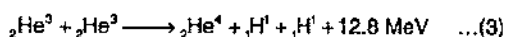
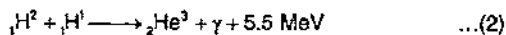
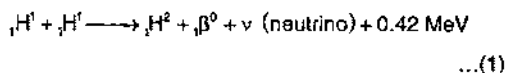
However, the very high temperature and pressure required for fusion cannot be easily created and maintained and as such it has not been possible as yet to use fusion for power generation.

CONCEPTUAL NOTE(S)

- For the fusion to take place, the component nuclei must be brought to within a distance of 10^{-14} m . For this they must be imparted high energies to overcome the repulsive force between nuclei. This is possible when temperature is enormously high.
- The principle of hydrogen bomb is also based on nuclear fusion. To start a fusion bomb very high temperature is required. This is achieved by incorporating an atom bomb within the nuclear bomb.
- The source of energy of sun and other stars is nuclear fusion (or thermo-nuclear reactions). There are two possible cycles:

(i) Proton-Proton cycle

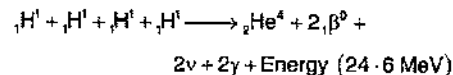
In 1938, Hans Bethe suggested that the stellar energy is produced by thermonuclear reactions in which protons are combined and transformed into helium nuclei. This is known as proton-proton cycle and is applicable for relatively low stellar temperature. The cycle is



The reactions (1) and (2) occur twice to yield two

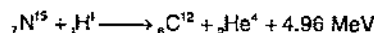
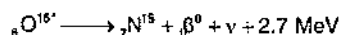
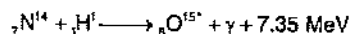
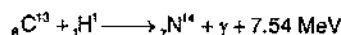
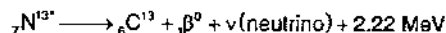
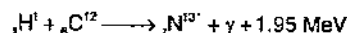
${}_2He^3$ nuclei.

Net result is

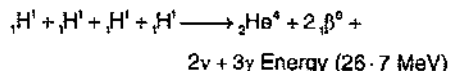


(ii) Carbon-Nitrogen cycle

For the main sequence stars with extremely high temperatures, Bethe suggested an alternative to proton-proton cycle called the Carbon-Nitrogen cycle. The cycle is



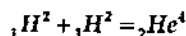
Net result is



For sun, both the cycles occur with equal probability. Stars with masses between 0.4 to 2.5 solar mass produce energy mainly by C-N cycle rather than P-P cycle. Stars with masses 0.4 solar mass or lower (which constitute the bulk of stellar population in our galaxy) mainly derive their energy from P-P cycle.

Illustration 53

It is proposed to use the nuclear fusion reaction:



in a nuclear reactor of 200 MW rating. If the energy from above reaction is used with a 25% efficiency in the reactor, how many grams of deuterium will be needed per day. (The masses of ${}_1H^2$ and ${}_2He^4$ are 2.0141 and 4.0026 u respectively).

Solution

Energy released in the nuclear fusion is

$$Q = \Delta mc^2 = \Delta m(931) \text{ MeV}$$

$$\Rightarrow Q = (2 \times 2.0141 - 4.0026) \times 931 \text{ MeV} = 23.834 \text{ MeV}$$

$$\Rightarrow Q = 23.834 \times 10^6 \text{ eV}$$

Since efficiency of reactor is 25%

$$\begin{aligned} \text{So effective energy used} &= \frac{25}{100} \times 23.834 \times 10^6 \times 1.6 \times 10^{-19} \text{ J} \\ &= 9.534 \times 10^{-13} \text{ J} \end{aligned}$$

Since the two deuterium nucleus are involved in a fusion reaction, therefore, energy released per deuterium is $\frac{9.534 \times 10^{-13}}{2}$

For 200 MW $\frac{\text{power per day}}{\text{number of deuterium nuclei required}}$

$$= \frac{200 \times 10^6 \times 86400}{\frac{9.534}{2} \times 10^{-13}} = 3.624 \times 10^{25}$$

Since 2g of deuterium constitute 6×10^{23} nuclei, therefore amount of deuterium required is

$$\frac{2 \times 3.624 \times 10^{25}}{6 \times 10^{23}} = 120.83 \text{ g/day}$$

Illustration 54

In the fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \longrightarrow {}^3_2\text{He} + {}^1_0\text{n}$, the masses of deuterium, helium and neutron expressed in amu are 2.015, 3.017 and 1.009 respectively. If 1 kg of deuterium undergoes complete fusion, find the amount of total energy released $1 \text{ amu} = 931.5 \text{ MeVc}^{-2}$.

Solution

$$\Delta m = 2(2.015) - (3.017 + 1.009) = 0.004 \text{ amu}$$

So, energy released is $\Delta E = (0.004 \times 931.5) \text{ MeV} = 3.726 \text{ MeV}$

$$\text{Energy released per deuteron} = \frac{3.726}{2} = 1.863 \text{ MeV}$$

$$\text{Number of deuterons in 1 kg} = \frac{6.02 \times 10^{26}}{2} = 3.01 \times 10^{26}$$

So, energy released per kg of deuterium fusion is

$$E = (3.01 \times 10^{26} \times 1.863) = 5.6 \times 10^{26} \text{ MeV}$$

$$\Rightarrow E \approx 9 \times 10^{13} \text{ J}$$

NUCLEAR HOLOCAUST

The estimate of after effects of the atomic (or nuclear) explosion is termed as nuclear holocaust. If a fusion bomb explodes, then a nuclear holocaust will not only destroy every form of life on earth but will also make this planet unfit for life for all times. The radioactive waste will hang like a cloud in earth's atmosphere and will absorb sun's radiations, thus causing a long nuclear winter. One can imagine this only by the mathematical figures quoted, according to which energy liberated by fission of 50 kg of U^{235} is equal to $4 \times 10^{15} \text{ J}$ which is the energy available from 20,000 tons of Trinitrotoluene (TNT).

USE OF RADIOISOTOPES

A. IN MEDICINE

Radioisotopes are extensively used in medicine:

- Radio iodine is used to determine the condition of human thyroid gland. Iodine-131 is administered orally to the patient. After a sufficient time, the activity is measured. From the observations it can be interpreted whether the gland is over-active, normal or under-active.
- Amounts of sodium and potassium in the body is measured by using Na-24 and K-42 as tracers.

- Radioactive isotopes are used to locate the position and extent of cancer.
- Radioisotopes are used in locating tumours within the brain.
- Radioactive Cr^{51} is used to locate the exact position where the haemorrhage might have taken place inside the body.
- Water contents of the body are measured by using deuterium and tritium as tracers.
- Radio gold is being used for the treatment of leukaemia.

B. IN INDUSTRY

- Radioactive Carbon-14 is used to study wear and tear of the position of an engine. C-14 is mixed with the ring. After some time, the engine oil is analysed to detect the presence of any radiation in it. In case of wear and tear the radiations are found to be there.
- Radio Cobalt is used for testing fields and castings by taking their photographs with γ -rays.

C. AS TRACERS

The radioactive isotope has identical chemical properties as another stable isotope. Therefore, by mixing it with stable isotope we can trace the presence or distribution of the element in a biological or physical system by detecting the radiation emitted by radio-isotope of that element. The radioisotope in such a case is said to be a Tracer.

Thus, a radioactive tracer is a radioisotope which, when mixed with a chemically similar element or artificially attached to a biological system, can be traced by radiation detecting devices.

Following are the few examples of radioisotopes acting as tracers:

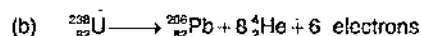
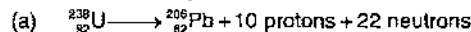
- Phosphorous-32 mixed with phosphorous manure has been used to study the process of extracting food from soil by various plants.
- Radio carbon is being used for research in photosynthesis in plants.
- The progress and absorption of sodium chloride in the body can be studied by feeding the person with radio-isotope sodium-24 alongwith sodium chloride.
- With the aid of radioisotopes, the rate, place and sequence of formation of the organic constituents of a living body can be studied.

ICE III

BASED ON BINDING ENERGY & NUCLEAR REACTIONS

(Solutions on page 4.104)

- Consider two decay reactions.



Are both the reactions possible?

Given : Average binding energy of ${}_{92}^{238}\text{U} = 7.57 \text{ MeV}$, that of ${}_{82}^{206}\text{Pb} = 7.83 \text{ MeV}$ and that of ${}^4_2\text{He} = 7 \text{ MeV}$ per nucleon.

2. Find the minimum kinetic energy of an α -particle to cause the reaction ${}^{14}_7\text{N} + {}^4_2\text{He} \rightarrow {}^{17}_8\text{O} + {}^1_1\text{H}$. The masses of ${}^{14}_7\text{N}$, ${}^4_2\text{He}$, ${}^1_1\text{H}$ and ${}^{17}_8\text{O}$ are respectively 14.00307u , 4.00260u , 1.00783u and 16.99913u .
3. In a neutron induced fission of ${}_{92}^{235}\text{U}$ nucleus, usable energy of 185 MeV is released. If a ${}_{92}^{235}\text{U}$ reactor is continuously operating it at a power level of 100 MW , find the time it takes for 1 kg of uranium to be consumed in this reactor.
4. Find the increase in mass of water when 1 kg of water absorbs $4.2 \times 10^3 \text{ J}$ of energy to produce a temperature rise of 1 K .
5. ${}^{212}_{83}\text{Bi}$ decay as per following equation.

$${}^{212}_{83}\text{Bi} \longrightarrow {}^{208}_{81}\text{Tl} + {}^4_2\text{He}$$

The kinetic energy of α particle emitted is 6.802 MeV . Calculate the kinetic energy of Tl recoil atoms.
6. Neon-23 decays in the following way

$${}^{23}_{10}\text{Ne} \longrightarrow {}^{23}_{11}\text{Na} + {}^0_{-1}\text{e} + \bar{\nu}$$

Find the minimum and maximum kinetic energy possessed by the beta particle (${}^0_{-1}\text{e}$). The atomic masses of ${}^{23}\text{Ne}$ and ${}^{23}\text{Na}$ are 22.9945u and 22.9898u , respectively.
7. The radionuclide ${}^{11}_6\text{C}$ decays according to ${}^{11}_6\text{C} \longrightarrow {}^{11}_5\text{B} + \text{e}^+ + \nu$. The maximum energy of the emitted positrons is 0.961 MeV .

Given that atomic mass of ${}^{11}_6\text{C} = m_c = 11.011434\text{u}$, atomic mass of ${}^{11}_5\text{B} = m_b = 11.009305\text{u}$, and the mass of positron $= m_p = 0.0005486\text{u}$, calculate disintegration energy Q and compare it with the maximum energy of the emitted positron given above. ($1\text{u} = 931 \text{ MeV}$).
8. It is proposed to use the nuclear fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \longrightarrow {}^4_2\text{He}$ in a nuclear reactor of 200 MW rating. If the energy from the above reaction is used with a 25% efficiency in the reactor, how many gram of deuterium fuel will be needed per day. Given that the masses of ${}^2_1\text{H}$ and ${}^4_2\text{He}$ are 2.0141 atomic mass units and 4.0026 atomic mass unit respectively.

9. Assuming the splitting of U^{235} nucleus liberates 200 MeV energy, find

- (a) the energy liberated in the fission of 1 kg of U^{235} and
- (b) the mass of coal with calorific value of 30 kJ/gm^{-1} which is equivalent to 1 kg of U^{235} .

10. In a nuclear reaction $\alpha + {}^7_3\text{N}^{14} \longrightarrow {}^8_4\text{O}^{17} + \text{p}$ when α -particles of $\text{KE } 7.7 \text{ MeV}$ were bombarded on nitrogen atom, protons were ejected with a kinetic energy of 5.5 MeV .

- (a) Find the Q -value of the reaction
- (b) Find the angle ϕ between the direction of motion of proton and α -particle.

Given that atomic mass of ${}^1_1\text{H} = 1.00814 \text{ amu}$, atomic mass of ${}^7_3\text{N}^{14} = 14.00752 \text{ amu}$, Atomic mass of ${}^8_4\text{O}^{17} = 17.00453$ and atomic mass of ${}^4_2\text{He} = 4.00388 \text{ amu}$.

11. 8 protons and 8 neutrons are separately at rest. How much energy will be released if we form ${}^{16}_8\text{O}$ nucleus?

Given : Mass of ${}^{16}_8\text{O}$ atom = 15.994915u

Mass of neutron = 1.008665u

Mass of hydrogen atom = 1.007825u

Solved Problems

Problem 1

A gas of hydrogen like ions is prepared in such a way that the ions are only in the ground state and the first excited state. A monochromatic light of wavelength $1216 \text{ \AA}$ is absorbed by the ions. The ions are lifted to higher excited states and emit radiations of six wavelengths, some higher, some lower or some greater than the incident wavelength.

- Find the principle quantum number of the final excited state.
- Identify the nuclear charge on the ions.
- Calculate the value of the maximum and minimum wavelengths.

Solution

- Energy corresponding to $\lambda = 1216 \text{ \AA}$ is,

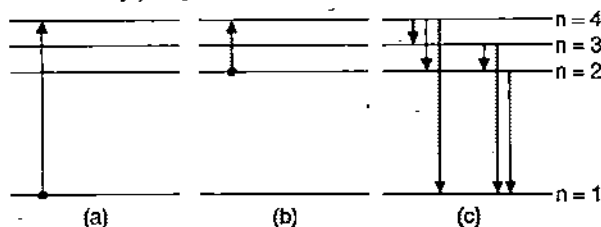
$$E = \frac{12375}{1216} \text{ eV} = 10.177 \text{ eV}$$

Now, total six wavelengths are obtained in the emission spectrum, hence from

$$\frac{n(n-1)}{2} = 6$$

We have $n = 4$, i.e., after excitation the single electron jumps to 3rd excited state or $n = 4$.

- Now it may jump either from $n = 1$ or $n = 2$.



If it jumps from $n = 1$, then in emission spectrum all the six photons have energy equal to or less than the energy of absorbed photon or the wavelength of emitted photon is either equal to or greater than the wavelength of absorbed photon. While in the question it is given

that the emitted wavelengths are either less than or greater than or smaller than the wavelength of absorbed photon. Which is possible only in the second case, i.e., when electron jumps from $n = 2$ to $n = 4$.

Hence, $E_4 - E_2 = 10.177 \text{ eV}$

$$\Rightarrow \frac{-13.6z^2}{4^2} - \left(\frac{-13.6z^2}{2^2} \right) = 10.177$$

Solving this, we get $z \approx 2$

- Maximum wavelength corresponds to minimum energy, i.e., a transition from $n = 4$ to $n = 3$. Thus,

$$\Delta E_{\min} = E_4 - E_3$$

$$\Rightarrow \Delta E_{\min} = \frac{(-13.6)(2)^2}{(4)^2} - \left[\frac{(-13.6)(2)^2}{(3)^2} \right]$$

$$\Rightarrow \Delta E_{\min} = 2.64 \text{ eV}$$

$$\Rightarrow \lambda_{\max} = \frac{hc}{\Delta E_{\min}} = \frac{12375}{2.64} = 4687 \text{ \AA}$$

Minimum wavelength corresponds to maximum energy, i.e., a transition from $n = 4$ to $n = 1$.

Hence, $\Delta E_{\max} = E_4 - E_1$

$$\Rightarrow \Delta E_{\max} = \frac{(-13.6)(2)^2}{(4)^2} - \left[\frac{(-13.6)(2)^2}{(1)^2} \right]$$

$$\Rightarrow \Delta E_{\max} = 51 \text{ eV}$$

$$\Rightarrow \lambda_{\min} = \frac{12375}{51} = 242 \text{ \AA}$$

Problem 2

A gas of hydrogen like atoms can absorb radiations of 68 eV . Consequently, the atoms emit radiations of only three different wavelengths. All the wavelengths are equal or smaller than that of the absorbed photon.

- Find the initial state of the gas atoms.
- Identify the gas atoms.
- Calculate the minimum wavelength of the emitted radiations.

- (d) Find the ionization energy and the respective wavelength for the gas atoms.

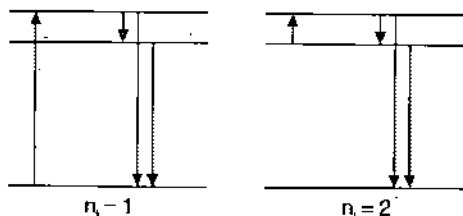
Solution

(a) Since, $N = \frac{n(n-1)}{2} = 3$

$\Rightarrow n = 3$

i.e., after excitation atom jumps to second excited state.

Hence $n_f = 3$. So n_i can be 1 or 2



If $n_i = 1$, then energy emitted is either equal to, greater than or less than the energy absorbed. Hence the emitted wavelength is either equal to, less than or greater than the absorbed wavelength. Hence $n_i \neq 1$.

If $n_i = 2$, then $E_e \geq E_a$ and hence $\lambda_e \leq \lambda_a$

So, $n_i = 2$

(b) Since, $E_3 - E_2 = 68 \text{ eV}$

$\Rightarrow (13.6)(Z^2)\left(\frac{1}{4} - \frac{1}{9}\right) = 68$

$\Rightarrow Z = 6$

(c) $\lambda_{\min} = \frac{12375}{E_3 - E_1} = \frac{12375}{(13.6)(6)^2\left(1 - \frac{1}{9}\right)} = 28.43 \text{ \AA}$

(d) Ionization energy is

$IE = (13.6)(6)^2 = 489.6 \text{ eV}$

$\Rightarrow \lambda = \frac{12375}{489.6} = 25.3 \text{ \AA}$

Problem 3

The energy levels of a hypothetical one electron atom are given by $E_n = -\frac{18}{n^2} \text{ eV}$, where $n = 1, 2, 3, \dots$

- Calculate the four lowest energy levels and construct the energy level diagram.
- Find the excitation potential of the stage $n = 2$.
- Find the wavelengths ($\text{\AA}$) which can be emitted when these atoms in the ground state are bombarded by electrons that have been accelerated through a potential difference of 16.2 V .
- Assuming these atoms to be in the ground state, will they absorb radiation having a wavelength of $2000 \text{ \AA}$?
- Also calculate the photoelectric threshold wavelength of this atom.

Solution

(a) Since, $E_n = -\frac{18}{n^2} \text{ eV}$, so we have

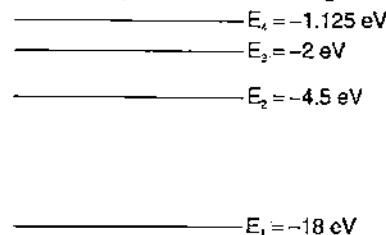
$E_1 = \frac{-18}{(1)^2} = -18 \text{ eV}$

$E_2 = \frac{-18}{(2)^2} = -4.5 \text{ eV}$

$E_3 = \frac{-18}{(3)^2} = -2 \text{ eV}$ and

$E_4 = \frac{-18}{(4)^2} = -1.125 \text{ eV}$

The energy level diagram is shown in figure.



(b) The excitation potential of stage $n = 2$ is

$E_2 - E_1 = 18 - 4.5 = 13.5 \text{ V}$

(c) Energy of the electron accelerated by a potential difference of 16.2 V is 16.2 eV . Since we observe that

$E_4 - E_1 = -1.125 - (-18) = 16.875 \text{ eV} > 16.2 \text{ eV}$

and $E_3 - E_1 = -2 - (-18) = 16 \text{ eV} < 16.2 \text{ eV}$

With this energy the electron will be able to excite the atom from $n=1$ to $n=3$, so we have the possible wavelengths corresponding to the transitions from $3 \rightarrow 1$, $3 \rightarrow 2$ and $2 \rightarrow 1$. Hence

$\lambda_{32} = \frac{12375}{E_3 - E_2} = \frac{12375}{-2 - (-4.5)} = 4950 \text{ \AA}$

$\lambda_{31} = \frac{12375}{E_3 - E_1} = \frac{12375}{16} = 773 \text{ \AA}$

and $\lambda_{21} = \frac{12375}{E_2 - E_1} = \frac{12375}{-4.5 - (-18)} = 917 \text{ \AA}$

(d) The energy corresponding to $\lambda = 2000 \text{ \AA}$ is given by

$E = \frac{12375}{2000} = 6.1875 \text{ eV}$

whereas the minimum excitation energy is 13.5 eV ($n=1$ to $n=2$).

Hence this is not possible.

(e) Threshold wavelength for photoemission to take place from such an atom is,

$\lambda_{\min} = \frac{12375}{18} = 687.5 \text{ \AA}$

Problem 4

Two hydrogen like atoms A and B are of different masses and each atom contains equal number of protons and neutrons. The difference in the energies between the first Balmer lines emitted by A and B is 5.667 eV . When the atom A and B , moving with the same velocity, strike a heavy target they rebound back with the same velocity. In the process, atoms B imparts twice momentum to the target than that A imparts. Identify the atoms A and B .

Solution

$$5.667 = 13.6(Z_B^2 - Z_A^2) \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$\Rightarrow Z_B^2 - Z_A^2 = 3 \quad \dots(1)$$

Applying Law of Conservation of Linear Momentum on atom A and heavy target, we get

$$m_A u = Mv_1 - m_A u$$

$$\Rightarrow 2m_A u = Mv_1$$

Similarly for atom B and heavy target, we get

$$2m_B u = Mv_2$$

$$\text{Given } Mv_2 = 2Mv_1$$

$$\Rightarrow m_B = 2m_A \quad \dots(2)$$

Since, both A and B contain equal number of protons and neutrons, so we have

$$\frac{m_A}{m_B} = \frac{2Z_A}{Z_B} = \frac{Z_A}{Z_B} \quad \dots(3)$$

From these equations we get

$$Z_A = 1 \text{ and } Z_B = 2$$

i.e., A is ${}_1H^2$ and B is ${}_2He^4$ (both having single electron).

Problem 5

Stopping potential of 24 kV , 10 kV , 110 kV and 115 kV are measured for photoelectrons emitted from a certain element when it is radiated with monochromatic X-ray. If this element is used as a target in an X-ray tube, what will be the wavelength of K_α line?

Solution

Stopping potentials are 24 kV , 10 kV , 110 kV and 115 kV

If the electrons are emitted from conduction band then the maximum kinetic energy of photoelectrons would be $115 \times 10^3 \text{ eV}$.

If they are emitted from next inner shell maximum kinetic energy of photoelectrons would be $110 \times 10^3 \text{ eV}$ and so on.

For photoelectrons of L shell it would be $100 \times 10^3 \text{ eV}$ and for K shell it is $24 \times 10^3 \text{ eV}$. Therefore, difference between energy of L shell and K shell is,

$$\Delta E = E_L - E_K = (100 - 24) \times 10^3 \text{ eV}$$

$$\Rightarrow \Delta E = 76 \times 10^3 \text{ eV}$$

The wavelength of K_α line (transition of electron from L shell to K shell) is,

$$\lambda_{K_\alpha} (\text{in } \text{\AA}) = \frac{12375}{\Delta E (\text{in eV})}$$

$$\Rightarrow \lambda_{K_\alpha} = \frac{12375}{76 \times 10^3} = 0.163 \text{ \AA}$$

Problem 6

For a certain hypothetical one-electron atom, the wavelength (in $\text{\AA}$) for the spectral lines for transitions originating at $n = p$ and terminating at $n = 1$ are given by

$$\lambda = \frac{1500p^2}{p^2 - 1} \text{ where } p = 2, 3, 4, \dots$$

- Find the wavelength of the least energetic and the most energetic photons in this series.
- Construct an energy level diagram for this element showing the energies of the lowest three levels.
- Calculate the ionization potential of this element.

Solution

$$(a) \text{ Since, } \lambda = \frac{1500p^2}{p^2 - 1}$$

$$\Rightarrow \lambda = 1500 \left(\frac{1}{1 - \frac{1}{p^2}} \right)$$

So, $\lambda_{\max}$ corresponds to least energetic photon with $p = 2$

$$\Rightarrow \lambda_{\max} = 1500 \left(\frac{1}{1 - \frac{1}{4}} \right) = 2000 \text{ \AA}$$

$\lambda_{\min}$ corresponds to most energetic photon with $p \rightarrow \infty$

$$\Rightarrow \lambda_{\min} = 1500 \text{ \AA}$$

$$(b) \lambda_{\infty \rightarrow 1} = 1500 \text{ \AA}$$

$$\Rightarrow E_\infty - E_1 = \frac{12375}{1500} \text{ eV} = 8.25 \text{ eV}$$

$$\Rightarrow E_1 = -8.25 \text{ eV} \quad \{\because E_\infty = 0\}$$

$$E_2 = -0.95 \text{ eV}$$

$$E_3 = -2.05 \text{ eV}$$

$$E_4 = -8.25 \text{ eV}$$

Further, $\lambda_{2 \rightarrow 1} = 2000 \text{ \AA}$

$$\Rightarrow E_2 - E_1 = \frac{12375}{2000} \text{ eV} = 6.2 \text{ eV}$$

$$\Rightarrow E_2 = -2.05 \text{ eV}$$

$$\text{Similarly } \lambda_{31} = 1500 \left(\frac{1}{1 - \frac{1}{9}} \right) = 1687.5 \text{ \AA}$$

$$\Rightarrow E_3 - E_1 = \frac{12375}{1687.5} \text{ eV} = 7.3 \text{ eV}$$

$$\Rightarrow E_3 = -0.95 \text{ eV}$$

(c) Ionization energy is

$$E_{i-\infty} = 8.25 \text{ eV}$$

So, Ionisation Potential equals 8.25 V

Problem 7

A radionuclide with half life 1620 sec is produced in a reactor at a constant rate 1000 nuclei per second. During each decay energy 200 MeV is released. If production of radio nuclides started at $t = 0$, calculate

(a) rate of release of energy at $t = 3240$ sec.

(b) total energy released upto $t = 405$ sec.

Solution

(a) Let N be the number of nuclei at time t , then net rate of increase of nuclei at instant t is,

$$\frac{dN}{dt} = \alpha - \lambda N$$

(where α = rate of production of nuclei)

$$\Rightarrow \int_0^N \frac{dN}{\alpha - \lambda N} = \int_0^t dt$$

$$\Rightarrow N = \frac{\alpha}{\lambda} (1 - e^{-\lambda t}) \quad \dots(1)$$

Rate of decay at this instant

$$R = \lambda N = \alpha (1 - e^{-\lambda t})$$

Hence, rate of release of energy at this time = R (energy released in each decay)

$$= \alpha (1 - e^{-\lambda t}) (200) \text{ MeVsec}^{-1}$$

Substituting the values, we get

$$\text{Rate of release of energy} = 1000 \left(1 - e^{-\frac{0.693}{1620} \times 3240} \right) (200)$$

$$\Rightarrow \left(\begin{array}{c} \text{Rate of Release} \\ \text{of Energy} \end{array} \right) = 1.5 \times 10^5 \text{ MeVsec}^{-1}$$

(b) Total number of nuclei decayed upto time t is $\alpha t - N$

$$\Rightarrow \left(\begin{array}{c} \text{Total number of} \\ \text{decayed nuclei} \end{array} \right) = \alpha t - \frac{\alpha}{\lambda} (1 - e^{-\lambda t})$$

Hence, total energy released upto this instant is

$$E = \left[\alpha t - \frac{\alpha}{\lambda} (1 - e^{-\lambda t}) \right] (200) \text{ MeV}$$

Substituting the values, we get

$$E = \left[1000 \times 405 - \frac{1000}{\frac{0.693}{1620}} \left(1 - e^{-\frac{0.693}{1620} \times 405} \right) \right] \times 200 \text{ MeV}$$

$$\Rightarrow E = 6.63 \times 10^6 \text{ MeV}$$

Problem 8

It is proposed to use nuclear reaction ${}_{84}\text{Po}^{210} \longrightarrow {}_{82}\text{Pb}^{206} + {}_2\text{He}^4$ to produce 2 kW electric power in a generator. The half life of polonium (Po^{210}) is 138.6 days. Assuming efficiency of the generator be 10%, calculate

(a) how many grams of (Po^{210}) are required per day at the end of 1386 days.

(b) initial activity of the material

Mass of nuclei: $\text{Po}^{210} = 209.98264 \text{ amu}$,

$\text{Pb}^{206} = 205.97440 \text{ amu}$,

${}_2\text{He}^4 = 4.00260 \text{ amu}$ (1 amu = 931 MeV)

Solution

(a) $\Delta m = 0.00564 \text{ amu} = 5.25 \text{ MeV} = 8.4 \times 10^{-13} \text{ J}$

$$\text{Since, } \lambda = \frac{0.693}{t_{1/2}} = 0.005 \text{ per day}$$

Let m g of Po^{210} are required per day for the reactor, then

$$n = \frac{(6.02 \times 10^{23})m}{210}$$

$$\Rightarrow \left(-\frac{dN}{dt} \right) = \lambda N = \frac{0.005 \times 6.02 \times 10^{23} \times m}{210} \text{ per day}$$

So, energy produced per day is

$$E = \frac{0.005 \times 6.02 \times 10^{23} \times m}{210} \times 8.4 \times 10^{-13} \text{ J}$$

$$\Rightarrow E = (12 \times 10^6) m \text{ J}$$

Now, 10% of $(12 \times 10^6) m$ equals $2 \times 10^3 \times 24 \times 3600 \text{ J}$

$$\Rightarrow m = \frac{2 \times 10^3 \times 24 \times 3600}{1.2 \times 10^6} = 144 \text{ g}$$

$$(b) R = \lambda N = (0.005) \left(\frac{144}{210} \right) (6.02 \times 10^{23})$$

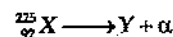
$$\Rightarrow R = 2.064 \times 10^{21} \text{ per day}$$

$$\text{Now, } R = R_0 \left(\frac{1}{2} \right)^n \quad \left\{ \because n = \frac{1386}{138.6} = 10 \right\}$$

$$\Rightarrow R_0 = (2)^{10} R = 2.11 \times 10^{24} \text{ per day}$$

Problem 9

Suppose a nucleus initially at rest undergoes α decay according to equation



At $t = 0$, the emitted α particle enters in a region of space where a uniform magnetic field $\vec{B} = B_0 \hat{i}$ and electric field

$\vec{E} = E_0 \hat{i}$ exist. The α particle enters in the region with velocity $\vec{v} = v_0 \hat{j}$ from $x = 0$. At time $t = \sqrt{3} \times 10^7 \frac{m_\alpha}{q_\alpha E_0}$ sec, the particle was observed to have speed twice the initial speed v_0 , then find

- the velocity of α particle at time t .
- the initial velocity v_0 of the α particle
- the binding energy per nucleon of α particle.

Given that $m(Y) = 221.03u$, $m(\alpha) = 4.003u$, $m(n) = 1.009u$, $m(p) = 1.008u$ mass on α particle $m_\alpha = \frac{2}{3} \times 10^{-26}$ kg charge on α particle $q_\alpha = 3.2 \times 10^{-19}$ C and $1u = 931 \text{ MeV } c^{-2}$

Solution

- Magnetic force on α particle, (at $t = 0$)

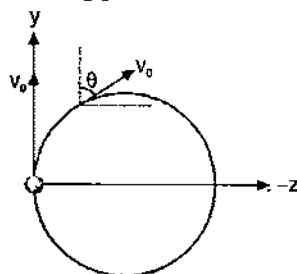
$$\vec{F}_m = q(\vec{v} \times \vec{B}) = q_\alpha [(v_0 \hat{j}) \times (B_0 \hat{i})]$$

$$\Rightarrow \vec{F}_m = -q_\alpha v_0 B_0 \hat{k}$$

Force due to electric field (at any time t)

$$\vec{F}_e = q\vec{E} = q_\alpha E_0 \hat{i}$$

Hence, the particle will move in a circular path in y - z plane due to magnetic field and at the same time it will move along x -direction. The resultant path is therefore, a helix with increasing pitch.



Hence, velocity of particle at any time t can be written as,

$$\vec{v} = \left(\frac{q_\alpha E_0 t}{m_\alpha} \right) \hat{i} + v_0 \cos \theta \hat{j} - v_0 \sin \theta \hat{k}$$

$$\text{where } \theta = \omega t = \frac{B_0 q_\alpha}{m_\alpha} t$$

- Speed of particle at any time t is

$$v = \sqrt{\left(\frac{q_\alpha E_0 t}{m_\alpha} \right)^2 + v_0^2} \quad \{\because \sin^2 \theta + \cos^2 \theta = 1\}$$

Given $v = 2v_0$ at $t = (\sqrt{3} \times 10^7) \frac{m_\alpha}{q_\alpha E_0}$, so, we get

$$(2v_0)^2 = (\sqrt{3} \times 10^7)^2 + v_0^2$$

$$\Rightarrow v_0 = 10^7 \text{ ms}^{-1}$$

- When an α -particle is emitted with velocity v_0 from a stationary nucleus X , decay product (nucleus Y) recoils. Then by Law of Conservation of Linear Momentum, we have

$$m_y v_y = m_\alpha v_0$$

$$\Rightarrow v_y = \frac{m_\alpha v_0}{m_y} = \left(\frac{4.003}{221.03} \right) (10^7)$$

$$\Rightarrow v_y = 1.81 \times 10^5 \text{ ms}^{-1}$$

Total energy released during α -decay of nucleus X is

$$E = \text{K.E. of nucleus } Y + \text{K.E. of } \alpha\text{-particle}$$

$$\Rightarrow E = \frac{1}{2} m_y v_y^2 + \frac{1}{2} m_\alpha v_0^2$$

$$\Rightarrow E = \frac{1.66 \times 10^{-27}}{2 \times 1.6 \times 10^{-13}} [(221.03)(1.81 \times 10^5)^2 + (4.003)(10^7)^2]$$

$$\Rightarrow E = 2.11 \text{ MeV}$$

$$\text{Hence, Mass lost during } \alpha\text{-decay is } \frac{2.11}{931.5} u = 0.0023u$$

Mass of nucleus X is

$$m_x = (m_y + m_\alpha + 0.0023)u$$

$$\Rightarrow m_x = 225.0353u$$

Mass defect in nucleus

$$X, \Delta m = 92m_p + (225 - 92)m_n - m_x = 1.898u$$

So, binding energy per nucleon is

$$\frac{BE}{A} = \frac{1.898 \times 931.5}{225} \text{ MeV} = 7.86 \text{ MeV}$$

Problem 10

A radionuclide consists of two isotopes. One of the isotopes decays by α -emission and other by β -emission with half lives $T_1 = 405 \text{ s}$, $T_2 = 1620 \text{ s}$, respectively. At $t = 0$, probabilities of getting α and β -particles from the radionuclide are equal. Calculate their respective probabilities at $t = 1620 \text{ s}$. If at $t = 0$, total number of nuclei in the radio nuclide are N_0 . Calculate the time t when total number of nuclei remained undecayed becomes equal to $\frac{N_0}{2}$.

Given, $\log_{10} 2 = 0.3010$, $\log_{10} 5.94 = 0.7742$ and $x^4 + 4x - 2.5 = 0$, $x = 0.594$.

Solution

At $t = 0$, probabilities of getting α and β particles are same. This implies that initial activity of both is equal, say R_0 .

Activity after $t = 1620 \text{ s}$ is

$$R_1 = R_0 \left(\frac{1}{2} \right)^{\frac{1620}{405}} = \frac{R_0}{16}$$

$$\text{and } R_2 = R_0 \left(\frac{1}{2} \right)^{\frac{1620}{1620}} = \frac{R_0}{2}$$

$$\text{Total activity } R = R_1 + R_2 = \frac{9}{16} R_0$$

$$\text{Probability of getting } \alpha \text{ particles is } \frac{R_1}{R} = \frac{1}{9} \text{ and}$$

$$\text{Probability of getting } \beta \text{ particles is } \frac{R_2}{R} = \frac{8}{9}$$

$$\text{Since, } R_{01} = R_{02}$$

$$\Rightarrow \frac{N_{01}}{T_1} = \frac{N_{02}}{T_2}$$

$$\Rightarrow \frac{N_{01}}{N_{02}} = \frac{1}{4}$$

Let N_0 be the total number of nuclei at $t=0$ then,

$$N_{01} = \frac{N_0}{5} \text{ and } N_{02} = \frac{4N_0}{5}$$

$$\text{Given, that } N_1 + N_2 = \frac{N_0}{2}$$

$$\Rightarrow \frac{N_0}{5} \left(\frac{1}{2} \right)^{\frac{t}{405}} + \frac{4N_0}{5} \left(\frac{1}{2} \right)^{\frac{t}{1620}} = \frac{N_0}{2} \quad \dots(1)$$

$$\text{Let } \left(\frac{1}{2} \right)^{\frac{t}{1620}} = x, \text{ then the equation (1) becomes}$$

$$x^4 + 4x - 2.5 = 0$$

$$\Rightarrow x = 0.594$$

$$\Rightarrow \left(\frac{1}{2} \right)^{\frac{t}{1620}} = 0.594$$

Solving, we get

$$t = 1215 \text{ s}$$

Problem 11

If the wavelength of the n th line of Lyman series is equal to the de-Broglie wavelength of electron in initial orbit of a hydrogen like element ($Z=11$). Find the value of n .

Solution

n^{th} line of Lyman series means transition from $(n+1)^{\text{th}}$ state to first state.

$$\frac{1}{\lambda} = RZ^2 \left(1 - \frac{1}{(n+1)^2} \right) \quad \dots(1)$$

de-Broglie wavelength in $(n+1)^{\text{th}}$ orbit is

$$\lambda = \frac{h}{mv} = \frac{hr}{mvr} = \frac{(2\pi)(hr)}{(n+1)h} = \frac{2\pi r}{(n+1)}$$

$$\Rightarrow \frac{1}{\lambda} = \frac{(n+1)}{2\pi r} \quad \dots(2)$$

Equating equations (1) and (2), we get

$$\left(\frac{n+1}{2\pi r} \right) = RZ^2 \left(\frac{n(n+2)}{(n+1)^2} \right) \quad \dots(3)$$

$$\text{Since, } r \propto \frac{n^2}{Z}$$

$$\Rightarrow r = \frac{(n+1)^2}{11} r_0$$

Substituting in equation (3), we get

$$\frac{11}{2\pi r_0} = \frac{R(11)^2(n)(n+2)}{(n+1)}$$

$$\Rightarrow (n+1) = (1.09 \times 10^7)(11)(2\pi) \times (0.529 \times 10^{-10})(n^2 + 2n)$$

Solving this equation, we get

$$n = 24$$

Problem 12

A radioactive element decays by β -emission. A detector records n beta particles in 2 seconds and in next 2-seconds it records $\frac{3}{4}n$ beta particles. Find mean life correct to nearest whole number. Given $\log_e 2 = 0.6931$, $\log_e 3 = 1.0986$.

Solution

Let n_0 be the number of radioactive nuclei at time $t=0$. Number of nuclei decayed in time t are given by $n_0(1 - e^{-\lambda t})$, which is also equal to the number of beta particles emitted during the same interval of time. For the given condition.

$$n = n_0(1 - e^{-2\lambda}) \quad \dots(1)$$

$$\left(n + \frac{3}{4}n \right) = n_0(1 - e^{-4\lambda}) \quad \dots(2)$$

Dividing equation (2) by (1), we get

$$1.75 = \frac{1 - e^{-4\lambda}}{1 - e^{-2\lambda}}$$

$$\Rightarrow 1.75 - 1.75e^{-2\lambda} = 1 - e^{-4\lambda}$$

$$\Rightarrow 1.75e^{-2\lambda} - e^{-4\lambda} = \frac{3}{4} \quad \dots(3)$$

Let us take $e^{-2\lambda} = x$

Then the above equation becomes

$$x^2 - 1.75x + 0.75 = 0$$

$$\Rightarrow x = \frac{1.75 \pm \sqrt{(1.75)^2 - (4)(0.75)}}{2}$$

$$\Rightarrow x = 1 \text{ and } \frac{3}{4}$$

$$\Rightarrow e^{-2\lambda} = 1 \text{ or } e^{-2\lambda} = \frac{3}{4}$$

But $e^{-2\lambda} = 1$ is not accepted because which means $\lambda = 0$.

$$\text{Hence, } e^{-2\lambda} = \frac{3}{4}$$

$$\Rightarrow -2\lambda \log_e(e) = \log_e(3) - \log_e(4) = \log_e(3) - 2\log_e(2)$$

$$\Rightarrow \lambda = \log_e(2) - \frac{1}{2} \log_e(3)$$

Substituting the given values, we get

$$\lambda = 0.6931 - \frac{1}{2} \times (1.0986) = 0.14395 \text{ s}^{-1}$$

So, mean life

$$t_m = \frac{1}{\lambda} = 6.947 \text{ s}$$

Problem 13

Nuclei of a radioactive element A are being produced at a constant rate α . The element has a decay constant λ . At time $t = 0$, there are N_0 nuclei of the element.

- Calculate the number N of nuclei of A at time t .
- if $\alpha = 2\lambda N_0$, calculate the number of nuclei of A after one half-life of A and also the limiting value of N as $t \rightarrow \infty$.

Solution

- Let at time t , number of radioactive nuclei are N

Net rate of formation of nuclei of A

$$\frac{dN}{dt} = \alpha - \lambda N$$

$$\Rightarrow \frac{dN}{\alpha - \lambda N} = dt$$

$$\Rightarrow \int_{N_0}^N \frac{dN}{\alpha - \lambda N} = \int_0^t dt$$

Solving this equation, we get

$$N = \frac{1}{\lambda} (\alpha - (\alpha - \lambda N_0) e^{-\lambda t}) \quad \dots(1)$$

- Substituting $\alpha = 2\lambda N_0$ and $t = t_{1/2} = \frac{\log_e(2)}{\lambda}$ in equation (1), we get,

$$N = \frac{3}{2} N_0$$

- Substituting $\alpha = 2\lambda N_0$ and $t \rightarrow \infty$ in equation (1), we get

$$N = \frac{\alpha}{\lambda} = 2N_0$$

$$\Rightarrow N = 2N_0$$

Problem 14

A nucleus at rest undergoes a decay emitting an α -particle of de-Broglie wavelength, $\lambda = 5.76 \times 10^{-15} \text{ m}$. If the mass of the daughter nucleus is 223.610 amu and that of the α -particle is 4.002 amu. Determine the total kinetic energy in the final state. Hence obtain the mass of the parent nucleus in amu. (1 amu = 931.470 MeV c^{-2})

Solution

Given mass of α -particle, $m = 4.002 \text{ amu}$ and mass of daughter nucleus,

$$M = 223.610 \text{ amu}$$

de-Broglie wavelength of α -particle,

$$\lambda = 5.76 \times 10^{-15} \text{ m}$$

So, momentum of α -particle is

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{5.76 \times 10^{-15}} \text{ kgms}^{-1}$$

$$\Rightarrow p = 1.151 \times 10^{-19} \text{ kgms}^{-1}$$

By Law of Conservation of Linear Momentum, this should also be equal to the linear momentum of the daughter nucleus (in opposite direction).

Let K_1 and K_2 be the kinetic energies of α -particle and daughter nucleus. Then total kinetic energy in the final state is

$$K = K_1 + K_2 = \frac{p^2}{2m} + \frac{p^2}{2M}$$

$$\Rightarrow K = \frac{p^2}{2} \left(\frac{1}{m} + \frac{1}{M} \right)$$

$$\Rightarrow K = \frac{p^2}{2} \left(\frac{M+m}{Mm} \right)$$

Since, 1 amu = $1.67 \times 10^{-27} \text{ kg}$

Substituting the values, we get

$$K = 10^{-12} \text{ J}$$

$$\Rightarrow K = \frac{10^{-12}}{1.6 \times 10^{-13}} = 6.25 \text{ MeV}$$

$$\Rightarrow K = 6.25 \text{ MeV}$$

$$\text{Mass defect, } \Delta m = \frac{6.25}{931.470} = 0.0067 \text{ amu}$$

$$\therefore \begin{pmatrix} \text{mass of} \\ \text{parent} \\ \text{nucleus} \end{pmatrix} = \begin{pmatrix} \text{mass} \\ \text{of} \\ \alpha\text{-particle} \end{pmatrix} + \begin{pmatrix} \text{mass of} \\ \text{daughter} \\ \text{nucleus} \end{pmatrix} + \begin{pmatrix} \text{mass} \\ \text{defect} \\ (\Delta m) \end{pmatrix}$$

$$\Rightarrow m_{\text{parent}} = (4.002 + 223.610 + 0.0067) \text{ amu}$$

$$\Rightarrow m_{\text{parent}} = 227.62 \text{ amu}$$

Hence, mass of parent nucleus is 227.62 amu.

Problem 15

A hydrogen-like atom (described by the Bohr model) is observed to emit six wavelengths, originating from all possible transitions between a group of levels. These levels have energies between -0.85 eV and -0.544 eV (including both these values).

- Find the atomic number of the atom.
- Calculate the smallest wavelength emitted in these transitions.

(Take $hc = 1240$ eV-nm, ground state energy of hydrogen atom $= -13.6$ eV)

Solution

- Total 6 lines are emitted. Therefore,

$$\frac{n(n-1)}{2} = 6$$

$$\Rightarrow n = 4$$

So, transition is taking place between m^{th} energy state and $(m+3)^{\text{th}}$ energy state.

$$E_m = -0.85 \text{ eV}$$

$$\Rightarrow -13.6 \left(\frac{z^2}{m^2} \right) = -0.85$$

$$\Rightarrow \frac{z}{m} = 0.25 \quad \dots(1)$$

Similarly, $E_{m+3} = -0.544$ eV

$$\Rightarrow -13.6 \frac{z^2}{(m+3)^2} = -0.544$$

$$\Rightarrow \frac{z}{(m+3)} = 0.2 \quad \dots(2)$$

Solving equations (1) and (2) for z and m , we get

$$m = 12 \text{ and } z = 3$$

- Smallest wavelength corresponds to maximum difference of energies which is obviously $(E_{m+3} - E_m)$.

$$\Rightarrow \Delta E_{\text{max}} = -0.544 - (-0.85) = 0.306 \text{ eV}$$

$$\Rightarrow \lambda_{\text{min}} = \frac{hc}{\Delta E_{\text{max}}} = \frac{1240}{0.306} = 4052.3 \text{ nm}$$

Problem 16

A small quantity of solution containing Na^{24} radio nuclide (half-life = 15 hour) of activity 1.0 microcurie is injected into the blood of a person. A sample of the blood of volume 1 cm^3 taken after 5 hour shows an activity of 296 disintegrations per minute. Determine the total volume of the blood in the body of the person. Assume that the radioactive solution mixes uniformly in the blood of the person.

(1 curie $= 3.7 \times 10^{10}$ disintegrations per second)

Solution

λ is the disintegration constant, then

$$\lambda = \frac{0.693}{t_{1/2}} = \frac{0.693}{15} \text{ h}^{-1}$$

$$\Rightarrow \lambda = 0.0462 \text{ h}^{-1}$$

Let R_0 be the initial activity then

$$R_0 = 1 \text{ microcurie} = 3.7 \times 10^4 \text{ dps}$$

Let r be the activity in 1 cm^3 of blood at $t = 5$ hr, then

$$r = \frac{296}{60} \text{ disintegration per second}$$

$$\Rightarrow r = 4.93 \text{ disintegration per second, and}$$

R be the activity of whole blood at time $t = 5$ hr

Total volume of blood should be

$$V = \frac{R}{r}$$

$$\Rightarrow V = \frac{R_0 e^{-\lambda t}}{r}$$

Substituting the values, we get

$$V = \left(\frac{3.7 \times 10^4}{4.93} \right) e^{-(0.0462)(5)} \text{ cm}^3$$

$$\Rightarrow V = 5.95 \times 10^3 \text{ cm}^3$$

$$\Rightarrow V = 5.95 \text{ lt.}$$

Problem 17

The element curium $^{248}_{96}\text{Cm}$ has a mean life of 10^{13} seconds. Its primary decay modes are spontaneous fission and α -decay, the former with a probability of 8% and the latter with a probability of 92%. Each fission releases 200 MeV of energy. The masses involved in decay are as follows:

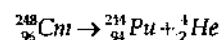
$$^{248}_{96}\text{Cm} = 248.072220 \text{ u},$$

$^{244}_{94}\text{Pu} = 244.064100 \text{ u}$ and $^4_2\text{He} = 4.002603 \text{ u}$. Calculate the power output from a sample of 10^{20} Cm atoms.

$$(1 \text{ u} = 931 \text{ MeV } c^{-2})$$

Solution

The reaction involved in α -decay is



Mass defect

$$\Delta m = \text{mass of } ^{248}_{96}\text{Cm} - \text{mass of } ^{244}_{94}\text{Pu} - \text{mass of } ^4_2\text{He}$$

$$\Rightarrow \Delta m = (248.072220 - 244.064100 - 4.002603) \text{ u}$$

$$\Rightarrow \Delta m = 0.005517 \text{ u}$$

Therefore, energy released in α -decay will be

$$E_{\alpha} = (0.005517 \times 931) \text{ MeV} = 5.136 \text{ MeV}$$

Similarly, $E_{\text{fission}} = 200 \text{ MeV}$

{given}

Mean life is

$$t_{\text{mean}} = 10^{13} \text{ s} = \frac{1}{\lambda}$$

So, disintegration constant $\lambda = 10^{-13} \text{ s}^{-1}$

Rate of decay at the moment when number of nuclei are 10^{20}

$$-\frac{dN}{dt} = \lambda N = (10^{-13})(10^{20})$$

$$\Rightarrow -\frac{dN}{dt} = 10^7 \text{ disintegration per second}$$

Of these disintegrations, 8% are in fission and 92% are in α -decay.

Therefore, energy released per second is

$$P = \frac{E}{t} = (0.08 \times 10^7 \times 200 + 0.92 \times 10^7 \times 5.136) \text{ MeV}$$

$$P = \frac{E}{t} = 2.074 \times 10^8 \text{ MeV}$$

So, power output (in watt) is

$$P = \text{energy released per second (Js}^{-1}\text{)}$$

$$\Rightarrow P = (2.074 \times 10^8)(1.6 \times 10^{-13})$$

So, power output is $P = 3.32 \times 10^{-5} \text{ W}$

Problem 18

A neutron of kinetic energy 65 eV collides inelastically with a singly ionized helium atom at rest. It is scattered at an angle of 90° with respect of its original direction.

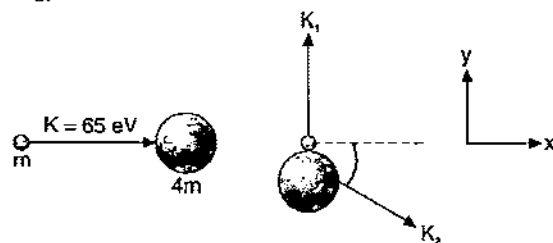
- Find the allowed values of the energy of the neutron and that of the atom after the collision.
- If the atom gets de-excited subsequently by emitting radiation, find the frequencies of the emitted radiation.

[Given : Mass of He atom = $4 \times$ (mass of neutrons)]

Ionization energy of H atom = 13.6 eV]

Solution

- Let K_1 and K_2 be the kinetic energies of neutron and helium atom after collision and ΔE be the excitation energy.



Applying Law of Conservation of Linear Momentum along x-direction, we get

$$p_i = p_f$$

$$\Rightarrow \sqrt{2Km} = \sqrt{2(4m)K_2} \cos \theta \quad \dots(1)$$

Similarly, Applying Law of Conservation of Linear momentum along y-direction, we get

$$\sqrt{2K_1m} = \sqrt{2(4m)K_2} \sin \theta \quad \dots(2)$$

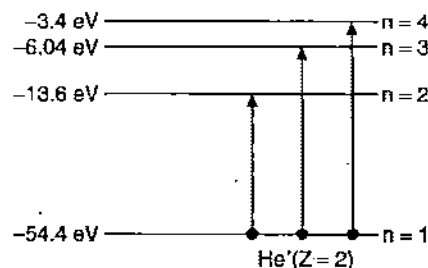
Squaring and adding equations (1) and (2), we get

$$K + K_1 = 4K_2 \quad \dots(3)$$

$$\Rightarrow 4K_2 - K_1 = K = 65 \text{ eV} \quad \dots(4)$$

Now, during collision, electron can be excited to any higher energy state. Applying Law of Conservation of Energy, we get

$$K = K_1 + K_2 + \Delta E$$



$$\Rightarrow 65 = K_1 + K_2 + \Delta E \quad \dots(5)$$

ΔE can have the following values,

$$\Delta E_1 = \{-13.6 - (-54.4)\} \text{ eV} = 40.8 \text{ eV}$$

Substituting in (5), we get

$$K_1 + K_2 = 24.2 \text{ eV} \quad \dots(6)$$

Solving (4) and (6), we get

$$K_1 = 6.36 \text{ eV and } K_2 = 17.84 \text{ eV}$$

Similarly, when we put $\Delta E = \Delta E_2$

$$\Rightarrow \Delta E = \{-6.04 - (-54.4)\} \text{ eV}$$

$$\Rightarrow \Delta E = 48.36 \text{ eV}$$

Put in equation (5), we get

$$K_1 + K_2 = 16.64 \text{ eV} \quad \dots(7)$$

Solving (4) and (7), we get

$$K_1 = 0.312 \text{ eV and } K_2 = 16.328 \text{ eV}$$

Similarly, when we put

$$\Delta E = \Delta E_3 = \{-3.4 - (-54.4)\} = 51 \text{ eV}$$

Put in equation (5), we get

$$K_1 + K_2 = 14 \text{ eV} \quad \dots(8)$$

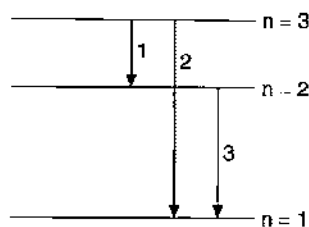
Now, solving (4) and (8), we get

$$K_1 = -1.8 \text{ eV and } K_2 = 15.8 \text{ eV}$$

But since the kinetic energy cannot have the negative values, the electron will not jump to third excited state i.e., $n = 4$.

Therefore, the allowed values of K_1 (KE of neutron) are 6.36 eV and 0.312 eV and of K_2 (KE of the atom) are 17.84 eV and 16.328 eV and the electron can jump upto second excited state only ($n = 3$).

- Possible emission lines are only three as shown in figure. The corresponding frequencies are



$$\nu_1 = \frac{(E_3 - E_2)}{h}$$

$$\Rightarrow \nu_1 = \frac{\{-6.04 - (-13.6)\} \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$\Rightarrow \nu_1 = 1.82 \times 10^{15} \text{ Hz}$$

$$\nu_2 = \frac{(E_3 - E_1)}{h}$$

$$\Rightarrow \nu_2 = \frac{\{-6.04 - (-54.4)\} \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$\Rightarrow \nu_2 = 11.67 \times 10^{15} \text{ Hz and } \nu_3 = \frac{(E_2 - E_1)}{h}$$

$$\Rightarrow \nu_3 = \frac{\{-13.6 - (-54.4)\} \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$\Rightarrow \nu_3 = 9.84 \times 10^{15} \text{ Hz}$$

Hence, the frequencies of emitted radiations are

$$1.82 \times 10^{15} \text{ Hz, } 11.67 \times 10^{15} \text{ Hz and } 9.84 \times 10^{15} \text{ Hz}$$

Problem 19

A particle of charge equal to that of an electron, $-e$, and mass 0.208 times of the mass of the electron (called a mu-meson) moves in a circular orbit around a nucleus of charge $+3e$. (Take the mass of the nucleus to be infinite). Assuming that the Bohr model of the atom is applicable to this system.

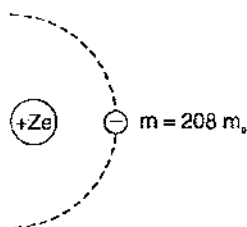
- Derive an expression for the radius of the n th Bohr orbit.
- Find the value of n for which the radius of the orbit is approximately the same as that of the first Bohr orbit for the hydrogen atom.
- Find the wavelength of the radiation emitted when the mu-meson jumps from the third orbit to the first orbit.

(Rydberg's constant $= 1.097 \times 10^7 \text{ m}^{-1}$)

Solution

If we assume that mass of nucleus is very much mass of mu-meson, then nucleus will be assumed to be at rest, only mu-meson is revolving round it.

- In n th orbit the necessary centripetal force to the mu-meson will be provided by the electrostatic force between the nucleus and the mu-meson.



$$\text{Hence, } \frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{(Ze)(e)}{r^2} \quad \dots(1)$$

Further, it is given that Bohr model is applicable to this system also. Hence,

$$\text{Angular momentum in } n\text{th orbit is } L = \frac{nh}{2\pi}$$

$$\Rightarrow mvr = n \frac{h}{2\pi} \quad \dots(2)$$

$$\Rightarrow v = \frac{nh}{2\pi mr}$$

Substituting in (1), we get

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z}$$

Substituting $Z = 3$ and $m = 208 m_e$, we get

$$r_n = \frac{n^2 h^2 \epsilon_0}{624 \pi m_e e^2}$$

- The radius of the first Bohr orbit for the hydrogen atom is

$$r_1 = \frac{h^2 \epsilon_0}{\pi m_e e^2}$$

Equating this with the radius calculated in part (a), we get

$$n^2 \approx 624$$

$$\Rightarrow n \approx 25$$

- Kinetic energy of atom is

$$K = \frac{1}{2} mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$$

$$\text{and the potential energy is } U = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

$$\Rightarrow \text{Total energy } E_n = -\frac{Ze^2}{8\pi\epsilon_0 r}$$

Substituting value of r , calculated in part (a), we get

$$E_n = \frac{1872}{n^2} \left(\frac{m_e e^4}{8\epsilon_0^2 h^2} \right)$$

But $\left(\frac{m_e e^4}{8\epsilon_0^2 h^2} \right)$ is the ground state energy of hydrogen atom and hence is equal to -13.6 eV

$$\Rightarrow E_n = \frac{-1872}{n^2} (13.6) \text{ eV} = -\frac{25459.2}{n^2} \text{ eV}$$

$$\Rightarrow E_3 - E_1 = -25459.2 \left(\frac{1}{9} - \frac{1}{1} \right) = 22630.4 \text{ eV}$$

So, the corresponding wavelength, is

$$\lambda (\text{in } \text{\AA}) = \frac{12375}{22630.4}$$

$$\Rightarrow \lambda = 0.546 \text{ \AA}$$

Practice Exercise

Single Correct Choice Type Questions

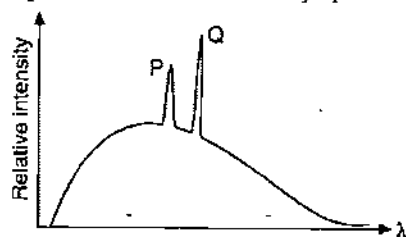
This section contains Single Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONLY ONE is correct.

- A radioactive substance is being produced at a constant rate of 200 nuclei. The decay constant of the substance is 1 s^{-1} . Assuming that initially there are no nuclei present, the time (in second) after which the number of nuclei will become 100 is
(A) 1 s (B) $\log_e(2)$ s
(C) $\frac{1}{\log_e(2)}$ s (D) 2 s
- Radioactivity is
(A) spontaneous process
(B) irreversible process
(C) self disintegration process
(D) all the above
- There are two radio nuclei A and B out of which A is an alpha emitter and B is a beta emitter. Their disintegration constants are in the ratio of 1:2. The ratio of number of atoms of A and B at any time t so that probabilities of getting alpha and beta particles are same at that instant is
(A) 1:2 (B) 1:e
(C) 2:1 (D) e:1
- The magnetic field at the centre (at nucleus) of the hydrogen like atoms (atomic number = z) due to the motion of electron in n th orbit is proportional to
(A) $\frac{n^3}{z^5}$ (B) $\frac{n^4}{z}$
(C) $\frac{z^2}{n^3}$ (D) $\frac{z^3}{n^5}$
- The half life of radon is 3.8 days. After how many days will $\frac{1}{10}$ th of the radon sample remain behind
(A) 1.262 days (B) 12.62 days
(C) 126.2 days (D) 1262 days
- α -particle emitted during various radioactive processes have same
(A) speed (B) momentum
(C) kinetic energy (D) specific charge
- In the beta decay, the mass number and atomic number of the daughter nuclei respectively
(A) increase by 2
(B) decrease by 1
(C) do not change at all
(D) mass number remains same but atomic number increases by 1
- Half life of a radioactive substance A is two times the half life of another radioactive substance B. Initially the number of nuclei of A and B are N_A and N_B respectively. After three half lives of A number of nuclei of both are equal. Then the ratio $\frac{N_A}{N_B}$ is
(A) $\frac{1}{3}$ (B) $\frac{1}{6}$
(C) $\frac{1}{4}$ (D) $\frac{1}{8}$
- The angular momentum of an electron in an orbit is quantized because it is a necessary condition for the compatibility with
(A) the wave nature of electron.
(B) particle nature of electron.
(C) Pauli's exclusion behaviour.
(D) None of these.
- In a sample of a radioactive substance, the fraction of the initial number of nuclei will remain undecayed after

a time $t = \frac{T}{2}$, where T is the half life of radioactive substance is

- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{4}$
(C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{1}{\sqrt{2}-1}$

11. In a characteristic X-ray spectra of some atom superimposed on continuous X-ray spectra



- (A) P represents K_{α} line
(B) Q represents K_{β} line
(C) Q and P represent K_{α} and K_{β} lines respectively
(D) Position of K_{α} and K_{β} depend on the particular atom

12. In X-ray tube when the accelerating voltage V is halved, the difference between the wavelengths of K_{α} line and minimum wavelength of continuous X-ray spectrum
(A) remains constant
(B) becomes more than two times
(C) becomes half
(D) becomes less than two times

13. The minimum kinetic energy of an electron, hydrogen ion, helium ion required for ionization of a hydrogen atom is E_1 in case electron is collided with hydrogen atom. It is E_2 when hydrogen ion is collided and E_3 when helium ion is collided. Then
(A) $E_1 = E_2 = E_3$ (B) $E_1 < E_2 < E_3$
(C) $E_1 > E_2 > E_3$ (D) $E_1 > E_2 > E_3$

14. The half life of ^{215}At is $100 \mu\text{s}$. The time taken for the radioactivity of a sample of ^{215}At to decay to $\frac{1}{16}$ th of its initial value is
(A) $400 \mu\text{s}$ (B) $6.3 \mu\text{s}$
(C) $40 \mu\text{s}$ (D) $300 \mu\text{s}$

15. The rate of decay of a radioactive element
(A) increases with increase in time
(B) decreases with increase in time
(C) remains constant with increase in time
(D) decreases exponentially with time

16. The total energy E of a sub-atomic particle of rest mass m_0 moving at non-relativistic speed v is

- (A) $E = m_0 c^2$ (B) $E = \frac{1}{2} m_0 v^2$
(C) $E = m_0 c^2 + \frac{1}{2} m_0 v^2$ (D) $E = m_0 c^2 - \frac{1}{2} m_0 v^2$

17. An electron is lying initially in the $n=4$ excited state. The electron de-excites itself to go to $n=1$ state directly emitting a photon of frequency ν_{41} . If the same electron first de-excites to $n=3$ state by emitting a photon of frequency ν_{43} and then goes from $n=3$ to $n=1$ state by emitting a photon of frequency ν_{31} , then

- (A) $\nu_{41} = \nu_{43} + \nu_{31}$ (B) $\nu_{41} = \nu_{43} - \nu_{31}$
(C) $\nu_{43} = \nu_{41} + 2\nu_{31}$ (D) Data Insufficient

18. The transition from the state $n=4$ to $n=3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition

- (A) $2 \rightarrow 1$ (B) $3 \rightarrow 2$
(C) $4 \rightarrow 2$ (D) $5 \rightarrow 4$

19. When an electron has a transition from the state $(n+1)$ to state n where n is quite large, then the frequency of the emitted radiation (ν) will vary with n as

- (A) $\nu \propto \frac{1}{n^3}$ (B) $\nu \propto \frac{1}{n^2}$
(C) $\nu \propto \frac{1}{n}$ (D) $\nu \propto n$

20. The spin angular momentum of an electron is equal to

- (A) $\frac{h}{\pi}$ (B) $\frac{h}{\sqrt{2}\pi}$
(C) $\frac{h}{2\pi}$ (D) $\frac{2h}{\sqrt{2}\pi}$

21. In Hydrogen spectrum the wavelength of H_{α} line is 656 nm , whereas in the spectrum of a distant galaxy, H_{α} line wavelength is 706 nm . Estimated speed of galaxy with respect to earth is

- (A) $2 \times 10^8 \text{ ms}^{-1}$ (B) $2 \times 10^7 \text{ ms}^{-1}$
(C) $2 \times 10^6 \text{ ms}^{-1}$ (D) $2 \times 10^5 \text{ ms}^{-1}$

22. If R be the Rydberg's constant, the energy of an electron in the ground state of the H atom is given by

- (A) $\frac{Rh}{c}$ (B) $-\frac{1}{Rhc}$
(C) $-Rhc$ (D) $\frac{hc}{R}$

23. The activity of a radioactive substance is R_1 at time t_1 and R_2 at time $t_2 (> t_1)$. Its decay constant is λ . Then

- (A) $R_1 t_1 = R_2 t_2$ (B) $R_2 = R_1 e^{\lambda(t_1 - t_2)}$
 (C) $\frac{R_1 - R_2}{t_2 - t_1} = \text{constant}$ (D) $R_2 = R_1 e^{\lambda(t_2 - t_1)}$

24. In the PROBLEM 23, number of atoms decayed between time interval t_1 and t_2 are

- (A) $\frac{\log_e(2)}{\lambda} (R_1 - R_2)$ (B) $R_1 e^{-\lambda t_1} - R_2 e^{-\lambda t_2}$
 (C) $\lambda (R_1 - R_2)$ (D) $\left(\frac{R_1 - R_2}{\lambda} \right)$

25. Half-life of a radioactive substance A is 4 days. The probability that a nucleus will decay in two half-lives is

- (A) $\frac{1}{4}$ (B) $\frac{3}{4}$
 (C) $\frac{1}{2}$ (D) 1

26. An artificial satellite moving in a circular orbit around the earth has a total (kinetic + potential) energy $-E_0$. Its potential energy is

- (A) $-E_0$ (B) $1.5E_0$
 (C) $2E_0$ (D) E_0

27. The de-Broglie wavelength of electron in ground state of an hydrogen atom is

- (A) $0.53 \text{ \AA}$ (B) $1.06 \text{ \AA}$
 (C) $1.52 \text{ \AA}$ (D) $3.33 \text{ \AA}$

28. The difference between n th and $(n+1)$ th Bohr's radius of H atom is equal to it's $(n-1)$ th Bohr's radius. The value of n is

- (A) 1 (B) 2
 (C) 3 (D) 4

29. The angular momentum of an electron in the hydrogen atoms is $\frac{3h}{2\pi}$, where, h is the Planck's constant. The

kinetic energy of this electron is

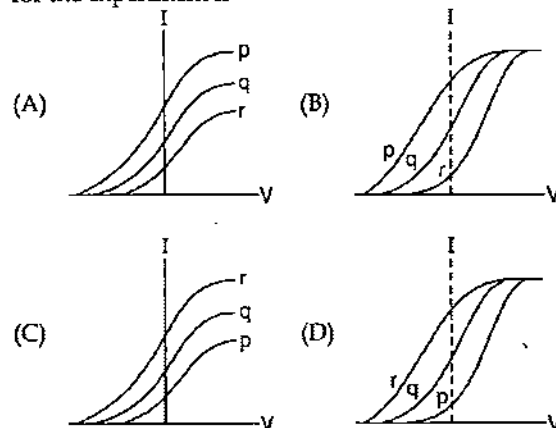
- (A) 1.51 eV (B) 3.4 eV
 (C) 4.35 eV (D) 6.8 eV

30. A γ -ray photon is emitted

- (A) after ionization of an atom
 (B) due to conversion of a neutron into a proton in the nucleus
 (C) after de-excitation of a nucleus
 (D) due to conversion of a proton into a neutron in the nucleus

31. Photoelectric effect experiments are performed using three different metal plates p , q and r having work functions $\phi_p = 2 \text{ eV}$, $\phi_q = 2.5 \text{ eV}$ and $\phi_r = 3 \text{ eV}$,

respectively. A light beam containing wavelengths of 550 nm, 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I - V graph for the experiment is



32. Two radioactive materials X_1 and X_2 have decay constants 10λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $\frac{1}{e}$ after a time

- (A) $\frac{1}{10\lambda}$ (B) $\frac{1}{11\lambda}$
 (C) $\frac{11}{10\lambda}$ (D) $\frac{1}{9\lambda}$

33. A potential difference of 20 kV is applied across an X-ray tube. The minimum wavelength of X-rays generated is

- (A) $0.26 \text{ \AA}$ (B) $0.62 \text{ \AA}$
 (C) $0.16 \text{ \AA}$ (D) $0.50 \text{ \AA}$

34. White X-rays are called 'white' due to the fact that

- (A) they are electromagnetic radiations having nature same as that of white light.
 (B) they are produced most abundantly in X-ray tubes.
 (C) they have a continuous wavelength range.
 (D) they can be converted to visible light using coated screens and photographic plates are affected by them just like light.

35. The half life period of a radioactive element X is same as the mean life time of another radioactive element Y. Initially both of them have the same number of atoms. Then,

- (A) X and Y have the same decay rate initially.
 (B) X and Y decay at the same rate always.
 (C) Y will decay at a faster rate than X.
 (D) X will decay at a faster rate than Y.

36. It has been observed that the most stable nuclei have generally the neutron to proton ratio as
(A) 1:2 (B) 2:1
(C) 1:4 (D) 1:1
37. Select the wrong statement.
(A) Radioactivity is a statistical process
(B) Radioactivity is a spontaneous process
(C) Radioactivity is the neutral characteristics of few elements
(D) Radioactive elements cannot be produced in the laboratory
38. Which of the following statements is correct?
(A) Beta rays are same as cathode rays.
(B) Gamma rays are high energy neutrons.
(C) Alpha particles are singly ionised helium atoms.
(D) Protons and neutrons have exactly the same mass.
39. The approximate value of quantum number n for the circular orbit of hydrogen 0.0001 mm in diameter is
(A) 1000 (B) 60
(C) 10000 (D) 31
40. Which of the following is its own antiparticle?
(A) photon (B) electron
(C) proton (D) π -meson
41. In Bohr's Model of hydrogen atom. The ratio between the period of revolution of an electron in orbit of $n=1$ to the period of revolution of the electron in the orbit $n=2$ is
(A) 1:2 (B) 2:1
(C) 1:4 (D) 1:8
42. K_α wavelength emitted by an atom of atomic number $Z=11$ is λ . Find the atomic number for an atom that emits K_α radiation with wavelength 4λ .
(A) $Z=6$ (B) $Z=4$
(C) $Z=11$ (D) $Z=44$
43. X-rays are produced in an X-ray tube operating at a given accelerating voltage. The wavelength of the continuous X-rays has values from
(A) 0 to ∞
(B) $\lambda_{\min}$ to ∞ where $\lambda_{\min} > 0$
(C) 0 to $\lambda_{\max}$ where $\lambda_{\max} < \infty$
(D) $\lambda_{\min}$ to $\lambda_{\max}$ where $0 < \lambda_{\min} < \lambda_{\max} < \infty$
44. The half-life of the radioactive radon is 3.8 days. The time, at the end of which $1/20^{\text{th}}$ of the radon sample will remain undecayed, is (given $\log_{10} e = 0.4343$)
(A) 3.8 day (B) 16.5 day
(C) 33 day (D) 76 day
45. In a hydrogen atom, the binding energy of the electron in the ground state is E_1 , then the frequency of revolution of the electron in the n^{th} orbit is
(A) $\frac{E_1 n}{h}$ (B) $\frac{E_1}{nh}$
(C) $\frac{2E_1 n}{h}$ (D) $\frac{2E_1}{nh}$
46. Magnetic moment due to the motion of the electron in n^{th} energy state of hydrogen atom is proportional to
(A) n^0 (B) n
(C) n^3 (D) n^5
47. A radioactive isotope is being produced at a constant rate X . Half life of the radioactive substance is Y . After some time the number of radioactive nuclei become constant. The value of this constant is
(A) XY (B) $\frac{X}{Y}$
(C) $\frac{XY}{\log_e(2)}$ (D) $(XY)\log_e(2)$
48. Which one of the following statements is wrong in the context of X-rays generated from an X-ray tube?
(A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases
(B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target
(C) Intensity of the characteristic X-rays depends on the electrical power given to the X-ray tube
(D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube
49. Assuming that all laws of thermodynamics can be applied to a nucleus, the α decay of a nucleus may be regarded as an
(A) isothermal process (B) isobaric process
(C) adiabatic process (D) isochoric process
50. Consider the fission reaction
$${}^{236}_{92}\text{U} \longrightarrow X^{117} + Y^{117} + n + n$$

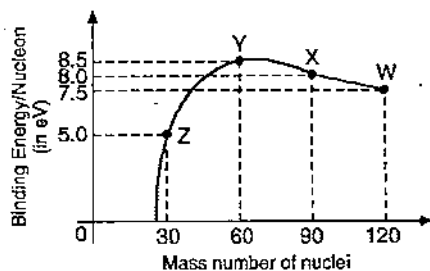
i.e. two nuclei of same mass number 117 are found plus two neutrons. The binding energy per nucleon of X and Y is 8.5 MeV whereas of U^{236} is 7.6 MeV. The total energy liberated is
(A) 2000 MeV (B) 200 MeV
(C) 20 MeV (D) 2 MeV
51. The activity of a radioactive sample is measured as 9750 counts (minute) $^{-1}$ at $t=0$ and 975 counts (minute) $^{-1}$ at $t=5$ minute. The decay constant is nearly
(A) 0.922 min^{-1} (B) 0.691 min^{-1}
(C) 0.461 min^{-1} (D) 0.230 min^{-1}

52. Magic numbers are
 (A) 2, 8, 18, 28, 50, 164....
 (B) 2, 8, 18, 50, 82, 164....
 (C) 2, 8, 20, 28, 50, 82, 126....
 (D) 2, 20, 50, 82, 164, 224....
53. The half life of radium is 1600 years. The fraction of the sample of radium that would remain after 6400 years is
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
 (C) $\frac{1}{8}$ (D) $\frac{1}{16}$
54. A and B are isotopes. B and C are isobars. If d_A , d_B and d_C be the densities of nuclei A, B and C respectively then
 (A) $d_A > d_B > d_C$ (B) $d_A < d_B < d_C$
 (C) $d_A = d_B = d_C$ (D) $d_A = d_B < d_C$
55. If u denotes 1 atomic mass unit. One atom of an element has mass exactly equal to Au , where A is mass number of element.
 (A) $A = 1$
 (B) $A = 12$
 (C) $A = 16$
 (D) A can take up any integral value from 1 to 110
56. The activity of a sample of radioactive material is R_1 at time t_1 and R_2 at time t_2 ($t_2 > t_1$). If mean life of the radioactive sample is T , then
 (A) $R_1 t_1 = R_2 t_2$ (B) $\frac{R_1 - R_2}{t_2 - t_1} = \text{constant}$
 (C) $R_2 = R_1 \exp\left(\frac{t_1 - t_2}{T}\right)$ (D) $R_2 = R_1 \exp\left(\frac{t_1}{T t_2}\right)$
57. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Which of the following statements is true?
 (A) its kinetic energy increases and its potential and total energies decrease
 (B) its kinetic energy decreases, potential energy increases and its total energy remains the same
 (C) its kinetic and total energies decrease and its potential energy increases
 (D) its kinetic, potential and total energies decrease
58. At time $t = 0$, N_1 nuclei of decay constant λ_1 and N_2 nuclei of decay constant λ_2 are mixed. The decay rate of mixture is
 (A) $-N_1 N_2 e^{-(\lambda_1 + \lambda_2)t}$
 (B) $-\left(\frac{N_1}{N_2}\right) e^{-(\lambda_1 + \lambda_2)t}$
 (C) $-(N_1 \lambda_1 e^{-\lambda_1 t} + N_2 \lambda_2 e^{-\lambda_2 t})$
 (D) $-N_1 \lambda_1 N_2 \lambda_2 e^{-(\lambda_1 + \lambda_2)t}$
59. If an electron has, orbital angular momentum quantum number $l = 7$, then it will have an orbital angular momentum equal to
 (A) $7\left(\frac{h}{2\pi}\right)$ (B) $42\left(\frac{h}{2\pi}\right)$
 (C) $\sqrt{7}\left(\frac{h}{2\pi}\right)$ (D) $\sqrt{56}\left(\frac{h}{2\pi}\right)$
60. The radius of the first orbit of hydrogen is 0.528 Å. The radius of the second orbit of hydrogen is
 (A) 4.752 Å (B) 2.112 Å
 (C) 0.071 Å (D) 0.142 Å
61. The speed of the electron in the first orbit (ground state) of the hydrogen atom in terms of velocity of light c is
 (A) $\frac{c}{2}$ (B) $\frac{c}{11}$
 (C) $\frac{c}{137}$ (D) $\frac{c}{274}$
62. A hydrogen atom emits a photon corresponding to an electron transition from $n = 5$ to $n = 1$. The recoil speed of hydrogen atom is almost
 (A) 10^{-4} ms^{-1} (B) $2 \times 10^{-2} \text{ ms}^{-1}$
 (C) 4 ms^{-1} (D) $8 \times 10^2 \text{ ms}^{-1}$
63. A beam of electron is used in an YDSE experiment. The slit width is d . When the velocity of electron is increased, then
 (A) no interference is observed
 (B) fringe width increases
 (C) fringe width decreases
 (D) fringe width remains same
64. Masses of two isobars ${}_{29}\text{Cu}^{64}$ and ${}_{30}\text{Zn}^{64}$ are 63.9298 u and 63.9292 u respectively. It can be concluded from these data that :
 (A) both the isobars are stable
 (B) Zn^{64} is radioactive, decaying to Cu^{64} through β -decay
 (C) Cu^{64} is radioactive, decaying to Zn^{64} through γ -decay
 (D) Cu^{64} is radioactive, decaying to Zn^{64} through β -decay
65. The ratio of molecular mass of two radioactive substances is $3/2$ and the ratio of their decay constant is $\frac{4}{3}$. Then the ratio of their initial activity per mole will be

- (A) 2 (B) $\frac{8}{9}$
(C) $\frac{4}{3}$ (D) $\frac{9}{8}$
66. The electronic transition in Li^{2+} ion that emits radiation of wavelength same as the wavelength of second Balmer line of H -atom is
(A) $4 \rightarrow 3$ (B) $6 \rightarrow 3$
(C) $12 \rightarrow 6$ (D) $12 \rightarrow 9$
67. The ratio between total acceleration of the electron is singly ionized helium atom and hydrogen atom (both in ground state) is
(A) 1 (B) 8
(C) 4 (D) 16
68. A radioactive substance X decays into another radioactive substance Y . Assuming that initially only X was present, λ_x and λ_y be the disintegration constants of X and Y , N_x and N_y be the number of nuclei of X and Y at any time t , then the number of nuclei N_y will be maximum when
(A) $\frac{N_x}{N_x - N_y} = \frac{\lambda_x}{\lambda_x - \lambda_y}$ (B) $\frac{N_y}{N_x - N_y} = \frac{\lambda_y}{\lambda_x - \lambda_y}$
(C) $\lambda_y N_x = \lambda_x N_y$ (D) $\lambda_y N_y = \lambda_x N_x$
69. The ratio of the maximum wavelength of the Lyman series in hydrogen spectrum to the maximum wavelength in the Paschen series is
(A) $\frac{3}{105}$ (B) $\frac{6}{15}$
(C) $\frac{52}{7}$ (D) $\frac{7}{108}$
70. The SI units of decay constant is
(A) m^{-1} (B) mm^{-1}
(C) s^{-1} (D) year
71. The velocity of an electron in the first orbit of H atom is v . The velocity of an electron in the 2nd orbit of He^+ is
(A) $2v$ (B) v
(C) $\frac{v}{2}$ (D) $\frac{v}{4}$
72. Electrons with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is
(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$
(C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$
73. A radioactive sample S_1 having an activity of $5 \mu Ci$ has twice the number of nuclei as another sample S_2 which has an activity of $10 \mu Ci$. The half lives of S_1 and S_2 can be
(A) 20 years and 5 years, respectively
(B) 20 years and 10 years, respectively
(C) 10 years each
(D) 5 years each
74. In the options given below, let E denote the rest mass energy of a nucleus and n a neutron. The correct option is
(A) $E(^{236}_{92}U) > E(^{137}_{53}I) + E(^{97}_{39}Y) + 2E(n)$
(B) $E(^{236}_{92}U) < E(^{137}_{53}I) + E(^{97}_{39}Y) + 2E(n)$
(C) $E(^{236}_{92}U) < E(^{140}_{56}Ba) + E(^{94}_{36}Kr) + 2E(n)$
(D) $E(^{235}_{92}U) < E(^{140}_{56}Ba) + E(^{94}_{36}Kr) + E(n)$
75. The potential difference across the Coolidge tube is 20 kV and 10 mA current flows through the voltage supply. Only 0.5% of the energy carried by the electrons striking the target is converted into X-rays. The power carried by X-ray beam is P .
(A) $P = 0.1 W$ (B) $P = 1 W$
(C) $P = 2 W$ (D) $P = 10 W$
76. For large principal quantum number n the frequency of revolution of electron is equal to the frequency of transition of the electron between two adjacent orbits. This frequency f is proportional to
(A) $\frac{1}{n^2}$ (B) $\frac{1}{n^3}$
(C) $\frac{1}{n}$ (D) $\frac{1}{n^6}$
77. A metal of atomic number Z is used as a target in a Coolidge tube. Let ν be the frequency of the K_α line. For a number of values of Z and ν which plot gives a straight line.
(A) ν vs Z (B) ν vs $\frac{1}{Z}$
(C) $\sqrt{\nu}$ vs Z (D) ν vs $\sqrt{Z}$
78. Solar constant of the sun is $\sigma = 8.106 \times 10^4 J min^{-1} m^{-2}$ and average sun earth distance is $1.5 \times 10^8 km$. The yearly loss in the mass of the sun is
(A) $13.8 \times 10^{17} kg$ (B) $1.38 \times 10^{19} kg$
(C) $1.38 \times 10^{17} kg$ (D) $13.8 \times 10^{20} kg$
79. Which of the following processes represents a γ -decay?
(A) $^A X_Z + \gamma \longrightarrow ^A X_{Z-1} + a + b$

- (B) ${}^A_ZX + {}^1_0n \rightarrow {}^{A-3}_{Z-2}X + c$
 (C) ${}^A_ZX \rightarrow {}^A_ZX + f$
 (D) ${}^A_ZX + e^- \rightarrow {}^A_{Z-1}X + g$

80. Binding energy per nucleon vs mass number curve for nuclei is shown in the figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is



- (A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$
 (C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

81. Packing fraction is given by the relation

- (A) $\frac{A}{M-A}$ (B) $\frac{A-M}{A}$
 (C) $\frac{M}{M-A}$ (D) $\frac{M-A}{A}$

82. In the following nuclear reaction, Y stands for
 ${}^{27}_{13}\text{Al} + {}^4_2\text{He} \rightarrow {}^{30}_{15}\text{P} + Y$

- (A) 1_0n (B) ${}^0_{-1}e$
 (C) ${}^1_1\text{H}$ (D) ${}^0_{+1}e$

83. If 1 mg of U^{235} is completely annihilated, the energy liberated is

- (A) $9 \times 10^{10} \text{ J}$ (B) $9 \times 10^{19} \text{ J}$
 (C) $9 \times 10^{18} \text{ J}$ (D) $9 \times 10^{17} \text{ J}$

84. The fraction of the initial number of active nuclei which remains undecayed after half of a half-life of the radioactive sample is

- (A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$
 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2} - 1$

85. The instantaneous concentration N , the initial concentration N_0 , the radioactive constant λ can be written as

- (A) $N = N_0 e^{\lambda t}$ (B) $N = N_0 e^{-\lambda t}$
 (C) $N = N_0 e^{-\lambda}$ (D) $N = N_0 e^{\lambda}$

86. How many α particles are emitted in the decay
 ${}^{235}_{92}\text{X} \rightarrow {}^{219}_{88}\text{Y}$

- (A) 4 (B) 5
 (C) 6 (D) no comments

87. In the above decay, how many β particles are given out?
 (A) 4 (B) 5
 (C) 6 (D) no comments

88. An electron jumps from the fourth orbit to the 2nd orbit of the hydrogen atom. If $R = 10^7 \text{ m}^{-1}$, the frequency of the emitted radiation will be

- (A) $\frac{3}{16} \times 10^{15} \text{ Hz}$ (B) $\frac{3}{16} \times 10^{15} \text{ Hz}$
 (C) $\frac{9}{16} \times 10^{15} \text{ Hz}$ (D) $\frac{3}{4} \times 10^{15} \text{ Hz}$

89. The recoil speed of a hydrogen atom after it emits a photon in going from $n=5$ state to $n=1$ state is

- (A) 4.718 ms^{-1} (B) 7.418 ms^{-1}
 (C) 4.178 ms^{-1} (D) 7.148 ms^{-1}

90. A hydrogen atom emits a photon of energy 12.1 eV. Its orbital angular momentum changes by ΔL . Then ΔL equals

- (A) $1.05 \times 10^{-34} \text{ Js}$ (B) $2.11 \times 10^{-34} \text{ Js}$
 (C) $3.16 \times 10^{-34} \text{ Js}$ (D) $4.22 \times 10^{-34} \text{ Js}$

91. The Bohr radius of the fifth electron of phosphorus atom (atomic number 15) acting as a dopant in silicon (relative dielectric constant 12) is

- (A) 380.9 pm (B) 390.8 pm
 (C) 930.8 pm (D) 830.9 pm

92. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector?

- (A) 2 photon of energy 10.2 eV
 (B) 2 photon of energy 1.4 eV
 (C) One photon of energy 10.2 eV and an electron of energy 1.4 eV
 (D) One photon of energy 10.2 eV and another photon of energy 1.4 eV

93. The half-life of ${}^{131}\text{I}$ is 8 days. Given a sample of ${}^{131}\text{I}$ at time $t=0$, we can assert that

- (A) no nucleus will decay before $t=4$ days
 (B) no nucleus will decay before $t=8$ days
 (C) all nuclei will decay before $t=16$ days
 (D) a given nucleus may decay at any time after $t=0$

94. The radius of second orbit of an electron in hydrogen atom is $2.116 \text{ \AA}$. The de-Broglie wavelength associated with this electron in this orbit would be
(A) $1.058 \text{ \AA}$ (B) $2.116 \text{ \AA}$
(C) $6.64 \text{ \AA}$ (D) $13.28 \text{ \AA}$
95. During a negative beta decay
(A) an atomic electron is ejected
(B) an electron which is already present within the nucleus is ejected
(C) a neutron in the nucleus decays emitting an electron
(D) a part of the binding energy of the nucleus is converted into an electron
96. Consider the spectral line resulting from the transition $n=2 \rightarrow n=1$ in the atoms and ions given below. The shortest wavelength is produced by
(A) hydrogen atom
(B) deuterium atom
(C) singly ionized helium
(D) doubly ionized lithium
97. The angular momentum (L) and radius (r) of a hydrogen atom are related to each other as
(A) $Lr = \text{constant}$ (B) $Lr^2 = \text{constant}$
(C) $Lr^4 = \text{constant}$ (D) $Lr^{-\frac{1}{2}} = \text{constant}$
98. The energy of a photon is equal to the kinetic energy of a proton. The energy of the photon is E . Let λ_1 be the de-Broglie wavelength of the proton and λ_2 be the wavelength of the photon. The ratio $\frac{\lambda_1}{\lambda_2}$ is proportional to
(A) E^0 (B) $E^{1/2}$
(C) E^{-1} (D) E^{-2}
99. The total energy of the electron in the hydrogen atom in the ground state is -13.6 eV . The kinetic energy of the electron is
(A) 0 (B) 13.6 eV
(C) 6.8 eV (D) -13.6 eV
100. Imagine an atom made up of a proton and a hypothetical particle of double the mass of electron but having the same charge as the electron. Apply the Bohr Atom Model and consider all possible transitions of this hypothetical particle to the first excited level. The longest wavelength photon that will be emitted has wavelength λ (given in terms of Rydberg constant R for the Hydrogen atom) equal to
(A) $\frac{9}{5R}$ (B) $\frac{36}{5R}$
(C) $\frac{18}{5R}$ (D) $\frac{4}{R}$
101. The binding energy of the electron with $n \rightarrow \infty$ in H atom is
(A) ZERO (B) infinite
(C) 13.6 eV (D) 10.2 eV
102. Energy levels A, B, C of a certain atom correspond to increasing values of energy i.e., $E_A < E_B < E_C$. If $\lambda_1, \lambda_2, \lambda_3$ are the wavelengths of radiations corresponding to the transitions C to B, B to A and C to A respectively, which of the following statements is correct?
(A) $\lambda_3 = \lambda_1 + \lambda_2$ (B) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$
(C) $\lambda_1 + \lambda_2 + \lambda_3 = 0$ (D) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$
103. If elements of quantum number greater than n were not allowed, the number of possible elements in nature would have been
(A) $\frac{1}{2}n(n+1)$ (B) $\left\{ \frac{n(n+1)}{2} \right\}^2$
(C) $\frac{1}{6}n(n+1)(2n+1)$ (D) $\frac{1}{3}n(n+1)(2n+1)$
104. The radius R of a nucleus changes with the nucleon number A of nucleus as
(A) $R \propto A^{\frac{2}{3}}$ (B) $R \propto A^{\frac{1}{3}}$
(C) $R \propto A^0$ (D) $R \propto A$
105. The net force between two nucleons 1 fm apart is F_1 if both are protons, F_2 if both are neutrons, and F_3 if one is a neutron and the other is a proton.
(A) $F_1 < F_2 < F_3$ (B) $F_2 < F_1 < F_3$
(C) $F_1 < F_2 = F_3$ (D) $F_1 = F_2 < F_3$
106. The half life of the radioactive substance is 40 days. The substance will disintegrate completely in
(A) 40 days (B) 400 days
(C) 4000 days (D) infinite time
107. A radioactive nuclide is produced at the constant rate of n per second (say, by bombarding a target with neutrons). The expected number N of nuclei in existence t seconds after the number is N_0 is given by
(A) $N = N_0 e^{-\lambda t}$ (B) $N = \frac{n}{\lambda} + N_0 e^{-\lambda t}$
(C) $N = \frac{n}{\lambda} + \left(N_0 - \frac{n}{\lambda} \right) e^{-\lambda t}$ (D) $N = \frac{n}{\lambda} + \left(N_0 + \frac{n}{\lambda} \right) e^{-\lambda t}$
(where λ is the decay constant of the sample)

108. The absorption coefficient of X-rays for a given wavelength is larger for
(A) lithium (B) lead
(C) aluminium (D) copper

109. For X-rays the wavelength λ (in Å) in terms of energy E (in keV) is given by

(A) $\lambda = \frac{6.20}{E}$ (B) $\lambda = \frac{12.40}{E}$
(C) $\lambda = \frac{3.10}{E}$ (D) $\lambda = \frac{24.80}{E}$

110. As per Bohr Model, the minimum energy (in eV) required to remove an electron from the ground state of doubly ionised Li atom ($Z=3$) is
(A) 1.51 (B) 13.6
(C) 40.8 (D) 122.4

111. The K_{α} X-ray emission line of tungsten occurs at $\lambda = 0.021$ nm. The energy difference between K and L levels in this atoms is about
(A) 0.51 MeV (B) 1.2 MeV
(C) 59 keV (D) 13.6 eV

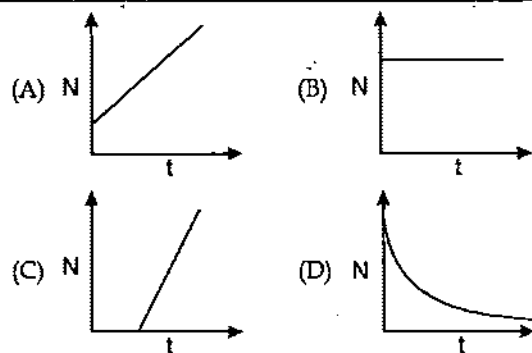
112. If the atom ${}_{100}\text{Fm}^{257}$ follows the Bohr's model and the radius of last orbit of ${}_{100}\text{Fm}^{257}$ is n times the Bohr radius, then find n
(A) 100 (B) 200
(C) 4 (D) $\frac{1}{4}$

113. A nucleus with mass number 220 initially at rest emits an α -particle. If the Q value of the reaction is 5.5 MeV, calculate the kinetic energy of the α -particle
(A) 4.4 MeV (B) 5.4 MeV
(C) 5.6 MeV (D) 6.5 MeV

114. The radioactivity of a sample is R_1 at time t_1 and R_2 at time t_2 . If the half life of the sample be T , the number of atoms that have disintegrated in time $(t_2 - t_1)$ is proportional to
(A) $R_1 t_1 - R_2 t_2$ (B) $(R_1 - R_2)^{-1}$
(C) $\frac{(R_1 - R_2)}{T}$ (D) $(R_1 - R_2)T$

115. In the given reaction, the radioactive radiations are emitted in the sequence as
 ${}_Z X^A \longrightarrow {}_{Z+1} Y^A \longrightarrow {}_{Z-1} T^{A-4} \longrightarrow {}_{Z-1} T^{A-4}$
(A) α, β, γ (B) β, α, γ
(C) γ, α, β (D) α, γ, β

116. The graph between the instantaneous concentration (N) of a radioactive element and time (t) is



117. The energy associated with a thermal neutron is of the order of
(A) 10 KeV (B) 1 KeV
(C) 0.1 MeV (D) 0.01 MeV

118. The ionisation potential of hydrogen atom is 13.6 volts. The energy of the atom in $n=2$ state will be
(A) -10.2 eV (B) -6.4 eV
(C) -4.4 eV (D) -3.4 eV

119. A radioactive nuclide can decay simultaneously by two different processes which have individual decay constants λ_1 and λ_2 respectively. The effective decay constant of the nuclide is λ given by
(A) $\lambda = \sqrt{\lambda_1 \lambda_2}$ (B) $\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$
(C) $\lambda = \frac{1}{2}(\lambda_1 + \lambda_2)$ (D) $\lambda = \lambda_1 + \lambda_2$

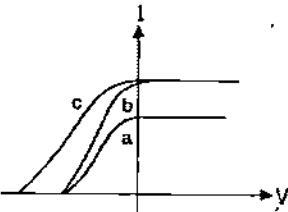
120. Masses of two isobars ${}^{64}_{29}\text{Cu}$ and ${}^{64}_{30}\text{Zn}$ are 63.9298 u and 63.9292 u respectively. It can be concluded from this data that
(A) both the isobars are stable.
(B) ${}^{64}\text{Zn}$ is radioactive, decaying to ${}^{64}\text{Cu}$ through β^- decay.
(C) ${}^{64}\text{Cu}$ is radioactive, decaying to ${}^{64}\text{Zn}$ through γ -decay.
(D) ${}^{64}\text{Cu}$ is radioactive, decaying to ${}^{64}\text{Zn}$ through β^- decay.

121. A fraction f_1 of a radioactive sample decays in one half life, and a fraction f_2 decays in one mean life.
(A) $f_1 > f_2$
(B) $f_1 < f_2$
(C) $f_1 = f_2$
(D) Data insufficient to arrive at a conclusion

122. The half life of ${}^{131}\text{I}$ is 8 days. Given a sample of ${}^{131}\text{I}$ at time $t=0$. We can assert that
(A) no nucleus will decay before $t=4$ days
(B) no nucleus will decay before $t=8$ days

- (C) all nuclei will decay before $t = 16$ days
(D) a given nucleus may decay at any time after $t = 0$
123. In the process of fission, the binding energy per nucleon
(A) increases
(B) decreases
(C) remains unchanged
(D) increases for mass number $A < 56$ nuclei but decreases for mass number $A > 56$ nuclei
124. A proton, a deuteron and an alpha particle are accelerated through potentials of V , $2V$ and $4V$ respectively. Their velocities will bear a ratio
(A) $1:1:1$ (B) $1:\sqrt{2}:1$
(C) $\sqrt{2}:1:1$ (D) $1:1:\sqrt{2}$
125. Let ν_1 be the frequency of the series limit of the Lyman series, ν_2 be the frequency of the first line of the Lyman series, and ν_3 be the frequency of the series limit of the Balmer series.
(A) $\nu_1 - \nu_2 = \nu_3$ (B) $\nu_2 - \nu_1 = \nu_3$
(C) $\nu_3 = \frac{1}{2}(\nu_1 + \nu_2)$ (D) $\nu_1 + \nu_2 = \nu_3$
126. The wavelength of the first spectral line in the Balmer series of hydrogen atom is $6561 \text{ \AA}$. The wavelength of the second spectral line in the Balmer series of singly-ionized helium atom is
(A) $1215 \text{ \AA}$ (B) $1640 \text{ \AA}$
(C) $2430 \text{ \AA}$ (D) $4687 \text{ \AA}$
127. ^{22}Ne nucleus, after absorbing energy, decays into two α -particles and an unknown nucleus. The unknown nucleus is
(A) Nitrogen (B) Carbon
(C) Oxygen (D) Boron
128. According to Bohr's theory of hydrogen atom, the product of the binding energy of the electron in the n th orbit and its radius in the n th orbit
(A) is inversely proportional to n^3
(B) is proportional to n^2
(C) has a constant value $7.2 \text{ eV \AA}$
(D) has a constant value of $10.2 \text{ eV \AA}$
129. If the half life and the mean life of the radioactive element is denoted by t and T respectively then
(A) $t = T$ (B) $t = \frac{1}{T}$
(C) $t < T$ (D) $t > T$
130. Let the potential energy of a hydrogen atom in the ground state be zero. Then its energy in the first excited state will be
(A) 10.2 eV (B) 13.6 eV
(C) 23.8 eV (D) 27.2 eV
131. In an excited state of hydrogen like atom an electron has a total energy of -3.4 eV . If the kinetic energy of the electron is E and its de Broglie wavelength is λ , then
(A) $E = 6.8 \text{ eV}$, $\lambda \sim 6.6 \times 10^{-10} \text{ m}$
(B) $E = 3.4 \text{ eV}$, $\lambda \sim 6.6 \times 10^{-10} \text{ m}$
(C) $E = 3.4 \text{ eV}$, $\lambda \sim 6.6 \times 10^{-11} \text{ m}$
(D) $E = 6.8 \text{ eV}$, $\lambda \sim 6.6 \times 10^{-11} \text{ m}$
132. The radioactivity of a substance is measured in terms of disintegration per second. 3×10^{10} disintegration per second is
(A) 1 eV (B) 1 MeV
(C) 1 rutherford (D) 1 curie
133. The shortest wavelength of the Brackett series of a hydrogen like atom having atomic number z , is the same as the shortest wavelength of the Balmer series of hydrogen atom. The value of z is
(A) 2 (B) 3
(C) 4 (D) 6
134. The shortest wavelength in the Lyman series is $912 \text{ \AA}$. Then the longest wavelength in this series is
(A) $1216 \text{ \AA}$ (B) $1824 \text{ \AA}$
(C) $2434 \text{ \AA}$ (D) $3648 \text{ \AA}$
135. The maximum angular speed of the electron of a hydrogen atom in a stationary orbit is
(A) $6.2 \times 10^5 \text{ rads}^{-1}$ (B) $4.1 \times 10^{16} \text{ rads}^{-1}$
(C) $2.4 \times 10^{10} \text{ rads}^{-1}$ (D) $9.2 \times 10^6 \text{ rads}^{-1}$
136. A hydrogen atom is in an excited state of principle quantum number n . It emits a photon of wavelength λ when returns to the ground state. The value of n is
(A) $\sqrt{\lambda R(\lambda R - 1)}$ (B) $\sqrt{\frac{\lambda R}{\lambda R - 1}}$
(C) $\sqrt{\frac{(\lambda R - 1)}{\lambda R}}$ (D) $\sqrt{\lambda(R - 1)}$
137. A freshly prepared radioactive source of half-life 2 h emits radiation of intensity which is 64 times the permissible safe level. The minimum time after which it would be possible to work safely with this source is
(A) 6 h (B) 12 h
(C) 24 h (D) 128 h

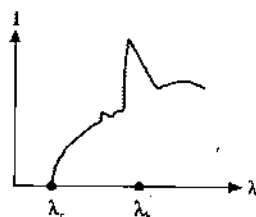
138. Consider α -particles, β -particles and γ -rays each having an energy of 0.5 MeV. In increasing order of penetrating powers, the radiations are
(A) α, β, γ (B) α, γ, β
(C) β, γ, α (D) γ, β, α
139. The equation:
 $4 {}^1_1\text{H}^+ \longrightarrow {}^4_2\text{He}^{2+} + 2e^- + 26 \text{ MeV}$
represents
(A) β -decay (B) γ -decay
(C) fusion (D) fission
140. If N_1 atoms of a radioactive element emit N_2 beta particles per second, then the decay constant of the element (in s^{-1}) is
(A) $\frac{N_2}{N_1}$ (B) $\frac{N_1}{N_2}$
(C) $N_2 \log_e(2)$ (D) $N_1 \log_e(2)$
141. The decay constant of a radioactive sample is λ . The half-life and mean-life of the sample are respectively given by
(A) $\frac{1}{\lambda}$ and $\frac{(\log_e 2)}{\lambda}$ (B) $\frac{(\log_e 2)}{\lambda}$ and $\frac{1}{\lambda}$
(C) $\lambda(\log_e 2)$ and $\frac{1}{\lambda}$ (D) $\frac{\lambda}{(\log_e 2)}$ and $\frac{1}{\lambda}$
142. Fast neutrons can easily be slowed down by
(A) the use of lead shielding
(B) passing them through heavy water
(C) elastic collisions with heavy nuclei
(D) applying a strong electric field
143. The electric potential between a proton and an electron is given by $V = V_0 \log_e \left(\frac{r}{r_0} \right)$, where r_0 is a constant. Assuming Bohr's model to be applicable, write variation of r_n with n , n being the principal quantum number.
(A) $r_n \propto n$ (B) $r_n \propto \frac{1}{n}$
(C) $r_n \propto n^2$ (D) $r_n \propto \frac{1}{n^2}$
144. After 280 days, the activity of a radioactive sample is 6000 dps. The activity reduces to 3000 dps after another 140 days. The initial activity of the sample in dps is
(A) 6000 (B) 9000
(C) 3000 (D) 24000
145. If elements with principal quantum number $n > 4$ were not allowed in nature, the number of possible elements would be
(A) 60 (B) 32
(C) 4 (D) 64
146. The binding energy of an electron in the ground state of He is equal to 24.6 eV. The energy required to remove both the electrons is
(A) 24.6 eV (B) 38.2 eV
(C) 49.2 eV (D) 79 eV
147. The binding energies of nuclei X and Y are E_1 and E_2 respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then
(A) $Q > E_2 - 2E_1$ (B) $Q < 2E_1 - E_2$
(C) $Q = E_2 - 2E_1$ (D) $Q = 2E_1 - E_2$
148. The shortest wavelength of X-rays emitted from an X-ray tube depends on
(A) the current in the tube
(B) the voltage applied to the tube
(C) the nature of the gas in tube
(D) the atomic number of the target material
149. Beta rays emitted by a radioactive material are
(A) electromagnetic radiations
(B) the electrons orbiting around the nucleus
(C) charged particles emitted by the nucleus
(D) neutral particles
150. A radioactive nucleus is being produced at a constant rate α per second. Its decay constant is λ . If N_0 are the number of nuclei at time $t = 0$, then maximum number of nuclei possible are
(A) $\frac{\alpha}{\lambda}$ (B) $N_0 + \frac{\alpha}{\lambda}$
(C) N_0 (D) $\frac{\lambda}{\alpha} + N_0$
151. The X-ray beam coming from an X-ray tube will be
(A) monochromatic
(B) having all wavelengths smaller than a certain maximum wavelength
(C) having all wavelengths larger than a certain minimum wavelength
(D) having all wavelengths lying between a minimum and a maximum wavelength
152. A radioactive isotope A decays into another isotope B which has a half-life equal to half the half-life of A. Both isotopes emit α -particles during their decay B decays into a stable nucleus. If a sample consists initially of atoms of A only, then the net activity of the sample initially
(A) decreases with time
(B) increases with time
(C) remains constant
(D) any of the above may be true

153. The probability of survival of a radioactive nucleus for one mean life is
 (A) $1 - \frac{1}{e}$ (B) $\frac{1}{e}$
 (C) $\frac{\log_e(2)}{e}$ (D) $1 - \frac{\log_e(2)}{e}$
154. The binding energy per nucleon of ${}_1H^2$ and ${}_2He^4$ are 1.1 eV and 7 MeV respectively. The energy released in the process ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4$ is
 (A) 20.8 MeV (B) 16.6 MeV
 (C) 25.2 MeV (D) 23.6 MeV
155. In a hydrogen atom, the electron is in n th excited state. It comes down to first excited state by emitting ten different wavelengths. The value of n is
 (A) 6 (B) 7
 (C) 8 (D) 9
156. There are two radioactive substances A and B such that decay constant of B is twice that of A. Initially both have equal number of nuclei. After n half lives of A, the rate of disintegration of both are equal. The value of n is
 (A) 16 (B) 4
 (C) 2 (D) 1
157. The figure shows the variation of photocurrent with anode potential for a photosensitive surface for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves a, b and c respectively

 (A) $f_a = f_b$ and $I_a \neq I_b$ (B) $f_a = f_c$ and $I_a = I_c$
 (C) $f_a = f_b$ and $I_a = I_b$ (D) $f_b = f_c$ and $I_b = I_c$
158. The ratio of the energies of the hydrogen atom in the first to the second excited state
 (A) 4:1 (B) 1:4
 (C) 4:9 (D) 9:4
159. The ratio of the wavelength of first line of Lyman series to the first line of Balmer series is
 (A) 1:4 (B) 5:27
 (C) 27:20 (D) 20:27
160. A hydrogen atom and a Li^{++} ion are both in the second excited state. If L_H and L_{Li} are their respective electronic angular momenta and E_H and E_{Li} their respective energies then,
 (A) $L_H > L_{Li}$ and $|E_H| > |E_{Li}|$
 (B) $L_H = L_{Li}$ and $|E_H| < |E_{Li}|$
 (C) $L_H = L_{Li}$ and $|E_H| > |E_{Li}|$
 (D) $L_H < L_{Li}$ and $|E_H| < |E_{Li}|$
161. The potential energy associated with an electron in the orbit
 (A) increases with the increase in radii of the orbit
 (B) decreases with the increase in the radii of the orbit
 (C) remains the same with the change in the radii of the orbit
 (D) None of the above
162. Suppose a radioactive substance disintegrates completely in 10 days. Each day it disintegrates at a rate which is twice the rate of the previous day. The percentage of the material left to be disintegrated after passing of 9 days is
 (A) 10 (B) 20
 (C) 25 (D) 50
163. Mirror nuclei are those in which the number of protons in the parent nuclei
 (A) is less than the daughter nuclei by one
 (B) is more than the daughter nuclei by two
 (C) equal to the neutron number
 (D) exceeds the number of neutrons in the daughter nuclei by unity by one positron emission
164. Half life period and mean life time of a radioactive element are
 (A) inversely proportional to each other
 (B) directly proportional to each other
 (C) equal to each other
 (D) not related to each other
165. If P_α , P_β and P_γ be the penetrating powers of α , β and γ radiations respectively then
 (A) $P_\alpha = P_\beta = P_\gamma$ (B) $P_\alpha > P_\beta > P_\gamma$
 (C) $P_\alpha < P_\beta < P_\gamma$ (D) $P_\alpha = P_\beta < P_\gamma$
166. A radioactive nuclide emits an α , β and a γ ray in close succession, the atomic mass of the end product is reduced by
 (A) 8 a.m.u. (B) about 4 a.m.u.
 (C) 2 a.m.u. (D) 1 a.m.u.
167. A star initially has 10^{40} deuterons. It produces energy via the processes ${}_1H^2 + {}_1H^2 \rightarrow {}_1H^3 + p$ and ${}_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + n$. If the average power radiated by the star is 10^{16} W, the deuteron supply of the star is exhausted in a time of the order of
 (A) 10^6 s (B) 10^8 s
 (C) 10^{12} s (D) 10^{16} s

- (The masses of nuclei are: $m(H^2) = 2.014 \text{ u}$, $m(p) = 1.007 \text{ u}$, $m(n) = 1.008 \text{ u}$, $m(He^4) = 4.001 \text{ u}$)
168. The potential difference applied to an X-ray tube is 5 kV and the current through it is 3.2 mA. The number of electrons striking the target per second is
(A) 2×10^{16} (B) 5×10^5
(C) 1×10^{17} (D) 4×10^{15}
169. The wavelength λ_a of X-rays produced by the X-ray tube is $0.76 \text{ \AA}$. The atomic number of the anode material of the tube is
(A) 30 (B) 40
(C) 50 (D) 60
170. In Rutherford experiment the number of α -particles scattered at angle 90° is 25. How many particles are scattered at an angle 60°
(A) 100 (B) 85
(C) 70 (D) 55
171. The kinetic energy of an electron in the first orbit of H atom is 13.6 eV. The total energy of an electron in the second orbit of He^+ is
(A) 13.6 eV (B) 3.4 eV
(C) -13.6 eV (D) -3.4 eV
172. If the elements with principal quantum number $n > 4$ were not allowed in nature, the possible number of elements would be
(A) 4 (B) 32
(C) 60 (D) 64
173. The ratio of areas between the electron orbits for the first excited state to the ground state for the hydrogen atom is
(A) 2:1 (B) 4:1
(C) 8:1 (D) 16:1
174. The wavelength of the first line of the Balmer series in the hydrogen atom spectrum is
(A) 6563 $\text{\AA}$ (B) 6365 $\text{\AA}$
(C) 6563 m (D) 6563 cm
175. An electron with kinetic energy $E \text{ eV}$ collides with a hydrogen atom in the ground state. The collision is observed to be elastic for
(A) $0 < E < \infty$ (B) $0 < E < 10.2 \text{ eV}$
(C) $0 < E < 13.6 \text{ eV}$ (D) $0 < E < 3.4 \text{ eV}$
176. The angular momentum of the electron in hydrogen atom in the ground state is
(A) $2h$ (B) $\frac{h}{2}$
(C) $\frac{h}{2\pi}$ (D) $\frac{h}{4\pi}$
177. An electron in the ground state of hydrogen has an angular momentum L_1 , and an electron in the first excited state of lithium has an angular momentum L_2 .
(A) $L_1 = L_2$ (B) $L_1 = 4L_2$
(C) $L_2 = 2L_1$ (D) $L_1 = 2L_2$
178. A certain radioactive element disintegrates for an interval of time equal to its mean life. The fraction of the original amount that remains undecayed is
(A) e (B) $\frac{1}{e}$
(C) e^2 (D) $\frac{1}{e^2}$
179. If N denotes the concentration of a radioactive element, then the rate of change of concentration with time (t) can be written as
(A) $\frac{dN}{dt} \propto N$ (B) $-\frac{dN}{dt} \propto N$
(C) $-\frac{dN}{dt} \propto e^N$ (D) $-\frac{dN}{dt} \propto e^{-N}$
180. In the sun about 4 billion kg of matter is converted to energy each second. The power output of the sun in watt is
(A) 3.6×10^{26} (B) 0.36×10^{26}
(C) 36×10^{26} (D) 0.036×10^{26}
181. Order of magnitude of density of uranium nucleus is ($m_p = 1.67 \times 10^{-27} \text{ kg}$)
(A) 10^{23} kgm^{-3} (B) 10^{17} kgm^{-3}
(C) 10^{14} kgm^{-3} (D) 10^{11} kgm^{-3}
182. The electron emitted in beta radiation originates from
(A) inner orbits of atoms
(B) free electrons existing in nuclei
(C) decay of a neutron in a nucleus
(D) photon escaping from the nucleus
183. Electrons with energy 80 keV are incident on the tungsten target of an X-ray tube. K shell electrons of tungsten have -72.5 keV energy. X-rays emitted by the tube contain only
(A) a continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $\sim 0.155 \text{ \AA}$.
(B) a continuous X-ray spectrum (Bremsstrahlung) with all wavelengths.

- (C) the characteristic X-ray spectrum of tungsten.
(D) a continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of $\sim 0.155 \text{ \AA}$ and the characteristic X-ray spectrum of tungsten.

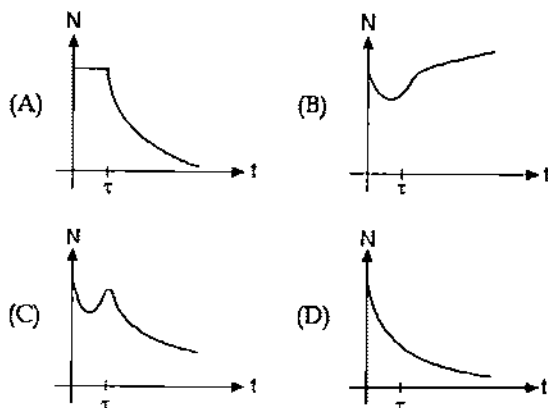
184. The intensity of X-rays from a Coolidge tube is plotted against wavelength λ as shown. The minimum wavelength found is λ_c and the wavelength of the K_α line is λ_k . As the accelerating voltage is increased



- (A) $(\lambda_k - \lambda_c)$ increases (B) $(\lambda_k - \lambda_c)$ decreases
(C) λ_k increases (D) λ_k decreases
185. The quantity $\frac{e^2}{2\hbar\epsilon_0 c}$ has a value

- (A) $\frac{1}{137} \text{ ms}^{-1}$ (B) $\frac{2}{137} \text{ ms}^{-1}$
(C) $\frac{1}{137}$ (D) $\frac{2}{137}$

186. A radioactive sample consists of two distinct species having equal number of atoms initially. The mean lifetime of one species is τ and that of the other is 5τ . The decay products in both cases are stable. A plot is made of the total number of radioactive nuclei as a function of time. Which of the following figures best represents the form of this plot?

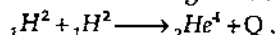


187. The largest wavelength in the ultraviolet region of the hydrogen spectrum is 122 nm . The smallest wavelength in the infrared region of the hydrogen spectrum (to the nearest integer) is
(A) 802 nm (B) 823 nm
(C) 1882 nm (D) 1648 nm

188. If a star can convert all the He nuclei completely into oxygen nuclei. The energy released per oxygen nuclei is [Mass of the helium nucleus is 4.0026 amu and mass of oxygen nucleus is 15.9994 amu]

- (A) 7.6 MeV (B) 56.12 MeV
(C) 10.24 MeV (D) 23.4 MeV

189. Consider the following reaction



If $m({}_1\text{H}^2) = 2.0141 \text{ u}$; $m({}_2\text{He}^4) = 4.0024 \text{ u}$.

The energy Q released (in MeV) in this fusion reaction is

- (A) 12 (B) 6
(C) 24 (D) 48

190. For uranium nucleus how does its mass vary with volume?

- (A) $m \propto V$ (B) $m \propto \frac{1}{V}$
(C) $m \propto \sqrt{V}$ (D) $m \propto V^2$

191. A radioactive element X converts into another stable element Y . If half life of X is 2 hr , initially only X to be present and after time t , the ratio of atoms of X and Y is found to be $1:4$, then time t in hr is

- (A) 2 (B) 4
(C) between 4 and 6 (D) 6

192. A radioactive material of half-life T was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are A_1 and A_2 respectively then their age difference equals

- (A) $\frac{T}{\log_e 2} \left| \log_e \left(\frac{2A_1}{A_2} \right) \right|$ (B) $T \left| \log_e \left(\frac{A_1}{A_2} \right) \right|$
(C) $\frac{T}{\log_e 2} \left| \log_e \left(\frac{A_2}{2A_1} \right) \right|$ (D) $T \left| \log_e \left(\frac{A_2}{2A_1} \right) \right|$

193. de-Broglie wavelength of an electron in the n th Bohr orbit is λ_n and the angular momentum is J_n , then

- (A) $J_n \propto \lambda_n$ (B) $\lambda_n \propto \frac{1}{J_n}$
(C) $\lambda_n \propto J_n^2$ (D) $\lambda_n \propto \frac{1}{J_n^2}$

194. When an electron in the hydrogen atom in ground state absorbs a photon of energy 12.1 eV , its angular momentum

- (A) decreases by $2.11 \times 10^{-34} \text{ Js}$
(B) decreases by $1.055 \times 10^{-34} \text{ Js}$
(C) increases by $2.11 \times 10^{-34} \text{ Js}$

(D) increases by 1.055×10^{-34} Js

195. A radioactive nucleus A finally transforms into a stable nucleus B. Then A and B can be

- (A) isotones (B) isotopes
(C) isobars (D) None of these

196. Number of nuclei of a radioactive substance at time $t=0$ are 1000 and 900 at time $t=2$ s. Then number of nuclei at time $t=4$ s will be

- (A) 700 (B) 790
(C) 800 (D) 810

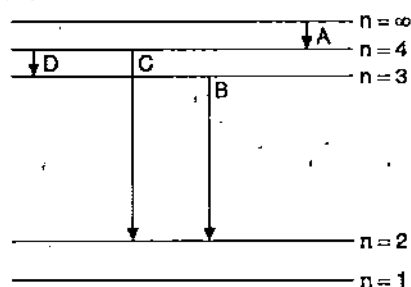
197. If ${}_{92}\text{U}^{238}$ changes to ${}_{82}\text{Pb}^{210}$ by a series of α and β decays, the number of α and β decays undergone is

- (A) 5 and 7 (B) 7 and 5
(C) 7 and 7 (D) 7 and 9

198. In hydrogen and hydrogen like atoms the ratio of difference of energies $E_{2n} - E_n$ and $E_{4n} - E_{2n}$ varies with atomic number z and principle quantum number n as

- (A) $\frac{z^2}{n^2}$ (B) $\frac{z^4}{n^4}$
(C) $\frac{z}{n}$ (D) $\left(\frac{z}{n}\right)^0$

199. Consider the electronic energy level diagram of H-atom. The photons associated with shortest and longest wavelengths would be emitted from the atom by the transitions labelled



- (A) D and C respectively
(B) C and A respectively
(C) C and D respectively
(D) A and C respectively

200. When a hydrogen atom emits a photon during the transition $n=5$ to $n=1$, its recoil speed is approximately

- (A) 0.1 mms^{-1} (B) 3 mms^{-1}
(C) 4 ms^{-1} (D) 800 ms^{-1}

201. A potential difference of 10^3 V is applied across an X-ray tube. The ratio of the de-Broglie wavelength of the incident electrons to the shortest wavelength of X-rays produced is

$$\left(\frac{e}{m} = 1.8 \times 10^{18} \text{ Ckg}^{-1} \text{ for an electron} \right)$$

- (A) $\frac{1}{10^4}$ (B) $\frac{1}{100}$
(C) $\frac{1}{20}$ (D) 1

202. During a nuclear fusion reaction

- (A) a heavy nucleus breaks into two fragments by itself
(B) a light nucleus bombarded by thermal neutrons breaks up
(C) a heavy nucleus bombarded by thermal neutrons breaks up
(D) two light nuclei combine to give a heavier nucleus and possibly other products

203. Four physical quantities are listed in Column I. Their values are listed in Column II in a random order

Column I		Column II
A	Thermal energy of air molecules at room temperature.	(i) 0.02 eV
B	Binding energy of heavy nuclei per nucleon.	(ii) 2 eV
C	X-ray photon energy.	(iii) 10 keV
D	Photon energy of visible light.	(iv) 7 MeV

The correct matching of columns I and II is given by

- | | A | B | C | D |
|-----|----|-----|-----|-----|
| (A) | i | iv | iii | ii |
| (B) | i | iii | ii | iv |
| (C) | ii | i | iii | iv |
| (D) | ii | iv | i | iii |

204. Of the various series of the hydrogen spectrum, the one which lies wholly in the ultra violet region is

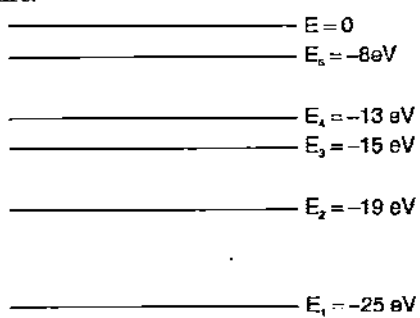
- (A) Lyman series (B) Balmer series
(C) Paschen series (D) Brackett series

Practice Exercise

Multiple Correct Choice Type Questions

This section contains Multiple Correct Choice Type Questions. Each question has four choices (A), (B), (C) and (D), out of which ONE OR MORE is/are correct.

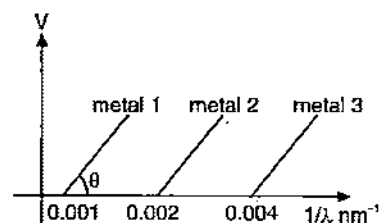
- The momentum of a single photon of red light of frequency 400×10^{12} Hz moving through free space is
(A) ZERO (B) 8.8×10^{-28} kgms⁻²
(C) 1.65×10^{-6} MeVc⁻¹ (D) Data Insufficient
- The radius of the orbit of an electron in a Hydrogen like atom is $4.5a_0$, where a_0 is the Bohr radius. Its orbital angular momentum is $\frac{3h}{2\pi}$. It is given that h is Planck constant and R is Rydberg constant. The possible wavelength(s), when the atom de-excites, is (are)
(A) $\frac{9}{32R}$ (B) $\frac{9}{16R}$
(C) $\frac{9}{5R}$ (D) $\frac{4}{3R}$
- H^+ , He^+ and O^{++} all having the same kinetic energy pass through a region in which there is a uniform magnetic field perpendicular to their velocity. The masses of H^+ , He^+ and O^{++} are 1 u, 4 u and 16 u respectively.
(A) H^+ will be deflected the most.
(B) O^{++} will be deflected the most.
(C) He^+ and O^{++} will be deflected equally.
(D) All will be deflected equally.
- Which of the following statement(s) is (are) correct?
(A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons
(B) The rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons
(C) In nuclear fission, energy is released by fusing two nuclei of medium mass (approximately 100 amu)
(D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus
- An electron makes transition inside a hydrogen atom. The orbital angular momentum of the electron may change by
(A) h (B) $\frac{h}{\pi}$
(C) $\frac{h}{2\pi}$ (D) $\frac{h}{4\pi}$
- An electron is excited from a lower energy state to a higher energy state in a hydrogen atom. Which of the following decrease in the excitation?
(A) potential energy (B) angular speed
(C) kinetic energy (D) angular momentum
- In Bohr's model of the hydrogen atom, let R , V , T and E represent the radius of the orbit, speed of the electron, time period of revolution of electron and the total energy of the electron respectively. The quantities proportional to the quantum number n are
(A) VR (B) RE
(C) $\frac{V}{E}$ (D) $\frac{T}{R}$
- Let m_p be the mass of a proton, m_n the mass of a neutron, M_1 the mass of a $^{20}_{10}\text{Ne}$ nucleus and M_2 the mass of a $^{40}_{20}\text{Ca}$ nucleus. Then,
(A) $M_2 = 2M_1$ (B) $M_2 > 2M_1$
(C) $M_2 < 2M_1$ (D) $M_1 < 10(m_n + m_p)$
- The potential difference applied to an X-ray tube is increased. As a result, in the emitted radiation
(A) the intensity increases
(B) the minimum wavelength increases
(C) the intensity remains unchanged
(D) the minimum wavelength decreases

10. The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$ where, n_1 and n_2 are the principal quantum numbers of the two states. Assume the Bohr Model to be valid. The time period of the electron in initial state is eight times that in the final state. The possible values of n_1 and n_2 are
 (A) $n_1 = 4, n_2 = 2$ (B) $n_1 = 8, n_2 = 2$
 (C) $n_1 = 8, n_2 = 1$ (D) $n_1 = 6, n_2 = 3$
11. The wavelength of K_α X-rays for lead isotopes Pb^{208} , Pb^{206} , Pb^{204} are λ_1 , λ_2 and λ_3 respectively. Then
 (A) $\lambda_1 = \lambda_2 = \lambda_3$ (B) $\lambda_1 > \lambda_2 > \lambda_3$
 (C) $\lambda_1 < \lambda_2 < \lambda_3$ (D) $\lambda_2 = \sqrt{\lambda_1 \lambda_3}$
12. A radioactive sample has initial concentration N_0 of nuclei.
 (A) The number of undecayed nuclei present in the sample decays exponentially with time.
 (B) The activity (R) of the sample at any instant is directly proportional to the number of undecayed nuclei present in the sample at that time.
 (C) The number of decayed nuclei grows linearly with time.
 (D) The number of decayed nuclei grows exponentially with time.
13. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{Ke^2}{3r^3}$. If m be the mass of electron, then application of Bohr's theory to hydrogen atom in this case shows that
 (A) energy in the n th orbit is proportional to n^6
 (B) energy is proportional to m^{-3}
 (C) energy the n th orbit is proportional to n^{-2}
 (D) energy is proportional to m^3
14. The mass number of a nucleus is
 (A) always less than its atomic number
 (B) always more than its atomic number
 (C) sometimes equal to its atomic number
 (D) sometimes more than and sometimes equal to its atomic number
15. A muonic hydrogen atom is a hydrogen atom with electron replaced by a muon whose mass is 212 times the mass of an electron, then
 (A) Bohr radius of the muonic atom is 250 fm
 (B) Ground state energy of muonic atom is 2883 eV
 (C) angular momentum in ground state is $\frac{h}{2\pi}$
 (D) angular momentum in ground state is $\frac{106h}{\pi}$
16. Consider an atom whose energy level diagram is shown in figure.

 Suppose an atom starts at level 3 then
 (A) Shortest wavelength photon the atom can emit is 1.24×10^{-7} m
 (B) Longest wavelength photon that it can absorb is 6.2×10^{-7} m
 (C) Lowest frequency photon that can ionize the atom is 3.62×10^{15} Hz
 (D) The number of ways of de-excitation of atom to ground state is 3
17. A positronium atom consists of a positron and electron revolving around their common centre of mass. Then compared to hydrogen atom the positronium atom has
 (A) ground state energy half of hydrogen atom
 (B) Rydberg constant half of hydrogen atom
 (C) radius of first orbit of electron double that in case of hydrogen atom
 (D) velocity of electron in first orbit same as in case of hydrogen atom
18. The ground state and first excited state energies of hydrogen atom are -13.6 eV and -3.4 eV respectively. If potential energy in ground state is taken to be zero. Then
 (A) potential energy in the first excited state would be 20.4 eV
 (B) total energy in the first excited state would be 23.8 eV
 (C) kinetic energy in the first excited state would be 3.4 eV
 (D) total energy in the ground state would be 13.6 eV
19. When a hydrogen atom is excited from ground state to first excited state then
 (A) its kinetic energy increases by 10.2 eV.
 (B) its kinetic energy decreases by 10.2 eV.
 (C) its potential energy increases by 20.4 eV.
 (D) its angular momentum increases by 1.05×10^{-34} Js
20. An electron orbiting in a circular orbit around the nucleus of an atom
 (A) has a magnetic dipole moment

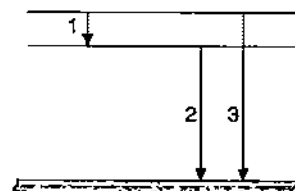
- (B) exerts an electric force on the nucleus equal to that on it by the nucleus
(C) does produce a magnetic induction at the nucleus
(D) has a net energy inversely proportional to its distance from the nucleus
21. In the Bohr model of the hydrogen atom
(A) the radius of the n th orbit is proportional to n^2
(B) the total energy of the electron in the n th orbit is inversely proportional to n
(C) the angular momentum of the electron in an orbit is an integral multiple of $\frac{h}{\pi}$
(D) the magnitude of the potential energy of the electron in any orbit is greater than its kinetic energy
22. The electron in a hydrogen atom makes a transition from 2nd excited state to the ground state. Then
(A) its K.E. increases and total energy decreases
(B) both its K.E. and total energy increases
(C) frequency of emitted photons may be 4.6×10^{14} Hz
(D) frequency of emitted photons must be 2.9×10^{15} Hz
23. The energy, the magnitude of linear momentum and orbital radius of an electron in a hydrogen atom corresponding to the quantum number n are E , P and r respectively. Then according to Bohr's theory of hydrogen atom
(A) PEr is proportional to $\frac{1}{n}$
(B) $\frac{P}{E}$ is proportional to n
(C) Er is constant for all orbits
(D) Pr is proportional to n
24. If potential energy in hydrogen atom with electron in ground state is taken to be 13.6 eV, then
(A) potential energy in the first excited state would be 34 eV
(B) total energy in the first excited state would be 37.4 eV
(C) kinetic energy in the first excited state would be 44.2 eV
(D) total energy in the ground state would be 27.2 eV
25. If electron of the hydrogen atom is replaced by another particle of same charge but of double the mass, then select the correct option(s).
(A) Bohr radius will increase to double value
(B) Ionisation energy of the atom will be doubled
(C) Speed of the new particle in a given state will be one fourth of what electron will possess in the same orbit
(D) Gap between energy levels will now be doubled
26. From the following equations pick out the possible nuclear fusion reactions
(A) ${}_6\text{C}^{13} + {}_1\text{H}^1 \longrightarrow {}_6\text{C}^{14} + 4.3 \text{ MeV}$
(B) ${}_6\text{C}^{12} + {}_1\text{H}^1 \longrightarrow {}_7\text{N}^{13} + 2 \text{ MeV}$
(C) ${}_7\text{N}^{14} + {}_1\text{H}^1 \longrightarrow {}_8\text{O}^{15} + 7.3 \text{ MeV}$
(D) ${}_{92}\text{U}^{235} + {}_0\text{n}^1 \longrightarrow {}_{54}\text{Xe}^{140} + {}_{36}\text{Sr}^{94} + {}_0\text{n}^1 + \gamma + 200 \text{ MeV}$
27. An electron in hydrogen atom first jumps from second excited state to ground state and then from first excited state to ground state. Let the ratio of wavelength, momentum and energy of photons emitted in these two cases be a , b and c respectively, then
(A) $c = \frac{1}{a}$ (B) $a = \frac{9}{4}$
(C) $b = \frac{5}{27}$ (D) $c = \frac{5}{27}$
28. A hydrogen like atom of atomic number Z is in an excited state of quantum number $2n$. It can emit a maximum energy photon of 204 eV. It makes a transition to quantum state n , a photon of energy 40.8 eV is emitted, then
(A) $Z = 2$ (B) $Z = 4$
(C) $n = 1$ (D) $n = 2$
29. Which of the following physical quantities in hydrogen atom are independent of the quantum number n . The symbols have their usual meanings.
(A) vr (B) $\frac{E}{v^2}$
(C) Er (D) v^2r
30. Select the correct statement(s).
(A) X-rays travel faster than ultraviolet rays in vacuum.
(B) Balmer series of H -spectrum is found in visible region.
(C) The characteristic X-rays are produced due to jumping of electrons from higher to lower shell.
(D) In photoelectric emission process, the maximum energy of the photoelectrons must increase with increasing intensity of incident light.
31. Total energy of electron in the first orbit of hydrogen atom is equal to the
(A) total energy of electron in 2nd orbit of He^+
(B) total energy of electron in 3rd orbit of He^+
(C) total energy of electron in 2nd orbit of Li^{++}
(D) total energy of electron in 3rd orbit of Li^{++}

32. Which of the following transitions in He^+ ion will give rise to a spectral line having the same wavelength as some spectral line in the hydrogen atom?
 (A) $n=4$ to $n=2$ (B) $n=6$ to $n=2$
 (C) $n=6$ to $n=3$ (D) $n=8$ to $n=4$
33. When atomic number Z is doubled in atoms, which of the following statements are consistent with Bohr's theory?
 (A) Energy of a state is doubled.
 (B) Radius of an orbit is doubled.
 (C) Velocity of electrons in an orbit is doubled.
 (D) Radius of an orbit is halved.
34. Let A_n be the area enclosed by the n^{th} orbit in a hydrogen atom. The graph of $\log_e \left(\frac{A_n}{A_1} \right)$ against $\log_e n$
 (A) will pass through the origin.
 (B) will have a slope of four units at all points.
 (C) will be a monotonically increasing nonlinear curve.
 (D) will be a straight line.
35. A particular hydrogen like atom has its ground state total energy -54.4 eV , then
 (A) its atomic number is 2
 (B) it can absorb a photon of 40.8 eV
 (C) in its ground state it cannot emit photon
 (D) for its ground state its potential energy is -108.8 eV and kinetic energy is $+54.4 \text{ eV}$
36. If the potential energy of the electron in the first allowed orbit in hydrogen atom is E then its
 (A) ionisation potential is $-\frac{E}{2}$
 (B) kinetic energy is $-\frac{E}{2}$
 (C) total energy is $\frac{E}{2}$
 (D) None of these
37. Hydrogen atoms absorb radiations of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which two wavelengths are shorter than λ_0 . Choose the correct alternative(s).
 (A) The final excited state of the atoms is $n=4$
 (B) The initial state of the atoms may be $n=2$
 (C) The initial state of the atoms may be $n=3$
 (D) There are three transitions belonging to Lyman series
38. For the electron in the n^{th} allowed orbit is
 (A) the linear momentum is proportional to $\frac{1}{n}$
 (B) the radius is proportional to n

- (C) the kinetic energy is proportional to $\frac{1}{n^2}$
 (D) the angular momentum is proportional to n
39. Whenever hydrogen atom emits a photon in the Balmer series
 (A) it may emit another photon in the Balmer series.
 (B) it must emit another photon in the Lyman series.
 (C) the second photon, if emitted, will have a wavelength of about 122 nm .
 (D) it may emit a second photon but the wavelength of this photon cannot be predicted.
40. The graph between $\frac{1}{\lambda}$ and stopping potential (V) of three metals having work function ϕ_1 , ϕ_2 and ϕ_3 in an experiment of photoelectric effect is plotted as shown in the figure. Which of the following statement(s) is/are correct? [Here λ is the wavelength of the incident ray]



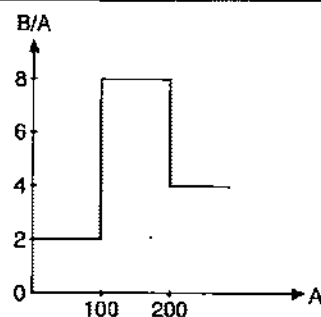
- (A) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 1 : 2 : 4$.
 (B) Ratio of work functions $\phi_1 : \phi_2 : \phi_3 = 4 : 2 : 1$.
 (C) $\tan \theta$ is directly proportional to $\frac{hc}{e}$, where h is Planck's constant and c is the speed of light.
 (D) The violet colour light can eject photoelectrons from metals 2 and 3.
41. The wavelengths and frequencies of photons in transitions 1, 2 and 3 for hydrogen like atom are λ_1 , λ_2 , λ_3 , ν_1 , ν_2 and ν_3 respectively. Then



- (A) $\nu_3 = \frac{\nu_1 \nu_2}{\nu_1 + \nu_2}$ (B) $\nu_3 = \nu_1 + \nu_2$
 (C) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ (D) $\lambda_3 = \lambda_1 + \lambda_2$
42. In a Young's double slit experiment, the separation between the two slits is d and the wavelength of the light is λ . The intensity of light falling on slit 1 is four

times the intensity of light falling on slit 2. Choose the correct choice(s).

- (A) If $d = \lambda$, the screen will contain only one maximum.
- (B) If $\lambda < d < 2\lambda$, at least one more maximum (besides the central maximum) will be observed on the screen.
- (C) If the intensity of light falling on slit 1 is reduced so that it becomes equal to that of slit 2, the intensities of the observed dark and bright fringes will increase.
- (D) If the intensity of light falling on slit 2 is increased so that it becomes equal to that of slit 1, the intensities of the observed dark and bright fringes will increase.



43. Assume that the nuclear binding energy per nucleon $\left(\frac{B}{A}\right)$ versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below.

- (A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy.
- (B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy.
- (C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments.
- (D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments.

Practice Exercise

Reasoning Based Questions

This section contains Reasoning type questions, each having four choices (A), (B), (C) and (D) out of which ONLY ONE is correct. Each question contains STATEMENT 1 and STATEMENT 2. You have to mark your answer as

Bubble (A) If both statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1.

Bubble (B) If both statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1.

Bubble (C) If STATEMENT 1 is TRUE and STATEMENT 2 is FALSE.

Bubble (D) If STATEMENT 1 is FALSE but STATEMENT 2 is TRUE.

- Statement-1:**
If the accelerating potential in an X-ray tube is increased, the wavelengths of the characteristic X-rays do not change.

Statement-2:
When an electron beam strikes the target in an X-ray tube, part of the kinetic energy is converted into X-ray energy.
- Statement-1:**
According to classical theory, the proposed path of an electron in Rutherford atom model will be parabolic.

Statement-2:
According to electromagnetic theory, an accelerated particle continuously emits radiation.
- Statement-1:**
An electron in hydrogen atom passes from $n=4$ to $n=1$ level. The maximum number of photons that can be emitted is 6.

Statement-2:
No. of photons emitted can never be more than 5.
- Statement-1:**
The wavelength of first Balmer line of deuterium is slightly more than that of hydrogen.

Statement-2:
In the centre of mass of an atom reference frame both nucleus and electron are non-stationary.
- Statement-1:**
In a hydrogen atom energy of emitted photon corresponding to transition from $n=2$ to $n=1$ is much greater as compared to transition from $n \rightarrow \infty$ to $n=2$.

Statement-2:
Wavelength of photon is directly proportional to the energy of emitted photon.
- Statement-1:**
Magnetic moment of an atom is due to both, the orbital motion and spin motion of every electron.

Statement-2:
A charged particle produces a magnetic field.
- Statement-1:**
Total energy in an orbit is negative in an atom.

Statement-2:
Electron is bounded by electrostatic attraction between electron and nucleus.
- Statement-1:**
Between any two given energy levels, the number of absorption transition is always less than number of emission transition.

Statement-2:
Absorption transition starts from the lowest energy level only and may end at any higher level. But emission transitions may start from any higher energy level and end at any energy level below it.

9. **Statement-1:**
Total energy of revolving electron in any stationary orbit is negative.
Statement-2:
Energy is a scalar quantity and it can take positive and negative value.
10. **Statement-1:**
In outermost stationary orbit, energy of electron is least negative.
Statement-2:
In outermost orbit, electron is at maximum distance from nucleus.
11. **Statement-1:**
Speed of electron in the n^{th} orbit of hydrogen atom is $v_H \propto \frac{1}{n}$.
Statement-2:
Speed of electron in n^{th} orbit of hydrogen like atom is $v_{H \text{ like}} \propto \frac{Z^2}{n}$.
12. **Statement-1:**
Atomic hydrogen gas excites to third excited state. The number of spectral lines in emission spectrum obtained is 6.
Statement-2:
Number of spectral lines in emission spectrum can never be less than 6.
13. **Statement-1:**
In a hydrogen atom energy of emitted photon corresponding to transition from $n=2$ to $n=1$ is much greater as compared to transition from $n \rightarrow \infty$ to $n=2$.
Statement-2:
Wavelength of photon is directly proportional to the energy of emitted photon.
14. **Statement-1:**
The difference between the wavelengths of series limit of the Lyman series of spectral lines and that (the series limit) of the Paschen series is equal to the wavelength of a spectral line of the Lyman series (for the hydrogen atom).
Statement-2:
The wave number of an atomic transition can be calculated from the formula $\bar{\nu} = R \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$, where $\bar{\nu}$ is the wave number, R is the Rydberg's constant, m and n are the quantum numbers of the initial and final states.
15. **Statement-1:**
The ionising power of β -particle is less compared α -particle but their penetrating power is more.
Statement-2:
The mass of β -particles is less than the mass of α -particle.
16. **Statement-1:**
A certain radioactive substance has a half life period of 30 days. The disintegration constant is 0.0231 day^{-1} .
Statement-2:
Decay constant varies inversely as half life.
17. **Statement-1:**
Half life of a certain radio-active element is 100 days. After 200 days fraction left undecayed will be 50%.
Statement-2:
 $\frac{N}{N_0} = \left(\frac{1}{2} \right)^{t/t_{1/2}}$, where symbols have usual meaning.
18. **Statement-1:**
 β -particles emitted from radioactive nuclei has continuous energy ranging from zero to a certain maximum value.
Statement-2:
In β -decay a neutron is converted into a proton, an electron and an antineutrino. The total energy emitted in β -decay is shared by β -particle and antineutrino.
19. **Statement-1:**
 γ -photons are emitted during annihilation process of electron and positron.
Statement-2:
High energy photons are emitted due to conversion of mass into energy.
20. **Statement-1:**
25% of the radioactive nucleus remains active after 200 days for an element of half life 100 days.
Statement-2:
 $N = N_0 \left(\frac{1}{2} \right)^{t/T}$ where symbols have usual meaning.
21. **Statement-1:**
If binding energy per nucleon increases after a nuclear reaction then the reaction is exothermic.
Statement-2:
If there is decrease in mass in nuclear reaction then the reaction is exothermic.

22. *Statement-1:*

If the current in the filament of electron gun in a X-ray tube is increased, the penetration power of X-rays is increased.

Statement-2:

Increasing current increases the number of electrons emitted by the electron gun.

23. *Statement-1:*

It is necessary to keep high vacuum in Coolidge tube to produce X-rays.

Statement-2:

High vacuum is kept in Coolidge tube so that the electron emitting from the filament of the tube may not lose their energy in colliding with the atom of the gas in the tube.

24. *Statement-1:*

If maximum frequency of Balmer and Paschen series is f_1 and f_2 respectively, then frequency of first line of Balmer series is $(f_1 - f_2)$.

Statement-2:

Difference of energy level between two orbits is constant for an atom.

25. *Statement-1:*

The wavelength of the K_α line from an element and that of the K_β line satisfy the relation $\lambda_{K_\alpha} > \lambda_{K_\beta}$.

Statement-2:

The energy separation between the K_α levels is smaller than that of the K_β levels.

26. *Statement-1:*

Radioactivity is independent of the physical and chemical conditions of substance.

Statement-2:

Radioactivity is a property of the nucleus.

27. *Statement-1:*

The wave number corresponding to the transition between the atomic levels $n=3$ and $n=2$ of a hydrogen atom i.e. $\bar{\nu}_{32}$, is related to the wave-numbers $\bar{\nu}_{21}$ and $\bar{\nu}_{31}$ for a hydrogen atom by the relation $\bar{\nu}_{31} = \bar{\nu}_{21} + \bar{\nu}_{32}$.

Statement-2:

The wave-number $\bar{\nu}_{mn} = \frac{1}{\lambda_{mn}}$ for a transition is given by

the expression $\frac{1}{\lambda_{mn}} = R \left(\frac{1}{n^2} - \frac{1}{m^2} \right)$, where R is the

Rydberg constant and m, n are integers representing the initial and final principal quantum numbers.

28. *Statement-1:*

When light is passed through a sample of hydrogen atoms in ground state, then wavelengths of absorption lines are same as wavelengths of lines of Lyman series in emission spectrum.

Statement-2:

In ground state hydrogen atom will absorb only those radiation which will excite to higher energy level.

Practice Exercise

Linked Comprehension Type Questions

This section contains Linked Comprehension Type Questions or Paragraph based Questions. Each set consists of a Paragraph followed by questions. Each question has four choices (A), (B), (C) and (D), out of which only one is correct. (For the sake of competitiveness there may be a few questions that may have more than one correct options)

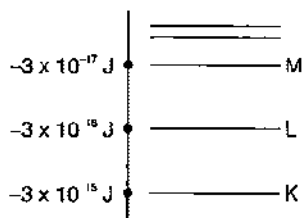
Comprehension 1

In a mixture of $H-He^+$ gas (He^+ is singly ionized He atom), H atoms and He^+ ions are excited to their respective first excited states. Subsequently, H atoms transfer their total excitation energy to He^+ ions (by collisions). Assume that the Bohr model of atom is exactly valid. Based on above information, answer the following questions.

- The quantum number n of the state finally populated in He^+ ions is
(A) 2 (B) 3
(C) 4 (D) 5
- The wavelength of light emitted in the visible region by He^+ ions after collisions with H atoms is
(A) 6.5×10^{-7} m (B) 5.6×10^{-7} m
(C) 4.8×10^{-7} m (D) 4×10^{-7} m
- The ratio of the kinetic energy of the $n=2$ electron for the H atom to that of He^+ ion is
(A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) 1 (D) 2

Comprehension 2

Simplified picture of electron energy levels in a certain atom bombarded with high energy electrons is shown in the figure. The impact of one of these electrons has caused the complete removal of K -level and is filled by an electron from the L -level with a certain amount of energy being released during the transition. This energy may appear



as X-ray or may all be used to eject an M -level electron from the atom. Based on above information, answer the following questions.

- The minimum potential difference through which electron may be accelerated from rest to cause the ejection of K -level electron from the atom is
(A) 1.875×10^4 V (B) 3×10^{15} V
(C) 3.33×10^{14} V (D) 10 V
- Energy released when L -level electron moves to fill the vacancy in the K -level is
(A) 3×10^{-15} J (B) 3×10^{-17} J
(C) 2.7×10^{-15} J (D) 2.7×10^{-16} J
- Kinetic energy of the electron emitted from the M -level is
(A) 3×10^{-16} J (B) 3×10^{-17} J
(C) 2.7×10^{-15} J (D) 2.67×10^{-15} J

Comprehension 3

When a particle is restricted to move along x -axis between $x=0$ and $x=a$, where a is of nanometer dimension, its energy can take only certain specific values. The allowed energies of the particle moving in such a restricted region, correspond to the formation of standing waves with nodes at its ends $x=0$ and $x=a$. The wavelength of this standing wave is related to the linear momentum p of the particle according to the de-Broglie relation. The energy of the particle of mass m is related to its linear momentum as $E = \frac{p^2}{2m}$. Thus, the energy of the particle can be denoted by a quantum number n taking values 1, 2, 3,.... ($n=1$, called the ground state) corresponding to the number of loops in the standing wave.

Advanced JEE Physics

Optics & Modern Physics

Use the model described above to answer the following three questions for a particle moving in the line $x=0$ to $x=a$. [Take $h=6.6 \times 10^{-34}$ Js and $e=1.6 \times 10^{-19}$ C].

7. The allowed energy for the particle for a particular value of n is proportional to

- (A) a^{-2} (B) $a^{\frac{3}{2}}$
(C) a^{-1} (D) a^2

8. If the mass of the particle is $m=1 \times 10^{-30}$ kg and $a=6.6$ the energy of the particle in its ground state is closest to

- (A) 0.8 meV (B) 8 meV
(C) 80 meV (D) 800 meV

9. The speed of the particle that can take discrete values is proportional to

- (A) $n^{\frac{3}{2}}$ (B) n^{-1}
(C) $n^{\frac{1}{2}}$ (D) n

Comprehension 4

Scientists are working hard to develop nuclear fusion reactor. Nuclei of heavy hydrogen, ${}^2_1\text{H}$ known as deuterium and denoted by D can be thought of as a candidate for fusion reactor. The D-D reaction is ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + n + \text{energy}$. In the core of fusion reactor, a gas of heavy hydrogen is fully ionized into deuterium nuclei and electrons. This collection of ${}^2_1\text{H}$ nuclei and electrons is known as plasma. The nuclei move randomly in the reactor core and occasionally come close enough for nuclear fusion to take place. Usually, the temperatures in the reactor core are too high and no material wall can be used to confine the plasma. Special techniques are used which confine the plasma for a time t_0 before the particles fly away from the core. If n is the density (number/volume) of deuterons, the product nt_0 is called Lawson number. In one of the criteria, a reactor is termed successful if Lawson number is greater than $5 \times 10^{14} \text{ scm}^{-3}$. It may be helpful to use the following : Boltzmann constant $k=8.6 \times 10^{-5} \text{ eV K}^{-1}$; $\frac{e^2}{4\pi\epsilon_0} = 1.44 \times 10^{-9} \text{ eVm}$. Based on above information, answer the following questions.

10. In the core of nuclear fusion reactor, the gas becomes plasma because of
(A) strong nuclear force acting between the deuterons
(B) Coulomb force acting between the deuterons
(C) Coulomb force acting between deuterium-electron pairs
(D) the high temperature maintained inside the reactor core

11. Assume that two deuterium nuclei in the core of fusion reactor at temperature T are moving towards each other, each with kinetic energy 1.5 kT , when the separation between them is large enough to neglect Coulomb potential energy. Also neglect any interaction from other particles in the core. The minimum temperature T required for them to reach a separation of $4 \times 10^{-15} \text{ m}$ is in the range
(A) $1 \times 10^9 \text{ K} < T < 2 \times 10^9 \text{ K}$
(B) $2 \times 10^9 \text{ K} < T < 3 \times 10^9 \text{ K}$
(C) $3 \times 10^9 \text{ K} < T < 4 \times 10^9 \text{ K}$
(D) $4 \times 10^9 \text{ K} < T < 5 \times 10^9 \text{ K}$

12. Results of calculations for four different designs of a fusion reactor using D-D reaction are given below. Which of these is most promising based on Lawson criterion?

- (A) Deuteron density $= 2 \times 10^{12} \text{ cm}^{-3}$, confinement time $= 5 \times 10^{-3} \text{ s}$
(B) Deuteron density $= 8 \times 10^{14} \text{ cm}^{-3}$, confinement time $= 9 \times 10^{-1} \text{ s}$
(C) Deuteron density $= 4 \times 10^{23} \text{ cm}^{-3}$, confinement time $= 1 \times 10^{-11} \text{ s}$
(D) Deuteron density $= 1 \times 10^{24} \text{ cm}^{-3}$, confinement time $= 4 \times 10^{-12} \text{ s}$

Comprehension 5

The mass of a nucleus ${}^A_Z\text{X}$ is less than the sum of the masses of $(A-Z)$ number of neutrons and Z number of protons in the nucleus. The energy equivalent to the corresponding mass difference is known as the binding energy of the nucleus. A heavy nucleus of mass M can break into two light nuclei of masses m_1 and m_2 only if $(m_1 + m_2) < M$. Also two light nuclei of masses m_3 and m_4 can undergo complete fusion and form a heavy nucleus of mass M' only if $(m_3 + m_4) > M'$. The masses of some neutral atoms are given in the table below

${}^1_1\text{H}$	1.007825 u	${}^2_1\text{H}$	2.014102 u
${}^6_3\text{Li}$	6.015123 u	${}^7_3\text{Li}$	7.016004 u
${}^{152}_{64}\text{Gd}$	151.919803 u	${}^{206}_{82}\text{Pb}$	205.974455 u
${}^3_1\text{H}$	3.016050 u	${}^4_2\text{He}$	4.002603 u
${}^{70}_{30}\text{Zn}$	69.925325 u	${}^{82}_{34}\text{Se}$	81.916709 u
${}^{209}_{83}\text{Bi}$	208.980388 u	${}^{210}_{84}\text{Po}$	209.982876 u

(1 u = 932 MeV c^{-2})

Based on above information, answer the following questions.

13. The correct statement is
(A) The nucleus ${}^6_3\text{Li}$ can emit an alpha particle

- (B) The nucleus ${}^{210}_{84}\text{Po}$ can emit a proton
 (C) Deuteron and alpha particle can undergo complete fusion.
 (D) The nuclei ${}^{70}_{30}\text{Zn}$ and ${}^{82}_{34}\text{Se}$ can undergo complete fusion
14. The kinetic energy (in keV) of the alpha particle, when the nucleus ${}^{210}_{84}\text{Po}$ at rest undergoes alpha decay, is
 (A) 5319 (B) 5422
 (C) 5707 (D) 5818

Comprehension 6

A dense collection of equal number of electrons and positive ions is called neutral plasma. Certain solids containing fixed positive ions surrounded by free electrons can be treated as neutral plasma. Let N be the number density of free electrons, each of mass m . When the electrons are subjected to an electric field, they are displaced relatively away from the heavy positive ions. If the electric field becomes zero, the electrons begin to oscillate about the positive ions with a natural angular frequency ω_p , which is called the plasma frequency. To sustain the oscillations, a time varying electric field needs to be applied that has an angular frequency ω , where a part of the energy is absorbed and a part of it is reflected. As ω approaches ω_p , all the free electrons are set to resonance together and all the energy is reflected. This is the explanation of high reflectivity of metals. Based on above information, answer the following questions.

15. Taking the electronic charge as e and the permittivity as ϵ_0 , use dimensional analysis to determine the correct expression for ω_p

- (A) $\sqrt{\frac{Ne}{m\epsilon_0}}$ (B) $\sqrt{\frac{m\epsilon_0}{Ne}}$
 (C) $\sqrt{\frac{Ne^2}{m\epsilon_0}}$ (D) $\sqrt{\frac{m\epsilon_0}{Ne^2}}$

16. Estimate the wavelength at which plasma reflection will occur for a metal having the density of electrons $N \approx 4 \times 10^{27} \text{ m}^{-3}$. Take $\epsilon_0 \approx 10^{-11}$ and $m \approx 10^{-30}$, where these quantities are in proper SI units
 (A) 800 nm (B) 600 nm
 (C) 300 nm (D) 200 nm

Comprehension 7

The β -decay process, discovered around 1900, is basically the decay of a neutron (n). In the laboratory, a proton (p) and an electron (e^-) and observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two body decay process, it was predicted theoretically that the kinetic energy of the electron should be

a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e., $n \rightarrow p + e^- + \bar{\nu}_e$, around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino ($\bar{\nu}_e$) to be massless and possessing negligible energy and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is $0.8 \times 10^6 \text{ eV}$. The kinetic energy carried by the proton is only the recoil energy. Based on above information, answer the following questions.

17. What is the maximum energy of the anti-neutrino?
 (A) ZERO
 (B) Much less than $0.8 \times 10^6 \text{ eV}$
 (C) Nearly $0.8 \times 10^6 \text{ eV}$
 (D) Much larger than $0.8 \times 10^6 \text{ eV}$
18. If the anti-neutrino had a mass of $3 \text{ eV}c^{-2}$ (where c is the speed of light) instead of zero mass, what should be the range of the kinetic energy, K , of the electron?
 (A) $0 \leq K \leq 0.8 \times 10^6 \text{ eV}$
 (B) $3 \text{ eV} \leq K \leq 0.8 \times 10^6 \text{ eV}$
 (C) $3 \text{ eV} \leq K < 0.8 \times 10^6 \text{ eV}$
 (D) $0 \leq K < 0.8 \times 10^6 \text{ eV}$

Comprehension 8

A radionuclide with half life T is produced in a reactor at a constant rate P nuclei per second. During each decay, energy E_0 is released. If production of radionuclide is started at $t=0$. Based on above information, answer the following questions.

19. The number of nuclei in the radionuclide at any instant t is given by

- (A) $N = \frac{PT}{\log_e 2} \left(1 - e^{-\frac{t \log_e 2}{T}} \right)$
 (B) $N = PT \left(1 - e^{-\frac{t}{T}} \right)$
 (C) $N = PT \log_e 2 \left(1 - e^{-\frac{t \log_e 2}{T}} \right)$
 (D) $N = PT \left(1 - e^{-\frac{t}{2T}} \right)$

20. The rate of release of energy as a function of time is

- (A) $\frac{PE_0}{\log_e 2} \left(1 - e^{-\frac{t \log_e 2}{T}} \right)$ (B) $PE_0 \left(1 - e^{-\frac{t \log_e 2}{T}} \right)$
 (C) $PE_0 \log_e 2 \left(1 - e^{-\frac{t \log_e 2}{T}} \right)$ (D) $PE_0 \left(1 - e^{-\frac{t}{2T}} \right)$

21. Total energy released upto time t is

- (A) $PE_0 - \frac{PTE_0}{\log_e 2} \left(1 - e^{-\frac{t \log_e 2}{T}} \right)$

- (B) $PtE_0 - PTE_0 \left(1 - e^{-\frac{t \log_e 2}{T}}\right)$
 (C) $PtE_0 - PTE_0 \log_e 2 \left(1 - e^{-\frac{t \log_e 2}{T}}\right)$
 (D) None of these

Comprehension 9

Two radioactive nuclei A and B, both convert into a stable nucleus C. The nucleus A converts into C after emitting two α -particles and three β -particles, the nucleus B converts into C after emitting one α -particle and five β -particles. A time $t=0$, nuclei of A are $4N_0$ and that of B are N_0 . The half-life of A for converting to C is 1 min and that of B is 2 min. Initially number of nuclei of C are zero. Based on above information, answer the following questions.

22. If atomic numbers and mass numbers of A and B are Z_1, Z_2, A_1 and A_2 respectively, then
 (A) $Z_1 - Z_2 = 6$
 (B) $A_1 - A_2 = 4$
 (C) Both (A) and (B) are correct
 (D) Both (A) and (B) are wrong
23. The number of nuclei of C, when number of nuclei of A and B are equal is
 (A) $2N_0$ (B) $3N_0$
 (C) $\frac{9N_0}{2}$ (D) $\frac{5N_0}{2}$
24. The rate of disintegrations of A and B are equal at time t_0 (in minute), then
 (A) $t_0 = 4$ (B) $t_0 = 6$
 (C) $t_0 = 8$ (D) $t_0 = 2$

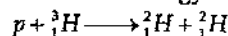
Comprehension 10

The electric current in an X-ray tube (from the target to the filament) operating at 40 kV is 10 mA. Assume that on an average, 1% of the total kinetic energy of the electrons hitting the target is converted into X-rays. Based on above information, answer the following questions.

25. The total power emitted as X-rays is
 (A) 40 W (B) 400 W
 (C) 4 W (D) 0.4 W
26. The heat produced in the target every second is
 (A) 29.6 J (B) 396 J
 (C) 3600 J (D) 360 J
27. Minimum wavelength of X-ray produced is nearly
 (A) 0.3 Å (B) 3 Å
 (C) 30 Å (D) 300 Å

Comprehension 11

Nuclear reactions are performed for artificial transmutation of elements for there are two types of nuclear reactions, exoergic and endoergic. In exoergic reactions energy is released. In endoergic reactions energy has to be supplied for the reaction to proceed. In exoergic reactions nuclear energy is converted into kinetic energy. In endoergic reactions, energy input is required in the form of kinetic energy to be converted into nuclear binding energy. The minimum energy required for the reaction to take place is called threshold energy. Consider the reaction given by



Given that the atomic masses are

$$m({}^1_1\text{H-atom}) = 1.007825 \text{ amu (u)}$$

$$m({}^3_1\text{H-atom}) = 3.016049 \text{ amu (u)}$$

$$m({}^2_1\text{H-atom}) = 2.014102 \text{ amu (u)}$$

Based on above information, answer the following questions.

28. Protons are incident on ${}^3_1\text{H}$ at rest. The threshold energy for the reaction is
 (A) 5.4 MeV (B) 10 MeV
 (C) 2 MeV (D) 8 MeV
29. When ${}^3_1\text{H}$ are incident on protons, then threshold energy is
 (A) 10 MeV (B) 8 MeV
 (C) 16 MeV (D) 20 MeV
30. Which of the following statement is correct?
 (A) Less energy is required for nuclear reaction if light part is at rest and heavy particle is incident
 (B) More energy is required for nuclear reaction when heavy particle is incident and light particle is at rest
 (C) Threshold energy does not depend on which particle is at rest
 (D) Threshold energy does not depend on Q-value of reaction

Comprehension 12

Consider a different atomic model in which electron revolves around the nucleus (proton) at a separation r under the action of force which is different from the electrostatic force of attraction. The potential energy between an electron and the proton due to this force is given by $U = -\frac{k}{r^4}$ where k is a constant. Based on above information, answer the following questions.

31. The radius of n th Bohr's orbit is
 (A) $r = \frac{\pi}{nh} \sqrt{km}$ (B) $r = \frac{2\pi}{nh} \sqrt{km}$
 (C) $r = \frac{4\pi}{nh} \sqrt{km}$ (D) $r = \frac{8\pi}{nh} \sqrt{km}$

32. The velocity in the n th orbit is given by
 (A) $v = \frac{nh}{8\pi^2 m \sqrt{k m}}$ (B) $v = \frac{n^2 h}{8\pi^2 m \sqrt{k m}}$
 (C) $v = \frac{nh^2}{4\pi^2 m \sqrt{k m}}$ (D) $v = \frac{n^2 h^2}{8\pi^2 m \sqrt{k m}}$

33. The kinetic energy of the electron in n th orbit is given by

(A) $KE = \frac{n^4 h^4}{64\pi^4 m^2 k}$ (B) $KE = \frac{n^4 h^4}{128\pi^4 m^2 k}$
 (C) $KE = \frac{n^4 h^4}{16\pi^4 k^2}$ (D) $KE = -\frac{n^2 h^2}{16k^2 \pi^4 m}$

34. The potential energy of the electron in the n th orbit is given by

(A) $PE = -\frac{n^4 h^4}{64k\pi^4 m^2}$ (B) $PE = -\frac{n^4 h^4}{128k\pi^4 m^2}$
 (C) $PE = -\frac{n^2 h^2}{128k\pi^4 m^2}$ (D) $PE = -\frac{n^4 h^4}{256k\pi^4 m^2}$

35. The total energy of the electron in the n th orbit is given by

(A) $TE = -\frac{n^4 h^4}{128\pi^4 m^2 k}$ (B) $TE = \frac{n^4 h^4}{128\pi^4 m^2 k}$
 (C) $TE = \frac{n^4 h^4}{256k\pi^4 m^2}$ (D) $TE = -\frac{n^4 h^4}{256k\pi^4 m^2}$

Comprehension 13

All nuclei consist of two type of particles i.e., protons and neutrons. Nuclear force is the strongest force and the stability of nucleus is determined by the neutron-proton ratio or mass defect or Binding energy per nucleons or packing fraction. The shape of nucleus is calculated by quadrupole moment. The spin of nucleus depends on even or odd mass number. The volume of nucleus depends on the mass number. The whole mass of atom (nearly 99%) is centred at the nucleus and the magnetic moment of nucleus is measured in terms of the nuclear magnetons. Based on above information, answer the following questions.

36. The correct statement(s) about nuclear force is/are
 (A) charge dependent
 (B) short ranges forces
 (C) non conservative force
 (D) spin dependent force

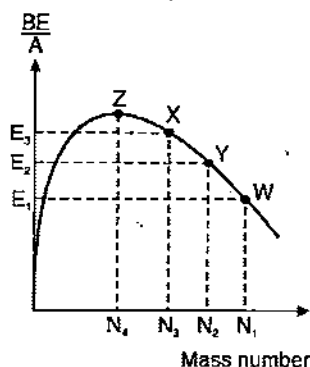
37. Binding energy per nucleon is maximum
 (A) for lighter order element (low mass number)
 (B) for heavier order element (high mass number)
 (C) for middle order element
 (D) equal for all order elements

38. Volume (V) of the nucleus is related with mass number (A) as

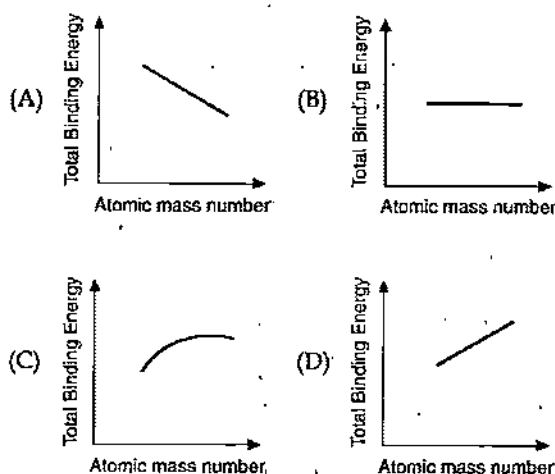
(A) $V \propto A^2$ (B) $V \propto A^{\frac{1}{3}}$
 (C) $V \propto A^{\frac{2}{3}}$ (D) $V \propto A$

Comprehension 14

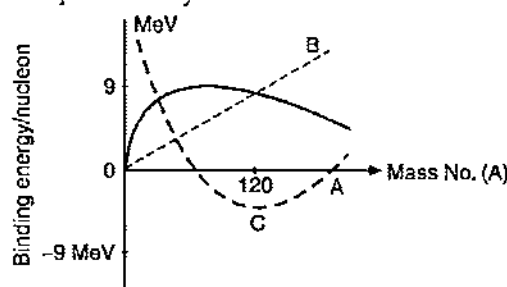
Consider the nuclear fission reaction $W \rightarrow X + Y$. Based on the graph given showing binding energy per nucleon vs number of nucleons, answer the following.



39. The Q value of the reaction is
 (A) $E_1 N_1 - (E_2 N_2 + E_3 N_3)$ (B) $(E_2 N_2 + E_3 N_3 - E_1 N_1)$
 (C) $E_2 N_2 + E_1 N_1 - E_3 N_3$ (D) $E_1 N_1 + E_3 N_3 - E_2 N_2$
40. If M_W is the mass of W , M_X is mass of X and M_Y is mass of Y nucleus, choose the correct statement.
 (A) $\frac{M_W}{N_1} > \frac{M_Y}{N_2} > \frac{M_X}{N_3}$ (B) $\frac{M_W}{N_1} < \frac{M_Y}{N_2} < \frac{M_X}{N_3}$
 (C) $\frac{M_W}{N_1} < \frac{M_Y}{N_2} > \frac{M_X}{N_3}$ (D) $\frac{M_W}{N_1} > \frac{M_Y}{N_2} < \frac{M_X}{N_3}$
41. The graph representing the relationship between atomic mass number A and the total binding energy (BE) of the nucleus, for nuclei heavier than Z is



42. When a slow neutron is captured by a U^{235} nucleus, a fission results which releases 200 MeV of energy. If the output of a nuclear reactor is 1.6 MW, the rate of nuclei (per sec) undergoing fission is
 (A) 8×10^{-3} (B) 1×10^{13}
 (C) 1×10^{23} (D) 5×10^{16}
43. The binding energy per nucleon versus mass number is best represented by the curve



- (A) A (B) B
 (C) C (D) None of these

Comprehension 15

In a nuclear fusion reactor, the reaction occurs in two stages.

Stage-I: Two deuterium (${}_1D^2$) nuclei fuse to form a tritium (${}_1T^3$) nucleus with a proton as a by-product. The reaction may be represented as $D(D, p)T$.

Stage-II: A tritium nucleus fuses with another deuterium nucleus to form a helium (${}_2He^4$) nucleus with a neutron as a by product. The reaction is represented as $T(D, n)\alpha$.

Given that $m({}_1D^2) = 2.014102u$ (atom)
 $m({}_1T^3) = 3.016049u$ (atom)
 $m({}_2He^4) = 4.002603u$ (atom)
 $m({}_1H^1) = 1.007825u$ (atom)
 $m({}_0n^1) = 1.008665u$

Based on above information, answer the following questions.

44. The energy released in the Stage-II of fusion reaction, is
 (A) 4.033 MeV (B) 17.587 MeV
 (C) 40.33 MeV (D) 1.7587 MeV
45. The energy released in the combined reaction per deuterium, is given by
 (A) 4.207 MeV (B) 5.207 MeV
 (C) 6.207 MeV (D) 7.207 MeV
46. The percentage of the mass energy of the initial deuterium is released is
 (A) 0.184% (B) 0.284%

- (C) 0.384% (D) 0.484%

Comprehension 16

Many unstable nuclei can decay spontaneously to a nucleus of lower mass but different combination of nucleons. The process of spontaneous emission of radiation is called Radioactivity in which the decay rate is actually exponentially decrease with time. Also, radioactive decay is a statistical process which is independent of all external conditions. Based on above information, answer the following questions.

47. If T_H is the half life and T_M is the mean life, which of the following statements is correct ?
 (A) $T_M > T_H$
 (B) Both are equal
 (C) $T_M < T_H$
 (D) Nothing can be said
48. If n is the number of α -particles being emitted per second by N atoms of a radioactive element, then half life of element, in second, will be
 (A) $\frac{n}{N}$ s (B) $\frac{N}{n}$ s
 (C) $\frac{0.693}{n}$ s (D) $\frac{0.693N}{n}$ s
49. The activity of radioactive substance of decay constant λ is R_1 at time t_1 and R_2 at time $t_2 (> t_1)$, then
 (A) $R_1 t_1 = R_2 t_2$ (B) $\frac{R_1 - R_2}{t_2 - t_1} = \text{constant}$
 (C) $R_2 = R_1 e^{-\lambda(t_2 - t_1)}$ (D) $R_1 t_2 = R_2 t_1$

Comprehension 17

An electron orbits a stationary nucleus of charge $+ze$ where z is a constant and e is the magnitude of electronic charge. It requires 47.2 eV to excite the electron from the second Bohr orbit to third Bohr orbit. Based on above information, answer the following questions.

50. The value of z is
 (A) 5 (B) 4
 (C) 3 (D) 2
51. The radius of first Bohr orbit is
 (A) $0.53 \text{ \AA}$ (B) $0.106 \text{ \AA}$
 (C) $0.318 \text{ \AA}$ (D) 0.53 nm

Comprehension 18

In quantum mechanics, some quantities are discrete and cannot be continuous. One of these quantities is the energy. Energy can only take certain values, like $E_1, E_2, E_3, E_4, \dots$, which are called energy levels. The energy cannot take any

values between E_1 and E_2 or E_2 and E_3 or E_3 and E_4 , etc. Certain transitions from one energy level to another result in the emission of a photon of radiation, whereas others can only take place if a photon is absorbed. The energy levels in a newly discovered gas are expressed as

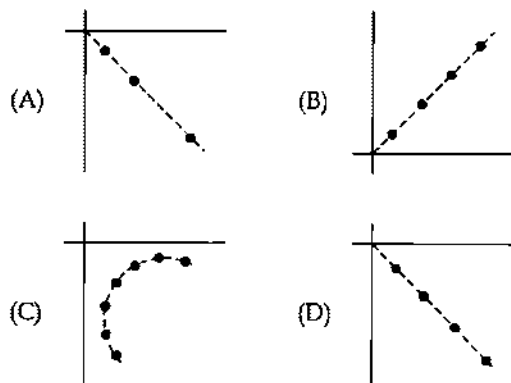
$$E_n = -\frac{E_1 z^2}{n^2}$$

where, $-E_1 z^2$ is the ground state energy. Taking $z=1$ for simplicity, but not assuming that the gas is hydrogen, an experiment is designed to measure the energy as a function of the level. The results obtained in the experiment are given in the table.

n	E_n (eV)
2	-144
3	-64
4	-36

Based on the information given, answer the following questions.

52. The ionization energy of the gas is
 (A) 244 eV (B) 576 eV
 (C) 144 eV (D) +13.6 eV
53. The ground state energy is
 (A) -144 eV (B) +144 eV
 (C) -244 eV (D) None of these
54. The graph of E_n versus $\frac{1}{n^2}$ is best represented as



Comprehension 19

An electron is orbiting in a circular orbit of radius r under the influence of a constant magnetic field of strength B . Assuming that Bohr's postulate regarding the quantisation of angular momentum holds good for this electron of charge e and mass m , answer the following questions.

55. The radius of n th orbit of the electron will be

(A) $\sqrt{\frac{2nh}{\pi Be}}$ (B) $\sqrt{\frac{nh}{2\pi Be}}$

(C) $\sqrt{\frac{neh}{2\pi Be}}$ (D) $\sqrt{\frac{2neh}{\pi Be}}$

56. The kinetic energy of the electron in n th orbit is

(A) $nh\left(\frac{Be}{m}\right)$ (B) $2nh\left(\frac{Be}{m}\right)$
 (C) $\frac{nh}{2}\left(\frac{Be}{m}\right)$ (D) $\frac{nh}{4}\left(\frac{Be}{m}\right)$

57. Potential energy of interaction between the magnetic moment of the orbital current due to the electron moving in its orbit and the magnetic field B is

(A) $\frac{nhBe}{2m}$ (B) $\frac{nhBe}{m}$
 (C) $\frac{2nhBe}{m}$ (D) $\frac{nhBe}{4m}$

58. Total magnetic flux due to the magnetic field B passing through the n th orbit is

(A) $\frac{\pi nh}{2e}$ (B) $\frac{\pi nh}{e}$
 (C) $\frac{4\pi nh}{e}$ (D) $\frac{2\pi nh}{e}$

Comprehension 20

A radioactive sample has mass m , decay constant λ and molecular weight M . If Avogadro number is N_A then answer the following questions.

59. The initial number of nuclei present is

(A) $\frac{m}{N_A}$ (B) $\frac{M}{m}N_A$
 (C) $\frac{m}{M}N_A$ (D) MN_A

60. The number of decayed nuclei after time t is

(A) $\frac{m}{M}N_A e^{-\lambda t}$ (B) $\frac{m}{M}e^{-\lambda t}$
 (C) $\frac{m}{M}N_A (1 - e^{-\lambda t})$ (D) $\frac{m}{M}(1 - e^{-\lambda t})$

61. The activity of sample after time t is

(A) $\left(\frac{mN_A\lambda}{M}\right)e^{-\lambda t}$ (B) $\frac{m}{M}N_A\lambda(1 - e^{-\lambda t})$
 (C) $\frac{M}{m}N_A\lambda e^{-\lambda t}$ (D) None of these

Comprehension 21

In atom X , a single electron orbits around a stationary nucleus of charge $+Ze$ where Z is a constant and e is the magnitude of the electronic charge. It requires 47.2 eV to excite the electron from the second Bohr orbit to the third

Bohr orbit. Take ionization energy of the hydrogen atom 13.6 eV.

62. The value of Z is.
 (A) 3 (B) 4
 (C) 5 (D) 6
63. Kinetic energy of electron in ground state of atom X is
 (A) 122.4 eV (B) 170 eV
 (C) 340 eV (D) 680 eV
64. The wavelength of electromagnetic radiation required to remove the electron from 3rd Bohr orbit to infinity is
 (A) 329 Å (B) 428 Å
 (C) 546 Å (D) 636 Å

Comprehension 22

The nuclei of a radioactive element X are being produced at a constant rate α and this element decays to a stable nucleus Y with a decay constant λ and half-life $T_{1/2}$. At time $t = 0$, there are N_0 nuclei of the element X . Based on the information given, answer the following questions.

65. The number N_X of nuclei of X at time $t = T_{1/2}$ is

(A) $\frac{\alpha - \lambda N_0}{2\lambda}$ (B) $(2\lambda N_0 - \alpha) \frac{1}{\lambda}$
 (C) $\left(\lambda N_0 + \frac{\alpha}{2}\right) \frac{1}{\lambda}$ (D) $\frac{\alpha + \lambda N_0}{2\lambda}$

66. The number N_Y of nuclei of Y at time t is

(A) $\alpha t - \left(\frac{\alpha - \lambda N_0}{\lambda}\right) e^{-\lambda t} + \left(\frac{\alpha - \lambda N_0}{\lambda}\right)$
 (B) $\alpha t + \left(\frac{\alpha - \lambda N_0}{\lambda}\right) e^{-\lambda t} - \left(\frac{\alpha - \lambda N_0}{\lambda}\right)$
 (C) $\alpha t + \left(\frac{\alpha - \lambda N_0}{\lambda}\right) e^{-\lambda t}$
 (D) $\alpha t - \left(\frac{\alpha - \lambda N_0}{\lambda}\right) e^{-\lambda t}$

67. The number N_Y of nuclei of Y at $t = T_{1/2}$ is

(A) $\frac{\alpha}{\lambda} \log_e(2) + \frac{3}{2} \left(\frac{\alpha - \lambda N_0}{\lambda}\right)$
 (B) $\frac{\alpha}{\lambda} \log_e(2) + \frac{1}{2} \left(\frac{\alpha - \lambda N_0}{\lambda}\right)$
 (C) $\frac{\alpha}{\lambda} \log_e(2) - \frac{1}{2} \left(\frac{\alpha - \lambda N_0}{\lambda}\right)$
 (D) $\frac{\alpha}{\lambda} \log_e(2) - 2 \left(\frac{\alpha - \lambda N_0}{\lambda}\right)$

Comprehension 23

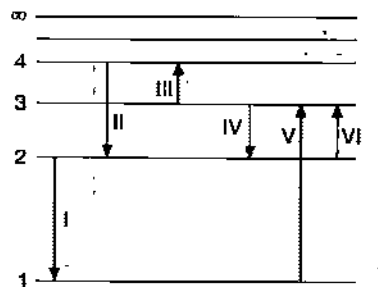
A gas of identical hydrogen like atoms has some atoms in lowest (ground) energy level A and some atoms in

particular upper (excited) energy level B and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 2.7 eV. Subsequently the atoms emit radiation of only six different photon energies. Some of emitted photons have energy 2.7 eV. Some have more and some have less than 2.7 eV. Based on the information given, answer the following questions.

68. The principal quantum number of initially excited level B is
 (A) 1 (B) 2
 (C) 3 (D) None of these
69. The magnitude of ionisation energy of gas atoms is
 (A) 13.6 eV (B) 12.7 eV
 (C) 54.4 eV (D) None of these
70. The maximum and minimum energies of the emitted photons
 (A) 12.75 eV and 0.66 eV
 (B) 13.6 eV and 12.75 eV
 (C) 13.6 eV and 0.66 eV
 (D) None of these

Comprehension 24

The figure shown an energy level diagram for the hydrogen atom. Several transitions are marked as I, II, III, IV, V, & VI. The diagram is only indicative and not to scale. Based on the information given, answer the following questions.



71. A Balmer series photon is absorbed in
 (A) II (B) III
 (C) IV (D) VI
72. The wavelength of the radiation involved in transition II is
 (A) 291 nm (B) 364 nm
 (C) 487 nm (D) 652 nm
73. The transition that occur, when a hydrogen atom is irradiated with radiation of wavelength 1025 Å
 (A) I (B) II
 (C) IV (D) V

Practice Exercise

Matrix Match Type Questions

Each question in this section contains statements given in two columns, which have to be matched. The statements in COLUMN-I are labelled A, B, C and D, while the statements in COLUMN-II are labelled p, q, r, s (and t). Any given statement in COLUMN-I can have correct matching with **ONE OR MORE** statement(s) in COLUMN-II. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following examples:

If the correct matches are $A \rightarrow p, s$ and t ; $B \rightarrow q$ and r ; $C \rightarrow p$ and q ; and $D \rightarrow s$ and t ; then the correct darkening of bubbles will look like the following :

	p	q	r	s	t
A	☒	☐	☐	☒	☒
B	☐	☒	☒	☐	☐
C	☒	☒	☐	☐	☐
D	☐	☐	☐	☒	☒

1. Some laws/processes are given in COLUMN-I. Match these with the physical phenomena given in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Nuclear fusion	(p) Converts some matter into energy
(B) Nuclear fission	(q) Generally possible for nuclei with low atomic number
(C) β -decay	(r) Generally possible for nuclei with higher atomic number
(D) Exothermic nuclear reaction	(s) Essentially proceeds by weak nuclear forces.

2. Some laws/processes are given in COLUMN-I. Match these with the physical phenomena given in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Transition between two atomic energy levels	(p) Characteristic X-rays
(B) Electron emission from a material	(q) Photoelectric effect
(C) Mosley's Law	(r) Hydrogen spectrum
(D) Change of photon energy into kinetic energy of electrons	(s) β -decay

3. Match the quantities in COLUMN-I with their dependence on the principal quantum number n and the atomic number Z in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Angular speed	(p) $\frac{n^3}{Z^2}$
(B) Time period	(q) n
(C) Angular momentum	(r) $\frac{Z^2}{n^3}$
(D) Magnetic moment	(s) $\frac{Z^3}{n^5}$
(E) Magnetic Field	

4. The energy, the magnitude of linear momentum, magnitude of angular momentum and orbital radius of an electron in a hydrogen atom corresponding to the quantum number n are E , p , L and r respectively. Then according to Bohr's theory of hydrogen atom match the expressions in COLUMN-I with statement in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Epr	(p) is independent of n
(B) $\frac{p}{E}$	(q) is directly proportional to n
(C) Er	(r) is inversely proportional to n
(D) pr	(s) is directly proportional to L

5. If f_1 is the maximum frequency of emitted photon of Lyman series, f_2 is minimum frequency of the emitted photon of Lyman series and f_3 is maximum frequency of the emitted photon of Balmer series, then match the relations in given in COLUMN-I with the corresponding relations in COLUMN-II.

COLUMN-I	COLUMN-II
(A) f_1	(p) greater than f_3
(B) f_2	(q) is negative
(C) $(f_2 - f_1)$	(r) less than f_2
(D) $(f_1 - f_2)$	(s) is equal to f_3

6. Match the quantities in COLUMN-I with their respective values in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Pair Production	(p) Few MeV
(B) Inverse photoelectric effect	(q) 20 KeV
(C) De-excitation of Be^{14} atom from second excited state	(r) 54 eV
(D) K_α X-ray photons of molybdenum $Z = 42$	(s) 0.1 eV

7. Match the processes given in COLUMN-I with their characteristics in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Alpha decay	(p) Mono energetic particles are emitted
(B) Beta decay	(q) Poly energetic particles are emitted
(C) Positron emission	(r) Angular momentum is conserved
(D) Electron capture	(s) Can take place inside and outside nucleus

8. Match the quantities in COLUMN-I with their respective values in COLUMN-II.

COLUMN-I	COLUMN-II
(A) 1 rutherford	(p) 1 disintegration/sec
(B) 1 becquerel	(q) 3.7×10^{10} dis/sec
(C) 1 curie	(r) 10^6 dis/sec
(D) Activity of 1 g Ra^{226}	(s) 10^{10} dis/sec

9. Match the processes given in COLUMN-I with their characteristics in COLUMN-II.

COLUMN-I	COLUMN-II
(A) α -decay	(p) Atomic number of the product nucleus decreases
(B) β^+ -decay	(q) Atomic number of the product nucleus increases
(C) β^- -decay	(r) Atomic number of the product nucleus not necessarily changes
(D) Electron capture	(s) some mass is converted into energy

10. Match the processes given in COLUMN-I with their characteristics in COLUMN-II.

COLUMN-I	COLUMN-II
(A) α -decay	(p) Mass number decreases
(B) β^- -decay	(q) Atomic number decreases
(C) β^+ -decay	(r) Mass number does not change
(D) γ -decay	(s) Chemical symbol of nucleus changes
	(t) Energy is released

11. Match the reactions/processes given in COLUMN-I with their characteristics given in COLUMN-II.

COLUMN-I	COLUMN-II
(A) Spontaneous radioactive decay of an uranium nucleus initially at rest as given by reaction ${}_{92}^{238}\text{U} \longrightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He} + \dots$	(p) Number of protons is increased
(B) Fusion reaction of two hydrogen nuclei as given by reaction ${}_1^1\text{H} + {}_1^1\text{H} \longrightarrow {}_2^2\text{H} + \dots$	(q) Momentum is conserved
(C) Fission of U^{235} nucleus initiated by a thermal neutron as given by reaction ${}_0^1\text{n} + {}_{92}^{235}\text{U} \longrightarrow {}_{56}^{144}\text{Ba} + {}_{36}^{89}\text{Kr} + 3{}_0^1\text{n}$	(r) Mass and energy are inter convertible
(D) β^- -decay (negative beta decay)	(s) Charge is conserved
	(t) Angular momentum is conserved

12. Match COLUMN-I of the nuclear processes with COLUMN-II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists.

COLUMN-I	COLUMN-II
(A) Alpha decay	(p) ${}_{8}^{15}\text{O} \longrightarrow {}_7^{15}\text{N} + \dots$
(B) β^- -decay	(q) ${}_{92}^{238}\text{U} \longrightarrow {}_{90}^{234}\text{Th} + \dots$
(C) Fission	(r) ${}_{83}^{209}\text{Bi} \longrightarrow {}_{82}^{208}\text{Pb} + \dots$
(D) Proton emission	(s) ${}_{94}^{239}\text{Pu} \longrightarrow {}_{57}^{140}\text{La} + \dots$

Practice Exercise

Integer Answer Type Questions

In this section the answer to each of the question is a four digit integer, ranging from 0 to 9999. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answer to question number X (say) is 6092, then the correct darkening of bubbles will look like the following :

X.	0	1	2	3	4	5	6	7	8	9
1	0	1	2	3	4	5	6	7	8	9
2	0	1	2	3	4	5	6	7	8	9
3	0	1	2	3	4	5	6	7	8	9
4	0	1	2	3	4	5	6	7	8	9
5	0	1	2	3	4	5	6	7	8	9
6	0	1	2	3	4	5	6	7	8	9
7	0	1	2	3	4	5	6	7	8	9
8	0	1	2	3	4	5	6	7	8	9
9	0	1	2	3	4	5	6	7	8	9

- How many head-on, elastic collisions must a neutron have with deuterium nucleus to reduce its energy from 1 MeV to 0.025 eV.
- The binding energy of an electron in the ground state of He atom is equal to $E_0 = 24.6$ eV. Find the energy required, in eV, to remove both electrons from the atom.
- Find the quantum number n corresponding to n th excited state of He^+ ion if on transition to the ground state the ion emits two photons in succession with wavelengths 108.5 nm and 30.4 nm. The ionization energy of the hydrogen atom is 13.6 eV.
- From what material (Z value) is the anode of an X-ray tube made, if the K_α line wavelength of the characteristic spectrum is 0.76 Å?
- A ${}^7\text{Li}$ target is bombarded with a proton beam current of 10^{-4} A for 1 hour to produce ${}^7\text{Be}$ of activity 1.8×10^8 disintegrations per second. Assuming that one ${}^7\text{Be}$ radioactive nucleus is produced by bombarding 1000 protons, determine its half-life, in days, to the nearest three digit integer.
- At time $t=0$, activity of a radioactive substance is 1600 Bq, at $t=8$ s activity remains 100 Bq. Find the activity, in Bq, at $t=2$ s.
- A ${}^{118}\text{Cd}$ radionuclide goes through the transformation chain.

$${}^{118}\text{Cd} \xrightarrow{30 \text{ min}} {}^{118}\text{In} \xrightarrow{45 \text{ min}} {}^{118}\text{Sn} \text{ (stable)}$$
The half lives are written below the respective arrows. A time $t=0$ only Cd was present. Find the percentage of nuclei transformed into stable over 60 minutes.
- The work functions of Silver and Sodium are 4.6 and 2.3 eV, respectively. Find the ratio of the slope of the stopping potential versus frequency plot for Silver to that of Sodium.
- A freshly prepared sample of a radioisotope of half-life 1386 s has activity 10^3 disintegrations per second. Given that $\log_2 2 = 0.693$, find the fraction of the initial number of nuclei (expressed in nearest, integer percentage) that will decay in the first 80 s after preparation of the sample.
- A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-

- space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^Z$ (where $1 < A < 10$). Find the value of Z .
- The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 s. The mass of an atom of this radioisotope is 10^{-25} kg. Find the mass (in mg) of the radioactive sample.
 - A proton is fired from very far away towards a nucleus with charge $Q = 120e$, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. Find the de-Broglie wavelength (in units of fm) of the proton at its start.
Take the proton mass, $m_p = \left(\frac{5}{3}\right) \times 10^{-27}$ kg,
 $\frac{h}{e} = 4.2 \times 10^{-25}$ JsC⁻¹, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ mF⁻¹ and
1 fm = 10^{-15} m.
 - A stream of α -particles is incident on a sample of hydrogen gas. What should be the minimum kinetic energy of α -particles, in eV, to ionize the hydrogen atoms.
 - Hydrogen gas in the atomic state is excited to an energy level such that the electrostatic potential energy of H-atom becomes -1.7 eV. Now a photoelectric plate having work function $W = 2.3$ eV is exposed to the emission spectra of this gas. Assuming all the transitions to be possible, find the minimum de-Broglie wavelength of the ejected photoelectrons in Å to the nearest integer.
 - A hydrogen like atom (described by the Bohr model) is observed to emit six wavelengths, originating from all possible transitions between a group of levels. These levels have energies between -0.85 eV and -0.544 eV (including both these values).
(a) Find the atomic number of the atom.
(b) Calculate the smallest wavelength emitted in these transitions, in Å.
(Take ground state energy of hydrogen atom to be -13.6 eV).
 - A neutron with an energy of 4.6 MeV collides with protons and is retarded. Assuming that upon each collision neutron is deflected by 45° find the number of collisions which will reduce its energy to 0.23 eV.
 - Emission spectrum of hydrogen atom has two lines of Balmer series with wavelength 4102 Å and 4861 Å. To what series does a spectral line belong if its wave number is equal to the difference of wave numbers of the above two lines? What is the wavelength of this line in Å? Given, $R = 1.097 \times 10^7$ m⁻¹.
 - A hydrogen like atom (atomic number Z) is in a higher excited state of quantum number n . The excited atom can make a transition to the first excited state by successively emitting two photons of energy 10.2 eV and 17 eV respectively. Alternately, the atom from the same excited state can make a transition to the second excited state by successively emitting two photons of energies 4.25 eV and 5.95 eV respectively. Determine the values of n and Z .
(Ionization energy of H-atom = 13.6 eV)
 - The ionization energy of a hydrogen like Bohr atom is 4 rydberg.
(a) What is the wavelength of the radiation emitted, in Å, when the electron jumps from the first excited state to the ground state?
(b) What is the atomic number Z of the atom?
(c) Also the radius of the first orbit for this atom as a multiple of the Bohr's radius a_0 is $\frac{a_0}{*}$, where $*$ is not readable. Find $*$.
 - A doubly ionized lithium atom is hydrogen like with atomic number 3. Find the wavelength of the radiation to the nearest three digit integer, in Å, required to excite the electron in Li^{++} from the first to the third Bohr orbit. The ionization energy of the hydrogen atom is 13.6 eV.
 - The electric current in an X-ray tube operating at 40 kV is 10 mA. Assume that on an average 1% of the total kinetic energy of the electrons hitting the target are converted into X-rays.
(a) What is the total power emitted as X-rays, in watt?
(b) How much heat, in joule is produced in the target every second?
 - The mean lives of a radioactive substance are 1200 yr and 600 yr for α -emission and β -emission respectively. Find out the time, in year, during which three fourth of a sample will decay if it is decaying both by α -emission and β -emission simultaneously.
Given $\log_e(4) \approx 1.4$

Answers

In Chapter Exercise (ICE) & Practice Exercise Sets

ICE I	BASED ON ATOMIC STRUCTURE & PROPERTIES
1. (a) 113.74 Å (b) 3	14. (a) 1.587 Å (b) $1.095 \times 10^9 \text{ ms}^{-1}$
2. 122.4 eV	15. $\frac{neh}{4\pi m}, \frac{\mu_0 \pi m^2 e^7}{8 \epsilon_0 h^6 n^3}$
3. 793.3 Å	16. $\left(\frac{n^2 h^2}{8am\pi^2} \right)^{1/4}$
4. 2 eV, 0.754 V	17. (a) 42 (b) 0.163 Å
5. $7.52 \times 10^{10} \text{ Hz}$	18. First Orbit: KE = 13.60 eV, PE = 0 eV, TE = 13.60 eV Second Orbit: KE = 3.40 eV, PE = 20.40 eV, TE = 23.80 eV
6. 15865 volt	19. 0.55 eV
7. 0.31 Å	20. $\sqrt{2}$
8. (a) He (b) 54.4 eV (c) 10.2 eV	21. (a) 3.4 eV (b) 6.63 Å
9. 12.1 eV	
10. -26.9 eV, -12 eV	
11. 10.2 V, 2, 10.2 eV, 51 eV	
12. $3.1 \times 10^9 \text{ ms}^{-1}$	
13. (a) 15.6 eV (b) 2335 Å (c) 12.52 V (d) $1.01 \times 10^7 \text{ m}^{-1}$ (e) (i) 6 eV, (ii) 0.7 eV	

ICE II

BASED ON NUCLEUS PROPERTIES & RADIOACTIVITY

1. 81
2. $1.55 \times 10^9 \text{ J}$
3. $1.4 \times 10^{17} \text{ s}$
4. $3.81 \times 10^{-4} \text{ year}^{-1}$ and $3.14 \times 10^{-2} \text{ year}^{-1}$
5. (a) 0.113 min^{-1}
(b) 6.132 min
6. $1.26 \times 10^{-11} \%$
7. (a) $1.42 \times 10^{14} \text{ per year}$
(b) $1.41 \times 10^{14} \text{ per year}$
(c) $5.4 \times 10^7 \text{ years}$
8. When $\lambda_1 \gg \lambda_2$, $N_2 \approx N_0 e^{-\lambda_2 t}$
When $\lambda_1 \ll \lambda_2$, $N_2 \approx \frac{\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t}$
9. 0.39
10. (a) 5724 years
(b) $12.58 \text{ dismin}^{-1} \text{g}^{-1}$
11. $1.12 \times 10^{-6} \text{ g}$
12. 1.88×10^{-3}
13. 0.205 mCi

ICE III

BASED ON BINDING ENERGY & NUCLEAR REACTIONS

1. No, first reaction is not possible.
2. 1.54 MeV
3. 8.78 day
4. $4.7 \times 10^{-14} \text{ kg}$
5. 0.1308 MeV
6. $K_{\min} = 0$, $K_{\max} = 4.4 \text{ MeV}$
7. 0.961 MeV
8. 120.35 g
9. (a) $8.09 \times 10^{13} \text{ J}$
(b) $2.7 \times 10^6 \text{ kg}$
10. (a) -1.18 MeV
(b) 43°
11. 127.6 MeV

ANSWERS

SINGLE CORRECT CHOICE TYPE QUESTIONS

1. B	2. D	3. C	4. D	5. B
6. D	7. D	8. D	9. A	10. A
11. C	12. D	13. B	14. A	15. D
16. C	17. A	18. D	19. A	20. C
21. B	22. A	23. D	24. D	25. B
26. C	27. D	28. D	29. A	30. C
31. A	32. D	33. B	34. C	35. C
36. D	37. D	38. A	39. D	40. A
41. D	42. A	43. B	44. B	45. D
46. B	47. C	48. B	49. C	50. B
51. C	52. C	53. D	54. C	55. B
56. C	57. A	58. D	59. D	60. B
61. C	62. C	63. C	64. D	65. C
66. C	67. B	68. D	69. D	70. C
71. B	72. A	73. A	74. A	75. B

Advanced JEE Physics

Optics & Modern Physics

76. B	77. C	78. C	79. C	80. C
81. D	82. A	83. A	84. C	85. B
86. A	87. A	88. C	89. C	90. B
91. A	92. C	93. D	94. C	95. C
96. D	97. D	98. B	99. B	100. C
101. A	102. B	103. D	104. B	105. C
106. D	107. C	108. B	109. B	110. D
111. C	112. D	113. B	114. D	115. B
116. D	117. D	118. D	119. D	120. D
121. B	122. D	123. A	124. D	125. A
126. A	127. B	128. C	129. C	130. C
131. B	132. D	133. A	134. A	135. B
136. B	137. B	138. A	139. C	140. A
141. B	142. B	143. A	144. D	145. A
146. D	147. C	148. B	149. C	150. A
151. C	152. B	153. B	154. D	155. A
156. D	157. A	158. D	159. B	160. B
161. A	162. D	163. D	164. B	165. C
166. B	167. C	168. A	169. B	170. A
171. C	172. C	173. D	174. A	175. B
176. C	177. C	178. B	179. B	180. A
181. B	182. C	183. D	184. A	185. C
186. D	187. B	188. C	189. C	190. A
191. C	192. C	193. A	194. C	195. B
196. D	197. C	198. D	199. C	200. C
201. D	202. D	203. A	204. A	

ANSWERS

MULTIPLE CORRECT CHOICE TYPE QUESTIONS

1. BC	2. AC	3. AC	4. AD	5. BC
6. BC	7. ACD	8. CD	9. AD	10. AD
11. AD	12. ABD	13. AB	14. CD	15. ABC
16. ABC	17. ABCD	18. ABCD	19. BCD	20. ABCD
21. AD	22. AC	23. ABCD	24. ABD	25. BD
26. BC	27. ACD	28. BD	29. BCD	30. BC
31. AD	32. AD	33. CD	34. ABD	35. BCD
36. ABC	37. ABD	38. ACD	39. BC	40. AC
41. BC	42. AB	43. BD		

ANSWERS

REASONING BASED QUESTIONS

1. B	2. D	3. C	4. D	5. C
6. C	7. A	8. A	9. B	10. B
11. C	12. C	13. C	14. D	15. B
16. A	17. D	18. A	19. A	20. D
21. B	22. D	23. A	24. A	25. B
26. A	27. A	28. A		

ANSWERS

LINKED COMPREHENSION TYPE QUESTIONS

1. C	2. C	3. A	4. A	5. C
6. D	7. A	8. B	9. D	10. D
11. A	12. B	13. C	14. A	15. C
16. B	17. C	18. D	19. A	20. B
21. A	22. B	23. C	24. B	25. C
26. B	27. A	28. A	29. C	30. C
31. C	32. D	33. B	34. D	35. C
36. B	37. C	38. D	39. B	40. A
41. C	42. D	43. A	44. B	45. D
46. C	47. A	48. C	49. C	50. A
51. B	52. B	53. D	54. A	55. B
56. C	57. A	58. B	59. C	60. C
61. A	62. C	63. C	64. A	65. D
66. B	67. C	68. B	69. A	70. A
71. D	72. C	73. D		

ANSWERS

MATRIX MATCH TYPE QUESTIONS

1. A → (p, q) B → (p, r) C → (p, s) D → (p, q, r)	2. A → (p, r) B → (p, q, s) C → (p) D → (q)	3. A → (r) B → (p) C → (q) D → (q) E → (s)	4. A → (p, s) B → (q, s) C → (q, s) D → (s)	5. A → (p) B → (p) C → (q) D → (s)
6. A → (p) B → (q) C → (r) D → (q)	7. A → (p, r) B → (q, r, s) C → (q, r) D → (p, r)	8. A → (r) B → (p) C → (q) D → (q)	9. A → (p, s) B → (p) C → (q, s) D → (p)	10. A → (p, q, s, t) B → (r, s, t) C → (q, r, s, t) D → (r, t)
11. A → (q, r, s, t) B → (q, r, s, t) C → (q, r, s, t) D → (p, q, r, s, t)	12. A → (q) B → (p) C → (s) D → (r)			

ANSWERS**INTEGER/ANSWER TYPE QUESTIONS**

1. 8	2. 79	3. 5	4. 41
5. 100	6. 800	7. 31	8. 1
9. 4	10. 8	11. 1	12. 7
13. 68	14. 4	15. (a) 4, (b) 40441	16. 24
17. 26206	18. 6	19. (a) 300, (b) 2, (c) 2	20. 114
21. (a) 4, (b) 396	22. 560		

Solutions

In Chapter Exercise (ICE)-I

(based on atomic structure & properties)

1. (a) $E_3 - E_1 = \frac{(-13.6)(3)^2}{(3)^2} - \left[\frac{(-13.6)(3)^2}{(1)^2} \right] = 108.8 \text{ eV}$

$$\Rightarrow \lambda = \frac{12375}{108.8} \text{ Å} = 113.74 \text{ Å}$$

(b) Number of lines in emission spectrum is

$$N = \frac{n(n-1)}{2}$$

$$\Rightarrow N = \frac{(3)(3-1)}{2} = 3$$

2. Since, $E_n = -\frac{(13.6)Z^2}{n^2} \text{ eV}$

So, ionisation energy, $n=1$, for Li^{++} ($Z=3$) is

$$IE = \frac{(13.6)(3)^2}{(1)^2} \text{ eV}$$

$$\Rightarrow IE = 122.4 \text{ eV}$$

3. Energy of electron in ground state of hydrogen atom is -13.6 eV . Earlier it had a kinetic energy of 2 eV

Therefore, energy of photon released during formation of hydrogen atom,

$$\Delta E = 2 - (-13.6) = 15.6 \text{ eV}$$

$$\Rightarrow \lambda = \frac{12375}{\Delta E} = \frac{12375}{15.6} = 793.3 \text{ Å}$$

4. For shorter wavelength, we have

$$\Delta E = E_4 - E_3 = \frac{(-13.6)(3)^2}{(4)^2} - \left[\frac{(-13.6)(3)^2}{(3)^2} \right] = 5.95 \text{ eV}$$

Using Einstein's Photo-Electric Equation, we get

$$W = E - K_{\max} = (5.95 - 3.95) \text{ eV} = 2 \text{ eV}$$

For longer wavelength, we have

$$\Delta E = E_5 - E_4 = \frac{(-13.6)(3)^2}{(5)^2} - \left[\frac{(-13.6)(3)^2}{(4)^2} \right] = 2.754 \text{ eV}$$

Again using Einstein's Photo-Electric Equation, we get

$$K_{\max} = E - W = 0.754 \text{ eV}$$

Hence stopping potential is 0.754 V

5. Using the equation, $\sqrt{f} = a(Z - b)$

Since, $b = 1$, so we get

(b = 1)

$$\frac{f_n}{f_m} = \left(\frac{Z_{L_A} - 1}{Z_{C_u} - 1} \right)^2$$

$$\Rightarrow f_n = f_m \left(\frac{Z_{L_A} - 1}{Z_{C_u} - 1} \right)^2$$

$$\Rightarrow f_n = 1.88 \times 10^{15} \left(\frac{57-1}{29-1} \right)^2$$

$$\Rightarrow f_n = 7.52 \times 10^{16} \text{ Hz}$$

6. Since, $\lambda = \frac{12375}{V}$, so we have

$$\lambda_1 - \lambda_2 = \frac{12375}{V_1} - \frac{12375}{V_2} = 12375 \left(\frac{1}{V} - \frac{1}{1.5V} \right)$$

where $\lambda_1 - \lambda_2 = 26 \text{ pm} = 0.26 \text{ Å}$

$$\Rightarrow 0.26 = (12375) \left(\frac{1}{3V} \right)$$

$$\Rightarrow V = 15865 \text{ volt}$$

7. Cutoff wavelength $\lambda_{\min}$ is given by,

$$\lambda_{\min} (\text{in Å}) = \frac{12375}{V (\text{in volt})} = \frac{12375}{40 \times 10^3} = 0.31 \text{ Å}$$

8. (a) As the atoms finally emit radiation of only 3 different photon energies final excited state corresponds to $n=3$.

So, the initial excited state corresponds to $n=2$

$$\Rightarrow Z^2 (13.6) \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{12375}{1654}$$

$$\Rightarrow Z = 2$$

Therefore, it is helium atom.

(b) Ionization energy is

$$IE = Z^2 (13.6 \text{ eV}) = (2)^2 (13.6 \text{ eV})$$

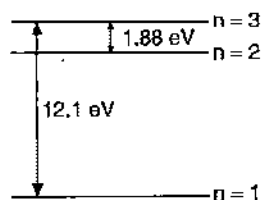
$$\Rightarrow IE = 54.4 \text{ eV}$$

(c) $N = \frac{n(n-1)}{2} = 6$

$$\text{So, } E = E_4 - E_2 = (13.6)(4) \left(\frac{1}{4} - \frac{1}{16} \right) = 10.2 \text{ eV}$$

9. Energy of photon corresponding to $\lambda = 6563 \text{ Å}$ is

$$\Delta E = \frac{12375}{6563} \text{ eV} = 1.88 \text{ eV}$$



This is the difference in energy between $n=3$ and $n=2$. Hence the single electron in the hydrogen atom should excite at least upto $n=3$ and for this the minimum energy of the striking electron should be 12.1 eV.

10. Given $E_0 = 0$

$$\lambda_1 = 460 \text{ \AA}$$

$$\Rightarrow E_3 - E_1 = \frac{12375}{460} = 26.9 \text{ eV}$$

$$\text{Since, } E_0 = 0$$

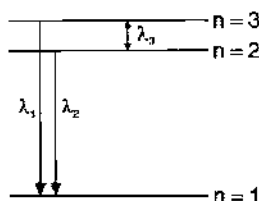
$$\Rightarrow 0 - E_1 = 26.9 \text{ eV}$$

$$\Rightarrow E_1 = -26.9 \text{ eV}$$

$$\text{Further, } \lambda_3 = 1035 \text{ \AA}$$

$$\Rightarrow E_3 - E_2 = \frac{12375}{1035} \text{ \AA} = 12 \text{ eV}$$

$$\Rightarrow E_2 = -12 \text{ eV}$$



11. When hydrogen atom is excited, then we have

$$eV = E_0 \left(\frac{1}{1} - \frac{1}{n^2} \right) \quad \dots(1)$$

When ion is excited, then

$$eV = E_0 Z^2 \left(\frac{1}{2^2} - \frac{1}{n_1^2} \right) \quad \dots(2)$$

Wavelength of emitted light is

$$\frac{hc}{\lambda_1} = E_0 \left(\frac{1}{1} - \frac{1}{n^2} \right) \quad \dots(3)$$

$$\frac{hc}{\lambda_2} = E_0 Z^2 \left(\frac{1}{1} - \frac{1}{n_1^2} \right) \quad \dots(4)$$

Further it is given that

$$\frac{\lambda_1}{\lambda_2} = \frac{5}{1} \quad \dots(5)$$

Solving the above equations, we get

$$Z = 2, n = 2, n_1 = 4 \text{ and } V = 10.2 \text{ V}$$

Energy of emitted photon by the hydrogen atom is

$$\Delta E = E_2 - E_1 = 10.2 \text{ eV}$$

and by the ion is

$$\Delta E' = E_4 - E_1 = (13.6)(2)^2 \left(1 - \frac{1}{16} \right) = 51 \text{ eV}$$

12. Energy of photon of the first line of Lyman series is

$$E = E_2 - E_1 = (13.6)(2)^2 \left(1 - \frac{1}{4} \right) = 40.8 \text{ eV}$$

Energy required to ionize the hydrogen atom is 13.6 eV. Therefore, kinetic energy of electron emitted from the hydrogen atom is

$$K = (40.8 - 13.6) \text{ eV} = 27.2 \text{ eV}$$

$$\Rightarrow K = 4.352 \times 10^{-18} \text{ J}$$

$$\text{Since, } K = \frac{1}{2}mv^2 = 4.352 \times 10^{-18}$$

$$\Rightarrow v = \sqrt{\frac{2 \times 4.352 \times 10^{-18}}{9.1 \times 10^{-31}}}$$

$$\Rightarrow v = 3.1 \times 10^8 \text{ ms}^{-1}$$

13. (a) From figure, we observe that

$$\left(\begin{array}{l} \text{Ionisation} \\ \text{Potential} \end{array} \right) = 15.6 \text{ eV}$$

$$(b) \lambda_{\min} = \frac{12375}{5.3} = 2335 \text{ \AA}$$

$$(c) \Delta E_{31} = -3.08 - (-15.6) = 12.52 \text{ eV}$$

Therefore, excitation potential for state $n=3$ is 12.52 V

$$(d) \frac{1}{\lambda_{31}} = \frac{\Delta E_{31}}{12375} \text{ \AA}^{-1} = \frac{12.52}{12375} \text{ \AA}^{-1}$$

$$\Rightarrow \frac{1}{\lambda_{31}} \approx 1.01 \times 10^7 \text{ m}^{-1}$$

$$(e) (i) E_2 - E_1 = 10.3 \text{ eV} > 6 \text{ eV}$$

Hence, the striking electron cannot excite the hypothetical atoms. So the electron will keep its energy with itself.

$$\Rightarrow K_{\min} = 6 \text{ eV}$$

$$(ii) E_2 - E_1 = 10.3 \text{ eV} < 11 \text{ eV}$$

So, the electron can excite the atom.

$$\Rightarrow K_{\min} = (11 - 10.3) \text{ eV} = 0.7 \text{ eV}$$

14. (a) $z = 3$ for Li^{+2} . Further we know that $r_n = \frac{n^2}{z} a_0$

Substituting, $n = 3$, $z = 3$ and $a_0 = 0.529 \text{ \AA}$, we get r_3 for Li^{+2}

$$r_3 = \frac{(3)^2}{(3)} (0.529) \text{ \AA} = 1.587 \text{ \AA}$$

- (b) $z = 2$ for He^+ . Also we know that

$$v_n = \frac{Z}{n} v_1$$

Substituting $n = 4$, $z = 2$ and $v_1 = 2.19 \times 10^6 \text{ ms}^{-1}$, we get for He^+ ,

$$v_4 = \left(\frac{2}{4} \right) (2.19 \times 10^6) \text{ ms}^{-1}$$

$$\Rightarrow v_4 = 1.095 \times 10^6 \text{ ms}^{-1}$$

15. Magnetic moment $\mu = NiA = \left(\frac{e}{T} \right) (\pi r^2)$

$$\Rightarrow \mu = \left(\frac{e}{2\pi r/v} \right) (\pi r^2) = \frac{evr}{2} \quad \dots(1)$$

$$\text{We know that } mvr = \frac{nh}{2\pi} \quad \dots(2)$$

Solving equations (1) and (2)

$$\mu = \frac{neh}{4\pi m}$$

$$\text{Magnetic induction, } B = \frac{\mu_0 i}{2r} = \frac{\mu_0 e}{2rT}$$

$$\Rightarrow B = \frac{\mu_0 ev}{(2r)(2\pi r)} = \frac{\mu_0 ev}{4\pi r^2} \quad \dots(3)$$

From Newton's Second Law, we have

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}$$

$$\Rightarrow v^2 = \frac{e^2}{4\pi\epsilon_0 mr} \quad \dots(4)$$

Solving these equations, we get

$$B = \frac{\mu_0 \pi m^2 e^7}{8\epsilon_0 h^4 n^5}$$

16. The force at a distance r is,

$$F = -\frac{dU}{dr} = -2ar$$

Suppose r be the radius of n th orbit. Then the necessary centripetal force is provided by the above force. Thus,

$$\frac{mv^2}{r} = 2ar \quad \dots(1)$$

Further, the quantization of angular momentum gives,

$$mvr = \frac{nh}{2\pi} \quad \dots(2)$$

Solving equations (1) and (2) for r , we get

$$r = \left(\frac{n^2 h^2}{8am\pi^2} \right)^{1/4}$$

17. (a) Given that

$$2 \left\{ \frac{12375 \times 10^{-10}}{34.3 \times 10^3} \right\} = \frac{1}{1.09 \times 10^7 (Z-1)^2 \left(1 - \frac{1}{4} \right)}$$

$$\Rightarrow Z = 42$$

- (b) Energy of K_α line = $100 - 24 = 76$ keV

$$\Rightarrow \lambda_{K\alpha} = \frac{12375}{76 \times 10^3} \text{ \AA}$$

$$\Rightarrow \lambda_{K\alpha} = 0.163 \text{ \AA}$$

18. Since, $E_1 = -13.60$ eV

Also, we know that

$$TE = -KE = \frac{PE}{2}$$

$$\Rightarrow K_1 = -E_1 = 13.60 \text{ eV}$$

$$\Rightarrow U_1 = 2E_1 = -27.20 \text{ eV}$$

$$\text{Further, } E_n = -\frac{13.6Z^2}{n^2}$$

$$\Rightarrow E_2 = \frac{E_1}{(2)^2} = -3.40 \text{ eV}$$

$$\Rightarrow K_2 = 3.40 \text{ eV and}$$

$$\Rightarrow U_2 = -6.80 \text{ eV}$$

Now $U_1 = 0$, i.e., potential energy has been increased by 27.20 eV. So, we will increase U and E in all energy states by 27.20 eV while kinetic energy will remain unchanged. So, we have

Orbit	K(eV)	U(eV)	E(eV)
First	13.60	0	13.60
Second	3.40	20.40	23.80

19. Wavelengths corresponding to minimum wavelength ($\lambda_{\min}$) or maximum energy will emit photoelectrons having maximum kinetic energy.

($\lambda_{\min}$) belonging to Balmer series and lying in the given range (450 nm to 750 nm) corresponds to transition from ($n = 4$ to $n = 2$). Here,

$$E_4 = -\frac{13.6}{(4)^2} = -0.85 \text{ eV}$$

$$\text{and } E_2 = -\frac{13.6}{(2)^2} = -3.4 \text{ eV}$$

$$\Rightarrow \Delta E = E_4 - E_2 = 2.55 \text{ eV}$$

So, $K_{\max} = \text{Energy of photon} - \text{work function}$

$$\Rightarrow K_{\max} = 2.55 - 2 = 0.55 \text{ eV}$$

20. For $0 \leq x \leq 1$, $PE = E_0$

$$\Rightarrow (\text{Kinetic energy } K_1) = (\text{Total energy}) - (PE)$$

$$\Rightarrow K_1 = 2E_0 - E_0 = E_0$$

$$\Rightarrow \lambda_1 = \frac{h}{\sqrt{2mE_0}} \quad \dots(1)$$

For $x > 1$, $PE = 0$

$$\Rightarrow \text{Kinetic energy } K_2 = \text{Total energy} = 2E_0$$

$$\Rightarrow \lambda_2 = \frac{h}{\sqrt{2m(E_0)}} \quad \dots(2)$$

From equations (1) and (2), we get

$$\frac{\lambda_1}{\lambda_2} = \sqrt{2}$$

21. (a) Kinetic energy of electron in the orbits of hydrogen and hydrogen like atoms = [Total energy]

So, Kinetic energy = 3.4 eV

- (b) The de-Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2Km}}$$

where K is the kinetic energy of an electron
Substituting the values, we get

$$\lambda = \frac{(6.6 \times 10^{-34} \text{ Js})}{\sqrt{2(3.4 \times 1.6 \times 10^{-19} \text{ J})(9.1 \times 10^{-31} \text{ kg})}}$$

$$\Rightarrow \lambda = 6.63 \times 10^{-10} \text{ m}$$

$$\Rightarrow \lambda = 6.63 \text{ \AA}$$

Solutions

In Chapter Exercise (ICE)-II

(based on nucleus properties & radioactivity)

1. In 1 second 90% of the nuclei have remained undecayed, so in another 1 second 90% of 90 i.e., 81 nuclei will remain undecayed.

2. Since, $N = \frac{10^{-3}}{210} \times 6.02 \times 10^{23} = 2.87 \times 10^{18}$

During one mean life period 63.8% nuclei are decayed.
Hence, energy released is

$$E = 0.638 \times 2.87 \times 10^{18} \times 5.3 \times 1.6 \times 10^{-13} \text{ J} = 1.55 \times 10^6 \text{ J}$$

3. $N = \frac{2}{238} \times 6.02 \times 10^{23} = 5.06 \times 10^{21}$

Since, $R = \lambda N$

$$\Rightarrow \lambda = \frac{R}{N} = \frac{2.5 \times 10^4}{5.06 \times 10^{21}} = 4.94 \times 10^{-18} \text{ s}^{-1}$$

$$\Rightarrow t_{1/2} = \frac{0.693}{\lambda} = 1.4 \times 10^{17} \text{ s}$$

4. $\frac{R_1}{R_2} = \frac{\lambda_1 N}{\lambda_2 N} = \frac{1.2}{98.8}$

$$\Rightarrow \lambda_2 = 82.33 \lambda_1$$

...(1)

Further, $\lambda = \frac{0.693}{21.8} \text{ year}^{-1} = 0.0318 \text{ year}^{-1}$

...(2)

Also, $\left(-\frac{dN}{dt}\right) = \left(-\frac{dN_1}{dt}\right) + \left(-\frac{dN_2}{dt}\right)$

$$\Rightarrow \lambda N = \lambda_1 N + \lambda_2 N$$

$$\Rightarrow \lambda = \lambda_1 + \lambda_2$$

...(3)

Solving equations (1), (2) and (3), we get

$$\lambda_1 = 3.81 \times 10^{-4} \text{ year}^{-1} \text{ and } \lambda_2 = 3.14 \times 10^{-2} \text{ year}^{-1}$$

5. (a) $R = R_0 e^{-\lambda t}$

$$\Rightarrow 2700 = 4750 e^{-\lambda t}$$

$$\Rightarrow \lambda = 0.113 \text{ min}^{-1}$$

(b) $t_{1/2} = \frac{0.693}{\lambda} = 6.132 \text{ min}$

6. $R_0 = \lambda N$

$$\Rightarrow R_0 = \frac{0.693}{14.3 \times 3600 \times 24} \times 6.02 \times 10^{23} \text{ per sec}$$

$$\Rightarrow R_0 = 3.37 \times 10^{17} \text{ per sec}$$

After 70 hours activity

$$R = R_0 e^{-\lambda t}$$

$$\Rightarrow R = (3.37 \times 10^{17}) e^{-\left(\frac{0.693}{14.3 \times 24}\right)(70)} = 2.92 \times 10^{17} \text{ per sec}$$

In fruits, the activity was observed $1 \mu\text{Ci}$ or 3.7×10^4 per sec. Therefore, percentage of activity transmitted from root to the fruit is

$$\left(\frac{\text{Percentage of Activity Transmitted}}{\text{Activity Transmitted}}\right) = \frac{3.7 \times 10^4}{2.92 \times 10^{17}} \times 100 = 1.26 \times 10^{-11} \%$$

7. (a) $N = \frac{1}{109} \times 6.02 \times 10^{23}$

$$\Rightarrow R = \lambda N = \frac{0.693}{2.7 \times 10^7} \times \frac{1}{109} \times 6.02 \times 10^{23}$$

$$\Rightarrow R = 1.42 \times 10^{14} \text{ per year}$$

(b) After 2 year, $R = R_0 e^{-\lambda t} = (1.42 \times 10^{14}) e^{-[0.693/2.7 \times 10^7](2)}$

$$\Rightarrow R = 1.41 \times 10^{14} \text{ per year}$$

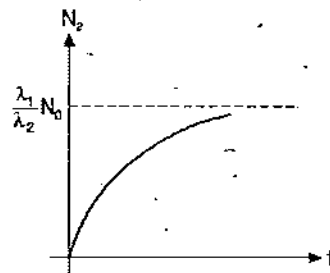
(c) After 2 half lives activity reduces to 25% of the original value, so we have

$$t = 2t_{1/2} = 5.4 \times 10^7 \text{ years}$$

8. Since, $N_2 = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t})$

When $\lambda_1 \gg \lambda_2$ we have, $N_2 \approx N_0 e^{-\lambda_1 t}$

Physically this means that parent nuclei practically instantly transform into daughter nuclei, which then decay According to the Law of Radioactive Decay with a certain decay constant.



$$N_2 \approx \frac{\lambda_1}{\lambda_2} N_0 e^{-\lambda_2 t}$$

When $\lambda_1 \ll \lambda_2$, then $N_2 \approx \frac{\lambda_1}{\lambda_2} N_0 e^{-\lambda_2 t}$

i.e., N_2 versus t graph in this case is as shown in the figure.

9. Probability of a nucleus to decay in time t is

$$\frac{N}{N_0} = 1 - e^{-\lambda t} = 1 - e^{-(\lambda/10)(5)} = 0.39$$

10. (a) $R = R_0 e^{-\lambda t}$

$$\Rightarrow 7.3 = 9.3 e^{-\lambda(4500-2500)}$$

$$\Rightarrow \lambda = 1.21 \times 10^{-4} \text{ year}^{-1}$$

$$\Rightarrow t_{1/2} = \frac{0.693}{\lambda} = 5724 \text{ years}$$

- (b) Further applying, $R = R_0 e^{-\lambda t}$

$$\Rightarrow R_0 = R e^{\lambda t} = (7.3) e^{(1.21 \times 10^{-4})(4500)}$$

$$\Rightarrow R_0 = 12.58 \text{ dismin}^{-1} \text{g}^{-1}$$

$$11. N = \frac{R}{\lambda} = \frac{5 \times 10^{-3} \times 3.7 \times 10^{10}}{\left(\frac{0.693}{138 \times 24 \times 3600}\right)} = 3.2 \times 10^{16}$$

$$\Rightarrow m = \left(\frac{3.2 \times 10^{16}}{6.02 \times 10^{23}}\right) (210) \text{ g}$$

$$\Rightarrow m = 1.12 \times 10^{-6} \text{ g}$$

12. Let x the desired ratio, then, mass of Co^{58} in 1 g be x

$$\Rightarrow N = \frac{x}{58} \times 6.02 \times 10^{23}$$

$$\text{Given, } 2.2 \times 10^{12} = \lambda N = \frac{0.693}{71.3 \times 24 \times 3600} \times \frac{x}{58} \times 6.02 \times 10^{23}$$

$$\Rightarrow x = 1.89 \times 10^{-3}$$

13. Since, $R = \lambda N = \frac{N \log_e(2)}{t_{1/2}}$

$$\Rightarrow R \propto \frac{N}{t_{1/2}}$$

$$\text{Given, } \frac{N_1}{N_2} = \frac{4}{1} \text{ and } \frac{(t_{1/2})_1}{(t_{1/2})_2} = \frac{14}{25}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{N_1}{N_2} \times \frac{(t_{1/2})_2}{(t_{1/2})_1} = \frac{4}{1} \times \frac{25}{14} = \frac{100}{14}$$

$$\Rightarrow R_1 = \frac{100}{114} \times 3 = 2.63 \text{ mCi}$$

$$\text{and } R_2 = 3 - 2.63 = 0.37 \text{ mCi}$$

After 60 days, we have

$$R'_1 = R_1 e^{-\lambda_1 t} = (2.63) e^{-\frac{0.693}{14} \times 60} = 0.135 \text{ mCi}$$

$$R'_2 = R_2 e^{-\lambda_2 t} = (0.37) e^{-\frac{0.693}{25} \times 60} = 0.07 \text{ mCi}$$

So, total activity is

$$R' = R'_1 + R'_2 = 0.205 \text{ mCi}$$

Solutions

In Chapter Exercise (ICE)-III

(based on binding energy & nuclear reactions)

1. The reaction is possible (spontaneously) if the binding energy of products is larger than that of ${}^{238}_{92}\text{U}$.

(a) Total binding energy of ${}^{238}_{92}\text{U} = 238 \times 7.57 = 1801.66 \text{ MeV}$
and binding energy of ${}^{206}_{82}\text{Pb} = 206 \times 7.83 = 1612.981 \text{ MeV}$
the binding energy of products is less than that of ${}^{238}_{92}\text{U}$.

Note that the protons and neutrons are free and do not have any binding energy. Hence, this reaction is not possible spontaneously. It can take place only if $1801.66 - 1612.98 = 188.68 \text{ MeV}$ of energy is supplied from outside.

(b) In the second case binding energy of products is larger than the binding energy of the parent nucleus. Hence, reaction is possible spontaneously.

2. Since, the masses are given in atomic mass units, so, we shall first calculate the mass difference between reactants and products in the same units and then multiply the mass difference by 931.5 MeV u^{-1} . Thus, we have

$$Q = (14.00307\text{u} + 4.00260\text{u} - 1.00783\text{u} - 16.99913\text{u}) \left(931.5 \frac{\text{MeV}}{\text{u}} \right)$$

$$\Rightarrow Q = -1.20 \text{ MeV}$$

Q value is negative, so the reaction is endothermic.

Hence, the minimum kinetic energy of α -particle to initiate this reaction is given by

$$K_{\min} = |Q| \left(\frac{m_{\alpha}}{m_{\text{N}}} + 1 \right)$$

$$\Rightarrow K_{\min} = (1.20) \left(\frac{4.00260}{14.00307} + 1 \right)$$

$$\Rightarrow K_{\min} = 1.54 \text{ MeV}$$

3. Total energy released is

$$\Delta E = \frac{1}{235} \times 6.02 \times 10^{26} \times 185 \times 1.6 \times 10^{-13} \text{ J} = 7.58 \times 10^{13} \text{ J}$$

$$\text{Power } P = 100 \times 10^6 = 10^8 \text{ Js}^{-1}$$

Therefore, time

$$t = \frac{7.58 \times 10^{13}}{10^8} \text{ sec} = 8.78 \text{ day}$$

$$4. \quad m = \frac{E}{c^2} = \frac{4.2 \times 10^3}{(3 \times 10^8)^2} \text{ kg}$$

$$\Rightarrow m = 4.7 \times 10^{-14} \text{ kg}$$

5. By Law of Conservation of Linear Momentum, we have

$$M_a v_a = mv$$

$$\Rightarrow v = \frac{M_a v_a}{m}$$

Kinetic energy of Tl atom is

$$(KE)_{\text{Tl}} = \frac{1}{2} M v^2 = \frac{1}{2} M \left(\frac{M_a^2 v_a^2}{m^2} \right) = \left(\frac{1}{2} M_a v_a^2 \right) \left(\frac{M_a}{m} \right)$$

$$\Rightarrow (KE)_{\text{Tl}} = (KE)_a \left(\frac{M_a}{m} \right)$$

$$\Rightarrow (KE)_{\text{Tl}} = 6.082 \times \frac{4}{208} \text{ MeV}$$

$$\Rightarrow (KE)_{\text{Tl}} = 0.1308 \text{ MeV}$$

6. Please observe that here, atomic masses are given (not the nuclear masses), but still we can use them for calculating the mass defect because mass of electrons get cancelled both sides. Thus,

$$\text{Mass defect } \Delta m = (22.9945 - 22.9898) = 0.0047\text{u}$$

$$\Rightarrow Q = (0.0047\text{u})(931.5 \text{ MeV u}^{-1})$$

$$\Rightarrow Q = 4.4 \text{ MeV}$$

Hence, the energy of beta particles can range from 0 to 4.4 MeV.

$$7. \quad Q = [m'_c - (m'_b + m_p)]c^2$$

where m'_c and m'_b are the nuclear mass of ${}^{12}\text{C}$ and ${}^{11}\text{B}$, so

$$m_b = m'_b + 5m_e$$

$$m_p = m'_p + 6m_e$$

$$\Rightarrow Q = [(m_c - 6m_e) - (m_b - 5m_e + m_p)]c^2$$

$$\Rightarrow Q = [m_c - m_b - 2m_p]c^2 \quad \left\{ \because \begin{array}{l} \text{mass of} \\ \text{electron} \end{array} = \begin{array}{l} \text{mass of} \\ \text{positron} \end{array} \right\}$$

$$\Rightarrow Q = (11.011434 - 11.009305 - 2 \times 0.0005486)931$$

$$\Rightarrow Q = 0.961 \text{ MeV}$$

The disintegration energy is equal to the maximum energy of the emitted photon.

$$8. \quad \Delta m = 2(\text{mass of } {}_1\text{H}^2) - (\text{mass of } {}_2\text{He}^4) = 0.0256u$$

$$\Rightarrow \Delta E = 0.0256 \times 931.5 \text{ MeV} = 23.85 \text{ MeV}$$

Total energy required per day is

$$E = 200 \times 10^8 \times 24 \times 3600 \text{ J} = 1.728 \times 10^{13} \text{ J}$$

Let m be the mass of deuterium required. Then energy required for reactor is

$$E = \left(\frac{25}{100} \right) \left(\frac{m/2}{2} \right) (6.02 \times 10^{23}) (23.85 \times 1.6 \times 10^{-13})$$

This should be equal to $1.728 \times 10^{13} \text{ J}$

$$\Rightarrow m = \frac{4 \times 100 \times 1.728 \times 10^{13}}{25 \times 6.02 \times 10^{23} \times 23.85 \times 1.6 \times 10^{-13}} \text{ g}$$

$$\Rightarrow m = 120.35 \text{ g}$$

$$9. \quad (a) \quad \text{Total number of atoms in 1 kg of } \text{U}^{238} \text{ is}$$

$$N = \frac{1}{238} \times 6.02 \times 10^{26} = 2.53 \times 10^{24}$$

$$\Rightarrow \text{Total energy released, } \Delta E = (200 \times 2.53 \times 10^{24}) \text{ MeV}$$

$$\Rightarrow \Delta E = 8.09 \times 10^{13} \text{ J}$$

$$(b) \quad m = \frac{8.09 \times 10^{13}}{30 \times 10^3} \text{ g} = 2.7 \times 10^9 \text{ g}$$

$$\Rightarrow m = 2.7 \times 10^6 \text{ kg}$$

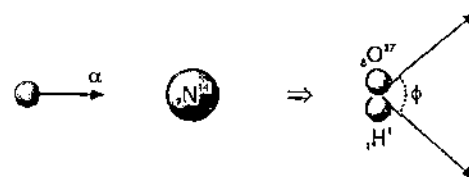
$$10. \quad (a) \quad Q\text{-value} = (\Delta m)931.5 \text{ MeV} = -1.18 \text{ MeV}$$

(b) By Law of Conservation of Linear Momentum, we have

$$P_\alpha^2 = P_O^2 + P_H^2 + 2P_O P_H \cos \phi$$

$$\Rightarrow 2m_\alpha K_\alpha = 2m_O K_O + 2m_H K_H$$

$$+ 2\sqrt{(2m_O K_O)(2m_H K_H)} \cos \phi \quad \dots (1)$$



By Law of Conservation of Energy, we have

$$K_O = Q + K_\alpha - K_H = (-1.18 + 7.7 - 5.5) \text{ MeV}$$

$$\Rightarrow K_O = 1.02 \text{ MeV}$$

$$\Rightarrow \cos \phi = \frac{m_O K_O - m_\alpha K_\alpha - m_H K_H}{2\sqrt{m_\alpha m_H K_\alpha K_H}} = 0.73$$

$$\Rightarrow \phi \approx 43^\circ 18'$$

$$11. \quad \text{Since, } \Delta E = (8m_p + 8m_n - m_O) \times 931.5 \text{ MeV}$$

$$\Rightarrow \Delta E = 127.6 \text{ MeV}$$

Solutions

Single Correct Choice Type Questions

1. [B]

Let N be the number of nuclei at any time t . Then

$$\frac{dN}{dt} = 200 - \lambda N$$

$$\Rightarrow \int_0^N \frac{dN}{200 - \lambda N} = \int_0^t dt$$

$$\Rightarrow N = \frac{200}{\lambda} (1 - e^{-\lambda t})$$

Given that $N = 100$ and $\lambda = 1 \text{ s}^{-1}$

$$\Rightarrow 100 = 200(1 - e^{-t})$$

$$\Rightarrow e^{-t} = \left(\frac{1}{2}\right)$$

$$\Rightarrow t = \log_e(2) \text{ s}$$

3. [C]

$$\frac{\lambda_A}{\lambda_B} = \frac{1}{2}$$

Probabilities of getting α and β -particles are same. So, rate of disintegration are equal.

$$\Rightarrow \lambda_A N_A = \lambda_B N_B$$

$$\Rightarrow \frac{N_A}{N_B} = \frac{\lambda_B}{\lambda_A} = 2$$

4. [D]

$$B_n = \frac{\mu_0 I_n}{2r_n}$$

$$\Rightarrow B_n \propto \frac{I_n}{r_n} \propto \frac{f}{r_n}$$

$$\Rightarrow B_n \propto \left(\frac{v_n}{r_n}\right) \propto \frac{v_n}{(r_n)^2}$$

$$\Rightarrow B_n \propto \left(\frac{Z}{n^2}\right) \propto \frac{Z^3}{n^5}$$

5. [B]

$$N = N_0 e^{-\lambda t}$$

$$t_1 = \frac{0.693}{\lambda}$$

$$\Rightarrow \lambda = \frac{0.693}{3.8}$$

$$\text{Since } \frac{N}{N_0} = \frac{1}{10}$$

$$\Rightarrow \frac{1}{10} = e^{-\frac{0.693}{3.8} t}$$

$$\Rightarrow \log 10 = \frac{0.693}{3.8} t$$

$$\Rightarrow 2.3 = \frac{0.693}{3.8} t$$

$$\therefore t = 12.62 \text{ days}$$

8. [D]

Three half-lives of A is equivalent to six half-lives of B. Hence, we have

$$N_A \left(\frac{1}{2}\right)^3 = N_B \left(\frac{1}{2}\right)^6$$

$$\Rightarrow \frac{N_A}{N_B} = \frac{1}{8}$$

9. [A]

$$mvr = \frac{nh}{2\pi}$$

$$\Rightarrow \frac{h}{mv} = \frac{(2\pi r)}{n}$$

where, $\frac{h}{mv}$ = de-Broglie wavelength

10. [A]

Fraction of nuclei which remain undecayed is

$$f = \frac{N}{N_0} = \frac{N_0 e^{-\lambda t}}{N_0} = e^{-\lambda t}$$

$$\Rightarrow f = e^{-\left(\frac{\log_e(2)}{t}\right) \left(\frac{t}{2}\right)} = e^{-\frac{1}{2} \log_e(2)}$$

$$\Rightarrow f = \frac{1}{e^{\log_e \sqrt{2}}} = \frac{1}{\sqrt{2}}$$

11. [C]

Since, $E_{K_a} < E_{K_b}$

$$\Rightarrow \lambda_{K_a} > \lambda_{K_b}$$

12. [D]

Since, $\Delta\lambda = \lambda_{K_a} - \lambda_{min}$... (1)

When V is made half, λ_{min} becomes two times, however λ_{K_a} remains the same.

$$\Rightarrow \Delta\lambda' = \lambda_{K_a} - 2\lambda_{min}$$

From (1), we have

$$\lambda_{min} = \lambda_{K_a} - \Delta\lambda$$

$$\Rightarrow \Delta\lambda' = \lambda_{K_a} - 2\lambda_{K_a} + 2\Delta\lambda$$

$$\Rightarrow \Delta\lambda' = 2\Delta\lambda - \lambda_{K_a}$$

$$\Rightarrow \Delta\lambda' < 2\Delta\lambda$$

13. [B]

Assuming that ionization occurs as a result of a completely inelastic collision, we can write

$$mv_0 = (m + m_H)V$$

where m is the mass of incident particle, m_H the mass of hydrogen atom, v_0 the initial velocity of incident particle and V the final common velocity of the particle after collision. Prior to collision, the KE of the incident particle was

$$E_0 = \frac{mv_0^2}{2}$$

The total kinetic energy after collision

$$E = \frac{(m + m_H)V^2}{2} = \frac{m^2 v_0^2}{2(m + m_H)}$$

The decrease in kinetic energy must be equal to ionization energy, so we have

$$E_i = E_0 - E = \left(\frac{m_H}{m + m_H} \right) E_0$$

$$\Rightarrow \frac{E_i}{E_0} = \frac{m_H}{m + m_H}$$

i.e., the greater the mass m , the smaller the fraction of initial kinetic energy that will be used for ionization.

14. [A]

$$\frac{1}{16} = \left(\frac{1}{2} \right)^4 = \left(\frac{1}{2} \right)^{\left(\frac{\text{Time lapsed}}{T_{1/2}} \right)}$$

$$\Rightarrow \frac{\text{Time lapsed}}{T_{1/2}} = 4$$

$$\Rightarrow \text{Time lapsed} = 400 \mu s$$

16. [C]

$$E = mc^2$$

$$\Rightarrow E = m_0 c^2 \left(1 - \frac{v^2}{c^2} \right)^{-\frac{1}{2}}$$

Since $\frac{v}{c} \ll 1$

$$\Rightarrow E \approx m_0 c^2 \left(1 + \frac{v^2}{2c^2} \right)$$

$$\Rightarrow E = m_0 c^2 + \frac{1}{2} m_0 v^2$$

17. [A]

According to Ritz Combination Principle

$$v_{m \rightarrow n} = v_{m \rightarrow i} + v_{i \rightarrow n} \quad [\text{for } m < i < n]$$

e.g., $v_{4 \rightarrow 1} = v_{4 \rightarrow 3} + v_{3 \rightarrow 1}$ or

$$v_{4 \rightarrow 1} = v_{4 \rightarrow 2} + v_{2 \rightarrow 1}$$

19. [A]

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n^2} - \frac{1}{(n+1)^2} \right]$$

$$\Rightarrow \frac{1}{\lambda} = R_H \left[\frac{(2n+1)}{n^2(n+1)^2} \right]$$

$$\Rightarrow \frac{1}{\lambda} = R_H \frac{2n}{n^4} \quad \because \text{for large } n, 2n+1 \approx 2n, n^2(n+1)^2 \approx n^4$$

$$\Rightarrow \frac{1}{\lambda} = R_H \frac{2}{n^3}$$

$$\Rightarrow v = \frac{c}{\lambda} = \frac{2R_H c}{n^3}$$

21. [B]

$$\frac{\Delta\lambda}{\lambda} = \mp \frac{v}{c}$$

(-) sign to be used for approaching

(+) sign to be used for receding

$$\Rightarrow \frac{\Delta\lambda}{\lambda} = \frac{706 - 656}{656} = + \frac{v}{c}$$

$$\Rightarrow v = 2 \times 10^7 \text{ ms}^{-1}$$

23. [D]

Let R_0 be the initial activity. Then

$$R_1 = R_0 e^{-\lambda t_1} \quad \text{and} \quad R_2 = R_0 e^{-\lambda t_2}$$

$$\Rightarrow \frac{R_2}{R_1} = e^{\lambda(t_1 - t_2)}$$

$$\Rightarrow R_2 = R_1 e^{\lambda(t_1 - t_2)}$$

24. [D]

$$R_1 = \lambda N_1$$

$$\Rightarrow N_1 = \frac{R_1}{\lambda}$$

$$\text{and } R_2 = \lambda N_2$$

$$\Rightarrow N_2 = \frac{R_2}{\lambda}$$

So, number of atoms decayed is

$$N = N_1 - N_2$$

$$\Rightarrow N = \left(\frac{R_1 - R_2}{\lambda} \right)$$

25. [B]

After two half lives $\frac{1}{4}$ th fraction of nuclei will remain undecayed.

Or, $\frac{3}{4}$ th fraction will decay. Hence, the probability that a nucleus

decays in two half lives is $\frac{3}{4}$.

26. [C]

The necessary centripetal force to the satellite is provided by the gravitational force of attraction.

$$\text{i.e. } \frac{mV^2}{r} = \frac{GMm}{r^2}$$

$$\frac{1}{2}mV^2 = \frac{GMm}{2r} = \text{KE}$$

PE of the satellite is given by

$$\text{PE} = -\frac{GMm}{r}$$

Now $\text{KE} + \text{PE} = E_0$

$$\Rightarrow \frac{GMm}{2r} - \frac{GMm}{r} = E_0$$

$$\frac{GMm}{2r} = E_0$$

$$\text{or } -\frac{GMm}{r} = 2E_0$$

Which is equal to PE.

27. [D]

$$\lambda = 2\pi r$$

where r = Radius of first orbit = $0.53 \text{ \AA}$

$$\Rightarrow \lambda = 2\pi(0.53) \text{ \AA} = 3.33 \text{ \AA}$$

28. [D]

Since, $r_n \propto n^2$

Given, $r_{n+1} - r_n = r_{n-1}$

$$\Rightarrow (n+1)^2 - n^2 = (n-1)^2$$

$$\Rightarrow n = 4$$

29. [A]

$$\text{Since, } \frac{3h}{2\pi} = n\left(\frac{h}{2\pi}\right)$$

$$\Rightarrow n = 3$$

$$\Rightarrow K_n = \frac{K_1}{(3)^2} = \frac{13.6}{9} = 1.51 \text{ eV}$$

31. [A]

$$E_1 = \frac{1240}{550} = 2.25 \text{ eV}$$

$$E_2 = \frac{1240}{450} = 2.75 \text{ eV}$$

$$E_3 = \frac{1240}{350} = 3.54 \text{ eV}$$

E_1 cannot emit photoelectrons from q and r plates. E_2 cannot emit photoelectrons from r .

Further, work function of p is least and it can emit photoelectrons from all three wavelengths. Hence magnitude of its stopping potential and saturation current both will be maximum.

32. [D]

$$N_1 = N_0 e^{-10\lambda t}$$

$$N_2 = N_0 e^{-\lambda t}$$

$$\Rightarrow \frac{N_1}{N_2} = \frac{1}{e} = e^{(-10\lambda + \lambda)t}$$

$$\Rightarrow 9\lambda t = 1$$

$$\Rightarrow t = \frac{1}{9\lambda}$$

33. [B]

$$\lambda = \frac{hc}{eV_0}$$

$$\Rightarrow \lambda = \frac{6.626 \times 10^{-24} \times 3 \times 10^8}{(20 \times 10^3)(1.6 \times 10^{-19})}$$

$$\Rightarrow \lambda = 0.62 \text{ \AA}$$

35. [C]

$$\frac{0.693}{\lambda_x} = \frac{1}{\lambda_y}$$

$$\Rightarrow \lambda_y > \lambda_x$$

$$\text{Since, } \left(-\frac{dN}{dt}\right)_x = \lambda_x N$$

$$\text{And } \left(-\frac{dN}{dt}\right)_y = \lambda_y N$$

$$\Rightarrow \left(-\frac{dN}{dt}\right)_y > \left(-\frac{dN}{dt}\right)_x$$

Decay rate of $Y >$ Decay rate of X

39. [D]

$$\frac{0.0001}{2} \times 10^{-3} = (0.5 \times 10^{-10}) n^2$$

$$\left\{ \begin{array}{l} \text{Because radius of } n\text{th orbit is equal to} \\ r_n = n^2 r_0 \text{ where } r_0 = 0.529 \text{ \AA} \end{array} \right\}$$

$$\Rightarrow n^2 = 1000$$

$$\Rightarrow n = 31$$

42. [A]

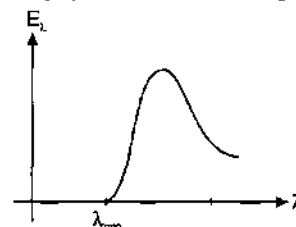
$$\frac{1}{\lambda} \propto (Z-1)^2$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \left(\frac{Z_2-1}{Z_1-1}\right)^2 \quad \text{or} \quad \frac{1}{4} = \left(\frac{Z_2-1}{11-1}\right)^2$$

Solving this, we get, $Z_2 = 6$

43. [B]

The continuous X-ray spectrum is shown in figure.



All wavelengths $> \lambda_{\min}$ are found, where

$$\lambda_{\min} = \frac{12375}{V \text{ (in volt)}} \text{ \AA}$$

Here, V is the applied voltage.

44. [B]

$$\text{Using } N = N_0 e^{-\lambda t} \text{ where } \lambda = \frac{\log_e 2}{t_{1/2}} = \frac{\log_e (2)}{3.8}$$

$$\Rightarrow \frac{N_0}{20} = N_0 e^{-\frac{\log_e(2)}{36} t}$$

Solving this equation with the help of given data we find $t = 16.5$ days.

45. [D]

The binding energy is numerically equal to the kinetic energy of the electron, so

$$\frac{1}{2}mv^2 = E, \quad \dots(1)$$

$$\text{Since, } mvr = \frac{nh}{2\pi} \quad \dots(2)$$

Dividing Equation (1) by Equation (2), we get

$$\frac{v}{2\pi r} = \frac{2E_1}{nh}$$

$$\Rightarrow f = \frac{2E_1}{nh} \quad \left\{ \because \frac{v}{2\pi r} = \frac{\omega}{2\pi} = f \right\}$$

46. [B]

$$\frac{\text{Magnetic moment}}{\text{Angular momentum}} = \frac{e}{2m}$$

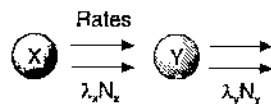
$\Rightarrow$ Magnetic moment $\propto$ Angular Momentum

$$\Rightarrow M \propto n \quad \left\{ \because L = n \frac{h}{2\pi} \right\}$$

47. [C]

Number of radionuclei become constant, when $X = \lambda N$

$$\Rightarrow N = \frac{X}{\lambda}$$



$$\Rightarrow N = \frac{X}{\lambda} = \frac{XY}{\log_e(2)}$$

48. [B]

Cut-off wavelength depends on the applied voltage not on the atomic number of the target. Characteristic wavelengths depend on the atomic number of target.

49. [C]

Since decay is to be regarded as a statistical spontaneous process, hence α decay can be regarded as an Adiabatic process.

50. [B]

Total initial binding energy is 236×7.6

$$\Rightarrow U_i = 1793.6 \text{ MeV}$$

Total final binding energy is $2(117)(8.5)$

$$\Rightarrow U_f = 1989 \text{ MeV}$$

$$\Rightarrow \Delta U = 195.4 \text{ MeV}$$

51. [C]

$$R = R_0 e^{-\lambda t}$$

$$\Rightarrow \lambda = \frac{1}{t} \log_e \left(\frac{R_0}{R} \right) = \frac{2.3}{t} \log_{10} \left(\frac{R_0}{R} \right)$$

$\therefore$ Decay constant

$$\lambda = \frac{1}{5} \times 2.3 \log_{10} \frac{9750}{975} = \frac{2.3}{5} = 0.461 \text{ min}^{-1}$$

53. [D]

$$\frac{N}{N_0} = \left(\frac{1}{2} \right)^t$$

$$\Rightarrow \frac{N}{N_0} = \frac{1}{16}$$

54. [C]

Since nuclear density is independent of mass of nucleus, hence all possess equal density.

56. [C]

If R_0 be the initial activity of the sample, then $R_1 = R_0 e^{-\lambda t_1}$ and

$$R_2 = R_0 e^{-\lambda t_2}$$

$$\text{where } \lambda = \frac{1}{T} \quad \left\{ \because \text{Mean life} = T = \frac{1}{\lambda} \right\}$$

$$\Rightarrow \frac{R_2}{R_1} = \frac{e^{-\lambda t_2}}{e^{-\lambda t_1}} = e^{\lambda(t_1 - t_2)}$$

$$\Rightarrow R_2 = R_1 \exp \left(\frac{t_1 - t_2}{T} \right)$$

58. [D]

Rate of decay of sample 1, $R_1 = \lambda_1 N_1 e^{-\lambda_1 t}$ and for sample 2, $R_2 = \lambda_2 N_2 e^{-\lambda_2 t}$.

When they are mixed, we have $R = R_1 + R_2$

$$\Rightarrow R = \lambda_1 N_1 e^{-\lambda_1 t} + \lambda_2 N_2 e^{-\lambda_2 t}$$

$$\Rightarrow R = \lambda_1 N_1 \lambda_2 N_2 e^{-(\lambda_1 + \lambda_2)t}$$

59. [D]

$$L = \sqrt{n(n+1)} \frac{h}{2\pi}$$

$$\Rightarrow L = \sqrt{56} \left(\frac{h}{2\pi} \right)$$

60. [B]

$$R_n = n^2 R_0$$

$$\Rightarrow R_2 = 4(0.528 \text{ Å})$$

$$\Rightarrow R_2 = 2.112 \text{ Å}$$

61. [C]

Speed of electron in the n th orbit of H like atom is

$$v_n = \frac{1}{137} \left(\frac{cZ}{n} \right)$$

62. [C]

$$\text{Since } \Delta p = \frac{h}{\lambda}$$

$$\Rightarrow m_e v = h R_H \left(\frac{1}{1} - \frac{1}{25} \right)$$

$$\Rightarrow v = \frac{h R_H}{m_e} \left(\frac{24}{25} \right)$$

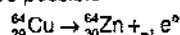
$$\Rightarrow v = 4 \text{ ms}^{-1}$$

Advanced JEE Physics

63. [C]
As velocity (or momentum) of electron is increased, the wavelength $\left(\lambda = \frac{h}{p}\right)$ will decrease.

Hence, fringe width will decrease ($\omega \propto \lambda$).

64. [D]
In beta decay, atomic number is increased by 1 whereas the mass number remains the same. Therefore, following equation can be possible



65. [C]
Activity $R = \lambda N$
Number of nuclei (N) per mole are equal for both the substances, so we have

$$R \propto \lambda$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{\lambda_1}{\lambda_2} = \frac{4}{3}$$

66. [C]
 $R_1 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R_2 \left(\frac{9}{n_1^2} - \frac{9}{n_2^2} \right)$
Substituting $n_1 = 6$, $n_2 = 12$ makes both sides equal

67. [B]
 $a = \frac{v^2}{r}$
 $\Rightarrow a \propto \frac{(z)^2}{\left(\frac{1}{z}\right)} \quad (\text{for } n=1)$
 $\Rightarrow a \propto z^3$
 $\Rightarrow \frac{a_1}{a_2} = \left(\frac{2}{1}\right)^3 = 8$

68. [D]
Net rate of formation of Y at any time t is

$$\frac{dN_y}{dt} = \lambda_x N_x - \lambda_y N_y$$

N_y is maximum when $\frac{dN_y}{dt} = 0$

$$\Rightarrow \lambda_x N_x = \lambda_y N_y$$

69. [D]
Maximum wavelength of Lyman series will correspond to the transition of electron from $n=2$ to $n=1$ and maximum wavelength of Paschen series will correspond to $n=4$ to $n=3$.

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\left(\frac{1}{9} - \frac{1}{16}\right)}{\left(\frac{1}{1} - \frac{1}{4}\right)} = \frac{7}{108}$$

71. [B]
 $v_n = \alpha \left(\frac{cZ}{n} \right)$ where $\alpha = \frac{e^2}{2\hbar c}$
is the fine structure constant $\left(\alpha \approx \frac{1}{137} \right)$

$$v_{H\alpha} = \alpha \left(\frac{c(2)}{2} \right) = \alpha c$$

$$\text{and } v_H = \alpha \left(\frac{c(1)}{1} \right) = \alpha c = v_{H\alpha}$$

72. [A]
Momentum of striking electrons

$$p = \frac{h}{\lambda}$$

Kinetic energy of striking electrons

$$K = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$$

This is also, maximum energy of X-rays photons.

$$\frac{hc}{\lambda_0} = \frac{h^2}{2m\lambda^2}$$

$$\Rightarrow \lambda_0 = \frac{2m\lambda^2 c}{h}$$

73. [A]
Activity of $S_1 = \frac{1}{2}$ (activity of S_2)

$$\Rightarrow \lambda_1 N_1 = \frac{1}{2} (\lambda_2 N_2)$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{N_2}{2N_1}$$

$$\Rightarrow \frac{T_1}{T_2} = \frac{2N_1}{N_2}$$

$$\left\{ T = \text{half life} = \frac{\log_e 2}{\lambda} \right\}$$

$$\text{Given } N_1 = 2N_2$$

$$\Rightarrow \frac{T_1}{T_2} = 4$$

74. [A]
Rest mass of parent nucleus should be greater than the rest mass of daughter nuclei.

75. [B]
Total power drawn by Coolidge tube is VI.
 $\Rightarrow P_{\text{total}} = VI = 200 \text{ W}$
Since only 0.5% is carried by X-rays, so

$$P = \left(\frac{0.5}{100} \right) P_{\text{total}}$$

$$\Rightarrow P = \frac{1}{200} (200) \text{ W}$$

$$\Rightarrow P = 1 \text{ W}$$

76. [B]
 $f_n = \frac{v_n}{2\pi r_n} = \frac{\frac{2\hbar e_0 c}{n}}{2\pi \frac{n^2 \hbar^2 e_0}{\pi m e^2 Z}}$
 $\Rightarrow f_n = \left(\frac{m e^4 Z^2}{4\epsilon_0^2 \hbar^3} \right) \frac{1}{n^3}$

78. [C]
Total energy radiated per minute from sun is
 $E_{\text{radiated}} = \sigma (4\pi R_{\text{sun}}^2)$
Energy radiated annually is given by

$$E_{\text{total}} = 24 \times 60 \times 365 \times E_{\text{released}}$$

$$\Rightarrow \text{Annual loss of mass} = \Delta m = \frac{E_{\text{total}}}{c^2} = 1.38 \times 10^{17} \text{ kg}$$

80. [C]

If it was A



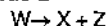
$$\text{Reactant: } R = 60 \times 8.5 = 510 \text{ MeV}$$

$$\text{Product: } P = 2 \times 30 \times 5 = 300 \text{ MeV}$$

$$\Delta E = -210 \text{ MeV}$$

ENDOTHERMIC

If it was B



$$R = 120 \times 7.5 = 900 \text{ MeV}$$

$$P = 90 \times 8 + 30 \times 5 = 870 \text{ MeV}$$

$$\Delta E = -30 \text{ MeV}$$

ENDOTHERMIC

If it was C



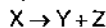
$$R = 120 \times 7.5 = 900 \text{ MeV}$$

$$P = 2 \times 60 \times 8.5 = 1020 \text{ MeV}$$

$$\Delta E = 120 \text{ MeV}$$

EXOTHERMIC

If it was D



$$R = 90 \times 8.0 = 720 \text{ MeV}$$

$$P = 60 \times 8.5 + 30 \times 5.0 = 660 \text{ MeV}$$

$$\Delta E = -60 \text{ MeV}$$

ENDOTHERMIC

81. [D]

$$f = \frac{\Delta m}{A} = \frac{M - A}{A}$$

83. [A]

$$E = mc^2$$

$$\Rightarrow E = (10^{-6})(9 \times 10^{16})$$

$$\Rightarrow E = 9 \times 10^{10} \text{ J}$$

84. [C]

$$\text{Since } \frac{N}{N_0} = \left(\frac{1}{2}\right)^n$$

$$\text{where } n = \frac{\text{Time lapsed}}{\text{Half life}}$$

$$\text{According to the problem } n = \frac{T_{1/2}}{T_{1/2}} = \frac{1}{2}$$

$$\Rightarrow \frac{N}{N_0} = \left(\frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}}$$

86. [A]

$$\Delta A = 235 - 219 = 16$$

$$\Rightarrow \text{Number of } \alpha\text{-particles emitted} = \frac{16}{4} = 4$$

87. [A]

Charge number decreases by 4, but actually it must decrease by 8 (due to emission of 4α -particles). Hence 4β particles must have also been emitted.

88. [C]

$$\frac{1}{\lambda} = R_H \left(\frac{1}{4} - \frac{1}{16} \right)$$

$$\text{Since } c = v\lambda$$

$$\Rightarrow v = \frac{c}{\lambda}$$

$$\Rightarrow v = 3 \times 10^8 \times 10^7 \times \left(\frac{3}{16} \right)$$

$$\Rightarrow v = \frac{9}{16} \times 10^{15} \text{ Hz}$$

89. [C]

$$E = R_H hc \left(1 - \frac{1}{25} \right)$$

Momentum of photon emitted is

$$p = \frac{E}{c} = R_H h \left(\frac{24}{25} \right)$$

Recoil momentum of H-atom will also be p.

$$\Rightarrow mv = p$$

$$\Rightarrow v = \frac{p}{m} = \frac{(1.097 \times 10^7)(6.626 \times 10^{-34})24}{(25)(1.67 \times 10^{-27})}$$

$$\Rightarrow v = 4.178 \text{ ms}^{-1}$$

90. [B]

$$\text{Since } E = -\frac{13.6}{n^2} \text{ eV}$$

$$\Rightarrow E_1 = -13.6 \text{ eV}$$

$$\Rightarrow E_2 = -3.4 \text{ eV}$$

$$\Rightarrow E_3 = -1.50 \text{ eV}$$

$$\Rightarrow E_4 = -0.85 \text{ eV}$$

From above we can see that

$$E_3 - E_1 = 12.1 \text{ eV}$$

i.e. the electron must be making a transition from $n=3$ to $n=1$ level.

$$\Rightarrow \Delta L = (3-1) \frac{h}{2\pi} = \frac{h}{\pi}$$

$$\Rightarrow \Delta L = \frac{6.626 \times 10^{-34}}{3.14}$$

$$\Rightarrow \Delta L = 2.11 \times 10^{-34} \text{ Js}$$

91. [A]

For the fifth electron to act as a dopant it must lie in the valence shell of P.

So $n=3$. (TRICKY HINT!)

$$Z=15$$

{for Phosphorus}

$$\text{Since } r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z} = \frac{n^2}{Z} r_0$$

$$\text{where } r_0 = 0.529 \text{ \AA} = 52.9 \text{ pm}$$

$$\Rightarrow r_3 = \frac{9}{15} (52.9 \text{ pm}) \quad \dots (1)$$

When phosphorus acts as dopant in silicon, the expression for Bohr radius is

$$r'_n = \frac{n^2 h^2 \epsilon}{\pi m e^2 Z} = k r_n$$

$$\Rightarrow r'_n = 12 r_n \quad \left(\because \epsilon = k \epsilon_0 \right)$$

$$\Rightarrow r'_3 = 12 r_3 = 12 \left[\frac{9}{15} (52.9) \text{ pm} \right]$$

$$\Rightarrow r'_3 = 380.88 \text{ pm}$$

92. [C]

The first photon will excite the hydrogen atom (in ground state) in first excited state (as $E_2 - E_1 = 10.2 \text{ eV}$). Hence, during de-excitation a photon of 10.2 eV will be released. The second photon of energy 15 eV can ionise the atom.

Hence the balance energy i.e., $(15 - 13.6) \text{ eV} = 1.4 \text{ eV}$ is retained by the electron.

Therefore, by the second photon an electron of energy 1.4 eV will be released.

93. [D]

Number of nuclei decreases exponentially

$$N = N_0 e^{-\lambda t} \text{ and}$$

$$\text{Rate of decay} \left(-\frac{dN}{dt} \right) = \lambda N$$

Therefore, decay process lasts upto $t \rightarrow \infty$. Therefore, a given nucleus may decay at any time after $t = 0$.

94. [C]

$$mvr = n \frac{h}{2\pi} = 2 \left(\frac{h}{2\pi} \right)$$

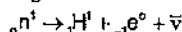
$$\Rightarrow mvr = \frac{h}{\pi}$$

So, de-Broglie wavelength is

$$\lambda = \frac{h}{mv} = \pi r = (3.14)(2.116 \text{ \AA}) = 6.64 \text{ \AA}$$

95. [C]

Following nuclear reaction takes place



$\bar{\nu}$ is antineutrino.

96. [D]

Shortest wavelength will correspond to maximum energy. As value of atomic number (Z) increases, the magnitude of energy in different energy states gets increased. Value of Z is maximum for doubly ionised lithium atom ($Z=3$) among the given elements. Hence, wavelength corresponding to this will be least.

97. [D]

$$\text{Since, } r = \frac{\epsilon_0 n^2 h^2}{e^2 \pi m}$$

$$L = n \left(\frac{h}{2\pi} \right)$$

$$\Rightarrow nh = 2\pi L$$

$$\Rightarrow r = \frac{\epsilon_0 (2\pi L)^2}{e^2 \pi m}$$

$$\Rightarrow L r^{-\frac{1}{2}} = \text{constant}$$

98. [B]

$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{h}{\sqrt{2mE}}}{\frac{h}{\sqrt{2mE}}} \text{ or } \frac{\lambda_1}{\lambda_2} \propto E^{1/2}$$

99. [B]

Since $K.E. = -T.E.$

$$\Rightarrow K.E. = +13.6 \text{ eV}$$

100. [C]

$$E_n = -\frac{me^4}{8n^2 h^2 \epsilon_0^2}, m = \text{mass of electron}$$

$$\Rightarrow E_n = -\frac{m_{\text{hypothetical}} e^4}{8n^2 h^2 \epsilon_0^2}$$

$$\Rightarrow E_n = -2 \left(\frac{Rhc}{n^2} \right)$$

The longest wavelength (or minimum energy) photon corresponds to transition between adjacent states.

i.e. $n=3$ to $n=2$

$$\Rightarrow \frac{hc}{\lambda_{\text{max}}} = E_3 - E_2 = -2Rhc \left(\frac{1}{9} - \frac{1}{4} \right)$$

$$\Rightarrow \frac{hc}{\lambda_{\text{max}}} = 2Rhc \left(\frac{5}{36} \right)$$

$$\Rightarrow \lambda_{\text{max}} = \frac{18}{5R}$$

102. [B]

According to Ritz Combination Principle

$$E_{C \rightarrow A} = E_{C \rightarrow B} + E_{B \rightarrow A}$$

$$\Rightarrow \frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$

$$\Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

103. [D]

For each principal quantum number n , number of electrons permitted equals the number of elements corresponding to the quantum number.

$$\Rightarrow \left(\begin{array}{c} \text{Total Number} \\ \text{of Elements} \end{array} \right) = \sum 2n^2 = \frac{n(n+1)(2n+1)}{3}$$

104. [B]

$$\frac{4}{3} \pi R^3 \propto A$$

$$\Rightarrow R \propto A^{\frac{1}{3}}$$

105. [C]

The nuclear force of interaction between any pair of nucleons is identical i.e. force between two neutrons (F_2) equals the force between neutron and proton (F_3). However, between two protons net force is equal to the resultant of nuclear force between them (attractive in nature) and electrostatic force between them (repulsive in nature). Hence F_1 is a value lesser than F_2 and F_3 . So $F_1 < F_2 = F_3$.

107. [C]

$$\frac{dN}{dt} = n - \lambda N$$

Because the population N is simultaneously increasing at rate n and decreasing due to decay at rate λN .

$$\Rightarrow \int_{N_0}^N \frac{dN}{n - \lambda N} = \int_0^t dt$$

$$\Rightarrow \frac{1}{\lambda} \ln \left(\frac{n - \lambda N_0}{n - \lambda N} \right) = t$$

$$\Rightarrow N = \frac{n}{\lambda} + \left(N_0 - \frac{n}{\lambda} \right) e^{-\lambda t}$$

OBJECTIVE TRICK

At $t = 0$, $N = N_0$ which is satisfied by (C) only.

108. [B]

$$I = I_0 e^{-\mu x}$$

Where I_0 is intensity at $x = 0$.

I is intensity at a distance x and

$\Rightarrow \mu$ is absorption coefficient; $[\mu] = L^{-1}$

μ is maximum for lead as lead has maximum ability to absorb radiations in a minimum distance.

109. [B]

$$\lambda = \frac{hc}{E}$$

$$\Rightarrow \lambda = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{1.6 \times 10^{-19} E(\text{in keV})}$$

$$\Rightarrow \lambda = \frac{1.242 \times 10^{-9}}{E(\text{in keV})} \text{ m}$$

$$\Rightarrow \lambda = \frac{12.42}{E(\text{in keV})} \text{ \AA}$$

110. [D]

$$E_n = \frac{Z^2}{n^2} (13.6 \text{ eV})$$

$$\Rightarrow E_n = 9(13.6 \text{ eV})$$

$$\Rightarrow E_n = 122.4 \text{ eV}$$

111. [C]

$$\lambda_{K\alpha} = 0.021 \text{ nm} = 0.21 \text{ \AA}$$

Since, $\lambda_{K\alpha}$ corresponds to the transition of an electron from L -shell to K -shell, therefore,

$$E_L - E_K (\text{in eV}) = \frac{12375}{\lambda(\text{in \AA})} = \frac{12375}{0.21} \approx 58928 \text{ eV}$$

$$\Rightarrow \Delta E \approx 59 \text{ keV}$$

112. [D]

$$(r_m) = \left(\frac{m^2}{z} \right) (0.53 \text{ \AA}) = (n \times 0.53) \text{ \AA}$$

$$\Rightarrow \frac{m^2}{z} = n$$

$$m = 5 \text{ for } {}_{100}\text{Fm}^{257} \text{ (the outermost shell) and } z = 100$$

$$\Rightarrow n = \frac{(5)^2}{100} = \frac{1}{4}$$

113. [B]

$$\text{Given that } K_1 + K_2 = 55 \text{ MeV} \quad \dots(1)$$

From Conservation of Linear Momentum,

$$P_1 = P_2$$

$$\Rightarrow \sqrt{2K_1(216 \text{ m})} = \sqrt{2K_2(4 \text{ m})}$$

$$\text{as } P = \sqrt{2Km}$$

$$\Rightarrow K_2 = 54K_1 \quad \dots(2)$$

Solving equations (1) and (2), we get $K_2 = KE$ of α -particle = 5.4 MeV.

114. [D]

$$R_1 = \lambda N_1 \text{ and } R_2 = \lambda N_2 \text{ (}\lambda \text{ is same for a given sample)}$$

$$\text{As } N_2 < N_1$$

Number of atoms disintegrated in time $(t_2 - t_1)$ is

$$N_1 - N_2 = \frac{R_1}{\lambda} - \frac{R_2}{\lambda} = \left(\frac{R_1 - R_2}{\lambda} \right)$$

$$N_1 - N_2 = \frac{(R_1 - R_2)T}{0.693} \propto (R_1 - R_2)T$$

117. [D]

$$E = \frac{1}{2} k_B T$$

$$\text{at } T = 300 \text{ K}$$

$$\Rightarrow E = \frac{1}{2} (1.38 \times 10^{-23})(300)$$

$$\Rightarrow E = \frac{1}{2} \frac{(1.38 \times 10^{-23})(300) \text{ eV}}{1.6 \times 10^{-19}}$$

$$\Rightarrow E \approx 0.01 \text{ eV}$$

118. [D]

$$E = -\frac{13.6}{n^2} \text{ eV}$$

$$\Rightarrow E_2 = -3.4 \text{ eV}$$

120. [D]

Expected mass of Cu must be less than that of zinc. So it is unstable and radioactive, decaying to Zn through β decay.

121. [B]

f_1 is the fraction decayed in one half life $T_{1/2}$ and f_2 is the fraction decayed in one mean life (T).

$$T_{1/2} = \frac{0.693}{\lambda} = 0.693T$$

$$\Rightarrow T > T_{1/2}$$

Further, since the fraction decayed is $(1 - e^{-\lambda t})$

So,

$$f_1 = 1 - e^{-\lambda T_{1/2}} = (1 - e^{-0.693}) = \frac{1}{2} \text{ and}$$

$$f_2 = 1 - e^{-\lambda T} = 1 - e^{-1} = 0.632$$

$$\Rightarrow f_2 > f_1$$

However, if we had been asked about the fraction of sample undecayed, then we must give our answer in the light of formula

$\frac{N}{N_0} = e^{-\lambda t}$ where N is number of undecayed nuclei in sample at time t .

124. [D]

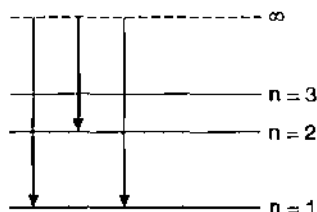
$$v = \sqrt{\frac{2qV}{m}}$$

125. [A]

Series limit (i.e. shortest wavelength) of Lyman implies transition from $\infty \rightarrow 1$

First line (or longest wavelength) of Lyman series implies transition from $2 \rightarrow 1$

Series limit of Balmer implies transition from $\infty \rightarrow 2$



According to Ritz Combination Principle

$$E_{\infty \rightarrow 1} = E_{\infty \rightarrow 2} + E_{2 \rightarrow 1}$$

$$\Rightarrow h\nu_1 = h\nu_2 + h\nu_3$$

$$\Rightarrow \nu_1 - \nu_2 = \nu_3$$

126. [A]

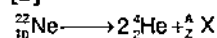
$$\frac{1}{\lambda_1} = R \left[\frac{1}{4} - \frac{1}{9} \right]$$

$$\frac{1}{\lambda_2} = 4R \left[\frac{1}{4} - \frac{1}{16} \right]$$

$$\frac{\lambda_2}{\lambda_1} = \frac{5}{27}$$

$$\Rightarrow \lambda_2 = 1215 \text{ \AA}$$

127. [B]



$$\Rightarrow A = 14$$

$$\Rightarrow Z = 6$$

i.e. ${}_{6}^{14}\text{C}$ is the unknown nucleus.

128. [C]

$$E_n \propto \frac{1}{n^2} \text{ and } r_n \propto n^2$$

So, $E_n r_n$ is independent of n

$$\Rightarrow E_{r1} = (13.6 \text{ eV})(0.53 \text{ \AA})$$

$$\Rightarrow E_{r1} = 7.2 \text{ eV\AA}$$

$$\Rightarrow E_{r1} = \text{constant}$$

129. [C]

$$t = \frac{0.693}{\lambda}$$

$$T = \frac{1}{\lambda}$$

$$\Rightarrow t = 0.693 T$$

$$\Rightarrow T > t$$

130. [C]

Since we assume the potential energy to be zero in the ground state. So,

$$\text{Total Energy} = U + K = \text{Kinetic Energy} \quad (\because U = 0)$$

$$\Rightarrow \text{Total Energy} = 13.6 \text{ eV (in ground state)}$$

If the potential energy is not assigned a zero value, then total energy is -13.6 eV .

So, we conclude that making potential energy zero increases the value of total energy by $13.6 - (-13.6) = 27.2 \text{ eV}$.

Now actual energy in second orbit $= -3.4 \text{ eV}$

Hence new value is $(-3.4 + 27.2) \text{ eV} = 23.8 \text{ eV}$

131. [B]

$$\text{Since, } E = -\frac{13.6}{n^2}$$

$$\Rightarrow -3.4 = -\frac{13.6}{n^2}$$

$$\Rightarrow n = 2$$

$$\text{Since, } \lambda = \frac{h}{mv}$$

Velocity of electron in second orbit is

$$v_2 = \frac{c}{2(137)} = \frac{c}{274}$$

$$\Rightarrow \lambda = \frac{6.626 \times 10^{-34}}{(9.1 \times 10^{-31}) \left(\frac{3 \times 10^8}{274} \right)}$$

$$\Rightarrow \lambda \sim 6.6 \times 10^{-10} \text{ m}$$

Further Kinetic Energy $= -(\text{Total Energy})$

$$\Rightarrow E = -(-3.4 \text{ eV}) = 3.4 \text{ eV}$$

133. [A]

Shortest wavelength of Brackett series corresponds to the transition of electron between $n_1 = 4$ and $n_2 \rightarrow \infty$ and the shortest wavelength of Balmer series corresponds to the transition of electron between $n_1 = 2$ and $n_2 \rightarrow \infty$. So,

$$(Z^2) \left(\frac{13.6}{16} \right) = \left(\frac{13.6}{4} \right)$$

$$\Rightarrow Z^2 = 4 \text{ or } Z = 2$$

134. [A]

The longest wavelength in a series is obtained when a transition takes place between the lowest consecutive levels. Here transition must take place from $n = 2$ to $n = 1$

135. [B]

Maximum angular speed will be in its ground state, so, we have

$$\omega_{\max} = \frac{v_1}{r_1} = \frac{2.2 \times 10^6 \text{ ms}^{-1}}{0.529 \times 10^{-10} \text{ m}}$$

$$\Rightarrow \omega_{\max} = 4.1 \times 10^{16} \text{ rads}^{-1}$$

136. [B]

$$\frac{hc}{\lambda} = Rhc \left(1 - \frac{1}{n^2} \right)$$

$$\Rightarrow n = \sqrt{\frac{\lambda R}{\lambda R - 1}}$$

137. [B]

$$\text{From } R = R_0 \left(\frac{1}{2} \right)^n$$

$$\text{We have, } 1 = 64 \left(\frac{1}{2} \right)^n$$

or $n = 6 = \text{number of half-lives.}$

$$\therefore t = n \times t_{1/2} = 6 \times 2 = 12 \text{ h}$$

138. [A]

Penetrating power is maximum for γ -rays, then of β -particles and then α -particles because basically it depends on the velocity. However, ionization power is in reverse order.

139. [C]

β and γ -decay take place from a radioactive nucleus.

Hence, options (A) and (B) are wrong. During fusion process two or more lighter nuclei combine to form a heavy nucleus.

140. [A]

Activity of a radioactive substance $R = \lambda N$

$$\Rightarrow \lambda = \frac{R}{N}$$

Since, $R = N_2$ particles per second

and $N = N_1$

$$\Rightarrow \lambda = \frac{N_2}{N_1}$$

142. [B]

Heavy water is used as moderators in nuclear reactors to slow down the neutrons.

143. [A]

$$U = eV = eV_0 \log_e \left(\frac{r}{r_0} \right)$$

$$\left| f \right| = \left| -\frac{dU}{dr} \right| = \frac{eV_0}{r}$$

This force will provide the necessary centripetal force.

$$\text{Hence, } \frac{mv^2}{r} = \frac{eV_0}{r}$$

$$\Rightarrow v = \sqrt{\frac{eV_0}{m}} \quad \dots(1)$$

Moreover,

$$mvr = \frac{nh}{2\pi} \quad \dots(2)$$

Dividing equation (2) by (1), we have

$$mr = \left(\frac{nh}{2\pi} \right) \sqrt{\frac{m}{eV_0}}$$

$$\Rightarrow r_0 \propto n$$

144. [D]

Activity reduces from 6000 dps to 3000 dps in 140 days. It implies that half-life of the radioactive sample is 140 days. In

280 days (or two half-lives) activity will remain $\frac{1}{4}$ th of the

initial activity. Hence, the initial activity of the sample is

$$4 \times 6000 \text{ dps} = 24000 \text{ dps}$$

145. [A]

The maximum number of electrons in an orbit are $2n^2$. If $n > 4$, is not allowed, then the number of maximum electrons that can be in first four orbits are:

$$2(1)^2 + 2(2)^2 + 2(3)^2 + 2(4)^2 = 2 + 8 + 18 + 32 = 60$$

Therefore, possible elements can be 60.

146. [D]

When one electron is removed, the remaining atom is hydrogen like atom whose energy in first orbit is

$$E_1 = -(2)^2 (13.6 \text{ eV}) \quad (Z = 2)$$

$$\Rightarrow E_1 = -54.4 \text{ eV}$$

Therefore, to remove the second electron an additional energy of 54.4 eV is required. Thus, to remove both the electrons $(24.6 + 54.4) \text{ eV} = 79 \text{ eV}$ energy is required.

147. [C]

During fusion, binding energy of daughter nucleus is always greater than the total binding energy of the parent nuclei. The difference of binding energies is the energy released. Hence,

$$Q = E_2 - 2E_1$$

148. [B]

Shortest wavelength or cut-off wavelength depend only upon the voltage applied to the Coolidge tube.

149. [C]

Beta particles are fast moving electrons which are emitted by the nucleus.

150. [A]

Maximum number of nuclei will be present when

Rate of decay = Rate of formation

$$\Rightarrow \lambda N = \alpha$$

$$\Rightarrow N = \frac{\alpha}{\lambda}$$

152. [B]

At time $t = 0$, A gets converted to B which has a higher activity. Therefore, activity increases.

153. [B]

$$P(\text{survival}) = \frac{N(t)}{N_0} = \frac{N_0 e^{-\lambda t}}{N_0} = e^{-\lambda t}$$

For, $t = \frac{1}{\lambda}$, we have

$$P(\text{survival}) = \frac{1}{e}$$

154. [D]

Energy released is given by

$$\Delta E = \left(\text{Total Binding Energy of } {}^4_2\text{He} \right) - 2 \left(\text{Total Binding Energy of } {}^2_1\text{H} \right)$$

$$\Rightarrow \Delta E = (4)(7) - 2(1.1)(2) = 23.6 \text{ MeV}$$

155. [A]

Number of possible emission lines are $\frac{n(n-1)}{2}$ when electron jumps from n th state to ground state. In this question this value is

$$N = \frac{(n-1)(n-2)}{2}$$

$$\Rightarrow 10 = \frac{(n-1)(n-2)}{2}$$

Solving this, we get
 $n = 6$

156. [D]

Let $\lambda_A = \lambda$ and $\lambda_B = 2\lambda$

Initially rate of disintegration of A is λN_0 and that of B is $(2\lambda)N_0$.

After one half-life of A, rate of disintegration of A will become $\frac{\lambda N_0}{2}$ and that of B would also be $\frac{\lambda N_0}{2}$, because

$$(\text{Half-life of B}) = \frac{1}{2}(\text{Half-life of A})$$

So, after one half-life of A or two half-lives of B, we have

$$\left(-\frac{dN}{dt}\right)_A = \left(-\frac{dN}{dt}\right)_B$$

$$\Rightarrow n = 1$$

157. [A]

Saturation current is proportional to intensity while stopping potential increases with increase in frequency.

Hence, $f_a = f_b$ while $I_a < I_b$

158. [D]

First excited state is $n = 2$ and second excited state is $n = 3$.

Also,

$$E_n \propto \frac{1}{n^2}$$

$$\Rightarrow \frac{E_2}{E_3} = \frac{9}{4}$$

159. [B]

$$\frac{1}{\lambda_B} = R_H \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$\frac{\lambda_A}{\lambda_B} = \frac{5}{36} = \frac{5}{27}$$

$$\frac{\lambda_A}{\lambda_B} = \frac{5}{3} = \frac{5}{27}$$

160. [B]

Second excited state implies $n = 3$

$$\Rightarrow L_H = n \left(\frac{h}{2\pi} \right) = 3 \left(\frac{h}{2\pi} \right)$$

$$\Rightarrow L_U = n \left(\frac{h}{2\pi} \right) = 3 \left(\frac{h}{2\pi} \right)$$

$$|E_H| = \frac{Z^2}{n^2} (13.6) \text{ eV}$$

$$\Rightarrow |E_H| = \frac{1^2}{9} (13.6) \text{ eV}$$

$$\Rightarrow |E_U| = \frac{3^2}{3^2} (13.6) \text{ eV}$$

$$\Rightarrow |E_U| = 13.6 \text{ eV}$$

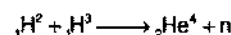
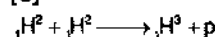
$$\Rightarrow |E_U| = 9|E_H|$$

$$\Rightarrow |E_U| > |E_H|$$

162. [D]

On the last day we have 100% decay i.e. on the ninth day 50% decay must be there or 50% must be left.

167. [C]



$$\Delta m = m({}_2\text{He}^4) + m(\text{p}) + m(\text{n}) - 3m({}_1\text{H}^2)$$

$$\Rightarrow \Delta m = [4.001 + 1.007 + 1.008 - 3(2.014)] \text{ u}$$

$$\Rightarrow \Delta m = -0.026 \text{ u}$$

$$\Rightarrow |\Delta E| = c^2 |\Delta m|$$

$$\Rightarrow \Delta E = (9 \times 10^{16}) (0.026 \times 1.67 \times 10^{-27})$$

$$\Rightarrow \Delta E = (931.5) (0.026) \text{ MeV}$$

$$\Rightarrow \Delta E = 3.87 \times 10^{-12} \text{ J}$$

As each reaction involves 3 deuterons, so total number of reactions involved in the process = $\frac{10^{40}}{3}$. If each reaction produces an energy ΔE , then

$$E_{\text{total}} = \frac{10^{40}}{3} \Delta E = 1.29 \times 10^{28} \text{ J}$$

$$E_{\text{total}} = Pt$$

Time of exhaustion of the star

$$t = \frac{1.29 \times 10^{28}}{10^{16}}$$

$$\Rightarrow t = 1.29 \times 10^{12} \text{ s}$$

168. [A]

$$I = \frac{q}{t} = \frac{ne}{t}$$

$$\Rightarrow 3.2 \times 10^{-3} = \left(\frac{n}{t} \right) (1.6 \times 10^{-19})$$

$$\Rightarrow \left(\frac{n}{t} \right) = 2 \times 10^{16}$$

169. [B]

$$\frac{1}{\lambda} = Z^2 R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For K_α line, $n_1 = 1$ and $n_2 = 2$

$$\Rightarrow \frac{1}{\lambda} = Z^2 R_H \left(\frac{3}{4} \right)$$

$$\Rightarrow Z = \sqrt{\frac{4}{3\lambda R_H}}$$

$$\Rightarrow Z = 39.9 \approx 40$$

170. [A]

Since in Rutherford experiment

$$N \propto \frac{1}{\sin^4 \left(\frac{\phi}{2} \right)}$$

$$\Rightarrow \frac{N_{90}}{N_{45}} = \frac{\sin^4(30)}{\sin^4(45)}$$

$$\Rightarrow \frac{N_{90}}{N_{80}} = \frac{16}{1}$$

$$\Rightarrow N_{80} = 100$$

171. [C]

Total Energy = $-(K.E.) = -13.6 \text{ eV}$

172. [C]

For quantum number n we have $2n^2$ electrons.

For $n = 1$ we have 2 elements

For $n = 2$ we have 8 elements

For $n = 3$ we have 18 elements

For $n = 4$ we have 32 elements

So, total number of elements is

$$2 + 8 + 18 + 32 = 60$$

173. [D]

$$R_n \propto n^2$$

$$\Rightarrow A_n \propto n^4$$

Hence

$$\frac{A_2}{A_1} = \frac{n_2^4}{n_1^4} = \frac{16}{1}$$

174. [A]

$$\frac{1}{\lambda} = R_H \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$\Rightarrow \lambda = \frac{36}{5R_H} = 6563 \text{ Å}$$

175. [B]

For an elastic collision to take place there must be no loss in the energy of electron. The hydrogen atom will absorb energy from the colliding electron only if it can go from ground state to first excited state i.e. from $n=1$ to $n=2$ state. For this Hydrogen atom must absorb energy

$$E_2 - E_1 = -3.4 - (-13.6) = 10.2 \text{ eV}.$$

So, if the electron possesses energy less than 10.2 eV it would never lose it and hence collision would be elastic.

177. [C]

$$L_1 = (1) \frac{h}{2\pi} \quad \dots(1)$$

In the first excited state of Li

$$L_2 = (2) \frac{h}{2\pi} \quad \dots(2)$$

$$\Rightarrow \frac{L_2}{L_1} = 2$$

178. [B]

$$N = N_0 e^{-\lambda t}$$

For $t = T_{av} = \frac{1}{\lambda}$, we have

$$N = \frac{N_0}{e}$$

180. [A]

$$\frac{m}{t} = 4 \times 10^9 \text{ kgs}^{-1}$$

$$E = mc^2$$

$$\Rightarrow \frac{E}{t} = \left(\frac{m}{t} \right) c^2$$

$$\Rightarrow \frac{E}{t} = 4 \times 10^9 \times 9 \times 10^{16}$$

$$\Rightarrow \frac{E}{t} = 3.6 \times 10^{26} \text{ Js}^{-1}$$

$$\Rightarrow \frac{E}{t} = 3.6 \times 10^{26} \text{ W}.$$

181. [B]

$$\rho = \frac{A(1.67 \times 10^{-27})}{\frac{4}{3}\pi R^3}$$

Since, $R = R_0 A^{\frac{1}{3}}$ where $R_0 = 1.1 \text{ fm}$

$$\Rightarrow \rho = \frac{A(1.67 \times 10^{-27})}{\frac{4}{3}\pi R_0^3 A}$$

$$\Rightarrow \rho = 2.9 \times 10^{17} \text{ kgm}^{-3}$$

183. [D]

$$\lambda_0 = \frac{hc}{E} = 1.55 \times 10^{-11} \text{ m}$$

$$\Rightarrow \lambda_0 = 0.155 \text{ Å}$$

is the minimum wavelength of continuous X-rays which carry energy equivalent to energy of incident electrons.

As this energy of incident radiation is more than that of K-shell electrons, the characteristic X-rays appear as peaks on the continuous spectrum.

184. [A]

λ_c decreases with increase in accelerating voltage in accordance with the expression given by

$$\lambda_c = \frac{hc}{eV}$$

Wavelength for K_α line is not affected as it is due to the electronic transition between $n=2$ and $n=1$ in the target element. Hence $(\lambda_K - \lambda_c)$ increase with increase in the accelerating voltage.

185. [C]

$$\frac{e^2}{2\hbar\epsilon_0 c} = \text{Fine structure constant } (\alpha) = \frac{1}{137}$$

$$\text{and } [\alpha] = M^0 L^0 T^0$$

187. [B]

The series in U-V region is Lyman series. Longest wavelength corresponds to minimum energy which occurs in transition from $n=2$ to $n=1$.

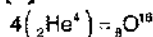
$$\Rightarrow 122 = \frac{1/R}{\left(\frac{1}{1^2} - \frac{1}{2^2} \right)} \quad \dots(1)$$

The smallest wavelength in the infrared region corresponds to maximum energy of Paschen series.

$$\Rightarrow \lambda = \frac{1/R}{\left(\frac{1}{3^2} - \frac{1}{\infty}\right)} \quad \dots(2)$$

Solving equations (1) and (2), we get
 $\lambda = 823.5 \text{ nm}$

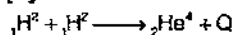
188. [C]



Mass defect, $\Delta m = \{4(4.0026) - 15.9994\} \text{ amu} = 0.011 \text{ amu}$

$\therefore$ Energy released per oxygen nuclei
 $= (0.011)(931.48) \text{ MeV} = 10.24 \text{ MeV}$

189. [C]



$$\Rightarrow \Delta m = m({}_2\text{He}^4) - 2m({}_1\text{H}^2)$$

$$\Rightarrow \Delta m = 4.0024 - 2(2.0141)$$

$$\Rightarrow \Delta m = -0.0258 \text{ u}$$

Since, $Q = c^2 \Delta m$

$$\Rightarrow Q = (0.0258)(931.5) \text{ MeV}$$

$$\Rightarrow Q = 24 \text{ MeV}$$

190. [A]

Nuclear density is constant hence, mass $\propto$ volume or $m \propto V$.

191. [C]

Let N_0 be the number of atoms of X at time $t = 0$. Then at $t = 4 \text{ h}$ (two half lives), we have

$$N_x = \frac{N_0}{4} \text{ and } N_y = \frac{3N_0}{4}$$

$$\Rightarrow \frac{N_x}{N_y} = \frac{1}{3}$$

and at $t = 6 \text{ h}$ (three half-lives), we have

$$N_x = \frac{N_0}{8} \text{ and } N_y = \frac{7N_0}{8}$$

$$\Rightarrow \frac{N_x}{N_y} = \frac{1}{7}$$

The given ratio $\frac{1}{4}$ lies between $\frac{1}{3}$ and $\frac{1}{7}$

Therefore, t lies between 4 h and 6 h

192. [C]

Since, $A_1 = \lambda N_0 e^{-\lambda t_1}$

$$\Rightarrow t_1 = \frac{1}{\lambda} \log_e \left(\frac{\lambda N_0}{A_1} \right) \quad \dots(1)$$

Since, $A_2 = (\lambda)(2N_0)e^{-\lambda t_2}$

$$\Rightarrow t_2 = \frac{1}{\lambda} \log_e \left(\frac{2\lambda N_0}{A_2} \right) \quad \dots(2)$$

$$\Rightarrow t_1 - t_2 = \frac{1}{\lambda} \log_e \left(\frac{A_2}{2A_1} \right)$$

$$\Rightarrow t_1 - t_2 = \frac{T}{\log_e(2)} \log_e \left(\frac{A_2}{2A_1} \right)$$

193. [A]

$$\lambda_n = \frac{(2\pi r)}{n}$$

$$\Rightarrow \lambda_n \propto n$$

$$\text{and } J_n \propto n$$

$$\Rightarrow \lambda_n \propto J_n$$

$$\left\{ \begin{array}{l} \because r \propto n, \\ \therefore J_n = \frac{nh}{2\pi} \end{array} \right.$$

194. [C]

After absorbing a photon of energy 12.1 eV electron jumps from ground state ($n=1$) to second excited state ($n=3$). Therefore change in angular momentum is

$$\Delta L = L_3 - L_1$$

$$\Rightarrow \Delta L = 3\left(\frac{h}{2\pi}\right) - \frac{h}{2\pi} = \frac{h}{\pi}$$

$$\Rightarrow \Delta L = \frac{6.6 \times 10^{-34}}{3.14} \text{ Js} = 2.11 \times 10^{-34} \text{ Js}$$

195. [B]

A and B can be isotopes if number of β -decays is two times the number of α -decays.

196. [D]

In 2s only 90% nuclei are left behind. So, in the next 2s 90% of 900 i.e., 810 nuclei will be left.

197. [C]

Let number of α -decays be x and number of β -decays be y . Then

$$92 - 2x + y = 85$$

$$\Rightarrow 2x - y = 7$$

$$\text{and } 238 - 4x = 210$$

$$\Rightarrow x = 7$$

Substituting this value in equation (1), we get

$$y = 7$$

198. [D]

$$\frac{E_{2n} - E_n}{E_{4n} - E_{2n}} = \frac{\frac{E_1}{4n^2} - \frac{E_1}{n^2}}{\frac{E_1}{16n^2} - \frac{E_1}{4n^2}} = 4 = \text{constant}$$

199. [C]

$$\text{Since, } \Delta E = K \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = \frac{hc}{\lambda}$$

In the given diagram, emission from $n=4 \rightarrow n=2$ would give photon of maximum energy (shortest wavelength) and $n=4 \rightarrow n=3$ transition would give photon of minimum energy (longest wavelength).

200. [C]

Energy of photon is

$$E = 13.6 \left(1 - \frac{1}{25} \right) \text{ eV}$$

$$\Rightarrow E = 13 \text{ eV}$$

Since, momentum is conserved, so we have

$$\frac{E}{c} = mv$$

$$\Rightarrow v = \frac{E}{mc}$$

$$\Rightarrow v = \frac{13 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27} \times 3 \times 10^8} = 4 \text{ ms}^{-1}$$

201. [D]

For incident electron, we have

$$\lambda_1 = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2\text{meV}}}$$

For shortest wavelength of X-rays, we have

$$\lambda_2 = \frac{hc}{eV}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{1}{c} \sqrt{\frac{eV}{2m}}$$

Substituting the values, we get

$$\frac{\lambda_1}{\lambda_2} = 1$$

Solutions

Multiple Correct Choice Type Questions

1. [B, C]

For a photon
 $pc = hv$

$$\Rightarrow p = \frac{hv}{c}$$

$$\Rightarrow p = 8.8 \times 10^{-29} \text{ kgms}^{-2}$$

$$\Rightarrow p = 1.65 \times 10^{-8} \text{ MeVc}^{-1}$$

2. [A, C]

$$\text{Since } L = \frac{nh}{2\pi} = \frac{3h}{2\pi}$$

$$\Rightarrow n = 3$$

$$\text{Since, } r_n = a_0 \frac{n^2}{Z}$$

$$\Rightarrow 4.5a_0 = a_0 \frac{n^2}{Z}$$

$$\Rightarrow Z = 2$$

Possible transitions are $3 \rightarrow 2$, $3 \rightarrow 1$ and $2 \rightarrow 1$

For $3 \rightarrow 2$, we have

$$\frac{1}{\lambda} = R(2)^2 \left(\frac{1}{4} - \frac{1}{9} \right) = 4R \left(\frac{9-4}{36} \right) = \frac{5R}{9}$$

$$\Rightarrow \lambda = \frac{9}{5R}$$

For $3 \rightarrow 1$, we have

$$\frac{1}{\lambda} = R(2)^2 \left(1 - \frac{1}{9} \right) = 4R \left(\frac{8}{9} \right) = \frac{32R}{9}$$

$$\Rightarrow \lambda = \frac{9}{32R}$$

For $2 \rightarrow 1$, we have

$$\frac{1}{\lambda} = R(2)^2 \left(1 - \frac{1}{4} \right) = 4R \left(\frac{3}{4} \right) = 3R$$

$$\Rightarrow \lambda = \frac{1}{3R}$$

3. [A, C]

$$r = \frac{mv}{qB} = \frac{\sqrt{2mE}}{qB}$$

$$\Rightarrow r \propto \frac{\sqrt{m}}{q}$$

$$\Rightarrow \text{Deflection} \propto \frac{q}{\sqrt{m}}$$



4. [A, D]

In nuclear fusion two or more lighter nuclei are combined to form a relatively heavy nucleus and thus, releasing the energy.

5. [B, C]

Let electron jump from $n_1 \rightarrow n_2$.

$$\text{So } \Delta L = (n_2 - n_1) \frac{h}{2\pi}$$

(According to Bohr's Quantisation Rule)

Since, n_1 and n_2 are integers ($n_1, n_2 > 1$), so $n_2 - n_1$ is also an integral value and hence ΔL must be an integral multiple of $\frac{h}{2\pi}$. This is satisfied by OPTION (C) and OPTION (B).

6. [B, C]

$$U \propto \frac{1}{r} \propto \frac{1}{n^2}$$

$$K \propto \frac{1}{r} \propto \frac{1}{n^2}$$

$$v \propto \frac{1}{n} \text{ and}$$

$$L \propto n$$

7. [A, C, D]

In Bohr's model of Hydrogen atom

$$R \propto n^2$$

$$V \propto \frac{1}{n}$$

$$T \propto n^3 \text{ and } E \propto \frac{1}{n^2}$$

$$\Rightarrow VR \propto n$$

$$\Rightarrow TE \propto n$$

$$\Rightarrow \frac{T}{R} \propto n$$

$$\Rightarrow \frac{V}{E} \propto n$$

8. [C, D]

For ${}^{20}_{10}\text{Ne}$ nucleus to exist

$$M_1 < 10m_p + 10m_n$$

$$M_1 < 10(m_p + m_n)$$

Further since ${}^{40}_{20}\text{Ca}$ has 20 more nucleons and thus it requires more energy to hold all of them together and hence $M_1 \neq 2M_2$ instead $M_1 > 2M_2$ because some additional mass defect must occur to provide an additional B.E. to ${}^{40}_{20}\text{Ca}$ nucleus.

9. [A, D]

$$\lambda_m (\text{in } \text{\AA}) = \frac{12375}{V (\text{in volts})}$$

With increase in V , λ_m will decrease. With decrease in λ_m energy of emitted photons will increase. And hence intensity will increase even if number of photons emitted per second are constant. Because intensity is basically energy per unit area per unit time.

10. [A, D]

$$T \propto n^3$$

$$\Rightarrow \frac{T_1}{T_2} = 8$$

satisfied by both (A) and (D)

11. [A, D]

Characteristic X-rays depend upon the atomic number Z , so

$$\lambda_1 = \lambda_2 = \lambda_3$$

$$\text{OR } \lambda_2 = \sqrt{\lambda_1 \lambda_3}$$

12. [A, B, D]

$$N = N_0 e^{-\lambda t}$$

N = Number of undecayed nuclei in the sample at time t .

Total number of undecayed nuclei equals $(N_0 - N)$

$$\Rightarrow (N_0 - N) = N_0 (1 - e^{-\lambda t})$$

which is growing exponentially with time.

$$\text{Activity } R = -\lambda N = \frac{dN}{dt}$$

13. [A, B]

$$\text{Since, } |F| = \frac{dU}{dr} = \frac{Ke^2}{r^4} \quad \dots(1)$$

$$\Rightarrow \frac{Ke^2}{r^4} = \frac{mv^2}{r} \quad \dots(2)$$

According to Bohr's Quantisation Rule, we have

$$mvr = \frac{nh}{2\pi} \quad \dots(3)$$

From (2) and (3), we get

$$r = \left(\frac{4\pi^2 e^2 K}{h^2} \right) \frac{m}{n^2} = K_1 \left(\frac{m}{n^2} \right) \quad \dots(4)$$

Since total energy E is half the potential energy, so we have

$$E = -\frac{Ke^2}{6r^3} = -\frac{Ke^2}{6 \left(\frac{K_1 m}{n^2} \right)^3} = -\frac{Ke^2 n^6}{6K_1^3 m^3}$$

So, total energy is $E \propto n^6$ and $E \propto m^{-3}$

14. [C, D]

In case of ${}^1_1\text{H}$, mass number and atomic number are equal and in case of ${}^2_1\text{H}$, mass number is greater than its atomic number.

15. [A, B, C]

Since Bohr's radius is given by

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m e^2 Z}$$

$$\Rightarrow r \propto \frac{1}{m}$$

$$\Rightarrow \frac{r}{r_0} = \frac{m_0}{m_\mu}$$

$$\Rightarrow r = \frac{r_0}{212} = \frac{0.53 \text{ \AA}}{212} = 250 \text{ fm}$$

Since, $F \propto m$

$$\Rightarrow E = E_0 \left(\frac{m_\mu}{m_0} \right)$$

$$\Rightarrow E = (13.6 \text{ eV})(212)$$

$$\Rightarrow E = 2883 \text{ eV}$$

Angular momentum is given as

$$L = mvr = \frac{nh}{2\pi}$$

For ground state $n = 1$, so we have

$$L = \frac{h}{2\pi}$$

16. [A, B, C]

$$\text{Since, } \frac{hc}{\lambda} = \Delta E$$

$$\Rightarrow \lambda = \frac{hc}{\Delta E}$$

$$\lambda_{\min} = \frac{hc}{\Delta E_{\max}} \quad \text{and} \quad \lambda_{\max} = \frac{hc}{\Delta E_{\min}}$$

Emission will take place when transition takes place from $n = 3$ to $n = 1$

$$\Rightarrow \lambda_{\min} = \frac{12400}{10(\text{eV})} = 1240 \text{ \AA} = 1.24 \times 10^{-7} \text{ m}$$

Absorption will take place when transition takes place from $n = 3$ to $n = 4$.

$$\Rightarrow \lambda_{\max} = \frac{12400}{2(\text{eV})} = 6200 \text{ \AA} = 6.2 \times 10^{-7} \text{ m}$$

Lowest frequency photon that can ionise the atom will have to remove the electron from $n = 3$ state, so we have $E = 15 \text{ eV}$

$$\Rightarrow 15 \times 1.6 \times 10^{-19} = 6.63 \times 10^{-34} \nu$$

$$\Rightarrow \nu = 3.62 \times 10^{15} \text{ Hz}$$

The total number of ways of de-excitation of atom to ground state is 2.

17. [A, B, C, D]

Since reduced mass is

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m}{2}$$

Now, energy is directly proportional to mass, so

$$E_0 \propto m$$

$$\Rightarrow E \propto \mu \propto \frac{m}{2}$$

$$\Rightarrow \frac{E}{E_0} = \frac{1}{2}$$

Since, $E_0 = R_0 hc$

So, Rydberg constant is

$$R_0 = \frac{E_0}{hc} \propto m$$

$$\Rightarrow R \propto \mu$$

$$\Rightarrow \frac{R}{R_0} = \frac{\mu}{m} = \frac{1}{2}$$

Radius of orbit is

$$r = \frac{n^2 \hbar^2 \epsilon_0}{\pi m e^2 Z}$$

So, for $n=1$, $Z=1$, we have

$$r_0 = \frac{\hbar^2 \epsilon_0}{\pi m e^2} \propto \frac{1}{m}$$

$$\Rightarrow r \propto \frac{1}{\mu}$$

$$\Rightarrow \frac{r}{r_0} = \frac{m}{\mu} = 2$$

Velocity of electron in first orbit is

$$v = \frac{e^2}{2\hbar \epsilon_0}$$

which is independent of mass, hence for positronium atom and hydrogen atom velocity of electron is same in both cases.

18. [A, B, C, D]

Under normal conditions, the total energy (E), kinetic energy (K) and potential energy (U) are

$$\left. \begin{array}{l} E_1 = -13.6 \text{ eV} \\ K_1 = 13.6 \text{ eV} \\ U_1 = -27.2 \text{ eV} \end{array} \right\} \text{ for ground state}$$

$$\left. \begin{array}{l} E_2 = -3.4 \text{ eV} \\ K_2 = 3.4 \text{ eV} \\ U_2 = -6.8 \text{ eV} \end{array} \right\} \text{ for first excited state}$$

If PE in ground state is taken to be zero, then KE remains unchanged, however new PE and TE are increased by 27.2 eV.

So, for ground state, we have

$$E_1 = (-13.6 + 27.2) \text{ eV} = 13.6 \text{ eV}$$

$$K_1 = 13.6 \text{ eV (same)}$$

$$U_1 = (-27.2 + 27.2) \text{ eV} = 0 \text{ eV}$$

For first excited state, we have

$$E_2 = (-3.4 + 27.2) \text{ eV} = 23.8 \text{ eV}$$

$$K_2 = 3.4 \text{ eV (same)}$$

$$U_2 = (-6.8 + 27.2) \text{ eV} = 20.4 \text{ eV}$$

19. [B, C, D]

Ground state is $n=1$

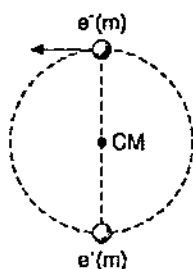
So, first excited state is $n=2$

$$KE = \frac{1}{4\pi\epsilon_0} \frac{e^2}{2r}$$

(for $Z=1$)

$$\Rightarrow KE = \frac{14.4 \times 10^{-10}}{2r} \text{ eV}$$

Since, $r = 0.53n^2 \text{ \AA}$



$$\Rightarrow (KE)_1 = \frac{14.4 \times 10^{-10}}{2 \times 0.53 \times 10^{-10}} \text{ eV} = 13.58 \text{ eV}$$

$$\Rightarrow (KE)_2 = \frac{14.4 \times 10^{-10}}{2 \times 0.53 \times 10^{-10} \times 4} \text{ eV} = 3.39 \text{ eV}$$

$\Rightarrow$ KE decreases by 10.2 eV

$$\text{Now } PE = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} = \frac{14.4 \times 10^{-10}}{r} \text{ eV}$$

$$\Rightarrow (PE)_1 = -\frac{14.4 \times 10^{-10}}{0.53 \times 10^{-10}} \text{ eV} = -27.1 \text{ eV}$$

$$\Rightarrow (PE)_2 = -\frac{14.4 \times 10^{-10}}{0.53 \times 10^{-10} \times 4} = -6.79 \text{ eV}$$

$\Rightarrow$ PE increases by 20.4 eV

$$\text{Now Angular momentum is } L = mvr = \frac{nh}{2\pi}$$

$$\Rightarrow L_2 - L_1 = \frac{h}{2\pi} \left(\frac{6.6 \times 10^{-34}}{6.28} \right) = 1.05 \times 10^{-34} \text{ Js}$$

21. [A, D]

$$r_n \propto \frac{n^2}{Z} \text{ and } |PE| = 2(KE)$$

22. [A, C]

$$\text{Since, } v = CRZ^2 \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$\Rightarrow v_{3 \rightarrow 2} = (3 \times 10^8)(1.1 \times 10^7)(1) \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$\Rightarrow v_{3 \rightarrow 2} = (3.3 \times 10^{15}) \left(\frac{5}{36} \right)$$

$$\Rightarrow v_{3 \rightarrow 2} = \frac{16.5}{36} \times 10^{15} = 4.6 \times 10^{14} \text{ Hz}$$

Similarly for transition from 3 to 1, we have

$$v_{3 \rightarrow 1} = (3 \times 10^8)(1.1 \times 10^7) \left(1 - \frac{1}{9} \right)$$

$$\Rightarrow v_{3 \rightarrow 1} = 2.9 \times 10^{15} \text{ Hz}$$

Please note that (D) is incorrect as it says "must be" because from second excited state i.e., $n=3$, photons are emitted for electron transitions from $3 \rightarrow 2$, $2 \rightarrow 1$, $3 \rightarrow 1$.

23. [A, B, C, D]

$$E \propto \frac{1}{r^2} \propto \frac{1}{n^2} \quad \dots(1)$$

$$P \propto v \propto \frac{1}{n} \quad \dots(2)$$

$$r \propto n^2 \quad \dots(3)$$

$$\Rightarrow PE \propto \frac{1}{n}$$

$$\Rightarrow \frac{P}{E} \propto n$$

$$\Rightarrow Er \propto n^2$$

$$\Rightarrow Pr \propto n$$

24. [A, B, D]

Actually, in the ground state, we have

$$KE = +13.6 \text{ eV}$$

$$PE = -27.2 \text{ eV and}$$

$$TE = -13.6 \text{ eV}$$

In the first excited state, we have

$$KE = +3.4 \text{ eV}, PE = -6.8 \text{ eV}, TE = -3.4 \text{ eV}$$

Now PE and TE both will be increased by

$$13.6 - (-27.2) = 40.8 \text{ eV}$$

KE remains unchanged being independent of reference.

25. [B, D]

$$\text{Since, } r \propto \frac{1}{m}$$

$$\text{and } E_n \propto m$$

For an orbit, $mvr = \frac{nh}{2\pi}$ is a constant, because $n = \text{constant}$

$$\Rightarrow v \propto m$$

26. [B, C]

In fusion reaction, two or more lighter nuclei combine to form a comparatively heavier nucleus.

27. [A, C, D]

From $n = 3$ to $n = 2$

$$\frac{1}{\lambda} \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\text{Since, } E = \frac{hc}{\lambda}$$

$$\Rightarrow E \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

For a photon of energy E , momentum p is

$$p = \frac{E}{v}, \text{ where } v \text{ is the speed of light.}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\frac{1}{1^2} - \frac{1}{2^2}}{\frac{1}{2^2} - \frac{1}{3^2}} = \frac{3}{5}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = a = \frac{27}{5}$$

$$\text{Further, } \frac{E_1}{E_2} = \frac{\frac{hc}{\lambda_1}}{\frac{hc}{\lambda_2}} = \frac{\lambda_2}{\lambda_1} = \frac{5}{27}$$

$$\Rightarrow c = \frac{5}{27} = \frac{1}{a}$$

$$\text{So, } b = \frac{p_1}{p_2} = \frac{E_1}{E_2} = c = \frac{5}{27}$$

28. [B, D]

$$\Delta E = 204 = 13.6 Z^2 \left(\frac{1}{1} - \frac{1}{4n^2} \right)$$

$$\Rightarrow 40.8 = 13.6 Z^2 \left(\frac{1}{n^2} - \frac{1}{4n^2} \right)$$

Satisfied for, $Z = 4$ and $n = 2$

29. [B, C, D]

$$v \propto \frac{1}{n}$$

$$E \propto \frac{1}{n^2}$$

$$r \propto n^2$$

$$\Rightarrow \frac{E}{v^2}, Er, v^2r \text{ are independent of } n.$$

$$\left\{ \because v = \left(\frac{e^2}{2\hbar\epsilon_0} \right) \frac{Z}{n} \right\}$$

30. [B, C]

Intensity of incident light may increase by decreasing distance of source without increasing frequency.

31. [A, D]

$$\text{Since, } E_n = -(13.6) \left(\frac{Z^2}{n^2} \right) \text{ eV}$$

$$E_1 = -13.6 \text{ eV for } Z = 1 \text{ and } n = 1$$

Similarly for $Z = n$, we have

$$E = E_1 = -13.6 \text{ eV}$$

i.e., for second orbit of He^+ and for third orbit of Li^{++} , we have energy equal to -13.6 eV

32. [A, D]

Let the transition in He^+ ion be from level $a \rightarrow b$ and that in H atom be from $p \rightarrow q$. Then $\lambda_1 = \lambda_2$ gives

$$4 \left(\frac{1}{b^2} - \frac{1}{a^2} \right) = \left(\frac{1}{q^2} - \frac{1}{p^2} \right)$$

$$\Rightarrow \frac{4}{b^2} - \frac{4}{a^2} = \frac{1}{q^2} - \frac{1}{p^2}$$

This equation will be satisfied when

$$a = 2p \text{ and } b = 2q$$

So, a and b are even integers greater than 1, satisfied by (A) and (D).

33. [C, D]

$$E = E_0 \frac{Z^2}{n^2}, r = a_0 \frac{n^2}{Z}, v = v_0 \frac{Z}{n}$$

where E_0 is energy of electron in ground state, a_0 is radius of ground state orbit and v_0 is velocity of electron in ground state.

34. [A, B, D]

$$r = a_0 \frac{Z^2}{n^2}$$

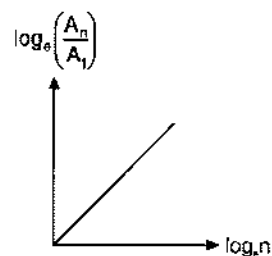
$$\Rightarrow A = \pi r^2 = \pi \frac{a_0^2 n^4}{Z^2}$$

$$\Rightarrow A_n = A_1 n^4$$

$$\Rightarrow \log_e A_n = \log_e A_1 + \log_e n^4$$

$$\Rightarrow \log_e A_n - \log_e A_1 = 4 \log_e n$$

$$\Rightarrow \log_e \left(\frac{A_n}{A_1} \right) = 4 \log_e n$$



So, the graph will be a straight line passing through origin having a slope of four units at all the points.

35. [B, C, D]

$$E_n = \frac{-13.6 Z^2}{n^2} \text{ eV}$$

$$\text{Since, } E_1 = -54.4 \text{ eV}$$

$$\Rightarrow \frac{-13.6 Z^2}{(1)^2} = -54.4$$

$$\Rightarrow Z^2 = 4$$

$$\Rightarrow Z = 2$$

Also, 40.8 eV is the difference between two energy levels $n = 2$ and $n = 1$.

Also the electron cannot fall from the ground state and hence cannot emit photon. So,

$$-E = K \text{ and } E = \frac{U}{2}$$

38. [A, C, D]

Since, $p = mv$ where $v \propto \frac{1}{n}$

$$\Rightarrow p \propto \frac{1}{n}$$

Since, $KE \propto \frac{1}{r}$ and $r \propto n^2$

$$\Rightarrow KE \propto \frac{1}{n^2}$$

$$\text{Further } L = \frac{nh}{2\pi}$$

$$\Rightarrow L \propto n$$

39. [B, C]

Any transition in the Balmer series must end up at $n=2$. This must be followed by the transition from $n=2$ to $n=1$ by emitting a photon of energy 10.2 eV. This 10.2 eV photon corresponds to a wavelength of about 122 nm, which belongs to the Lyman series.

40. [A, C]

From the relation,

$$eV = \frac{hc}{\lambda} - \phi \quad \text{or} \quad V = \left(\frac{hc}{e} \right) \left(\frac{1}{\lambda} \right) - \frac{\phi}{e}$$

This is equation of straight line.

$$\text{Slope is } \tan \theta = \frac{hc}{e}$$

$$\Rightarrow \phi_1 : \phi_2 : \phi_3 = \frac{hc}{\lambda_{01}} : \frac{hc}{\lambda_{02}} : \frac{hc}{\lambda_{03}} = \frac{1}{\lambda_{01}} : \frac{1}{\lambda_{02}} : \frac{1}{\lambda_{03}} = 1:2:4$$

$$\frac{1}{\lambda_{01}} = 0.001 \text{ nm}^{-1} \Rightarrow \lambda_{01} = 10000 \text{ \AA}$$

$$\frac{1}{\lambda_{02}} = 0.002 \text{ nm}^{-1} \Rightarrow \lambda_{02} = 5000 \text{ \AA}$$

$$\frac{1}{\lambda_{03}} = 0.004 \text{ nm}^{-1} \Rightarrow \lambda_{03} = 2500 \text{ \AA}$$

Violet colour has wavelength 4000 \AA.

So violet colour can eject photoelectrons from Metal -1 and Metal -2.

41. [B, C]

According to Ritz Combination Principle, we have

$$E_3 = E_1 + E_2$$

$$\Rightarrow h\nu_3 = h\nu_1 + h\nu_2$$

$$\Rightarrow \nu_3 = \nu_1 + \nu_2$$

$$\Rightarrow \frac{c}{\lambda_3} = \frac{c}{\lambda_1} + \frac{c}{\lambda_2} \quad \left\{ \because \nu = \frac{c}{\lambda} \right\}$$

$$\Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

42. [A, B]

For $d = \lambda$, there will be only one, central maxima.

For $\lambda < d < 2\lambda$, there will be three maximas on the screen corresponding to path difference, $\Delta x = 0$ and $\Delta x = \pm \lambda$.

43. [B, D]

In fusion two or more lighter nuclei combine to make a comparatively heavier nucleus.

In fission, a heavy nucleus breaks into two or more comparatively lighter nuclei.

Further, energy will be released in a nuclear process if total binding energy increases.

Solutions

Reasoning Based Questions

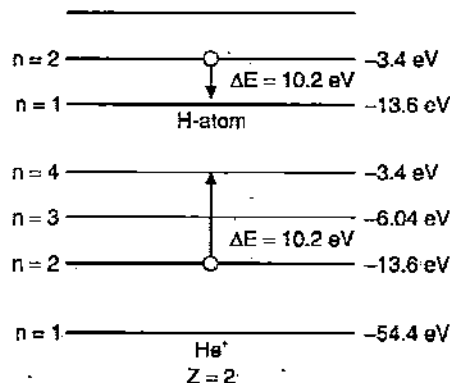
1. [B]
Cut-off wavelength depends on the accelerating voltage, not the characteristic wavelengths. Further, approximately 2% kinetic energy of the electrons is utilised in producing X-ray. Rest 98% is lost in heat.
2. [D]
According to classical electromagnetic theory, an accelerated charge continuously emits radiation. As electrons revolving in circular paths are constantly experiencing centripetal acceleration, hence, they will be losing their energy continuously and the orbital radius will go on decreasing and form spirals and finally the electron will fall on the nucleus.
4. [D]
 $\frac{1}{\lambda} \propto R \propto \mu$, where μ is the reduced mass of the system.
 $\Rightarrow \mu_D > \mu_H$
 $\Rightarrow R_D > R_H$
 $\Rightarrow \lambda_D < \lambda_H$
 In the centre of mass frame both the nucleus and the electron revolve about the common axis passing through the centre of mass.
5. [C]
Since $\lambda = \frac{hc}{\Delta E}$, so the wavelength will be inversely proportional to the energy difference between the levels. The energy difference is more when the transition takes place from $n \rightarrow \infty$ to $n=2$ than when the transition takes place from $n=2$ to $n=1$.
7. [A]
Both Statement-1 & Statement-2 are correct & Statement-2 is the correct explanation of Statement-1.
8. [A]
Both Statement-1 and Statement-2 are true and Statement-2 is the correct explanation of Statement-1.
9. [B]
Total energy is negative because electron is bound to the atom due to coulomb attraction and in the bound system energy is negative.
10. [B]
At outermost orbit total energy is zero and electron is free from the influence of the nucleus of the atom.
11. [C]
Speed of electron in H like atom is $v_{H \text{ like}} = \left(\frac{e^2}{2h\epsilon_0} \right) \frac{Z}{n}$
12. [C]
No. of lines in emission spectrum is $N = \frac{n(n-1)}{2}$
 $N = \frac{4(4-1)}{2} = 6$
 and this depends on number of energy levels available for transition.
13. [C]
Lyman series - its energy is in the ultra violet region
Balmer series - its energy is in visible region
17. [D]
After 200 days the number of un-decayed nuclei in the sample will be $\frac{1}{4}$ the initial number of un-decayed nuclei in the sample initially.
22. [D]
Statement-1 is false, the penetration power depends upon accelerating potential.
Statement-2 is true, increasing current increases the temperature of filament causing it to emit more electrons.
24. [A]
 $h\nu_1 = 0 - E_2$ and
 $h\nu_2 = 0 - E_3$
 $\Rightarrow h(\nu_1 - \nu_2) = E_3 - E_2$
 So, Statement-1 and Statement-2 both are correct and Statement-2 correctly explains Statement-1.

Solutions

Linked Comprehension Type Questions

1. [C]

Energy given by H-atoms in transition from $n=2$ to $n=1$ is equal to energy taken by He^+ atom in transition from $n=2$ to $n=4$.



2. [C]

Visible light lies in the range, $\lambda_1 = 4000 \text{ \AA}$ to $\lambda_2 = 7000 \text{ \AA}$. Energy of photons corresponding to these wavelengths (in eV) would be

$$E_1 = \frac{12375}{4000} = 3.09 \text{ eV}, E_2 = \frac{12375}{7000} = 1.77 \text{ eV}$$

From energy level diagram of He^+ atom we can see that in transition from $n=4$ to $n=3$ energy of photon released will lie between E_1 and E_2 .

$$\Delta E_{43} = -3.4 - (-6.04) = 2.64 \text{ eV}$$

Wavelength of photon corresponding to this energy,

$$\lambda = \frac{12375}{2.64} \text{ \AA} = 4687.5 \text{ \AA}$$

$$\lambda = 4.68 \times 10^{-7} \text{ m}$$

3. [A]

Kinetic energy $K \propto Z^2$

$$\Rightarrow \frac{K_H}{K_{\text{He}^+}} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

4. [A]

$$\text{eV} = 3 \times 10^{-15}$$

$$\Rightarrow V = \frac{3 \times 10^{-15}}{1.6 \times 10^{-19}} = 1.875 \times 10^4 \text{ V}$$

5. [C]

$$\Delta E = 3 \times 10^{-15} - 0.3 \times 10^{-15} \text{ J}$$

$$\Rightarrow \Delta E = 2.7 \times 10^{-15} \text{ J}$$

6. [D]

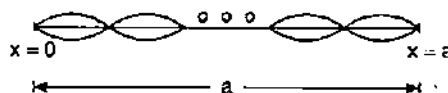
The difference of the energy will be gained by the emitted electron as kinetic energy, so

$$KE = 2.7 \times 10^{-15} - 3 \times 10^{-17}$$

$$\Rightarrow KE = 2.67 \times 10^{-15} \text{ J}$$

7. [A]

$$a = \frac{n\lambda}{2}$$



$$\Rightarrow \lambda = \frac{2a}{n} = \frac{h}{p} = \frac{h}{\sqrt{2Em}} \quad \dots(1)$$

$$\Rightarrow \sqrt{E} \propto \frac{1}{a}$$

$$\Rightarrow E \propto \frac{1}{a^2}$$

8. [B]

From equation (1), we get

$$E = \frac{n^2 h^2}{8a^2 m}$$

In ground state $n=1$

$$\Rightarrow E_1 = \frac{h^2}{8ma^2}$$

Substituting the values, we get

$$E_1 = 8 \text{ meV}$$

9. [D]

From equation (1), we get

$$p \propto n$$

$$\Rightarrow mv \propto n$$

$$\Rightarrow V \propto n$$

11. [A]

From conservation of mechanical energy, we have

$$U_i + K_i = U_f + K_f$$

$$\Rightarrow 0 + 2(1.5 \text{ kT}) = \frac{1}{4\pi\epsilon_0} \frac{(e)(e)}{d} + 0$$

Substituting the values, we get

$$T = 1.4 \times 10^9 \text{ K}$$

12. [B]

As given in the paragraph, a reactor is termed successful, if

$$n t_0 > 5 \times 10^{14} \text{ s cm}^{-3}$$

13. [C]

$$m({}_1^2\text{H}) + m({}_2^4\text{He}) = 2.014102 + 4.002603$$

$$\Rightarrow m({}_1^2\text{H}) + m({}_2^4\text{He}) = 6.016705 \text{ u}$$

$$\text{Since, } m({}_3^6\text{Li}) = 6.015123 \text{ u}$$

$$\Rightarrow m_1 + m_2 > M$$

So, (A) is incorrect.

$$m({}_1^1\text{H}) + m({}_{83}^{209}\text{Bi}) = 1.007825 + 208.980388$$

$$m({}_1^1\text{H}) + m({}_{83}^{209}\text{Bi}) = 209.988213 \text{ u}$$

$$\text{Since, } m({}_{84}^{210}\text{Po}) = 209.982876 \text{ u}$$

$$\Rightarrow m_1 + m_2 > M$$

So, B is incorrect

$$m({}_1^2\text{H}) + m({}_2^4\text{He}) = 2.014102 + 4.002603$$

$$\Rightarrow m({}_1^2\text{H}) + m({}_2^4\text{He}) = 6.016705 \text{ u}$$

$$\text{Since, } m({}_3^6\text{Li}) = 6.015123 \text{ u}$$

$$\Rightarrow (m_3 + m_4) > M'$$

So, (C) is correct and hence deuteron and alpha particle can go complete fusion.

$$m({}_{30}^{70}\text{Zn}) + m({}_{34}^{82}\text{Se}) = 69.925325 + 81.916709$$

$$\Rightarrow m({}_{30}^{70}\text{Zn}) + m({}_{34}^{82}\text{Se}) = 151.842034 \text{ u}$$

$$\text{Since, } m({}_{62}^{152}\text{Gd}) = 151.919803 \text{ u}$$

$$\Rightarrow m_3 + m_4 < M'$$

So, (D) is incorrect.

14. [A]

$${}_{84}^{213}\text{Po} = {}_{82}^{206}\text{Pb} + {}_2^4\text{He} + \Delta E$$

$$m({}_{82}^{206}\text{Pb}) = 205.974455 \text{ u}$$

$$m({}_2^4\text{He}) = 4.002603 \text{ u}$$

$$\Rightarrow m({}_{82}^{206}\text{Pb}) + m({}_2^4\text{He}) = 209.977058 \text{ u}$$

$$\text{Now, } \Delta m = 209.977058 - 209.982876$$

$$\Rightarrow \Delta m = 0.005818 \text{ u}$$

$$\Rightarrow Q = \Delta E = 0.005818 \times 931.5$$

$$\Rightarrow Q = 5.419467 \text{ MeV} = 5419.467 \text{ keV}$$

$$\Rightarrow Q = 5419.5 \text{ keV}$$

By Law of Conservation of Momentum, we have

$$0 = p_\alpha - p_{\text{lead}}$$

$$\Rightarrow p_\alpha = p_{\text{lead}}$$

$$\Rightarrow \sqrt{2m_\alpha E_\alpha} = \sqrt{2m_{\text{Pb}} E_{\text{Pb}}}$$

$$\Rightarrow 4E_\alpha = 206E_{\text{Pb}}$$

$$\Rightarrow E_\alpha = \frac{103}{2} E_{\text{Pb}}$$

$$\text{Now, since } E_\alpha = \left(\frac{m_{\text{Pb}}}{m_{\text{Pb}} + m_\alpha} \right) Q$$

$$\Rightarrow E_\alpha = \left(\frac{206}{206 + 4} \right) Q$$

$$\Rightarrow E_\alpha = \frac{103}{105} (5.422) = 5319 \text{ MeV}$$

15. [C]

$$\sqrt{\frac{m^3 \times C^2 \times \text{Nm}^{-2}}{\text{Kg} \times C^2}}$$

$$\{N = \text{kg ms}^{-2}\}$$

$$= \sqrt{\text{s}^{-2}}$$

16. [B]

$$\lambda = \frac{2\pi c}{\omega}$$

$$\Rightarrow \lambda = \frac{2\pi c}{\sqrt{\frac{Ne^2}{m\epsilon_0}}} \approx 600 \text{ nm}$$

17. [C]

Maximum energy of the antineutrino will be nearly $0.8 \times 10^6 \text{ eV}$.

18. [D]

Minimum kinetic energy of electron can be zero or greater than zero. But maximum kinetic energy will be less than $0.8 \times 10^6 \text{ eV}$.

19. [A]

$$\text{Since, } \frac{dN}{dt} = P - \lambda N$$

$$\Rightarrow \frac{dN}{P - \lambda N} = dt, \text{ where } \lambda = \frac{\log_e 2}{T}$$

$$\Rightarrow N = \frac{PT}{\log_e 2} \left(1 - e^{-\frac{\log_e 2}{T} t} \right)$$

20. [B]

$$\text{Since, } A = \lambda N = P \left(1 - e^{-\frac{\log_e 2}{T} t} \right)$$

$$\Rightarrow \text{Rate of energy release is } AE_0 \therefore PE_0 \left(1 - e^{-\frac{\log_e 2}{T} t} \right)$$

21. [A]

Energy released upto time t is

$$E_{\text{released}} = (Pt - N)E_0$$

22. [B]

$$Z_1 - (2)(2) + (3)(1) = Z_2 - (2)(1) + (5)(1) = Z_3$$

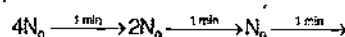
$$\Rightarrow Z_1 - Z_2 = 4$$

$$A_1 - (4)(2) = A_2 - (1)(4) = A_3$$

$$\Rightarrow A_1 - A_2 = 4$$

23. [C]

For A



$$\frac{N_0}{2} \xrightarrow{1 \text{ min}} \frac{N_0}{4}$$

For B

$$N_0 \xrightarrow{2 \text{ min}} \frac{N_0}{2} \xrightarrow{2 \text{ min}} \frac{N_0}{4}$$

After 4 minute, we have

$$N_A = N_B = \frac{N_0}{4}$$

$$\Rightarrow N_C = (4N_C + N_0) - \left(\frac{N_0}{4} + \frac{N_0}{4}\right) = \frac{9N_0}{2}$$

24. [B]

Given $R_A = R_B$

$$\Rightarrow \lambda_A N_A = \lambda_B N_B$$

$$\Rightarrow \left(\frac{\log_e 2}{T_A}\right) (4N_0 e^{-\lambda_A t_0}) = N_0 \left(\frac{\log_e 2}{T_B}\right) e^{-\lambda_B t_0}$$

$$\Rightarrow t_0 = 6 \text{ min}$$

25. [C]

$$P = (1\%) \times 40 \times 10^3 \times 10 \times 10^{-3}$$

$$\Rightarrow P = 4 \text{ W}$$

26. [B]

$$\text{Total power is } P = VI = 40 \times 10^3 \times 10 \times 10^{-3}$$

$$\Rightarrow P_{\text{total}} = 400 \text{ W}$$

So, heat produced per second is

$$P_{\text{total}} - P_{\text{emitted}} = 400 - 4 = 396 \text{ W}$$

27. [A]

$$\lambda_{\text{min}} = \frac{12400}{40 \times 1000} = 0.3 \text{ \AA}$$

28. [A]

If $x + X \longrightarrow Y + y$

For the above nuclear reaction, threshold energy is given as

$$E_{\text{th}} = -Q \left(1 + \frac{m_x}{m_X}\right)$$

The Q-value of reaction is given by

$$Q = (1.007825 + 3.016049 - 2 \times 2.014102) \times 931.5$$

$$\Rightarrow Q = -4.033 \text{ MeV}$$

When protons are incident on ${}^3\text{H}$, then

$$x = {}^1_1\text{H} \text{ and } X = {}^3_1\text{H}$$

$$K_{\text{th}} = 4.033 \text{ MeV} \left(1 + \frac{1.007825}{3.016049}\right)$$

$$\Rightarrow K_{\text{th}} = 5.381 \text{ MeV}$$

29. [C]

When ${}^3\text{H}$ is incident on protons

$$x = {}^3_1\text{H} \text{ and } X = {}^1_1\text{H}$$

$$\Rightarrow K_{\text{th}} = (4.033 \text{ MeV}) \left(1 + \frac{3.016049}{1.007825}\right)$$

$$\Rightarrow K_{\text{th}} = 16.10 \text{ MeV}$$

30. [C]

From above calculations we observe that less energy is required for a nuclear reaction, which a light particle is incident on a heavy target than if a heavy particle is incident on a light target.

31. [C]

$$F = -\frac{dU}{dr}$$

$$\Rightarrow F = \frac{4k}{r^5}$$

$$\Rightarrow \frac{mv^2}{r} = \frac{4k}{r^5}$$

$$\Rightarrow mv^2 = \frac{4k}{r^4} \quad \dots(1)$$

Now, according to Bohr's Quantisation rule, we have

$$v = \frac{nh}{2\pi mr} \quad \dots(2)$$

$$\Rightarrow m \left(\frac{n^2 h^2}{4\pi^2 m^2 r^2} \right) = \frac{4k}{r^4}$$

$$\Rightarrow \frac{n^2 h^2}{4\pi^2 m} = \frac{4k}{r^2}$$

$$\Rightarrow r^2 = \frac{16\pi^2 mk}{n^2 h^2}$$

$$\Rightarrow r = \frac{4\pi}{nh} \sqrt{mk} \quad \dots(3)$$

32. [D]

Substituting (3) in (2), we get

$$v = \frac{nh}{2\pi m \left(\frac{4\pi}{nh} \sqrt{mk} \right)}$$

$$\Rightarrow v = \frac{n^2 h^2}{8\pi^2 m \sqrt{mk}}$$

33. [B]

$$KE = \frac{1}{2} mv^2 = \frac{1}{2} m \left(\frac{n^4 h^4}{64\pi^4 m^2 k} \right)$$

$$\Rightarrow KE = \frac{n^4 h^4}{128\pi^4 m^2 k}$$

34. [D]

$$PE = U = -\frac{k}{r^4}$$

$$\Rightarrow U = -\frac{k}{\left(\frac{16\pi^2 mk}{n^2 h^2} \right)^2} = -\frac{kn^4 h^4}{256\pi^4 m^2 k^2}$$

$$\Rightarrow U = -\frac{n^4 h^4}{256\pi^4 m^2 k}$$

35. [C]

$$TE = KE + PE$$

$$\Rightarrow TE = \frac{n^4 h^4}{128\pi^4 m^2 k} - \frac{n^4 h^4}{256\pi^4 m^2 k}$$

$$\Rightarrow TE = \frac{n^4 h^4}{256\pi^4 m^2 k}$$

37. [C]
B.E. per nucleon for intermediate nucleus is more than lighter or heavier nuclei.

38. [D]
 $V_{Nu} = \frac{4}{3} \pi r_{Nu}^3$
Since, $r_{Nu} = r_0 A^{\frac{1}{3}}$
 $\Rightarrow V_{Nu} = \left(\frac{4}{3} \pi r_0^3 \right) A$
 $\Rightarrow V_{Nu} \propto A$

39. [B]
 $Q = (N_2 E_2 + N_3 E_3) - N_1 E_1$

42. [D]
Since, $1 \text{ kW} = 3.1 \times 10^{13} \frac{\text{fission}}{\text{sec}}$
 $\Rightarrow 1.6 \times 10^3 \text{ kW} = 1.6 \times 10^3 \times 3.1 \times 10^{13}$
 $= 5 \times 10^{16} \text{ fission per second}$

44. [B]
The reaction for the second stage is given by
 ${}_1^3\text{T} + {}_1^2\text{D} \longrightarrow {}_2^4\text{He} + {}_0^1\text{n} + E_2$
 $\Rightarrow \Delta m = (3.016049 + 2.014102) - (4.002603 + 1.008665)$
 $\Rightarrow \Delta m = 0.01888 \text{ u}$
 $\Rightarrow E_2 = 0.01888 \times 931 = 17.587 \text{ MeV}$

45. [D]
Total energy released $E = E_1 + E_2 = 4.033 + 17.587$
 $\Rightarrow E = 21.62 \text{ MeV}$
So, energy released / deuteron is $\frac{21.62}{3} = 7.207 \text{ MeV}$

46. [C]
 $\left(\frac{\% \text{ of Rest Mass of}}{\text{deuterium released}} \right) = \frac{7.207 \times 100}{2.014102 \times 931.4} = 0.384\%$

47. [A]
 $T_M = \frac{1}{\lambda}$ and $T_H = \frac{0.693}{\lambda}$
 $\Rightarrow T_M > T_H$

48. [C]
 $A = \frac{dN}{dt} = \lambda N$
 $\Rightarrow n = \lambda N = \frac{0.693}{T} N$
 $\Rightarrow T = \frac{0.693}{n}$

49. [C]
 $R_1 = R_0 e^{-\lambda t_1}$ and $R_2 = R_0 e^{-\lambda t_2}$
 $\Rightarrow \frac{R_1}{R_2} = e^{\lambda(t_2 - t_1)}$
 $\Rightarrow R_2 = R_1 e^{-\lambda(t_2 - t_1)}$

50. [A]
Since, $\frac{1}{\lambda} = RZ^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$
 $\Rightarrow \frac{hc}{\lambda} = (Rhc) Z^2 \left(\frac{1}{4} - \frac{1}{9} \right)$
where $Rhc = 13.6 \text{ eV}$
Also, given that $\frac{hc}{\lambda} = 47.2 \text{ eV}$
 $\Rightarrow 47.2 = 13.6 Z^2 \left(\frac{5}{36} \right)$
 $\Rightarrow Z^2 = 25$
 $\Rightarrow Z = 5$

51. [B]
Since, $r_n = (0.53) \frac{n^2}{Z} \text{ \AA}$
 $\Rightarrow r_1 = \frac{0.53}{5} \text{ \AA}$
 $\Rightarrow r_1 = 0.106 \text{ \AA}$

52. [B]
Since, $E_n = -\frac{E_1}{n^2}$
 $\Rightarrow E_1 = 4E_2 = 4(-144) = -576 \text{ eV}$
So, ionization energy = 576 eV

53. [D]
 $E_1 = -576 \text{ eV}$

54. [A]
Since, $E_n = -\frac{E_1 Z^2}{n^2}$
So, graph of E_n vs $\frac{1}{n^2}$ is a straight line passing through origin with negative slope. The distance between successive points is non-uniform.

55. [B]
 $\frac{mv^2}{r} = eVB$... (1)

$\Rightarrow \frac{v}{r} = \frac{eB}{m}$... (2)

According to Bohr Quantisation rule,

$mvr = \frac{nh}{2\pi}$
 $\Rightarrow vr = \frac{nh}{2\pi m}$... (3)

From (2) & (3), we get

$r = \sqrt{\frac{nh}{2\pi Be}}$... (4)

56. [C]
Kinetic energy, $K = \frac{1}{2} mv^2$
 $\Rightarrow K = \frac{1}{2} m \left(\frac{n^2 h^2}{m^2 r^2} \right)$
 $\Rightarrow K = \frac{1}{2} nh \left(\frac{Be}{m} \right)$... (5)

57. [A]

$$\text{Since, } T = \frac{2\pi r}{v}$$

$$\Rightarrow i = \frac{e}{T} = \frac{e^2 B}{2\pi m}$$

$$\text{Since, Area} = \pi r^2 = \pi \left(\frac{nh}{Be} \right)$$

$$\therefore r^2 = \frac{nh}{Be}$$

$$\Rightarrow M = iA = \left(\frac{e^2 B}{2\pi m} \right) \left(\frac{\pi nh}{Be} \right) = \frac{neh}{2m}$$

$$\text{Since, PE} = U = MB \sin(90^\circ)$$

$$\Rightarrow U = \frac{nhBe}{2m}$$

58. [B]

$$\phi = \pi r^2 B$$

$$\Rightarrow \phi = \pi \left(\frac{nh}{Be} \right) B = \frac{\pi nh}{e}$$

59. [C]

$$N_0 = \frac{mN_A}{M}$$

60. [C]

The number of undecayed nuclei after time t is $N = N_0 e^{-\lambda t}$

So, the number of decayed nuclei is

$$N' = N_0 - N = N_0 (1 - e^{-\lambda t})$$

61. [A]

$$\text{Activity} = A = -\frac{dN}{dt}$$

$$\Rightarrow A = A_0 e^{-\lambda t} = \lambda N_0 e^{-\lambda t}$$

62. [C]

$$\Delta E = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\Rightarrow 47.2 = 13.6 Z^2 \left(\frac{1}{4} - \frac{1}{9} \right)$$

$$\Rightarrow Z = 5$$

63. [C]

$$\text{Total energy TE} = 13.6(5)^2 = 340 \text{ eV}$$

$$\text{Since, TE} = \text{KE}$$

$$\Rightarrow \text{KE} = 340 \text{ V}$$

54. [A]

$$\frac{hc}{\lambda} = \frac{13.6(Z)^2}{n^2}$$

$$\Rightarrow \lambda = \frac{hc n^2}{(13.6 \text{ eV}) Z^2}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8 \times 9}{13.6 \times 25 \times 1.6 \times 10^{-19}}$$

$$\Rightarrow \lambda = 329 \text{ \AA}$$

65. [D]

$$\frac{dN_x}{dt} = \alpha - \lambda N_x$$

$$\Rightarrow N_x = \frac{1}{\lambda} [\alpha - (\alpha - \lambda N_0) e^{-\lambda t}] \quad \dots(1)$$

At $t = T_{1/2} = \frac{\log_e 2}{\lambda}$, we have

$$N_x = \frac{1}{\lambda} [\alpha - (\alpha - \lambda N_0) e^{-\lambda \left(\frac{\log_e 2}{\lambda} \right)}]$$

$$\Rightarrow N_x = \frac{1}{\lambda} [\alpha - (\alpha - \lambda N_0) e^{-\log_e 2}]$$

Since, $e^{\log_e x} = x$

$$\Rightarrow e^{-\log_e 2} = e^{\log_e \left(\frac{1}{2} \right)} = \frac{1}{2}$$

$$\Rightarrow N_x = \frac{1}{\lambda} \left[\alpha - \left(\frac{\alpha - \lambda N_0}{2} \right) \right]$$

$$\Rightarrow N_x = \frac{\alpha + \lambda N_0}{2\lambda}$$

66. [B]

Further, due to the decay of X to the stable nucleus Y , we have

$$\frac{dN_y}{dt} = \lambda N_x$$

Substituting value of N_x from equation (1) and solving, we get

$$\Rightarrow N_y = \alpha t + \left(\frac{\alpha - \lambda N_0}{\lambda} \right) e^{-\lambda t} - \left(\frac{\alpha - \lambda N_0}{\lambda} \right) \quad \dots(2)$$

67. [C]

Substituting the value of $T_{1/2} = \frac{\log_e(2)}{\lambda}$ in equation (2), we get

$$N_y = \frac{\alpha}{\lambda} \log_e(2) + \left(\frac{\alpha - \lambda N_0}{2\lambda} \right) - \left(\frac{\alpha - \lambda N_0}{\lambda} \right)$$

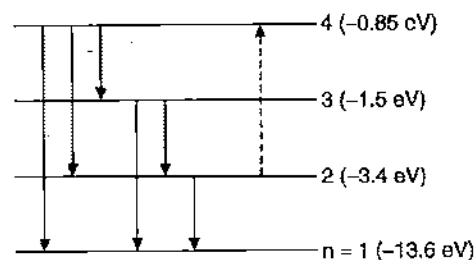
$$\Rightarrow N_y = \frac{\alpha}{\lambda} \log_e(2) - \frac{1}{2} \left(\frac{\alpha - \lambda N_0}{\lambda} \right)$$

68. [B]

$$E_2 - E_4 = -2.5 \text{ eV} < 2.7 \text{ eV}$$

So, the electron will be making a transition to $n = 4$.

$$\Rightarrow n = 2$$



69. [A]

$$E_n = -\frac{13.6}{n^2} Z^2$$

$$\Rightarrow E_B = E_2 = -3.4 Z^2$$

$$\text{and } E_C = E_4 = -0.85 Z^2$$

$$\text{Now, } E_C - E_B = 2.7 \text{ eV}$$

$$\Rightarrow Z = 1$$

The ionisation energy is $IE = 13.6 \text{ eV}$

70. [A]

$$E_{\max} = E_4 - E_1 = 12.75 \text{ eV}$$

$$\text{and } E_{\min} = E_4 - E_3 = 0.66 \text{ eV}$$

$$\Rightarrow -0.85 + 3.4 - 2.55 \text{ eV} = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{12400}{2.55} = 4862 \text{ \AA} = 486.2 \text{ nm}$$

72. [C]

$$E_4 - E_2 = \frac{hc}{\lambda}$$

$$\Rightarrow -\frac{13.6 \times (1)^2}{16} - \frac{(-13.6) \times (1)^2}{4} = \frac{hc}{\lambda}$$

73. [D]

$$\Delta E = \frac{hc}{\lambda} = \frac{12400}{1025} = 12.09$$

Solutions

Matrix Match Type Questions

1. A → (p, q)
B → (p, r)
C → (p, s)
D → (p, q, r)

2. A → (p, r)
B → (p, q, s)
C → (p)
D → (q)

3. A → (r)
B → (p)
C → (q)
D → (q)
E → (s)

4. A → (p, s)
B → (q, s)
C → (q, s)
D → (s)

$$E \propto \frac{1}{n^2}$$

$$\Rightarrow p \propto \frac{1}{n}$$

$$\Rightarrow r \propto n^2$$

$$\Rightarrow \frac{E}{p} \propto \frac{1}{n} \text{ and } Epr \propto \frac{1}{n}$$

$$\Rightarrow \frac{r}{p} \propto n^3 \text{ and } Er = \text{constant}$$

5. A → (p)
B → (p)
C → (q)
D → (s)

$$f \propto \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\Rightarrow f = k \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \text{ where } k \text{ is a constant}$$

For f_1 , we have $n_1 = 1$ and $n_2 \rightarrow \infty$

$$\Rightarrow f_1 = k$$

For f_2 , we have $n_1 = 1$ and $n_2 = 2$

$$\Rightarrow f_2 = \frac{3k}{4}$$

For f_3 , we have $n_1 = 2$ and $n_2 \rightarrow \infty$

$$\Rightarrow f_3 = \frac{k}{4}$$

$$\Rightarrow f_1 - f_2 = f_3$$

6. A → (p)
B → (q)
C → (r)
D → (q)

$$\gamma \rightarrow e^- + e^+$$

For pair production, we have

$$E = 2m_0c^2 = 2 \times 0.51 \text{ MeV} = 1.02 \text{ MeV}$$

Inverse photoelectric effect is X-ray production and energy involved in it is of order of tens of KeV

For de-excitation of Be^{14} from first excited state, we have

$$E = \frac{Z^2}{n^2} \times 13.6 = \frac{4^2}{2^2} \times 13.6$$

$$\Rightarrow E \approx 54.4 \text{ eV} = 55 \text{ eV}$$

For K_α X-ray photon of molybdenum, we have

$$E(K_\alpha) = h\nu = \frac{h \times 3}{4} cR(Z-1)^2$$

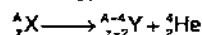
$$\Rightarrow E(K_\alpha) = \frac{3}{4} (hcR)(Z-1)^2 = \frac{3}{4} E_0 (Z-1)^2$$

$$\Rightarrow E(K_\alpha) = \frac{3}{4} \times 13.6 \times (42-1)^2 = 17.146 \times 10^3 \text{ eV}$$

$$\Rightarrow E(K_\alpha) \approx 17 \text{ KeV}$$

7. A → (p, r)
B → (q, r, s)
C → (q, r)
D → (p, r)

For α -decay, we have



$$Q = (KE)_Y + (KE)_\alpha$$

Since, $(KE)_Y \ll (KE)_\alpha$, we have

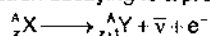
$$Q \approx (KE)_\alpha$$

So, kinetic energy of all the emitted α - particles is equal to Q i.e., mono-energetic α - particles are emitted.

Angular momentum is conserved in α - decay

For β^- Decay, we have

a neutron decaying to a proton, so



$$Q = (KE)_Y + (KE)_{\bar{\nu}} + E_{e^-}$$

Since, $(KE)_Y \ll (KE)_{\bar{\nu}}$,

$$\Rightarrow Q \approx (KE)_{\bar{\nu}} + E_{e^-}$$

Also note that,

(a) $E_{\bar{\nu}}$ is the energy of anti-neutrino and E_{e^-} takes on

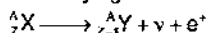
values from zero to maximum. Hence poly-energetic particles are emitted. i.e., poly-energetic antineutrinos are emitted.

(b) Due to emission of antineutrinos spin angular momentum is conserved.

(c) Neutron can decay in free space i.e., outside nucleus

For Positron emission, we have

a proton decaying to a neutron, so



Same explanation as above in case of Beta decay can be applied here too.

Proton cannot decay in free space i.e., outside nucleus because rest mass of proton is less than that of neutron.

For Electron capture, we have



$$Q = (KE)_Y + E_{\nu}$$

Since, $(KE)_Y \ll E_{\nu}$,

$$\Rightarrow Q \approx E_{\nu}$$

All the neutrinos emitted are of equal energies and their energies are approximately equal to Q . That is mono-energetic neutrinos are emitted. So, for electron capture we have that

(a) angular momentum is conserved.

(b) it cannot take place outside nucleus i.e., in free space.

8. $A \rightarrow (r)$

$B \rightarrow (p)$

$C \rightarrow (q)$

$D \rightarrow (q)$

9. $A \rightarrow (p, s)$

$B \rightarrow (p)$

$C \rightarrow (q, s)$

$D \rightarrow (p)$

10. $A \rightarrow (p, q, s, t)$

$B \rightarrow (r, s, t)$

$C \rightarrow (q, r, s, t)$

$D \rightarrow (r, t)$

11. $A \rightarrow (q, r, s, t)$

$B \rightarrow (q, r, s, t)$

$C \rightarrow (q, r, s, t)$

$D \rightarrow (p, q, r, s, t)$

12. $A \rightarrow (q)$

$B \rightarrow (p)$

$C \rightarrow (s)$

$D \rightarrow (r)$

In alpha decay, charge number decreases by 2 and mass number decreases by 4.

In β^- - decay, charge number decreases by 1 and mass number remains same.

In proton emission, charge no, decreases by 1 and mass no, decreases by 1.

Solutions

Integer Answer Type Questions

1. From our knowledge of Collision Theory, the fraction of kinetic energy lost by neutron is

$$\frac{\Delta K}{K_i} = \frac{4m_1 m_2}{(m_1 + m_2)^2} = \frac{4(1)(2)}{(1+2)^2} = \frac{8}{9}$$

where K_i is the initial kinetic energy of neutron and ΔK is the energy loss.

After first collision $\Delta K_1 = \frac{8}{9} K_0$

After second collision $\Delta K_2 = \frac{8}{9} K_1$ and so on

So, total energy loss is

$$\Delta K = \Delta K_1 + \Delta K_2 + \dots + \Delta K_n = \frac{8}{9} (K_0 + K_1 + \dots + K_{n-1})$$

where, $K_i = K_0 - \Delta K_i = \frac{K_0}{9}$

$$K_2 = \frac{K_1}{9} = \left(\frac{1}{9}\right)^2 K_0$$

$$\Rightarrow K_{n-1} = \left(\frac{1}{9}\right)^{n-1} K_0$$

$$\Rightarrow \Delta K = \frac{8}{9} K_0 \left[1 + \frac{1}{9} + \dots + \left(\frac{1}{9}\right)^{n-1} \right]$$

$$\Rightarrow \Delta K = \frac{8}{9} K_0 \left(\frac{1 - \frac{1}{9^n}}{1 - \frac{1}{9}} \right) = 1 - \frac{1}{9^n}$$

Since, $K_0 = 10^6$ eV and $\Delta K = (10^6 - 0.025)$ eV

$$\Rightarrow \frac{1}{9^n} = \frac{K_0 - \Delta K}{K_0} = \frac{0.025}{10^6}$$

$$\Rightarrow 9^n = 4 \times 10^7$$

Taking log both sides, we get
 $n = 8$

2. After removing the first electron it will become He^+ ion. The ionization energy of single electron in He^+ ion ($Z = 2$) is

$$IE = 13.6(Z^2) = 54.4 \text{ eV}$$

Therefore, total energy required to remove both the electrons is given by

$$E = (24.6 + 54.4) \text{ eV} = 79 \text{ eV}$$

$$3. E_n - E_1 = \frac{12375}{1085} + \frac{12375}{304}$$

$$\Rightarrow (13.6)(Z^2) \left(1 - \frac{1}{n^2} \right) = 52.1 \text{ eV}$$

$$\Rightarrow 1 - \frac{1}{n^2} = 0.96$$

$$\{\because Z = 2\}$$

$$\Rightarrow n \approx 5$$

$$4. \text{ Since, } \frac{1}{\lambda} = R(Z-1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\Rightarrow \frac{1}{0.76 \times 10^{-10}} = (1.09 \times 10^7)(Z-1)^2 \left(\frac{3}{4} \right)$$

$$\Rightarrow Z-1 \approx 40$$

$$\Rightarrow Z \approx 41$$

5. At time t , let say there are N atoms of ${}^7\text{Be}$ (radioactive). Then net rate of formation of ${}^7\text{Be}$ nuclei at this instant is,

$$\frac{dN}{dt} = \frac{10^{-4}}{1.6 \times 10^{-19} \times 1000} - \lambda N$$

$$\Rightarrow \frac{dN}{dt} = 6.25 \times 10^{11} - \lambda N$$

$$\Rightarrow \int_0^{N_0} \frac{dN}{6.25 \times 10^{11} - \lambda N} = \int_0^{3600} dt$$

where N_0 are the number of nuclei at $t = 1 \text{ hr}$ or 3600 second.

$$\Rightarrow -\frac{1}{\lambda} \log_e \left(\frac{6.25 \times 10^{11} - \lambda N_0}{6.25 \times 10^{11}} \right) = 3600$$

Since, $\lambda N_0 = \text{Activity of } {}^7\text{Be at } t = 1 \text{ hr} = 1.8 \times 10^8$ disintegrations/second

$$\Rightarrow -\frac{1}{\lambda} \log_e \left(\frac{6.25 \times 10^{11} - 1.8 \times 10^8}{6.25 \times 10^{11}} \right) = 3600$$

$$\Rightarrow \lambda = 8 \times 10^{-8} \text{ sec}^{-1}$$

Therefore, half life

$$t_{1/2} = \frac{0.693}{8 \times 10^{-8}} = 8.66 \times 10^6 \text{ sec} = 100.26 \text{ days}$$

$$\Rightarrow t_{1/2} \approx 100 \text{ days}$$

6. Since, $R = R_0 \left(\frac{1}{2}\right)^n$

where, n is the number of half lives.

Given, $R = \frac{R_0}{16}$

$\Rightarrow \frac{R_0}{16} = R_0 \left(\frac{1}{2}\right)^n$

$\Rightarrow n = 4$

Four half lives are equivalent to 8 s. Hence, 2 s is equal to one half life. So in one half life activity will remain half of 1600 Bq i.e., 800 Bq.

7. At time $t = t$

$N_1 = N_0 e^{-\lambda_1 t}$

and $N_2 = \frac{N_0 \lambda_1}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t})$

$\Rightarrow N_3 = N_0 - N_1 - N_2$

$\Rightarrow N_3 = N_0 \left[1 - e^{-\lambda_1 t} - \frac{\lambda_1}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}) \right]$

$\Rightarrow \frac{N_3}{N_0} = 1 - e^{-\lambda_1 t} - \frac{\lambda_1}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t})$

Since, $\lambda_1 = \frac{0.693}{30} = 0.0231 \text{ min}^{-1}$

$\lambda_2 = \frac{0.693}{45} = 0.0154 \text{ min}^{-1}$

and $t = 60$ minutes

$\Rightarrow \frac{N_3}{N_0} = 1 - e^{-0.0231 \times 60} - \frac{0.0231}{0.0154 - 0.0231} (e^{-0.0231 \times 60} - e^{-0.0154 \times 60})$

$\Rightarrow \frac{N_3}{N_0} = 1 - 0.25 + 3(0.25 - 0.4) = 0.31$

$\Rightarrow \% \text{age} = \frac{N_3}{N_0} \times 100\% = 31\%$

8. $V = \frac{hf}{e} - \frac{\phi}{e}$

Slope $= \frac{h}{e}$

It is same for both, so

Ratio = 1

9. Since, $N = N_0 e^{-\lambda t}$

$\Rightarrow \frac{N}{N_0} = e^{-\lambda t}$

$\Rightarrow \frac{N}{N_0} = e^{\frac{\log_e 2 \times 60}{1386}}$

$\Rightarrow \frac{N}{N_0} = e^{\frac{0.693 \times 60}{1386}}$

$\Rightarrow \frac{N}{N_0} = e^{-0.04}$

$\Rightarrow \frac{N}{N_0} = \left(\frac{1}{e}\right)^{0.04}$

Fraction of nuclei decayed $= 1 - \frac{N}{N_0} = 1 - \left(\frac{1}{e}\right)^{0.04} = 0.04 = 4\%$

10. $\frac{hc}{\lambda} - \phi = eV_0$

$\Rightarrow V_0 = \frac{ne}{4\pi\epsilon_0 r}$

$\Rightarrow \frac{1240}{200} \text{ eV} - 4.7 \text{ eV} = \left(\frac{xne}{4\pi\epsilon_0 r}\right) \text{ eV}$

$\Rightarrow 6.2 - 4.7 = \frac{9 \times 10^9 \times n \times 1.6 \times 10^{-19}}{10^{-2}}$

$\Rightarrow n = \frac{1.5 \times 10^{-2}}{9 \times 1.6 \times 10^{-30}} \approx 10^8$

11. $\frac{dN}{dt} = -\lambda N = -10^{10}$

$\Rightarrow N = 10^{10} \times \left(\frac{1}{\lambda}\right)$

$\Rightarrow m_{\text{total}} = N \times m_i$

$\Rightarrow m_{\text{total}} = 10^{10} \times 10^9 \times 10^{-25} \times 10^6 \text{ mg}$

$\Rightarrow m_{\text{total}} = 1 \text{ mg}$

12. $\frac{120e \times e}{4\pi\epsilon_0 (10 \times 10^{-15})^2} = \frac{p^2}{2m} = \frac{h^2}{\lambda^2 \times 2m}$

$\Rightarrow \lambda^2 = \frac{h^2}{2m} \times \frac{4\pi\epsilon_0 \times 10^{-14}}{120e^2}$

$\Rightarrow \lambda = \frac{h}{e} \times \sqrt{\frac{4\pi\epsilon_0 \times 10^{-14}}{2 \times 120 \times m}}$

$\Rightarrow \lambda = 4.2 \times 10^{-15} \times \sqrt{\frac{10^{-14} \times 3}{9 \times 10^9 \times 2 \times 120 \times 5 \times 10^{-27}}}$

$\Rightarrow \lambda = 7 \times 10^{-15} \text{ m}$

$\Rightarrow \lambda = 7 \text{ fm}$

13. Let m_1 and m_2 be the mass of α -particle and hydrogen atom.

By Law of Conservation of Momentum, we get

$m_1 u_i = (m_1 + m_2) v$

where u_i is the initial velocity of the incident α -particle and v is the final common velocity (or velocity of centre of mass) of the particles.

By Law of Conservation of Energy, we get

$\frac{1}{2} m_1 v_i^2 = \frac{1}{2} (m_1 + m_2) v^2 + \Delta E_0$

where ΔE_0 is the Ionization Energy

The loss in KE of the α -particles must be gained by the atom as ionisation energy, so we have

$\frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) v_i^2 = \Delta E_0$

$\Rightarrow K_i = \frac{1}{2} m_1 v_i^2 = \left(\frac{m_1 + m_2}{m_2} \right) \Delta E_0$

$\Rightarrow K_i = \left(1 + \frac{m_1}{m_2} \right) \Delta E_0$

$\Rightarrow K_i = \left(1 + \frac{4}{1} \right) (13.6) \text{ eV}$

$\Rightarrow K_i = 68 \text{ eV}$

14. $U = -1.7 \text{ eV}$

$$\Rightarrow E = \frac{U}{2} = -0.85 \text{ eV} = \frac{-13.6}{n^2}$$

$$\Rightarrow n = 4$$

Ejected photoelectron will have minimum de-Broglie wavelength corresponding to transition from $n = 4$ to $n = 1$, so we have

$$\Delta E = E_4 - E_1 = -0.85 - (-13.6) = 12.75 \text{ eV}$$

Using Einstein's Photo-Electric Equation, we get

$$K_{\max} = \Delta E - W = 10.45 \text{ eV}$$

$$\Rightarrow \lambda = \sqrt{\frac{150}{10.45}} \text{ \AA} \quad (\text{for an electron})$$

$$\Rightarrow \lambda = 3.8 \text{ \AA} \approx 4 \text{ \AA}$$

15. (a) Since, $N = \frac{n(n-1)}{2} = 6$

$$\Rightarrow n = 4$$

i.e., if $n_1 = n$, then $n_2 = n - 4$

$$\Rightarrow \frac{(13.6)Z^2}{n^2} = 0.544 \quad \dots(1)$$

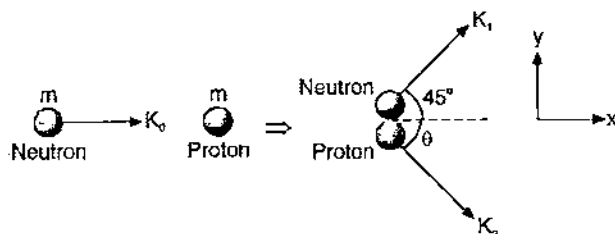
$$\Rightarrow \frac{(13.6)Z^2}{(n-4)^2} = 0.85 \quad \dots(2)$$

Solving equations (1) and (2), we get

$$Z = 4 \text{ and } n = 20$$

(b) $\lambda_{\min} = \frac{12375}{-0.544 - (0.85)} = 40441 \text{ \AA}$

16. Mass of neutron $\approx$ mass of proton $= m$



Applying Law of Conservation of Linear Momentum along y-direction, we get

$$\sqrt{2mK_1} \sin 45^\circ = \sqrt{2mK_2} \sin \theta \quad \dots(1)$$

along x-direction, we get

$$\sqrt{2mK_0} - \sqrt{2mK_1} \cos 45^\circ = \sqrt{2mK_2} \cos \theta \quad \dots(2)$$

Squaring and adding equations (1) and (2), we get

$$K_2 = K_1 + K_0 - \sqrt{2K_0K_1} \quad \dots(3)$$

By Law of Conservation of Energy, we get

$$K_2 = K_0 - K_1 \quad \dots(4)$$

Solving equations (3) and (4), we get

$$K_1 = \frac{K_0}{2}$$

$$\Rightarrow \Delta K = K_0 - K_1 = \frac{K_0}{2}$$

i.e., after each collision energy remains half. Therefore, after n collisions, we have

$$K_n = K_0 \left(\frac{1}{2}\right)^n$$

$$\Rightarrow 0.23 = (4.6 \times 10^5) \left(\frac{1}{2}\right)^n$$

$$\Rightarrow 2^n = \frac{4.6 \times 10^5}{0.23}$$

Taking log both sides, we get

$$n \approx 24$$

17. Given, $\lambda_1 = 4102 \text{ \AA}$, $\lambda_2 = 4861 \text{ \AA}$

For Balmer Series, we have

$$\Rightarrow \frac{1}{\lambda_1} = R \left(\frac{1}{2^2} - \frac{1}{n_1^2} \right)$$

Substituting values of λ_1 and R , we get

$$n_1 = 6$$

Similarly, for Balmer Series, we have

$$\frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{n_2^2} \right)$$

Substituting values of λ_2 and R , we get

$$n_2 = 4$$

Now difference of wave numbers of above two lines is

$$\frac{1}{\lambda_1} - \frac{1}{\lambda_2} = \frac{1}{\lambda} = R$$

$$\Rightarrow \frac{1}{\lambda_1} - \frac{1}{\lambda_2} = \frac{1}{\lambda} = R \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

Transition $n_2 \rightarrow n_1$ or $6 \rightarrow 4$ corresponds to second line of Brackett series, whose wavelength is given by

$$\frac{1}{\lambda} = R \left(\frac{1}{4^2} - \frac{1}{6^2} \right)$$

Substituting the values, we get

$$\lambda = 26206 \text{ \AA}$$

18. From the given conditions

$$E_n - E_2 = (10.2 + 17) \text{ eV} = 27.2 \text{ eV} \quad \dots(1)$$

$$\text{and } E_n - E_3 = (4.25 + 5.95) \text{ eV} = 10.2 \text{ eV} \quad \dots(2)$$

Equation (1) - (2) gives

$$E_3 - E_2 = 17 \text{ eV}$$

$$\Rightarrow Z^2 (13.6) \left(\frac{1}{4} - \frac{1}{9} \right) = 17$$

$$\Rightarrow Z^2 (13.6) \left(\frac{5}{36} \right) = 17$$

$$\Rightarrow Z^2 = 9$$

$$\Rightarrow Z = 3$$

From equation (1), we get

$$Z^2 (13.6) \left(\frac{1}{4} - \frac{1}{n^2} \right) = 27.2$$

$$\Rightarrow (3)^2 (13.6) \left(\frac{1}{4} - \frac{1}{n^2} \right) = 27.2$$

$$\Rightarrow \frac{1}{4} - \frac{1}{n^2} = 0.222$$

$$\Rightarrow \frac{1}{n^2} = 0.0278$$

$$\Rightarrow n^2 = 36$$

$$\Rightarrow n = 6$$

19. (a) 1 rydberg $= 2.2 \times 10^{-18} \text{ J} = Rhc$

Ionisation energy is given as 4 rydberg, so

$$IE = 8.8 \times 10^{-18} \text{ J} = \frac{8.8 \times 10^{-18}}{1.6 \times 10^{-19}} = 55 \text{ eV}$$

⇒ Energy in first orbit is

$$E_1 = -55 \text{ eV}$$

Energy of radiation emitted when electron jumps from first excited state ($n=2$) to ground state ($n=1$) is

$$E_{21} = \frac{E_1}{(2)^2} - E_1 = -\frac{3}{4}E_1 = 41.25 \text{ eV}$$

So, wavelength of photon emitted in this transition is

$$\lambda = \frac{12375}{41.25} = 300 \text{ Å}$$

(b) Let Z be the atomic number of given element. Then

$$E_1 = (-13.6)(Z^2)$$

$$\Rightarrow -55 = (-13.6)(Z^2)$$

$$\Rightarrow Z = 2$$

(c) Since, $r = \frac{a_0}{Z}$, so the radius of first orbit of this atom is

$$r_1 = \frac{a_0}{2}$$

$$\Rightarrow Z = 2$$

$$20. E_n = -\frac{(13.6)Z^2}{n^2} \text{ eV}$$

Substituting $z=3$, we get

$$E_n = -\frac{122.4}{n^2} \text{ eV}$$

$$\Rightarrow E_1 = -\frac{122.4}{(1)^2} = -122.4 \text{ eV}$$

$$\text{and } E_3 = -\frac{122.4}{(3)^2} = -13.6 \text{ eV}$$

$$\Rightarrow \Delta E = E_3 - E_1 = 108.8 \text{ eV}$$

The corresponding wavelength is

$$\lambda = \frac{12375}{\Delta E(\text{in eV})} \text{ Å} = \frac{12375}{108.8} \text{ Å} = 113.74 \text{ Å}$$

$$\Rightarrow \lambda \approx 114 \text{ Å}$$

21. Total number of electrons striking the target per second

$$n = \frac{10 \times 10^{-3}}{1.6 \times 10^{-16}} = 6.25 \times 10^{16}$$

Kinetic energy of an electron is

$$K = 40 \times 10^3 \text{ eV} = 40 \times 10^3 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-15} \text{ J}$$

∴ Total energy of electrons striking the target (per second) is

$$E = 6.25 \times 10^{16} \times 6.4 \times 10^{-15} = 400 \text{ J}$$

(a) Total power emitted as X-rays is 1% of 400 W. So

$$P = 4 \text{ W}$$

(b) Heat produced per second is

$$H = (400 - 4) \text{ Js}^{-1} = 396 \text{ Js}^{-1}$$

22. When an substance decays by α and β emission simultaneously, the average disintegration constant λ_{av} is given by

$$\lambda_{av} = \lambda_\alpha + \lambda_\beta$$

where λ_α = disintegration constant for α -emission only

λ_β = disintegration constant for β -emission only

Mean life is given by $T_m = \frac{1}{\lambda_{av}}$

$$\Rightarrow \lambda_{av} = \lambda_\alpha + \lambda_\beta$$

$$\Rightarrow \frac{1}{T_m} = \frac{1}{T_\alpha} + \frac{1}{T_\beta} = \frac{1}{1200} + \frac{1}{600}$$

$$\Rightarrow T_m = 400 \text{ yr}$$

$$\text{Since, } t = \frac{1}{\lambda} \log_e \left(\frac{N_0}{N_t} \right) = T_m \log_e \left(\frac{N_0}{N_t} \right)$$

$$\Rightarrow t = 400 \log_e \left(\frac{100}{25} \right) = 400 \log_e (4)$$

$$\Rightarrow t = 400 \times 1.4 = 560 \text{ year}$$